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Functional Genetic Variants in DC-SIGNR Are Associated with Mother-to-Child Transmission of HIV-1

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Abstract: **BACKGROUND:** Mother-to-child transmission (MTCT) is the main cause of HIV-1 infection in children worldwide. Given that the C-type lectin receptor, dendritic cell-specific ICAM-grabbing non-integrin-related (DC-SIGNR, also known as CD209L or liver/lymph node-specific ICAM-grabbing non-integrin (L-SIGN)), can interact with pathogens including HIV-1 and is expressed at the maternal-fetal interface, we hypothesized that it could influence MTCT of HIV-1. **METHODS AND FINDINGS:** To investigate the potential role of DC-SIGNR in MTCT of HIV-1, we carried out a genetic association study of DC-SIGNR in a well-characterized cohort of 197 HIV-infected mothers and their infants recruited in Harare, Zimbabwe. Infants harbouring two copies of DC-SIGNR H1 and/or H3 haplotypes (H1-H1, H1-H3, H3-H3) had a 3.6-fold increased risk of in utero (IU) ($P = 0.013$) HIV-1 infection and a 5.7-fold increased risk of intrapartum (IP) ($P = 0.025$) HIV-1 infection after adjusting for a number of maternal factors. The implicated H1 and H3 haplotypes share two single nucleotide polymorphisms (SNPs) in promoter region (p-198A) and intron 2 (int2-180A) that were associated with increased risk of both IU ($P = 0.045$ and $P = 0.003$, respectively) and IP ($P = 0.025$, for int2-180A) HIV-1 infection. The promoter variant reduced transcriptional activity in vitro. In homozygous H1 infants bearing both the p-198A and int2-180A mutations, we observed a 4-fold decrease in the level of placental DC-SIGNR transcripts, disproportionately affecting the expression of membrane-bound isoforms compared to infant noncarriers ($P = 0.011$). **CONCLUSION:** These results suggest that DC-SIGNR plays a crucial role in MTCT of HIV-1 and that impaired placental DC-SIGNR expression increases risk of transmission.

Text: Without specific interventions, the rate of HIV-1 mother-to-child transmission (MTCT) is approximately 15-45% [1]. UNAIDS estimates that last year alone, more than 400,000 children were infected worldwide, mostly through MTCT and 90% of them lived in sub-Saharan Africa. In the most heavily affected countries, such as Zimbabwe, HIV-1 is responsible for one third of all deaths among children under the age of five. MTCT of HIV-1 can occur during pregnancy (in utero, IU), delivery (intrapartum, IP) or breastfeeding (postpartum, PP). High maternal viral load, low CD4 cells count, vaginal delivery, low gestational age have all been identified as independent factors associated with MTCT of HIV-1 [1]. Although antiretrovirals can reduce MTCT to 2%, limited access to timely diagnostics and drugs in many developing world countries limits the potential impact of this strategy. A better understanding of the mechanisms acting at the maternal-fetal interface is crucial for the design of alternative interventions to antiretroviral therapy for transmission prevention.

Dendritic cell-specific ICAM-grabbing non-integrin-related (DC-SIGNR, also known as CD209L or liver/lymph node-specific ICAM-grabbing non-integrin (L-SIGN)) can interact with a plethora of pathogens including HIV-1 and is expressed in placental capillary endothelial cells [2]. DC-SIGNR is organized in three distinct domains, an N-terminal cytoplasmic tail, a repeat region containing seven repeat of 23 amino acids and a C-terminal domain implicated in pathogen binding. Alternative splicing of DC-SIGNR gene leads to the production of a highly diversified isoforms repertoire which includes membrane-bound and soluble isoforms [3]. It has been proposed that interaction between DC-SIGNR and HIV-1 might enhance viral transfer to other susceptible cell types [2] but DC-SIGNR can also internalize and mediate proteasome-dependant degradation of viruses [4] that may differently affect the outcome of infection.

Given the presence of DC-SIGNR at the maternal-fetal interface and its interaction with HIV-1, we hypothesized that it could influence MTCT of HIV-1. To investigate the potential role of DC-SIGNR in MTCT of HIV-1, we carried out a genetic association study of DC-SIGNR in a well-characterized cohort of HIV-infected mothers and their infants recruited in Zimbabwe, and identified specific DC-SIGNR variants associated with increased risks of HIV transmission. We further characterized the functional impact of these genetic variants on DC-SIGNR expression and show that they affect both the level and type of DC-SIGNR transcripts produced in the placenta.

Samples consisted of stored DNA extracts obtained from 197 mother-child pairs co-enrolled immediately postpartum in the ZVITAMBO Vitamin A supplementation trial (Harare, Zimbabwe) and followed at 6 weeks, and 3-monthly intervals up to 24 months. The ZVITAMBO project was a randomized placebocontrolled clinical trial that enrolled 14,110 mother-child pairs, between November 1997 and January 2000, with the main objective of investigating the impact of immediate postpartum vitamin A supplementation on MTCT of HIV-1. The samples used in the present study were from mother-child pairs randomly assigned to the placebo group of the ZVITAMBO project. Antiretroviral prophylaxis for HIV-1-positive antenatal women was not available in the Harare public-sector during ZVITAMBO patient recruitment. The samples were consecutively drawn from two groups: 97 HIV-1-positive mother/HIV-1-positive child pairs and 100 HIV-1-positive mother/HIV-negative child pairs. Mother's serological status was determined by ELISA and confirmed by Western Blot. Infants were considered to be infected if they were HIV-1 seropositive at 18 months or older and had two or more positive HIV-1-DNA polymerase chain reaction (PCR) results at earlier ages. 100 infants were considered to be uninfected as they were ELISA negative at 18 months or older and had two DNA PCR negative results from samples collected at a younger age. Of the 97 HIV-1-infected infants, 57 were infected IU, 11 were infected IP, and 17 were infected PP as determined by PCR analyses of blood samples collected at birth, 6 weeks, 3 and 6 months of age and according to the following definitions adapted from Bryson and colleagues [5] . Briefly, infants who were DNA PCR positive at birth were infected IU. Infants with negative PCR results from sample obtained at birth but who become positive by 6 weeks of age were infected IP. Infants with negative PCR results at birth and 6 weeks of age but who subsequently became DNA PCR positive were considered to be infected during the PP period. In the analysis comparing the 3 different modes of MTCT, 12 HIV-1-infected infants were excluded because the PCR results were not available at 6 weeks of age. Full methods for recruitment, baseline characteristics collection, laboratory procedures have been described elsewhere [6] .

The nucleotide sequence variation of the entire promoter, coding and part of 3'-UTR regions of DC-SIGNR gene in the study population was determined previously [7] . Haplotype reconstruction was performed using Bayesian statistical method implemented in PHASE [8] , version 2.1.1, using single nucleotide polymorphism (SNP) with a minimum allele frequency (MAF) of 2%. We applied the algorithm five times, using different randomly generated seeds, and consistent results were obtained across runs (Figure 1). Fifteen haplotype-tagged SNPs (htSNPs) were identified by the HaploBlockFinder software [9] with

a MAF 5%. These htSNPs were genotyped in the 197 infants by direct PCR sequencing analysis as we have described previously [7]. The DC-SIGNR exon 4 repeat region genotype was determined by PCR amplification followed by migration in 1.5% agarose gels [10]. DNA sequences in the promoter region were analysed with the TESS interface (<http://www.cbil.upenn.edu/tess>) for putative transcription factors binding sites using the TRANSFAC database.

Luciferase reporter assays using pGL2-Basic vector were performed in order to investigate the functional effect of mutations on DC-SIGNR promoter activity. Genomic DNA from subjects homozygous for the promoter variants and WT was amplified from nucleotide position 2715 to 21 and cloned between the BglII and HindIII multiple cloning sites in the pGL2-Basic vector which harbours a reporter firefly luciferase gene downstream (Invitrogen Canada inc, Burlington, Canada). All recombinant clones were verified by DNA sequencing. The firefly luciferase test reporter vector was co-transfected at a ratio of 10:1 with the constitutive expressor of Renilla luciferase, phRL-CMV (Promega, Madison, WI, USA). We cultured HeLa cells in 6 wells plates (2610 5 cells) and transfected them the following day using lipofectamine (Invitrogen) according to the manufacturer. Cells were lysed and luciferase assays were performed using 20 µg of protein extract according to the manufacturer (Promega) at 44 h post-transfection. Firefly luciferase activity was normalized to Renilla luciferase activity. 0 µg, 0.5 µg or 1 µg CMV-Tat vector was transfected with LTR-Luc as a positive control in these experiments. We carried out luciferase assays in triplicate in three independent experiments. Results are expressed as mean ± standard error of the mean (S.E.M).

First-term placental tissues were obtained from abortions following voluntary interruption of pregnancy at CHUM Hôpital Saint-Luc (Montreal, Canada). Tissues from 3 H1 (associated with MTCT of HIV-1) and 3 H15 (wild-type) homozygous haplotypes were used to analyse possible differences in isoform expression. Total placental RNAs were extracted by MasterPure DNA and RNA Extraction Kit (Epicentre Biotechnologies, Madison, WI, USA) according to the manufacturer. Fragments corresponding to the DC-SIGNR coding region were reverse transcribed (RT) and then amplified by nested PCR with the following primers; RT primers RR, first PCR RF and RR and second PCR RcF and RcR according to Liu and colleagues [11]. 1 µg of total RNA was reverse transcribed with Expand RT (Roche Applied Science, Indianapolis, IN, USA) according to the manufacturer and were PCR-amplified with DNA Platinum Taq Polymerase (Invitrogen). Major PCR products from the second PCR reaction were gel extracted with the Qiagen Gel Extraction Kit (Qiagen Canada

inc, Mississauga, ON, Canada) and cloned using the TOPO TA Cloning Kit for sequencing (Invitrogen). For each placenta, 15 different clones were randomly selected and amplified with M13 primers and sequenced with ABI PRISM 3100 capillary automated sequencer (Applied Biosystems, Foster City, CA, USA). Sequences were analysed and aligned with GeneBank reference sequence NM_014257 using Lasergene software (DNA Stars, Madison, WI, USA).

Quantitative expression of DC-SIGNR isoforms 1,5 mg of placental RNA was reverse transcribed using 2.5 mM of Oligo dT 20 and Expand RT in 20 ml volume according to the manufacturer (Roche Applied Science). 15 ng of total cDNA in a final volume of 20 ml was used to perform quantitative real-time PCR using Universal Express SYBR GreenER qPCR Supermix (Invitrogen) on a Rotor Gene Realtime Rotary Analyser (Corbett Life Science, Sydney, Australia). Samples from 2 subjects in each group were used because RNA quality of others was not suitable for a qRT-PCR analysis. Amplification of all DC-SIGNR isoforms was performed using an exon 5 specific primer pair (Table S1). Membrane-bound isoforms were amplified using primers specific for exon 3, corresponding to the common trans-membrane domain of DC-SIGNR. Primers were targeted to the exon-exon junction and RNA extracts were treated with DNase (Fermantas International inc, Burlington, ON, Canada) to avoid amplification of contaminant DNA. Standard curves (50-500 000 copies per reaction) were generated using serial dilution of a full-length DC-SIGNR or commercial GAPDH (Invitrogen) plasmid DNA. All qPCR reactions had efficiencies ranging from 99% to 100%, even in the presence of 20 ng of non-specific nucleic acids, and therefore could be compared. The copy number of unknown samples was estimated by placing the measured PCR cycle number (crossing threshold) on the standard curve. To correct for differences in both RNA quality and quantity between samples, the expression levels of transcripts were normalised to the reference GAPDH gene transcripts. GAPDH primer sequences were kindly provided by A. Mes-Masson at the CHUM. The results are presented as target gene copy number per 10 5 copies of GAPDH. The ratio of membrane-bound isoforms was calculated as E3/E5. Soluble isoforms were calculated by subtracting membrane-bound from total isoforms. We carried out qPCR assays in triplicate in three independent experiments. Results are expressed as mean $\pm$ S.E.M.

Statistical analysis was performed using the GraphPad PRISM 5.0 for Windows (GraphPad Software inc, San Diego, CA, USA). Differences in baseline characteristics and genotypic frequencies of haplotypes or htSNPs were compared between groups using the χ^2 analysis or Fisher's exact test. Logistic regression analysis was used to estimate odds ratios (OR) for

each genotype and baseline risk factors. Multiple logistic regression was used to define independent predictors identified as significant in the crude analysis. ORs and 95% confidence interval were calculated with the exact method. Comparisons of continuous variables between groups were assessed with the unpaired two-tailed Student's t test when variables were normally distributed and with the Mann-Whitney U test when otherwise. Differences were considered significant at $P \leq 0.05$.

Written informed consent was obtained from all mothers who participated in the study and the ZVITAMBO trial and the investigation reported in this paper were approved by The

We carried out an association study of DC-SIGNR polymorphism in 197 infants born to untreated HIV-1-infected mothers recruited in Harare, Zimbabwe. Among them, 97 infants were HIV-1-infected and 100 infants remained uninfected. Of the 97 HIV-1-infected infants, 57 were infected IU, 11 were infected IP, and 17 were infected PP. Timing of infection was not determined for 12 HIV-1-infected infants. Baseline characteristics of mothers and infants are presented in Table 1. Maternal age and CD4 cell count, child sex, mode of delivery, duration of membrane rupture and gestational age were similar among all groups. However, maternal viral load $\geq 29\,000$ copies/ml was associated with increased risk in both IU and PP with odds ratios (OR) of 3.64 (95% CI = 1.82-7.31, $P = 0.0002$) and 4.45 (95% CI = 1.50-13.2, $P = 0.0045$) for HIV-1 transmission, respectively.

Fifteen haplotype-tagged SNPs (htSNPs) corresponding to the 15 major DC-SIGNR haplotypes (Figure 1) described among Zimbabweans [7] were genotyped in our study samples (Tables S2 and S3). H1 (31%) and H3 (11%) were the most frequent haplotypes observed (Figure 1). Being homozygous for the H1 haplotype was associated with increased risk of both IU (OR: 4.42, $P = 0.022$) and PP (OR: 7.31, $P = 0.016$) HIV-1 transmission (Table 2). Infants harbouring two copy combinations of H1 and/ or H3 haplotypes (H1-H1, H1-H3 or H3-H3) had increased risk of IU (OR: 3.42, $P = 0.007$) and IP (OR: 5.71, $P = 0.025$) but not PP ($P = 0.098$) HIV-1 infection compared to infant noncarriers (Table 2). The latter associations remained significant after adjustment was made for the maternal viral load for both IU (OR: 3.57, 95% CI = 1.30-9.82, $P = 0.013$) and IP (OR: 5.71, 95% CI = 1.40-23.3, $P = 0.025$) HIV-1 transmission. The H1 and H3 haplotypes share a cluster of mutations (p-198A, int2-391C, int2-180A, ex4RPT, int5+7C) (Figure 1). Of these, the p-198A and int2-180A variants were significantly associated with MTCT of HIV-1 (Table S2). In the unadjusted regression analysis, homozygous infants for the p-198A and int2-

180A variants had increased risk of IU (OR: 2.07 $P = 0.045$, OR: 3.78, $P = 0.003$, respectively) and IP (OR: 2.47, $P = 0.17$, O.R: 5.71, $P = 0.025$, respectively) HIV-1 infection compared to heterozygote infants or noncarriers (Table 3) . When adjustment was made for maternal factors, only the association with the int2-180A variant remained significant for IU (OR: 3.83, 95% CI = 1.42-10.4, $P = 0.008$) and IP (O.R: 5.71, 95% CI = 1.40-23.3, $P = 0.025$) HIV-1 transmission. Thus, infants homozygous for DC-SIGNR variant int2-180A contained in H1 and H3 haplotypes were 4-fold to 6-fold more likely to be infected by HIV-1 during pregnancy or at delivery, respectively.

Alternative splicing of the DC-SIGNR gene in the placenta produces both membrane-bound and soluble isoform repertoires [3] . The relative proportion of membrane bound and soluble DC-SIGNR could plausibly influence the susceptibility to HIV-1 infection [11] . We therefore hypothesized that the DC-SIGNR mutations associated with MTCT of HIV-1 would have an impact on both the level of DC-SIGNR expression and in the isoform repertoire produced. We investigated DC-SIGNR transcript expression in first-term placentas obtained after elective abortion.

We cloned DC-SIGNR from placental tissues by RT-PCR from 3 homozygous H1 samples containing both the DC-SIGNR p-198AA and int2-180AA variants associated with HIV-1 transmission and 3 homozygous wild-type (WT) (p-198CC, int2-180GG) samples. Fifteen clones per sample were randomly selected for sequencing. As expected, we found an extensive repertoire of DC-SIGNR transcripts in all samples with 9 to 16 different isoforms per individual. A total of 65 distinct transcripts were identified (Figure S1), of which 3 were full-length transcripts. 64 of the sequenced clones contained a total of 69 amino acid substitutions with 3 new C termini and 2 premature stop codons. However, the diversity was mostly attributable to the entire deletion of exon 2 or exon 3 or to variations in the length of the neck region (exon 4) of DC-SIGNR. The deletion of exon 3 eliminates the trans-membrane domain of the protein and leads to the expression of soluble DC-SIGNR isoforms [3] . Interestingly, the abundance of membrane-bound isoforms in placental tissues of the H1 homozygotes appears to be lower than that observed in samples from WT individuals (Figure S1). The deletion of exon 3 was confirmed by sequencing and we hypothesize that the skipping of exon 3, could be due to the presence of the int2-180A mutation observed in infants with the H1 haplotype. In fact, this intron mutation is located 180 bp downstream from exon 3 and potentially modifies splicing events (Figure 2A). We confirmed that the variation in transcript proportions seen between the two groups was also reflected at the level of mRNA expression in the placenta. To quantify membrane-

bound vs soluble isoforms in placental samples from homozygous H1 and WT infants, we amplified the exon 5 (E5) sequence present in all DC-SIGNR isoforms (total transcripts). We then amplified exon 3 (E3) which is deleted in the soluble forms and then calculated the E3:E5 ratio. We found that placental tissues from homozygous H1 infants express a significantly lower proportion of membrane-bound DC-SIGNR (18%) compared to that in WT individuals (36%) ($P = 0.004$) (Figure 2B) suggesting that exon 3 skipping happens more frequently in presence of the DC-SIGNR int2-180A variant associated with MTCT of HIV-1.

The DC-SIGNR int2-180A variant is always transmitted with the promoter mutation p-198A (Figure 1). In the unadjusted regression analysis, the p-198A variant was significantly associated with IU but not with IP and PP HIV-1 transmission (Table 3). Computational transcription factor binding site analysis predicts Table 1. Baseline characteristics of mother and infants risk factors for intrauterine (IU), intrapartum (IP) and postpartum (PP) mother-to-child HIV-1 transmission. Figure 3A). The luciferase activity of the p-198A variant construct was significantly lower than that of the WT p-198C promoter construct (p-198C/A ratio = 2, $P = 0.006$) (Figure 3B) suggesting that DC-SIGNR p-198A affects promoter activity. The other promoter mutants (p-577C and p-323A) observed in the Zimbabwean population did not affect DC-SIGNR transcription in this assay (Figure S2). To determine the net impact of the DC-SIGNR p-198A mutation on DC-SIGNR expression in the placenta, we quantitated the absolute number of total and membrane-bound DC-SIGNR transcripts in the H1 homozygote and wild-type placental samples as described earlier. The total number of DC-SIGNR transcripts was determined to be 6856213 (DC-SIGNR copies/6S.E.M per 10⁵ GAPDH copies) in the placental samples from homozygous H1 infants and was 4-fold lower compared to that found in placentas from WT individuals (27816638, $P = 0.011$) (Figure 3C). As suggested earlier, the int2-180A mutation might induce exon 3 skipping leading to a lower production of membrane-bound DC-SIGNR. Although, the decrease in the total number of DC-SIGNR transcripts in H1 homozygous placental samples containing both the p-198AA and int2-180AA variants affected the proportion of membrane-bound and soluble isoforms, the effect of these mutations was more pronounced on the membrane-bound isoforms with an 8-fold decrease (H1 = 117636.2 vs WT = 9906220.6, $P = 0.003$) compared to a 3-fold decrease in total soluble isoforms (H1 = 5686181.9 vs WT = 19256495.3, $P = 0.03$) (Figure 3C). Therefore, DC-SIGNR p-198A and int2-180A mutations associated with MTCT of HIV-1 significantly decreased the level of total placental DC-SIGNR transcripts, disproportionately affecting the membrane-bound isoform production. Table 3. Associations between infant DC-SIGNR promoter p-198 and intron 2 (int2)-180 variants and intrauterine (IU), intrapartum (IP) and postpartum (PP) mother-to-child HIV-1 transmission.

Our genetic results, supported by expression assay in placenta, suggest the involvement of DC-SIGNR in MTCT of HIV-1. Homozygosity for the haplotype H1 was associated with IU transmission in the unadjusted regression analysis. However, the association disappeared after adjustment was made for the maternal factors presumably because of the small number of H1 homozygote infants analysed in each groups. H1 and H3 were the most frequent haplotypes observed in the study population and they share a cluster of mutations (Figure 1). Grouping haplotypes H1 and H3 increased the power of the study and permitted the identification of specific DC-SIGNR mutations associated with MTCT of HIV-1. Indeed, two mutations shared by haplotypes H1 and H3 were associated with vertical transmission of HIV-1. The int2-180A was associated with a 4-fold increased risk of IU and 6fold increased risk of IP after adjustment for the maternal factors. Although the p-198A variant was associated with IU transmission, the association disappeared after adjustment was made for the maternal viral load. Nevertheless, we showed that this mutation reduces DC-SIGNR transcriptional activity in vitro and produces lower level of DC-SIGNR transcripts in placental tissues in combination with the int2-180A variant. Since int2-180A is always transmitted with p-198A on the MTCT associated combined haplotypes H1/H3, whereas p-198A is carried on other nonassociated haplotypes (Figure 1) , we can speculate that the p-198A mutation alone may have a minor effect in vivo whereas in combination with the int2-180A variant, they both act to reduce the level of placental DC-SIGNR expression resulting in an increased risk of MTCT of HIV-1.

The majority of IU transmission occurs during the last trimester of pregnancy (reviewed in [12]). Full-term placenta samples were not available for the current study and the expression assays were performed on first-term placental tissues. A previous study looking at DC-SIGNR placental isoforms repertoire in full-term placenta samples demonstrated similar diversity of DC-SIGNR transcripts as in the first-term placental tissues studied herein [3] . However, since levels of DC-SIGNR expression have never been compared between the different terms of pregnancy, it is not known whether DC-SIGNR expression varies during the course of pregnancy. Nevertheless, it is reasonable to assume that the inter-individual differences in both DC-SIGNR isoform repertoire and transcript levels observed between the H1 and WT homozygous infants would be reflected throughout the pregnancy. To date, most studies have focused on the potential role of DC-SIGNR in trans infection of HIV-1 in vitro [2, 10] . However, the multiple mechanisms involved in trans infection and redundancy among C-type lectin functions make it difficult to determine the actual participation of DC-SIGNR in this mode of infection in vivo [13, 14] . The strong

correlation we observed between MTCT of HIV-1 and DC-SIGNR genetic variants producing low levels of DC-SIGNR in the placenta suggested that mechanisms other than DC-SIGNR-mediated trans infection might operate during vertical transmission of HIV-1. For example, DC-SIGNR has also been shown to function as a HIV-1 antigen-capturing receptor [15]. Chan and colleagues recently demonstrated that DC-SIGNR transfected CHO cells diminish SARS-CoV titers by enhanced capture and degradation of the virus in a proteasome-dependent manner [4]. Since endothelial cells express MHC-I and II, degraded viral antigens could then be presented to immune cells to elicit an adaptive immune response [16, 17]. The HIV-1 coreceptor CCR5, but not CD4, is co-expressed with DC-SIGNR on placental and blood-brain barrier (BBB) endothelial cells [18, 19]. HIV-1 gp120 binding to CCR5 receptor on endothelial cells compromises BBB integrity and enhances monocytes adhesion and transmigration across the BBB [20, 21]. It is thus possible that reduced expression of DC-SIGNR, particularly the membranebound isoforms, on placental capillary endothelial cells might favour HIV-1 binding to CCR5 receptor, instead of DC-SIGNR receptor, facilitating the migration of maternal HIV-1-infected cells across the placental barrier resulting in IU transmission of HIV-1.

The int2-180A variant contained in the H1 and H3 haplotypes was associated with IP transmission suggesting that DC-SIGNR also affect transmission of HIV-1 during delivery. Little is known about the mechanisms underlying transmission of HIV-1 during delivery. Passage through the birth canal could potentially expose infants through a mucosal portal entry (presumably ophthalmic, skin, or gastrointestinal), whereas placental insult during delivery (physical or inflammatory) may enhance transplacental passage of maternal HIV-1-infected cells into foetal circulation [22, 23]. Such process called microtransfusion has been proposed in regards to the results obtain in a Malawian cohort. Kweik and colleagues found a significant association between levels of maternal DNA in umbilical cord blood and IP transmission of HIV-1 suggesting that passage of maternal infected cells through the placenta is likely to occur during delivery [22]. Thus, in a similar fashion as suggested earlier for IU transmission, the relatively lower level of DC-SIGNR in the placenta of homozygous infants harbouring the int2-180A variant could promote HIV-1 binding to CCR5 receptor on endothelial cells affecting the placental barrier integrity and facilitating the passage of maternal infected cells in foetal circulation during delivery.

Beside DC-SIGNR, other HIV-1 receptors are known to influence MTCT of HIV-1 (reviewed in [24]). Genetic variants in CCR5 have been shown to influence vertical transmission of HIV-1. CCR5 promoter variants resulting in higher expression of the receptor were associated

with increased risk of MTCT of HIV-1 among sub-Saharan Africans [25, 26] . The 32-pb deletion polymorphism in CCR5 has been shown to protect from vertical transmission of HIV-1 [27] , but this variant is virtually absent among African populations [28] . High copy numbers of CCL3L1, a potent HIV-1 suppressive ligand for CCR5, are associated with higher chemokine production and lower risk of MTCT of HIV-1 among South African infants [29, 30] . Mannose-binding lectin (MBL) is an innate immune receptor synthesised in the liver and secreted in the bloodstream in response to inflammation signal. MBL promotes pathogen elimination by opsonization and phagocytosis, and reduced expression of MBL resulting from polymorphism in coding and non-coding regions has been associated with an increased risk of MTCT of HIV-1 [31, 32] .

In this study, we demonstrate for the first time, the potential functional impact of DC-SIGNR mutations on its expression in the placenta and in vertical transmission of HIV-1. We believe that the presence of DC-SIGNR at the placental endothelial cell surface may protect infants from HIV-1 infection by capturing virus and promoting its degradation/presentation. However, in placenta containing low levels of DC-SIGNR, HIV-1 would preferentially bind CCR5 on endothelial cells resulting in a loss of placental barrier integrity and enhanced passage of maternal HIV-1-infected cells in foetal circulation leading to MTCT of HIV-1. This mechanism may also apply to other vertically-transmitted pathogens known to interact with DC-SIGNR such as HIV-2, hepatitis C and dengue viruses and warrant further investigation.

Associations between child DC-SIGNR exon 4 repeated region genotypes and mother-to-child HIV-1 transmission. CI, Confidence interval; N, number; NA; not applicable; OR, odds ratio a P-value as determined by the Chi-square test. b Comparison between genotype and all others. Found at: doi:10.1371/journal.pone.0007211.s003 (0.05 MB DOC) Figure S1 DC-SIGNR transcripts repertoire in placenta. Major RT-PCR products from RNA extract from 3 homozygous H1 and 3 homozygous WT placenta samples were purified, cloned and sequenced. Sequences were analysed according to NCBI reference sequence NM_014257. CT; cytoplasmic tail, TM; trans-membrane domain; WT; wild-type Found at: doi:10.1371/journal.pone.0007211.s004 (0.11 MB DOC) Figure S2 Effect of DC-SIGNR promoter variant on transcriptional activity in luciferase reporter assay in vitro in transfected HeLa cells. Relative luciferase expression from pGL2-Basic, parental vector without promoter. Expression DC-SIGNR promoter constructs, spanning p-577C variant or p-323A variant were calculated relatively to this value. Data are presented in mean values $\pm$ S.E.M of three independent experiments performed in triplicate. One-way ANOVA

test followed by the Dunnett test for multiple comparison was used to compare the relative luciferase expression of the p-557C and p-323A variant reporters against the wild-type (WT) construct (not significant). 0 mg, 0.5 mg or 1 mg CMV-Tat vector was transfected with LTR-Luc as a positive control in these experiments.

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Role of S-Palmitoylation on IFITM5 for the Interaction with FKBP11 in Osteoblast Cells

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Abstract: Recently, one of the interferon-induced transmembrane (IFITM) family proteins, IFITM3, has become an important target for the activity against influenza A (H1N1) virus infection. In this protein, a post-translational modification by fatty acids covalently attached to cysteine, termed S-palmitoylation, plays a crucial role for the antiviral activity. IFITM3 possesses three cysteine residues for the S-palmitoylation in the first transmembrane (TM1) domain and in the cytoplasmic (CP) loop. Because these cysteines are well conserved in the mammalian IFITM family proteins, the S-palmitoylation on these cysteines is significant for their functions. IFITM5 is another IFITM family protein and interacts with the FK506-binding protein 11 (FKBP11) to form a higher-order complex in osteoblast cells, which induces the expression of immunologically relevant genes. In this study, we investigated the role played by S-palmitoylation of IFITM5 in its interaction with FKBP11 in the cells, because this interaction is a key process for the gene expression. Our investigations using an established reporter, 17-octadecynoic acid (17-ODYA), and an

inhibitor for the S-palmitoylation, 2-bromopalmitic acid (2BP), revealed that IFITM5 was S-palmitoylated in addition to IFITM3. Specifically, we found that cysteine residues in the TM1 domain and in the CP loop were S-palmitoylated in IFITM5. Then, we revealed by immunoprecipitation and western blot analyses that the interaction of IFITM5 with FKBP11 was inhibited in the presence of 2BP. The mutant lacking the S-palmitoylation site in the TM1 domain lost the interaction with FKBP11. These results indicate that the S-palmitoylation on IFITM5 promotes the interaction with FKBP11. Finally, we investigated bone nodule formation in osteoblast cells in the presence of 2BP, because IFITM5 was originally identified as a bone formation factor. The experiment resulted in a morphological aberration of the bone nodule. This also indicated that the S-palmitoylation contributes to bone formation.

Text: The interferon-induced transmembrane (IFITM) protein family (also known as the Frigilis family in mice) is a part of the dispanin family [1] and is composed of double-transmembrane α -helices connected by a cytoplasmic (CP) loop and extracellular (EC) amino-and carboxyl-terminal polypeptide sequences (Figure 1-A) . The IFITM proteins are evolutionarily conserved in vertebrates [2] . Recent genomic research has revealed that there are 5 IFITM members in humans (IFITM1, 2, 3, 5 and 10) and 7 members in mice (IFITM1, 2, 3, 5, 6, 7, and 10). These proteins play roles in diverse biological processes, such as germ cell maturation during gastrulation (IFITM1-3) [3] [4] [5] , cell-to-cell adhesion (IFITM1) [6] [7] [8] , antiviral activity (IFITM1-3) [9] [10] [11] [12] [13] [14] [15] [16] [17] , and bone formation (IFITM5) [18] [19] [20] [21] [22] , although the detailed functions of IFITM6, 7, and 10 are unknown at present. In particular, IFITM3 has been a target of intensive studies on its activity against influenza A (H1N1) virus infection and internalization [9] [10] [11] [12] [13] [14] .

In 2010, Dr. Yount and co-workers reported that the antiviral activity of IFITM3 is dependent on S-palmitoylation on the protein [10] . The S-palmitoylation [23] is a post-translational modification on proteins by C 16 saturated-fatty acids (palmitic acids) covalently attached to certain cysteine residues via a thioester linkage (Figure 1-B) . The modification is reversibly catalyzed by protein acyltransferases and acylprotein thioesterases, and confers unique properties to the protein, such as membrane binding and targeting, immunoreactivity,

Amino-acid sequence alignment of IFITM5, IFITM1, IFITM2, and IFITM3 derived from mice. The conserved residues are highlighted in black. The three conserved cysteines are highlighted in red and numbered based on the sequence of IFITM5 (top) and IFITM3 (bottom). The residues unique in IFITM5 are highlighted in gray. The first and the second transmembrane domains, the extracellular sequences, and the cytoplasmic loop are indicated by arrows and denoted as TM1 and TM2, EC, and the CP loop, respectively. The TM domains were predicted by SOSUI. The aspartates at the C-terminal region in IFITM5 are shown in blue. B) The schematic illustration of the protein S-palmitoylation. The C 16 - palmitic acid is attached to cysteine via a thioester linkage. The palmitoylation and depalmitoylation are catalyzed by protein acyltransferases and acylprotein thioesterases, respectively. In this study, hydroxylamine, NH_2OH , was used to reduce the thioester linkage. C) The amino acid sequence identity (similarity) among IFITM5, IFITM1, IFITM2, and IFITM3 is summarized. doi: 10.1371/journal.pone.0075831.g001 and protein-protein interaction. The authors revealed that IFITM3 is S-palmitoylated on three membrane proximal cysteines, Cys71 and Cys72 in the first transmembrane (TM1) domain, and Cys105 in the CP loop (Figure 1-A) [10] . In addition, IFITM3 lacking the S-palmitoylation is not clustered in the cell membrane and significantly diminishes the antiviral activity. Moreover, the cysteines in IFITM2, Cys70, Cys71, and Cys104 are also palmitoylated in the same manner, which affects the intracellular localization [24] . A recent study has revealed that murine IFITM1 has four cysteine residues (Cys49, Cys50, Cys83, and Cys103) for the S-palmitoylation, which is required for the antiviral activity and the protein stability [25] . The other IFITM family members also possess these cysteines (Figure 1-A) , and thus the role of the S-palmitoylation on the cysteines should be significant for the functions of IFITM proteins.

Here, we focused on IFITM5, which is also known as bone-restricted IFITM-like (BRIL) protein [18] . Among the IFITM family proteins, IFITM5 is unique. (i) Expression of IFITM5: Unlike the other IFITM family proteins, the expression of IFITM5 is not induced by interferons because the region upstream of the *ifitm5* gene lacks the interferon regulatory elements [26] . Furthermore, the expression of IFITM5 is mostly restricted to osteoblast cells [18, 19, 27] , while the other IFITM proteins are expressed ubiquitously (ii). Amino-acid sequence similarity: The amino acid sequence of IFITM5 is relatively dissimilar to IFITM1-3 proteins (~ 65% similarity), while IFITM1-3 proteins share ~ 85% similarity with each other (Figure 1 -C). In addition, IFITM5 has an aspartate-rich domain in the C-terminal region, which could be involved in calcium binding (Figure 1 -A) [26] . (iii) Role of IFITM5 in bone formation: The expression of IFITM5 is associated with mineralization during the bone formation process in osteoblast cells [18] [19] [20] [21] . Previous studies have confirmed

the expression of IFITM5 in bone tissues in mice, rats, humans and tammar wallabies [2] . The ifitm5-gene knockout mice have smaller bones [19] . Moreover, the knockdown of the ifitm5 gene by small hairpin RNA induces a decrease in bone nodule formation, whereas overexpression of the gene in UMR106 cells has been shown to increase calcium uptake and bone nodule formation [18] . (iv) Role of IFITM5 for immune activity: Recent studies have revealed that IFITM5 interacts with the FK506-binding protein 11 (FKBP11) to form IFITM5-FKBP11-CD81-the prostaglandin F2 receptor negative regulator (FPRP) complex [28] . When the complex is formed, the expressions of 5 interferon-induced genes are induced, including bone marrow stromal cell antigen 2 (Bst2), interferon inducible protein 1 (Irgm), interferoninduced protein with tetratricopeptide repeats 3 (Ifit3), b(2)microglobulin (B2m), and MHC class I antigen gene. Consequently, these results indicate that IFITM5 is involved not only in the bone formation but also in the immune system activity.

In this study, we investigated the S-palmitoylation of IFITM5 and its role in the interaction with FKBP11 in mouse osteoblast cells. Cells transfected by a plasmid DNA encoding mouse IFITM5 were grown in the presence of an established chemical reporter, 17-octadecynoic acid (17-ODYA) [29, 30] , or an inhibitor for the S-palmitoylation, 2-bromopalmitic acid (2BP) [31] . The biochemical assays using these compounds revealed that the wild-type IFITM5 is S-palmitoylated. To identify the Spalmitoylation site in IFITM5, we prepared cysteine-substituted mutants, IFITM5-C86A, -C52A/C53A, and -C52A/53A/86A (Cys-less). The chemical reporter assay suggested that at least two out of three cysteines in IFITM5 are S-palmitoylated. The interaction of IFITM5 with FKBP11 was examined by immunoprecipitation assay, resulting in the loss of the interaction in the presence of 2BP. The same result was obtained in the two mutants, C52A/C53A and Cys-less. These results suggested that the S-palmitoylation on Cys52 and/or Cys53 in the TM1 domain of IFITM5 is necessary for the interaction with FKBP11. On the other hand, Cys86 in the CP loop of IFITM5 was S-palmitoylated but not involved in the interaction. Because this interaction is important for the immunologically relevant gene expression, it was indicated that the role of the S-palmitoylation is to promote the interaction of IFITM5 with FKBP11 and to regulate the immune activity in the osteoblast cells. The possible interaction mechanism and the effect of the S-palmitoylation on the bone nodule formation will be discussed.

For mammalian cell expression, plasmid vectors of wild-type IFITM5 (IFITM5-WT) and FLAG-fused FKBP11 (FKBP11-FLAG) were constructed by inserting the cloned genes into a pBApo-CMV Neo expression vector (Takara Bio, Shiga, Japan). The details of the

recombinant DNA constructs were the same as described previously [19]. The genes of IFITM5 mutants (IFITM5-C86A, -C52A/53A, and -C52A/C53A/C86A (Cys-less)) were prepared using a QuikChange site-directed mutagenesis kit (Stratagene, La Jolla, CA). The plasmid vectors of FLAG-fused IFITM5-WT, -C52A/53A, and Cys-less were constructed by inserting the cloned genes into the pBApo-CMV Neo expression vector. For *E. coli* cell expression, the plasmid vector of IFITM5-WT was constructed by inserting the cloned gene into a pET22b (Novagen, Madison, WI) expression vector. The forward primer 5'-GGAATTCCATATGGACACTTCATATCCCCGTG-3' and the reverse primer 5'-CCGCTCGAGGTTATAGTCCTCCTCATCAAATTGG-3' were used to amplify the gene encoding the entire IFITM5 from the plasmid vector for mammalian cell expression described above. The underlined letters denote an NdeI and an XhoI cleavage site, respectively. The plasmids of IFITM5 mutants were prepared using a QuikChange site-directed mutagenesis kit. The sense and anti-sense primers used were 5'-GGCAGTATGGCTCCAAAGCCAAGGCGTACAACATCCTGG CTGC-3' and 5'-GCAGCCAGGATGTTGTACGCCTTGGCTTTGGAGCCATACT GCC-3' for IFITM5-C86A; and 5'-GCACGATGTACCTGAATCTGGCGGCGCTTGGATTCCTGG CGC-3' and 5'-GCGCCAGGAATCCAAGCGCCGCCAGATTCAGGTACATCG TGC-3' for IFITM5-C52A/C53A, respectively (Sigma-Aldrich, St. Louis, MO).

Osteoblast-like MC3T3 cells were provided by the RIKEN, Cell Bank (RCB 1126). The procedures for cell culture, transfection, and protein expression were the same as reported previously. When necessary, 2-bromopalmitic acid (2BP; Wako, Osaka, Japan) and 17-octadecynoic acid (17-ODYA; Sigma-Aldrich) were dissolved in 99.5% dimethyl sulfoxide (DMSO; Wako) and added to differentiation medium at concentrations of 100 μ M and 50 μ M in less than 0.1% DMSO, respectively [30, 31].

Wild-type and mutant IFITM5 proteins were also produced using an *E. coli* recombinant expression system. *E. coli* BL21(DE3) cells transformed by the expression plasmid were grown at 37°C in LB medium containing 50 μ g/mL ampicillin. After four-hour induction by 1 mM isopropyl β -D-thiogalactopyranoside (IPTG), cells were harvested by centrifugation (6,400 $\times$ g for 10 min at 4°C). The cells were suspended in 50 mM Tris-HCl buffer (pH 8) and disrupted by a French press (Ohtake, Tokyo, Japan) (100 MPa $\times$ 4 times). The crude membrane fraction was collected by ultracentrifugation (178,000 $\times$ g for 90 min at 4°C). The collected fraction was solubilized with 1.5% n-dodecyl- β -D-maltopyranoside (DDM) (Dojindo Lab, Kumamoto, Japan) in 50 mM Tris-HCl, pH 8, containing 0.3 M NaCl and 5 mM imidazole. After the ultracentrifugation, the supernatant was incubated with Ni²⁺-NTA

agarose resin (Qiagen, Hilden, Germany). The resin was applied to a chromatography column and washed with 50 mM imidazole containing 50 mM Tris-HCl (pH 8), 0.3 M NaCl and 0.1% DDM. The DDM-solubilized IFITM5 was collected by elution with the same buffer containing 0.3 M imidazole. The sample media were replaced by the appropriate buffer solution by two passages over a PD-10 column (GE Healthcare UK, Ltd., Amersham Place, England).

The experimental details are described in previous reports [19, 28] . Briefly, total proteins were extracted from the osteoblast cells which co-expressed IFITM5 and FKBP11-FLAG using a total protein extraction kit (BioChain Institute Inc., Newark, CA). Then, the cell lysate was incubated with anti-FLAG M2 agarose gel (Sigma-Aldrich) at 4°C for 2 h. To recover FKBP11-FLAG, 500 ng/ μ L 3 $\times$ FLAG peptide (Sigma-Aldrich) dissolved in Tris-buffered saline was added to the collected gel at 4°C for 1 h. The recovered proteins and the cell lysate containing total proteins were analyzed by SDS-PAGE (15% ePAGE; ATTO, Tokyo, Japan) and western blot. The anti-IFITM5 polyclonal antibody, which was prepared from the amino-terminal peptide sequence (TSYPREDPRAPSSRC), and anti-FLAG monoclonal antibody (Sigma-Aldrich) were used as primary antibodies. The HRP-conjugated goat anti-rabbit IgG (H+L) (Zymed Laboratories, San Francisco, CA) and goat anti-mouse IgG (H+L) (Sigma-Aldrich) antibodies were used as secondary antibodies for the anti-IFITM5 and anti-FLAG primary antibodies, respectively. The proteins were detected by chemiluminescent reaction (Merck-Millipore, Billerica, MA).

The cell lysate extracted from the osteoblast cells metabolically labeled by 17-ODYA was incubated with anti-FLAG M2 agarose gel to obtain purified FLAG-fused IFITM5 proteins. The 17-ODYA-labeled proteins were chemically labeled with azide-PEG 3 -5(6)-carboxytetramethylrhodamine (TAMRA-azide; Click Chemistry Tools, Scottsdale, AZ) with reference to previous studies [10, 29, 30, 32] and the manufacturer's guide. The proteins separated by SDS-PAGE were visualized using a 532-nm laser for excitation and the fluorescence by TAMRA (565 nm) was detected using a 575nm long-path filter (Typhoon FLA 9000; GE Healthcare).

The subcultured osteoblast MC3T3 cells were seeded at a density of 5,000 cells/cm² in 40 mm dishes and cultured in α -Modified Eagle's Medium (α -MEM; Sigma-Aldrich) containing 10% (v/v) fetal bovine serum (FBS; Nichirei Biosciences Inc., Tokyo, Japan). On the next day, this was replaced with differentiation medium, containing 2 mM glycerophosphate and 50

µg/mL sodium ascorbate at final concentrations, to induce osteoblast differentiation. When necessary, 100 µM 2BP in less than 0.1% DMSO, or 0.1% DMSO alone was added to the differentiation medium at final concentrations. All cultures were incubated at 37°C in a humidified atmosphere containing 5% CO₂ for 27 days. Mineralized nodules were stained with Alizarin Red S (Sigma-Aldrich). The standard staining procedure was used. The mineralized nodules were checked every three days.

To identify the S-palmitoylation on IFITM5, the osteoblast cells harboring the plasmid DNA encoding IFITM5-WT were cultured in the absence and presence of 2BP, which inhibits the S-palmitoylation (Figure 2-A) [31]. Then, the cell lysate containing total protein was extracted for use in the SDS-PAGE and western blot analyses. For purposes of comparison, *E. coli* cells were also cultured in the absence of 2BP and the cell lysate was extracted. Figure 2-B shows the results of the western blot assay for IFITM5-WT expressed in the osteoblast and the *E. coli* cells. In the osteoblast cells, IFITM5-WT exhibited a single band near the 17.4 kDa molecular-mass marker (see lane 1) in the absence of 2BP. However, in the presence of 2BP (see lane 4), the band appeared at a lower position than that in the absence of 2BP (lane 1). These results suggested that IFITM5-WT has high and low molecular-mass forms in the absence and presence of 2BP, respectively. The S-palmitoylation is a reversible reaction, and therefore is depalmitoylated by a strong reductant such as hydroxylamine [10]. Following hydroxylamine treatment (see lane 2), the band appeared at the same position as in the presence of 2BP (lane 4). In prokaryote *E. coli* cells, the post-translational modification S-Palmitoylation on IFITM5 PLOS ONE | www.plosone.org does not occur. Hence, the band was also observed at the same lower position (see lane 3). In the case of IFITM3, the palmitoylation was also reported to induce a change in mobility on electrophoresis, just as in our present results [10].

For direct observation of the S-palmitoylation, an established chemical reporter, 17-ODYA (Figure 2-C), was used. The osteoblast cells harboring the plasmid encoding IFITM5-WT were cultured in the presence of 17-ODYA to label the protein metabolically. Following the extraction and the purification of the cell lysate, the labeled IFITM5-WT was ligated with TAMRA-azide according to the Cu(I)-catalyzed [3+2] azidealkyne cycloaddition method [10, 29, 30, 32]. An in-gel fluorescence image of the 17-ODYA-TAMRA-labeled IFITM5-WT (see lane 2 in Figure 2-D) showed that IFITM5 was S-palmitoylated in the osteoblast cells. The FLAG-tag attached to IFITM5 has no influence on the modification and chemical labeling (lanes 1 and 5). In addition, after the hydroxylamine treatment (see lane 6), the fluorescence became weak because of the dissociation of 17-ODYA from IFITM5, which

was the same mechanism as the dissociation of the palmitic acid from IFITM5 by reduction as described above (lane 2 of Figure 2-B) .

Therefore, we concluded that the IFITM5 expressed in the native osteoblast cells is S-palmitoylated. In addition, the bands corresponding to the high and the low molecular-mass forms shown in western blot analysis were tentatively assigned to the S-palmitoylated and the depalmitoylated forms, respectively.

As described above in the Introduction, cysteine residues are the substrate for S-palmitoylation. IFITM5 possesses three cysteines, Cys52 and Cys53 in the TM1 domain, and Cys86 in the CP loop (Figure 1-A) . All of these cysteines are highly conserved among the mammalian IFITM family proteins (Figure 3-A) . To identify the modification site in IFITM5, we prepared cysteine-substituted mutants, IFITM5-C52A/C53A, -C86A, and -C52A/C53A/C86A (Cys-less). The osteoblast cells harboring each plasmid were cultured in the absence of 2BP, and then the cell lysate was extracted. Figure 3 -B shows the results of the western blot detecting the expression of all the mutants in the osteoblast cells. In the C52A/C53A and Cys-less mutants (see lanes 2 and 4), the low molecular-mass form was detected. This result indicates that either Cys52 or Cys53 is involved in the S-palmitoylation. In addition, as shown in Figure 2 -D, strong and weak fluorescence were detected in the C52A/ C53A mutant in the absence and presence of hydroxylamine (lanes 3 and 7) , respectively, but not in the Cys-less mutant (lanes 4 and 8) . These results suggested that the rest of the cysteine in the C52A/C53A mutant, Cys86, is S-palmitoylated and the Cys-less mutant completely lost the S-palmitoylation because all the cysteines were substituted. Therefore, we concluded that Cys86, plus one or two other cysteine residues in IFITM5, i.e., Cys52 and/or Cys53, are S-palmitoylated. In addition, it was found that the S-palmitoylation on the TM1 domain has a major effect on the mobility in the gel (lower panel of Figure 2 -D and Figure 3-B) . Therefore, we hereafter refer to the high and low molecular-mass forms as the TM1palmitoylated and the TM1-depalmitoylated forms, respectively. Finally, we reassigned the bands shown in the western blot analysis as follows: IFITM5-WT is fully palmitoylated, the C86A mutant is partially palmitoylated at Cys52 and/or Cys53, the C52A/C53A mutant is partially palmitoylated at Cys86, and the Cys-less mutant is completely depalmitoylated.

Previous studies have revealed that IFITM5 interacts with FKBP11 [19] . FKBP11 belongs to the FK506-binding protein family and has a transmembrane domain. The interaction

between IFITM5 and FKBP11 is important for the immune activity because formation of the IFITM5-FKBP11-CD81-FPRP complex induces the expression of interferon-induced genes namely, the Bst2, Irgm, Ifit3, B2m, and MHC class I antigen gene [28]. To investigate the effect of the S-palmitoylation on the interaction of IFITM5 with FKBP11, we carried out an immunoprecipitation assay. The osteoblast cells co-transfected by the plasmids encoding IFITM5-WT and FKBP11-FLAG were cultured in the absence and the presence of 2BP. Then, the extracted cell-lysate was incubated with anti-FLAG agarose gel. The gel was washed several times. Finally, the proteins were competitively eluted by the addition of FLAG peptide. If IFITM5 interacted with FKBP11, it was expected that IFITM5 The conserved cysteines are highlighted in orange and numbered. In the lower panel, the numbers given in parenthesis correspond to the residual number for IFITM2. For the calculation of probability, a total of 23 IFITM2, 23 IFITM3, and 17 IFITM5 sequences derived from mammalian species in the Kyoto Encyclopedia of Genes and Genomes (KEGG) database were used. Sequence alignment was carried out using CLUSTALW. Sequence logos were generated using WEBLOGO 3. B) Western blot for the wild-type and cysteine-substituted mutants of IFITM5 expressed in the osteoblast cells. For detection, the anti-IFITM5 antibody was used as a primary antibody. The upper arrow indicates that C52 and/or C53 in the TM1 domain is S-palmitoylated (lanes 1 and 3). The C52A/C53A (lane 2) and Cys-less (lane 4) mutants are partially and completely depalmitoylated. The experiment was carried out 2 times. doi: 10.1371/journal.pone.0075831.g003 would be obtained during this step and detected by immunoblotting. Figure 4 -A shows the results of the western blot for the co-immunoprecipitation of IFITM5-WT with FKBP-FLAG. The band corresponding to FKBP11 appeared in all the lanes (upper panel). Lanes 1 and 2 are controls to ensure that IFITM5 and FKBP11 are both contained in the cell lysate before the immunoprecipitation. The controls also ensured that IFITM5 was S-palmitoylated in the absence of 2BP (see lane 1), whereas IFITM5 was not S-palmitoylated in the presence of 2BP (see lane 2). After the immunoprecipitation, a single band corresponding to the S-palmitoylated IFITM5 appeared in the absence of 2BP (see lane 3), indicating the interaction of the S-palmitoylated IFITM5 with FKBP11. However, in the presence of 2BP, no band corresponding to IFITM5 appeared (see lane 4), indicating that the two molecules do not interact with each other. These results suggest that the S-palmitoylation on IFITM5 contributes to the interaction with FKBP11.

Next, we further investigated the relationship between the S-palmitoylation and the interaction with FKBP11 by using the IFITM5 mutants described above. The osteoblast cells cotransfected by the plasmids encoding IFITM5 mutants (C52A/ C53A, C86A, and Cys-less) and FKBP11-FLAG were cultured. The immunoprecipitation assay was carried out in

the same way as described above. Figure 4 -B shows the results of the western blot for the co-immunoprecipitation of the wild-type and the IFITM5 mutants with FKBP11. Figure 4 -C shows the results of the control experiment using the cell lysate before the immunoprecipitation. As described in the previous section 3-3, the band corresponding to FKBP11 appeared in all the lanes (upper panels) because the immunoprecipitation was carried out using the anti-FLAG agarose gel. In the lower panel of Figure 4 -B, single bands were observed for the IFITM5-WT and -C86A mutant (lanes 1 and 3) but not for the -C52A/C53A and Cys-less mutants (lanes 2 and 4) . This result indicates that the wild-type and the C86A mutant interact with FKBP11, whereas the other two mutants do not. Interestingly, this tendency mirrored the trend for the S-palmitoylation profiles, which means that Cys52 and/or Cys53 in the TM1 domain of the IFITM5-WT and -C86A mutants is S-palmitoylated, whereas these residues are not S-palmitoylated in the C52A/C53A and Cys-less mutants (see Figures 2-D, 3 -B and the lower panel of Figure 4 -C). Because the S-palmitoylation contributes to the IFITM5-FKBP11 interaction, as described in the previous section 3-3 (also in Figure 4-A) , the results of Figure 4 -B suggest that the mutants which lost the S-palmitoylation site(s), Cys52 and/or Cys53, are not able to interact with FKBP11. In other words, the S-palmitoylation on these cysteines is necessary for the interaction of IFITM5 with FKBP11.

As described above in the Introduction, previous studies have revealed that IFITM5 also contributes to bone formation [18] [19] [20] [21] . Therefore, we investigated the influence of Spalmitoylation on the bone nodule formation in osteoblast cells, in which native IFITM5 is expressed. Figure 5 shows the time-dependent nodule formation in the absence and the presence of 2BP (Figure 5-A and -B) . Figure 5 -C shows the results of the control trial to verify the effect of DMSO, which was used as the solvent for 2BP, on the nodule formation. The mineralized nodule was stained with Alizarin Red, which reacts with deposited calcium. In Figure 5 -D, the area of the mineralized nodule was plotted against experimental time. In the absence of 2BP (Figure 5-A, -C, and -D) , the mineralization was started 15 days after the initiation of the cell differentiation (Day 0). On the other hand, in the presence of 2BP (Figure 5-B and -D) , the nodule was formed on Day 12. The halftime for the maximum mineralization in the presence of 2BP was estimated to be 7 days earlier than that in the absence of 2BP (Figure 5-D) . In addition, differences in the form of the mineralized nodules were observed. Figure 5 -E shows an enlarged view of each nodule on Day 21. The stained nodules were diffused in the presence of 2BP (panel b), whereas in the absence of 2BP the nodules formed a large cluster (panels a and c). Therefore, our observations in this study suggested that the S-palmitoylation affects the bone nodule formation in the osteoblast cells.

In this study, we confirmed the S-palmitoylation on IFITM5 in the osteoblast cells, which was the same as that previously reported for IFITM3 and IFITM2. As reported previously, in IFITM3 and IFITM2, which share 85% sequence similarity (Figure 1-C) , two cysteines in the TM1 domain (Cys71 and Cys72 for IFITM3, Cys70 and Cys71 for IFITM2) and one cysteine in the CP loop (Cys105 for IFITM3, Cys104 for IFITM2) are all S-palmitoylated in cells [10, 24] . On the other hand, although IFITM5 shares 68% and 66% sequence similarity to IFITM3 and IFITM2, respectively, more than one cysteine in the TM1 domain (Cys52 or Cys53) and one cysteine in the CP loop (Cys86) are S-palmitoylated. Taking into account the high conservation of three cysteines in the IFITM proteins (Figures 1-A and 3-A) , all the cysteines in IFITM5 may be involved in the S-palmitoylation just as in the case of IFITM3 and IFITM2 [10, 24] .

The roles of the S-palmitoylation on IFITM3 have been studied intensively, and the S-palmitoylation has been shown to be crucial for the correct positioning in the membrane and the resistance to viral infection and internalization [10] (the roles are summarized in Figure 6 -A and discussed in detail below). A recent study has revealed that the S-palmitoylation on IFITM2 is also important for the protein clustering in the membrane [24] . However, we do not know the role of the S-palmitoylation of IFITM5 for the clustering in the membrane at present because we have not yet succeeded in obtaining a proper antibody for immunohistochemistry, despite our allocating much time to the search and considering a considerable number of antibodies.

Dr. Hanagata and co-workers previously reported that IFITM5 lacking the TM1 domain and the CP loop, which and IFITM5 (lower panels), the anti-FLAG and the anti-IFITM5 antibodies were used as primary antibodies, respectively. Arrows indicate the existence of each protein and the S-palmitoylation on IFITM5. A) Western blot for the co-immunoprecipitation of the wild-type IFITM5 with the FLAG-fused FKBP11 (FKBP11-FLAG) in the osteoblast cells in the absence and the presence of 2BP (denoted as "-" and "+", respectively). Lanes 1 and 2 are the results for the control trials used to verify the existence of IFITM5 and FKBP11 before the immunoprecipitation, and Lanes 3 and 4 show the results after the immunoprecipitation. The experiment was repeated 3 times. B) Western blot for the co-immunoprecipitation of the wild-type and the cysteine-substituted mutants of IFITM5 with FKBP11-FLAG in the osteoblast cells. The band corresponding to FLAG peptide is not shown because of the smaller molecular-mass of FLAG peptide relative to FKBP11-FLAG.

C) The control experiment of Figure 4 -B used to verify that IFITM5 and FKBP11 were both present in the cell lysate before the immunoprecipitation. The experiment was repeated 2 times. A) The functional mechanism of IFITM3 is summarized from previous studies. (i) IFITM3 is S-palmitoylated at Cys71, Cys72, and Cys105, (ii) which induces clustering and correct positioning in the membrane, (iii) resulting in the antiviral activity against influenza virus. B) The functional mechanism of IFITM5 is summarized by combining the results from the present and the previous studies. (i) Cys86, plus one or two other cysteine residues in IFITM5, i.e., Cys52 and/or Cys53, are S-palmitoylated (ii). The S-palmitoylation allows IFITM5 to interact with FKBP11 in the osteoblast cells (iii). The dissociation of CD9 from the FKBP11-CD81-FPRP/CD9 complex is induced by formation of the IFITM5-FKBP11-CD81-FPRP complex and leads to the immunologically relevant gene expression. IFITM5 also contributes to the bone formation, but it is unknown which states as described in (i)-(iii) are important for the bone formation at present. At present, no interactive protein has been identified in IFITM3 and IFITM2. On the other hand, IFITM5 interacts with the partner protein, FKBP11, and the S-palmitoylation clearly makes a significant contribution to the interaction. Therefore, IFITM5 forms a hetero-oligomer in the cell membrane for its physiological function.

contain the relevant modification sites, lost the ability to interact with FKBP11 [19] . In the present study, we determined that the S-palmitoylation on Cys52 and/or Cys53 in the TM1 domain is necessary for the interaction. From these results, we speculate that Cys52 and Cys53 face toward the interaction surface with FKBP11, and therefore IFITM5 and FKBP11 interact with each other through the palmitic acid(s) attached to the cysteine(s) (summarized in Figure 6 -B, discussed in detail later). Our investigation revealed that Cys86 is involved in the S-palmitoylation but does not contribute to the interaction with FKBP11. We speculate that some other residues in the CP loop located near the TM1 domain make some contribution to the interaction.

Previous investigations also revealed that IFITM5 expressed in the heterologous fibroblast NIH3T3 cells exhibited direct interactions with CD81, the B cell receptor-associated protein 31 (BCAP31), and the hydroxysteroid (17-beta) dehydrogenase 7 (HSD17b7). These three proteins bind to the IFITM5 without the S-palmitoylation (low molecular-mass form; see Figure 3 -b in ref [19] . and Figure 1 -B in ref [28] .). In the fibroblast cells, the S-palmitoylation on IFITM5 is insufficient [19] . These interactions are not observed in the native osteoblast cells, and therefore are nonspecific. Taking these facts into

consideration, we speculate that the S-palmitoylation on IFITM5 promotes the specific interaction with FKBP11 in the osteoblast cells.

The role played by the S-palmitoylation of IFITM5 in immune activity of the osteoblast cells will be discussed by combining the results from the present and the previous studies. A specific interaction between IFITM5 and FKBP11 should be necessary to form the IFITM5-FKBP11-CD81-FPRP complex. CD81, also known as TAPA-1, is a member of the tetraspanin membrane protein family and a component of the B-cell coreceptor complex which mediates the B-cell signaling for immune responses. When forming this complex, CD9, a partner protein with CD81, dissociates from the FKBP11-CD81-FPRP/CD9 complex and consequently induces the osteoblast-specific expression of the interferon-induced genes, *Bst2*, *Irgm*, *Ifit3*, *B2m*, and the MHC class I antigen gene [28]. If the S-palmitoylation-mediated specific interaction of IFITM5 with FKBP11 were lost, the IFITM5-FKBP11-CD81-FPRP complex would not be formed, and consequently the interferon-induced gene expression would be inhibited because CD9 would remain associated with the FKBP11-CD81-FPRP/CD9 complex. In this respect, we speculate that IFITM5 is involved in the immune system activity in the osteoblast cells and the interaction of the S-palmitoylated IFITM5 with FKBP11 regulates the immune activity.

In addition, it was suggested that the S-palmitoylation on IFITM5 contributes to the bone nodule formation, including morphology and time for mineralization, in the osteoblast cells (Figure 5). It is difficult to conclude at present that the lack of the S-palmitoylation on IFITM5 causes the diffusion of the bone nodules (panel b of Figure 5 -E); we can say, however, that IFITM5 will probably not be S-palmitoylated in the cells in the presence of 2BP. While 2BP is commonly used as an inhibitor of palmitoylation, it also targets many metabolic enzymes [33, 34]. Thus, it is also difficult to interpret the results of the long-term incubation of the osteoblast cells in the presence of 2BP. In any case, these are interesting and key observations in terms of clarifying the role played by the S-palmitoylation of IFITM5 in bone formation, and further studies are required. Figure 6 describes a possible mechanism of the interaction of IFITM5 with FKBP11 and the role of IFITM5 in the osteoblast cell function by means of a comparison with IFITM3. In the case of IFITM3, as shown in Figure 6 -A, the following are observed. (i) The three cysteines are all S-palmitoylated (ii). The S-palmitoylation leads to the clustering and the correct positioning of IFITM3 molecules in the membrane (iii). The S-palmitoylation and the following clustering are crucial for the resistance to the influenza virus. When IFITM3 lacks the S-palmitoylation,

the IFITM3 molecules do not cluster, which leads to the significant decrease in the antiviral activity.

On the other hand, Figure 6 -B shows that the following observations are made in the case of IFITM5. (i) Cys86, plus one or two other cysteine residues in IFITM5, i.e., Cys52 and/or Cys53, are S-palmitoylated (ii). The S-palmitoylated IFITM5 is able to interact specifically with FKBP11. The interaction is presumed to be mediated by the palmitic acid(s) attached to the cysteine(s) facing toward the interaction surface on FKBP11. Cys86 is involved in the S-palmitoylation but not in the interaction of IFITM5 with FKBP11. At present, however, little is known about the role of the S-palmitoylation of IFITM5 for the localization in the membrane. When the S-palmitoylation affects the localization of IFITM5 as in the case of IFITM3 [10] , the S-palmitoylated IFITM5 molecules should be localized in the membrane or the depalmitoylated molecules should be delocalized. The loss of the interaction between IFITM5 and FKBP11 could be due to a relocation of the depalmitoylated IFITM5 that prevents its association with FKBP11 (iii). The Spalmitoylated IFITM5 interacts with the FKBP11-CD81-FPRP/CD9 complex through FKBP11, which induces the dissociation of CD9 from the complex and the expression of 5 immunologically relevant genes. Finally, IFITM5 forms the IFITM5-FKBP11-CD81-FPRP complex. It is unknown at present which of the three states (i)~(iii) illustrated in Figure 6 -B is important for the bone mineralization of the osteoblast cells. The lack of the S-palmitoylation influences the interaction with FKBP11, which could account for the following complex formation and gene expression. In addition, the bone nodule formation is also affected. Note that the role of the Spalmitoylation has been involved in the bone formation [35] . It is indicated that the S-palmitoylation on IFITM5 plays roles not only for the regulation of the immune activity but also for the bone formation.

In conclusion, we have revealed the S-palmitoylation on IFITM5 and its role in the interaction with FKBP11. Not only the immune activity but also the bone mineralization in the osteoblast cells is affected by the S-palmitoylation. In general, the functional role of the S-palmitoylation is different for each protein [36] . For many proteins, the palmitoylation and depalmitoylation cycle is constitutive and regulated by enzymes. Based on the present results, it is difficult to address (i) whether the S-palmitoylation on IFITM5 is constitutive or regulated, or (ii) when and where IFITM5 is S-palmitoylated in the osteoblast cells. Further studies are required and are currently underway.

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--- START OF ARTICLE 2 ---

First Complete Genome Sequence of a French Bovine coronavirus Strain

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Abstract: We sequenced the first Bovine coronavirus (BCoV) complete genome sequence from France. This BCoV was directly sequenced from a fecal sample collected from a calf in Normandy in 2014.

Text: B ovine coronavirus (BCoV) belongs to the Nidovirales order, the Coronaviridae family, the Coronavirinae subfamily, and the Betacoronavirus (<https://talk.ictvonline.org/ICTV/proposals/2008.085-122V.v4.Coronaviridae.pdf>). Its genome is a single-stranded, linear, and nonsegmented RNA of around 31 kb. BCoV is responsible for respiratory and enteric diseases in cattle, particularly during winter (1, 2) . To date, the 19 complete BCoV genome sequences available in GenBank databases (consulted on 17 January 2017) originated from the United States or Asia. Here, we report the first complete genome sequence of a BCoV detected in France.

The BCoV/FRA-EPI/CAEN/2014/13 strain was obtained from a fecal sample collected from a 1-week-old calf in Normandy in 2014. The presence of BCoV in the fecal sample was assessed using an in-house reverse transcription-PCR (RT-PCR) targeting the M gene (3). A cDNA library was synthesized using SuperScript III (Invitrogen, Carlsbad, CA, USA) and hexamers. The complete genome sequencing of overlapping PCR products was carried out in both directions, using original primers and Sanger's dideoxy sequencing. Sequencing reactions were performed as previously described (3). Sequences were assembled and annotated using the Geneious software (version 5.1.6). We obtained a sequence counting 30,847 nucleotides. The orf1ab, HE, S, ns5, E, M, and N genes of the obtained BCoV were submitted to a Blastn analysis. According to these analyses, the orf1ab (20kb nucleotides, located at the 5' side of the genome) gene is closely related to the Dromedary camel coronavirus (DcCoV) HKU23-23-362F strain from the United Arab Emirates (accession no. KF906251), with a nucleotide identity of 99.19%. Conversely, the NS2, HE, S, ns5, and M genes are closely related to the BCoV Bubalus/Italy/179/07-11 strain (accession no. EU019216), with nucleotide identities of 99.88%, 99.45%, 99.02%, 98.79%, and 99.28%, respectively. The E gene is closely related to the Chinese Bovine coronavirus strain BCV-AKS-01 (accession no. KU886219), with a nucleotide identity of 100%. Finally, the highest Blastn score for the N gene was found with the American enteric BCoV-ENT (accession no. AF391541), associated with a nucleotide identity of 100%.

Multiple-sequence alignment, including 20 BCoVs and 10 clade A betacoronaviruses closely related to BCoV from North America, two DcCoVs from the United Arab Emirates, and two Human coronavirus OC43 (HCoV-OC43) strains from France, was performed using the Muscle algorithm implemented in MEGA7 (4, 5). The phylogenetic analysis on the orf1ab confirms that BCoV/FRA-EPI/CAEN/2014/13 is closely related to the Dromedary camel coronavirus (DcCoV) HKU23-23-362F. The orf1ab gene of these two viruses together clustered separately from that of BCoV and BCoV-like viruses from North America and Asia. This finding also confirms the results from our previous analysis on partial genomes in which nsP12, S, and N genes of American and Asian BCoVs group together in a cluster tentatively named C 1. The nsP12 and N coding regions of BCoVs from France and DcCoVs from the United Arab Emirates clustered together in C 2. The DcCoV S gene individualized from both HCoV-OC43 and BCoV S genes. Potential recombination events could be at the origin of DcCoV.

Accession number(s). The complete genome sequence sequence of the BCoV/FRA-EPI/CAEN/2014/13 isolate has been deposited in GenBank under the accession number KX982264.

--- END OF ARTICLE ---

--- START OF ARTICLE 3 ---

Species-specific clinical characteristics of human coronavirus infection among otherwise healthy adolescents and adults

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Abstract: Human coronavirus (HCoV) is a known cause of influenza-like illness (ILI). In a multisite, observational, longitudinal study of ILI among otherwise healthy adolescents and adults, 12% of subjects were PCR-positive for HCoV. The distribution of species was as follows: HCoV-OC43 (34%), HCoV-229E (28%), HCoV-NL63 (22%), and HCoV-HKU1 (16%). We did not observe species-specific differences in the clinical characteristics of HCoV infection, with the exception of HCoV-HKU1, for which the severity of gastrointestinal symptoms trended higher on the fourth day of illness.

Text: Clinical manifestations of human coronavirus (HCoV) infection range from a mild, self-limiting illness of the upper respiratory tract to an acute respiratory distress syndrome with a high mortality rate.

Highly virulent species of HCoV were responsible for outbreaks of severe acute respiratory syndrome (SARS) and Middle East respiratory syndrome (MERS); case-fatality rates ranged from 14% to 45%. [1] [2] [3] By contrast, other HCoV species (HCoV-HKU1, HCoV-OC43, HCoV-NL63, and HCoV-229E) are much more prevalent, much less severe, and common causes of influenza-like illness (ILI). [4] [5] [6] [7] [8] [9] [10] [11] Five previous studies have described the species-specific clinical characteristics of HCoV infection among adults. 6, 7, [10] [11] [12] In two of these studies, a significant proportion of the study population had underlying medical conditions. 6, 7 Herein, we describe, among a cohort of otherwise healthy adolescents and adults with influenza-like illness (ILI), the species-specific prevalence and severity of symptoms associated with HCoV infection. 13 Patients 0-65 years of age and presenting for care <72 hours after onset of ILI symptoms were recruited for study participation. ILI was defined as a temperature $\geq 100.4^{\circ}\text{F}$ and sore throat or one of the following respiratory symptoms: cough, sputum production, shortness of breath, or chest pain. Both inpatient and outpatient subjects were eligible to participate. Patients with underlying medical conditions (eg, diabetes, chronic obstructive pulmonary disease, severe asthma), women with a high-risk or complicated pregnancy, and patients with a poorly controlled psychiatric disorder were excluded. Information on patient demographics and presence/severity of symptoms at the time of enrollment was collected by in-person interview. Participants were then instructed on the use of a daily diary to record the presence/severity of symptoms for 7 days following initial symptom onset. Symptom severity was rated on an ordinal scale from 0 (none) to 3 (severe). Symptom severity scores were quantified using the following five measures: (i) individual symptom score for 20 symptoms, (ii) the upper respiratory symptom score, calculated as the sum of severity scores for earache, runny nose, sore throat, and sneezing, (iii) the lower respiratory symptom score, calculated as the sum of severity scores for cough, difficulty breathing, hoarseness, and chest discomfort, (iv) the gastrointestinal symptom score, calculated as the sum of severity scores for diarrhea, vomiting, anorexia, nausea, and (Table 1) .

There was season-to-season variability in the leading causes of

The findings of our study, conducted over a 5-year period at five geographically dispersed sites in the USA, demonstrate that human coronavirus (HCoV) is an important cause of influenza-like illness (ILI) ranged from 4% to 22%. [8] [9] [10] [11] 14 Additionally, we found HCoV-OC43

to be the most common species among adults, as has been reported elsewhere. 8, 9, 11, 12, 14 HCoV-OC43 and HCoV-229E were the most common strains in alternate seasons, reflecting a season-to-season variability of HCoV strain circulation that has been reported in other multiyear studies. 4 8 The mechanisms by which this particular species elicits these symptoms are not known.

The strengths of this study of HCoV in otherwise healthy adolescents and adults include its multisite and multiyear design, the use of a multiplex diagnostic panel, the prospective collection of symptom data, and the use of a symptom severity scale similar to what has been employed previously. 15 One important limitation of this study was our selective recruitment of individuals who had presented to a healthcare facility for care of an ILI. Therefore, our cases are not representative of HCoV infection in the community, where individuals with mild, self-limiting illness due to HCoV opt not to seek medical care for the management of their ILI.

In summary, we have shown that HCoV is a significant cause of ILI among otherwise healthy adolescents and adults presenting for medical evaluation. Although there were differences in species distribution by age group, we did not detect any differences between species with respect to the clinical spectrum of disease.

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--- START OF ARTICLE 4 ---

One step closer to an experimental infection system for Hepatitis B Virus? --- the identification of sodium taurocholate cotransporting peptide as a viral receptor

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3562259/>

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Abstract: Following the successful cloning of receptor for SARS coronavirus a few years ago, Dr. Wenhui Li and colleagues raised attention again by publishing a possible receptor for hepatitis B virus in eLife. We will briefly review the significance of this finding and the future prospects of hepatitis B research.

Text: Among the five hepatotropic hepatitis viruses, only hepatitis B virus (HBV) and its satellite hepatitis D virus (HDV) still wait for the development of an in vitro infection system in cell culture. One hepatocellular carcinoma (HCC) cell line, HepaRG, can be infected at a modest efficiency after weeks of culture and induced differentiation [1] . Even primary human hepatocytes rapidly lose the capacity for HBV infection after brief cell culture. The HBV infection demands both intracellular and cell-surface factors. The intracellular requirements appear less stringent, as after transfection of HBV DNA into many HCC cell lines or mouse liver, which cannot be infected naturally, the viral genome is expressed and replicates actively. Thus, the failure of HBV infection is considered largely to be due to strict restriction on the interaction between HBV virions and the cell membrane.

The molecules on the cell membrane needed for HBV infection can be divided into two classes: low affinity and high affinity molecules. Among others, the heparan sulfates in the membrane proteins mediate the broad, but less specific, virus-cell interaction. However, the high affinity membrane partners for HBV remain elusive (the carboxypeptidase D found for duck hepatitis B virus may be the only serious contender [2]).

HBV envelope protein, namely the surface antigens, plays an essential role in the infection process. Both genetic and functional examination identified one domain in the Nterminus of HBV preS1 (amino acids 1-47) necessary for infection. This domain has been shown to function as a direct mediator for HBV by binding presumably cellular corresponding receptor(s) [3] . More importantly, the myristoylated peptide is shown to effectively block HBV infection in primary human hepatocytes and in the human hepatocytechimera mouse at a nanomolar concentration [4] . In fact, a clinical trial testing the efficacy of this peptide in preventing HBV infection has been ongoing [5] . Clearly, this preS1 peptide can be a useful probe to pull out the interacting cellular factors, including specific viral receptors.

Yan et al. have taken a reasonable approach to fish out possible HBV receptor(s) [6] . They engineered the first 2-47 amino acid peptide from PreS1 to increase its capacity to be cross-linked with proteins interacting with the cell membrane, without affecting its binding specificity. In order to obtain sufficient materials after cross-linking, they adopted the Tupaia hepatocytes, instead of human hepatocytes, for the experiments. The strategies actually brought down many membrane proteins, but in comparison with the negative control (homologous peptide without specific binding), they identified one cellular protein, NTCP (sodium taurocholate cotransporting peptide) by LC/MS/MS. The same protein was pulled down from human hepatocytes as well. The authors further produced HCC cell lines stably expressing NTCP and subsequently infected them with HBV or HDV.

Immunofluorescence staining clearly demonstrated the expression of HBV and HDV proteins in these cell lines, suggestive of a successful viral infection. In addition, they documented a 2-4-fold increase of viral RNA and DNA after infection in the cell line by real-time PCR. They also showed a Southern blot supporting the presence of HBV covalently closed circular DNA in the infected cell, a well-recognized marker for productive HBV infection. Finally, they identified a stretch of 10 amino acids in the NTCP transmembrane domain, as the motif directly interacting with the PreS1 peptide.

NTCP is a transmembrane protein, usually located in the lateral surface (canalicular) of hepatocytes, which mediates bile acid transport [7] . Geographically, it is a good candidate for an HBV receptor. Moreover, the authors could convert the cell lines previously non-permissible to HBV infection to permissible by over-expression of NTCP, again supporting its possible role in the HBV infection process. This can be a critical and long-awaited

discovery toward understanding HBV receptors and establishing an experimental HBV infection system.

Looking forward, we need to understand how NTCP interacts with both HBV envelope proteins and with other cellular proteins, especially through the motif embedded in the cell membrane. NTCP itself is not sufficient to allow HBV infection, as the majority of HepaRG cells were found to express NPCT but not to be infected [8] . NTCP might initiate or mediate molecular interactions that can overcome the cell-surface restrictions for viral entry. Such cooperative cellular or viral factors have to be discovered and demonstrated to enhance the efficiency of viral infection, at a level comparable to a natural one (hundreds or thousands fold viral amplification). For example, the authors can use the NTCP-expressing cell lines as the starting materials to systemically identify other factors (maybe carboxypeptidase D) and make these cell lines more productive and permissive to HBV infection. In the near future, standard virological assays for HBV infections, including Northern or Western blots, are expected to demonstrate the successful HBV infections in vitro.

The HBV research community has searched for HBV receptors for decades. Many candidates have been discovered and then discarded. The current study, however, took advantage of a well-documented viral peptide required for HBV entry in combination with a state-of-the-art proteomics platform. As a Chinese proverb says "a thousand-mile journey starts from one incremental step". As such, the identification of NTCP as a potential viral receptor for HBV may serve as an important initial step for this journey, leading to the development of an HBV infection system to facilitate the HBV research and hepatitis B treatment.

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--- START OF ARTICLE 5 ---

Development of an ELISA-array for simultaneous detection of five encephalitis viruses

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3305475/>

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Abstract: Japanese encephalitis virus(JEV), tick-borne encephalitis virus(TBEV), and eastern equine encephalitis virus (EEEV) can cause symptoms of encephalitis. Establishment of accurate and easy methods by which to detect these viruses is essential for the prevention and treatment of associated infectious diseases. Currently, there are still no multiple antigen detection methods available clinically. An ELISA-array, which detects multiple antigens, is easy to handle, and inexpensive, has enormous potential in pathogen detection. An ELISA-array method for the simultaneous detection of five encephalitis viruses was developed in this study. Seven monoclonal antibodies against five encephalitis-associated viruses were prepared and used for development of the ELISA-array. The ELISA-array assay is based on a "sandwich" ELISA format and consists of viral antibodies printed directly on 96-well microtiter plates, allowing for direct detection of 5 viruses. The developed ELISA-array proved to have similar specificity and higher sensitivity compared with the conventional ELISAs. This method was validated by different viral cultures and three chicken eggs inoculated with infected patient serum. The results demonstrated that the developed ELISA-array is sensitive and easy to use, which would have potential for clinical use.

Text: Japanese encephalitis virus(JEV), tick-borne encephalitis virus(TBEV), eastern equine encephalitis virus (EEEV), sindbis virus(SV), and dengue virus(DV) are arboviruses and cause symptoms of encephalitis, with a wide range of severity and fatality rates [1] . Establishment of an accurate and easy method for detection of these viruses is essential for the prevention and treatment of associated infectious diseases. Currently, ELISA and IFA are the methods which are clinically-available for the detection of encephalitis viral antigens, but they could only detect one pathogen in one assay [2, 3] .

There are a variety of different methods available for identifying multiple antigens in one sample simultaneously, such as two-dimensional gel electrophoresis , protein chip, mass spectrometry, and suspension array technology [4] [5] [6] . However, the application of these techniques on pathogen detection is still in an early phase, perhaps due to the complicated use and high cost.

Antibody arrays for simultaneous multiple antigen quantification are considered the most accurate methods [7] [8] [9] [10] . Liew [11] validated one multiplex ELISA for the detection of 9 antigens; Anderson [12] used microarray ELISA for multiplex detection of antibodies to tumor antigens in breast cancer, and demonstrated that ELISA-based array assays had the broadest dynamic range and lowest sample volume requirements compared with the other assays.

However, the application of ELISA-based arrays is currently limited to detection of cancer markers or interleukins; no detection of pathogens has been reported. In this study, we developed an ELISA-based array for the simultaneous detection of five encephalitis viruses. Seven specific monoclonal antibodies were prepared against five encephalitis viruses and used to establish an ELISA-array assay. The assay was validated using cultured viruses and inoculated chicken eggs with patient sera. The results demonstrated that this method combined the advantage of ELISA and protein array (multiplex and ease of use) and has potential for the identification of clinical encephalitis virus.

Monoclonal antibodies were prepared from hybridoma cell lines constructed by Prof. Zhu et al. Purification was conducted by immunoaffinity chromatography on protein G affinity sepharose [13] . Specific monoclonal antibodies (4D5 against JEV, 2B5 against TBEV, 1F1 against SV, 2B8 against serotype 2 DV, 4F9 against serotype 4 DV, 4E11 against EEEV, and 2A10 against Flavivirus) were selected for this study. All of the antibodies were raised according to standard procedures.

Using 4D5, 2B5, 1F1, 2B8, 4F9, and 4E11 as capture antibodies, detection antibodies (2A10, 1 F1, and 4E11) were coupled to biotin-NHS ester(Pierce, Germany) at 4°C for 3 h according to the manufacturer's instructions. Unincorporated biotin was removed by Desalt spin column (Pierce). Immunologic reactions were reported by Streptavidin-HRP

(CWBIO, Beijing, China) and Super Signal ELISA Femto Maximum sensitive substrate. Purified goat-anti mouse antibody was used as a positive control.

JEV and DV were cultured in C6/36 cells; SV, TBEV, and EEEV were cultured in BHK-21 cells. The culture of TBEV and EEEV was conducted in biosafety level 3 facility, however, JEV, DV and SV were conducted in biosafety level 2 facility. Viral titers were determined by the 50% tissue culture infectious dose (TCID₅₀) method. All the cultures were inactivated by 0.025% β -propionolactone at 4°C overnight, then 37°C for 1 h to decompose β -propionolactone.

Antibodies were spotted using a BIODOT machine (BD6000; California, USA) on ELISA plates (30 nl/dot). The plates were blocked with 3% BSA-PBS in 37°C for 1 h, followed by washing 3 times with PBS containing 0.1% Tween-20 for 2 min each. Then, the plates were dried, sealed, and stored at 4°C before use [11].

When spotting, different spotting buffers and concentrations of capture monoclonal antibodies were evaluated to optimize the ELISA-array assay. The optimization was evaluated by dot morphology and signal intensity. The tested spotting buffers included 1 × phosphate buffer saline (PBS), PBS +20% glycerol, and 1 × PBS + 20% glycerol+0.004% Triton-X100. A range of monoclonal antibody concentrations (0.0125, 0.025, 0.05, 0.1, and 0.2 mg/ml) were compared.

Following a double antibody sandwich format, printed plates were incubated sequentially with inactivated viral cultures, biotin-labeled detecting antibody, HRP-labeled avidin, and substrate, followed by signal evaluation.

Antigen binding was performed in PBS(containing 0.1% Tween-20 and 5% FCS) at 37°C for 2 h, followed by washing 3 times(1 × PBS containing 0.1% Tween-20). Incubation of ELISA plates with biotinylated detecting antibody cocktails was performed in PBS (containing 0.1% Tween-20 and 5% FCS) at 37°C for 2 h. After washing, specific binding of the detecting antibodies was reported by streptavidin-HRP and stained with Super Signal ELISA Femto Maximum sensitive substrate (Thermo scientific, Rockford, USA) [11, 14, 15]. Visualization of the plate was performed in AE 1000 cool CCD image analyzer(Beijing BGI

GBI Biotech Company., LTD, China). The signal intensity and background of each spot was read out and recorded with "Monster" software. The positive signals were defined as a signal value > 400 and a signal value (sample)/signal value (negative) > 2.

The identical antibodies used in the ELISA-array format were also tested in a conventional ELISA format to determine the difference in sensitivity and specificity of the two methods. The conventional ELISAs were performed at the same time as the ELISA-array assays to ensure similar reaction conditions. The conventional ELISAs were performed in an identical manner to the ELISA-array, except that antibodies were coated at a concentration of 2 µg/mL in PBS (pH 7.4), and substrate TMB was used instead of Super Signal ELISA Femto Maximum sensitive substrate [16, 17] .

Three serum samples were collected from patients with nervous system symptoms and histories of tick bites. The serum samples were treated with penicillin and streptomycin, then inoculated into the allantoic cavities of chicken eggs. 3 days later, the liquid was collected and divided into two portions (one for inactivation and one for RNA extraction). The RNA and inactivated samples were stored at -70°C before use.

RNA was extracted from the inoculated chicken eggs using a RNeasy mini kit (Qiagen Inc., Valencia, CA, USA) according to the manufacturer's instructions. All RNA extraction procedures were conducted at BSL-3 facilities. The primers and probes were used as previously described [18] . The real-time RT-PCR was conducted with a Quti-teck q-RT-PCR Kit (Qiagen Inc.,). The reaction consisted of 10 µL of 2 × reaction buffer (0.2 µL reverse transcription enzyme, and 250 nmol/l primers and probes). RNA and deionized water were added to a final volume of 20 µl. PCR was performed with a LightCycler 2.0 (Roche, Switzerland) [19] .

Optimization of the ELISA-array assay

The spotted array layout is depicted in Figure 1 and the efficacy of three different spotting buffers on the quality of the printed ELISA-arrays were investigated by spot morphology observation and signal intensity comparison.

The spotting concentration of the capture antibodies varied from 0.2 to 0.0125 mg/ml (each was serially diluted 2-fold). The efficacy of the spotting concentration of the capture antibodies was evaluated by virus culture detection, the proper spotting concentration was determined by a combination of minimized cross reaction and higher signal intensity. Figure 1 illustrates the array layout and Figure 2 demonstrates the result of the three spotting buffers and spot concentration of antibody 2B5 by TBE virus culture detection. Cross reaction detection was also conducted by applying JEV, YF, and DV cultures.

Spot morphology observation (Figures 2a, b , and 2c) demonstrated that spotting buffer containing PBS with 20% glycerol produced tailed spot morphology; buffers containing PBS alone and PBS with 20% glycerol +0.004% Triton-X100 gave good spot morphology (round and full). Buffers containing PBS with 20% glycerol and PBS with 20% glycerol+0.004% Triton-X100 produced higher signal intensities than PBS alone. Thus, PBS with 20% glycerol+0.004% Triton-X100 was adopted as the optimized spotting buffer for subsequent experiments. Simultaneously, the spot concentration evaluation suggested that 0.05 mg/ml was optimal. At this concentration, the signal intensity was higher and the cross-reaction did not appear (Figure 2d). Consequently, spotting concentration optimization of other capture antibodies (4D5, 1F1, 4E11, and 2B8) demonstrated that 0.05 mg/ml was also suitable(data not shown).

The optimized ELISA array layout is shown in Figure 3 , which was applied in the following experiments.

Successful detection of viral pathogens requires a test with high sensitivity and specificity. To evaluate the performance of the designed antibody arrays, the specificity and sensitivity of the individual analytes were examined. By testing serially-diluted viral cultures, including DV-2, DV-4, JEV, TBE, SV, and EEEV, the sensitivity of ELISAarray and the identical conventional ELISA were compared (Table 1). The detection limit of the two methods was compared and demonstrated. The cross-reactivity test was conducted using BHK-21 and vero cell lysate, Yellow fever virus (YFV) cultures (5×10^5 TCID₅₀ /ml, West Nile virus(WNV) cultures(2×10^6 TCID₅₀ /ml), and Western equine encephalitis virus(1×10^7 TCID₅₀ /ml). The results demonstrated that neither the ELISA-array nor traditional ELISA displayed cross-reactivity.

Equal volumes of cultured TBEV, JEV, DV-2, DV-4, SV, and EEEV were prepared for single sample detection; two or three of the cultures were mixed for multiplex detection. A cocktail of biotin conjugated antibody (2A10, 4E11, and 1F1) was used in all tests. The results demonstrated that for all virus combinations, each virus was detected specifically, with no false-positive or-negative results (Figures 4 and 5) .

Chicken eggs inoculated with infected human serum were used for validation of the ELISA-array assay. All samples showed high reaction signals with capture antibody 2B5, which was specific for TBEV (Figure 6b). The ELISA-array assay suggested that the three patients were all infected with TBEV.

To verify the results tested by ELISA-array, RNA extracted from chicken eggs was applied to a real time-RT-PCR assay using primers and probes targeting TBEV. The results were also positive (Figure 6a) . The consensus detection results confirmed that the ELISAarray assay was reliable.

To be widely used in the clinical setting, the detection system should be easy to use and can be performed by untrained staff with little laboratory and experimental experience. Moreover, when the volume of the clinical samples is limited and an increasing number of pathogens per sample needs to be tested, the detecting system should be high-throughput to allow detection of multiple pathogens simultaneously [6, 20, 21] . Multiple detection, easy to use, and affordability are requirements for detection methods in the clinical setting. Thus, an ELISA-array, which combines the advantages of ELISA and protein array, meets the above requirements.

It has been reported that an ELISA-array has been used in the diagnosis of cancer and auto-allergic disease [7, 12] ; however, No study has reported the detection of viral pathogens. In this study, we developed a multiplex ELISA-based method in a double-antibody sandwich format for the simultaneous detection of five encephalitis-associated viral pathogens.

The production of a reliable antibody chip for identification of microorganisms requires careful screening of capture of antibodies [14] . Cross-reactivity must be minimized and

the affinity of the antibody is as important as the specificity. First, we prepared and screened 23 monoclonal antibodies against eight viruses and verified the specificity and affinity to the target viruses by an immunofluorescence assay. Then, the antibodies were screened by an ELISA-array with a double-antibody sandwich ELISA format. The antibodies which produced cross-reactivity and low-positive signals were excluded. Finally, six antibodies were selected as capture antibodies. Another monoclonal antibody, 2A10, which could specifically react with all viruses in the genus Flavivirus was used for detecting antibody against DV, JEV, and TBEV. For the detection of EEEV and SV, although the detecting and trapping antibodies were the same (1F1 and 4E11, respectively), the antibodies produced excellent positive signals. The epitope was not defined; however, we suspect that the antibodies both target the surface of the virions. As one virion exits as, many with the same epitope appear, thus no interference occurred using the same antibody in the double-antibody sandwich format assay.

Currently, the availability of antibodies suitable for an array format diagnostic assay is a major problem. In the ELISA-array assay, this problem exists as well. Because of the limitation of available antibodies, this assay could only detect 5 pathogens. In the future, with increasing numbers of suitable antibodies, especially specific antibodies against Flavivirus, this ELISAarray might be able to test more pathogens and be of greater potential use. To make the assay more amenable to multiple virus detection, the assay protocol was optimized. In addition to the dotting buffer, the capture antibody concentration and the different virus inactivation methods (heating and β -propiolactone) were also compared and evaluated. Heat inactivation was performed by heating the viral cultures at 56°C for 1 h, and β -propiolactone inactivation was performed by adding β -propiolactone into the retains better antigenicity than the heat-inactivation method. Thus, β -propiolactone treatment was chosen as the virus-inactivation method. A conventional ELISA is a standard method in many diagnostic laboratories. We compared the ELISA-array with a conventional ELISA and confirmed that the advantage of the ELISA-array was evident with comparable specificity and higher sensitivity than ELISA. The time required for the ELISA-array is significantly less than for conventional ELISA (4 h vs. a minimum of 6 h, respectively). Furthermore, less IgG is required for printing than for coating ELISA plates. Coating of a single well in microtiter plate requires 100 μ l of a 1 μ g/ml antibody solution, which is equivalent to 100 ng of IgG. For the ELISA-array, only 30 nl of a 50 μ g/ml antibody solution is required for each spot, which is equivalent to 1.5 ng of IgG. With the characteristics of ease of use, sensitivity, specificity, and accuracy, the ELISA-array assay would be widely accepted for clinical use.

--- END OF ARTICLE ---

--- START OF ARTICLE 6 ---

Changes in pulmonary tuberculosis prevalence: evidence from the 2010 population survey in a populous province of China

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Abstract: **BACKGROUND:** This paper reports findings from the prevalence survey conducted in Shandong China in 2010, a province with a population of 94 million. This study aimed to estimate TB prevalence of the province in 2010 in comparison with the 2000 survey; and to compare yields of TB cases from different case finding approaches.

METHODS: A population based, cross-sectional survey was conducted using multi-stage random cluster sampling. 54,279 adults participated in the survey with a response rate of 96%. Doctors interviewed and classified participants as suspected TB cases if they presented with persistent cough, abnormal chest X-ray (CXRAY), or both. Three sputum specimens of all suspected cases were collected and sent for smear microscopy and culture. **RESULTS:** Adjusted prevalence rate of bacteriologically confirmed cases was 34 per 100,000 for adults in Shandong in 2010. Compared to the 2000 survey, TB prevalence has declined by 80%. 53% of bacteriologically confirmed cases did not present persistent cough. The yield of bacteriologically confirmed cases was 47% by symptom screening and 95% by CXRAY. Over 50% of TB cases were among over 65's. **CONCLUSIONS:** The

prevalence rate of bacteriologically confirmed cases was significantly reduced compared with 2000. The survey raised challenges to identify TB cases without clear symptoms.

Text: China, with an estimated prevalence of all TB cases of 108 per 100,000 in 2010, has the second highest TB burden in the world, accounting for 13% of all cases worldwide [1]. The World Health organization (WHO) estimated that China had reached the targets of 85% treatment success by 1993 and 70% case detection rate by 2005 [2]. National TB prevalence surveys were conducted in China in 1979 China in , 1990 China in , 2000 , and 2010 [4]. Survey results provide more accurate estimates for TB prevalence rates than the WHO estimates and can be used to assess the likelihood of China achieving global targets for TB prevalence.

Shandong province has a population of 94 million. It is a relatively developed province with a per capita GDP 1.6 times of the national average in 2010 [5]. The prevalence rate of TB in Shandong was lower compared with the average rate of China in 2000 [3]. Population representative samples were drawn in Shandong in the surveys of 2000 and 2010 using similar methods. The study aimed to estimate the TB prevalence in Shandong based on the 2010 survey, and compare results of the two cross sectional surveys.

The target population of the TB prevalence survey was residents of 15 years old or above who had lived in the selected clusters for more than 6 months. A population based, cross-sectional survey was conducted using multistage random cluster sampling method.

The survey employed the same sampling methods as the China national survey in 2010, which was similar to the sampling methods used in 2000 [6]. The design of the surveys was in accordance with WHO recommendations [7]. The design effect factor due to cluster sampling was estimated at 1.28 [8]. A sample size of 52500 adults (≥ 15 years old), in 35 clusters, was calculated based on detecting a change of 20% in prevalence rate of TB smear positive cases compared with the rate of the 2000 survey (95 per 100,000), with a probability greater than 95% and 95% power, accounting for 90% response rate of participants [9].

A stratified multi stage random sampling was used to select the 35 clusters within 17 prefectures in Shandong province. The number of clusters was randomly allocated in proportion to the provincial population at the prefectural, county/district and township levels. A cluster was defined as a community (a village in the rural area or a resident community in an urban area) with a population of 1250 to 1750 adults (i.e., those of 15 years or older). If the community contained less than 1250 adult residents, the neighboring community to the north was annexed. If the community or combined communities containing more than 1750 adults, we randomly selected households and then included all adults in the household for the survey until the total number of selected adults reached 1750. Military barracks and prisons located in the cluster were excluded [7] .

The survey was conducted from March to June 2010 by survey teams consisting of clinicians, public health doctors, radiologists, laboratory technicians and nurses. Local media was used to promote awareness of the survey. Community workers conducted a house-to-house census to update the database of residents, inform survey participants and obtain informed consent. The study did not involve children under 15 years old. Written informed consent was obtained from all participants of 16 years old or above. While from those of 15 years old, written informed consents were obtained from their parents or guardians. All documents were properly stored in the Shandong Chest Hospital. Ethical approvals for the study and consent procedures were obtained from the Institutional Review Board (IRB) of Shandong Chest Hospital (NIH register number IRB00006010).

Those who agreed to participate in the survey were invited to the county TB dispensary, where they completed a consultation with a trained clinical TB doctor regarding any symptoms suggestive of TB, such as persistent cough (lasting two weeks or longer), haemoptysis, weight loss and fever. All participants had a chest X-ray (CXRAY) taken that then were reviewed by a panel of radiologists. Those with symptoms or CXRAY films suggestive of TB were classified as suspected TB cases. All suspected cases were asked to produce three sputum samples, one at the time of consultation, another at night and the third in the early morning of the following day. Identified suspects completed an additional questionnaire regarding their social-economic situation, smoking status, and the presence of TB related symptoms in the preceding six months (cough, fever, weight loss, chest pain and haemoptysis).

Sputum smears were conducted in local TB dispensaries. All sputum samples were cultured using the Löwenstein-Jensen medium in the provincial laboratory within 24 hours using cold chain transportation. Samples were excluded from TB when non-tuberculosis bacilli were identified from the culture. All sputum smear and culture were conducted strictly according to the national TB laboratory external quality control measure, which is consistent with the WHO TB prevalence survey guideline [7]. TB classification was made according to the China national TB guideline [10]. A positive smear had at least one acid fast bacillus identified during examination of at least 100 fields. Participants with positive sputum smear specimens were classified as sputum positive cases. Those with positive smear or culture sputum specimens were classified as sputum bacteriologically confirmed cases. Those being culture negative with abnormal CXRAY suggestive of TB and having been ruled out from other diseases by clinicians and radiologists were classified as CXRAY suggestive bacteriologically negative cases. Due to resource limitations the recommendation of broad-spectrum antimicrobial agents to confirm the diagnosis of negative TB cases was not applied in this survey [11]. Newly diagnosed cases were distinguished from previously diagnosed cases through checks during the interviews and against the TB registration system. Initial diagnosis was made by a group of local clinicians and radiologists. Subsequently, samples and CXRAY films of all suspected and confirmed cases were re-assessed by a group of senior clinicians and radiologists at provincial and national levels. CXRAY films of 100% of those scored as abnormal and 10% random sampling of those scored as normal were transferred for independent reading. The provincial laboratory team randomly examined one slide from the three samples of sputum positive cases, all three samples of CXRAY suggestive TB cases, and randomly selected 10% of the non-TB cases.

Prevalence estimates of sputum positive, bacteriologically confirmed and all TB cases were calculated. In all analyses, population weightings were employed to adjust for the stratified multi-stage sampling design effect [8]. The survey results in 2010 and 2000 were standardized against the age structures of China's census population in 2010. The 2000 TB prevalence survey included all age groups [12]. The WHO recommended method was used to enable comparison between the two surveys that prevalence rates of child TB were assumed to reduce to the same extent as adult TB from 2000 to 2010 [13]. Subgroup analysis in gender, age groups and urban/rural residence were conducted. Case identification rate was calculated as the number of cases identified by a screening method over all suspected cases found by the method. Yields of the symptom consultation and CXRAY were calculated as a proportion of the total number of bacteriologically confirmed cases.

The survey selected 17 urban clusters and 18 rural clusters. It covered a total population of 89,093, of which 56,671 were eligible for the survey (Figure 1). The response rate ranged from 95% to 97% in different clusters. 54,279 participants attended clinical consultation and were examined by CXRAY. Among them, 47% were males. The average age was 46 years with 14% of 65 years and older. A total of 572 suspected TB cases were found. Of these, 264 (46%) were identified based on CXRAY abnormalities, 228 (40%) were based on persistent cough, 80 (14%) were based on both. The survey diagnosed 172 new cases, including 19 new bacteriologically confirmed cases (including 11 sputum and culture positive cases, and 8 sputum negative but culture positive cases) and 153 CXRAY suggestive bacteriologically negative cases. The survey also identified 11 existing cases registered on the national TB program. In addition, the survey found four cases with culture positive non-tuberculosis bacilli, and excluded them from TB patients.

All participants of the survey were first screened by symptoms and CXRAY. Those who had symptoms of consistent cough or haemoptysis, or CXRAY abnormalities were then screened by smear and culture. Case identification rates of new bacteriologically confirmed cases from the suspected cases were significantly higher with CXRAY as a primary tool (Figure 1 , 3.8%, $P = 0.012$) and further increased by both symptom screen of persistent cough and CXRAY (10%, $P < 0.001$) compared with symptom screen alone (0.4%). The same pattern of case identification rate was observed in the sputum positive cases (7.5%, 1.9% and 0% respectively). The proportion reporting persistent cough was not significantly higher among bacteriologically confirmed cases compared with other suspects ($P = 0.565$). The symptom consultation alone identified 308 suspects, including 6 (1.9%) sputum smear positive TB and 9 (2.9%) bacteriologically confirmed TB. Among the 344 suspects with CXRAY abnormalities, 11 (3.2%) had sputum positive TB and 18 (5.2%) had bacteriologically confirmed TB. The yield of bacteriologically confirmed cases was 47.4% by screening consultation and 94.7% by CXRAY. In the population of over 65 years old, symptom consultation and the CXRAY identified 174 and 182 suspected cases respectively, yielding 5 (2.9%) and 9 (4.9%) of bacteriologically confirmed cases. Yields of bacteriologically confirmed cases were 55.6% by symptom consultation and 100% by CXRAY among over 65's.

Of the 512 suspected cases that completed the additional questionnaire, 42% were farmers and 31% were current smokers (Table 1) . Per capita household income of

bacteriologically confirmed cases was less than 50% of that of the non-TB cases ($P < 0.05$). Though smoking rate was higher among TB cases compared with non-TB cases, no significant differences were found ($P > 0.05$). Of the ten bacteriologically confirmed cases not presenting with persistent cough at the prevalence survey, one coughed for two days, one had chest pain, and the other eight had no symptoms of TB in the last six months.

The crude prevalence rate in Shandong in 2010 of sputum positive cases was 22.1 (95% CI: 9.6-34.6), bacteriologically confirmed cases was 36.8 (95% CI: 17.8-55.8), and all cases were 337.1 (95% CI: 254.1-420.0) per 100,000 in adult population (Table 2). The adjusted prevalence rates of the whole population in Shandong were 17.8 (95% CI: 8.3-17.5), 27.8 (95% CI: 14.8-28.0) and 239.4 (95% CI: 179.9-298.9) per 100,000 in 2010. A remarkable decline of 82.0%, 80.2% and 31.4% was observed in TB prevalence rates of sputum positive, bacteriologically confirmed, and all cases, respectively, compared to the adjusted rates in 2000 [12] . Large declines were observed in males between 40 and 65 years old, and in females over 60 years old (Figure 2) .

The adjusted prevalence rates in the adult population were 21.4 (95% CI: 10.0-32.8), 33.5 (95% CI: 17.8-49.2) and 285.8 (95% CI: 254.2-356.4) for sputum positive cases, bacteriologically confirmed cases and all cases, respectively. Significant differences regarding adjusted TB prevalence rates were observed between males and females, over 65's and 15 to 64 years old, in rural and urban areas (Table 2 , $P < 0.001$). The male to female ratios were 5.5 in sputum positive cases and 2.8 in bacteriologically confirmed cases, while the ratios climbed to 6.0 and 4.1, respectively, among those over 65 years. The majority of TB patients, 54.5% of sputum positive cases and 47.3% of bacteriologically confirmed cases, were from people 65 years or older. The ratio between over 65's and 15 to 64 years old was 8.4 in sputum positive cases and 5.9 in bacteriologically confirmed cases. The ratio between rural and urban areas was 2.7 in sputum positive cases and 4.8 in bacteriologically confirmed cases.

The most striking finding was that a large proportion of TB patients did not present consistent cough. Passive case finding is the routine practice in developing countries where sputum microscopy is performed to identify TB cases among people with persistent cough. A large proportion of TB cases may be missed using this method as 53% of bacteriologically confirmed cases and 45% sputum positive cases in this study had no persistent cough but were identified through abnormal CXRAY. Nearly half of

bacteriologically confirmed cases reported no symptoms in the last six months. This finding, although initially surprising, is consistent with reports from Vietnam (47% of bacteriologically confirmed cases not presenting persistent cough) [14], Myanmar (38%) and Ethiopia (48%) [13]. CXRAY was sensitive in detecting TB cases, as yields of bacteriologically confirmed cases were much higher by CXRAY compared with by symptom screening, as reported in Vietnam [15] and some high HIV prevalence settings [16, 17]. CXRAY, though expensive at the initial installment, may improve TB case finding due to its short turnover time and high throughput [18]. Our findings suggest that the strategy of case finding using CXRAY followed by sputum or culture as the primary and secondary screening tests could be more effective, especially among the population of over 65 year olds, as the yields were higher in over 65's compared with the general Table 2 Prevalence rates of sputum positive TB cases, bacteriologically confirmed TB cases and all cases in Shandong, China, 2010 No population. Although using CXRAY to examine everyone is not feasible, it can be used in routine elder physical examinations. The China public health package now covers free CXRAY for elders, as well annual employee body examinations provided free CXRAY.

In this survey, only one sputum positive patient had been detected and treated by the national program, though specific clinical consultation was conducted to identify any patients who have been diagnosed and treated for TB before. This may reflect the difference between the active case finding approach in the survey and the passive casing finding approach in practice. Nevertheless, it indicated that a large proportion of bacteriologically confirmed TB cases are missed by the national TB program.

Another notable change is the sharp decline of the proportion of sputum positive cases, which accounted for 30.5% of all cases in the 2000 survey but was reduced to 6.6% in the 2010 survey. The proportion of notified sputum cases out of all TB cases in Shandong also declined from 80.9% in 2005 to 64.6% in 2010 [19].

The prevalence rate of bacteriologically confirmed cases has reduced by 80% in the last decade in Shandong, compared with a national decline of 45% (from 216/ 100,000 in 2000 to 119/ 100,000 in 2010) [4]. The rapid decline of TB prevalence rate of bacteriologically confirmed cases in the recent decade may be attributed to China's strengthened public health system following the outbreak of severe acute respiratory syndrome in 2003 [2]. Another reason may be due to improved reporting of TB cases in the online communicable

disease reporting system, and the improved collaboration between public hospitals and TB dispensaries [20] . Other factors such as social economic development may also have played an important role in the reduction of TB prevalence, as found in a study of TB notification rates trends in 134 countries [21] .

The adjusted prevalence rate of bacteriologically confirmed cases in Shandong was lower than the WHO estimates for China in 2010 [1] . But the national prevalence rates of bacteriologically confirmed cases, 119/100,000 in 2010 [4] , was higher than the WHO estimate, 108/ 100,000, even the survey did not collect negative and extra-pulmonary TB cases. Vietnam reported similar findings in its 2006 survey [14] . One reason is that prevalence surveys results are based on active case finding while WHO estimates are based on notification rates from passive case finding. A re-evaluation of the reported TB prevalence in China is needed based on the recent survey.

CXRAY suggestive bacteriologically negative cases may be smear or culture negative TB cases if they had any TB symptoms, while some may be caused by suboptimal smear or culture. As reported in China's previous surveys [3, 22] , including these cases as TB cases may result in an over-estimate of all pulmonary cases [23] .

The survey revealed that over half of the TB patients were 65 years and older in Shandong, while the over 65's were more likely to present with abnormal CXRAY and persistent cough. Similar trends have been documented in other developed cities such as Hong Kong and Singapore [24] . These high rates may reflect the higher TB rates in the past and decline in immunity in the over 65's. How to treat elders with TB and other complications such as diabetes remains an ongoing challenge in China and similar settings.

The survey results can be generalized to the Shandong population of 94 million or similar international settings with middle income and middle TB prevalence levels. The patterns of the TB epidemic found in Shandong, i.e., the proportion of patients with symptoms, ratios between urban and rural areas, men and women, were similar to those found in the national survey [4] . However, the prevalence rates cannot be extrapolated to western provinces in China with a higher TB prevalence. For logistical reasons, the eligible population did not include adults staying in the sampled clusters less than 6 months, which was the same practice in the 2000 survey. However, shortterm migrants may have a

potentially higher prevalence of TB than the general population [25] . This may result in a lower estimate of the true prevalence rate. The survey did not collect social-economic indicators, smoking status and HIV status of all participants, so comparisons between TB cases and all non-TB patients are not available. However, the HIV prevalence in Shandong China is below 0.01%, and would not significantly alter the TB prevalence rate. In addition, the survey did not evaluate child TB and extra pulmonary TB. Discussions of using CXRAY as a screening tool was on the technical aspect, but not on costing side as we did not conduct any cost effectiveness analysis or the social willingness to pay for such a strategy in similar settings.

This study has shown that the prevalence of bacteriologically confirmed TB in Shandong has reduced substantially over the last decade. Importantly, the majority of these cases did not present with persistent cough and the proportion of sputum positive cases has declined sharply. Further studies are recommended to assess the feasibility of adopting CXRAY in the existing health care services to detect TB cases and the cost effectiveness of such intervention.

The authors declare that they have no competing interests.

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--- START OF ARTICLE 7 ---

Design, Synthesis, Evaluation and Thermodynamics of 1-Substituted Pyridylimidazo[1,5-a]Pyridine Derivatives as Cysteine Protease Inhibitors

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Abstract: Targeting papain family cysteine proteases is one of the novel strategies in the development of chemotherapy for a number of diseases. Novel cysteine protease inhibitors derived from 1-pyridylimidazo[1,5-a]pyridine representing pharmacologically important class of compounds are being reported here for the first time. The derivatives were initially designed and screened in silico by molecular docking studies against papain to explore the possible mode of action. The molecular interaction between the compounds and cysteine protease (papain) was found to be very similar to the interactions observed with the respective epoxide inhibitor (E-64c) of papain. Subsequently, compounds were synthesized to validate their efficacy in wet lab experiments. When characterized kinetically, these compounds show their $K(i)$ and $IC(50)$ values in the range of 13.75 to 99.30 μM and 13.40 to 96.50 μM , respectively. The thermodynamics studies suggest their binding with papain hydrophobically and entropically driven. These inhibitors also inhibit the growth of clinically important different types of Gram positive and Gram negative bacteria having $MIC(50)$ values in the range of 0.6–1.4 $\mu g/ml$. Based on Lipinski's rule of Five, we also propose these compounds as potent antibacterial prodrugs. The most active antibacterial compound was found to be 1-(2-pyridyl)-3-(2-hydroxyphenyl)imidazo[1,5-a]pyridine (3a).

Text: Cysteine-protease inhibitors (CPI) have gained considerable attention over the last couple of decades and many classes of compounds are currently in human clinical trials for a number of diseases. Interest in papain family cysteine proteases as chemotherapeutic targets is derived from the recognition that they are critical to the life cycle or pathogenicity of many microorganisms. The cysteine proteases from *Streptococcus* sp. (streptopain) [1], *Staphylococcus* sp. (staphopain) [2], *Plasmodium falciparum* (falcipain-1, -2, and -3) and *Trypanosoma cruzi* (cruzipain) [3] are some of the most widely studied members of papain family which have been reported to be linked with severity of infection and various pathological conditions caused by these microorganisms.

The activation of the kallikrein-kinin pathway, which could be activated by more than sixteen bacterial proteases, is a mechanism that some pathogens exploit to ensure that

there is a supply of nutrients to the site of infection by increasing vascular permeability. This has been shown to occur in infections with several microbial species, including *Pseudomonas*, *Serratia*, *Clostridium*, *Candida*, *Bacteroides*, *Porphyromonas* and *Staphylococcus* sp. [4] . Many bacteria secrete several nonspecific proteases e.g. *Pseudomonas*, *Serratia*, *Streptococcus*, *Staphylococcus* and *Bacteroides* sp. have potent metallo-, cysteine and serine proteases with broad ranges of activities [5] . The critical role of bacterial proteases in virulence was successfully demonstrated by eliminating the proteaseencoding gene in *P. gingivalis* [6] .

Recently described cystatin superfamily of proteins comprises both eukaryotic and prokaryotic cysteine protease inhibitors [7] . Human cystatins C, D and S, rat cystatins A and S, chicken cystatin and oryza cystatin have been reported to inhibit the replication of certain viruses and bacteria [8] although it has not yet been directly demonstrated that these effects are due to the protease inhibitory capacity of the cystatins [9] . The key role of cysteine proteases in microbial infections, coupled with the relative lack of redundancy compared to mammalian systems has made microbial proteases attractive targets for the development of novel chemotherapeutic approaches [10, 11] .

Imidazopyridine ring systems represent an important class of compounds not only for their theoretical interest but also from a pharmacological point of view. They have been shown to possess a broad range of useful pharmacological activities [12] including antigastric, antisecretory, local anesthetic, antiviral, antianxiety, antibacterial, antifungal, antihelminthic, antiprotozoal, anticonvulsant, gastrointestinal, antiulcer (Zolmidine), anxiolytic (Alpidem), hypnotic (Zolpidem) and immunomodulatory [13] . The nature and the position of the substituents on the pyridinic moiety influence these pharmacological activities. These imidazopyridine heterocyclic structures form part of the skeleton of natural alkaloids, neuromuscular blocking agents [14] , reversible inhibitors of the H^+ , K^+ -ATPase enzymes with a potent antisecretory activity, and are known to be sedative hypnotics of the nervous system [15] . In this study, we have proposed kinetically and thermodynamically characterized 1-substituted pyridylimidazo[1,5-a]pyridine derivatives as a potent and novel cysteine protease inhibitors which also acts as antibacterial agents.

The crystal structure of papain was extracted from Protein Data Bank (PDB code: 1PE6) [16] . All the water molecules and heteroatoms were removed and hydrogen atoms were added to the protein. CharMm forcefield [17] was applied and the structure was subjected to

energy minimization for 1000 steps using steepest descent method. The chemical structures of all the synthesized compounds were generated using chemdraw and were subsequently converted into 3D format using CORINA. A series of docking experiments were carried out with all the designed 1substituted pyridylimidazo[1,5-a]pyridine derivatives against papain using AutoDock Tools 4.0 [18] for possible cysteine-protease inhibitory activities. The compounds were selected on the basis of their binding energies and those reflecting good binding affinity were further analyzed on in silico platform. As a parameter for the molecular docking, the Lamarckian genetic algorithm, a combination between the genetic algorithm and the local search Pseudo-Solis and Wets algorithm, was employed. A grid box of 60660660 Å was generated around active site of papain making sure those inhibitors can freely rotate inside the grid. The number of docking runs was set to 10. Each docking was repeated five times, having in the end a total of 50 docking runs, to check the precision of results. The finally obtained docked complexes were subsequently visualized using PyMol [19] . The work was further authenticated in the wet lab after its detailed analysis on in silico platform.

The designed derivatives were filtered by Lipinski's "Rule of five" that sets the criteria for drug-like properties. Drug likeness is a property that is most often used to characterize novel lead compounds [20] . According to this rule, poor absorption is expected if MW .500, log P.5, hydrogen bond donors .5, and hydrogen bond acceptors .10 [21] . In silico absorption, distribution, metabolism and excretion (ADME) properties of these derivatives were also predicted using following online bioinformatics tools.

N <http://www.organic-chemistry.org>. N <http://mobyli.rpbs.univ-paris-diderot.fr/cgi-bin/portal.py?Form=admetox> N <https://secure.chemsilico.com/pages/submit.php>

The above study gave us an idea about the existence of possible mutagenic and tumorigenic properties in synthesized compounds. The result obtained helped us to screen out the synthesized compounds for their further usage as potent leads.

Based on the results of docking studies, ten derivatives of 1pyridylimidazo[1,5-a]pyridine were synthesized according to Siddiqui et al., 2006 [22] which are named as follows: 1-

The capacity of the 1-pyridylimidazo[1,5-a]pyridine derivatives to inhibit cysteine proteases was tested using papain as the model enzyme. The proteolytic activity of the reaction mixtures was determined using Bz-DL-Arg-pNA as the chromogenic substrate [23]. To solutions of active papain (final concentration: 0.05 mM) were added concentrated solutions of the different derivatives to final concentrations of 0.2 mM. After incubation for 30 min at 37°C, the substrate solution was added and after a further incubation for 20 min the reaction was stopped by the addition of 5% trichloric acid (TCA) acidified with 2.25% HCl and the absorbance of the reaction mixture was determined at a wavelength of 410 nm by Microplate Manager 4.0 (Bio-Rad laboratories). The same procedure was used at 32°C and 42°C for thermodynamics studies. The kinetic parameters for the substrate hydrolysis were determined by measuring the initial rate of enzymatic activity. The inhibition constant K_i was determined by Dixon method [24] and also by the Lineweaver-Burk equation. The K_m value was calculated from the double-reciprocal equation by fitting the data into the computer software Origin 6.1. The Lineweaver-Burk plot was used to determine the types of inhibition. For the kinetic analysis and rate constant determinations, the assays were carried out in triplicate, and the average value was considered throughout this work. Temperature dependence of the inhibition constants was used to determine the thermodynamic parameters. Changes in enthalpy (ΔH) were determined from the Van't Hoff plots by using the equation,

Where ΔH is enthalpy change, R is gas constant, ΔS is entropy change and T is the absolute temperature. The entropy change was obtained from the equation,

The assay was done at different temperatures (32°C, 37°C, 42°C) calculating various K_i of 1-pyridylimidazo[1,5-a]pyridine derivatives with papain as model enzyme.

The disk diffusion method [25] was used for the preliminary antibacterial evaluation of 1-pyridylimidazo[1,5-a]pyridine derivatives. The MIC 50 of these derivatives, showing inhibition in the preliminary tests, were further determined by the microtitre plate technique using micro dilution method [26]. In brief, the bacterial strains (*S. aureus*, *P. vulgaris*, Group D Streptococci, *Bacillus* sp., *E. coli*, *P. aeruginosa* and *S. morganii*) were grown and diluted to 2.6×10^5 colony-forming units (CFU)/ml in sodium phosphate buffer (SPB) containing 0.03% Luria-Bertani (LB) broth. The synthesized derivatives were dissolved in DMSO and their serial dilution was performed in 50 mL of LB medium in 96-well microtitre plate to achieve the required concentrations (0.1-10 mg/ml) with bacterial

inoculums (5610 4 CFU per well). DMSO was taken as negative control and Ceftriaxone and clotrimazole were taken as positive control. After incubation at 37°C overnight, the MICs were taken as the lowest inhibitor concentration at which the bacterial growth was inhibited. The average of three values was calculated and that was the MIC for the test material and bacterial strain.

For the agar plate count method [27], 25 mL aliquots of bacteria at 1610 5 CFU/ml in SPB containing 0.03% LB broth were incubated with 25 mL of diluted compounds for 2 h at 37°C. The mixtures of bacteria and compounds were serially diluted 10-fold with SPB and plated on LB plates that were incubated at 37°C overnight. Bacterial colonies were enumerated the following day.

After having determined the MICs, bacterial strains from the wells of the microtitre plate with no visible bacterial growth were removed for serial sub cultivation of 2 ml into another new microtitre plate containing 100 ml of broth per well and further incubated for 24 h. The lowest concentration with no visible growth was defined as MBC [28], indicating 99.5% killing of the original inoculum. The absorbance of each well was measured at a wavelength of 620 nm by Microplate Manager 4.0 (Bio-Rad laboratories) and compared with a blank. Solvent (DMSO) was used as a negative control. Three replicates were done for each compound and experiment was repeated two times.

Bacteria use their cysteine proteases for pathogenicity as could be depicted from the structure of Cif homolog in *Burkholderia pseudomallei* (CHBP) which reveals a papain-like fold and a conserved Cys-His-Gln catalytic triad [29]. It has been proven that bacterial pathogens have a unique papain-like hydrolytic activity to block the normal host cell cycle progression as the core of an avirulence (Avr) protein (AvrPphB) from the plant pathogen *Pseudomonas syringae*, resembles the papain-like cysteine proteases. The similarity of this AvrPphB protein with papain includes the catalytic triad of Cys-98, His-212, and Asp-227 in the AvrPphB active site [30].

Turk et al. have proposed, on the basis of kinetic and structural studies, that papain has seven subsites at the active site but only five subsites are important which can bind to an amino acid residue of the substrate [31]. A variety of intermediates are generated when papain reacts with substrate or an inhibitor [2]. Like serine proteases, cysteine proteases

tend to have relatively shallow, solvent-exposed active sites that can accommodate short substrate/inhibitor segments of protein loops (e.g. from endogenous inhibitors such as cystatins) or strands. The inhibitor Table 3 . Name, Structure, IC₅₀ & K_i of 1-substituted pyridylimidazo[1,5-a]pyridine derivatives against cysteine protease papain.

Type of inhibition K_i (mM) IC₅₀ (mM)

Non-Competitive 13.7 13.4 compound bound to protease with a combination of hydrogen bonds and hydrophobic interactions. As a part of our investigation in developing novel and efficient cysteine protease inhibitors, ten 1-substituted pyridylimidazo [1,5a] pyridine derivatives (3a-j) were primarily designed and screened on the basis of their docking energies against papain to elucidate their possible mode of action. It was found that these compounds were specific inhibitors of cysteine protease, papain and didn't show inhibition against other types of proteases like serine, aspartic or metalloproteases. They are specific for CA clan of cysteine protease and didn't show any significant inhibition against other clans of cysteine proteases.

These new compounds were devised based on the knowledge of ability of a protein to alter its conformation to accommodate a binding ligand and enabled us to directly compare the relative positions of the residue in the binding pocket. Molecular docking study provided the structural insight into the binding of these compounds (3a-j) (Figure 1) within the active site of papain which mainly consist of a catalytic triad of Cys 25, His 159 and Asp 175 [32] . Moreover, role of other residues present in the active site of papain, playing important role in the accommodation of compounds have also been revealed. Initially, docking was performed with all the designed compounds (3a-j) against papain, a known cysteine protease enzyme and in this context, we observed very interesting results where our proposed inhibitors (3a-j) take advantage of aromatic and hydrophilic residues by making a variety of interactions with target enzyme. Although, compounds 3e-j gave significant results when docked with papain but during evaluation of antibacterial properties in wet lab experiments, they gave insignificant results (data not shown). Therefore, only four compounds were considered for discussion and further experiments like kinetic and thermodynamic studies to characterize these compounds as potent pro-inhibitors, were performed (3a-d).

The findings of the above study have shown that the molecular interactions between the compounds 3a-d and papain were very similar to the interactions observed for E-64c, a derivative of naturally occurring epoxide inhibitor (E-64c) (Figure 1) of cysteine proteases [31, 32] , with papain; especially with regard to the hydrogen bonding and hydrophobic interactions of the ligands with conserved residues in the catalytic binding site (Figure 2 A-D). Several papain residues participated in hydrophobic interactions with compounds 3a-d, including Gln19, Cys25, Gly66 and Asp158. The pyridine moieties of compounds 3a-d interact with S2 site of papain which includes (Tyr61, Asn64, Gly65 & Tyr67) amino acids (Figure 2 A-D) . The active site residues that were found to be key player in the interaction of compounds within the active site (mostly through hydrophobic interactions) were Cys25, Tyr61, His159 and Trp177, while Trp177, Gln19 were found to be making hydrogen bonds only with compound 3a. Besides this many other residues were also found to be actively involved (Table 1) . Furthermore, the binding energies for the compound 3a, 3b, 3c and 3d with papain were found to be 26.12, 25.76, 26.84 and 25.62 Kcal/mol respectively, which were in great agreement with our wet lab experiments; shall be discussed later (Table 1) . This confirmed the accuracy of our docking protocol. Since, the binding energy is a direct measure of strength of interaction and our compounds 3a-d showed stronger binding within the active site of papain in comparison to the inhibitor E-64c (DG: 24.04 Kcal/mol), therefore, the results suggest that these 1-substituted pyridylimidazo[1,5-a]pyridine derivatives (3ad) could be potent inhibitors of papain like cysteine proteases.

The in silico interaction of compounds 3a-d with papain, which were observed as discussed above, was validated with wet lab Table 5 . Prediction of antibacterial compounds as drugs (<http://www.organic-chemistry.org>). Table 2) . Interestingly, the observed in silico binding energies for the compounds 3a-d against papain were found to be in great agreement (standard error 62 Kcal/mol) with the value of free energy of binding (DG) observed during thermodynamics studies (Table 1 and 2) . Similarly, enthalpy change (DH) of the binding was negative whereas entropy (DS) change of the binding was positive which indicated the exothermic and entropically driven nature of binding. This pattern of temperature dependence is characteristic of hydrophobic interaction [33] . As discussed earlier that all the compounds (3a-d) were found to interact with the active site residues of papain through hydrophobic interactions at most instances during in silico studies, the same was observed by the analysis of Van't Hoff plots for all the proposed inhibitors at three different temperatures (32uC, 37uC and 42uC) in wet lab experiments (Figure 3) . This proves the importance of these types of interactions in the positioning of compounds within the active site. Hence, thermodynamics as well as in silico study reveals that hydrophobic interactions favor binding of these proposed inhibitors with papain like

cysteine proteases. Further wet lab results proposed the non competitive interaction of compounds (3a, 3c & 3d) with papain except for compound 3b which showed competitive interaction. In sum up, the above results of molecular docking studies and thermodynamic analysis of compounds 3a-d with papain showed that these compounds have the potential to be novel and unique cysteine protease inhibitors.

In the current study, the cysteine protease inhibitory activity of synthesized derivatives of 1-substituted pyridylimidazo[1,5-a] pyridine (3a-d)) was also performed against papain and the inhibition constants (K_i) for the above said enzyme were observed to be 13.70, 23.20, 90.00 and 99.30 mM for compounds 3a, 3b, 3c and 3d respectively (Table 3) . Furthermore, the calculated IC 50 values were also found to be 13.40, 21.17, 94.50 and 96.50 mM for compounds 3a, 3b, 3c and 3d respectively (Table 3) . Except compound 3b, rest of the compounds showed non competitive, reversible inhibitions but all the compounds irrespective of types of binding, showed hydrophobic and entropically driven interaction. These derivatives (3a-j) were eventually evaluated for their antibacterial activities against seven clinically important microbes (*S. aureus*, *P. vulgaris*, Group D Streptococci, *Bacillus* sp., *E. coli*, *P. aeruginosa* and *S. morganii*). Here, we are showing the data of only four compounds (3a-d) because of their significant results (Table 4). All the compounds strictly followed the pattern of antiprotease activity towards bacterial growth except *P. vulgaris* and *E. coli* at one instance each (Table 4) . Since compound 3c & 3d do not have much difference in their IC50 values (3c-94.5 mM and 3d-96.5 mM) against cysteine protease, papain and hence in antibacterial activity in all instances except one. It might be random due to so close in IC50 values. Compounds 3c & 3d are having much difference in their IC50 values (3b-21.17 mM and 3c-94.5 mM) and they showed exact pattern for their antibacterial activity for all microbes except for *E. coli* at one instance. Although, *E. coli* does contain six major cysteine proteases but none belong to the CA clan of papain. It is argued that these compounds also inhibited the cysteine proteases of other clan than papain but with low efficacy.

Since, pyridylimidazo[1,5-a]pyridine derivatives is absolutely new scaffold towards antibacterial agents and hence, not any standard compound(s) of same scaffold is available for reference. So, Clotrimazole (1-[(2-chlorophenyl)(diphenyl)methyl]-1H-imidazole), an imidazole derivatives and Ceftriaxone (third-generation cephalosporin antibiotic with broad spectrum activity against Gram-positive and Gram-negative bacteria) have been used as positive control whereas DMSO has been used as negative control. All the above mentioned bacterial species have been shown to secrete certain cysteine

proteases which play very important role in the pathogenecity of different diseases caused by these microorganisms. The minimum inhibitory concentration (MICs) of compounds (3a-d) (Table 4) against all tested bacteria except *E. coli* and *P. vulgaris*, were observed to be in great agreement with their respective inhibition constant (K_i)/IC 50 values against papain (Table 3) which clearly indicates that these compounds have the potential to inhibit the papain like cysteine proteases of these pathogens. The partition coefficient (logP) is a well-established measure of the compound's lipophilicity. The distribution of calculated logP (cLogP) values of a majority of drugs in the market is in the range of zero to five. All the compounds studied except 3d, showed good agreement with the criteria laid down for the prediction of a compound to be a potential drug (Table 5). All the compounds do not show any threat against toxicity risk assessment except compound 3d which showed threat as tumorigenic effect due to the presence of isobutyl group. Among all the tested compounds, compound 3a was the most potent whose MIC was the lowest among all the tested compounds and showed maximum drug score and positive values for drug likeness.

In summary, the results of the present study have established that 1-substituted pyridylimidazo[1,5-a]pyridine derivatives could be candidate for novel and potent inhibitors of papain like cysteine proteases, which play deleterious role in the progression of different diseases caused by diverse microorganisms. Therefore, this group of compounds could be the subject of future research to confront the challenges with resistant microorganisms that is a major threat globally.

File S1 Types of inhibitions with K_i (Compounds 3a-3d).

(DOC)

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--- START OF ARTICLE 8 ---

RNAi Therapeutic Platforms for Lung Diseases

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3816685/>

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Abstract: RNA interference (RNAi) is rapidly becoming an important method for analyzing gene functions in many eukaryotes and holds promise for the development of therapeutic gene silencing. The induction of RNAi relies on small silencing RNAs, which affect specific messenger RNA (mRNA) degradation. Two types of small RNA molecules, i.e. small interfering RNAs (siRNAs) and microRNAs (miRNAs), are central to RNAi. Drug discovery studies and novel treatments of siRNAs are currently targeting a wide range of diseases, including various viral infections and cancers. Lung diseases in general are attractive targets for siRNA therapeutics because of their lethality and prevalence. In addition, the lung is anatomically accessible to therapeutic agents via the intrapulmonary route. Recently, increasing evidence indicates that miRNAs play an important role in lung abnormalities, such as inflammation and oncogenesis. Therefore, miRNAs are being targeted for therapeutic purposes. In this review, we present strategies for RNAi delivery and discuss the current state-of-the-art RNAi-based therapeutics for various lung diseases.

Text: traditional surgery, Bivas-Benita et al. reported that no mortality occurred as a result of the use of the endotracheal technique. Endotracheal applications are currently being used by many practitioners in the pulmonary field [22, 34] ; this is useful for studying pulmonary drug delivery in mice. However, the approach is more complex in humans because an artificial path for the delivery of drugs into the lungs is used. Therefore, the method is being used in animal models to test and evaluate its reliability for possible clinical applications. Intratracheal route: under anesthesia, the trachea is exposed surgically, and a tube or needle is inserted through an incision made between the tracheal rings. Complications, such as vascular injury and air leakage, are possible due to the tracheotomy. (b) Endotracheal route: siRNAs are sprayed directly from the mouth into the lungs using a MicroSprayer[®] aerolizer (Penn-Century, Philadelphia, PA, USA) and a laryngoscope. It is important to maintain a clear view of the trachea during the procedure.

Intranasal delivery is another common method of pulmonary drug application in animal studies. In many studies, in vivo success has been demonstrated in delivering siRNAs to the lungs intranasally [22, 35, 36]. An experimental setup of intranasal delivery by spray or droplet is simple and painless for the animal. Although the success in delivering siRNAs intranasally in rodents cannot be completely extrapolated to human use because of the significant differences in lung anatomy [37], this approach has potential for the clinical application of siRNAs. Phase II clinical trials have been initiated for the treatment of respiratory syncytial virus (RSV) infection, making use of intranasal application of naked chemically modified siRNA molecules that target viral gene products [17, 38] (see Section 3.1.1. for details).

Intranasal entry has long been used to administer small molecules, such as proteins, for systemic delivery. Because the nasal mucosa is highly vascularized, delivery of a thin epithelium of medication across the surface area can result in rapid absorption of the medication into the blood. Therefore, siRNAs administered intranasally might be deposited in the nose, and some of them may be unable to reach the lower respiratory tract. In fact, it has been reported that intranasal application of unformulated siRNAs resulted in lower delivery efficiency and homogeneous pulmonary distribution than that achieved with intratracheal application [31]. The intranasal method is suitable for some lung diseases, such as upper respiratory infection by RSV, and it also has potential for systemic delivery rather than pulmonary delivery of siRNAs. Therefore, it is important to consider the route of administration in animal studies when assessing the delivery and therapeutic efficacy of a formulation for pulmonary delivery. Careful choice of efficient delivery in response to the condition of lung diseases is necessary.

The use of aerosols to deliver medication to the lungs has a long history. Administration by inhalation is a popular and non-invasive method of delivering agents into the lungs. There are several inhalation devices available for the delivery of drugs into the lungs. Metered dose inhalers (MDIs) and dry powder inhalers (DPIs) are the most common modes of inhaled delivery. MDIs are the most commonly used inhalers for several lung diseases, such as asthma, bronchitis, and chronic obstructive pulmonary disease (COPD), and a spacer is an external device that is attached to an MDI to allow for better drug delivery by enhanced actuation and inhalation coordination. For most MDIs, the propellant is one or more gases called chlorofluorocarbons (CFCs). Although CFCs in drugs are safe for patients to inhale, they are harmful to the environment. Therefore, further development of inhalable siRNAs may not be the best way forward. DPIs are devices that deliver

medication to the lungs in the form of dry powder. The use of DPIs has already shown promise for the in vivo delivery of therapeutic macromolecules such as insulin [39] and low-molecular-weight heparin [40] ; thus, it could be a better device for delivering siRNAs to the lungs. The advantages of DPIs are improved stability and sterility of biomolecules over liquid aerosols and propellant-free formation.

Although drugs are commonly delivered to the lungs by inhalation, most in vivo studies using siRNAs have relied on intratracheal or intranasal delivery. The reason could be the difficulty in formulating inhalable siRNAs and maintaining the stability during the delivery process. A suitable carrier is also needed to protect nucleic acids from degradation due to shear force and increased temperature during the drying process. The use of spray-drying as a technique for engineering dry powder formulations of siRNA nanoparticles, which might enable the local delivery of biologically active siRNA directly to the lung tissue, has been demonstrated [24, 25] . In the future, the technique is desirable to estimate the in vivo study on siRNA therapy for inhalation. In the long term, we anticipate that there will be more sophisticated devices for clinical use and that those currently being developed will be more suitable.

There are two main barriers to efficient pulmonary siRNA delivery to the cells of the lung. The first is the complex, branched anatomy of the lungs and biomechanical barriers, such as the mucus layer covering the airway cells [41, 42] (Figure 2) . A remarkable feature of the respiratory tract is its high degree of branching. Airway consists of respiratory bronchioles, alveolar ducts, and alveolar sacs. All of these structures bear alveoli, the tiny air sacs in which the gas exchange takes place. It is generally acknowledged that the critical factor for efficient siRNA delivery depends on the properties of RNAi drug particles in terms of size, charge, shape, velocity and density. For efficient pulmonary siRNA delivery, the particles must be deposited in the lower respiratory tract. Deposition in the airway is affected by the particle size and patient's pulmonary function. A particle size between 1-5 μm is found to be the most appropriate for deposition at the lower respiratory tract [23] . In addition, the presence of mucus and surfactant proteins, the mucociliary clearance actions, and phagocytosis by macrophages present major barriers to targeted pulmonary delivery. Therefore, delivery systems usually require delivery vectors, and these vectors need to be designed in order to maximize the siRNA deposition to the diseased area of the respiratory tract. Besides, the extracellular barriers to siRNA delivery also depend on physiological features of the respiratory tract, which may change with the disease stage and characteristics of the patient. At the active stage of lung disease, the physiological

conditions of the airways might change and have significant impact on the efficiency of the pulmonary delivery system. During infection, inflammation, and allergic reaction, there is an increase in mucus secretion along with the impaired mucociliary clearance [43]. Moreover, asthma and COPD are both chronic inflammatory conditions of the lung associated with structural "remodeling" that is inappropriate to the maintenance of normal lung function [44]. The airway wall thickness, the high viscosity, and the composition of the mucus layer might be altered in patients who have inflammatory lung diseases. Figure 2 .

Extracellular barriers to pulmonary siRNA delivery. The anatomical feature of the respiratory tract is its high degree of branching. The mucus lines the respiratory epithelium from the nasal cavity to the terminal bronchioles. The deposited particles on the ciliated epithelial cells are rapidly cleared by the mucociliary clearance actions. Mucus and mucociliary clearance of mucus-trapped particles is a pulmonary defense mechanism as a physiological barrier. In the alveolar, clara cells and type II alveolar cells secrete on the surface of the alveolar epithelium, forming a thin layer of pulmonary surfactants. The surfactants act as the main barrier for siRNA delivery because they reduce the transfection efficiency. In addition, the macrophages located in the alveoli rapidly engulf the foreign particles by phagocytosis. The particles taken up into the macrophages are subsequently degraded inside the cells. These factors present major barriers to targeted pulmonary delivery.

The second is the airway cell membrane and its intracellular barriers (Figure 3). For efficient gene silencing in the lungs, siRNAs must be delivered to their site of action, be stable, enter the target cells, and be present in the cytoplasm at sufficient concentration. Once the siRNAs reach the target cells, they must be trafficked into the cytoplasm and taken up by Argonaute (Ago)2/RNA-induced silencing complex (RISC), which degrades mRNAs and, subsequently, suppresses the sequence-specific gene expression. For efficient endocytosis to occur, particles should be under 150 nm in size. Particles within this size range could also avoid macrophage uptake and delayed lung clearance [45]. The physicochemical properties of siRNAs also play a significant role in crossing the biological membrane. Despite their small size, the negative charge and chemical degradability of siRNA molecules prevent them from readily crossing biological membranes. Therefore, efficient siRNA delivery approaches need to overcome this limitation by facilitating cellular uptake. One of the main functions of a delivery vector is to facilitate the cellular uptake of siRNAs [46]. The electrostatic complexation of siRNA molecules with cationic lipids and polymers helps to mask their net negative charge. The positively charged siRNA carrier complex interacts with anionic proteoglycans on the cell membrane, forms an endocytic vesicle, and enters the cells by endocytosis [47]. After cellular internalization, the siRNA

carrier complex in endocytic vesicles is transported along microtubules to lysosomes that are co-localized with the microtubule-organizing center. To avoid lysosomal degradation, siRNAs must escape from the endosome into the cytoplasm, where they can associate with the RNAi machinery. Endosomal escape is a major barrier for efficient siRNA delivery [48, 49]. The endosomal entrapment and lysosomal degradation of siRNA and carriers contribute to the low transfection efficiency and is a major difficulty for delivery vectors. An ideal delivery agent should protect siRNAs from enzymatic degradation, facilitate cellular uptake, and promote endosomal escape inside the cells with negligible toxicity.

Multiple approaches for the delivery of siRNAs have been reported, ranging from the relatively simple direct administration of saline-formulated siRNAs to lipid-based and polymer-based nanoparticle approaches and siRNA conjugation and complexation approaches [50]. The negative charge and chemical degradability of siRNAs under physiologically relevant conditions make its delivery a major challenge. Accordingly, the delivery of siRNAs usually requires a vector or carriers for their transfection into the target cells. In general, both viral and non-viral vectors are being assessed for siRNA delivery to lung cells. Some viral vectors, such as retroviruses and adenoviruses, have been demonstrated to mediate gene silencing in an in vitro lung model [51] and to induce RNAi in a range of animal tissues [52]. Recently, Guo et al. showed that lentivirus-mediated siRNA was used to specifically knock down the expression of nuclear protein 1 (NUPR1) in vivo, which resulted in inhibited tumor growth [53]. However, viral-based delivery has several disadvantages. The immune response to viruses not only impedes gene delivery but also has the potential to cause severe complications [54]. Recent well-documented cases, such as the death of Jesse Gelsinger due to complications related with an adenoviral delivery vector, highlight this problem [55]. In addition, some viral vectors may insert their genome at random positions in the host chromosome, which eventually restrict the gene function [56].

Intracellular barriers to pulmonary siRNA delivery. Barriers to cellular internalization are dependent on the surface properties of siRNA and carriers (e.g., charge and size). After siRNAs are successfully taken into the target cells by endocytosis, the main barriers for delivering siRNAs to its site of action are the endosomal entrapment and lysosomal degradation of siRNA and carriers. To direct target-gene silencing, the siRNAs need to escape from the endosome into the cytoplasm, where they associate with the Ago2/RNA-induced silencing complex (RISC) to direct the cleavage of mRNAs bearing complementary binding sites.

As an alternative to viral vectors, non-viral vectors, including lipid and polymer-based vectors, have been generally used for the delivery of siRNAs to the lungs due to their reduced toxicity [57]. Ongoing research into the transfection of primary cells and whole organisms with siRNA using non-viral transfection agents has produced some promising results. Lipid-based delivery vectors are successfully used to deliver siRNA in vitro and in vivo [58]. Cationic lipids are composed of positively charged head, a linker and hydrophobic. In general, lipid-based complexes are easy to formulate and good transfection efficacy is achieved due to interaction with negative charged cell membrane. Many commercial siRNA transfection agents are lipid-based delivery system, some of which are also employed for pulmonary delivery-DharmFECT [30], Oligofectamine [59], Lipofectamine [60] and TransIT-TKO [35]. Similarly, cationic polymers have also been assessed for siRNA delivery to lung cells. Cationic polymer polyethylenimine (PEI) is widely used for siRNA delivery [13, 61]. PEI is considered as the gold standard for in vitro gene delivery and its transfection efficiency depends on the molecular weight and degree of branching.

On the other hand, lipid-based vectors can also induce toxicity and non-specific activation of inflammatory cytokine and interferon responses [62, 63]. Although polymer-based vectors elicit a relatively less strong immune response than lipid-based vectors, effective siRNA delivery to a local area in lung diseases requires more attention to the development of non-toxic delivery vectors. An important point for siRNA-mediated inhibition of gene expression is whether the observed effects are specific rather than due to off-target effects and free from potential interferon responses [64, 65]. Interestingly, some studies have shown that it was possible to administer "naked siRNAs" to mice and down-regulate an endogenous or exogenous target without inducing an interferon response [66].

The term "naked siRNAs" refers to the delivery of siRNAs without any delivery vectors. Naked siRNAs are degraded by serum endonucleases and are usually removed by glomerular filtration, resulting in a short plasma half-life of < 10 min. Thus, some studies of systemic delivery of naked siRNAs have failed to achieve the downregulation of the targeted gene [67, 68]. In contrast, there have also been some successes of locally delivering naked siRNAs to the lungs [15, 16, 20, 31]. A few of them reported that the use of delivery vectors showed no significant difference in gene silencing efficiency compared to that of naked siRNAs [16, 35]. Indeed, in one clinical trial, the delivery of naked siRNAs for the treatment of RSV has been used [17, 38]. This successful evidence can be because that naked siRNAs for clinical applications are highly chemically modified to prevent

nuclease-induced degradation and presumably minimize immune stimulatory effects. Although it is unclear how the naked siRNAs cross the cell membrane, gain access to the cytoplasm, and remain intact to perform their biological action, both animal and human trials have been conducted successfully, showing the efficacy of naked siRNAs (ALN-RSV01) that were administered intranasally. This explanation has not been confirmed, but the physiological damage of respiratory epithelial cells caused by viral infection may have possibly influenced the mystery. The active change in airway epithelial cell membrane caused by infectious disease might affect cellular internalization. Naked siRNA delivery has some advantages, such as simple formation and the absence of toxicity or inflammatory responses that are usually associated with delivery vectors. Nevertheless, the advantage of naked siRNAs over delivery vectors in the treatment of lung diseases is controversial [69, 70]. Further in vivo investigations about both naked siRNAs and non-viral vectors are required.

Lung disease is a major cause of death, and diminished quality of life is responsible for the suffering of many patients. Various lung diseases make life extremely difficult for the patients, and severe cases of these lung diseases can result in death. The high death rates associated with lung cancer are partially due to the fact that it is unfortunately difficult to cure. Above all, COPD is the fourth-leading cause of death in most industrialized countries and is predicted to become third by 2020 [71]. Therefore, decisive action is needed to stem the rising health and economic burden this represents. Chronic lung diseases, such as COPD and asthma, are disorders of the airways largely related to the presence of persistent inflammation. The approval of inhaled corticosteroids pioneered a new generation of therapy in treating chronic inflammatory diseases. This was the first time that an anti-inflammatory product was available to reduce the characteristic lung inflammation in airways and the associated obstruction. Corticosteroids are still an important therapeutic intervention. However, they are used with limitations in COPD and moderate to severe asthma. Likewise, the treatment of various refractory lung diseases also depends on systemic corticosteroid therapy. Many of these patients also suffered various side effects from systemic corticosteroid use, such as weight gain and uncontrolled hyperglycemia. Treatment of lung disease using cell-specific targeting as well as RNAi techniques represents a novel strategy and could possibly provide new opportunities in nanomedicine. Pulmonary applications of siRNA in in vivo conditions are frequently studied and often result in clinical trials [57, 72]. The findings of recent clinical studies of pulmonary RNAi therapeutics are discussed.

Since the discovery of RNAi, the therapeutic potential of siRNAs has been rapidly recognized. In 2004, the first human clinical trial of RNAi-based therapy was initiated for the treatment of age-related macular degeneration with a siRNA targeting VEGF-receptor 1 delivered intravitreally [73] . Many studies have been conducted over the past few years that involve the delivery of siRNAs to the lungs for the treatment of various lung diseases. Delivery to the lungs will be most important to moving siRNA technology into the clinic. A number of siRNA-based therapies are being evaluated in clinical trials for the treatment of different conditions, including lung diseases such as asthma and RSV infection. Table 1 is a summary of clinical trials of siRNA-based therapeutics [74] .

SiRNA shows potential for the treatment of various pulmonary viral infections, and it has been reported that siRNA-based therapeutics can also be used in the treatment of influenza [13] , parainfluenza virus [35] , severe acute respiratory syndrome (SARS) [14] , and RSV [35] . Above all, RSV is the most promising therapeutic target of siRNAs.

RSV is a common cause of serious respiratory infections in infants and children. It also produces significant morbidity and mortality in adult immunocompromised or elderly populations [75] . An RSV vaccine is not available, and the only approved antiviral therapy for RSV is undesirable for pediatric patients due to its potential teratogenicity and limited effectiveness. Thus, a safe and efficacious RSV therapy has long been awaited for both pediatric and adult patients. RNAi-based therapy has shown promising effects in murine models of RSV infection [35] . The siRNA, ALN-RSV01, is directed against the mRNA encoding the N-protein of RSV that exhibits specific in vitro and in vivo anti-RSV activity. It is delivered without a delivery vector as a nasal spray and targets the upper respiratory tract instead of the lower lung area. ALN-RSV01 has undergone complete phase I intranasal and inhalation studies in healthy adults and has been found to be generally well tolerated [38] . Additionally, ALN-RSV01 has been evaluated in a randomized, double-blind, placebo-controlled phase II trial in lung transplant patients with RSV respiratory tract infection [76] . The administration of ALN-RSV01 to RSV infected lung transplant patients was safe and well tolerated and associated with a statistically significant improvement in symptoms. Based on these results, a larger multinational, randomized, double-blind Phase IIb trial of ALN-RSV01 has been initiated in lung transplant patients to confirm and extend these findings.

Cancer is a major target of RNAi-based therapy, as oncogenes, mutated tumor suppressor genes, and several other genes contributing to tumor progression are potentially important targets for gene silencing by RNAi. Lung cancer is one of the most frequent tumors worldwide with regard to incidence rates and mortality. Patients with lung cancer are commonly diagnosed at an advanced stage of the disease and have limited therapeutic options. Although the knowledge regarding the genetic and molecular basis of lung cancer has regularly increased, the median survival rates of individuals with advanced lung cancer are still poor.

RNAi-based therapy is an attractive strategy for the development of more effective anticancer therapies with reduced treatment-related toxicity. The major advantage of RNAi therapeutics in cancer might be the simultaneous targeting of multiple genes belonging to different cellular pathways that are involved in tumor progression. The simultaneously inhibition of several genes would also minimize the risk of drug resistance normally encountered with small molecule-based therapies, involving siRNAs and miRNAs. There have already been significant improvements in siRNAs for primary or metastatic lung cancer treatment by targeting oncogenes such as Akt1 [9] , Wilms tumor 1 (WT1) [12] , overexpressed genes such as the insulin-like growth factor receptor 1 (IGF-1R) [77] , NUPR1 [53] and EZH2 [78] . Some of these studies have successfully shown the efficacy of RNAi-based therapy through intrapulmonary administration of siRNAs with non-viral vectors. Although strategies to minimize off-target and nonspecific immune stimulatory effects must be devised, these data suggest that the silencing of the target gene with siRNAs is an attractive strategy for the prevention and treatment of primary and metastatic lung cancer. There are currently some clinical trials in progress estimating the safety and efficacy of siRNA-based drugs for cancer treatment. Atu027, a siRNA-lipoplex targeted against protein kinase N3 (PKN3), prevented lung metastasis in a phase I trial of various cancer models [79] . PKN3 is a downstream effector of the phosphoinositide 3-kinase (PI3K) signaling pathway [80] , which regulates diverse cellular responses, including development, growth, and survival [81] . Recently, PKN3 has also been considered as a suitable therapeutic target for modulating tumor angiogenesis because loss of function analysis with Atu027 in cultured primary endothelial cells showed an essential role of PKN3 for endothelial tube formation and migration [79] . Atu027 can be considered as a potential siRNA for preventing lung metastasis and might be suitable for preventing hematogenous metastasis combined with conventional cancer therapy.

Inflammatory lung disease, also called COPD, includes a wide range of lung ailments. These related diseases include asthma, pulmonary fibrosis, and chronic bronchitis. They are influenced by a combination of environmental, genetic, and epigenetic components [82]. COPD is a chronic inflammatory disease of the airways. This disease is hallmarked by airflow that is not fully reversible. Systemic and local airway inflammation has been implicated in the pathogenesis of COPD [83]. COPD is mainly associated with tobacco smoking, and recent studies investigating the pathophysiology of emphysema have demonstrated that cigarette smoke can cause cells to enter cellular senescence.

Smoking might cause cells to senesce due to DNA damage through increased cell turnover, which in turn leads to accelerated telomere shortening [84]. Lately, a lot of studies have investigated the role of cellular senescence in the development and progression of COPD [85]. Although several medication classes, including inhaled corticosteroids, are used for COPD treatment, none of these medications have been shown to significantly improve long-term lung function during the progression of the disease. Current interventions that have been shown to improve mortality in COPD are cessation of smoking and delivery of supplemental oxygen when hypoxemia is present.

Many people are developing COPD, and the cause of this condition is complicated and not thoroughly understood. One key factor is genetic susceptibility. Some studies have shown a large genetic contribution to the variability in pulmonary function and COPD [86, 87]. Polymorphisms in multiple genes have been reported to be associated with COPD [87], such as transcription factor [e.g. nuclear factor-kappa B (NFkB)] [88], extracellular matrix (e.g., matrix metalloproteinase-12 (MMP-12)) [89, 90], cytokines [e.g. tumor necrosis factor (TNF)- α] [91], chemokines [e.g. interleukins (IL)-8, IL-8 receptor and chemokine receptor (CCR)1] [92, 93], and apoptosis (e.g., caspase-3 and vascular endothelial growth factor (VEGF)) [94, 95]. Many of these have been identified as possible targets for therapeutic intervention using molecule inhibitors or antagonists. Although several new treatments that target the inflammatory process are now in clinical development, such as TNF- α inhibitors and I-kappaB kinase complex 2 (IKK2) inhibitors [96, 97], clinical trials with siRNAs have never been performed in COPD. The delay of drug development for COPD might be due to the relatively recent emergence of research addressing the molecular basis of COPD. Furthermore, more research is needed to understand the essential molecular mechanisms about the pathogenesis of COPD and to develop monitoring techniques to support the development of RNAi therapies. Currently, no available treatments reduce the progression of COPD or suppress the inflammation in small airways

and lung parenchyma. The RNAi-based approach for the key molecules also has potential implications for the treatment of COPD.

Asthma is also a chronic inflammatory disease of the airways characterized by variable and recurring symptoms and reversible airflow obstruction. The World Health Organization estimates that 300 million people are currently affected and that, by the year 2025, another 100 million will be affected by the disease [98]. Inhaled corticosteroids are very effective in mild asthma because they improve symptoms and decrease exacerbations. However, in moderate and severe asthma, inhaled corticosteroids have important therapeutic limitations. Although corticosteroids remain an important therapeutic intervention for inflammatory lung diseases, their use is not always completely effective and is associated with side effects. Due to such limitations, it is clear that there is a need for new types of medications that can treat and improve the prognosis of moderate to severe asthma.

Many target genes have been identified that participate in the pathogenesis of asthma. The most promising targets include genes coding for cytokines (IL-4, IL5, and IL-13), cytokine and chemokine receptors (IL-4 receptor and CCR3), and tyrosine kinases [splen tyrosine kinase (Syk) and LCK/YES-related novel tyrosine kinase (Lyn)], as well as for transcription factors [signal transducers and activators of transcription 1 (STAT1), STAT6, GATA3, and NFkB] that are involved in asthma [19, 99, 100]. The genes that have been assessed as siRNA targets for the treatment of asthma in preclinical models are reported [101]. Currently, in a clinical trial for asthma, Excellair™ (ZaBeCor, Bala Cynwyd, PA, USA), a siRNA that targets Syk, is being used. The kinase is involved in signaling from a B cell receptor and is a key regulator of downstream signaling cascades that ultimately lead to the activation of several pro-inflammatory transcription factors. It has been reported that antisense oligonucleotides administered by aerosol were potent to decrease Syk expression, mediator release from alveolar macrophages, and Syk-dependent pulmonary inflammation [102]. Moreover, inhibition of inflammatory mediators was shown in a study using siRNA targeting Syk in airway epithelial cells [103]. Following the successful results of the company's Phase I clinical trial, a Phase II trial for its asthma drug candidate Excellair™ has already been initiated. Some of the current treatments for asthma and other inflammatory conditions, such as TNF- α inhibitors or leukotriene inhibitors, inhibit only one of the mediators of inflammation. In contrast, siRNA targeting Syk seeks to inhibit an initial signaling step of inflammation and, thereby, prevent the release of multiple inflammatory mediators. Overall, recent progress of siRNAs to the lungs has also improved the

therapeutic feasibility of RNAi for inflammatory lung diseases. The rapid progress will put siRNA-based therapeutics on a fast track to the clinic.

MiRNAs are small endogenous noncoding RNAs that regulate gene expression by repressing translation or promoting the degradation of their target mRNA. MiRNAs regulate gene expression by binding to the 3' untranslated region (UTR) of their target mRNAs and mediating mRNA degradation or translational inhibition. In the human genome, transcripts of approximately 60% of all mRNAs are estimated to be targeted by miRNAs [104]. According to their function, miRNAs play an important role in cellular processes as development, proliferation, and apoptosis of pulmonary pathologies [105]. A growing number of miRNAs have been shown to be involved in different lung diseases. This evidence makes miRNAs a promising technology for current and future therapeutic development. We discuss the role of some miRNAs in various lung diseases as well as the possible future of these discoveries in clinical applications. Table 2 shows the summary of miRNAs in therapeutic development. At this point, a miRNA-based therapy has already entered a phase II clinical trial.

There is evidence that upregulation or downregulation of miRNAs is critical for lung homeostasis and, thus, may contribute to the development of pathological pulmonary conditions. Many studies have focused on the role of miRNAs in inflammatory lung diseases, such as COPD [116, 117], pulmonary fibrosis [118] [119] [120] [121], and asthma [122] [123] [124] [125] (Table 3). [130] [117, 129] The pathogenesis of COPD is attributed to not only chronic inflammation in the airways but also systemic inflammation [131]. Cigarette smoking is the main risk factor for the development of COPD.

Smoking has been shown to cause biological change in the gene expression of the lungs [132], and there are some reports about smoking-related miRNAs [117, 129, 130]. However, there are few reports that focus on the miRNAs related to the pathogenesis of this disease with systemic inflammatory components. Recent study on pulmonary fibroblasts of COPD patients presents less expression of miR-146a after stimulation with proinflammatory cytokines when compared with non-COPD subjects with similar smoking histories [127]. The downregulation of miR-146a resulted in a prolonged mRNA half-life of cyclooxygenase-2, thus increasing prostaglandin E2 in fibroblasts from COPD subjects.

Moreover, Ezzie et al. researched the difference of miRNA profiles expressed in the lungs of smokers with and without COPD. They concluded that miR-223 and miR-1274a were the most affected miRNAs in subjects with COPD [126]. Yet, COPD is a complex, multi-component, and heterogeneous disorder with a number of different pathological processes and subgroups with their own characteristics and natural history [133]. A better understanding of the complexity of the disease and potential clinical relevance of the identified miRNAs is needed.

Pulmonary fibrosis can be caused by an identifiable irritation to the lungs, but, in many cases, the cause is unknown, and the therapeutic possibilities are limited. Cigarette smoking is one of the most recognized risk factors for the development of pulmonary fibrosis. This disorder is mainly accompanied by increased expression of the key fibrotic mediator transforming growth factor β (TGF- β) and other cytokines produced at the lesion of active fibrosis [128]. Recently, it was reported that miRNAs may play an important regulatory role in the pulmonary fibrotic change in the lungs. The downregulation of let-7d in idiopathic pulmonary fibrosis (IPF) resulted in increased collagen deposition and alveolar septal thickening [119]. In addition, Liu et al. reported that the oncogenic miR-21 was found to be upregulated in IPF patients and in the murine lungs with bleomycin-induced fibrosis [118]. Although these miRNAs may be potential therapeutic targets because their expression is related to the regulation of TGF- β , the factor is necessary but not sufficient for pathologic fibrosis of the lungs. Pulmonary fibrosis is also a complicated illness that can have many different causes.

Focus on the role of miRNAs in asthma has recently increased. Asthma is an inflammatory disease of the airway that is characterized by an abnormal response of T helper-2 (Th2)-type CD4⁺T lymphocytes against inhaled allergens [134]. In a different asthmatic mouse model, there was an observed increase in the expression of miR-21 in the lungs [123]. This report might contribute to the understanding of the inflammatory mechanism in the airway through the inhibition of IL-12, favoring the Th2 lymphocyte response. A toll-like receptor 4 (TLR4)-induced Th2 lymphocyte induces high expression of miR-126, and selective blockade of miR-126 suppressed the asthmatic phenotype [124]. In addition, airway remodeling is a characteristic feature of asthma and has important functional implications. Rodriguez et al. have shown that miR-155 is related to the development of inflammatory infiltration into the lung and airway remodeling [122]. Thus, some studies present a functional connection between miRNA expression and asthma pathogenesis and suggest that targeting miRNAs in the airways may lead to anti-inflammatory treatments for allergic

asthma. Despite the evidence from experimental models, the expression profiling of miRNAs in airway biopsies from patients with mild asthma before and after treatment with inhaled corticosteroids and in healthy volunteers revealed no differences in miRNA expression [135]. Further investigations about the role of miRNAs related to asthma pathogenesis are required.

Although the basic evidence of miRNA biology is still providing new insights, applications of miRNA-based therapy for inflammatory lung diseases are less advanced than those for lung cancer [136]. One reason for this could be that the disease heterogeneity is caused by the effects of many environmental air pollutants, including smoke and volatile organic compounds. The presence of several risk factors makes the understanding of the pathogenesis of inflammatory lung diseases complicated. Understanding the role that miRNAs play in the modulation of gene expression, leading to sustain the pathogenesis of lung diseases, is important for the development of new therapies that focus on the prevention of disease progression and symptom relief.

Given the significant roles that miRNAs play in multiple pathways of lung carcinogenesis, increasing efforts are dedicated to the research and development of miRNA-based therapies, including restoring functions of tumor suppressive miRNAs or inhibiting oncogenic miRNAs. The development of miRNA-based therapies for lung cancer is growing prosperously with the help of new RNAi technologies. Compared to siRNA-based therapies, which are already in clinical trials, miRNAs are less toxic and have the potential to target multiple genes. The difficulty associated with miRNA delivery is mainly equal to that of siRNAs. The critical problems for the development of this therapy are effective delivery into target sites, potency of the therapy, and elimination of off-target effects [137].

There are two strategies as the therapeutic applications of miRNAs for lung cancer [138]. One strategy is miRNA replacement therapy, which involves the re-introduction of a tumor suppressor miRNA mimic to restore a loss of the function. MiRNA mimics are synthetic RNA duplexes designed to mimic the endogenous functions of miRNAs with chemical modifications for stability and cellular uptake. The concept of miRNA replacement therapy is most exemplified by the let-7 miRNA. let-7 is a tumor-suppressor miRNA in non-small-cell lung cancer that inversely correlates with the expression of the RAS oncoprotein, a key cancer gene [139]. Intranasal administration of let-7 mimic into mouse models of lung cancer significantly reduced tumor growth, suggesting that miRNA replacement therapy is

indeed promising [106, 140, 141] . Another miRNA that shows the value of miRNA replacement is provided by miR-34a [107, 142] . Local and/or systemic delivery of a synthetic miR-34a mimic led to accumulation of miR-34a in the tumor tissue and inhibition of lung tumor growth. Lately, Ling et al. also showed that tumor suppressor miR-22 exhibited anti-lung cancer activity through post-transcriptional regulation of ErbB3 [143] . Thus, therapeutic miRNA mimics have a powerful potential by attacking multiple genes relevant to several diseases. However, it is necessary to pay attention to the potential toxicity in normal tissues under conditions in which the therapeutic delivery of miRNA mimics will lead to an accumulation of exogenous miRNAs in normal cells [138] . Although the assumptions are well founded, there is still insufficient evidence for toxicity caused by miRNA mimics. Indeed, several in vivo studies failed to reveal side effects caused by the miRNA mimics and suggested that delivery of miRNA mimics to normal tissues was well tolerated [107, 141] . It will be important to research miRNA mimic-induced effects in normal cells and to carefully assess toxicity before using them in clinical practice.

The second strategy is directed toward a gain of function and aims to inhibit oncomiRs by using anti-miRNAs. Chemical modifications, such as 2'-O-methyl-group and locked nucleic acid (LNA), would increase oligo stability against nucleases [144] . Antisense oligonucleotides contained in these modifications are termed antagomirs or "LNA-antimiRs" [144, 145] . They are oligonucleotides with sequences complementary to the endogenous miRNA and inhibit the specific miRNA function. An LNA-antimiR against miR-122 has been shown to effectively silence miR-122 in non-human primates [145] , and the findings support the potential of these compounds as a new class of therapeutics. Moreover, it has also been reported that anti-miR-150 delivered into lung tumor xenografts in mice led to inhibited tumor growth [146] . Relative to studies on miRNA mimics, studies with antisense oligonucleotides have shown effective evidence with naked oligonucleotides. This illustrates the potential of chemical modifications of oligonucleotides to improve their stability, resistance to RNase, and pharmacologic properties. Therefore, inhibition of miRNA function by chemically modified antimiR oligonucleotides has become an important and widely used approach. Recent data from the first phase II study in patients with chronic HCV infection treated with the LNA-modified antimiR-122 showed that this compound was well tolerated and provided continuing viral suppression.

An increasing number of studies have examined the therapeutic potential of miRNAs. Recently, the evidence of roles for miRNAs in determining drug resistance has emerged

[147] . Cytotoxic and molecular target drugs have been widely used in the treatment of advanced lung cancer; unfortunately, many cases are still refractory to chemotherapy. In this situation, combining miRNA mimics or antimiR with chemotherapy may potentiate the efficacy of the cancer treatment in the future. In addition, miRNAs related with cancer stem cells may significantly broaden the field of miRNA-based therapy and suggest that miRNAs can be potential tools to kill cancer cells associated with therapy resistance, recurrence, and metastasis [108, 148] . Hence, the main challenge is the successful delivery and chemical modifications of the therapeutic miRNAs to the target tissue without harming normal tissues.

RNAi-based approaches provide a promising therapeutic modality for the treatment of various lung diseases. One of the greatest challenges in RNAi-based therapy continues to be the delivery method of the therapeutic siRNAs and miRNAs to the target cells. Pulmonary delivery applications are very attractive, since they tend to be non-invasive, are locally restricted, and can be administered by the patient. A realistic therapeutic intervention, such as aerosolization, can enhance drug delivery to the site of action and decrease systemic exposure of the patient to the therapy, thereby reducing off-target effects. The advancement of pulmonary siRNA delivery to the clinic illustrates that RNAi-based therapy holds a central place in the future treatment of lung diseases. On the other hand, miRNAs have the opportunity to target multiple genes in a fine-tuned manner, and the miRNA-based therapy will provide an attractive anti-tumor and anti-inflammatory approach for various lung diseases. In particular, anti-miRNA therapy by chemically modified antimiR oligonucleotides has become a potential therapy for lung diseases because the oligonucleotides can be successfully delivered without delivery vectors. Increased evidence has indicated that miRNAs fulfill causative roles in a variety of lung diseases and have prompted investigations into their potential as therapeutic targets. Further understanding of the detailed mechanisms of RNAi-based therapy and investigations of more effective delivery methods are required for future development. These novel approaches could open new avenues for various lung diseases and improve the clinical outcome of the patients.

--- END OF ARTICLE ---

--- START OF ARTICLE 9 ---

Complete Genome Sequence of a Nephropathogenic Infectious Bronchitis Virus Strain Isolated in China

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3795213/>

SHA: f2df4fc3c60338755fd23da3d7e01c0455e20745

Authors: Yang, Jing-tian; Ma, Bing-cun

Date: 2013-10-10

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Abstract: Infectious bronchitis virus (IBV) causes tremendous economic losses to the poultry industry. Here, we report the complete genome analysis results for a new natural recombination nephropathogenic IBV strain named SAIBK, which was isolated in the Sichuan province of China in 2005.

Text: tagious and acute disease in domestic chickens, belongs to group III of the genus Coronavirus in the family Coronaviridae (1) . It is an enveloped, unsegmented, positive-sense, single-stranded RNA (ssRNA) virus and has a genome of approximately 27.6 kb (2) . Recently, many epidemiological analysis reports have suggested that nephropathogenic IBVs have become increasingly prevalent (3) (4) (5) (6) in China. In this work, the complete genome sequence of an isolate named SAIBK was analyzed and recombination was detected between SAIBK and some previously reported IBVs.

A rapid amplification of cDNA ends (RACE) kit (TaKaRa, Japan) was used to obtain the 5' and 3' ends of the genome. Other parts were amplified by 19 primers with overlap between each fragment and were cloned into the pMD19-T vector (TaKaRa, Japan). All fragments were sequenced three times by Sangon Biotech (Shanghai, China). The sequenced fragments were assembled using the SeqMan software program (DNASTar, Inc.). Sequence alignment was conducted and a phylogenetic tree was constructed using the software

program MEGA5 (7). Recombination analysis was performed using the RDP 4.14 (8) and SimPlot 3.5.1 (9) software programs.

The complete genome of the SAIBK strain is 27,534 nucleotides (nt) in length, including the poly(A) tail. It has a classical IBV genome organization with 10 open reading frames (ORFs):

The genome sequence of SAIBK shows the highest identity (94.3%) to the Chinese IBV strain SC021202 (GenBank accession no. EU714029) and the lowest identity (85.8%) to two Chinese IBV strains, BJ (GenBank accession no. AY319651) and DY07 (GenBank accession no. HM245923). It has lower nucleotide identities of 88.1%, 87.9%, and 87.7% to the most popularly used IBV vaccine strains, H120, H52, and M41, respectively.

Phylogenetic analysis of the complete genome results indicated that the SAIBK strain clusters into the same branch as the IBV YN strain (GenBank accession no. JF893452) and the SC021202 strain (GenBank accession no. EU714029). The S1 subunit of the IBV genome is the major determinant of serotype (10) (11) (12) (13) , and S1 analysis indicated that the SAIBK strain has a 4/91-like serotype.

The employed recombination detection methods revealed that SAIBK is a chimera virus, with recombination by the SC021202 strain as a major parent and the H120 vaccine strain as a minor parent. The first and second recombination regions were located at positions 7231 to 9126 and 13437 to 14473 in genes 1a and 1b, respectively. There were two other recombination regions detected at positions 951 to 1067 and 5393 to 5605 of SAIBK, which were recombined with the SC021202 strain as a major parent and the H52 vaccine strain as a minor parent. The recombination detection results suggested that SAIBK is possibly a chimera virus derived from the popularly used vaccine strains H120 and H52 and the field strain SC021202, and the SC021202 strain was isolated from chickens vaccinated with H120 in the Sichuan province of China in 2003 (14) . This result revealed that the field IBVs in Sichuan Province have undergone genetic recombination and are possibly emerging as new mutant strains, such as SAIBK.

Nucleotide sequence accession number. The complete genome sequence of the SAIBK isolate was submitted to GenBank and assigned the accession no. DQ288927.

--- END OF ARTICLE ---

--- START OF ARTICLE 10 ---

Preparation for Possible Sustained Transmission of 2019 Novel Coronavirus

Lessons From Previous Epidemics

<https://jamanetwork.com/journals/jama/fullarticle/2761285>

February 11, 2020

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COVID-19 Update From China

Transmissibility and severity are the 2 most critical factors that determine the effect of an epidemic. Neither the 2009 pandemic influenza A(H1N1) virus ([H1N1]pdm09) pandemic or the severe acute respiratory syndrome coronavirus (SARS-CoV) or the Middle East respiratory syndrome coronavirus (MERS-CoV) epidemics had the combination of both high transmissibility and severity. Control strategies are driven by this combination. R_0 , the basic reproduction number, is a commonly used measure of transmissibility and is defined as the number of additional persons one case infects over the course of their illness. An R_0 of less than 1 indicates the infection will die out “eventually.” An R_0 of greater than 1 indicates the infection has the potential for sustained transmission.

For example, influenza A(H1N1)pdm09, first identified in southern California on April 15, 2009, was highly transmissible. By May 5, 2009, influenza A(H1N1)pdm09 had spread to 41

US states and 21 countries.¹ While influenza A(H1N1)pdm09 was highly transmissible, it was not severe. Initial estimates of the R_0 of influenza A(H1N1)pdm09 were 1.7.² Although an estimated 201 200 respiratory deaths due to influenza A(H1N1)pdm09 occurred during the first year of the pandemic, the number of deaths per population was 30 times lower than that seen during the 1968 influenza pandemic, 1000 times less than the 1918 pandemic, and even less than typical seasonal influenza epidemics (estimated by the World Health Organization [WHO] to be 250 000 to 500 000 per year, although estimation methods differ).³ Influenza A(H1N1)pdm09 was highly transmissible but not severe.

SARS-CoV (2003) and MERS-CoV (2012-current) cause severe disease, but despite the initial R_0 estimations of greater than 2.0 for SARS-CoV (indicating sustained and even worldwide transmission could occur), and some large outbreaks, neither were as transmissible as initial concerns suggested. SARS-CoV caused 8098 reported cases and 774 deaths (case-fatality rate, 9.6%) in 37 countries before the epidemic was controlled. Control was thought to have been possible because a high proportion of cases were severe, making it easier to rapidly identify and isolate infected individuals. In addition, the virus was present at lower levels in upper airway secretions. There was no secondary transmission in the United States from the 8 imported cases, although in Toronto, Canada, a single importation is thought to have led to about 400 cases and 44 deaths. Later estimates of R_0 were less than 1, indicating that SARS-CoV may not have been capable of sustained transmission, especially in the setting of control measures.⁴

Similarly, MERS-CoV appears to have high severity and low transmissibility. Since 2012, MERS-CoV has caused 2494 reported cases and 858 deaths (case-fatality rate, 34%) in 27 countries. MERS-CoV has also caused some rapid outbreaks, mainly in hospitals in Saudi Arabia, Jordan, and South Korea, but estimates of MERS-CoV R_0 are less than 1, and thus far it has been contained.⁵

Can a respiratory virus that is both transmissible and severe be contained? In preparation for an influenza pandemic, the US Department of Health and Human Services' Pandemic Influenza Plan included a combination of nonpharmaceutical (border and school closing, infection control measures) and pharmaceutical (antiviral prophylaxis, vaccines) interventions meant to be used in combination to interrupt or slow influenza transmission. Despite implementation of some of these interventions, influenza A(H1N1)pdm09 spread to 120 countries in 3 months.

With the emergence of MERS-CoV in the Middle East, a preparedness plan was developed that included a surveillance plan, laboratory testing, and contact tracing guidance. Infection control guidance was developed for use in health care settings and traveler guidance was developed for the public.⁶ The US Centers for Disease Control and Prevention (CDC) distributed MERS-CoV polymerase chain reaction test kits to state health departments. Two cases were imported into the United States. Contacts were traced, including household, hospital, and airline contacts. No secondary cases were identified in the United States. MERS-CoV was thought to be severe and control measures relied on recognition of suspect cases. However, during a hospital outbreak in Jeddah, Saudi Arabia, among hospitalized patients only 5 of 53 (9%) health care–associated cases had documented presence in the same room as a patient with MERS.⁵ Despite the high case-fatality rate (an important measure of severity), MERS cases can be asymptomatic and mild (25% in one outbreak). Although it is not known how often asymptomatic or mildly symptomatic patients transmit MERS, initiating comprehensive measures such as isolating patients suspected of having or having been exposed to the virus and using personal protective equipment when caring for them may be extremely difficult because so many patients have mild and nonspecific symptoms.

Is the world ready for a respiratory virus with high transmissibility and severity? After a new influenza virus (H7N9) was identified in China in 2013, a series of modeling articles described the effect of, and level of preparedness for, a severe, single-wave pandemic in the United States.⁷ In scenarios that used clinical attack rates (the proportion of individuals who become ill with or die from a disease in a population initially uninfected) of 20% to 30% (for comparison the clinical attack rate was 20% in the first year of the 2009 H1N1 pandemic), depending on severity there would be an estimated 669 000 to 4.3 million hospitalizations and an estimated 54 000 to 538 000 deaths without any interventions in the United States. The models suggested that without a vaccine, school closures would be unlikely to affect the pandemic, an estimated 35 000 to 60 000 ventilators would be needed, up to an estimated 7.3 billion surgical masks or respirators would be required, and perhaps most important, if vaccine development did not start before the virus was introduced, it was unlikely that a significant number of hospitalizations and deaths could be averted due to the time it takes to develop, test, manufacture, and distribute a vaccine.

It is impossible to know what will happen so early in this novel 2019 coronavirus (2019-nCoV) epidemic. The scope, morbidity, and mortality will depend on the combination of severity and transmissibility. Numerous experts have “nowcasted” how many cases have occurred and forecasted how many cases will likely occur. A recent study suggests rapid person to person transmission can occur.⁸ Disease modelers have estimated R_0 to be 2.2.⁹ The University of Hong Kong estimates the outbreak could infect more than 150 000 persons per day in China at its peak.

Is 2019-nCoV infection severe? To date approximately 14% of cases of 2019-nCoV have been described as severe by WHO, with a case-fatality rate of 2.1%.¹⁰ Estimates of severity are usually higher in the beginning of an epidemic due to the identification of the most severely affected cases and decline as the epidemic progresses. However, because many infected persons have not yet recovered and may still die, the case-fatality rate and severity could be underestimated. On January 30, 2020, WHO officially declared the 2019-nCoV epidemic as a Public Health Emergency of International Concern, indicating its concern that countries aside from China could be affected by 2019-nCoV.

In preparing for possible sustained transmission of 2019-nCoV beyond China, applicable lessons from previous experiences with epidemics/pandemics of respiratory viruses should be carefully considered to better control and mitigate potential consequences. Influenza preparedness plans have been developed that aim to stop, slow, or limit the spread of an influenza pandemic to the United States. These plans address limiting domestic spread and mitigating disease but also sustaining infrastructure and reducing the adverse effects of the pandemic on the economy and society. These plans would be useful to enact during the 2019-nCoV epidemic should the United States experience sustained transmission. Countries have been successful in the past and there is nothing yet to predict that this time it is likely to be worse. Effective prevention and control will not be easy if there is sustained transmission and will require the full attention of public health, federal and local governments, the private sector, and every citizen.

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February 12, 2020

Understanding R and Disease Control

Oz Mansoor | Public Health Physician, Wellington

The message, that we need to prepare for a pandemic is vital. But the article misreports some key ideas. Firstly, SARS was not controlled "because a high proportion of cases were severe." While that helped, it was because cases were not infectious before some days after symptom onset (usually in the second week of illness). This gave more time for case identification and isolation. And most cases did not pass on infection to anybody, but a few spread to many. When all such individuals were identified and isolated, spread stopped.

Unfortunately, the new virus appears to be spreading from people much earlier in the course of illness, and even with mild symptoms - which was never documented for SARS. However, it is not clear that it is any different or better at spread between people, and perhaps with the same pattern of most cases not causing further spread.

Secondly, the R_0 , the basic reproduction number, is correctly described as the average number of infections each case causes. But it lacks two key ideas: 1) the 0 after the R implies the native state, which is a fully susceptible population and without any control measures. R is the effective number and can include the impact of control measures.

To claim that it was the lack of transmissibility, rather than the control measures that ended SARS, is not based on any evidence. And it ignores the heroic efforts of affected countries.

Elimination of SARS demonstrated the potential of globally coordinated collective action, as well as the damage caused by ignorance and prejudice. Most seem to have already forgotten the lessons of SARS. **CONFLICT OF INTEREST:** Worked for WHO/WPRO in SARS response [READ MORE](#)

February 24, 2020

COVID 19: a global presence and not only a new pathogen?

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In the winter season there comes the time of upper and lower respiratory tract infections characterised by cough, dyspnea and eventually fever (influenza-like illness). Some of the patients, especially older people living alone affected by the disease, may need hospitalization and eventually intensive care. In many of the cases who are hospitalized nasal and/or tracheal fluid are examined for viral or bacterial agents. Only in less than 50% of the cases influenza viruses are considered to be the cause of the disease. In the rest of the cases diagnostic procedure for human coronaviruses is not performed routinely. One of the four different Human Coronaviruses (HuCoV: 229E, NL 63,0C43 and HKU1) can however be found in up to 30% of patients negative for influenza viruses (1). Chinese scientists in Wuhan, who had to deal with an increasing number of acute respiratory tract diseases resembling viral pneumonia, performed deep sequencing analysis from samples taken from the lower respiratory tract and found a "novel" coronavirus. The sequence of the complete genome was made public. At the same time, however, the notice from Wuhan brought to mind the SARS- and MERS-epidemics. The measures taken by the Chinese- and WHO-authorities are now well known.

Recently about 150 new cases have been identified in northern Italy and health authorities are still looking for case 0 (the source). Is it possible that COVID-19 was already existent in Italy -- and not only in Italy but possibly everywhere in the world -- and that newly available nucleotide sequence allows now to find the cause of previously undefined influenza-like illness?

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Exhaled breath condensate sampling is not a new method for detection of respiratory viruses

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Abstract: **BACKGROUND:** Exhaled breath condensate (EBC) sampling has been considered an inventive and novel method for the isolation of respiratory viruses. **METHODS:** In our study, 102 volunteers experiencing upper airway infection were recruited over the winter and early spring of 2008/2009 and the first half of the winter of 2009/2010. Ninety-nine EBCs were successfully obtained and screened for 14 commonly circulating respiratory viruses. To investigate the efficiency of virus isolation from EBC, a nasal swab was taken in parallel from a subset of volunteers. The combined use of the ECoVent device with the RTube™ allowed the registration of the exhaled volume and breathing frequency during collection. In this way, the number of exhaled viral particles per liter air or per minute can theoretically be estimated. **RESULTS:** Viral screening resulted in the detection of 4 different viruses in EBC and/or nasal swabs: Rhinovirus, Human Respiratory Syncytial Virus B, Influenza A and Influenza B. Rhinovirus was detected in 6 EBCs and 1 EBC was Influenza B positive. We report a viral detection rate of 7% for the EBCs, which is much lower than the detection rate of 46.8% observed using nasal swabs. **CONCLUSION:** Although very promising, EBC collection using the RTube™ is not reliable for diagnosis of respiratory infections.

Text: Human respiratory tract infections represent the most commonly encountered infections worldwide. In the majority of cases, the etiology of these infections remains undetermined due to rapid convalescence after infection. Respiratory tract infections in healthy adults can be caused by a variety of pathogens and the detection of these agents is

currently based on their isolation from nasal swabs (NS), bronchoalveolar lavages (BAL), nasopharyngeal aspirates and sputum samples. The acquisition of these specimens by semi-invasive and invasive techniques is often unpleasant for the patient. Therefore, exhaled breath condensate (EBC) analysis has recently been explored as a new and non-invasive method to monitor lung inflammation and pulmonary disease such as chronic obstructive pulmonary disease (COPD), asthma, cystic fibrosis, lung cancer etc. EBCs mainly consist of water vapour but a small fraction contains respiratory droplets derived from the airway lining fluid [1, 2]. This observation has created a growing interest in the use of EBC as a new sampling method for the screening of respiratory viruses infecting the upper airways. At first, investigators suspected that turbulence of the inhaled air was responsible for the aerosolisation of the respiratory fluid. However, the effect of the turbulent airflow is limited to the upper airways since the turbulent airflow becomes laminar as it reaches the smaller bronchial airways and alveoli. Recently, the bronchiole fluid film burst model has been described [3]. This model suggests that aerosols are produced during inhalation by the bursting of fluid bubbles present in the bronchioles.

The aim of this study was to investigate whether the EBC collection method was suited for the efficient condensation of aerosolised virus particles during normal breathing and to explore the isolation of respiratory viruses in the condensate. Therefore we screened the EBC samples with virus specific PCR assays targeting 14

In this study, 102 EBCs were collected from otherwise healthy volunteers showing respiratory or flu-like symptoms (defined in Table 1), using a commercially available condenser (RTube™, Respiratory Research Inc., Charlottesville, Virginia, USA). The patient was instructed to breath orally at tidal volumes into a mouthpiece attached to a condenser for 10 minutes. No nose clips were used during collection and saliva contamination was avoided by the presence of a one-way valve and the T-shaped section of the mouthpiece.

In a first part of the study that started during the winter and spring of 2008/2009, 70 EBC samples were collected from patients who voluntary presented themselves to our laboratory. The majority of these volunteers were students that responded to the information leaflet, distributed in the university buildings of the Catholic University of Leuven. The samples were collected with the aluminium cooler sleeve chilled at -80°C.

In the fall and first half of the winter of 2009/2010, 32 condensates were collected from patients who presented themselves to their general practitioner. Due to practical circumstances, the condensates were collected with the cooler chilled at -20°C. For 13 out of 32 collections, the RTube™ was connected by a custom made connectingpiece to the ECoVent (Jaeger, Germany). This device registers ventilatory parameters such as the exhaled volume, breathing frequency and tidal volume. Additionally, a NS was obtained in parallel with the condensate collection from each patient.

All EBCs were immediately stored at -20°C. Nasal swabs (NS) were refrigerated. After viral DNA and RNA extraction, EBC samples and nasal swabs were stored at -80°C. Three specimens were excluded from the study due to incorrect condensate collection. A short questionnaire was used to document the date of birth, the severity of respiratory complaints and to record the days of symptomatic illness from all volunteers. This study was approved by the Medical Ethics Committee of the University Hospital of Leuven and informed consents were received from all participants.

Viral DNA and RNA were isolated with the QIAamp MinElute Virus kit (Qiagen, Westburg, The Netherlands) according to the instruction manual. EBC extracts were eluted in 60 µl elution buffer and NS extracts in 110 µl elution buffer.

The breath condensates were screened for 11 respiratory RNA viruses (CoV NL63, E229 and OC43, RV, HMPV, InfA&B and PIV1-4) [4] [5] [6] [7] using a OneStep RT-PCR Kit (Qiagen, Westburg, The Netherlands) in a 50 µl reaction containing 10 µl of the extracted RNA, 0.6 µM of forward and reverse primers (Table 2), 1.5 µl One Step Enzyme Mix, 10 µl 5 × One Step RT-PCR Buffer and 400 µM of each dNTP. For adenovirus screening, a DNA PCR was carried out for which the amplification reaction mix contained 0.5 µM forward primer (AdFW) and reverse primer (AdRV), 0.4 mM dNTPs, 10 µl Buffer C and 1 U Taq polymerase in a final volume of 50 µl. The PCR primers used were located in conserved regions of the genomes of the respiratory pathogens (Table 2). The reactions were carried out in a T3000 Thermocycler 48 (Westburg, Leusden, The Netherlands) with an initial reverse transcription step for RNA viruses at 50°C for 30 min, followed by PCR activation at 95°C for 30 s, 45 cycles of amplification followed by a final extension step for 10 min at 72°C. The DNA amplification program was initiated with a denaturation step at 94°C for 3 min, followed by 45 cycles of 94°C for 30 s, 55°C for 30 s and a final extension step at 72°C for 1 min. The amplicons were subjected to a 6% polyacrylamide gel and visualised under UV light by

staining with ethidium bromide. PCR products were purified using the Invitex MSB Spin PCRapace Kit and cycle sequenced in forward and reverse direction using the ABI PRISM Big-Dye Termination Cycle Sequencing Ready Reaction kit (Applied Biosystems, Foster City, CA, USA). Sequence analysis was performed with the ABI3130 Genetic Analyser (Applied Biosystems, Foster City, CA, USA). Consensus sequences were obtained using the SeqMan II software (DNASTAR, Madison, Wis.). For samples from HRSV was detected using a RT-PCR assay as previously described [8, 9]. In brief, a multiplex mix was prepared in a final volume of 25 µl using 5 µl extracted RNA, 12.5 µl of Eurogentec One-Step Reverse Transcriptase qPCR Master Mix containing ROX as a passive reference, 0.125 µl Euroscript + RT & RNase inhibitor (Eurogentec, Seraing, Belgium) 200 nM of HRSV-A and -B specific forward and reverse primers and 100 nM of HRSV-A and -B MGB probes. cRNA standards were constructed using the MEGAscript T7 kit (Ambion, Austin, TX, USA) and spectrophotometrically quantified.

The viral load of RV positive samples were quantified by qRT-PCR as described in the manuscript published by Lu and coworkers [10]. The Eurogentec One-Step Reverse Transcriptase qPCR kit was used for preparation of the master mix as described above. The primerset

HRSV-AF F 669-695 ctgtgatagattccaacaaaagaaca [8, 9] HRSV-AF F 718-745 agttacacctgcattaacactaaattcc [8, 9] HRSV-BN N 435-458 ggctccagaatataggcatgattc [8, 9] HRSV-BN N 480-508 tgggtattacaagaagagcagctatacacagt [8, 9] MGB probes and probe, located in 5'UTR, were added to a final concentration of 1 µM and 0.1 µM, respectively. cRNA standards were constructed based on the PCR product of sample 1 using the MegaScript kit (Ambion, Austin, TX, USA). Quantification was performed with a spectrophotometer at 260 nm and converted to the molecule number [11]. Tenfold serial dilutions, allowing detection in a range of 8.6×10^6 to 8.6×10^2 RNA copies were used. The RT-PCR assays were carried out on a ABI PRISM 7500 Sequence Detection System (Applied Biosystems, Foster City, CA, USA). An initial reverse transcription step was performed at 48°C for 30 min, followed by a denaturation step at 95°C for 10 min. Finally, an amplification step of 45 cycles at 95°C for 15 sec and 1 min at 60°C was completed. (37.5%) men, with a median age of 29 (range 9 -46 years). Age and gender was missing for 2 participants of the second group. In total, 52% of the participants were between 20-30 years old. Only 6% were younger than 20 years old and 3% were older than 70 years. In totality, 80 patients (78.4%) were already feeling ill for 1 to 7 days at the day the sample was obtained. Seven volunteers (6.8%) were symptomatic for 8 to 14 days and 9 participants

(8.8%) were already ill for more than 14 days at the day of sample collection. Data on the duration of symptoms was lacking for 6 patients. Almost all volunteers experienced at least 2 symptoms except for two patients (Table 1) . Forty-seven (46.1%) volunteers complained about a constant runny or stuffy nose, 43 (42.2%) had frequent sneezing events and 38 (37.3%) participants had a serious sore throat (Table 1) .

In a first part of the study, we collected 70 EBCs. Screening of the EBCs for 14 respiratory viruses (Table 2) , showed 5 RV (7.1%) positive samples (Table 3) . In a second part, we collected 32 EBCs from patients that presented themselves to their general practitioner. Two of these EBCs were positive for one of the 14 investigated respiratory viruses, 1 for RV and 1 for InfB. To inspect the detection rate of respiratory viruses in the condensate, a NS was taken from this second group of volunteers for comparison. In 15 out of 32 NS (46.8%), one or more viral pathogens were isolated. Viral screening of the NS resulted in the detection of RV, InfA (subtype H1N1) and HRSV-B. Quantification of the HRSV-B viral load demonstrated for samples 72 and 101 viral titers of 8.0×10^4 RNA copies/ml and 6.8×10^7 RNA copies/ml respectively. The RV RT-PCR assay did not allow the quantification of all samples that tested positive for RV by PCR (Table 3) . Presence of the same pathogen in both the EBC and the NS was confirmed for only 1 sample: sample 71, which tested positive for RV in both the EBC and the NS. For sample 81, RV was detected in the NS and analysis of the EBC demonstrated an InfB infection.

For EBC samples that were collected in the fall and winter of 2009/2010, measurements with the ECoVent in (Table 3 , sample 81) was positive for InfB when using the RTube™ in combination with the EcoVent. In theory, the viral generation rate (number of viral RNA copies exhaled per minute) can be predicted by quantification of the exhaled viral load. Then, an estimation of the RNA copies per litre exhaled air or per minute can be calculated. Quantification of the exhaled InfB would allow us to predict the generation rate for this virus. Due to insufficient sample volume, we could not determine the number of RNA copies in the sample.

Collection of exhaled breath condensates is a novel and non-invasive method for obtaining samples of the upper respiratory tract. The collection of EBC is easy to perform and can be conducted in a home environment. This method is much more agreeable for the patient when compared to the unpleasant and invasive collection of nasal swabs, BAL, aspirates, etc. This aspect renders the method very attractive for routine laboratory diagnostics of

viral infections. Most studies that perform breath analyses for viral detection use modified face masks, with a removable central region in electret or a removable Teflon filter on which exhaled particles impact [12] [13] [14] . With the RTube™ collection device, aerosolized particles of the airway lining fluid are precipitated into a condensate when the breath is cooled which serves as an immediate starting point for molecular testing.

Until now, this is the study with the largest subset of volunteers that investigated EBC as a specimen for the detection of respiratory viruses. Previous studies reported the inclusion of a limited subset of participants and investigated the presence of a limited number of viruses in the breath samples. The study performed by Fabian and colleagues, included 12 volunteers [12] . Huynh and co-workers recruited 9 volunteers for exhaled breath sampling [13] . In the study by Stelzer-Braid et al., 50 EBCs were analysed [14] and St-George et al. report the participation of 12 adults [15] . These studies have focused on the detection of InfA and -B, PIV1-3, HRSV and HMPV, while we have screened the samples for a panel of 14 commonly circulating respiratory viruses. Based on the analysis of 99 EBCs (3 EBCs were excluded), our results support the exhalation of RV and InfB in 7% of our samples. Since many of the volunteers had already been experiencing symptoms for 1 to 7 days, we initially presumed that they were already recovering from the infection and were no longer exhaling the virus. For common cold infections it is suggested that a person may already be infectious for 1 or 2 days before experiencing any symptoms. However, in a second part of our study we started collecting EBCs in parallel with nasal swabs from patients presenting themselves to their medical doctor, 1 to 3 days after onset of symptoms. Only for 1 condensate the same pathogen was detected in both the EBC and the NS. The detection rate for respiratory viral pathogens in the NS was 46.8% which is much higher than the 7% detection rate in the EBCs. The low detection of virus positive condensates can therefore not be attributed to the fact that volunteers were no longer infectious. The discrepant detection rate between samples may also be explained by different severity of respiratory infection, since comparator samples were of different parts of the respiratory tract. Patients that delivered a positive NS may have possibly suffered from an upper airway infection whereas EBC positive volunteers may have experienced a more advanced, lower respiratory tract infection. However, the effect of nasal inhalation on EBC collection, guiding formed particles in the upper respiratory tract to the lower compartments, instead of oral inhalation was not investigated. Patients with positive EBC samples were experiencing symptoms for maximum two days at the time of collection. However, this was not different for 7 patients with positive NS. Six patients that provided positive NS were experiencing symptoms for a longer period at the time of collection (Table 3) . In the group of volunteers that provided an EBC negative or EBC and NS negative sample, the

manifestation of symptoms were reported ranging from 1 day to more than two weeks. When reported symptoms were compared between EBC positive patients (7) and NS positive patients (15), 27% and 33% in the positive NS group experienced shivering and muscle pain whereas this symptom was not indicated by any patient of the EBC positive group. In all groups fever, headache, watering eyes, stuffed nose, frequent sneezing, sore throat and coughing were reported.

Volunteers were not diagnosed with other pathogens before participation in the study. Since we did not test these samples for other than viral pathogens, we can not exclude the possibility that some of the negative NS are positive for bacteria or other pathogens causing respiratory illness. Recently, one study reported a detection rate of 5% for influenza in EBC [15]. This is in the same range of the detection rate that we report for respiratory viruses in general. Other studies with a limited number of patients, describe a markedly higher sensitivity of 33 to 36% [12] [13] [14] but the higher percentage may be due to the low number of participants subjects were included [12]. Remarkably, the studies reporting this higher detection rate used collections masks, while the study using the RTube™ reported comparable findings. Face masks consist of electret which trap viruses based on permanently charged fibres [13]. In addition, the Teflon filter has 2 µm pores which will retain all larger particles. Possibly, the lower detection rate can partly be explained by the fact that the RTube™ is manufactured in polypropylene and does not possess a virus attracting and filtering feature like the aforementioned materials.

The qRT-PCR developed by Lu and coworkers for the detection of RV, did not allow the assessment of the viral load present in the EBC samples [10]. Also for 4 NS, the viral titer remained undetermined, probably due to the limited sensitivity of the assay. For diagnosis, more sensitive methods might be necessary to detect respiratory viruses present in EBC since it is unpredictable how diluted the viral particles in the specimen are. Recently, nested qRT-PCR assays have been developed to allow a more sensitive detection of viruses in aerosols [16].

Also person-dependent factors, such as the number of particles produced, the exhaled volume and the age of the patient, have been suggested to play an important role for exhalation of viral particles. The participants that were recruited in the study of Fabian and coworkers were 12 years of age and older [12]. For hospitalized children a much higher rate of virus positive samples is reported [14]. In our study, the majority of volunteers were

between 20 and 30 years old. Only two children less than 10 years and 3 elderly people (> 70 years) were included. One of the children tested positive for InfA in the NS, but the infection was not confirmed in the EBC.

For influenza, an exhaled generation rate of <3.2 to 20 influenza RNA copies per minute was predicted by quantifying the virus aerosols that impacted on a removable Teflon filter of a collection mask [12]. We used the RTube™ in combination with the ECoVent, that allowed the registration of additional ventilation parameters such as breathing frequency and exhaled volume. In this way, when the number of RNA copies in the EBC is quantified, the amount of viral particles that are exhaled per litre or per minute can be estimated. Unfortunately, we were not able to predict a virus generation rate for InfB since viral load remained undetermined.

Although an inventive, new and promising method, EBC collected by the RTube™ does not appear to be appropriate for diagnosis of respiratory infections. Nonetheless, this method may provide an alternative for current sample procurement for epidemiological studies of circulating viruses. This technique also confirms the observation that viruses are able to disseminate through normal breathing, particularly RV.

In addition, EBC collection from patients during respiratory infections may be further investigated for biomarker patterns. In calves that were experimentally infected with bovine RSV, an increase in leukotriene B₄, indicating oxidative stress, was observed. This increased level was also associated with the development of bronchial hyperresponsiveness [17]. In humans, a transiently elevated H₂O₂ level was observed during common cold infection. This marker returned to baseline values when volunteers recovered from infection. H₂O₂ has also been recognized as an interesting marker in asthma, where it is associated with chronic lower airway inflammation [18]. In InfA infected volunteers, an increased CO level was observed during upper respiratory infection. This observation might imply that CO is an indicator of airway inflammation or represents one of the host defence mechanisms against viral infection [19]. Therefore, a better identification of the biomarker signature in condensates of individuals experiencing a viral infection might imply interesting findings towards the identification of markers reflecting inflammation or antiviral protection. This may contribute to the biomarker profiles established for diseases like asthma and COPD, for which viral infections are suggested to trigger or exacerbate symptoms [20].

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Community-acquired pneumonia in children — a changing spectrum of disease

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5608782/>

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Abstract: Pneumonia remains the leading cause of death in children outside the neonatal period, despite advances in prevention and management. Over the last 20 years, there has been a substantial decrease in the incidence of childhood pneumonia and pneumonia-associated mortality. New conjugate vaccines against *Haemophilus influenzae* type b and *Streptococcus pneumoniae* have contributed to decreases in radiologic, clinical and complicated pneumonia cases and have reduced hospitalization and mortality. The importance of co-infections with multiple pathogens and the predominance of viral-associated disease are emerging. Better access to effective preventative and management strategies is needed in low- and middle-income countries, while new strategies are needed to address the residual burden of disease once these have been implemented.

Text: Pneumonia has been the leading cause of death in children younger than 5 years for decades. Although there have been substantial decreases in overall child mortality and in pneumonia-specific mortality, pneumonia remains the major single cause of death in children outside the neonatal period, causing approximately 900,000 of the estimated 6.3

million child deaths in 2013 [1] . Substantial advances have occurred in the understanding of risk factors and etiology of pneumonia, in development of standardized case definitions, and in prevention with the production of improved vaccines and in treatment. Such advances have led to changes in the epidemiology, etiology and mortality from childhood pneumonia. However in many areas access to these interventions remains sub-optimal, with large inequities between and within countries and regions. In this paper we review the impact of recent preventative and management advances in pneumonia epidemiology, etiology, radiologic presentation and outcome in children.

The overall burden of childhood pneumonia has been reduced substantially over the last decade, despite an increase in the global childhood population from 605 million in 2000 to 664 million in 2015 [2] . Recent data suggest that there has been a 25% decrease in the incidence of pneumonia, from 0.29 episodes per child year in low-and middle-income countries in 2000, to 0.22 episodes per child year in 2010 [3] . This is substantiated by a 58% decrease in pneumonia-associated disability-adjusted life years between 1990 and 2013, from 186 million to 78 million as estimated in the Global Burden of Disease study [1] . Pneumonia deaths decreased from 1.8 million in 2000 to 900,000 in 2013 [1] . These data do not reflect the full impact of increasingly widespread use of pneumococcal conjugate vaccine in low-and middle-income countries because the incidence of pneumonia and number of deaths are likely to decrease still further as a result of this widespread intervention [4] .

Notwithstanding this progress, there remains a disproportionate burden of disease in low-and middle-income countries, where more than 90% of pneumonia cases and deaths occur. The incidence in high-income countries is estimated at 0.015 episodes per child year, compared to 0.22 episodes per child year in low-and middle-income countries [3] . On average, 1 in 66 children in high-income countries is affected by pneumonia per year, compared to 1 in 5 children in low-and middle-income countries. Even within low-and middleincome countries there are regional inequities and challenges with access to health care services: up to 81% of severe pneumonia deaths occur outside a hospital [5] . In addition to a higher incidence of pneumonia, the case fatality rate is estimated to be almost 10-fold higher in low-and middle-income countries as compared to high-income countries [3, 5] .

Childhood pneumonia can also lead to significant morbidity and chronic disease. Early life pneumonia can impair longterm lung health by decreasing lung function [6] . Severe or recurrent pneumonia can have a worse effect on lung function; increasing evidence suggests that chronic obstructive pulmonary disease might be related to early childhood pneumonia [7, 8] . A meta-analysis of the risk of long-term outcomes after childhood pneumonia categorized chronic respiratory sequelae into major (restrictive lung disease, obstructive lung disease, bronchiectasis) and minor (chronic bronchitis, asthma, abnormal pulmonary function) groups [9] . The risk of developing at least one of the major sequelae was estimated as 6% after an ambulatory pneumonia event and 14% after an episode of hospitalized pneumonia. Because respiratory diseases affect almost 1 billion people globally and are a major cause of mortality and morbidity [10] , childhood pneumonia might contribute to substantial morbidity across the life course.

Chest radiologic changes have been considered the gold standard for defining a pneumonia event [11] because clinical findings can be subjective and clinical definitions of pneumonia can be nonspecific. In 2005, to aid in defining outcomes of pneumococcal vaccine studies, the World Health Organization's (WHO) standardized chest radiograph description defined a group of children who were considered most likely to have pneumococcal pneumonia [12] . The term "end-point consolidation" was described as a dense or fluffy opacity that occupies a portion or whole of a lobe, or the entire lung. "Other infiltrate" included linear and patchy densities, peribronchial thickening, minor patchy infiltrates that are not of sufficient magnitude to constitute primary end-point consolidation, and small areas of atelectasis that in children can be difficult to distinguish from consolidation. "Primary end-point pneumonia" included either end-point consolidation or a pleural effusion associated with a pulmonary parenchymal infiltrate (including "other" infiltrate).

Widespread use of pneumococcal conjugate vaccination and Haemophilus influenzae type B conjugate vaccination has decreased the incidence of radiologic pneumonia. In a review of four randomized controlled trials and two case-control studies of Haemophilus influenzae type B conjugate vaccination in high-burden communities, the vaccination was associated with an 18% decrease in radiologic pneumonia [13] . Introduction of pneumococcal conjugate vaccination was associated with a 26% decrease in radiologic pneumonia in California between 1995 and 1998 [14] . In vaccine efficacy trials in low-and middle-income countries, pneumococcal conjugate vaccination reduced radiologic

pneumonia by 37% in the Gambia [15] , 25% in South Africa [16] and 26% in the Philippines [17] .

The WHO radiologic case definition was not intended to distinguish bacterial from viral etiology but rather to define a sub-set of pneumonia cases in which pneumococcal infection was considered more likely and to provide a set of standardized definitions through which researchers could achieve broad agreement in reporting chest radiographs. However, despite widespread field utilization, there are concerns regarding inter-observer repeatability. There has been good consensus for the description of lobar consolidation but significant disagreement on the description of patchy and perihilar infiltrates [18, 19] . In addition, many children with clinically severe lung disease do not have primary end-point pneumonia: in one pre-pneumococcal conjugate vaccination study, only 34% of children hospitalized with pneumonia had primary end-point pneumonia [20] . A revised case definition of "presumed bacterial pneumonia" has been introduced, and this definition includes pneumonia cases with WHO-defined alveolar consolidation, as well as those with other abnormal chest radiograph infiltrates and a serum C-reactive protein of at least 40 mg/L [21, 22] . This definition has been shown to have greater sensitivity than the original WHO radiologic definition of primary end-point pneumonia for detecting the burden of pneumonia prevented by pneumococcal conjugate vaccination [23] . Using the revised definition, the 10-valent pneumococcal conjugate vaccine (pneumococcal conjugate vaccination-10), had a vaccine efficacy of 22% in preventing presumed bacterial pneumonia in young children in South America [22] , and pneumococcal conjugate vaccination-13 had a vaccine efficacy of 39% in preventing presumed bacterial pneumonia in children older than 16 weeks who were not infected with human immunodeficiency virus (HIV) in South Africa [21] . Thus there is convincing evidence that pneumococcal conjugate vaccination decreases the incidence of radiologic pneumonia; however there is no evidence to suggest that pneumococcal conjugate vaccination modifies the radiologic appearance of pneumococcal pneumonia.

Empyema is a rare complication of pneumonia. An increased incidence of empyema in children was noted in some high-income countries following pneumococcal conjugate vaccination-7 introduction, and this was attributed to pneumococcal serotypes not included in pneumococcal conjugate vaccination-7, especially 3 and 19A [24] . In the United States, evidence from a national hospital database suggests that the incidence of empyema increased 1.9-fold between 1996 and 2008 [25] . In Australia, the incidence rate ratio increased by 1.4 times when comparing the pre-pneumococcal conjugate

vaccination-7 period (1998 to 2004) to the post-pneumococcal conjugate vaccination-7 period (2005 to 2010) [26] . In Scotland, incidence of empyema in children rose from 6.5 per million between 1981 and 1998, to 66 per million in 2005 [27] . These trends have been reversed since the introduction of pneumococcal conjugate vaccination-13. Data from the United States suggest that empyema decreased by 50% in children younger than 5 years [28] ; similarly, data from the United Kingdom and Scotland showed substantial reduction in pediatric empyema following pneumococcal conjugate vaccination-13 introduction [29, 30] .

Several national guidelines from high-income countries, as well as the WHO recommendations for low-and middleincome countries, recommend that chest radiography should not be routinely performed in children with ambulatory pneumonia [31] [32] [33] . Indications for chest radiography include hospitalization, severe hypoxemia or respiratory distress, failed initial antibiotic therapy, or suspicion for other diseases (tuberculosis, inhaled foreign body) or complications. However, point-of-care lung ultrasound is emerging as a promising modality for diagnosing childhood pneumonia [34] .

In addition to the effect on radiologic pneumonia, pneumococcal conjugate vaccination reduces the risk of hospitalization from viral-associated pneumonia, probably by reducing bacterial-viral co-infections resulting in severe disease and hospitalization [35] . An analysis of ecological and observational studies of pneumonia incidence in different age groups soon after introduction of pneumococcal conjugate vaccination-7 in Canada, Italy, Australia, Poland and the United States showed decreases in all-cause pneumonia hospitalizations ranging from 15% to 65% [36] . In the United States after pneumococcal conjugate vaccination-13 replaced pneumococcal conjugate vaccination-7, there was a further 17% decrease in hospitalizations for pneumonia among children eligible for the vaccination, and a further 12% decrease among unvaccinated adults [28] .

A systematic review of etiology studies prior to availability of new conjugate vaccines confirmed *S. pneumoniae* and *H. influenzae* type B as the most important bacterial causes of pneumonia, with *Staphylococcus aureus* and *Klebsiella pneumoniae* associated with some severe cases. Respiratory syncytial virus was the leading viral cause, identified in 15-40% of pneumonia cases, followed by influenza A and B, parainfluenza, human metapneumovirus and adenovirus [37] .

More recent meta-analyses of etiology data suggest a changing pathogen profile, with increasing recognition that clinical pneumonia is caused by the sequential or concurrent interaction of more than one organism. Severe disease in particular is often caused by multiple pathogens. With high coverage of pneumococcal conjugate vaccination and Haemophilus influenzae type B conjugate vaccination, viral pathogens increasingly predominate [38] . In recent case-control studies, at least one virus was detected in 87% of clinical pneumonia cases in South Africa [39] , while viruses were detected in 81% of radiologic pneumonia cases in Sweden [40] . In a large multi-center study in the United States, viral pathogens were detected in 73% of children hospitalized with radiologic pneumonia, while bacteria were detected in only 15% of cases [41] . A meta-analysis of 23 case-control studies of viral etiology in radiologically confirmed pneumonia in children, completed up to 2014, reported good evidence of causal attribution for respiratory syncytial virus, influenza, metapneumovirus and parainfluenza virus [42] . However there was no consistent evidence that many other commonly described viruses, including rhinovirus, adenovirus, bocavirus and coronavirus, were more commonly isolated from cases than from controls. Further attribution of bacterial etiology is difficult because it is often not possible to distinguish colonizing from pathogenic bacteria when they are isolated from nasal specimens [43] .

Another etiology is pertussis. In the last decade there has also been a resurgence in pertussis cases, especially in high-income countries [44] . Because pertussis immunity after acellular pertussis vaccination is less long-lasting than immunity after wild-type infection or whole-cell vaccination, many women of child-bearing age have waning pertussis antibody levels. Their infants might therefore be born with low transplacental anti-pertussis immunoglobulin G levels, making them susceptible to pertussis infection before completion of the primary vaccination series [45] . In 2014, more than 40,000 pertussis cases were reported to the Centers for Disease Control and Prevention in the United States; in some states, population-based incidence rates are higher than at any time in the last 70 years [44] . In contrast, most low- and middle-income countries use whole-cell pertussis vaccines and the numbers of pertussis cases in those countries were stable or decreasing until 2015 [46] . However recent evidence from South Africa (where the acellular vaccine is used) shows an appreciable incidence of pertussis among infants presenting with acute pneumonia: 2% of clinical pneumonia cases among infants enrolled in a birth cohort were caused by pertussis [39] , and 3.7% of infants and young children presenting to a tertiary academic hospital had evidence of pertussis infection [47] .

Similarly, childhood tuberculosis is a major cause of morbidity and mortality in many low- and middle-income countries, and *Mycobacterium tuberculosis* has increasingly been recognized as a pathogen in acute pneumonia in children living in high tuberculosis-prevalence settings. Postmortem studies of children dying from acute respiratory illness have commonly reported *M. tuberculosis* [48, 49] . A recent systematic review of tuberculosis as a comorbidity of childhood pneumonia reported culture-confirmed disease in about 8% of cases [50] . Because intrathoracic tuberculosis disease is only culture-confirmed in a minority of cases, the true burden could be even higher; tuberculosis could therefore be an important contributor to childhood pneumonia incidence and mortality in high-prevalence areas.

Childhood pneumonia and clinically severe disease result from a complex interaction of host and environmental risk factors [37] . Because of the effectiveness of pneumococcal conjugate vaccination and *Haemophilus influenzae* type B conjugate vaccination for prevention of radiologic and clinical pneumonia, incomplete or inadequate vaccination must be considered as a major preventable risk factor for childhood pneumonia. Other risk factors include low birth weight, which is associated with 3.2 times increased odds of severe pneumonia in low- and middle-income countries, and 1.8 times increased odds in high-income countries [51] . Similarly, lack of exclusive breastfeeding for the first 4 months of life increases odds of severe pneumonia by 2.7 times in low- and middle-income countries and 1.3 times in high-income countries. Markers of undernutrition are strong risk factors for pneumonia in low- and middle-income countries only, with highly significant odds ratios for underweight for age (4.5), stunting (2.6) and wasting (2.8) . Household crowding has uniform risk, with odds ratios between 1.9 and 2.3 in both low- and middle-income countries and high-income countries. Indoor air pollution from use of solid or biomass fuels increases odds of pneumonia by 1.6 times; lack of measles vaccination by the end of the first year of age increases odds of pneumonia by 1.8 times [51] . It is estimated that the prevalence of these critical risk factors in low- and middle-income countries decreased by 25% between 2000 and 2010, contributing to reductions in pneumonia incidence and mortality in low- and middle-income countries, even in countries where conjugate vaccines have not been available [3] .

The single strongest risk factor for pneumonia is HIV infection, which is especially prevalent in children in sub-Saharan Africa. HIV-infected children have 6 times increased odds of developing severe pneumonia or of death compared to HIV-uninfected children [52] . Since the effective prevention of mother-to-child transmission of HIV, there is a

growing population of HIV-exposed children who are uninfected; their excess risk of pneumonia, compared to HIV unexposed children, has been described as 1.3-to 3.4-fold higher [53] [54] [55] [56] [57] .

The pneumococcal conjugate vaccination and Haemophilus influenzae type B conjugate vaccination have been effective tools to decrease pneumonia incidence, severity and mortality [58, 59] . However, equitable coverage and access to vaccines remains sub-optimal. By the end of 2015, Haemophilus influenzae type B conjugate vaccination had been introduced in 73 countries, with global coverage estimated at 68%. However, inequities are still apparent among regions: in the Americas coverage is estimated at 90%, while in the Western Pacific it is only 25%. By 2015, pneumococcal conjugate vaccination had been introduced into 54 countries, with global coverage of 35% for three doses of pneumococcal conjugate vaccination for infant populations [60] . To address this issue, the WHO's Global Vaccine Access Plan initiative was launched to make life-saving vaccines more equitably available. In addition to securing guarantees for financing of vaccines, the program objectives include building political will in low-and middle-income countries to commit to immunization as a priority, social marketing to individuals and communities, strengthening health systems and promoting relevant local research and development innovations [61] .

Maternal vaccination to prevent disease in the youngest infants has been shown to be effective for tetanus, influenza and pertussis [62] . Influenza vaccination during pregnancy is safe, provides reasonable maternal protection against influenza, and also protects infants for a limited period from confirmed influenza infection (vaccine efficacy 63% in Bangladesh [63] and 50.4% in South Africa [64]). However as antibody levels drop sharply after birth, infant protection does not persist much beyond 8 weeks [65] . Recently respiratory syncytial virus vaccination in pregnancy has been shown to be safe and immunogenic, and a phase-3 clinical trial of efficacy at preventing respiratory syncytial virus disease in infants is under way [66] . Within a decade, respiratory syncytial virus in infancy might be vaccine-preventable, with further decreases in pneumonia incidence, morbidity and mortality [67] .

Improved access to health care, better nutrition and improved living conditions might contribute to further decreases in childhood pneumonia burden. The WHO Integrated Global Action Plan for diarrhea and pneumonia highlights many opportunities to protect,

prevent and treat children [68] . Breastfeeding rates can be improved by programs that combine education and counseling interventions in homes, communities and health facilities, and by promotion of baby-friendly hospitals [69] . Improved home ventilation, cleaner cooking fuels and reduction in exposure to cigarette smoke are essential interventions to reduce the incidence and severity of pneumonia [70, 71] . Prevention of pediatric HIV is possible by providing interventions to prevent mother-to-child transmission [72] . Early infant HIV testing and early initiation of antiretroviral therapy and cotrimoxazole prophylaxis can substantially reduce the incidence of community-acquired pneumonia among HIV-infected children [73] . Community-based interventions reduce pneumonia mortality and have the indirect effect of improved-careseeking behavior [58] . If these cost-effective interventions were scaled up, it is estimated that 67% of pneumonia deaths in low-and middle-income countries could be prevented by 2025 [58] .

Case management of pneumonia is a strategy by which severity of disease is classified as severe or non-severe. All children receive early, appropriate oral antibiotics, and severe cases are referred for parenteral antibiotics. When implemented in highburden areas before the availability of conjugate vaccines, case management as part of Integrated Management of Childhood Illness was associated with a 27% decrease in overall child mortality, and 42% decrease in pneumonia-specific mortality [74] . However the predominance of viral causes of pneumonia and low case fatality have prompted concern about overuse of antibiotics. Several randomized controlled trials comparing oral antibiotics to placebo for non-severe pneumonia have been performed [75] [76] [77] and others are ongoing [78] . In two studies, performed in Denmark and in India, outcomes of antibiotic and placebo treatments were equivalent [76, 77] . In the third study, in Pakistan, there was a non-significant 24% vs. 20% rate of failure in the placebo group, which was deemed to be non-equivalent to the antibiotic group [75] . Furthermore, because WHO-classified non-severe pneumonia and bronchiolitis might be considered within a spectrum of lower respiratory disease, many children with clinical pneumonia could actually have viral bronchiolitis, for which antibiotics are not beneficial [79] . This has been reflected in British [33] and Spanish [31] national pneumonia guidelines, which do not recommend routine antibiotic treatment for children younger than 2 years with evidence of pneumococcal conjugate vaccination who present with non-severe pneumonia. The United States' national guidelines recommend withholding antibiotics in children up to age 5 years presenting with non-severe pneumonia [32] . However, given the high mortality from pneumonia in low-and middle-income countries, the lack of easy access to care, and the high prevalence of risk factors for severe disease, revised World Health Organization

pneumonia guidelines still recommend antibiotic treatment for all children who meet the WHO pneumonia case definitions [80] .

Use of supplemental oxygen is life-saving, but this is not universally available in low-and middle-income countries; it is estimated that use of supplemental oxygen systems could reduce mortality of children with hypoxic pneumonia by 20% [81] . Identifying systems capacity to increase availability of oxygen in health facilities, and identifying barriers to further implementation are among the top 15 priorities for future childhood pneumonia research [82] . However, up to 81% of pneumonia deaths in 2010 occurred outside health facilities [5] , so there are major challenges with access to health services and health-seeking behavior of vulnerable populations. Identifying and changing the barriers to accessing health care is an important area with the potential to impact the survival and health of the most vulnerable children [82] .

Much progress has been made in decreasing deaths caused by childhood pneumonia. Improved socioeconomic status and vaccinations, primarily the conjugate vaccines (against *Haemophilus influenzae* and pneumococcus), have led to substantial reductions in the incidence and severity of childhood pneumonia. Stronger strategies to prevent and manage HIV have reduced HIV-associated pneumonia deaths. However, despite the substantial changes in incidence, etiology and radiology globally, there remain inequities in access to care and availability of effective interventions, especially in low-and middle-income countries. Effective interventions need to be more widely available and new interventions developed for the residual burden of childhood pneumonia.

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Immunomodulatory Activity and Protective Effects of Polysaccharide from *Eupatorium adenophorum* Leaf Extract on Highly Pathogenic H5N1 Influenza Infection

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Abstract: The development of novel broad-spectrum, antiviral agents against H5N1 infection is urgently needed. In this study, we evaluated the immunomodulatory activities and protective effect of *Eupatorium adenophorum* polysaccharide (EAP) against the highly pathogenic H5N1 subtype influenza virus. EAP treatment significantly increased the production of IL-6, TNF- α , and IFN- γ both in vivo and in vitro as measured by qPCR and ELISA. In a mouse infection model, intranasal administration of EAP at a dose of 25 mg/kg body weight prior to H5N1 viral challenge efficiently inhibited viral replication, decreased lung lesions, and increased survival rate. We further evaluated the innate immune recognition of EAP, as this process is regulated primarily Dectin-1 and mannose receptor (MR). These results indicate that EAP may have immunomodulatory properties and a potential prophylactic effect against H5N1 influenza infection. Our investigation suggests an alternative strategy for the development of novel antiinfluenza agents and benefits of *E. adenophorum* products.

Text: Highly pathogenic H5N1 subtype influenza virus can be transmitted directly from poultry to human and cause acute respiratory infections. Pandemic influenza virus H5N1 posed a worldwide threat to the public health because of rapid spread and high pathogenicity [1, 2]. The symptoms in animals or humans infected with H5N1 include fever, encephalitis, pneumonia, and severe acute respiratory syndrome (SARS) [3, 4]. The World Health Organization reported 622 human cases of highly pathogenic H5N1 influenza virus infection, including 371 deaths (a mortality rate >50%), from 2003 to 2013 (http://www.who.int/influenza/human_animal_interface/H5N1_cumulative_table_archives/en/index.html). Currently, the most effective preventive measure against the influenza virus is vaccination. Several antiinfluenza medications have been widely used, including zanamivir (Relenza) and oseltamivir (Tamiflu).

Unfortunately, their benefits have been significantly restricted by drug-resistance and frequent antigenic mutation [5, 6] . Therefore, the development of novel antiinfluenza agents against the H5N1 subtype is very important.

The invasive plant *Eupatorium adenophorum*, native to Central America, has a strong ability to adapt to different environments all over the world. This plant first invaded southern Yunnan Province (China) in the 1940s from Burma and Vietnam, and quickly spread across southwestern China throughout the 1950s [7, 8] . Over the past 50 years, *E. adenophorum* has seriously impacted the ecological environment in China's middle subtropical zones, including Yunnan, Guizhou, Sichuan, and Guangxi Provinces, by encroaching farmlands, pasture fields, and forests [7] . Manual, chemical, or biological control of *E. adenophorum* has hindered its comprehensive development and utilization for economic benefit. Many bioactive components isolated from *E. adenophorum* have shown antimicrobial activity and immunomodulating 2 Evidence-Based Complementary and Alternative Medicine properties [9] . In a recent study, the anti-inflammatory properties of ethanolic leaf extract was evaluated [10] . However, there have been few reports addressing the bioactivity of *E. adenophorum* polysaccharide (EAP).

The immunomodulating properties and therapeutic potential of a large number of botanical polysaccharides have been reported [11] . Several polysaccharides from *Cordyceps militaris*, *Portulaca oleracea*, *Gracilaria lemaneiformis*, *Gyrodinium impudium*, and *Panax ginseng* have been described as efficacious antiinfluenza agents against H1N1 and H3N2 strains [12] [13] [14] [15] . In recent reports, polysaccharidebased adjuvants enhanced the immunogenicity and improved the protective efficacy of H5N1 vaccines in animal infection models [16, 17] . However, to our knowledge there have not been any reports regarding the treatment with EAP against highly pathogenic H5N1 influenza.

In the present study, we investigated the potential effect of EAP against H5N1 influenza infection in a mouse model. Immune enhancement effects and the innate immune recognition of EAP were also evaluated. Our results suggest the anti-H5N1 effects of EAP offer an alternative strategy for developing antiinfluenza agents and the utilization of *E. adenophorum* products.

Virus. The H5N1 influenza virus (A/bar-headed goose/ Qinghai/1/2010) used in this study was isolated from Qinghai Lake in May 2010. This isolate is highly pathogenic in poultry, mouse, and Madin-Darby canine kidney (MDCK) cells. The virus was propagated in MDCK cells at 37 °C for 48 h, and the viral supernatant was harvested, aliquoted, and stored at –80 °C. Viral titers were determined by plaque assay as described previously [18] .

Animal and Cells. 8-10-week-old Female BALB/c mice were obtained from Vital River Laboratories (Beijing, China), and the original breeding pairs were purchased from Charles River (Beijing, China). Mice were raised in independent ventilated cages (IVC) and received pathogen-free food and water. Animal treatments were governed by the Regulations of Experimental Animals of Beijing Authority, and approved by the Animal Ethics Committee of the China Agriculture University.

The mouse leukemic monocyte macrophage Raw 264.7 cell line, human lung adenocarcinoma epithelial A549 cell line, and Madin-Darby canine kidney (MDCK) cell lines were provided by the Cell Resource Center of Peking Union Medical College. The cells were cultured and maintained according to the supplier's recommendations.

Yunnan province, China. The leaves were sliced and dried in shade. 100 g dried materials were powdered in a mixer and then filtered with 40 meshes. Leaf powder was extracted by ultrasonic treatment with 1000 mL of distilled water for 45 min. The supernatant was collected and the precipitate resuspended in 1000 mL of distilled water and again extracted by ultrasonic treatment for 30 min. The resulting supernatant was combined with that obtained from the first ultrasonic treatment. The final aqueous fraction was evaporated to dryness in a rotary evaporator. The residue obtained was dissolved in distilled water and kept frozen at 4 °C.

The extract was centrifuged at 3000 g/min for 25 min and concentrated under 80 °C for 8 h to prepare polysaccharide. The supernatant was then deproteinized using the Sevag method, and dialyzed against water for 48 h. The final liquid was mixed with three-fold volume of 95% ethanol (v/v) and centrifuged at 3000 g/min for 10 min. The precipitates were successively washed with absolute ethanol, ether, and dried under vacuum at 40 °C to obtain the crude polysaccharide (yield = 1.2%). EAP content was determined by the phenol-H₂SO₄ method [19] .

Vitro. 2.5 mL A549 and Raw 264.7 cells (4×10^5 /mL) per well were plated in 6-well plates and cultured at 37 ° C under 5% CO₂ for 24 h. Media was removed and 2.5 mL culture medium containing different concentrations of EAP (50, 100, 200 g/mL) was added to each well. Controls were treated with phosphate-buffered saline (PBS). Cells were collected 36 h after treatment for RNA extraction and quantitative polymerase chain reaction (qPCR).

Assay. Mice were administrated EAP at a dose of 5, 10, 25, or 50 mg/kg body weight, intranasally once daily for 5 days before the challenge. Control mice were administered PBS using the same schedule. Influenza virus stocks were diluted in PBS. Mice were anesthetized with Zotile (Virbac, France) intramuscularly at 15 mg/kg (body weight) and then infected intranasally with 120 plaqueforming units (PFU) of H5N1 influenza virus in 50 L. The lung tissue of five mice per group was collected on day 0 before challenge for qPCR and ELISA. Lung tissue from another five mice on day 3 postinfection was collected for plaque assay and qPCR. Ten mice per group were observed for survival for 14 days and body weights recorded.

2.6. Plaque Assay. MDCK cells were cultured in DMEM (Hyclone Laboratories, Logan, UT, USA) containing 10% FBS (Hyclone Laboratories), 100 U/mL penicillin, and 100 g/mL streptomycin (Invitrogen, San Diego, CA, USA). Lung tissue supernatant was diluted 10-fold and added to a cell monolayer covered by semisolid agar containing 0.5 g/mL of trypsin TPCK (Sigma-Aldrich, St. Louis, MO, USA). Plates were incubated at 37 ° C, 5% CO₂ for 60-72 h and stained with 1% crystal violet.

Total RNA from 1×10^6 cells or 10 mg lung tissue were prepared by Trizol (Invitrogen) according to the manufacturer's instructions. DNaseI-treated RNA (0.2 g) was reverse transcribed into cDNA using random primers. The expression of the hemagglutinin (HA) gene of H5N1 influenza virus was detected by qPCR using the Power SYBR Green PCR Master Mix kit (Applied Biosystems, Foster City, CA, USA). The following primers AGG CAC CA-3' 5'-CTC CTT AAT GTC ACG CAC GAT TTC-3' h IL-6 5'-CCT TCG GTC CAG TTG CCT TCT-3' 5'-CCA GTG CCT CTT TGC TGC TTT C-3' h IFN were used: forward primer, 5'-CGC AGT ATT CAG AAG AAG CAAGAC-3'; and reverse primer, 5'-TCC ATA AGG ATA GAC CAG CTA CCA-3'. The reaction was run on an ABI 7500 thermal cycler with an initial denaturation step at 95 ° C for 10 min, followed by 40 cycles of 95 ° C for 15 s, 56 ° C for 30 s, and 72 ° C for 40 s. The

copy number of the HA gene was calculated by 7500 software v2.0 (Applied Biosystems) using an HA-containing plasmid of known concentration as a standard.

Relative qPCR was performed for other eight genes: hactin, h IL-6, h IFN-, and hTNF-for A549 cells; mactin, mTLR-2, mTLR-4, mDectin-1, mMR, mL-6, mIFN-, and mTNF-for Raw264.7 cells. The sequences of primers were shown in Table 1 . The reaction was run with 95 ° C for 10 min, followed by 40 cycles of denaturation at 95 ° C for 15 sec, annealing at 52 ° C for 30 s, and extension at 72 ° C for 40 s. The fold change in gene expression was normalized to controls (naive mice) by $2^{-\Delta\Delta CT}$ using -actin as an internal standard [20] .

2.8. ELISA. IL-6, TNF-, and IFN-levels in lung were tested with ELISA kits (Boster, Wuhan, China) according to the manufacturer's protocol. One gram of lung tissue from each mouse was ground in 1 mL PBS and centrifuged for 20 min at 5000 rpm. The supernatants were collected and diluted 10fold for ELISA. 2.10. Statistical Analysis. The statistical analysis was performed using one-way ANOVAs with SPSS 12.0 (SPSS Taiwan Corp., Taiwan), and < 0.05 was considered significant.

Many botanical polysaccharides exhibit an immunomodulatory effect [11] . To determine the immunomodulatory properties of EAP, we investigated the potential effect of the polysaccharides on A549 and Raw264.7 cells. Cells were treated with various concentrations of EAP (50, 100, 200 g/mL) for 36 h. The mRNA levels of IL-6, TNF-, and IFN- were detected by qPCR. Figure 1 shows the immunomodulatory activities of EAP in vitro. Various concentrations of EAP triggered a strong secretion of IL-6, TNF-, and IFN-in a dosedependent manner both in A549 cells (Figures 1(a)-1(c)) and Raw264.7 cells (Figures 1(d) -1(f)) compared with the PBS treatment group.

To test whether EAP could protect H5N1 infected mice, mice were treated with EAP at a dose of 5, 10, 25, or 50 mg/kg body weight intranasally once daily for 5 days prior to viral challenge with 120 PFU. Ten mice per group were monitored for 14 days for the survival rate. As shown in Figure 2 (a), all mice receiving PBS died at day 11. Mice administrated 25 mg/kg EAP had a survival rate of 50% at day 14, which was significantly higher than those receiving PBS (by log rank analysis). EAP treatment of 10 mg/kg and 50 mg/kg also appeared to have a survival advantage, but not statistically significant. This result suggests

that the protective effect of EAP against H5N1 infection requires a moderate dose. EAP treatment also alleviated weight loss in infected mice (Figure 2(b)).

To determine the viral load in the lung of the infected mice, plaque assays and qPCR were performed. The pulmonary viral titers in the EAP (25 mg/kg) group were significantly lower than the titers in the mice that received PBS at day 3 postinfection (Figures 2(c) and 2(d)). These data clearly indicate that intranasal administration of EAP controls H5N1 viral replication and improves survival rates in a mouse model. The protective effect of EAP against H5N1 virus is likely due to its immunomodulatory properties. To detect IL-6, TNF-, and IFN-expression, lungs of five mice per group were collected at day 0 before infection and tested by qPCR and ELISA. The mRNA levels in the EAP group (25 mg/kg) were significantly higher than those in the PBS control (naive mice) (Figures 3(a)-3(c)). Soluble cytokine levels at day 0 were measured by ELISA, and results were consistent with the qPCR results, even though IFN-production in the EAP group was not significantly higher than that of the PBS group (= 0.0599) (Figures 3(g)-3(i)). These results suggest that EAP increases the IL-6, TNF-, and IFN-production.

IL-6, TNF-, and IFN-expression at day 3 postinfection was determined by qPCR. In contrast, TNF-mRNA levels following EAP (25 mg/kg) treatment were significantly lower than those in the PBS group (Figure 3(e)), while IL-6 and IFN-expression were only slightly lower (not significant) (Figures 3(d) and 3(f)). These results may be explained by a higher viral load, and the more severe inflammatory response in PBS treated mice.

Excessive inflammation can cause severe lung lesions during H5N1 influenza infection. To evaluate histopathological changes in the lungs of infected mice, tissues of each group at day 3 postinfection were examined. The lungs of PBS treated mice exhibited a severe inflammation response, characterized by interstitial edema, inflammatory cellular infiltration around small blood vessels, alveolar lumen flooded with edema fluid mixed with exfoliated alveolar epithelial cells, and a thickening of alveolar walls (Figures 4(c) and 4(d)). The lungs of EAP (25 mg/kg) treated mice exhibited milder lesions than those receiving PBS, characterized by signs of bronchopneumonia with interstitial edema, and inflammatory cell infiltration around small blood vessels (Figures 4(a) and 4(b)). Viral loads and inflammatory cytokine production in the lung were correlated; suggesting that EAP treatment reduces lung lesions in H5N1 infected mice.

Polysaccharides derived from many plants enhance the secretion of cytokines and chemokines, such as TNF-, IL-6, IL-8, and IL-12 [11] . This immunomodulatory effect is mediated mainly through recognition of polysaccharide polymers by several pattern recognition receptors (PRRs). To determine which receptor contributes directly to the innate immune recognition of EAP, Toll-like receptor 2 (TLR2), TLR4, Dectin-1, and mannose receptor (MR) were examined by qPCR both in vivo and in vitro. Mice were treated with EAP at a dose of 25 mg/kg body weight intranasally once daily for 5 days, with control mice receiving PBS. Lung total RNA was prepared for qPCR. The expression of Dectin-1 and MR in EAP treated mice was significantly elevated compared with controls, while expression of TLR2 and TLR4 were slightly higher, but not statistically significant (Figure 5(a)). In vitro assay showed similar trends. As shown in Figure 5 (b), Raw264.7 cells were treated with 200 g/mL EPA for 36 h before qPCR. Dectin-1 and MR levels were significantly higher, while expression of TLR2 and TLR4 did not change. These data suggest that EAP recognition occurred mainly via the Dectin-1 and MR pathway.

In this study, we evaluated the immunomodulatory activities and protective effect of EAP against H5N1 influenza infection in a mouse model. To our knowledge, these findings are the first to show the anti-H5N1 effect of EAP. Intranasal administration of EAP prior to H5N1 viral challenge improved survival rates of infected mice with a corresponding reduction of pulmonary viral load. The anti-H5N1 effect was very likely due to the innate immune recognition of EAP and the secretion of innate immune mediators (IL-6, TNF and IFN-) before infection. Furthermore, the effect of EAP on PRR expression (including TLR2, TLR4, Dectin-1, and MR) was determined both in vivo and in vitro. These results suggest that the innate immune recognition of EAP was dependent upon the activation of the Dectin-1 and MR pathways. Our data demonstrate the feasibility of using EAP as a novel immunomodulatory agent against influenza infection. Unfortunately, the sugar composition of EAP has not been characterized.

The emergence of new drug-resistant strains resulting from antigenic drift limits the therapeutic benefits of vaccination and antiviral agents in controlling influenza [6, 21, 22] . Thus, development of novel broad-spectrum antiinfluenza strategies is urgently needed. Most botanical polysaccharides are ideal candidates for novel immunomodulatory agents due to their nontoxic properties and fewer side effects compared with bacterially derived polysaccharides. A number of polysaccharides isolated from plant and fungi exhibit effective antiviral benefits against influenza A virus (including H1N1 and H3N2 subtypes) [12] [13] [14] [15] . The use of polysaccharides as immunomodulatory agent in anti-H5N1

studies is rare. In this paper, our data show the immunomodulatory activities of EAP both in vivo and in vitro. EAP treatment elevated the production of IL-6, TNF-, and IFN and provides a survival advantage in H5N1 infected mice. The survival rate following EAP pretreatment (25 mg/kg body weight) was significantly higher than in mice receiving PBS (50% to 0%).

In previous reports, high levels of proinflammatory cytokines and chemokines (including TNF-, IL-6 and IFN-) were detected during H5N1 infection [23, 24]. This "cytokine storm" leads to the severe respiratory symptoms and host immune injury. Thus, H5N1-induced cytokine storms are hypothesized to be the main cause of mortality, and the use of anti-inflammatory agents may therefore provide a therapeutic effect [25, 26]. However, it is unclear whether the lack of proinflammatory cytokines (such as TNF and IL-6) facilitates viral clearance. Interestingly, knockout 8

Evidence-Based Complementary and Alternative Medicine mice deficient in TNF-, TNF-receptor, IL-6, MIP-1, and IL-1R or steroid-treated, wild-type mice did not have a survival advantage compared with wild-type mice following H5N1 influenza infection [27, 28]. Interestingly, prophylactic treatment of TLR3 agonist PolyI:LC, which strongly upregulates cytokine production, provides protection against H1N1 and H5N1 infections [29, 30]. These conflicting studies may be explained in that the inflammatory response helps clear the virus, while aggravating host pathological damage. Elevated production of cytokines, such as IL-6, TNF-, and IFN are very important for viral clearance in the early stage of infection by activating the innate immune system. Once the viral infection has triggered a cytokine storm due to the high viral load, the inflammatory response causes severe pathological injury or even death. In this case, receiving an immunomodulator alone cannot help animal to survive [25]. This likely explains why immunomodulator treatment prior to viral infection results in a better survival rate [26, 30]. In our study, treatment of EAP shortly after infection or 24 h postinfection did not provide a survival advantage (data not show).

The antiinfluenza properties of IL-6, TNF-, and IFN have been discussed in many studies, despite their participation in cytokine storms triggered by influenza infection. IL-6 plays an important role in protecting against influenza A virus as it is required for viral clearance and essential for animal survival [31]. TNF- has been reported to exert a defensive effect against influenza infection in vitro [32]. IFN-treatment in the early stages of influenza infection improves the survival rate in mouse models [33]. In addition, high levels of IFN-secretion

stimulated by ginseng polysaccharides provide an antiinfluenza effect in vivo [12] . In this report, intranasal administration of EAP before H5N1 challenge elevates expression of IL-6, TNF-, and IFN compared with mice receiving PBS. The high levels of these mediators contribute to the viral clearance and antiviral response. Pulmonary viral titers following EAP treatment were lower at day 3 postinfection. In contrast, IL-6 and IFN-mRNA levels were slightly lower, while TNF-production was significantly lower than that of PBS group. Regarding the excessive inflammation induced by H5N1 virus, massive secretion of mediators contributes to lung injury rather than an antiviral response. Therefore, the timing of EAP treatment as a prophylactic agent is very important.

The immunomodulatory activities of botanical polysaccharides are thought to be mediated by several PRRs [11] . In this study, we examined the mRNA levels of TLR2, TLR4, Dectin-1, and MR after EAP treatment. EAP was found to upregulate Dectin-1 and MR mRNA expressions significantly both in vivo and in vitro. Our hypothesis is that the innate immune recognition of EAP is driven mainly via a Dectin-1 and MR dependent pathway. Binding to these receptors, EAP may activate complex intracellular signaling pathways, and increase cytokine production, leading to an antiviral response. Thus, the protection against H5N1 by EAP treatment is less likely to cause drug resistance, and may represent a broad-spectrum antiinfluenza effect.

In conclusion, our study demonstrates that EAP leaf extract is a prophylactic and immune enhancement agent against H5N1 influenza virus infection. Treatment with EAP effectively inhibits H5N1 viral replication and improves animal survival. This approach offers an alternative strategy for antiinfluenza immunomodulatory agent development, and benefits the utilization of *E. adenophorum* products.

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A 3-year prospective study of the epidemiology of acute respiratory viral infections in hospitalized children in Shenzhen, China

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Abstract: **BACKGROUND:** The epidemiology of local viral etiologies is essential for the management of viral respiratory tract infections. Limited data are available in China to describe the epidemiology of viral respiratory infections, especially in small–medium cities and rural areas. **OBJECTIVES:** To determine the viral etiology and seasonality of acute respiratory infections in hospitalized children, a 3-year study was conducted in Shenzhen, China. **METHODS:** Nasopharyngeal aspirates from eligible children were collected. Influenza and other respiratory viruses were tested by molecular assays simultaneously. Data were analyzed to describe the frequency and seasonality. **RESULTS:** Of the 2025 children enrolled in the study, 971 (48·0%) were positive for at least one viral pathogen, in which 890 (91·7%) were <4 years of age. The three most prevalent viruses were influenza A (IAV; 35·8%), respiratory syncytial virus (RSV; 30·5%) and human rhinovirus (HRV; 21·5%). Co-infections were found in 302 cases (31·1%), and dual viral infection was dominant. RSV, HRV and IAV were the most frequent viral agents involved in co-infection. On the whole, the obvious seasonal peaks mainly from March to May were observed with peak strength varying from 1 year to another. **CONCLUSIONS:** This study provides a basic profile of the epidemiology of acute respiratory viral infection in hospitalized children in Shenzhen. The spectrum of viruses in the study site is similar to that in other places, but the seasonality is closely related to geographic position, different from that in big cities in northern China and neighboring Hong Kong.

Text: Acute respiratory tract infections (ARTIs) are a persistent and pervasive public health problem in both developed and developing countries. They cause a great burden of disease worldwide. Especially in developing countries including China, ARTIs, mainly pneumonia, are the leading cause of death among children under the age of 5 years. 1,2 A great variety

of pathogens can cause ARTIs, and viruses have been considered as the predominant pathogens in this children population. 3, 4 The most frequently reported viruses include respiratory syncytial virus (RSV), influenza viruses A and B (IAV, IBV), parainfluenza viruses (PIVs), human rhinovirus (HRV) and adenovirus (ADV), which are responsible for most episodes of ARTIs in children. 1 In the past decade, several new viruses associated with ARTIs such as human metapneumovirus (HMPV), novel strains of coronaviruses (SARS-CoV, HCoV-NL63 and HKUI), human bocavirus (BOV), WU polyomavirus (WUPoV) and KI polyomavirus (KIPoV) have been discovered in human respiratory tract specimens. Among them, some have been identified to be causative pathogens of ARTIs. 1, 4, 5 Currently, there are no approved vaccines or medications available for most of the respiratory viruses. 1 A better understanding of the epidemiology of viral respiratory tract infections in children plays a key role for the prevention, control and treatment of ARTIs. Studies showed that many viral respiratory infections exhibited predictable seasonal variations. However, the epidemiological profiles of viral respiratory infections from different climate zones or different countries in the same climate zone may be varied. [6] [7] [8] [9] [10] [11] [12] China is a large country crossing three climate zones, and great differences in climate are found from region to region. A better understanding of the epidemiology of ARTIs in different regions could be helpful to develop effective surveillance, prevention and treatment strategies. Although some studies on the epidemiology of ARTIs have recently been reported in big cities such as Beijing, Shanghai and Hong Kong, [13] [14] [15] [16] the epidemic characteristics of viruses in ARTIs are still not well established all around China, especially in other cities and rural areas.

Shenzhen is the largest migratory city of China with high population density and population mobility. It is located in southern China at 22°27'0" - 22°52'0" N and 113°46'0" - 114°37'0" E, immediately north of Hong Kong, with a typical subtropical monsoon climate. The annual average temperature and relative humidity of Shenzhen are about 23°C (12-33°C) and 77%, respectively. The purpose of this study is to investigate the prevalence, seasonality and clinical characteristics of acute viral respiratory infections in hospitalized children in Shenzhen and to provide insights into etiologies of ARTIs in local infants and children.

A consecutive 3-year prospective study from July 2007 to June 2010 was conducted in Shenzhen, a coastal city neighboring Hong Kong. Four hospitals including a children's hospital were chosen for the study. Selected patients with ARTIs admitted to the pediatric wards were enrolled. The inclusion criteria were as follows: less than 14 years old, acute fever ($T \geq 38^{\circ}\text{C}$), with any one of respiratory symptoms (such as sore throat, cough,

wheezing and dyspnoea/ tachypnoea), normal or low leukocyte count, the onset of illness within 3 days before hospitalization. The diagnosis of pneumonia was based on the guideline of the management of childhood community acquired pneumonia (CAP) issued by the Chinese Medical Association in 2006.¹⁷ In the guideline, the clinical symptoms and signs for the diagnosis of childhood CAP include fever, cough, tachypnoea (defined according to different age), difficulty breathing and/or lower chest wall indrawing. X-ray evaluation has been carried out when necessary. The study protocol was approved by the medical ethical committees of the hospitals. Written informed consent was obtained from the parents or legal guardians of the children.

Nasopharyngeal aspirates (NPA) were obtained by trained personnel following standard operating procedures within 24 hour after admission. The specimens were transported immediately to the laboratory by sterile viral transport media, then divided into aliquots and immediately frozen at À80°C until further processing.

Total viral nucleic acids (DNA and RNA) were extracted from 200 µl of NPA specimen using the AxyPrep Body Fluid DNA/RNA Miniprep Kit (Axygen, Union City, CA, USA), according to the manufacturer's instructions. Purified DNA and RNA were stored at À80°C in aliquots for further PCR analysis.

For each specimen, assays for ten common and newly identified viruses were performed. Briefly, WUPoV and BOV were tested using monoplex PCRs described previously.^{18, 19} Other viruses were tested using the Luminex platform and multiplex xTAG™ respiratory viral panel assay (RVP Assay) according to the manufacturer's instructions.²⁰ All multiple infection samples were retested. If there was discordance between two tests, the sample was confirmed by monoplex PCR.

Statistical Package for the Social Sciences (SPSS) for Windows version 11.0 (SPSS Inc. Chicago, IL, USA) was used. For comparison of categorical data, chi-square or Fisher's exact test was used. All tests were two-tailed, and a P value below 0.05 was considered statistically significant.

A total of 2025 specimens were obtained from 2025 eligible patients ranging from 15 days to 14 years old with a median age of 12 months, in which 89.6% of patients were < 4 years old. There were 964 (47.6%) females and 1061 (52.4%) males included. Of all hospitalized children enrolled in this study, 84.0% involved lower respiratory infection and 16.0% had upper respiratory tract infection (Table 1) . Among 971 positive cases, 572 (58.9%) were diagnosed as pneumonia.

About 971 of the 2025 cases (48.0%) were positive for at least one viral pathogen. Among them, IAV, RSV, HRV and PIVs were detected in 348 (35.8%), 296 (30.5%), 209 (21.5%) and 169 (17.4%) cases, respectively. Single infection was observed in 669 (68.9%) cases, and multiple infection was found in 302 (31.1%). Our results also showed that RSV, IAV and HRV were the main pathogens in single viral infection cases (Table 1) . The monthly positive rates varied from 32.5% to 75.0% with a mean of 45.1% (Figure 1). In the year 2009, when influenza A (H1N1) was pandemic worldwide, the positive rate started to increase in March and the highest positive rate 75.0% was observed in May.

Among the 971 positive cases, a total of 1335 viral pathogens were detected. The most frequently detected pathogen was IAV (26.1%, 348/1335), followed by RSV (22.2%, 296/1335), HRV (15.7%, 209/1335), PIV1 and PIV3 (12.7%, 169/1335) (

About 302 co-infection cases were identified, accounting for 14.9% of all 2025 hospitalized children. During the H1N1 outbreak from March to August 2009, co-infection cases and co-infection rate increased significantly (Figure 2). 143 of 302 (47.4%) co-infection cases were detected during that time. Among them, 121 cases were involved in IAV infection, including 90 dual infection cases. Of all co-infection cases, 247 (81.8%), 49 (16.2%) and 5 (1.7%) were infected with two, three and four potential viral pathogens, respectively. One multiple infection with five viruses was detected in a RSV IAV HRV BOV PIV3 ADV HMPV WUPoyV PIV1 IBV Total A Total cases 163 172 85 55 51 49 37 26 24 7 669 Bronchitis 12 19 7 7 5 2 2 3 5 0 62 Bronchiolitis 45 19 20 7 13 9 8 2 5 0 128 Pneumonia 93 97 55 28 24 28 25 18 14 5 387 URTI 13 37 3 13 9 10 2 3 0 2 *Case number and percentage in all enrolled children. **Incidence rate in this age group significantly higher than the other age groups.

***No significant difference between these three age groups. †No significant differences between these two age groups, but significantly lower than the other age groups. 6-month-old infant. IAV, RSV and HRV were the three most frequently found viruses in co-infection and detected in 176, 133 and 124 cases with co-infection rates of 50.6%, 44.9% and 59.3%, respectively (Table 2) . Various multiple infection patterns were observed in the study. A total of 152 (50.3%) co-infection cases involved at least two viruses of RSV, HRV and IAV. Co-infection rate of each individual virus detected varied significantly. The lowest and highest co-infection rates were observed in WUPoV (33.3%) and IBV (66.7%), respectively. 91.4% (276/302) of co-infection cases were tested in the age group of 4 years old or younger (Table 2) , but among all age groups, no statistical difference in co-infection rate was found ($\chi^2 = 1.83$, $P = 0.8721$). Gender-specific difference in co-infection rate was not observed ($\chi^2 = 2.17$, $P = 0.1404$). There was no significant difference in co-infection rates between PICU and non-PICU cases. Similarly, no significant difference in clinical symptoms was observed between co-infection and single cases (data not shown).

In general, respiratory viruses were detected more often in the period of March to May than in other months (55.4% and 40.6%, respectively, $\chi^2 = 28.06$, $P = 0.0000$), and obvious seasonal peaks were observed during those months with peak strength varying from 1 year to another. A weaker seasonal peak could also be distinguished in some winter months in different years (Figure 1) .

The seasonality profile of each individual virus detected was diversified. A seasonal distribution of IAV can be observed from late spring to summer (mainly March to May) and sometimes in fall (October, November or December). A wide seasonal peak of IAV infection was detected from March to August 2009 (Figure 3A). Although RSV was tested almost a whole year, two yearly peaks were identified. One was found in November and/or December and the other stronger one was found in March to May of the year. The peak duration in 2009 was longer than those in other years. The seasonal trends of HRV and PIVs were similar to that of RSV, but the peaks of these three viruses fluctuated and shifted mildly (Figure 3B). Although IBV and ADV had a low detection rate in the study, similar seasonality was observed and their infection peaks were mainly in midwinter. Peaks in spring and summer were also observed in some years (Figure 3C).

Our investigation did not find regular seasonality in BOV infections. A sudden increase in BOV infection was recorded in April and May 2010. Although the positive rate of HMPV

infection was only 4.8%, regular seasonality was observed from March to May of each year. Of 39 patients with WUPoyV infection, 36 were detected after July 2008. Our data implied that peak months of WUPoyV infection were from March to May (Figure 3D).

The positive rates of viral infections in male and female were 52.5% and 47.5%, respectively. No significant gender difference was revealed ($\chi^2 = 0.012$, $P = 0.9118$). The distribution of viral agents and infection patterns in different age groups are shown in Table 2 .

Of all 971 positive children, 890 (91.7%) were 4 years old or younger. The positive rate in this age group was significantly higher than that in children more than 4 years old ($\chi^2 = 8.26$, $P = 0.0041$). Children under 6 months were the most susceptible to respiratory viral pathogens with a positive rate of 14.8% (Table 2) .

Very few long-term prospective studies were performed for viral etiologies of ARTIs among hospitalized children. In this present study, the infection frequency, seasonality, co-infection pattern and clinical features of viral respiratory infections were investigated based on prospective analysis of three consecutive year's data from hospitalized children with ARTIs. Our results provided a distinctive epidemiological profile of viral respiratory infections in hospitalized children with ARTIs in the study areas, which was different from those in the big cities in northern China such as Beijing and Shanghai and also different from that in adjacent Hong Kong.

Overall, 48.0% of our cases were positive for respiratory virus infections, which resembled the latest study in the same city. 21 A similar incidence rate has been obtained in neighboring regions 13, 22 and other cities such as Rome 23 and Milan, 24 but it was different from other studies. [10] [11] [12] In China, the overall positive rate reported varied from 27.3 to 74.8% depending on different areas and detection methods. 15, 16, [25] [26] [27] [28] [29] [30] [31] The rate of respiratory viral infections varied worldwide, and many factors such as geographic distribution, study design and detection protocols could lead to these variations. 1, 7, 8, 32 In our study, leukocyte count was used as an indicator of inclusion criteria and it probably affected the positive rate. Viruses not considered in the study, for example coronaviruses, would underestimate the positive rate. Most studies showed that RSV or HRV was the most prevalent viruses in children with viral respiratory

tract infection. 1 In this study, IAV was the most frequently detected respiratory virus, followed by RSV and HRV. IAV (H1N1) outbreak in 2009 could explain this shift. Data showed that about 60% of IAV infections were detected during the outbreak period. Studies showed that the H1N1 outbreak could change viral distribution patterns. 24, 29, 33 Regardless of the IAV (H1N1) outbreak, RSV and HRV were the two most common viral pathogens in ARTIs, which was consistent with most previous studies. 1, 10, 15, 16, 22, [25] [26] [27] [28] [29] Our study further confirmed the importance of RSV and HRV in children with ARTIs, especially in children < 4 years of age. 10, 14, 23 Our results also showed that 12.7% of viral pathogens detected were PIV1 and PIV3, which implied that PIVs played an important role in children with ARTIs. Similar findings were obtained in the studies conducted in Shanghai, 14, 34 Changsha, 26 Harbin, 30 Hong Kong 13 and Rome. 23 The prevalence of PIV3 was twofold higher than that of PIV1, particularly in infants, which was similar with other reports, 25, 26, 30, 35 implying that infants could be more vulnerable to infection with PIV3 than PIV1.

HMPV has been proven to be one of the main viral pathogens responsible for ARTIs in children. 5 The positive rate found in the study was consistent with previously published results. 10, 36, 37 In China, the infection rate of HMPV varied from 3.2 to 10.6%. 22, 26, 28, 29, 31 The seasonality of HMPV in this study was mainly from March to May, similar to that in Hong Kong, 36 but different from other places. 5, 37 In our study, 4.9% of cases were positive for BOV, which coincided with 5.0% in Hong Kong 38 and higher than Guangzhou 39 and eastern Guangdong. 22 Our result suggested that BOV might be present throughout the year with no seasonal distribution. However, seasonal distribution was noted from September to February in Hong Kong 38 and May and June in Guangzhou. 39 The use of multiple PCR made it possible to simultaneously detect a broad spectrum of viruses with excellent sensitivity, at the same time, with increased viral detection rate and co-infection rate for ARTIs. 12,40 Among our positive cases, co-infection rate was 31.1%, which was similar to 27.9% reported by Do et al. 10 Co-infection rate reported elsewhere varied widely from 25.4 to 47.9%. 40 The relatively lower co-infection rates ranging from 0.24 to 26.9% were reported in the studies conducted in various cities of China. 22, [25] [26] [27] [28] [29] [30] [31] In most of these studies, immunofluorescence kits were used to test a lower number of respiratory viruses. It was worth to note that in the study by Peng et al. 32 in Wuhan, China, 69.5% of co-infection rate was reported with immunofluorescence kit. These variations might be attributed to geographic differences, diagnostic methods for viral agents and study design. 12, 32, 34, 41, 42 Pathogens in those negative patients need to be further investigated as only ten common and newly identified viruses were included in

our study, which might underestimate positive rate or coinfection rate. It was notable that the correlation between co-infection rate and positive rate was not observed.

Of multiple infections, dual infection was predominant in this study whether or not considering the IAV (H1N1) outbreak in 2009, which was consistent with previous studies. 28, 32, 42, 43 Similar with the studies conducted in the cities of Guangzhou and Wuhan, China, 28, 29 our study showed that IAV, RSV and HRV were the main viruses involved in multiple infections. High co-infection rate between these three viruses could be explained from the overlap of their seasonal distributions. A variety of predominant multiple infection patterns between respiratory viruses were observed in different studies. 12, 32, 42, 43 For example, it was shown in Martin et al.'s study 43 that ADV and coronaviruses were the most common co-infection pattern.

Our study showed that RSV and HRV were the two most viruses involved in multiple infection, followed by IAV and PIVs, regardless of IAV infection in the H1N1 outbreak period. It was difficult to explain the variations of coinfection patterns based only on seasonal distribution. A recent study suggested that co-infection patterns were not random and certain pathogens had higher frequency of coinfection. 41 As molecular assays only detect nucleic acid and positive result does not mean the presence of the pathogen, when studying co-infection patterns of respiratory viruses, the ability to differentiate the real causative pathogens needs to be solved first. Viral load detection could provide some clues for solving this issue. 43, 44 Although high co-infection rates have been reported in various studies, the associations among multiple infections, hospitalization rate and severity of ARTIs were still not clear with inconsistent results in different studies. 42, 43, 45 Our data suggested that multiple infection had less association with the severity of disease, consistent with Peng et al.'s study. 32 The relationship between co-infection rates and age group was also investigated in our study, and little correlation was observed. Several previous studies observed that co-infection rates were more frequent in a certain age group, but results were varied. 32, 43 In contrast to temperate region, where most viruses had winter-spring seasonality, the respiratory viral infections in tropical and subtropical regions appeared mainly to be spring-summer seasonality. 9 In this study, due to the high detection rate and similar seasonality of RSV, HRV, IAV, PIV and HMPV, an overall spring-summer seasonality of viral respiratory infections in children was concluded. Studies conducted in Hong Kong showed that a clear seasonal peak was from April to September, 36, 46 with a longer duration than our study. The overall seasonality in this study was also different from the studies conducted in northern or central cities of China,

in which the seasonality of most viruses presented in autumn-winter and/or winter-spring. 15, [25] [26] [27] 30 The winter-spring seasonality was also observed in Guangzhou, a city about 150 kilometers north of Shenzhen. 28 Different seasonal onset and duration were observed in various studies conducted in (sub-) tropical regions. In these studies, ambient temperature, humidity and rainfall were widely used to explain these differences in seasonality, but inconsistent results were observed. 9, 46, 47 Although most studies demonstrated that the seasonality of viral respiratory infections was correlated with increased rainfall, effects of climate factors such as humidity and temperature on the seasonality were complex and interactive. 9, 46, 48 The study areas have four indistinct seasons, and the coldest month usually emerges in January (average 12°C). During the period from March to May, the weather featured warm ambient temperature (average 18-25°C), high relative humidity (average 85%-90%) and increasing rainfall. These meteorological conditions were perhaps conducive to viral survival. 9, 48 In addition, intensive temperature fluctuations during seasonal alternation could increase the susceptibility to infections. 49 As reported in other studies in temperate, tropical and subtropical regions, viral infection rates in children population showed an inverse correlation with age, with younger individuals experiencing higher viral infection rates. 3, 4, 6, 9, 24 Our results suggested that children younger than 4 years of age, particularly <6 months, were at higher risk of hospitalization for ARTIs, compared with older children. This was particularly substantiated in RSV infection. Our presumption was supported by other studies. 14,25-28 Of course, this speculation needed to be validated by the population-based study. The findings reported elsewhere suggested that more males than females were affected by ARTIs, which were not observed in our study.

Notably, our study occurred over a span of 3 years, which included the IAV (H1N1) outbreak in 2009. The impact of the outbreak on the results should be considered. Data showed that the detection rate of IAV increased significantly and co-infection rate during outbreak months was much higher than average co-infection rate. Unfortunately, we did not type these influenza strains based on the original study design. It was most likely that these strains contributed to the relatively high proportion of IAV. Relatively higher single and multiple infections of RSV, HRV and PIVs were also observed during the outbreak of IAV. Increased susceptible population and awareness, intensive testing and altered patient and physician behavior could lead to these increases. These factors could partly explain the relatively high proportion of pneumonia cases in the study. Furthermore, studies showed that the outbreak of IAV (H1N1) could increase the risk of other viral infections such as RSV and HRV. 24, 33 Other limitations also existed in this study. First, molecular methods allowed the detection of only viral nucleic acid even without virus replication, which

complicates the interpretation of positive detection results. Second, the subtype identification of some common respiratory viruses such as IAV and HRV was not performed in our study, particularly during the IAV (H1N1) outbreak in 2009.

In summary, despite those aforementioned limitations, this three consecutive years' surveillance would provide a basic profile of the spectrum, seasonality, age and gender distribution, co-infection patterns as well as clinical association of viral respiratory infections in hospitalized children in the study sites. It could help the prediction, prevention and control of ARTIs in children.

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Spread from the Sink to the Patient: In Situ Study Using Green Fluorescent Protein (GFP)-Expressing *Escherichia coli* To Model Bacterial Dispersion from Hand-Washing Sink-Trap Reservoirs

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Abstract: There have been an increasing number of reports implicating Gammaproteobacteria as often carrying genes of drug resistance from colonized sink traps to vulnerable hospitalized patients. However, the mechanism of transmission from the

wastewater of the sink P-trap to patients remains poorly understood. Herein we report the use of a designated hand-washing sink lab gallery to model dispersion of green fluorescent protein (GFP)-expressing *Escherichia coli* from sink wastewater to the surrounding environment. We found no dispersion of GFP-expressing *E. coli* directly from the P-trap to the sink basin or surrounding countertop with coincident water flow from a faucet. However, when the GFP-expressing *E. coli* cells were allowed to mature in the P-trap under conditions similar to those in a hospital environment, a GFP-expressing *E. coli*-containing putative biofilm extended upward over 7 days to reach the strainer. This subsequently resulted in droplet dispersion to the surrounding areas (<30 in.) during faucet operation. We also demonstrated that P-trap colonization could occur by retrograde transmission along a common pipe. We postulate that the organisms mobilize up to the strainer from the P-trap, resulting in droplet dispersion rather than dispersion directly from the P-trap. This work helps to further define the mode of transmission of bacteria from a P-trap reservoir to a vulnerable hospitalized patient. **IMPORTANCE** Many recent reports demonstrate that sink drain pipes become colonized with highly consequential multidrug-resistant bacteria, which then results in hospital-acquired infections. However, the mechanism of dispersal of bacteria from the sink to patients has not been fully elucidated. Through establishment of a unique sink gallery, this work found that a staged mode of transmission involving biofilm growth from the lower pipe to the sink strainer and subsequent splatter to the bowl and surrounding area occurs rather than splatter directly from the water in the lower pipe. We have also demonstrated that bacterial transmission can occur via connections in wastewater plumbing to neighboring sinks. This work helps to more clearly define the mechanism and risk of transmission from a wastewater source to hospitalized patients in a world with increasingly antibiotic-resistant bacteria that can thrive in wastewater environments and cause infections in vulnerable patients.

Text: Despite early reports (1) (2) (3) (4) (5) , the premise that hand-wash sink traps can act as reservoirs of bacteria that cause nosocomial infections has been frequently overlooked. There has recently been an alarming increase in sink-related outbreaks worldwide, with many reports establishing an observational link (6) (7) (8) (9) (10) (11) (12) (13) . A sink often operates as an open conduit to wastewater in a patient care area that is often in the same room as the patient.

Health care establishments often invest in desperate interventions to deal with nosocomial outbreaks. The preferred method for addressing most of the environment-related transmission is to employ enhanced cleaning using chemical and physical agents (14, 15) .

Unfortunately, routine approaches are inefficient in completely eliminating drug-resistant Gammaproteobacteria in an inaccessible microbiologically active area such as a sink trap (6, (16) (17) (18) (19) (20) . The wet, humid, and relatively protected environment in a sink trap favors the formation of rich stable microbial communities (16, 21, 22) . These communities will be exposed to liquids and waste that are discarded in a sink and may include antimicrobials, discarded beverages, soap, presumably pathogenic bacteria from health care workers' hands, and other items. In short, sink traps could serve as a breeding ground for opportunistic and highly antimicrobial-resistant bacteria that cannot be easily cleaned or removed (23) (24) (25) (26) (27) (28) .

There are many reports of a genetic association between pathogens found in sink traps and those found in patients (29, 30) . However, surprisingly little work has been done to understand the microscale transmission dynamics. It was previously demonstrated using a suspension of fluorescent particles (Glo Germ; Glo Germ Co., Moab, UT) that material injected into the P-trap gets dispersed around a hand-washing sink (6) . This result however has not been replicated hitherto in the follow-up studies. Dispersion has never been investigated with living organisms. Ultimately, many details remain unaddressed surrounding the spread of Enterobacteriaceae in sink-trap wastewater systems: (i) can organisms grow retrograde from the P-trap water to the sink strainer, (ii) can organisms spread from one sink to another along the internal surfaces of pipes with shared drainage systems, and (iii) which portion of a colonized drain pipe results in dispersion into the sink bowl during a hand-washing event? We aim to better understand the dispersion dynamics of Gammaproteobacteria living in the wastewater of a sink strainer and P-trap into an area where patients and health care workers could be exposed. To study this dynamic, we used a surrogate organism that could be easily tracked while remaining in the Enterobacteriaceae family, where some of the most concerning threats in antimicrobial resistance are developing (30) .

Growth and colonization of GFP-expressing *E. coli* in the P-trap. In the first 14 days following the installation of the P-trap with established green fluorescent protein (GFP)-expressing *Escherichia coli* and just water running from the faucet, GFP-expressing *E. coli* was not detected in the tailpipe beyond 1.5 in. above the liquid level in the P-trap. GFP-expressing *E. coli*, however, was found to be viable in the P-trap without any nutrients added. A nutrient regimen was then instituted to understand the influence of nutrients on mobility and upward growth. The addition of tryptic soy broth (TSB) promoted GFP-expressing *E. coli* growth as early as day 1, with growth observed in the tailpipe 2 in. above

the liquid surface in the P-trap (Table 1) . On day 7, the strainer (j8 in. above the liquid in the P-trap) was found to be colonized with GFP-expressing *E. coli*. This translates to an average growth rate of 1 in./day along the length of the tailpipe with the addition of nutrients and without faucet operation. GFP-expressing *E. coli* was not detected in the faucet water.

Sink-to-sink transmission of bacteria. In these experiments, a flanking sink (sink 5) was the only P-trap inoculated with GFP-expressing *E. coli* and therefore was the sole source for transmission to the connected sinks. Starting with a lower inoculum concentration (10³ CFU/ml) in sink 5, on day 7, GFP-expressing *E. coli* was detected in the sink 2 and sink 3 P-traps (Fig. 1a) . With inoculum concentrations of 10⁶ CFU/ml and 10¹⁰ CFU/ml in sink 5, all of the sink P-traps in the sink gallery with the exception of sink 1 were found to be colonized with GFP-expressing *E. coli* after 7 days (Fig. 1b and c) . Faucet water and aerators tested negative for GFP-expressing *E. coli*. Irrespective of the starting inoculum concentration, on day 7 the highest level of colonization was recorded in the sink 3 P-trap. After day 7, when the nutrient regimen (described previously) was followed for an additional 7 days in each of the sinks in the sink gallery with an inoculum concentration of 10¹⁰ CFU/ml, GFP-expressing *E. coli* was detected in the strainers of sinks 2 and 3 on day 14. This finding validated the upward growth and growth rate in the tailpipe when nutrients were added. Nonfluorescent colonies were occasionally observed in the P-trap water samples collected from the sinks, which were subsequently identified to be *Pseudomonas* sp. or *Stenotrophomonas maltophilia*, and fluorescent colonies were confirmed to be *E. coli*.

Dispersion of microspheres from sinks. In the first dispersion experiment, when fluorescent microspheres were inoculated into the offset drain tailpiece only 4 in. below the strainer, no microspheres were detected on the polyester sheets placed on the counter space.

However, when the sink bowl was coated with the microspheres, polyester sheets overlaid on the counter space captured the dispersed microspheres caused by the faucet operation. Dispersion was observed on almost all zones of the sink counter space (Fig. 2) . Relatively higher levels of dispersion were observed along the major and minor axes of the elliptical sink bowl (zones 2, 5, 6, 9, 11, and 12) . Anterior corners of the sink counter space (zones 4 and 7), which were most distant from the impact of water in the sink bowl, received the lowest dispersion.

Dispersion of GFP-expressing *E. coli* from sinks. Initially the P-trap alone was inoculated with GFP-expressing *E. coli* and carefully installed, keeping the tailpipe and strainer free of GFP-expressing *E. coli* before operating the faucets. No fluorescent CFU were observed on the plates placed on the counter or attached to the bowl surface after faucet operation. Similarly, no fluorescent CFU were detected when GFP-expressing *E. coli* was inoculated into the offset drain tailpiece only 4 in. below the strainer. Interestingly, when there was conspicuous water backup over the strainer as a result of a higher water flow rate from the faucet than the drainage rate from the P-trap, dispersal was detected on the plates attached to the bowl surface.

The dispersion pattern recorded when the sink bowl was coated with GFP-expressing *E. coli* was comparable to the pattern recorded when the sink bowl was coated with fluorescent microspheres (Fig. 2) . Dispersion was significantly higher along the axes (zones 6, 9, 11, and 12) and lower at the corners of the sink counter space (zones 4, 7, and 10).

In contrast, dispersion of GFP-expressing *E. coli* caused by the faucet operation was much more extensive when the strainer was allowed to be colonized with GFP-expressing *E. coli* prior to the dispersion experiment. In addition to the sink counter space, we measured dispersion to the sink bowl, faucet, faucet handles, splatter shields, and the extended counter surface. Dispersion of GFP-expressing *E. coli* was highest on the plates attached to the sink bowl (Fig. 3b) . Further, dispersion was greater along the

"Δ" and "ε" denote the absence and presence of GFP-expressing *E. coli*, respectively.

minor axis of the sink bowl (Fig. 3b , zones B3, B4, and B10) than along the major axis of the sink bowl, associated with a shorter distance from the strike point of the faucet water to the bowl along this axis. The next highest CFU count from the dispersal was recorded on the counter area near the faucets (Fig. 3a , zones 12 and 11). A similar pattern of higher dispersion near the faucets and lower dispersion at the corners of the counter space (Fig. 3a , zones 4, 7, and 10) was also observed using microspheres. Dispersion was also recorded in other zones of the counter space, on the Plexiglas splatter shields, faucets, faucet handles, and extended surface (Fig. 3c). There were no GFP-expressing *E. coli* CFU recorded on plates placed beyond 30 in. from the strainer, demarcating the range of

dispersion under these experimental conditions. Table 2 gives a summary of the total distribution loads recorded using fluorescent microspheres and GFP-expressing *E. coli* across each experiment. The loads of dispersion on the sink counter were comparable when the sink bowl was coated with microspheres or GFP-expressing *E. coli* before the faucet operation. Although the dispersion load on the sink counter was lower when the sink strainer was colonized, it is interesting to note that the sink bowl received the highest dispersion.

To mimic dispersion in a hospital setting, we first investigated whether GFP-expressing *E. coli* would establish consistent colonization in a sink trap as many other Gammaproteobacteria implicated in nosocomial outbreaks have done (6, 28). Many recent reports demonstrate that P-traps become colonized with highly consequential Gammaproteobacteria, which then results in nosocomial transmission (29, 31, 32). The retained water in a sink P-trap is present to provide a water barrier to prevent off-gassing of sewer smell, but it may inadvertently provide favorable conditions for pathogenic and opportunistic antibiotic-resistant microorganisms to survive and develop resilient biofilms (3, 33). However, the mechanism of dispersal of the bacteria in the P-trap to patients or the surrounding health care area had not been fully elucidated. We began with the hypothesis that the bacteria originate from the P-trap via droplet creation when the water from the faucet hits the P-trap water, thus contaminating the sink bowl and the surrounding area. The finding supporting this theory had been previously reported using Glo Germ particles (6). However, in the present study with careful attention to avoid strainer and tail piece contamination, the dispersal directly from the sink P-trap with either microspheres or GFP-expressing *E. coli* could not be reproduced as previously reported (6).

Rather this work demonstrates a different, more staged mode of transmission from a P-trap reservoir to the sink and surrounding environment. GFP-expressing *E. coli* in the P-trap alone was sustained for 14 days but did not grow or mobilize up the tailpipe to the strainer with just intermittent water exposure. However, when nutrients were subsequently added to the system, the organisms rapidly grew up the tailpipe to the strainer at approximately an inch per day. In a real-world setting, motility of bacteria inside the tailpipe is restricted to relatively sporadic and brief wetting events in which swimming is an opportunity to colonize new surfaces. It is assumed that once established, the biofilm promotes the upward growth of GFP-expressing *E. coli* in the tailpipe at an accelerated rate. The nutrient regimen that promoted colonization in our model reflects our and others' observations of items

commonly disposed of in hospital sinks (intravenous fluids, feeding supplements, and leftover beverages) (5, 32) .

Transmission of bacteria between sinks via a common pipe was a key finding in this study as this highlights the concept that premise plumbing may be a more continuous system with shared microbiology than a single isolated sink. The sink gallery used in this study provided a unique in situ advantage to investigate sink-to-sink transmission of bacteria through common drains. The two possible mechanisms for P-trap strainers becoming colonized are seeding of organisms from above and retrograde spread of organisms along common pipes in a hospital wastewater infrastructure. Here we demonstrate that it is possible for GFP-expressing *E. coli* to contaminate adjacent P-traps with just time and water given a standard U.S. code piping rise of 0.25 ft/ft. Sink-to-sink or retrograde transmission may explain the recurrence of pathogen colonization following intervention strategies like disinfection or replacement of plumbing (23) . Sink 3 was lowest on the slope in the drain line (see Fig. 4) with arguably the most opportunity for reflux and retrograde wetting. Sink 1, on the other hand, was farthest away from the source (sink 5), and its P-trap had the greatest incline in the drain line connecting the sinks, which could perhaps contribute to the reasons there was no GFP-expressing *E. coli* colonization detected in it after 7 days. There has been more investigation about microbiologic dynamics of infectious viral particles such as those of severe acute respiratory syndrome (SARS) and Ebola viruses through premise plumbing systems (34) (35) (36) . However, the microbiology, sustainability, and dynamics might be very different, although the backflow and inoculation issues could have some parallels when comparing viruses to bacteria. As Enterobacteriaceae can either multiply or remain viable for long periods of time in biofilms coating the interior of P-traps and the connected plumbing, it may not be sustainable to target any intervention limited to a single isolated sink as a source of a particular pathogen.

Data from different dispersal experiments suggest that although P-traps can act as the source or the reservoir of pathogens, the physical presence of the organism in the sink bowl or colonization of the strainer is necessary for the dispersal to occur. Colonization of strainers or drains reported in earlier studies (7, 10, 13, 24, 37) was perhaps a result of ascending biofilm growth from the P-trap to the strainer or introduction through contaminated fluids. Many of the studies used swab samples, which likely sampled the strainer rather than P-trap water (17, 20) . Once the strainer was colonized, the water from the faucet resulted in GFP-expressing *E. coli* dispersion in the bowl and to the surrounding surfaces of up to 30 in. The range of dispersal (6) . Greater dispersal near the faucet may be

attributed to the specific designs of the sink bowl and faucet in this study, which determine the contact angle of water impact. This is an important finding since many sinks in hospitals are similar in design, with faucet handles representing a high-touch surface for the sink users (38) . It can also be concluded from the dispersion experiments that secondary and successive dispersals would likely increase the degree and the scope of dispersion. There are several limitations to this work. First the use of similar sink bowls across these sinks only examines the dispersion pattern of this particular sink design. Similarly the sink-to-sink transmission may not be applicable to all wastewater plumbing systems as the fixtures on the pipe are very close together, unlike most layouts in health care settings. However, we speculate that transmission could occur on larger systems over greater time scales, especially if heavy nutrient and contamination loads were also included. GFP-expressing *E. coli* is a laboratory surrogate, and the putative biofilms established in the short time frame of our experiments are unlikely to be as complex or stable as biofilms developed in a hospital wastewater system over many years. However, to address the monomicrobial dominance of the GFP-expressing *E. coli* added to the system, we kept the system open, and other environmental organisms were able to cocolonize in an attempt to mimic the hospital system. Another limitation was the need to add nutrients to the drain to ensure rapid and robust colonization. We are not clear how widespread the practice of disposing of dextrose-containing intravenous fluids or leftover beverages in the hand-wash sinks is; however, we have observed this practice, and anecdotally it appears to be relatively common in the United States. We also did not completely characterize the droplet sizes, nor do we demonstrate air sampling to understand if the dispersion is only droplet or if there are also aerosols that contain GFP-expressing *E. coli*. This would require additional testing and is planned as future work.

In summary, this work for the first time better models the mechanisms of spread of multidrug-resistant pathogens arising from the sink drain and infecting patients. Droplet dispersion from the P-trap does not happen directly. Rather it is a multistage process: dispersal originates from the strainer and/or the bowl after growth of the biofilm up from the microbial reservoir of the P-trap. We also demonstrate sink-to-sink transmission via a common sanitary pipe. This work could have implications for patient safety, infection control, and interventions as well as the design of future hospital plumbing systems to eliminate this mode of transmission to vulnerable hospitalized patients.

Sink gallery design. A dedicated sink gallery was set up to simulate hospital hand-washing sinks. The gallery was comprised of five sink modules assembled next to each other (Fig. 4)

. The five hand-wash sink stations were identical in bowl designs and dimensions and were modeled from the most common intensive care unit hand-washing sink type in the acute care hospital at the University of Virginia Medical Center. Partitions made of 24-in.-high Plexiglas sheet were installed between the sinks to prevent splatter and cross contamination. Each sink module was built with Corian integrated sink/countertops without an overflow and fitted with an 8-in. centerset 2-handle gooseneck faucet (Elkay, Oak Brook, IL). The drain line (Dearborn Brass-Oatey, Cleveland, OH). All of the fixtures were made of brass with chrome plating. Each of the sink P-traps was connected to a 3-in. common cast-iron pipe sloping into a T-joint leading into the building sanitary line located behind sink 3 (Fig. 4) .

Inoculation, growth, and establishment of GFP-expressing *E. coli* in sink P-traps. For the GFP-expressing *E. coli* strain (ATCC 25922GFP), the green fluorescent protein (GFP) gene is contained on a plasmid that also contains an ampicillin resistance gene. A single isolated colony of GFP-expressing *E. coli* grown from a $\Delta 80^{\circ}\text{C}$ stock was inoculated into 5 ml tryptic soy broth (TSB) (Becton, Dickinson and Company, Sparks, MD) containing 100 g/ml ampicillin (ATCC medium 2855). The inoculum concentration and method varied for each experiment. For establishment of GFP-expressing *E. coli* in sink P-traps, new autoclaved P-traps were filled with 100 ml 0.1x TSB and inoculated with 10^3 CFU/ml GFP-expressing *E. coli*. Following inoculation, both ends of the P-traps were covered with perforated Parafilm (Bemis, Inc., Oshkosh, WI) and allowed to incubate at room temperature ($22 \pm 2^{\circ}\text{C}$) for 14 days to facilitate adherent bacterial growth. The medium in the P-trap was decanted and replaced with fresh 0.1x TSB every 48 h. An aliquot of decanted medium and a swab sample from the inner surface of the P-trap were plated on tryptic soy agar (Becton, Dickinson and Company, Sparks, MD) plates containing 100 g/ml ampicillin (TSA) to monitor the growth of GFP-expressing *E. coli* in the P-traps. TSA plates were incubated overnight at 37°C , and CFU fluorescing under UV light were enumerated. All preparatory culturing of GFP-expressing *E. coli* took place in a separate room from the sink gallery to avoid unintentional contamination.

Installation of P-traps colonized with GFP-expressing *E. coli*. After the 14-day incubation, P-traps were fastened into the plumbing of the sinks (Fig. 5a) . The remainder of the drain line was either autoclaved (strainer, tailpipe, and trap arms) prior to installation or surface disinfected (sink bowl, countertop, and faucets) with Caviwipes-1 (Meterx Research, Romulus, MI), maintaining at least 1 min of contact time. After the P-trap was installed, a daily regimen was followed in which 25 ml of TSB followed by 25 ml of 0.9% NaCl solution

(saline) were added in the ratio 1:3 via the strainer (Fig. 5b) to mimic the potential nutrient exposure in the hospital.

Sampling and enumeration of GFP-expressing *E. coli*. To monitor the growth of GFP-expressing *E. coli* in the plumbing, sampling ports were drilled along the length of the tailpiece (between the P-trap and the strainer) and the trap arm (between the P-trap and the common line). These holes were fitted with size 00 silicone stoppers (Cole-Parmer, Vernon Hills, IL) (Fig. 5a). Sterile cotton swabs (Covidien, Mansfield, MA) presoaked in saline were inserted through these sampling ports, and samples were collected by turning the swab in a circular motion on the inner surface (j20 cm²) of the tailpipes. Sample swabs were pulse-vortexed in 3 ml saline, and serial dilutions were plated on TSA. The strainer, faucet aerator, and bowl surface were sampled with presoaked swabs and processed as described earlier.

Sink-to-sink transmission of bacteria. To investigate sink-to-sink transmission of bacteria, a distal sink (sink 5) (Fig. 4) was fitted with a P-trap inoculated with GFP-expressing *E. coli*. The effects of different inoculum concentrations of GFP-expressing *E. coli*-10³, 10⁶, and 10¹⁰ CFU/ml (colonized for 14 days)-were investigated. Identification to the species level of fluorescent and nonfluorescent colonies identified from mixed pipe cultures was performed using a matrix-assisted laser desorption-ionization (MALDI)-time of flight (MALDI-TOF) mass spectrometer (Vitek-MS; bioMérieux, Durham, NC). The wastewater paths of sinks 1 to 4 were either autoclaved (strainer, tailpipe, P-traps, and trap arms) prior to installation or surface disinfected (sink bowl, countertop, and faucets) with Caviwipes-1 (Meterx Research, Romulus, MI). Faucets on each of the five sinks were turned on simultaneously for 1 min, supplying water at a flow rate of 8 liters/min, once every 24 h for 7 days. No additional feed to any of the sinks was added during this period of 7 days. On days 0 and 7, P-traps on each of the five sinks were unfastened, and swab samples from the P-trap were collected and processed as described earlier.

Dispersion measured using fluorescent microspheres. Fluoresbrite YO carboxylate microspheres (Polysciences, Inc.) which had a 1-μm diameter and maximum excitation and emission of 529 nm and 546 nm, respectively, were used as a tracer in the preliminary experiments to understand droplet dispersion from the hand-wash sinks.

To test whether microspheres could be dispersed from below the sink strainer, a 1-ml suspension of microspheres (10^{10} particles) was injected through a strainer attached to a Hert 4.5-in. offset drain tailpiece typically used for wheelchair-accessible sinks (American Standard, model 7723018.002) (Fig. 5c). The vertical distance between the strainer and microsphere suspension injected into the tailpipe was 4 in. Counter space around the sink bowl was thoroughly wiped with alcohol wipes (Covidien Webcol 6818; Kendall), and polyester sheets precut to appropriate shapes were placed on the counter to cover the entire sink counter and labeled according to position (Fig. 6a). The faucet was turned on for 5 min at a water flow rate of 1.8 to 3.0 liters/min. Polyester sheets were harvested and immediately analyzed using a ChemiDoc MP system (Bio-Rad Laboratories, Inc.) with an exposure time of 5 s. Fluorescent microspheres were enumerated from the digital micrographs using the Image Lab Software (Bio-Rad Laboratories, Inc.).

To test whether microspheres could be dispersed from the surface of the sink bowl, the sink bowl was evenly coated with a 20-ml microsphere suspension (10^{10} particles/ml) using a disposable swab (Sage Products, Inc., Cary, IL), and the dispersion experiment was repeated following the protocol described above. To ascertain there was no nonspecific background fluorescence in the sink and/or the water from the faucet, a control using the same protocol but without the fluorescent microspheres was performed before each experiment. Dispersion measured using GFP-expressing *E. coli*. Dispersion using GFP-expressing *E. coli* was investigated in three experiments. To test whether live organisms in the P-trap could be dispersed by running water, 10^{10} CFU/ml GFP-expressing *E. coli* in saline was added to an autoclaved P-trap and fitted into the drain line that was preautoclaved (strainer, tailpipe, and trap arms). Similarly, to test whether live organisms could be dispersed from the tailpieces of wheelchair-accessible sinks, a suspension of 10^{10} CFU/ml GFP-expressing *E. coli* was added with a syringe through the strainer into the Hert 4.5-ft offset drain tailpiece (Fig. 5c). Just as in the microsphere dispersion experiment, the vertical distance between the strainer and GFP-expressing *E. coli* suspension injected into the tailpipe was 4 in.

We next tested whether live organisms from the surface of the sink bowl could be dispersed by running water. The sink bowl surface was evenly coated with an approximately 20-ml suspension of 10^{10} CFU/ml GFP-expressing *E. coli*.

Finally, to mimic all of these conditions, a P-trap colonized with GFP-expressing *E. coli* (for 14 days) was installed, and a nutrient regimen (Fig. 5b) was followed for 14 days to intentionally promote the GFP-expressing *E. coli* colonization in the attached tailpipe and strainer. On day 15, the dispersion experiment was performed.

Before each of the GFP-expressing *E. coli* dispersion experiments, the counter space was thoroughly disinfected with Caviwipes-1. TSA plates were then positioned on the sink counter surrounding the bowl and an extension platform (Fig. 6b). Additional plates were attached to the sink bowl, faucets, Plexiglas partitions, and faucet handles using adhesive tape. TSA plates were also placed 3 m away from the sink as negative controls. The faucet was turned on for 5 min with a water flow rate of 1.8 to 3.0 liters/min. Lids of the TSA plates were removed only during faucet operation. Swab samples from the faucet aerators before and after operation were collected and plated on TSA. Prior to each dispersion experiment, 50 ml water from the faucet was also collected, and aliquots were plated to assess for the presence of GFP-expressing *E. coli* in source water and ensure cross contamination of GFP-expressing *E. coli* had not occurred. A control dispersion experiment was also performed using the same protocol prior to GFP-expressing *E. coli* inoculation in each case. Dispersion per defined area (CFU per square centimeter) was deduced by dividing the CFU counts in the TSA plate by the surface area of the TSA plate.

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The impact of rapid molecular diagnostic testing for respiratory viruses on outcomes for emergency department patients

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Abstract: **OBJECTIVE:** To determine whether rapid polymerase chain reaction (PCR) testing for influenza and respiratory syncytial viruses (RSV) in emergency departments (EDs) is associated with better patient and laboratory outcomes than standard multiplex PCR testing. **DESIGN, SETTING:** A before-and-after study in four metropolitan EDs in New South Wales. **PARTICIPANTS:** 1491 consecutive patients tested by standard multiplex PCR during July–December 2016, and 2250 tested by rapid PCR during July–December 2017. **MAIN OUTCOME MEASURES:** Hospital admissions; ED length of stay (LOS); test turnaround time; patient receiving test result before leaving the ED; ordering of other laboratory tests. **RESULTS:** Compared with those tested by standard PCR, fewer patients tested by rapid PCR were admitted to hospital (73.3% v 77.7%; $P < 0.001$) and more received their test results before leaving the ED (67.4% v 1.3%; $P < 0.001$); the median test turnaround time was also shorter (2.4 h [IQR, 1.6–3.9 h] v 26.7 h [IQR, 21.2–37.8 h]). The proportion of patients admitted to hospital was also lower in the rapid PCR group for both children under 18 (50.6% v 66.6%; $P < 0.001$) and patients over 60 years of age (84.3% v 91.8%; $P < 0.001$). Significantly fewer blood culture, blood gas, sputum culture, and respiratory bacterial and viral serology tests were ordered for patients tested by rapid PCR. ED LOS was similar for the rapid (7.4 h; IQR, 5.0–12.9 h) and standard PCR groups (6.5 h; IQR, 4.2–11.9 h; $P = 0.27$). **CONCLUSION:** Rapid PCR testing of ED patients for influenza virus and RSV was associated with better outcomes on a range of indicators, suggesting benefits for patients and the health care system. A formal cost–benefit analysis should be undertaken.

Text: The health and economic burdens associated with acute respiratory infections by influenza and respiratory syncytial viruses (RSV) are significant in Australia and overseas. 1-3 Polymerase chain reaction (PCR) testing is effective for confirming respiratory viral infections. 4 Multiplex PCR can detect numerous respiratory viruses, including influenza and parainfluenza viruses, RSV, adenovirus, rhinovirus, human metapneumovirus, enterovirus, bocavirus and coronavirus with very high sensitivity and specificity. 5 Although the results of standard multiplex PCR are accurate and comprehensive, it has traditionally been performed in a central laboratory with a lengthy turnaround time, which may be inconvenient in settings where test results are urgently required, including emergency departments (EDs).

Rapid, easy-to-use PCR-based respiratory virus diagnostic tests have been introduced in recent years; 6,7 the GeneXpert system (Cepheid), for instance, was introduced in New South Wales in July 2017. Rapid PCR tests were expected to facilitate timely and appropriate initiation of treatment, improve outbreak prevention and infection control measures, and expedite the assessment of patients in EDs.

In this study, we analysed routinely collected data to determine whether rapid PCR testing for influenza and RSV infections in EDs is associated with improved patient and laboratory outcomes. We compared data for patients tested for influenza A/B viruses and RSV immediately after rapid PCR diagnosis was introduced (July-December 2017) with data for patients tested with a standard multiplex PCR system during July-December 2016.

We undertook a before-and-after study in four metropolitan public hospital EDs in Sydney, NSW: three general hospitals (EDs A, B and C; 76 228, 54 443 and 50 025 annual ED presentations respectively) and one children's hospital (ED D; 36 700 annual ED presentations; all data for January-December 2016 8). The four hospitals were served by a single pathology laboratory provider.

We analysed data for all patients tested for influenza virus or RSV. During July-December 2016, patients were tested with the standard PCR system, a central laboratory-based multiplex PCR test for sixteen respiratory viruses (including RSV and influenza viruses A and B), available as a referral test at the central laboratory in Hospital B. During July-December 2017, patients were tested with the rapid PCR system, a hospital laboratory-based test specific for RSV and influenza viruses A and B. Hospitals A, B and D have onsite laboratories that perform rapid PCR testing; Hospital C sends samples to the nearby Hospital A.

All tests were conducted in virology laboratories by trained staff, and test results were entered into laboratory information system datasets. We obtained relevant patient characteristics and The proportion of patients admitted to hospital was also lower in the rapid PCR group for both children under 18 (50.6% v 66.6%; $P < 0.001$) and patients over 60 years of age (84.3% v 91.8%; $P < 0.001$). Significantly fewer blood culture, blood gas, sputum culture, and respiratory bacterial and viral serology tests were ordered for patients

tested by rapid PCR. ED LOS was similar for the rapid (7.4 h; IQR, 5.0-12.9 h) and standard PCR groups (6.5 h; IQR, 4.2-11.9 h; $P = 0.27$).

Conclusion: Rapid PCR testing of ED patients for influenza virus and RSV was associated with better outcomes on a range of indicators, suggesting benefits for patients and the health care system. A formal cost-benefit analysis should be undertaken.

The known: Rapid polymerase chain reaction (PCR) testing for influenza and respiratory syncytial viruses (RSV) was introduced in New South Wales in July 2017. Its impact on outcomes for emergency department (ED) patients has not been investigated.

The new: Compared with standard PCR testing, rapid PCR was associated with significantly fewer hospital admissions, more rapid test turnaround, more patients receiving test results before leaving the ED, and reduced numbers of some other common microbiology tests. It did not significantly affect ED length of stay.

The implications: Rapid PCR testing of ED patients for major respiratory viruses can benefit patients and reduce resource use.

MJA 210 (7) ■ 15 April 2019 317 laboratory data by linking the ED and laboratory information system datasets. Detailed information about the datasets and the linkage process have been described previously. 9

The primary outcomes were admission to hospital, ED length of stay (LOS), test turnaround time, and the patient receiving their test result before leaving the ED. ED LOS was defined as the time from arrival in the ED to discharge or admission to hospital. Turnaround was defined as the time from when the sample was received by the laboratory to when the test result was available in hospital electronic records. As secondary outcomes, we compared ordering of typical biochemistry and haematology tests (full blood count; electrolyte, urea, creatinine levels; liver function test; blood gas analysis; C-reactive protein level) and microbiology tests (blood culture; urine microscopy, culture and sensitivity analysis; sputum culture; respiratory bacteria and virus serology) during the two study periods.

Analyses were conducted in Stata 15 (StataCorp). Descriptive statistics are reported (medians with interquartile ranges [IQRs], means with standard deviations [SDs], numbers with proportions). Baseline characteristics were compared in χ^2 tests (categorical outcomes) or Wilcoxon rank-sum tests (continuous outcomes). Logistic regression analyses of binary outcomes (eg, hospital admission: yes/no) and quantile regression analyses of continuous outcomes (eg, ED LOS) were undertaken. All analyses were adjusted for baseline characteristics.

Sensitivity analyses limited to data for children (under 18 years of age) or older patients (over 60 years of age) were conducted. $P < 0.05$ (2-tailed) was deemed statistically significant.

Ethics approval for the investigation was granted by the Human Research Ethics Committee of the South Eastern Sydney Local Health District (reference, HREC/16/POWH/412).

We analysed data for 3741 patients presenting to the four EDs during two periods: 1491 consecutive patients during July-December 2016 (standard PCR) and 2250 during July-December 2017 (rapid PCR). Baseline characteristics for the two groups were similar in terms of sex, triage category, and arrival day of the week, but differed significantly for age, arrival time, and mode of arrival (Box 1

The overall numbers of tests per patient were similar in the standard PCR (mean, 7.2 tests; SD, 3.8) and rapid PCR groups (mean, 7.1 tests; SD, 3.4). The mean number of microbiology tests per patient was significantly lower for the rapid PCR group (1.5 tests; SD, 1.8) than for the standard PCR group (2.0 tests; SD, 2.1; $P < 0.001$ after controlling for baseline characteristics).

The 16 265 biochemistry/haematology and microbiology tests comprised 71.1% of the 22 876 other tests (that is, not including PCR virus testing) ordered for patients in the study. After adjusting for baseline characteristics, the proportions of patients for whom full blood count, electrolyte/urea/creatinine levels, liver function, or C-reactive protein were

assessed were similar, as were the proportions for urine microscopy, culture and sensitivity tests. Significantly fewer blood culture, blood gas, sputum culture, and respiratory bacterial and viral serology tests were ordered for patients in the rapid PCR group (Box 4). Information, figure 1).

ED LOS was similar for the standard PCR and rapid PCR groups in both age-based subgroups (Supporting Information, figure 2A). The differences in test turnaround time identified in the main analysis were also evident for each age-based subgroup (Supporting Information, figure 2B).

In this before-and-after study, we found that rapid PCR testing of ED patients for major respiratory viruses was associated with significantly fewer admissions to hospital, more rapid delivery of test results, more patients receiving their test results before leaving the ED, and less frequent ordering of some common microbiology tests.

Other studies have also reported that hospital admission numbers were significantly lower when rapid influenza virus testing was used in EDs. An analysis of outcomes for more than 300 adults at a tertiary care centre in New York found that early diagnosis of respiratory infections was associated with significantly fewer hospitalisations of influenza-positive patients. 7 In a small Irish study (73 patients), the hospital admission rate for obstetric patients declined from 88% to 45% after on-site rapid influenza PCR testing was introduced. 10 The differences in clinical setting and patient group may explain the smaller decline in our study (from 78% to 67%). Non-PCR-based rapid diagnostic tests for respiratory viruses have also been associated with lower hospital admission rates. 11, 12 The main reason for fewer hospital admissions of patients tested by rapid PCR may be that the earlier availability of results enables clinicians to quickly diagnose or exclude respiratory infections and to make timely and informed decisions about whether to discharge the patient or admit them to hospital. When standard 2 Primary outcomes for 3741 patients at four metropolitan emergency departments (EDs) tested for influenza and respiratory syncytial viruses by standard or rapid polymerase chain reaction (PCR)

After adjusting for baseline characteristics (Box 1): * P = 0.012; ** P < 0.001. ♦

MJA 210 (7) ■ 15 April 2019 PCR was used, in contrast, our findings suggest that these decisions were made before the test results were available. The possible benefits of not admitting patients to hospital, beyond those for individual patient management, include better infection control and outbreak prevention, as well as reduced demands on hospital resources. 13, 14 The impact of rapid PCR testing on outbreak prevention and infection control measures should be evaluated. Rapid influenza virus testing may also have practical implications for hospital bed management. 10, 15 ED LOS was similar in our study before and after the introduction of rapid PCR methods. This finding was not unexpected, as test turnaround time is not the only rate-limiting factor for decision making in EDs. 16 Before rapid PCR methods were introduced, the long turnaround time of multiplex PCR did not necessarily extend a patient's stay in the ED, as they were usually admitted to hospital or discharged home before the results were available. Consequently, more rapid delivery of test results alone would not reduce ED LOS.

Reports on the effect of rapid influenza virus testing and LOS have been conflicting. While evidence for an association between rapid testing and shorter overall inpatient LOS has been reported, 6,11 findings regarding ED LOS are inconsistent. 7, 17, 18 For example, a Cochrane review based on three randomised controlled trials did not find a statistically significant association of rapid viral diagnosis with lower mean ED LOS. 18 In a study in children, ED LOS was actually 26 minutes longer with rapid PCR; inpatient LOS did not differ between the two groups, but was significantly shorter when the analysis was limited to patients with positive test results. 6 We found that ordering of some other microbiology tests, including blood culture, sputum culture, and respiratory bacterial and virus serology, was significantly less frequent for patients tested by rapid PCR. The effect of PCR-based rapid testing on the ordering of other laboratory tests has not previously been reported, although some studies of antigen-based point-of-care testing have examined this outcome. 12 Consistent with our finding, several investigators have reported fewer blood culture tests 19, 20 and basic biochemistry and haematology tests, including full blood count, 20,21 C-reactive protein testing, 21 and urinalysis, 20,21 when point-of-care testing was used.

The higher rate of positive results for patients tested by rapid PCR than for those tested by standard PCR may reflect a higher prevalence of influenza during 2017 than in 2016.

The rapid PCR system in our study accurately detects influenza viruses A/B and RSV but, unlike the standard multiplex PCR, cannot detect other clinically relevant respiratory viruses, such as rhinovirus, coronavirus, human metapneumovirus, parainfluenza virus, adenovirus, enterovirus, and bocavirus. If infection with other respiratory viruses is suspected, patients may therefore need further investigations. Standard multiplex PCR can provide broader information, as it can detect multiple respiratory viruses in a single run, although the long turnaround time restricts its suitability for urgent clinical decision making. Improving the turnaround time of multiplex PCR analysis may achieve better outcomes.

The strengths of our study include its relatively large sample size; further, unlike many similar investigations, ours was a multicentre study in four hospital EDs, enhancing the generalisability of our findings. However, our analyses were not adjusted for comorbid conditions, as this information was not available. Because our study was an uncontrolled before-and-after study, our results cannot be interpreted as indicating causal relationships. As with all non-randomised trials, we could not fully account for all potential confounding variables, nor for temporal trends and other unmeasured factors, such as changes in clinician testing practices or differences in the prevalence and severity of disease during the two study periods. ²² For example, a shortage of inpatient beds caused by a high prevalence of influenza could influence decisions in a busy ED about admitting patients to hospital. However, we attempted to reduce seasonal effects by selecting similar time frames for the two study periods, to reduce selection bias by including all ED patients tested for influenza virus and RSV, and to control for differences in baseline patient characteristics by applying multivariate modelling. As medications data were not available to us, we were unable to assess the impact of rapid PCR testing on antibiotic and antiviral drug use. Similarly, the cost-benefit balance of rapid testing was not evaluated because relevant data were not available. The cost per patient of rapid PCR testing is generally higher than for central laboratory testing, but our findings suggest potential savings through lower numbers of hospital admissions and reduced resource use. This question could be evaluated in a further study.

Rapid PCR testing for influenza virus and RSV infections in patients attending EDs was associated with significant improvements in a range of patient and laboratory outcomes, suggesting potential benefits for both the patients and the health care system. A cost-benefit analysis could examine the impact of rapid PCR testing on bed management and antimicrobial drug prescribing.

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Techniques to Study Antigen-Specific B Cell Responses

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Abstract: Antibodies against foreign antigens are a critical component of the overall immune response and can facilitate pathogen clearance during a primary infection and also protect against subsequent infections. Dysregulation of the antibody response can lead to an autoimmune disease, malignancy, or enhanced infection. Since the experimental delineation of a distinct B cell lineage in 1965, various methods have been developed to understand antigen-specific B cell responses in the context of autoimmune diseases, primary immunodeficiencies, infection, and vaccination. In this review, we summarize the established techniques and discuss new and emerging technologies for probing the B cell response in vitro and in vivo by taking advantage of the specificity of B cell receptor (BCR)-associated and secreted antibodies. These include ELISPOT, flow cytometry, mass cytometry, and fluorescence microscopy to identify and/or isolate primary antigen-specific B cells. We also present our approach to identify rare antigen-specific B cells using magnetic enrichment followed by flow cytometry. Once these cells are isolated, in vitro proliferation assays and adoptive transfer experiments in mice can be used to further characterize antigen-specific B cell activation, function, and fate. Transgenic mouse

models of B cells targeting model antigens and of B cell signaling have also significantly advanced our understanding of antigen-specific B cell responses in vivo.

Text: In his Nobel lecture in 1908, Paul Ehrlich likened the antibody-antigen interaction to a lock and key. He reasoned that antitoxins (antibodies) contained in a solution in the serum of immunized animals must be identical to a cellular receptor "for a really well-made key will not open different locks at the same time" (1) . It took almost five decades before immunofluorescence microscopy was used to confirm the cellular origin of antibodies (2) . Major strides in the B cell and antibody field followed in the 1970s with the development of hybridoma technology to produce monoclonal antibodies and the discovery that somatic rearrangement during B cell differentiation was responsible for antibody diversification (3, 4) . The subsequent explosion of available monoclonal antibodies led to revolutionary diagnostic, therapeutic, and research reagents to distinguish different types of immune cells (5) . Together, these discoveries have allowed us to probe humoral immunity at the level of the antigen-specific B cell.

Methods to probe the antigen-specific B cell response have advanced our understanding of how to harness the remarkable breadth of the B cell repertoire and the exquisite specificity of the individual B cell in developing (1) vaccine candidates that elicit protective antibodies; (2) antibodies that prevent disease when given prophylactically; and (3) antibodies that can be given as therapy after the onset of disease. Many of the vaccines currently available were originally developed empirically either by inactivating, attenuating, or administering a subunit of the pathogen. However, vaccine development against pathogens that are traditionally difficult to vaccinate against may rely on a deeper investigation of the B cell response to the antigens exposed on the surface of these pathogens.

For HIV-1, the discovery of broadly neutralizing antibodies (bnAbs) that protect against infection across diverse viral isolates has intensified efforts to understand the developmental pathway of the rare B cells that produce these antibodies (6) (7) (8) (9) . Insights into the ontogeny of these rare B cells could allow the design of a step-wise vaccine regimen that stimulates the germ-line precursor to expand and mature to produce circulating bnAbs which could protect against HIV acquisition (10, 11) . For RSV, stabilized versions of the fusion (F) protein in the pre-fusion conformation have led to insights in the B cell's response to infection and has generated potentially safer and more efficacious

vaccine candidates (12, 13) . Influenza also performs fusion through the stem region of the hemagglutinin protein, and the identification of B cells that target this relatively conserved site has spurred research on the development of a universal influenza vaccine (14) (15) (16) . Like RSV, HIV, and influenza, the fusion proteins of EBV and CMV exist in a pre-fusion conformation, and stabilization in their pre-fusion states could greatly accelerate vaccine development against these pathogens (17-19). Rare memory B cells producing antibodies specific for the EBV fusion machinery have been isolated; these can neutralize both B cell and epithelial cell infection (20). A new paradigm in malaria vaccine development is also emerging with the discovery of IgM+ and IgD+ memory B cells targeting the Merozoite Surface Protein 1, that rapidly respond to malaria re-infection (21). Further, highly potent neutralizing antibodies targeting a novel and conserved site on the Circumsporozoite Protein have been isolated from B cells (22). Together, these examples demonstrate the importance of studying antigen-specific humoral responses to infectious diseases. The solutions to the crystal structures of surface proteins for a variety of pathogens, the conformational stabilization of these antigens, and the application of the methods summarized in this review, to probe antigen-specific B cell responses, have created new opportunities for systematic and rational vaccine design for HIV, RSV, EBV, malaria, and many other pathogens.

The study of B cell responses has not only informed vaccine design but has also advanced our understanding of antibody-mediated autoimmune diseases, such as rheumatoid arthritis and systemic lupus erythematosus (23, 24). Up to 20% of mature, naïve B cells have receptors with the capacity to bind self-antigens (25). Although these cells are potentially pathogenic, the deletion of B cells with high affinity to self-antigen through apoptosis, anergy of B cells with low affinity to self-antigen, and the absence of T cell help combine together to protect against autoimmune disease in mice (26). The study of autoantigen-specific B cells and a detailed analysis of B cell subsets with pathogenic potential in humans could lead to a better understanding of how to prevent and treat autoimmune diseases.

Although the term antigen-specific B cell is used throughout this mini-review to denote the analysis of B cells based on binding between the B cell receptor (BCR) and a specific antigen used as bait, it is important to keep in mind that BCRs within the polyclonal B cell repertoire exhibit a spectrum of polyreactivity. On one end of the spectrum, a highly polyreactive BCR is able to bind multiple structurally unrelated antigens with physiologically relevant affinities. The frequency of polyreactivity in the normal adult

human B cell repertoire has been estimated to be 4% of naïve B cells, 23% of IgG⁺ memory B cells, and 26% of intestinal IgA⁺ and IgG⁺ plasmablasts (27-29). On the other end of the spectrum, a mono reactive BCR is activated only when it encounters a single cognate antigen. Although there are exceptions, the accumulation of somatic hypermutations within the variable regions of the BCR during the process of affinity maturation is generally thought to lead to increased affinity and specificity for the cognate antigen (30, 31).

Several general techniques are commonly used to identify antigen-specific B cells (Table 1). The B cell enzyme linked immunospot (ELISPOT) technique relies on the principle of capturing the secreted antibody in the vicinity of each cell. In the B cell ELISPOT, antibody secreting B cells (ASCs) present in a sample or differentiated in vitro are added to plates coated with the antigen of interest. Antigen-specific antibodies will bind in close proximity to the location of the individual B cells producing those antibodies. Enzyme or fluorescent labeled secondary antibodies are then used to visualize spots of antibody secretion and binding to plate-bound antigen at the location of the ASCs. Each spot corresponds to antibody produced from a single antigen-specific B cell and therefore the technique is extremely sensitive. Secondary antibodies conjugated to combinatorial colored beads can also be used to detect the antibodies secreted from individual B cells with the advantage of multiplexing the assay (32). One limitation of the assay is its requirement for antibody secretion by B cells thereby limiting the assay to only a subset of B cells in the repertoire, namely ASCs (33). Memory B cells can be stimulated in vitro to differentiate into ASCs prior to addition to the antigen-coated plate (34) . Further, the antigenspecific B cells identified by ELISPOT are generally not available for downstream analysis.

Limiting dilution is another technique that has been used to isolate antigen-specific B cells. In this approach, primary cells can be diluted serially until individual B cells are separated in microwell plates (36) . The B cells can then be cultured and expanded ex vivo and/or immortalized using EBV such that each well contains a monoclonal antibody (3, 37, 38) . Antigen-specific B cells can be selected by screening the culture supernatants for monoclonal antibodies that bind an antigen of interest. Although antibodies can be sequenced and cloned, the requirement for an ex vivo culture prior to selection precludes determination of the transcriptional profile of the original B cell in this approach. This technique can potentially be time-consuming and laborious, but the use of microfluidics and robotics has greatly improved the throughput for selecting antigen-specific B cells (39) . Advances in single cell next generation sequencing technology have allowed high throughput transcriptional profiling and sequencing of paired immunoglobulin heavy and

light chains (40) . In this approach, antigen specificity can be tested after monoclonal antibodies are cloned and produced using the sequencing data. This method can be useful in identifying antigen-specific B cells that have undergone clonal expansion after vaccination or acute infection (41) . Flow cytometry is the most common method used for single cell analysis and isolation (39) . Flow cytometry-based analysis of antigen-specific B cells is dependent on labeling antigen with a fluorescent tag to allow detection. Fluorochromes can either be attached covalently via chemical conjugation to the antigen, expressed as a recombinant fusion protein, or attached non-covalently by biotinylating the antigen. After biotinylation, fluorochrome-conjugated streptavidin is added to generate a labeled tetramer of the antigen. Biotinylation of the antigen at a ratio ≤ 1 biotin to 1 antigen is important, since each streptavidin has the potential to bind four biotins. If the ratio of biotin to antigen is $>1:1$, then clumping and precipitation of the antigen out of solution can occur as soon as streptavidin is added. Alternatively, site directed biotinylation can be accomplished by adding either an AviTag or BioEase tag to the recombinant antigen prior to expression (77, 78) . When site-specific biotinylation is utilized, researchers must keep in mind that the tag may occlude an epitope from recognition by B cells which can be problematic for vaccine antigens. Further, for proteins that oligomerize, multiple tags may be incorporated, possibly resulting in aggregation.

Another important consideration is the potential for confounding by B cells in the repertoire that bind to the fluorochrome, streptavidin, or any linkers rather than to the antigen of interest. Binding between fluorochromes, linkers, or streptavidin and BCRs from humans and mice never exposed to these antigens are generally of low affinity, and these BCRs are generally expressed by naïve and potentially polyreactive B cells (62, 79, 80) . Dual labeling, in which the same antigen is separately labeled with two different fluorochromes, can be used to identify double positive B cells and remove confounding by B cells that bind the fluorochrome (12, 42) . However, even when tetramers are utilized for dual labeling, streptavidin-specific B cells will contaminate the double positive population. To fully remove confounding from the fluorochrome, streptavidin, and linkers, a "decoy" tetramer can be used to identify these contaminating B cells (21, 26). In this approach, the same fluorochrome used to identify antigen-specific B cells is conjugated to a different fluorochrome such that the emission spectrum is altered by fluorescence resonance energy transfer (FRET) (26). Decoy-binding B cells can therefore be excluded from the true antigen-specific B cells. Notably, it is critical to use the same source of fluorochrome conjugated streptavidin in the tetramer and decoy reagent, because conjugation methods, recombinant streptavidin, and protein fluorochromes like R-phycoerythrin vary enough from company to company to alter some of the epitopes available for B cells to bind.

One weakness of the flow cytometric approach is the reliance on antigens that can be readily conjugated to a fluorochrome or biotinylated. In addition to recombinant proteins and synthesized peptides, labeled polysaccharides, lipids, haptens, virus-like particles, and pseudo viruses have also been used to identify antigen-specific cells by flow cytometry (33, [43] [44] [45] [46] [47] [48] [49] [50] [51] [52] [53] [54] [55] [56] [57] [58] [59] . Further, epitope-specific B cells have been identified by screening bacteriophage-displays or microarray peptide libraries with polyclonal antibodies targeting the native antigen to select conformational epitopes that can be fused to fluorescent proteins for use in flow cytometry (47, 60) .

With technologic advancements increasing the number of simultaneously measurable parameters, antigen-specific B cells can be further characterized by cell surface markers and intracellular staining. Additionally, the immunoglobulin capture assay is a flow cytometry-based adaptation of the ELISPOT assay in which a streptavidin-conjugated anti-CD45 antibody carrying four biotinylated anti-IgG antibodies is used to simultaneously bind plasmablasts and capture secreted antibody followed by fluorescent-labeled antigen to detect antigen-specific plasmablasts (61) . The mean fluorescence intensity measured by flow cytometry and normalized to the level of BCR expression also provides a measure of the relative amount of antigen binding to a B cell and can be used as a rough surrogate for binding affinity (79, 81, 82) . Preincubation of B cells with increasing concentrations of a monomeric antigen prior to labeling with tetrameric antigen can also be used to further quantify binding affinity. Cells expressing high affinity BCRs will bind monomeric antigen at low concentrations, whereas low affinity BCRs will require higher concentrations of monomeric antigen to compete with and inhibit tetramer binding (26). Individual cells can also be isolated by fluorescence activated cell sorting (FACS) for downstream analysis, including BCR sequencing and cloning, BCR affinity measurement, in vitro proliferation, and transcriptional profiling.

Methods have recently been developed to further improve the sensitivity for detecting rare antigen-specific B cells. Magnetic nanoparticles conjugated to antibodies targeting the fluorochrome on the antigen of interest, allow for the enrichment of antigen-specific B cells prior to flow cytometry (20, 26, 80, 83) . This approach is particularly useful for detecting rare antigen-specific naïve B cells, autoreactive B cells, memory B cells, and plasmablasts (21, 26, 47, 50) . The magnetic enrichment strategy allows for the analysis of significantly

more cells in a shorter period of time by concentrating the cells of interest prior to flow cytometry (Figure 1) . Notably, as with any method that seeks to identify a population of cells at a very low frequency, the background and noise inherent in the detection system is magnified with respect to the signal of interest, especially when that signal is weak.

Therefore, to detect the antigen-specific population of interest, the following considerations are critical: (1) Using decoys to exclude B cells of unwanted specificities;

(2) careful design of flow cytometry panels to avoid emission spillover into the channel for the antigen of interest; and (3) choosing the brightest fluorochromes, like R-phycoerythrin or allophycocyanin.

In vivo methods to probe antigen-specific B cell responses in the presence of other antigen-presenting cells and T cell helpers, have increased our mechanistic understanding of the humoral immune response during vaccination, infection, and autoimmunity.

Adoptively transferred B cells can be distinguished from recipient lymphocytes by taking advantage of mouse strains with allelic variations in CD45 or mice devoid of B cells. The adoptively transferred B cells can come from wildtype mice or from mice expressing transgenic BCRs (Table 2) , and antigen-specific B cells can be analyzed using the techniques described above.

Microscopy is another general technique that has been used to identify antigen-specific cells in vivo and offers the advantage of direct visualization. In the first reported application of this technique to demonstrate the cellular origin of antibodies in 1955, fluorescein-conjugated antibodies against ovalbumin and human immunoglobulin were used to stain tissue sections of the spleen from hyperimmune rabbits (2) . Since then, other groups have fluorescently labeled antigens to localize antigen-specific B cells by microscopy (62, 65) . Advances in laser capture dissection microscopy, already used in the T cell field, also provide an opportunity for isolating individual antigen-specific B cells for downstream analysis, including sequencing and cloning of the BCR or transcriptional profiling (66) . However, antigen staining of BCRs in situ can be challenging depending on the binding of antigens from pathogens to other cellular receptors or an alteration of BCR specificity during tissue fixation or processing. Two-photon or multiphoton microscopy has the ability to resolve images at greater depths and with less photobleaching than confocal microscopy (67, 68) . As a result, this technology has allowed real-time imaging in living, intact lymphoid tissues of mice, permitting the direct in vivo observation of immune cell

interactions. The dynamic movements and interactions of antigen-specific B cells can be studied in vivo by combining an adoptive transfer of individual B cells (isolated by limiting dilution or FACS) with two-photon microscopy (63, 69, 70) .

Humanized mouse models are powerful tools for translating experiments in mice to applications in humans. Transgenic mice that produce humanized cytokines by knock-in replacement can be used to support human hematopoietic stem cells (104) . Transgenic mice with complete humanization of the mouse immunoglobulin loci provide an opportunity for recapitulating the breadth of the human B cell repertoire and serve as a valuable tool for therapeutic antibody discovery (71) . However, one caveat is that the allele frequencies found in the B cell repertoires of these mouse models may not necessarily recapitulate those found in humans (72) . Mass cytometry has the potential to provide further high-dimensional analysis of antigen-specific B cells. In this method, heavy metal ion tags rather than fluorochromes are used to label cells. Since data is collected as time-offlight mass spectrometry, up to 42 unique parameters can be simultaneously measured from a single sample without significant spillover between channels or the need for compensation. Mass cytometry with heavy metal-labeled tetramers can be constructed using streptavidin (73) . Mass cytometry with metal-labeled peptide-MHC tetramers has been used successfully to identify and characterize antigen-specific T cells, but to our knowledge has not yet been applied to antigen-specific B cells (73, 74) . One limitation of this approach is that cells are unavailable for downstream analysis since they are vaporized by a plasma torch to atomize the ion tags. However, by simultaneously detecting many more surface markers and intracellular cytokines, transcription factors, and detecting more signaling molecules from individual cells than previously possible with traditional fluorescent labels, the application of mass cytometry with dimensionality reduction algorithms could help dissect the complexity of the B cell compartment, provide a higher resolution view of B cell development, and reveal novel subsets of antigen-specific B cells involved in mediating autoimmune diseases or protection against infection.

On the horizon, single cell RNA-sequencing (RNA-seq) technologies have the potential to revolutionize the study of antigen-specific immune cells (75, 76) . The ability to generate a library of tetramers with unique barcodes could allow the simultaneous examination of gene expression profiles from a large number of cells with different antigen specificities in a single experiment. Combining barcoded tetramers with oligonucleotide-conjugated antibodies and RNA-seq to simultaneously measure the protein and gene expression of antigen-specific cells could further increase the amount of unbiased multi-omic

information about individual antigen-specific cells in normal and disease states and aid the rational design of vaccines and therapeutics (105) (106) (107) .

The ongoing analysis of antigen-specific B cell responses has led to the development of new diagnostic, therapeutic, and research reagents. Methods for studying antigen-specific B cell responses are being increasingly applied to tackle diseases like HIV, RSV, and autoimmune diseases, in which the immune response either fails to protect or clear disease, or where it enhances disease or is responsible for the disease itself. Considerable opportunities exist on the horizon for applying these methods to a myriad of diseases in which B cells play an active role.

JB and JT reviewed the literature, generated figures and tables, and wrote the manuscript.

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Demographic Variations of MERS-CoV Infection among Suspected and Confirmed Cases:
An Epidemiological Analysis of Laboratory-Based Data from Riyadh Regional Laboratory

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Abstract: Introduction. Middle East respiratory syndrome coronavirus was first recognized in September 2012 in Saudi Arabia. The clinical presentations of MERS and non-MERS SARI are often similar. Therefore, the identification of suspected cases that may have higher chances of being diagnosed as cases of MERS-CoV is essential. However, the real challenge is to flag these patients through some demographic markers. The nature of these markers has not previously been investigated in Saudi Arabia, and hence, this study aims to identify them. METHODS: It was a surveillance system-based study, for which data from a total of 23,646 suspected patients in Riyadh and Al Qassim regions were analyzed from January 2017 until December 2017 to estimate the prevalence of MERS-CoV among suspected cases and to determine potential demographic risk factors related to the confirmation of the diagnosis. RESULTS: Of 23,646 suspected cases, 119 (0.5%) were confirmed by laboratory results. These confirmed cases (67.2% of which were males) had a mean age of 43.23 years (SD $\pm$ 22.8). Around 42.2% of the confirmed cases were aged between 41 and 60 years and about 47% of confirmed cases had their suspected specimen tested in the summer. The study identified three significant and independent predictors for confirmation of the disease: an age between 41 and 60 years, male gender, and summer season admission. CONCLUSION: The study provides evidence that the MERS-CoV epidemic in the subject regions has specific characteristics that might help future plans for the prevention and management of such a contagious disease. Future studies should aim to confirm such findings in other regions of Saudi Arabia as well and explore potential preventable risk factors.

Text: A respiratory viral disease caused by the Middle East Respiratory Syndrome CoronaVirus (MERS-CoV) was first isolated in 2012, in a 60-year-old man who died in Jeddah, KSA due to severe acute pneumonia and multiple organ failure [1] . Since then, 27 countries have reported the presence of this virus, including the 12 countries of the Eastern Mediterranean region. Several outbreaks have occurred in multiple countries including Saudi Arabia, the United Arab Emirates and the Republic of Korea [2] . Recent fatality rate (CFR) of 21% [5, 6] . Very limited evidence is available for exploring the epidemiology of this virus among the pediatric population [7] . The literature shows that MERS-CoV infects males more than females [8, 9] . The case fatality rate of men (52%) is higher than that of women (23%) [10] . Males with a history of serious medical conditions are highly susceptible to this infection. Moreover, the mean age of infection in adults is 60 years [10] . The mode of transmission is not entirely understood yet [2] ; however, human-to-human [11] and zoonotic sources of transmission [12] have been documented in many studies. Dromedary camels are the major animal source of MERS-CoV transmission to humans. Interhuman transmission of the virus did not occur easily, but it is seen mainly in patients' families and

healthcare settings [2] . Clinical pictures of this infection varied from asymptomatic to mild respiratory symptoms to severe respiratory distress and death [2] . Severe ailment can often cause respiratory catastrophes that need mechanical ventilation and support in ICUs across different healthcare settings [4] . Studies have suggested an incubation period of 16 days with a mean of 5-6 days [12, 13] , while the median time until death is 11-13 days (range 5-27 days) among severely ill patients [13] . The gold standard test for the detection of this virus is real-time reverse-transcription polymerase chain reaction (rRT-PCR) assays [14] .

There is no specific treatment for MERS-CoV. Like most viral infections, the treatment options are supportive and symptomatic [2] . At present, no vaccine exists for preventing the infections of MERS-CoV. The CDC indicated that preventative actions should be taken for any type of respiratory illness [4] . Such actions include washing hands with water and soap for around 20 seconds or using hand sanitizers with alcohol if no water is available. One must cover their nose and mouth during instances of sneezing and coughing with a tissue and avoid touching the mouth, nose, or eyes with their hands until washed properly. Repeatedly touched surfaces, such as door knobs, should be disinfected and cleaned regularly. Intimate personal contact, e.g., kissing, and sharing cups or eating utensils must also be avoided [15] .

Many studies have been conducted in recent years in Saudi Arabia to combat this deadly disease. A large multicentre study showed that it is nearly impossible to differentiate between patients of MERS-CoV and non-MERS-CoV just on the basis of clinical presentation [16] . Another cohort study, which was hospital-based (17 cases vs. 82 controls), found that there were statistically significant differences in terms of gender, clinical, and radiographic presentations [17] . Similarly, two more single-centre case control studies reported that the presenting symptoms of MERS-CoV infection were not specific [18, 19] .

Physicians and public health practitioners need to identify suspected cases which have higher chances of diagnosis as confirmed cases prior to laboratory testing (which usually takes between 12 and 24 hours). Identification of a confirmed case is necessary to implement preventive strategies to combat the spread of the disease to family members and hospital healthcare workers [20] . Mild symptomatic cases, which result in a positive PCR, may be isolated at home. Severe to moderate cases should be admitted to and

isolated in a hospital until they improve and then be discharged for isolation at home for an extended period. Both mild and severe cases are retested after 7 days, and the test is subsequently repeated after every 3 days until a negative result is obtained [20] .

Identifying suspected cases which may have higher chances of getting diagnosed as a confirmed case and implementing strict procedures on them might offer the best solution.

One challenge is to flag these patients by some demographic markers, as the clinical presentation of MERS-CoV infected patients were non-specific. Therefore, we aimed to identify some demographic markers specific to confirmed cases of MERS-CoV. The nature of these markers has not been investigated in Saudi Arabia, and hence this study aims to identify them.

A cross-sectional study was conducted at the regional laboratory and blood bank, located at Shumaisi Hospital in Riyadh, KSA. The laboratory has received the Central Blood Banks and Reference Laboratories Accreditation Program Saudi Central Board for Accreditation of Healthcare Institution (CBAHI) 2018 [21] .

Technique. Data were collected during the period of January 2017 to December 2017. All patients in Riyadh and Al-Qassim regions who had their samples tested at Riyadh regional lab during the study period were considered as suspected cases.

The study had two aims: descriptive and analytical. For the descriptive aim, we estimated the prevalence of MERS-CoV. For the analytical aim, a binary logistic regression model was developed. In this model, we included the risk factors of gender, age, seasons, nationality, healthcare status (yes/no), hospitals, and area of residence. Data were cross-checked with a labcomputerized database. Further data were collected on demographic characteristics (age and sex), underlying nationality, and health care status.

We collected data from 25,400 cases, of which 23,646 suspected cases of MERS-CoV were included in the final analysis. Data were cleaned, entered, stored, and managed with an excel database and IBM SPSS Version 25. The statistical analyses consisted of descriptive

counts and percentages. For those continuously scaled items, nonparametric statistics (medians, interquartile ranges, minimum, and maximum) were used to describe the distribution. A logistic regression analysis was used to identify predictors of confirmation of infection within the suspected cases groups. At first, univariate analyses were conducted to estimate the unadjusted contribution and to determine the significant risk factors. It was followed by a multivariate logistic regression analysis to estimate the independent contribution of each covariate. To determine significant factors, a p value below 0.05 and a 95% confidence interval were considered.

A confirmed case is defined as a suspected case with laboratory confirmation of MERS-CoV infection [20] .

A total of 23,646 of MERS-CoV suspected cases were included in this study, of which 52.3% were males (n = 12376) and 47.7% were females (n = 11270). The age of individuals with suspected cases ranged between 0 to 92 years with a mean age of 43. The adjusted odds of MERS-CoV remained significant among different age groups; the odds of patients aged between 20-40 years increased threefold (A.O.R: 3.11, 95% CI: 1.104-8.76, P value = 0.032), whereas in the age group of 41-60 years, it increased further to a risk that was six times higher

This is a cross-sectional study about the epidemiological analysis of MERS-CoV infection. Laboratory-based data was conducted in Riyadh over a one-year period (2017). A total of 23,646 suspected cases were included in the results. Of the total suspected cases, 119 cases had been confirmed via laboratory results. All the confirmed cases are reported to MOH through HESN (health electronic surveillance networks) and to the World Health Organization (WHO) through the International Health Regulations (IHR), National Focal Point of Saudi Arabia. We found that MERS-CoV infection was found significantly in people aged between 41 and 60 years and was reported most commonly during the summer season. The odds of infection among males were found to be twice as high as that of females with suspected cases.

During the study period, i.e., the year 2017, only 119 confirmed cases were reported, which means that the number of MERS-CoV infection cases has decreased in Riyadh and Al-Qassim regions in comparison to that of the last three years. From 2015 to 2016, there was

a 25.4% decrease, whereas from 2016 to 2017, it decreased by 48.7%, which translates into a 50% decrease between the two periods. This also complements the findings reported by Da'ar and Ahmed in their paper [23]. The predominance of infection in males was also observed in another study performed in KSA (2015), which reported the percentage of confirmed cases among males to be 66%, compared with 34% among females [24].

It is worth mentioning that Saudi Arabia defines age categories differently from the WHO (children: 0-14, adult: otherwise) [20]. However, unlike the classification used in Saudi Arabia, we have followed the WHO categorization of age to differentiate between children/adolescents (0 to 19 years) and adults (20 years and older) as indicated in WHO reports for age-standardized population and in infectious diseases [25]. This categorization was also followed by Aly and his collaborators in their recent paper published in 2017 [14]. Adults were further subcategorized into three groups according to the age distribution of the study population using the following two cutoff points (age of 41 and age of 60) [14].

These data agreed with a previous surveillance study, which stated that the majority of confirmed cases of MERS-CoV were reported among people aged 40 and above [24]. In 2016, only 9 of 552 cases (1.6%) of MERS-CoV infection were found among pediatric patients. Moreover, the study which was conducted in King Fahad Medical City in Riyadh (KFMC) between January 2012 and December 2013 did not report any MERS-CoV cases among children [26]. The study which was conducted across the Gulf countries for four years by Mahmoud Aly et al. between 2012 and 2016 suggests that the prevalence and distribution of MERS-CoV were the highest-risk in elderly aged 60 years or above [14]. Similar to our results, this study also reported the highest number of confirmed cases during the summer season [14].

Among confirmed cases, only 25.2% were healthcare workers, whereas around 75% were non-healthcare workers.

This is in agreement with the study done by Ahmad to estimate the survival rate in MERS-CoV globally prior to 26 January 2017; 86.9% were not health-care workers compared with 13.1% confirmed cases of healthcare workers [27]. Similarly, other studies also reported a lower prevalence in healthcare workers [28] [29] [30].

Our data reported a higher prevalence of infection among Saudi nationals as compared with non-Saudi. Another study also showed similar results but with a much higher percentage among Saudis, which may be due to the fact that it included Saudis from all regions [29]. There is no finding basis for comparison as such, because our study was focused on the Riyadh and Al Qassim regions only.

In our study, we detected a low prevalence (0.5%). The low positive predictive value of our lab results is not related to the low sensitivity and specificity of the lab assay. The estimated analytical sensitivity and specificity of the Real Star kit from Altona was reported to be 100% with no cross reactivity with other respiratory pathogens [31]. Moreover, this low predictive value in the lab results is related to the high burden of false positive cases referred to the lab. In fact, this research is just the starting point to shed the light on more factors that might help in putting more descriptive criteria to lower the financial and human resources burden.

To the best of our knowledge, no one has developed a logistic regression that focuses on demographic risk factors such as sex, age, and seasons prior to our study. However, it is worth mentioning that Ahmed et al. developed a risk prediction model that encompasses risk factors such as chest pain, leukopenia, and elevated aspartate aminotransferase (AST) [21]. However, further investigations are needed to confirm our findings.

One of the major strengths of our study is that it is a comprehensive regional study which included all the suspected cases of MERS-CoV in the Riyadh and Al-Qassim regions. Secondly, the external validity of our study is also expected to be high, as it covers the two regions completely, meaning that the records of all suspected cases in these two main regions in Saudi Arabia were included. Thirdly, the quality of the data is considered to be high, given that the contagious and life-threatening nature of this disease has led to strict obedience to rules which are enforced in a timely manner, thus ensuring accurate reporting of suspected cases. In addition to this, quality assurance policies are implemented at HESN in order to maintain the highest level of validity and reliability of the data collection process. The variables available for suspected cases were limited to demographics, which limited the scope of our research, but they provided valuable information to form a basis for future studies of a broader scope. Variables such as primary/secondary infections are vital pieces of information, but due to the limitation of the data available, we could not determine their effects.

According to our knowledge, this is one of the few studies that have specifically investigated MERS-CoV risk factors in the Riyadh and Al-Qassim areas (two major regions in KSA). Given that all suspected and confirmed cases were included in this study, we assume that our results are generalizable for both the regions with confidence. It must be noted that the comparative group of this study is different from that of the previous ones, as we compared those with confirmed MERS-CoV with those with suspected MERS-CoV who have passed all stages of screening at the hospital, whereas other studies were hospital but not lab-based with an aim of identifying factors that help in suspecting rather than confirming cases. It might be the reason why we have found some significant demographic factors unlike other reports.

In conclusion, this research is about predictors for the confirmation of diagnosis among suspected cases only, meaning that the factors we found can help in identifying suspected cases that may have a higher chance of testing positive. It will help primary healthcare professionals to develop a better screening tool for suspected cases, as currently only a small minority of suspected cases are confirmed positive via lab results, consequently resulting in a lot of resources being spent to test thousands of samples, just for the identification of a few cases. The three factors we identified are important because, for example, a female, aged 18, presenting in winter will be less likely to be diagnosed than a male, aged 45, presenting in the summer, or, to give another example, a 60-year-old male who is presenting MERS-CoV signs with a negative lab result may need retesting.

Our study covered two main regions in Saudi Arabia and provides evidence that the MERS-CoV epidemic in these two regions has specific characteristics that might help future plans for prevention and management of such contagious diseases. Our results showed that only a minority of suspected cases are actually diagnosed with the disease, meaning that the procedures being implemented seemed to be highly sensitive but not highly specific. The majority of confirmed cases were male, aged 41 to 60 years, and presented to healthcare facilities in the summer. Future studies should aim to confirm such findings in other regions in Saudi Arabia, to explore potential preventable risk factors and go deeper to know the underlying factors that make male aged 41-60 more susceptible than others.

e laboratory data used to support the findings of this study were provided by Riyadh Regional Laboratory under license and are not freely available. However, access to data will be considered from the corresponding author upon request.

e authors declare that they have no competing interests.

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--- START OF ARTICLE 19 ---

Gemcitabine and Nucleos(t)ide Synthesis Inhibitors Are Broad-Spectrum Antiviral Drugs that Activate Innate Immunity

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5923505/>

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Abstract: Nucleoside analogs have been frequently identified as antiviral agents. In recent years, gemcitabine, a cytidine analog in clinical use for the treatment of many solid tumors, was also shown to have antiviral activity against a broad range of viruses. Nucleoside analogs generally interfere with cellular nucleos(t)ide synthesis pathways, resulting in the depletion or imbalance of (d)NTP pools. Intriguingly, a few recent reports have shown that some nucleoside analogs, including gemcitabine, activated innate immunity, inducing the expression of interferon-stimulated genes, through nucleos(t)ide synthesis inhibition. The precise crosstalk between these two independent processes remains to be determined.

Nonetheless, we summarize the current knowledge of nucleos(t)ide synthesis inhibition-related innate immunity and propose it as a newly emerging antiviral mechanism of nucleoside analogs.

Text: Nucleoside analogs have been historically used for anti-cancer chemotherapy because they inhibit cellular DNA/RNA polymerases [1] . More recently, nucleoside analogs have expanded their therapeutic applications and are being used to develop antiviral drugs against a wide range of serious and life-threatening viruses. Some nucleoside analog drugs targeting specific viral polymerases (acyclovir for herpesviruses, zidovudine for human immunodeficiency virus (HIV), and sofosbuvir for hepatitis C virus (HCV)) have been successful in clinical trials [2] [3] [4] [5] and are currently in use for the treatment of virus-infected patients. Another class of nucleoside analog drugs such as ribavirin, more broadly-acting on various viruses, has been used in conjunction with IFN- α [6] . Importantly, extensive studies on the antiviral action of ribavirin have established the underlying molecular framework of nucleoside analogs.

The primary mechanism to explain the antiviral effect of nucleoside analogs is based on their direct action on viral polymerization. Nucleoside analogs are transported into the cells and phosphorylated by the consecutive action of viral or cellular kinases, eventually generating nucleotide triphosphates. Mature nucleotide analogs, which are similar to physiological nucleotides, can directly incorporate into the growing viral genome during polymerization, resulting in the termination of chain reaction or the accumulation of mutations (Figure 1). Alternatively, nucleotide analogs can bind to the nucleotide-binding region on viral polymerases and block the entry of incoming natural nucleotides. The other mechanism is based on the modulation of cellular nucleos(t)ide synthesis. There have been accumulating reports that nucleoside analogs act as antiviral agents by interfering with host nucleos(t)ide synthesis pathways [7] [8] [9] [10] . By targeting metabolic enzymes(s), nucleoside analogs block the natural flow of nucleos(t)ide synthesis and consequently cause the depletion or imbalance of (d)NTP pools. As viral replication is highly dependent on the availability of host nucleotides, a nucleotide-defective condition decreases the efficiency of viral replication. A more recently proposed mechanism has been based on the observations that a few nucleoside analogs activate innate immunity, especially involving the upregulation of interferon-stimulated genes (ISGs). Importantly, this phenomenon is usually mediated by the inhibition of nucleotide synthesis, suggesting a potential crosstalk between nucleotide biosynthesis and innate immunity. However, the precise mechanism of this crosstalk remains to be elucidated.

There is now an increasing number of nucleoside analogs with antiviral activity toward a wide range of viruses. They have been well-summarized in a previous report [1] . In the present review, we focus more on gemcitabine as a nucleoside analog, which is clinically relevant and whose broad-spectrum antiviral activity has been recently reported by many groups including our group. More importantly, we summarize inhibitors of the purine/pyrimidine biosynthesis pathways that induce innate immunity and propose possible mechanisms of action for these inhibitors. can directly incorporate into the growing viral genome during polymerization, resulting in the termination of chain reaction or the accumulation of mutations (Figure 1). Alternatively, nucleotide analogs can bind to the nucleotide-binding region on viral polymerases and block the entry of incoming natural nucleotides. The other mechanism is based on the modulation of cellular nucleos(t)ide synthesis. There have been accumulating reports that nucleoside analogs act as antiviral agents by interfering with host nucleos(t)ide synthesis pathways [7] [8] [9] [10] . By targeting metabolic enzymes(s), nucleoside analogs block the natural flow of nucleos(t)ide synthesis and consequently cause the depletion or imbalance of (d)NTP pools. As viral replication is highly dependent on the availability of host nucleotides, a nucleotide-defective condition decreases the efficiency of viral replication. A more recently proposed mechanism has been based on the observations that a few nucleoside analogs activate innate immunity, especially involving the upregulation of interferonstimulated genes (ISGs). Importantly, this phenomenon is usually mediated by the inhibition of nucleotide synthesis, suggesting a potential crosstalk between nucleotide biosynthesis and innate immunity. However, the precise mechanism of this crosstalk remains to be elucidated.

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Gemcitabine is a cytidine analog that has been clinically used for the treatment of various cancers [11, 12] . However, in recent years, the antiviral activity of gemcitabine has also been reported against a broad range of RNA viruses, including Middle East respiratory syndrome coronavirus (MERS-CoV), severe acute respiratory syndrome coronavirus (SARS-CoV), Zika virus (ZIKV), HCV, poliovirus (PV), influenza A virus (IAV), HIV, and enteroviruses (EV) [13] [14] [15] [16] [17] [18] .

The antiviral activities of gemcitabine against the abovementioned viruses are summarized in Table 1 . MERS-CoV and SARS-CoV belong to the family of Coronaviridae and are causative agents of severe viral respiratory illness in humans. To efficiently select appropriate antiviral drug Figure 1 . The mechanism of antiviral effect of nucleos(t)ide analogs. Nucleos(t)ide synthesis inhibition-related innate immunity, a newly emerging antiviral mechanism of nucleoside analogs, was highlighted by yellow boxes.

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The antiviral activities of gemcitabine against the abovementioned viruses are summarized in Table 1 . MERS-CoV and SARS-CoV belong to the family of Coronaviridae and are causative agents of severe viral respiratory illness in humans. To efficiently select appropriate antiviral drug candidates, Dyall et al. screened 290 FDA-approved drugs in virus-infected Vero E6 cells and identified gemcitabine as one of drugs with antiviral activity against both MERS-CoV and SARS-CoV (EC 50 of 1.2 μ M and 4.9 μ M, respectively) [13] . More recently, gemcitabine was shown to effectively suppress ZIKV infection and replication in human retinal pigment epithelium (RPE) cells, particularly at non-cytotoxic concentrations (EC 50 of 0.01 μ M vs. CC 50 of > 10 μ M) [14] . ZIKV, a member of the Flaviviridae family, can infect pregnant women and cause congenital abnormalities such as microcephaly in infants, which has attracted increasing public attention as well as extensive research and development into possible treatments. Effective antiviral activities of gemcitabine were also found for the replication of HCV in Huh-7 cells and the infection of HIV in U373-MAGI-CXCR4 CEM cells, with estimated EC 50 s of 12 nM and 16.3 nM,

respectively [17, 19] , which were lower concentrations than those used in cancer therapy [20] . In the case of HIV, the combination of gemcitabine with decitabine, another nucleoside analog in clinical use for cancer therapy, synergistically reduced HIV infectivity by increasing the viral mutation frequency [21] . In a follow up study, Clouser et al. further reported the antiviral effect of gemcitabine against HIV-related retrovirus, murine leukemia virus (MuLV), in vitro (EC 50 of 1.6 nM) and even in murine AIDS model [17] . A significant antiviral effect of gemcitabine on IAVs was also reported for RPE cells by Denisova et al. (EC 50 of 0.068 μ M) [16] . They also tested whether gemcitabine had an antiviral effect on several other viruses of different families and found its strong inhibitory effect on Sindbis virus and herpes simplex virus-1 (HSV-1) (>2 log reduction in virus titer) but relatively weak effects on Semliki forest virus and human echovirus 6, and minimal effects on Bunyamwera virus, measles virus (MeV), and vaccinia virus [16] . The antiviral effect of gemcitabine on EVs, initially performed on Cocksackievirus B3 (CVB3), was found from screening FDA-approved drugs in CVB3 replicon-harboring Vero cells by our group (EC 50 of 0.4 μ M) [18] . Its broad-spectrum antiviral activity on EVs was further identified by observing a similar inhibitory effect on enterovirus 71 (EV71) and human rhinoviruses (HRVs) (EC 50 s of 1 and 1-5 μ M, respectively). In the case of HRV, the antiviral effect of gemcitabine was further confirmed in a virus-infected mouse model [22] . In this study, intranasal administration of gemcitabine significantly lowered the pulmonary viral load and inflammation by decreasing proinflammatory cytokines, including TNF- α and IL-1 β , and the number of lung infiltrating lymphocytes. More recently, Zhang et al. also identified gemcitabine as the best anti-PV inhibitor from a screen of FDA-approved drugs in PV replicon-harboring HeLa cells (EC 50 of 0.3 μ M) [15] . As previously mentioned, accumulating evidence has definitively demonstrated that gemcitabine is an effective broad-spectrum inhibitor of RNA viruses and has a therapeutic potential for the treatment of various virus-associated diseases. Moreover, it is possible that gemcitabine is effective for other untested RNA viruses. Because gemcitabine is a deoxycytidine analog that interferes with DNA as well as RNA synthesis, DNA viruses may not be the exception. Consistent with this possibility, there has been a report that the infection of HSV-1, which is a representative DNA virus classified into the Herpesviridae family, was strongly affected by gemcitabine [16] . Most of the abovementioned viruses have, at best, limited prophylactic or therapeutic drugs as possible treatments. This is especially true for newly emerging or re-emerged viruses involving serious illnesses, such as MERS-CoV, SARS-CoV, and ZIKV, which are major threats to public health and which urgently need an effective treatment during their early stages of infection. In this regard, repurposing of gemcitabine for the treatment of patients infected with these deadly viruses is a realistic approach. Importantly, it is noteworthy that ZIKV was the most strongly affected by gemcitabine, with a low nanomolar EC 50 , which was lower than that used in cancer therapy [14, 20] . Even

for other viruses with a relatively high EC 50 , there is an option to treat patients with a combination of gemcitabine with other antiviral agents. In this manner, an effective antiviral treatment may be achieved by the synergistic action of two antivirals with much lower doses for each drug, which minimizes deleterious side effects when used clinically. As an example, the synergistic antiviral effect of gemcitabine in combination with ribavirin, an antiviral drug currently being used against a few RNA viruses, was reported against EVs such as CVB3 and EV71 [18] . As previously mentioned, the combination of gemcitabine with decitabine synergistically suppressed HIV infectivity both in vitro and in vivo [17, 21] . However, the actual use of gemcitabine in virus-infected patients necessitates prior in vivo animal studies and clinical trials. Even though most antiviral data have originated from in vitro studies, two recent studies have reported the antiviral effects of gemcitabine in murine models [17, 22] . More extensive analyses of gemcitabine in animal models in the near future will accelerate its therapeutic applications in clinical trials.

Most studies regarding the antiviral activity of gemcitabine lack experimental evidence of the mode of action. However, our group has recently reported that gemcitabine had an anti-EV effect by targeting the salvage pathway of pyrimidine biosynthesis [23] . Moreover, gemcitabine strongly induced the expression of several ISGs including CXCL10, IRF7, IRF9, IFIT1, and DDX58, which were the major effectors in the innate immunity that defended the host against the virus infection. These results were consistent with a previous report that gemcitabine stimulated the production of IFN- β and IFN- γ in IAV-infected RPE cells [16] . Importantly, the activation of ISGs was well-correlated with the inhibition of pyrimidine biosynthesis, suggesting a link between pyrimidine biosynthesis and innate immunity. Similar phenomena in terms of ISG activation have been previously reported with a few compounds out of several purine or pyrimidine biosynthesis inhibitors that had antiviral activity, as summarized in Table 2 [6, 10, [23] [24] [25] [26] [27] [28] [29] [30] [31] [32] [33] [34] [35] [36] [37] [38] [39] [40] [41] [42] . Regarding purine biosynthesis inhibitors, ribavirin and mycophenolic acid (MPA) are inhibitors of inosine-5 -monophosphate (IMP) dehydrogenase (IMPDH), which is a key enzyme of the purine biosynthesis pathway. These inhibitors have been successfully used as clinical antiviral or immunosuppressant agents for decades. Both have antiviral activities against viruses such as HCV, hepatitis E virus (HEV), MERS-CoV, dengue virus, yellow fever, hepatitis B virus, West Nile virus (WNV), Chikungunya virus (CHIKV), and IAV [24] [25] [26] [27] [28] [29] [30] , majorly through the inhibition of the purine biosynthesis pathway, with the antiviral activity against HCV and HEV shown to involve the stimulation of ISGs [10, 30] . For the antiviral activity of ribavirin against HCV, ribavirin specifically induced the expression of IRF7, IRF9, and ISG15 mRNAs, which are known to be important for anti-HCV immune responses [6] . ISG activation

occurred through an undefined mechanism that was different from the classical IFN signaling, intracellular dsRNA sensing pathway, Toll-like receptor and nuclear factor B pathways. More importantly, ribavirin-induced ISG activation and antiviral activity were suppressed using supplemented guanosine, a natural analog of ribavirin, suggesting IMPDH inhibition-mediated ISG activation as an alternative innate immunity pathway. Like ribavirin, MPA remarkably induced the expression of several ISGs, including IRF1, IRF9, ISG15, IFI6, IRF7, CXCL10, IFIT2, and IFITM3 mRNAs in naïve or HEV-infected Huh-7 cells, and the induction of ISGs was at least partially abrogated by the use of supplemented guanosine [10]. Mechanistically, the induction of ISGs by MPA was independent of the classical JAK/STAT system, which is similar to that observed with ribavirin [30]. Similar results were obtained with several IMPDH1 or IMPDH2 inhibitors, with various affinities, that were custom-designed and synthesized [10].

As shown in Table 2, most pyrimidine biosynthesis inhibitors target dihydroorotate dehydrogenase (DHODH), an essential enzyme in de novo pyrimidine synthesis. Lucas-Hourani et al. identified DD264 as an interferon-sensitive response element (ISRE)-stimulating compound from high-throughput screening, and further analyses suggested that it was a DHODH inhibitor with a strong antiviral activity against various viruses including MeV, CHIKV, and WNV [37]. DD264 enhanced the expression of several ISGs, which were almost completely suppressed by the addition of supplemented uridine, indicating DHODH inhibition-mediated ISG activation. Moreover, the antiviral activity of and ISG activation by DD264 required the interferon regulatory factor 1 (IRF1) transcription factor, a master regulator of antiviral gene expression [37], which was consistent with the observation that the anti-HCV activity of MPA was partially mediated by IRF1 [30]. In this study, similar results were shown with brequinar, another well-known DHODH inhibitor. FA-613 is also an antiviral compound, which inhibits the pyrimidine biosynthesis pathway, probably via targeting DHODH and inducing the expression of ISGs such as IFNB1, CXCL10, ISG15, and CCL5 [38]. However, whether ISG activation is mediated by pyrimidine biosynthesis inhibition remains to be determined.

The mechanism of nucleotide synthesis inhibitor-induced ISG activation is still presently unclear. Nevertheless, there has been accumulating evidence showing that nucleotide synthesis inhibitor-induced ISG activation is independent of the classical JAK/STAT-mediated IFN signal [6, 10, 23]. First, Wang et al. clearly showed that ISG activation and anti-HEV activity induced by MPA or brequinar was not mediated by JAK [10]. Second, IRF7 induction by ribavirin was not affected by knockdown of STAT1, while that of IFN- α was

strongly affected under the same conditions [6] . Third, our recent study with gemcitabine further confirmed IFN signal-independent ISG activation by parallel studies comparing the effects of gemcitabine and IFN- α . In our study, the phosphorylation of STAT1 at Tyr701, which was dramatically triggered by IFN- α , did not occur when treated with gemcitabine [23] . Moreover, the upregulation of DDX58 mRNAs induced by gemcitabine was not affected by IRF9 knockdown, which was contrary to the result that IFN- α -induced upregulation of DDX58 mRNAs was significantly suppressed under the same conditions. Consistent with above observations, there have been some reports that ISGs was induced in the absence of JAK1 or STAT1 activation [43, 44] .

Despite limited data, we speculate the scenario of ISG activation that is independent of JAK/STAT-mediated IFN signal. Purine or pyrimidine biosynthesis inhibitors could interfere with the metabolic pathway through targeting some key enzymes such as IMPDH and DHODH, leading to the depletion or imbalance of the (d)NTP pool. Inactivation of metabolic enzyme(s) itself or consequently altered nucleos(t)ide pools might trigger a signal, which is ultimately delivered to certain cis-acting elements on the promoter of a subset of ISGs, possibly through the relay of kinases and transcription factors. Based on the previously mentioned reports, this signal is less likely to be dependent on STAT1/2-IRF9 (IFN-stimulated gene factor 3; ISGF3), at least for gemcitabine, which is the major transcriptional complex in the IFN-induced JAK/STAT pathway. It should also be considered that Thomas et al. excluded the involvement of an intracellular double-stranded RNA sensing pathway, Toll-like receptor and nuclear factor κ B pathways, as well as a classical IFN signal in the activation of ISGs induced by ribavirin [6] . Despite the consensus of ISG activation, each purine/pyrimidine biosynthesis inhibitor seems to induce distinct sets of ISGs, at least with different patterns [10] . Targeting an enzyme in which pathways (purine or pyrimidine synthesis) or steps (early/late and de novo/salvage) produce different levels of intermediates and nucleos(t)ides will consequently result in diverse outcomes of ISG activations. There might be more than one signaling pathway involved. The synergistic antiviral activity of gemcitabine and ribavirin observed in our study might be explained by the possible existence of two separate signaling pathways that mediate each inhibition of nucleotide synthesis toward ISG activation. Systematic analyses of signaling kinases, IRFs, and STATs using siRNA knockdown and/or pharmacological inhibition and metabolic analyses of corresponding intermediates and nucleos(t)ides should therefore clarify the underlying molecular mechanisms of ISG activation by purine/pyrimidine biosynthesis inhibitors.

As newly emerging or re-emerged viruses such as SARS-CoV, MERS-CoV, and ZIKV have become a major threat to public health, the need for broad-spectrum antiviral drug has increased. In this regard, nucleoside analogs that directly target viral RNA-dependent RNA polymerase and present a high barrier to the development of resistant viruses have been considered advantageous. Moreover, recent discovery of a new antiviral mode of nucleoside analogs acting through innate immunity strengthens the molecular basis for their therapeutic application as broad-spectrum antiviral drugs.

Nucleoside analogs probably induce different subsets of ISGs, at least with a different pattern, leading to various combinations of ISGs and resulting antiviral outcomes. Moreover, according to Schoggins et al., different viruses are affected by distinct subsets of ISGs and some ISGs such as IRF1, MB21D1, HPSE, DDX58, MDA, and IFITM3 act broadly on various viruses [45]. Thus, more systematic analyses on the subsets of ISGs induced by antiviral nucleoside analogs are required for the identification of better antiviral drugs that can be used broadly or specifically. Given the clinical side effects of IFN treatment, nucleotide analogs that differ from IFN in the activation of subsets of ISGs need to be considered as alternatives. Nevertheless, nucleoside analogs interfering with the host nucleotide synthesis pathway suggest possible side effects in their clinical applications. Careful evaluation of clinical safety is required and their application for the urgent measure of patients infected with deadly viruses would be worth being primarily considered.

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Identification and characterisation of the CD40-ligand of *Sigmodon hispidus*

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Abstract: Cotton rats are an important animal model to study infectious diseases. They have demonstrated higher susceptibility to a wider variety of human pathogens than other rodents and are also the animal model of choice for pre-clinical evaluations of some vaccine candidates. However, the genome of cotton rats remains to be fully sequenced, with much fewer genes cloned and characterised compared to other rodent species. Here we report the cloning and characterization of CD40 ligand, whose human and murine counterparts are known to be expressed on a range of cell types including activated T cells and B cells, dendritic cells, granulocytes, macrophages and platelets and exerts a broad array of immune responses. The cDNA for cotton rat CD40L we isolated is comprised of 1104 nucleotides with an open reading frame (ORF) of 783bp coding for a 260 amino acid protein. The recombinant cotton rat CD40L protein was recognized by an antibody against mouse CD40L. Moreover, it demonstrated functional activities on immature bone marrow dendritic cells by upregulating surface maturation markers (CD40, CD54, CD80, and CD86), and increasing IL-6 gene and protein expression. The availability of CD40L gene identity could greatly facilitate mechanistic research on pathogen-induced-immunopathogenesis and vaccine-elicited immune responses.

Text: The cotton rat (*Sigmodon hispidus*) was first used in polio research in the 1930s [1], and throughout the last century, it has proven to be an excellent model for biomedical research [2, 3, 4]. Historically in biomedical research, the mouse has been exploited as the default animal model. This is in part due to its well defined immunological and genetic information, costeffectiveness, and abundant inbred strains and research reagents. However, the use of mice as models to study infectious diseases has its limitation since mice are not naturally infected by most human pathogens. On the other hand, cotton rat is susceptible to many human pathogens and is the ideal model of choice for measles (paramyxovirus) [5], herpes simplex (oral a1111111111 a1111111111 a1111111111 a1111111111 a1111111111 and ophthalmic) [6], influenza (orthomyxovirus) [7, 8], HIV-1 [9], RSV (respiratory syncytial virus) [10], adenovirus [11, 12], human parainfluenza [13], and human metapneumovirus [14]. This model has been valuable for adenovirus-based gene replacement therapy research [15, 16], and was also proven to be indispensable in pre-clinical evaluation of the prophylactic antibodies (RespiGam 1 [17], and Synagis 1 [18]

. Indeed, the cotton rat model was found to be valuable in terms of its biological and immunological relevance, it was deemed unnecessary to test the adenovirus-based gene therapy and the Synagis 1 prophylactic treatment against RSV disease in non-human primate prior to the human trials [19, 20] .

A number of methods and reagents have been developed for the analysis of immune responses in cotton rats over the last decade. Up to date, more than 200 genes encoding cytokines, chemokines, cell surface markers and regulatory molecules have been cloned, with various related research reagents being commercially available. As a result, the use of cotton rats in pathogenesis studies addressing mechanistic questions has significantly increased. Nevertheless, the gene encoding CD154 and CD40 ligand (CD40L), remains elusive.

CD40L plays a critical role in orchestrating immune responses against pathogens. Depending on the post-translational modification, the murine CD40L is a 32-39 kDa type II membrane glycoprotein that was initially identified as a surface marker exclusive to activated CD4 + T cells [21, 22] . It is a member of the TNF superfamily consisting of a sandwiched extracellular structure composed of a β -sheet, α -helix loop, and a β -sheet, allowing for the trimerization of CD40L, an additional feature of the TNF family of ligands [23] . Since its initial discovery, CD40L has been shown to be not only expressed on CD4+ T cells, but on dendritic cells (DCs) [24] , B cells [25] , and platelets [26] .

It has been shown that upon interacting with its receptor, CD40, CD40L induces profound effects on T cells, DCs, B cells, endothelial cells, as well as many cells of the hematopoietic and non-hematopoietic systems. Moreover, when CD40L engages CD40 on the surface of DCs, it promotes cytokine production, the induction of cell surface co-stimulatory molecules, and facilitates the cross-presentation of antigen by these cells [27] , enabling DCs to mature and effectively induce the activation and differentiation of T cells. When CD40L engages CD40 on the surface of B cells, it promotes germinal center formation, immunoglobulin (Ig) isotype switching, somatic hypermutation to enhance antigen affinity, and lastly, the formation of long-lived plasma cells and memory B cells [28] . Various studies have been conducted to utilize gene delivery of CD40L to DCs and tumor cells for tumor immunotherapy. It was found that expression of CD40L in a small proportion of tumor cells was sufficient to generate a long-lasting systemic anti-tumor immune response in mice that was shown to be dependent on cytotoxic T lymphocytes [29, 30] .

Here we report the successful cloning of the gene encoding cotton rat CD40L (crCD40L); we also expressed and purified the CD40L produced in mammalian cells. Further characterisation of the recombinant cotton rat CD40L revealed its functional activities in promoting DC maturation and cytokine production. [6] [7] weeks old cotton rats were obtained from an inbred colony maintained at Envigo (USA). All animal experiments were conducted in accordance with Institutional Care and Use Committee (IACUC) of Health Canada Ottawa Animal Care Committee which approved this study. The rats were housed 3 animals per cage in Allentown NexGen individually ventilated cages with free access to food and water. These cages provided a floor space of 142 in 2 / 916 cm². Body weight and any sign of distress were monitored daily. If anything associated the animal health was observed, a full examination would be conducted. As In this study spleen cells from normal, healthy animals were isolated, we did not observe any adverse reaction. To isolate splenocytes from the animals, isoflourane was used to put the animals to sleep via inhalation with oxygen for euthanasia.

The spleens from three naïve cotton rats were removed aseptically and snap frozen in liquid nitrogen. The spleens were homogenized individually with a TissueLyser II (Qiagen) and total RNA extracted using the RNeasy Mini kit (Qiagen) with on-column DNase digestion according to the user's manual. The 3' RACE system (Life Technologies) was then used with to amplify the 3' portion of the cotton rat CD40L from the total RNA according to the manufacturer's instructions. A schematic of the 3' RACE procedure used is provided in S1 Fig. A gene specific primer (5'-GGACTCTATTATGTCTACACCCAAGTCACCTTCTG -3') was derived from a consensus sequence aligning the rat (*Rattus norvegicus* UniProt: Q9Z2V2), mouse (*Mus musculus* UniProt: P27548), and golden hamster (*Mesocricetus auratus* NCBI Reference Sequence: XM_005084522.3) CD40L sequences obtained from the National Center for Biotechnology Information (NCBI). Following first strand cDNA synthesis, the 3' portion of the cotton rat CD40L mRNA was PCR amplified using the consensus sequence derived gene specific primer and the abridged universal amplification primer with an annealing temperature at 56 °C. The reverse complementary sequence of this primer was then used as a reverse primer with the forward primer (5'-GATAGAAACATACAGCCAACCTTCTCCAGATC -3') to amplify the 5' portion of the cotton rat CD40L mRNA with an annealing temperature of 57 °C.

All amplified fragments were sequenced with BigDye Terminator v.3.1 Cycle Sequencing kit (ThermoFisher cat # 4336917). Briefly, samples were amplified in a PTC-200 thermal cycle (MJ Research) with the following program: 26 cycles of 1 °C/S to 96 °C, 96 °C for 10 seconds, 1 °C/S to 50 °C, 50 °C for 5 seconds, 1 °C/S to 60 °C, 60 °C for 4 minutes. The samples were cleaned using DyeEx 2.0 Spin kit (Qiagen cat # 63204) and loaded onto a 3130xl Genetic Analyzer (Applied Biosystems). Raw sequencing data was edited by the instrument's software (ThermoFisher 3130xl Genetic Analyzer Data Collection Software v3.0), and then imported into GeneCodes Sequencer v4.6.1 sequencing analysis software for further editing. The final sequenced contigs are then imported to NCBI BLAST (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>) to confirm the identity.

Putative conserved domains, trimer interface, and receptor binding sites were determined by performing a standard protein BLAST (blastp algorithm; <https://blast.ncbi.nlm.nih.gov>). The sequences producing significant alignments were imported into Geneosis software, (Auckland, New Zealand). Multiple alignment was conducted as previously described [31], with phylogenetic analysis using Geneosis Pro 5.6.7.

Once the mRNA sequence was confirmed, a construct was designed beginning with a kozak sequence (5'-CACCGCCGCCACC-3'), followed by a secretion signal consisting of 23 amino acid (aa) (MLLAVLYCLLWSFQTSAGHFPR) from the human tyrosinase signal peptide as previously described [32]. This is followed by six histidine residues to facilitate protein purification. Following this sequence, a 27-aa fragment from the bacteriophage T4 fibrin trimerization motif was added [33] and finally connected to the full length 783bp open reading frame (ORF) of the cotton rat CD40L sequence at the C terminus. This construct was synthesized and cloned into pUC57 (Biobasic, Markham, ON).

Generation of a recombinant vaccinia virus expressing cotton rat CD40L protein construct was achieved using a vaccinia virus E3L and K3L double deletion mutant virus as the parental virus and taterapoxvirus K3L as the positive selection marker (Jingxin Cao, unpublished information). Briefly, the recombination plasmid vector for expression of the CD40L construct gene consists of the homologous flanking vaccinia DNA sequences targeting vaccinia A45R gene (SOD homolog); the CD40L construct gene driven by a modified vaccinia H5 promoter (Vaccine 1996, 14:1451), and taterapoxvirus 037 gene driven by vaccinia K3L promoter as the positive selection marker. The recombination vector was transfected into a HeLa PKR knockout cells infected with a vaccinia virus with both E3L

and K3L genes deleted. Selection and purification of the recombinant vaccinia virus expressing the CD40L was done in BHK21 cells.

Expression of the CD40L protein was confirmed by Western blotting using His-tag Ab. Cell monolayers were lysed in sample buffer and homogenized using QIAshredder columns (Qiagen). Western blotting was performed using 4 to 15% TGX gel and Tris/Glycine/SDS running buffer (Bio-Rad Laboratories Inc.), and the protein samples were transferred to Immobilon-FL PVDF membranes (Millipore). Protein was detected with Tetra-HIS Ab (Qiagen) and goat anti-mouse IRDye-800CW (LiCor). Membranes were developed using the Odyssey system (LiCor).

The vaccinia virus carrying the crCD40L gene was propagated in BHK21 cells. The cells were collected and washed with PBS once and then lysed with a denaturing buffer (10 mM Tris-HCl, 100 mM sodium phosphate, 6 M guanidine hydrochloride, 10 mM reduced glutathione, pH 8.0) and disrupted by sonication on ice using a Branson sonifier 150 (ThermoFisher, Waltham, MA) at level 1 for two 10sec bursts with 1min rest on ice between. After separation of cell debris, the supernatant was added to a slurry of Ni-NTA resin (Qiagen, Mississauga, ON, Canada) (10 mL resin bed) and stirred at room temperature for 30 min before loading into a column. The column was purified using an AKTA purifier (Amersham Biosciences) with Unicorn 5.3 software (Amersham Biosciences). Refolding was accomplished under oxidative conditions with a gradient of denaturing buffer to buffer B (buffer B: 10 mM Tris-HCl, 100 mM sodium phosphate, pH 7.8) over 10 column volumes (CVs). The column was then washed with three CVs of buffer B + 60 mM imidazole (pH 7.8) to remove unspecific binding. The protein was eluted off the column with buffer B + 250 mM imidazole (pH 7.8). The resulting protein was dialysed against PBS pH 7.5 and then confirmed by western blot.

96-well plates were coated with either recombinant mouse CD40L (R&D Systems) or the recombinant crCD40L protein 2ug/ml in 100µl PBS. Plates were washed with wash buffer (PBS-0.1% tween-20) and then blocked with 200µl/well blocking buffer (PBS containing 0.1% Tween 20 and 3% IgG Free BSA) for 1 hour at 37 °C. Plates were washed with wash buffer and incubated at 37 °C for 1 hour with 100µl/well goat anti-mouseCD40L (R&D Systems) 2ug/ml in blocking buffer. Plates were subsequently washed and incubated at 37 °C for 1 hour with 100µl/well with rabbit anti-goat IgG HRP conjugate (Zymed). Plates were washed again and incubated for 10 min in the dark with 100µl/well 3,3',5'-

tetramethylbenzidine substrate (New England Bio Labs). The reaction was stopped with Stop solution (New England Bio Labs) and absorbance was read at 450nm on a BioTek Synergy 2 plate reader.

Primary bone marrow cells from Balb/c mice (Chicago, IL) were thawed and cultured in dendritic cell medium from manufacture (Cell Biologics M7711) supplemented with GMCSF (Cell Biologics) without IL-4 at 4×10^5 cells/well in a volume of 200 μ l. The cells were treated with 0.5 μ g/ml recombinant mouse CD40L (Preprotech, Montreal, QC) or the recombinant crCD40L protein at 0.5 μ g/ml, 5 μ g/ml, or 50 μ g/ml. Forty hours later, flow cytometry was performed on a BD LSRFortessa cell analyser after 2×10^5 cells/tube were stained using CD11c-PE-CF594, CD54-FITC, CD40-BV786, CD80-BV421, and CD86-BV711 antibodies. All antibodies were purchased from BD Biosciences. The resulting spectra were analysed using FACS-Diva version 8.0.1 software.

To assess IL-6 mRNA production of immature bone marrow murine DCs in response to targeting by recombinant crCD40L, quantitative real-time PCR was conducted on an ABI Prism 7500 Fast Sequence detection system (Applied Biosystems). TaqMan assay reagent kits (Applied Biosystems) were used that contain pre-standardized primers and TaqMan MGB probes for IL-6 and 18S which were used to normalize the data. Total RNA was isolated from 8×10^5 stimulated bone marrow DCs using the RNeasy Mini Kit (Qiagen) according to manufactures instructions. The isolated RNA was used to make cDNA using the Superscript III First-Strand Synthesis System for RT-PCR (Invitrogen) according to manufacturer's instructions. The cDNA was then subjected to quantitative PCR using the TaqMan Fast Advanced Master Mix (Applied Biosystems) according to manufactures instructions. Samples were run in duplicate and C_t values were obtained. Fold change over unstimulated DCs was calculated using the $2^{-\Delta\Delta CT}$ method of relative quantification [34], using 18S as the housekeeping reference gene. To investigate IL-6 secretion by murine bone marrow DCs, supernatant from forty hour stimulated cultures were collected and assayed using the Mouse IL-6 DuoSet ELISA Kit (R & D Systems) following the manufacturer's protocol.

The complete mRNA sequence of CD40L was obtained in two steps (Fig 1). A sequence corresponding to nucleotides 535 through to the poly-A tail was obtained using the 3' RACE kit and mRNA as starting material, which was isolated from cotton rat splenocytes and a rodent consensus sequence as a primer. This portion of the sequence has the 3' un-

translated region of the mRNA as well as the stop codon. The 5' end of the protein was obtained in the next step by PCR amplification of the cDNA obtained in the first step with the 3' RACE kit and the reverse complement of the consensus sequence primer and a second consensus sequence primer designed to bind to the beginning of the CD40L mRNA. The 783bp ORF encodes 260aa followed by a stop codon.

Comparison of the sequenced CD40L gene revealed that the crCD40L coding sequence shares 93%, 89%, and 83%, identity with golden hamster, rat, and mouse, respectively. At the amino acid (aa) level, the corresponding identities are 91%, 82%, and 82%, Fig 2a. At both the mRNA and aa levels, the crCD40L shared the closest similarity with *Peromyscus maniculatus bairdii* (or deer mouse) at 93% and 92% respectively. When sequence homology analysis is performed, crCD40L clusters with other members of the Cricetidae family Fig 2b. We next examined the functional domains in crCD40L in comparison with other known CD40L. As shown in Fig 3a, crCD40L has a putative tumor necrosis factor (TNF) superfamily Using EZmol software [35] , we predicted folding of the protein as shown in Fig 3b. The cotton rat CD40L cDNA that we have isolated was a 1104 nucleotide sequence with a poly-A tail containing an ORF of 783bp which coded for a 260 aa protein. The homology of cotton rat CD40L, at both the amino acid and nucleic acid level, is closer to members of the Cricetidae family (hamster and deer mouse) than to those of the Muridae family (rat and mouse) as shown in Fig 2b. As with other known CD40L proteins, there is a putative TNF superfamily domain, a transmembrane domain, trimerization sites, and receptor binding sites [36] .

TNF superfamily members include TNF (TNF-alpha), LT (lymphotoxin-alpha, TNF-beta), CD40 ligand, Apo2L (TRAIL), Fas ligand, and osteoprotegerin (OPG) ligand, among others [37] . The TNF superfamily is composed of 19 ligands and 29 receptors, in which each has vastly diversified roles in the body and exhibit pro-inflammatory activity, partly via activation of NF-kB [37] . Members of this family generally have an intracellular N-terminal domain, a short transmembrane segment, an extracellular stalk, and a globular TNF-like extracellular domain of about 150 residues [23] . They initiate apoptosis by binding to related receptors, some of which have intracellular death domains [38] . These proteins typically form homo-or hetero-trimeric complexes and bind one elongated receptor molecule along each of three clefts formed by neighboring monomers of the trimer and ligand trimerization is for receptor binding [23, 39] . All seven known conserved residues that constitute the trimer interface on the conserved TNF domain [23, 40] , were mapped to the putative crCD40L protein sequence. Additionally, all six known conserved receptor

binding sites on the conserved TNF domain [23, 40] , were mapped to the crCD40L protein sequence.

In order to further evaluate the crCD40L deduced sequence, the full 783bp ORF of the crCD40L was cloned into a vaccinia virus vector. The crCD40L construct was designed to carry a secretion signal, histidine tag, and a trimerization motif (Fig 4a) . Selection and purification of the recombinant vaccinia virus expressing the CD40L construct was conducted in BHK21 cells. Western blot with anti-histidine antibody (Ab) was used to confirm expression of the CD40L protein construct Fig 4b and S2 Fig. The resulting 36 kDa protein product was found in both the cell lysate and supernatant (faint band-48 hours only). Since the highest expression was found in the cell lysate, it was used for further purification of the protein. It should be noted that the protein was only able to be detected under reducing conditions. Under non-reducing conditions, the protein was unable to be detected by the anti-histidine Ab, even in the cell lysate (data not shown). This indicates that the histidine tag is folded within the trimer and is unavailable in the native form for purification. This is an additional reason for the need to purify the protein from the cell lysate under harsh denaturing conditions followed by protein refolding. The reason we utilized a mammalian expression system to produce the protein rather than a bacterial system is to facilitate its proper folding into its native structure, trimerization, and glycosylation. The aa backbone predicts a protein of 29 kDa, yet initial studies of the CD40L protein suggested a molecular mass of 39 kDa, and on most cell types the molecular mass of CD40L is 32-33kDa, consistent with extensive post-translation modification [36] .

The BHK21 cells expressing the crCD40L construct were collected and lysed with 6 M guanidine hydrochloride with reduced glutathione and sonication. The lysate was loaded on the nickel column and the washed with denaturing buffer as described in materials and methods. The bound proteins were refolded on the column with gradient buffer exchange, to allow slow refold the protein, given that CD40L biological activity is dependent on a homo-trimer configuration [23] . The resulting bound protein was subsequently eluted with imidazole. The resulting fractions that showed a peak were pooled and dialysed against PBS.

The purified protein was confirmed in ELISA. Since the cotton rat CD40L protein sequence shared 82% identity with the mouse CD40L protein sequence, an Ab known to detect mouse CD40L was used to identify the purified crCD40L protein. The purified recombinant

crCD40L was used as a coating antigen in a concentration gradient manner, and was detected with an Ab generated against the mouse CD40L at all concentrations (Fig 5) . Uncoated controls were performed in parallel and were negative for CD40L in ELISA. We measured the overall strength of the antigen-antibody complex in the presence of 6M urea [41] . The avidity of the cotton rat CD40L for the anti-mouse CD40L antibody was decreased in the presence of 6M urea at all concentrations. Clearly, as the antibody used was raised against mouse CD40L, the crCD40L is detected by mouse CD40L. crCD40L was expressed in vaccinia virus and purified from infected BHK21 cell lysate on a nickel column. The purified protein was detected by ELISA using a mouse antibody against CD40L in a concentration gradient dependent manner. The avidity of the mouse CD40L antibody to the cotton rat CD40L protein was evaluated in the presence of 6M urea. The difference between the untreated and 6M urea treated for each group was calculated using students t-test $p < 0.001$, $p < 0.0001$ (n = 2). Data shown is a representative experiment of three separate experiments where two (n = 2) technical replicates are conducted in each experiment. The no-coating and noprimary antibody negative controls gave average OD values of 0.56 and 0.107 respectively.

<https://doi.org/10.1371/journal.pone.0199067.g005> addition of urea treatment would substantially weaken the interaction between the antibody and crCD40L.

Since the cotton rat CD40L protein sequence shared 82% identity with the mouse CD40L protein sequence with similar functional domains, we evaluated the biological activity of the recombinant crCD40L on immature murine bone marrow DCs. We conducted experiments based on known functional activities of CD40L in other animal species. Specifically, maturation of immature DCs after exposure to antigen is known to play a crucial role in their immunity-stimulating function [36] , while trimeric recombinant CD40L has been shown to stimulate DC immunomodulating functions [42] . When CD40L engages CD40 on the surface of DCs, it promotes cytokine production, the induction of cell surface co-stimulatory molecules, and facilitates the cross-presentation of antigen by these cells [27] . In addition, CD11c is a DC integrin marker and upon stimulation, is down-regulated [43] . Intracellular adhesion marker CD54, along with co-stimulatory markers CD40, CD80, and CD86 are all upregulated upon stimulation with CD40L [44, 45] . Moreover, mouse I-A d major histocompatibility complex is also up-regulated upon stimulation with CD40L [45] . When our recombinant crCD40L was used to stimulate immature murine bone marrow DCs, we observed similar results to that when murine CD40L is used (Tables 1 and 2) . CD11c was down regulated in both median fluorescence intensity (Table 1) and the

percentage of positive cells (Table 2). The co-stimulatory molecules CD54, CD40, CD80, and CD86 were all up-regulated in both median fluorescence intensity (Table 1) and the percentage of positive cells (Table 2). The Mouse I-A d major histocompatibility complex was upregulated in median fluorescence intensity (Table 1) but not up-regulated in terms of the overall percentage of positive cells (Table 2) . We speculate this to be due to the species incompatibility since we are stimulating mouse bone marrow cells with cotton rat CD40L. Nevertheless, the crCD40L was able to promote up-regulation of key co-stimulatory markers on immature DCs promoting DC maturation. The gating strategy used for the flow cytometry analysis is provided in S3 Fig along with overlapping histograms of the intracellular adhesion marker and co-stimulatory markers. CD40-induced activation of cytokine gene expression in DCs by CD40L is an important process in the initiation of primary immune responses and is critical for DC maturation and the generation of antigen-specific T cell responses [46] . IL-6 is a highly pleiotropic cytokine in that it stimulates the activation, proliferation, and survival of T cells, and furthermore, modifies DC function and survival [47] [48] [49] [50] . We tested if the recombinant crCD40L could induce IL-6 gene expression (Fig 6a) and production of the cytokine (Fig 6b) by immature murine bone marrow DCs. The results indicate that a significant increase in both IL-6 gene expression and cytokine production in immature murine bone marrow DCs was observed forty hours after stimulation with the crCD40L. Collectively, the observation that both the upregulation of immature DC cell surface maturation markers and increased IL-6 gene expression and cytokine production provide strong evidence of the biological activity of crCD40L.

In summary, the cotton rat CD40L cDNA that we isolated was a 1104 nucleotide sequence with a poly-A tail containing an ORF of 783 bp which coded for a 260 aa protein. The recombinant cotton rat CD40L was recognized by an Ab against mouse CD40L in direct ELISA, and showed biological activity by upregulating maturation markers (CD40, CD54, CD80, and CD86) as well as I-A d on immature bone marrow murine DCs and moreover, inducing upregulation of IL-6 gene and cytokine expression in these cells.

The isolation of the cotton rat CD40L sequence and availability of CD40L has the potential to positively impact basic immunological research and vaccine development, given the critical importance of this protein in orchestrating immune responses [51, 52] .

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--- START OF ARTICLE 21 ---

Bioinformatics analysis of rabbit haemorrhagic disease virus genome

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Abstract: **BACKGROUND:** Rabbit haemorrhagic disease virus (RHDV), as the pathogen of Rabbit haemorrhagic disease, can cause a highly infectious and often fatal disease only affecting wild and domestic rabbits. Recent researches revealed that it, as one member of the Caliciviridae, has some specialties in its genome, its reproduction and so on. **RESULTS:** In this report, we firstly analyzed its genome and two open reading frameworks (ORFs) from this aspect of codon usage bias. Our researches indicated that mutation pressure rather than natural is the most important determinant in RHDV with high codon bias, and the codon usage bias is nearly contrary between ORF1 and ORF2, which is maybe one of factors regulating the expression of VP60 (encoding by ORF1) and VP10 (encoding by ORF2). Furthermore, negative selective constraints on the RHDV whole genome implied that VP10 played an important role in RHDV lifecycle. **CONCLUSIONS:** We conjectured that VP10 might be beneficial for the replication, release or both of virus by inducing infected cell apoptosis initiated by RHDV. According to the results of the principal component analysis for ORF2 of RSCU, we firstly separated 30 RHDV into two genotypes, and the ENC values indicated ORF1 and ORF2 were independent among the evolution of RHDV.

Text: Synonymous codons are not used randomly [1]. The variation of codon usage among ORFs in different organisms is accounted by mutational pressure and translational selection as two main factors [2, 3]. Levels and causes of codon usage bias are available

to understand viral evolution and the interplay between viruses and the immune response [4]. Thus, many organisms such as bacteria, yeast, *Drosophila*, and mammals, have been studied in great detail up on codon usage bias and nucleotide composition [5]. However, same researches in viruses, especially in animal viruses, have been less studied. It has been observed that codon usage bias in human RNA viruses is related to mutational pressure, G +C content, the segmented nature of the genome and the route of transmission of the virus [6]. For some vertebrate DNA viruses, genome-wide mutational pressure is regarded as the main determinant of codon usage rather than natural selection for specific coding triplets [4]. Analysis of the bovine papillomavirus type 1 (BPV1) late genes has revealed a relationship between codon usage and tRNA availability [7]. In the mammalian papillomaviruses, it has been proposed that differences from the average codon usage frequencies in the host genome strongly influence both viral replication and gene expression [8]. Codon usage may play a key role in regulating latent versus productive infection in Epstein-Barr virus [9]. Recently, it was reported that codon usage is an important driving force in the evolution of astroviruses and small DNA viruses [10, 11]. Clearly, studies of synonymous codon usage in viruses can reveal much about the molecular evolution of viruses or individual genes. Such information would be relevant in understanding the regulation of viral gene expression.

Up to now, little codon usage analysis has been performed on Rabbit haemorrhagic disease virus (RHDV), which is the pathogen causing Rabbit haemorrhagic disease (RHD), also known as rabbit calicivirus disease (RCD) or viral haemorrhagic disease (VHD), a highly infectious and often fatal disease that affects wild and domestic rabbits. Although the virus infects only rabbits, RHD continues to cause serious problems in different parts of the world. RHDV is a single positive stranded RNA virus without envelope, which contains two open reading frames (ORFs) separately encoding a predicted polyprotein and a minor structural protein named VP10 [12]. After the hydrolysis of self-coding 3C-like cysteinase, the polyprotein was finally hydrolyzed into 8 cleavage products including 7 nonstructural proteins and 1 structural protein named as VP60 [13, 14]. Studies on the phylogenetic relationship of RHDVs showed only one serotype had been isolated, and no genotyping for RHDV was reported. It reported that the VP10 was translated with an efficiency of 20% of the preceding ORF1 [15]. In order to better understand the characteristics of the RHDV genome and to reveal more information about the viral genome, we have analyzed the codon usage and dinucleotide composition. In this report, we sought to address the following issues concerning codon usage in RHDV: (i) the extent and causes of codon bias in RHDV; (ii) A possible genotyping of RHDV; (iii) Codon usage bias as a factor reducing the expression of VP10 and (iiii) the evolution of the ORFs.

The 30 available complete RNA sequences of RHDV were obtained from GenBank randomly in January 2011. The serial number (SN), collection dates, isolated areas and GenBank accession numbers are listed in Table 1 .

To investigate the characteristics of synonymous codon usage without the influence of amino acid composition, RSCU values of each codon in a ORF of RHDV were calculated according to previous reports (2 Sharp, Tuohy et al. 1986) as the followed formula:

Where g_{ij} is the observed number of the i th codon for j th amino acid which has n_i type of synonymous codons. The codons with RSCU value higher than 1.0 have positive codon usage bias, while codons with value lower than 1.0 has relative negative codon usage bias. As RSCU values of some codons are nearly equal to 1.0, it means that these codons are chosen equally and randomly.

The index GC3s means the fraction of the nucleotides G+C at the synonymous third codon position, excluding Met, Trp, and the termination codons.

The ENC, as the best estimator of absolute synonymous codon usage bias [16] , was calculated for the quantification of the codon usage bias of each ORF [17] . The predicted values of ENC were calculated as $ENC = 2 + s + 29$

where s represents the given (G+C) 3 % value. The values of ENC can also be obtained by EMBOSS CHIPS program [18] .

Analyses were conducted with the Nei-Gojobori model [19] , involving 30 nucleotide sequences. All positions containing gaps and missing data were eliminated. The values of d_n , d_s and ω (d_n/d_s) were calculated in MEGA4.0 [20] .

Multivariate statistical analysis can be used to explore the relationships between variables and samples. In this study, correspondence analysis was used to investigate the major

trend in codon usage variation among ORFs. In this study, the complete coding region of each ORF was represented as a 59 dimensional vector, and each dimension corresponds to the RSCU value of one sense codon (excluding Met, Trp, and the termination codons) [21] .

Correlation analysis was used to identify the relationship between nucleotide composition and synonymous codon usage pattern [22] . This analysis was implemented based on the Spearman's rank correlation analysis way. All statistical processes were carried out by with statistical software SPSS 17.0 for windows.

The values of nucleotide contents in complete coding region of all 30 RHDV genomes were analyzed and listed in Table 2 and Table 3 . Evidently, (C+G)% content of the ORF1 fluctuated from 50.889 to 51.557 with a mean value of 51.14557, and (C+G)% content of the ORF2 were ranged from 35.593 to 40.113 with a mean value of 37.6624, which were indicating that nucleotides A and U were the major elements of ORF2 against ORF1. Comparing the values of A 3 %, U 3 %, C 3 % and G 3 %, it is clear that C 3 % was distinctly high and A 3 % was the lowest of all in ORF1 of RHDV, while U 3 % was distinctly high and C 3 % was the lowest of all in ORF2 of Table 2 Identified nucleotide contents in complete coding region (length > 250 bps) in the ORF1 of RHDV (30 isolates) genome Table 4 . Most preferentially used codons in ORF1 were C-ended or G-ended codons except Ala, Pro and Ser, however, A-ended or G-ended codons were preferred as the content of ORF2. In addition, the d_n , d_s and $\omega(dN/dS)$ values of ORF1 were separately 0.014, 0.338 and 0.041, and the values of ORF2 were 0.034, 0.103 and 0.034, respectively. The ω values of two ORFs in RHDV genome are generally low, indicating that the RHDV whole genome is subject to relatively strong selective constraints.

COA was used to investigate the major trend in codon usage variation between two ORFs of all 30 RHDV selected for this study. After COA for RHDV Genome, one major trend in the first axis ($f' 1$) which accounted for 42.967% of the total variation, and another major trend in the second axis ($f' 2$) which accounted for 3.632% of the total variation. The coordinate of the complete coding region of each ORF was plotted in Figure 1 defining by the first and second principal axes. It is clear that coordinate of each ORF is relatively isolated. Interestingly, we found that relatively isolated spots from ORF2 tend to cluster into two groups: the ordinate value of one group (marked as Group 1) is

To estimate whether the evolution of RHDV genome on codon usage was regulated by mutation pressure or natural selection, the A%, U%, C%, G% and (C+G)% were compared with A 3 %, U 3 %, C 3 %, G 3 % and (C 3 +G 3)%, respectively (Table 5). There is a complex correlation among nucleotide compositions. In detail, A 3 %, U 3 %, C 3 % and G 3 % have a significant negative correlation with G%, C%, U% and A% and positive correlation with A%, U%, C% and G%, respectively. It suggests that nucleotide constraint may influence synonymous codon usage patterns. However, A 3 % has non-correlation with U% and C%, and U 3 % has noncorrelation with A% and G%, respectively, which haven't indicated any peculiarity about synonymous codon usage. Furthermore, C 3 % and G 3 % have non-correlation with A%, G% and U%, C%, respectively, indicating these data don't reflect the true feature of synonymous codon usage as well. Therefore, linear regression analysis was implemented to analyze the correlation between synonymous codon usage bias and nucleotide compositions. Details of correlation analysis between the first two principle axes (f' 1 and f' 2) of each RHDV genome in COA and nucleotide contents were listed in Table 6 . In surprise, only f2 values are closely related to base nucleotide A and G content on the third codon position only, suggesting that nucleotide A and G is a factor influencing the synonymous codon usage pattern of RHDV genome. However, f' 1 value has non-correlation with base nucleotide contents on the third codon position; it is observably suggest that codon usage patterns in RHDV were probably influenced by other factors, such as the second structure of viral genome and limits of host. In spite of that, compositional constraint is a factor shaping the pattern of synonymous codon usage in RHDV genome. Figure 1 A plot of value of the first and second axis of RHDV genome in COA. The first axis (f' 1) accounts for 42.967% of the total variation, and the second axis (f' 2) accounts for 3.632% of the total variation. Table 5 Summary of correlation analysis between the A, U, C, G contents and A 3 , U 3 , C 3 , G 3 contents in all selected samples

There have been more and more features that are unique to RHDV within the family Caliciviridae, including its single host tropism, its genome and its VP10 as a structural protein with unknown function. After we analyzed synonymous codon usage in RHDV (Table 2) , we obtained several conclusions and conjectures as followed.

4.1 Mutational bias as a main factor leading to synonymous codon usage variation ENC-plot, as a general strategy, was utilized to investigate patterns of synonymous codon usage. The ENC-plots of ORFs constrained only by a C 3 +G 3 composition will lie on or just below the curve of the predicted values [18] . ENC values of RHDV genomes were plotted against its corresponding (C 3 +G 3) %. All of the spots lie below the curve of the predicted values,

as shown in Figure 2 , suggesting that the codon usage bias in all these 30 RHDV genomes is principally influenced by the mutational bias.

As we know, the efficiency of gene expression is influenced by regulator sequences or elements and codon usage bias. It reported that the RNA sequence of the 3terminal 84 nucleotides of ORF1 were found to be crucial for VP10 expression instead of the encoded peptide. VP10 coding by ORF2 has been reported as a low expressive structural protein against VP60 coding by ORF1 [5] . And its efficiency of translation is only 20% of VP60. According to results showed by Table 4 , it revealed the differences in codon usage patterns of two ORFs, which is a possible factor reducing the expression of VP10.

Although VP10 encoded by ORF2, as a minor structural protein with unknown functions, has been described by LIU as a nonessential protein for virus infectivity, the ω Figure 2 Effective number of codons used in each ORF plotted against the GC3s. The continuous curve plots the relationship between GC3s and ENC in the absence of selection. All of spots lie below the expected curve.

value of ORF2 suggests VP10 plays an important role in the certain stage of whole RHDV lifecycle. After combining with low expression and ω value of VP10, we conjectured that VP10 might be beneficial for the replication, release or both of virus by inducing infected cell apoptosis initiate by RHDV. This mechanism has been confirmed in various positive-chain RNA viruses, including coxsackievirus, dengue virus, equine arterivirus, footand-mouth disease virus, hepatitis C virus, poliovirus, rhinovirus, and severe acute respiratory syndrome [23] [24] [25] [26] [27] [28] [29] , although the details remain elusive.

As preceding description, ENC reflects the evolution of codon usage variation and nucleotide composition to some degree. After the correlation analysis of ENC values between ORF1 and ORF2 (Table 7) , the related coefficient of ENC values of two ORFs is 0.230, and p value is 0.222 more than 0.05. These data revealed that no correlation existed in ENC values of two ORFs, indicating that codon usage patterns and evolution of two ORFs are separated each other. Further, this information maybe helps us well understand why RSCU and ENC between two ORFs are quite different.

Interestingly, we found that relatively isolated spots from ORF2 tend to cluster into two groups: the ordinate value of one group (marked as Group 1) is positive value and the other one (marked as Group 2) is negative value. And all of those strains isolated before 2000 belonged to Group 2, including Italy-90, RHDV-V351, RHDV-FRG, BS89, RHDV-SD and M67473.1. Although RHDV has been reported as only one type, this may be a reference on dividing into two genotypes.

In this report, we firstly analyzed its genome and two open reading frameworks (ORFs) from this aspect of codon usage bias. Our researches indicated that mutation pressure rather than natural is the most important determinant in RHDV with high codon bias, and the codon usage bias is nearly contrary between ORF1 and ORF2, which is maybe one of factors regulating the expression of VP60 (encoding by ORF1) and VP10

(encoding by ORF2). Furthermore, negative selective constraints on the RHDV whole genome implied that VP10 played an important role in RHDV lifecycle. We conjectured that VP10 might be beneficial for the replication, release or both of virus by inducing infected cell apoptosis initiate by RHDV. According to the results of the principal component analysis for ORF2 of RSCU, we firstly separated 30 RHDV into two genotypes, and the ENC values indicated ORF1 and ORF2 were independent among the evolution of RHDV. All the results will guide the next researches on the RHDV as a reference.

--- END OF ARTICLE ---

--- START OF ARTICLE 22 ---

Obesity and risk of respiratory tract infections: results of an infection-diary based cohort study

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Abstract: **BACKGROUND:** Respiratory tract infections (RTIs) are a major morbidity factor contributing largely to health care costs and individual quality of life. The aim of the study was to test whether obesity ($\text{BMI} \geq 30 \text{ kg/m}^2$) is one of the risk factors underlying frequent RTIs in the German adult population. **METHODS:** We recruited 1455 individuals between 18 to 70 years from a cross-sectional survey on airway infections in Germany and invited them to self-report in diaries incident RTIs experienced during three consecutive winter/spring seasons. RTIs reported in these 18 months and summary measures adding-up individual RTIs were the outcomes of interest. **RESULTS:** Compared to individuals with normal weight, obese individuals reported a consistently higher frequency of upper and lower RTIs and predominantly fell in the upper 10% group of a diary sumscores adding-up 10 different RTI symptoms over time. Obesity was associated both with lower RTIs ((adjusted)OR = 2.02, 95%CI = 1.36–3.00) and upper RTIs ((adjusted)OR = 1.55, 95%CI = 1.22–1.96). Adjusting for demographic and lifestyle variables did only marginally affect ORs. Stratified analyses suggested a stronger association for women and effect modifications by sports activity and dietary habits. **CONCLUSIONS:** We confirm the association of obesity with infection burden and present evidence for putative interaction with sports activity and dietary patterns. **ELECTRONIC SUPPLEMENTARY MATERIAL:** The online version of this article (10.1186/s12889-018-5172-8) contains supplementary material, which is available to authorized users.

Text: Frequent and severe respiratory tract infections (RTIs) constitute an important morbidity factor in our society and a considerable cost burden in terms of medical treatment and time of work-loss [1, 2]. RTIs are divided into upper RTIs (URTIs) including common cold, pharyngitis, otitis, sinusitis, laryngotracheitis, epiglottitis and lower RTIs (LRTIs) including bronchitis, pneumonia and bronchiolitis [3]. Individual exposure to infectious agents and host factors such as smoking [4, 5] and vitamin D status [6, 7] are believed to contribute to observed differences in RTI risk. In addition, the role of overweight (body mass index (BMI) = $25.0\text{--}29.9 \text{ kg/m}^2$) and in particular obesity ($\text{BMI} \geq 30 \text{ kg/m}^2$) in predisposition to RTIs is increasingly discussed [8] [9] [10] [11] [12] [13]. This growing

interest is driven by the rising number of overweight and obese individuals worldwide [14] and the emerging knowledge of notable immunological imbalances in association with obesity [15]. Most of the studies targeting adults explored the association of obesity with specific RTIs and their outcomes. Thus, obesity was associated with non-allergic rhinitis [8] and influenza like-illness [9]. Moreover, two population-based studies which investigated the role of obesity as risk factor for community acquired pneumonia (CAP) in the general population resulted in controversial findings [10, 11]. Two recent Danish population-based studies reported an excess of a large spectrum of RTIs including pneumonia among obese people [12, 13]. The overall aim of our study targeting the adult population in South Baden, Germany, is to identify risk factors for the susceptibility to RTIs. Here we present data on the role of obesity as contributing factor to a high RTI burden in the German society and explore effect modification by gender, sports activity and nutritional patterns.

Study participants (n = 1455) were recruited from the airway infection susceptibility (AWIS) cross sectional study querying RTI burden in an adult population in South-Baden, Germany [16]. The study protocol was approved by community officials and the Ethics Committee of the University of Freiburg (Ref. No. 258/11_120365). Based on the RTI history-score individuals of putative low, medium and high risk of future RTIs were invited to the actual sub-cohort. The RTI history score is summarizing information on the frequency and severity of RTIs and antibiotics use over the past two years, selfassessed RTI susceptibility, and occurrence of selected severe infections [16]. Study participants were requested to fill-in an additional questionnaire (baseline questionnaire) on lifestyle factors and co-morbidities and to complete monthly diaries registering the monthly occurrence and the duration (< 2 weeks, > 2 weeks) of RTIs, namely sinusitis, rhinitis, otitis media, pharyngitis/laryngitis, tonsillitis, influenza-like illness, bronchitis, pneumonia, pleurisy and other acute RTIs, from the beginning of November to the end of April of three seasons: 2012/13, 2013/14 and 2014/15. Furthermore, the intake of antibiotics, doctor visits, hospitalisation for RTIs and the impact of RTI symptoms on their daily activities were queried. Further recruitment details into the AWIS study and the present sub-cohort are presented under Additional files 1 and 2. Informed consent was obtained from all individual participants included in the study.

In order to describe the association between obesity and RTIs, different outcome indicators were considered: outcomes at the level of each month ["any RTI", "any URTI" (sinusitis, rhinitis, otitis media, pharyngitis/laryngitis and tonsillitis), "any LRTI" (bronchitis, pneumonia and pleurisy), "≥3 RTIs", "any long lasting infection" (> 2 weeks)]; at the level of

each winter season ("≥4 months with infections", "≥3 long lasting infections"); and at the individual level (i.e. are defined once per individual and covering the overall study period). The ten specific RTI symptom categories were considered with the binary symptom indicators "infection reported" or "no infection reported" for each month.

When counting the episodes for the outcome indicator "≥3 long lasting infections", different infection symptoms were counted as separate episodes, even if they overlapped in time. However, within one symptom category at least one month without this specific infection was required to call it a new episode. We also calculated a monthly diary RTI score, averaging the ten RTI symptom categories with the coding "0" for "no infection reported", "1" for "reported infection with duration < 2 weeks", and "2" for "reported infection present with duration >2 weeks". Missing values for individual infection items were treated as zero. If an individual RTI symptom was reported, but information on duration was missing, it was counted as "reported infection with duration < 2 weeks". If all items were missing, no diary score was computed. The diary RTI score at the monthly level was expanded to a score at the seasonal level by averaging over the six months (November-April) of each season, and to an overall score at the individual level by averaging over all available months. The respective upper 10% of these diary scores within each month, season and overall served as additional outcome indicators.

Further variables considered in the study were age, gender, self-reported weight and height for BMI calculation (BMI was categorized as < 30 (non-obese), $25 \leq \text{BMI} < 30$ (overweight) and ≥ 30 (obese)), educational level, contact with children, comorbidities, removed immunological organs, smoking status, sports activity and dietary intake patterns. Details on these variables are described in the Additional file 1 and supplementary information on dietary intake patterns is presented in Additional file 3.

Statistical analysis was performed using Stata (version 14 STATSCorp, USA). Descriptive statistics: Monthly prevalences of individual RTI symptoms were computed by taking the average over all subjects available at each month and then averaging over all 18 months covered. Prevalences at the seasonal level were computed accordingly averaging over all three seasons covered. The corresponding confidence intervals (CIs) and p-values are based on a generalised linear model with identity link and binomial type variance together with robust variance estimates. The frequency of long lasting infections among all months with infections was analysed accordingly. However, due to the limited number of cases for

tonsillitis and otitis media we determined the monthly frequency of long-lasting infections by pooling the data over all seasons and for pneumonia by pooling all indicated months.

At the monthly level ORs were computed using a logistic regression model with a random intercept applied to the individual data for each month taking the 18 months as a categorical covariate into account in addition to the obesity status indicator. Due to its small prevalence, pleurisy was not considered as single outcome in these analyses. Outcomes at the seasonal level were analysed accordingly with the individual data for each winter season and taking into account the three seasons as a categorical covariate. Outcomes at the individual level were analysed using a logistic regression model. Results are ORs and 95% CIs. Adjusted ORs are based on including age groups and education as simultaneous categorical covariates. Furthermore, in order to study the stability of the obesity-RTI association with respect to potential confounders, ORs were adjusted by respective variables. Subjects with incomplete covariate data were excluded from multivariate analyses.

Effect modification by a binary variable was assessed by fitting an overall model with the corresponding interactions parametrized so that we could directly read off the two subgroup-specific ORs. Effect modification by sports activity and nutrition patterns was explored among those representing the lower and upper third of respective scores.

The study population comprised 1455 individuals (931 female and 524 male) with a median age of 51.08 years. Based on BMI calculated from self-reported weight and height, 2.1% of the population was underweight ($\text{BMI} < 18.5 \text{ kg/m}^2$), 54% had a normal weight ($18.5 \text{ kg/m}^2 \leq \text{BMI} < 25 \text{ kg/m}^2$), 31.1% was overweight, and 12.8% was considered obese (Table 1). In women, the distribution was 2.8%, 60.21%, 25.0%, and 12.1% and in men 0.76%, 43.1%, 41.8%, and 14.3%, respectively. The study participants were mainly of medium and high educational level, non- or ex-smokers, moderately affected by selected co-morbidities and they reported rather infrequent contact to small children. Further information on the study population and completed diaries is reported in Table 1 and Additional file 4.

Missing rates of single items in the returned diaries were limited and ranged from 1.2% for rhinitis and pharyngitis/laryngitis to 2.6% for other acute respiratory infections. Study participants reported most frequently rhinitis (26.6%), followed by influenza-like illness

(11.4%) and pharyngitis/laryngitis (10.5%), whereas pleurisy (0.10%) was rarely experienced. Any URTI (31.5%) was more frequent than any LRTI (7.9%). Apart from the LRTIs bronchitis, pneumonia and pleurisy, which more men than women reported, all other RTIs were more prevalent among women (Table 2). Seasonal patterns of reported infections show a February peak for two of the three assessed infection seasons (2012/13 and 2014/15, see Additional file 5). Respiratory infections with a high fraction of long duration were almost exclusively LRTIs, namely pneumonia (59%), followed by bronchitis (48.2%). Men were overrepresented among those with long-lasting RTIs (Table 2) .

Compared to normal weight individuals, overweight and obese people consistently had a higher prevalence (Table 3) for the single RTIs, URTIs, LRTIs, as well as the other outcome parameters we looked at with other acute infections and pneumonia as the exceptions. For pneumonia, only obese subjects had a higher prevalence. The overweight group was typically falling in between the groups with normal weight and obesity (Table 3). The strongest association was seen for pneumonia and bronchitis, and accordingly, any LRTI was more strongly associated with obesity than any URTI. Long-lasting RTIs, frequent RTIs and high diary scores were also more strongly associated with obesity than the individual symptoms. Adjustments by age and education did only marginally change these estimates. Among subjects with an infection, long lasting infections were again associated with obesity, reaching significance for any RTI, rhinitis, pharyngitis/laryngitis, influenza-like illness, and bronchitis (Table 3) .

For a better understanding of the robustness of the relationship between RTI burden and obesity, the effect of adjusting for putative confounders was explored (Additional file 6). The studied demographic and lifestyle variables (age, gender, education level, smoking status, contact to children, asthma, sports activity, dietary patterns and previous removal of immune organs) did only marginally affect ORs. However, adjustment for asthma, chronic obstructive pulmonary disease (COPD) or a summary score covering all queried co-morbidities weakened the relationship between obesity and all outcomes considerably. Adjustment for vitamin D levels among those for which serum was available (n = 508), had only a slight effect on the magnitude of the association between obesity and RTI outcomes.

The association between obesity and RTI outcomes was more prominent for women than for men and reached statistical significance only for the former (Table 4). For most outcomes this interaction was not significant, with the individual level diary score as an

exception. When looking at sports activity, for most outcomes the association with obesity was confined to those physically more active and not seen for those reporting little sports activity (Table 5). For all outcomes the association was less pronounced in the latter group (compare the ratios of ORs in Table 5), a difference that reached significance for all outcomes except those with low prevalence. Typically the prevalence of an outcome was only increased in the small group of people with obesity and higher sports activity whereas all other groups presented rather similar patterns. Similarly, the prevalence of outcomes was increased among people with obesity and a more favourable nutritional pattern, but comparable among the other groups (Table 6). The interaction reaches significance for the majority of outcomes.

RTIs constitute an important morbidity factor considering the high health care costs, the time lost from work, and the impaired quality of life among those recurrently affected [1, 2, 17] . Obesity belongs to one of the host risk factors for RTI and has possibly an emerging role due to the dramatically increasing prevalence of obesity worldwide. In the present study, we report on the association of obesity with individual RTIs as well as with a diary score summarising different incident RTI symptoms over a period of 18 months. Our investigation could demonstrate an association between obesity and RTIs confirming previous findings on influenza-like illness [9] , bronchitis [18] and pneumonia [10, 12] . We also saw an association between obesity and rhinitis, sinusitis and pharyngitis/laryngitis. An elevated risk for sinusitis among obese was also reported in a populationbased cohort of Danish women [13] . None of the two Danish population-based studies [12, 13] used ORs of monthly prevalence, but hazard ratios (HRs), as they could identify events on a daily basis. The HR of 1.6 [12] for the association with RTIs and the HR of 1.48 [13] for the association with URTIs are, however, of similar magnitude to the risk estimates which we observed. Mechanistically, excess adiposity might weigh down host defence as several mouse as well as human studies have suggested [19, 20] . The here observed associations were more prominent for LRTIs compared to URTIs, but evident for both, and more pronounced when considering long lasting or frequent RTIs compared to single symptoms. Based on the infection diary data, we generated a RTI diary score summing-up all ten symptoms and allowing to average per month, per whole season or over the whole period of three years. Considering the upper ten percentile of the distribution of such scores as an outcome, associations were typically stronger than when considering single symptoms, and interactions were more pronounced. Moreover, the results of the seasonal score were very similar or even stronger than those of the three-years score, arguing for the adequacy to query six months infectious events in future studies to identify the infection-prone subgroup of the population. Lifestyle habits seem to contribute to an individual's risk for RTI.

Among them, cigarette smoking has been reported as a major environmental risk factor for recurrent and severe RTIs [4, 5]. Frequent contact to small children [21, 22], vitamin D deficiency [23, 24], and lack of physical activity [25, 26] constitute other exposures associated with heightened RTI risks. Moreover, higher levels of education were associated with a lower risk of CAP [27]. Based on those previous findings we investigated their role as possible confounders. The association between obesity and RTIs remained nearly unchanged after adjustment for age, gender, educational status, contact to children, smoking status, sports activity and nutrition scores, suggesting that the association is not markedly confounded by the effects of these factors on both BMI and the risk of infections. Also additional adjustment by measured serum vitamin D in a subgroup for which measurements were available did not change the risk estimates considerably. This supports arguments that the observed associations between obesity and RTI burden are due to physiological differences in the immune responsiveness between obese and non-obese individuals rather than lifestyle differences. In addition, some chronic diseases, foremost asthma and COPD, are associated with both an increased risk of RTIs and obesity [28] [29] [30] [31] [32]. Considering these associations we investigated the effect of asthma, COPD and a comorbidity scoresummarizing the other chronic conditionson the relationship between obesity and individual RTIs and the RTI diary score. Adjusting for these conditions individually and even more so in a combined fashion resulted in a considerable attenuation of the association between obesity and considered RTI outcomes. Hence part of the association between infections and obesity might be explainable by associations of co-morbidities with both. We see a gender difference in the observed associations with more noticeable findings for women. A significantly increased risk for combined RTIs was also restricted to women in a Danish blood donor cohort [12]. Several lines of research support this notion: Szabova et al. and Ilavska et al. reported gender-dependent effects of obesity on the immune system [33, 34]. The effect of BMI on a variety of immune parameters including those with relevance for immune defence was much more apparent in women than in men [34]. NK cells (CD3-/CD16+/CD56+), represent first-line cells for the clearing of virus-infected cells. Reduced levels of these cells reported for obese women, but not for respective men, might underlie the gender effect seen in our study. We also investigated a potential effect modification by sports activity and nutrition. Interestingly, an association between obesity and RTIs was evident only for those obese individuals who reported a higher level of sports activity. Thus, only the group of obese people who engaged in more intensive sports activity reported RTIs more frequently whereas obese people with low sports activity and non-obese with low or high sports activity showed comparable lower prevalences for most outcomes. We hypothesize that oxidative stress induced by vigorous aerobic as well as anaerobic sports activity is exacerbated in people with obesity, but not in normal weight individuals. Evidence

supporting this has been previously published [35] . An imbalanced oxidative stress status may have negative consequences on mounting an appropriate immune response towards respiratory pathogens. Excessive reactive oxygen species (ROS) was shown to hinder T cell responses to viral infection [36] and ROS accumulation was detected in autophagy-deficient effector T cells rendering them incapable of controlling viral infections [37] .

A similar surprising result was found when studying the effect modification by dietary patterns. Here we queried the participants' dietary habits and classified them as adhering to a more favourable or more unfavourable dietary pattern according to Winkler et al. [38] . Aware of the limitations of a one-time assessment of a habitual diet, we found a more pronounced relationship between obesity and infections among obese people who reported an apparent healthier diet. Thus, again only the group of obese individuals who presumably eat a healthier diet showed an increased risk of RTIs. The question arises as to whether misreporting of dietary habits among these individuals with and without RTIs may explain the puzzle. One can imagine that obese individuals may have an increased perception of RTI related symptoms experiencing the contradiction between living a healthy lifestyle and being affected by excess weight and frequent infections. On the other hand the inconspicuous results from the non-obese population with respect to favourable and unfavourable diet pattern would somewhat argue against this explanation. Alternatively, among the group of people with obesity a genetically defined subgroup may exist predisposing to both, excess body weight and proneness to infections.

As strengths of our study we count 1) its sample size, allowing for the analysis of effect modification, 2) its prospective design involving 18 months infection diaries for the exploration of the relationship between BMI and subsequent RTI frequency and severity, 3) the comprehensive information on lifestyle and co-morbidities allowing to study the interplay of such factors on their effect on infections, and 4) the wide range of outcome indicators considered. The uniformity of the results with respect to these outcomes also suggests that in the field of airway infection morbidity, studies may be comparable despite the fact that they often concentrate on different RTI outcomes. In line with the majority of epidemiological studies in this area of research, our study suffers from some limitations, including the reliance on self-reported outcomes and exposure data with the risk of misclassification. However, we found -for instance -a good agreement between BMI derived from self-reported weight and height data and BMI calculated from measured values available for a sub-cohort (n = 508). Moreover, differential misclassification which would substantially bias the relationship between obesity and RTIs is rather unexpected in

this setting. The disproportional selection of women into the study may negatively impact the generalizability of some of our results.

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A Global Champion for Health—WHO's Next?

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Abstract: In this month's editorial, the PLOS Medicine Editors propose ideal qualities for the World Health Organization's next Director General, for whom the selection process is now underway.

Text: response to the Ebola outbreak [1] . Reformation of WHO to ready it to lead responses to future health emergencies is one area of active debate.

Chan will step down from WHO on June 30, 2017 after more than a decade in the post. The process for choosing WHO's next leader has begun, promising to be protracted and rigorous as befits the importance of the role. Factoring in the many influential stakeholders in the process of appointing Chan's successor, however, transparency of the selection

process may be one area unlikely to attract plaudits. Although too soon to speculate about the identity of WHO's next Director-General, it is worth reflecting on what qualities an incoming leader should bring to WHO and how that person might need to conceive changes in the structure and behavior of the organization against a landscape of important and evolving threats to the health of the fastgrowing global population.

Instead of electing a new Director-General, Lorenz Von Seidlein of Mahidol University, Thailand, argued that "the problems. . .are now so deeply ingrained that replacing the WHO with new, more appropriate organizations is the logical solution. . .at a fraction of current cost, free of cumbersome, archaic obligations and entitlements and [with] an ability to respond to new problems." This viewpoint is indicative of the strength of feeling that WHO's deficiencies have come to evoke in some of those committed to the cause of improving the health of people in low-income and middle-income countries. But this perception acknowledges that an accountable global body will always be needed to promote, set standards in, and evaluate progress toward better health for people in all countries. The next Director-General will need to heed critics of the organization and craft a process of streamlining and restructuring to produce a new WHO that is demonstrably effective in leading responses to threats to health, and efficient in doing so. As Gostin commented to PLOS Medicine, "WHO urgently needs a bold reform agenda to fix long-standing problems recognized by every independent group that has evaluated the Organization." Political machinations and the enemy within, bureaucracy, are likely to impede reform. For example, WHO's regional and country offices are seen by some as unaccountable, yet the agency of the future will need to be connected and responsive to the resources and needs of all constituent countries. As Gostin also noted, "[WHO] has failed to include civil society in its governance, unlike. . .newer organizations."

WHO's next Director-General should be a proven leader and advocate, perhaps from a lowincome or middle-income country. The new recruit will be greeted by a full in-tray, and featuring prominently are likely to be the constraints imposed by WHO's current funding mechanisms. A substantial proportion of WHO's existing budget is earmarked for specific projects, leaving the organization with little financial flexibility to respond to unanticipated demands. However, any improved funding mechanism is likely to follow, and be dependent on, organizational reform. According to Kruk, "WHO is both essential and hamstrung. . .the election of the Director-General should be a moment for member countries and other funders to reflect on whether they want an implementation agency for their favored health agenda, or an independent institution with the intelligence, agility, and operational capacity

to tackle the coming global health challenges." Above all, the incoming leader of WHO will need to be open-minded and creative. More than one of the experts we contacted emphasized the fluid nature of the threats to human health to which WHO should shape the world's response. WHO must be able to lead responses in some areas of global health, but, in other areas, working together with more nimble and focused organizations will be pragmatic. Large-scale infectious disease outbreaks are continuing, and noncommunicable diseases, including cancer, dementia, and mental illnesses, are growing in prevalence and increasing demand for treatment and care. The resources and ingenuity of researchers and clinicians will need to be harnessed, and interventions adapted to new settings, with much greater dynamism. The secular issues of population ageing, conflict, climate change, migration, and others will produce health problems that only an organization with a global reach, responsible to all, can hope to meet. We look forward to welcoming a new leader for WHO with the energy and vision to remold the organization to meet the health needs of the world's people and societies for the 21st century.

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A focus reduction neutralization assay for hepatitis C virus neutralizing antibodies

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1852297/>

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Abstract: BACKGROUND/AIM: The role of humoral immunity in hepatitis C virus (HCV) infection is poorly understood. Nevertheless, there is increasing interest in characterizing the neutralizing antibodies in the serum of HCV-infected patients. Focus reduction assays have been widely used to evaluate neutralizing antibody responses against a range of non-cytopathic viruses. Based on the recent development of a HCV cell culture system using the genotype 2 JFH-1-strain, we developed a focus reduction assay for HCV-neutralizing antibodies. **METHODS:** The focus reduction assay was based on a standard microneutralization assay in which immunostained foci on tissue culture plates are counted. The neutralizing anti-HCV antibodies titers of purified serum immunoglobulin samples from seventy-seven individuals were determined using a 50% focus reduction neutralization assay. Each titer was determined as the log value of the reciprocal antibody dilution that reduced the number of viral foci by 50%. IgG antibodies were first purified from each serum in order to avoid the facilitating effect of HDL on HCV entry. **RESULTS:** The assay's cut-off using an ELISA and RNA HCV-negative samples was found to be 1.25 log, corresponding to a dilution of 1:18. The assay was compared with a commercial HCV ELISA and exhibited specificity and sensitivity values of 100% and 96.5%, respectively, and good reproducibility (with intra-assay and inter-assay coefficients of variation of 6.7% and 12.6%, respectively). The assay did not show any cross-reactivity with anti-HIV, anti-HBs or heterophile antibody-positive samples. The neutralizing antibodies titers were 2.13 log (1:134) for homologous samples from HCV genotype 2 infected patients harboring the same genotype as JFH-1 and 1.93 log (1:85) for heterologous samples from patients infected by genotypes other than type 2. These results confirm the presence of broadly cross-neutralizing antibodies already reported using the HCV pseudoparticles system. **CONCLUSION:** This study presents a simple, specific and reproducible cell culture-based assay for determination of HCV-neutralizing antibodies in human sera. The assay should be an important tool for gauging the relationship between the neutralizing antibodies response and viral load kinetics in acutely or chronically infected patients and for investigating the possible eradication or prevention of HCV infection by neutralizing antibodies.

Text: Hepatitis C virus (HCV, a member of the Flaviviridae family) is an enveloped, positive-stranded RNA virus that preferentially replicates in hepatocytes. At least 170 million people worldwide are persistently infected with hepatitis C virus. Chronic HCV infection is associated with a significant risk of progression to cirrhosis and hepatocellular carcinoma [1]. Antiviral therapy with pegylated alpha-interferon and ribavirin (the current best therapeutic regimen) is only successful in about 50% of all treated patients.

Better knowledge of the viral and host factors that determine HCV clearance or persistence during the acute stage of infection is needed in order to improve antiviral therapy and to develop efficient vaccines. Studies focusing on innate and cellular immune responses have shown that a sufficiently large HCV inoculum is able to evade, subvert or circumvent the host's defences. At present, the chimpanzee is the only reliable experimental animal model in which the initial post-HCV infection events and the efficacy of vaccine candidates can be evaluated [2] . It has been shown that HCV-specific T-cell immunity is important in the control of HCV infection [3, 4] . Several studies have indicated a role for humoral immunity in the acute stage of HCV infection but this aspect remains poorly characterized. The E1 and E2 glycoproteins are thought to be the viral attachment proteins and thus the main targets for HCV-neutralizing antibodies; identification of protective epitopes conserved across different strains of HCV is therefore a major challenge in vaccine design. A number of antibodies capable of blocking E2 binding to cells or cell receptors have been described, [5] [6] [7] [8] some of which neutralize HCV entry in animal or cellular models [9, 10] . Cell entry has been shown to involve several surface molecules (notably including the tetraspanin CD81 and the SR-BI receptor [11, 12]), although further studies are needed to better understand how viral entry occurs and how it might be neutralized. Detection of neutralizing antibodies in human blood had been problematical until an efficient and reliable cell culture system for HCV became available. Hence, the development of an in vitro neutralization assay for HCV could be extremely valuable for characterizing the humoral immune response to HCV and for evaluating the potential of passive and active immunization against hepatitis C. Recent studies using an in vitro neutralization assay system (based on infectious retroviral pseudoparticles (HCVpp) bearing HCV envelope glycoproteins) have confirmed that HCV-infected patient sera can indeed neutralize infection [13, 14] . However, it has also been shown that the neutralizing activity of antibodies from HCV-infected patients is attenuated by a factor present in human serum, identified as the highdensity lipoprotein (HDL) fraction [11, 13, 15] . HDL facilitation of HCVpp entry is a post-binding event [16] , suggesting that HDLs favour internalization of virions and thus the latter's escape from neutralizing antibodies.

Recently, an HCV cell culture model (HCVcc) has been developed [17] [18] [19] , allowing the production of virus particles that can be efficiently propagated in cell culture. Some preliminary neutralization assays have been carried out by these authors. In this study, we describe how we set up a standardized focus reduction neutralization assay based on HCVcc.

Focus reduction assays have been widely used to evaluate the neutralizing antibody responses to viruses that can form foci in infected cells. Following the recent development of the HCVcc model, the principle of the focus reduction assay has been applied to HCV-neutralizing antibodies detection. The JFH-1 HCV 2a viral strain was grown on a Huh-7 human hepatoma cell line. After three days of infection and cell permeabilization, detection of the HCV foci was carried out using an inactivated HCVpositive patient serum primary antibody and a peroxidase-coupled, Fc-specific anti-human IgG-antibody. The reaction was revealed with DAB peroxidase substrate. The viral foci were thus stained brown, making them easy to count (see Fig. 1a). It has been recently shown that the neutralizing activity of HCV antibodies is attenuated by a serum factor associated with the HDL fraction. Hence, HDLs were able to facilitate HCVpp and HCVcc entry via a mechanism which depended on the expression of the scavenger receptor BI (SR-BI) and its selective lipid-uptake function [11, 15, 16, 20] . In view of the role of HDL in HCV entry, immunoglobulins were purified from each serum sample prior to determination of the neutralizing antibody titer (see Fig. 1b).

The specificity of the HCV neutralization assay was assessed by testing 20 anti-HCV-ELISA-negative samples, including five positive for hepatitis B virus surface antibodies (anti-HBs) and five positive for heterophile antibodies. All samples tested negative with two commercial anti-HCV antibody detection assays (AxSYM[®] HCV Version 3.0, Abbott, Wiesbaden, Germany; Vitros[®] Anti-HCV reagent pack, Ortho-Clinical Diagnostic, High Wycombe, United Kingdom) and HCV-RNA-negative with a qualitative, commercial assay (Cobas Amplicor HCV test Version 2.0, Roche Diagnostics, Meylan, France).

These anti-HCV-negative samples were compared with 11 samples from patients chronically infected with HCV genotype 2. The neutralization titers of anti-HCV-negative serum samples are shown in Fig. 2 ., with a mean value of 1.083 ± 0.083 (corresponding to a dilution of 1:12). The assay's cut-off (determined as the mean value for negative samples plus two standard deviations) corresponded to a dilution of 1:18. The assay exhibited specificity and sensibility values of 100% and 96.5%, respectively. The assay did not show any cross-reactivity with anti-HIV, anti-HBs or heterophile antibody-positive samples (data not shown). Conversely, the chronically HCV genotype 2-positive samples displayed strong reactions, with a mean value of 2.128 ± 0.365 (corresponding to a dilution of 1:134) ($p < 0.001$).

Inter-assay variability was determined by testing one HCV genotype 2 sample in 10 consecutive experiments ($n = 10$), whereas intra-assay variability was evaluated by testing the same sample 10 times ($n = 10$) in the same experiment, whilst running the dilution series. The intra-assay and inter-assay coefficients of variation (CV) of the log neutralization titers were 6.7% and 12.6%, respectively.

Fifty-seven HCV-positive antibodies samples were evaluated using the HCV focus reduction neutralization assay. The genotypes were distributed as follows; for types 1a, 1b, 2, 3, 4 and 5, we studied 11, 11, 11, 12, 10 and 2 samples, respectively. The mean values of the different genotypes is shown in Fig. 3 . and Table 1 . The mean log neutralization titers for genotypes 1a, 2 and 3 are very similar (2.046 ± 0.671 for genotype 1a, 2.128 ± 0.365 for genotype 2 and 2.148 ± 0.478 for genotype 3). The mean average values are lower for genotype 1b (1.747 ± 0.462) and genotype 4 (1.786 ± 0.236). Strikingly, very high heterologous titers were observed for five patients -three infected with HCV genotype 1a and two infected with HCV genotype 3 (see Fig. 3a). There were too few genotype 5 samples to compare with the other genotypes but the corresponding results nevertheless indicate that the neutralization assay is suitable for this genotype. The two The distribution of the log neutralization titers across all the HCV ELISA and RNA-positive samples as a function of the HCV genotype is shown in Fig. 3b . More than 60% of the neutralizing antibodies titers fell in the range from 1.7 to 2.69 log titers, corresponding to dilutions of 1:50 and 1:500, respectively. Overall, 3.5% of the samples displayed a titer greater than log 3.0 (1:1000) and, conversely, 3.5% displayed a titer below the cut-off value, i.e. log 1.25 (1:10). Thus, of 57 HCV-infected patients, only two did not test positive for neutralizing antibodies in this assay (the titers were 0.960 and 0.932, respectively).

The role of neutralizing antibodies during acute and chronic viral infection remains an important question and has generated controversial results. Initially, the presence of neutralizing antibodies was shown to control the HCV load and to contribute to viral eradication in patients capable of clearing the infection [13] . In other studies, the appearance of neutralizing antibodies was delayed and restricted to IgG1 antibodies in patients who develop a chronic infection [2, 21] . The chimpanzee model has been critical for the study of HCV transmission and host immune responses; however, neutralizing antibodies were not detected in some animals that resolved their infection -suggesting a minimal role in viral clearance, as also observed in human studies [14, 15] . Experimentally

infected chimpanzees and naturally infected humans can be re-infected with homologous and heterologous HCV strains, suggesting that the humoral immunity that develops after spontaneous resolution of acute hepatitis C is not sterilizing [22] [23] [24] . During chronic infection in humans, the presence and/or production of neutralizing antibodies do not suffice for curing the infection but could regulate the spread of the virus. Thus, it can be postulated that during chronic infection, viral mutants can continuously escape the renewed production of neutralizing antibodies.

Retroviral pseudoparticles have been used to develop a very interesting tool for measuring neutralizing antibodies in vitro [14] . The assay has demonstrated the presence of HCV-neutralizing antibodies in human sera with relatively high titers ($>1:320$) and broadly neutralizing activity against different HCV genotypes. However, this model does not represent genuine HCV virions; in particular, the budding of retroviral particles is thought to be very different and may involve a variety of cellular pathways. Characterization of infectious retroviral pseudotype particles bearing HCV glycoproteins have been shown to be very heterogeneous, and so it is possible that these pseudoparticles may not be as relevant as the native HCV virions [25] .

The recent development of a cell culture model for HCV enables the production of native HCV virions that can be efficiently propagated in cell culture [17] [18] [19] . This cell culture system has allowed us to develop a neutralization assay for evaluating the level and the proportion of HCV-neutralizing antibodies in chronically infected HCV patients. We analysed a number of parameters (such as practicability, reproducibility and specificity) and tested the effect of a range of variables (viral inoculum size, incubation time, fixation and permeabilization methods, blocking and revelation reagents) on these parameters (data not shown). Overall, the neutralization assay described in this study performs similarly to standardized neutralization assays for many other viruses [26] [27] [28] .

The assay relies on the ability of the specific JFH-1 genotype 2 viral strain to replicate and multiply on a Huh-7 human hepatoma cell line in a cell culture model, enabling the rapid detection of viral foci after 72 hours of infection. Moreover, no secondary foci were detectable at this time point. Fixation with paraformaldehyde and permeabilization with Triton X-100 were chosen in order to preserve antigenicity and prevent the cell monolayer from detaching during washes. Development with DAB peroxide substrate made it easy to count specifically coloured viral foci. The viral inoculum size is an important parameter; it

has to be low enough to enable good assay sensitivity but high enough to produce a statistically significant number of foci, i.e. allowing the reduction in the number of foci (and thus the effect of neutralization) to be monitored. Thus, 100 FFUs were used as the inoculum in this neutralization assay.

In order to test different human samples, we had to take into account the ability of HDL to facilitate HCVcc entry via a mechanism which depends on expression of the scavenger receptor BI [11, 15, 16, 20] . Given HDL's role in HCV entry, immunoglobulins were purified from each serum sample prior to determination of the neutralizing antibodies titer; this frees the assay of the risk of non-specific neutralization activity of the serum via the effects of HDL, the complement system and/or serum amyloid A protein (SAA) [29] .

The HCV neutralization assay exhibited good reproducibility, for both duplicate assays and independent tests. As expected, the intra-assay coefficient of variation (CV) was lower than the interassay CV. The test also showed good specificity, since there was no interaction with anti-HIV, anti-HBV or heterophile antibodies. Very low titers were found with HCV ELISA and RNA-negative samples, and the assay's cut-off was determined as the mean titer for negative samples plus two standard deviations (1.25 log, corresponding to a dilution of 1:18).

Given that only the JFH-1 strain of HCV genotype 2a was available for the assay, we evaluated the neutralization titer of sera from patients chronically infected with other HCV genotypes, i.e. 1, 2, 3, 4 and 5. Most of these sera were detected as positive by the neutralization assay, except for two sera from HCV genotype 1-infected patients. These two samples presented a high specific antibody ratio according to the ELISA but only very low inhibition by neutralization assay (far below the cut-off, in fact). We conclude that either the samples lacked neutralizing antibodies or that any such antibodies that were present did not cross-neutralize with HCV genotype 2a.

The sensitivity was 100% -not only for genotype 2 (the genotype of the strain used for the assay) but also for other HCV genotypes (except genotype 1). HCV genotype 5 antibodies were also measured but there were too few samples for accurate testing. Moreover, the positive sera (96.5%) had comparable and significantly high titers (1.99 ± 0.63), whatever the genotype. This finding suggests that most neutralizing antibodies are cross-reactive.

Another possibility is that most of the patients had been previously infected by a genotype 2 strain. However, this is unlikely because few genotype 2 strains are circulating in France [30]. As expected for a neutralization test, the assay presented in the present study appeared to be very specific (independently of the genotype) and usable in most circumstances. For most viral infections, neutralization assays such as that described in this study are used as reference assays. Thus, we are confident that as other HCVcc genotypes become available, these assays will replace the pseudoparticle assay in the near future because they are probably more relevant. Our assay is somewhat time-consuming and could be simplified by using one dilution to count the foci; however, this type of "short cut" would make it difficult to extrapolate to the dilution neutralizing 50% of the inoculum. Another approach would consist in using recombinant HCV capable of expressing reporter genes (such as luciferase) in order to use a single dilution and obtain a quantitative result [31]. However, further neutralization studies using other genotypes are needed in order to complete our observations and to char-

A simple, specific and reproducible cell culture-based neutralization assay was developed for the determination of neutralizing anti-HCV antibodies in human sera. This test should be an important tool for gauging the relationship between the neutralizing response and viral load kinetics in acutely and chronically infected patients.

The Huh-7 human hepatoma cells [32] were grown in Dulbecco's minimum essential medium (Invitrogen) supplemented with 10% fetal bovine serum. All cell cultures were maintained in 5% CO₂ at 37°C.

The plasmid pJFH-1 containing the full-length cDNA of the JFH-1 isolate (which belongs to subtype 2a (GenBank accession no. AB047639)), was a gift from Dr Wakita (Department of Microbiology, Tokyo Metropolitan Institute for Neuroscience, Tokyo, Japan) and has been described previously [17]. To generate genomic HCV RNA, the plasmid pJFH-1 was linearized at the 3' end of the HCV cDNA and used as a template for in vitro transcription, as described previously [33]. Viral stocks were obtained by harvesting cell culture supernatants and freezing them at -80°C. Virus titration was performed on Huh-7 cells with 6-well microtiter plates (Corning, NY) 72 hours after incubation, by immunostaining the cells with antibodies from a HCV-positive patient serum that had previously been inactivated at 56°C (see the section on the virus neutralization assay). The viral titer was

determined in triplicate from the mean number of foci and expressed as focus forming units/mL (FFU/mL).

Seventy-seven human serum samples were tested. Collection of the sera was approved by the local Ethics Committee and informed consent had been obtained from the donors. Fifty-seven of these samples were obtained from chronically infected HCV patients. The presence of HCV antibodies was determined and confirmed using two third-generation HCV EIA assays (AxSYM[®] HCV Version 3.0, Abbott, Wiesbaden, Germany and Vitros[®] Anti-HCV reagent pack, Ortho-Clinical Diagnostic, High Wycombe, United Kingdom). HCV RNA was determined with a qualitative commercial assay (Cobas Amplicor HCV test Version 2.0, Roche Diagnostics, Meylan, France) and HCV genotyping was performed by direct sequencing, as described elsewhere [34]. The genotypes were distributed as follows: 11, 11, 11, 12, 10 and 2 samples of types 1a, 1b, 2, 3, 4 and 5, respectively. A set of 20 anti-HCV-negative serum samples was used to evaluate the assay's specificity, including five serum samples with positive hepatitis B virus surface antibody (anti-HBs) status and five sera from Epstein-Barr virus-infected patients that had tested positive for heterophile antibodies. All serum samples had been stored at -80°C upon collection and had not been thawed until the time of assay.

Serum immunoglobulins G (IgG) fraction was purified using protein G-Sepharose (GE Healthcare, Orsay, France)

The HCV focus reduction neutralization assay was performed in 96-well microtiter plates. Serial dilutions of purified IgG (10 µg) ranging from 1:10 to 1:1,280 were established. Each dilution was tested twice. 25 µL of each sample was mixed with 25 µL of virus (100 FFU) in 96well microtiter plates and incubated for 1 hour at 37°C, 5% CO₂. A volume of 100 µL of Huh-7 cell suspension (10,000 cells/well) in culture medium was added and incubated for 5 hours at 37°C, 5% CO₂. After 5 hours, the supernatants were removed and 100 µL of culture medium were added to the monolayers. After 72 hours, the cells were fixed with paraformaldehyde and permeabilized with 0.5% Triton X-100. Primary antibody (a HCVpositive patient serum inactivated at 56°C) was diluted to 1:500 prior to use and then incubated for 1 h at room temperature. A peroxidase-coupled, Fc-specific anti-human IgG antibody (Sigma, Saint Quentin Fallavier, France) diluted to 1:200 was dispensed onto the cell monolayer and incubated for 30 min at room temperature. The reaction was developed with DAB peroxidase substrate (Sigma, Saint Quentin Fallavier, France) and stopped after

10 min of incubation with distilled water. The number of HCV foci in each dilution was determined. Controls were included in each assay (non-neutralized virus, purified IgG from each patient at a 1:10 dilution). The dilution that neutralized 50% of the virus was calculated by curvilinear regression analysis using XLSTAT 2006 software (Addinsoft SARL, Paris, France) [35]. Each titer was determined as the log value of the reciprocal antibody dilution that reduced the number of viral foci by 50%.

Titers were expressed as logarithmic values and means $\pm$ standard deviation were calculated. Student's t-test was used to compare data between groups. p values below 0.05 were considered to be significant.

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Relationship between hepcidin and oxidant/antioxidant status in calves with suspected neonatal septicemia

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Abstract: AIM: This study has been conducted for the purpose of determining serum hepcidin, total antioxidant status (TAS), total oxidant status (TOS), and Fe levels in calves

with suspected neonatal septicemia before and after treatment and the clinical significance of hepcidin in calves with suspected neonatal septicemia. **MATERIALS AND METHODS:** The study material consisted of 15 calves of different ages and sexes brought to the Training, Research and Application Center at the Kafkas University Faculty of Veterinary Medicine with suspected neonatal septicemia. 8.5 mL of blood was drawn from the jugular vein of each animal into coagulant tubes before and after treatment for one-off biochemical analyses and centrifuged. After this, the serum was separated. Hepcidin, TAS, TOS, and Fe levels in the serum were measured. **RESULTS:** While pre-treatment hepcidin levels were 58.42 ± 3.46 ng/mL, post-treatment levels were 46.87 ± 2.98 ng/mL ($p < 0.05$). Pre-treatment Fe levels were 60.13 ± 7.27 µg/dl, while post-treatment levels were 83.1 ± 8.09 µg/dl ($p < 0.05$). The changes in the TAS and TOS levels were also found to be statistically significant. **CONCLUSION:** In light of the fact that hepcidin plays a role function in the regulation of Fe as well as the fact that Fe is a significant nutritional source for many microorganisms, it was concluded that hepcidin may play a significant role in nutritional immunity and the pathogenesis of diseases.

Text: Neonatal calf septicemia causes high morbidity and mortality and is one of the leading and most significant difficulties in raising cattle. Calf septicemia is the main cause of death in the neonatal period [1] . Its etiology involves bacteria (commonly *Escherichia coli*), viruses (rota and coronavirus), parasites, and other factors. As the disease progresses quickly and is lethal, diagnosis and treatment should be initiated as quickly as possible [2] .

Hepcidin is a low molecular weight, antimicrobial peptide hormone and was first discovered in human urine [3] . It is produced by the liver as a firstline response to inflammatory reactions and high Fe concentrations [4, 5] . Hepcidin plays a fundamental role in the regulation of Fe metabolism [6] , which is a part of foundational cellular functions and thus of vital importance. On the other hand, by participating in redox reactions leading to the production of reactive oxygen species (ROSs), Fe also causes oxidative stress. Therefore, Fe has been regarded as a potentially toxic element to cells [7] . Fe also plays an important role in pathogenesis of bacterial infections as bacteria utilize Fe for survival, growth and proliferation; therefore, it is of paramount importance to control the Fe metabolism [6] . It is well known that the abundance of Fe suppresses defense system leading host vulnerable to infections. There is a significant relationship between Hepcidin, Fe metabolism, inflammation, and the immune system. The fact that hepcidin plays an active role in the regulation of Fe release from macrophages and in the control of excessive

Fe absorption from the duodenum is well documented [6] . Hepcidin is a part of the natural defense mechanism, thus it limits the amount of Fe that can be utilized by pathogens [8] . In inflammatory conditions, hypoferremia is an important first-line protective mechanism in response to infections [9] . Fe also participates in redox reactions, causing the production of ROS, and thus leading to oxidative stress [7] . Free radicals play a significant role in the pathogenesis of many diseases [10] . Newborns are subject to oxidative stress during birth. It is also reported that in livestock diseases, especially enteritis and pneumonia, antioxidant capacity is efficacious [11] .

This study was designed to determine the clinical significance of hepcidin in calves with suspected neonatal septicemia by evaluating serum hepcidin, total antioxidant status (TAS), total oxidant status (TOS), and Fe levels in calves suspected of neonatal septicemia before and after treatment.

This study was conducted after obtaining approval from the Mehmet Akif Ersoy University Animal Experiments Local Ethics Committee (MAKU-HADYEK-Submission: 2014/77).

The study consisted of 15 calves with suspected neonatal septicemia aged between 1 and 10 days old admitted to the Teaching Hospital of Veterinary Medicine. Suspected septicemia was diagnosed based on clinical (diarrhea, weakness in or absence of sucking reflex, the calf being in a supine position on the ground or being unable to stand, severe dehydration, abnormal rectal temperature [hypo-or hyperthermia], mucosal hyperemia, and full sclera) and hematological (increase in white blood cell [WBC] count) examinations; the animals were suspected to have septicemia [12, 13] . The animals were given standard treatment (antibiotic, nonsteroidal anti-inflammatory drugs, vitamin C, fluid therapy, and intestinal astringent). For determination of serum hepcidin, TAS, TOS, Fe levels, and hematological parameters; blood samples were taken before and after treatment in all cases. 8.5 mL of blood was taken from the jugular vein of each animal into coagulant tubes for biochemical analysis, and 3 mL blood was taken into EDTA tubes for hematological analysis. Samples were centrifuged at 3000 rpm for 10 min, and the serum was harvested and kept at -20°C until the analysis. Serum hepcidin (Mybiosource[®]), TAS (Rel Assay Diagnostics[®]), and TOS (Rel Assay Diagnostics[®]) were determined using commercial ELISA kits, and Fe value was measured spectrophotometrically. Hematological (WBC, lymphocyte [LYM], red blood cells [RBC], mean corpuscular volume

(MCV), and hematocrit [HCT]) analysis was performed on blood counter (VG-MS4e[®], Melet Schloesing, France).

The results were evaluated using the t-test in the SPSS[®] (SPSS 20, USA) statistical package program to determine the differences between values before and after treatment.

Calves with suspected septicemia exhibited clinical signs of loss of appetite, fatigue, indifference to surroundings, reduced/absence of sucking reflex, cool extremities, inability to stand, diarrhea, eye sinking into their sockets, and hyperemia in the conjunctiva. The average body temperature, heart rate, and respiratory rates of the animals were $37.18 \pm 0.13^{\circ}\text{C}$, $104 \pm 4.33/\text{min}$, and $28.86 \pm 0.75/\text{min}$ pre-treatment; and $38.54 \pm 0.1^{\circ}\text{C}$, $107.53 \pm 2.20/\text{min}$ and $26.40 \pm 0.36/\text{min}$ post-treatment, respectively.

The changes in hepcidin, TAS, TOS and Fe levels in the calves with suspected septicemia before and after treatment are given in Table- 1. After treatment, serum hepcidin and TOS levels were significantly lower than before treatment in calves. On contrary, serum TAS and Fe levels were significantly higher than before treatment (Table-1).

The treatment of calves resulted in significant changes in the hematological parameters that were examined except for RBC. The WBC count, LYM count, MCV and HCT significantly changed after treatment when compared to values obtained before treatment (Table-2).

This study aimed to determine the clinical importance or use of hepcidin by comparing the values of serum hepcidin, TAS, TOS and Fe levels in calves with suspected neonatal septicemia before and after treatment.

Clinicians rely on clinical and laboratory examinations of patients to form a working diagnosis, so hematological and serum biochemical parameters are usually used for this purpose [14] . The hematological parameters (WBC, HCT, LYM, and MCV) evaluated in this study were comparable with those reported by others in neonatal calves with diarrhea and suspected septicemia [15] [16] [17] . Treatment significantly corrected to normal values the hematological parameters that were examined with the exception of RBC. Pretreatment

leukocyte count was high because of the inflammation that occurred in the organism, and that the HCT levels were high due to the dehydration that occurred due to diarrhea. Hepcidin is controlled by the presence of inflammation in the body, Fe storage, and erythropoietic activity in the bone marrow and plays a primary role in the homeostasis of Fe [4]. The increase in tissue and plasma Fe levels stimulates the synthesis of hepcidin and reduces Fe release and enteric Fe absorption from macrophages and hepatocytes [18]. Increased hepcidin concentrations during inflammation and infection reduce serum Fe levels by decreasing Fe release from macrophages and hepatocytes, and thus Fe required for microorganisms and tumor cells is restricted [19].

Serum hepcidin levels in calves with suspected septicemia were significantly high before treatment when compared to after treatment; also Fe levels were lower before treatment when compared to after treatment in this study. This situation could be related to the interaction between hepcidin and Fe and also gives credence to the role of hepcidin in the hemostasis of Fe during inflammation and infection. As in our study, Fe levels are well known to decrease in diarrheic calves when compared to healthy calves [20, 21]. Although no study exists reporting hepcidin concentration in diseased calves, studies in human subjects show that cord blood hepcidin levels might be an important indicator in diagnosing early-onset of neonatal sepsis. The cord blood hepcidin levels of neonatal infants with sepsis varied between 118.1 and 8400 ng/mL and were significantly higher than the healthy infants [22]. A similar result was reported that hepcidin concentrations in neonatal infants with sepsis were significantly higher than in healthy infants [23]. These findings along with our results add credence to the idea that hepcidin-Fe interaction may play a role in the pathogenesis of septicemia.

The production of free oxygen species causes alterations in protein, lipid, and DNA during oxidative stress and leads to the development of lesions in the organs [24]. Free iron has toxic characteristics as it catalyses the production of ROSs [25] and thus causes oxidative stress [26]. The role of Fe in the development of oxidative stress may once more show the importance of hepcidin, as an important Fe regulator, with regard to enhancing antioxidant capacity through inhibiting utilization of Fe by the organism as well as the host cells.

The antioxidant and oxidative system are in a constant state of balance in the organism. Any event breaking up this balance in favor of the oxidative stress molecules will cause cell damage [27, 28]. The host cells initiate the antioxidant system in case of exposure to

oxidative stress [27] . Kabu et al. [16] reported TOS and TAS values in neonatal calves with diarrhea as $13.47 \pm 0.81 \mu\text{mol H}_2\text{O}_2/\text{L}$ and $0.51 \pm 0.02 \text{ mmol Trolox-equivalent/L}$, respectively, and treatment of these calves caused changes in these values of $11.21 \pm 0.26 \mu\text{mol H}_2\text{O}_2/\text{L}$ and $0.55 \pm 0.02 \text{ mmol Trolox-equivalent/L}$, respectively. Studies also reported that parameters used for oxidative stress (malondialdehyde) were higher [29] and antioxidant parameters (superoxide dismutase [21] , TAS) were lower in diarrheic calves [29] . Similarly, in our study, TAS level was significantly lower and TOS level was significantly higher in diarrheic calves before treatment, and treatment caused corrections in these parameters. Decrease in TAS and increase in TOS levels demonstrated that oxidative stress was evident in the diseased calves in our study. Increased TOS and hepcidin levels before treatment are thought that associated with inflammation. After treatment increased TAS and decreased hepcidin levels support this opinion.

Hepcidin may play an important part in non-specific immunity and is a key molecule that plays a role in the pathogenesis of diseases by enhancing the development of antioxidant system. However, more detailed studies are needed on the role of hepcidin in the pathogenesis of septicemia.

This work was carried out in collaboration between all authors. EEE, HME and AHK: Designed the experimental procedures. EEE, EG and MK: Conducted the research work. EEE, AHK, MO and AK: Helped in laboratory analysis. All authors read and approved the final manuscript.

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Acute Hemorrhagic Encephalitis Responding to Combined Decompressive Craniectomy, Intravenous Immunoglobulin, and Corticosteroid Therapies: Association with Novel RANBP2 Variant

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Abstract: **BACKGROUND:** Acute hemorrhagic encephalomyelitis (AHEM) is considered as a rare form of acute disseminated encephalomyelitis characterized by fulminant encephalopathy with hemorrhagic necrosis and most often fatal outcome. **OBJECTIVE:** To report the association with Ran Binding Protein (RANBP2) gene variant and the response to decompressive craniectomy and high-dose intravenous methylprednisolone (IVMP) in life-threatening AHEM. **DESIGN:** Single case study. **CASE REPORT:** A 6-year-old girl known to have sickle cell disease (SCD) presented an acquired demyelinating syndrome (ADS) with diplopia due to sudden unilateral fourth nerve palsy. She received five pulses of IVMP (30 mg/kg/day). Two weeks after steroid weaning, she developed right hemiplegia and coma. Brain magnetic resonance imaging showed a left frontal necrotic-hemorrhagic lesion and new multifocal areas of demyelination. She underwent decompressive craniotomy and evacuation of an ongoing left frontoparietal hemorrhage. Comprehensive investigations ruled out vascular and infectious process. The neurological deterioration stopped concomitantly with combined neurosurgical drainage of the hematoma, decompressive craniotomy, IVMP, and intravenous immunoglobulins (IVIG). She developed during the following months Crohn disease and sclerosing cholangitis. After 2-year follow-up, there was no new neurological manifestation. The patient still suffered right hemiplegia and aphasia, but was able to walk. Cognitive/behavioral abilities significantly recovered. A heterozygous novel rare missense variant (c.4993A>G, p.Lys1665Glu) was identified in RANBP2, a gene associated with acute necrotizing encephalopathy. RANBP2 is a protein playing an important role in the energy homeostasis of neuronal cells. **CONCLUSION:** In any ADS occurring in the context of SCD and/or autoimmune condition, we recommend to slowly wean steroids and to closely monitor the patient after weaning to quickly treat any recurrence of neurological symptom with IVMP. This case report, in addition to others, stresses the likely efficacy of combined craniotomy, IVIG, and IVMP treatments in AHEM. RANBP2 mutations may sensitize the brain to inflammation and predispose to AHEM.

Text: Acute hemorrhagic encephalomyelitis (AHEM) or acute hemorrhagic leukoencephalitis is considered a rare and extremely severe form of acute disseminated encephalomyelitis (ADEM). AHEM is characterized by an acute and rapidly progressive encephalopathy including hemorrhagic necrosis of the parenchyma of the central nervous system. It is usually fatal (1) (2) (3) . Many treatment options have been used including intravenous (IV) steroids, intravenous immunoglobulins (IVIG), and plasmapheresis (4) . There have been few reports of survival following early intervention with high-dose corticosteroid therapy and/or decompressive craniotomy (5) (6) (7) (8) (9) .

RANBP2, a nuclear pore protein, has numerous roles in the cell cycle. RANBP2 is associated with microtubules and mitochondria suggesting roles in intracellular protein trafficking or energy maintenance and homeostasis of neuronal cells. RANBP2 mutations have been reported in acute necrotizing encephalopathy (ANE) which could present with coma, convulsions, and encephalopathy. The hallmark of ANE is multiple, symmetric brain lesions located in the thalami bilaterally, putamina, deep periventricular white matter, cerebellum, and brainstem. It could be triggered by a viral infection in previously healthy children (10) .

We report a new case of AHEM associated to a Ran Binding Protein (RANBP)-2 variant and responsive to combined craniectomy, intravenous methylprednisolone (IVMP), and IVIG as inaugural manifestation of multisystemic autoimmunity in a girl with sickle cell disease (SCD).

A 6-year-old girl known for SCD treated on folic acid and hydroxyurea was admitted for new-onset diplopia [day 0 (D0): refers to the start of the diplopia] 6 weeks after respiratory tract infection due to rhinovirus. She was diagnosed with a fourth nerve palsy secondary to an acquired demyelinating syndrome. The initial brain magnetic resonance imaging (MRI) performed at D5 after onset of neurological symptom showed left midbrain and pontine edema with expansion of the brainstem, right caudate nucleus, and scattered supratentorial white matter foci of high T2/FLAIR signal (Figure 1). Brain MR angiography (MRA) showed a normal appearing circle of Willis. The cerebrospinal fluid (CSF) obtained by lumbar puncture was normal (WBC 1 cells/ μ L, RBC 0 cells/ μ L, glucose 2.9 mmol/L, protein 0.18 g/L, and absent oligoclonal bands). The infectious workup including blood bacterial culture, CSF bacterial and viral cultures, nasopharyngeal aspirate (tested for

Influenza A, Influenza B, Parainfluenza 1-2-3, Respiratory Syncytial Virus, Adenovirus, Coronavirus 229E, Coronavirus OC43, Metapneumovirus, Enterovirus, and Rhinovirus), and serologies for Epstein-Barr virus, Mycoplasma pneumoniae, HTLV I, HTLV II, HIV1, and Lyme disease were negative. Bartonella Henselae IgG was positive (1:1,280) reflecting a previously acquired common and self-limited infection in our area. Antinuclear antibodies (ANA) were positive (1:160). B12 and folate levels were normal. Smooth muscle antibodies were negative. Anti-mitochondrial antibodies were positive. Sedimentation rate was 65 mm/h. She was treated with five doses of IVMP (30 mg/kg/day) followed by 9 days of oral prednisone (1 mg/kg/day). At discharge, her neurological exam was significant only for vertical diplopia.

She presented 1 month later with 5 days of upper respiratory tract infection symptoms, fever, headache, and a rapidly progressive right-hand weakness (D30) with normal alertness. She had normal blood pressure (120/81 mmHg). She was started on cefotaxime, vancomycin, and acyclovir. White cell count was 13.4×10^9 /L, hemoglobin was 7.8 g/L, and platelets were 239×10^9 /L. While in the MRI machine (D30) she deteriorated with vomiting and reduced level of consciousness (Glasgow Coma Scale dropped from 15 to 8 over 30 min). Brain MRI showed a rapid progression over a few sequences of an active bleed involving both superficial and deep gray matter as well as subcortical white matter of the left hemisphere anterior quadrant. Brain MRA was normal (Figures 2A-F). The patient was immediately brought out of the magnet and her physical exam demonstrated unequal dilated pupils. She received IV mannitol and hypertonic saline for the management of acute intracranial hypertension/ herniation and was taken for surgery. She underwent left frontotemporoparietal decompressive craniotomy, evacuation of left frontoparietal intracerebral hemorrhage, and insertion of an external ventricular drain (EVD). Upon opening the skull, there was significant dural tension, and on opening the dura mater, there was a large amount of bleeding, in addition to brain swelling and necrosis. Estimated blood loss was 3.5 L. She received 8 units of packed red blood cells, 3 units of cryoprecipitate, 6 units of fresh frozen plasma, and 3 units of platelets. Coagulation profile showed international normalization ratio = 3.38, prothrombin time = 51.2 s, and partial thromboplastin time = 122 s. An intraventricular pressure monitor was inserted. She returned with stable vitals to PICU. At D31, the CT scan showed extensive multi-compartmental bleed involving the left frontoparietal lobes, the interhemispheric fissure, and the left hemispheric arachnoid spaces. New white matter lesions were detected in the left posterior parietal and occipital lobes and in the left caudate head. MRI at D33 showed interval worsening with disseminated gray and white matter non-hemorrhagic lesions in the right cerebral and both cerebellar hemispheres, bilateral deep gray nuclei, as well as new

necrotic non-hemorrhagic lesions in the left hemisphere (Figures 2G-I) . She was started on IVMP (30 mg/kg/ day for 5 days) and IVIG (1 g/kg/day for 2 days). Repeat MRI at D9 showed no new parenchymal hemorrhage and partial resolution of the non-hemorrhagic lesions (Figure 3) . Prednisolone was tapered course over 6 weeks. At discharge (D71), she was able to say a few words and had better power of her right side. Brain MRI performed 3 months later showed complete resolution of the non-hemorrhagic non-necrotic lesions, mainly seen in the right cerebral hemisphere and the cerebellum.

Brain biopsy of the hematoma, some small vessels, cortex, and white matter showed necrotic area, reactive and non-specific findings which could be entirely explained by compressive changes adjacent to a hematoma. There was diffuse microglial activation and signs of early microinfarcts. Blood, CSF and urine culture, and PCR (HSV1/2) were negative for bacteria and for viruses. CSF obtained through craniotomy and EVD performed at D32 showed elevated proteins 2.56 g/L, glucose 3.6 mmol/L, white blood cells 9 cells/ μ L, and red blood cells 1,341 cells/ μ L. ANA and anti-DNA antibody were negative.

Anti-extractable nuclear antigens (SSA-RO, SSB-LA, smith, RNP) were negative. Serum autoimmune antibodies panel (NMO, NMDAR, AMPA I/II, GAB, MAG, VGCC, MOG, YO, HU, RI) were negative but GAD antibody was slightly positive, possibly due to the IVIG infusion. EBV showed no signs of recent infection.

After discharge, the patient was started on regular transfusion exchange. Six months later, the patient was diagnosed to have Crohn's disease and primary sclerosing cholangitis. Two years later, the patient still suffers right hemiparesis but is able to walk without support. She presents an expressive aphasia. Her intellectual abilities are average, or below the mean but in the normal range, except for the speed of information processing, verbal working memory, and some elaborated executive functions.

A gene panel (Table 1) targeting inflammatory disorders and post-infectious necrotic encephalopathies found a heterozygous RANBP2 missense mutation (NM_006267.4, c.4993A>G, p.Lys1665Glu). This mutation has not been previously reported in the HGMD database. This variant has been observed at a frequency of <0.01% across the entire Broad ExAC dataset of individuals without severe childhood onset disease (6/117,118 alleles). Analysis of amino acid conservation indicates that the wild-type amino acid Lys1665 is

conserved in 59 of 60 mammals examined, including 12 of 12 primates, and in 25 of 34 nonmammalian vertebrates increasing the likelihood that a change at this position might not be tolerated. In silico tools predict that this variant is damaging (SIFT and Align GVGD).

Several differential diagnoses of acute encephalopathy in a patient with sickle cell anemia can be considered. An infectious encephalitis, including herpes encephalitis, was ruled out by blood and CSF bacterial and viral cultures and negative HSV I/ II PCR.

Nasopharyngeal aspirate was negative for viruses. Some infections have been previously associated with necrotizing encephalitis such as Influenza A (11) . SCD patients are prone to ischemic or hemorrhagic strokes (12) . Primary hemorrhagic stroke is uncommon in pediatric SCD. Most cases were from adults and have been described in the context of previous ischemic stroke, aneurysms, low hemoglobin, acute chest syndrome, and hypertransfusions. Moreover, although hemorrhagic stroke has been described in SCD patients receiving transfusion or corticosteroids, it was in the context of elevated blood pressure which was not present in our case (13) . This was ruled out as the MRI findings were not consistent with a specific vascular territory and normal arterial and venous flows were shown on vascular imaging. Another differential is posterior reversible encephalopathy syndrome which has been reported in SCD patients (13) (14) (15) (16) . However, it is unlikely in our case due to the severity of the brain injury and the absence of classic precipitating factors of posterior reversible encephalopathy syndrome such as high blood pressure. Macrophage activation syndrome could also lead to acute necrotic brain injury. However, it is associated to high ferritin and low triglycerides at the time of the encephalopathy, other multisystemic injuries, typical neuropathological findings, and recurrence over time, which were not noted in our patient (17) . Parvovirus B19 has been described to cause encephalopathy in sickle cell patients. It is associated with aplastic anemia. It caused punctate areas of hemorrhages in the basal ganglia, periventricular white matter, and mainly along the posterior parietal cortex. This was attributed to parvovirus B19-induced vasculitis (18) . In our patient, there was no sign of aplasia or any neuroradiological finding of parvovirus B19 infection. Finally, acute encephalitis has been observed in SCD patients in the context of arterial hypoxemia from fat embolism, pulmonary embolism, sudden anemia, or acute chest syndrome due to pneumonia (19) . This was ruled out as the patient did not have clinical or radiological signs of acute chest syndrome or embolism and there was no arterial hypoxemia.

Acute hemorrhagic encephalomyelitis has been described in pediatric patients following ADEM or ADEM-like episodes (20, 21) . AHEM is the most plausible diagnosis in our

patients based on the clinical and radiological presentation, the preceding ADEM-like episode, and the exclusion of other etiologies of acute encephalopathy. Other patients with AHEM have been described in the SCD context (7, 19) . Many treatment options have been used to treat AHEM; of these, IV steroids have been associated with survival following aggressive, high-dose corticosteroid therapy (5) (6) (7) (8) (9) (22) (23) (24) (25) .

Autosomal dominant mutations (with incomplete penetrance) in RANBP2 have been associated with susceptibility to infection-induced necrotizing encephalopathy (26, 27) . Previously healthy patients with pathogenic mutations in RANBP2 can present acutely with encephalopathy and convulsions in the context of an infection, with brain imaging revealing involvement of the brainstem, thalamus, putamina, cerebellum and external capsules, and claustrum (10) . Our patient has a similar presentation and imaging features as infection-induced necrotizing encephalopathy, including bilateral thalamic involvement. The rare heterozygous previously unreported variant we identified in RANBP2 affects a very conserved amino acid and is predicted deleterious using in silico tools (a prediction tool performing a fast bioinformatics analysis which can predict the pathogenicity of a variant based on the change to an amino acid). It is possible that this variant is pathogenic and responsible for the clinical phenotype. There is an overlap between the diagnostic criteria of AHEM and those of acute hemorrhagic encephalopathy (25, 26) making possible that both entities might be part of the same pathophysiological continuum. RANBP2 is a protein playing an important role in the energy homeostasis of neuronal cells (28) . Hence, RANBP2 dysfunction might make neuronal cells much vulnerable to energy failure and necrosis when exposed to inflammatory or other stresses, such as those implicated in AHEM.

This study was carried out in accordance with the recommendations of our institutional ethic committee. Written informed consent was obtained from all the participants for the publication.

All authors participated in gathering the data, designing the article, and discussing and editing the manuscript.

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New Isoxazolidine-Conjugates of Quinazolinones—Synthesis, Antiviral and Cytostatic Activity

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6273226/>

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Abstract: A novel series of (3-diethoxyphosphoryl)isoxazolidines substituted at C5 with various quinazolinones have been synthesized by the 1,3-dipolar cycloaddition of N-methyl-C-(diethoxyphosphoryl)nitron with N3-substitued 2-vinyl-3H-quinazolin-4-ones. All isoxazolidines were assessed for antiviral activity against a broad range of DNA and RNA viruses. Isoxazolidines trans-11f/cis-11f (90:10), trans-11h and trans-11i/cis-11i (97:3) showed weak activity (EC(50) = 6.84, 15.29 and 9.44 μ M) toward VZV (TK(+)) strain) which was only one order of magnitude lower than that of acyclovir used as a reference drug. Phosphonates trans-11b/cis-11b (90:10), trans-11c, trans-11e/cis-11e (90:10) and trans-11g appeared slightly active toward cytomegalovirus (EC(50) = 27–45 μ M). Compounds containing benzyl substituents at N3 in the quinazolinone skeleton exhibited slight

antiproliferative activity towards the tested immortalized cells with IC(50) in the 21–102 μ M range.

Text: Nitrogen-containing heterocycles form the core of natural products (e.g., alkaloids) and they are also present in many pharmacophores as well as in numerous marketed drugs. Among them, quinazolines and quinazolinones have drawn special attention due to the broad spectrum of biological activities of their derivatives, including sedative [1] [2] [3], anticancer [4] [5] [6] [7], antiviral [8] [9] [10] [11] [12], antibacterial [13] [14] [15], antifungal [15, 16], anti-inflammatory [15, [17] [18] [19] and antifibrotic [20, 21] activities. Several reviews focused on the synthetic strategies and biological activities of these compounds have been published [22] [23] [24] [25] [26] [27] [28] [29]. The significant impact of various functional groups installed into quinazoline/quinazolinone frameworks on pharmacological properties have been proven.

In the last decades several compounds containing the quinazolin-4-one framework, which exhibited promising anticancer as well as antiviral properties, have been obtained (Figure 1). Furthermore, some biologically active substituted quinazolin-4(3H)-ones were isolated from various fungi and bacteria species. For example, 2-(4-hydroxybenzyl)quinazolin-4(3H)-one (1) was found in an entomopathogenic fungus *Isaria farinosa* and its strong inhibitory properties on the replication of tobacco mosaic virus (TMV) [30] were recognised, whereas its 2-(4-hydroxybenzoyl) analogue 2 present in fungus from *Penicillium* genus appeared only slightly active toward TMV [30]. Moreover, compound 1 exhibited significant cytotoxicity toward various cancer cell lines [31, 32]. Quinazolinone 3 isolated from *Streptomyces* sp. appeared cytotoxic against Vero cells [33]. Very recently synthetic pyridine-containing analogue 4 and its 3-substituted derivatives 5 and 6 have been obtained and their slight activity against influenza A virus was revealed [34]. On the other hand, various 2,3-disubstituted quinazolin-4(3H)-ones, including compounds 7-10, have been found to possess antitumor activity [35]. slight activity against influenza A virus was revealed [34]. On the other hand, various 2,3-disubstituted quinazolin-4(3H)-ones, including compounds 7-10, have been found to possess antitumor activity [35]. In continuation of our studies on antiviral and cytostatic activity of isoxazolidine analogues of C-nucleoside analogues, we designed a new series of compounds of the general formula 11 containing a substituted quinazolinone moiety as a false nucleobase at C5 in the isoxazolidine ring and the diethoxyphosphoryl function attached at C3. Our synthetic strategy to compounds trans-11/cis-11 relies on the 1,3-dipolar cycloaddition of N-methyl-

C-(diethoxyphosphoryl)nitron 12 [36] with 2-vinyl-3H-quinazolin-4-ones 13 substituted at N3 (Scheme 1). Scheme 1. Retrosynthesis of (isoxazolidinyl) phosphonates trans-11/cis-11.

2-Vinyl-3H-quinazolin-4-ones 13 modified at N3 with substituted benzyl groups were synthesized from commercially available 2-aminobenzamide (14) by acylation with 3-chloropropionyl chloride followed by cyclization and dehydrohalogenation to prepare 2-vinyl-3H-quinazolin-4-one (13a) as a key intermediate [37] and a subsequent reaction with substituted benzyl bromides 13b-i [38] (Scheme 2). Moreover, compounds 13j (R = Me) and 13k (R = Et) were also obtained with intention to determine the influence of the benzyl substituent on biological activity of the designed isoxazolidines trans-11/cis-11. In the ¹H-NMR spectra of compounds 13a-k characteristic signals for vinyl protons were observed in the 6.94-5.59 ppm (three doublets of doublets). In continuation of our studies on antiviral and cytostatic activity of isoxazolidine analogues of C-nucleoside analogues, we designed a new series of compounds of the general formula 11 containing a substituted quinazolinone moiety as a false nucleobase at C5 in the isoxazolidine ring and the diethoxyphosphoryl function attached at C3. Our synthetic strategy to compounds trans-11/cis-11 relies on the 1,3-dipolar cycloaddition of N-methyl-C-(diethoxyphosphoryl)nitron 12 [36] with 2-vinyl-3H-quinazolin-4-ones 13 substituted at N3 (Scheme 1). slight activity against influenza A virus was revealed [34]. On the other hand, various 2,3-disubstituted quinazolin-4(3H)-ones, including compounds 7-10, have been found to possess antitumor activity [35]. In continuation of our studies on antiviral and cytostatic activity of isoxazolidine analogues of C-nucleoside analogues, we designed a new series of compounds of the general formula 11 containing a substituted quinazolinone moiety as a false nucleobase at C5 in the isoxazolidine ring and the diethoxyphosphoryl function attached at C3. Our synthetic strategy to compounds trans-11/cis-11 relies on the 1,3-dipolar cycloaddition of N-methyl-C-(diethoxyphosphoryl)nitron 12 [36] with 2-vinyl-3H-quinazolin-4-ones 13 substituted at N3 (Scheme 1). Scheme 1. Retrosynthesis of (isoxazolidinyl) phosphonates trans-11/cis-11.

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mixtures of isoxazolidine cycloadducts were then subjected to purification on silica gel columns. However, attempts to isolate pure diastereoisomers were fruitful for trans-11a (R = H), trans-11c (R = 2-NO₂-C₆H₄-CH₂), trans-11g (R = 3-F-C₆H₄-CH₂), trans-11h (R = 4-F-C₆H₄-CH₂) and trans-11j (R = Me) only. The relative configurations of isoxazolidines trans-11a and cis-11a were established based on our previous studies on stereochemistry of cycloaddition of N-methyl-C-(diethoxyphosphoryl)nitron (12) with various vinyl aryls [39, 40] since similar ¹H-NMR spectral patterns for the respective series of trans- and cis-isoxazolidines were observed. Since for compound trans-11a all necessary coupling constants were successfully extracted from the ¹H- and ¹³C-NMR spectra, detailed conformational analysis was performed based on these data {J(H₃-H_{4α}) = 9.3 Hz [41], J(H₃-H_{4β}) = 8.3 Hz, J(H_{4α}-P) = 9.9 Hz Scheme 3. Synthesis of Isoxazolidines cis-11a-k and trans-11a-k. Reaction and conditions: a. toluene, 70 °C, 24 h. The relative configurations of isoxazolidines trans-11a and cis-11a were established based on our previous studies on stereochemistry of cycloaddition of N-methyl-C-(diethoxyphosphoryl)nitron (12) with various vinyl aryls [39, 40] since similar ¹H-NMR spectral patterns for the respective series of trans and cis-isoxazolidines were observed. Since for compound trans-11a all necessary coupling constants were successfully extracted from the ¹H- and ¹³C-NMR spectra, detailed conformational analysis was performed based on these data {J(H₃-H_{4α}) = 9.3 Hz [41], J(H₃-H_{4β}) = 8.3 Hz, J(H_{4α}-P) = 9.9 Hz [42, 43], J(H_{4β}-P) = 16.9 Hz, J(H_{4α}-H₅) = 6.2 Hz, J(H_{4β}-H₅) = 8.3 Hz, J(CCCP) = 8.5 Hz [44, 45]} and revealed that isoxazolidine ring in trans-11a adopts a 3E conformation in which the diethoxyphosphoryl group resides in the equatorial position of the isoxazolidine ring while a quinazolinone substituent is located pseudoequatorially (Figure 2). On the other hand, cis configuration of the minor isomer was established from the corresponding couplings [J(H₃-H_{4α}) = 9.0 Hz, J(H₃-H_{4β}) = 6.5 Hz, J(H_{4α}-P) = 11.2 Hz, J(H_{4β}-P) = 20.0 Hz, J(H_{4α}-H₅) = 9.1 Hz, J(H_{4β}-H₅) = 3.9 Hz, J(CCCP) = 7.3 Hz] indicating the 2E conformation of the isoxazolidine ring (Figure 2). The additional arguments to support our assignments follow from shielding of the CH₃CH₂OP protons observed for the cis isomer (Δδ ca. 0.1 ppm) when compared with the trans-11a. Furthermore, it was found that on a ¹H-NMR spectrum taken on the 83:17 mixture of cis and trans-11a, the H-N proton in the quinazolinone ring of cis-11a was significantly deshielded (Δδ = 0.7 ppm) when compared with the trans isomer, highly likely, as a result of the hydrogen bond formation with the phosphoryl oxygen amide, a phenomenon spatially achievable in the cis isomer only.

Since introduction of various substituents at N3 of quinazolinone moiety has no influence on the stereochemical outcome of the cycloaddition therefore configuration of the all major isoxazolidines 11 were assigned as trans, thereby minor ones as cis. Figure 2). The

additional arguments to support our assignments follow from shielding of the CH₃CH₂OP protons observed for the cis isomer ($\Delta\delta$ ca. 0.1 ppm) when compared with the trans-11a. Furthermore, it was found that on a ¹H-NMR spectrum taken on the 83:17 mixture of cis- and trans-11a, the H-N proton in the quinazolinone ring of cis-11a was significantly deshielded ($\Delta\delta$ = 0.7 ppm) when compared with the trans isomer, highly likely, as a result of the hydrogen bond formation with the phosphoryl oxygen amide, a phenomenon spatially achievable in the cis isomer only.

Since introduction of various substituents at N3 of quinazolinone moiety has no influence on the stereochemical outcome of the cycloaddition therefore configuration of the all major isoxazolidines 11 were assigned as trans, thereby minor ones as cis. Ganciclovir, cidofovir, acyclovir, brivudin, zalcitabine, zanamivir, alovudine, amantadine, rimantadine, ribavirin, dextran sulfate (molecular weight 10,000, DS-10000), mycophenolic acid, Hippeastrum hybrid agglutinin (HHA) and Urtica dioica agglutinin (UDA) were used as the reference compounds. The antiviral activity was expressed as the EC₅₀: the compound concentration required to reduce virus plaque formation (VZV) by 50% or to reduce virus-induced cytopathogenicity by 50% (other viruses).

Several isoxazolidines trans-11/cis-11 were able to weakly inhibit the replication of TK + and TK – VZV strains with EC₅₀ values in the range of 6.84-100 μ M (Table 2). Among them, phosphonates Ganciclovir, cidofovir, acyclovir, brivudin, zalcitabine, zanamivir, alovudine, amantadine, rimantadine, ribavirin, dextran sulfate (molecular weight 10,000, DS-10000), mycophenolic acid, Hippeastrum hybrid agglutinin (HHA) and Urtica dioica agglutinin (UDA) were used as the reference compounds.

The antiviral activity was expressed as the EC 50 : the compound concentration required to reduce virus plaque formation (VZV) by 50% or to reduce virus-induced cytopathogenicity by 50% (other viruses).

Several isoxazolidines trans-11/cis-11 were able to weakly inhibit the replication of TK + and TKV ZV strains with EC 50 values in the range of 6.84-100 μ M (Table 2) . Among them, phosphonates trans-11f/cis-11f (90:10) (R = 2-F-C₆H₄-CH₂) (EC 50 = 6.84 μ M), trans-11h (R = 4-F-C₆H₄-CH₂) (EC 50 = 15.29 μ M), trans-11i/cis-11i (97:3) (R = 2,4-diF-C₆H₃-CH₂) (EC 50 = 9.44 μ M) were the most active toward TK + VZV Oka strain, while exhibiting no

activity toward TK⁺VZV strain. The activity of these isoxazolidines trans-11/cis-11 against TK⁺ + VZV Oka strain was 8-to 22-folds lower than that of the reference drug acyclovir. On the other hand, the EC₅₀ values for the TK⁺VZV 07-1 strain (which is an acyclovir resistant strain) of the phosphonates trans-11e/cis-11e (90:10) (R = 4-NO₂-C₆H₄-CH₂) (EC₅₀ = 42.87 μM) and trans-11k/cis-11k (97:3) (R = Et) (EC₅₀ = 41.57 μM) were comparable to that of acyclovir (EC₅₀ = 39.69 μM). These derivatives showed similar EC₅₀'s for TK⁺ and TK⁺VZV strains and therefore their potency against TK⁺ + VZV was approximately 50-fold lower compared to acyclovir.

Furthermore, compounds trans-11b/cis-11b (90:10) (R = C₆H₅-CH₂), trans-11c (R = 2-NO₂-C₆H₄-CH₂), trans-11e/cis-11e (90:10) (R = 4-NO₂-C₆H₄-CH₂) and trans-11g (R = 3-F-C₆H₄-CH₂) showed some activity against human cytomegalovirus (EC₅₀ = 27-45 μM), although they were less active than ganciclovir and cidofovir used as the reference compounds (Table 3). None of the phosphonate derivatives here described showed activity against the other tested DNA and RNA viruses.

The 50% cytostatic inhibitory concentration (IC₅₀) causing a 50% decrease in cell proliferation was determined against murine leukemia L1210, human lymphocyte CEM, human cervix carcinoma HeLa and immortalized human dermal microvascular endothelial cells (HMEC-1). Isoxazolidines trans-11a (R = H) and trans-11j (R = Me) did not inhibit cell proliferation at the highest tested concentration (i.e., 250 μM), whereas trans-11k/cis-11k (97:3) (R = Et) appeared slightly cytostatic towards the tested cell lines (IC₅₀ = 85-101 μM). On the other hand (Table 4, entries b to i), compounds having benzyl substituents at N3 in the quinazolinone moiety showed lower IC₅₀ values (IC₅₀ = 21-102 μM) thereby indicating that installation of functionalized benzyl groups was profitable for inhibitory properties.

Table 4. Inhibitory effect of the tested compounds against the proliferation of murine leukemia (L1210), human T-lymphocyte (CEM), human cervix carcinoma (HeLa) and immortalized human dermal microvascular endothelial cells (HMEC-1).

To the solution of 2-vinyl-3H-quinazolin-4-one (13a, 1.00 mmol) in acetonitrile (15 mL) potassium carbonate (3.00 mmol) was added. After 15 min the respective benzyl bromide (1.10 mmol) was added and the reaction mixture was stirred under reflux for 4 h. A solvent was removed and the residue was extracted with water (3 × 10 mL). An organic layer was dried (MgSO₄), concentrated and the crude product was purified on a silica gel column with a methylene chloride: hexane mixture (7:3, v/v) followed by crystallisation

(chloroform-petroleum ether) to give pure quinazolinones 13b-e and 13g-i. 133.57, 128.60, 128.28, 128.25, 127.67, 126.60, 123.81, 123.61, 115.59, 68.32 (s, N-CH₂). Anal. Calcd. for C₁₇H₁₂N₂O: 133.57, 128.60, 128.28, 128.25, 127.67, 126.60, 123.81, 123.61, 115.59, 68.32 (s, N-CH₂). To the solution of 2-vinyl-3H-quinazolin-4-one (13a, 1.00 mmol) in acetonitrile (15 mL) potassium carbonate (3.00 mmol) was added. After 15 min. iodomethane (2.00 mmol) or iodoethane (1.10 mmol) was added and the reaction mixture was stirred at 60 °C for 5 h. The solvent was removed and a residue was extracted with water (3 × 10 mL). Organic layer was dried (MgSO₄), concentrated and the crude product was purified on a silica gel column with methylene chloride:hexane mixture (7:3, v/v) followed by crystallization (chloroform : petroleum ether) to give pure quinazolinones 13j [35] or 13k.

3-Methyl-2-vinylquinazolin-4(3H)-one (13j). Amorphous solid, m.p. = 122 °C-124 °C (reference [35] m.p. = 123 °C-125 °C).

A solution of the nitron 12 (1.0 mmol) and the respective vinyl quinazolinone (1.0 mmol) in toluene (2 mL) was stirred at 70 °C until the disappearance (TLC) of the starting nitron. All volatiles were removed in vacuo and crude products were subjected to chromatography on silica gel columns with a chloroform/methanol (100:1, 50:1, 20:1, v/v) mixtures as eluents.

Diethyl trans-(2-methyl-5-(4-oxo-3,4-dihydroquinazolin-2-yl)isoxazolidin-3-yl)phosphonate (trans-11a). Yellowish oil; IR (film, cm⁻¹) ν max : 3085, 2980, 2929, 2782, 1687, 1610, 1469, 1331, 1132, 1098, 1052, 13 Diethyl trans-(2-methyl-5-(3-(3-nitrobenzyl)-4-oxo-3,4-dihydroquinazolin-2-yl)isoxazolidin-3-yl)-phosphonate (trans-11d). Data noted below correspond to a 92:8 mixture of trans-11d and cis-11d. A yellowish oil; IR (film, cm⁻¹) ν max : 3070, 2982, 2930, 2910, 1620, 1574, 1531, 1497, 1415, 1298, 1103, 1025, 774

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Epidemiology of HBoV1 infection and relationship with meteorological conditions in hospitalized pediatric patients with acute respiratory illness: a 7-year study in a subtropical region

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Abstract: **BACKGROUND:** Human bocavirus 1 (HBoV1) is an important cause of acute respiratory illness (ARI), yet the epidemiology and effect of meteorological conditions on infection is not fully understood. To investigate the distribution of HBoV1 and determine the effect of meteorological conditions, hospitalized pediatric patients were studied in a subtropical region of China. **METHODS:** Samples from 11,399 hospitalized pediatric patients (≤ 14 years old), with ARI were tested for HBoV1 and other common respiratory pathogens using real-time PCR, between July 2009 and June 2016. In addition, local meteorological data were collected. **RESULTS:** Of the 11,399 patients tested, 5606 (49.2%) were positive for at least one respiratory pathogen. Two hundred forty-eight of 11,399 (2.2%) were positive for HBoV1 infection. Co-infection was common in HBoV1-positive patients (45.2%, 112/248). A significant difference in the prevalence of HBoV1 was found in patients in different age groups ($p < 0.001$), and the peak prevalence was found in patients aged 7–12 months (4.7%, 56/1203). Two HBoV1 prevalence peaks were found in summer (between June and September) and winter (between November and December). The prevalence of HBoV1 was significantly positively correlated with mean temperature and negatively correlated with mean relative humidity, and the mean temperature in the preceding month had better explanatory power than the current monthly temperature. **CONCLUSIONS:** This study provides a better understanding of the characteristics of HBoV1 infection in children in subtropical regions. Data from this study provide useful information for the future control and prevention of HBoV1 infections.

Text: Human bocavirus 1 (HBoV1), which belongs to family Parvoviridae, was firstly identified in respiratory secretions of children with respiratory tract disease in 2005 [1, 2] .

HBoV1 has been confirmed as an important respiratory pathogen and is found in respiratory infections in children and adults worldwide. The prevalence of HBoV1 nucleic acid detection varies from 1.5 to 33% in patients with acute respiratory illness (ARI), according to different studies [3] [4] [5] [6] [7] . Serological and nucleic acid test results are generally consistent [8] [9] [10] [11] , showing HBoV1 infection is very common. HBoV1 can cause both upper respiratory illness (URI) and lower respiratory illness (LRI) [12] [13] [14] [15] [16] [17] [18] . Infection with HBoV1 can lead to development of a cough, rhinitis, fever and other common clinical symptoms [15, 19] . In some cases, it can cause respiratory distress, hypoxia, wheezing and other severe respiratory symptoms [18, 20] . Clinical diagnosis is mainly pneumonia, bronchitis, pneumothorax, mediastinal emphysema and otitis media and other complications [18] [19] [20] [21] [22] . In some cases, patients develop severe respiratory injury symptoms, which can be fatal [21, 23] . HBoV1 can be detected in fecal samples [24] , blood samples [25, 26] , urine [27, 28] , cerebrospinal fluid [29] [30] [31] , river water [32] and sewage [33, 34] , indicating that HBoV1 may be associated with a variety of diseases. Current in vitro studies modeling tissue-like airway epithelial cells cultures show HBoV1 infection can lead to disruption of the tight-junction barrier, loss of cilia and epithelial cell hypertrophy [35] [36] [37] , similar to lung injury tissue changes in vivo. There is currently no vaccine or specific treatment for this virus; prevention and treatment of HBoV1-related diseases still require further research. The prevalence of respiratory viruses is associated with many factors, including local climate, which may impact the survival and spread of the viruses [38] . Studying the epidemiology of HBoV1 and its relationship with meteorological conditions will improve diagnosis, treatment, control and prevention of this virus.

In this study, we investigated the epidemiology of HBoV1 infection in children (≤ 14 years old) hospitalized with ARI in a subtropical region in China over a 7-year period. In addition, we collected climate data to determine if there was a relationship between HBoV1 prevalence and meteorological conditions. This study will add to existing epidemiological data on HBoV1 and its relationship with climate conditions in subtropical regions and will play a positive role in HBoV1 control and prevention.

The study sites were three tertiary hospitals in Guangzhou, southern China (Longitude: E112°57' to E114°03'; Latitude N22°26' to N23°56'). Inclusion criteria were pediatric patients (≤ 14 years old) who presented with at least two of the following symptoms: cough, pharyngeal discomfort, nasal obstruction, rhinitis, dyspnea or who were diagnosed with pneumonia by chest radiography during the previous week. Chest radiography was

conducted according to the clinical situation of the patient. Throat swab samples were collected from the enrolled patients between July 2009 and June 2016 for routine screening for respiratory viruses, *Mycoplasma pneumoniae* (MP), and *Chlamydia pneumoniae* (CP). The samples were refrigerated at 2-8°C in viral transport medium, transported on ice and analyzed immediately or stored at – 80°C before analysis, as described previously [15, 39] .

Meteorological data for Guangzhou, were collected from July 2009 to June 2016, from the China Meteorological Administration, including the monthly mean temperature (°C), mean relative humidity (%), rainfall (mm), mean wind speed (m/s), mean air pressure (hPa), mean vapor pressure (hPa), sunshine duration (h).

Real-time PCR for HBoV1 and common respiratory pathogen detection DNA and RNA were extracted from the respiratory samples using the QIAamp DNA Mini Kit and QIAamp Viral RNA Mini Kit (Qiagen, Shanghai, China), respectively, in accordance with the manufacturer's protocols. Taqman real-time PCR for HBoV1 was designed based on the conserved region of the NP1 gene, as described previously [15] . Common respiratory pathogens, including respiratory syncytial virus (RSV), influenza A virus (InfA), influenza B virus (InfB), four types of parainfluenza (PIV1-4), adenovirus (ADV), enterovirus (EV), human metapneumovirus (HMPV), four strains of human coronavirus (HCoV-229E, OC43, NL63 and HKU1), human rhinovirus (HRV), MP and CP were detected simultaneously as previously reported [40] .

Data were analyzed using Chi-squared test and Fisher's exact test in SPSS 19.0 (SPSS Inc., Chicago, IL, USA). Correlation with climate data was analyzed using multiple linear regression analysis. All tests were two-tailed and a p value < 0.05 was considered as statistically significant.

Eleven thousand three hundred ninety-nine pediatric patients (≤ 14 years old) hospitalized with ARI were enrolled in the study between July 2009 and June 2016. The male-to-female ratio was 1.82:1 (7361:4038) and the median age was 1.75 years (interquartile range 0.75-3.83). Overall, 86.5% (9857/11399) of patients were under the age of 5 years. All the 11,399 patients were tested for all 18 pathogens mentioned, and 5606 (49.2%) were positive for one or more of those pathogens (Table 1) , and had a median age of 1.50 years (interquartile

range 0.67-3.00). The male-to-female ratios were 1.94: 1 (3698:1908) in pathogen-positive patients and 1.72: 1 (3663:2130) in pathogen-negative patients ($p = 0.002$).

Two hundred forty-eight of 11,399 patients (2.2%) tested positive for HBoV1 infection. Of the HBoV1-positive patients, 112 (45.2%) were co-infected with other pathogens, most frequently with RSV (11.7%, 29/248) (Table 1). The median age was 1 year (interquartile range 0.75-1.83). The male-to-female ratio was 2.54:1 (178:70) in HBoV1-positive patients and 1.81:1 (7183:3968) in HBoV1-negative patients ($p = 0.019$).

To clarify the age distribution of HBoV1, patients were divided into seven age groups; 0-3 months, 4-6 months, 7-12 months, 1-2 years, 3-5 years, 6-10 years and 11-14 years old. There was a significant difference in the prevalence of HBoV1 in patients in different age groups ($p < 0.001$) and the peak prevalence was found in patients aged 7-12 months (4.7%, 56/1203) (Fig. 1) .

In this study, we monitored the prevalence of HBoV1 in patients (≤ 14 years old) hospitalized with ARI from July

We collected meteorological data for Guangzhou, including monthly mean temperature, mean relative humidity, rainfall, mean wind speed, mean air pressure, mean vapor pressure and sunshine duration for a 7-year period, to explore the correlation between meteorological conditions and prevalence of HBoV1. Guangzhou, which is located in southern China (longitude 112°57' to 114°3', latitude 22°26' to 23°56'), has a maritime subtropical monsoon climate. Between July 2009 and June 2016, the mean temperature was $21.8 \pm 5.8^{\circ}\text{C}$ (mean $\pm$ standard deviation), humidity was $77.2 \pm 7.3\%$, sunshine duration was 132.7 ± 59.5 h, wind speed was 2.2 ± 0.6 m/s, rainfall was 175.2 ± 165.9 mm, air pressure was 1005.6 ± 6.0 hPa and vapor pressure was 21.3 ± 7.4 hPa. Between 2009 and 2016, the mean temperature from May to September was greater than 25°C (Fig. 3) .

For multiple linear regression analysis of HBoV1 prevalence and meteorological conditions correlation, independent variables of mean air pressure (adjusted $R^2 = 0.793$, $p < 0.001$) and mean vapor pressure (adjusted $R^2 = 0.929$, $p < 0.001$), which linearly associated with mean temperature, and rainfall (adjusted $R^2 = 0.278$, $p < 0.001$), which strongly correlated

with mean relative humidity, were excluded. The independent variables for the final multiple linear regression analysis included mean temperature, mean relative humidity, mean wind speed and sunshine hours. The effect of temperature had a delay therefore mean temperature in the preceding month (mean temperature 1 month before) was also included as an independent variable in the analysis (Table 2) . Both regression models were established ($p < 0.001$) and the adjusted R^2 values were 0.373 and 0.231 in the mean temperature in the preceding month model and the current monthly temperature model, respectively. HBoV1 prevalence was positively correlated with temperature (coefficient = 0.259 in the current temperature model ($p = 0.002$), coefficient = 0.328 in mean temperature in the preceding month model ($p < 0.001$)). Conversely, HBoV1 prevalence was negatively correlated with relative humidity (coefficient = -0.126 in the current temperature model ($p = 0.024$), coefficient = -0.083 in the temperature delay model ($p = 0.039$)) (Table 2).

ARI is one of the most common human diseases, predominantly caused by different respiratory viruses [41, 42] . One of these viruses, HBoV1 infection, causes global epidemics, has a high public health burden and circulates with different patterns in different areas [3] [4] [5] [6] [7] 43] . In general, the prevalence of viruses varies because of factors such as Multiple linear regression analysis was performed using HBoV1 monthly prevalence as the dependent variable, monthly mean temperature (or mean temperature in the preceding month), mean relative humidity, mean wind speed and sunshine duration as the independent variables Data captured in bold are highly significant geographical location, climatic conditions, population and social activity [38] . Epidemiology of HBoV1 in temperate regions has been described in more detail and a high incidence of infection has been observed in children under the age of 2 years in winter and spring [15, 16, 39, 44] .

To describe the epidemiology of HBoV1 in Guangzhou, we collected throat swabs from 11,399 children (≤ 14 years old), hospitalized with ARI and monitored HBoV1 and other common respiratory pathogens over a 7-year period (Table 1).

In the current study, 86.5% (9857/11399) of patients were under the age of 5 years, with a median age of 1.75 years, indicating that infants and young children were most at risk of ARI, consistent with previous reports [45, 46] . Overall, 49.2% (5606/11399) of patients tested positive for one or more respiratory pathogens, 2.2% (248/11399) of patients were tested with HBoV1 infection (Table 1) . A higher prevalence of HBoV1 was detected in male

patients compared with female patients ($p = 0.019$), consistent with previous reports [15, 16, 39, 44] .

Co-infection with HBoV1 and other pathogens is common [14, 15] . In our study, 45.2% (112/248) of HBoV1-positive patients also tested positive for other pathogens (Table 1) . This may be partly caused by coinciding epidemics of HBoV1 and other pathogens. In our study, the HBoV1 seasonal distribution and total positive pathogen distribution were consistent, confirming this inference (Fig. 2) . Current research shows that HBoV1 infection can lead to the collapse of the first line of defense of airway epithelium [35] [36] [37] , which may lead to a higher susceptibility to other pathogens, explaining the high rate of co-infection. Whether co-infection leads to more severe disease is currently unknown and more research is needed to determine this. The characteristics of the HBoV1 infection are likely to be a good model for studying the effects of co-infections.

In this study, there was a significant difference in prevalence of HBoV1 in patients of different ages ($p < 0.001$). The majority of HBoV1 infections occurred in patients under 2 years old and the peak frequency of HBoV1 infection occurred in patients aged 7-12 months (Fig. 1) , consistent with previous serological and epidemiological reports on the virus [8-11, 15, 16, 39, 44] . This might be because children's immune systems are still under development and maternal antibodies gradually disappear in this age group. The distribution of HBoV1 in patients of different ages will provide important reference for future vaccines and new drug research and development, as well as providing important data for disease prevention and control.

Many factors affect the epidemiology of pathogens, such as geographical location and local climate. Guangzhou, a central city and main transport hub in southern China, is located in a subtropical region.

Guangzhou is hot and has high annual rainfall, long summers, short winters and the annual precipitation and high temperature are almost in the same period (Fig. 3) . In this study, two HBoV1 peaks were observed (Fig. 2) . The large prevalence peaks of HBoV1 infection occurred between June and September of each year, which are the summer months in Guangzhou, with mean temperatures of higher than 25°C (Fig. 3) . Small peaks of HBoV1 infection occurred in winter, between November and December of each year. This seasonal

distribution is similar to the prevalence in subtropical regions reported previously [47] , but different from the HBoV1 epidemics in temperate regions, which mostly occur in winter and spring [15, 16, 39, 44] , as well as from tropical regions, such as India, where no obvious epidemic season has been found [48] .

To analyze the correlation between HBoV1 prevalence and meteorological conditions, multiple linear regression analysis was performed, with HBoV1 monthly prevalence as the dependent variable and mean temperature (or mean temperature in the preceding month), mean relative humidity, mean wind speed and sunshine duration as the independent variables (Table 2) . Both regression models were established ($p < 0.001$) and the adjusted R^2 value (0.373) of the temperature dorp 1 month model was greater than the adjusted R^2 value (0.231) of the current monthly temperature model, indicating that the temperature dorp 1 month model had better explanatory power than the current monthly temperature model. Both of the models showed that the prevalence of HBoV1 was significantly correlated with temperature and relative humidity (Table 2). In detail, HBoV1 prevalence was positively correlated with temperature, that is consistent with previous reports [47, 49] . Conversely, HBoV1 prevalence was negatively correlated with relative humidity, this was different from a previous report in Suzhou [47] , which may be related to Guangzhou high humidity (mean monthly relative humidity was $77.2 \pm 7.3\%$) (Fig. 3) . It is common for pathogen prevalence to fluctuate over time because of a variety factors. In this study, HBoV1 prevalence was relatively low in 2013 to 2014. It might be partly related to the relatively higher mean relative humidity during this period (Fig. 3) . Climate conditions may impact the survival and spread of respiratory viruses, however no significant linear relationship between HBoV1 infection and wind speed or sunshine duration were found in this study ($p > 0.05$) (Table 2) .

Some limitations of this study should be noted. First, because our study mainly focused on HBoV1 circulation in hospitalized patients with ARI, HBoV1 in outpatients and the asymptomatic population were not included. Second, many factors can affect virus epidemics, meteorological data analysis alone may not serve as a final conclusive interpretation. Third, the study was only conducted in three hospitals and may not be representative of the overall population.

Our study has provided a better understanding of the epidemiology of HBoV1 in subtropical regions, specifically correlations with climate data; these data will be helpful for future control and prevention of HBoV1 infections.

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Whole genome sequencing and phylogenetic analysis of human metapneumovirus strains from Kenya and Zambia

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Abstract: **BACKGROUND:** Human metapneumovirus (HMPV) is an important cause of acute respiratory illness in young children. Whole genome sequencing enables better identification of transmission events and outbreaks, which is not always possible with sub-genomic sequences. **RESULTS:** We report a 2-reaction amplicon-based next generation sequencing method to determine the complete genome sequences of five HMPV strains, representing three subgroups (A2, B1 and B2), directly from clinical samples. In addition to reporting five novel HMPV genomes from Africa we examined genetic diversity and sequence patterns of publicly available HMPV genomes. We found that the overall nucleotide sequence identity was 71.3 and 80% for HMPV group A and B, respectively, the diversity between HMPV groups was greater at amino acid level for SH and G surface

protein genes, and multiple subgroups co-circulated in various countries. Comparison of sequences between HMPV groups revealed variability in G protein length (219 to 241 amino acids) due to changes in the stop codon position. Genome-wide phylogenetic analysis showed congruence with the individual gene sequence sets except for F and M2 genes. CONCLUSION: This is the first genomic characterization of HMPV genomes from African patients.

Text: Human metapneumovirus (HMPV) is a single-stranded RNA virus in the family Paramyxoviridae and closely related to human respiratory syncytial virus (RSV) [1] . HMPV causes respiratory disease similar to RSV, ranging from mild upper respiratory infection to bronchiolitis and pneumonia [2] . HMPV infections are seasonal and coinfection with other respiratory pathogens is common [1] . The HMPV genome is approximately 13 kb and comprises eight open reading frames (ORFs) encoding nucleoprotein (N), phosphoprotein (P), matrix protein (M), fusion glycoprotein (F), transcription enhancer protein (M2), small hydrophobic protein (SH), attachment glycoprotein (G), and large polymerase protein (L) [3] . The membrane glycoproteins F and G sequences are used to define two major genotypes or groups, A and B, which are further classified into four subgroups (A1, A2, B1, and B2). HMPV A2, the most frequently observed subgroup, is further divided into two proposed sub-lineages (A2a and A2b) [3] .

HMPV is reported to have an important contribution to acute respiratory infections (ARI) in Africa. For instance, HMPV-associated hospitalization was estimated at 6.5 per 1000 person years in infants in Soweto, South Africa [4] ; at 4% in hospitalized children with severe ARI during a 2-year period in Cameroon [5] ; and in rural western Kenya, incidence of HMPV associated with ARI cases in outpatient clinic visits was estimated at 0.43 per 100 person-years among outpatients [6] . In Kilifi coastal Kenya, between January 2007 to December 2011, children under 6 months of age accounted for 44% of HMPV positive cases, while 74% were children under 1 year, and 1.3% (2/160) were children > 36 months [7] . In Dadaab and Kakuma refugee camps in Kenya, HMPV was detected in 5.7% hospitalizations, and virus-positive crude hospitalization rate (per 1000 children < 5 years old) was 4 for HMPV [8] . In Mali, contribution of HMPV to pneumonia had a population attributable fraction of 9% (95% CI: 7-11%) [9] ; while in Morocco [10] , 8.9% of children < 5 years admitted with severe pneumonia were infected with HMPV. HMPV prevalence and incidence elsewhere globally, is indicated in Additional file 4: Table S1 . Of note is that the variations in incidence rates could be attributed to study population, seasonality and even detection methods. Nonetheless, genomic epidemiology of HMPV in Africa is inadequately

reported, and comparison of genetic similarity and differences between African and global strains is not documented.

Genome sequences provide valuable resources for characterizing viral evolution and disease epidemiology, and for identifying transmission events and outbreaks, which is not always possible with sub-genomic fragments [11] [12] [13] . The increased number of phylogenetically informative variant sites obtained from full genomes may allow better linking of cases and aid public health interventions in real time during epidemics [14, 15] . PCR approaches for targeted whole genome sequencing, in contrast to random amplification, can preferentially amplify the target virus over host or environmental nucleic acids [16, 17] potentially focusing sequencing on the virus of interest. To date, the largest dataset of HMPV whole genomes (n = 61) sequenced from any tropical country is from three Peruvian cities, Lima, Piura and Iquitos [18] . In Africa, apart from one metapneumovirus genome identified from a wild mountain gorilla in Rwanda (GenBank accession number HM197719), there are no HMPV genomes reported according to the NIAID Virus Pathogen Database and Analysis Resource (ViPR, <http://www.viprbrc.org/>, accessed April 30, 2019). This has led to limited understanding of the genetic and genomic diversity of HMPV in the continent.

This work describes a whole genome sequencing (WGS) approach for HMPV from a small number of HMPV positive clinical samples collected at Kilifi County Hospital in Kilifi, Kenya and University Teaching Hospital in Lusaka, Zambia. The genomes were generated by sequencing overlapping PCR amplicons spanning the entire genome. These are the first reported complete genome sequences of locally circulating HMPV strains obtained directly from clinical samples in Africa. We also combined the new genomes with publicly available sequences to examine patterns in global HMPV genetic diversity.

Whole genome sequencing was successful for all 5 clinical samples that were attempted. A single genomic sequence was obtained from each sample, and the length of the 5 new HMPV genomes ranged from 13,097 to 13,134 nt (> 95% length coverage). Sequencing and data assembly parameters, including coverage depth are shown in Table 1 .

Sequence annotation of the full-length genomes using Geneious R8.1.5 (<https://www.geneious.com>) identified the expected eight coding ORFs and non-coding

genomic regions. The overall nucleotide identity (i.e., identical sites averaging over all sequence pairs and excluding positions containing gaps) between all 143 genome sequences analyzed (5 new genomes plus 138 from ViPR) was 58.2%. Nucleotide sequence identity was 71.3% within HMPV-A and 80% within HMPV-B. Intr subgroup, A1, A2, B1 and B2 genomes shared 92.1% (10 sequences), 76.8% (88 sequences), 91% (24 sequences) and 89.6% (21 sequences) amino acid sequence identity.

For the 143 HMPV genomes, we checked sequence conservation at transcriptional control regions, at the termini of each gene, as well as the lengths of intergenic sequences between gene boundaries. The length of the F-M2 intergenic region was different between group A and B viruses, that is, 13 nt and 2 nt, respectively. The SH-G and G-L intergenic regions were the longest, up to 125 nt and to 190 nt, respectively. Consensus nucleotides (9 to 19 length) at the putative start and end regions flanking the ORF of the viral genes are shown in Fig. 1. The gene-start and -end regions of N and P were conserved (> 90% average pairwise identity) in both HMPV groups, and the M2 and M gene-start and -end were also conserved in HMPV group A and B, respectively. The putative ATG start codon was consistently located at positions 14-16 upstream of a gene start motif (consensus: GG/AGAC/TAAA/GTnnnnATG), except for the internal M2-2. An additional ATG start codon upstream of the gene-start motif was observed in the SH gene for the B1 and B2 strains. In five of the eight annotated genes (N, P, F, M2, and G (B1 and B2 strains only)), the intergenic regions were short and the ORFs for these 5 genes terminated within the positioned gene-end motifs.

We combined the five genome sequences from Kenya and Zambia with available global sequences, aligned individual genes and calculated the percent nucleotide (nt) and amino acid (aa) identity (Table 2).

The coding sequences of N, M, F, M2-1, M2-2, and L genes were conserved at nucleotide and amino acid levels, by sharing > 85% between-subgroup nucleotide identity and 90% protein identity (Table 3). The nucleoprotein gene was the most conserved among all subgroups at the nt and aa levels. SH and G glycoprotein genes were more divergent between the HMPV subgroups at the nucleotide level with 76 and 63% identity, respectively. The SH protein length was variable between group A and B strains due to a nucleotide substitution (CAA → TAA) at gene position 532 in group B, resulting in protein lengths of 178 and 180 aa, respectively. The predicted G protein length also varied among

the different HMPV subgroups, between 219 and 241 aa, due to different positions of the Stop codon. Amino acid sequence diversity for G and SH glycoproteins is depicted in Fig. 2 and Additional file 2: Figure S2 , respectively. The diversity of the complete nucleotide sequences of SH and G genes is depicted in phylogenetic trees in Fig. 3 .

We evaluated phylogenetic classification and relationship between the 5 new genomes obtained in this study and previously published genomes (Fig. 3) . Full genome Figure S3 . There was phylogenetic congruence with the individual gene sequence sets as with the full genome dataset, except for F and M2 gene (Additional file 3: Figure S3).

Variant or drifted viral strains may lower the sensitivity of detection resulting in a decreased quantitation of the viral load and underestimation of disease incidence [19] . We checked the new HMPV genomes for nucleotide differences in the genomic regions targeted by our diagnostic rRT-PCR primers and probes (Additional file 7: Table S4) used for HMPV detection. Up to eight primer-and probetemplate mismatches were identified (Fig. 4) : one mismatch in the forward primer region in HMPV group A (F gene-based rRT-PCR assay, Fig. 4a); one mismatch in each of the forward and probe target regions in group B (F gene-based rRT-PCR assay, Fig. 4b) ; and 5 different mismatches with the N-gene based rRT-PCR assay (Fig. 4c) . Note, the F gene-based rRT-PCR assays are different or specific to the two HMPV groups.

HMPV causes respiratory illness presenting as mild upper respiratory tract infection or life-threatening severe bronchiolitis and pneumonia primarily in children, sometimes adults as well as immunocompromised individuals [2] . However, HMPV genome sequence data from Africa is sparse and information on genome-wide diversity is limited. In the present study, the whole genome sequences of five HMPV strains from Kenya and Zambia were determined and compared with the genomes published previously from around the world. Comparative sequence analysis indicated fairly conserved positioning of the gene-start and -end regions as well as translational start and -end codons. Variation in genestart and -end sequences can have significant impact on transcription initiation and termination efficiency so that there is more selective pressure preventing changes in these regions [20] , and this likely explains our observation. The additional ATG start codon found upstream of the gene-start motif of the SH gene was consistent with a previous report [21] , though its role in gene expression is yet to be identified.

These observed sequence conservation in N, M, F, M2-1, M2-2, and L genes is not unusual and is suggestive of functional and structural constraints on diversity, but less expected of the F gene because of its status as a neutralization and protective antigen, similar to its close 'relative' RSV [22] . It has also been suggested that the low diversity in F gene might make a substantial contribution to cross-neutralization and cross-protection between the HMPV subgroups [21] . The relatively high frequency of amino acid diversity in G (and to a lesser extent SH) could be attributable to selective pressure for amino acid change coming from host immunity; and the ability of the protein to tolerate substitutions, which might be due to its proposed extended, unfolded nature [22] . The phylogenetic incongruence observed between whole genome tree and the F and G gene trees, is as reported previously for HMPV [23] , and could be attributed to differential rates of evolution, selection pressure or past recombination events [24] . The prevalence of HMPV in hospitalized pediatric population in Kilifi county in coastal Kenya has been reported [7, 25] . However, it is notable that in recent years, HMPV has been detected at low prevalence in Kilifi (unpublished observations from hospital-based pneumonia surveillance). Whether this low prevalence is due to reduced virus transmission, or decreased sensitivity of our HMPV molecular diagnostic assay due to progressive primer/probe mismatches, is yet to be established.

We present the first full genome sequences of circulating HMPV strains from sub-Saharan Africa. A limitation of our sequencing method, as is common with amplicon sequencing protocols [26, 27] , was absent coverage at the 3' leader and 5' trailer regions not captured by these primers. Our results demonstrate the application of amplicon sequencing to generate full length HMPV genomes directly from clinical samples. The observed diversity of the individual genes is comparable to that described previously [20] [21] [22] . This method and data provide a useful reference for design of local molecular diagnostics and for studies aimed at understanding HMPV epidemiology and evolution in Africa.

Nasopharyngeal and oropharyngeal (NP-OP) swab samples were collected from children (1-59 months) hospitalized with pneumonia, four of whom were enrolled in the PERCH study [18] in 2012. The fifth sample was collected from a child enrolled in the routine pneumonia surveillance study at Kilifi County Hospital, Kenya, in 2015. The samples were tested for HMPV by multiplex semi-quantitative real-time reverse transcription PCR (rRT-PCR) assays. The rRT-PCR primers and probes used, cycling conditions and assay set up have been described elsewhere [28, 29] . Fusion (F) and glycoprotein (G) encoding genes of the HMPV positive samples were amplified in a one-step RT-PCR assay (OneStep RT-PCR kit, QIAGEN), as described previously [7] . Partial G or F nucleotide sequences were

analyzed by maximum likelihood (ML) phylogenetic trees using IQ-TREE [30] , together with reference strains of HMPV subgroups (accession numbers AF371337.2, FJ168779, AY297749, AY530095, JN184401 and AY297748). Five HMPV positive samples from the Kenya and Zambia study sites, belonging to the A2a (n = 1), A2b (n = 2), B1 (n = 1) and B2 (n = 1) genetic subgroups based on their G and F gene sequences, were selected for whole genome sequencing. Data on age, sex and clinical assessment information collected at the time of sample collection, for the five selected samples, are shown in Table 3 .

The sequencing protocol consisted of four steps as follows: (i) primer design, (ii) preparation of primer mixes, (iii) cDNA and PCR (iv) Illumina sequencing and data analysis.

All human metapneumovirus (HMPV) full genome sequences were retrieved from GenBank (January 2018) using the query (txid162145 (Organism) AND 12000(SLEN): 14000(SLEN) NOT patent). Sequence entries with gaps larger than 6 nt were excluded to generate a set of yielding 178 genomes. All possible 23 nt sequences were generated from the genomes dataset and trimmed to a final calculated melting temperature (T_m) of 47.9-49.5°C. Sequences with homology to rRNA sequences, with GC content outside < 0.3 or > 0.75 or with a single nucleotide fractional content of > 0.6 were discarded. The primer set was then made nonredundant yielding 60,746 potential primers. All potential primers were mapped against the 178 HMPV full genomes and the number of perfect matches (frequency score) was determined as a measure of primer sequence conservation. To select primers, the HMPV genome sequences were divided into amplicons with 222 nt overlap spanning the virus genome. Potential primers that mapped within the terminal 5' and 3' 222 nt of each amplicon were identified and the sequence with the highest frequency score was selected, and primers mapping to the reverse bins were reverse complemented. In this manner, 24 primers were selected for each of the 4 HMPV genotype representative genomes (GenBank accession number HMPV A1: AF371337, HMPV A2: FJ168779; HMPV B1: AY525843, and HMPV B2: FJ168778). Because of conservation between genotypes, there was primer redundancy which was removed. The final set of 65 primer sequences, their lengths, calculated T_m , fractional GC content and mapping position on the HMPV genome are presented in Additional file 5: Table S2 . The primers were computationally tested against each of the 4 HMPV subgroups. A graphical representation of the primer target sites is presented in Additional file 1: Figure S1 .

Amplification was performed in two reactions. To avoid generating small products from adjacent forward and reverse primers, amplicons were assigned to alternate Table 3).

Bootstrap support values (evaluated by 1000 replicates) are indicated along the branches. Genetic subgroups A1, A2a, A2b, B1, and B2, are indicated. Multiple sequence alignment was done using MAFFT and the ML phylogeny inferred using GTR + Γ nucleotide substitution model and ultrafast bootstrap approximation in IQ-TREE. The genotype B2 Sabana strain sequence (GenBank accession number HM197719) reported from a wild mountain gorilla in Rwanda is marked in blue. The scaled bar indicates nucleotide substitutions per site reactions, with reaction 1 containing primers for amplicons 1,3,5,7,9,11; reaction 2 containing primers for amplicons 2,4,6,8,10,12. Each reverse transcription used Forward Primer Mixes (FPMs) made with 3.0 μ l of each reverse primer (100 pmol/ μ l) plus water to 200 μ l to generate a primer concentration of 24 pmol/ μ l. Two microlitre of the FPM is then used in a 20 μ l reverse transcription reaction (2.4 pmol/ μ l final concentration in reaction or 2.4 μ M/primer). For PCR amplification, each amplicon reaction used a separate PCR Primer Mix (PPM) containing 1.5 μ l of each 100 pmol/ μ l forward primer and 1.5 μ l of each reverse primer (5.3-5.5 pmol/ μ l total primer in the PPM). 2 μ l PPM was used per 25 μ l PCR reaction = 0.5 pmol/ μ l in reaction (= 500 nM).

Viral nucleic acids were extracted from the original samples using QIAamp Viral RNA Mini kit (QIAGEN). RNA (5 μ l) was reverse transcribed into cDNA using SuperScript III (200 U, Invitrogen), RT buffer (1X final concentration, Invitrogen), and 2 μ l of FPM in 20 μ l reactions. An aliquot of cDNA (5 μ l) was amplified in 35 cycles using Phusion Highfidelity PCR kit (New England Biolabs) and 2 μ l of PPM in a 25 μ l reaction. The PCR mixture was incubated at 98°C for 30 s, followed by 35 cycles of 98°C for 10 s, 43°C for 30 s, and 72°C for 90s and a final extension of 72°C for 10 min. Expected PCR products for each amplicon were approximately 1500 bp. PCR products from the two reactions for each sample were pooled for Illumina library preparation. Fig. 4 Mismatches between the rRT-PCR diagnostic primers and probes and their expected binding sites in the five genomes from Kenya and Zambia. 'Fwd primer' = Forward primer and 'Rev primer' = Reverse primer. Two rRT-PCR assays were used for HMPV detection. The colored bars in the figure indicate nucleotide differences (mismatches) between (a) three HMPV-A genomes and HMPV-A specific primers and probes targeting fusion gene, (b) two HMPV-B genomes and HMPV-B specific primers and probes also targeting fusion gene, and (c) all five genomes reported here and specific primers and probes targeting nucleoprotein gene. The sequences of the rRT-PCR primers and probes checked against the African HMPV genomes are listed in Additional file 7: Table

S4 Illumina sequencing and data analysis Libraries were prepared using Nextera XT kit (Illumina) and pair-end sequencing (2 × 300 base pairs) with the MiSeq Reagent V3 kit (Illumina), following the manufacturer's instructions. The Nextera enzyme mix was used to simultaneously fragment input DNA and tag with universal adapters in a single tube reaction, followed by 12-cycle PCR reaction for dual indexing. Agencourt AMPure XP beads (Beckman Coulter) were used for all purification steps and libraries were quantified and quality-checked using the Qubit (Thermo Fisher) and

Bioanalyzer (Agilent). Adapter trimming, quality filtering, kmer normalization of sequencing reads, de novo assembly, calculation of mean genome coverage was as previously described [31] .

A dataset of HMPV genome sequences was retrieved from ViPR in order to infer relationship between HMPV viruses from Kenya and Zambia and viral populations sampled globally. The dataset included 138 sequence entries (> 13,000 nt) that included date (year) and location of sample Table S3). Sequence alignment was done using MAFFT v.7.221 [32] using the parameters 'localpair -maxiterate 1000'. IQ-TREE was used to infer maximum likelihood (ML) trees of the complete genome and individual genes under general time-reversible (GTR) substitution model with gamma-distributed among-site rate heterogeneity. A summary of the methodology outlined here is depicted in Fig. 5 .

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Viral Respiratory Tract Infections in Adult Patients Attending Outpatient and Emergency Departments, Taiwan, 2012–2013: A PCR/Electrospray Ionization Mass Spectrometry Study

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Abstract: Viral etiologies of respiratory tract infections (RTIs) have been less studied in adult than in pediatric populations. Furthermore, the ability of PCR/electrospray ionization mass spectrometry (PCR/ESI-MS) to detect enteroviruses and rhinoviruses in respiratory samples has not been well evaluated. We sought to use PCR/ESI-MS to comprehensively investigate the viral epidemiology of adult RTIs, including testing for rhinoviruses and enteroviruses. Nasopharyngeal or throat swabs from 267 adults with acute RTIs (212 upper RTIs and 55 lower RTIs) who visited a local clinic or the outpatient or emergency departments of a medical center in Taiwan between October 2012 and June 2013 were tested for respiratory viruses by both virus isolation and PCR/ESI-MS. Throat swabs from 15 patients with bacterial infections and 27 individuals without active infections were included as control samples. Respiratory viruses were found in 23.6%, 47.2%, and 47.9% of the 267 cases by virus isolation, PCR/ESI-MS, and both methods, respectively. When both methods were used, the influenza A virus (24.3%) and rhinoviruses (9.4%) were the most frequently identified viruses, whereas human coronaviruses, human metapneumovirus (hMPV), enteroviruses, adenoviruses, respiratory syncytial virus, and parainfluenza viruses were identified in small proportions of cases (<5% of cases for each type of virus). Coinfection was observed in 4.1% of cases. In the control group, only 1 (2.4%) sample tested positive for a respiratory virus by PCR/ESI-MS. Patients who were undergoing steroid treatment, had an active malignancy, or suffered from chronic obstructive pulmonary disease (COPD) were at risk for rhinovirus, hMPV, or parainfluenza infections, respectively. Overall, immunocompromised patients, patients with COPD, and patients receiving dialysis were at risk for noninfluenza respiratory virus infection. Rhinoviruses (12.7%), influenza A virus (10.9%), and parainfluenza viruses (7.3%) were the most common viruses involved in the 55 cases of lower RTIs. The factors of parainfluenza infection, old age, and immunosuppression were independently associated with lower RTIs. In conclusion, PCR/ESI-MS improved the diagnostic yield for viral RTIs. Non-influenza respiratory virus infections were associated with patients with comorbidities and with lower RTIs. Additional studies that delineate the clinical need for including non-influenza respiratory viruses in the diagnostic work-up in these populations are warranted.

Text: Viral respiratory tract infections (RTIs) in humans occur throughout the year and represent a major cause of clinical visits worldwide. In the past, the viral causes of RTIs were largely unknown, primarily due to the insensitivity of culture-based methods for the detection of viruses or to the narrow spectrum of viral detection using singleplex nucleic acid tests (NATs). Recently, the development of multiplex respiratory NATs has allowed for the simultaneous, rapid, and sensitive detection of multiple viruses, which facilitates comprehensive studies regarding the epidemiology of viral RTIs. Currently, the viral epidemiology of RTIs has been studied more extensively among pediatric populations compared with adult populations throughout the world. 1 Similarly, most studies describing the viral etiology of respiratory illness in Taiwan, a subtropical country in Eastern Asia, were limited to pediatric populations. [2] [3] [4] Thus, studies among adult patients are lacking, particularly regarding infections due to fastidious or newly identified viruses, such as human metapneumovirus (hMPV) and human coronavirus (hCoV). Overlapping clinical presentations shared by different respiratory viruses make differential diagnoses difficult to perform based solely on the clinical parameters. 5 Moreover, effective antiviral agents are currently restricted to influenza virus infections. Hence, a better understanding of the epidemiology of adult viral RTIs would aid the future design of diagnostic strategies, infection control, and patient management.

Among the various multiplex NATs, multilocus polymerase chain reaction coupled with electrospray ionization mass spectrometry (PCR/ESI-MS) can simultaneously identify and subtype multiple respiratory viruses. [6] [7] [8] [9] Despite the diagnostic potential, the ability of PCR/ESI-MS to detect human enterovirus and rhinovirus in respiratory samples from patients with RTIs has not been well evaluated. Previous PCR/ESI-MS studies in patients with RTIs did not include these 2 viruses in the diagnostic panels. [6] [7] [8] [9] Here, we expanded upon these previous studies utilizing PCR/ESI-MS for respiratory virus detection. We aimed to comprehensively investigate the epidemiology of adult viral RTIs using PCR/ESI-MS and compare the diagnostic performance between PCR/ESI-MS and conventional culture methods for identifying multiple, clinically relevant, respiratory viruses, including enterovirus and rhinovirus.

To conduct a comprehensive epidemiologic study that included patients with and without comorbidity, we enrolled adults (of at least 18 yr of age) with acute RTIs within 7 days of onset who were treated at a local outpatient clinic of YC hospital or the outpatient or

emergency departments of National Cheng-Kung University Hospital (NCKUH), a university-affiliated medical center in southern Taiwan, between October 2012 and June 2013. Acute RTI was defined as the simultaneous occurrence of at least 1 respiratory symptom or sign (new or worsening cough, sputum production, sore throat, nasal congestion, rhinorrhea, dyspnea, wheezing, or injected tonsils) and at least 1 of the following symptoms: fever, chills, and cough. Lower RTI (LRTI) was defined as the presence of acute RTI and a new infiltrate on chest radiograph. For patients experiencing more than 1 episode of RTI, the most recent episode was counted as separate only if the patient fully recovered from the previous episode and there was a least a 3-week interval between the onset of the 2 episodes. Clinical, laboratory, and radiological data and the contact history of each patient were retrieved. Comorbidities were assessed in all patients based on the Charlson comorbidity index (CCI). 10 Steroid use was defined as the receipt of corticosteroid treatment (10 mg prednisolone or an equivalent daily dosage) for more than 2 weeks. An immunocompromised state was diagnosed if the patients met one of the following conditions: corticosteroid treatment, solid organ or hematopoietic stem cell recipient, or chemotherapy for an underlying malignancy during the past 6 months.

Nasopharyngeal or throat swabs were obtained from all patients and collected in transport medium, as previously described. 11 for virus detection and identification by both virus isolation and PCR/ESI-MS. Clinical specimens were stored at 48C and transported to the study sites within 24 hours of collection. Throat swabs from 42 cases without respiratory infections during the month prior to enrollment were included as control samples for PCR/ESI-MS analysis, including 15 patients with exclusively bacterial infections (documented cases of bacteremia or urinary tract infection) who were admitted to NCKUH and 27 individuals without active infections. These subjects without active infections included 10 patients with stable chronic diseases followed up in NCKUH clinics and 17 healthy individuals whose medical information was collected using a clinical questionnaire.

The study was approved by the Institutional Review Board (B-ER-101-031) of the study hospital, and all patients provided informed consent.

Respiratory specimens were inoculated onto appropriate tissue cultures (Madin-Darby canine kidney, MRC-5, A549, and rhabdomyosarcoma) to isolate human influenza virus, parainfluenza virus, genus Enterovirus, cytomegalovirus (CMV), adenovirus, respiratory

syncytial virus (RSV), herpes simplex viruses 1 and 2 (HSV-1 and -2), and varicella zoster virus (VZV). The isolation and identification of viruses were performed using a previously described method 11 and enteroviruses were identified by a immunofluorescence assay using a Chemicon Pan EV mix that cross-reacts with rhinovirus (Light Diagnostics, Chemicon [Millipore], MA). 11, 12 Virus Detection and Identification by PCR/ESI-MS Total nucleic acids were extracted from 700 mL of swab samples using a nucleic acid autoextractor (MagNA Pure Compact Instrument, Mannheim, Germany), and the eluate was stored at À80°C until analysis. During the analyses, the extracted nucleic acids were added to both a PLEX-ID Respiratory Virus assay plate and a PLEX-ID Broad Viral I assay plate (PLEX-ID, Abbott Laboratories, Abbott Park, Illinois). The PLEX-ID Respiratory Virus assay detects human adenovirus, hCoV, hMPV, influenza A and B, parainfluenza types 1 to 3, and RSV, 6 whereas the PLEX-ID Broad Viral I assay detects human adenovirus, enterovirus, rhinovirus, BK and JC polyomavirus, parvovirus B19, HSV-1 and -2, VZV, Epstein-Barr virus (EBV), CMV, and human herpesvirus (HHV)-8. 13, 14 In this study, respiratory viruses refer to adenovirus, hCoV, hMPV, influenza, parainfluenza, RSV, enterovirus, and rhinovirus. Nucleic acid amplification and analyses of PCR products were conducted using the PCR/ESI-MS platform (PLEX-ID, Abbott Laboratories) following the manufacturer's instructions, with test turnaround time from sample to result within 6 to 8 hours. 8, 13 The PCR/ESI-MS analyses included automated PCR desalting, ESI-MS signal acquisition, spectral analysis, and data reporting. Organism identification was based on the total mass and base compositions of the PCR amplicons compared with those in the molecular signature database established by the PLEX-ID manufacturer. 6, 8, 13, 14 Samples in which PCR/ESI-MS results disagreed with culture results at the species level were reexamined by a second molecular method. For enteroviruses, rhinovirus was differentiated from enterovirus using a conventional PCR sequencing analysis with the previously described primers (Rhinovirus s1 and as) and a BLAST search. 15

All analyses were performed with the Statistical Package for the Social Sciences version 17.0 (SPSS Inc, Chicago, IL). Continuous variables were expressed as mean values AE standard deviations and were compared using the analysis of variance test. Categorical variables were compared using the Fisher exact test or χ^2 test. All biologically plausible variables with a P value 0.10 in the univariate analysis were considered for inclusion in the logistic regression model for the multivariate analysis. A P value less than 0.05 was considered statistically significant, and all tests were 2-tailed.

During the 9-month study period, a total of 267 episodes of acute RTIs from 263 patients were recorded, including 96 episodes at a local clinic and 171 episodes at NCKUH (19 outpatient and 152 in the emergency departments). For convenience, each episode was counted as 1 case. Overall, 123 (46.1%) cases were male patients, and 152 (56.9%), 60 (22.5%), and 55 (20.6%) patients were 18 to 39, 40 to 59, and ≥60 years of age, respectively. Two-hundred and twelve (79.4%) patients presented with upper RTIs (URTIs), and 55 (20.6%) cases presented with LRTIs. Compared with patients attending the local clinic, patients attending the medical care center were older and had more comorbidities (Table 1). The detailed demographic data of the 267 RTI cases and 42 control cases are presented in Table 1 .

All 267 respiratory samples from each RTI case were examined for viruses by both virus isolation and PCR/ESI-MS, and the results are presented in Table 2 . For virus isolation, respiratory viruses were detected in 63 (23.6%) cases, including influenza A (48 cases, 18.0%), enterovirus (13, 4.9%), and parainfluenza virus (2, 0.7%), and no coinfection was detected. Virus isolation identified additional parainfluenza type 3 and enterovirus infections that were not found by PCR/ESI-MS in 2 samples.

By PCR/ESI-MS, respiratory viruses were detected in 126 cases (47.2%). Influenza A (65 cases, 24.3%) was the most frequently identified virus, among which 36 (13.5%) cases were subtyped as pandemic H1N1/09 virus, 28 (10.5%) cases as seasonal H3N2 virus, and 1 case as influenza A matching both pandemic H1N1 and seasonal H3N2. Genus Enterovirus (34, 12.7%) was the second-most frequently detected virus, including rhinovirus (25, 9.4%), enterovirus (8, 3.0%), and 1 culture-negative case matching for both rhinovirus and enterovirus. hCoV (13, 4.9%), hMPV (10, 3.7%), adenovirus (6, 2.2%), RSV (6, 2.2%), and parainfluenza (4, 1.5%) were detected in small proportions of cases. Simultaneous detection of more than 1 respiratory virus was observed in 11 (4.1%) patients, and rhinovirus (5 cases) was most likely to be codetected with another respiratory virus (Table 2). Of note, 4 cultivated viruses identified as enterovirus because of reactivity with the Chemicon Pan EV mix were characterized as rhinovirus by PCR/ESI-MS. Further PCR-sequencing analysis of the 4 clinical specimens confirmed the existence of rhinoviruses but not enteroviruses. PCR/ESI-MS identified additional respiratory viruses in 65 culture-negative samples, mostly rhinovirus (21 samples), and a second respiratory virus in 3 culture-positive influenza A samples. Overall, the positive detection rates for any respiratory virus by culture, PCR/ESI-MS, and both methods were 23.6%, 47.2%, and 47.9% (128/267), respectively. Of 61 specimens positive by both methods, PCR/ESI-MS and

culture methods reached levels of agreement of 100% at the species level for influenza and parainfluenza and 100% at the genus level for the genus Enterovirus. In the control group, only 1 (2.4%) healthy individual tested positive for a respiratory virus (rhinovirus) by PCR/ESI-MS.

With respect to herpesviruses, PCR/ESI-MS identified EBV, HSV-1, CMV, and VZV in 128 (47.9%), 25 (9.4%), 7 (2.6%), and 2 (0.7%) samples from RTI cases, with similar detection rates observed in the control group. There was no detection of polyomavirus, parvovirus B19, HSV-2, or HHV-8 virus in samples from cases with RTIs or the control group.

Cases that tested positive for any respiratory virus either by culture or by PCR/ESI-MS were analyzed. The positive detection rates declined with age: 55.3%, 41.7%, and 34.5% in the 18-39, 40-59, and ≥60-year-old groups, respectively ($P = 0.02$) (Figure 1A). A higher positivity rate was observed in patients with URTIs than that in patients with LRTIs (50.5% vs. 38.2%, $P = 0.10$) (Table 3 and Figure 1B). There were similar distributions of respiratory viruses in cases from the local clinical and the medical center (Table 2), and between patients from the 3 age groups (Figure 1A). Of 128 cases with identifiable respiratory viruses, non-influenza virus infection was more common in patients with LRTIs than those with URTIs (81.0% [17/21] vs. 48.6% [52/107], $P = 0.007$). Rhinovirus (12.7%), influenza A (10.9%), and parainfluenza (7.3%) were the 3 leading respiratory viruses involved in 55 cases of LRTIs, and parainfluenza was more frequently observed in the LRTI group than in the URTI group (Table 3 and Figure 1B). There was no seasonal variation in any individual respiratory virus over the 9-month period.

Of 128 patients with identifiable respiratory viruses, univariate analysis revealed that patients with 1 of the following conditions were more likely to have non-influenza respiratory virus infections: immunocompromised state, chronic obstructive pulmonary disease (COPD), and chronic renal failure receiving dialysis (OR 5.4, 95% CI 1.2-25.5, $P = 0.02$). Multivariate analysis demonstrated that steroid use was an independent risk factor for rhinovirus infection (OR 15.3, 95% CI 1.5-154.7, $P = 0.02$), active malignancy was an independent risk factor for hMPV infection (OR 29.3, 95% CI 2.4-358.1, $P = 0.008$), and COPD was an independent risk factor for parainfluenza infection (OR 229.2, 95% CI 10.5-5020.8,

While comparing the URTI and LRTI groups, factors found to be associated with LRTI by univariate analysis included old age (≥ 60 years), a high comorbidity index, congestive heart failure, COPD, malignancy, immunocompromised state, and detection of parainfluenza or EBV, whereas detection of influenza A was less frequently associated with LRTI. Codetection of respiratory virus was not associated with the development of LRTI. By multivariate analysis, only old age, immunocompromised state, and detection of parainfluenza remained 3 independent factors associated with LRTI (Table 3) .

Among the 117 episodes of single respiratory virus infections, arthralgia was more frequently observed in influenza A infections than in non-influenza infections (66.1% [39/59] vs. 46.6% [27/58], $P = 0.033$); for these 2 types of infections, the other examined symptoms, including sore throat, rhinorrhea, cough, purulent sputum, wheezing, dyspnea, and headache, were detected at similar frequencies.

Of 55 cases of LRTIs, coinfection with bacterial pathogens by sputum culture or blood culture was found in 3 (8.8%) of 34 patients who tested positive for respiratory viruses and in 2 (9.5%) of 21 patients who tested negative for respiratory viruses. Four of 6 cases of influenza A LRTI had received oseltamivir. Two patients died of pneumonia and the worsening of an underlying malignancy; 1 of these patients tested positive for hMPV, and the other patient tested negative for a respiratory virus. Four

Our study of the viral epidemiology of adult acute RTI using PCR/ESI-MS technology has 3 major advantages. First, we expanded on previous studies utilizing PCR/ESI-MS for respiratory virus detection. The PLEX-ID Broad Viral I assay, which targets enterovirus, rhinovirus, herpesviruses, JC and BK polyomaviruses, and parvovirus B19, and the PLEX-ID Respiratory Virus assay tests were both adopted for the detection of multiple clinically relevant respiratory viruses. Second, 2 control groups (patients with exclusively bacterial infections and individuals without active infections) were enrolled to eliminate false-positive artifacts of NATs and estimate the prevalence of detectable asymptomatic carriers of respiratory viruses. Third, this study enrolled immunocompetent and immunocompromised patients visiting a local clinic or a medical center who presented with an URTI or LRTI, which reflects the true viral epidemiology of adult RTIs.

By supplementing the conventional culture method with PCR/ESI-MS, a 2-fold increase in the respiratory virus detection rate was achieved, from 23.6% by culture alone to 47.9% by a combination of both methods. Diagnostic gain was observed for both culturable viruses, especially rhinovirus, and fastidious viruses. Although we did not compare an alternative NAT due to sample volume limitations, it has been reported that PCR/ ESI-MS has a high sensitivity (92.9-100%) and specificity (99-100%) for variable respiratory virus detection relative to immunologic and PCR-based methods as gold standard assays, with the exception of parainfluenza (sensitivity 63.4%).⁶ Coincidentally, we found that parainfluenza type 3 was 1 of only 2 viruses that were not detected by PCR/ESI-MS. The potential causes contributing to the lower detection rate for parainfluenza remain to be explored.

The positive detection rate (47.2%) for respiratory viruses by PCR/ESI-MS in the present study was similar to those of parallel adult surveillance programs using NATs (43.2-57%).^{5,16-18} but notably higher than an earlier study using the Ibis T5000 biosensor system (the prototype of PCR-ESI/ MS) using the respiratory virus surveillance II kit (35.9%), likely because the kit was not designed for the detection of enterovirus and rhinovirus.⁸ Enterovirus and rhinovirus, both members of the Enterovirus genus, contributed to 13.1% of RTI cases in our study and 9.8-17.8% of adult cases in other studies.^{5, 16, 17} Considering their prevalence, enterovirus and rhinovirus should be included in the diagnostic panels of respiratory viruses if comprehensive viral detection is indicated.

The codetection rate (4.1%) was within the range of 2.0-7.2% that has been reported elsewhere.^{5, 16, 17} and rhinovirus was the virus most frequently involved in coinfections, probably due to its high prevalence throughout the year.¹⁸ Influenza A and rhinovirus were the 2 most frequently detected respiratory viruses, whereas hCoV, hMPV, enterovirus, adenovirus, RSV, and parainfluenza were detected in small proportions of cases. This finding is similar to the viral epidemiology of adult RTIs observed by other study groups.^{5, 16, 17} The similar distributions of viruses between cases from a local clinic and a medical center and between patients of the 3 age groups suggest that individuals of all age groups are susceptible to multiple respiratory viruses that simultaneously circulate in the community. A lower positive detection rate was observed in the elderly population, probably because older adult patients shed lower titers of viruses.¹⁹ However, the roles of EBV, HSV-1, and CMV in adult RTIs remain incompletely²⁰ Moreover, the univariate association between EBV and LRTIs observed in this study may have been caused by the confounding factor of age, particularly given that old age was identified as an independent

factor for EBV detection (data not shown). The lack of detection of BK and JC polyomavirus or parvovirus B19 implies that these viruses play a minor role in adult RTIs and that oropharyngeal cells are not involved in BK and JC polyomavirus persistence. 21 Furthermore, the low positive detection rate for respiratory viruses in the control group suggests a low possibility of false-positive artifacts in PCR/ESI-MS or a lower rate of asymptomatic colonization of respiratory viruses. In addition to the advantage of sensitive detection, PCR/ESI-MS possesses the capability of simultaneous subtype identification of respiratory viruses. 22 In this study, influenza A viruses were subtyped as pandemic H1N1 influenza A and seasonal H3N2 influenza. In Europe, both viruses cocirculated in the community in the 2012-2013 influenza season. 23 In the genus Enterovirus, acid-labile rhinovirus can be differentiated from enterovirus using an acid lability test. 24 while PCR/ESI-MS can rapidly differentiate the 2 species in a single test, as demonstrated in our study. The 13 hCoVs were subtyped as hCoV-OC43, -229E, and -HKU1, which was further validated by conventional PCR-sequencing assays (data not shown). The newly identified HCoV-NL63 was not detected during the study period, and a low detection rate (<1%) was reported in China. 16 Our understanding of the roles of non-influenza respiratory viruses in patients with comorbidities or LRTIs has been strengthened in our study. Patients who were undergoing steroid treatment, had an active malignancy, or suffered from COPD were at risk for rhinovirus, hMPV, or parainfluenza infections, respectively. Overall, immunocompromised patients, those with COPD, and patients receiving dialysis were at risk for non-influenza respiratory virus infection. Non-influenza virus infections were also more frequently involved in LRTIs than in URTIs. Among LRTIs, rhinovirus and parainfluenza were ranked as the first-and third-most common pathogens, respectively, and parainfluenza was an independent factor associated with LRTIs, a finding consistent with prior reports that both viruses are significant causes of LRTIs. 18, [25] [26] [27] On the other hand, despite an increasing role of non-influenza respiratory viruses, currently available antiviral agents and vaccines primarily target influenza infection. Although viral RTI is a self-limited illness, as observed in the majority of our patients with LRTIs who recovered from illness without the aid of antiviral agents, a definite etiological diagnosis can help to reduce the unwarranted use of anti-influenza agents or antimicrobials and/or unnecessary hospitalizations, and provide useful information for the control of RTIs. However, we observed that clinical differentiation of influenza infection from other respiratory virus infections is difficult due to overlapping symptoms, as described previously. 5 Collectively, the association of non-influenza virus infection with patients with comorbidities or LRTIs reported here suggests that a complete respiratory viral panel would be appropriate in the diagnostic work-up for RTIs in these populations. The additional costs incurred by the use of a complete panel of PCR/ESI-MS-based assessments or other molecular tests would likely be offset by the accompanying reductions in unnecessary antimicrobial therapy

and/or hospitalization. 18 Our study has some limitations. First, parainfluenza type 4 and 3 newly identified respiratory viruses, human bocavirus, human polyomavirus KI and WU polyomavirus were not included in the panels. [28] [29] [30] [31] and their roles in adult RTIs in Taiwan are unclear. Second, although certain risk factors for specific virus infections, such as hMPV or parainfluenza infections, have been identified, these associations should be re-examined in additional largescale clinical studies, and the clinical impact and underlying mechanisms of these associations should be explored. Similarly, more control cases may be needed to better estimate the prevalence of asymptomatic carriers of respiratory viruses. Third, only 3 seasons were covered, and the seasonality of viral respiratory infections could not be demonstrated.

In conclusion, compared with virus isolation, PCR/ESI-MS produced a greater diagnostic yield for viral RTIs, with a low possibility of false-positive artifacts. Non-influenza respiratory virus infection was significantly associated with patients with comorbidities and with LRTIs. Additional studies to delineate the clinical need for and economic benefits of including non-influenza respiratory viruses in the diagnostic work-up in these populations are warranted.

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A super-spreading ewe infects hundreds with Q fever at a farmers' market in Germany

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1618839/>

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Abstract: **BACKGROUND:** In May 2003 the Soest County Health Department was informed of an unusually large number of patients hospitalized with atypical pneumonia. **METHODS:** In exploratory interviews patients mentioned having visited a farmers' market where a sheep had lambed. Serologic testing confirmed the diagnosis of Q fever. We asked local health departments in Germany to identify notified Q fever patients who had visited the farmers market. To investigate risk factors for infection we conducted a case control study (cases were Q fever patients, controls were randomly selected Soest citizens) and a cohort study among vendors at the market. The sheep exhibited at the market, the herd from which it originated as well as sheep from herds held in the vicinity of Soest were tested for *Coxiella burnetii* (*C. burnetii*). **RESULTS:** A total of 299 reported Q fever cases was linked to this outbreak. The mean incubation period was 21 days, with an interquartile range of 16–24 days. The case control study identified close proximity to and stopping for at least a few seconds at the sheep's pen as significant risk factors. Vendors within approximately 6 meters of the sheep's pen were at increased risk for disease compared to those located farther away. Wind played no significant role. The clinical attack rate of adults and children was estimated as 20% and 3%, respectively, 25% of cases were hospitalized. The ewe that had lambed as well as 25% of its herd tested positive for *C. burnetii* antibodies. **CONCLUSION:** Due to its size and point source nature this outbreak permitted assessment of fundamental, but seldom studied epidemiological parameters. As a consequence of this outbreak, it was recommended that pregnant sheep not be displayed in public during the 3rd trimester and to test animals in petting zoos regularly for *C. burnetii*.

Text: Q fever is a worldwide zoonosis caused by *Coxiella burnetii* (*C. burnetii*), a small, gram-negative obligate intracellular bacterium. *C. burnetii* displays antigenic variation with an infectious phase I and less infectious phase II. The primary reservoir from which human infection occurs consists of sheep, goat and cattle. Although *C. burnetii* infections in animals are usually asymptomatic, they may cause abortions in sheep and goats [1]. High concentrations of *C. burnetii* can be found in birth products of infected mammals [2]. Humans frequently acquire infection through inhalation of contaminated aerosols from parturient fluids, placenta or wool [1]. Because the infectious dose is very low [3] and *C. burnetii* is able to survive in a spore-like state for months to years, outbreaks among

humans have also occurred through contaminated dust carried by wind over large distances [4] [5] [6] .

C. burnetii infection in humans is asymptomatic in approximately 50% of cases. Approximately 5% of cases are hospitalized, and fatal cases are rare [1] . The clinical presentation of acute Q fever is variable and can resemble many other infectious diseases [2] . However, the most frequent clinical manifestation of acute Q fever is a self-limited febrile illness associated with severe headache. Atypical pneumonia and hepatitis are the major clinical manifestations of more severe disease. Acute Q fever may be complicated by meningoencephalitis or myocarditis. Rarely a chronic form of Q fever develops months after the acute illness, most commonly in the form of endocarditis [1] . Children develop clinical disease less frequently [7, 8] . Because of its non-specific presentation Q fever can only be suspected on clinical grounds and requires serologic confirmation. While the indirect immunofluorescence assay (IFA) is considered to be the reference method, complement fixation (CF), ELISA and microagglutination (MA) can also be used [9] . Acute infections are diagnosed by elevated IgG and/or IgM anti-phase II antibodies, while raised anti-phase I IgG antibodies are characteristic for chronic infections [1] .

In Germany, acute Q fever is a notifiable disease. Between 1991 and 2000 the annual number of cases varied from 46 to 273 cases per year [10] . In 2001 and 2002, 293 and 191 cases were notified, respectively [11, 12] .

On May 26, 2003 the health department of Soest was informed by a local hospital of an unusually large number of patients with atypical pneumonia. Some patients reported having visited a farmers' market that took place on May 3 and 4, 2003 in a spa town near Soest. Since the etiology was unclear, pathogens such as SARS coronavirus were considered and strict infection control measures implemented until the diagnosis of Q fever was confirmed.

An outbreak investigation team was formed and included public health professionals from the local health department, the local veterinary health department, the state health department, the National Consulting Laboratory (NCL) for *Coxiellae* and the Robert Koch-Institute (RKI), the federal public health institute. Because of the size and point source appearance of the outbreak the objective of the investigation was to identify etiologic

factors relevant to the prevention and control of Q fever as well as to assess epidemiological parameters that can be rarely studied otherwise.

On May 26 and 27, 2003 we conducted exploratory interviews with patients in Soest hospitalized due to atypical pneumonia.

Attending physicians were requested to test serum of patients with atypical pneumonia for *Mycoplasma pneumoniae*, *Chlamydia pneumoniae*, *Legionella pneumophila*, *Coxiella burnetii*, Influenza A and B, Parainfluenza 1-3, Adenovirus and Enterovirus. Throat swabs were tested for Influenza virus, Adenovirus and SARS-Coronavirus. Laboratory confirmation of an acute Q fever infection was defined as the presence of IgM antibodies against phase II *C. burnetii* antigens (ELISA or IFA), a 4-fold increase in anti-phase II IgG antibody titer (ELISA or IFA) or in anti phase II antibody titer by CF between acute and convalescent sera. A chronic infection was confirmed when both anti-phase I IgG and anti-phase II IgG antibody titers were raised.

Because patients with valvular heart defects and pregnant women are at high risk of developing chronic infection [13, 14] we alerted internists and gynaecologists through the journal of the German Medical Association and asked them to send serum samples to the NCL if they identified patients from these risk groups who had been at the farmers' market during the outbreak.

The objective of the first case control study was to establish whether there was a link between the farmers' market and the outbreak and to identify other potential risk factors. We conducted telephone interviews using a standardised questionnaire that asked about attendance at the farmers' market, having been within 1 km distance of one of 6 sheep flocks in the area, tick bites and consumption of unpasteurized milk, sheep or goat cheese. For the purpose of CCS1 we defined a case (CCS1 case) as an adult resident of the town of Soest notified to the statutory sur-veillance system with Q fever, having symptom onset between May 4 and June 3, 2003. Exclusion criterion was a negative IgM-titer against phase II antigens. Two controls per case were recruited from Soest inhabitants by random digit dialing.

We calculated the attributable fraction of cases exposed to the farmers' market on May 4 (AFE) as $(OR-1)/OR$ and the attributable fraction for all cases due to this exposure as:

The farmers' market was held in a spa town near Soest with many visitors from other areas of the state and even the entire country. To determine the outbreak size we therefore asked local public health departments in Germany to ascertain a possible link to the farmers' market in Soest for all patients notified with Q-fever. A case in this context ("notified case") was defined as any person with a clinical diagnosis compatible with Q fever with or without laboratory confirmation and history of exposure to the farmers' market.

Local health departments also reported whether a notified case was hospitalized. To obtain an independent, second estimate of the proportion of hospitalizations among symptomatic patients beyond that reported through the statutory surveillance system we calculated the proportion of hospitalized patients among those persons fulfilling the clinical case definition (as used in the vendors' study (s.b.)) identified through random sampling of the Soest population (within CCS2 (s.b.)) as well as in two cohorts (vendors' study and the 9 sailor friends (see below)).

The objective of CCS2 was to identify risk factors associated with attendance of the farmers' market on the second day. We used the same case definition as in CCS1, but included only persons that had visited the farmers' market on May 4, the second day of the market. We selected controls again randomly from the telephone registry of Soest and included only those persons who had visited the farmers' market on May 4 and had not been ill with fever afterwards. Potential controls who became ill were excluded for analysis in CCS2, but were still fully interviewed. This permitted calculation of the attack rate among visitors to the market (see below "Estimation of the overall attack rate") and gave an estimate of the proportion of clinically ill cases that were hospitalized (s.a.).

In the vendors' study we investigated whether the distance of the vendor stands from the sheep pen or dispersion of *C. burnetii* by wind were relevant risk factors for acquiring Q fever. We obtained a list of all vendors including the approximate location of the stands from the organizer. In addition we asked the local weather station for the predominant wind direction on May 4, 2003. Telephone interviews were performed using standardized questionnaires. A case was defined as a person with onset of fever between May 4 and

June 3, 2003 and at least three of the following symptoms: headache, cough, dyspnea, joint pain, muscle pain, weight loss of more than 2 kg, fatigue, nausea or vomiting.

The relative distance of the stands to the sheep pen was estimated by counting the stands between the sheep pen and the stand in question. Each stand was considered to be one stand unit (approximately 3 meters). Larger stands were counted as 2 units. The direction of the wind in relation to the sheep pen was defined by dividing the wind rose (360°) in 4 equal parts of 90°. The predominant wind direction during the market was south-south-east (Figure 1). For the purpose of the analysis we divided the market area into 4 sections with the sheep pen at its center. In section 1 the wind was blowing towards the sheep pen (plus minus 45°). Section 4 was on the opposite side, i.e. where the wind blew from the sheep pen towards the stands, and sections 2 and 3 were east and west with respect to the wind direction, respectively. Location of the stands in reference to the sheep pen was thus defined in two ways: as the absolute distance to the sheep pen (in stand units or meters) and in reference to the wind direction.

We identified a small cohort of 9 sailor friends who visited the farmers' market on May 4, 2003. All of these were serologically tested independently of symptoms. We could therefore calculate the proportion of laboratory confirmed persons who met the clinical case definition (as defined in the cohort study on vendors).

The overall attack rate among adults was estimated based on the following sources:

(1) Interviews undertaken for recruitment of controls for CCS2 allowed the proportion of adults that acquired symptomatic Q fever among those who visited the farmers' market on the second day;

Attributable fraction AFE Number of cases exposed All cases = *

(2) Interviews of cases and controls in CCS2 yielded information about accompanying adults and how many of these became later "ill with fever";

(3) Results of the small cohort of 9 sailor friends (s.a.);

(4) Results from the cohort study on vendors.

Local health departments that identified outbreak cases of Q fever (s.a. "determination of outbreak size and descriptive epidemiology") interviewed patients about the number of persons that had accompanied them to the farmers' market and whether any of these had become ill with fever afterwards. However, as there was no differentiation between adults and children, calculations to estimate the attack rate among adults were performed both with and without this source.

To count cases in (1), (3) and (4) we used the clinical case definition as defined in the cohort study on vendors.

For the calculation of the attack rate among children elicited in CCS2 was the same for all visitors. The number of children that visited the market could then be estimated from the total number of visitors as estimated by the organizers. We then estimated the number of symptomatic children (numerator). For this we assumed that the proportion of children with Q fever that were seen by physicians and were consequently notified was the same as that of adults. It was calculated as:

Thus the true number of children with Q fever was estimated by the number of reported children divided by the estimated proportion reported. Then the attack rate among children could be estimated as follows:

Because this calculation was based on several assumptions (number of visitors, proportion of adult visitors and clinical attack rate among adults) we performed a sensitivity analysis where the values of these variables varied.

Serum was collected from all sheep and cows displayed in the farmers' market as well as from all sheep of the respective home flocks (70 animals). Samples of 25 sheep from five

other flocks in the Soest area were also tested for *C. burnetii*. Tests were performed by ELISA with a phase I and phase II antigen mixture.

We conducted statistical analysis with Epi Info, version 6.04 (CDC, Atlanta, USA). Dichotomous variables in the case control and cohort studies were compared using the Chi-Square test and numerical variables using the Kruskal-Wallis test. P-values smaller than 0.05 were considered statistically significant.

The outbreak investigation was conducted within the framework of the Communicable Diseases Law Reform Act of Germany. Mandatory regulations were observed.

Patients at the local hospital in Soest reported that a farmers' market had taken place on May 3 and 4, 2003 in a spa town close to the town of Soest. It was located in a park along the main promenade, spanning a distance of approximately 500 meters. The market attracted mainly three groups of people: locals, inhabitants of the greater Soest region, patients from the spa sanatoria and their visiting family or friends. Initial interviewees mentioned also that they had spent time at the sheep pen watching new-born lambs that had been born in the early morning hours of May 4, 2003. The ewe had eaten the placenta but the parturient fluid on the ground had merely been covered with fresh straw.

Overall 171 (65%) of 263 serum samples submitted to the NCL were positive for IgM anti-phase II antibodies by ELISA. Results of throat swabs and serum were negative for other infectious agents. (Figure 2). If we assume that symptom onset in cases was normally distributed with a mean of 21 days, 95% of cases (mean $\pm$ 2 standard deviations) had their onset between day 10 and 31. The two notified cases with early onset on May 6 and 8, respectively, were laboratory confirmed and additional interviews did not reveal any additional risk factors. Of the 298 cases with known gender, 158 (53%) were male and 140 (47%) were female. Of the notified cases, 189 (63%) were from the county of Soest, 104 (35%) were from other counties in the federal state of Northrhine Westphalia and 6 (2%) were from five other federal states in Germany (Figure 3). Only eight (3%) cases were less than 18 years of age, the mean and median age was 54 and 56 years, respectively (Figure 4). 75 (25%) of 297 notified cases

were hospitalized, none died. Calculation of the proportion of cases hospitalized through other information sources revealed that 4 of 19 (21%; 95% CI = 6-46%;

(1/5 (CCS2), 2/11 (vendors study) and 1/3 (sailor friends)) clinically ill cases were hospitalized.

Laboratory confirmation was reported in 167 (56%) outbreak cases; 66 (22%) were confirmed by an increase in anti-phase II antibody titer (CF), 89 (30%) had IgM antibodies against phase II antigens, 11 (4%) were positive in both tests and one was confirmed by culture. No information was available as to whether the 132 (44%) cases without laboratory confirmation were laboratory tested.

18 patients with valvular heart defects and eleven pregnant women were examined. None of them had clinical signs of Q fever. Two (11%) of 18 cardiological patients and four (36%) of 11 pregnant women had an acute Q fever infection. During childbirth strict hygienic measures were implemented. Lochia and colostrum of all infected women were tested by polymerase chain reaction and were positive in only one woman (case 3; Table 1). Serological follow-up of the mothers detected chronic infection in the same woman (case 3) 12 weeks after delivery. One year follow-up of two newborn children (of cases 1 and 3) identified neither acute nor chronic Q fever infections.

We recruited 20 cases and 36 controls who visited the farmers' market on May 4 for the second case control study. They did not differ significantly in age and gender (OR for male sex = 1.7; 95%CI = 0.5-5.3; p = 0.26; p-value for age = 0.23). Seventeen (85%) of 20 cases indicated that they had seen the cow (that also was on display at the market next to the sheep) compared to 7 (32%) of Geographical location of Q fever outbreak cases notified to the statutory surveillance system Figure 3 Geographical location of Q fever outbreak cases notified to the statutory surveillance system. or directly at the gate of the sheep pen compared to 8 (32%) of 25 controls (OR = 5.0; 95%CI = 1.2-22.3; p = 0.03). Touching the sheep was also significantly more common among cases (5/20 (25%) CCS2 cases vs. 0/22 (0%) controls; OR undefined; lower 95% CI = 1.1; p = 0.02). 17 (85%) of 20 CCS2 cases, but only 6 (25%) of 24 controls stopped for at least a few seconds at or in the sheep pen, the reference for this variable was "having passed by the pen without stopping" (OR = 17.0;

95%CI = 3.0-112.5; $p < 0.01$). Among CCS2 cases, self-reported proximity to or time spent with/close to the sheep was not associated with a shorter incubation period.

We were able to contact and interview 75 (86%) of 87 vendors, and received second hand information about 7 more (overall response rate: 94%). Forty-five (56%) were male and 35 (44%) were female. 13 (16%) met the clinical case definition. Of the 11 vendors who worked within two stand units of the sheep pen, 6 (55%) became cases compared to only 7 (10%) of 70 persons who worked in a stand at a greater distance (relative risk (RR) = 5.5 (95%CI = 2.3-13.2; $p = 0.002$); Figure 1). Of these 7 vendors, 4 had spent time within 5 meters of the pen on May 4, one had been near the pen, but at a distance of more than 5 meters, and no information on this variable was available for the remaining 2. In the section of the market facing the wind coming from the pen (section 4, Figure 1), 4 (9%) of 44 vendors became cases, compared to 2 (13%) of 15 persons who worked in section 1 ($p = 0.6$). Among 22 persons who worked in stands that were perpendicular to the wind direction, 7 (32%) became cases. (Table 3). In all scenarios the AR among adults was significantly higher than that among children (Figure 5).

In total, 5 lambs and 5 ewes were displayed on the market, one of them was pregnant and gave birth to twin lambs at 6:30 a.m. on May 4, 2003 . Of these, 3 ewes including the one that had lambed tested positive for *C. burnetii*. The animals came from a flock of 67 ewes, of which 66 had given birth between February and June. The majority of the births (57 (86%)) had occurred in February and March, usually inside a stable or on a meadow located away from the town. Six ewes aborted, had stillbirths or abnormally weak lambs. Among all ewes, 17/67 (25%) tested positive for *C. burnetii*.

The percentage of sheep that tested positive in the other 5 sheep flocks in the region ranged from 8% to 24% (8%; 12%; 12%; 16%; 24%).

We have described one of the largest Q fever outbreaks in Germany which, due to its point-source nature, provided the opportunity to assess many epidemiological features of the disease that can be rarely studied otherwise.

In 1954, more than 500 cases of Q fever were, similar to this outbreak, linked to the abortion of an infected cow at a farmers' market [15] . More recently a large outbreak occurred in Jena (Thuringia) in 2005 with 322 reported cases [16] associated with exposure to a herd of sheep kept on a meadow close to the housing area in which the cases occurred.

The first case control study served to confirm the hypothesis of an association between the outbreak and the farmers' market. The fact that only attendance on the second, but not the first day was strongly associated with illness pointed towards the role of the ewe that had given birth. Persons accompanying notified cases (source 5) were a mixture of adults and children and are therefore listed separately.

in the early morning hours of May 4, 2005 . This strong association and the very high attributable fraction among all cases suggested a point source and justified defining cases notified through the reporting system as outbreak cases if they were clinically compatible with Q fever and gave a history of having visited the farmers' market. The point-source nature of the outbreak permitted calculation of the incubation period of cases which averaged 21 days and ranged from 2 to 48 days with an interquartile range of 16 to 24 days. This is compatible with the literature [1] . An additional interview with the two cases with early onset (2 and 4 days after attending the market on May 4, Attack rates among adults and children in a most likely scenario and 8 other scenarios Figure 5 Attack rates among adults and children in a most likely scenario and 8 other scenarios. Most likely scenario: 3000 visitors, 83% adult visitors and 20% clinical attack rate among adults. Scenarios 1-8 varied in the assumptions made for "number of visitors", "proportion of adult visitors" and "attack rate among adults" (see Table 3). Displayed are attack rates and 95% confidence intervals. respectively) could not identify any other source of infection. A short incubation period was recently observed in another Q fever outbreak in which the infectious dose was likely very high [17] .

The second case control study among persons who visited the market on May 4 demonstrated that both close proximity to the ewe and duration of exposure were important risk factors. This finding was confirmed by the cohort study on vendors which showed that those who worked in a stand close to (within 6 meters) the sheep pen were at significantly higher risk of acquiring Q fever. The study failed to show a significant role of the location of the stand in reference to the wind direction, although we must take into

account that the wind was likely not always and exactly as reported by the weather station. However, if the wind had been important at all more cases might have been expected to have occurred among vendors situated at a greater distance to the sheep.

According to statutory surveillance system data, the proportion of clinical cases hospitalized was 25%, similar to the proportion of 21% found in persons pooled from the other studies conducted. Several publications report lower proportions than that found in this investigation: 4% (8/ 191) [7] , 5% [1] and 10% (4/39) [5] , and there was at least one study with a much higher proportion (63% (10/ 16)) [18] . It is unlikely that hospitals reported cases with Q fever more frequently than private physicians because the proportion hospitalized among Q fever patients identified through random telephone calls in the Soest population or those in the two cohorts was similar to that of notified cases. Thus reporting bias is an unlikely explanation for the relatively high proportion of cases hospitalized. Alternative explanations include overly cautious referral practices on the part of attending physicians or the presumably high infectious dose of the organism in this outbreak, e.g. in those cases that spent time in the sheep pen.

The estimated attack rate among adults in the four studies varied between 16% and 33%. The estimate of 23% based on the random sample of persons visiting the market on the second day would seem most immune to recall bias, even if this cannot be entirely ruled out. The estimation based on information about persons accompanying the cases may be subject to an overestimation because these individuals presumably had a higher probability of being close to the sheep pen, similar to the cases. On the other hand the estimate from the cohort study on vendors might be an underestimate, since the vendors obviously had a different purpose for being at the market and may have been less interested in having a look at the sheep. Nevertheless, all estimates were independent from each other and considering the various possible biases, they were remarkably similar. In comparison, in a different outbreak in Germany, in which inhabitants of a village were exposed to a large herd of sheep (n = 1000-2000) [5, 7] the attack rate was estimated as 16%. In a similar outbreak in Switzerland several villages were exposed to approximately 900 sheep [19] . In the most severely affected village, the clinical attack rate was 16% (estimated from the data provided) [19] . It is remarkable that in the outbreak described here, the infectious potential of one pregnant ewe -upon lambing -was comparable to that of entire herds, albeit in different settings.

Our estimate of the proportion of serologically confirmed cases that became symptomatic (50% (3/6)) is based on a very small sample, but consistent with the international literature. In the above mentioned Swiss outbreak, 46% of serologically positive patients developed clinical disease [7] .

Only approximately half of all symptomatic cases were reported to the statutory surveillance system. Patients who did not seek health care due to mild disease as well as underdiagnosis or underreporting may have contributed to the missing other half. Our estimated 3% attack rate among children is based on a number of successive assumptions and must therefore be interpreted with caution. Nevertheless, sensitivity analysis confirmed that adults had a significantly elevated attack rate compared to children. While it has been suggested that children are at lower risk than adults for developing symptomatic illness [7, 8] few data have been published regarding attack rates of children in comparison to adults.

The estimated *C. burnetii* seroprevalence in the sheep flocks in the area varied from 8% to 24%. The 25% seroprevalence in the flock of the exhibited animals together with a positive polymerase chain reaction in an afterbirth in June 2003 suggested a recent infection of the flock [20] . Seroprevalence among sheep flocks related to human outbreaks tend to be substantially higher than those in flocks not related to human outbreaks. The median seroprevalence in a number of relevant studies performed in the context of human outbreaks [7, 20, 21] , was 40% compared to 1% in sheep flocks not linked to human outbreaks [20] .

This outbreak shows the dramatic consequences of putting a large number of susceptible individuals in close contact to a single infected ewe that (in such a setting) can turn into a super-spreader upon lambing. There is always a cultural component in the interaction between people and animals, and these may contribute to outbreaks or changing patterns of incidence. During the past decades urbanization of rural areas and changes in animal husbandry have occurred [20] , with more recent attempts to put a "deprived" urban population "in touch" with farm animals. Petting zoos, family farm vacations or the display of (farm) animals at a market such as this may lead to new avenues for the transmission of zoonotic infectious agents [20, [22] [23] [24] . While not all eventualities can be foreseen, it is important to raise awareness in pet and livestock owners as well as to strengthen recommendations where necessary. This outbreak led to the amendment and extension of

existing recommendations [25] which now forbid the display of sheep in the latter third of their pregnancy and require regular testing of animals for *C. burnetii* in petting zoos, where there is close contact between humans and animals.

Due to the size and point source nature this outbreak permitted reassessment of fundamental, but seldom studied epidemiological parameters of Q fever. It also served to revise public health recommendations to account for the changing type and frequency of contact of susceptible humans with potentially infectious animals.

Abbreviations AFE = attributable fraction of cases exposed

The author(s) declare that they have no competing interests.

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Prediction of lncRNA-protein interactions using HeteSim scores based on heterogeneous networks

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5473862/>

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Abstract: Massive studies have indicated that long non-coding RNAs (lncRNAs) are critical for the regulation of cellular biological processes by binding with RNA-related proteins. However, only a few experimentally supported lncRNA-protein associations have been reported. Existing network-based methods are typically focused on intrinsic features of lncRNA and protein but ignore the information implicit in the topologies of biological networks associated with lncRNAs. Considering the limitations in previous methods, we propose PLPIHS, an effective computational method for Predicting lncRNA-Protein Interactions using HeteSim Scores. PLPIHS uses the HeteSim measure to calculate the relatedness score for each lncRNA-protein pair in the heterogeneous network, which consists of lncRNA-lncRNA similarity network, lncRNA-protein association network and protein-protein interaction network. An SVM classifier to predict lncRNA-protein interactions is built with the HeteSim scores. The results show that PLPIHS performs significantly better than the existing state-of-the-art approaches and achieves an AUC score of 0.97 in the leave-one-out validation test. We also compare the performances of networks with different connectivity density and find that PLPIHS performs well across all the networks. Furthermore, we use the proposed method to identify the related proteins for lncRNA MALAT1. Highly-ranked proteins are verified by the biological studies and demonstrate the effectiveness of our method.

Text: most commonly used approach is guilt-by-association (GBA) 19 , which provides the central top-down principle for analyzing gene networks in functional terms or assessing their quality in encoding functional information. New emerged methods, including the Katz method 20 , Combining dATa Across species using Positive-Unlabeled Learning Techniques(CATAPULT) 19 , Random Walk with Restart (RWR) 21 , and lncRNA-protein Interaction prediction based on Heterogeneous Network model (LPIHN) 22 , have extended the association from just direct protein interactions to more distant connections in various ways. The KATZ measure 20 is a weighted sum of the number of paths in the network that measures the similarity of two nodes. CATAPULT 19 is a supervised machine learning method that uses a biased support vector machine where the features are derived from walks in a heterogeneous gene-trait network. RWR 21 is a method for prioritization of candidate genes by use of a global network distance measure, random walk analysis, for definition of similarities in protein-protein interaction networks and it add weight to the assumption that phenotypically similar diseases are associated with disturbances of subnetworks within the larger protein interactome that extend beyond the disease proteins themselves. LPIHN 22 is a network-based method by implement a random walk on a heterogeneous network. PRINCE is a global method based on formulating constraints on the prioritization function that relate to its smoothness over the network and usage of prior

information. Compared with LPIHN and RWR, PRINCE propagates information in a smaller network but contains more connotative meaning when build the initial probability values and has made great performance in gene prioritization ²³ and disease identification ²⁴ .

However, many existing network-based methods simply view objects in heterogeneous networks as the same type and do not consider the subtle semantic meanings of different paths. In this paper, we adopt a method named HeteSim, which is a path-based measure to calculate the relevance between objects in heterogeneous network ²⁵ . The basic idea is that similar objects are more likely to be related to some other objects. Considering the relatedness of heterogeneous objects is path-constrained, HeteSim gives a uniform and symmetric measure for arbitrary paths to evaluate the relatedness of heterogeneous object pair (same or different types) with one single score. Due to the relevance path not only captures the semantics information but also constrains the walk path, the score is also a path-based similarity measure.

An example of HeteSim score is illustrated in (Fig. 1). The number of paths from A to C and B to C is 3 and 2, respectively. The walk count between A and C is larger than B and C, which might indicate that A is more closer to C than B. But the connectivity between B and C is more intense than A and C in the sight of HeteSim score, since most edges starting from B are connected with C, when A only has a small part of edges connected with C.

Here, we propose a method named PLPIHS (Fig. 2) to predict lncRNA-Protein interactions using HeteSim scores. We first construct a heterogeneous network consisting of a lncRNA-lncRNA similarity network, a lncRNA-protein association network and a protein-protein interaction network. Then, we use the HeteSim measure to calculate the score for each lncRNA-protein pair in the network. A SVM classifier is built based on the scores of different paths. We compare our PLPIHS with PRINCE, RWR and LPIHN and find that PLPIHS outperforms the other methods in many performance measures.

Validation measures. LOOCV(Leave-One-Out Cross Validation) ²⁶ is implemented on the verified lncR-NA-protein associations to evaluate the performance of LPIHN ²² . We leave a known lncRNA-protein pair in turn as the test sample and all the other known lncRNA-protein pairs are regarded as training samples. In order to improve the accuracy of PLPIHS, we remove all connected lncRNAs and proteins while in each validation round. Receiver

Operating Characteristic(ROC) curve 27 is used to evaluate the prediction performance, which plots true-positive rate (TPR, sensitivity or recall) versus false-positive rate (FPR, 1-specificity) at different rank cutoffs. When varying the rank cutoffs of successful prediction, we can obtain the corresponding TPR and FPR. In this way, ROC curve is drawn and the area under the curve(AUC) is calculated as well. For a rank threshold, sensitivity(SEN) 28 and specificity(SPE) 29 These measurements are also used to assess the capability of PLPIHS during the preprocessing procedure.

Affection of network preprocessing characteristics. In this paper, we only have two kinds of objects, lncRNA and protein. Thus, the paths from a lncRNA to a protein in our heterogeneous network with length less than six is listed in Table 1 . In order to pick out the most efficient paths, we compared the performances of these 14 paths under different combinations (Fig. 3) . We can see that all paths achieve a favorable status except path 1'~2'. Path 1'~14' obtains the best performance across all measures, which means that the path with length greater than three contains more significant meanings. The constant factor β is used to control the influence of longer paths. The longer the path length is, the smaller the inhibiting factor is. Path length equals 3 matches with constant β , path length equals 4 matches with constant β^2 and path length equals 5 matches with constant β^3 . Table 2 shows that β has tiny impact on the final results and $\beta = 0.2, 0.4$ and 0.7 achieved the best AUC score and the others are not far behind yet.

To further verify the dependability of our method, we compare the three networks of different connectivity density under different cutoff value 0.3, 0.5 and 0.9 (see lncRNA-Protein associations). The results are shown in Fig. 4 . There are tiny performance differences between different sparse networks. The AUC score of the 0.5 network is higher than that of others while the 0.9 network outperforms others in ACC, SEN, MCC and F1-Score. This suggests that PLPIHS performs well across networks with different densities. Table 1 . The paths from a lncRNA to a protein in our heterogeneous network with length less than six.

the RWR method, there is only one restart probability r and it's effects is very slight, which is proved by experiments. The parameter r is set as 0.5 in this comparison. In order to calculate the performance of the different methods, we use a leave-one-out cross validation procedure. We extract 2000 lncRNA-protein associations from the 0.9 network as positive samples, the same number of negative samples are chosen randomly from the

0.3 network as well, avoiding the error caused by imbalance dataset. The gold set which containing 185 lncRNA-protein interactions downloaded from NPinter database has been included in positive pairs as well. In the lncRNA protein prioritization, each lncRNA-protein interaction is utilized as the test set in turn and the remaining associations are used as training data. The whole experiment will be repeated 4000 times to testing each lncRNA-protein pairs in the dataset. ROC curve is drawn based on true positive rate (TPR) and false positive rate (FPR) at different thresholds. The AUC score is utilized to measure the performance. AUC = 1 demonstrates the perfect performance and AUC = 0.5 demonstrates the random performance. The ROC curve of PLPIHS, LPIHS, PRINCE and RWR are plotted in Fig. 5. The results show that the AUC score of PLPIHS in 0.3 network is 96.8%, which is higher than that of PRINCE, LPIHS and RWR, achieving an AUC value of 81.3%, 88.4% and 79.2%, respectively. Similarly, PLPIHS outperforms other methods in 0.5 network and 0.9 network as well.

Performance evaluation by independent test. For further validation, we also randomly selected 2000 lncRNA-protein associations from the rest of positive samples in 0.9 network and the same number of negative interactions are picked out from the remaining negative samples of 0.3 network to generate the independent test data set. Since the existing network based methods is not suitable for independent test, we only evaluate the performance for the proposed PLPIHS. The independent test results are shown in Fig. 6, an AUC score of 0.879 is achieved by PLPIHS, illustrating the effectiveness and advantage of the proposed approach. Case Studies. By applying the proposed PLPIHS method, novel candidate lncRNA-related proteins are predicted using LOOCV. We applied PLPIHS onto the 2000 known lncRNA-protein associations, which includes 1511 lncRNAs and 344 proteins to infer novel lncRNA-protein interactions. As a result, an area under the ROC curve of 0.9669, 0.9705 and 0.9703 (Fig. 5) is achieved using the three networks of different connectivity density, which demonstrate that our proposed method is effective in recovering known lncRNA-related proteins.

To further illustrate the application of our approach, a case study of lncRNA MALAT1 (ensemble ID: ENSG00000251562) is examined. MALAT1 is a long non-coding RNA which is over-expressed in many human oncogenic tissues and regulates cell cycle and survival [31]. MALAT1 have been identified in multiple types of physiological processes, such as alternative splicing, nuclear organization, epigenetic modulating of gene expression. A large amount of evidence indicates that MALAT1 also closely relates to

various pathological processes, including diabetes complications, cancers and so on 32, 33 .

MALAT1 is associated with 68 proteins in NPInter 3.0 34 . We construct the interaction networks of lncRNA MALAT1 by using the prediction results of these four methods (Fig. 7) . Among the 68 known lncRNA-protein interactions, PLPIHS wrongly predicts 6 interactions, while 13 associations are predicted mistakenly by PRINCE and RWR method and 15 lncRNA-protein pairs are falsely predicted by the LPIHN method.

We manually check the top 10 proteins in the ranked list under 0.5 network (Table 3) .Three of the top 10 predicted proteins have interactions with MALAT1, and most of them had high ranks in the predicted protein lists. For example, In the investigation of colorectal cancer (CRC), MALAT1 could bind to SFPQ, thus releasing PTBP2 from the SFPQ/PTBP2 complex and the interaction between MALAT1 and SFPQ could be a novel therapeutic target for CRC 35 . MALAT1 interacts with SR proteins (SRSF1, SRSF2, SRSF3 and SRSF5) and regulates cellular levels of phosphorylated forms of SR proteins 36 . And it is also as target of TARDBP to play the biological performance and found that TDP-43 bound to long ncRNAs in highly sequence-specific manner in tissue from subjects with or without FTLD-TDP, the MALAT1 ncRNA recruits splicing factors to nuclear speckles and affects phosphorylation of serine/arginine-rich splicing factor proteins 37, 38 . All these results indicate that our proposed method is effective and reliable in identifying novel lncRNA-related proteins.

lncRNAs are involved in a wide range of biological functions through diverse molecular mechanisms often including the interaction with one or more protein partners 12, 13 . Only a small number of lncRNA-protein interactions have been well-characterized. Computational methods can be helpful in suggesting potential interactions for possible experimentation 25 . In this study, we use HeteSim measure to calculate the relevance between lncRNA and protein in a heterogeneous network. The importance of inferring novel lncRNA-protein interactions by considering the subtle semantic meanings of different paths in the heterogeneous network have been verified 39 . We first construct a heterogeneous network consisting of a lncRNA-lncRNA similarity network, a lncRNA-protein association network and a protein-protein interaction network. Then, we use the HeteSim measure to calculate a score for each lncRNA-protein pairs in each path. Finally, a SVM classifier is used to combine the scores of different paths and making predictions. We compare the proposed PLPIHS with PRINCE, RWR and LPIHN and find that PLPIHS obtain

an AUC score of 0.9679 in 0.3 network, which is significantly higher than PRINCE, RWR and LPIHN (0.813, 0.884 and 0.7918, respectively). We also compare the performance of these four methods in networks of different connectivity density. As a result, PLPIHS outperforms the other method across all the networks. Moreover, when analysing the predicted proteins interacted with lncRNA MALAT1, PLPIHS successfully predicts 63 out of 68 associations, while PRINCE, RWR and LPIHN retrieve much lower interactions of 57, 57 and 53, respectively. And the top-ranked lncRNA-protein interactions predicted by our method are supported by existing literatures. The results highlight the advantages of our proposed method in predicting possible lncRNA-protein interactions.

Methods lncRNA-Protein associations. All human lncRNA genes and protein-coding genes are downloaded from the GENCODE Release 24 9 . A total of 15941 lncRNA genes and 20284 protein-coding genes are extracted. To obtain genome-wide lncRNA and protein-coding gene associations, we combine three sources of data:

- Co-expression data from COXPRESdb 40 . Three preprocessed co-expression datasets (Hsa.c4-1, Hsa2.c2-0 and Hsa3.c1-0) including pre-calculated pairwise Pearson's correlation coefficients for human were collected from COXPRESdb. The correlations are calculated as follows:

where $C(l, p)$ is the overall correlation between gene l (lncRNA) and protein-coding gene p , $C_d(l, p)$ is the correlation score between l and p in dataset d , D is the number of gene pairs (l and p) with positive correlation scores. Gene pairs with negative correlation scores are removed.

- Co-expression data from ArrayExpress 41 and GEO 42 . We obtained the co-expression data from the work of Jiang et al. 43 . RNA-Seq raw data of 19 human normal tissues are obtained from ArrayExpress (E-MTAB-513) and GEO (GSE30554). TopHat and Cufflinks with the default parameters are used to calculate the expression values. Pearson's correlation coefficients are used to evaluate the co-expression of lncRNA-protein pairs.
- lncRNA-protein interaction data. We download known lncRNA-protein interaction dataset from Protein-protein interactions. We obtain the protein-protein interaction (PPI) data from STRING database V10.0 45 , which contains weighted protein interactions derived from computational prediction methods, high-throughput experiments, and text mining. The

confidence scores are computed by combining the probabilities from the different evidence channels, correcting for the probability of randomly observing an interaction.

The HeteSim measure. The HeteSim measure is a uniform and symmetric relevance measure. It can be used to calculate the relatedness of objects with the same or different types in a uniform framework, and it is also a path-constrained measure to estimate the relatedness of object pairs based on the search path that connects two objects through a sequence of node types [39]. Further, the HeteSim score has some good properties (i.e., selfmaximum and symmetric), which have achieved positive performance in many studies [25]. In this study, we use HeteSim scores to measure the similarities between lncRNAs and proteins.

Definition 1 Transition probability matrix [39] L and P are two kinds of object in the heterogeneous network, $(L|P)$ $n \times m$ is an adjacent matrix between L and P , then the normalized matrix of $L|P$ along the row vector is defined as

$\frac{L|P}{L|P + 1}$ **Definition 2** Reachable probability matrix [39] In a heterogeneous network, the reachable probability matrix R for path $= + P_1 P_2 \dots P_n$ of length n , where P_i belongs to any objects in the heterogeneous network, can be expressed as

$\frac{P_1 P_2 \dots P_n}{P_1 P_2 \dots P_n + 1}$ Based on the definitions above, the steps of calculating HeteSim scores between two kinds of objects (lncRNA and protein) can be presented as follows:

- Split the path into two parts. When the length n of path is even, we can split it into $\frac{n}{2} = P_1 P_2 \dots P_{\frac{n}{2}}$

Otherwise, if n is odd, the path cannot be divided into two equal length paths. In order to deal with such problem, we need to split the path twice by setting $\frac{n-1}{2}$, respectively. Then, we can obtain a HeteSim score for each mid value, the final score will be the average of the two scores.

- Achieve the transition probability matrix and reachable probability matrix under the path L and R .
- Calculate the HeteSim score: where R^{-1} is the reverse path of R . An example of calculating HeteSim score is indicated in Fig. 8. We can see that there are three kinds of objects L , T and P in the network. The simplified steps of computing HeteSim score between l_3 and p_2 under the path $\pi = (LTP)$ is as follows:

- Split the path π into two components $\pi = LT$ ()

- Given the adjacent matrix I_{LT} and I_{TP} below, which means the interactions between lncRNAs and proteins, we can obtain the transition probability matrix T_{LT} and T_{TP} by normalizing the two matrix along the row vector. The PLPIHS method. Among a heterogeneous network, different paths can express different semantic meanings. For instance, a lncRNA and a protein is connected via 'lncRNA-lncRNA-protein' path or 'lncRNA-protein-protein' path representing totally different meanings. The former means that if a lncRNA is associated with a protein, then another lncRNA similar to the lncRNA will be potential associated with the protein. The latter shows that if a protein associated with a lncRNA, then another protein interacted with the protein will be likely associated with the lncRNA. Therefore, the potential information hidden in each path is extraordinary essential to be taken into account during prediction.

The PLPIHS framework is illustrated in Fig. 2. Firstly, we construct a heterogeneous network consisting of a lncRNA-lncRNA similarity network, a lncRNA-protein association network and a protein-protein interaction network. Three kinds of sparse networks are obtained from the heterogeneous network under different cutoff value 0.3, 0.5 and 0.9 (see lncRNA-Protein associations). The larger cutoff is, the network is more sparse. A total of 15941 lncRNAs genes and 20284 protein-coding genes are extracted as presented in Section 2.3. We randomly take out 1511 lncRNAs and 344 proteins to construct a smaller network for the following experiments in consideration of computing costs. The construction of the smaller heterogeneous networks under different cutoff values are shown in Table 4, where 'lnc2lnc' denotes the lncRNA-lncRNA network, 'lnc2code' denotes the lncRNA-protein network and 'code2code' denotes the protein-lncRNA network. Table 1. We use id to indicate the path combination, i.e., '1~2' represents path 'LLP' and path 'LPP'. Next, we calculate the heteSim score for each lncRNA-protein pair under each path. The results of different paths are used as different features. And we combine a constant factor β to inhibit the influence of longer paths. The longer the path length is, the smaller the

inhibiting factor is. Finally, a SVM classifier is built with these scores to predict potential lncRNA-protein associations. On the account of the HeteSim measure is based on the path-based relevance framework 39 , it can effectively dig out the subtle semantics of each paths.

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--- START OF ARTICLE 33 ---

Controlled efficacy trial confirming toltrazuril resistance in a field isolate of ovine *Eimeria* spp.

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6034276/>

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Abstract: **BACKGROUND:** Coccidiosis due to *Eimeria* spp. infections in lambs causes increased mortality and substantial production losses, and anticoccidials are important for control of the infection. Anticoccidial resistance has been reported in poultry and swine, and we recently described reduced toltrazuril efficacy in ovine *Eimeria* spp. in some Norwegian sheep farms using a newly developed faecal oocyst count reduction test (FOCRT). The aim of the present study was to use a controlled efficacy trial to assess the efficacy of toltrazuril against a field isolate suspected of being resistant. **METHODS:** Twenty lambs, 17–22 days old and raised protected against exposure to coccidia, were infected with a field isolate of 100,000 *Eimeria* spp. oocysts. This isolate was obtained from a farm

with a previously calculated drug efficacy of 56% (95% confidence interval: -433.9 to 96.6%). At day 7 post-infection, 10 of the lambs were orally treated with 20 mg/kg toltrazuril (Baycox Sheep vet., Bayer Animal Health), while the other 10 lambs (controls) were given physiological saline. Clinical examinations were conducted, and weight gains recorded. Daily faecal samples were scored for diarrhoea on a scale from 1 to 5, and oocyst excretion was determined using a modified McMaster technique. Oocysts were morphologically identified to species level. At 17–24 days post-infection, the lambs were euthanized and necropsied. RESULTS: The tested *Eimeria* isolate was resistant against toltrazuril, and resistance was seen in both pathogenic and non-pathogenic species. In addition, no significant differences in faecal score, growth, gross pathology or histological changes were identified between the two groups. The pathogenic *E. ovinoidalis* was the dominant species, and no significant difference in the individual prevalence of *E. ovinoidalis* post-treatment was found between treated (66.9%) and control lambs (61.9%). Other species identified included *E. crandallis/veybridgensis*, *E. parva*, *E. marsica*, *E. faurei*, *E. pallida*, *E. ahsata* and *E. bakuensis*. CONCLUSIONS: This study confirms toltrazuril resistance in ovine *Eimeria* spp.; in addition, the data support the use of FOCRT as an appropriate tool for field evaluation of anticoccidial efficacy. Due to limited anticoccidial treatment alternatives, these findings may have important implications for the sheep industry, particularly in northern Europe. ELECTRONIC SUPPLEMENTARY MATERIAL: The online version of this article (10.1186/s13071-018-2976-4) contains supplementary material, which is available to authorized users.

Text: Anticoccidial resistance (ACR), which develops mainly as a result of intensive long-term use of anticoccidial drugs, occurs widely in poultry production and has also been identified in *Cystoisospora suis* in piglets [1] [2] [3] [4] [5] . In addition, a field method for the evaluation of reduced anticoccidial efficacy (ACE) in ovine *Eimeria* spp., the faecal oocyst count reduction test (FOCRT), has recently been developed and indicated that the efficacy of toltrazuril is reduced in some Norwegian sheep flocks [6] .

Infections with *Eimeria* spp. may impact both animal welfare and productivity in the sheep industry, and controlling the infection is important to minimise mortality and morbidity, and to ensure that lamb growth is not compromised [7] [8] [9] . Suggested strategies to control ruminant coccidiosis include pasture management, adequate nutrition, and hygienic measures [10, 11] . However, these measures are often difficult to implement in practice, and the main control approach is often metaphylaxis with anticoccidials [12] [13] [14] [15] . Metaphylactic administration of a single oral dose of toltrazuril in the prepatent period has

been shown to be effective at reducing clinical signs and maintaining adequate lamb growth rates in different production systems [13, [15] [16] [17] [18] [19] . In contrast, treatment of clinical coccidiosis is considered inefficient due to the extensive intestinal damage already caused by the infection [20, 21] . Loss of sensitivity to toltrazuril, the only anticoccidial registered for use in sheep in the Nordic countries [22] [23] [24] , should therefore be a matter for serious concern for lamb production.

The World Association for the Advancement of Veterinary Parasitology guidelines for evaluation of ACE in mammals [25] , states that there is a need for verified methods for evaluation of ACE. Field methods for assessment of drug efficacy, such as the FOCRT [6] and the faecal egg count reduction test used to evaluate anthelmintic efficacy [26] , give only an indication of reduced efficacy, and need verification through controlled efficacy trials (CET) [27, 28] . In addition, due to the variation in pathogenicity between ovine *Eimeria* spp., the differentiation of species should be considered separately [25] .

The aim of the present study was to perform a CET in order to determine whether different species in a field isolate of ovine *Eimeria* spp. with suspected ACR, based on the FOCRT [6] , actually demonstrated resistance to toltrazuril.

A total of 20 lambs from 8 ewes of the Norwegian White Sheep breed ("Norsk kvit sau") was included in the study, which was approved by the Norwegian Animal Research Authority (ID: 11657). The ewes were synchronised using Chronogest® CR and PMSG® (MSD Animal Health, Buckinghamshire, UK) and served by natural mating. Lambs were either snatched at birth (n = 16) or delivered by caesarean section (n = 4) over a period of 6 days, and thereafter reared artificially. Individual ear tags were used for identification. Directly after birth, all lambs were washed with Optima pH 4 soap (Optima Produkter AS, Norheimsund, Norway) and dried before being placed in boxes with expanded metal floors, in groups of four. Infrared heaters were used during the whole trial. An overview of the study groups, including lamb age, birth weight and gender can be found in Additional file 1: Table S1 .

Lambs received ovine colostrum from ewes vaccinated against *Clostridium* spp. (Covexin-8, Zoetis) during the first 30 min of life, followed by colostrum from vaccinated cows (Covexin-8, Zoetis) during the next 24 h. To avoid cases of haemolytic anaemia, the cow-colostrum had previously been tested on naturally reared lambs. Lambs were then fed ad

libitum with a commercial milk replacer (Denkamilk, Denkavit, Fiskå, Mølle, Stavanger), using an automatic feeding system (Holm & Laue, Godkalven, Figgjo, Norway). The lambs had ad libitum access to water, hay and commercial lamb-starter concentrate (FORMEL lam vår, Felleskjøpet, Norway). To ensure that transmission of *Eimeria* to the lambs via contaminated colostrum and hay could not occur, both were frozen at -75°C for a minimum of 24 h, prior to provision to the lambs.

The field isolate of *Eimeria* spp. was obtained from one of the flocks (ID 35) participating in the recent FOCRT study [6]. According to the FOCRT results, toltrazuril had reduced efficacy against *Eimeria* in two flocks. However, neither of these flocks were available for the CET, due to geographical and practical reasons. Thus, treatment with toltrazuril in the selected flock had been found to have an efficacy of 56.0%, but the results were classified as inconclusive, due to the wide 95% confidence interval (CI) of -433.9 and 96.6% [6].

To obtain sufficient *Eimeria* oocysts of this mixed field isolate (named "NMBU ID 35") for the present study, faecal samples were obtained from 35 lambs in this flock 9 days after toltrazuril treatment (Baycox® Sheep vet., Bayer Animal Health, Oslo, Norway). Oocysts were isolated according to Jackson [29] with some modifications. Briefly, faeces were mixed 1:1 with water and filtered. The faecal mix filtrate was subsequently mixed 1:1 with saturated sugar-solution (density: 1.5 g/l) in a plastic container and left to float onto a glass slide. The slide was washed every second hour with deionized water for three consecutive days, and the washings collected. The washings were centrifuged at 2300× g for 20 min, the supernatant discarded and the sediment mixed 1:1 with deionized water in a glass flask with constant aeration. The oocysts in the flask were left to sporulate for 7 days at room temperature. Sporulated oocysts were stored for 18 days at 4°C. Based on morphology [30], as seen by light microscopy at 400× magnification (see also Faecal samples section), and classification of 300 oocysts, the field isolate consisted of *E. parva* (32%), *E. crandallis/ weybridgensis* (25%), *E. ovinoidalis* (24%), *E. faurei* (9%), *E. marsica* (8%), *E. pallida* (1%), *E. ahsata* (< 1%) and *E. bakuensis* (< 1%).

All lambs were infected (day 0) at 17-22 days of age, using an oesophageal tube. A dose of approximately 100,000 sporulated oocysts, diluted in water to a total volume of 5 ml, was given to each of the 20 lambs. Then, two randomly selected (coin toss) lambs from each group of four were orally treated (day 7) with 0.4 ml/kg toltrazuril (Baycox® Sheep vet. 50

mg/ml, Bayer Animal Health) and the remaining lambs (controls) were given 0.4 ml/kg of 0.9% NaCl (B. Braun Medical AS, Vestskogen, Norway).

Clinical examinations were performed daily throughout the trial. Rectal temperature was measured at days 0, 1, 2 and 7, and daily from day 14, and temperatures $> 40.5^{\circ}\text{C}$

were considered as fever. The lambs were weighed once a week using a calibrated weight (Kruuse, Drøbak Norway) with 0.1 kg sensitivity, until 14 days post-infection, and thereafter three times a week.

Two lambs (controls) were treated orally with trimethoprim/sulphamethoxazole (Bactrim, Roche, Etterstad, Norway) during the first three days of life due to suspected *Escherichia coli*-infection, from which both recovered within 48 h. Six lambs, two controls and four treated with toltrazuril, developed lameness due to interdigital abscessation, and *Streptococcus aureus* was detected in two lambs. Four lambs recovered without treatment, and two of the lambs recovered after treatment with benzylpenicillinprocaine (Penovet vet., Boehringer Ingelheim Vetmedica, Copenhagen, Denmark) administered intramuscularly for three days.

On clinical examination, special attention was paid to clinical signs associated with *Eimeria* spp. infections, i.e. dehydration, pyrexia, weakness, anorexia and, in particular, the presence of diarrhoea.

Severe haemorrhagic diarrhoea and dehydration in one lamb at day 17, led to euthanasia of that whole group of four lambs. At day 18, another lamb showed signs of haemorrhagic diarrhoea, and all lambs in this group were also euthanized. The remaining three groups were euthanized on days 21, 23, and 24.

Blood samples were drawn from v. jugularis using vacuette tubes (plain and EDTA-treated; BD, Franklin Lakes, USA) at 48 ± 2 h after birth and at days 0, 7 and at euthanasia. Haematology was performed using the ADVIA 120 Haematology system (Bayer Diagnostics, Leverkusen, Germany). Dehydration was considered with a haematocrit (hct)

of > 45.0% [31] . Whole blood tubes were centrifuged, and the serum removed and stored at -20°C until further analysis. Biochemical analysis was performed by ABX Pentra 400 (Horiba, Les Ulis, France), and included analysis of iron, total protein, albumin, urea, creatinine, gamma-glutamyl transferase, glutamate dehydrogenase and beta hydroxybutyric acid.

Individual faecal samples from each of the lambs were obtained daily from day 10 of life until the end of the experiment. Visual scoring of faecal consistency was performed on a scale from one to five (1: normal, pelleted; 2: soft; 3: liquid; 4: watery; 5: watery with blood and/or intestinal tissue) [32] . A score ≥ 3 was considered as diarrhoea.

Samples were collected using an in-house "faecal spoon" [6] and the faecal samples were put in zip-lock bags, which were vacuum packed (Fresh'n'easy, OBH Nordica, Sundbyberg, Sweden), stored at 4°C, and analysed within 37 days. The rate of oocyst excretion was determined using a modified McMaster technique with a theoretical sensitivity of 5 oocysts per gram (OPG) [6] . One hundred *Eimeria* oocysts from all samples ≥ 1000 OPG were examined by light microscopy at 400 $\times$ magnification and identified to species level, using morphological criteria [30] . However, due to their morphological similarity, oocysts of *E. crandallis* and *E. weybridgensis* were not differentiated.

Oocyst counts were analysed by the FOCRT [6] , which consists of a two-step procedure. First, timing of treatment and sampling was evaluated, followed by evaluation of treatment efficacy, by comparing post-treatment faecal samples from treated lambs with equivalent samples from untreated controls. Pre-treatment samples (sample 1) were obtained on day 7 (day of treatment), and post-treatment samples (sample 2) were obtained on days 14-18. The FOCRT was then run using the post-treatment oocyst counts for all five possible time intervals (7-11 days) between samples 1 and 2.

Faecal samples obtained at euthanasia were analysed for rotavirus, coronavirus, *Cryptosporidium* spp. and general bacteriology. Additional testing for *Cryptosporidium* spp. was performed in diarrhoeic lambs at the time of infection (day 0, n = 10). Faecal smears were analysed at the Norwegian Veterinary Institute in Oslo for *Cryptosporidium* by direct immunofluorescence analysis (Crypt-a-Glo™, Waterborne Inc., New Orleans, USA), whereas presence of rotavirus and coronavirus were tested by standard diagnostic

methods. Samples for bacteriological analyses were obtained from mid-jejunum and the colon spiral, spread on sheep blood agar plates, and incubated under anaerobic and aerobic conditions for 24-48 h at 37°C and 5% CO₂. In cases of haemorrhagic diarrhoea, additional samples were grown on bromothymol-blue lactose cysteine agar (brolactin/CLED agar) for potential identification of *Salmonella* [33].

Lambs were euthanized at days 17-24, by intravenous injection with pentobarbital (Euthasol vet., Virbac, Sollihøgda, Norway) at 140 mg/kg. Standard necropsy was performed immediately thereafter, with emphasis on the intestines.

Histological samples were taken from mid-jejunum, proximal and distal ileum, mid and base of caecum, colon spiral, and distal colon, in addition to heart, lung, liver and kidney. The samples were immersion-fixed in 4% formaldehyde, paraffin-embedded, and stained with haematoxylin and eosin (HE). Histological evaluation was performed by light microscopy and a blinded semi-quantitative evaluation (single evaluator) was done to assess intestinal pathology. Evaluation parameters included changes in: (i) villi, (ii) surface epithelium (atrophy/attenuation), (iii) degree of *Eimeria*-infection, (iv) hyperaemia, (v) oedema, (vi) infiltration of inflammatory cells and (vii) crypt abscesses, and were scored as follows: 0 = minimal; 1 = little; 2 = moderate; 3 = severe, including half-step grading. In addition, the presence of epithelial necrosis was graded as present (1) or absent (0). A total histology score was calculated for each tissue by summation of all parameters evaluated (i-vii).

Data were managed in Excel 2013 (Microsoft Inc., Redmond, USA), and subsequently analysed in R [34] and Stata 14 (Stata Statistical Software: Release 14. Stata-Corp LP, College Station, TX, USA). Evaluation of efficacy was performed according to the FOCRT [6]. For calculations of significance based on means, a t-test was used. $P < 0.05$ was considered significant.

Mean growth rates were above 300 g/day until days 14-16, whereupon mean growth rate decreased to around 0 g/day (Fig. 1). Growth rates increased again from day 21 onwards. The same pattern was observed in both treated and control lambs.

From day 15, both treated and control lambs had a mean faecal score of ≥ 3 , indicating diarrhoea. The maximum mean faecal score was seen at day 17 (3.9 ± 0.2) and day 18 (4.4 ± 0.3) in the treated and control groups, respectively. Haemorrhagic diarrhoea was seen from day 14, in two treated and five control lambs, and tenesmus was observed in two control lambs (day 17).

An increase in rectal temperature was seen from day 14, with maximum temperatures measured at day 18 ($40.4 \pm 0.4^\circ\text{C}$) and 16 ($40.9 \pm 0.4^\circ\text{C}$) in the treated and control groups, respectively. The mean duration of fever ($> 40.5^\circ\text{C}$) was 2.3 ± 0.5 days and 1.9 ± 0.4 days for the treated and control groups, respectively. For these parameters, no significant difference between groups were seen at any time.

At euthanasia, the mean hct was $39.2 \pm 1.7\%$ and $41.4 \pm 1.9\%$ in the treated and control groups, respectively. However, dehydration (hct $> 45.0\%$) was only seen in 3 lambs, of which one had been treated with toltrazuril. Mean total serum protein decreased in both groups from infection to euthanasia, but no significant differences between the groups were observed. Other biochemical parameters were within normal ranges (data not shown).

Oocyst excretion was first recorded in one treated lamb at day 10 (10 OPG), followed by oocyst excretion in all lambs in both groups from day 14 onwards. Peak oocyst excretion was seen in the treated group at day 20 (mean OPG: 5,438,500), and in the control group at day 21 after infection (mean OPG: 3,630,850) (Fig. 2) . Thereafter, oocyst excretion decreased. There was no significant difference in oocyst excretion and species distribution between the groups at any time. All species present in the field isolate were isolated from the faecal samples of all the 20 infected lambs. *E. ovinoidalis* was the most prevalent species in both treated and control lambs (Table 1) .

Efficacy, according to the FOCRT, was evaluated with confidence if the slope was ≥ 0.75 , and with caution if slope was ≥ 0.5 and < 0.75 [6] . The slope ranged from 1.24 to 1.69 for the total oocyst excretion in the control lambs.

Slopes, maximum likelihood estimates, and 95% CIs for the geometric mean efficacy of all oocysts, *E. ovinoidalis*, *E. crandallis/veybridgensis*, and the non-pathogenic *Eimeria* spp. are presented in Table 2 ; reduced efficacy of toltrazuril is apparent against both pathogenic and non-pathogenic species. The slope was ≥ 0.75 for all time intervals and species, except for four of the five time intervals of *E. crandallis/veybridgensis*.

Samples analysed for *Cryptosporidium* spp., *Salmonella*, coronavirus and rotavirus were all negative. Bacteriological analyses showed a mixed flora, dominated by coliforms and *Enterococcus* spp.

Gross pathological findings included diffused thickened and folded ileal mucosa (7 treated and 7 controls), and fibrinous ileal content in two lambs (one treated and one control). Nodular or plaque-like foci in the ileal mucosa were seen in 4 treated and 6 control lambs (Fig. 3a). The regional distal jejunal lymph nodes were moderately increased in size and oedematous in 5 treated and 6 control lambs. Finally, watery abomasal content was seen in > 50 % of the animals in both groups.

Microscopy evaluation showed lesions, mainly in the ileum, caecum and colon, with minor lesions in the jejunum (Fig. 3b-f). However, there were no significant differences with respect to histological scores between the treated and control groups in any of the intestinal segments. The highest calculated histological score was found in the proximal ileum and at the base of caecum (Fig. 4) . The mean score for each parameter can be found in Additional file 2: Table S2 . Varying quantities of intracellular *Eimeria* stages were observed in all intestinal segments, except from jejunum, and they were mostly located in the villus epithelium, with fewer parasites in the crypt epithelium and lamina propria, and few in the submucosa and lymphatic vessels. In both treated and control lambs, changes in the intestinal surfaces varied from light atrophy of the jejunal epithelium and blunting of affected ileal villi (Fig. 3b) , to areas of total flattening, attenuation of surface epithelium (Fig. 3e) and necrosis (Fig. 3d) . Patches of epithelial necrosis were found in all lambs. Infiltration of inflammatory cells included mostly monocytes and eosinophils, but also neutrophils and macrophages, and was found in both the lamina propria and submucosa. Different degrees of oedema, hyperaemia, and haemorrhage were seen in all tissue sections examined, and in both treated and control lambs. Crypt abscesses (Fig. 3b) were found in varying degree in all lambs, and contained inflammatory cells, debris and different stages of *Eimeria* spp.

As far as we know, this is the first report of experimentally confirmed toltrazuril resistance in a field isolate of ovine *Eimeria* spp. The results also support the use of FOCRT as a tool to evaluate ACE in the field. Although ten of the 20 lambs experimentally infected with *Eimeria* were metaphylactically treated with the recommended dose of 20 mg/kg toltrazuril (Baycox® Sheep vet., Bayer Animal Health), this treatment did not result in a significant reduction in oocyst excretion in the treated animals, compared with the controls. In addition, no significant differences were noted in clinical presentation, gross pathology, and histopathological findings. The speciation data showed that both pathogenic and non-pathogenic species of *Eimeria* in this isolate were resistant to toltrazuril. The lambs excreted high numbers of oocysts, as has previously been recorded in experimental infections with multiple *Eimeria* spp. [35]. Although oocyst excretion decreased from around day 20 after infection, the total duration of excretion could not be determined, as the lambs were euthanized. The excretion pattern noted here, with an exponential increase, a plateau phase, and a decline, has previously been noted in experimental infections [35] [36] [37]. However, due to continuous reinfection under natural field conditions, the duration of oocyst excretion may be longer [38, 39] than observed in the present study. This might also explain why the calculated slope seen for all species in this experimental study is higher than the slopes reported from the preceding field trial [6].

Multi-species resistance, as observed here, has also been noted in field isolates of avian *Eimeria* spp. [3, 40]. Notes: The estimates were based on post-treatment oocyst counts for five time intervals between sample 1 (day 7 after infection) and sample 2, and was calculated according to the FOCRT [6]. A slope ≥ 0.5 and < 0.75 was evaluated with caution, whereas a slope < 0.5 was interpreted as invalid. Four lambs were euthanized at day 17. Abbreviations: *E. ovi*, *E. ovinoidalis*; *E. c/w*, *E. crandallis/veybridgensis*; Non-pathogenic, all species except *E. ovinoidalis* and *E. crandallis/veybridgensis*.

Of particular importance in this study is that *E. ovinoidalis* was the dominant species excreted from infected lambs. As this species is one of the most pathogenic *Eimeria* spp. in sheep [41, 42], resistance against the most commonly used anticoccidial drug indicates that severe clinical coccidiosis may be expected to occur in resistant flocks. Although *E. ovinoidalis* was the dominant species excreted, the most prevalent species in the original field-isolate inoculum was *E. parva*. This could reflect similarities between *E. ovinoidalis* and *E. ninakholyakimovae* in goats, the latter of which develops macroschizonts in

endothelial cells, resulting in the release of thousands of merozoites [42, 43] . Thus, the extent of intracellular multiplication/ replication, which is presumably also related to the extent of pathogenicity associated with this species, is higher for *E. ovinoidalis* than for the other *Eimeria* species.

For *E. crandallis*/*weybridgensis*, the FOCRT calculations showed invalid results from three of the five sampling time points, probably due to the tests being performed too early in the infection. Excretion of *E. crandallis*/*weybridgensis* increased predominantly from day 16 onwards, and euthanasia was performed at days 17-24. Thus, the longer prepatent periods for these species compared with *E. ovinoidalis* [44] probably explain these results. This is an important finding, as the number of invalid farms tested in the FOCRT [6] might have been fewer should sample 2 have been collected 10-11 days after sample 1. These findings also highlight the fact that although *Eimeria* spp. are often considered as a relatively uniform group, they are in fact separate species with potentially important differences in biology and pathogenic potential. Two of the lambs were treated with trimethoprim/ sulpha during their first days of life, preparations that have been shown to be effective in treating ovine coccidiosis [45, 46] . However, withdrawal periods for comparable drugs licenced in cattle are 10-15 days for meat [47] , and these lambs were treated > 17 days prior to the experimental infection. In addition, these treated lambs were in the control group, and therefore this treatment should not have affected the results of the study.

Similar clinical signs as observed here might be caused by *Cryptosporidium* spp., coronavirus, rotavirus, and *Salmonella* spp., but none of these pathogens were detected. In addition, the findings of coliforms and *Enterococcus* spp. may be considered as normal intestinal flora of lambs [48] . The observed clinical signs were therefore almost certainly caused by *Eimeria* spp., particularly the two major pathogenic species, *E. ovinoidalis* and *E. crandallis* [35, 36] . Thickened ileal mucosa is often seen in lambs infected with *E. ovinoidalis* [49] . In addition, the histological changes, such as blunted villi and surface necrosis, as well as the presence of coccidia, hyperaemia, oedema, infiltration of inflammatory cells and crypt abscesses, are also in accordance with previous reports [42, 50, 51] .

To improve our study, an additional group of uninfected lambs might have been advantageous as this would have enabled better comparisons between weight gain and histopathological changes. However, this was not feasible at the time of the study.

Furthermore, due to the lack of defined cut-off values for ACE, it might have been advantageous to include an oocyst isolate from a non-suspected farm (i.e. a susceptible isolate) [25] . This would have enabled comparisons of different parameters, such as oocyst excretion, between treated and control lambs infected with susceptible or resistant *Eimeria* spp. However, due to lack of tools for selection of such susceptible ovine *Eimeria* isolates, we therefore chose to restrict our CET to treated and control lambs infected with isolate "NMBU ID 35" as a first step in the characterisation of anticoccidial resistance in ovine *Eimeria* spp.

Although the initial efficacy values have not been provided for toltrazuril by the manufacturer, several studies have investigated its effect on oocyst excretion. For example, its efficacy has been found to be 96.9-99.9% in the period from 7 to 98 days after first treatment, in a study in which the lambs were treated every 14 days [52] . Other studies have shown toltrazuril efficacies [either provided in the publication or calculated as $1 - (\text{mean OPG treated group}) / (\text{mean OPG control group})$ from data in the publication] ranging from 90.0 to 100.0% in the period from two to three weeks after treatment [13, 18, 19, [53] [54] [55] [56] . These efficacies are far higher than that calculated in the present study, and therefore the comparative data provides a further clear indication of resistance in the "NMBU ID 35" isolate.

Toltrazuril has been marketed for anticoccidial treatment in sheep since the 1980s, and its use has increased during recent years, both in Norway [57] and in the UK (Dr Gillian Diesel, personal communication). Extensive use of a drug over time may result in decreased efficacy, possibly due to the haploid stages of *Eimeria*, which immediately select for resistance [1, 5] . Since toltrazuril is the only registered anticoccidial for sheep in several countries, development of resistance in ovine *Eimeria* species may result in there being few treatment options available for sheep farmers, especially in northern Europe [22] [23] [24] . Diclazuril is an anticoccidial that has been registered for treatment of sheep in several countries, but as it may share a common mode of action to that of toltrazuril [58] , cross-resistance between these two triazine-derivates in ovine *Eimeria* spp. seems highly likely and should be investigated. Indeed, cross-resistance between diclazuril and toltrazuril was reported for an isolate of avian *Eimeria* spp. over 20 years ago [3] .

Our results indicate that there is a clear need for tools for evaluating ACE, such that inefficient treatments and, thus, the potential for reduced animal welfare and productivity

can be avoided. Such tools are available for poultry, using different metrics, such as oocyst index, body weight gain, relative weight gain, lesion scores and anticoccidial index [59] . However, such methods have not yet been established for use in ruminants [25] , with Fig. 4 Box-and-whisker plots with outliers illustrating the histology score. The score was a summation of all histological parameters evaluated (see text) in the 20 *Eimeria* spp. infected lambs, red: toltrazuril treated, and blue: controls the exception of the newly published FOCRT [6] . Although FOCRT may serve as a tool for field evaluation of ACE, there is a clear requirement for further testing of its use in different settings.

Confirmation of the spectre of resistance in ovine *Eimeria* species increases the urgency of identifying alternative treatments and optimising other control strategies. The anticoccidial effects of different plants and natural extracts, such as sainfoin (*Onobrychis viciifolia*), carob pods (*Ceratonia siliqua*), pomegranate (*Punica granatum*) peel extract, grape seed proanthocyanidin extracts, and different natural antioxidants, have been investigated in vivo and in vitro in different hosts [60] [61] [62] [63] [64] . However, none of these bioactive substances have, as yet, been brought to the market for the prevention of clinical coccidiosis. In addition, there are vaccines available for avian *Eimeria* spp. [65, 66] , and successful immunisation of goat kids with attenuated *Eimeria* spp. oocysts has been performed [67] .

Future studies are necessary in order to develop a commercial vaccine against ovine *Eimeria* spp. Therefore, current efforts should focus on identifying ACE, and maintaining the efficacy of toltrazuril in susceptible flocks. Management strategies that decrease the need for anticoccidials by reducing the infection pressure, possibly achieved by applying strict hygienic measures, and improved flock and pasture management should be actively encouraged by veterinarians and agricultural policy incentives [11] . Additionally, farmers should be informed about the importance of correct drenching techniques, including dosage estimation and drench gun calibration, as these have been shown to be inadequate in several farms [12] .

To our knowledge, this is the first report of ACR against toltrazuril in an ovine *Eimeria* field isolate, which included the highly pathogenic species, *E. ovinoidalis*. The results also support the use of FOCRT for field evaluation of ACE. However, the distribution and prevalence of ACR is unknown and further studies are warranted. In the future, difficulties

in managing coccidiosis without chemotherapy, due to few available treatment options, may severely affect both animal welfare and the economy of the sheep industry.

Additional file 1: Table S1 . Information about the 20 lambs infected with *Eimeria* spp. at day 0. (PDF 22 kb) Additional file 2: Table S2 . Histopathological findings from toltrazuril treated lambs and controls euthanized 17-24 days post-infection with 100,000 *Eimeria* oocysts. (PDF 118 kb) Abbreviations ACE: anticoccidial efficacy; ACR: anticoccidial resistance; CET: controlled efficacy trial; FOCRT: faecal oocyst count reduction test; hct: haematocrit; OPG: oocysts per gram

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Multimodal Imaging in an Unusual Cluster of Multiple Evanescent White Dot Syndrome

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Abstract: **OBJECTIVE:** To describe an unusual cluster of multiple evanescent white dot syndrome (MEWDS) encountered within a 3-month period. **METHODS:** This retrospective observation study is comprised of seven patients who presented with MEWDS in a 3-month period in central Israel. Data were collected from patients' medical records on clinical, multimodal imaging, and viral serology findings. **RESULTS:** Six women and one man of

mean age 31.5 ± 7.2 years. Three reported a precedent viral infection. All had unilateral decreased vision. Funduscopy revealed foveal granularity. **MAIN IMAGING FINDINGS:** Hyperfluorescent spots on blue autofluorescence (BAF), hypofluorescent spots on indocyanine green angiography, dark lesions on infrared photos, and ellipsoid zone irregularities on spectral domain optical coherence tomography (SD-OCT). Resolution of the spots on BAF correlated with anatomic (SD-OCT) and visual recovery. OCT angiography performed following the convalescence stage demonstrated intact retinal and choroidal flow. Serologic findings were inconclusive. **CONCLUSION:** We report a unique cluster of MEWDS patients presented in a short period of time. SD-OCT findings of ellipsoid zone disruption in combination with other multimodal imaging modalities are outlined meticulously. Recognizing these imaging features along with high index of clinical suspicion is important for the diagnosis of MEWDS. Serologic testing might be considered in future patients.

Text: Multiple evanescent white dot syndrome (MEWDS) was first described in 1984 as a rare, sudden onset of unilateral chorioretinopathy, with the predominant sign being multifocal yellow-white spots throughout the retina [1, 2]. The clinical spectrum of MEWDS has expanded over the years to include bilaterality and recurrences [3] or an atypical presentation involving the fovea without the white spots [4]. Symptoms include acute onset of decreased visual acuity unilaterally accompanied in most cases by photopsia and scotomata. A prodromal flu-like illness has been reported in up to 50% of cases [1]. One report described a patient with elevated levels of total serum IgG during the disease course and negative findings for IgM to herpes zoster, herpes simplex, mumps, and measles [5].

Although MEWDS is suspected to occur as a consequence of a viral-like infection in genetically susceptible individuals, its precise pathogenesis remains unknown. Recovery is gradual, over weeks to months, and the visual prognosis is very favorable [2]. Treatment is usually not required.

The incidence of MEWDS is unknown. Only small case series are reported in the literature [4] [5] [6] [7] [8] [9] [10] [11] [12]. One of the largest described 34 affected patients reviewed over several years' period [1, 13, 14].

The aim of the present report was to describe an unusual cluster of seven cases of MEWDS encountered within a 3month period, with an emphasis on the clinical presentation and multimodal imaging findings. The cluster prompted us to seek a common infectious association.

A retrospective observational study was conducted in seven patients who presented with MEWDS between July and September 2013 at two tertiary medical centers in central Israel. Data on background, clinical, and laboratory parameters were collected from the medical files. The study was approved by the institutional ethics review board.

All patients underwent a comprehensive ophthalmic examination and multimodal imaging tests, including blue autofluorescence (BAF), fluorescein angiography (FA) and/ or indocyanine green angiography (ICGA), infrared (IR) photography, and spectral domain optical coherence tomography (SD-OCT). Images were acquired with the HRA-2 and the Spectralis HRA + OCT devices (Heidelberg Engineering, Heidelberg, Germany) at the following wavelengths: BAFexcitation 488 nm, barrier cut-off 496 nm; IR-820 nm; ICGA-excitation 790 nm, emission 800 nm; and SD-OCTsuperluminescent diode light source 870 nm. The volume scan option was used to acquire the multiple SD-OCT scans (25-49 horizontal scans over a 6 mm region covering the area of pathology). Precise registration between findings seen on IR or BAF and SD-OCT was enabled by the dual-beam laser eye-tracking system, where one laser is used to image the retina and the other laser to perform the OCT scans. Accurate rescanning in areas of interest was ensured by the Spectralis follow-up function which automatically places subsequent scans on the same location as the previous ones.

OCT angiography images were acquired using the RTVue XR Avanti with AngioVue (Optovue Inc., Fremont, California, USA), with an A-scan-rate of 70 000 scans per second, a light source of 840 nm, and a bandwidth of 45 nm. Macular cubes (3 × 3 mm) were acquired, each cube consisting of 304 clusters of 2 repeated B-scans containing 304 A-scans each. Split-spectrum amplitude decorrelation technology was employed to improve the signal-to-noise ratio by splitting the spectrum to generate multiple repeat OCT frames from 2 original repeat OCT frames [15] .

Motion correction was performed using registration of 2 orthogonally captured imaging volumes. Automatic segmentation of the retinal layers was performed by the viewing software and was used to generate en face projection images after adjusting the level of the segmented layer on the B-scans.

Serology testing was performed for viruses commonly present at the time of the patients' presentation, namely, immunoglobulin IgG and IgM for herpes simplex virus (HSV) I-II, varicella zoster virus (VZV), West Nile virus, coxsackievirus, echovirus (subgroup of enterovirus), and corona virus.

Findings. There were one male and six female patients of mean age 31.5 ± 7.2 years (range 22-41 years). Table 1 summarizes the demographic data. Three patients reported a prodromal virus infection.

All patients presented with acute onset of unilateral decreased vision. The best corrected visual acuity at presentation ranged from 6/9 to 6/30 in the affected eye. None of the patients had signs of anterior or vitreous inflammation in the affected eye. Funduscopy findings at presentation included foveal granularity in six patients; in four patients (patients 1, 4, 5, and 6), it was the sole pathologic retinal finding (Figure 1); and in three patients (patients 2, 3, and 7), foveal granularity was associated with faint white retinal lesions (Figure 2), located mainly in the midperipheral retina extending to the periphery. Patient 6 had a swollen disc and mild signs of optic neuropathy (mild red desaturation, enlarged blind spot on visual field). Patient 6 underwent neurological evaluation due to initial presentation mimicking optic neuritis. Neurological evaluation including full neurological exam and neuroimaging excluded additional neurological deficit, before the diagnosis of MEWDS was established. The clinical findings are summarized in Table 2. 3.2. Multimodal Imaging Findings. Patients who underwent imaging less than 2 weeks from onset of symptoms had the most typical findings.

BAF revealed hyperautofluorescent lesions in the macula between and along the arcades in four patients (patients 1, 3, 6, and 7). IR photos showed dark lesions in similar, though not identical, locations (Figure 3). Patients 1 and 6, who underwent ICGA, had hypofluorescent lesions in numbers typically exceeding those detected by both clinical and other imaging modalities. B-scan SD-OCT through the fovea showed a disrupted inner

segment ellipsoid zone band of varied severity in all 7 affected eyes. The ellipsoid zone hyper reflective band on SD-OCT anatomically correlates to photoreceptors' inner segment, ellipsoid section densely packed with mitochondria [16] . The transient disruption of the foveal ellipsoid zone on SD-OCT corresponded to the clinically apparent foveal granularity. In patient 5, who presented with sole retinal finding of foveal granularity and mild optic disc leakage on FA, the SD-OCT finding of ellipsoid zone disruption was the main sign for diagnosis MEWDS (Figure 1). Foveal hyperreflectivity found in 3 patients (patients 1, 4, and 7) was noted extending into the inner retinal layers (Figure 4). The lesions identified on the BAF, IR, and ICGA images corresponded to the areas of disruption of the ellipsoid zone, on the SD-OCT scans (Figure 3). FA demonstrated nonspecific early punctate hyperfluorescent lesions, with slight staining during the early phase, in four patients (patients 2, 3, 6, and 7). These lesions did not correspond to the findings by either the clinical or other imaging modalities. No pathology was noted in the foveal area despite the presence of typical foveal granularity. Mild optic disc leakage was evident in four patients (patients 1, 4, 5, and 6).

During the course of the disease, the hyperautofluorescent areas decreased in number and faded without leaving hypoautofluorescent abnormalities. The resolution of the BAF lesions corresponded to the anatomic recovery observed on SD-OCT. The foveal hyperreflectivity disappeared as well (Figure 5). Figure 6 .

Four patients (patients 1, 4, 6, and 7) underwent serological testing with negative results except for a common result of elevated titer of IgG to VZV.

After 6 months of follow-up, the best corrected visual acuity ranged from 6/6 to 6/6.6 (Table 2).

Although MEDWS is traditionally considered as a rare syndrome [2] , we report an unusual cluster of seven patients who presented within a three-month period. All patients were otherwise healthy, and all presented with decreased vision in one eye. This cluster of cases could break to some measure the statistical improbability of the rarity of the disease. The atypical presentation in most of our patients could suggest that MEWDS is underdiagnosed. However, it may be in line with the speculation that sometimes atypical findings may simply reflect the moment in time in which the patients were examined and

are not a true atypical presentation [4] . In its original description by Jampol et al. [2] , MEWDS cases were unilateral with fundus presentation including numerous white dots scattered in the posterior pole and beyond the arcades. During the disease course, granularity appearance of the macula develops in most cases and, when seen, determines the diagnosis. The number of white spots is very variable, and in fact, they may be absent. Given that characteristic white dots were not present in four patients (patients 1, 4, 5, and 6), we were guided by other fundus features, in particular foveal granularity, symptoms, multimodal imaging, and clinical course. While the presumed pathogenesis of MEWDS involves a viral infection, only few reports to date have described a search for the pathogen [5, [17] [18] [19] . The present cluster of cases provided us with a unique opportunity to seek a common viral denominator. Serological testing yielded only an elevated titer of IgG to VZV, most often an indicative of past VZV infection or vaccination; thus, we could not make any generalization regarding these findings.

Multimodal imaging (BAF, SD-OCT, IR, FA, and ICGA) has proven to have high value in the challenging diagnosis of MEWDS. Most of the findings noted here have been described separately in earlier reports [7-9, 11, 12] . However, the present study offered two important advantages. We were able to examine all patients with simultaneously acquired imaging, and multiple correlations between the imaging findings and the clinical evaluation were possible. Moreover, the relatively large size of the cohort and the repeated scans allowed us to verify the imaging findings in this rare disease.

We observed corresponding locations of the dark spots on IR images, the hyperautofluorescent spots on the BAF images, and the foci of outer retinal pathology on SD-OCT images. Small hyperreflective points, located in the ganglion cell layer, the ellipsoid zone, and the choriocapillaris, have been noted and described on "en face" EDI SD-OCT [20] . However, we noted a unique finding of foveal hyperreflectivity extending into the inner retinal layers. Our finding reinforces a recently described finding in the literature [14] which is believed to be pathognomonic to MEWDS. During the disease course, both the IR and the BAF findings faded in concurrence with the anatomical resolution of the disruption in the ellipsoid zone and the foveal hyperreflective lesion on SD-OCT. Thus, IR images may provide an easy, widely available imaging modality for follow-up of patients with MEWDS. Although IR autofluorescent changes were recently described in patients with MEWDS [21, 22] , this modality is not widely available, whereas IR imaging is routinely performed. Furthermore, on the basis of our findings with multimodal imaging, we suggest that the diagnosis of MEWDS can be established with the simultaneous use of such

noninvasive techniques as BAF, IR, and SD-OCT. ICGA and FA may be reserved for secondary use, when findings are equivocal. OCTA is relatively new noninvasive imaging modality that demonstrates flow characteristics of the vascular network within the regional circulation to construct noninvasive images of the vascular network. En face images generated by OCTA also allow us to study the spatial relationships between vasculature and adjacent retinal/choroidal layers with greater precision than dye angiography, and OCTA findings demonstrated no flow impairment in the retinal and choroidal vasculature of the patients scanned after convalescence stage.

We cannot overestimate the role of multimodal imaging in these patients, since not too often, the diagnosis is mistaken for optic neuritis, and clinical findings are very subtle.

Limitations of the study were the variability in time from disease onset to serologic testing, making the IgM results hard to interpret. Therefore, we consider these tests inconclusive. Secondly, not all the patients had imaging with all modalities. In addition, future research is required using OCT angiography to study the nature of the dots in MEWDS patients and its correlation to other multimodal imaging modalities in the acute and convalescent stage.

In conclusion, we present a large unique cluster of patients who presented with MEWDS over a short period Figure 6 : OCTA images following convalescence stage of patients 7's right eye (a-b) and 6's left eye (c-d). The green and red lines represent the x and y axes. Patient 7 after recurrent episodes. 3 × 3 mm OCT angiogram of the choriocapillaris (a1), superficial layer (a2), and deep layer (a3) centered at the macula without any flow compromise. Corresponding x-axis OCT structural B-scan (b1) simultaneously obtained during the same scan as the OCT angiogram with flow overlay at the cross-section demonstrated by the green line in (a1). SD-OCT (b2) demonstrating normal anatomy of the outer retina 6 months after the first acute episode. Patient 6, 3× 3 mm OCT angiogram of the choriocapillaris (c1), superficial layer (c2), and deep layer (c3) centered at the macula without any flow compromise. 3 × 3 mm en face structural OCT (d1) of the choriocapillaris centered at the macula as in c1. This image was simultaneously obtained during the same scan as the OCT angiogram in (c). En face structural OCT of the deep (d2) and outer retina (d3). of time. To the best of our knowledge, such a cluster was not previously reported in the literature nor encountered by us at different seasons. The diagnosis was supported by the presence of key features of foveal granularity and disruption of the ellipsoid zone on OCT and their correlation with the hyperautofluorescent lesions identified on BAF. Attention

should also be addressed to the dark spots demonstrated on IR images, which may serve as an additional diagnostic clue provided by a noninvasive imaging modality. The disease course in our patients was typical for MEWDS, with almost complete recovery of visual acuity. The specific pathogenesis of MEWDS is unknown but is believed to be an inflammatory condition following a viral infection. We suggest continued serological testing in patients who meet the clinical criteria. The clinical signs of MEWDS are subtle, such that the diagnosis relies on a high index of suspicion.

The authors have no conflict of interest to declare.

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Etiology of respiratory tract infections in the community and clinic in Ilorin, Nigeria

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Abstract: **OBJECTIVE:** Recognizing increasing interest in community disease surveillance globally, the goal of this study was to investigate whether respiratory viruses circulating in the community may be represented through clinical (hospital) surveillance in Nigeria.

RESULTS: Children were selected via convenience sampling from communities and a tertiary care center (n = 91) during spring 2017 in Ilorin, Nigeria. Nasal swabs were

collected and tested using polymerase chain reaction. The majority (79.1%) of subjects were under 6 years old, of whom 46 were infected (63.9%). A total of 33 of the 91 subjects had one or more respiratory tract virus; there were 10 cases of triple infection and 5 of quadruple. Parainfluenza virus 4, respiratory syncytial virus B and enterovirus were the most common viruses in the clinical sample; present in 93.8% (15/16) of clinical subjects, and 6.7% (5/75) of community subjects (significant difference, $p < 0.001$). Coronavirus OC43 was the most common virus detected in community members (13.3%, 10/75). A different strain, Coronavirus OC 229 E/NL63 was detected among subjects from the clinic (2/16) and not detected in the community. This pilot study provides evidence that data from the community can potentially represent different information than that sourced clinically, suggesting the need for community surveillance to enhance public health efforts and scientific understanding of respiratory infections.

Text: Acute Respiratory Infections (ARIs) (the cause of both upper respiratory tract infections (URIs) and lower respiratory tract infections (LRIs)) are a major cause of death among children under 5 years old particularly in developing countries where the burden of disease is 2-5 times higher than in developed countries [1] . While these viruses usually cause mild cold-like symptoms and can be self-limiting, in recent years novel coronaviruses such as severe acute respiratory syndrome (SARS) and Middle East respiratory syndrome (MERS) have evolved and infected humans, causing severe illness, epidemics and pandemics [2] . Currently, the majority of all infectious disease outbreaks as recorded by the World Health Organization (WHO) occur in the continent of Africa where there is high transmission risk [3, 4] . Further, in developing areas (both rural and urban), there are increasing risk factors such as human-animal interfaces (due to residential-proximity to livestock). These changing epidemiological patterns have resulted in calls for improved ARI surveillance, especially in places of high transmission risk [5] .

Nigeria is one such place with high prevalence of many of the risk factors implicated in ARI among children including; age, sex, overcrowding, nutritional status, socio-economic status, and where study of ARIs is currently limited [6] . These broad risk factors alongside limited resources have indicated the need for community-based initiatives for surveillance and interventions [6, 7] . For ARI surveillance in particular, infections in the community are those that do not get reported clinically. Clinical data generally represents the most severe cases, and those from locations with access to healthcare institutions. In Nigeria, hospitals are visited only when symptoms are very severe. Thus, it is hypothesized that viral information from clinical sampling is insufficient to either capture disease incidence in

general populations or its predictability from symptoms [8] . Efforts worldwide including in East and Southern Africa have been focused on developing community-based participatory disease surveillance methods [9] [10] [11] [12] [13] . Community-based approaches have been shown useful for learning more about emerging respiratory infections such as assessing under-reporting [14] , types of viruses prevalent in communities [10] , and prediction of epidemics [15] .

Concurrently, advancements in molecular identification methods have enabled studies regarding the emergence and epidemiology of ARI viruses in many locations (e.g. novel polyomaviruses in Australia [16, 17] , human coronavirus Erasmus Medical Center (HCoV-EMC) in the Middle East and United Kingdom [18, 19] , SARS in Canada and China [20] [21] [22]), yet research regarding the molecular epidemiology of ARI viruses in Nigeria is limited. Diagnostic methods available and other constraints have limited studies there to serological surveys of only a few of these viruses and only in clinical populations [23, 24] . Thus, the utility of community-based surveillance may be appropriate in contexts such as in Nigeria, and the purpose of this pilot study was to investigate if clinical cases may describe the entire picture of ARI among children in Nigeria.

We performed a cross-sectional study in three community centers and one clinical in Ilorin, Nigeria. Ilorin is in Kwara state and is the 6th largest city in Nigeria by population [25] . Three Local Government Areas (Ilorin East, Ilorin South and Ilorin West LGAs) were the community sites and Children's Specialist Hospital, Ilorin the clinical site. Convenience sampling was used for the purposes of this pilot study, and samples were obtained from March 28 to April 5 2017. Inclusion criteria were: children less than 14 years old who had visible symptoms of ARI within the communities or those confirmed at the hospital with ARI. Exclusion criteria were: children who were 14 and above, not showing signs of ARI and subjects whose parents did not give consent. Twenty-five children with symptoms were selected each from the three community locations while 16 symptomatic children were sampled from the hospital. The total sample size ($n = 91$) was arrived at based on materials and processing cost constraints, as well as to provide enough samples to enable descriptive understanding of viral circulation patterns estimated from other community-based studies [10] .

Disease Surveillance and Notification Officers, who are employed by the State Ministry of Health and familiar with the communities in this study, performed specimen and data

collection. Symptoms considered were derived in accordance with other ARI surveillance efforts: sore throat, fever, cough, running nose, vomiting, body ache, leg pain, nausea, chills, shortness of breath [10, 26] . Gender and age, type of residential area (rural/urban), education level, proximity of residence to livestock, proximity to an untarred road and number of people who sleep in same room, were all recorded. The general difference between the two settings was that those from the hospital had severe illnesses, while those from the community were generally "healthy" but exhibiting ARI symptoms (i.e. mild illness).

Nasal swabs were collected from the subjects and stored in DNA/RNA shield (Zymo Research, Irvine, California). Collected samples were spinned and the swab removed. Residues containing the nasal samples were stored at -20 °C prior to molecular analysis.

Viral RNA was isolated using ZR Viral RNA™ Kit (Zymo Research, Irvine, California) per manufacturer instructions (<http://www.zymoresearch.com/downloads/dl/file/id/147/r1034i.pdf>). Real-time PCR (polymerase chain reaction), commonly used in ARI studies [10, 19, 27] , was then carried out using RV15 One Step ACE Detection Kit, catalogue numbers RV0716K01008007 and RV0717B01008001 (Seegene, Seoul, South Korea) for detection of 15 human viruses: parainfluenza virus 1, 2, 3 and 4 (PIV1-4), respiratory syncytial virus (RSV) A and B, influenza A and B (FLUA, FLUB), rhinovirus type A-C, adenovirus (ADV), coronavirus (OC 229 E/NL63, OC43), enterovirus (HEV), metapneumovirus (hMPV) and bocavirus (BoV).

Reagents were validated in the experimental location using an inbuilt validation protocol to confirm issues of false negative and false positive results were not of concern. Amplification reaction was carried out as described by the manufacturer: reverse transcription 50 °C-30', initial activation 94°-15', 45 cycles: denaturation 94°-30", annealing 60°-1' 30", extension 72°-1', final extension 72°-10', hold 4°. Visualization was performed using electrophoresis on a 2% agarose gel in TBE 1X with EtBr, in presence of RV15 OneStep A/B/C Markers; molecular weight marker. Specimen processing was not blinded as there was no risk of experimental bias. Standardized procedures were used for community and clinic sampling.

All statistical analyses were performed using R version 3.2.4. Univariate statistics [mean and 95% confidence interval (CI)] are described. Bivariate statistics (difference in proportions) were assessed using a two-proportion z-test. A p value < 0.001 was considered significant. No observations used in this study had any missing data for analyses in this study.

Basic participant demographics are summarized in Table 1. PCR results showed that ten different viruses (influenza A, coronavirus OC 229 E/NL63, RSVA, RSV B, parainfluenza 1-4) were detected. Figure 1 shows how these infections were distributed across virus types as well as in the community versus clinic samples. In sum, a total of 33 of the 91 subjects surveyed had one or more respiratory tract virus (36.3%, 95% CI 26.6-47.0%, Fig. 1). Furthermore, 10 of those cases were triple infections and 5 were quadruple infections (illustrated by color of bars in Fig. 1). Figure 2 indicates how frequently each pair of viruses were found in the same participant; co-infections were most common among enterovirus and parainfluenza virus 4 (Fig. 2) .

We also compared and contrasted the clinical and community results. Parainfluenza virus 4, respiratory syncytial virus B and enterovirus were the most common viruses found in the clinical sample. These three infections resulted in 41 viruses detected in 15 subjects clinically, and eight infections detected in five people in the community. Together they infected 94% (15/16, 95% CI 67.7-99.7%) of clinical subjects, and 7% (5/75, 95% CI 2.5-15.5%) in the community (significant difference, $p < 0.001$). The most common virus detected in community samples was Coronavirus OC43; this virus was detected in 13.3% (95% CI 6.9-23.6%) people in the community and not in any of the clinical samples. However a different strain, coronavirus OC 229 E/NL63 was detected in 12.5% of the clinical subjects (2/16, 95% CI 2.2-39.6%) and not detected in the community. Double, triple and quadruple infections were another common feature of note.

We identified ten different respiratory tract viruses among the subjects as shown in Fig. 1 . Samples collected from the Children's specialist hospital showed 100% prevalence rate of infection with one or more viruses. This might not be surprising, as the basic difference between the community and clinic samples was an increased severity of illness in the clinical sample. This may also explain the high level of co-infection found among the clinical subjects. The most prevalent virus in the clinical sample (coronavirus OC43) was not detected in the community sample. Further, there was a significant difference between

prevalence of the most common viruses in the clinical sample (parainfluenza virus 4, respiratory syncytial virus B and enterovirus) and their prevalence in the community. Finally, some of the viruses detected in this study have not been detected and implicated with ARIs in Nigeria. There is no report, to the best of our knowledge, implicating coronavirus in ARIs in Nigeria, and it was detected in 12 subjects in this study. Although cases of double and triple infections were observed in a study in Nigeria in 2011 [28] , as far as we are aware, reports of quadruple infections are rare and have not been reported in Nigeria previously.

Due to the unique nature of the data generated in this study and novelty of work in the setting, it is not possible to exactly compare results to other studies. For example, though we found a similar study regarding ARIs in clinical subjects in Burkina Faso [27] , due to the small sample size from this study it would not be feasible to infer or compare prevalence rates. Studies of ARI etiology have mostly been generally focused in areas of the world that are more developed [29] , and it is important to note that the availability of molecular diagnostic methods as employed in this study substantially improve the ability to detect viruses which hitherto have not been detected in Nigeria. Further, findings from this work also add to the growing body of research that shows value of community-data in infectious disease surveillance [8] . As most of the work to-date has been in higher resource areas of the world this study adds perspective from an area where healthcare resources are lower.

In conclusion, results of this study provide evidence for active community surveillance to enhance public health surveillance and scientific understanding of ARIs. This is not only because a minority of children with severe infection are admitted to the hospital in areas such this in Nigeria, but also findings from this pilot study which indicate that viral circulation in the community may not get detected clinically [29] . This pilot study indicates that in areas of Nigeria, etiology of ARIs ascertained from clinical samples may not represent all of the ARIs circulating in the community.

The main limitation of the study is the sample size. In particular, the sample is not equally representative across all ages. However, the sample size was big enough to ascertain significant differences in community and clinic sourced viruses, and provides a qualitative understanding of viral etiology in samples from the community and clinic. Moreover, the sample was largely concentrated on subjects under 6 years, who are amongst the groups at highest risk of ARIs. Despite the small sample size, samples here indicate that

circulation patterns in the community may differ from those in the clinic. In addition, this study resulted in unique findings. Given that resources are limited for research and practice, we hope these pilot results may motivate further systematic investigations into how community-generated data can best be used in ARI surveillance. Results of this study can inform a larger study, representative across demographic and locations to systematically assess the etiology of infection and differences in clinical and community cohorts. A larger study will also enable accounting for potential confounders such as environmental risk factors. Finally, while it may be intuitive, findings from this pilot study shed light on the scope of differences in ARI patterns including different types and strains of circulating viruses. Also, because PCR was used for viral detection, the study was limited to detection of viruses in the primer sets. Given that these are the most up-to-date and common viruses, this approach was deemed sufficient for this initial investigation.

The study was conceived by RC and OK. RC and OK, MO and TD were involved in the design of the study, which was conducted by MO and TD. RC and OK analyzed the data. RC and OK wrote and revised the manuscript. All authors read and approved the final manuscript.

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Aetiology of Acute Respiratory Tract Infections in Hospitalised Children in Cyprus

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Abstract: In order to improve clinical management and prevention of viral infections in hospitalised children improved etiological insight is needed. The aim of the present study was to assess the spectrum of respiratory viral pathogens in children admitted to hospital with acute respiratory tract infections in Cyprus. For this purpose nasopharyngeal swab samples from 424 children less than 12 years of age with acute respiratory tract infections were collected over three epidemic seasons and were analysed for the presence of the most common 15 respiratory viruses. A viral pathogen was identified in 86% of the samples, with multiple infections being observed in almost 20% of the samples. The most frequently detected viruses were RSV (30.4%) and Rhinovirus (27.4%). RSV exhibited a clear seasonality with marked peaks in January/February, while rhinovirus infections did not exhibit a pronounced seasonality being detected almost throughout the year. While RSV and PIV3 incidence decreased significantly with age, the opposite was observed for influenza A and B as well as adenovirus infections. The data presented expand our understanding of the epidemiology of viral respiratory tract infections in Cypriot children and will be helpful to the clinicians and researchers interested in the treatment and control of viral respiratory tract infections.

Text: Viral Respiratory tract infections (RTI) represent a major public health problem because of their world-wide occurrence, ease of transmission and considerable morbidity and mortality effecting people of all ages. Children are on average infected two to three times more frequently than adults, with acute RTIs being the most common infection in childhood [1, 2] . Illnesses caused by respiratory viruses include, among others, common colds, pharyngitis, croup, bronchiolitis, viral pneumonia and otitis media. Rapid diagnosis is important not only for timely therapeutic intervention but also for the identification of a beginning influenza epidemic and the avoidance of unnecessary antibiotic treatment [3, 4]

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RTIs are a major cause of morbidity and mortality worldwide. Acute RTI is most common in children under five years of age, and represents 30-50% of the paediatric medical admissions, as well as 20-40% of hospitalizations in children. Respiratory infections cluster during winter and early spring months. The leading viral agents include respiratory syncytial virus (RSV), influenza A and B (INF-A, INF-B) viruses, parainfluenza viruses (PIVs), and human adenoviruses (HAdVs). In addition, there is a continuously increasing list of

new respiratory viruses that contribute significantly to the burden of acute respiratory infections, such as the recently identified human metapneumovirus (HMPV) and human Bocavirus (HBoV) [5] .

Acute RTIs are classified as upper (URTIs) and lower RTI (LRTIs), according to the involved anatomic localization. URTIs cause non-severe but widespread epidemics that are responsible for continuous circulation of pathogens in the community. LRTIs have been classified as frank pneumonia and bronchiolitis with clinical, radiological and etiological features that usually overlap [6, 7] . Viruses are again the foremost agents of LRTIs often misdiagnosed as bacterial in origin and hence treated with antibiotics unnecessarily [8] .

The main aim of this study was to determine the aetiology of acute respiratory tract infections in Cypriot children and assess the epidemiology of the identified viral pathogens over three epidemic seasons.

The study was approved by the Cyprus National Bioethics Committee. Accordingly, written informed consent was obtained from parents prior to sample taking. Between November 2010 and October 2013, 485 nasopharyngeal swab samples were collected from children up to 12 years of age, who had been hospitalized with acute respiratory tract infection at the Archbishop Makarios III hospital, Nicosia. Clinical and demographic information including symptoms, duration of hospitalisation, diagnosis and treatment were recorded. Nasal swab samples were collected using the BD Universal Viral Transport Collection Kit. Viral RNA/DNA was extracted from 400 µl sample using the iPrep PureLink Virus Kit on an iPrep purification instrument (Invitrogen).

A set of four multiplex Real-Time RT-PCR assays was established and validated for the detection of the 15 most common respiratory viruses as follows: assay 1: influenzaviruses A and B, RSV, assay 2: parainfluenzaviruses 1-4, assay 3: HAdV, enteroviruses, HMPV and HBoV and assay 4: rhinoviruses and the human coronaviruses OC43, NL63 and 229E (Table 1) .

Published primer and probe sets were used as a basis for designing the assays, however, all primer/probe sequences were checked against newly build sequence alignments of all

viruses tested and were modified, if necessary, to account for possible sequence variations. For this purpose, all available complete genome sequences were obtained for each virus from GenBank, imported into the BioEdit Sequence Alignment Editor v7.1.7 and aligned using ClustalX. In case of mismatches between published primers/probe and target sequences, modifications were applied, as indicated in Table 1 . The alignments for the viruses, which necessitated changes to the primers/probe are available in Fasta-Format as supplement S1-S4 Files.

Primer concentrations and reaction conditions for the four assays were subsequently optimised for multiplexing. In order to assess the sensitivity and specificity of the assays, the laboratory enrolled for two consecutive years in Quality Control for Molecular Diagnostics (QCMD) external quality assessment schemes for all viruses, except Bocavirus, which was unavailable. In summary, the established assays were able to correctly identify all viruses tested, proving their suitability for diagnostic application.

A possible correlation of virus prevalence and age of infection was assessed using univariate analyses. The Fisher's exact test was used where cell counts below 5 were encountered; otherwise, the chi-squared test was performed. The same statistical tests were used to compare the frequency of subjects with single or multiple infections between age groups. In addition, Pearson correlation was used to examine co-infections of different viruses. All statistical analyses were performed using StataSE 12 (StatCorp. 2007. College Station, TX, USA).

The present study was a prospective investigation of children hospitalized with acute respiratory tract infections between November 2010 and October 2013 in Cyprus. The median age of the children was 15 months (range: 0-140 months) with 243 being male and 181 female (male/ female ratio 1.34). The age distribution is shown in Fig 1.

Out of the 424 samples analysed, 364 (85.8%) were positive for one or more viruses. Results are summarized in Table 2 .The most commonly detected viruses were RSV, which was found in 129 (30.4%) patients and rhinoviruses in 116 (27.4%) accounting together for almost 60% of all detections. With moderate frequency have been detected HAdV in 31(7.3%) patients, influenza A in 28 (6.6%), HBoV in 24 (5.7%), enteroviruses and PIV 3 in 23 (5.4%) of patients respectively, and Influenza B in 21 (5.0%). A low frequency was exhibited

by HMPV with 16 (3.8%) positive samples, human coronavirus OC43 with 13 (3.1%), PIV 1 with 12 (2.8%), PIV 4 with 9 (2.1%), PIV 2 with 7 (1.7%) and HCoV NL63 with 6 (1.4%). Coronavirus 229E could be detected only in a single sample.

Co-infections with two or more viruses were observed in 84 out of the 364 positive samples (see Table 2). Dual infections accounted for 17% of all positive samples and three viruses were detected in 2.7% of samples). A single patient sample displayed a quadruple infection being simultaneously positive for RSV, rhinovirus, HBoV and influenza B. Table 3 summarizes the frequency of each virus in single vs. multiple infections as well as the number of co-occurrences of viruses for each possible virus combination. In absolute terms the most common combination observed was RSV/rhinovirus. As a percentage, however, the virus appearing most often in co- infections was HBoV, which was found in more than 70% of cases together with another virus, followed by coronaviruses HCoV OC43 and HCoV NL63 with 61% and 67%, respectively. On the other hand, the viruses most rarely seen in co-infections were influenza viruses A and B as well as RSV. Pearson correlation coefficients were calculated to examine the likelihood of co-infections of different viruses. The results of the analysis are summarized in Table 1 in S1 Table. Significant correlation (P-value < 0.05) was seen mostly for co-infections with RSV, however correlations were very weak ($r < 0.3$) and negative. This finding can probably be explained by the fact that RSV infections occurred predominantly in the very young, where co-infections were less frequently observed. On the other hand, a significant positive correlation was observed for enterovirus and rhinovirus co-infection hinting maybe at similarities in circulation patterns and/or transmission modes.

Regarding seasonality, different patterns of circulations could be observed for RSV, rhinoviruses and influenzaviruses (A and B combined) (Fig 2) , with RSV and influenza exhibiting a clear seasonality with marked peaks in January/February, while rhinovirus infections did not exhibit a pronounced seasonality being detected almost throughout the year. However, as more than 100 different rhinovirus strains have been identified to be circulating worldwide in parallel and successively, a potential seasonality of individual rhinovirus serotypes may be masked by overlapping patterns [18, 19] .

The data was further analysed with regard to the age distribution of virus infection (see Table 2). In infants up to 3 months old, RSV was by far the most common pathogen (58.1%), followed by rhinovirus (20.3%) and PIV3 with 8.1% each. The incidence of RSV,

however, decreases significantly with increasing age ($p\text{-value} < 0.0001$) dropping to 13% in children older than 3 years old, while the reverse relationship is observed for Influenza A and B and HAdV. Rhinoviruses, HBoV and enteroviruses are most frequently observed in children from 4 months to 3 years of age. The age dependency of the virus incidence is visualized in Fig 3 for the seven most frequently observed viruses. The positivity rate also showed a trend according to the age group dropping from 90.5% in the under 3-month old to 78.3% in the 4-12 years old ($p\text{-value} = 0.020$). This may point to an increasing role of pathogens not included in the assays, such as bacterial infections in older children.

Regarding multiple infections, children less than 3 month of age and those older than 4 years had a significantly smaller risk to present with multiple infections as compared to the other two age groups ($p\text{-value} = 0.014$).

A reason for this could be that very young children have limited contact to others reducing thereby the chance for a co-infection, whereas children older than 3 years already established immunity to an increasing number of viruses encountered previously.

This study for the first time examined the aetiology of acute respiratory tract infections in hospitalised children in Cyprus. Four multiplex Real-Time RT-PCR assays were developed in order to detect the most common respiratory viral pathogens in a fast and cost-effective way. The high rate of positive samples (85.8%) is evidence of the high sensitivity of the Multiplex-assays used and that the range of viruses included in the analysis is comprehensive. Many previous studies have shown detection rates ranging from below 50% to 75% [20] [21] [22] [23] [24] .

The most common viruses detected were RSV and rhinovirus accounting for almost 60% of all cases. Both viruses were reported previously by others as the major aetiology for respiratory viral infections in young children with rhinoviruses being recognized increasingly for their role in lower respiratory tract infections [20, [25] [26] [27] [28] [29] [30] .

Our data support the results of similar studies performed in the Middle East region. A recently published study found that RSV was the most commonly detected virus in nasopharyngeal swabs from children presenting symptoms of RTIs and in addition to that it

also showed that RSV infections follow a similar circulation pattern peaking from December to March [31] . Another study has revealed that RSV and PIV3 incidence decreases significantly with age, whereas the opposite is observed for influenza and adenovirus infections, a trend that was also observed in our study [26] .

Mixed infections were observed in approximately 20% of all samples, which is in the middle of previously reported rates ranging from 10 to almost 40%. HBoV, HCoV and EV were found most frequently in co-infections. All three subtypes of HCoV were co-detected with several other viruses, while HBoV was co-detected mainly with HRV and RSV. In the case of EV infections, EV were almost predominantly associated with HRV. The rare presence of InfA and InfB viruses in multiple infections witnessed in our study was also observed elsewhere [32, 33] . Even though this study did not allow for investigating a possible association between multiple infections and disease severity, a review of the literature shows that such a potential association is still subject to controversy, since there are reports showing no relationship of multiple virus infection with respiratory illness severity on one hand or a significant association on the other. Studies have shown that viral co-infection was significantly associated with longer duration of illness symptoms, but with a decreased severity in hospitalized children regarding oxygen requirement and intensive care unit admission, whereas the findings of other studies have indicated that severe clinical phenotypes were more prevalent in co-infection patients, especially in RSV co-infections that may increase the severity of RSV associated disease in children [25, [34] [35] [36] [37] [38] [39] [40] .

Viral respiratory infections continue to be a worldwide health concern. As the clinical symptoms of patients with acute respiratory tract infections do usually not allow a discrimination of viral or bacterial aetiology, rapid and reliable diagnostic tools are required for better antibiotic stewardship and the implementation of appropriate infection control measures [4, 41] . The data presented expand our understanding of the epidemiology of viral respiratory tract infections in Cypriot children and will be helpful to the clinicians and researchers interested in the treatment and control of viral respiratory tract infections.

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The First Detection of Equine Coronavirus in Adult Horses and Foals in Ireland

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Abstract: The objective of this study was to investigate the presence of equine coronavirus (ECoV) in clinical samples submitted to a diagnostic laboratory in Ireland. A total of 424 clinical samples were examined from equids with enteric disease in 24 Irish counties between 2011 and 2015. A real-time reverse transcription polymerase chain reaction was used to detect ECoV RNA. Nucleocapsid, spike and the region from the p4.7 to p12.7 genes of positive samples were sequenced, and sequence and phylogenetic analyses were conducted. Five samples (1.2%) collected in 2011 and 2013 tested positive for ECoV. Positive samples were collected from adult horses, Thoroughbred foals and a donkey foal. Sequence and/or phylogenetic analysis showed that nucleocapsid, spike and p12.7 genes were highly conserved and were closely related to ECoVs identified in other countries. In contrast, the region from p4.7 and the non-coding region following the p4.7 gene had deletions or insertions. The differences in the p4.7 region between the Irish ECoVs and other ECoVs indicated that the Irish viruses were distinguishable from those circulating in other countries. This is the first report of ECoV detected in both foals and adult horses in Ireland.

Text: Equine coronavirus (ECoV) is a positive-stranded RNA virus and belongs to the species Betacoronavirus 1 in the genus Betacoronavirus [1, 2]. The clinical signs associated with ECoV infection during outbreaks in the USA [3] and Japan [4] [5] [6] were fever, anorexia, lethargy and diarrhoea. The same clinical signs were also recorded in an experimental challenge study using Japanese draft horses [7]. The main transmission route

is considered to be faecal-oral [7] and ECoV is usually detected in faecal samples. However, the molecular detection of ECoV in faeces from horses with diarrhoea, does not prove causation. Coronaviruses can cause both enteric and respiratory disease in many avian and mammalian species but ECoV is less likely to be found in respiratory secretions than in faeces [8, 9] .

Both molecular and seroepidemiology studies suggest that ECoV may be more prevalent in the USA than in other countries [10] . ECoV was detected in samples collected from equids in 48 states of the USA [11] . In central Kentucky, approximately 30% of both healthy and diarrheic Thoroughbred foals were infected with ECoV [12] . All of the qPCR positive foals with diarrhoea were co-infected with other pathogens such as rotavirus or *Clostridium perfringens*, suggesting that there was potential for ECoV to be over-diagnosed as a causative agent in complex diseases. In contrast in Japan, although an outbreak of diarrhoea occurred among ECoV-infected draft horses at one racecourse [4] [5] [6] , there have been no similar outbreaks subsequently, and all rectal swabs collected from diarrheic Thoroughbred foals were negative. Furthermore, only 2.5% of the rectal swabs collected from healthy foals in the largest Thoroughbred horse breeding region in Japan were positive for ECoV [13] . In France, 2.8% of 395 faecal samples and 0.5% of 200 respiratory samples collected in 58 counties tested positive for ECoV [9] . Similar to the reports from Japan and France, a low prevalence of ECoV was also observed in the UK [14] , Saudi Arabia and Oman [15] . The objective of this study was to investigate the presence of ECoV in clinical samples submitted to a diagnostic laboratory in Ireland. The samples were tested by real-time reverse transcription polymerase chain reaction (rRT-PCR) as it has been shown to be the most sensitive diagnostic method for ECoV [16] and is routinely employed as an alternative to virus isolation in diagnostic laboratories worldwide, both for timely diagnosis and in epidemiological studies [9, 10] . Virus isolation and biological characterisation were beyond the capacity of this study, which was similar in scope to that of the studies in horse populations in the USA, Europe and Asia [8, 9, 13, 14] . The rRT-PCR assay was performed as previously described using a primer set targeting the nucleocapsid (N) gene (ECoV-380f, ECoV-522r and ECoV-436p) [3] (Table 1) and AgPath-ID One-Step RT-PCR Kit (Thermo Fisher Scientific, MA, USA) according to the manufacturer's instructions. To prove that the extraction was successful and that there was no inhibition during rRT-PCR amplification, an internal positive control primer/probe (PrimerDesign, Southampton, UK) was added to the master mix. Thermal cycling conditions were; 48 • C for 10 min and 95 • C for 10 min, followed by 40 cycles at 94 • C for 15 s and 60 • C for 45 s. The SuperScript III One-Step RT-PCR System with Platinum Taq High Fidelity (Thermo Fisher Scientific, MA, USA) was used for sequencing analysis of two of the five ECoV samples identified. There was inadequate

viral nucleic acid in the other three samples for sequencing. The primer sets used to amplify the nucleocapsid (N) gene [4] , the partial spike (S) gene [9] , and the region from the p4.7 to p12.7 genes of non-structural proteins (Oue, personal communication) are shown in Table 1 . The RT-PCR products were sequenced commercially by GATC Biotech (Cologne, Germany). Sequence analysis was performed using the BLAST and CLUSTALW programs, and Vector NTI Advance 11.5 software (Thermo Fisher Scientific, MA, USA). Phylogenetic analysis of nucleotide sequences was conducted with MEGA software Version 5.2 [17] . A phylogenetic tree was constructed based on nucleotide sequences of the K2+G (N gene) and TN93 (S gene) using the maximum likelihood method. MEGA software was used to select the optimal substitution models. Statistical analysis of the tree was performed with the bootstrap test (1000 replicates) for multiple alignments. The complete genome sequences of NC99 (EF446615) [2] , Tokachi09 (LC061272), Obihiro12-1 (LC061273) and Obihiro12-2 (LC061274) [1] , the N (AB671298) and S (AB671299) genes of Obihiro2004, the N gene of Hidaka-No.61/2012 (LC054263) and Hidaka-No.119/2012 (LC054264) [13] , the S gene of ECoV_FRA_2011/1 (KC178705), ECoV_FRA_2011/2 (KC178704), ECoV_FRA_2012/1 (KC178703), ECoV_FRA_2012/2 (KC178702) and ECoV_FRA_2012/3 (KC178701) [9] were used in sequence and/or phylogenetic analysis.

The accession numbers registered in GenBank/EMBL/DDBJ are as follows: the complete sequences of the N gene; 11V11708/IRL (LC149485) and 13V08313/IRL (LC149486), the partial sequences of the S gene; 11V11708/IRL (LC149487) and 13V08313/IRL (LC149488) and the complete sequences from the p4.7 to p12.7 genes; 11V11708/IRL (LC149489) and 13V08313/IRL (LC149490). One six-week-old foal was the only clinical case on a public Thoroughbred stud farm with approximately 30 mares when it presented with diarrhoea. Recovery took over three weeks during which it received fluid therapy, probiotics, antiulcer medication and antibiotics. The second foal was a 14-day-old filly, which had been hospitalised with diarrhoea two days prior to sample collection. The foal responded well to supportive treatment and at the time of sample collection, the diarrhoea had resolved. The five ECoV positive samples tested negative for equine rotavirus.

The nucleotide sequences of the complete N gene, the partial S gene and the region from the p4.7 to p12.7 genes of two positive samples (11V11708/IRL/2011 and 13V08313/IRL/2013) were determined. The nucleotide identities of the N and S genes of the two Irish ECoVs were 99.8% (1338/1341 nucleotides) and 99.5% (650/653 nucleotides), respectively. The nucleotide identities of the N gene of the two Irish ECoVs and the ECoVs from other continents are summarised in Table 2 .

Phylogenetic analysis was performed for the nucleotide sequences of the complete N and partial S genes (Figure 1). The analysis for the N gene showed that Irish ECoVs were independently clustered although they were closely related to Japanese viruses identified after 2009. In the phylogenetic tree of the S gene, Irish ECoVs were closely related to all other ECoVs analysed.

The length of the region from the p4.7 to p12.7 genes in the two viruses was 544 base pairs. Compared with NC99, Irish ECoVs, had a total of 37 nucleotide deletions within p4.7 and the non-coding region following the p4.7 gene. Compared with Obihiro 12-1 and 12-2, Irish ECoVs had a three-nucleotide insertion. When compared with Tokachi09, the Irish ECoVs had a 148-nucleotide insertion (see Figure S1). The p12.7 gene of the two Irish ECoVs did not have deletions or insertions, and the nucleotide identities were 98.8-99.7% between these viruses and the other ECoVs (NC99, Tokachi09, Obihiro12-1 and Obihiro12-2).

This study provides the first report of ECoV circulating in Ireland, the third European country with a significant horse industry where the virus has been detected in horses with enteric disease. However, detection of ECoV in faeces samples from horses with enteric disease does not prove

This study provides the first report of ECoV circulating in Ireland, the third European country with a significant horse industry where the virus has been detected in horses with enteric disease. However, detection of ECoV in faeces samples from horses with enteric disease does not prove causation. In this study, 424 samples collected between 2011 and 2015 from equids with enteric disease were tested, and only five samples (1.2%) were positive for ECoV. The inclusion of an internal positive control in the rRT-PCR eliminated the possibility of false negative results due to the presence of PCR inhibitors but the high content of nucleases associated with faeces samples may have caused some RNA degradation. However, this low prevalence of ECoV is similar to that identified in France [9] and among Thoroughbred foals in Japan [13] .

Although ECoV has been identified on three continents, little is known about the genetic and pathogenic diversity in field viruses. In this study, sequence and phylogenetic analysis (Figure 1) demonstrated a high level of homology between viruses detected in a donkey

and a horse in two provinces in Ireland in different years. This suggests that Irish ECoVs may have low genetic diversity. Compared with the ECoVs of other countries, the N, S and p12.7 genes of the two Irish viruses were highly conserved. In contrast, the region from p4.7 and the non-coding region following the p4.7 gene had deletions or insertions (Figure S1). Because of polymorphism in this region, this region could be useful for epidemiological investigation [5] . The differences in the p4.7 region between the Irish ECoVs and other ECoVs indicated that the viruses in Ireland may be distinguishable from those circulating in other countries. The positive samples were collected in November (1), March (1) and April (3) in this study. Higher case numbers are identified in the USA during the colder months (October to April) [11] , and our results were consistent with the circulation period in USA. It has been reported that outbreaks mainly occurred among adult riding, racing and show horses in USA [11] . The choice of cases to include in the current study may not have been optimal for detection of ECoV as the majority of samples were from foals. However, two positive samples were collected from adult horses in a combined riding school/show jumping yard in the West of Ireland. At the time of sample collection in April 2013, the monthly mean temperatures were below long-term average and in parts of the West, were the coldest in 24 years [18] . Cold weather may have been a predisposing factor to the ECoV infection on the farm.

Two positive samples were collected from Thoroughbred foals. A faeces sample collected from one foal with severe watery diarrhoea and inappetence was positive for ECoV but a sample collected three days later tested negative. A potential difficulty in detecting ECoV from naturally infected horses has been noted previously as serial samples from seven sick horses in the USA suggested that ECoV only persisted for three to nine days in faeces [3] . In both cases, the diarrhoea may have been caused by other unidentified coinfecting pathogens as has been suggested by investigators in the USA [12] . This is the first report of ECoV detection in faeces samples from both foals and adult horses in Ireland. The viruses identified in Ireland are genetically closely related to the Japanese viruses and the results of this study give no indication of significant genetic or phenotypic diversity. In recent years, there has been an increase in awareness and testing for ECoV in the USA and elsewhere [10] . Horse breeding and racing activities in Ireland are the most prominent and important of any country on a per capita basis. There are over 50 Thoroughbred horses per 10,000 of population in Ireland, compared to between three and five for Great Britain, France and the USA [19] . Thus, an investigation of ECoV in Ireland is pertinent not only to increase awareness nationally of the epidemiology of the virus and promote discussion on its clinical importance, but also to inform the industry globally of the health status of Irish

horses. Ireland exports horses all over the world. By illustration, in 2016 the country was the second biggest seller of bloodstock at public auctions second only to the USA [19] .

Many questions remain with regard to the clinical significance of ECoV. The outbreak at a draft-horse racetrack in Japan in 2009 affected 132 of approximately 600 horses and resulted in non-starters and the implementation of movement restrictions [4] . However, draft horses appear to have a higher infection rate than other breeds and an outbreak of similar severity has not been reported in Thoroughbred racehorses [10, 20] . The much higher incidence of ECoV positive Thoroughbred foals identified in Kentucky compared to similar populations internationally suggests an increased susceptibility to ECoV infection in that population. In the past, specific environmental factors were associated with extensive reproductive loss in the Kentucky area and to a lesser extent in other states [21] , but predisposing regional factors such as differences in management, environment or husbandry have not been identified for ECoV. It has been suggested that ECoV is a coinfecting agent in foals with diarrhoea and clinical infections have predominantly been reported in adult horses with a mono-infection with ECoV [10] . There was no indication from the results of this study that coronavirus is a major cause of diarrhoea in Irish horses but the introduction of rRT-PCR as a routine diagnostic test will assist in elucidating the significance of this virus to the Irish breeding, racing and sports industries. The primary focus in future will be on testing adult horses that present with anorexia, lethargy, fever and changes in faecal character as a significant association has been demonstrated between this clinical status and molecular detection of ECoV in faeces [11] .

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Population-Based Pertussis Incidence and Risk Factors in Infants Less Than 6 Months in Nepal

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Abstract: **BACKGROUND:** Pertussis is estimated to cause 2 percent of childhood deaths globally and is a growing public health problem in developed countries despite high vaccination coverage. Infants are at greatest risk of morbidity and mortality. Maternal vaccination during pregnancy may be effective to prevent pertussis in young infants, but population-based estimates of disease burden in infants are lacking, particularly in low-income countries. The objective of this study was to estimate the incidence of pertussis in infants less than 6 months of age in Sarlahi District, Nepal. **METHODS:** Nested within a population-based randomized controlled trial of influenza vaccination during pregnancy, infants were visited weekly from birth through 6 months to assess respiratory illness in the prior week. If any respiratory symptoms had occurred, a nasal swab was collected and tested with a multitarget pertussis polymerase chain reaction (PCR) assay. The prospective cohort study includes infants observed between May 2011 and August 2014. **RESULTS:** The incidence of PCR-confirmed *Bordetella pertussis* was 13.3 cases per 1000 infant-years (95% confidence interval, 7.7–21.3) in a cohort of 3483 infants with at least 1 day of follow-up. **CONCLUSIONS:** In a population-based active home surveillance for respiratory illness, a low risk for pertussis was estimated among infants in rural Nepal. Nepal's immunization program, which includes a childhood whole cell pertussis vaccine, may be effective in controlling pertussis in infants.

Text: A resurgence of pertussis across age groups has occurred in several countries in recent years [1] . Middle-and high-income countries that use an acellular pertussis vaccine for the primary vaccination series have been particularly affected [2, 3] , and infants and adolescents have experienced the greatest increase [4] . Factors that may contribute to the increased risk of pertussis include rapidly waning immunity from those vaccinated with acellular vaccines [1, 5, 6] , asymptomatic transmission from individuals vaccinated with acellular vaccines [7] , genetic adaption of *Bordetella pertussis* [8] , vaccination delay or refusal [9] , improved surveillance and laboratory capabilities [2] , and overall increased

awareness of the continuing circulation of B pertussis [1] . Some countries experiencing epidemic pertussis, including the United States, United Kingdom, and Argentina, now recommend pertussis immunization in pregnancy and vaccination of close contacts [10, 11] to protect the youngest infants from pertussis before they can be vaccinated themselves [12] . Recent data from maternal vaccination trials demonstrate the ability of antibodies to be transferred from mothers to their infants in pregnancy and their persistence in infants [13] .

Global estimates of pertussis show the highest childhood burden in Southeast Asia [14] . In this region, maternal pertussis vaccination during pregnancy may be a way to protect infants, similar to the approach using tetanus toxoid vaccine. However, globally only 1 population-based estimate of pertussis in infants from birth has been conducted (Senegal) [15] , and surveillance and laboratory capabilities in Asia are lacking [16, 17] . The World Health Organization (WHO) recently recommended that countries using whole cell pertussis vaccines continue to do so in light of recent data indicating that acellular pertussis vaccines are less effective than whole cell pertussis vaccines [18] . Population-based data are needed, especially in low-income settings, to provide a more accurate estimate of the burden of pertussis in infants to inform childhood and maternal immunization policies [19, 20] .

We report on a prospective cohort study following infants weekly in their homes to monitor for pertussis disease from birth to age 6 months. The objective was to provide a population-based estimate of laboratory-confirmed pertussis incidence in infants less than 6 months of age in the Sarlahi District, Nepal.

The study was nested within 2 consecutive randomized controlled trials of maternal influenza vaccination during pregnancy set in the Sarlahi District, located in the central Terai (low-lying plains) region of Nepal [21] . At the start of the trial, prevalent pregnancies were identified through a census of all households in the catchment area. For the duration of the trial, field workers visited all households in the communities, every 5 weeks, where married women (15-40 years) resided, for surveillance of incident pregnancies. Once a pregnancy was identified, women provided consent and were enrolled. From April 25, 2011 through September 9, 2013, women between 17 and 34 weeks gestation were randomized and vaccinated with either an influenza vaccine or placebo. The study was a population-based prospective cohort of infants followed from birth through 6 months postpartum.

Approval for the study was obtained from the Institutional Review Boards at the Johns Hopkins Bloomberg School of Public Health, Cincinnati Children's Medical Center, the Institute of Medicine at Tribhuvan University, Kathmandu, and the Nepal Health Research Council. The trials are registered at Clinicaltrials.gov (NCT01034254).

At baseline, information was collected on household structure, socioeconomic status, and demographics. At enrollment, date of last menstrual period and pregnancy history data were collected. As soon as possible after delivery, the mother and infant were visited to collect detailed birth information including infant weight and breastfeeding status. From birth through 6 months, postpartum infants were visited weekly by a field worker, who recorded any infant respiratory symptoms in the past 7 days. If an infant had any of the following symptoms, a mid-nasal nylon flocked swab was collected: fever, cough, wheeze, difficulty breathing, or ear infection. Starting on August 17, 2012, new symptoms, more specific for pertussis, were added to the weekly morbidity visit: apnea, cyanosis, cough with vomit, or whoop/whooping cough. The swabs were stored for up to 1 week at room temperature in PrimeStore Molecular Transport Medium (Longhorn Diagnostics LLC, Bethesda, MD). In addition to these signs, mothers were asked which, if any, infant vaccinations were received in the past 7 days, including pertussis vaccination [22]. Mid-nasal swabs were also collected on a weekly basis from mothers from enrollment through 6 months postpartum who reported fever plus one additional morbidity (cough, sore throat, nasal congestion, or myalgia). All nasal swabs collected from infants were tested for *Bordetella pertussis*, *Bordetella parapertussis*, and *Bordetella bronchiseptica*. Only the nasal swabs of mothers whose infants tested positive for any of these pathogens were tested for the same pathogens.

Real-time polymerase chain reaction (PCR) testing was conducted at the University of Washington's Molecular Virology Laboratory according to previously published methods [23]. Two-target PCR was used to assess the presence of 3 *Bordetella* species: *B. pertussis*, *B. parapertussis*, and *B. bronchiseptica*. The amplified targets were chromosomal repeated insertion sequence IS481 (IS) and the polymorphic pertussis toxin *ptxA* promoter region (PT).

After amplification, the melting points of the amplicons were measured in an iCycler (Bio-Rad). A sample was interpreted as positive when the target(s) had a melting temperature within the species-specific acceptable range and a computed tomography ≤ 42 . A sample

was negative if none of the targets tested positive or a single positive target was not reproducible. Maternal nasal swabs were tested for those mothers whose infants tested positive for any *Bordetella* species

Polymerase chain reaction was also performed for several viral infections (influenza, rhinovirus [RV], respiratory syncytial virus [RSV], bocavirus [BoV], human metapneumovirus, coronavirus, adenovirus, and parainfluenza [1] [2] [3] [4]) as previously described [21] .

Of 3693 women enrolled, 3646 infants were live born to 3621 women (Supplementary Figure 1). Infants were included in this analysis if they were followed for any length of the follow-up period (0 to 180 days); median total follow-up was 146 days per infant (Supplementary Figure 2) . The final dataset consists of 3483 infants, contributing 1280 infant-years of observation, with at least 1 follow-up visit during the first 6 months. This includes infants from the entire trial period, both before and after more pertussis-specific additions to the weekly symptom questionnaire.

At baseline, data on household structure were gathered. At enrollment, women reported their literacy status (binary) and pregnancy history. The field workers identified their ethnicity into 2 broad groups (Pahadi, a group originating from the hills; or Madeshi, a group originating from north India) from names and observation. Women were categorized as nulliparous or multiparous. Responses to 25 questions about household construction, water and sanitation, and household assets were used to develop an index to measure the socioeconomic status of households. Binary variables for each of the 25 questions and a mean SES score were calculated for each household.

Gestational age was measured using a woman's report of date of last menstrual period during pregnancy surveillance. Birth weight was collected as soon as possible after birth using a digital scale (Tanita model BD-585, precision to nearest 10 grams). Birth weights collected >72 hours after birth were excluded from the analysis. Small for gestational age (SGA) was calculated using the sex-specific 10th percentile cutoff described by Alexander et al [24] and the INTERGROWTH-21 standards [25] . Women were asked within how many hours of birth breastfeeding was initiated and binary breastfeeding categories were created (≤ 1 hour versus > 1 hour postdelivery).

Incidence was calculated as the number of pertussis cases per 1000 infant-years at risk. Poisson exact 95% confidence intervals (CIs) were constructed. Characteristics of infant pertussis cases were compared with nonpertussis cases using bivariate Poisson regression. Characteristics of all pertussis respiratory episodes were compared with nonpertussis respiratory episodes; t tests were used for continuous predictors and Fisher's exact tests were used for categorical associations due to the low number of pertussis episodes. All statistical analyses were conducted in Stata/SE 14.1.

A total of 3483 infants had 4283 episodes of respiratory illness between May 18, 2011 and April 30, 2014. Thirty-nine percent (n = 1350) of infants experienced no respiratory episodes. The incidence of respiratory illness was 3.6 episodes per infant-year (95% CI, 3.5-3.7). Mean episode duration was 4.7 days (95% CI, 4.6-4.9). A total of 3930 (92%) episodes were matched to 1 or more pertussis-tested nasal swabs from 2026 infants (Supplementary Figure 1) .

Seventeen cases of B pertussis were identified from 19 nasal swabs (nasal swabs were positive on 2 consecutive weeks for 2 infants). The incidence of PCR-confirmed B pertussis was 13.3 cases per 1000-infant years (95% CI, 7.7-21.3). Five cases of B parapertussis were detected with an incidence of 3.9 cases per 1000 infant-years (95% CI, 1.3-9.1). No cases of B bronchiseptica were identified.

The average pertussis episode duration was 8 days (range, 2-33) (Table 1). Mean age of onset of symptoms was 83 days (range, 19-137) (median, 80; interquartile range, 63-109). The most common symptoms were cough, difficulty breathing, and cough with vomit. None of the additional symptoms related to pertussis that were added in year 2 (cyanosis, apnea, cough with vomit, and whoop) resulted in collection of nasal swabs based solely on these additional symptoms. Pertussis episodes were statistically significantly more likely to include difficulty breathing, cough with vomit, and whoop compared with other respiratory illness. Six infants had at least 1 pertussis vaccination before pertussis disease onset (three <2 weeks and three >2 weeks before pertussis illness) with a mean of 18 days from vaccination to illness compared with 49 days for nonpertussis episodes (P = .03). Five infants received their first pertussis vaccination postpertussis disease onset, whereas 6 infants received no pertussis vaccination in the first 180 days. Three fourths of pertussis episodes were coinfectd with at least 1 virus, with RV and BoV the most common. Cases

of pertussis were more likely to be infected with BoV than respiratory cases due to causes other than pertussis. The majority of cases occurred between February 2013 and January 2014 (Figure 1) .

No statistically significant differences between risk factors for pertussis and nonpertussis cases (Table 2) were documented. Given the low number of pertussis cases, the lack of a statistical association is not evidence of nonassociation. No deaths occurred in infants who had pertussis. Of the 8 mothers of B pertussis-positive infants who had a nasal swab collected (14 nasal swabs total) during their own follow-up, none were positive for any pertussis species.

The 5 B parapertussis cases were primarily male whose mothers were primiparous, literate, and Pahadi ethnicity (Supplementary Table 1). No mothers of infants who had B parapertussis had a nasal swab collected during follow-up.

The average B parapertussis episode duration was 4 days (Supplementary Table 2). Mean age of onset of symptoms was 58 days with a range of 7-95 days. The most common symptoms were cough and wheeze. Rhinovirus and RSV were the only coinfections observed. All B parapertussis cases occurred between September 2011 and February 2012 (Figure 1).

A low incidence of pertussis and generally mild clinical presentation were found in infants <6 months in Nepal. To our knowledge, this represents one of the first population-based active surveillance of PCR-confirmed pertussis among young infants in Asia. Acellular pertussis vaccine trials conducted in the 1990s found the average pertussis incidence in the whole cell vaccine groups ranged from 1 to 37 cases per 1000 infantyears [26] . Our finding of 13 B pertussis cases per 1000 infantyears was on the lower end of this range. In the United States in 2014, the estimated pertussis incidence in infants less than 6 months was 2 cases per 1000 infant-years [27] , much lower than observed in our study; however, this passive surveillance system likely vastly underestimates pertussis incidence. Thus, there is a need for active surveillance data such as ours. Furthermore, given our highly sensitive case detection method, many of our pertussis cases would likely not have been detected in the previous acellular pertussis vaccine trials. More stringent respiratory symptom criteria would have lowered our incidence estimate even further. The low

incidence was found in a population where pentavalent vaccine (Pentavac: Diphtheria, Tetanus, Pertussis [Whole Cell], Hepatitis-B and Haemophilus Type b Conjugate Vaccine; Serum Institute of India Pvt. Ltd), scheduled for administration at 6, 10, and 14 weeks, is received with significant delays (7% of infants received all 3 recommended pertussis vaccines by 6 months) [22] . These data support the WHO's recommendation that countries using whole cell pertussis vaccine continue to do so given that the majority of outbreaks have been concentrated in countries using the acellular pertussis vaccine [2] . Recent studies suggest that protection from acellular pertussis vaccine is not as strong or long lasting as that conferred by the whole cell pertussis vaccine [6, 28] .

Another contributing factor to the low pertussis incidence observed could be that surveillance was conducted during a period of low pertussis transmission. Pertussis is a cyclical disease, thought to peak every 2 to 4 years, and we may have captured the burden at a low circulation period [6] . We observed over 70% of our B pertussis cases over a 1-year period. This increase from earlier observation periods could indicate a temporary rise in pertussis consistent with its cyclical pattern or a true increase in the baseline burden. Previous research on pertussis seasonality has in different places and time periods demonstrated various periods of peak transmission or no discernable patterns [29, 30] . Although our data do not support a seasonal pattern, the numbers observed are too low to be conclusive.

Pertussis symptom duration and severity were mild compared with the classic pertussis case presentation. Only 3 of the 17 cases fulfilled the WHO criteria, which requires a minimum of 2 weeks of cough, whoop, or posttussive vomiting [31] . Studies on pertussis in infants have generally been clinic-based, hospital-based, or in an outbreak, which therefore required a certain severity of illness for parents to recognize a need for medical attention [29, 30, 32] . These study designs and passive surveillance efforts therefore may have missed milder pertussis cases [33] . Our study, which required only 1 respiratory symptom for a nasal swab to be collected, had increased sensitivity to detect a range of pertussis case presentations. An alternative explanation for the mild cases seen could be an increase in the proportion of mild compared with severe pertussis cases in Nepal.

Although cough, difficulty breathing, and cough with vomit were the most common symptoms, no symptom was present in all B pertussis cases. During an epidemic period in Washington state, among infants <1 year, who had a minimum of 14 days cough plus an

additional symptom, 82% had posttussive emesis, 29% had apnea, 26% had whoop, and 42% had cyanosis [32] . A study of US neonates with pertussis showed the symptom prevalence to be 97% for cough, 91% for cyanosis, 58% for apnea, and 3% for fever [34] . Our study found lower or equal symptom prevalence with the exception of fever. Fever prevalence was higher in our study, similar to that found in Peru [29] .

Although not statistically significant, infants with pertussis were more likely to have been born preterm, low birth weight, and SGA, and their mothers were more likely to be primiparous. These findings are similar to previous studies showing no difference in pertussis cases by sex [29, 35, 36] or crowding [35] but showing differences by birth weight [36] . Coinfections were common, consistent with findings from other hospital-based studies [33] . Codetection of B pertussis and B parapertussis with respiratory viruses may be due to asymptomatic pertussis carriage. The incidence of B parapertussis of 4 cases per 1000 person-years was comparable to that of 2 per 1000 person-years found in the Italian acellular pertussis vaccine trial in 1992-1993 [37] . The duration of illness was shorter for B parapertussis with a maximum duration of 6 days compared with a maximum of 33 days for B pertussis. A milder presentation is consistent with clinical knowledge of B parapertussis infection [37, 38] . Bordetella parapertussis cases occurred only during a 5-month period.

There were several study design limitations. We cannot be certain whether the reported symptoms were caused by pertussis, another organism, or whether symptoms were related to 2 or more etiologic agents. We were unable to perform multivariate regression modeling for characteristics associated with pertussis disease and pertussis cases due to the small number of cases we detected.

Infant respiratory symptoms were reported by parents, who may have missed signs that might have been observed by a healthcare worker. However, the criteria for collection of the nasal swab were broad and did not require sophisticated clinical skills. However, apnea and cyanosis may have been difficult for parents to identify. Although the criteria for specimen collection changed in year 2, no infant experienced a pertussis-specific symptom in isolation without also having one of the originally specified respiratory symptoms. These data support our assumption that we were unlikely to have missed pertussis cases in year 1 with our less sensitive respiratory symptom criteria.

Nasal swabs were collected in the mid-nasal region for influenza virus detection, which may have lowered the sensitivity of pertussis detection. In a field site, the acceptability of an additional nasopharyngeal swab would likely have increased the participant refusal rate. This would have decreased the generalizability of our results to the entire population. Although nasopharyngeal swabs or nasopharyngeal aspirates are the recommended specimen collection method [39], the nasopharyngeal region was established as the collection area of choice when the diagnostic measure was culture, which has low sensitivity. Recent data demonstrated the comparability of using mid-nasal versus nasopharyngeal swabs in PCR pertussis detection [40].

Strengths of the study included being a population-based, prospective study, with very low refusal rates. Risk factors, clinical symptoms, and coinfections were prospectively identified without the potential bias that may occur when these data are collected retrospectively or in clinical settings. The community-based design allows generalizability of these results to the entire population and not just those seeking care at a health facility or in an outbreak situation. The Sarlahi District is located in the Terai region where the majority of Nepalese reside, and it has similar demographics to the entire population of Nepal [41]. Sarlahi's location near sea level and on the border with India supports the generalizability of these results to many populations living on the Indian subcontinent. The weekly active surveillance with sensitive criteria for pertussis testing was able to detect mild and atypical pertussis cases, which may have been missed by previous traditional surveillance. The multitarget PCR method allowed highly sensitive and specific detection of 2 additional *Bordetella* species beyond the primary *B. pertussis* target.

We observed a low incidence of pertussis in infants in a whole cell vaccine environment. Pertussis cases were generally milder than expected compared with traditional pertussis clinical definitions. These data support clinicians considering pertussis in their differential diagnosis of infants with mild respiratory symptoms. Policymakers in Nepal will need to weigh the benefit of an additional prenatal pertussis vaccine or a switch to acellular primary pertussis vaccine with the low burden of pertussis in infants less than 6 months. Our study demonstrated that mid-nasal swabs were able to detect pertussis using a sensitive multitarget PCR. The less invasive mid-nasal nasal swab is an attractive alternative for pertussis nasal swab collection, and further research is needed to compare this collection site with nasopharyngeal swabs. In the future, this method may enhance population-based surveillance efforts.

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Estimating Sensitivity of Laboratory Testing for Influenza in Canada through Modelling

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2722738/>

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Abstract: **BACKGROUND:** The weekly proportion of laboratory tests that are positive for influenza is used in public health surveillance systems to identify periods of influenza activity. We aimed to estimate the sensitivity of influenza testing in Canada based on results of a national respiratory virus surveillance system. **METHODS AND FINDINGS:** The weekly number of influenza-negative tests from 1999 to 2006 was modelled as a function of laboratory-confirmed positive tests for influenza, respiratory syncytial virus (RSV), adenovirus and parainfluenza viruses, seasonality, and trend using Poisson regression. Sensitivity was calculated as the number of influenza positive tests divided by the number of influenza positive tests plus the model-estimated number of false negative tests. The sensitivity of influenza testing was estimated to be 33% (95%CI 32–34%), varying from 30–40% depending on the season and region. **CONCLUSIONS:** The estimated sensitivity of influenza tests reported to this national laboratory surveillance system is considerably less than reported test characteristics for most laboratory tests. A number of factors may explain this difference, including sample quality and specimen procurement issues as well

as test characteristics. Improved diagnosis would permit better estimation of the burden of influenza.

Text: Although influenza virus infection is associated with considerable morbidity and mortality [1] [2] [3] , laboratory confirmation of clinical illness is the exception rather than the rule. Clinicians do not routinely seek laboratory confirmation for several reasons: diagnosis will often not alter patient management, a paucity of real-time, accurate, inexpensive testing methods [4] and because influenza is not recognized as the etiology of the clinical presentation [5] . Accurate diagnosis of influenza-like illness, however, could improve clinical care through reduced use of antibiotics and ancillary testing, and more appropriate use of antiviral therapy [6] . Although rapid influenza tests such as point-of-care tests are purported to generate results in a timely fashion to influence clinical care, the performance characteristics of the currently available tests are sub-optimal [7] . New technologies with improved sensitivity such as reverse-transcriptase polymerase chain reaction (RT-PCR) [8] as well as the use of more effective collection systems such as the flocked nasopharyngeal swab compared to traditional rayon wound swabs, and the recommendation to collect more ideal specimens, such as nasopharyngeal swabs rather than throat swabs are likely to improve diagnostic sensitivity [9] [10] [11] [12] . The performance characteristics of currently available tests for influenza vary considerably and the overall sensitivities of these tests when used in routine practice are also dependent on the type of specimen collected, the age of the patient and point in their illness in which they are sampled [4, 9, [13] [14] [15] .

We sought to estimate the sensitivity of influenza testing based on results of a national respiratory virus surveillance system using a model-based method [1, 2, [16] [17] [18] .

Weekly respiratory virus identifications from September 1999 to August 2006 were obtained from the Respiratory Virus Detection Surveillance System (RVDSS), Public Health Agency of Canada [19, 20] . The RVDSS collects, collates, and reports weekly data from participating laboratories on the number of tests performed and the number of specimens confirmed positive for influenza, respiratory syncytial virus (RSV), para-influenza virus (PIV), and adenovirus. Specimens are generally submitted to laboratories by clinicians in the course of clinical care, and by clinicians participating in one of our national influenza surveillance programs, (FluWatch [20]). Indicators of influenza activity are reported year round on a weekly basis to the FluWatch program. The RVDSS is supplemented by case

reports of influenza positive cases [19, 21] . From the case reports, influenza A was confirmed in all age groups and sporadic cases were confirmed in the off-season months of June through September. Infants and children under the age of 5 years accounted for 25% of the influenza A positive tests, and persons over the age 65 years another 35%. Unfortunately, FluWatch surveillance data does not provide the total number of tests by age. Testing practices are known to be varied [22, 23] . The predominant testing methods used for influenza detection varied considerably by province or laboratory and over time. For the 2005/06 season a survey of laboratory techniques in current use indicated that culture accounted for 44% of the diagnostic tests with RT-PCR, rapid antigen tests and direct fluorescent-antibody assay (DFA) accounting for 21%, 19%, and 16% respectively [23] .

The weekly number of tests negative for influenza was modelled, using Poisson regression, as a function of viral identifications for influenza, RSV, adenovirus and PIV as well as a baseline consisting of seasonality, trend and holiday variables. The estimated baseline implicitly accounts for influenza tests on specimens taken from patients with respiratory infections due to respiratory pathogens other than the four viruses captured in the RVDSS, as long as both the testing behaviour of clinicians and respiratory illnesses caused by other respiratory pathogens follow a consistent seasonal pattern as prescribed by the model (see below,

The Poisson regression model with a linear link function was estimated using SAS [24] PROC GENMOD: Coefficients b_5 to b_9 are multipliers. The weekly number of influenza negative tests estimated to be falsely negative is given by $b_5 \text{ InflA}_w + b_6 \text{ InflB}_w$. The weekly number of influenza negative tests attributed to RSV is given by $b_7 \text{ RSV}_w$, and similarly for adenovirus and PIV. For each positive influenza A test, an additional b_5 tests above baseline were performed and found to be negative. By specifying a linear link, a value of 0.33, say, for coefficient b_5 , means that for every test for which influenza A was confirmed, 0.33 additional tests, on average, were performed on truly influenza A positive specimens and found to be negative which corresponds to a sensitivity of 75%.

Sensitivity was calculated as the number of influenza positive tests divided by the number of influenza positive tests plus the model-estimated number of false negative tests, or equivalently, the estimates of sensitivity for influenza A and B are given by $1/(1+b_5)$ and $1/(1+b_6)$ respectively. The false negative rate is 1 minus sensitivity. While the null value for

b 5 is zero, which indicates no statistical association between the number of influenza positive tests and the number of influenza negative tests, the corresponding null value for sensitivity is 1.

For each test confirmed positive for RSV, on average b 7 tests were performed for influenza and found to be negative for influenza. These b 7 tests are attributed to an RSV infection, however the number of influenza-negative tests that actually tested positive for RSV is unknown. If all specimens had been tested for the same viruses (panel tests), 1/b 7 would correspond to the sensitivity for RSV testing, and the sensitivity for adenovirus and PIV given by 1/b 8 and 1/b 9 respectively. Some laboratories are known to test for viruses sequentially [22] , and so 1/b 7 -1/b 9 were not interpreted as estimates of the sensitivity for other viruses. Sequential testing may occur if a rapid test for influenza is negative and the laboratory then performs PCR or culture testing. Similarly in young children with a respiratory illness in the winter, rapid tests for RSV infection may be performed first, and only specimens with negative results submitted for subsequent testing for influenza or other respiratory viruses [25] . By contrast, many laboratories conduct panel tests for multiple viruses for ease of handling, decreased patient sampling, and recognition that coinfection can occur. Either form of sequential testing would not bias the estimate of sensitivity applicable to test results reported to RVDSS, though significant use of rapid antigen tests in the laboratories reporting to RVDSS would reduce the overall sensitivity. As a single specimen may undergo multiple tests, the false-negative rate applicable to a specimen that has undergone multiple tests would be expected to be much lower than the system average for individual tests. Parameters b 1. to b 4 account for trends and the seasonality of truly negative specimens (patients presenting with other acute respiratory infections).

Over 50,000 tests for influenza were reported to the RVDSS each year, peaking in 2004/05 at 101,000. Overall 10% of the influenza tests were positive for influenza, ranging from 4% to 13% depending on the season. The proportion positive for RSV, parainfluenza and adenovirus averaged 9%, 3% and 2% respectively. As seen in Figure 1 , no virus was identified in 75% of specimens submitted for testing (white area under the curve). Even for the winter months of December through April, one of these 4 viruses was identified on average in no more than 30% of the specimens. The strong and consistent synchronization of negative tests with influenza positive tests, as seen in Figure 1 , is suggestive that false negative results contributed to the large number of negative tests during periods of influenza activity.

The sensitivity for influenza A testing averaged 33.7% (with model-estimated 95% confidence intervals of 33.3-34.1) for the 1999/2000-2005/06 period. Influenza B testing had a similar estimated sensitivity at 34.7 (95% CI 33.4-36.1). Estimated sensitivities varied somewhat from season to season, generally ranging from 30%-40% (Table 1) , and provincial level estimates, as well, were within a similar range. Stratifying by province or season produced similar estimates for the sensitivity of influenza A testing: 32% (95% CI 30-34) and 36% (95% CI 33-41) respectively. Estimates of sensitivity based on test results reported to the RVDSS for individual laboratories with sufficient data to fit the model showed significant variation, with estimates of sensitivity ranging from 25-65%. As expected, laboratories using primarily rapid antigen tests had lower estimated sensitivities, and laboratories that used PCR methods had higher sensitivity estimates. However, information on testing procedures is limited primarily to the 2005/06 survey. As well, additional irregularities were noticed in the laboratory data and not all laboratories provided sufficient data to fit the model. Figure 2 illustrates a good model fit where the weekly number of influenza negative tests is well explained by the model covariates, with a few exceptions. Firstly, it is evident that additional specimens were tested during the SARS period, as indicated by the period where the number of weekly influenza negative tests exceeded the expected number, or equivalently, a period of successive positive residuals. Residuals typically capture random variation; hence represent tests that can not be allocated based on the specified model. In addition to the SARS period, testing appears to have been elevated for a number of weeks in January 2000 during the peak of the 1999/2000 A/ Sydney/05/97 (H3N2) season in which respiratory admissions were unusually elevated [26, 27] , and in December 2003, when an elevated risk of paediatric deaths associated with the A/Fujian/411/02 (H3N2) strain [28] was identified in the US. As these periods corresponded to a period of heightened public awareness due to severe influenza outbreaks, parameter estimation was repeated without these data points. Exclusion of these data points did not alter the sensitivity estimate for influenza.

The attribution of influenza negative test results to influenza and other viruses is illustrated in Figure 3 . The baseline curve is the model estimate of the number of tests that were likely truly negative for all four viruses tested. A reduction in specimen collection and testing, primarily for viruses other than influenza, is also evident over the Christmas period (Figure 3) .

The weekly proportion of tests confirmed positive for influenza peaked each season at 15 to 30%. Accounting for the model estimated false negative rate suggests that during periods of peak influenza activity, 40-90% of tests were performed on specimens taken from persons recently infected with influenza. Influenza was confirmed in only 14% of specimens sent for testing over the winter period, whereas the sensitivity estimate would imply that up to 40% of influenza tests could be attributed to an influenza infection. The corresponding figures for the whole year indicate that 10% of specimens were confirmed positive for influenza and 30% of influenza tests could be model-attributed to an influenza infection annually.

Despite a relatively large number of tests in the off-season, the number of influenza positive tests was almost negligible; suggesting that the false positive rate applicable to RVDSS influenza testing is minimal.

The model estimated sensitivity based on influenza test results reported to the RVDSS of 30-40% is much lower than the standard assay sensitivities documented in the literature. Standard sensitivities for diagnostic procedures used by participating laboratories ranged from 64% for rapid antigen tests to 95% for RT-PCR tests, averaging 75% for the study period [23] . As performance characteristics of specific tests are generally based on high quality specimens, the difference of approximately 40% is likely linked to any one of many operational procedures that affects the quality of the specimen and its procurement. Unlike validation studies, our samples are taken from a variety of clinical settings and processed with a variety of procedures across the country. As well, variation in the indications for diagnostic testing may vary across the country.

As there are many other respiratory pathogens that are not routinely tested for, or reported to the RVDSS, including human metapneumovirus (hMPV), coronaviruses, and rhinoviruses for which patients may seek medical care and present with influenza like illness [29] [30] [31] [32] , a large proportion of negative test results was expected. The overall model fit, and the general consistency of the sensitivity estimates, suggests that these many respiratory viruses were reasonably accounted for by the seasonal baseline and that the strong association between the number of influenza positive and influenza negative tests on a weekly basis is indicative of a significant number of false negative results, rather than the activity of another virus or viruses exactly synchronous with influenza. The latter would bias the estimated sensitivity of the system downwards. However, to significantly and

consistently bias the estimate, the degree of synchronization would have to be fairly strong, persist over the whole study period, and occur in all provinces. Synchronization was not observed among the RVDSS viruses (influenza A, influenza B, RSV, adenovirus and PIV), and elsewhere other viruses such as rhinovirus, coronavirus and hMPV accounted for only a small proportion of the viral identifications and were not found to be synchronized with influenza [33] . As well, patients may present for care due to a secondary bacterial infection. While any specimen would likely test negative as the virus, at this point, is likely not detectable, the model would statistically attribute a negative test in this case to the primary infection; one of the four RVDSS viruses or to the seasonal baseline that represents other respiratory infections, depending on the level of viral activity at the time of the test. This is not considered a source of bias.

The large variation in false negative rates estimated for individual laboratories reporting to the RVDSS suggests that standardization of sample procurement, testing and reporting procedures would likely reduce the overall false negative rate. The accuracy of diagnostic tests is known to be affected by the quality of the specimen [10, 11] , its handling, the timing of collection after symptom onset, and the age of the patient [14, 15] . Even with the most sensitive molecular methodologies, yield was shown to be strongly related to the time since onset of symptoms [9, 14] , with a 3-fold decline in proportion positive within 3 to 5 days after onset of symptoms for both RT-PCR and culture procedures. For most laboratory tests, specimen procurement within 72 hours of from the onset of symptoms is recommended [6] , yet patients often present much later in the course of illness. Estimates of the median time since onset of symptoms suggest a delay of 3 and 5 days for outpatient and inpatients respectively [15] , however these estimates are limited to patients with laboratory confirmed influenza. In addition, there are inherent differences in the performance characteristics of the currently used diagnostic tests [4, 6, 8, [34] [35] [36] [37] [38] . Lack of standardization between diagnostic tests and algorithms used in different laboratories reporting to the RVDSS adds to this complexity. The routine use of RT-PCR testing has only recently become available in Canada (only 20% of tests used RT-PCR methods as of 2005/06 [23]), but increased use of this modality is expected to improve accuracy.

Population or system level sensitivity estimates that include the effects of sample quality are limited. Grijalva and colleagues [39] estimated the diagnostic sensitivity in a capture recapture study of children hospitalized for respiratory complications at 69% for a RT-PCR

based system and 39% for a clinical-laboratory based system (passive surveillance of tests performed during clinical practice, and using a variety of commercially available tests).

Though the expected proportion of influenza tests that were due to influenza infections is unknown and variable, our model estimate of 30% appears plausible. Cooper and colleagues [33] attributed 22% of telephone health calls for cold/flu to influenza over two relatively mild years, and elsewhere 20% of admissions for acute respiratory infections (including influenza) in adults aged 20-64 years were attributed to influenza, and 42% for seniors [1] .

While there are limitations with this approach, there are no other simple alternatives to assist in the interpretation of the RVDSS data. It would have been helpful to analyze data based on each specimen sent for testing. With only the number of weekly tests and number of positive results, we were unable to calculate the number of specimens that were actually found to be negative for all four viruses, or to estimate the extent of co-infection. Coinfection, which was not accounted for in our model, could result in an underestimation of the number of falsely negative tests, as the attribution of an influenza negative test that was actually coinfecting with influenza and another respiratory virus would have to be split between the viruses. With auxiliary information associated with each specimen, model estimates of false negative rates based on, for example, test type, time since onset of symptoms, age of the patient, or clinical presentation would have allowed us to explore the reasons for the high false negative rates. As the false negative rate appears to be laboratory dependant (data not shown), this estimated range is applicable only to the RVDSS for the study period. A significant reduction in the false negative rate is anticipated as methods become standardized and with the uptake of the new RT-PCR methods. As positive results, particularly for culture, are often obtained a week or more after the specimen was received, some positive results may have been reported in a different week than the test. Multiple test results for a single specimen may have also contributed to reporting irregularities. These irregularities would tend to bias the estimated parameter towards zero, and hence the estimated sensitivity towards 1. Considering the overall model fit and the relative severity of influenza [1] , we conclude that our estimate of sensitivity may be slightly over-estimated (number of false negatives under-estimated).

Poor test sensitivity contributes to the chronic underestimation of the burden of influenza in the general population. Since estimates of the burden of illness drive planning for

preventive and therapeutic interventions, it is important to improve all aspects leading to improved diagnostic accuracy. We have illustrated a simple method that uses the surveillance data itself to estimate the system wide sensitivity associated with the weekly proportion of tests confirmed positive. Although our estimate of sensitivity is only applicable to the interpretation of the RVDSS data over the study period, similar estimates for specific cohorts or laboratory procedures may help guide further investigation into the reasons for the large number of false negative test results. The capacity for improved diagnostic accuracy will ultimately improve our understanding of the epidemiology of influenza.

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Can't RIDD off viruses

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Abstract: The mammalian genome has evolved to encode a battery of mechanisms, to mitigate a progression in the life cycle of an invasive viral pathogen. Although apparently disadvantaged by their dependence on the host biosynthetic processes, an immensely faster rate of evolution provides viruses with an edge in this conflict. In this review, I have discussed the potential anti-virus activity of inositol-requiring enzyme 1 (IRE1), a well characterized effector of the cellular homeostatic response to an overloading of the

endoplasmic reticulum (ER) protein-folding capacity. IRE1, an ER-membrane-resident ribonuclease (RNase), upon activation catalyses regulated cleavage of select protein-coding and non-coding host RNAs, using an RNase domain which is homologous to that of the known anti-viral effector RNaseL. The latter operates as part of the Oligoadenylate synthetase OAS/RNaseL system of anti-viral defense mechanism. Protein-coding RNA substrates are differentially treated by the IRE1 RNase to either augment, through cytoplasmic splicing of an intron in the Xbp1 transcript, or suppress gene expression. This referred suppression of gene expression is mediated through degradative cleavage of a select cohort of cellular RNA transcripts, initiating the regulated IRE1-dependent decay (RIDD) pathway. The review first discusses the anti-viral mechanism of the OAS/RNaseL system and evasion tactics employed by different viruses. This is followed by a review of the RIDD pathway and its potential effect on the stability of viral RNAs. I conclude with a comparison of the enzymatic activity of the two RNases followed by deliberations on the physiological consequences of their activation.

Text: Establishment of infection by a virus, even in permissive host cells, is beset with a plethora of challenges from innate-antiviral and cell-death pathways. Therefore, the host response to a virus infection might prove to be inhibitory for the viral life cycle in a direct or an indirect manner. The direct mechanism involves expression of multiple anti-viral genes that have evolved to recognize, react, and thereby rid the infected host of the viral nucleic acid (Zhou et al., 1997; Thompson et al., 2011) . On the other hand the pathways, e.g., those that culminate in initiating an apoptotic death for the host cell, indirectly serve to limit the spread of virus (Roulston et al., 1999) . A major difference between these two mechanisms is that while the former response is transmissible to neighboring uninfected cells through interferon (IFN) signaling, the latter is observed mostly in cis. Recent reports, however, have demonstrated transmission of an apoptotic signal between cells that are in contact through gap junctions, although such a signaling from an virus infected host cell to an uninfected one is not known yet (Cusato et al., 2003; Udawatte and Ripps, 2005; Kameritsch et al., 2013) . Successful viral pathogens, through a process of active selection, have evolved to replicate and simultaneously evade or block either of these host responses. The viral nucleic acids which could be the genome (positive-sense singlestranded RNA virus) or RNA derived from transcription of the genome [negative-stranded single-sense RNA or double-stranded RNA (dsRNA) or DNA virus], offer critical targets for both detection and eradication. The viral nucleic acid targeting armaments in the host arsenal include those that recognize the associated molecular patterns like toll-like receptors (TLRs), DDX58 (or RIG-1), IFIH1 (or MDA5), IFIT proteins [IFN-stimulated genes (ISG)56 and ISF54], etc. (Aoshi et al., 2011; Bowzard et al., 2011; Jensen and

Thomsen, 2012) . This is followed by IFN signaling and expression or activation of factors that target the inducer for degradation or modification like OAS/ribonuclease L (RNaseL) system, APOBEC3, MCPIP1, the ZC3HAV1/exosome system and RNAi pathways (Gao et al., 2002; Sheehy et al., 2002; Guo et al., 2007; Daffis et al., 2010; Sidahmed and Wilkie, 2010; Schmidt et al., 2012; Cho et al., 2013a; Lin et al., 2013) . In this review we focus on two proteins containing homologous RNase domains, RNaseL with a known direct antiviral function and Inositolrequiring enzyme 1 (IRE1 or ERN1) which has an RNaseL-like RNase domain with a known role in homeostatic response to unfolded proteins in the endoplasmic reticulum (ER) and a potential to function as an antiviral (Figure 1 ; Tirasophon et al., 2000) .

In mammalian cells the tell-tale signs of RNA virus infection, like the presence of cytosolic RNA having 5 -ppp or extensive (>30 bp) dsRNA segments are detected by dedicated pathogen associated molecular pattern receptors (PAMPs) or pattern recognition receptors (PRRs) in the host cell, like RIG-1, MDA5, and the IFIT family of proteins (Aoshi et al., 2011; Bowzard et al., 2011; Vabret and Blander, 2013) . The transduction of a signal of this recognition results in the expression of IFN genes the products www.frontiersin.org FIGURE 1 | Schematic representation of the ribonuclease activity of IRE1 and RNaseL showing cross-talk between the paths catalysed by the enzymes. The figure shows activation of RNase activity following dimerization triggered by either accumulation of unfolded proteins in the ER-lumen or synthesis of 2-5A by the enzyme OAS, respectively, for IRE1 and RNaseL. The cleavage of Xbp1u by IRE1 releases an intron thus generating Xbp1s. The IRE1 targets in RIDD pathway or all RNaseL substrates are shown to undergo degradative cleavage. The cleavage products generated through degradation of the respective substrate is shown to potentially interact with RIG-I thereby leading to Interferon secretion and trans-activation of Oas genes through Interferon signaling. Abbreviations: RIG-I = retinoic acid inducible gene-I, Ifnb = interferon beta gene loci, IFN = interferons, ISG = interferon-sensitive genes, 2-5A = 2 -5 oligoadenylates.

of which upon secretion outside the cell bind to cognate receptors, initiating further downstream signaling (Figure 1 ; Randall and Goodbourn, 2008) . The genes that are regulated as a result of IFN signaling are termed as IFN-stimulated or IFN-regulated genes (ISGs or IRGs; Sen and Sarkar, 2007; Schoggins and Rice, 2011) . Oligoadenylate synthetase or OAS genes are canonical ISGs that convert ATP into 2 -5 linked oligoadenylates (2-5A) by an unique enzymatic mechanism (Figure 1 ; Hartmann et al., 2003) . Further, they are RNA-binding proteins that function like PRRs, in a way that the 2-

5A synthesizing activity needs to be induced through an interaction with dsRNA (Minks et al., 1979; Hartmann et al., 2003) . In a host cell infected by an RNA virus, such dsRNA is present in the form of replication-intermediates (RI), which are synthesized by the virus-encoded RNA-dependent RNA polymerases (RdRp) and subsequently used by the same enzyme to synthesize more genomic RNA, through asymmetric transcription (Weber et al., 2006) . However, the replications complexes (RCs) harboring these RI molecules are found secluded inside host-membrane derived vesicles, at least in positive-strand RNA viruses, a group which contains many human pathogens (Uchil and Satchidanandam, 2003; Denison, 2008) . Reports from different groups suggest OAS proteins to be distributed both in the cytoplasm as well as in membrane-associated fractions, perhaps indicating an evolution of the host anti-viral methodologies towards detection of the membrane-associated viral dsRNAs (Marie et al., 1990; Lin et al., 2009) . DNA viruses on the other hand, produce dsRNA by annealing of RNA derived from transcription of both strands in the same viral genomic loci, which are probably detected by the cytoplasmic pool of OAS proteins (Jacobs and Langland, 1996; Weber et al., 2006) . Post-activation the OAS enzymes synthesize 2-5A molecules in a non-processive reaction producing oligomers which, although potentially ranging in size from dimeric to multimeric, are functionally active only in a trimeric or tetrameric form (Dong et al., 1994; Sarkar et al., 1999; Silverman, 2007) . These small ligands, which bear phosphate groups (1-3) at the 5 end and hydroxyl groups at the 2 and 3 positions, serve as co-factor which can specifically interact with and thereby allosterically activate, existing RNaseL molecules (Knight et al., 1980; Zhou et al., 1997; Zhou et al., 2005; Sarkar et al., 1999) . As part of a physiological control system these 2-5A oligomers are quite unstable in that they are highly susceptible to degradation by cellular 5'-phosphatases and PDE12 (2'-phosphodiesterase; Silverman et al., 1981; Johnston and Hearl, 1987; Kubota et al., 2004; Schmidt et al., 2012) . Viral strategies to evade or overcome this host defense mechanism ranges from preventing IFN signaling which would hinder the induction of OAS expression or thwarting activation of expressed OAS proteins by either shielding the viral dsRNA from interacting with it or modulating the host pathway to synthesize inactive 2-5A derivatives (Cayley et al., 1984; Hersh et al., 1984; Rice et al., 1985; Maitra et al., 1994; Beattie et al., 1995; Rivas et al., 1998; Child et al., 2004; Min and Krug, 2006; Sanchez and Mohr, 2007; Sorgeloos et al., 2013) . Shielding of viral RNA from interacting with OAS is possible through enclosure of dsRNA replication intermediates in membrane enclosed compartments as observed in many flaviviruses (Ahlquist, 2006; Miller and Krijnse-Locker, 2008; Miorin et al., 2013) .

RNaseL is a 741 amino acid protein containing three predominantly structured region, an N-terminal ankyrin repeat domain (ARD), a middle catalytically inactive pseudo-kinase (PK)

and a C-terminal RNase domain (Figure 2A ; Hassel et al., 1993; Zhou et al., 1993) . The activity of the RNase domain is negatively regulated by the ARD, which is relieved upon binding of 2-5A molecules to ankyrin repeats 2 and 4 followed by a conformational alteration (Figure 1 ; Hassel et al., 1993; Tanaka et al., 2004; Nakanishi et al., 2005) . In support of this contention, deletion of the ARD has been demonstrated to produce constitutively active RNaseL, although with dramatically lower RNase activity (Dong and Silverman, 1997) . However, recent reports suggest that while 2-5A links the ankyrin repeats from adjacent molecules leading to formation of dimer and higher order structures, at sufficiently high in vitro concentrations, RNaseL could oligomerize even in the absence of 2-5A . Nonetheless, in vivo the RNaseL nuclease activity still seems to be under the sole regulation of 2-5A (Al-Saif and Khabar, 2012) . In order to exploit this dependence, multiple viruses like mouse hepatitis virus (MHV) and rotavirus group A (RVA) have evolved to encode phosphodiesterases capable of hydrolysing the 2 -5 linkages in 2-5A and thereby attenuate the RNaseL cleavage activity (Zhao et al., 2012; Zhang et al., 2013) . In addition to 5 -phosphatases and 2 -phosphodiesterases to reduce ClustalW alignment of primary sequence from a segment of the PK domain indicating amino acid residues which are important for interacting with nucleotide cofactors. The conserved lysine residues, critical for this interaction (K599 for IRE1 and K392 in RNaseL) are underlined. (C) Alignment of the KEN domains in RNaseL and IRE1. The amino acids highlighted and numbered in IRE1 are critical for the IRE1 RNase activity (Tirasophon et al., 2000) .

the endogenous 2-5A levels, mammalian genomes encode posttranscriptional and post-translation inhibitors of RNaseL activity in the form of microRNA-29 and the protein ABCE1 (RNaseL inhibitor or RLI), respectively (Bisbal et al., 1995; Lee et al., 2013) . Direct inhibition of RNaseL function is also observed upon infection by Picornaviruses through, either inducing the expression of ABCE1 or exercising a unique inhibitory property of a segment of the viral RNA (Martinand et al., 1998 (Martinand et al., , 1999 Townsend et al., 2008; Sorgeloos et al., 2013) .

Once activated by 2-5A, RNaseL can degrade single-stranded RNA irrespective of its origin (virus or host) although there seems to exist a bias towards cleavage of viral RNA (Wreschner et al., 1981a; Silverman et al., 1983; Li et al., 1998) . RNA sequences that are predominantly cleaved by RNaseL are U-rich with the cleavage points being typically at the 3' end of UA or UG or UU di-nucleotides, leaving a 5'-OH and a 3'-monophosphate in the cleavage product (Floyd-Smith et al., 1981; Wreschner et al., 1981b) . A recent report shows a more general consensus of 5'-UNN-3' with the cleavage point between the second

and the third nucleotide (Han et al., 2014) . Cellular targets of RNaseL include both ribosomal RNA (rRNA) and mRNAs, the latter predominantly representing genes involved in protein biosynthesis (Wreschner et al., 1981a; Al-Ahmadi et al., 2009; Andersen et al., 2009) . Additionally, RNaseL activity can also degrade specific ISG mRNA transcripts and thereby attenuate the effect of IFN signaling (Li et al., 2000) . Probably an evolution towards insulating gene expression from RNaseL activity is observed in the coding region of mammalian genes where the UU/UA dinucleotide frequency is rarer (Bisbal et al., 2000; Khabar et al., 2003; Al-Saif and Khabar, 2012) . Perhaps not surprisingly, with a much faster rate of evolution, similar observations have been made with respect to evasion of RNaseL mediated degradation by viral RNAs too (Han and Barton, 2002; Washenberger et al., 2007) . Moreover, nucleoside modifications in host mRNAs, rarely observed in viral RNAs, have also been shown to confer protection from RNaseL (Anderson et al., 2011) . In addition to directly targeting viral RNA, the reduction in functional ribosomes and ribosomal protein mRNA affects viral protein synthesis and replication in an indirect manner. Probably, as a reflection of these effects on cellular RNAs, RNaseL is implicated as one of the factors determining the anti-proliferative effect of IFN activity . The anti-viral activity of RNaseL extends beyond direct cleavage of viral RNA, through stimulation of RIG-I by the cleavage product (Malathi et al., , 2007 (Malathi et al., , 2010 . A global effect of RNaseL is observed in the form of autophagy induced through c-jun N-terminal kinase (JNK) signaling and apoptosis, probably as a consequence of rRNA cleavage (Li et al., 2004; Chakrabarti et al., 2012; Siddiqui and Malathi, 2012) . RNaseL has also been demonstrated to play a role in apoptotic cell death initiated by pharmacological agents extending the physiological role of this pathway beyond the boundary of being only an anti-viral mechanism (Castelli et al., 1997 (Castelli et al., , 1998 .

The ER serves as a conduit for maturation of cellular proteins which are either secreted or destined to be associated with a membrane for its function. An exclusive microenvironment (high Calcium ion and unique ratio of reduced to oxidized glutathione) along with a battery of ER-lumen resident enzymes (foldases, chaperones, and lectins) catalyse/mediate the necessary folding, disulfide-bond formation, and glycosylation reactions (Schroder and Kaufman, 2005) . A perturbation of the folding capacity, due to either physiological disturbances or virus infection, can lead to an accumulation of unfolded proteins in the ER lumen, which signals an unfolded protein response (UPR). UPR encompasses a networked transcriptional and translational gene-expression program, initiated by three ER-membrane resident sensors namely IRE1 or ERN1, PKR-like ER Kinase (PERK or EIF2AK3) and activating transcription factor 6 (ATF6; Hetz, 2012) . IRE1 is a type I single-pass trans-membrane protein in which, similar to what is observed with RNaseL, the

N-terminal resident in the ER lumen serves as sensor and the cytosolic C-terminal as the effector (Figure 1 ; Chen and Brandizzi, 2013) . The IRE1 coding gene is present in genomes ranging from yeast to mammals and in the latter is ubiquitously expressed in all tissues (Tirasophon et al., 1998) . Signal transduction by stimulated IRE1 initiates multiple gene regulatory pathways with either pro-survival or pro-apoptotic consequences (Kaufman, 1999) . During homeostasis or unstressed conditions the sensor molecules are monomeric, a state maintained co-operatively by the " absence" of unfolded proteins and the "presence" of HSPA5 (GRP78 or Bip, an ERresident chaperone) molecules bound to a membrane-proximal disordered segment of the protein in the ER-lumen-resident Nterminus (Credle et al., 2005) . Accumulated unfolded proteins in the lumen triggers coupling of this domain from adjacent sensor molecules through a combination of (a) titration of the bound HSPA5 chaperone molecules and (b) direct tethering by malformed protein molecules (Shamu and Walter, 1996; Credle et al., 2005; Aragon et al., 2009; Korennykh et al., 2009) . Abutting of the luminal domains juxtapose the cytosolic C-terminal segments, leading to an aggregation of the IRE1 molecules into distinct ER-membrane foci (Kimata et al., 2007; Li et al., 2010) . The C-terminal segment has a serine/threonine kinase domain and a RNase domain homologous to that of RNaseL (Figure 1 ; Tirasophon et al., 1998 Tirasophon et al., , 2000 . A trans-autophosphorylation by the kinase domain allosterically activates the RNase domain (Tirasophon et al., 2000; Lee et al., 2008; Korennykh et al., 2009) . In fact, exogenous over-expression of IRE1 in mammalian cells lead to activation suggesting that, under homeostatic conditions, the non-juxtaposition of cytosolic domains maintains an inactive IRE1 (Tirasophon et al., 1998) . Once activated, IRE1 performs cleavage of a variety of RNA substrates mediated by its RNase domain, in addition to phosphorylating and thereby activating JNK (Cox and Walter, 1996; Urano et al., 2000) . Depending on the RNA substrate, the cleavage catalyzed by IRE1 RNase produces differential consequence. Although scission of the Xbp1 mRNA transcript at two internal positions is followed by splicing of the internal segment through ligation of the terminal cleavage products, that in all other known IRE1 target RNA is followed by degradation (Figure 1 ; Sidrauski and Walter, 1997; Calfon et al., 2002) . The latter mode of negative regulation of gene expression is termed as the regulated IRE1-dependent decay (RIDD) pathway (Hollien and Weissman, 2006; Oikawa et al., 2007; Iqbal et al., 2008; Lipson et al., 2008) . Gene transcripts regulated by RIDD pathway includes that from IRE1 (i.e., selftranscripts), probably in a negative feedback loop mechanism (Tirasophon et al., 2000) . In addition to protein coding RNA, RIDD pathway down-regulates the level of a host of microRNA precursors (pre-miRNAs) and can potentially cleave in the anticodon loop of tRNA Phe (Korennykh et al., 2011; Upton et al., 2012) .

The IRE1 RNase domain cleaves the Xbp1u (u for unspliced) mRNA transcript at two precise internal positions within the open reading frame (ORF) generating three segments, the terminal two of which are ligated by a tRNA ligase in yeast and by an unknown ligase in mammalian cells, to produce the Xbp1s (s for spliced) mRNA transcript (Figure 1 ; Yoshida et al., 2001) . The Xbp1s thus generated has a longer ORF, which is created by a frame-shift in the coding sequence downstream of the splice site (Cox and Walter, 1996; Calfon et al., 2002) . A similar dual endonucleolytic cleavage is also observed to initiate the XRN1 and Ski2-3-8 dependent degradation of transcripts in the RIDD degradation pathway (Hollien and Weissman, 2006) . The RIDD target transcript genes are predominantly those that encode membrane-associated or secretory proteins and which are not necessary for ER protein folding reactions (Hollien and Weissman, 2006) . The cleavage of Xbp1 and the RIDD-target transcripts constitute homeostatic or pro-survival response by IRE1 since XBP1S trans-activates genes encoding multiple chaperones (to fold unfolded proteins) and the ERAD pathway genes (to degrade terminally misfolded proteins) whereas RIDD reduces flux of polypeptides entering the ER lumen (Lee et al., 2003; Hollien and Weissman, 2006) . On the other hand, cleavage of pre-miRNA transcripts which are processed in the cell to generate CASPASE-2 mRNA (Casp2) controlling miRNAs, constitutes the pro-apoptotic function of IRE1 (Upton et al., 2012) . Another pro-apoptotic signal from IRE1 emanates from signaling through phosphorylation of JNK1 (Urano et al., 2000) . Although in the initial phase RIDD activity does not cleave mRNAs encoding essential ER proteins, at later stages of chronic UPR such transcripts are rendered susceptible to degradation promoting apoptosis induction (Han et al., 2009; Bhattacharyya et al., 2014) .

Infection of mammalian cells by a multitude of viruses induce an UPR which is sometimes characterized by suppression of signaling by one or more of the three sensor(s; Su et al., 2002; Tardif et al., 2002; He, 2006; Yu et al., 2006 Yu et al., , 2013 Medigeschi et al., 2007; Zhang et al., 2010; Merquiol et al., 2011) . Among these at least two viruses from diverse families, HCMV (a DNA virus) and hepatitis C virus (a hepacivirus), interfere with IRE1 signaling by different mechanism (Tardif et al., 2004; Stahl et al., 2013) . An observed inhibition of any cellular function by a virus infection could suggest a potential anti-virus function for it, which the virus has evolved to evade through blocking some critical step(s). In both the cases mentioned above, stability of the viral proteins seems to be affected by ERAD-mediated degradation, although other potential anti-viral effect of IRE1 activation are not clear yet (Isler et al., 2005; Saeed et al., 2011) . Interestingly, host mRNA fragments produced following IRE1 activation during bacterial infection, has been shown to activate RIG-I signaling (Figure 1 ; Cho et al., 2013b) . Theoretically, other functions of IRE1 can also have anti-viral effect necessitating its inhibition for uninhibited viral replication. It is,

however, still not clear whether IRE1 is able to cleave any viral RNA (or mRNA) in a manner similar to that of other RIDD targets (Figure 1) . The possibilities of such a direct anti-viral function are encouraged by the fact that all these viruses encode at least one protein which, as part of its maturation process, requires glycosylation and disulfide-bond formation. Such a necessity would entail translation of the mRNA encoding such a protein, which in case of positive-sense single-stranded RNA viruses would mean the genome, in association with the ER-membrane (Figure 1 ; Lerner et al., 2003) . Additionally for many RNA viruses, replication complexes are housed in ER-derived vesicular structures (Denison, 2008; den Boon et al., 2010) . Considering the proximity of IRE1 and these virus-derived RNAs it is tempting to speculate that probably at some point of time in the viral life cycle one or more virus-associated RNA would be susceptible to cleavage by IRE1. However, studies with at least two viruses have shown that instead of increasing viral titre, inhibiting the RNase activity of activated IRE1 has an opposite effect (Hassan et al., 2012; Bhattacharyya et al., 2014) . This implies potential benefits of IRE1 activation through one or more of the following, (a) expression of chaperones or other pro-viral molecules downstream of XBP1 Supregulation or JNK-activation, (b) cleavage of potential anti-viral gene mRNA transcripts by RIDD activity. However, the mode of protection for the viral RNA from RIDD activity is still not clear. It is possible that the viral proteins create a subdomain within the ER membrane, which through some mechanism excludes IRE1 from diffusing near the genomic RNA, thereby protecting the replication complexes (Denison, 2008) . It is therefore probably not surprising that single-stranded plus-sense RNA viruses encode a polyprotein, which produces replication complexes in cis, promoting formation of such subdomains (Egger et al., 2000) . The fact that IRE1 forms bulky oligomers of higher order probably aggravates such an exclusion of the activated sensor molecules from vicinity of the viral replication complexes. The UPR signaling eventually attenuate during chronic ER-stress and since that is what a virus-induced UPR mimics, probably the viral RNA needs protection only during the initial phase of UPR activation (Lin et al., 2007) . Since the choice of RIDD target seems to be grossly driven towards mRNAs that encode ER-transitory but are not ER-essential proteins, it is also possible that one or more viral protein have evolved to mimic a host protein the transcript of which is RIDD-resistant (Hollien and Weissman, 2006) . Most of the RIDD target mRNA are observed to be ER-membrane associated, the proximity to IRE1 facilitating association and cleavage (Figure 1 ; Hollien and Weissman, 2006) . Although ER-association for an mRNA is possible without the mediation of ribosomes, Gaddam and co-workers reported that continued association with polysomes for a membrane-bound mRNA can confer protection from IRE1 cleavage (Cui et al., 2012; Gaddam et al., 2013) . This would suggest important implications for the observed refractory nature of Japanese encephalitis virus (JEV) and influenza virus RNA to RIDD cleavage (Hassan et al., 2012; Bhattacharyya et al., 2014) . In contrast to Influenza virus,

flaviviruses (which include JEV) do not suppress host protein synthesis implying the absence of a global inhibition on translation as would be expected during UPR (Clyde et al., 2006; Edgil et al., 2006) . Therefore, a continued translation of viral RNA in spite of UPR activation can in principle confer protection from the pattern of RNA cleavage observed in the RIDD pathway.

IRE1 and RNaseL, in addition to biochemical similarities in protein kinase domain and structural similarities in their RNase domain, share the functional consequences of their activation in initiating cellular apoptosis through JNK signaling (Table 1 and Figure 2 ; Liu and Lin, 2005; Dhanasekaran and Reddy, 2008) . Though initial discoveries were made in the context of homeostatic and anti-viral role for the former and latter, differences between the pathways are narrowed by further advances in research. In the same vein, while inhibition of IRE1 signaling in virus infected cells indicates a potential anti-viral role, www.frontiersin.org (Tirasophon et al., 1998; Dong and Silverman, 1999; Papa et al., 2003; Lin et al., 2007) Nature of RNase substrates Both 28S rRNA and mRNAs IRE1 β can cleave both 28S rRNA and mRNA while IRE1 α substrates include only mRNAs (Iwawaki et al., 2001) Dissimilarities

Cleavage substrates Beside 28S rRNA, predominantly cleaves mRNAs encoding ribosomal proteins (Andersen et al., 2009) Xbp1u and other mRNAs in addition to microRNA precursors which are targeted as part of the RIDD pathway Selection of cleavage site Cleaved between 2nd and 3rd nucleotide positions of UN/N sites (Han et al., 2014) RNA sequence with the consensus of 5'-CUGCAG-3' in association with a stem-loop (SL) structure essential for recognition of Xbp1u and other mRNAs (Oikawa et al., 2010) association of RNaseL mutations with generation of prostate cancer extends the ambit of influence of this anti-viral effector to more non-infectious physiological disorders (Silverman, 2003) . Biochemically, the similarity in their RNase domains does not extend to the choice of either substrates or cleavage point, which are downstream of UU or UA in RNaseL and downstream of G (predominantly) for IRE1 (Figure 2C ; Yoshida et al., 2001; Hollien and Weissman, 2006; Upton et al., 2012) . Further, while RNaseL cleaves predominantly in single-stranded region, IRE1 seems to cleave equally well in single-and double-stranded region (Upton et al., 2012) . However, a recent report suggested a consensus cleavage site with the sequence UN/N, in RNaseL targets and in those mRNAs that are cleaved by IRE1 as part of the RIDD pathway (Han et al., 2014) . Access to potential cleavage substrate for RNaseL is conjectured to be facilitated through its association with polyribosomes, while no such association is known for IRE1 (Salehzada et al., 1991) .

Possibilities exist that IRE1 would have preferential distribution in the rough ER which, upon activation, would give it ready access to mRNAs for initiating the RIDD pathway.

In the context of a virus infection, the pathway leading from both these proteins have the potential to lead to cell death. Notwithstanding the fact that this might be an efficient way of virus clearance, it also portends pathological outcomes for the infected organism. Future research would probably lead to design of drugs targeting these proteins based on the structural homology of their effector domains, regulating the pathological denouement of their activation without compromising their anti-viral or potential anti-viral functions.

--- END OF ARTICLE ---

--- START OF ARTICLE 41 ---

Nearly Complete Genome Sequence of an Echovirus 30 Strain from a Cluster of Aseptic Meningitis Cases in California, September 2017

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6953510/>

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Abstract: We report the nearly complete genome sequence of a human enterovirus, a strain of echovirus 30, obtained from a cerebrospinal fluid specimen from a teenaged patient with aseptic meningitis in September 2017.

Text: Echoviruses are members of the Enterovirus B species of the Enterovirus (EV) genus in the Picornaviridae family of nonenveloped, single-stranded, positive-sense RNA viruses. Echoviruses were named from the acronym enteric cytopathic human orphan virus at the time of their discovery in the 1950s but were later found to be associated with respiratory illness, hand-foot-and-mouth disease, and aseptic meningitis, similar to other enteroviruses (1) .

According to the California Code of Regulations, meningitis cases are reportable to the California Department of Public Health (CDPH) within 1 day of identification of etiology (2) . In the fall of 2017, a cluster of aseptic meningitis cases from a northern California high school were reported to the CDPH. The Viral and Rickettsial Disease Laboratory (VRDL) at the CDPH detected EV from 19 of 30 patients (63%) by real-time reverse transcription-PCR (RT-PCR), as previously described (3) . We generated and analyzed partial capsid (viral protein 1 [VP1]) sequences using methods developed by Minnaar et al. (4) . Fifteen of 19 (79%) EV-positive patients were confirmed to have echovirus 30 (E-30), using cerebrospinal fluid (CSF) samples. This cluster of E-30 meningitis cases is similar to previously reported E-30 aseptic meningitis cases (5, 6) in symptoms and epidemiology.

Here, we report a nearly complete genome sequence from one of the E-30-positive CSF specimens. The CSF was processed by centrifugation, 0.45- μ m filtration, and nuclease treatment prior to extraction using the NucliSENS easyMAG system (bioMérieux, Durham, NC) (7). The extracted nucleic acids were then treated with DNase to yield RNA, which was subjected to random reverse transcription and PCR (7) . The next-generation sequencing (NGS) library was prepared using a Nextera XT kit and sequenced on a MiSeq platform 300-cycle paired-end run (Illumina, San Diego, CA). The NGS data were analyzed using an in-house Centers for Disease Control and Prevention (CDC) pipeline which involves the removal of host sequences using bowtie2/2.3.3.1, primer removal, low-quality (below Q20) and read length ($\geq$ 50 nucleotides) filtering using cutadapt 1.18, read duplication removal using a Dedup.py script, de novo assembly using SPAdes 3.7 default parameters, and BLAST search of the resultant contigs (8) . There were a total of 141,329 postprocessing FASTQ reads. The final consensus genome was inspected and annotated using Geneious v10.0.9 (9) . The contig was built from 15,712 reads, assembled to an E-30 reference genome (GenBank accession number JX976773), and deemed nearly complete by comparison to the reference, and the termini were determined as part of the protocol (7).

The total GC content is 48.3% for 7,155 bases. The average read coverage was 260-fold for the E-30 genome.

The genome sequence was designated E-30 USA/2017/CA-RGDS-1005. Its VP1 sequence was confirmed by the CDC Picornavirus Laboratory to be nearly identical to those of E-30s identified in an aseptic meningitis outbreak that occurred in the fall of 2017 in Nevada; it also has greater than 99% nucleotide identity to the VP1 sequences of E-30 strains from the southern United States identified by the CDC in May 2017 (GenBank accession numbers MG584831 and MG584832), as measured using the online version of blastn (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>). The genome sequence of E-30 USA/2017/CA-RGDS-1005 shares less than 89% nucleotide identity (NI) and less than 98% amino acid identity (AI) with other publicly available E-30 sequences. The sequence contains the complete protein-coding region, with short sections in the untranslated regions (UTRs) missing due to a lack of read coverage (approximately 182 and 90 nucleotides of the 5' and 3' UTRs, respectively). The enterovirus polyprotein can be divided into one structural (P1-capsid) and two nonstructural (P2 and P3) regions. The polyprotein regions of the E-30 genome reported here share 96%, 88%, and 84% NI (P1, P2, and P3, respectively) with other E-30 sequences in GenBank.

Data availability. The E-30 sequence of USA/2017/CA-RGDS-1005 has been deposited in GenBank under the accession number MK238483. The quality-filtered FASTQ reads have been deposited in the Sequence Read Archive with the run accession number SRR10082176.

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The findings and conclusions in this report are those of the author(s) and do not necessarily represent the official position of the Centers for Disease Control and Prevention.

--- END OF ARTICLE ---

--- START OF ARTICLE 42 ---

iNR-Drug: Predicting the Interaction of Drugs with Nuclear Receptors in Cellular Networking

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Abstract: Nuclear receptors (NRs) are closely associated with various major diseases such as cancer, diabetes, inflammatory disease, and osteoporosis. Therefore, NRs have become a frequent target for drug development. During the process of developing drugs against these diseases by targeting NRs, we are often facing a problem: Given a NR and chemical compound, can we identify whether they are really in interaction with each other in a cell? To address this problem, a predictor called “iNR-Drug” was developed. In the predictor, the drug compound concerned was formulated by a 256-D (dimensional) vector derived from its molecular fingerprint, and the NR by a 500-D vector formed by incorporating its sequential evolution information and physicochemical features into the general form of pseudo amino acid composition, and the prediction engine was operated by the SVM (support vector machine) algorithm. Compared with the existing prediction methods in this area, iNR-Drug not only can yield a higher success rate, but is also featured by a user-friendly web-server established at <http://www.jci-bioinfo.cn/iNR-Drug/>, which is particularly useful for most experimental scientists to obtain their desired data in a timely manner. It is anticipated that the iNR-Drug server may become a useful high throughput

tool for both basic research and drug development, and that the current approach may be easily extended to study the interactions of drug with other targets as well.

Text: With the ability to directly bind to DNA (Figure 1) and regulate the expression of adjacent genes, nuclear receptors (NRs) are a class of ligand-inducible transcription factors. They regulate various biological processes, such as homeostasis, differentiation, embryonic development, and organ physiology [1] [2] [3] . The NR superfamily has been classified into seven families: NR0 (knirps or DAX like) [4, 5] ; NR1 (thyroid hormone like), NR2 (HNF4-like), NR3 (estrogen like), NR4 (nerve growth factor IB-like), NR5 (fushi tarazu-F1 like), and NR6 (germ cell nuclear factor like). Since they are involved in almost all aspects of human physiology and are implicated in many major diseases such as cancer, diabetes and osteoporosis, nuclear receptors have become major drug targets [6, 7] , along with G protein-coupled receptors (GPCRs) [8] [9] [10] [11] [12] [13] [14] [15] [16] [17] , ion channels [18] [19] [20] , and kinase proteins [21] [22] [23] [24] . Identification of drug-target interactions is one of the most important steps for the new medicine development [25, 26] . The method usually adopted in this step is molecular docking simulation [27] [28] [29] [30] [31] [32] [33] [34] [35] [36] [37] [38] [39] [40] [41] [42] [43] . However, to make molecular docking study feasible, a reliable 3D (three dimensional) structure of the target protein is the prerequisite condition. Although X-ray crystallography is a powerful tool in determining protein 3D structures, it is time-consuming and expensive. Particularly, not all proteins can be successfully crystallized. For example, membrane proteins are very difficult to crystallize and most of them will not dissolve in normal solvents. Therefore, so far very few membrane protein 3D structures have been determined. Although NMR (Nuclear Magnetic Resonance) is indeed a very powerful tool in determining the 3D structures of membrane proteins as indicated by a series of recent publications (see, e.g., [44] [45] [46] [47] [48] [49] [50] [51] and a review article [20]), it is also time-consuming and costly. To acquire the 3D structural information in a timely manner, one has to resort to various structural bioinformatics tools (see, e.g., [37]), particularly the homologous modeling approach as utilized for a series of protein receptors urgently needed during the process of drug development [19, [52] [53] [54] [55] [56] [57] . Unfortunately, the number of dependable templates for developing high quality 3D structures by means of homology modeling is very limited [37] .

To overcome the aforementioned problems, it would be of help to develop a computational method for predicting the interactions of drugs with nuclear receptors in cellular networking based on the sequences information of the latter. The results thus obtained can

be used to pre-exclude the compounds identified not in interaction with the nuclear receptors, so as to timely stop wasting time and money on those unpromising compounds [58] .

Actually, based on the functional groups and biological features, a powerful method was developed recently [59] for this purpose. However, further development in this regard is definitely needed due to the following reasons. (a) He et al. [59] did not provide a publicly accessible web-server for their method, and hence its practical application value is quite limited, particularly for the broad experimental scientists; (b) The prediction quality can be further enhanced by incorporating some key features into the formulation of NR-drug (nuclear receptor and drug) samples via the general form of pseudo amino acid composition [60] .

The present study was initiated with an attempt to develop a new method for predicting the interaction of drugs with nuclear receptors by addressing the two points.

As demonstrated by a series of recent publications [10, 18, [61] [62] [63] [64] [65] [66] [67] [68] [69] [70] and summarized in a comprehensive review [60] , to establish a really effective statistical predictor for a biomedical system, we need to consider the following steps: (a) select or construct a valid benchmark dataset to train and test the predictor; (b) represent the statistical samples with an effective formulation that can truly reflect their intrinsic correlation with the object to be predicted; (c) introduce or develop a powerful algorithm or engine to operate the prediction; (d) properly perform cross-validation tests to objectively evaluate the anticipated accuracy of the predictor; (e) establish a user-friendly web-server for the predictor that is accessible to the public. Below, let us elaborate how to deal with these steps.

The data used in the current study were collected from KEGG (Kyoto Encyclopedia of Genes and Genomes) [71] at <http://www.kegg.jp/kegg/>. KEGG is a database resource for understanding high-level functions and utilities of the biological system, such as the cell, the organism and the ecosystem, from molecular-level information, especially large-scale molecular datasets generated by genome sequencing and other high-throughput experimental technologies. Here, the benchmark dataset can be formulated as

where is the positive subset that consists of the interactive drug-NR pairs only, while the negative subset that contains of the non-interactive drug-NR pairs only, and the symbol represents the union in the set theory. The so-called "interactive" pair here means the pair whose two counterparts are interacting with each other in the drug-target networks as defined in the KEGG database [71]; while the "non-interactive" pair means that its two counterparts are not interacting with each other in the drug-target networks. The positive dataset contains 86 drug-NR pairs, which were taken from He et al. [59]. The negative dataset contains 172 non-interactive drug-NR pairs, which were derived according to the following procedures: (a) separating each of the pairs into single drug and NR; (b) re-coupling each of the single drugs with each of the single NRs into pairs in a way that none of them occurred in ; (c) randomly picking the pairs thus formed until reaching the number two times as many as the pairs in . The 86 interactive drug-NR pairs and 172 non-interactive drug-NR pairs are given in Supplementary Information S1, from which we can see that the $86 + 172 = 258$ pairs in the current benchmark dataset are actually formed by 25 different NRs and 53 different compounds.

Since each of the samples in the current network system contains a drug (compound) and a NR (protein), the following procedures were taken to represent the drug-NR pair sample.

First, for the drug part in the current benchmark dataset, we can use a 256-D vector to formulate it as given by

where D represents the vector for a drug compound, and d_i its i -th ($i = 1, 2, \dots, 256$) component that can be derived by following the "2D molecular fingerprint procedure" as elaborated in [10]. The 53 molecular fingerprint vectors thus obtained for the 53 drugs in are, respectively, given in Supplementary Information S2.

The protein sequences of the 25 different NRs in are listed in Supplementary Information S3. Suppose the sequence of a nuclear receptor protein P with L residues is generally expressed by

where $1R$ represents the 1st residue of the protein sequence P , $2R$ the 2nd residue, and so forth. Now the problem is how to effectively represent the sequence of Equation (3) with a

non-sequential or discrete model [72]. This is because all the existing operation engines, such as covariance discriminant (CD) [17, 65, [73] [74] [75] [76] [77] [78] [79], neural network [80] [81] [82], support vector machine (SVM) [62] [63] [64] [83], random forest [84, 85], conditional random field [66], nearest neighbor (NN) [86, 87]; K-nearest neighbor (KNN) [88] [89] [90], OET-KNN [91] [92] [93] [94], and Fuzzy K-nearest neighbor [10, 12, 18, 69, 95], can only handle vector but not sequence samples. However, a vector defined in a discrete model may completely lose all the sequence-order information and hence limit the quality of prediction. Facing such a dilemma, can we find an approach to partially incorporate the sequence-order effects? Actually, one of the most challenging problems in computational biology is how to formulate a biological sequence with a discrete model or a vector, yet still keep considerable sequence order information. To avoid completely losing the sequence-order information for proteins, the pseudo amino acid composition [96, 97] or Chou's PseAAC [98] was proposed. Ever since the concept of PseAAC was proposed in 2001 [96], it has penetrated into almost all the areas of computational proteomics, such as predicting anticancer peptides [99], predicting protein subcellular location [100] [101] [102] [103] [104] [105] [106], predicting membrane protein types [107, 108], predicting protein submitochondria locations [109] [110] [111] [112], predicting GABA(A) receptor proteins [113], predicting enzyme subfamily classes [114], predicting antibacterial peptides [115], predicting supersecondary structure [116], predicting bacterial virulent proteins [117], predicting protein structural class [118], predicting the cofactors of oxidoreductases [119], predicting metalloproteinase family [120], identifying cysteine S-nitrosylation sites in proteins [66], identifying bacterial secreted proteins [121], identifying antibacterial peptides [115], identifying allergenic proteins [122], identifying protein quaternary structural attributes [123, 124], identifying risk type of human papillomaviruses [125], identifying cyclin proteins [126], identifying GPCRs and their types [15, 16], discriminating outer membrane proteins [127], classifying amino acids [128], detecting remote homologous proteins [129], among many others (see a long list of papers cited in the References section of [60]). Moreover, the concept of PseAAC was further extended to represent the feature vectors of nucleotides [65], as well as other biological samples (see, e.g., [130] [131] [132]). Because it has been widely and increasingly used, recently two powerful soft-wares, called "PseAAC-Builder" [133] and "propy" [134], were established for generating various special Chou's pseudo-amino acid compositions, in addition to the web-server "PseAAC" [135] built in 2008.

According to a comprehensive review [60], the general form of PseAAC for a protein sequence P is formulated by

where the subscript i is an integer, and its value as well as the components $(1, 2, \dots, 20)$ will depend on how to extract the desired information from the amino acid sequence of P (cf. Equation (3)). Below, let us describe how to extract useful information to define the components of PseAAC for the NR samples concerned. First, many earlier studies (see, e.g., [136] [137] [138] [139] [140] [141]) have indicated that the amino acid composition (AAC) of a protein plays an important role in determining its attributes. The AAC contains 20 components with each representing the occurrence frequency of one of the 20 native amino acids in the protein concerned. Thus, such 20 AAC components were used here to define the first 20 elements in Equation (4); i.e., $(1) (1, 2, \dots, 20) f_i$ (5) where f_i is the normalized occurrence frequency of the i -th type native amino acid in the nuclear receptor concerned. Since AAC did not contain any sequence order information, the following steps were taken to make up this shortcoming.

To avoid completely losing the local or short-range sequence order information, we considered the approach of dipeptide composition. It contained $20 \times 20 = 400$ components [142]. Such 400 components were used to define the next 400 elements in Equation (4); i.e., $(2) (1, 2, \dots, 400) f_{ij}$

where f_{ij} is the normalized occurrence frequency of the j -th dipeptides in the nuclear receptor concerned. To incorporate the global or long-range sequence order information, let us consider the following approach. According to molecular evolution, all biological sequences have developed starting out from a very limited number of ancestral samples. Driven by various evolutionary forces such as mutation, recombination, gene conversion, genetic drift, and selection, they have undergone many changes including changes of single residues, insertions and deletions of several residues [143], gene doubling, and gene fusion. With the accumulation of these changes over a long period of time, many original similarities between initial and resultant amino acid sequences are gradually faded out, but the corresponding proteins may still share many common attributes [37], such as having basically the same biological function and residing at a same subcellular location [144, 145]. To extract the sequential evolution information and use it to define the components of Equation (4), the PSSM (Position Specific Scoring Matrix) was used as described below.

According to Schaffer [146], the sequence evolution information of a nuclear receptor protein P with L amino acid residues can be expressed by a $20 \times L$ matrix, as given by

where (7) were generated by using PSI-BLAST [147] to search the UniProtKB/Swiss-Prot database (The Universal Protein Resource (UniProt); <http://www.uniprot.org/>) through three iterations with 0.001 as the E-value cutoff for multiple sequence alignment against the sequence of the nuclear receptor concerned. In order to make every element in Equation (7) be scaled from their original score ranges into the region of $[0, 1]$, we performed a conversion through the standard sigmoid function to make it become

Now we extract the useful information from Equation (8)

Moreover, we used the grey system model approach as elaborated in [68] to further define the next 60 components of Equation (4) ($1, 2, \dots, 20$)

In the above equation, w_1 , w_2 , and w_3 are weight factors, which were all set to 1 in the current study; $f_j(1)$ has the same meaning as in Equation (5)

where α

and

Combining Equations (5), (6), (10) and (12), we found that the total number of the components obtained via the current approach for the PseAAC of Equation (4)

and each of the 500 components is given by (1)

Since the elements in Equations (2) and (4) are well defined, we can now formulate the drug-NR pair by combining the two equations as given by (19) where G represents the

drug-NR pair, $\bar{x}$ the orthogonal sum, and the $256 + 500 = 756$ components are defined by Equations (2) and (18). For the sake of convenience, let us use x_i ($i = 1, 2, \dots, 756$) to represent the 756 components in Equation (19); i.e., (20) To optimize the prediction quality with a time-saving approach, similar to the treatment [148] [149] [150], let us convert Equation (20) to

where the symbol $\bar{\cdot}$ means taking the average of the quantity therein, and SD means the corresponding standard derivation.

In this study, the SVM (support vector machine) was used as the operation engine. SVM has been widely used in the realm of bioinformatics (see, e.g., [62] [63] [64] [151] [152] [153] [154]). The basic idea of SVM is to transform the data into a high dimensional feature space, and then determine the optimal separating hyperplane using a kernel function. For a brief formulation of SVM and how it works, see the papers [155, 156]; for more details about SVM, see a monograph [157].

In this study, the LIBSVM package [158] was used as an implementation of SVM, which can be downloaded from <http://www.csie.ntu.edu.tw/~cjlin/libsvm/>, the popular radial basis function (RBF) was taken as the kernel function. For the current SVM classifier, there were two uncertain parameters: penalty parameter C and kernel parameter γ . The method of how to determine the two parameters will be given later.

The predictor obtained via the aforementioned procedure is called iNR-Drug, where "i" means identify, and "NR-Drug" means the interaction between nuclear receptor and drug compound. To provide an intuitive overall picture, a flowchart is provided in Figure 2 to show the process of how the predictor works in identifying the interactions between nuclear receptors and drug compounds.

To provide a more intuitive and easier-to-understand method to measure the prediction quality, the following set of metrics based on the formulation used by Chou [159] [160] [161] in predicting signal peptides was adopted. According to Chou's formulation, the sensitivity, specificity, overall accuracy, and Matthew's correlation coefficient can be respectively expressed as [62, [65] [66] [67] Sn 1

where N_{11} is the total number of the interactive NR-drug pairs investigated while N_{10} the number of the interactive NR-drug pairs incorrectly predicted as the non-interactive NR-drug pairs; N_{01} the total number of the non-interactive NR-drug pairs investigated while N_{00} the number of the non-interactive NR-drug pairs incorrectly predicted as the interactive NR-drug pairs.

According to Equation (23) we can easily see the following. When $0 \leq N_{10} \leq N_{11}$ meaning none of the interactive NR-drug pairs was mispredicted to be a non-interactive NR-drug pair, we have the sensitivity $S_n = 1$; while $N_{10} = N_{11}$ meaning that all the interactive NR-drug pairs were mispredicted to be the non-interactive NR-drug pairs, we have the sensitivity $S_n = 0$. Likewise, when $0 \leq N_{01} \leq N_{00}$ meaning none of the non-interactive NR-drug pairs was mispredicted, we have the specificity S_p we have $MCC = 0$ meaning total disagreement between prediction and observation. As we can see from the above discussion, it is much more intuitive and easier to understand when using Equation (23) to examine a predictor for its four metrics, particularly for its Mathew's correlation coefficient. It is instructive to point out that the metrics as defined in Equation (23) are valid for single label systems; for multi-label systems, a set of more complicated metrics should be used as given in [162].

How to properly test a predictor for its anticipated success rates is very important for its development as well as its potential application value. Generally speaking, the following three cross-validation methods are often used to examine the quality of a predictor and its effectiveness in practical application: independent dataset test, subsampling or K-fold (such as five-fold, seven-fold, or 10-fold) crossover test and jackknife test [163]. However, as elaborated by a penetrating analysis in [164], considerable arbitrariness exists in the independent dataset test. Also, as demonstrated in [165], the subsampling (or K-fold crossover validation) test cannot avoid arbitrariness either. Only the jackknife test is the least arbitrary that can always yield a unique result for a given benchmark dataset [73, 74, 156, [166] [167] [168]. Therefore, the jackknife test has been widely recognized and increasingly utilized by investigators to examine the quality of various predictors (see, e.g., [14, 15, 68, 99, 106, 107, 124, 169, 170]). Accordingly, in this study the jackknife test was also adopted to evaluate the accuracy of the current predictor.

As mentioned above, the SVM operation engine contains two uncertain parameters C and γ . To find their optimal values, a 2-D grid search was conducted by the jackknife test on the

benchmark dataset . The results thus obtained are shown in Figure 3 , from which it can be seen that the iNR-Drug predictor reaches its optimal status when $C = 2^3$ and 9×2^3 . The corresponding rates for the four metrics (cf. Equation (23)) are given in Table 1 , where for facilitating comparison, the overall accuracy Acc reported by He et al. [59] on the same benchmark dataset is also given although no results were reported by them for Sn, Sp and MCC. It can be observed from the table that the overall accuracy obtained by iNR-Drug is remarkably higher than that of He et al. [59] , and that the rates achieved by iNR-Drug for the other three metrics are also quite higher. These facts indicate that the current predictor not only can yield higher overall prediction accuracy but also is quite stable with low false prediction rates.

As mentioned above (Section 3.2), the jackknife test is the most objective method for examining the quality of a predictor. However, as a demonstration to show how to practically use the current predictor, we took 41 NR-drug pairs from the study by Yamanishi et al. [171] that had been confirmed by experiments as interactive pairs. For such an independent dataset, 34 were correctly identified by iNR-Drug as interactive pairs, i.e., $S_n = 34 / 41 = 82.92\%$, which is quite consistent with the rate of 79.07% achieved by the predictor on the benchmark dataset via the jackknife test as reported in Table 1 .

It is anticipated that the iNR-Drug predictor developed in this paper may become a useful high throughput tool for both basic research and drug development, and that the current approach may be easily extended to study the interactions of drug with other targets as well. Since user-friendly and publicly accessible web-servers represent the future direction for developing practically more useful predictors [98, 172] , a publicly accessible web-server for iNR-Drug was established.

For the convenience of the vast majority of biologists and pharmaceutical scientists, here let us provide a step-by-step guide to show how the users can easily get the desired result by using iNR-Drug web-server without the need to follow the complicated mathematical equations presented in this paper for the process of developing the predictor and its integrity.

Step 1. Open the web server at the site <http://www.jci-bioinfo.cn/iNR-Drug/> and you will see the top page of the predictor on your computer screen, as shown in Figure 4 . Click on

the Read Me button to see a brief introduction about iNR-Drug predictor and the caveat when using it.

Step 2. Either type or copy/paste the query NR-drug pairs into the input box at the center of Figure 4 . Each query pair consists of two parts: one is for the nuclear receptor sequence, and the other for the drug. The NR sequence should be in FASTA format, while the drug in the KEGG code beginning with the symbol #. Examples for the query pairs input and the corresponding output can be seen by clicking on the Example button right above the input box. Step 3. Click on the Submit button to see the predicted result. For example, if you use the three query pairs in the Example window as the input, after clicking the Submit button, you will see on your screen that the "hsa:2099" NR and the "D00066" drug are an interactive pair, and that the "hsa:2908" NR and the "D00088" drug are also an interactive pair, but that the "hsa:5468" NR and the "D00279" drug are not an interactive pair. All these results are fully consistent with the experimental observations. It takes about 3 minutes before each of these results is shown on the screen; of course, the more query pairs there is, the more time that is usually needed.

Step 4. Click on the Citation button to find the relevant paper that documents the detailed development and algorithm of iNR-Durg.

Step 5. Click on the Data button to download the benchmark dataset used to train and test the iNR-Durg predictor.

Step 6. The program code is also available by clicking the button download on the lower panel of Figure 4 .

--- END OF ARTICLE ---

--- START OF ARTICLE 43 ---

Characterization of a New Member of Alphacoronavirus with Unique Genomic Features in Rhinolophus Bats

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6521148/>

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Abstract: Bats have been identified as a natural reservoir of a variety of coronaviruses (CoVs). Several of them have caused diseases in humans and domestic animals by interspecies transmission. Considering the diversity of bat coronaviruses, bat species and populations, we expect to discover more bat CoVs through virus surveillance. In this study, we described a new member of alphaCoV (BtCoV/Rh/YN2012) in bats with unique genome features. Unique accessory genes, ORF4a and ORF4b were found between the spike gene and the envelope gene, while ORF8 gene was found downstream of the nucleocapsid gene. All the putative genes were further confirmed by reverse-transcription analyses. One unique gene at the 3' end of the BtCoV/Rh/YN2012 genome, ORF9, exhibits ~30% amino acid identity to ORF7a of the SARS-related coronavirus. Functional analysis showed ORF4a protein can activate IFN- β production, whereas ORF3a can regulate NF- κ B production. We also screened the spike-mediated virus entry using the spike-pseudotyped retroviruses system, although failed to find any fully permissive cells. Our results expand the knowledge on the genetic diversity of bat coronaviruses. Continuous screening of bat viruses will help us further understand the important role played by bats in coronavirus evolution and transmission.

Text: Members of the Coronaviridae family are enveloped, non-segmented, positive-strand RNA viruses with genome sizes ranging from 26-32 kb [1]. These viruses are classified into two subfamilies: Letovirinae, which contains the only genus: Alphaletovirus; and Orthocoronavirinae (CoV), which consists of alpha, beta, gamma, and deltacoronaviruses

(CoVs) [2, 3] . Alpha and betacoronaviruses mainly infect mammals and cause human and animal diseases. Gamma-and delta-CoVs mainly infect birds, but some can also infect mammals [4, 5] . Six human CoVs (HCoVs) are known to cause human diseases. HCoV-HKU1, HCoV-OC43, HCoV-229E, and HCoV-NL63 commonly cause mild respiratory illness or asymptomatic infection; however, severe acute respiratory syndrome coronavirus (SARS-CoV) and

All sampling procedures were performed by veterinarians, with approval from Animal Ethics Committee of the Wuhan Institute of Virology (WIVH5210201). The study was conducted in accordance with the Guide for the Care and Use of Wild Mammals in Research of the People's Republic of China.

Bat fecal swab and pellet samples were collected from November 2004 to November 2014 in different seasons in Southern China, as described previously [16] .

Viral RNA was extracted from 200 μ L of fecal swab or pellet samples using the High Pure Viral RNA Kit (Roche Diagnostics GmbH, Mannheim, Germany) as per the manufacturer's instructions. RNA was eluted in 50 μ L of elution buffer, aliquoted, and stored at -80 °C. One-step hemi-nested reverse-transcription (RT-) PCR (Invitrogen, San Diego, CA, USA) was employed to detect coronavirus, as previously described [17, 18] .

To confirm the bat species of an individual sample, we PCR amplified the cytochrome b (Cytob) and/or NADH dehydrogenase subunit 1 (ND1) gene using DNA extracted from the feces or swabs [19, 20] . The gene sequences were assembled excluding the primer sequences. BLASTN was used to identify host species based on the most closely related sequences with the highest query coverage and a minimum identity of 95%.

Full genomic sequences were determined by one-step PCR (Invitrogen, San Diego, CA, USA) amplification with degenerate primers (Table S1) designed on the basis of multiple alignments of available alpha-CoV sequences deposited in GenBank or amplified with SuperScript IV Reverse Transcriptase (Invitrogen) and Expand Long Template PCR System (Roche Diagnostics GmbH, Mannheim, Germany) with specific primers (primer sequences are available upon request). Sequences of the 5' and 3' genomic ends were obtained by 5'

and 3' rapid amplification of cDNA ends (SMARTer Viruses 2019, 11, 379 3 of 19 RACE 5'/3' Kit; Clontech, Mountain View, CA, USA), respectively. PCR products were gel-purified and subjected directly to sequencing. PCR products over 5kb were subjected to deep sequencing using Hiseq2500 system. For some fragments, the PCR products were cloned into the pGEM-T Easy Vector (Promega, Madison, WI, USA) for sequencing. At least five independent clones were sequenced to obtain a consensus sequence.

The Next Generation Sequencing (NGS) data were filtered and mapped to the reference sequence of BatCoV HKU10 (GenBank accession number NC_018871) using Geneious 7.1.8 [21] . Genomes were preliminarily assembled using DNASTar lasergene V7 (DNASTar, Madison, WI, USA). Putative open reading frames (ORFs) were predicted using NCBI's ORF finder (<https://www.ncbi.nlm.nih.gov/orffinder/>) with a minimal ORF length of 150 nt, followed by manual inspection. The sequences of the 5' untranslated region (5'-UTR) and 3'-UTR were defined, and the leader sequence, the leader and body transcriptional regulatory sequence (TRS) were identified as previously described [22] . The cleavage of the 16 nonstructural proteins coded by ORF1ab was determined by alignment of aa sequences of other CoVs and the recognition pattern of the 3C-like proteinase and papain-like proteinase. Phylogenetic trees based on nt or aa sequences were constructed using the maximum likelihood algorithm with bootstrap values determined by 1000 replicates in the MEGA 6 software package [23] . Full-length genome sequences obtained in this study were aligned with those of previously reported alpha-CoVs using MUSCLE [24] . The aligned sequences were scanned for recombination events by using Recombination Detection Program [25] . Potential recombination events as suggested by strong p-values ($<10^{-20}$) were confirmed using similarity plot and bootscan analyses implemented in Simplot 3.5.1 [26] . The number of synonymous substitutions per synonymous site, Ks, and the number of nonsynonymous substitutions per nonsynonymous site, Ka, for each coding region were calculated using the Ka/Ks calculation tool of the Norwegian Bioinformatics Platform (<http://services.cbu.uib.no/tools/kaks>) with default parameters [27] . The protein homology detection was analyzed using HHpred (<https://toolkit.tuebingen.mpg.de/#/tools/hhpred>) with default parameters [28] .

A set of nested RT-PCRs was employed to determine the presence of viral subgenomic mRNAs in the CoV-positive samples [29] . Forward primers were designed targeting the leader sequence at the 5'-end of the complete genome, while reverse primers were designed within the ORFs. Specific and suspected amplicons of expected sizes were purified and then cloned into the pGEM-T Easy vector for sequencing.

Bat primary or immortalized cells (Rhinolophus sinicus kidney immortalized cells, RsKT; Rhinolophus sinicus Lung primary cells, RsLu4323; Rhinolophus sinicus brain immortalized cells, RsBrT; Rhinolophus affinis kidney primary cells, RaK4324; Rousettus leschenaultii Kidney immortalized cells, RlKT; Hipposideros pratti lung immortalized cells, HpLuT) generated in our laboratory were all cultured in DMEM/F12 with 15% FBS. Pteropus alecto kidney cells (Paki) was maintained in DMEM/F12 supplemented with 10% FBS. Other cells were maintained according to the recommendations of American Type Culture Collection (ATCC, www.atcc.org).

The putative accessory genes of the newly detected virus were generated by RT-PCR from viral RNA extracted from fecal samples, as described previously [30]. The influenza virus NS1 plasmid was generated in our lab [31]. The human bocavirus (HBoV) VP2 plasmid was kindly provided by prof. Hanzhong Wang of the Wuhan Institute of Virology, Chinese Academy of Sciences. SARS-CoV ORF7a was synthesized by Sangon Biotech. The transfections were performed with Lipofectamine 3000 Reagent (Life Technologies). Expression of these accessory genes were analyzed by Western blotting using an mAb (Roche Diagnostics GmbH, Mannheim, Germany) against the HA tag.

The virus isolation was performed as previously described [12]. Briefly, fecal supernatant was acquired via gradient centrifugation and then added to Vero E6 cells, 1:10 diluted in DMEM. After incubation at 37°C for 1 h the inoculum was replaced by fresh DMEM containing 2% FBS and the antibiotic-antimycotic (Gibco, Grand Island, NY, USA). Three blind passages were carried out. Cells were checked daily for cytopathic effect. Both culture supernatant and cell pellet were examined for CoV by RT-PCR [17].

Apoptosis was analyzed as previously described [18]. Briefly, 293T cells in 12-well plates were transfected with 3 µg of expression plasmid or empty vector, and the cells were collected 24 h post transfection. Apoptosis was detected by flow cytometry using by the Annexin V-FITC/PI Apoptosis Detection Kit (YEASEN, Shanghai, China) following the manufacturer's instructions. Annexin-V-positive and PI-negative cells were considered to be in the early apoptotic phase and those stained for both Annexin V and PI were deemed to undergo late apoptosis or necrosis. All experiments were repeated three times. Student's t-test was used to evaluate the data, with $p < 0.05$ considered significant.

HEK 293T cells were seeded in 24-well plates and then co-transfected with reporter plasmids (pRL-TK and pIFN- β IFN-or pNF- κ B-Luc) [30] , as well as plasmids expressing accessory genes, empty vector plasmid pcAGGS, influenza virus NS1 [32] , SARS-CoV ORF7a [33] , or HBoV VP2 [34] . At 24 h post transfection, cells were treated with Sendai virus (SeV) (100 hemagglutinin units [HAU]/mL) or human tumor necrosis factor alpha (TNF- α ; R&D system) for 6 h to activate IFN β or NF- κ B, respectively. Cell lysates were prepared, and luciferase activity was measured using the dual-luciferase assay kit (Promega, Madison, WI, USA) according to the manufacturer's instructions.

Retroviruses pseudotyped with BtCoV/Rh/YN2012 RsYN1, RsYN3, RaGD, or MERS-CoV spike, or no spike (mock) were used to infect human, bat or other mammalian cells in 96-well plates. The pseudovirus particles were confirmed with Western blotting and negative-staining electromicroscopy. The production process, measurements of infection and luciferase activity were conducted, as described previously [35, 36] .

The complete genome nucleotide sequences of BtCoV/Rh/YN2012 strains RsYN1, RsYN2, RsYN3, and RaGD obtained in this study have been submitted to the GenBank under MG916901 to MG916904.

The surveillance was performed between November 2004 to November 2014 in 19 provinces of China. In total, 2061 fecal samples were collected from at least 12 Rhinolophus bat species (Figure 1A). CoVs were detected in 209 of these samples (Figure 1B and Table 1). Partial RdRp sequences suggested the presence of at least 8 different CoVs. Five of these viruses are related to known species: Mi-BatCoV 1 (>94% nt identity), Mi-BatCoV HKU8 [37] (>93% nt identity), BtRf-AlphaCoV/HuB2013 [11] (>99% nt identity), SARSr-CoV [38] (>89% nt identity), and HKU2-related CoV [39] (>85% nt identity). While the other three CoV sequences showed less than 83% nt identity to known CoV species. These three viruses should represent novel CoV species. Virus isolation was performed as previously described [12] , but was not successful. While the other three CoV sequences showed less than 83% nt identity to known CoV species. These three viruses should represent novel CoV species. Virus isolation was performed as previously described [12] , but was not successful.

We next characterized a novel alpha-CoV, BtCoV/Rh/YN2012. It was detected in 3 *R. affinis* and 6 *R. sinicus*, respectively. Based on the sequences, we defined three genotypes, which represented by RsYN1, RsYN3, and RaGD, respectively. Strain RsYN2 was classified into the RsYN3 genotype. Four full-length genomes were obtained. Three of them were from *R. sinicus* (Strain RsYN1, RsYN2, and RsYN3), while the other one was from *R. affinis* (Strain RaGD). The sizes of these 4 genomes are between 28,715 to 29,102, with G+C contents between 39.0% to 41.3%. The genomes exhibit similar structures and transcription regulatory sequences (TRS) that are identical to those of other alpha-CoVs (Figure 2 and Table 2). Exceptions including three additional ORFs (ORF3b, ORF4a and ORF4b) were observed. All the 4 strains have ORF4a & ORF4b, while only strain RsYN1 has ORF3b.

The replicase gene, ORF1ab, occupies~20.4 kb of the genome. The replicase gene, ORF1ab, occupies~20.4 kb of the genome. It encodes polyproteins 1a and 1ab, which could be cleaved into 16 non-structural proteins (Nsp1-Nsp16). The 3'-end of the cleavage sites recognized by 3C-like proteinase (Nsp4-Nsp10, Nsp12-Nsp16) and papain-like proteinase (Nsp1-Nsp3) were confirmed. The proteins including Nsp3 (papain-like 2 protease, PL2pro), Nsp5 (chymotrypsin-like protease, 3CLpro), Nsp12 (RdRp), Nsp13 (helicase), and other proteins of unknown function (Table 3). The 7 concatenated domains of polyprotein 1 shared <90% aa sequence identity with those of other known alpha-CoVs (Table 2), suggesting that these viruses represent a novel CoV species within the alpha-CoV. The closest assigned CoV species to BtCoV/Rh/YN2012 are BtCoV-HKU10 and BtRf-AlphaCoV/Hub2013. The three strains from Yunnan Province were clustered into two genotypes (83% genome identity) correlated to their sampling location. The third genotype represented by strain RaGD was isolated to strains found in Yunnan (<75.4% genome identity). We then examined the individual genes (Table 2). All of the genes showed low aa sequence identity to known CoVs. The four strains of BtCoV/Rh/YN2012 showed genetic diversity among all different genes except ORF1ab (>83.7% aa identity). Notably, the spike proteins are highly divergent among these strains. Other structure proteins (E, M, and N) are more conserved than the spike and other accessory proteins. Comparing the accessory genes among these four strains revealed that the strains of the same genotype shared a 100% identical ORF3a. However, the proteins encoded by ORF3as were highly divergent among different genotypes (<65% aa identity). The putative accessory genes were also BLASTed against GenBank records. Most accessory genes have no homologues in GenBank-database, except for ORF3a (52.0-55.5% aa identity with BatCoV HKU10 ORF3) and ORF9 (28.1-32.0% aa identity with SARS-CoV ORF7a). We analyzed the protein homology with HHpred software. The results showed that ORF9s and SARS-CoV ORF7a are homologues (possibility: 100%, E value <10⁻⁴⁸). We further screened the genomes for

potential recombination evidence. No significant recombination breakpoint was detected by bootscan analysis.

To confirm the presence of subgenomic RNA, we designed a set of primers targeting all the predicted ORFs as described. The amplicons were firstly confirmed via agarose-gel electrophoresis and then sequencing (Figure 3 and Table 2). The sequences showed that all the ORFs, except ORF4b, had preceding TRS. Hence, the ORF4b may be translated from bicistronic mRNAs. In RsYN1, an additional subgenomic RNA starting inside the ORF3a was found through sequencing, which led to a unique ORF3b.

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Phylogenetic trees were constructed using the aa sequences of RdRp and S of BtCoV/Rh/YN2012 and other representative CoVs (Figure 4) . In both trees, all BtCoV/Rh/YN2012 were clustered together and formed a distinct lineage to other known coronavirus species. Two distinct sublineages were observed within BtCoV/Rh/YN2012. One was from Ra sampled in Guangdong, while the other was from Rs sampled in Yunnan. Among the strains from Yunnan, RsYN2 and RsYN3 were clustered together, while RsYN1 was isolated. The topology of these four strains was correlated to the sampling location. The relatively long branches reflect a high diversity among these strains, indicating a long independent evolution history.

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The Ka/Ks ratios (Ks is the number of synonymous substitutions per synonymous sites and Ka is the number of nonsynonymous substitutions per nonsynonymous site) were calculated for all genes. The Ka/Ks ratios for most of the genes were generally low, which indicates these genes were under purified selection. However, the Ka/Ks ratios of ORF4a, ORF4b, and ORF9 (0.727, 0.623, and 0.843, respectively) were significantly higher than those of other ORFs (Table 4). For further selection pressure evaluation of the ORF4a and ORF4b gene, we sequenced another four ORF4a and ORF4b genes (strain Rs4223, Rs4236, Rs4240, and Ra13576 was shown in Figure 1B).

As SARS-CoV ORF7a was reported to induce apoptosis, we conducted apoptosis analysis on BtCoV/Rh/YN2012 ORF9, a ~30% aa identity homologue of SARS-CoV ORF7a. We transiently transfected ORF9 of BtCoV/Rh/YN2012 into HEK293T cells to examine whether this ORF9 triggers apoptosis. Western blot was performed to confirm the expression of ORF9s and SARS-CoV ORF7a (Figure S1). ORF9 couldn't induce apoptosis as the ORF7a of SARS-CoV Tor2 (Figure S2). The results indicated that BtCoV/Rh/YN2012 ORF9 was not involved in apoptosis induction.

To determine whether these accessory proteins modulate IFN induction, we transfected reporter plasmids (pIFN β -Luc and pRL-TK) and expression plasmids to 293T cells. All the cells over-expressing the accessory genes, as well as influenza virus NS1 (strain PR8), HBoV VP2, or empty vector were tested for luciferase activity after SeV infection. Luciferase

activity stimulated by SeV was remarkably higher than that without SeV treatment as expected. Influenza virus NS1 inhibits the expression from IFN promoter, while HBoV VP2 activate the expression. Compared to those controls, the ORF4a proteins exhibit an active effect as HBoV VP2 (Figure 5A). Other accessory proteins showed no effect on IFN production (Figure S3). Expression of these accessory genes were confirmed by Western blot (Figure S1). was remarkably higher than that without SeV treatment as expected. Influenza virus NS1 inhibits the expression from IFN promoter, while HBoV VP2 activate the expression. Compared to those controls, the ORF4a proteins exhibit an active effect as HBoV VP2 (Figure 5A). Other accessory proteins showed no effect on IFN production (Figure S3). Expression of these accessory genes were confirmed by Western blot (Figure S1). Samples were collected at 6 h postinfection, followed by dual-luciferase assay. The results were expressed as the firefly luciferase value normalized to that of Renilla luciferase. (B) ORF3a protein activate NF- κ B. 293T cells were transfected with 100 ng pNF- κ B-Luc, 10 ng pRL-TK, empty vector (500 ng), an NS1-expressing plasmid (500 ng), a SARS-CoV ORF7a-expressing plasmid (500 ng), or ORF3a-expressing plasmids (500 ng). After 24 h, the cells were treated with TNF- α . Dual-luciferase activity was determined after 6 h. The results were expressed as the firefly luciferase activity normalized to that of Renilla luciferase. The experiments were performed three times independently. Data are representative of at least three independent experiments, with each determination performed in triplicate (mean $\pm$ SD of fold change). Asterisks indicate significant differences between groups (compared with Empty vector-NC, $p < 0.05$, as determined by student t test).

NF- κ B plays an important role in regulating the immune response to viral infection and is also a key factor frequently targeted by viruses for taking over the host cell. In this study, we tested if these accessory proteins could modulate NF- κ B. 293T cells were co-transfected with reporter Samples were collected at 6 h postinfection, followed by dual-luciferase assay. The results were expressed as the firefly luciferase value normalized to that of Renilla luciferase. (B) ORF3a protein activate NF- κ B. 293T cells were transfected with 100 ng pNF- κ B-Luc, 10 ng pRL-TK, empty vector (500 ng), an NS1-expressing plasmid (500 ng), a SARS-CoV ORF7a-expressing plasmid (500 ng), or ORF3a-expressing plasmids (500 ng). After 24 h, the cells were treated with TNF- α . Dual-luciferase activity was determined after 6 h. The results were expressed as the firefly luciferase activity normalized to that of Renilla luciferase. The experiments were performed three times independently. Data are representative of at least three independent experiments, with each determination performed in triplicate (mean $\pm$ SD of fold change). Asterisks indicate significant

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To understand the infectivity of these newly detected BtCoV/Rh/YN2012, we selected the RsYN1, RsYN3 and RaGD spike proteins for spike-mediated pseudovirus entry studies. Both Western blot analysis and negative-staining electron microscopy observation confirmed the preparation of BtCoV/Rh/YN2012 successfully (Figure S5). A total of 11 human cell lines, 8 bat cells, and 9 other mammal cell lines were tested, and no strong positive was found (Table S2) .

In this study, a novel alpha-CoV species, BtCoV/Rh/YN2012, was identified in two *Rhinolophus* species. The 4 strains with full-length genome were sequences. The 7 conserved replicase domains of these viruses possessed $<90\%$ aa sequence identity to those of other known alpha-CoVs, which defines a new species in accordance with the ICTV taxonomy standard [42] . These novel alpha-CoVs showed high genetic diversity in their structural and non-structural genes. Strain RaGD from *R. affinis*, collected in Guangdong province, formed a divergent independent branch from the other 3 strains from *R. sinicus*, sampled in Yunnan Province, indicating an independent evolution process associated with geographic isolation and host restrain. Though collected from same province, these three virus strains formed two genotypes correlated to sampling locations. These two genotypes had low genome sequence identity, especially in the S gene and accessory genes. Considering the remote geographic location of the host bat habitat, the host tropism, and the virus diversity, we suppose BtCoV/Rh/YN2012 may have spread in

these two provinces with a long history of circulation in their natural reservoir, *Rhinolophus* bats. With the sequence evidence, we suppose that these viruses are still rapidly evolving.

Our study revealed that BtCoV/Rh/YN2012 has a unique genome structure compared to other alpha-CoVs. First, novel accessory genes, which had no homologues, were identified in the genomes. Second, multiple TRSs were found between S and E genes while other alphacoronavirus only had one TRS there. These TRSs precede ORF3a, ORF3b (only in RsYN1), and ORF4a/b respectively. Third, accessory gene ORF9 showed homology with those of other known CoV species in another coronavirus genus, especially with accessory genes from SARSr-CoV.

Accessory genes are usually involved in virus-host interactions during CoV infection [43]. In most CoVs, accessory genes are dispensable for virus replication. However, an intact 3c gene of feline CoV was required for viral replication in the gut [44] [45] [46]. Deletion of the genus-specific genes in mouse hepatitis virus led to a reduction in virulence [47]. SARS-CoV ORF7a, which was identified to be involved in the suppression of RNA silencing [48], inhibition of cellular protein synthesis [49], cell-cycle blockage [50], and apoptosis induction [51, 52]. In this study, we found that BtCoV/Rh/YN2012 ORF9 shares ~30% aa sequence identity with SARS-CoV ORF7a. Interestingly, BtCoV/Rh/YN2012 and SARSr-CoV were both detected in *R. sinicus* from the same cave. We suppose that SARS-CoV and BtCoV/Rh/YN2012 may have acquired ORF7a or ORF9 from a common ancestor through genome recombination or horizontal gene transfer. Whereas, ORF9 of BtCoV/Rh/YN2012 failed to induce apoptosis or activate NF- κ B production, these differences may be induced by the divergent evolution of these proteins in different pressure.

Though different BtCoV/Rh/YN2012 ORF4a share <64.4% amino acid identity, all of them could activate IFN- β . ORF3a from RsYN1 and RaGD upregulated NF- κ B, but the homologue from RsYN2 downregulated NF- κ B expression. These differences may be caused by amino acid sequence variations and may contribute to a viruses' pathogenicity with a different pathway.

Though lacking of intestinal cell lines from the natural host of BtCoV/Rh/YN2012, we screened the cell tropism of their spike protein through pseudotyped retrovirus entry with human, bat and other mammalian cell lines. Most of cell lines screened were

unsusceptible to BtCoV/Rh/YN2012, indicating a low risk of interspecies transmission to human and other animals. Multiple reasons may lead to failed infection of coronavirus spike-pseudotyped retrovirus system, including receptor absence in target cells, failed recognition to the receptor homologue from non-host species, maladaptation in non-host cells during the spike maturation or virus entry, or the limitation of retrovirus system in stimulating coronavirus entry. The weak infectivity of RsYN1 pseudotyped retrovirus in Huh-7 cells could be explained by the binding of spike protein to polysaccharide secreted to the surface. The assumption needs to be further confirmed by experiments.

Our long-term surveillances suggest that Rhinolophus bats seem to harbor a wide diversity of CoVs. Coincidentally, the two highly pathogenic agents, SARS-CoV and Rh-BatCoV HKU2 both originated from Rhinolophus bats. Considering the diversity of CoVs carried by this bat genus and their wide geographical distribution, there may be a low risk of spillover of these viruses to other animals and humans. Long-term surveillances and pathogenesis studies will help to prevent future human and animal diseases caused by these bat CoVs.

Supplementary Materials: The following are available online at <http://www.mdpi.com/1999-4915/11/4/379/s1>, Figure S1 : western blot analysis of the expression of accessory proteins. Figure S2 : Apoptosis analysis of ORF9 proteins of BtCoV/Rh/YN2012. Figure S3 : Functional analysis of ORF3a, ORF3b, ORF4b, ORF8 and ORF9 proteins on the production of Type I interferon. Figure S4 : Functional analysis of ORF3b, ORF4a, ORF4b, ORF8 and ORF9 proteins on the production of NF- κ B. Figure S5 : Characteristic of BtCoV/Rh/YN2012 spike mediated pseudovirus. Table S1 : General primers for AlphaCoVs genome sequencing. Table S2 : Primers for the detection of viral subgenomic mRNAs. Table S3

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In Vitro Antiviral Activity of Circular Triple Helix Forming Oligonucleotide RNA towards Feline Infectious Peritonitis Virus Replication

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3950953/>

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Abstract: Feline Infectious Peritonitis (FIP) is a severe fatal immune-augmented disease in cat population. It is caused by FIP virus (FIPV), a virulent mutant strain of Feline Enteric Coronavirus (FECV). Current treatments and prophylactics are not effective. The in vitro antiviral properties of five circular Triple-Helix Forming Oligonucleotide (TFO) RNAs (TFO1 to TFO5), which target the different regions of virulent feline coronavirus (FCoV) strain FIPV WSU 79-1146 genome, were tested in FIPV-infected Crandell-Rees Feline Kidney (CRFK) cells. RT-qPCR results showed that the circular TFO RNAs, except TFO2, inhibit FIPV replication, where the viral genome copy numbers decreased significantly by 5-fold $\log_{10}$ from 10^{14} in the virus-inoculated cells to 10^9 in the circular TFO RNAs-transfected cells. Furthermore, the binding of the circular TFO RNA with the targeted viral genome segment was also confirmed using electrophoretic mobility shift assay. The strength of binding kinetics between the TFO RNAs and their target regions was demonstrated by NanoITC assay. In conclusion, the circular TFOs have the potential to be further developed as antiviral agents against FIPV infection.

Text: Feline Infectious Peritonitis Virus (FIPV) is an enveloped virus with a nonsegmented, positive sense, single-stranded RNA genome. FIPV is grouped as feline coronavirus (FCoV), under the family Coronaviridae. FCoV is divided into two biotypes, namely, Feline Enteric Coronavirus (FECV), a ubiquitous enteric biotype of FCoV, and FIPV, a virulent biotype of FCoV [1]. The relationship between these two biotypes still remains unclear. Two hypotheses have been proposed, (i) internal mutation theory and (ii) circulating high virulent-low virulent theory. Internal mutation theory stated that the development of FIP is due to the exposure of cat to variants of FCoV which have been mutated by gaining the ability to replicate within the macrophages [2], while the circulating high virulent-low

virulent theory explains the existence of both distinctive pathogenic and benign lineages of viruses within the cat population [3] .

Study has shown that about 40-80% of cats are detected with FECV shedding in their faeces [4] . About 12% of these FECV-positive cats have developed immune-mediated fatal FIP disease [4] . The prevalence of FIP among felines is due to continual cycles of infection and reinfection of FECV and indiscernible clinical symptoms of infected cats with FECV at an early stage before the progressive development of FIPV.

Vaccination against FIPV with an attenuated, temperature-sensitive strain of type II FIPV induces low antibody titre in kittens that have not been exposed to FCoV. However, there is considerable controversy on the safety and efficacy of this vaccine, since the vaccine contains type 2 strain, whereas type 1 viruses are more prevalent in the field [4] . In addition, antibodies against FIPV do not protect infected cats but enhance the infection of monocytes and macrophages via a mechanism known as Antibody-Dependent Enhancement [1] . Besides vaccines, several antiviral drugs such as ribavirin, 2 BioMed Research International interferons, and immunosuppressive drugs have been used as treatments for FIPV-infected cats, mainly to suppress the inflammatory and detrimental immune response [5] [6] [7] [8] . However, those treatments were ineffective. Hence, there is still significant unmet medical need to develop effective treatments and prophylactics for FIPV infection.

Triple Helix Forming Oligonucleotide (TFO) is defined as homopyrimidine oligonucleotides, which can form a sequence-specific triple helix by Hoogsteen bonds to the major groove of a complementary homopyrimidinehomopurine stretch in duplex DNA [9] . Furthermore, double helical RNA or DNA-RNA hybrids can be targeted as a template for triple helix formation, once the strand composition on the stabilities of triple helical complexes is determined [10] . Hence, TFO has been used to impede gene expressions by transcription inhibition of viral genes or oncogenes [11] [12] [13] [14] [15] [16] . The main purpose of this study is to develop and evaluate the in vitro antiviral properties of circular TFO RNAs against FIPV replication.

serotype II strain WSU 79-1146 (ATCC no. VR-1777) was grown in CRFK cells. A serial 10-fold dilution of FIPV was prepared from the working stock. Confluent 96-well plate was

inoculated with 100 L of each virus dilution/well. The plate was incubated in a humidified incubator at 37 ° C, 5% CO₂. Cytopathic effects (CPE) development was observed. The results were recorded after 72 hours and the virus tissue culture infective dose 50 (TCID₅₀) was calculated using Reed and Muench's method [17].

Oligonucleotide RNA. The Triple Helix Forming Oligonucleotides (TFOs) were designed based on the genome sequence of FIPV serotype II strain WSU 79-1146 (Accession no: AY994055) [18]. TFOs, which specifically target the different regions of the FIPV genome, and one unrelated TFO were constructed (Table 1). The specificity of the TFOs was identified using BLAST search in the NCBI database. The designed linear TFOs were synthesized by Dharmacon Research (USA), whereby the 5 and 3 ends of the linear TFOs were modified with phosphate (PO₄) group and hydroxide (OH) group, respectively. These modifications were necessary for the circularization of linear TFO. The process of circularization, using the T4 RNA ligase 1 (ssRNA ligase) (New England Biolabs Inc., England), was carried out according to the manufacturer's protocol. After ligation, the circular TFO RNAs were recovered by ethanol precipitation and the purity of the circular TFO RNAs was measured using spectrophotometer.

Denaturing of urea polyacrylamide gel electrophoresis was performed as described before [19] with modification. Briefly, 20% of denatured urea polyacrylamide gel was prepared and polymerized for 30 minutes. Then, the gel was prerun at 20 to 40 V for 45 minutes. Five L of TFO RNA mixed with 5 L of urea loading buffer was heated at 92 ° C for 2 minutes and immediately chilled on ice. It was run on the gel at 200 V for 45 minutes. Finally, the gel was stained with ethidium bromide (Sigma, USA) and viewed with a Bio-Rad Gel Doc XR system (CA, USA). (EMSA). The target regions of the FIPV genome were synthesized by Dharmacon Research (USA) (Table 1). Each TFO RNA was mixed with the target region in 1X binding buffer containing 25 mM Tris-HCl, 6 mM MgCl₂, and 10 mM NaCl in a final volume of 10 L and subsequently incubated at 37 ° C for 2 hours. The sample was run on 15% native polyacrylamide gel at 80 V, in cool condition. The stained gel was viewed by a Bio-Rad Gel Doc XR system.

Regions. The binding strength was measured using a nano Isothermal Titration Calorimeter (ITC) (TA instruments, Newcastle, UK). The RNA sample mixtures, consisting of circular TFOs (0.0002 mM), were incubated with their respective synthetic target regions (0.015 mM) using 1X binding buffer as the diluent. The experiment was run at 37 ° C with 2

L/injection, for a total of 25 injections. Data was collected every 250 seconds and analyzed using the NanoAnalyze software v2.3.6 provided by the manufacturer.

This experiment was conducted in CRFK cells, where 3×10^4 cell/well was seeded in 96-well plate to reach 80% confluency 24 hours prior to transfection. One hundred nM of TFO RNAs was separately transfected into the CRFK cells using a HiPerFect Transfection Reagent (Qiagen, Germany), as per the manufacturer's protocol. The plate was incubated at 37 ° C with 5% CO₂ for 6 hours. Then, the cultures were infected with 100TCID₅₀ of FIPV serotype II strain WSU 79-1146 for 1 hour at 37 ° C (100 L/well). Finally, the viral inoculum was replaced by fresh maintenance media (MEM containing 1% FBS and 1% pen/strep). Virus-infected and uninfected cells were maintained as positive and negative controls, respectively. The morphology of the cultures was recorded 72 hours after infection and samples were harvested at this time point and stored at -80 ° C prior to RNA extraction.

Inhibition. Different concentrations of circular TFO1 RNA (25 nM, 50 nM, 100 nM, and 500 nM) were transfected into CRFK cells. The plate was incubated for 6 hours followed by virus inoculation for 1 hour at 37 ° C with 5% CO₂. The cells were processed as described above.

Madin-Darby Canine Kidney (MDCK) cell (ATCC no. CCL-34), at a concentration of 4×10^4 cell/well, was seeded in 96-well plate to reach 80% confluency 24 hours prior to transfection. Transfection was performed the same as before. One hundred nM of circular TFO RNA was transfected into MDCK cells. Following 6 hours

ORF1a/1b and 530-541

ORF1a/1b and 7399-7411

ORF1a/1b and 14048-14061

- * Highlighted in bold indicated the binding region. * * Unrelated circular TFO. [20, 21] , respectively. The reverse transcriptase quantitative real-time PCR (RT-qPCR) was performed using a Bio-Rad CFX96 real-time system (BioRad, USA). The reaction was amplified in a final volume of 25 L using a SensiMix SYBR No-ROX One-Step Kit (Bioline, UK), which consisted of 12.5 L 2X SensiMix SYBR No-Rox One-

Step reaction buffer, 10 M forward and reverse primers, 10 units RiboSafe RNase inhibitor, and 5 L template RNA. Absolute quantification approach was used to quantify qPCR results where a standard curve of a serial dilution of virus was plotted before the quantification. Amount of the virus in the samples was quantified based on this standard curve.

Analysis. Data statistical analysis was performed using SPSS 18.0. Data were represented as mean $\pm$ SE of three independent tests. One-way ANOVA, Tukey post hoc test was used to analyze the significant level among the data. ≤ 0.05 was considered significant. genome, which play important roles in viral replication, were selected as the target binding sites for the triplex formation. The target regions were 5 untranslated region (5 UTR), Open Reading Frames (ORFs) 1a and 1b, and 3 untranslated region (3 UTR) (Table 1). The TFOs were designed in duplex, as they can bind with the single stranded target region and reshape into triplex. Both ends of the duplex TFOs were ligated with a linker sequence or clamps (C-C) to construct circular TFO RNA.

Denaturing PAGE assay was carried out after the ligation process to determine the formation of the circular TFO. As shown in Figure 1 , the circular TFO RNAs migrated faster than the linear TFO RNAs, when subjected to 20% denaturing PAGE.

Target Region. The binding ability was determined using Electrophoretic Mobility Shift Assay (EMSA) [23] . The appearance of the slow mobility band indicates the successful hybridization of circular TFO RNA with its target region. The binding ability of different TFO RNAs (TFO1 to TFO5) against their target regions was determined by EMSA (Figure 2) . TFO1, TFO3, TFO4, and TFO5 showed slow mobility band, while TFO2 showed the lack of an upward shifted band. This indicates the possession of triplex binding ability for all circular TFO RNAs, except TFO2.

TFO RNA. Study on the interaction and hybridization of TFO towards its target region is crucial, since the stronger the binding is, the more stable the triplex structure forms. As shown in supplementary Figure 1 (Table 3) .

The antiviral effect of circular TFO RNAs was investigated by RT-qPCR assay at 72 hours after transfection. The results showed viral RNA genome copy numbers of 3.65×10^9 , 3.22×10^{14} , 5.04×10^9 , 5.01×10^9 , 4.41×10^9 , and 3.96×10^{14} in cells treated with TFO1, TFO2, TFO3, TFO4, TFO5, and TFO7, respectively. The data analyzed by one-way ANOVA, Tukey post hoc test showed significant high viral RNA genome copy number of 4.03×10^{14} for virus inoculated cells as compared to circular TFO1, TFO3, TFO4, and TFO5 treatments (≤ 0.05). The viral RNA copies of circular TFO2, linear TFO3 and TFO4, and unrelated circular TFO7 RNAs transfected cells also showed high viral RNA copy numbers which did not show significant differences to the infected cells (≥ 0.05) (Figure 3). The morphological changes of the cells were also captured 72 hours after transfection. The cells transfected with circular TFO1, TFO3, TFO4, and TFO5 appeared to be in good condition following virus inoculation, while the cells transfected with circular TFO2 and linear TFO3 and TFO4 showed visible cytopathic effect (CPE), the same as virus inoculated cells (supplementary Figure 2) . Furthermore, cells transfected with TFO only remain viable indicating that TFO treatment is generally not toxic to the cells. Hence, these results illustrated the capacity of circular TFO RNAs (except TFO2) to inhibit FIPV replication.

Concentrations on FIPV Replication. Circular TFO1 was used to examine the dose-response relationship as a representative to other TFOs. The experimental conditions were identical to that of the previous experiment, except for TFO1 concentrations of 25 nM, 50 nM, 100 nM, and 500 nM. There was no significant reduction in viral RNA genome copies using the concentration of 25 nM TFO1. The other concentrations caused significant reductions in copy numbers as compared to the virus-infected cells. However, no significant difference was detected in copy numbers from all of these concentrations (Figure 4).

The specificity of the TFO towards FIPV was tested, using TFO1 and TFO5, as the proper representatives of TFOs, on influenza A virus H1N1 New Jersey 8/76. The analyzed data using one-way ANOVA, Tukey post hoc test did not show significant reductions in the copies of viral RNA for both TFOs compared to the influenza virus inoculated cells (≥ 0.05) (supplementary Figure 3). Complex structure G4/Cir4 Figure 2 : EMSA analysis. EMSA

analysis illustrated the binding of circular TFO 1, 3, 4, and 5 to the target regions as evidenced by upward band shift. Binding of each circular TFO except circular TFO2 to its respective target forms a complex that migrates slower than unbound TFO. G1 to G5 represent the target region for circular TFO1 to TFO5 and Cir1 to Cir5 represent the circular TFO1 to TFO5, respectively. in the replication process [24] . Meanwhile, the ORF1a/1b of FIPV are translated into polyproteins that are cleaved into nonstructural proteins which assemble into replicationtranscription complexes together with other viral proteins [24] . Hence, the development of molecular therapy targeting these critical regions may provide the possibility to inhibit FIPV replication.

Development of antiviral therapies against FIPV using siRNA [25] and viral protease inhibitors [26] Figure 4 : TFO1 dose-response study for inhibiting FIPV replication. The concentrations of 50 nM and higher showed significant antiviral effects. 50 nM of circular TFO1 RNA was able to reduce viral copy number by 5-fold log 10 from 10^{14} to 10^9 , while 100 and 500 nM showed 4-fold reduction. Data are averages of 3 independent tests (mean $\pm$ SE). * Significantly different from FIPV-infected group.

as potential new treatments against FIPV infection. In this study, circular Triple Helix Forming Oligonucleotide (TFO) RNAs, specifically targeting the short regions of viral genome for triplex formation, were designed and evaluated. TFO1 and TFO2 targeted the 5' and 3' UTRs of the viral genome, respectively. TFO3 to TFO5 targeted different regions of the ORF1a/1b on FIPV genome. Prior to in vitro antiviral study, the ligated circular TFOs were evaluated using PAGE analysis. All of the circularised TFO showed faster migration pattern compared to the linear TFO; however, only slight variation was detected for some of the TFO (Figure 1). The reason for this is not clear but probably due to the differences in length and the tertiary structures of the TFOs leading to differences in the migration rate. EMSA was used to show the binding capability of each circular TFO towards the target region in the FIPV genome except for TFO2 which showed lack of formation of complex structure upon hybridization (Figure 2) . The EMSA result also concurred with the antiviral study, where all circular TFOs (except TFO2) were able to demonstrate a significant reduction in the viral RNA genome copy numbers by 5-fold log 10 from 10^{14} in virus inoculated cells to 10^9 in TFO-transfected cells (Figure 3). However, no antiviral properties were detected from the linear TFOs and unrelated circular TFO7 RNA, confirming that the antiviral activity is associated with specific binding of circular TFOs towards targeted regions.

Furthermore, the binding of the circular TFO to the target region was confirmed by nanoITC analysis; where the low value and high stability allowed TFOs to compete effectively with the target regions for inhibiting transcription in cell-free systems. Since, TFO1 shows the lowest value (Table 3), the antiviral properties of this TFO were evaluated in dose-response study. As shown in Figure 4, 50 and 100 nM of TFO1 showed similar antiviral effects indicating the potential therapeutic application of TFO1 on FIPV replication. However, increasing the concentration of TFO1 to 500 nM failed to reduce the viral load further probably due to inefficiency of the transfection reagent to transfect the TFO into the cells. In addition, the virus has fast replication rate upon in vitro infection, where previous study on the growth of FIPV in CRFK cells showed that by 2 hours approximately 67% of FIPV 79-1146 were internalized by CRFK cells by endocytosis increasing to more than 70% at 3 hours [27, 28]. The above finding probably also explained the reason why no antiviral effect was detected when the transfection of the TFO was performed on virus-infected cells (data not shown).

The antiviral properties, as demonstrated by the circular TFOs, were probably associated with the binding of the TFO to the target region, based on both the Watson-Crick and Hoogsteen hydrogen bonds, which enhance the stability in terms of enthalpy, which is brought about by joining together two out of three strands of the triple helix in the proper orientation [29]. Therefore, the triplex formation is tightly bonded and not easy to detach. Furthermore, the circular TFOs were designed in such way that the presence of hydrogen bonding donors and acceptors in the purines is able to form two hydrogen bonds, while the pyrimidine bases can only form one additional hydrogen bond with incoming third bases [30]. However, there are various factors that may limit the activity of TFOs in cells like intracellular degradation of the TFO and limited accessibility of the TFO to the target sites which can prevent triplex formation [31]. These findings may also explain the inability of the designed TFO1 to inhibit further virus replication in dose-response study (Figure 4).

Various molecular-based therapies against infectious diseases and cancer have been developed and tested. However, only the siRNA-based therapy has been studied extensively as a novel antiviral and anticancer therapy [32, 33]. Recently, McDonagh et al. [25] developed siRNA with antiviral activity against the FIPV 79-1146, where the designed siRNA was able to reduce the copy number of viral genome compared with virus-infected cells. The potential therapeutic application of TFOs, such as linear TFO conjugated with psoralen to inhibit the transcription of human immunodeficiency provirus [13] and TFO to inhibit the transcription of 1(I) collagen in rat fibroblasts [14], has also been reported. In

addition, short TFO conjugated with daunomycin targeting the promoter region of oncogene has been designed and evaluated on human cancer cells [31] . These studies indicated the flexibility of using TFO-based oligonucleotides as a potential molecular-based therapy. In this study, we demonstrated short circular TFO RNAs between 28 and 34 mers (Table 1) , which are able to inhibit FIPV replication by binding to specific target regions of the FIPV genome. All designed circular TFOs (except TFO2) showed significant inhibitory effects against FIPV replication. The TFOs that formed triplex structures showed antiviral effects towards FIPV replication. The reason why TFO2 failed to show any interaction with the target region or antiviral activity is probably due to the length of TFO2 (i.e., 24 mers), which might be insufficient to a triplex formation upon hybridization (Figure 2), be effective enough to suppress viral RNA transcription, and eventually inhibit virus replication. Nevertheless, the inability of TFO2 to show antiviral effect due to failure in the formation of functional tertiary structure of the triplex formation cannot be ruled out. In vitro antiviral study which showed no antiviral property for unrelated TFO (TFO7) and also inability of circular TFO1 and TFO5 to inhibit influenza A virus H1N1 infected cells confirms the specificity of the TFOs' activity.

In conclusion, the circular TFO RNA has the potential to be developed as a therapy against FIPV in cats. However, further studies on TFO specificity, actual mechanism of circular TFO RNA in the transcription alteration consequence of inhibiting the viral transcription process, and in vivo animal studies are important for this approach to work as a therapy in the future.

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Inhibitory Effect and Possible Mechanism of Action of Patchouli Alcohol against Influenza A (H2N2) Virus

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Abstract: In the present study, the anti-influenza A (H2N2) virus activity of patchouli alcohol was studied in vitro, in vivo and in silico. The CC(50) of patchouli alcohol was above 20 μ M. Patchouli alcohol could inhibit influenza virus with an IC(50) of 4.03 ± 0.23 μ M. MTT assay showed that the inhibition by patchouli alcohol appears strongly after penetration of the virus into the cell. In the influenza mouse model, patchouli alcohol showed obvious protection against the viral infection at a dose of 5 mg/kg/day. Flexible docking and molecular dynamic simulations indicated that patchouli alcohol was bound to the neuraminidase protein of influenza virus, with an interaction energy of -40.38 kcal mol⁽⁻¹⁾. The invariant key active-site residues Asp151, Arg152, Glu119, Glu276 and Tyr406 played important roles during the binding process. Based on spatial and energetic criteria, patchouli alcohol interfered with the NA functions. Results presented here suggest that patchouli alcohol possesses anti-influenza A (H2N2) virus properties, and therefore is a potential source of anti-influenza agents for the pharmaceutical industry.

Text: The influenza virus, which is one of the main causes of acute respiratory infections in humans, can lead to annual epidemics and infrequent pandemics. The two influenza pandemics of the 20th century, "Asian Influenza (1957/H2N2)" and "Hong Kong Influenza (1968/H3N2)" resulted in the deaths of an estimated 2-3 million people globally [1, 2]. Today, their descendants continue to cause the majority of influenza infections in humans [3]. So far as it is learned that the most effective antiviral drug is the neuraminidase (NA) inhibitor, which target the NA glycoproteins of influenza A and B virus [4, 5].

The release of new virions from the infected cell is a key step in the influenza life cycle and need neuraminidase (NA) to cleave the α -ketosidic linkage between terminal sialic acid and an adjacent sugar residue [6]. The NA inhibitors were designed to prevent the key step by blocking the active site of enzyme and thus allow sufficient time for the host immune systems to remove infected viruses [7]. Consistent efforts have been devoted to the

development of NA inhibitors, using the crystal structure of the N2 sub-type NA protein [8] [9] [10] [11] [12] [13] [14] [15] . Indeed, oseltamivir (Tamiflu) is the representative NA inhibitor that has proven to be uniquely applicable oral drug in clinical practice for the treatment of influenza infection [4, 8, 9] . However, with an increase in medical use, the oseltamivir-resistant strains have been found and probably lead to a large scale outbreak of novel pandemic flu [16, 17] .

Patchouli alcohol (Figure 1) has been well known for over a century. It is a major constituent of the pungent oil from the East Indian shrub *Pogostemon cablin* (Blanco) Benth, and widely used in fragrances. Patchouli oil is an important essential oil in the perfume industry, used to give a base and lasting character to a fragrance [16, 17] . The essential oil is very appreciated for its characteristic pleasant and long lasting woody, earthy, and camphoraceous odor, as well as for its fixative properties, being suitable for use in soaps and cosmetic products [16, 17] . The aerial part of *Pogostemon cablin* has widely been used for the treatment of the common cold and as an antifungal agent in China [16, 17] . Moreover, the plant is widely used in Traditional Chinese Medicine as it presents various types of pharmacological activity according to the composition of the oil [16, 17] . Patchouli alcohol, as the major volatile constituent of patchouli oil, has been found to strongly inhibit H1N1 replication and weakly inhibit B/Ibaraki/2/85 replication [18] . To the best of our knowledge, the anti-influenza virus (H2N2) activities of patchouli alcohol have not been evaluated yet. Therefore, the aim of the present study was to evaluate the anti-influenza A virus (H2N2) activity of patchouli alcohol by MTT assay and mouse influenza model. On such basis, explicitly solvated docking and molecular dynamic (MD) methods were applied to investigate the binding mode involving patchouli alcohol with influenza virus NA protein. We anticipate that the insight into the understanding of inhibiting mechanism will be of value in the rational design of novel anti-influenza drugs.

First the efficacy of patchouli alcohol on influenza A (H2N2) virus replication and cell viability were examined. CC 50 was used to express the cytotoxicity of patchouli alcohol on MDCK. The CC 50 of patchouli alcohol was above 20 mM, which indicated that patchouli alcohol did not affect the growth of MDCK (Table 1) . Thus, it seems that the antiviral effects of patchouli alcohol were not due to the cytotoxicity. Moreover, patchouli alcohol was found to inhibit influenza A (H2N2) virus with an IC 50 of $4.03 \pm 0.23 \mu\text{M}$. Based on the IC 50 and CC 50 values, the selectivity index (SI) was calculated as >4.96 . It is reported that a SI of 4 or more is appropriate for an antiviral agent [18] , suggesting that patchouli alcohol can be judged to have anti-influenza A (H2N2) virus activity.

Until now, it has been found that patchouli alcohol showed dose-dependent anti-influenza virus (A/PR/8/34, H1N1) activity, with an IC₅₀ value of 2.635 μ M. Furthermore, it showed weak activity against B/Ibaraki/2/85 (IC₅₀ = 40.82 μ M) [19]. With the addition of the above H2N2 inhibitory activity, we have a comprehensive view of the anti-influenza activity of patchouli alcohol.

Cells were pretreated with patchouli alcohol prior to virus infection (pretreatment cells), viruses were pretreated prior to infection (pretreatment virus), and patchouli alcohol was added during the adsorption period (adsorption) or after penetration of the viruses into cells (replication). Experiments were repeated independently three times and data presented are the average of three experiments. The symbols * indicated very significant difference $p < 0.01$ with respect to other mode (pretreatment virus, adsorption and pretreatment cell).

As shown in Figure 2, patchouli alcohol showed anti-influenza A (H2N2) virus activity in a time-dependent manner. It showed best antiviral activity when added at a concentration of 8 μ M during the replication period with inhibition of the viral replication of $97.68\% \pm 2.09\%$ for influenza A (H2N2) at 72 h. However, no significant effect was detected when patchouli alcohol was used for pretreatment of cells or viruses or when patchouli alcohol was only added during the adsorption phase. These results suggested that the inhibition of influenza A (H2N2) virus by patchouli alcohol appears to occur much more strongly after penetration of the virus into the cell. Besides, biochemical studies have indicated that the bioactivity of NA protein is essential determinant after the replication of influenza A (H2N2) virus [20] [21] [22]. Hence, we conclude that the function of NA protein may be suppressed by patchouli alcohol.

To evaluate the toxicity of patchouli alcohol, the mean value of body weight of mice in each group was statistically analyzed. The mean weights of mice administered at the 2 mg/kg/dose oseltamivir, 2 mg/kg/dose patchouli alcohol and 10 mg/kg/dose of patchouli alcohol one time daily for 7 days were not significantly different compared with the normal control mice, showing no toxicity of patchouli alcohol and oseltamivir within the testing concentration ($P > 0.05$). Physiological status was observed in virus infection mice. Three days after viral infection, some mice, especially mice in the H2N2 infected control group showed changes in behavior, such as a tendency to huddle, diminished vitality, and ruffled

fur, etc. In the mouse influenza model, viral infection leads to loss of body weight and high mortality. Therefore, the efficacy of patchouli alcohol and oseltamivir were evaluated on the basis of survival rate measured for 15 days post-infection, for treated infected animals relative to untreated infected (control) animals. A comparison of efficacy of patchouli alcohol and oseltamivir in vivo mouse influenza model (oral treatment) showed that at a dose of 5 mg/kg/day, patchouli alcohol showed obvious protection against the influenza virus, as the mean day to death was detected as 11.8 ± 1.1 (Table 2) . When the dose was lowered to 1 mg/kg/day, patchouli alcohol showed weaker protection (measured by Survivors/total) than that of 5 mg/kg/day, the mean day to death was 7.5 ± 1.8 . Whereas oseltamivir at this dose level (1 mg/kg/day) showed 50% protection (measured by survivors/total) against the influenza virus. In the H2N2 infected control group, there were no survivors. In view of both in vitro and in vivo data, we conclude that patchouli alcohol could be used in the treatment of human influenza virus infections.

Based on the above experiment data, patchouli alcohol is determined to be bound within NA protein. As the total energies and backbone root-mean-square-deviations (RMSD) in Figure 3 indicate, the energy-minimized patchouli alcohol-NA complex has been in equilibrium since about 0.5 ns, and then retains quite stable in the last 19.5 ns. It is consistent with the previous MD results of other NA inhibitors [23] [24] [25] [26] [27] [28] . Accordingly, the geometric and energetic analyses were made on the average structures of 0.5~20.0 ns MD trajectories, where the system has been already at equilibrium. The interaction energy (E_{inter}) of patchouli alcohol with NA was calculated at $-40.38 \text{ kcal mol}^{-1}$, where the vdW rather than electrostatic interactions were found to play a dominant role, contribute to about 72% ($-29.18 \text{ kcal mol}^{-1}$). As shown in Figure 4, the patchouli alcohol was bound at the active site which also bound to oseltamivir and zanamivir [28] . As Figure 5 shows, the oxygen atom of patchouli alcohol was oriented towards the sidechains of residues Glu119 and Tyr406, with one H-bond formed with each residue. The values of distances in Figure 6 further reveal that the docked complex remains rather stable throughout the simulation, with the average distances of Glu119:OE2patchouli alcohol:O and Tyr406:OH -patchouli alcohol:O less than $2.8 \text{ \AA}$. The sum contributions (E_{sum}) of residues Glu119 and Tyr406 amounted to -8.46 and $-7.37 \text{ kcal mol}^{-1}$, respectively (Table 3) . Besides, patchouli alcohol was stabilized by residues Arg118, Asp151, Arg152, Trp178, Ala246, Glu276, Arg292, Asn294 and Gln347, especially residues Asp151, Arg152 and Glu276 (Figure 5 and Table 3). As a matter of fact, residues Asp151, Arg152, Glu119, Glu276 and Tyr406 of the NA protein have already received enough attention from rational drug designs [14, 30, 31] . The catalytic residues Asp151, Arg152 and Glu276 are crucial to the NA functions and the residues Glu119 and Tyr406 are important to stabilize the NA

active sites [32, 33] . It suggests that the NA functions will be affected by the presence of patchouli alcohol, consistent with the above experiments. Patchouli alcohol matches with the NA active site and has an acceptable interaction energy. Considering the obvious structure discrepancies against current NA inhibitors, it represents an ideal lead compound for the designs of novel anti-influenza agents.

Patchouli alcohol and oseltamivir were obtained from Sigma Chemical Co. (St. Louis, MO, USA, purity > 99%) and was stored in glass vials with Teflon sealed caps at -20 ± 0.5 °C in the absence of light.

MDCK (Madin-Darby canine kidney) was purchased from Harbin Veterinary Research Institute (Harbin, Heilongjiang, China). The cells were grown in monolayer culture with Eagle's minimum essential medium (EMEM) supplemented with 10% fetal calf serum (FCS), 100 U/mL penicillin and 100 µg/mL streptomycin. The monolayers were removed from their plastic surfaces and serially passaged whenever they became confluent. Cells were plated out onto 96-well culture plates for cytotoxicity and anti-influenza assays, and propagated at 37 °C in an atmosphere of 5% CO₂ .

The influenza strain A/Leningrad/134/17/1957 H2N2) was purchased from National Control Institute of Veterinary Bioproducts and Pharmaceuticals (Beijing, China). Virus was routinely grown on MDCK cells. The stock cultures were prepared from supernatants of infected cells and stored at -80 °C.

The cellular toxicity of patchouli alcohol on MDCK cells was assessed by the MTT method. Briefly, cells were seeded on a microtiter plate in the absence or presence of various concentrations (20 µM -0.0098 µM) of patchouli alcohol (eight replicates) and incubated at 37 °C in a humidified atmosphere of 5% CO₂ for 72 h. The supernatants were discarded, washed with PBS twice and MTT reagent (5 mg/mL in PBS) was added to each well. After incubation at 37 °C for 4 h, the supernatants were removed, then 200 µL DMSO was added and incubated at 37 °C for another 30 min. After that the plates were read on an ELISA reader (Thermo Molecular Devices Co., Union City, USA) at 570/630 nm. The mean OD of the cell control wells was assigned a value of 100%. The maximal non-toxic concentration (TD₀) and 50% cytotoxic concentration (CC₅₀) were calculated by linear regression analysis of the dose-response curves generated from the data.

Inhibition of virus replication was measured by the MTT method. Serial dilution of the treated virus was adsorbed to the cells for 1 h at 37 °C. The residual inoculum was discarded and infected cells were added with EMEM containing 2% FCS. Each assay was performed in eight replicates. After incubation for 72 h at 37 °C, the cultures were measured by MTT method as described above. The concentration of patchouli alcohol and oseltamivir which inhibited virus numbers by 50% (IC₅₀) was determined from dose-response curves.

Cells and viruses were incubated with patchouli alcohol at different stages during the viral infection cycle in order to determine the mode of antiviral action. Cells were pretreated with patchouli alcohol before viral infection, viruses were incubated with patchouli alcohol before infection and cells and viruses were incubated together with patchouli alcohol during adsorption or after penetration of the virus into the host cells. Patchouli alcohol was always used at the nontoxic concentration. Cell monolayers were pretreated with patchouli alcohol prior to inoculation with virus by adding patchouli alcohol to the culture medium and incubation for 1 h at 37 °C. The compound was aspirated and cells were washed immediately before the influenza A (H2N2) inoculum was added. For pretreatment virus, Influenza A (H2N2) was incubated in medium containing patchouli alcohol for 1h at room temperature prior to infection of MDCK cells. For analyzing the anti-influenza A (H2N2) inhibition during the adsorption period, the same amount of influenza A (H2N2) was mixed with the drug and added to the cells immediately. After 1 h of adsorption at 37 °C, the inoculum was removed and DMEM supplemented with 2 % FCS were added to the cells. The effect of patchouli alcohol against influenza A (H2N2) was also tested during the replication period by adding it after adsorption, as typical performed in anti-influenza A (H2N2) susceptibility studies. Each assay was run in eight replicates.

Kunming mice, weighing 18-22 g (6 weeks of age) were purchased from Harbin Veterinary Research Institute Animal Co., Ltd. (Harbin, Heilongjiang, China) . First, the toxicity of patchouli alcohol and oseltamivir was assessed in the healthy mice by the loss of body weight compared with the control group (2% DMSO in physiological saline). The mice were orally administered with 10 mg/kg/dose patchouli alcohol, 2 mg/kg/dose patchouli alcohol or 2 mg/kg/dose oseltamivir (dissolved in 2% DMSO in physiological saline) one time daily for 7 days. The weight of mice was determined daily. We conducted procedures according to Principle of Laboratory Animal Care (NIH Publication No. 85 -23, revised 1985) and the guidelines of the Peking University Animal Research Committee.

Kunming mice were anesthetized with isoflurane and exposed to virus (A/Leningrad/134/17/1957) by intranasal instillation. Drugs were prepared in 2% DMSO in physiological saline and administered 4 h prior to virus exposure and continued daily for 5 days. All mice were observed daily for changes in weight and for any deaths. Parameters for evaluation of antiviral activity included weight loss, reduction in mortality and/or increase in mean day to death (MDD) determined through 15 days.

The N2 sub-type neuraminidase crystal structure (PDB code 1IVD) was obtained from the RCSB Protein Data Bank [34]. For convenience, the structure is named as NA hereafter. Geometry and partial atomic charges of the patchouli alcohol (Figure 1) were calculated with the Discover 3.0 module (Insight II 2005) [35] by applying the BFGS algorithm [36] and the consistent-valence force-field (CVFF), with a convergence criterion of $0.01 \text{ kcal mol}^{-1} \text{ \AA}^{-1}$. The docking and molecular dynamics (MD) simulations were performed by the general protocols in the Insight II 2005 software packages, consistent with the previous literatures [24, 26, 28, 35, [37] [38] [39]. During the MD simulations, the canonical ensemble (NVT) was employed at normal temperature (300 K). The MD temperature was controlled by the velocity scaling thermostat [40]. Integrations of the classical equations of motion were achieved using the Verlet algorithm. The systems were solvated in a large sphere of TIP3P water molecules [40] with the radius of $35.0 \text{ \AA}$, which is enough to hold the ensembles [40]. The MD trajectories were generated using a 1.0-fs time step for a total of 20.0 ns, saved at 5.0-ps intervals. The interaction energies of patchouli alcohol with NA and the respective residues at the NA active site were calculated by the Docking module [35], over the 0.5~20.0 ns MD trajectories.

All results are expressed as mean values $\pm$ standard deviations (SDs) ($n = 3$). The significance of difference was calculated by one-way analysis of variance, and values $p < 0.001$ were considered to be significant.

In conclusion, patchouli alcohol possesses anti-influenza A (H2N2) virus activity via interference with the NA function that cleaves the α -glycosidic bond between sialic acid and glycoconjugate. Our results provide the promising information for the potential use of patchouli alcohol in the treatment of influenza A (H2N2) virus infectious disease. Further mechanistic studies on the anti-influenza A virus activity are needed to support this point of view.

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Isothermal Amplification Using a Chemical Heating Device for Point-of-Care Detection of HIV-1

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Abstract: **BACKGROUND:** To date, the use of traditional nucleic acid amplification tests (NAAT) for detection of HIV-1 DNA or RNA has been restricted to laboratory settings due to time, equipment, and technical expertise requirements. The availability of a rapid NAAT with applicability for resource-limited or point-of-care (POC) settings would fill a great need in HIV diagnostics, allowing for timely diagnosis or confirmation of infection status, as well as facilitating the diagnosis of acute infection, screening and evaluation of infants born to HIV-infected mothers. Isothermal amplification methods, such as reverse-transcription, loop-mediated isothermal amplification (RT-LAMP), exhibit characteristics that are ideal for POC settings, since they are typically quicker, easier to perform, and allow for integration into low-tech, portable heating devices. **METHODOLOGY/SIGNIFICANT FINDINGS:** In this study, we evaluated the HIV-1 RT-LAMP assay using portable, non-instrumented nucleic acid amplification (NINA) heating devices that generate heat from the exothermic reaction of calcium oxide and water. The NINA heating devices exhibited stable temperatures throughout the amplification reaction and consistent amplification results between three

separate devices and a thermocycler. The performance of the NINA heaters was validated using whole blood specimens from HIV-1 infected patients. **CONCLUSION:** The RT-LAMP isothermal amplification method used in conjunction with a chemical heating device provides a portable, rapid and robust NAAT platform that has the potential to facilitate HIV-1 testing in resource-limited settings and POC.

Text: HIV-1 diagnostic tests are held to a high standard of performance, as diagnosis has a direct impact on patient care and reduction of transmission. Despite technological advances in the field of HIV diagnostics and the high sensitivity and specificity associated with most HIV diagnostic tests that are currently available, it is estimated that approximately 20% of HIV-infected individuals living in the United States remain undiagnosed [1] . Furthermore, testing sites have reported as many as 35 to 50% of individuals with an initial positive test result will not return for a confirmatory diagnosis if follow-up laboratory testing is required [2] . Rapid HIV antibodybased tests, which can be performed with minimal training and typically provide results in under 30 minutes [3] , have facilitated HIV testing at the point-of-care and subsequently increased the numbers of individuals aware of their serostatus [4] . Rapid tests are currently a key component of HIV screening at the point-of-care (POC), significantly expanding the diagnostic capabilities of testing sites in developed countries, as well as resource-limited settings.

Despite the advances made by the widespread availability of rapid tests, all antibody-based tests for the detection of HIV exhibit some limitations. HIV-specific antibody typically begins to appear around three weeks post-infection, allowing for detection by most antibody-based assays within 3-6 weeks [3, 5] . The window of time prior to or during early seroconversion may lead to false-negative test results in recently infected individuals. Additionally, accurate diagnosis of infants born to HIV-infected mothers can be challenging if based solely on antibody positivity, since vertically transferred maternal antibodies may persist for 12-18 months after birth [6, 7] . For confirmatory diagnosis of early HIV infection or infant diagnosis, nucleic acid amplification tests (NAAT) are preferred, as HIV-1 RNA can be detected as early as 10-12 days post infection and HIV-1 DNA and/or RNA are definitive indicators of active infection [5] . In their current form, however, NAAT's are not feasible for POC testing, because they are timeconsuming, expensive, and technically complicated. To date, the Aptima HIV-1 RNA assay (Gen-Probe, Inc., <http://www.fda.gov/BiologicsBloodVaccines/BloodBloodProducts/ApprovedProducts/LicensedProductsBLAs/BloodDonorScreening/InfectiousDisease/UCM080466>) is the only

FDA-approved NAAT for the diagnosis or confirmation of HIV-1 infection and it is only suitable for laboratory testing.

To meet the needs of HIV-1 diagnosis at the POC, a rapid NAAT that can be performed with minimal training, limited equipment, and a relatively short turnaround time (,1 hour) is desirable [8] . The development of a rapid NAAT has proven to be especially challenging since the technology involved in simplifying the test procedure often equates to increased equipment and material costs [8] . Additionally, the reduction in technical complexity should not compromise test sensitivity and specificity. For increased applicability at the POC, an increasing number of novel isothermal amplification techniques have been developed [9] . Isothermal amplification is an attractive alternative to traditional PCR or RT-PCR since thermal cycling is not required, allowing for greater versatility in terms of heating or amplification devices. One such amplification method, termed Loop-Mediated Isothermal Amplification (LAMP) [10] , has been optimized for the detection of DNA and/or RNA (RT-LAMP) from a wide range of bacterial and viral pathogens [11, 12, 13, 14, 15, 16, 17, 18, 19] , including HIV [20, 21] .

LAMP or RT-LAMP exhibits several characteristics that are ideal for integration into a rapid nucleic-acid based diagnostic test. The amplification reaction requires six primers specific for eight separate regions within the target sequence, contributing to the high specificity of the amplification method. Amplified material can typically be detected within 15-60 minutes when incubated at a constant reaction temperature of 60-65°C [22] . LAMP has also proven to be less sensitive to biological inhibitors than PCR [23, 24] , which enables direct amplification from clinical specimens, thereby eliminating the need for an additional nucleic acid extraction step. Direct amplification from plasma, whole blood, and oral fluid has previously been demonstrated for HIV-1 [20, 21, 25] . Lastly, immediate visual detection of amplified products is facilitated by the large amount of DNA that is generated by each reaction. Several groups have incorporated fluorescent detection methods into the LAMP assay for real-time or immediate naked-eye detection [15, 17, 21, 22, 26] .

The simplicity and isothermal nature of the LAMP procedure opens the door for the evaluation of low-tech integrated devices or novel heating elements, which are appropriate for low-resource settings, where costly equipment and electricity cannot be obtained. In this study, the HIV-1 RT-LAMP assay was evaluated using portable, non-instrumented nucleic acid amplification (NINA) devices that generate heat from the exothermic reaction

of calcium oxide and water [27, 28] . We demonstrated the temperature stability of the NINA heating devices and feasibility for POC testing of whole blood specimens from HIV-1 infected individuals.

Prototype NINA heaters were designed and provided by Program for Appropriate Technology in Health (PATH, Seattle, WA), as described [27, 28] . Briefly, an amplification temperature of approximately 60°C was provided by the exothermic reaction of calcium oxide (CaO; Sigma-Aldrich, St. Louis, MO) and water. The heating devices, containing the chemical reaction, were designed using thermally insulated, stainless-steel canisters with plastic screw-top lids (Fig. 1) . The lids were modified to contain three sample wells that fit standard 200 µl PCR tubes and were filled with a proprietary phase-change material (PCM) that was used to buffer the heat derived from the exothermic reaction, thereby providing a constant temperature. Lastly, plastic caps containing foam insulation were designed to fit on the top of the canister lids. The thermal profiles of the sample wells were measured and recorded using a digital thermometer (DaqPRO 5300 Data recorder; OMEGA Engineering, Inc., Stamford, CT).

DNA and RNA linearity panels were prepared to determine the sensitivity of the HIV-specific RT-LAMP assay. A DNA panel was generated from DNA extracted from the human monocytic cell line OM-10.1 [29] , using a QIAamp DNA blood mini kit (QIAGEN, Valencia, CA). Cell count was used to quantify the input DNA copy number, as a single integrated provirus is contained in each cell [29] . The extracted DNA was diluted tenfold in RNase-free water to create a linearity panel, ranging from 10⁵ copies/ml to 10³ copies/ml. An RNA linearity panel was obtained commercially (PRD801; SeraCare Life Sciences, Milford, MA) and ranged from 2.9610⁶ copies/ml to 8 copies/ml, as determined by Roche AMPLICOR HIV MONITOR TM v 1.5, Bayer VERSANT HIV-1 RNA bDNA 3.0 Assay, bioMérieux NucliSensH HIV-1 QT, and Abbott Real Time HIV-1 m2000 TM . RNA was extracted from the panel members using a Viral RNA mini kit (QIAGEN). Negative controls included DNA extracted from PBMC infected with HIV-2 SLRHC [30] and RNA extracted from HIV-2 NIH-Z purified virus (Advanced Biotechnologies Inc., Columbia, MD).

Whole blood from HIV-1 infected individuals was collected as part of a separate, IRB-approved study [31] , or obtained commercially (SeraCare Life Sciences). All HIV-positive samples were confirmed using the following tests: Genetic Systems HIV-1/ HIV-2 plus O EIA (Bio-Rad Laboratories, Redmond, WA), GS HIV-1 Western blot (Bio-Rad Laboratories),

Aptima HIV-1 RNA assay (Gen-Probe, Inc., San Diego, CA), and Amplicor HIV-1 DNA assay (Roche Diagnostics, Branchburg, NJ). Viral and proviral loads are unknown, since the samples were tested with qualitative, nucleic acid-based assays. All clinical specimens evaluated in this study were obtained from individuals infected with subtype B HIV-1 virus. As a negative control, HIV-1 seronegative blood samples (SeraCare Life Sciences) were included in every experiment involving whole blood. A positive control included HIV-1 seronegative blood spiked with 5610 6 virus particles/ml of HIV-1 BaL (Advanced Biotechnologies Inc.).

HIV-1-specific RT-LAMP primers were designed to recognize a conserved sequence within the reverse transcriptase (RT) gene. The six primers required for the RT-LAMP reaction, forward outer (F3), backward outer (B3), forward inner (FIP), backward inner (BIP), and the loop primers (LoopF and LoopB), were designed using the PrimerExplorer V4 software (Eiken Chemical Co. Ltd.; [http:// primerexplorer.jp/e/](http://primerexplorer.jp/e/)). The LAMP primers and amplification cycle have been described in detail by Nagamine et al. [32] . Additional modifications included a linker sequence of four thymidines inserted between the F2 and F1c sequences of the FIP primer, as described [20] , and the addition of the fluorescent molecule HEX to the 59 end of the LoopF primer. The labeled primer, along with a quencher probe, allowed for immediate visual detection of amplified products [21] . The quencher probe consisted of the complementary sequence of the LoopF primer with Black Hole Quencher-1 (BHQ-1) added to the 39 end. The HIV-1 HXB2 sequence (GenBank accession number AF033819) was used as the reference for generating the RT-LAMP primers. The sequences of the HIV-1 RT-specific primers and quencher are listed in Table 1 .

The RT-LAMP reaction was performed using the following reaction mix: 0.2 mM (final concentration) of each F3 and B3 primers, 1.6 mM of each FIP and BIP primers, 0.8 mM of each LoopF and HEX-LoopB primers, 0.8 M betaine (Sigma-Aldrich), 10 mM MgSO₄ , 1.4 mM dNTPs, 16 ThermoPol reaction buffer (New England Biolabs, Ipswich, MA), 16 U Bst DNA polymerase (New England Biolabs) and 2 U AMV reverse transcriptase (Invitrogen, Carlsbad, CA). The reaction was carried out in a total volume of 25 ml for amplification of extracted nucleic acid, 10 ml of which constituted the sample. For amplification of whole blood specimens, a 100 ml reaction volume was used to facilitate visual detection of amplified products. Whole blood was added directly into the reaction at a total volume of 40 ml, following a 1:4 dilution with red blood cell lysis buffer (2.5 mM KHCO₃ , 37.5 mM NH₄ Cl, and 0.025 mM EDTA), as previously described [21] . The reaction mixture was incubated at 60°C for 60 minutes, using a GeneAmpH PCR System (Applied Biosystems,

Foster City, CA) or the NINA heaters. For reactions amplified in the thermalcycler, an additional two minute heating step of 80uC was added at the end of the amplification cycle to terminate the reaction.

The reaction tubes were evaluated for the presence of amplification, following addition of the quencher probe at a 2:1 ratio of quencher to labeled-primer, as previously described [21]. Amplification was determined visually by observing fluorescence in the reaction tubes, using the UV lamp from a ChemiDoc XRS system (Bio-Rad Laboratories, Hercules, CA). Amplification was confirmed by electrophoresis using a 1.2% agarose gel containing SYBRH Safe gel stain (Invitrogen), which was subsequently visualized using the ChemiDoc XRS system.

To compare temperature and amplification consistency, three NINA heaters were tested in parallel. The heating reaction was initiated by adding 18 g of CaO to each NINA canister, followed by 6 ml of water. The lid of each canister was then sealed to contain the exothermic reaction. After adding 200 ml of water to each of the sample wells, temperature recording was initiated. Reaction tubes were added to the sample wells once each reaction chamber reached a temperature of 58.5uC. For all samples incubated in the NINA heater, 15 ml of mineral oil was added to the reaction tube during the reaction mix preparation. The samples were incubated in the heaters for a total of 60 minutes. All reactions were carried out in a temperature-controlled laboratory with an ambient temperature of 28uC, unless otherwise stated. Following the amplification reaction, the samples were incubated for two minutes in a heat block set to 80uC. After each amplification cycle, the temperature profile of each device was analyzed by calculating the temperature mean, standard deviation, median, minimum, and maximum from the data provided by the DaqPRO 5300.

The stability of the NINA heaters at extreme low and high temperatures was evaluated by placing the canisters in a refrigerator set to 4uC or a 37uC incubator during the length of the amplification reaction. The temperature profiles were recorded and compared to those of reactions that occurred at the laboratory room temperature of 28uC.

To determine the sensitivity of RT-LAMP reaction using RTspecific primers, DNA and RNA linearity panels were tested in a thermalcycler. The limit of detection for HIV-1 DNA was 10 copies/reaction. For the RNA linearity panel, the sample containing 1700 copies/reaction

was detected in all of the three replicates, while the sample containing 140 copies/reaction was detected in three out of five replicates (60%). For both DNA and RNA linearity panels, the two samples nearest the limit of detection were chosen to further evaluate the performance consistency between the thermalcycler and NINA heaters. In terms of positivity, the amplification results were consistent between all three heaters and the thermalcycler (Table 2) . Since the RT-LAMP assay requires a constant temperature of 60°C for the length of the amplification reaction, the temperature profiles of the sample wells were compared over the course of the incubation and between all three NINA heaters. A representative temperature profile is displayed in Figure 2 , showing a steady reaction temperature at or close to 60°C for length of amplification reaction. During the 60 minute incubation, the average temperature for each device was 60.2, 59.8, and 59.7 (Table 3). The minimum temperature achieved during the reaction reflects the fact that the temperature of the sample port dropped temporarily after the sample tubes are added to the device, as shown in Figure 2 . The maximum temperature of the devices deviated from the desired reaction temperature of 60°C by less than one degree.

The ability of the NINA heaters to maintain a steady reaction temperature in a wide range of ambient temperatures is essential for POC testing, whether referring to an air-conditioned laboratory or high-temperature field site. To evaluate the performance of the NINA heaters at extreme low or high temperatures, the canisters were placed in a 4°C refrigerator or a 37°C incubator for the length of the amplification reaction. The limit of detection for the DNA and RNA linearity panels was similar to the results obtained in our temperature-controlled laboratory (28°C; Table 2). The greatest degree of temperature variation of the sample wells was observed at the ambient temperature of 4°C (Table 3). The average temperature was approximately two degrees lower than the desired reaction temperature of 60°C. Additionally, the temperature of the devices tended to decline from their steady state during the last 20 minutes of the reaction (data not shown). The temperature profiles at the ambient temperature of 37°C, however, were similar to those at 28°C.

Whole blood samples from HIV-1 infected individuals were added directly into the RT-LAMP reaction and tested in the NINA heaters. Positivity of the clinical specimens was consistent between the thermalcycler and devices (Table 4). Amplification consistency was most evident with two of the patient samples (patient #4 and #5) that were only positive in one of the three replicates, regardless of the heating device that was used. All HIV-negative blood samples, included in each reaction, were negative (data not shown). A representative

experiment using the NINA heaters is displayed in Figure 3 , showing detection by agarose gel and visual identification of fluorescence in the reaction tubes.

In this study, we demonstrate the performance of portable, inexpensive, non-instrumented nucleic acid (NINA) heaters for amplification of HIV-1 using RT-LAMP. The isothermal amplification reaction coupled with a device that generates heat from an exothermic chemical reaction, as opposed to grid electricity or battery power, comprises a point-of-care NAAT that is practical for use in resource-limited settings. The heating devices require minimal training and technical expertise to operate and take approximately 10-15 minutes to reach a reaction temperature of 60°C once the chemical reaction has been initiated [27, 28] . Furthermore, the temperature of the sample wells remain relatively stable at the desired reaction temperature of 60°C throughout the amplification reaction, as demonstrated by the heating profiles and the consistency in amplification between the devices and thermalcycler.

Since point-of-care testing may refer to an air-conditioned laboratory or a field site with high temperatures and humidity, the stability of the temperature generated by the heating devices must be reliable. Though the temperature profiles at a representative cold temperature of 4°C indicated a loss in reaction temperature towards the end of the 60 minute incubation, the temperature fluctuations were not significant enough to affect the amplification reaction. Regardless, this thermal effect could be mitigated with small modifications to the device to reduce heat loss at lower temperatures. It should be possible to extend the temperature range of the NINA heaters to 4°C and below by either adding a larger quantity of heating mixture, better insulation, or both. Of greater concern is the performance of the NINA heaters in high temperature field sites, where temperature control is not an option.

We demonstrate no difference in the temperature stability of the NINA heaters and amplification consistency at an ambient temperature of 37°C as compared to our temperature-controlled laboratory.

For increased applicability for use at the POC, several modifications can be made to the NINA heaters. The prototype devices evaluated in this study contained only three sample wells; however, up to 16 sample wells can be added to the lid of the insulated canisters for

a larger testing volume. In this study, samples were removed from the NINA heaters after the amplification reaction and heated for an additional two minutes in an 80uC heat block to terminate the reaction. While the additional heating step is not necessary to observe the amplified products from extracted nucleic acid, the short, high-temperature incubation facilitates the visual observation of the fluorescent label in the whole blood samples. Modifications may be made to the whole blood sample preparation method to eliminate the need for the heating step. Alternatively, a second temperature-moderating compartment can be added to the alternate end of the NINA canisters, so the samples can be removed from the amplification compartment and reinserted into the 80uC compartment. Lastly, the DaqPRO data recorder was used in this study for validation purposes only and would not be necessary for the final POC product.

The feasibility of using LAMP as a diagnostic method in resource-limited settings has been demonstrated for tuberculosis [33] . To reduce hands-on time and preparation error, the authors describe the use of reaction tubes pre-prepared with lyophilized reaction mix. For POC use, limited sample manipulation and reagent preparation is desired and, therefore, it is anticipated that the test procedure of the end product will include reconstituting the amplification reagents in water and adding the sample directly into the reaction tube. We demonstrate the use of the NINA heaters for amplification directly from whole blood specimens, eliminating the need for a time-consuming, nucleic acid extraction procedure and reducing the volume of sample needed for the amplification reaction. A total volume of 10 ml of whole blood was added to each reaction tube, which can easily be obtained by finger-stick in settings where venipuncture is not feasible. Additionally, our fluorescent detection method enables immediate visualization of amplified products in the absence of specialized equipment. To avoid cross-contamination of amplified material, it is preferred that the reaction tubes remain closed post-amplification. Future modifications will include optimizing the labeled primer/quencher sequences so that all components can be added into the reaction mix prior to amplification. Due to availability, the Bio-Rad ChemiDoc system was used as the UV source in this study; however, an inexpensive keychain light would be more suitable for naked-eye detection at the POC. For sensitive and specific detection of diverse HIV-1 isolates, including non-B subtypes, identification of the optimal primer set/sets is a key step in the development of the RT-LAMP assay. Although all experiments performed in this study involved subtype B standards and specimens, ongoing research involves the continued development and optimization of RT-LAMP primers based on regions of the HIV-1 genome that are conserved among diverse subtypes. Future studies will include large-scale evaluation of clinical specimens with the optimized RT-LAMP assay and NINA device. In summary, the RT-LAMP isothermal amplification method used in

conjunction with a simplified, chemical heating device exhibits characteristics that are ideal for a rapid NAAT for POC testing. The simplified, portable assay has the potential to fill an important gap in HIV-1 diagnostics, providing immediate knowledge or confirmation of HIV-1 infection status at the POC.

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Viral and bacterial co-infection in severe pneumonia triggers innate immune responses and specifically enhances IP-10: a translational study

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Abstract: Mixed viral and bacterial infections are widely described in community-acquired pneumonia; however, the clinical implications of co-infection on the associated immunopathology remain poorly studied. In this study, microRNA, mRNA and cytokine/chemokine secretion profiling were investigated for human monocyte-derived macrophages infected in-vitro with Influenza virus A/H1N1 and/or Streptococcus pneumoniae. We observed that the in-vitro co-infection synergistically increased interferon- γ -induced protein-10 (CXCL10, IP-10) expression compared to the singly-

infected cells conditions. We demonstrated that endogenous miRNA-200a-3p, whose expression was synergistically induced following co-infection, indirectly regulates CXCL10 expression by targeting suppressor of cytokine signaling-6 (SOCS-6), a well-known regulator of the JAK-STAT signaling pathway. Additionally, in a subsequent clinical pilot study, immunomodulators levels were evaluated in samples from 74 children (≤ 5 years-old) hospitalized with viral and/or bacterial community-acquired pneumonia. Clinically, among the 74 cases of pneumonia, patients with identified mixed-detection had significantly higher (3.6-fold) serum IP-10 levels than those with a single detection ($P = 0.03$), and were significantly associated with severe pneumonia ($P < 0.01$). This study demonstrates that viral and bacterial co-infection modulates the JAK-STAT signaling pathway and leads to exacerbated IP-10 expression, which could play a major role in the pathogenesis of pneumonia.

Text: Scientific RepoRts | 6:38532 | DOI: 10.1038/srep38532 pathogenesis of several diseases and has been suggested as a potential biomarker of viral infection 10, 11, late-onset bacterial infection in premature infants 12, and a promising biomarker of sepsis and septic shock 13, 14. Combined analysis of IP-10 and IFN- γ has also been reported as a useful biomarker for diagnosis and monitoring therapeutic efficacy in patients with active tuberculosis [15] [16] [17], and both remain detectable in the urine of patients with pulmonary diseases in the absence of renal dysfunction 18.

With airway epithelial cells 19, resident alveolar macrophages (AMs) and blood monocytes-derived macrophages (recruited into tissues under inflammatory conditions 20, 21) represent a major line of defense against both pneumococcal (through their high phagocytic capacity [22] [23] [24]) and influenza infection 25, 26. So far, no studies have yet focused on the intracellular mechanisms that regulate IP-10 in human blood leukocytes during mixed IAV and SP infection. Several studies indicated that host non-coding small RNAs (including microRNAs) may function as immunomodulators by regulating several pivotal intracellular processes, such as the innate immune response 27 and antiviral activity 28, 29; both of these processes are closely related to toll-like receptor (TLR) signaling pathways.

In this study, we firstly investigated the in vitro intracellular mechanisms that mediate the innate immune response in IAV and/or SP infected human monocyte-derived macrophages (MDMs). Using this approach, we observed that mixed-infection of MDMs induces a

synergistic production of IP-10 which can be related to a miRNA-200a/JAK-STAT/SOCS-6 regulatory pathway. Subsequently, in a retrospective analysis of clinical samples collected from children ≤ 5 years-old hospitalized with pneumonia, we confirmed that serum IP-10 level could be related to both viral and/or bacterial etiologies and disease severity.

Characteristics of MDMs infected by IAV and/or SP. Initially, we investigated in vitro the impact of single and mixed IAV and SP infection on MDMs. Firstly, active replication of IAV was assessed by qRT-PCR and quantification of new infectious viral particles in the cell supernatants (Fig. 1a,b). IAV titer increased over time after single infection with IAV and correlated with increased production of negative-strand IAV RNA. Maximum viral replication was observed at 18-24 hours post-infection, after which time both RNA replication and the quantity of infectious particles decreased. In this in vitro model, subsequent challenge of IAV-infected MDMs with SP had no significant impact on the production of new infectious viral particles (Fig. 1b) . Together, these results indicate permissive and productive infection of MDMs by IAV. Secondly, we evaluated whether MDMs are permissive for both IAV and SP infection. The presence of pneumococci within IAV-and SP-infected primary MDMs was confirmed at 8 h post-infection (Fig. 1c) , suggesting that MDMs are permissive for viral and bacterial co-infection in the early steps of infection. Importantly, confocal co-detection of mixed IAV and SP was only effective following 8 h post-infection due to the bactericidal impact of SP internalization within human macrophages (after 24 h, data not shown). Thirdly, we evaluated the impact of single and mixed infection with IAV and SP on MDM viability. Mixed infection significantly decreased cell viability ($65.2 \pm 4.5\%$ total cell death at 48 hours post-infection; $P < 0.0001$) compared to single SP and IAV infection ($39.6 \pm 1.7\%$ and $17.4 \pm 1.1\%$ total cell death, respectively; Fig. 1d). Taken together, these results confirmed human MDMs are permissive to mixed viral and bacterial infection. mRNA, microRNA and protein expression profiling reveal an overall induction of the host innate immune response following IAV and/or SP infection of MDMs. To investigate the innate immune response orchestrated by IAV-and SP-infected human MDMs, we firstly evaluated the expression of 84 genes involved in the innate and adaptive immune responses (Table S1) ; the major differentially-expressed genes are summarized in Fig. 2a . Expression profiling indicated an overall induction of genes related to the JAK-STAT, NF- κ B and TLR signaling pathways. Indeed, all interferon-stimulated genes (ISGs) screened, including CXCL10 (fold-change [FC] = 240.9), CCL-2 (FC = 34.2) and MX-1 (FC = 151.4) were upregulated following mixed infection compared to uninfected cells, most of which are closely related to STAT-1 (FC = 52.3), IRF-7 (FC = 6.8) and IFNB1 (FC = 5.2) also found upregulated in mixed infected cells. Secondly, we investigated the endogenous microRNA expression profiles of IAV-and SP-infected

MDMs. A selection of microRNAs that were found to be differentially-expressed under different infection conditions are shown in Fig. 2b and Table S2 . MiRNA-200a-3p was overexpressed after both single IAV (FC = 6.9), single SP (FC = 3.7) and mixed IAV/SP infection (FC = 7.3), indicating this miRNA may play a role in the innate immune response to viral and bacterial co-infection. Similar miRNA-200a-3p dysregulation profiles were obtained following IAV and/or SP infections of human macrophages-like (THP-1 monocytes-derived macrophages) or primary MDMs (data not shown). Thirdly, the secreted levels of various antiviral, pro-inflammatory and immunomodulatory cytokines/chemokines were assayed in IAV-and SP-infected-THP-1 and primary MDM cell supernatants. We observed a remarkable correlation between the mRNA and protein expression profiles of single or mixed infected MDMs especially regarding CXCL-10 and IP-10 expression. Indeed, the level of IP-10 was synergistically increased in the supernatant of IAV-infected THP-1 MDMs exposed to SP (mean: $30,589 \pm 16,484$ pg ml⁻¹) compared to single IAV infection ($1,439 \pm 566.5$ pg ml⁻¹) and single SP infection ($4,472 \pm 2,001$ pg ml⁻¹ ; $P \leq 0.05$; Fig. 2c) at 24 hours after infection. In those cells, IP-10 expression reduced over time (48 to 72 hours), coinciding with a significant higher proportion of necrotic and apoptotic cells (Fig. 1d) . The synergistic expression of IP-10 was similarly observed at 24 hours post-infection using primary MDMs (Fig. 2d) . Significantly increased secretion of the other tested cytokines and chemokines was not observed post-infection, even in mixed infected MDMs (Fig. S1) . Interestingly, a significant production of IP-10 was also observed in supernatants of primary human airway epithelial cells (HAEC) mixed-infected by IAV and SP compared to the single infections (Fig. 2e) . Taken together, the mRNA and protein profiling results suggested that mixed viral and bacterial infection of MDMs induces a synergistic pro-inflammatory response related to the type-1 interferon and JAK-STAT signaling pathways, with IP-10 as signature of IAV/SP co-infection. Among all microRNAs screened, miR-200a-3p was the most Scientific RepoRts | 6:38532 | DOI: 10.1038/srep38532 overexpressed in IAV/SP co-infection of human MDMs. In the remainder of this study, we decided to investigate the interconnection between miR-200a-3p expression and the innate immune response.

Endogenous miRNA-200a-3p expression correlates with CXCL10 (IP-10) induction following mixed IAV and SP infection of human MDMs. Using a specific Taqman probe assay targeting miR-200a-3p, we confirmed a significant upregulation of miR-200a-3p following mixed IAV and SP infection of human MDMs (Fig. 3a) . In this experiment, a more marked up-regulation of miR-200a-3p was observed following IAV+ SP compared to results obtained previously (Fig. 2b) . This discrepancy has been attributed to the use of two different approaches to quantify miR-200a-3p expression. The use of a target-specific stem-loop

reverse transcription primer in Fig. 3a allows a better sensitivity of miR-200a-3p detection compared to the non-specific fluorescent dye used in Fig. 2b . As the general trend was suggestive of a synergistic induction of miR-200a-3p in response to mixed infection (Fig. 3a) , we hypothesized microRNA-200a-3p may play a role in the regulation of CXCL10 (IP-10), which was also synergistically upregulated in mixed-infected MDMs (Fig. 2c and d) and primary HAEC (Statistical analyses were performed using two-way ANOVA with Tukey's post-hoc test; *P < 0.05; **P < 0.01; ***P < 0.001; ****P < 0.0001.

Scientific RepoRts | 6:38532 | DOI: 10.1038/srep38532 CXCL10 (Fig. 3d) . These results suggested miR-200a-3p indirectly regulates CXCL10 and led us to hypothesize that miR-200a-3p controls a potential repressor of the JAK-STAT signaling pathway. . At 18 h after transfection, the MDMs were singly or mixed infected as described previously. At 8 h post-IAV and/or SP infection, total mRNA was extracted and amplified by PCR using specific primers for the indicated genes. Values represent median $\pm$ IQR (a, c) or mean $\pm$ SEM (d, e) of three biological replicates. Statistical analyses were performed using a Kruskal-Wallis test (non-parametric, one-way ANOVA with Dunn's post-hoc test) for data presented in (a, c). An ordinary two-way ANOVA (with Tukey's post-hoc multiple comparison test) was used for data presented in (d, e). *P < 0.05; **P < 0.01; ***P < 0.001; ****P < 0.0001. MiRNA-200a-3p indirectly regulates IP-10 expression by targeting SOCS6. As shown in Fig. 2a , several JAK-STAT signaling pathway genes were deregulated in mixed IAV-and SP-infected human MDMs; therefore, we hypothesized that miR-200a-3p directly regulates a regulator of the JAK-STAT signaling pathway. Predictive target analysis indicated that the 3' UTR of suppressor of cytokine signaling-6 (SOCS6) may be targeted by miR-200a-3p (Fig. 3b) . SOCS proteins constitute a class of negative regulators of JAK-STAT signaling pathways that are induced by both cytokines and TLR signaling. MiRNA-200a-3p was not predicted to target any of the other six members of the SOCS gene family. Transfection of human MDMs with MIM-200a downregulated SOCS6 (FC = 0.57) while inhibition of miR-200a-3p (INH-200a) upregulated SOCS6 (FC = 1.55), confirming that miR-200a-3p effectively regulates the expression of SOCS6 (Fig. 3e) . Moreover, SOCS6 was synergistically downregulated in IAV-or IAV/SP-infected MDMs overexpressing miRNA-200a (Fig. 3e) , suggesting that both infection and miR-200a-3p negatively regulate the expression of SOCS6. Finally, western blotting confirmed that expression of SOCS-6 sharply reduced following infection, especially after mixed IAV and SP infection (Fig. 3f) .

These results indicate miR-200a-3p is strongly induced in response to mixed viral and bacterial co-infection, which in turn leads to downregulation of the JAK-STAT regulator SOCS-6 at both the mRNA and protein levels and subsequent upregulation of IP-10.

analyses demonstrated mixed IAV and SP infection of human MDMs and HAEC induced significant production of IP-10. As blood leukocytes and respiratory tract epithelial cells actively contribute to inflammation during pneumonia, we hypothesized the level of IP-10 in serum of patient with pneumonia may be both indicative of mixed respiratory infection and disease severity. As part of a prospective, hospital-based, multicenter case-control study on the etiology of pneumonia among children under 5-years-old, a total of 74 patients (44 male, 30 female) were included in this pilot evaluation. According to WHO guidelines, retrospective analysis indicated 44 (59.5%) children had clinical signs of non-severe pneumonia and 30 (40.5%) children had signs of severe pneumonia. The main patient characteristics at inclusion are shown in Table 1 . Patients with severe pneumonia had significantly more recorded episodes of dyspnea ($P < 0.001$), cyanosis ($P = 0.03$), lower chest indrawing ($P < 0.001$), dullness to percussion ($P < 0.001$) and lethargy ($P < 0.001$) during chest examination than patient with non-severe pneumonia. Moreover, pleural effusions were significantly more observed among critically ill patients and the duration of hospitalization was significantly longer for the children with severe pneumonia than for those with non-severe pneumonia ($P = 0.0015$). Two deaths occurred within the group of children retrospectively defined with severe pneumonia. Evaluation of the systemic inflammatory response of the 74 cases is shown in Table 2 . Serum level of CRP, IP-10, PCT, G-CSF, IL-6, IL-8 and MIP-1 β were significantly more elevated in serum samples from critically ill patients. Patients with severe pneumonia had significantly higher (4.2-fold) serum IP-10 levels than those with a non-severe pneumonia ($P < 0.001$) suggesting IP-10 as a promising prognostic marker in pneumonia. Diagnostic accuracy measures for predicting pneumonia severity using blood-based biomarkers are summarized in Table S3 . Briefly, in this study, the optimal IP-10 cut-off value for identifying patient with severe pneumonia was 4,240 pg ml⁻¹ , with an area under the receiver operating characteristic curve of 0.69 (95% CI, 0.57 to 0.82, $P < 0.001$). Defining as positive a serum IP-10 level above this cut-off resulted in a sensitivity of 63.3%, specificity of 63.6% and a positive likelihood ratio of 1.74. Prognostic values of IP-10 were closed to procalcitonin (PCT; AUC = 0.70; 95% IC, 0.58 to 0.82, $P < 0.001$) and IL-6 (AUC = 0.70; 95% IC, 0.58-0.83, $P < 0.001$).

Multiplex PCR-based screening of respiratory and blood samples reveal a high variety of pathogen associations (Table 3) . Respiratory viruses were detected in the nasal aspirates

(NAs) of 63/74 patients (85.1%). Etiological bacteria of pneumonia (*S. pneumoniae*, n = 19; *S. aureus*, n = 1; or *H. influenzae* type B, n = 7) were identified via real-time PCR in the blood samples of 27/74 (36.5%) of the patients. Multiplex PCR assays allowed the identification of respiratory bacteria in the blood of 19 patients with negative blood culture results. Among the 74 cases PCR-positive for respiratory pathogens, a single virus or bacteria were detected in the NAs of 7 (9.4%) and 3 (4.0%) patients, respectively; these 10/74 (13.5%) cases were defined as the single infection group. The mixed infection group included the 62/74 (83.8%) cases in which (1) multiple viruses and/or bacteria were identified in NAs (38/74; 51.3%) without any bacteria identified in blood samples or (2) one or more viruses and/or bacteria were identified in NAs and associated with a blood bacteremia (24/74; 32.4%). We evaluated whether IP-10 serum level could correlate with the viral and bacterial etiologies of pneumonia. Patients with mixed infection had significant higher (3.6-fold) IP-10 serum level than patient with single detection ($P = 0.03$; Table 4). A stratified analysis reveals that the highest IP-10 serum level was observed among patients with both several respiratory pathogens identified (mixed-detection group) and severe pneumonia (14,427 pg ml⁻¹ , IQR (3,981-82,994). In detail, a remarkable IP-10 serum level (142,531 pg ml⁻¹), representing 33-fold higher above cut-off value predicting pneumonia severity was observed in patient with hRV in NA co-detected with *S. pneumoniae* (serotype 14) in pleural effusion and blood. In concordance with our in-vitro model of co-infection, a significant IP-10 level (90,338 pg ml⁻¹) was quantified in blood sample of patient with severe bacteremic pneumococcal (serotype 14) pneumonia with a positive co-detection of Influenza B virus in NA. Taken together, these results suggest that high serum IP-10 levels are significantly associated with mixed viral and bacterial detection and also related to pneumonia pathogenesis.

This study provides additional in vitro and clinical data to improve our understanding of the immunopathology of mixed viral and bacterial pneumonia (Fig. 4) .

The in vitro model of influenza and pneumococcal superinfection of human MDMs demonstrated that mixed infection synergistically induced release of the pro-inflammatory chemokine IP-10, strongly suggesting human Scientific RepoRts | 6:38532 | DOI: 10.1038/srep38532 blood leukocytes contribute to the immunopathology of pneumonia. Additionally, transcriptomics and omics analyses provided new data on the inflammatory pathways that are activated during mixed infection and related to synergistic induction of the pro-inflammatory chemokine IP-10 in mixed infected cells. Our observations are consistent with a recent study describing IP-10 induction as host-proteome signature of

both viral and bacterial infections 30 . Of the differentially-expressed genes observed in mixed infected MDMs, the transcription factors STAT-1 and IRF-7 appear to play crucial roles in the regulation of interferon-stimulated genes including CXCL10 (IP-10). By focusing on the intracellular mechanisms that regulate inflammatory pathways, we demonstrated a novel role for miRNA-200a-3p in the regulation of CXCL10 (IP-10). These observations are consistent with previous reports showing that RNA virus infection upregulates miR-155 in macrophages and dendritic cells and also regulates suppressor of cytokine signaling 1 (SOCS1), suggesting the existence of a miRNA/JAK-STAT/SOCS regulatory pathway during viral infection 29 . Our study suggests co-infection leads to overexpression of miR-200a-3p, which in turn targets and downregulates the JAK-STAT regulator SOCS-6 and consequently increases CXCL10 (IP-10) expression. Interestingly, a complementary in-silico approach reveals that several microRNAs that were found dysregulated in our experiments of IAV and SP co-infection of MDMs or HAEC, might target several genes of SOCS family and play similar role than miR-200a-3p. Indeed, miRNA-142-3p might target SOCS4, 5, 6 mRNA while miRNA-194-5p might target SOCS2, 3, 4, 5 and 7 mRNA. These observations underline that intra-cellular regulation of IP-10 is not limited to the contribution of a sole microRNA. A complex inter-relationship between numerous host microRNAs and inhibitors of the JAK-STAT signaling pathway occur to control host innate inflammatory response against viral and/or bacterial infections. Clinically, the majority of pediatric CAP cases in this study were associated with both positive viral and/or bacterial detection. Respiratory microorganisms were detected in 97% of cases; 51.3% of which were viral-viral, viral-bacterial or bacterial-bacterial co-detected only in nasal aspirates, 32.4% of which co-detected in both nasal aspirates and blood samples. These data are consistent with previous etiological studies of pediatric CAP 3,31-33 . *S. pneumoniae* was the major bacteria identified in blood (19/74; 25.7%) and mainly co-detected with respiratory viruses in NAs (16/19; 84.2%). We observed a very high diversity of viral and bacterial associations in biological samples from children with pneumonia. In comparison with IAV and SP14 combination evaluated in-vitro, no pneumonia cases were singly influenza and pneumococcus infected, and no similar co-detection with those two pathogens has been clinically observed. Nevertheless, Influenza B (IVB) virus was identified in 5 patients and two of them had a positive SP co-detection in blood (one non-typable strain and one serotype 14 using our molecular typing test). IVB and SP14 combination seems to be the nearest pathogen co-detection to that in-vitro investigated. Clinically, this co-detection was associated with both a very high IP-10 expression and a very severe pneumonia case definition. Interestingly, our translational pilot evaluation reveals IP-10 expression can be induced by several different viral and/or bacterial combinations. As immune response to each pathogen is different, further in-vitro investigations using different pathogens

associations are needed to better characterize the mechanisms involved in the immunopathology of pneumonia.

In this cohort, highest serum IP-10 levels were identified among patients with both several pathogen detected and severe pneumonia, suggesting a significant role of IP-10 on pneumonia pathogenesis. Indeed, high plasma levels of IP-10 have previously been reported in patients with sepsis 12 , and were associated with high mortality rate, especially among patients with CAP 34 . Additionally, the IP-10-CXCR3 axis has been related to acute immune lung injury and lymphocyte apoptosis during the development of severe acute respiratory syndrome (SARS) 35, 36 . Moreover, an in vivo study that modeled influenza and pneumococcal superinfection in mice indicated that pro-inflammatory chemokines, including IP-10, play a crucial role in influenza-induced susceptibility to lung neutrophilia, severe immunopathology and mortality 37 . In this study, markedly elevated IP-10 (92,809 pg ml⁻¹) combined with the highest PCT level (74.4 pg ml⁻¹) were quantified in the serum sample of a child who died, in whom *S. pneumoniae* (serotype 9 V) was identified in the blood (PCR and blood culture) and co-detected with *Haemophilus influenzae* type B in nasal aspirate. These observations suggest an interrelationship between co-detection, elevated serum IP-10 and the pathogenesis of pneumonia.

Several limitations of this pilot translational study need to be acknowledged before concluding mixed infection is related to elevated IP-10 and disease severity. Indeed, although viral shedding (e.g., of HRV and HBoV) is common in asymptomatic children, we were unable to evaluate the levels of immunomodulators in the serum samples of a control group. Moreover, although the samples were collected within the first 24 hours after admission, only a single blood sample was processed for each patient. Therefore, a larger, longitudinal study on the etiology and severity of pneumonia will be necessary to confirm these results. In conclusion, the present findings suggest that mixed respiratory infections and IP-10 may play major, interconnected roles in the pathogenesis of pneumonia. Clinically, assessment and monitoring of induced IP-10 serum level may assist clinicians to improve diagnosis and patient management of severe community-acquired pneumonia.

Viral and bacterial strains. The 10 ng ml⁻¹ M-CSF (Miltenyi Biotec). THP- 1 MDMs were obtained by culturing cells with 10 ng ml⁻¹ phorbol myristate acetate (PMA; Invivogen, Toulouse, France) for 72 hours. Human airway epithelial cells (HAEC, bronchial cell type) originated from a 54-years old woman with no pathology reported (batch number

MD056501) were provided by Mucilair (Epithelix, Geneva, Switzerland). Sterility, tissue integrity (TEER), mucus production and cilia beating frequency have been certified by the company. Gene expression profiling. Total cellular mRNA was purified using the RNeasy kit (Qiagen, Hilden, Germany). Reverse-transcription of total mRNA was performed using the RT 2 First Strand Kit (SABiosciences, Hilden, Germany). The expression of 84 genes involved in the human innate and adaptive immune responses was evaluated using the RT 2 profiler [™] PCR Array (SABiosciences) according to the manufacturer's recommendations. The $\Delta\Delta$ Ct method was applied to calculate the fold changes in gene expression for each gene relative to uninfected control cells using the web-based RT 2 profiler PCR Array Data Analysis software (SABiosciences).

MicroRNA profiling array. Total cellular microRNAs were purified using the miRNeasy Mini kit (Qiagen) and reverse-transcribed using the miScript Reverse Transcription kit (Qiagen). The profiling of 84 miRNAs was performed using the Human Immunopathology miScript miRNA PCR Array kit (Qiagen) according to the manufacturer's instructions. Data were analyzed using the miScript miRNA PCR array data analysis web portal.

In silico miRNA target prediction. MiRNA target genes were retrieved and compiled using TargetScan 38 and microRNA.org resource 39 . The interactions between miRNAs and intracellular pathways were predicted using DIANA-miRPath v2.0 40 .

THP-1 MDMs were seeded in 24-well plates (0.5×10^6 per well) in triplicate, exposed to Influenza A H1N1 (A/Solomon islands/3/2006) virus (IAV) under serum-free conditions for 1 hour and then cultured for 3 hours in fresh RPMI-1640 containing 2% FBS. Streptococcus pneumoniae (SP) serotype 14 was added at 4 hours after IAV infection. Gentamicin ($10 \mu\text{g ml}^{-1}$) was added 2 hours after SP infection (i.e. 6 hours post-influenza infection) and maintained in the culture media throughout the experiment to kill extracellular bacteria and limit bacterial growth. Cell viability was determined by flow-cytometry using the FITC/Annexin V apoptosis detection kit (BD Biosciences), according to the manufacturer's instructions. #4427975) . In this assay, fold changes have been defined by the $\Delta\Delta$ Ct method using control RNU-44 and -48 as reference microRNAs. Total mRNA was purified from transfected and infected MDMs using the RNeasy kit (Qiagen) and specific primers were used to amplify transforming growth factor beta-2 (TGFB2; F: 5' - CCATCCCGCCCACTTTCTAC-3' , R: 5' -AGCTCAATCCGTTGTTTCAGGC-3'), SOCS6 (F: 5' - AAGAATTCATCCCTTGGATTAGGTAAC-3' , R: 5' -CAGACTGGAGGTCGTGGAA-3') 41 43 , and

3) absence of wheezing at auscultation, and, 4) first symptoms appearing within the last 14 days, and 5) radiological confirmation of pneumonia as per WHO guidelines 44 . Based on these primary criteria defining pneumonia cases, all 74 cases were retrospectively re-evaluated according to the WHO "Pocket book of hospital care for children" 45 criteria to evaluate pneumonia severity. Cases that died during the study, or who had at least one additional clinical signs including central cyanosis, dullness to percussion during chest examination, prostration/lethargy, pleural effusion observed on chest radiography were retrospectively included in the severe pneumonia group. Patients without any of these additional clinical signs were included in the non-severe pneumonia group. Table 4 . a IP-10 values are expressed in pg ml⁻¹ . IP-10 concentration differences between groups were compared using unpaired Mann-Whitney tests; significant changes ($P < 0.05$) are in bold. Clinical and molecular analysis. Nasopharyngeal aspirates (NAs) and whole blood samples were collected from children within 24 hours of admission. Whole blood samples were used for complete blood counts, blood culture and multiplex real-time PCR to identify *Staphylococcus aureus*, *Streptococcus pneumoniae* and *Haemophilus influenzae* type B 46 . *S. pneumoniae* serotypes were defined using a 11 multiplex real-time PCR assay targeting the 40 most frequently represented serotypes or serogroups according to protocol developed by Messaoudi et al. 47 . Serum C-reactive protein (CRP; AssayPro, St. Charles, Missouri, United States) and Procalcitonin (PCT; VIDAS B.R.A.H.M.S; bioMérieux) were quantified from whole-blood samples. Multiplex real-time non quantitative PCR (Fast-Track Diagnostic, Sliema, Malta) was used to detect 19 viruses and five bacteria in the respiratory specimens (NAs and pleural effusions). Mixed detection was defined as 1) PCR-positive for multiple viruses in NAs, 2) positive blood culture or PCR-positive for multiple bacteria in blood or 3) PCR-positive for one or multiple viruses in NAs and one or multiple bacteria in blood (identified by PCR and blood culture).

Ethical approval. The study protocol, informed consent statement, clinical research form, any amendments and all other study documents were submitted to and approved by the Ethical Committee of the Instituto de Investigaciones en Ciencias de la Salud, the Universidad Nacional de Asunción (IICS-UNA) and the Hospital Pediátrico Niños de Acosta Ñu. Informed consent was obtained from all subjects involved in this study. The clinical investigation was conducted according to the principles expressed in the Declaration of Helsinki.

Statistical analysis. The Chi-square test and Fisher's exact test were used to compare categorical variables; continuous variables and non-normally distributed data were

compared using the Mann-Whitney U-test; normally distributed data were compared using unpaired t-tests. Comparative analyses between experimental conditions (i.e., MOCK, IAV, SP or IAV + SP) were performed using one-way ANOVA with Tukey's post-hoc test or Kruskal-Wallis analysis with Dunn's post-hoc tests. Receiver operating curve (ROC) analysis was used to determine the optimal cut-off value for IP-10 to differentiate between non-severe and severe pneumonia cases. $P < 0.05$ was considered statistically significant. All statistical analyses were performed using GraphPad Prism (La Jolla, California, United States).

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In Vitro Bactericidal Activity of 4- and 5-Chloro-2-hydroxy-N-[1-oxo-1-(phenylamino)alkan-2-yl]benzamides against MRSA

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Abstract: A series of nine substituted 2-hydroxy-N-[1-oxo-1-(phenylamino)alkan-2-yl]benzamides was assessed as prospective bactericidal agents against three clinical isolates of methicillin-resistant *Staphylococcus aureus* (MRSA) and *S. aureus* ATCC 29213 as the reference and quality control strain. The minimum bactericidal concentration was determined by subculturing aliquots from MIC determination onto substance-free agar

plates. The bactericidal kinetics of compounds 5-chloro-2-hydroxy-N-[(2S)-3-methyl-1-oxo-1-[[4-(trifluoromethyl)phenyl]amino]butan-2-yl]benzamide (1f), N-[(2S)-1-[(4-bromophenyl)amino]-3-methyl-1-oxobutan-2-yl]-4-chloro-2-hydroxybenzamide (1g), and 4-chloro-N-[(2S)-1-[(3,4-dichlorophenyl)amino]-3-methyl-1-oxobutan-2-yl]-2-hydroxybenzamide (1h) was established by time-kill assay with a final concentration of the compound equal to 1x, 2x, and 4x MIC; aliquots were removed at 0, 4, 6, 8, and 24 h time points. The most potent bactericidal agent was compound 1f exhibiting remarkable rapid concentration-dependent bactericidal effect even at 2x MIC at 4, 6, and 8 h (with a reduction in bacterial count ranging from 3.08 to 3.75 log(10) CFU/mL) and at 4x MIC at 4, 6, 8, and 24 h (5.30 log(10) CFU/mL reduction in bacterial count) after incubation against MRSA 63718. Reliable bactericidal effect against other strains was maintained at 4x MIC at 24 h.

Text: The antibiotic resistance of invasive pathogens has become one of the most challenging and persistent health problems [1] . Methicillin-resistant *Staphylococcus aureus* (MRSA) has become the most common clinically relevant multiresistant pathogen [2] causing both healthcare-associated and community-acquired bloodstream infections with mortality rates up to 40% [3] .

The prevalence of MRSA is increasing worldwide and, according to the latest information of the European Centre for Disease Prevention and Control from 2012 [4] , can be considered alarming in some European countries, especially in Portugal and Romania, where $\geq 50\%$ of all *S. aureus* isolates from invasive infections were identified as MRSA in 2012 (although, e.g., in Romania the prevalence of MRSA was 25-50% in 2010), followed by Italy, Greece, and Poland with 25-50% isolates being MRSA in 2012 (for comparison, in Poland MRSA isolates constituted 10-25% from all *S. aureus* isolates in 2010).

The treatment failure of vancomycin, the therapeutic anti-MRSA agent of choice, due to the strains with elevated vancomycin minimum inhibitory concentration (MIC) values (i.e., the lowest concentration of an antimicrobial that will inhibit the visible growth of a microorganism) within the susceptible range was described previously [5, 6] . Thus, the emergence of MRSA (and vancomycin-resistant *S. aureus* in the recent years as well [7]) makes the discovery of new molecular scaffolds a priority, and the current situation even necessitates the reengineering and repositioning of some old drug families to achieve adequate control of these bacteria [8] . However, for the treatment of *S. aureus*

bloodstream infections, bactericidal antimicrobial agents are considered to be superior to bacteriostatic drugs [9] . This fact should be considered during the development of effective and safe treatment options for MRSA infections.

The history of clinical usage of salicylanilides (2-hydroxy-N-phenylbenzamides) dates back to the 1940s in therapy of tinea capitis, followed by the discovery of their anthelmintic properties in the mid 1950s [10] . Nowadays, salicylanilides (SALs) are a class of aromatic compounds possessing a wide range of interesting pharmacological activities, such as anthelmintic [11] , antibacterial [12, 13] , antimycobacterial [13] , antifungal [14] , and antiviral [15, 16] , among others. Despite being studied since the 1960s, the mechanism of action responsible for biological activities of these compounds has not been explained so far. SALs have been found to inhibit the two-component regulatory systems (TCS) of bacteria [17] . The latest studies specified them also as selective inhibitors of interleukin-12p40 production that plays a specific role in initiation, expansion, and control of cellular response to tuberculosis [18] . Furthermore, salicylanilides have been recognised as inhibitors of some bacterial enzymes, such as sortase A from *S. aureus* [19] , d-alanine-d-alanine ligase [20] , or transglycosylases from *S. aureus* (but not from *M. tuberculosis*) [12] . These enzymes participate in secretion of various proteins or in biosynthesis of bacterial cell wall. Recently, salicylanilides-like derivatives were described to inhibit two enzymes essential for mycobacteria: (i) methionine aminopeptidase, catalyzing a key step of the posttranslational modification of nascent proteins, and (ii) isocitrate lyase, which is essential for the metabolism of fatty acids [21] . Thus, SALs seem to be promising candidates for development of new antibacterial agents with a novel mechanism of action. Such new agents could be a solution to the resistance challenges.

This study is a follow-up paper to a recently published article [13] . The synthesis of the series of novel derivatives of salicylamides, 4-and 5-chloro-2-hydroxy-N-[1-oxo-1-(phenylamino)alkan-2-yl]benzamides, called diamides due to their skeleton (for general structure see Table 1), was described previously [13, 22] , and their antimycobacterial and antibacterial activities against various bacterial species were reported [13] . As these compounds expressed very significant antibacterial activity with low MIC values against clinical isolates of MRSA as representatives of multidrugresistant bacteria, we decided to extend the knowledge about the antibacterial properties of these compounds against MRSA.

The aim of the current study was to assess the overall in vitro bactericidal activity of nine newly synthesized diamides in dependence on time and concentration against clinical isolates of MRSA as representatives of multidrug-resistant bacteria. To the best of our knowledge, this is the first study dealing with the evaluation of novel microbiological characteristics of SAL analogues and revealing their bactericidal effect.

The synthetic pathway of the series of novel diamides was described recently [13, 22], and their structures (see Table 1) were confirmed by IR, NMR, and MS spectrometry, and the purity of the compounds was checked by CHN analysis [13, 22]. [27]; and MRSA SA 3202 [27] (National Institute of Public Health, Prague, Czech Republic) both of human origin. Suspected colonies were confirmed by PCR; a 108 bp fragment specific for *S. aureus* was detected [28]. All isolates were tested for the presence of the *mecA* gene encoding methicillin resistance [29]. These three clinical isolates were classified as vancomycin-susceptible (but with higher MIC of vancomycin equal to 2 g/mL (VA2-MRSA) within the susceptible range for MRSA 63718) methicillin-resistant *S. aureus* (VS-MRSA). For the MICs of vancomycin, see Table 1. Vancomycin-susceptible methicillin-susceptible *Staphylococcus aureus* (VS-MSSA) ATCC 29213, obtained from the American Type Culture Collection, was used as the reference and quality control strain. The bacteria were stored at -80°C and were kept on blood agar plates (Columbia agar base with 5% ovine blood) between experiments. (MBCs). The MBCs (i.e., the lowest concentrations of antibacterial agents required to kill a particular bacterium) were determined by subculturing aliquots (20 L) from wells with no visible bacterial growth and from control wells of MIC determination onto substance-free Mueller-Hinton agar (MHA) plates. The plates were incubated aerobically at 37°C for 24 h for colony count. The MBC was defined as the lowest concentration of substance, which produced $\geq 99.9\%$ killing Table 1 : Chemical structures and in vitro MIC and MBC [g/mL] values of tested 5- and 4-chloro-2-hydroxy-N-[1-oxo-1-(phenylamino)alkan-2-yl]benzamides (bactericidal effect of individual compounds against particular strains marked in bold). after 24 h of incubation as compared to the colony count of the starting inoculum [30]. To ensure reproducibility, each MBC assay was performed in at least triplicate on separate occasions.

[illegible]

5-Cl 4-CF₃ (S)-CH(CH₃)₂ 4 2 2 2 4 4 8 4 1g 4-Cl 4-Br (S)-CH(CH₃)₂ 8 4 4 4 1 6 8 8 8 1h 4-Cl 3,4-Cl (S)-CH(CH₃)₂ 2 2 1 1 1 4 1 4 2 1i 4-Cl 3,4-Cl (S)-benzyl 1 1 0.5 0.5 8 1 8 1 AMP - - - >16 >16 >16 0.25 >16 >16 >16 0.25 CPX - - - >16 >16 >16 0.5 >16 >16 >16 0.5 VAN - - - 2 1 1 1 2 1 1 1

Time-kill assays were performed by the broth macrodilution method according to previously described methodology [30] with some modifications. Briefly, flasks containing sterile fresh Mueller-Hinton broth (MHB) with the appropriate antimicrobial agent were inoculated with the test organism in logarithmic growth phase to obtain the starting inoculum with the concentration of approximately 7.5×10^6 CFU/mL (actual inoculum concentrations ranged from 0.9×10^5 to 2.9×10^6 CFU/mL) and a final concentration of the antibiotic equal to 1x, 2x, and 4x MIC in 10 mL volume. For the determination of viable counts, aliquots were removed at 0, 4, 6, 8, and 24 h time points after inoculation, serially diluted in sterile phosphate buffered saline, and aliquots (20 L) were plated on MHA plates in duplicate. Colony counts were performed on plates yielding 6 to 60 colonies, and the mean was calculated. Antimicrobial carry-over was controlled by dilution and visual inspection of the distribution of colonies on the plates with observation of possible inhibition of growth at the site of the initial streaks. The plates were incubated at 37 ° C for 24 to 48 h, and the number of colonies was determined. To ensure reproducibility, each time-kill experiment was carried out in duplicate on separate occasions with results presented as the mean of all experiments. The growth control without the addition of antimicrobial agents and the control containing DMSO without any antimicrobial agent to exclude antibacterial activity of this solvent were included. Time-kill curves were constructed by plotting the log₁₀ CFU per millilitre versus time (over 24 h), and the change in bacterial concentration was determined. The results were analysed by evaluating the numbers of strains that yielded $\Delta(\log_{10} \text{ CFU/mL})$ values of -1 (corresponding to 90% killing), -2 (99% killing), and -3 (99.9% killing) at 4, 6, 8, and 24 h compared to counts at 0 h. Bactericidal activity was defined as a reduction of at least 99.9% ($\geq 3 \log_{10}$) of the total count of CFU/mL in the original inoculum.

Diamides seem to be promising candidates for antibacterial agents with very strong anti-MRSA activity, as it was published recently [13]. In the present study the series of nine newly synthesized diamides was evaluated as prospective bactericidal agents against representatives of multidrugresistant bacteria, three clinical isolates of MRSA, and *Staphylococcus aureus* ATCC 29213 (methicillin-susceptible) as the reference and quality control strain. Since SALs and their analogues are known as compounds with

bacteriostatic effect [31] , this is the first study where SAL-like compounds were considered as prospective bactericidal agents and the dependence of bactericidal effect of these compounds on time and concentration was evaluated. Thus, absolutely novel microbiological characteristics of these compounds were revealed in the present study.

Recently MIC values of diamides expressed as molar concentrations in mol/L were published [13] . To allow comparison with MBC values of the present study, MICs in g/mL were calculated and are recorded in Table 1 along with the activity of reference antibacterial drugs, ampicillin, ciprofloxacin, and vancomycin. Potential bactericidal activity of diamides was assessed using MBC assay [26] . MBC values of all tested compounds are recorded in Table 1 as well.

Based on the obtained results, all compounds assessed as active according to MIC values in our previous study (1f-i) showed low or moderate MBC values against all four strains. The MBC values of these compounds did not exceed the highest tested drug concentration and ranged from 1 to 16 g/mL. In all cases, there were comparable MBC values for the clinical isolates of MRSA and the *S. aureus* reference strain.

Bactericidal activity is defined as a ratio of MBC to MIC of ≤ 4 [32] . Table 1 bactericidal activity is expressed in bold.

As mentioned above, SALs are known to exhibit a bacteriostatic effect [31] , so it was very interesting to discover that diamides possess bactericidal activity. The amide bond (-CONH-) can cause interactions with a variety of enzymes [33] ; therefore the presence of two amide bonds could be responsible for the bactericidal effect of diamides against MRSA. The activity of SALs and their analogues results from multiple mechanisms, which are still under investigation; for example, it was found that SALs are capable of inhibiting transglycosylases in later stages of *S. aureus* (including MRSA) cell wall biosynthesis [12] . These enzymes catalyse the step prior to the transpeptidation in the peptidoglycan biosynthesis and are responsible for polymerization of lipid II, which occurs at the outer face of the membrane [12] . Since antibacterial agents targeting cell wall biosynthesis act as bactericidal agents [30, 34] , the failure in the cell wall biosynthesis due to the inhibition of transglycosylases could be responsible for bactericidal activity of diamides against MRSA.

Based on these findings, antibacterial active diamides with bactericidal effect against all four tested strains as prospective bactericidal agents were chosen for subsequent timekill curve studies to determine the real dependence of bactericidal effect on concentration over time.

1-oxobutan-2-yl}-2-hydroxybenzamide (1h) were tested in time-kill studies at 1x, 2x, and 4x MIC against all MRSA isolates and the *S. aureus* reference strain. The antibacterial effect of DMSO [35] used as the solvent of the tested compounds was excluded in this assay, as time-kill curves of this solvent were identical or very similar to those of the growth control. The extent of bacterial killing was estimated by the number of these strains showing a decrease ranging from 1 to 3 log₁₀ CFU/mL in viable cell count at different times after incubation. A summary of these data is presented in Table 2 . Based on these data it can be concluded that the bactericidal potency of tested diamides against all four strains decreased as follows: 1f > 1h > 1g. No bactericidal activity (i.e., ≥ 3 log₁₀ CFU/mL decrease) was observed at 1x MIC for any strain and time after incubation tested. At 4x MIC from the four strains, compounds 1f, 1g, and 1h killed 2, 1, and 2 strains, respectively, at 8 h after incubation and 4, 2, and 2 strains, respectively, at 24 h after incubation.

The findings of time-kill studies for each of the four staphylococci strains at exposure to compounds 1f, 1g, and 1h are summarized in Table 3 . Bactericidal activity (i.e., ≥ 3 log₁₀ CFU/mL decrease) is expressed in bold.

For compound 1f rapid concentration-dependent antibacterial effect was recorded against clinical isolate of MRSA 63718. Time was not the predictive factor influencing the antibacterial activity because log₁₀ differences in CFU/mL from the starting inoculum were the same for 4x MIC (with the highest efficiency with a reduction in bacterial count of 5.30 log₁₀ CFU/mL) or very similar for 2x MIC (with a moderate regrowth after 24 h causing a loss of bactericidal activity) over 24 h. The bactericidal effect was maintained even at 2x MIC at 4 h after incubation for this strain (reduction of 3.08 log₁₀ CFU/mL). For the remaining strains, clinical isolates of MRSA SA 630, MRSA SA 3202, and *S. aureus* ATCC 29213, reliable bactericidal effect was recorded at 4x MIC at 24 h after incubation for all these strains with a reduction in bacterial count of 3.22, 3.30, and 3.65 log₁₀ CFU/mL, respectively.

For compound 1g bactericidal effect against MRSA 63718 was noticed at 2x MIC at 6 and 8 h after incubation and at 4x MIC at 4, 6, and 8 h after incubation with a reduction in bacterial count ranging from 3.10 to 3.58 log 10 CFU/mL. The most effective killing was achieved at 6 h for both concentrations. As in the case of compound 1f, a regrowth was observed after 24 h after incubation. For the remaining isolates of MRSA, SA 630 and SA 3202, bactericidal effect occurred only at 4x MIC at 24 h after incubation with a reduction in bacterial count of 3.38 and 4.01 log 10 CFU/mL, respectively. The highest bactericidal effect was recorded for MRSA SA 3202 at 4x MIC at 24 h after incubation. A reduction consistent with bacteriostatic effect (0.03 to 2.37 log 10 CFU/mL) was observed at other concentrations over time for both isolates. No bactericidal effect was observed for the *S. aureus* reference strain; compound 1g demonstrated a pattern of bacteriostatic activity against this strain with a reduction in bacterial count ranging from 0.07 to 2.33 log 10 CFU/mL at 4x MIC over time. In other cases, a slight increase in bacterial counts (i.e., overgrowth) compared with the starting inoculum was observed with values ranging from 0.10 to 1.57 log 10 CFU/mL for this reference strain.

For compound 1h bactericidal effect against MRSA 63718 was maintained at 4x MIC at 6 and 8 h after incubation with a reduction in bacterial count of 3.54 and 3.31 log 10 CFU/mL, respectively. The same as for 1g, the most potent bactericidal effect was maintained at 6 h after incubation. Regrowth at 24 h after incubation causing a loss of bactericidal activity was recorded similarly as with previous compounds. The reason for regrowth of the test organism at 24 h in the experiment is unknown. Most probably, selection of resistant mutants is responsible for this phenomenon [30] ; degradation of the drug in the growth medium is not assumed, as regrowth was

Number of strains showing the following log 10 CFU/mL decrease at the designated incubation time not observed for any other tested strain. For MRSA SA 630 concentration-dependent killing was recorded at 4x MIC at 6, 8, and 24 h after incubation with log 10 differences in CFU/mL from the starting inoculum being very similar over time (ranging from 3.18 to 3.39 log 10 CFU/mL). For MRSA SA 3202 reliable bactericidal effect was maintained only at 4x MIC at 24 h after incubation with a reduction in bacterial count of 3.02 log 10 CFU/mL. As for compound 1g, bacteriostatic activity against *S. aureus* reference strain was observed with a reduction in bacterial count ranging from 0.34 to 2.62 log 10 CFU/mL at 2x and 4x MIC. Overgrowth (values ranging from 0.04 to 1.43 log 10 CFU/mL) was recorded at

1x MIC for this strain. It is of note that in all staphylococci strains with similar MICs and MBCs for compounds 1g and 1h the responsiveness to antibacterial activity of these compounds varied with clinical strains of MRSA being effectively killed and the reference strain remaining unaffected at 4x MIC.

There is a discrepancy between bactericidal results of MBC assay compared with time-kill kinetics. This difference could be caused by comparing microtiter (MBC assay) to macrobroth (time-kill assay) dilutions [36] . Moreover, although time-kill assays are more labour intensive and time consuming than MBC assays, they are recognised to provide a greater degree of characterisation of the cell eradication potential of antibacterial agents [37] .

Concerning antibacterial effect, it is not generally important if the antibacterial agent is also bactericidal at higher concentrations, because the inhibition of bacterial proliferation usually achieves a therapeutic effect; the patient's immune system is capable of coping with the infection then [34] . However, bactericidal therapy could produce a better treatment result by rapid reduction of the bacterial load [38] . Moreover, in the case of an immune system disorder (e.g., immunosuppressive therapy, AIDS patients, etc.) bactericidal agents are unequivocally indicated. Considering steadily escalating numbers of immunocompromised patients with endocarditis, meningitis, or osteomyelitis in recent years, it is necessary to achieve bacterial killing and broaden the spectrum of antimicrobial agents with bactericidal active compounds [30] .

The clinical outcome of MRSA bacteraemia is significantly influenced by vancomycin MIC. Treatment failure exceeding 60% for *S. aureus* with vancomycin MIC of 4 g/mL resulted in the change of susceptibility breakpoint from 4 g/mL to 2 g/mL by the Clinical and Laboratory Standards Institute (CLSI) in 2006 [23] as well as by the US Food and Drug Administration (FDA) in 2008 [39] . It has been recommended that for infections caused by MRSA strains with elevated vancomycin MICs (2 g/mL), alternative therapy should be considered [40] . It is of note that based on time-kill assays in the present study, all tested diamides (particularly compound 1f exhibiting rapid bactericidal concentration-dependent effect even at 2x MIC) were most effective against isolate MRSA 63718, which is the strain with elevated vancomycin MIC of 2 g/mL. The activity against the remaining isolates with vancomycin MIC of 1 g/mL was lower.

Considering the emergence of decreasing vancomycin susceptibility of MRSA isolates and thus the therapeutic efficacy of vancomycin therapy, our aim was to determine the potential bactericidal role of novel antibacterial compounds against MRSA in vitro. Based on the obtained results, diamides can be suitable candidates for such novel bactericidal active compounds presenting a promising starting point for further investigations to ascertain real in vivo activity and the exact mechanism of action.

The present study is the first evidence of bactericidal effect of SAL analogues. Against other strains, reliable bactericidal effect was maintained at 4x MIC at 24 h after incubation. Considering the necessity to broaden the spectrum of bactericidal agents, diamides from the current study with a novel mechanism of action could present a very promising and interesting solution to this challenge for the future.

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Exploring the Innate Immunological Response of an Alternative Nonhuman Primate Model of Infectious Disease; the Common Marmoset

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Abstract: The common marmoset (*Callithrix jacchus*) is increasingly being utilised as a nonhuman primate model for human disease, ranging from autoimmune to infectious disease. In order to fully exploit these models, meaningful comparison to the human host response is necessary. Commercially available reagents, primarily targeted to human cells, were utilised to assess the phenotype and activation status of key immune cell types and cytokines in naive and infected animals. Single cell suspensions of blood, spleen, and lung were examined. Generally, the phenotype of cells was comparable between humans and marmosets, with approximately 63% of all lymphocytes in the blood of marmosets being T cells, 25% B-cells, and 12% NK cells. The percentage of neutrophils in marmoset blood were more similar to human values than mouse values. Comparison of the activation status of cells following experimental systemic or inhalational infection exhibited different trends in different tissues, most obvious in cell types active in the innate immune response. This work significantly enhances the ability to understand the immune response in these animals and fortifies their use as models of infectious disease.

Text: The common marmoset (*Callithrix jacchus*), a New World monkey (NWM) species is a small, arboreal nonhuman primate (NHP), native to the Atlantic Coastal Forest in Northeast Brazil and parts of South East Brazil. In recent years the common marmoset has become more widely used in applied biomedical research, and an increasing body of evidence suggests the physiological and immunological responses to biological insults are similar between marmosets and humans [1] . In the field of infectious disease, the marmoset is primarily being investigated as an alternative NHP model to complement the more traditionally used Old World monkeys (OWM) (e.g., rhesus and cynomolgus macaques). Evolutionarily, both NWM and OWM sit within the simiiformes infraorder of the suborder Haplorhini of primates [2] . Marmosets sit within the family Callitrichidae of the Platyrrhini parvorder, while OWM sit within the Cercopithecidae family of the Catarrhini Parvorder. Marmosets therefore are separated from Old World monkeys by one ancestral step and are a lower order primate.

Marmosets have been used to model the infection syndrome caused by a number of public health pathogens including Lassa virus [3] , Hepatitis C virus [4] , Dengue virus [5] , Herpesvirus [6] , Junin virus [7] Rift Valley Fever [8] , and SARS [9] . Marmosets have also been used to model a number of biodefense pathogens including Eastern Equine Encephalitis virus [10] , *Bacillus anthracis* [11] , *Francisella tularensis* [12, 13] , *Burkholderia pseudomallei* [14] , Marburg haemorrhagic fever virus [15, 16] , Ebola haemorrhagic fever virus [16] , and Variola virus [17] . The utility of marmosets to assess

medical countermeasures has also been demonstrated; a vaccine has been tested for Lassa fever [18] and the efficacy of ciprofloxacin and levofloxacin has been tested as postexposure therapies for anthrax and tularemia, respectively [19, 20] .

In order to exploit these models fully and to allow meaningful comparison with the human condition, the response of the immune system to infection/therapy needs to be 2 Journal of Immunology Research characterised and understood. Generally, NHPs have a close molecular, immunological, reproductive, and neurological similarity with humans making them ideal surrogates for humans and the study of infectious diseases. There is a high level of gene homology between humans and NHPs which underlies physiological and biochemical similarities. Similarities at the genetic level extend to the phenotypical level making NHPs well suited to modelling pathophysiological responses in man [21] . Immunologically, there is a high degree of homology between humans and marmosets [22] . The similarity of various immunological factors produced by humans and marmosets has been investigated at both the genetic and protein levels. There is at least 95% homology between human costimulatory molecules (e.g., CD80, CD86 etc.) and those of marmosets [23] . Also the immunoglobulin and T-cell receptor repertoire of humans and marmosets show at least 80% homology [24, 25] .

Currently, the availability of commercial reagents specifically designed for the marmoset is limited although a number of antibodies designed for use with human samples have been shown to cross-react with leucocytes from marmoset blood [26] [27] [28] . However, these reagents have not been exploited to investigate the immune response to infectious disease. To date, investigation of the immune response in marmosets has primarily been achieved using pathogen-specific antibodies to determine the serological response using ELISA such as in the smallpox, Dengue, Rift Valley Fever, and Herpes models [5, 6, 8, 17] or by immunohistochemistry to identify, for example, CD8+, CD3+, CD20+ cells, and IL-6 in the smallpox model [17] ; neutrophils and macrophages in the Herpes model [6] ; or CD3+ and CD20+ cells in the Lassa model [3] .

The work presented here focuses on understanding the immune profile of the naive marmoset as well as identifying and quantifying the immune response to infectious disease. The aim of this work is to determine key changes and identify correlates of infection or protection.

Healthy sexually mature common marmosets (*C. jacchus*) were obtained from the Dstl Porton Down breeding colony and housed in vasectomized male and female pairs. The Dstl colony was established during the 1970s and is a closed colony with a stable genotype. Animals included in these studies were mixed sex pairs, between 18 months and 5 years old and weighing between 320 g to 500 g. All animals were allowed free access to food and water as well as environmental enrichment. All animal studies were carried out in accordance with the UK Animals (Scientific Procedures) Act of 1986 and the Codes of Practice for the Housing and Care of Animals used in Scientific Procedures 1989. Animals were challenged with an intracellular pathogen by either the subcutaneous or inhalational route and were humanely killed at various time points after challenge. Prior to the infection study, animals were bled to determine baseline immunological parameters. Studies were performed to establish infection models in order to evaluate the efficacy of suitable therapies for transition ultimately to the clinic.

Populations. Blood and tissue samples were homogenised to provide single cell suspensions [12] . Red blood cells were lysed, and the mixed leucocyte population was washed and stained with various combinations of the following fluorescent antibody stains: CD3 (SP34-2), CD8 (LT8), CD11c (SHCL3), CD14 (M5E2), CD16 (3G8), CD20 (Bly1), CD45RA (5H9), CD54 (HCD54), CD56 (B159), CD69 (FN50), CD163 (GHI/61), and MCHII (L243) (BD Bioscience, Insight Bioscience, AbD serotec). Samples were fixed in 4% paraformaldehyde for 48 hrs at 4 °C and analysed by flow cytometry (FACScanto II BD) within 72 hours of staining.

Levels of circulating cytokines and chemokines were also quantified in the blood of marmosets from the Dstl colony using human multiplex kits available commercially (BD cytokine flex beads and the Luminex system). These systems show significant cross-reactivity with the marmoset suggesting a high degree of conservation between the two species for IL-6, MIP-1 , MIP-1 , and MCP-1 [29] . However, for other cytokines that are pivotal in the innate response, TNF and IFN reagents were obtained from U-CyTech Biosciences and Mabtech AB, respectively, due to a lack of cross-reactivity observed within the kit obtained from BD [13] .

In order to fully characterise the immune response to infectious agent in the marmoset, single cell suspensions of lung and spleen tissue were also examined in conjunction with the traditionally used blood cells. These tissue homogenates are of particular interest in

relation to target sites of infection: the lung as the site of initial infection following an inhalational challenge and the spleen as a representative organ following a parental challenge. Cell types targeted during this analysis include cells important in the innate response (e.g., neutrophils, macrophages, and NK cells) and the adaptive response (T and B cells) with a view to determine the response to infection and vaccination and to derive immune correlates of infection/protection. Dapi was included as a nuclear marker to ensure that the initial gating included only intact cells. Basic cell types in blood were easily identified by measuring size (forward) and granularity (side) scatter (Figure 1(a)). Identification of cell types in tissue samples was more difficult as the scatter profiles are less clearly compartmentalized. The common leukocyte antigen (CD45) normally used to locate all leukocytes in human samples also worked well in marmoset blood but failed to provide relevant information in the tissue samples. Confirmation of neutrophil identification was done by nuclear morphology and macrophages were identified by their adherent nature in initial experiments (data not shown). Neutrophils were stained as CD11c dim CD14⁻ and macrophages as CD11c + CD14⁺ regardless of tissue origin (Figure 1(b)). Figure 1 shows the basic division of lymphocytes between T, B, and NK cells from a healthy blood sample.

Using this approach, the percentage of NK cells, B-cells, total T-cells, CD8⁺ T-cells, neutrophils, and monocytes was determined in the blood of naive marmosets (Figure 2 (a), Table 1); approximately 63% of all lymphocytes were T cells, 25% B cells, and 12% NK cells. The variability of the data is depicted in Figure 2 (a) with the greatest variability observed in the proportion of neutrophils. There were no obvious differences attributable to age or sex of the animals. This analysis was also applied to lung and spleen homogenates from naive marmosets (Figures 2(b) and 2(c)). Greater variability was observed in the data relating to the identification of cell types in tissue samples, attributed to the inherent difficulties in identifying cell types in tissue homogenates by size and granularity and also the smaller cohort of animals. As expected, low numbers of neutrophils are found in naive spleen or lung tissue (8% both). Healthy mouse spleens typically have approximately 1-2% granulocytes [30] .

Understandably, there are few reports on the typical cell percentages expected in healthy human individuals for these tissues. However, it is reported that B cells are more prevalent in the spleens of humans at a ratio of 5 to 4 B to T cells than in the lungs which have a ratio of 1 to 8 B to T cells [34] . In marmoset data reported here, a ratio of 2 to 3 B to T cells in the

spleen and 1 to 6 B to T-cells in the lungs was observed compared to a ratio of 3 to 2 B to T cells in mouse spleens [30] .

Upon comparison, the marmoset data is generally consistent with previously reported data which is only available for marmoset blood samples [27] and information available for human blood [32, 33] (Table 1). However, one report found the proportion of CD8+ T-cells was almost three times greater in marmosets than humans, 61% to 21% respectively [35] compared to the 30% observed in this study and the work previously reported by Brok et al. [27] . Brok's study involved a small number of animals (eight) and also used a different CD8+ clone to identify cells. Contrastingly, in mice, differences are observed in the proportion of both B cells and neutrophils [31] , although these differences are highly strain specific. C57BL/6J mice are reported to have 67% B cells and BALB/C mice 46%; both of which are consistently higher than the percentage found in marmosets and humans of approximately 25% (Table 1) [27, 31] . The proportion of neutrophils found in the blood of C57BL/6J mice at 13% is lower than the 35% found in marmosets and the 40-75% expected for healthy human blood. This is encouraging as neutrophils play a pivotal role in the innate response to infection [36] . A cross-species comparison suggests that monocytes comprise 3% of leukocytes (Table 1) . Levels of circulating cytokines and chemokines (IL-6, IL-1 , MIP-1 , MCP-1, Rantes, TNF , and IFN) were also quantified in the blood, lung, and spleen of naïve marmosets from the Dstl colony. None of these cytokines were detected in blood samples from uninfected animals; however low levels of MIP-1 , MCP-1, and Rantes were found in spleen and lung tissue.

Preliminary investigation of the immune response has supported the development of marmoset model of infection at Dstl. The levels of different cell types were measured at specific times after challenge with inhalational *F. tularensis*, *B. pseudomallei*, and Marburg virus [13] [14] [15] . Following challenge with *F. tularensis*, increasing levels of NK cells, neutrophils, T cells, and macrophages were observed, peaking at 48 hours after challenge before rapidly declining. This study also demonstrated the importance of investigating the immunological response in key target organs, as an increase in CD8+ T cells and T cells was observed in the spleen and lungs but not in the blood. Increasing levels of various cytokines, MCP-1, MIP-1 , MIP-1 , IL-6, and IL-1 , were observed in Table 1 : Comparison of the percentages of different cell types observed in the blood from healthy marmosets, mice, and humans.

Identification markers Marmoset (present data)

Marmoset [27] Mouse 4 [30, 31] Human Asian [32] Human Caucasian [33] Number the lungs, spleen, and blood as the disease progressed (TNF and IFN were not measured in this study). Following inhalational challenge of marmosets with *B. pseudomallei*, an increase in the number of neutrophils was observed in the blood at 36 hours after challenge, followed by a rapid decline that was associated with an influx of neutrophils into the lung at 46 hours after challenge. A subsequent decline in the number of neutrophils in the lung was associated with the increased number in the spleen of animals that exhibited severe disease and were humanely killed. There was a gradual increase in the number of macrophages in the spleen as the disease progressed with numbers of macrophages peaking in the blood and lungs at 36 hours after challenge. A rapid decline in the number of macrophages in the lungs and blood was observed by 46 hours after challenge.

The levels of various cell types and cytokines were also measured in the blood of animals following inhalational challenge with Marburg virus [15]. In these animals a general increase in the numbers of T cells, NK cells, macrophages IFN-, IL-1, and MCP-1 was observed with time (TNF was not measured).

In order to gain more information from these acute bacterial infection models, we have sought out other markers from the literature. Primarily this was from marmoset models of autoimmune disorders such as rheumatoid arthritis and multiple sclerosis where the cross-reactivity of human antibodies was investigated, as well as the functionality of cells [37] [38] [39] [40]. More recent work at Dstl has reported further cross-reactivity between marmoset cells and human cytokines to induce activity in marmoset T cells [36, 41]. These studies, combined with increasing information available on the cross-reactivity of human antibodies to various NHPs (e.g., NIH NHP reagent resource, <http://www.nhpreagents.org/NHP/default.aspx>), has expanded the ability to assess activation markers for disease. Detection of the following cell surface markers with human antibodies was trialed: CD54 (ICAM-1) associated with cellular adhesion, inflammation, and leukocyte extravasation; CD69 the early activation marker; CD16 as a macrophage activation marker; CD163 the alternative macrophage activation marker; and MHC class II (HLA-DR). CD56 was originally included to identify NK cells; however, it was noted that its expression on T cells was upregulated during disease and that cells defined as CD3+ CD16- CD56+ have been shown to be functionally cytotoxic in marmosets [37, 42].

These markers have been used to expand on our previously published work to determine changes in the activation status of basic cell types in response to an acute bacterial infection. Animals were challenged with bacteria at a comparable dose either by inhalation (= 22) or by a systemic route (= 12) and humanely killed once they had reached a humane endpoint (between day 4 and day 5 after challenge). Figure 3 illustrates the cellular activity in representative tissues following inhalational (Figures 3(b) and 3(e)) or systemic challenge (Figures 3(c) and 3(f)) and in naïve samples (Figures 3(a) and 3(d)). Naïve T and NK cells appear to have similar resting activation states regardless of origin, whereas neutrophils and macrophages have differential expression of activation, for example, CD16. In response to disease, the proportions of the cell types appear to remain relatively constant; however, the activation markers provide more detailed information and show involvement of all the cell types explored. Extensive activation was to be expected considering that the samples were taken at the humane endpoint. There is also extensive variation between The response to infection within the lungs has similarities across disease routes in terms of neutrophil reduced expression of CD16 and CD54 and macrophage increased expression of CD16 and reduction in MHCII. Unexpectedly, the T and NK cells appear to be more actively involved in systemic disease, indicating that the disease develops a pneumonic element regardless of initial route of infection.

Levels of circulating cytokines and chemokines (IL-6, IL-1 , MIP-1 , MCP-1, Rantes, TNF , and IFN) were also quantified in the lung and spleen samples. All of the cytokines (with the exception of Rantes) were expressed at high levels (ng/mg) in all samples, which was expected as the animals had succumbed to terminal disease.

The work presented here adds significant relevant information to the marmoset models of infection and to the understanding of the immune response in these animals. This work extends marmoset immunology from autoimmune disorders into the field of infectious diseases; this coupled with an increase in the information available on crossreactivity of human reagents to a variety of NHPs increases the utility/application of marmosets as models of human disease. In conclusion, the immune response in marmosets to infectious disease can be characterised in terms of the phenotype and activation status of all the major immune cells and key cytokine and chemokine expression. This can aid in the identification of correlates of infection or protection in medical countermeasures

assessment studies. This information can also potentially be used for pivotal studies to support licensure of products under the FDA Animal Rule.

This, in conjunction with the small size of marmosets, their immune response to infection that is comparable to humans, and the ability to house more statistically relevant numbers within high containment, makes the marmoset an appropriate animal model for biodefense-related pathogens.

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Pretreatment Hepatitis C Virus NS5A/NS5B Resistance-Associated Substitutions in Genotype 1 Uruguayan Infected Patients

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6112080/>

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Abstract: Hepatitis C Virus (HCV) infection treatment has dramatically changed with the advent of direct-acting antiviral agents (DAAs). However, the efficacy of DAAs can be attenuated by the presence of resistance-associated substitutions (RASs) before and after treatment. Indeed, RASs detected in DAA treatment-naïve HCV-infected patients could be

useful for clinical management and outcome prediction. Although the frequency of naturally occurring HCV NS5A and NS5B RASs has been addressed in many countries, there are only a few reports on their prevalence in the South American region. The aim of this study was to investigate the presence of RASs to NS5A and NS5B inhibitors in a DAA treatment naïve cohort of Uruguayan patients infected with chronic hepatitis C and compare them with reports from other South American countries. Here, we found that naturally occurring substitutions conferring resistance to NS5A and NS5B inhibitors were present in 8% and 19.2%, respectively, of treatment-naïve HCV genotype 1 infected patients. Importantly, the baseline substitutions in NS5A and NS5B herein identified differ from the studies previously reported in Brazil. Furthermore, Uruguayan strains subtype 1a clustered within all major world clades, showing that HCV variants currently circulating in this country are characterized by a remarkable genetic diversity.

Text: Hepatitis C Virus (HCV) infection treatment has dramatically improved thanks to the introduction of direct-acting antiviral agents (DAAs). These antivirals have significantly increased response rates (up to 98%) and greatly reduced treatment duration [1] . Currently available DAAs are classified into four categories given their molecular targets in the HCV replication cycle: (1) NS3/4A protease inhibitors (PIs) bind to the active site of the NS3/4A protease; (2) NS5A inhibitors interact with domain 1 of the NS5A dimer, although the exact mechanism of NS5A inhibition remains to be fully elucidated; (3) nucleos(t)ide analog NS5B polymerase inhibitors are incorporated into the nascent RNA chain resulting in chain termination by compromising the binding of the incoming nucleotide; (4) nonnucleoside NS5B polymerase inhibitors interact with either the thumb 1, thumb 2, palm 1, or palm 2 domain of NS5B and inhibit polymerase activity by allosteric mechanisms [2] [3] [4] . However, the extreme mutation and high replication rates of HCV, together with the immune system pressure, lead to a remarkable genetic variability that can compromise the high response rates to DAAs due to the preexistence of resistance-associated substitutions (RASs) [5, 6] .

Each drug or class of DAA is characterized by specific resistance profiles. The likelihood that a DAA will select for and allow outgrowth of viral populations carrying RASs depends on the DAA's genetic barrier to resistance (the number and type of mutations needed to generate an amino acid substitution that confers resistance), the viral fitness (replicative capacity) of the resistant variant, and viral genotypes and subtypes [7, 8] .

The prevalence of RASs in treatment-naïve patients has been broadly reported worldwide [9] [10] [11] [12] [13] [14] [15] [16] . However, apart from Brazil and Argentina, this issue has not been fully addressed in other South American countries yet [9, [17] [18] [19] . The lack of information in relation to preexisting baseline RASs, added to the high cost of these new drugs, are the major limiting factors for the broad implementation of these new therapies in Uruguay as well as in other Latin American countries (low-or lower-middle income) [20] .

In this study, we explored the presence of resistance variants to NS5A and NS5B inhibitors in a DAA treatment naïve cohort of Uruguayan patients chronically infected with hepatitis C. Here, we aimed to contribute to the knowledge of the circulation of HCV resistant variants in the South American region.

Samples. Serum samples were obtained from 31 patients with serological markers for HCV, which were recruited between 2015 and 2017 at the Gastroenterology Clinic from Hospital de Clínicas, Montevideo, Uruguay. HCV infection was confirmed by Abbott realtime HCV (Abbott Molecular Inc., Des Plaines, USA). Patients selected for this study were both chronically infected with HCV genotype 1 and DAA treatment-naïve at the time of blood extraction. Written informed consent was obtained from all patients. The studies have been performed according to the World Medical Association Declaration of Helsinki and approved by the appropriate institutional board (Hospital de Clínicas ethical committee).

2.2. RNA Extraction, cDNA Synthesis, and NS5A and NS5B Amplification. Viral RNA was extracted from 140 µl of serum using the QIAamp Viral RNA mini kit (QIAGEN, Hilden, Germany) according to the manufacturer's protocol. The viral RNA was heated at 65°C for 5 min and used as a template for a reverse transcription reaction. The reverse transcription reaction mixture contained 5 µl of the RNA template, 1 µl of random hexamer 100 ng/µl (Invitrogen Life Technologies, Carlsbad, CA, USA), 1 µl of dNTP mix (10 mM each), 4 µl of 5X first-strand buffer, 2 µl of 0.1 M DTT, 1 µl of SuperScript II reverse transcriptase (200 U/µl) (Invitrogen Life Technologies, Carlsbad, CA, USA), and 1 µl (40 U/µl) RNaseOUT (Invitrogen Life Technologies, Carlsbad, CA, USA). The reverse transcription was performed at 42°C for 50 min, and then the reverse transcriptase enzyme was inactivated at 70°C for 15 min. PCR amplification of NS5A and NS5B genome regions was performed using primers and conditions previously described [10] . Amplicons were purified using the Illustra GFX PCR DNA and Gel Band Purification Kit (GE Healthcare Life Science, Buckinghamshire, UK) according to the manufacturer's protocol.

2.3. NS5A and NS5B Sequencing. The purified product was then sequenced using the same sets of primers used for PCR amplification. Bidirectional Sanger sequencing was performed by MacroGen Korea (<http://www.macrogen.com>).

2.4. NS5A and NS5B Genotype Determination. HCV NS5A and NS5B consensus sequences obtained from Uruguayan patients were aligned with sequences from HCV representing all genotypes and main subtypes isolated in different geographic regions of the world. These sequences were obtained from Los Alamos HCV sequence database and from the NIAID Virus Pathogen Database and Analysis Resource (ViPR) [21, 22]. For strains included in these studies, see Supplementary Material Table S1. Sequences were aligned using the CLUSTAL W software [23]. Once aligned, the best evolutionary model that described our sequence data was assessed using ModelGenerator program [24]. Using the GTR + G + I model (General time reversible + gamma + invariant sites), maximum likelihood phylogenetic trees were constructed for both NS5A and NS5B using the MEGA 5.0 software [25]. For NS5A, 953 nucleotides (positions 6367 to 7319, relative to HCV 1a reference strain, H77 NC_004102) were included in the phylogenetic analysis, whereas for NS5B, only 361 nucleotides corresponding to the Okamoto region (positions 8265 to 8625, relative to strain H77 NC_004102) were included. As a measure of the robustness of each node, we employed the bootstrapping method (1000 pseudoreplicates).

For NS5A 1a Uruguayan sequences (n = 20), a second alignment and maximum likelihood phylogenetic tree was generated in order to analyze HCV evolutionary relationships between Uruguayan, Brazilian, and worldwide strains. For non-Uruguayan strains included in this analysis, see Supplementary Material Table S2.

2.5. NS5A and NS5B Sequence Analysis. In order to properly identify substitution changes in NS5A and NS5B regions from HCV strains circulating in Uruguayan patients, we generated world consensus sequences for 1a and 1b subtypes using a wide range of NS5A and NS5B sequences from HCV strains isolated worldwide. For this purpose, NS5A gene sequences corresponding to subtypes 1a (n = 160) and 1b (n = 88) were retrieved from Los Alamos HCV sequence database and from the NIAID ViPR [21, 22]. Likewise, datasets of 150 and 124 NS5B sequences were generated for subtypes 1a and 1b, respectively. Using Seqman program, implemented in DNASTar 5.01 package (DNASTAR, Madison, USA), a world consensus nucleotide sequences were generated for each gene and subtype. Each Uruguayan sequence was subsequently aligned to the corresponding reference sequences, and then in silico

translated. The amino acid sequences obtained were compared in order to explore the presence of RASs as well as the presence of polymorphisms at a RAS position (RAPs) in Uruguayan HCV strains. RAPs are defined as any change from reference sequence for a specific genotype at a position associated with NS5A resistance [26] .

To study the genetic variability of NS5A and NS5B regions of HCV strains circulating in Uruguayan patients, sequences of these regions (accession numbers MH070029-MH070090) were aligned with corresponding sequences from 59 HCV strains isolated elsewhere, representing all genotypes and main subtypes (for strains included in these analyses, see Supplementary Material Table S1). Therefore, maximum likelihood phylogenetic trees were constructed. The results of these studies are shown in Figure 1 All strains in the phylogenies were assigned according to their genotype, and each cluster was supported by very high bootstrap values for both analyzed regions. Strains isolated from Uruguayan patients (n = 31) were assigned to genotype 1, 20 of which corresponded to subtype 1a and 11 to subtype 1b. The results of NS5A (Figure 1 (a)) and NS5B (Figure 1

Genotype 1b phylogenetic analyses were concordant for both genomic regions in all 31 sequences, suggesting no recombination events between these regions.

To further analyze the evolutionary relationships between the Uruguayan strains and those circulating in Brazil and elsewhere, a second maximum likelihood phylogenetic tree of HCV-1a sequences of NS5A partial region was built (Figure 2). As was previously described, two distinct 1a clades (clades 1 and 2) were observed. Brazilian sequences clustered in a large group of related sequences inside clade 1 [9] . Whereas NS5A Uruguayan strains (in red) did not cluster in a particular clade, rather, they grouped dispersedly within all major world clades.

With the purpose of studying the amino acid (AA) substitutions along the NS5A protein, Uruguayan HCV AA sequences were aligned with NS5A world consensus sequences (residues 23 to 354 relative to NS5A protein sequence). AA substitutions at positions previously found to be potentially associated with resistance to NS5A inhibitors, as well as polymorphisms at a RAS position, were identified. These results are summarized in Table 1

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RASs to NS5A inhibitors (L31M and L31V) were identified in 2 strains out of 25 (8%) fully sequenced samples. RAPs were found in 3 strains (subtype 1a): 2 exhibited the substitution H58P and 1 the substitution K24Q. Although these substitutions were not reported as resistant, some changes at these positions were previously described as RASs in subtype 1a, namely H58D and K24R [27, 28] . Finally, substitution E62D was found in one subtype 1a strain. This change is considered as a secondary substitution because, although it does not confer resistance by itself, when combined with a known RAS it does. In fact, it confers a higher level of resistance than the one achieved by the RAS alone [26] . In addition, several polymorphisms that have not been previously reported to be associated with a resistant phenotype were also detected (see Supplementary Material Table S3).

In order to study substitutions along NS5B protein, Uruguayan HCV AA sequences were aligned to the NS5B world consensus sequences. Almost full-length AA sequences were obtained in 26 out of 31 analyzed strains. 23 sequences span residues 36 to 539 whereas the remaining 3 span residues 36 to 557 of NS5B protein.

This issue limited our studies, since many of the described RASs are observed as of residue 553.

Importantly, RASs to NS5B inhibitors (Table 2) were observed in 5 strains out of 26 sequenced samples (19.2%). C451R was found in two isolates while A421V was found in only one. In 2 of the 3 strains for which we were able to obtain longer sequences, RASs S556G (subtype 1a) and Q556R (subtype 1b) were observed.

Finally, we found two RAPs: A421V (in 2 subtype 1b strains) and A553G (in 1 subtype 1a strain). Although A421V has been associated with resistance to beclabuvir (BCV) in patients infected with HCV subtype 1a, this resistant phenotype has not been proven in strains subtype 1b [29] . In position 553, the substitution reported as resistant was A553T [8] .

As was the case for NS5A, different polymorphisms not previously associated with a resistant phenotype were also detected in NS5B (see Supplementary Material Table S4).

The advent of DAAs therapies constitutes one of the major breakthroughs in HCV infected patients management. However, these new treatment options are far from being universally available, in particular for HCV infected patients relying on Latin American public healthcare systems. The main limiting factors for worldwide access to DAAs in our region concern the high cost, the inadequate management of public healthcare systems, the limited access of low-income or uninsured populations to healthcare providers, and the lack of accurate epidemiological information [20, [30] [31] [32] . In Uruguay, these therapies became recently available, and although some have been approved for their use by the public health authorities (Viekira pak and sofosbuvir/ledipasvir therapies), they are not currently financially covered, except in specific cases. Despite the high rates of viral response achieved with DAA-based treatments, still 1 to 10% of the patients fails to eliminate infection, and in these cases, baseline and emergent resistance variants turn out to be key factors contributing to treatment failure [5, 17, 33] .

Unfortunately, we are currently unable to properly assess the number of HCV infected people in Uruguay and even more to figure out the frequency and type of RASs circulating. These facts could compromise the effectiveness of these new therapies in our country.

We have previously reported that naturally occurring substitutions conferring resistance to NS3 inhibitors exist in a significant proportion of Uruguayan patients infected with HCV genotype 1, and we showed that this frequency seemed to be higher than in other South American countries (Brazil and Argentina) [34] . The present study describes the prevalence of baseline NS5A and NS5B RASs in HCV genotype 1 infected DAA-naïve patients in a Uruguayan cohort. The presence of substitutions conferring resistance to NS5A inhibitors has been widely reported both in therapy-naïve and in relapser patients from Europe [10, 33, [35] [36] [37] [38] , USA [37, 39, 40] , and Asia [41] [42] [43] . However, NS5A sequences from South America are poorly analyzed yet [9, 44] . Recent studies have revealed that the mean prevalence of NS5A genotype 1 baseline RASs to different inhibitors ranges from 6% to 16% using population sequencing or deep sequencing [27, 37, 45, 46] . Importantly, the prevalence and type of baseline NS5A RASs varies slightly by geographic regions. For instance, L31M was found in 2.2% of genotype 1a infected patients in Europe, in 4.1% of those in Oceania, and strikingly in no patient from the USA [27] . For this reason, we believe that there is a need to contribute data from our region, for which we still do not have enough information, apart from Brazil [9, 44] . The results of this study indicate the presence of DAA NS5A RASs in 2 HCV strains (8% of the patients enrolled in this study), with baseline RASs detected at position 31 (see Table 1). L31M substitution confers

resistance to daclatasvir (DCV), ledipasvir (LDV), and elbasvir (EBV) in both 1a and 1b subtypes [5, 6, 8, 28, 47, 48] , whereas substitution L31V does it to DCV in subtypes 1a and 1b, to LDV in subtype 1b, and to EBV in subtype 1a [5, 6, 28] . Given that both L31V and L31M are clinically relevant RASs, their detection at baseline may influence the choice of first-line treatment regimens [28] .

The substitutions H58P and K24Q found in two patients are considered as resistance-associated polymorphisms (RAPs). The RASs characterized at these positions were H58D and K24G/N/R [5, 6, 27, 28, 49, 50] . The substitution H58P was found as a baseline RAP in relapsers to LDV (HARVONI prescription, https://www.gilead.com/-/media/files/pdfs/medicines/liver-disease/harvoni/harvoni_pi.pdf?la=en). However, it is sometimes regarded as a RAS [10, 51] , despite conferring only 1.2 fold change in resistance in in vitro studies using the 1a replicon system [39] .

We did not find M28T/V, Q30R/H, or Y93H substitutions as there were previously reported in Brazil and worldwide [9, 27, 44] . The amino acid substitution E62H was found in one Uruguayan patient. Although this change does not confer resistance by itself but in combination with Q30R, it generates a high resistance level to DCV [52] .

The presence of baseline NS5A RASs impacts treatment outcome in some patient groups by affecting SVR rates. The detection of NS5A preexistent RASs may play a relevant role in the choice of first-line treatment regimens or in the simplification/shortening of recommended regimens, in order to bring SVR rates close to the highest achievable [27, 38, 41, 53] , in particular in countries such as Uruguay, where only two different DAA-containing treatment regimens are approved for their use.

Regarding NS5B gene, global analysis (with the exception of South America [17, 19]) revealed that NS5B DAA resistance substitutions are infrequent [14] . Our study showed the presence of NS5B inhibitors RASs in 5 out of 26 analyzed HCV infected Uruguayan patients naïve to treatment (19.2%). Substitutions found in this work were A421V and S556G associated in subtype 1a with resistance to BCV and dasabuvir (DSV), respectively [8, 28, 29, 54, 55] , and Q556R associated with resistance to DSV both in genotype 1a and 1b [12, 28] . Substitution C451R, observed in two Uruguayan patients, was reported previously in patients who failed to clear the infection after treatment with OBV/PTV/r + DSV

± RBV. In these cases, it appeared in combination with G558R (Trial Coral I-Cohort 2: <http://www.hcv-trials.com/showStudy.asp?Study=86>). RAPs in positions 421 and 553 (A421V in two subtype 1b isolates and A553G in one subtype 1b isolate) were also found. Although A421V has been associated with resistance to BCV in patients with subtype 1a, this phenotype has not been proven in strains of subtype 1b [29] . In position 553, the substitutions reported as resistant are A553T in subtype 1a [8] and A553V in subtype 1b [54] , conferring resistance to DSV.

In contrast to our results, Noble and coworkers (2016) reported the presence of V321A, A421G, M414V, Y448H, L159F, and C316N in Brazilian isolates [17] , yet none of these mutations were found in this study, probably due to the diversity found between Uruguayan and Brazilian strains (Figure 2). Nevertheless, substitution A421V was found in Brazil [17] , Argentina [19] , and Uruguay. The RAS S282T was detected neither in Brazilian reports nor in this current work (Uruguay) [17, 18, 56] . Our findings further confirm and complement previous studies which evidenced a low prevalence of this substitution in vivo, probably due to its low replicative fitness [14, 18, 57] . Despite our results, it is worth mentioning that the presence of baseline NS5B RASs conferring resistance to nucleotide or nonnucleoside NS5B inhibitors has not been shown to have any impact on virologic responses thus far [53, 58] .

These results show both diversity in the baseline polymorphisms found in different Latin American countries and in the evolutionary relationships of Uruguayan isolates (Figure 2). This fact could be linked not only to the isolates' geographic region and viral intrinsic characteristics but also to the genetic background of the host. It is worth mentioning that we live in a vast continent inhabited by populations with different genotypic characteristics that might, depending on the situation, require different approaches to treatment. Indeed, we have recently found that allele and genotype frequencies at IL28B locus of Uruguayan individuals closely resemble those of an admixed population rather than a uniformly European-descendant one [59] . Altogether, we believe that it could be important to carry out studies throughout the South American region in order to establish the prevalence of RASs in NS5A and NS5B in different countries. In fact, this will aid in understanding that not every treatment regimen might be adequate for every patient and country. The data we presented here might guide not only physicians in making therapeutic decisions but also public health authorities in approving more diverse treatment combinations. These treatment formulations would cover most of the circulating strains in our region, a region with an extremely diverse genetic background population.

To our knowledge, the present study revealed for the first time the presence of RASs in the NS5A and NS5B regions of HCV genotype 1 Uruguayan strains from patients who have not been previously treated with DAAs and is one of the few South American countries to report on this matter. It is currently unclear if preexisting viral variants with reduced susceptibility to DAAs are clinically relevant for the prediction of virologic treatment failure. However, individualized DAA therapy based on baseline resistance analysis may be beneficial for optimizing treatment efficacy in patients with HCV genotype 1 infection and risk factors for treatment failure. Therefore, the potential role of baseline resistance testing remains an area of critical research and clinical questions.

The data used to support the findings of this study are included within the article.

The authors declare that they have no conflicts of interest.

Fabián Aldunate and Natalia Echeverría contributed equally to this work.

Supplementary Material Table S1 : hepatitis C Virus NS5A and NS5B sequences used as representatives of each genotype to perform the phylogenetic analysis. Their corresponding genotype, country of isolation, and GenBank accession number are indicated. Supplementary Material Table S2 : hepatitis C Virus NS5A subtype 1a sequences used to reveal evolutionary relationships between Uruguayan strains and others isolated elsewhere. Their corresponding country of isolation and GenBank accession number are indicated. Supplementary Material Table S3 : amino acid substitutions in NS5A protein not previously associated with resistance to NS5A inhibitors. Supplementary Material Table S4 : amino acid substitutions in NS5B protein not previously associated with resistance to polymerase inhibitors. (Supplementary Materials)

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Efficient generation of recombinant RNA viruses using targeted recombination-mediated mutagenesis of bacterial artificial chromosomes containing full-length cDNA

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Abstract: **BACKGROUND:** Infectious cDNA clones are a prerequisite for directed genetic manipulation of RNA viruses. Here, a strategy to facilitate manipulation and rescue of classical swine fever viruses (CSFVs) from full-length cDNAs present within bacterial artificial chromosomes (BACs) is described. This strategy allows manipulation of viral cDNA by targeted recombination-mediated mutagenesis within bacteria. **RESULTS:** A new CSFV-BAC (pBeloR26) derived from the Riems vaccine strain has been constructed and subsequently modified in the E2 coding sequence, using the targeted recombination strategy to enable rescue of chimeric pestiviruses (vR26_E2gif and vR26_TAV) with potential as new marker vaccine candidates. Sequencing of the BACs revealed a high genetic stability during passages within bacteria. The complete genome sequences of rescued viruses, after extensive passages in mammalian cells showed that modifications in the E2 protein coding sequence were stably maintained. A single amino acid substitution (D3431G) in the RNA dependent RNA polymerase was observed in the rescued viruses vR26_E2gif and vR26, which was reversion to the parental Riems sequence. **CONCLUSIONS:** These results show that targeted recombination-mediated mutagenesis provides a powerful tool for expediting the construction of novel RNA genomes and should be applicable to the manipulation of other RNA viruses.

Text: Bacterial artificial chromosomes (BACs) are ideally suited for the stable maintenance of large DNA sequences derived from viral genomes [1] . A considerable number of BAC systems have been established for large DNA viruses; in particular many different herpesvirus genomes have been cloned into BACs (for review see [2]). The first BAC systems using RNA virus cDNAs were described for coronaviruses [3] [4] [5] [6] and recently the first BAC containing a full-length cDNA for a negative-stranded RNA virus was described [7] . Similarly, cDNAs corresponding to the full-length genomes of members of the Flaviviridae family (Japanese encephalitis virus [8] and Dengue virus [9]) have been inserted into BACs.

BACs containing full-length cDNAs of pestiviruses (also within the Flaviviridae), including bovine viral diarrhea virus (BVDV) and classical swine fever virus (CSFV) have recently been established [10, 11] . Infectious pestiviruses can be rescued using RNA transcripts derived from these BACs. The pestiviruses have single stranded positive sense RNA genomes, about 12.3 kb in length, which includes a single long open reading frame, encoding a large polyprotein, flanked by 5' and 3' untranslated regions (UTRs) that are critical for autonomous replication of the genome [12, 13] . The polyprotein is cleaved by cellular and viral proteases into four structural proteins (nucleocapsid protein C, envelope glycoproteins E_{ns} , E1 and E2) and eight nonstructural proteins (N_{pro} , p7, NS2, NS3, NS4A, NS4B, NS5A and NS5B). The availability of genetically defined and stable pestivirus BACs facilitates the functional study of viral proteins or RNA structures and also the development of new marker vaccine candidates. Several CSFV vaccines with marker properties based on chimeric pestiviruses have been developed over the years [14] . In particular, chimeric pestiviruses with substitution of the entire E2 protein have been described [15] [16] [17] but also mutants with more subtle modifications, such as the modification of the important TAV-epitope [18] within the CSFV-E2 protein [19, 20] are promising marker vaccine candidates.

Manipulation of BACs using traditional cloning procedures can be difficult (e.g. because of a lack of convenient restriction enzyme sites) and thus a range of methodologies that apply bacterial genetics, including homologous recombination (e.g. Red/ET homologous recombineering) within the *E. coli* host, have been developed (for review, see [21]). The use of homologous recombination allows site-directed mutagenesis of BACs [22] and, by employing a counterselection scheme, specific modifications can be obtained without leaving residual "foreign" sequences [23] . The main advantage of this method is that there are no target limitations (e.g. based on size or location) and no need for suitable restriction

sites. The integration of the modified sequence is performed in vivo (within *E. coli*) thereby potentially being more accurate than in vitro approaches like PCR-based methods. Although in vitro cloning approaches based on the use of high-fidelity polymerases for PCR amplification have significantly improved in recent years, the use of in vivo approaches should allow a more accurate method of mutagenesis due to the use of the cells own high-fidelity replication system which includes proof reading. Whereas BAC recombination has been commonly used for modifying DNA viruses, there are only very few reports about the use of this technology for RNA viruses [7, 24, 25] .

Here, a generally applicable strategy for the manipulation and rescue of chimeric pestiviruses from BACs is described as a model, and the flexibility of this approach is demonstrated by generating different modifications in the viral cDNA of the new CSFV-BAC, pBeloR26, derived from the modified live vaccine strain "C-strain Riems". The targeted recombination-mediated mutagenesis described here includes the substitution of the 9 amino acid (aa) linear TAV-epitope (TAVSPTTLR) present in the E2 protein with the corresponding region (TTVSTSTLA) of a heterologous pestivirus (border disease virus, BDV, strain "Gifhorn") and also the replacement of the entire CSFV E2 protein coding region with the whole E2 coding region from the same BDV, to generate marked vaccine viruses that can be discriminated using specific anti-E2 monoclonal antibodies. The genetic stabilities of both the BAC constructs (within *E. coli*) and the rescued viruses have also been assessed.

Porcine kidney (PK15) and sheep fetal thymoid (SFT-R) cells were grown at 37°C (with 5% (v/v) CO₂) in Dulbecco's minimal essential medium (DMEM) supplemented with 5% (v/v) pestivirus-free fetal calf serum. Virus from a bait containing the modified live vaccine CSFV "C-strain Riems" (Riemser Arzneimittel AG, Germany) was propagated once in PK15 cells and termed vRiemser. RNA obtained from BDV strain "Gifhorn" [26] was used for amplification of the Gifhorn E2-coding sequence.

Oligonucleotide primers used are listed in Additional file 1: Table S1 .

The BAC construct, pBeloR26, was constructed using the long RT-PCR method as previously described [11] using RNA derived from the "C-strain Riems". Briefly, full-length viral cDNAs flanked by NotI sites were amplified by long RT-PCR using primers

5'Cstrain_T7_Not1 (which includes a T7 promotor for in vitro transcription, a NotI site and a region corresponding to the first 44 nt of the genome) and 3'CSFV_Not1 (that contains a NotI site and sequence complementary to the 3'-terminal 35 nt of the genome that are conserved among many CSFVs including the Cstrain). The product (ca. 12.3 kbp) was digested with NotI and inserted into similarly digested pBeloBAC11 (New England Biolabs, GenBank accession U51113). All BACs were modified and maintained in *E. coli* DH10B cells (Invitrogen) grown at 37°C in LB medium containing chloramphenicol (Cam, 15 µg/ml). The electroporation of bacteria was performed in 0.1 cm cuvettes using 1 pulse at 1800 V, 25 µF and 200 Ω in a Gene Pulser Xcell (Bio-Rad). BACs to be used as templates for long PCR or for screening by restriction enzyme digestion were purified from 4 ml overnight cultures of *E. coli* DH10B using the ZR BAC DNA Miniprep Kit (Zymo Research). BACs required for direct genome sequencing were purified from 500 ml cultures using the Large-construct kit (Qiagen).

Modifications to the full-length CSFV cDNA were accomplished in *E. coli* DH10B (streptomycin resistant, Strep R) using the Counter Selection BAC Modification Kit (Gene Bridges, Heidelberg, Germany).

The Red/ET recombination involved three steps (i-iii).

Step i) the temperature-sensitive pRedET expression plasmid (Gene Bridges) was introduced into electroporation competent *E. coli* DH10B cells containing the parental BAC (phenotype Cam R, Strep R). The pRedET expresses the phage lambda proteins $\text{red}\alpha$, $\text{red}\beta$ and $\text{red}\gamma$, under control of the arabinose-inducible pBAD promoter, allowing homologous recombination to occur. Immediately after electroporation, pre-warmed LB medium without antibiotics (1 ml) was added to the cells which were then incubated at 30°C for 1 hour, prior to spreading onto agar plates containing Cam (15 µg/ml) and tetracycline (Tet) (3 µg/ml) and then incubated at 30°C overnight to maintain the pRedET. The presence of the pRedET plasmid (conferring Tet R) was verified by visual inspection of BAC-DNA preparations from the Cam R /Tet R colonies using agarose gel electrophoresis.

Step ii) counter-selection marker cassettes with an extra NotI site for screening purposes (rpsL-neo, 1325 bp) were amplified by PCR using primers with 30 nt or 50 nt extensions that were homologous to the target site in the BAC using the rpsL-neo plasmid (Gene Bridges)

as template and the Phusion hot start II HF DNA polymerase (Thermo Scientific) with cycling conditions as follows: 98°C for 30s, followed by 35 cycles of 98°C for 10s, 60°C for 20s, 72°C for 60s, and 1 cycle at 72°C for 4 min. The PCR products (ca. 1400 bp) were isolated on 1% (w/v) TBE agarose gels and purified using a GeneJET gel extraction kit (Thermo Scientific). Samples (30 µl), from an E. coli culture containing pRedET and the parental BAC grown overnight at 30°C in LB media (Cam, Tet), were used to inoculate 1.4 ml of fresh LB media with the same antibiotics to obtain exponentially growing bacteria at 30°C. Red/ET recombination proteins were induced by adding 50 µl of 10% (w/v) L-arabinose (Sigma). The PCR product (200 ng) containing the rpsL-neo cassette was introduced into these bacteria using electroporation (as above). Following electroporation, the cells were grown at 37°C for 70 min (to allow recombination) and then selected on plates containing Cam (15 µg/ml), Tet (3 µg/ml) and kanamycin (Kan, 15 µg/ml) overnight at 30°C to maintain the pRedET. Note, the rpsL cassette confers Streptomycin sensitivity (Strep S) onto the resistant DH10B strain and the neo confers Kanamycin resistance (Kan R). The correct phenotype (Cam R, Kan R, Tet R, Strep S) of the resulting colonies was confirmed by streaking the colonies onto plates containing Cam (15 µg/ml), Tet (3 µg/ml) and Kan (15 µg/ml) and grown at 30°C. Importantly, for the third step, the replacement of the rpsL-neo cassette (using counter-selection), the selected colonies were also streaked onto plates containing Cam (15 µg/ml) plus Strep (50 µg/ml) and shown to be Strep S indicating incorporation of a functional rpsL gene. The structures of the intermediate BACs were verified by restriction enzyme analysis and sequencing around the inserts.

Step iii) the replacement of the rpsL-neo selection cassettes from the intermediate constructs using linear DNA fragments was achieved through counter-selection and Red/ET recombination. Again, the homologous sequences at the ends of the DNA fragment were used for Red/ET mediated recombination events to replace the rpsL-neo cassette with the sequence of interest. Counterselection against the rpsL-neo cassette (phenotype Cam R, Kan R, Tet R, Strep S) was employed using media containing Cam (15 µg/ml) and Strep (50 µg/ml) to isolate the required derivatives (phenotype Cam R and Strep R).

Initially, the intermediate construct, pBeloR26_E2rpsLneo (Figure 1), was generated using Red/ET recombination by insertion of the rpsL-neo cassette with an extra NotI site for screening purposes which was amplified using primers Criems-TAVfor and Criems-TAVrev (Additional file 1: Table S1) in place of the TAVSPTTLR coding sequence (27 nt). Secondly, the rpsL-neo cassette in this intermediate construct was then replaced using counter-selection Red/ET recombination using a single-stranded oligonucleotide,

Riems_TAV_Gifhorn (Additional file 1: Table S1) with the same homology arms as used for the rpsL-neo cassette, to introduce the coding sequence for the BDV "Gifhorn" epitope sequence (TTVSTSTLA). The resulting construct was named pBeloR26_TAV (Figure 1). The initial intermediate construct (with rpsL-neo) was then used to produce the pBeloR26_E2gif construct (Figure 1). For this, the E2 coding sequence was amplified from cDNA prepared from BDV "Gifhorn" RNA using two different primer pairs, one set with 50 nt homology arms (Criems_E2_gifFlong/Criems_E2_gifRlong) and another with 30 nt homologous sequences (Criems_E2_gifF/Criems_E2_gifR).

For generation of BACs with substitution of the entire E2 coding sequences, PCR products consisting of the sequence of interest flanked with homology arms identical to the target area were generated by PCR (as for the rpsL-neo cassette). For making constructs with substitution of shorter sequences (e.g. the TAV-epitope), the recombination was achieved using synthetic single stranded oligonucleotides rather than PCR products. Pre-heating of single stranded oligonucleotides at 95°C for 2 min followed by snap-freezing, prior to electroporation, empirically showed the best results. In each case, the DNA molecules were introduced into E. coli containing the BAC derivatives including the rpsL-neo cassettes together with the pRedET plasmid by electroporation as described above. The structures of the modified BACs were verified by restriction enzyme analysis and subsequent full-genome sequencing (see below).

BAC DNA (1 µg) was linearized with NotI or 1 µl BAC DNA was used as template for long PCR amplification using primers 5'C-strain_T7_Not1 and 3'CSFV (Additional file 1: Table S1). Linearized BACs or PCR products were purified with the GeneJet PCR purification kit (Thermo Scientific) and transcribed in vitro using a Megascript T7 kit (Invitrogen). Viruses were rescued from RNA transcripts (1 to 5 µg) by electroporation of porcine (PK15) or ovine (SFT-R) cells essentially as described previously [24] . Cells were analysed using immunofluorescence microscopy (typically after 3 days) for the expression of NS3 and E2 proteins using specific monoclonal antibodies (mAbs), these were anti-NS3 (WB103/105, pan-pestivirus), anti-CSFV E2 (WH211, WH303, both CSFV specific) and anti-BDV E2 (WB166, BVDV/BDV specific) (AHVLA Scientific, United Kingdom) together with Alexa 488 conjugated goat antimouse IgG antibody (Molecular Probes, Invitrogen). The nuclei of cells were visualized using DAPI (Vector Laboratories) and images were recorded using a BX63 fluorescence microscope (Olympus). For peroxidase staining, cells were fixed and stained for the presence of pestivirus antigens using biotinylated pig anti-CSFV/BVDV polyclonal IgG followed by avidin-conjugated horseradish peroxidase (eBioscience) as previously

described [27] . The same staining procedure was also performed using the anti-E2 mAbs. Samples containing virus-positive cells were passaged onto new cells. Virus growth curves were generated as previously described [24] . Briefly, PK15 or SFT-R cells were infected at a multiplicity of infection (MOI) of 0.1 pfu/cell and grown for three days.

BAC DNAs (5 µg), purified using the Large-construct kit (Qiagen), or PCR products (1 µg) amplified from viral cDNA or from BACs using the long PCR method (as above) were consensus sequenced using a 454 FLX (Roche) or an Ion PGM (Life Technologies). Both Newbler (Roche) and the bwa.bwasw alignment algorithm [28] were used for mapping the reads to the expected sequence. A combination of Samtools [29] and LoFreq SNV-caller [30] was used for downstream single nucleotide variant (SNV) analysis. Finally, clone consensus sequences were aligned using MAFFT in the Geneious software platform (Biomatters).

Generation of a BAC containing full-length cDNA corresponding to the modified live vaccine "C-strain Riems"

BACs containing the full-length cDNA corresponding to the parental vRiems ("C-strain Riems") were constructed according to the method described previously for the "Paderborn" strain of CSFV [11] . BACs containing the complete CSFV cDNAs were identified by restriction Figure 1 Schematic representation of the CSFV genome organization and the BACs constructed and used in this study. Nucleotide (nt) and amino acid (aa) positions within R26 for the 5' and 3' termini together with the translational start and stop codons of the polyprotein coding region plus cleavage sites used to make the individual proteins (N pro , C, E rns , E1, E2, p7, NS2, NS3, NS4A, NS4B, NS5A and NS5B) are indicated. Insertion of the rpsL-neo in place of the TAV-epitope within CSFV E2 for the intermediate construct (R26_rpsLneo) and the subsequent replacement with the TTVSTSTLA sequence (R26_TAV) and the complete substitution of the E2 sequence (R26_E2gif) are shown. Names of BAC constructs begin with "pBelo" and rescued viruses with "v" (e.g. pBeloR26 and vR26). Cell culture passage no. of virus is indicated with "/P" (e.g. vR26/P-4).

digest analysis and following linearization by NotI, RNA transcripts were produced and electroporated into PK15 cells. This screening resulted in the identification of a BAC

containing a cDNA insert of 12316 nt, pBeloR26 (Figure 1) , which yielded infectious virus, termed vR26, that could be propagated in SFT-R cells (Figure 2 , upper panels) and in PK15 cells (Figure 3). The rescued vR26 displayed higher growth rate at the early stage (about 10fold difference in virus yield at 24 h) compared to the parental vaccine virus, but after 48 hours similar virus titres were obtained (Figure 3). Full-genome sequencing of the cloned BAC template, pBeloR26, revealed a number of differences throughout the genome when compared to the full-length consensus sequence of the cDNA used for the cloning procedure (see Table 1). These differences are non-representative variants within the cDNA. Overall, the BAC sequence differed from the cDNA sequence in 18 positions, 9 of these lead to predicted amino acid substitutions within the polyprotein; one in each of N pro , E ns , E1, E2 and NS3 and four amino acid substitutions in NS5B (Table 1) . When compared to the published reference sequence (GenBank accession AY259122.1), the pBeloR26 BAC sequence differed at an additional 11 positions, 1 of these lead to a predicted amino acid substitution and there was one large insertion (27 nt) in the hypervariable region of the 3'-UTR (Additional file 2: Table S2).

To determine the utility of the targeted recombination-mediated mutagenesis system for pestiviruses, two different modifications of the E2 protein coding sequence within pBeloR26 were generated using the Red/ET recombination methodology. Initially, the sequence encoding the linear TAV-epitope (TAVSPTTLR) within the CSFV-E2 was substituted with the sequence encoding the corresponding region (encoding TTVSTSTLA) from the BDV strain "Gifhorn" as described in the Materials and Methods section. More than 90% of the colonies obtained using this procedure contained the required BAC

Anti-CSFV E2 (WH211) Figure 2 Antibody reaction patterns of pestivirus infected cells. SFT-R cells were infected with vR26 and its two derivatives vR26_E2gif and vR26_TAV plus vGifhorn [26] . After 72 h, the cells were fixed and stained with monoclonal antibodies against the NS3 protein (WB103/105, left column), the CSFV E2 protein (WH303 and WH211, middle columns) and the BDV E2 protein (WB166, right column) as indicated and viewed using a fluorescence microscope. structure as determined by NotI digestions. The complete genome sequences of the CSFV cDNA within two selected BACs, designated pBeloR26_TAV have been verified (data not shown). In addition, the complete coding sequence (1119 nt) for the CSFV-E2 protein was substituted by the corresponding sequence from BDV "Gifhorn". Again more than 90% of the colonies obtained contained the required BAC and the same proportion of correctly recombined BACs was obtained using

either 30 nt or 50 nt homology arms. The chimeric BAC was designated, pBeloR26_E2gif and the complete virus genome sequence (cDNA) was verified (data not shown).

After electroporation with RNA transcripts derived from either pBeloR26_TAV or pBeloR26_E2gif a large number of CSFV NS3-positive cells could be observed (data not shown) and chimeric virus stocks, termed vR26_TAV and vR26_E2gif, were generated after further passages in cells. Cells infected with these viruses and with the parental vR26 and vGifhorn strains were all stained with mAbs directed against the NS3 protein (Figure 2). However, in contrast to the parental vR26 virus, the chimeric viruses rescued from the recombined BACs were not recognized by anti-E2 mAbs specific for the CSFV-E2 proteins (Figure 2) and thus, consistent with their structure, displayed the same antibody reaction pattern as vGifhorn. Two different anti-CSFV E2 mAbs, WH211 and WH303, were used for the staining and the latter has been shown previously to target the TAV-epitope [18] . As anticipated, cells infected with either the vGifhorn or with the chimeric vR26_E2gif could be shown to express the "Gifhorn" E2 protein using staining with an anti-BDV mAb (Figure 2). The presence of the BDV epitope TTVSTSTLA in vR26_TAV was insufficient to permit efficient recognition by this anti-BDV mab, although a weak signal was observed in some cells.

The BAC constructs pBeloR26 and pBeloR26_E2gif were analysed for the genetic stability of the cDNA to determine the suitability of the BAC vector for maintaining full-length pestivirus cDNAs. *E. coli* DH10B cells containing the BACs were passaged 15 times, by overnight growth, and the complete viral cDNAs within the BACs were sequenced after the 1st and the 15th passage. No mutations were observed within the 12316 nt virus cDNA sequences after this extensive propagation of the BACs in the bacterial host, indicating a highly stable system for the maintenance of complete pestivirus cDNA sequences.

The viruses, vR26 and vR26_E2gif, rescued from their respective BAC constructs, were also tested for their genetic stability within mammalian cells. Linearized BAC DNA was transcribed in vitro and the RNA was electroporated into PK15 cells. Three days after electroporation the cells were stained with the anti-NS3 antibody to detect the presence of replicating virus. Samples containing virus positive cells were passaged onto new cells, this process

*Nt position 10665 in vR26/P-12 is reverted from A to G as in the parental cDNA.

was repeated for 12 separate passages (each of three days). The virus titre (as TCID₅₀ /ml) was determined for each passage. Passage of the rescued vR26_E2gif chimeric virus in PK15 cells resulted in rapidly decreasing virus titres and was discontinued after the 2nd passage (Figure 4A). Instead, further passage of this chimeric virus was performed in ovine SFT-R cells (the preferred cell type for BDV) and resulted in much higher titers of the chimeric virus. Virus titers reached more than 10⁶ TCID₅₀ /ml after the 1st passage and remained stable for 12 passages (Figure 4A). The rescued vR26 was also efficiently propagated on the SFT-R cells but maintained a slightly lower titer than the vR26_E2gif chimeric virus (Figure 4A).

To check that the viruses retained their antibody reaction properties (Figure 2) after these passages, cells were infected with viruses from the 12th SFT-R cell culture passage (termed vR26/P-12 and vR26_E2gif/P-12) and stained with a polyclonal anti-pestivirus serum and with specific mAbs directed against the CSFV-E2 and BDV-E2 proteins (Figure 4B). Cells infected with either the vR26/P-12 or the chimeric vR26_E2gif/P-12 were each detected by the polyclonal anti-pestivirus serum as expected. The anti-CSFV-E2 mAb specifically detected cells infected with vR26/P-12 but not cells infected by the chimeric virus containing the BDV-E2 protein (consistent with the results shown in Figure 2). In contrast, the anti-BDV-E2 mAb specifically detected infection by the vR26_E2gif/P-12 and did not recognize cells infected with vR26/P-12. Each result is in accord with the structure of the viruses. The 4th passage of vR26 (vR26/P-4) displayed a slower growth rate than the virus obtained after 12 passages (see Figure 5A). It also had a reduced growth rate compared to both the vR26_E2gif/P-4 and vR26_E2gif/P-12. The fulllength sequence of pBeloR26 had revealed ten non-silent mutations compared to the reference sequence (AY25 9122.1) for this virus (Additional file 2: Table S2). Any of these mutations could be responsible for the impaired growth acting alone or in concert. For further investigation of this issue, full length cDNAs prepared from vR26/ P-4, vR26/P-12, vR26_E2gif/P-4 and vR26_E2gif/P-12 were deep-sequenced using both the 454 FLX and Ion PGM platforms for comparison and to determine the quasispecies distribution (Additional file 3: Figure S1 and Additional file 4: Figure S2). Sequencing data from both platforms revealed that both the vR26/P-12 and vR26_E2gif/P-12 were close to 100% changed at nt position A10665G compared to the BAC clones (resulting in the predicted amino acid substitution D3431G within the NS5B protein, the RNAdependent RNA polymerase, see Figure 5B). This adaptation is a reversion back to the consensus cDNA sequence of the parental vaccine virus, vRiemsers (Additional file 2:

Table S2). Additionally, vR26/P-4 and vR26_E2gif/P-4 already showed evidence for this reversion being present within the population. For vR26/P-4, the level of reversion was 57%, while for vR26_E2gif/P-4 the extent of change was 73% (see Figure 5B).

In this study, we have established the first BAC containing the full-length cDNA of a CSFV vaccine strain. The BAC differed from the parental cDNA sequence in 18 positions leading to 9 aa substitutions (Table 1). The method that has been used for the generation of pBeloR26 is based on full genome amplification of cDNA followed by direct cloning to obtain the BACs [11] . This approach results in cDNA clones that reflect the quasispecies composition of the parental viral RNA and thus it is not guaranteed to obtain cDNA clones corresponding to the consensus sequence of the cDNA used. However, it is possible to correct the mutations using the BAC recombination approach if a consensus clone is needed. To demonstrate the utility of the Red/ET mediated recombination method we have generated a series of modified BACs derived from this CSFV full-length cDNA. These include BACs with substitution of the linear TAV-epitope present in the E2 protein and also BACs with substitution of the complete E2 protein with heterologous pestivirus sequences. We have also used the same approach for a range of different targeted modifications within CSFV BACs including specific deletions and substitutions in the 5'UTR of CSFV [24] and for insertions of heterologous reporter sequences into CSFV replicons [25] . Using Red/ET recombination-mediated mutagenesis for the targeted design, the work can be expedited and focused, in principal, on any sequence within the viral genome and is not dependent on the use of internal restriction sites. The results demonstrate that Red/ ET recombination-mediated mutagenesis of pestivirus BAC cDNAs provides a useful tool for advancing the construction of modified pestiviruses. Cells infected with the parental vR26 virus were recognized by the two anti-E2 mAbs (WH211 and WH303) specific for the CSFV-E2 proteins, in contrast cells infected with the modified viruses vR26_TAV and vR26_E2gif, rescued from the recombined BACs, were not detected by these mAbs. Furthermore, as expected, cells infected with the vR26_E2gif were recognized by the anti-BDV mAb (WB166) whereas no staining was observed with this antibody in vR26 infected cells or in cells with vR26_TAV. The mAb WH303 recognizes the CSFV TAV-epitope [18] and the difference in 4 aa between the TAV-epitope and the corresponding sequence from BDV strain "Gifhorn" is enough to completely abolish the recognition by this mAb. The lack of staining of vR26_TAV infected cells by the WH211 indicated that the TAV-sequence is also important for the epitope recognized by this mAb. Thus, the chimeric pestiviruses, vR26_TAV and vR26_E2gif, containing heterologous E2 sequences can be readily discriminated from the vR26 using specific anti-E2 monoclonal antibodies. These new chimeric pestiviruses represents Cstrain based marked vaccine candidates with the characteristics desired for safe and

efficacious DIVA vaccines against CSFV. Indeed, vR26_E2gif vaccinated pigs could be efficiently discriminated from C-strain vaccinated pigs and from CSFV infected pigs using CSFV-E2 specific antibody ELISAs (Rasmussen et al., unpublished results).

Nucleotide sequence data for the pBeloR26 showed a number of changes from the published reference sequence for "C-strain Riems". Some of these differences are present in the cDNA derived from the vaccine stock at a detectable level whereas others may represent low-level variants within the cDNA or errors introduced by the RT-PCR amplification. Full-length sequencing revealed that no changes occurred in the cDNA during extensive propagation in *E. coli* DH10B of the pBeloR26 and the E2chimeric derivative, pBeloR26_E2gif, indicating a very high stability of these BAC-cloned CSFV cDNAs. This is essential if this system is to be useful for cloning and sequence manipulation, and contrasts with stability problems encountered with conventional plasmids containing fulllength pestivirus cDNAs [31] . The stability of these BACs is consistent with previous reports on the stability of BACs containing other viruses of the family Flaviviridae in *E. coli* [8, 10] .

Extensive passaging of the rescued vR26 and the chimeric virus derivative, vR26_E2gif, resulted in a change at nucleotide position A10665G (resulting in the predicted aa

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Which Kind of Provider's Operation Volumes Matters? Associations between CABG Surgical Site Infection Risk and Hospital and Surgeon Operation Volumes among Medical Centers in Taiwan

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Abstract: **BACKGROUND:** Volume-infection relationships have been examined for high-risk surgical procedures, but the conclusions remain controversial. The inconsistency might be due to inaccurate identification of cases of infection and different methods of categorizing service volumes. This study takes coronary artery bypass graft (CABG) surgical site infections (SSIs) as an example to examine whether a relationship exists between operation volumes and SSIs, when different SSIs case identification, definitions and categorization methods of operation volumes were implemented. **METHODS:** A population-based cross-sectional multilevel study was conducted. A total of 7,007 patients who received CABG surgery between 2006 and 2008 from 19 medical centers in Taiwan were recruited. SSIs associated with CABG surgery were identified using International Classification of Diseases, 9th Revision, Clinical Modification (ICD-9 CM) codes and a Classification and Regression Trees (CART) model. Two definitions of surgeon and hospital operation volumes were used: (1) the cumulative CABG operation volumes within the study period; and (2) the cumulative CABG operation volumes in the previous one year before each CABG surgery. Operation volumes were further treated in three different ways: (1) a continuous variable; (2) a categorical variable based on the quartile; and (3) a data-driven categorical variable based on k-means clustering algorithm. Furthermore, subgroup analysis for comorbidities was also conducted. **RESULTS:** This study showed that hospital volumes were not significantly associated with SSIs, no matter which definitions or categorization methods of operation volume, or SSIs case identification approaches were used. On the contrary, the relationships between surgeon's volumes varied. Most of the models demonstrated that the low-volume surgeons had higher risk than high-volume surgeons. **CONCLUSION:** Surgeon volumes were more important than hospital volumes in exploring the relationship between CABG operation volumes and SSIs in Taiwan. However, the relationships were not robust. Definitions and categorization methods of operation volume and correct identification of SSIs are important issues for future research.

Text: data, which should use hierarchical models, may result in biased estimation of the variation and also lead to incorrect conclusions.

SSIs following coronary artery bypass graft (CABG) procedures place a heavy burden on patients and healthcare systems. The total length of stay and expenditure for patients with SSIs after CABG surgery is significantly longer and higher than those without SSIs. [20, 21] In 2008, the Centers for Medicare & Medicaid of the United States of America implemented the "Never Event" policy, where hospitals would no longer receive higher payments for the additional costs associated with treating patients for certain healthcare-acquired infections, including those related to CABG.

In view of the accuracy of SSIs identification and the heterogeneity of definition and categorization methods, no existing studies have used different infection case identification nor definitions and categorization methods of operation volume simultaneously to explore the relationship between operation volumes and infection. The current study takes CABG SSIs as an example to examine whether a relationship exists between operation volumes and SSIs, given different SSI cases identification, operation volume definitions and categorization methods.

This retrospective and cross-sectional study adopted a multilevel design to examine the relationships between provider volumes and SSIs after adjusting for patient-, surgeon-, and hospital-level covariates.

We used data from the Taiwan National Health Insurance Research Database (NHIRD) from 2005 and 2008. The NHIRD, published by the Taiwan National Health Research Institute, includes all the original claims data and registration files for beneficiaries enrolled under the National Health Insurance (NHI) program. The database covers the 23 million Taiwanese enrollees (approximately 98% of the population) in the NHI program. It is a de-identified secondary database containing patient-level demographic and administrative information; however, treatment items are aggregated and without time-related and clinical information. The data is released for research purposes.

The protocol for the study was approved by the Institutional Review Board of the National Taiwan University Hospital (protocol #201001027R). The dataset we used in this study was secondary data; all information was de-identified by data owners.

In this study, we adopted the ICD-9-CM SSI codes (hereafter referred to as the ICD-9-CM based model) and the Classification and Regression Trees (CART) model, which was developed in our previous work [11] to identify SSI cases. As we mentioned above, the ICD-9-CM SSI codes were the most popular tool to identify the SSI cases in claims data. In the ICD-9-CM based model, SSI cases were divided into two categories: index hospitalization events and post-discharge events (i.e., SSIs that occurred within 1 year after discharge and required readmission to a hospital and/ or the use of ambulatory services). Following Wu et al [13], this study adopted the secondary ICD-9-CM diagnosis codes for index hospitalization events (ICD-9-CM code: 996.03, 996.61, 996.72, and 998.5), and the primary and secondary diagnosis codes for post-discharge events (ICD-9-CM code: 038.0-038.4) as the criteria for SSI identification, in order to avoid cases in which infection existed prior to hospitalization. If a case had an index hospitalization event or a post-discharge event, then he/ she will be identified as SSIs by the ICD-9-CM based model. In the CART model, we adopted the type of antibiotics, dose of cefazolin, length of stay, and number of vessels obstructed (as a proxy indicator of duration of operation) as the parameters to identify the SSIs, according to our previous findings. [11] In our previous work, we used the 2005-2008 National Health Insurance claims data and healthcare-associated infection surveillance data from two medical centers for model development and model verification. Infection cases based on surveillance were identified by infection control personnel if the patient met the Taiwan CDC's criteria, which are the same as those adopted in the U.S. CDC. They manually review medical records of all patients at risk for the specified healthcare-associated infection.

The classification algorithms, the multivariable regression model, and the data mining model were adopted to develop alternative models based on surrogate indicators to identify cases of CABG SSIs and to compare the performance among these models and the ICD-9-CM based model. For the classification algorithms, researchers build up several criteria, and if a case satisfies (or exceeds) a specific number of criteria, then it will be identified as a case of infection. For the multivariable regression model, researchers usually calculated a risk score by the logistic regression model, and the optimal cutoff point was determined according to the resulting receiver operating characteristic curve.

Concerning the data mining approach, which is widely used for predicting and classifying objects, the characteristics are: automatic discovery of patterns, prediction of likely

outcomes, creation of actionable information, and focus on large data sets and databases. The classification and regression tree (CART) model, which is the most popular approach as applied in our work, and the growing, stopping, and pruning of the tree were determined by Gini improvement measures. [22, 23] After referring to the literature and conferring with infectious disease specialists, we adopted the following seven parameters: type of antibiotic, doses of antibiotic, doses of cefazolin, use of second-line antibiotics, length of stay, and number of vessels obstructed. Additionally, cross-validation was also employed, where data from one medical center was used for model development, and another one was used for model validation.

The results of our previous work revealed that the CART model offered better performance than that of the other identification models or the ICD-9-CM based model, especially in the positive predictive value ($>70\%$), which was only found to be 20% in the ICD-9-CM based model. (Table 1) The findings also implied that the CART was a decidedly better tool for identifying cases of SSI in the Taiwan National Health Insurance database. Therefore, this study also adopted the CART model for identifying CABG SSIs.

To ensure homogeneity, current study analyzed 7,007 patients from 19 medical centers in Taiwan who underwent CABG surgery (ICD-9-CM procedure codes 36.1x-36.2x) between 2006 and 2008. CABG patients under the age of 18 years or over 85 years were excluded in this study. A total of 302 cases were identified as SSIs by ICD-9-CM based model, and a total of 107 cases were identified as SSIs by CART model.

In this study, we used the following two definitions to define operation volumes: (1) the cumulative operation volumes by each surgeon and hospital within the study period, which was the most common definition in the literature; and (2) following Yasunaga et al.'s study, [24] cumulative operation volumes by each surgeon and hospital in the previous one year for each surgery. However, our data was skewed, which did not follow a normal distribution. Therefore, we conducted the log transformations on operation volumes.

The current work treated operation volumes in three different ways: (1) a continuous variable; (2) a categorical variable based on the first and the third quartile as cutoff points (the most common method to categorize service/ operation volumes) [25] [26] [27] [28] ; and (3) a data-driven categorical variable based on k-means clustering algorithm. This

study categorized surgeon and hospital volumes into low, medium, and high volume groups by quartile method and kmeans clustering algorithm.

In the quartile method, the cut-off value (transformed by logarithm) of the first quartile (<25%) for hospital volumes was 5.65, and the third quartile (>75%) was 6.43. In terms of surgeon volumes, the first quartile was 4.38, and the third was 5.35, when we used the cumulative operation volumes within the study period as the definition. While the definition changed, first quartile (<25%) for hospital volumes was 4.66, and the third quartile (>75%) was 5.31. In terms of surgeon volumes, the first quartile was 3.40, and the third was 4.32.

K-means clustering is an unsupervised machine-learning algorithm introduced by MacQueen in 1960s. This method is not only a simple and very reliable method in categorization/ classification, but is also recognized as one of the top 10 algorithms in data mining. [29] This method has often been applied in many fields. [30] [31] [32] Yu and his colleagues even applied it to define the quality of CABG care, and to explore the relationship among patient's income status, the level of quality of care, and inpatient mortality. [33] The main idea of this method is to partition observed data points into k non-overlapping clusters by minimizing the within-group sum of squares. Each point is assigned to the mean of its cluster using the Euclidian distance. Firstly, k cluster centers were randomly generated. Previous studies usually divided surgeons and hospitals into low-, medium-, and high-volume groups; therefore, we also predetermined the surgeon and hospital service volumes into 3 groups ($k = 3$). Then, participants were assigned to the cluster with the shortest distance to these cluster centers. Finally, the cluster centers were recomputed using the new cluster assignment and these steps would be iterated until convergence was achieved. [34] The cut-off values of hospital volumes were 5.21 and 5.69, and for surgeon's volumes were 2.40 and 4.38 respectively, when cumulative operation volumes within the study period was used as the definition. Likewise, when cumulative operation volumes before each surgery was used as definition, the cut-off values were 4.11 and 4.89 for hospital volumes, and 2.64 and 3.91 for surgeon's volumes. All cutoff values were transformed by logarithm. The results of k-means clustering are demonstrated in Figs 1-4. As the results show, the operation volumes were divided into three groups separately. In addition to surgeon and hospital volumes and SSI, we collected patient-, surgeon-, and hospital-level data. Firstly, patient-level variables included age, gender, length of ICU stay, number of vessels obstructed that were involved in the surgical operation, and the presence of important underlying diseases (e.g. diabetes mellitus, chronic obstructive

pulmonary disease (COPD), heart failure, renal failure and renal insufficiency, which were associated with SSI).

[13] Secondly, the surgeon-level variables included age and gender. Thirdly, the hospital-level variables included hospital ownership and geographic location.

All statistical analyses of volume-infection relationship were performed using SAS (version 9.2, SAS Institution Inc., Cary, NC, USA). In statistical testing, a two-sided p value 0.05 was considered statistically significant. The distributional properties of continuous variables were expressed by mean $\pm$ standard deviation (SD), whereas categorical variables were presented by frequency and percentage. In univariate analysis, the potential three-level predictors of SSI were examined using chi-square test or two-sample t-test as appropriate. Next, to account for the correlations within surgeon (level-2) and hospital (level-3), multivariate analysis was conducted by fitting mixed-effects logistic regression models to each patient's data for estimating the effects of three-level predictors on the probability of post-operational SSI. Furthermore, subgroup analysis for comorbidities was also conducted. Table 2 shows that there were 7,007 patients with CABG performed by 199 surgeons in 19 hospitals during 2006-2008 in Taiwan. The majority of patients were male (77.5%), and the mean age of patients was 65.3 years. The average ICU stay was 6.05 days, the mean level of number of vessels obstructed was around 1.6, while 51.8% of patients had diabetes mellitus, 33.3% had heart failure, 14.1% had renal failure and renal insufficiency, and 22.0% had COPD. Three hundred and two patients (4.31%) were identified as having the ICD-9-CM SSI codes. However, identification by the CART model only revealed 107 infection cases, and 94 cases were identified in both models. Most cases received CABG surgery by male surgeons, with a mean age of 45.0 years, and the surgeon's average operation volumes within the study period was 151.64, while the average operation volumes before surgery was 52.18. More than half of the cases were performed with CABG in not-for-profit hospitals, and the hospitals' average operation volumes within the study period was 473.60, while the average operation volumes before each surgery was 158.79. Moreover, most of patients received their surgeries by high-volume surgeons and hospitals, when k-means algorithm was used for categorization, regardless of which definition of operation volumes were used. Table 3 shows the results of multilevel mixed-effect models, with the SSIs being identified by ICD-9-CM codes, and the operation volumes defined as the cumulative volumes within the study period. The results of Model 1 (continuous) reveal that the surgeon's volumes were negatively associated with SSIs, while hospital's volumes were not associated with surgical site infection SSIs. Model 2 (quartile) suggests that low-

volume surgeons had higher SSI risk (OR = 2.220, p-value = 0.022) than high-volume surgeons. There were also no associations between hospital's operation volumes and SSIs. Model 3 (k-means) shows that the association did not exist between hospital's/ surgeon's volumes and SSIs. Table 4 displays the results of multilevel mixed-effect models, in which the SSIs were identified by the CART model, and the operation volumes were also defined as the cumulative volumes within the study period. Model 1 again indicated a negative association between surgeon's volumes and SSIs, and hospital's volumes were not found to be associated with SSIs. In Model 2, the results showed that the relationship between hospital's/ surgeon's volumes and SSIs did not exist. In Model 3, results revealed low-volume surgeons had higher risk (OR = 1.691, p = 0.002) than high-volume surgeons. Table 5 displays the results of multilevel mixed-effect models, in which the SSIs were identified by ICD-9-CM codes, but the operation volumes were defined as the cumulative volume in the previous one year for each surgery. Model 1 also indicated a negative association between surgeon's volumes and SSIs, and hospital's volumes were not found to be associated with SSIs. In Model 2, the results showed that the relationship between hospital's/ surgeon's volumes and SSIs did not exist. In Model 3, results also revealed low-volume surgeons had higher risk (OR = 1.642, p = 0.040) than high-volume surgeons. Table 6 displays the results of multilevel mixed-effect models, in which the SSIs were identified by the CART model, and the operation volumes were also defined as the cumulative volume in previous one year for each surgery. In Model 1, different to the above findings, there was no association between hospital's/ surgeon's volumes and SSIs. In Model 2, the results showed that the relationship between hospital's/ surgeon's volumes and SSIs did not exist. In Model 3, results also revealed low-volume surgeons had higher risk (OR = 1.163, p = 0.020) than high-volume surgeons.

We further examined the associations of surgeon and hospital volumes with SSIs in stratification analyses by underlying diseases. When the operation volumes were defined as the cumulative operation volume within the study period, no relationships existed between hospital/ surgeon operation volumes and SSIs. (Table 7) However, when the operation volumes were defined as the cumulative operation volumes in the previous one year for each surgery, the results suggested that there was a negative association between surgeon volumes and SSIs in the diabetes group, except that the volumes were treated as continuous variable and the infection cases were identified by ICD-9 codes. In terms of hospital operation volumes, the association did not exist. (Table 8)

No studies have evaluated how different service/ operation volumes definitions and categorization methods affect volume-infection relationships. Moreover, several studies have pointed out the inappropriateness of identifying infection cases using the ICD-9-CM codes in claims data. Given these reasons, this study adopted two approaches to identifying SSIs, two definitions of operation volumes, and three methods for categorizing operation volumes to examine the relationships between operation volumes and SSIs. Our findings showed that the relationships between hospital volumes and SSIs did not exist, no matter which definitions, categorization methods, or SSIs case identification approaches were used. On the contrary, the relationships between surgeon volumes and SSIs were not robust in our data. It might be affected by different definitions and categorization methods of operation volumes, and also by different SSI cases identification approaches. In summary, most of the models demonstrated that the low-volume surgeons had higher risk than high-volume surgeons, and they also showed the risks were similar between medium-volume and high-volume surgeons. However, why did surgeon volume relate to SSIs, but hospital volume did not? Except for those issues we were concerned about in this study, there are some disagreements in the literature. Such as "Does provider volume really represent quality of care?" [12, 35] Or "Is provider volume the only one predictor for outcome of care?" [35, 36] These issues are worthy of further discussion, but are out of the scope of this study.

Service/ operation volumes are treated as a proxy indicator for experiences; previous studies used it to examine whether practice makes perfect or not. But, except for provider's experiences, SSIs are also impacted by many factors, such as environmental and clinical factors. Wu et al once used Taiwan 2001 NHI claims data to explore the relationship between provider CABG operation volumes and SSIs. [13] They found that hospital volumes had a greater effect than surgeon volumes and claimed that this may imply that hospital teamwork is more important than individual surgeon. However, our findings demonstrated that there was no relationship between hospital volumes and SSIs. Wu et al. adopted the cumulative operation volumes within the study period as the definition, and identified SSIs by ICD-9-CM codes. Except, there were two differences between our work and Wu et al., which were the length and year of the data; our data was longer and more updated than theirs. Moreover, it is worth noting that there was an outbreak of severe acute respiratory syndrome (SARS) in Taiwan in 2003, after which the hospital infection control system in Taiwan was reviewed and re-designed. Wu et al data was before SARS, so these efforts may also have improved the level of SSIs control in hospitals, leading to different findings in this study.

In addition, although most models revealed that there were negative relationships between surgeon's volumes and surgical site infection, the relationships were not robust. The results varied between different definitions and categorization method of operation volumes, and between SSIs identification approaches. Researchers need to consider how to identify SSIs correctly, how to choose optimal cut-off values, and how to decide on which definition is appropriate.

Finally, the results of stratification analyses showed that low-volume surgeon had higher risk than high-volume surgeon in the diabetes mellitus group, when the cumulative operation in the previous one year before surgery was used as definition. A large number of studies have indicated diabetes mellitus is associated with a higher risk of SSIs, [37] [38] [39] and the findings of this study suggest that CABG patients with diabetes mellitus should be cared for by experienced surgeons.

A multilevel analysis was applied to manage the nested factors, and two definitions of operation volume along with three different operation volume categorization methods were adopted to examine the relationship between volume and SSIs under two kinds of SSIs identification approaches. Nevertheless, the study suffered from several major limitations. First, the accuracy of SSIs identification was still an issue. Although the performance of the CART model to identify CABG SSIs was better than ICD-9-CM codes in Taiwan NHI claims data, it did not reach the perfect scenario. The accuracy of SSIs identification was still a challenge in our work. The second limitation relates to unmeasured variables, such as length of stay before operation, infection condition, hair removal, clinical information (e.g. blood glucose level, causative microorganism), time-related information (e.g. the duration of operation), the environment, surgical skills, use of post-operative drains, number of operations involved, and surgical site and wound care, etc. [40] Furthermore, information about type (elective or urgent) and incision site for surgery was not available in the Taiwan NHI claims data.

In conclusion, the findings of this study suggest that different definitions and categorization methods of operation volumes, and different SSIs identification approaches might lead to different findings, although surgeon volumes were more important than hospital volumes in exploring the relationships between CABG operation volumes and SSIs in Taiwan, but

they were still not robust. Definitions and categorization methods of operation volumes, and correct identification of SSIs are important issues for future research.

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The influenza pandemic preparedness planning tool InflaSim

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1832202/>

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Abstract: **BACKGROUND:** Planning public health responses against pandemic influenza relies on predictive models by which the impact of different intervention strategies can be evaluated. Research has to date rather focused on producing predictions for certain localities or under specific conditions, than on designing a publicly available planning tool which can be applied by public health administrations. Here, we provide such a tool which is reproducible by an explicitly formulated structure and designed to operate with an optimal combination of the competing requirements of precision, realism and generality. **RESULTS:** InflaSim is a deterministic compartment model based on a system of over 1,000 differential equations which extend the classic SEIR model by clinical and demographic parameters relevant for pandemic preparedness planning. It allows for producing time courses and cumulative numbers of influenza cases, outpatient visits, applied antiviral treatment doses, hospitalizations, deaths and work days lost due to sickness, all of which may be associated with economic aspects. The software is programmed in Java, operates

platform independent and can be executed on regular desktop computers. **CONCLUSION:** InflaSim is an online available software which efficiently assists public health planners in designing optimal interventions against pandemic influenza. It can reproduce the infection dynamics of pandemic influenza like complex computer simulations while offering at the same time reproducibility, higher computational performance and better operability.

Text: Preparedness against pandemic influenza has become a high priority public health issue and many countries that have pandemic preparedness plans [1] . For the design of such plans, mathematical models and computer simulations play an essential role because they allow to predict and compare the effects of different intervention strategies [2] . The outstanding significance of the tools for purposes of intervention optimization is limited by the fact that they cannot maximize realism, generality and precision at the same time [3] . Public health planners, on the other hand, wish to have an optimal combination of these properties, because they need to formulate intervention strategies which can be generalized into recommendations, but are sufficiently realistic and precise to satisfy public health requirements.

Published influenza models which came into application, are represented by two extremes: generalized but oversimplified models without dynamic structure which are publicly available (e.g. [4]), and complex computer simulations which are specifically adjusted to real conditions and/or are not publicly available (e.g. [5, 6]). The complexity of the latter simulations, however, is not necessary for a reliable description of infection dynamics in large populations [7] . A minimum requirement for a pandemic influenza planning tool is a dynamic modelling structure which allows investigation of time-dependent variables like incidence, height of the epidemic peak, antiviral availability etc. The tool should, on the other hand, be adjustable to local conditions to adequately support the pandemic preparedness plans of different countries which involve considerably different assumptions (Table 1) .

Here we describe a publicly available influenza pandemic preparedness planning tool [8] which is designed to meet the requirements in preparedness planning. It is based on an explicitly formulated dynamic system which allows addressing time-dependent factors. It is sufficiently flexible to evaluate the impact of most candidate interventions and to consider local conditions like demographic and economic factors, contact patterns or constraints within the public health system. In subsequent papers we will also provide

examples and applications of this model for various interventions, like antiviral treatment and social distancing measures.

The model is based on a system of 1,081 differential equations which extend the classic SEIR model. Demographic parameters reflect the situation in Germany in 2005, but can be adjusted to other countries. Epidemiologic and clinic values were taken from the literature (see Tables 1, 2, 3, 4, 5, 6 and the sources quoted there). Pre-set values can be varied by sliders and input fields to make different assumptions on the transmissibility and clinical severity of a new pandemic strain, to change the costs connected to medical treatment or work loss, or to simply apply the simulation to different demographic settings. Model properties can be summarized as follows. The mathematical formulation of this model is presented in detail in the online supporting material. The corresponding source code, programmed in Java, and further information can be downloaded from [8] .

According to the German National Pandemic Preparedness Plan [9] , the total population is divided in age classes, each of which is subdivided into individuals of low and high risk (Table 2) . Transmission between these age classes is based on a contact matrix (Table 3) which is scaled such that the model with standard parameter values yields a given basic reproduction number R_0 . Values for the R_0 associated with an influenza strain with pandemic potential are suggested to lie between 2 and 3 [10] . This value is higher than the effective reproduction number which has been estimated to be slightly lower than 2 [11, 12] . As a standard parameter, we use $R_0 = 2.5$ which means that cases infect on average 2.5 individuals if everybody is susceptible and if no interventions are performed.

Susceptible individuals who become infected, incubate the infection, then become fully contagious and finally develop protective immunity (Table 4) . A fraction of cases remains asymptomatic; others become moderately sick or clinically ill (i.e. they need medical help). Depending on the combination of age and risk group, a fraction of the clinically ill cases needs to be hospitalized, and an agedependent fraction of hospitalized cases may die from the disease (Table 5). This partitioning of the cases into four categories allows combining the realistic description of the transmission dynamics with an easy calculation of the resources consumed during an outbreak. The degree and duration of contagiousness of a patient depend on the course of the disease; the latter furthermore depends on the age of the patient (Table 5) . Passing through the incubation and contagious period is modelled in several stages which allows for realistic distributions of the sojourn times (Table 4). The

last two stages of the incubation period are used as early infectious period during which the patient can already spread the disease. Infectiousness is highest after onset of symptoms and thereafter declines geometrically (Table 6). Clinically ill patients seek medical help on average one day after onset of symptoms. Very sick patients are advised to withdraw to their home until their disease is over, whereas extremely sick patients need to be hospitalized and may die from the disease (Table 4) . After the end of their contagious period, clinically ill patients go through a convalescent period before they can resume their ordinary life and go back to work (Table 4) .

We provide some examples of model output of InflaSim [8] , version 2.0, by means of four sensitivity analyses; further investigations will be presented elsewhere. Figure 1 shows the graphical user interface of the software which is divided into input and output windows. The user may set new values in the input fields or move sliders to almost simultaneously obtain new results for the course of an epidemic in a given population. Figures 2A and 2B show pandemic waves which result from varying the basic reproduction number from 1.5 to 4.0. Using the standard parameter values as given in Tables 2, 3 , 4, 5, 6 and omitting all interventions in a town of 100,000 inhabitants results in a pandemic wave which lasts for about ten weeks (Figure 2A , with $R_0 = 2.5$). The peak of the pandemic wave is reached after six to seven weeks, with a daily incidence of up to 2,340 influenza patients seeking medical help, with up to 280 hospital beds occupied by influenza cases and with up to 14,000 out of 60,000 working adults unable to go to work because of illness or convalescence. These results depend on the assumptions concerning the yet unknown contagiousness and pathogenicity of the virus. Figures 2C and 2D show how the shape of the curves depends on the course of contagiousness: the pandemic wave proceeds relative slowly if the contagiousness does not change during the infectious period ($x_{50} = 50\%$), but proceeds quickly if the contagiousness is highest after onset of symptoms and decreases thereafter ($x_{50} > 50\%$).

The influenza pandemic preparedness planning tool InflaSim stands between simple spreadsheet models and sophisticated stochastic computer simulations. It describes a pandemic wave within a homogeneously mixing population like a town or city, but surprisingly produces the same dynamics as individual-based simulations which explicitly consider geographic spread through the US (cf. [6] and [5] with Figure 2 using $R_0 = 2$). Similar observations were made with a simple deterministic compartmental model [7] . Stochastic models are known to behave quasi-deterministically when the simulated population becomes very large.

A further reason for the congruence of complex stochastic and simple deterministic models must lie in the incredibly quick way in which pandemic influenza spreads geographically. Unless being controlled at the place of origin [12, 13], a pandemic starting in a far-off country will lead to multiple introductions [14] into the large industrialized nations where it can be expected to quickly spread to neighbouring towns and to rural areas. The large populations which have to be considered susceptible to a pandemic virus and the quick geographic spread tend to diminish the differences between the results of sophisticated individual-based and simple deterministic models.

However, a deterministic model like InflaSim cannot reliably represent effects originating from stochasticity, from effects in small populations, or from heterogeneities. Examples are: (i) a geographically limited spread and fairly effective control measures can imply that the epidemic affects only a small population and thus, may be strongly influenced by stochastic events [15] [16] [17]; (ii) transmission which predominantly occurs in households or hospitals, or which is driven by other substantial features of the contact network is not in agreement with the assumption of homogeneous mixing in the deterministic model cannot reliably predict the spread of infection [18] [19] [20] [21] [22] [23]. In particular, (iii) super-spreading events can substantially change the course of an epidemic compared to the deterministic prediction [24] [25] [26] [27]. Apart from such factors, the predictability of intervention success is generally subject to uncertainties in the choice of parameter values, Assumed scenarios and outcomes of pandemic preparedness plans. * Gross attack rate (i.e. clinically ill and moderately ill cases). A population of $N = 100,000$ inhabitants of Germany is subdivided according to age a and risk category r . We assume that all age groups are fully susceptible at begin of the outbreak. A fraction of $F_a = 6\%$ of all children (age < 20 years) are regarded as being under high risk ($r = r_1$) after an influenza infection whereby the remaining 94% are under low risk ($r = r_2$). The high risk fractions of working adults (ages 20-59) and elderly (ages 60+) are $F_a = 14\%$ and $F_a = 47\%$, respectively. Source: [9] demanding additional efforts like Bayesian approaches [28] to evaluate the reliability of predictions [29].

Pandemic preparedness plans must consider constraints and capacities of locally operating public health systems. The time-dependent solutions of InflaSim allow assessing peak values of the relevant variables, such as outpatients, hospitalizations and deaths. Various interventions may be combined to find optimal ways to reduce the total number of

cases, to lower the peak values or to delay the peak, hoping that at least part of the population may benefit from a newly developed vaccine.

Special care was taken when implementing a variety of pharmaceutical and non-pharmaceutical interventions which will be discussed in subsequent papers. Despite its comprehensible structure, the model does not suffer from over-simplifications common to usual compartment models. Instead of implicitly using exponentially distributed sojourn times, we have implemented realistically distributed delays. For example, the model considers that individuals may transmit infection before onset of symptoms, and that some cases may remain asymptomatic, but still infecting others. Such features have serious implications for the success of targeted control measures.

InfluSim is freely accessible, runs on a regular desktop computer and produces results within a second after changing parameter values. The user-friendly interface and the ease at which results can be generated make this program a useful public health planning tool. Although we have taken care of providing a bug-free program, including the source code, the user is encouraged to treat results with due caution, to test it, and to participate in bug-reports and discussions on the open-source platform [30] which also provides regular updates of InfluSim.

The author(s) declare that they have no competing interests.

ME developed the model, MS designed the software, HPD wrote the manuscript and SOB formulated the public The who-acquires-infection-from-whom matrix shows the frequency of contacts (per week per person) between different age classes.

Source: [38] . Distribution of sojourn times (the last two stages of the latent period are used as early infectious period with an average duration of $D_L = 0.5$ days). Sources: A [11] , B [39, 40] , C assumed, D [41] health requirements of the software. All authors read and approved the final manuscript.

Susceptible individuals $S_{a,r}$ are infected at a rate $\lambda_{a,r}(t)$ which depends on their age a and on time t . Infected individuals, $E_{a,r}$, incubate the infection for a mean duration D_E . To obtain a realistic distribution of this duration, the incubation period is modelled in n stages so that progression from one stage to the next one occurs at rate $\delta = n/D_E$.

The last l incubation stages are regarded as early infectious period during which patients may already spread the infection (this accounts for an average time of lD_E/n for the "early infectious period" which is about half a day for the standard set of parameters). After passing through the last incubation stage, infected individuals become fully contagious and a fraction of them develops clinical symptoms. The course of disease depends on the age a of the infected individual and on the risk category r to which he or she belongs: a fraction $c_{a,r}(A)$ becomes asymptomatic ($A_{a,r}$), a fraction $c_{a,r}(M)$ becomes moderately sick ($M_{a,r}$), a fraction $c_{a,r}(V)$ becomes very sick ($V_{a,r}$) and the remaining fraction $c_{a,r}(X)$ becomes extremely sick ($X_{a,r}$) and need hospitalization (i.e., $c_{a,r}(A) + c_{a,r}(M) + c_{a,r}(V) + c_{a,r}(X) = 1$ for each combination of a and r). A fraction $f_V(t)$ of all severe and a fraction $f_X(t)$ of all extremely severe cases who visit the doctor within D_T days after onset of symptoms are offered antiviral treatment, given that its supply has not yet been exhausted. As our model does not explicitly consider the age of the disease (which would demand partial differential equations), we use the contagious stages to measure time since onset and allow for treatment up to stage $m_{a,r}$. Sources: Contagiousness of asymptomatic cases: [11]; degree of contagiousness during the early infectious period and equality of the contagiousness of moderately and severely sick cases: assumed. Independent of age a and risk group r , a fraction $c_{a,r}(A) = 33\%$ of infections result in asymptomatic cases, a fraction $c_{a,r}(M) = 33.5\%$ become moderately sick and the remaining fraction develops severe disease. An age- and risk-dependent fraction $h_{a,r}$ of untreated patients with severe disease needs hospitalization. An age-dependent fraction d_a of hospitalized cases dies. Sources: fraction of asymptomatic cases: [11]; 50% of symptomatic cases see a doctor: [9]; hospitalizations per severe case: [9]; case fatality of hospitalized, but untreated patients calculated from [4].

(see below for details). This imposes some variability to the maximum time until which treatment can be given, which may even improve the realism of the model with respect to real-life scenarios. Antiviral treatment reduces the patients' contagiousness by f_I percent and it reduces hospitalization and death by f_H percent. Extremely sick patients, whose hospitalization is prevented by treatment, are sent home and join the group of treated very sick patients ($W_{a,r}$). The remaining duration of disease and contagiousness of treated

cases is reduced by f/D percent so that their rate of progressing from one stage to the next has To obtain a realistic distribution of this sojourn time, convalescence is modelled in j stages so that progression from one stage to the next occurs at rate $p = j/D$. Fully recovered patients who have passed through their last stage of convalescence join the group of healthy immunes I ; working adults will go back to work. Further interventions, describing the reduction of contacts, will be discussed after the presentation of the differential equations.

InfluSim user interface Figure 1 InfluSim user interface. $x_{50} = 95\%$ means that 95% of the cumulative contagiousness is concentrated during the first half of the contagious period, see Table 6). D : Cumulative number of deaths for values of x_{50} as in C. All other parameters as listed in Tables 2-6 . Hospitalized, but untreated cases

Contact matrix For the mixing of the age classes, we employ a whoacquires-infection-from-whom matrix which gives the relative frequency of contacts of infective individuals of age a_i with other people of age a_s . In this paper, we assume bi-directional contacts (e.g. children have the same total number of contacts with adults as adults with children). Multiplication of this matrix with an appropriate constant scaling factor κ (see below)

results in the matrix of crude contact rates .

In the absence of interventions, we have to multiply these contact rates with the contagiousness factors b_L , b_A , b_M and b_V to obtain the effective contact rates:

during the early infectious period, of asymptomatic cases, of moderately sick cases, of (untreated) very sick cases.

To assess the effect of day care centre and school closing on the transmission of an infectious disease, we have to first make an assumption on what fraction r_{sch} of the contacts among healthy children who are in the same age class occurs in day care centres and schools. The contact rates between very sick or hospitalized children (who do not

attend day care centre or school) and other children need, therefore, be reduced to (contact rate between healthy and very sick children in the same age class, i.e. $a_i = a_s$).

As very sick children have to be taken care of by adults at home or in hospital, their contact rate to adults increases by a factor F_{HC} (contact rate between very sick children of age a_i and adults of age a_s).

Contacts between very sick children and other children in a higher or lower age class remain unchanged: (contact rate between healthy children of age a_s and very sick children of a different age a_i).

Closing day care centres and schools at time t will not necessarily prevent all the contacts that would have happened with other children. During the closing of schools and day care centres, the contact rates between susceptible children of age a_s and infected children of age a_i who are in their late incubation period (), who are asymptomatic (), or who are moderately sick () are reduced by the factor r_{sch} if the children are in the same age class:

where $1_{sch}(t)$ is a function which indicates when schools and day care centres are opened or closed:

..., While day care centres and schools are closed, children (age a_i) need adult supervision at home. Their contact with susceptible adults (age a_s) increases by the "child care factor" F_{CC} :

Child care at home also increases the exposure of healthy children (age a_s) to contagious adults (age a_i):

Cancelling mass gathering events effects only the contacts of adults who are healthy enough to attend such events. Assuming that such an intervention at time t reduces contacts by a fraction r_{mass} , we get for all contacts between susceptible adults of age a_s and infectious adults of age a_i the following contact rates:

where $1_{\text{mass}}(t)$ is a function which indicates when mass gathering events are possible or when they are closed:

As contacts with adults who are too sick to attend such mass gathering events cannot be prevented by this measure it is .

During some time in the epidemic, the general population may effectively reduce contacts which can be a result of wearing facial masks, increasing "social distance", adopting improved measures of "respiratory hygiene" or simply of a general change in behaviour. This will be implemented in the program by reducing the contacts of susceptible individuals at that time t by factor $r_{\text{gen}}(t)$

while mass gathering events are forbidden while m mass gathering events are allowed. while the population reduces their contacts while the population behaves as usual.

The contact rates of cases in the late incubation period and that of asymptomatic cases remain unchanged:

for infected individuals in the late incubation period, for asymptomatic cases.

To allow for a contagiousness which changes over the course of disease, we multiply each contact rate with a weighting factor whereby k is the stage of contagiousness. This leads to the following contact rates:

for asymptomatic cases in For $x = 1$, contagiousness is equally high in all stages; for $x = 0$, only the first stage is contagious; for $0 < x < 1$, the contagiousness decreases in a geometric procession. We make the simplifying assumption that contagiousness does not change during the late incubation period for cases in stage $k = n-l, \dots, n$ of the incubation period.

At time $t = 0$ and in the absence of interventions, the next generation matrix has the following elements where ω is the fraction of untreated extremely severe cases who die from the disease (see below for details). The dominant eigenvalue of this matrix is called the basic reproduction number R_0 . If κ (which determines the value of the contact rates) is given, the eigenvectors of this matrix can numerically be calculated. The user-specified value of R_0 is now used to determine numerically the scaling factor κ . Let $\mathbf{v}$ be the eigenvector which has the largest eigenvalue R_0 .

)) –

Using the user-specified numbers of people N_a in the age classes and the fractions F_a of people under high risk within each age class (Table 2), we obtain the initial population sizes according to age and risk class: Using these initial values, the set of differential equations is solved numerically with a Runge-Kutta method with step-size control.

if and in treatment window otherwise 0 $\cup$

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Genome Sequences of Porcine Epidemic Diarrhea Virus: In Vivo and In Vitro Phenotypes

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4056290/>

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Date: 2014-06-12

Abstract: Since the outbreak of porcine epidemic diarrhea virus (PEDV) in May 2013, U.S. swine producers have lost almost five million baby pigs. In an attempt to understand the evolution of PEDV in the United States and possibly develop a control strategy, we compared the genome sequences of a PEDV strain isolated from an infected piglet against its in vitro adapted version. The original PEDV strain was grown in Vero cells and passed 10 times serially in a MARC145 cell line. The sequence analysis of the native PEDV strain and in vitro passaged virus shows that the cell culture adaptation specifically modifies PEDV spike protein whereas the open reading frame 1a/b (ORF1a/b)-encoded polyprotein, the nucleoprotein, NS3B (ORF3), and membrane and envelope proteins remain unchanged.

Text: highly contagious swine disease. While older pigs have a chance of survival, 80 to 100 percent of PEDV-infected piglets die within 24 h of being infected. PEDV spreads primarily through fecal-oral contact (1, 2) . Once the virus is internalized, it destroys the lining of piglets' intestines, making them incapable of digesting and deriving nutrition from milk and feed (1) . The virus causes diarrhea, vomiting, and death from dehydration and starvation (2) .

PEDV is a member of the Coronavirinae subfamily and belongs to the Alphacoronavirus genus. Its genomic size ranges from approximately 26 to 32 kb, which is relatively large for an RNA virus. Although vaccines for PEDV exist in China, Japan, and South Korea, there is no approved vaccine in the United States or Europe (3) . Furthermore, PEDV is still evolving within the U.S. swine population.

This report briefly describes the comparison of genome sequences of a PEDV strain isolated from small intestine samples of an infected piglet and its in vitro adapted version. The original PEDV strain was dubbed NPL-PEDV/2013, grown in Vero cells, and passed 10 times in a MARC145 cell line. The serial in vitro passage strain was named NPL-PEDV/2013/P10. The total viral RNA was extracted by TRIzol LS reagent and sequenced by Sanger dideoxy sequencing using a primer walking technique. The raw sequences were imported into the Geneious assembler (Biomatters, CA), assembled, annotated, and

compared against each other using USA/Colorado/2013 (GenBank accession no. KF272920) as a reference sequence.

The whole-genome sequences of NPL-PEDV/2013 and NPL-PEDV/2013/P10 contain 28,038 and 28,025 nucleotides (nt), respectively, including the 5' and 3' untranslated regions (UTR). The NPL-PEDV/2013 genome shares 99% identity with all the U.S. isolates sequenced to date and many Chinese isolates as well. The top three BLAST hits were against U.S. isolates, USA/Colorado/2013 (GenBank accession no. KF272920), IA1 (GenBank accession no. KF468753.1), and an isolate from Iowa, 13-019349 (GenBank accession no. KF267450.1). The NPL-PEDV/2013 isolate also shares 99% identity with the Chinese outbreak isolate AH2012 (GenBank accession no. KC210145).

When the NPL-PEDV/2013/P10 strain was compared against NPL-PEDV/2013, the open reading frame 1a/b (ORF1a/b) polyprotein, the nucleoprotein, NS3B, and membrane and envelope proteins were found to be 100% identical at the amino acid level. In contrast, the spike gene contains six nonsynonymous single nucleotide polymorphisms, resulting in amino acid (aa) substitutions in the following positions: 375 (F_iL), 486 (T_iP), 856 (D_iE), 1081 (A_iV), 1099 (A_iS), and 1253 (Y_iD). The S1 domain of spike protein contains 2 aa substitutions, whereas the S2 domain contains 4 aa substitutions. PEDV has been shown to use porcine aminopeptidase N (pAPN) as the major receptor for cell entry (4, 5). However, Vero and MARC145 cells lack pAPN, clearly indicating that other receptors or receptor-independent pathways may be used for entry (6). The spike protein in its trimeric conformation interacts with the cell receptor and contains numerous neutralizing antibody binding epitopes (7). Analysis of the spike by PeptideCutter (http://web.expasy.org/peptide_cutter/) shows that the native spike protein of NPL-PEDV/2013 has 63 trypsin and 2 chymotrypsin cleavage sites at 100% efficiency whereas NPL-PEDV/2013/P10 has lost one trypsin cleavage site but the number of chymotrypsin sites remain unchanged. This indicates that cell culture adaptation specifically modifies the PEDV spike protein; however, the immunological implications are unknown.

Nucleotide sequence accession numbers. The whole-genome sequences of the NPL-PEDV/2013 and NPL-PEDV/2013/P10 strains have been deposited at DDBJ/EMBL/GenBank under accession no. KJ778615 and KJ778616.

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Serological Assays Based on Recombinant Viral Proteins for the Diagnosis of Arenavirus Hemorrhagic Fevers

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Abstract: The family Arenaviridae, genus Arenavirus, consists of two phylogenetically independent groups: Old World (OW) and New World (NW) complexes. The Lassa and Lujo viruses in the OW complex and the Guanarito, Junin, Machupo, Sabia, and Chapare viruses in the NW complex cause viral hemorrhagic fever (VHF) in humans, leading to serious public health concerns. These viruses are also considered potential bioterrorism agents. Therefore, it is of great importance to detect these pathogens rapidly and specifically in order to minimize the risk and scale of arenavirus outbreaks. However, these arenaviruses are classified as BSL-4 pathogens, thus making it difficult to develop diagnostic techniques for these virus infections in institutes without BSL-4 facilities. To overcome these difficulties, antibody detection systems in the form of an enzyme-linked immunosorbent assay (ELISA) and an indirect immunofluorescence assay were developed using recombinant nucleoproteins (rNPs) derived from these viruses. Furthermore, several antigen-detection assays were developed. For example, novel monoclonal antibodies (mAbs) to the rNPs of Lassa and Junin viruses were generated. Sandwich antigen-capture (Ag-capture) ELISAs using these mAbs as capture antibodies were developed and

confirmed to be sensitive and specific for detecting the respective arenavirus NPs. These rNP-based assays were proposed to be useful not only for an etiological diagnosis of VHFs, but also for seroepidemiological studies on VHFs. We recently developed arenavirus neutralization assays using vesicular stomatitis virus (VSV)-based pseudotypes bearing arenavirus recombinant glycoproteins. The goal of this article is to review the recent advances in developing laboratory diagnostic assays based on recombinant viral proteins for the diagnosis of VHFs and epidemiological studies on the VHFs caused by arenaviruses.

Text: The virus family Arenaviridae consists of only one genus, but most viruses within this genus can be divided into two different groups: the Old World arenaviruses and the New World arenaviruses (also known as the Tacaribe complex) [1, 2]. The differences between the two groups have been established through the use of serological assays. Most of the arenaviruses cause persistent infection in rodents without any symptoms, and humans acquire a variety of diseases when zoonotically infected. Lymphocytic choriomeningitis virus (LCMV) is the only arenavirus to exhibit a worldwide distribution, and causes illnesses such as meningitis [3, 4]. Congenital LCMV infections have also been reported [4, 5]. Most importantly, viral hemorrhagic fever (VHF) can be caused by several arenaviruses. Lassa fever, caused by the Lassa virus (LASV), an Old World arenavirus, is one of the most devastating VHFs in humans [6]. Hemorrhaging and organ failure occur in a subset of patients infected with this virus, and it is associated with high mortality. Many cases of Lassa fever occur in Western Africa in countries such as Guinea, Sierra Leone, and Nigeria [7] [8] [9] [10] [11] [12] [13]. Tacaribe complex lineage B of the New World arenaviruses consists of the Junin virus (JUNV), Guanarito virus (GUNV), Sabia virus (SABV) and Machupo virus (MACV), the etiological agents of Argentine, Venezuelan, Brazilian, and Bolivian hemorrhagic fevers, respectively [14, 15]. Although genetically distinct from one another, they appear to produce similar symptoms, accompanied by hemorrhaging in humans [14, 15]. These pathogenic New World arenavirus species are closely associated with a specific rodent species [6].

Humans are usually infected with pathogenic arenaviruses through direct contact with tissue or blood, or after inhaling aerosolized particles from urine, feces, and saliva of infected rodents. After an incubation period of 1-3 weeks, infected individuals abruptly develop fever, retrosternal pain, sore throat, back pain, cough, abdominal pain, vomiting, diarrhea, conjunctivitis, facial swelling, proteinuria, and mucosal bleeding. Neurological problems have also been described, including hearing loss, tremors, and encephalitis. Because the symptoms of pathogenic arenavirus-related illness are varied and

nonspecific, the clinical diagnosis is often difficult [14, 16] . Human-to-human transmission may occur via mucosal or cutaneous contact, or through nosocomial contamination [14, 16] . These viruses are also considered to be potential bioterrorism agents [2] .

A number of arenavirus species have been recently discovered as a result of both rodent surveys and disease outbreaks [17] [18] [19] [20] [21] [22] [23] [24] [25] [26] . A novel pathogenic New World arenavirus, Chapare virus (CHPV), has been isolated from a fatal case of VHF in Bolivia [20] . In addition, five cases of VHF have been reported in South Africa, and a novel arenavirus, named Lujo virus, was isolated from a patient [17] . The Lujo virus is most distantly related to the other Old World arenaviruses [17] . To date, there is no information concerning the vertebrate host for the Chapare and Lujo viruses.

There is some evidence of endemicity of the Lassa virus in neighboring countries [27, 28] . However, as the magnitude of international trade and travel is continuously increasing, and the perturbation of the environment (due either to human activity or natural ecological changes) may result in behavioral changes of reservoir rodents, highly pathogenic arenaviruses could be introduced to virus-free countries from endemic areas. In fact, more than twenty cases of Lassa fever have been reported outside of the endemic region in areas such as the USA, Canada, Europe, and Japan [29] [30] [31] [32] [33] . It is of great importance to detect these pathogens rapidly and specifically in order to minimize the risk and scale of outbreaks of VHFs caused by arenaviruses. However, these arenaviruses are classified as biosafety level (BSL)-4 pathogens, making it difficult to develop diagnostic techniques for these virus infections in laboratories without BSL-4 facilities. To overcome these difficulties, we have established recombinant viral nucleoproteins (rNPs)-based serological assays, such as IgG-enzyme-linked immunosorbent assay (ELISA), indirect immunofluorescence assay (IFA), and antigen (Ag)-capture ELISA for the diagnosis of VHFs caused by highly pathogenic arenaviruses. Furthermore, virus neutralization assays using pseudotype virus-bearing arenavirus GPs have been developed. In this review, we describe the usefulness of such recombinant protein-based diagnostic assays for diagnosing VHFs caused by arenaviruses.

In outbreaks of VHFs, infections are confirmed by various laboratory diagnostic methods. Virus detection is performed by virus isolation, reverse transcription-polymerase chain reaction (RT-PCR), and antigen-capture ELISA. It has been shown that monoclonal

antibody panels against pathogenic arenaviruses are useful for detecting viral antigens on the virus-infected cells as well as for investigating of antigenic relationships of arenaviruses [34] [35] [36] . Detection of the virus genome is suitable for a rapid and sensitive diagnosis of VHF patients in the early stage of illness, and extensive reviews of such RT-PCR assays have been described [37, 38] . More recently, progress in the RT-PCR method covering genetic variations of the hemorrhagic fever viruses (HFVs) [39, 40] and a multiplexed oligonucleotide microarray for the differential diagnosis of VHFs have also been reported [41] . On the other hand, antibodies against these viruses can be detected by the indirect immunofluorescence assay (IFA), or IgG-and IgM-ELISA. An IFA detects the antibody in the serum, which is able to bind to the fixed monolayer of the virus-infected cells. Although the interpretation of immunofluorescence results requires experience, the assay has advantages over other methods, since each virus generates a characteristic fluorescence pattern that adds specificity to the assay compared to a simple ELISA readout. A serological diagnosis by the detection of specific IgM and IgG antibodies to the HFVs must be sensitive, specific and reliable, because a misdiagnosis can lead to panic in the general population. An IgM-specific ELISA is suitable for detecting recent infection, but the relevance of IgM testing for acute VHF depends on the virus and the duration of illness; specific IgM is not often present in the very early stage of illness, and patients who die of VHF often fail to seroconvert at all. An IgG-specific ELISA is efficacious, not only in the diagnosis of a large number of VHF cases, especially during convalescence, but also for epidemiological studies in the endemic regions. The detailed methods used for the IFA and IgG-and IgM-ELISAs for the diagnosis of VHF using authentic virus-antigens have been described in detail [42] [43] [44] [45] .

Arenaviruses have a bisegmented, negative-sense, single stranded RNA genome with a unique ambisense coding strategy that produces just four known proteins: a glycoprotein, a nucleoprotein (NP), a matrix protein (Z), and a polymerase (L) [46] . Of these proteins, the NP is the most abundant in virus-infected cells. Recombinant protein technology could meet the demand for a simple and reliable VHF test system, and recombinant NP (rNP) has been shown to be useful for serological surveys of IgM-and IgG antibodies against arenaviruses [47] [48] [49] [50] .

Recombinant baculoviruses that express the full-length rNP of arenaviruses have been generated [48, 50, 51] . The method used for the purification of arenavirus rNP from insect Tn5 cells infected with recombinant baculoviruses is effective and simple compared to those for Ebola, Marburg, and Crimean-Congo hemorrhagic fever virus rNPs [51] [52] [53]

[54] [55] . Most of the arenavirus rNPs expressed in insect cells using the recombinant baculoviruses are crystallized [56] and are solubilized in PBS containing 8M urea. Since the majority of Tn5 cellular proteins are solubilized in PBS containing 2M urea, the arenavirus rNPs in the insoluble fraction in PBS containing 2M urea can be solubilized by sonication in PBS containing 8M urea. After a simple centrifugation of the lysates in PBS containing 8M urea, the supernatant fractions can be used as purified rNP antigens without further purification steps [51] . The control antigen is produced from Tn5 cells infected with baculovirus lacking the polyhedrin gene (ΔP) in the same manner as the arenavirus rNPs (Figure 1).

Purified rNPs. The expression and purification efficiency of arenavirus rNP were analyzed by sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) after staining the gels with Coomassie blue. Purified NP antigens with approximate molecular weights of 62 kDa from Luna, LCM, Lassa, Lujo, Junin, Machupo, Guanarito, Sabia, and Chapare viruses and the purified negative control antigen (ΔP) are shown.

As described above, recombinant baculoviruses allow the delivery of rNP antigens without using infectious live arenaviruses. An ELISA plate coated with the predetermined optimal quantity of purified rNPs (approximately 100 ng/well) is used for the IgG-antibody detection assay. An advantage of using recombinant rNP for the IgG-ELISA is that it enables a direct comparison of antibody cross-reactivity among arenavirus rNPs, since antigen preparations of all arenavirus rNPs tested are performed using the same method [51] . Rabbit anti-sera raised against LCMV-rNP and LASV-rNP show cross-reactivity to LASV-rNP and LCMV-rNP, respectively, indicating that rabbit antibodies against rNPs of Old World arenaviruses cross-react with rNPs of other Old World arenaviruses (Table 1) [51] . Similarly, rabbit anti-sera generated against JUNV-NP show cross-reactivity to the LASV-rNP and LCMV-rNP, although the reaction is weak. However, rabbit anti-sera against LASV-NP and LCMV-NP show a negative reaction to the JUNV-rNP (Table 1) [51] , indicating that rabbit antibodies against JUNV (a pathogenic New World arenavirus) NP might cross-react with the Old World arenavirus NP, whereas antibodies against Old World arenavirus NPs may not be able to react with pathogenic New World arenavirus NPs.

The rNP-based IgG-ELISA has also been used for the characterization of a mouse monoclonal antibody (MAb). Nakauchi et al. [50] have investigated the cross-reactivity of MAbs against JUNV rNP to pathogenic New World arenavirus rNPs, as well as LASV rNP.

MAb C11-12 reacts at the same level with the rNPs of all of the pathogenic New World arenaviruses, including JUNV, GTOV, MACV, SABV, and CHPV, indicating that this MAb recognizes an epitope conserved among pathogenic New World arenaviruses. Another MAb, C6-9, reacts specifically with the rNP of JUNV, but does not react with those of the other pathogenic New World arenaviruses [50] . This indicates that MAb C6-9 recognizes a JUNV-specific epitope. None of these MAbs reacts with the rNP of the human pathogenic Old World arenavirus LASV. Thus, the MAb C11-12 is considered to be a broadly reactive MAb against New World arenaviruses, whereas MAb C6-9 is JUNV-specific. These findings have been confirmed by detailed epitope analyses using peptide mapping [50] . Similarly, the cross-reactivity of MAbs against LASV rNP has been analyzed [51] . MAb 4A5 cross-reacts with the Mopeia virus (MOPV) but not with the LCMV rNP. MAb 6C11 cross-reacts with LCMV rNP, while MAb 2-11 does not cross-react with LCMV rNP [51] . Table 1 . Anti-serum reactivity for rNPs of different arenaviruses in IgG ELISAs.

Reactivity for rNP from LASV LCMV JUNV anti-LASV NP

It is important to evaluate whether rNP-based ELISA is useful for the diagnosis of human VHF cases. The specificity of the LASV-rNP-based IgG ELISA has been confirmed by using sera obtained from Lassa fever patients [51] . The Lassa fever patients' sera show a highly positive reaction in the LASV-rNP-based IgG-ELISA, but sera from patients with Argentine hemorrhagic fever (AHF), which is caused by JUNV, do not. The serum from an AHF patient showed a highly positive reaction in the JUNV-rNP-based IgG-ELISA [49] . In addition, it was shown that, using sera obtained from AHF cases, the results of the JUNV rNP-based IgG ELISA correlate well with an authentic JUNV antigen-based IgG ELISA [49] . An IgM-capture ELISA using purified LASV-rNP as an antigen has been developed in the same way as in previous reports [54, 57] and detects an LASV-IgM antibody [58] . In addition, immunoblot assays based on N-terminally truncated LASV rNP have been developed for detecting IgG and IgM antibodies against LASV. These methods may provide a rapid and simple Lassa fever test for use under field conditions [47] .

An IFA using virus-infected cells is a common antibody test for VHF viruses [59] [60] [61] [62] [63] . To avoid the use of highly pathogenic viruses for the antigen preparation, mammalian cells expressing recombinant rNP have been developed [51, 57, [64] [65] [66] [67] [68] . Lassa virus NP antigen for IFA can be prepared simply as described [51] . Briefly, the procedure involves (1) transfecting HeLa cells with a mammalian cell expression vector

inserted with the cloned NP cDNA; (2) expanding the stable NP-expressing cells by antibiotic selection; (3) mixing the rNP-expressing cells with un-transfected HeLa cells (at a ratio of 1:1); (4) spotting the cell mixtures onto glass slides, then drying and fixing them in acetone.

In the IFA specific for LASV-NP, antibody positive sera show characteristic granular staining patterns in the cytoplasm (Figure 2) [69], thus making it easy to distinguish positive from negative samples. The specificity of the assay has also been confirmed by using sera obtained from Lassa fever patients [51]. In addition, an IFA using JUNV rNP-expressing HeLa cells has been developed to detect antibodies against JUNV, and the assay has been evaluated by using AHF patients' sera [70]. The LASV-rNP-based antibody detection systems such as ELISA and IFA are suggested to be useful not only for the diagnosis of Lassa fever, but also for seroepidemiological studies of LASV infection. In our preliminary study, approximately 15% of the sera collected from 334 Ghanaians and less than 3% of 280 Zambians showed positive reactions in the LASV-rNP-based IgG ELISA [58]. These results are in agreement with the fact that Lassa fever is endemic to the West African region, including Ghana, but less in the East African region.

For the diagnosis of many viral infections, PCR assays have been shown to have an excellent analytical sensitivity, but the established techniques are limited by their requirement for expensive equipment and technical expertise. Moreover, the high degree of genetic variability of the RNA viruses, including arenavirus and bunyavirus, poses difficulties in selecting primers for RT-PCR assays that can detect all strains of the virus. Since the sensitivity of the Ag-capture ELISA is comparable to that of RT-PCR for several virus-mediated infectious diseases, including Lassa fever and filovirus hemorrhagic fever [51, [71] [72] [73], the Ag-capture ELISA is a sophisticated approach that can be used for the diagnosis of viral infections. Ag-capture ELISAs detecting viral NP in viremic sera have been widely applied to detect various viruses, since they are the most abundant viral antigens and have highly conserved amino acid sequences [50, 51, 54, 71, 72, 74, 75]. Polyclonal anti-sera or a mixture of MAbs present in the ascetic fluids from animals immunized for HFVs have been used for capture-antibodies in the Ag-capture ELISA [36, [76] [77] [78] [79]. MAbs recognizing conserved epitopes of the rNP are also used as capture antibodies since they have a high specificity for the antigens, and an identification of the epitopes of these MAbs is of crucial importance for the assessment of the specificity and cross-reactivity of the assay system [50, 51, 53, 54, 71, 75]. In order to develop a sensitive diagnostic test for Lassa fever and AHF, rNPs of LASV and JUNV (see above) have

been prepared, and newly established MABs against them have been characterized and used for Ag-capture ELISAs [50, 51] . The Ag-capture ELISA using MAb 4A5 has been confirmed to be useful in the detection of authentic LASV antigen in sera serially collected from hamsters infected with LASV [51] . The sensitivity of the MAb 4A5-based Ag-capture ELISA was similar to that of conventional RT-PCR, suggesting that the Ag-capture ELISA can be efficiently used in the diagnosis of Lassa fever [51] . Therefore, the MAb 4A5-based Ag-capture ELISA is considered to be useful in the diagnosis of Lassa fever. Also, by using MABs raised against the rNP of JUNV, Ag-capture ELISAs specific for JUNV and broadly reactive to human pathogenic New World arenaviruses have been developed [50] . The Ag-capture ELISA using MAb E4-2 and C11-12 detected the Ags of all of the pathogenic New World arenaviruses tested, including JUNV. On the other hand, the Ag-capture ELISA using MAb C6-9 detects only the JUNV Ag. Considering that the symptoms of JUNV infection in humans are indistinguishable from those due to other pathogenic New World arenaviruses, the Ag capture ELISA using MAb C6-9 may be a useful diagnostic tool, especially for AHF [50] .

The virus neutralization assay is accepted as the "gold standard" serodiagnostic assay to quantify the antibody response to infection and vaccination of a wide variety of viruses associated with human diseases [80] [81] [82] [83] [84] [85] [86] . The presence of neutralizing antibodies is a reliable indicator of protective immunity against VHF [87] [88] [89] . The most direct method for detection of neutralizing antibodies against HFVs is by plaque reduction neutralization tests using infectious viruses. However, because of the high pathogenicity of HFVs to humans and the strict regulation of select agents, only a limited number of laboratories are able to perform such neutralization tests. For many HFVs, replication-incompetent pseudotyped virus particles bearing viral envelope protein (GP) have been shown to mimic the respective HFV infections, thus, neutralization assays using the pseudotypes may be advantageous in some laboratory settings for the detection of antibodies to HFVs without the need for heightened biocontainment requirements.

The VSV-based vector has already been used to generate replication-competent recombinant VSVs to study of the role of GPs of various viruses [90] [91] [92] . Recent advances in producing pseudotype virus particles have enabled the investigation of the virus cell entry, viral tropism, and effect of entry inhibitors, as well as measurement of the neutralization titers, by using human immunodeficiency virus-, feline immunodeficiency virus-, murine leukemia virus-, or VSV-based vectors [86, [93] [94] [95] [96] [97] [98] [99] [100] [101] [102] [103] . Pseudotypes based on VSV have advantages compared with other

pseudotypes based on retroviruses for the following reasons. First, the pseudotype virus titer obtained with the VSV system is generally higher than that of the pseudotyped retrovirus system [104] . Second, the infection of target cells with a VSV pseudotype can be readily detected as green fluorescent protein (GFP)-positive cells at 7-16 h post-infection because of the high level of GFP expression in the VSV system [104, 105] . In contrast, the time required for infection in the pseudotyped retrovirus system is 48 h [106, 107] , which is similar to the time required for infectious viruses to replicate to a level that results in plaque-forming or cytopathic effects in infected cells. A high-throughput assay for determining neutralizing antibody titers using VSV pseudotypes expressing secreted alkaline phosphatase [108, 109] or luciferase (Figure 3) has also been developed. We have recently developed a VSV-based pseudotype bearing Lassa virus GP (VSV-LAS-GP) for the detection of neutralizing antibodies in the sera obtained from a Lassa fever patient. An example of the LASV neutralization assay using the VSV pseudotype is shown (Figure 4). In the presence of serum from Lassa fever patients, the number of GFP-positive cells (infectivity of VSV-LAS-GP) is significantly reduced compared with the number in the absence of the patient's serum (Figure 4A). The control VSV pseudotype bearing VSV GP (VSV-VSV-G) is not neutralized by any sera. When the cut-off serum dilution is set at 50% inhibition of infectivity compared with the infectivity in the absence of the test serum, the neutralization titer of this patient's serum for VSV-LAS-GP is calculated to be 75 (Figure 4B). Likewise, a VSV-based pseudotype bearing the Junin virus GP has been developed for the detection of neutralizing antibodies from AHF patients' sera. The accuracy of the results of VSV-based neutralization assays has been confirmed by comparison with the results of the neutralization assay using live Junin virus [70] . The Lujo virus is a new member of the hemorrhagic fever-associated arenavirus family from Zambia and southern Africa, and the virus is classified as a BSL-4 pathogen [17] . The genome sequence analysis of the Lujo virus suggests that the virus is genetically distinct from previously characterized arenaviruses. In order to study the infectivity of this newly identified arenavirus, we have recently developed a luciferase-expressing VSV pseudotype bearing Lujo virus GPC (VSV-Lujo-GP). As shown in Figure 3 , infection with VSV-Lujo-GPC is specifically neutralized by rabbit anti-Lujo GPC serum. Thus, the VSV-Lujo-GP may be a useful tool not only for determining the neutralizing antibody titer within the serum, but also for exploring yet-to-be-defined cellular receptor(s) for Lujo virus infection or for screening inhibitors of the Lujo virus GP-mediated cell entry.

Hemorrhagic fever outbreaks caused by pathogenic arenaviruses result in high fatality rates. A rapid and accurate diagnosis is a critical first step in any outbreak. Serologic diagnostic methods for VHFs most often employ an ELISA, IFA, and/or virus neutralization

assay. Diagnostic methods using recombinant viral proteins have been developed and their utilities for diagnosing of VHF have been reviewed. IgG-and IgM-ELISAs and IFAs using rNPs as antigens are useful for the detection of antibodies induced in the patients' sera. These methods are also useful for seroepidemiological surveys for HFVs. Ag-capture ELISAs using MAbs to the arenavirus rNPs are specific for the virus species or can be broadly reactive for New World arenaviruses, depending on the MAb used. Furthermore, the VSV-based pseudotype system provides a safe and rapid tool for measuring virus neutralizing antibody titers, as well as a model to analyze the entry of the respective arenavirus in susceptible cells without using live arenaviruses. Recent discoveries of novel arenavirus species [17, 26, 110] and their potential to evolve predominantly via host switching, rather than with their hosts [110, 111] , suggest that an unknown pathogenic arenavirus may emerge in the future, and that the diagnostic methods for VHF caused by arenaviruses should thus be further developed and improved.

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Glycyrrhizin Exerts Antioxidative Effects in H5N1 Influenza A Virus-Infected Cells and Inhibits Virus Replication and Pro-Inflammatory Gene Expression

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Abstract: Glycyrrhizin is known to exert antiviral and anti-inflammatory effects. Here, the effects of an approved parenteral glycyrrhizin preparation (Stronger Neo-Minophafen C) were investigated on highly pathogenic influenza A H5N1 virus replication, H5N1-induced apoptosis, and H5N1-induced pro-inflammatory responses in lung epithelial (A549) cells. Therapeutic glycyrrhizin concentrations substantially inhibited H5N1-induced expression of the pro-inflammatory molecules CXCL10, interleukin 6, CCL2, and CCL5 (effective glycyrrhizin concentrations 25 to 50 µg/ml) but interfered with H5N1 replication and H5N1-induced apoptosis to a lesser extent (effective glycyrrhizin concentrations 100 µg/ml or higher). Glycyrrhizin also diminished monocyte migration towards supernatants of H5N1-infected A549 cells. The mechanism by which glycyrrhizin interferes with H5N1 replication and H5N1-induced pro-inflammatory gene expression includes inhibition of H5N1-induced formation of reactive oxygen species and (in turn) reduced activation of NFκB, JNK, and p38, redox-sensitive signalling events known to be relevant for influenza A virus replication. Therefore, glycyrrhizin may complement the arsenal of potential drugs for the treatment of H5N1 disease.

Text: Highly pathogenic H5N1 influenza A viruses are considered to be potential influenza pandemic progenitors [1] [2] [3] [4] [5] [6] . At least for the first wave of an H5N1 pandemic, no sufficient amounts of adequate vaccines will be available [1] [2] [3] [4] [6] [7] [8] . Therefore, antiviral therapy for influenza A viruses including highly pathogenic H5N1 virus strains remains of great importance for the first line defense against the virus [1] [2] [3] [4] [6, 9] .

The neuraminidase inhibitors oseltamivir and zanamivir as well as the adamantanes amantadin and rimantadin that interfere with the influenza M2 protein are licensed for the treatment of influenza [1] [2] [3] [4] [6] . However, the use of both drug classes is limited by the emergence of resistant virus strains. In seasonal influenza strains, the majority of H3N2 viruses and a great proportion of H1N1 viruses in humans are now considered to be amantadine- and rimantadine-resistant [10] [11] [12] [13] . Moreover, a drastic increase in oseltamivir-resistant H1N1 viruses has been reported during the 2007/2008 influenza season in the northern hemisphere [14] [15] [16] [17] . Preliminary data from the United States predict a further rise for the 2008/2009 season, possibly resulting in more than 90% of the circulating H1N1 strains to be oseltamivir resistant [14] .

H5N1 virus strains appear to be generally less sensitive to antiviral treatment than seasonal influenza A virus strains and treatment-resistant H5N1 strains emerge [1] [2] [3] [4] 6, [18] [19] [20] [21] . More-over, parenteral agents for the treatment of seriously ill patients are missing. Glycyrrhizin, a triterpene saponine, is a constituent of licorice root. It has been found to interfere with replication and/or cytopathogenic effect (CPE) induction of many viruses including respiratory viruses such as respiratory syncytial virus, SARS coronavirus, HIV, and influenza viruses [22] [23] [24] [25] [26] [27] [28] . Moreover, antiinflammatory and immunomodulatory properties were attributed to glycyrrhizin [26] . The severity of human H5N1 disease has been associated with hypercytokinaemia ("cytokine storm") [29, 30] . Delayed antiviral plus immunomodulator treatment reduced H5N1-induced mortality in mice [31] . Therefore, antiinflammatory and immunomodulatory effects exerted by glycyrrhizin may be beneficial for treatment of H5N1. Also, glycyrrhizin is a known antioxidant [26] and antioxidants were already shown to interfere with influenza A virus replication and virus-induced pro-inflammatory responses [32] [33] [34] .

Stronger Neo-Minophagen C (SNMC) is a glycyrrhizin preparation (available as tablets or parenteral formulation) that is approved in Japan for the treatment of chronic hepatic diseases and is marketed in Japan, China, Korea, Taiwan, Indonesia, India, and Mongolia. Here, we investigated the influence of SNMC on H5N1 replication, on H5N1-induced cytokine expression, on H5N1-induced cellular oxidative stress, and on critical H5N1-induced cellular signalling events in human pneumocytes (A549 cell line).

Glycyrrhizin (Stronger Neo Minophagen C) was obtained from Minophagen Pharmaceuticals Co., Ltd. (Tokyo, Japan).

The influenza strain A/Vietnam/1203/04 (H5N1) was received from the WHO Influenza Centre (National Institute for Medical Research, London, UK). The H5N1 influenza strain A/Thailand/ 1(Kan-1)/04 was obtained from Prof. Pilaipan Puthavathana (Mahidol University, Bangkok, Thailand).

Virus stocks were prepared by infecting Vero cells (African green monkey kidney; ATCC, Manassas, VA) and aliquots were stored at 280uC. Virus titres were determined as 50% tissue culture infectious dose (TCID 50 /ml) in confluent Vero cells in 96-well microtiter plates.

A549 cells (human lung carcinoma; ATCC: CCL-185, obtained from LGC Standards GmbH, Wesel, Germany) were grown at 37°C in minimal essential medium (MEM) supplemented with 10% FBS, 100 IU/ml of penicillin and 100 mg/ml streptomycin.

Human monocytes were isolated from buffy coats of healthy donors, obtained from Institute of Transfusion Medicine and Immune Haematology, German Red Cross Blood Donor Center, Johann Wolfgang Goethe-University, Frankfurt am Main. After centrifugation on Ficoll (Biocoll)-Hypaque density gradient (Biochrom AG, Berlin, Germany), mononuclear cells were collected from the interface and washed with PBS. Then, monocytes were isolated using magnetically labeled CD14 MicroBeads (Miltenyi Biotec GmbH, Bergisch Gladbach, Germany) following the manufacturer's instructions. Monocytes were cultivated in IMDM supplemented with 10% pooled human serum, 100 IU/ml of penicillin, and 100 mg/ml streptomycin.

The cellular viability was assessed on confluent cell layers with CellTiter-GloH Luminescent Cell Viability Assay (Promega GmbH, Mannheim, Germany) according to the manufacturers' protocol. Cell viability was expressed as percentage of non-treated control.

To determine intracellular NP localisation, H5N1-infected A549 were fixed 8 hours p.i. for 15 min with ice-cold acetone/ methanol (40:60, Mallinckrodt Baker B.V., Deventer, The Netherlands) and stained with a mouse monoclonal antibody (1 h incubation, 1:1000 in PBS) directed against the influenza A virus nucleoprotein (NP) (Millipore, Molsheim, France). An Alexa Fluor 488 goat anti-mouse IgG (H&L) (Invitrogen, Eugene, Oregon, USA) was used (1 h incubation, 1:1000 in PBS) as secondary antibody. Nuclei were stained using 4',6-diamidino-2-phenylindole (DAPI) (Sigma-Aldrich Chemie GmbH, Munich, Germany). Fluorescence was visualised using Olympus IX 1 fluorescence microscope (Olympus, Planegg, Germany).

For flow cytometric analysis, the same antibodies were used.

The cytopathogenic effect (CPE) reduction assay was performed as described before [34]. Confluent A549 cell monolayers grown in 96-well microtitre plates were infected with

influenza A strains at the indicated multiplicities of infection (MOIs). After a one hour adsorption period, cells were washed to remove non-detached virus. The virus-induced CPE was recorded at 24 h post infection (p.i.).

Unless otherwise stated, A549 cells were continuously treated with glycyrrhizin starting with a 1 h pre-incubation period. For time-of-addition experiments, glycyrrhizin was added exclusively during the 1 h pre-incubation period, exclusively during the 1 h adsorption period, or after exclusively after the wash-out of input virus.

Total RNA was isolated from cell cultures using TRI reagent (Sigma-Aldrich, Munich, Germany). Real time PCR for H5 was performed using described methods [35]. The following primers were used: sense 5'-acg tat gac tac ccg cag tat tca g-3'; antisense 5'-aga cca gcy acc atg att gc-3'; probe 6-FAM-tca aca gtg gcg agt tcc cta gca-TAMRA.

The fraction of cells with fractional DNA content ("sub-G1" cell subpopulation) indicates cytotoxicity. Sub-G1 cells are considered to be dead (usually apoptotic) cells. Cells were fixed with 70% ethanol for two hours at 220°C. The cellular DNA was stained using propidium iodide (20 mg/ml) and analysed by flow cytometry (FacsCalibur, BD Biosciences, Heidelberg, Germany).

Caspase activation was measured using the Caspase-Glo 8, 9, or 3/7 Assays (Promega, Mannheim, Germany) following the manufacturer's instructions.

Cell culture supernatants were collected and frozen at -80°C. Cytokines/chemokines were quantified by specific ELISA Duo Sets (R&D Systems GmbH, Wiesbaden, Germany) following the manufacturer's instructions.

NFκB activity was investigated in H5N1 (MOI 0.01)-infected cells by quantification of the NFκB subunits Rel A (p65) and NFκB1 (p50) from nuclear extracts using the TransAM™ transcription factor DNA-binding ELISAs (Active Motif, Rixensart, Belgium). Nuclear extract were prepared using the Nuclear Extract Kit (Active Motif, Carlsbad, CA, USA) following the manufacturer's instruction.

Cell culture supernatants were investigated for chemotactic activity by measurement of the activity to induce monocyte migration through membrane inserts in 24-well plates (pore size 8 µm; BD Biosciences, Heidelberg, Germany). Monocytes (1610 6 in 100 µl of IMDM with 10% pooled human serum) were added into the cell culture inserts (upper chamber) and cell culture supernatants (300 µl), were added to the lower chamber of the well. After a 48 h incubation period, cells were fixed with 4% paraformaldehyde and permeabilised with PBS containing 0.3% Triton X-100. Then, nuclei were stained with 49,6-diamidino-2phenylindole (DAPI). The upper side of the membrane was wiped with a wet swab to remove the cells, while the lower side of the membrane was rinsed with PBS. The number of cells at the lower side of each membrane was quantified by counting of cells from three randomly chosen sections (3.7 mm²) using an Olympus IX 1 fluorescence microscope (Olympus, Planegg, Germany).

Cells were lysed in Triton X-sample buffer and separated by SDS-PAGE. Nuclear extract were prepared using the Nuclear Extract Kit (Active Motif, Carlsbad, CA, USA) following the manufacturer's instruction. Proteins were detected using specific antibodies against bactin (Sigma-Aldrich Chemie GmbH, Munich, Germany), JNK, phosphorylated JNK, p38, or phosphorylated p38, (all purchased from New England Biolabs GmbH, Frankfurt am Main, Germany) and were visualised by enhanced chemiluminescence using a commercially available kit (Amersham, Freiburg, Germany).

Reactive oxygen species (ROS) were detected using the Image-iT LIVE Green Reactive Oxygen Species Kit (Molecular Probes, distributed by Invitrogen, Karlsruhe, Germany).

Two groups were compared by t-test. More groups were compared by ANOVA with subsequent Student-Newman-Keuls test.

The A549 cell line, derived from a human pulmonary adenocarcinoma, is an established model for type II pneumocytes [36], and commonly used for the investigation of the effect of influenza viruses on this cell type [see e.g. 6,37,38]. If not otherwise stated, glycyrrhizin was continuously present in cell culture media starting with a 1 h preinfection period. Glycyrrhizin 200 µg/ml (the maximum tested concentration) did not affect A549 cell viability (data not shown) but clearly decreased CPE formation in A549 cells infected with

the H5N1 influenza strain A/Thailand/1(Kan-1)/04 at MOIs of 0.01, 0.1 or 1 (Figure 1A). Similar results were obtained in A549 cells infected with strain A/Vietnam/1203/04 (H5N1) (Suppl. Figure 1A) . Staining of A549 cells for influenza A nucleoprotein 24 h after infection with strain H5N1 A/Thailand/1(Kan-1)/04 indicated that glycyrrhizin 200 mg/ml significantly reduces the number of influenza A nucleoprotein positive cells (Figure 1B) .

To examine the influence of glycyrrhizin on virus progeny, A549 cells were infected with the H5N1 influenza strain A/ Thailand/1(Kan-1)/04 at MOI 0.01 or MOI 1 and infectious virus titres were determined 24 h post infection (Figure 1C). While glycyrrhizin in concentrations up to 50 mg/ml did not affect H5N1 replication, moderate effects were exerted by glycyrrhizin 100 mg/ ml and more pronounced effects by glycyrrhizin 200 mg/ml (MOI 0.01: 13-fold reduction, MOI 1: 10-fold reduction). Next, influence of glycyrrhizin on H5N1 replication was confirmed by the detection of viral (H5) RNA using quantitative PCR. Only glycyrrhizin concentrations $\geq$ 100 mg/ml significantly reduced Figure 1B) or H5N1 A/Vietnam/1203/04-infected (Suppl. Figure 1C) A549 cells (MOI 0.01) 24 h post infection.

Time-of-addition experiments revealed that maximal effects were achieved when glycyrrhizin was continuously present starting with a 1 h pre-incubation period (Figure 1D). Addition of glycyrrhizin post infection showed reduced antiviral effects while pre-incubation alone or glycyrrhizin addition during the adsorption period did not significantly affect H5N1 replication.

For investigation of H5N1-induced cytokine expression, five pro-inflammatory genes were chosen that had been correlated to severity of influenza disease: CXCL10 (also known as interferon-inducible protein 10, IP-10), interleukin 6 (IL6), interleukin 8, (IL8; also known as CXCL8), CCL2 (also known as monocyte chemoattractant protein 1, MCP-1), and CCL5 (also known as RANTES). A549 cells were infected with H5N1 A/Thailand/ 1(Kan-1)/04 or H5N1 A/Vietnam/1203/04 at MOI 0.01, 0.1, or 1. Glycyrrhizin treatment was performed with 25, 50, 100, or 200 mg/ml. Cytokine expression was detected 24 h post infection by ELISA. Glycyrrhizin did not affect cytokine expression of noninfected cells (data not shown) but inhibited expression of all cytokines investigated in H5N1-infected cells in a dose-dependent manner (Figure 2, Figure 3A). Effects were more pronounced at lower MOIs. Notably, expression of all cytokines except IL8 was significantly inhibited after treatment with glycyrrhizin 50 mg/ml Figure 3A) although these glycyrrhizin concentrations had no effect on H5N1 replication in A549 cells (Figure 1, Figure S1).

Cytokine expression by influenza A virus-infected respiratory cells causes recruitment of peripheral blood monocytes into the lungs of patients where they differentiate to macrophages which are thought to contribute to influenza A virus pathogenicity [5, 39] . In a chemotaxis assay, the influence of glycyrrhizin was investigated on migration of monocytes towards supernatants of H5N1 A/Thailand/1(Kan-1)/04 (MOI 0.1)-infected A549 cells through 8 mm filters. Monocyte migration towards supernatants of H5N1-infected cells was strongly increased relative to migration towards supernatants of non-infected cells. Treatment of H5N1- infected cells with glycyrrhizin 100 mg/ml clearly suppressed chemoattraction activity of supernatants (Figure 3B).

Influenza viruses including H5N1 have been shown to induce caspase-dependent apoptosis in airway cells and this apoptosis has been correlated to the virus pathogenicity [40, 41] . Glycyrrhizin concentrations up to 200 mg/ml did not affect caspase activation in non-infected cells (Figure 4A-C) . Glycyrrhizin concentrations 100 mg/ml inhibited H5N1 A/Thailand/1(Kan-1)/04 (MOI 0.01)-induced activation of the initiator caspases 8 and 9 as well as of the effector caspases 3/7 in A549 cells as determined 24 h post infection (Figure 4A-C) . Lower glycyrrhizin concentrations did not affect H5N1-induced apoptosis. The detection of cells in sub-G1 phase resulted in similar findings (Figure 4D) .

Substances that inhibit H5N1-induced caspase 3 activation including caspase 3 inhibitors cause nuclear retention of RNP complexes [34, 42] . In accordance, glycyrrhizin also interfered with nuclear export RNP at MOI 1 (Figure S2) . Similar results were obtained in MOI 0.01 H5N1 A/Thailand/1(Kan-1)/04infected cells (Figure S3) .

Influence of glycyrrhizin on H5N1-induced activation of nuclear factor κ B (NF κ B), p38, and on H5N1-induced cellular reactive oxygen species (ROS) formation Activation of NF κ B, p38, and JNK have been associated with influenza A virus replication and virus-induced pro-inflammatory gene expression [34, [43] [44] [45] [46] [47] . While glycyrrhizin did not influence NF κ B activity in non-infected A549 cells in the tested concentrations (data not shown), glycyrrhizin inhibited NF κ B activation in H5N1-infected cells (Figure 5A) . Moreover, glycyrrhizin inhibited H5N1-induced phosphorylation of the MAPKs p38 and JNK (Figure 5B) .

In addition to their roles during influenza A virus replication and virus-induced cytokine/chemokine expression, NFkB, p38, and JNK are constituents of redox-sensitive signalling pathways [48] [49] [50] [51] . Antioxidants had been already found to interfere with influenza A virus-induced signalling through NFkB, p38, and JNK, with influenza A virus replication, and with influenza A virus-induced pro-inflammatory gene expression [32] [33] [34] . Since glycyrrhizin is known to exert antioxidative effects [26] we speculated that glycyrrhizin may interfere with H5N1-induced ROS formation. Indeed glycyrrhizin exerted clear antioxidative effects in H5N1 (MOI 0.01)-infected cells (Figure 5C) causing significant reduction of ROS formation already at a concentration of 25 mg/ml (Figure 5D).

Here, we show that glycyrrhizin inhibits the replication of highly pathogenic H5N1 influenza A virus, H5N1-induced apoptosis, and H5N1-induced expression of pro-inflammatory cytokines in lung-derived A549 cells. After intravenous administration, achievable plasma concentrations of glycyrrhizin have been described to be about 100 mg/ml [52] . Therefore, the glycyrrhizin concentrations found to interfere with H5N1 replication and H5N1-induced pro-inflammatory gene expression in the present report are in the range of therapeutic plasma levels. Notably, although higher glycyrrhizin concentrations were needed to interfere with SARS coronavirus replication [22] than with H5N1 replication, beneficial results were reported in glycyrrhizin (SNMC)-treated SARS patients in comparison to SARS patients who did not receive glycyrrhizin [23] . Notably, investigation of different glycyrrhizin derivatives against SARS coronavirus led to the identification of compounds with enhanced antiviral activity [53] . Therefore, glycyrrhizin might also serve as lead structure for the development of novel anti-influenza drugs.

Experimental results suggested that glycyrrhizin might be able to affect seasonal influenza A virus disease by antiviral and immunomodulatory effects [26, 27] . Mice were prevented from lethal H2N2 infection by glycyrrhizin although no influence on virus replication was detected. The mechanism was suggested to be induction of interferon- γ in T-cells by glycyrrhizin [54] . Moreover, glycyrrhizin was shown to influence seasonal influenza A virus replication through interaction with the cell membrane [25, 28] . However, these effects were observed only in concentrations ≥ 200 mg/ml when glycyrrhizin was added during the virus adsorption period. Since glycyrrhizin addition during the adsorption period did not influence H5N1 replication in our experiments it appears not likely that membrane effects contribute to anti-H5N1 effects detected here in lower concentrations.

Our results rather suggest that glycyrrhizin interferes with H5N1-induced oxidative stress. Influenza A virus (including H5N1) infection induces ROS formation. Antioxidants were found to inhibit influenza A virus replication and influenza A virus-induced pro-inflammatory gene expression [32] [33] [34] and glycyrrhizin is known to exert antioxidative effects [26] . Here, glycyrrhizin interfered with H5N1-induced activation of NFkB, p38, and JNK representing redox-sensitive signalling events [48] [49] [50] [51] involved in influenza A virus replication and influenza A virus-induced cellular cytokine/chemokine production [34, [43] [44] [45] [46] 55] . Glycyrrhizin 50 mg/ml significantly reduced H5N1-induced activation of NFkB. In addition, glycyrrhizin concentrations as low as 25 mg/ml effectively interfered with H5N1-induced ROS formation and with phosphorylation of the redox-sensitive MAPKs p38 and JNK. In our model, activation of p38 appears to be critical for H5N1-associated redox signalling since p38 inhibition had been shown before to mimic effects of the antioxidant N-acetyl-cysteine (NAC) [34] . Interestingly and in contrast to glycyrrhizin, NAC failed to inhibit H5N1 replication or H5N1-induced cytokine/chemokine expression in therapeutically relevant concentrations.

Glycyrrhizin diminished H5N1-induced cellular cytokine/ chemokine production in concentrations (#50 mg/ml) that did not interfere with H5N1 replication although redox-sensitive signalling pathways have been described to be involved in both processes. Therefore, H5N1-induced proinflammatory gene expression appears to be more sensitive to inhibition of ROS formation than H5N1 replication. Indeed, influenza viruses had been shown to induce cellular pathways through replication-dependent and -independent events [56] . In a previous report, we could show that similar glycyrrhizin concentrations like those investigated here interfered with H5N1-induced pro-inflammatory gene expression but not with H5N1 replication in human monocyte-derived macrophages [57] . In addition, other immunomodulatory treatment regimens that did not influence H5N1 replication reduced mortality in H5N1-infected mice [31, 58] . Therefore, glycyrrhizin represents a potential additional treatment option that interferes with both H5N1 replication and H5N1-induced expression of pro-inflammatory cytokines in lung cells.

Interference with immune responses may also result in the loss of control of virus replication by cytotoxic immune cells including natural killer cells and cytotoxic CD8 + T-lymphocytes. Global immunosuppressants like corticosteroids failed to protect from lethal influenza virus infection [59] . Moreover, antiviral drugs may interfere with cytotoxic cells that control virus replication as demonstrated for ribavirin that was shown to hamper NK

cell cytolytic activity [60] . In this context, glycyrrhizin had already been shown not to affect natural killer cell activity in the concentrations used here [57] .

In conclusion, we show in this report that therapeutic concentrations of glycyrrhizin (used as clinically approved parenteral preparation SNMC) interfere with highly pathogenic H5N1 influenza A virus replication and H5N1-induced proinflammatory gene expression at least in part through interference with H5N1-induced ROS formation and in turn reduced activation of p38, JNK, and NFkB in lung cells. Since we used the clinical formulation SNMC effects of other ingredients like glycine or cysteine cannot be excluded. Vaccines and antiviral agents will fail to meet global needs at least at the beginning of a severe influenza A virus pandemic [61] . Anti-inflammatory and immunomodulatory agents are considered to be important candidates as constituents of anti-influenza treatment strategies that may save lives in an influenza pandemic situation [61] . Therefore, glycyrrhizin may complement the arsenal of potential drugs for the treatment of H5N1-caused disease.

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Emergent severe acute respiratory distress syndrome caused by adenovirus type 55 in immunocompetent adults in 2013: a prospective observational study

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Abstract: **INTRODUCTION:** Since 2008, severe cases of emerging human adenovirus type 55 (HAdV-55) in immunocompetent adults have been reported sporadically in China. The clinical features and outcomes of the most critically ill patients with severe acute respiratory distress syndrome (ARDS) caused by HAdV-55 requiring invasive mechanical ventilation (IMV) and/or extracorporeal membrane oxygenation (ECMO) are lacking. **METHODS:** We conducted a prospective, single-center observational study of pneumonia with ARDS in immunocompetent adults admitted to our respiratory ICU. We prospectively collected and analyzed clinical, laboratory, radiological characteristics, sequential tests of viral load in respiratory tract and blood, treatments and outcomes. **RESULTS:** The results for a total of five consecutive patients with severe ARDS with confirmed HAdV-55 infection were included. All five patients were immunocompetent young men with a median age of 32 years. The mean time from onset to dyspnea was 5 days. Arterial blood gas analysis at ICU admission revealed profound hypoxia. Mean partial oxygen pressure/fraction of inspired oxygen was 58.1. Mean durations from onset to a single-lobe consolidation shown on chest X-rays (CXRs) and, from the first positive CXR to bilateral multilobar lung infiltrates, were 2 days and 4.8 days, respectively. The viral load was higher than 1×10^8 copies in three patients and was 1×10^4 in one patient. It was negative in the only patient who survived. The mean duration for noninvasive positive pressure ventilation (NPPV) failure and IMV failure were 30.8 hours and 6.2 days, respectively. Four patients received venovenous ECMO. Four (80%) of the five patients died despite receiving appropriate respiratory support. **CONCLUSIONS:** HAdV-55 may cause severe ARDS in immunocompetent young men. Persistent high fever, dyspnea and rapid progression to respiratory failure within 2 weeks, together with bilateral consolidations and infiltrates, are the most frequent clinical manifestations of HAdV-55-induced severe ARDS. Viral load monitoring may help predict disease severity and outcome. The NPPV and IMV failure rates were very high, but ECMO may still be the respiratory support therapy of choice. **TRIAL REGISTRATION:** Clinicaltrials.gov NCT01585922. Registered 20 April 2012

Text: Human adenoviruses (HAdVs) are notorious pathogens in people with compromised immune function and a frequent cause of outbreaks of acute respiratory disease among young children. Life-threatening adenoviral pneumonia has previously been documented among military trainees, patients with AIDS and transplant recipients [1] [2] [3] [4] [5] . Human adenovirus type 55 (HAdV-55), which is emerging as a highly virulent pathogen for acute fatal adenoviral pneumonia among immunocompetent adults in China, has gained increasing attention [6] . HAdV-55 is a newly identified, emergent acute respiratory disease

pathogen causing two recent outbreaks in China in 2006 [7] and in Singapore in 2005 [8] . In 2011, this pathogen apparently re-emerged in Beijing, China, causing several cases of severe community-acquired pneumonia [9] . This pathogen was fully characterized by whole-genome sequencing [10] . Comparative studies showed that the ability of HAdV to cause severe disease may relate to the serotypes of HAdVs. Severe adenoviral pneumonia induced by HAdV-55 has been reported to be more closely related to severe cases compared to other serotypes (HAdV-3, HAdV-7 and HAdV-14) [6] .

Current knowledge of HAdV-55-induced severe acute respiratory distress syndrome (ARDS) requiring invasive mechanical ventilation and/or extracorporeal membrane oxygenation (ECMO) support in immunocompetent adults is derived from single case reports or relatively small, single-center series. As a result, little information is available on HAdV-55 pneumonia complicated with severe ARDS, the frequency of which is expected to increase in the coming years. Here we describe the clinical features and outcomes of five prospective cases of HAdV-55 pneumonia complicated with severe ARDS in immunocompetent adults in our ICU.

Beginning in May 2012, a randomized trial of noninvasive positive pressure ventilation (NPPV) in ARDS patients was carried out in our center (ClinicalTrials.gov ID: NCT01585922). From May 2012 to April 2014, all adult patients with ARDS caused by pneumonia who were admitted to the respiratory ICU of Beijing Chao-Yang Hospital were prospectively enrolled. Severe ARDS was diagnosed according to the Berlin definition: (1) developing within 1 week of a known clinical insult or new or worsening respiratory symptoms; (2) bilateral opacities not fully explained by effusions, lobar and/or lung collapse, or nodules; (3) respiratory failure not fully explained by cardiac failure or fluid overload; (4) partial oxygen pressure/ fraction of inspired oxygen ($\text{PaO}_2 / \text{FiO}_2$) ≤ 100 mmHg with positive end-expiratory pressure (PEEP) ≥ 5 cmH₂O; and (5) a chest radiograph with three or four quadrants with opacities. Patients with HAdV-55 infection and severe ARDS who failed conventional NPPV and invasive mechanical ventilation (IMV) were included in the analysis. This study was approved by the Institutional Review Board of Beijing Chao-Yang Hospital (LLKYPJ2012031). Data were analyzed anonymously. Each patient gave written informed consent for their data to be used for research and publication.

Clinical information collected by investigators with a standardized data form included the following: demographic characteristics (age and sex), comorbidities, clinical symptoms (fever, cough, sputum, dyspnea, chest pain, rash, nausea, vomiting, abdominal pain, diarrhea and headache), signs (body temperature, heart rate, respiratory frequency, blood pressure and crackles in the lungs), laboratory tests (whole-blood cell count and blood chemistry) and microbiological findings and images of the lung (chest X-ray (CXR) and computed tomography). Concomitant medications, respiratory support, complications and outcomes were also recorded.

Patients' specimens, including sputum, whole blood and serum samples, were collected upon admission and during hospitalization. Microbiological tests were performed at the Department of Infectious Disease and Clinical Microbiology in our center, and the detection methods used were described in our previous report [6] . Common viruses causing respiratory illness were screened using a kit with 15 different viral assays. Serum samples were used for *Mycoplasma pneumoniae*, *Chlamydia pneumoniae* and *Legionella pneumophila* antibodies. All patients had their HAdV-55 infection confirmed by RT-PCR assay. Partial sequences of the hexon gene were analyzed to type the phylogeny of HAdV-55 strains. The adenoviral load was also performed on both respiratory specimens and blood by multiplex RT-PCR assay.

Viral pneumonia was diagnosed based on the presence of HAdV detected in sputum or throat swab samples by molecular methods.

Continuous variables were summarized as mean $\pm$ standard deviation (SD) or median (interquartile range).

During the study period, a total of eight patients diagnosed with HAdV infection and respiratory failure were admitted to our ICU, and seven of them received a diagnosis of ARDS. Five consecutive patients with severe ARDS with confirmed HAdV-55 infection were admitted to our ICU between April and July 2013. They were included in the analysis. The other two patients had mild ARDS and were infected with other types of HAdVs.

All five patients were immunocompetent young men with a median age of 32 years (range, 28 to 40 years). All of the patients shared a B blood type and came from the same city: Baoding city, Hebei province, northern China. All patients had no exposure to farm animals, corn or hay. Patient 3 had tuberculosis pleuritis and received antituberculosis therapy at ICU admission. His blood tests, including the T-SPOT tuberculosis assay (Oxford Immunotec, Marlborough, MA, USA) and antibody of *Mycobacterium tuberculosis*, were negative.

Flulike symptoms, such as fever, cough and little sputum, were commonly observed at the onset of illness. All patients presented with a high fever, with a mean body temperature of 39.5°C (range, 39.0°C to 40.0°C), which persisted for 8 days (range, 6 to 11 days). Productive cough was observed in two patients. Dull substernal chest pain and rash were also observed in two patients. All patients had dyspnea. The mean time from onset to dyspnea was 5 days (range, 1 to 10 days). After the onset of dyspnea, patients usually progressed to respiratory failure or hypoxemia. The mean time from onset to ICU admission was 9.6 days (range, 8 to 11 days) (Table 1) . All patients had tachypnea when admitted to the ICU, with a mean rate of 43 breaths per minute (range = 38 to 52). Arterial blood gas analysis at ICU admission revealed profound hypoxia, with a mean PaO₂ /FiO₂ of 58.1 (range = 49 to 62.5). White blood cell counts were low or in the normal range. All patients had elevated serum aspartate aminotransferase (AST), lactate dehydrogenase (LDH) and hydroxybutyrate dehydrogenase (HBDH) (Table 1) . At admission, all patients' levels of immunoglobulin (serum immunoglobulins G and M) and components C3 and C4 were in the normal range.

Four patients had lower than normal T-cell subset counts (Table 2) .

CXRs revealed multiple bilateral lobar or segment consolidation in the lungs of all five patients, and radiographic lesions progressed rapidly after ICU admission (Figure 1). Three patients were examined by highresolution computed tomography (HRCT). Unilateral or bilateral consolidations and infiltrates were found on HRCT scans of all three of these patients. Consolidations within a single lobe or several lobes with a clear border and air bronchogram were the most common findings on HRCT scans. Nodules, patches, pleural effusion, abscess and a cavity were also seen visualized by HRCT (Figure 2). The mean duration from onset to a single-lobe consolidation on CXRs was 2 days (range = 1 to 5

days). The mean duration from the first positive CXR to bilaterally multilobar lung infiltrates was 4.8 days (range = 4 to 7 days).

All patients had HAdV-55 viremia. In four of the five patients, it was first detected in endotracheal aspirate (ETA) samples. The time between initial ETA sample collection of adenoviruses and positive results for HAdV-55 nucleic acid in the blood was 1 to 10 days (Table 3). Virus DNA copies in ETAs were determined for all patients during their ICU stay. The viral load was higher than 1×10^8 copies in three patients and 1×10^4 in one patient. The viral load became negative in the only patient who survived. In the four patients who did not survive, DNA copies did not decrease, even with antiviral therapy (Figure 3).

Oxygenation was not maintained with conventional NPPV or IMV support in any of the patients. The mean duration until NPPV failure was 30.8 hours (range = 22 to 48 hours), and the mean time until IMV failure was 6.2 days (range 2 = to 13 days) (Table 1). Four patients received venovenous ECMO to maintain oxygen saturation, and one patient refused ECMO support and received high-frequency oscillatory ventilation instead. Table 4 gives the oxygenation data of patients before and after venovenous ECMO support.

All patients received antiviral therapy, including acyclovir (10 mg/kg, every 8 hours, intravenous drip), ganciclovir (5 mg/kg, every 12 hours, intravenous drip) and ribavirin (250 mg, twice daily, intravenous drip). Considering that bacterial coinfection may combine with a severe viral infection, broad-spectrum intravenous antibiotics were given to all patients. Tests for bacterial pathogens were negative for only one patient (Table 3). Four (80%) of the five patients died. Among the four patients receiving venovenous ECMO, only one patient survived. The other four patients died due to ARDS, *Aspergillus fumigatus* coinfection, septic shock and catheter-related bloodstream infection due to *Acinetobacter baumannii*, respectively.

To the best of our knowledge, this is the first cohort observational study on the clinical characteristics of patients with severe ARDS caused by emergent HAdV-55 infection and also the first on the evaluation of a viral load test for monitoring the reaction to therapy and for prediction of patient outcome. The following are the main findings of this study. (1) HAdV-55 may cause severe ARDS in immunocompetent young men with blood type B. All of our patients were from the same city of Hebei province, northern China. (2) Persistent

high fever, dyspnea and rapid progression to respiratory failure within 2 weeks, together with bilateral consolidations and infiltrates at the same time, are the most frequent clinical manifestations of severe HAdV-55-induced ARDS. (3) Viral load monitoring may help predict disease severity and patient outcome. (4) The NPPV and IMV failure rates were very high, and ECMO may be the last support method for this group of patients. (5) HAdV-55-induced severe ARDS has a very high mortality rate (80%) despite appropriate respiratory support.

Sporadic severe adenoviral infection in healthy adults has historically been described for serotype 4 [11], serotype 7 [4, 12] and, more recently, serotype 14 in the general population and in military trainees [13, 14]. HAdV-55 was first completely characterized in Shaanxi, China [7] and then reemerged in Hebei, a province close to Beijing, where it caused several cases of acute respiratory disease [9]. It was presumed that HAdV-55 was a recombinant form of the B2 species of HAdV-14 and HAdV-11 [7, 15] due to its sharing a hexon gene with the HAdV-11 and HAdV-14 chassis [16]. The results of our study show that HAdV-55, as an emerging pathogen among immunocompetent adults, may cause severe ARDS.

The prevalence of severe fatal adenoviral pneumonia induced by HAdV-55 in our study is somewhat similar to that described by Cao and colleagues [6]. All cases of reported HAdV-55 in our study were from the same city: Baoding, Hebei province, northern China. They occurred between April and July 2013, just partly overlapping or following the influenza epidemic. The patients with severe disease also came from the same region and were treated during a similar time period, which suggests that HAdV-55 may be an important viral pathogen derived from this region.

Our study results suggest that the following may be clinical features of ARDS caused by HAdV-55: persistent high fever, rapid progression of dyspnea, need for mechanical ventilation support, elevated AST level and rapid progression from unilateral infiltrates to bilateral consolidations. These clinical features are highly similar to those of ARDS caused by other types of HAdV described in previous reports [6, 9].

Recent studies have shown that the immune system plays a crucial role in the clearance of HAdV viremia and survival of the host [17]. Chen et al. reported that, in the acute phase of HAdV-55 infection, patients with severe disease may have high levels of dendritic cells and Th17 cells [18]. In our study, the only patient who recovered from severe infection had

higher T-cell counts. Three of the five patients had relatively low T-cell counts when admitted. Our results suggest that these three patients may have been relatively immunocompromised and that a lower T-cell count may be a risk factor for HAdV-55 infection in young adults. HAdV-55 DNA was previously reported in 41.2% of patients with severe infection [18]. In our study, HAdV-55 DNA was detected and monitored in all patients with severe ARDS. The initial, and trend of, viral load that presented as HAdV-55 DNA copies in the respiratory tract samples and blood may suggest the severity of infection and may predict both the reaction to therapy and patient outcome.

The use of mechanical ventilation and ECMO in patients with ARDS caused by HAdV-55 has not been detailed in previous studies. In our cohort, we found that severe HAdV-55 infection could cause a rapid progression of respiratory failure, with a very high failure rate for NPPV and IMV. This failure rate may be a result of the large area of consolidation that induced a severe shunt in the lung, which may lead to lack of response to positive pressure ventilation. For patients with severe ARDS, ECMO should be considered a better choice for oxygenation.

Our study has limitations. It is an observational study with no comparison group, so the difference between the severe and modest infections could not be clarified in terms of immune status, clinical features, radiological findings, viral load and treatment effects on respiratory support and antiviral therapy. Sequential dynamic analysis is needed to determine the relationship between HAdV-55 viremia and treatment response.

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--- START OF ARTICLE 58 ---

Diagnostic accuracy of C-reactive protein and procalcitonin in suspected community-acquired pneumonia adults visiting emergency department and having a systematic thoracic CT scan

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Abstract: **INTRODUCTION:** Community-acquired pneumonia (CAP) requires prompt treatment, but its diagnosis is complex. Improvement of bacterial CAP diagnosis by biomarkers has been evaluated using chest X-ray infiltrate as the CAP gold standard, producing conflicting results. We analyzed the diagnostic accuracy of biomarkers in suspected CAP adults visiting emergency departments for whom CAP diagnosis was established by an adjudication committee which founded its judgment on a systematic multidetector thoracic CT scan. **METHODS:** In an ancillary study of a multi-center prospective study evaluating the impact of systematic thoracic CT scan on CAP diagnosis, sensitivity and specificity of C-reactive protein (CRP) and procalcitonin (PCT) were evaluated. Systematic nasopharyngeal multiplex respiratory virus PCR was performed at inclusion. An adjudication committee classified CAP diagnostic probability on a 4-level Likert scale, based on all available data. **RESULTS:** Two hundred patients with suspected CAP were analyzed. The adjudication committee classified 98 patients (49.0 %) as definite CAP, 8 (4.0 %) as probable, 23 (11.5 %) as possible and excluded in 71 (35.5 %, including 29 patients with pulmonary infiltrates on chest X-ray). Among patients with radiological pulmonary infiltrate, 23 % were finally classified as excluded. Viruses were identified by PCR in 29 % of patients classified as definite. Area under the curve was 0.787 [95 % confidence interval (95 % CI), 0.717 to 0.857] for CRP and 0.655 (95 % CI, 0.570 to 0.739) for PCT to detect definite CAP. CRP threshold at 50 mg/L resulted in a positive predictive value of 0.76 and a negative predictive value of 0.75. No PCT cut-off resulted in satisfactory positive or negative predictive values. CRP and PCT accuracy was not improved by exclusion of the 25 (25.5 %) definite viral CAP cases. **CONCLUSIONS:** For patients with suspected CAP visiting emergency departments, diagnostic accuracy of CRP and PCT are insufficient to confirm the CAP diagnosis established using a gold standard that includes thoracic CT scan. Diagnostic accuracy of these biomarkers is also insufficient to distinguish bacterial CAP from viral CAP. **TRIAL REGISTRATION:** ClinicalTrials.gov registry

NCT01574066 (February 7, 2012) ELECTRONIC SUPPLEMENTARY MATERIAL: The online version of this article (doi:10.1186/s13054-015-1083-6) contains supplementary material, which is available to authorized users.

Text: Community-acquired pneumonia (CAP) is a frequently seen disease, with high morbidity and mortality, accounting for 600,000 hospitalizations each year. It represents the seventh leading cause of death in the USA [1] . CAP prognosis depends on the rapidity of specific treatment, which should ideally be initiated within four hours and no later than eight hours after diagnosis [2, 3] . CAP diagnosis is based on the clustering of non-specific pulmonary and general symptoms [4, 5] , an increase in biomarkers reflecting systemic inflammatory response syndrome (SIRS), and the presence of new parenchymal infiltrates on chest X-ray. However, CAP diagnosis remains uncertain in many cases with alternative diagnoses, such as cardiac failure, acute bronchitis, chronic obstructive pulmonary disease (COPD) exacerbations, pulmonary embolism, neoplasia, and sepsis [6, 7] .

Part of the uncertainty of CAP diagnosis may be due to the high rate of chest X-ray misdiagnosis [8, 9] ; over diagnosis of CAP is frequent when infiltrates of noninfectious origin coexist with pulmonary or general symptoms, and the diagnosis of CAP is often ignored when the lung infiltrates are at the limit of visibility or are hidden due to superposition [10] . We recently published a study in which thoracic CT scan was systematically performed in a population of clinically suspected CAP patients visiting the emergency department for CAP (the ESCAPED study) [11] . We showed that CAP diagnosis based on chest X-ray led to a false CAP diagnosis in many patients: among CAP suspected patients with radiological pulmonary infiltrate, CAP diagnosis was excluded in around 30 % of patients based on CT scan results; on the contrary, among patients without radiological pulmonary infiltrate, one-third had a pulmonary infiltrate on thoracic CT-scan. We also reported the isolation of viruses in one-third of patients [11, 12] .

Several attempts have been made to improve CAP diagnosis based on biomarkers, such as C-reactive protein (CRP) and procalcitonin (PCT); however, there are conflicting data on their reliability [13] [14] [15] [16] [17] . This could be due to the consideration of CAP diagnosis based on chest X-ray as establishing pulmonary infection. In the present study, we aimed to analyze CRP and PCT values in the population of the ESCAPED study reported above for whom CAP diagnosis was established by an adjudication committee which founded its judgment on all usual available data, systematic multidetector thoracic CT

scan performed at inclusion, and results from a day-28 follow-up. We also analyzed whether the viral etiology of definite CAP based on polymerase chain reaction (PCR) multiplex naso-pharyngeal swab interfered with the accuracy of the biomarkers.

Setting ESCAPED was a multicenter, prospective, interventional study, entitled "Early Thoracic CT-Scan for Community-Acquired Pneumonia at the Emergency Department (ESCAPED)" [11], conducted from November 2011 to January 2013, in four emergency departments (EDs) of four tertiary teaching hospitals in Paris, France, designed to measure the impact of thoracic CT scan on clinical decision. The study was sponsored and monitored by the Paris public health hospitals, and funded by the French Ministry of Health. The French health authorities (Agence nationale de sécurité des médicaments et produits de santé, ANSM) and the institutional review board for the protection of human subjects approved the study protocol and patient informed consent procedures. All enrolled patients provided written informed consent for inclusion. The protocol was registered in the clinicaltrial.gov website under the PACSCAN acronym, the French translation of the English ESCAPED acronym (NCT01574066). The Ethics Committee of Ile de France (Comité de Protection des Personnes. Paris N°2 011-oct-12749) approved the study protocol.

The primary objective was to compare CRP and PCT values in the four different categories of CAP level of certainty using the day-28 adjudication committee classification. The four categories were: 1) absence of CAP hereafter referred to as excluded CAP diagnosis; 2) possible CAP; 3) probable CAP; and 4) definite CAP. The secondary objectives were to assess whether CRP and PCT were associated with CAP diagnosis using sensitivity analyses in three successive subgroups chosen a priori; 1) when specifically considering patients classified as having excluded CAP diagnosis and definite CAP (i.e., the patients for whom the level of certainty was the highest); 2) when patients with excluded CAP diagnosis and diagnosed extra-pulmonary infectious disease (which may increase biomarker values) were not taken into account, in the excluded CAP group; and 3) when patients classified as viral CAP were not taken into account in the definite CAP group, as PCT has been reported to be lower in viral infections as compared to bacterial infections [18].

Consecutive adults ([19]).

Multidetector thoracic CT-scan was performed after chest X-ray, ideally within the four hours following inclusion. Chest X-ray and thoracic CT-scan were performed using a standardized protocol. The four levels of CAP probability according to CT scan were defined as definite (systematic alveolar condensation, alveolar condensation with peripheral and localized ground glass opacities, bronchiolar focal or multifocal micronodules), probable (peripheral alveolar condensation, retractile systematic alveolar condensation, or diffuse ground glass opacities), possible (pulmonary infarct), or excluded (pulmonary mass, other abnormalities, or normal images). Scan views were recorded on a DVD.

Based on data collected from baseline standardized case report forms, DVD recorded pictures of X-ray and CTscan, and blinded to local interpretations, an adjudication committee consisting of three independent senior experts in infectious diseases, pneumology and radiology retrospectively assigned the probability of CAP diagnosis using the same 4-level Likert scale, with all available data including patients' discharge summary, and follow-up data obtained by assistant investigators who contacted by phone either the patient, relatives or general practitioners at day 28. For this study, the gold standard of CAP was the diagnosis assessed by this adjudication committee. Alternative diagnoses were established for excluded CAP and classified as non-CAP pulmonary diseases and extra-pulmonary infectious diseases and others.

Blood samples were collected at inclusion in sodium heparin-treated tubes, centrifuged, and stored at -40°C until completion of the study. CRP and PCT concentrations were measured a posteriori on plasma collection (see Additional file 1 for methodology), except for patients in whom marker dosage was performed by the emergency practitioner on his own initiative.

Naso-pharyngeal swabs were collected at enrollment and placed in a Middle Virocult MWE (Sigma®) transport medium. Samples were kept at room temperature and sent to the virology laboratory of Bichat -Claude Bernard Hospital (Paris) as soon as possible after collection. The samples were not frozen and thawed. Multiplex PCR (RespiFinder-19 assay (Pathofinder®, Maastricht, Netherlands)) was performed on naso-pharyngeal swabs to detect 15 respiratory viruses -coronavirus 229E, NL63, OC43, human metapneumovirus (hMPV), influenza A, A (H1N1) pdm2009 and B viruses, parainfluenza viruses 1, 2, 3, and 4, respiratory syncytial virus (RSV) A and B, rhinovirus, adenovirus, and 4 intracellular bacteria -Bordetella pertussis, Chlamydomphila pneumoniae, Legionella pneumophila, Mycoplasma

pneumoniae, in one reaction. The multiplex PCR results were not available to the adjudication committee. Routine microbiological examinations were also performed at the discretion of the emergency physicians and included blood culture, sputum culture, and antigenuria (see Additional file 1 for methodology). CAP, classified as definite, was considered as being of viral origin when multiplex PCR was positive for at least one of the 15 respiratory viruses and no bacteria were found using PCR and routine bacterial microbiological samples (sputum, blood culture, antigenuria) when performed.

Baseline and follow-up characteristics were described by means and standard deviations (SD) or by median and interquartile range (IQR) for continuous variables normally distributed or with skewed distribution, respectively, and by percentages for categorical variables, for the total study population and for the study groups. We performed chi-square or Fisher exact tests when appropriate for qualitative variables, and the Student or Mann-Whitney tests for continuous variables with skewed distributions to compare baseline patient characteristics and study outcomes between study groups.

The distribution values of the biomarkers were determined in the different populations of patients using boxplots. The performances of CRP and PCT in predicting definite CAP were evaluated by sensitivity analysis (definite CAP vs excluded CAP). CRP was evaluated at several cut-off points of 20 mg/L, 30 mg/L, 50 mg/L, 70 mg/L, and 100 mg/L, values used in previous studies [15, 20, 21]. Several cut-off points for PCT were chosen at the level of 0.10 µg/L [18], and at the two levels for suspected bacterial infection as stated by the manufacturer, i.e., 0.25 µg/L and 0.50 µg/L. Sensitivities, specificities, positive predictive values (PPVs), negative predictive values (NPVs), and likelihood ratio were calculated. Receiver operating characteristic (ROC) curves were drawn, area under the curve AUC was computed and optimal cut-off was identified by the maximization of the Youden's index, comparing biomarker values in patients with excluded CAP and definite CAP. From these optimal cut-offs for CRP and PCT, sensitivity analyses were performed combining the CRP and PCT cut-offs.

A multivariate logistic regression model was built to identify factors associated with having definite CAP as compared to having an excluded CAP diagnosis. We excluded from the excluded CAP diagnosis group, patients with an extra-pulmonary infectious disease. All variables with a p value of < 0.25 in the bivariate analysis were entered into a multivariate logistic regression with a backward stepwise approach; the discrimination was evaluated

by the C-index and its 95 % confidence interval (95 % CI) and the calibration was evaluated by the Hosmer Lemeshow goodness-of-fit test.

All tests were two-sided, and p-values below 0.05 were considered to denote statistical significance. All statistical analyses were performed using SPSS statistical software version 21.0 (SPSS Inc., Chicago, IL, USA).

Two hundred patients with suspected CAP out of the 319 in the ESCAPED study were included in the present study, for which CRP and PCT assays and nasopharyngeal swab for multiplex PCR were available (Fig. 1) . Characteristics of the 200 patients (age, age more than 65, gender, probability of CAP diagnosis by adjudication committee) were not significantly different from those of the 119 other patients of the ESCAPED study and are summarized in Table 1 . CRP and PCT assays were performed based on the emergency practitioner's own initiative in 70 patients for CRP and 131 for PCT, or performed a posteriori on plasma samples of the remaining patients. Sex ratio was approximately 1. More than half of the patients (54 %) were 65 years of age or older. The

Pulmonary infiltrates were seen on chest X-ray in 127 (63.5 %) patients. Thoracic CT-scan excluded a CAP diagnosis in 16.5 % of these 127 patients; on the contrary, thoracic CT-scan revealed a parenchymal infiltrate in 27 % of the 73 patients without infiltrate on chest X-ray.

Based on all available data including multidetector CT scan results (but excluding PCR results), the adjudication

The CRP and PCT distributions in the 200 patients are presented in Fig. 2 A statistically significant difference between the two groups (excluded CAP vs definite CAP) was demonstrated for several cut-off points for CRP and PCT (Table 2). For CRP, the value of 50 mg/L resulted in a PPV of 0.76 and a NPV of 0.75. For PCT, no value resulted in a satisfactory PPV or NPV. For these two biochemical markers, the ability to predict CAP was evaluated by a ROC curve. The AUC was 0.787 (95 % CI 0.717-0.857), optimal cut-off = 45.9 mg/L for CRP (Fig. 3) and 0.655 (95 % CI 0.570-0.739), optimal cut-off = 0.13 µg/ L for PCT (Fig. 4) .

Sensitivity analyses for the combination of CRP and PCT, using these optimal cut-offs, resulted in a PPV of 0.74 and a NPV of 0.58. Use of the other PCT cut-offs did not result in better PPV or NPV (Table 2) .

The present study is novel as patients prospectively benefited from extensive investigation to determine the diagnosis of CAP in the ED, including both early multidetector thoracic CT-scan and day-28 adjudication committee. This led to the correction of CAP diagnosis previously based on chest X-ray in a high number of patients. In these extensively characterized patients, both CRP and PCT lacked operational precision to allow the decisionmaking process to rule out or confirm diagnosis of CAP even in selected subgroups.

The clinical characteristics of the patients included in this sub-study are consistent with those in the current literature. As previously reported, patients frequently had a history of respiratory disorders, cancer and congestive heart failure [21, 22] . The design of the ESCAPED study required exclusion of patients within the highest CRB 65 categories, which limited the inclusion of patients older than 65. This may explain why the mean age of our patients (64 years) falls within the lower values of those reported elsewhere [19] . Data to identify the microbial agent responsible for the disease were collected by the usual techniques and multiplex PCR. Viral identification using naso-pharyngeal PCR that revealed viral respiratory infection in approximately one-third of cases was concordant with values reported in the literature [23] . Therefore, we believe that our results can be extrapolated to most emergency patients suffering from CAP.

In the present study, patients were recruited on the basis of initial clinical assessment for the diagnosis of CAP. Therefore, we believe that the characteristics of the patients closely correspond to those that lead practitioners to consider a possible diagnosis of CAP. In these patients, the design of our study allowed us to confirm or refute CAP diagnosis with a high level of certainty. Results confirmed the poor predictive value of clinical symptoms (new onset of systemic features and symptoms of an acute lower respiratory tract illness) in identifying CAP patients [21] . Indeed, clinical presentation of excluded CAP patients was similar to that of definite CAP patients except for fever and cough that were more frequent in definite CAP patients. Furthermore, the design also revealed that the combination of clinical symptoms and chest X-ray results led to CAP misdiagnosis in a high number of

patients, including the 98 whose CAP diagnosis was excluded by the adjudication committee and who would have been considered as possible, probable or definite CAP without the use of the CT scan. This low specificity of clinical-standard radiological evaluation led to the consideration of either non-infectious pulmonary diseases (such as, cardiac failure, pulmonary embolism, pulmonary neoplasia or bronchitis) or extra-pulmonary infectious diseases as CAP. Of note, some of these diseases are also associated with increased biomarker values. This raises concerns about previous evaluations of biomarkers in CAP-suspected patients, which used clinical and standard radiological (chest X-ray) evaluations as the gold standard for CAP diagnosis [15] . The use of biomarkers has been advocated to improve diagnosis and management of patients with lower respiratory tract infections [14] . However, this issue is still unresolved [24] , with conflicting positions [14, 15, 25, 26] . In our study, while median values of both biomarkers did increase with level of certainty for CAP diagnosis, we were unable to establish discriminating values for PCT. Recent data suggested that CRP could be of more help in assisting in the diagnosis of lower respiratory tract infections (LRTI) [15, 27, 28] . In our study, although CRP seems more discriminating than PCT, neither the experimental exclusion of extra-pulmonary bacterial infections from the excluded CAP group, nor the exclusion of viral CAP from the definite CAP patients group, made possible the determination of a discriminant cutoff. The combination of CRP and PCT was not more discriminating than each biomarker separately. An operational algorithm has been released to assist physicians in prescribing antimicrobial therapy [14, 26, 29] . According to this strategy, a PCT concentration higher than 0.25 µg/L should prompt administration of antibiotics to patients with suspected LRTI. In our study, this value was associated with poor performance. Additionally, mean PCT levels remained above this threshold both in excluded CAP patients without infectious disorders and in definite CAP presumably related to virus. Therefore, the gold standard for the diagnosis of CAP may influence the performance and utility of PCT in this setting.

This study has some limitations. First, the adjudication committee was not blinded to the value of biomarkers measured at bedside in some patients (70 for CRP and 131 for PCT) and its CAP classification could thus have been influenced by these results. However, the lack of statistically significant differences in the mean CRP and PCT values in the definite CAP cases, whether or not these biomarkers were available for the adjudication committee, argues against a major impact of these results on adjudication committee classification. Second, another critical point is the prescription of antibiotic therapy (34 %) previous to inclusion. We cannot exclude that these previously-treated CAP patients may have altered biomarker performance and reduced the yield of bacterial cultures, although

such a population reflects the usual emergency department practice. Third, multiplex PCR was performed on naso-pharyngeal sampling and not on lower respiratory tract samples, which does not allow definite confirmation of the viral origin of CAP. However, a recent large study on CAP patients which reported a viral etiology of CAP at a comparable rate, did not find upper respiratory tract shedding in a control population without CAP explored during the same year and season [30] . Finally, even if multidetector thoracic CT scan is a better imaging examination than X-ray to explore the chest, only invasive local microbiological samples would have provided a diagnosis with certainty.

Given the diversity of the clinical and radiological CAP presentations, CAP diagnosis is often uncertain. In our population of patients treated in the emergency room with clinical symptoms evoking CAP, neither CRP nor PCT cut-off values carried sufficient weight to confirm or refute CAP diagnosis at bedside; this underlines that these biomarkers are telltales of the host inflammatory response to the intrusion of microorganisms independent of the site of infection. These results, based on a systematic thoracic CT scan evaluation of CAP-suspected patients, do not argue for the use of CRP and PCT in routine care to diagnose CAP with certainty in patients visiting the ED for suspected CAP.

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Pandemic Influenza Due to pH1N1/2009 Virus: Estimation of Infection Burden in Reunion Island through a Prospective Serosurvey, Austral Winter 2009

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Abstract: **BACKGROUND:** To date, there is little information that reflects the true extent of spread of the pH1N1/2009v influenza pandemic at the community level as infection often results in mild or no clinical symptoms. This study aimed at assessing through a prospective study, the attack rate of pH1N1/2009 virus in Reunion Island and risk factors of infection, during the 2009 season. **METHODOLOGY/PRINCIPAL FINDINGS:** A serosurvey was conducted during the 2009 austral winter, in the frame of a prospective population study. Pairs of sera were collected from 1687 individuals belonging to 772 households, during and after passage of the pandemic wave. Antibodies to pH1N1/2009v were titrated using the hemagglutination inhibition assay (HIA) with titers $\geq 1/40$ being considered positive. Seroprevalence during the first two weeks of detection of pH1N1/2009v in Reunion Island was 29.8% in people under 20 years of age, 35.6% in adults (20–59 years) and 73.3% in the elderly (≥ 60 years) ($P < 0.0001$). Baseline corrected cumulative incidence rates, were 42.9%, 13.9% and 0% in these age groups respectively ($P < 0.0001$). A significant decline in antibody titers occurred soon after the passage of the epidemic wave. Seroconversion rates to pH1N1/2009 correlated negatively with age: 63.2%, 39.4% and 16.7%, in each age group respectively ($P < 0.0001$). Seroconversion occurred in 65.2% of individuals who were seronegative at inclusion compared to 6.8% in those who were initially seropositive. **CONCLUSIONS:** Seroincidence of pH1N1/2009v infection was three times that estimated from clinical surveillance, indicating that almost two thirds of infections occurring at the community level have escaped medical detection. People under 20 years of age were the most affected group. Pre-epidemic titers $\geq 1/40$ prevented seroconversion and are likely protective against infection. A concern was raised about the long term stability of the antibody responses.

Text: In April 2009, the first cases of acute respiratory infections caused by a novel triple-reassortant influenza virus, pH1N1/ 2009v, occurred in Mexico and the United States [1] . The rapid spread of infection to other continents led the World Health Organization (WHO) to declare on 11 June 2009 that a pandemic of pH1N1/2009v influenza was under way, which raised major international concern about the risk of high morbidity and lethality and the potential for severe socio-economic impact. Actually, the potential impact of this first third-millennium influenza pandemic has been revisited downwards as morbidity and case-

fatality rates were less severe than initially anticipated [2] . Illness surveillance data do not allow to an accurate estimate of the true influenza infection rate, as a substantial proportion of infections are asymptomatic or mild [3] . Serological surveys can overcome this limitation, but must take into account that a significant proportion of the population that exhibited crossprotective antibody titers before circulation of the pH1N1/2009v [4] . This so-called "baseline immunity" has to be subtracted from the seroprevalence observed after the pandemic wave, to determine seroincidence in serosurveys [5] [6] [7] [8] . However, except for few studies [9] [10] [11] , most of these serosurveys did not use serial measurements in the same person, which allows for a better understanding of antibody kinetics and the dynamics of infection within individuals and communities.

Reunion Island (805,500 inhabitants) is a French overseas department located in the southwestern Indian Ocean, 700 km east of Madagascar and 200 km southwest of Mauritius. The first imported case of pH1N1/2009v was identified on 5 th July 2009 (week 29) in a traveller returning from Australia. The first case indicating community transmission was detected on 21 st July (week 30). pH1N1/2009v became the predominant circulating influenza virus within four weeks of its first detection, its activity peaked during week 35 (24) (25) (26) (27) (28) (29) (30) and ended at week 38 [12] . Contrary to initial fears, the health care system was not overwhelmed, as morbidity and mortality rates were lower than predicted [12] [13] [14] .

In order to assess at the community level, the actual magnitude of the pH1N1/2009v pandemic and the extent of the herd immunity acquired after passage of the epidemic wave, a prospective population serosurvey was conducted in Reunion Island during the passage of the epidemic wave in the 2009 austral winter season (July-December 2009): prevalence of infection was assessed on a weekly basis and seroconversion rates were measured using paired sera.

The CoPanFLu-RUN was part of the CoPanFLu international project, a consortium between the French National Institute of Health and Medical Research (INSERM), the Institute of Research for Development (IRD) and the Mérieux Foundation under the promotion of the School of Advanced Studies in Public Health (EHESP). To enable the rapid implementation of the study in anticipation of the imminent spread of the pandemic wave, we used a pre-existing sample of 2442 households established in October 2006 for the investigation of the Chikungunya outbreak (SEROCHIK) and updated in May 2008 throughout a follow-up

telephone survey (TELECHIK) on a basis of 1148 households [15, 16] . We took special attention to select households representing a wide range of geographic locations in order to minimize the repartition bias.

The inclusion phase started on July 21st (week 30) and was continued up to week 44, throughout the epidemic wave and beyond. A first serum sample (sample 1) was obtained from each household member. An active telephonic inquiry was then conducted twice a week to record symptoms compatible with influenza-like illness (ILI) occurring in households. Report of ILI (fever $\geq 37.8^{\circ}\text{C}$ associated with any respiratory or systemic symptom) led to three consecutive visits of a nurse to the incident case-dwelling (on day 0, +3 and +8 post-report) to record symptoms and collect nasal swabs from all family members (for qRT-PCR detection of pH1N1/2009v. At week 45, the active inquiry was discontinued and a second (post-epidemic) serum sample (sample 2) was obtained (weeks 45-52) to determine seroconversion rates. Sera were aliquoted and stored at -280°C .

The protocol was conducted in accordance with the Declaration of Helsinki and French law for biomedical research (Nu ID RCB AFSSAPS: 2009-A00689-48) and was approved by the local Ethics Committee (Comité de Protection des Personnes of Bordeaux 2 University). Every eligible person for participation was asked for giving their written informed consent.

Viral genome detection by RT-PCR. Viral RNA was extracted from 140 μL of nasal swab eluate using the QIAamp Viral RNA kit (Qiagen) and processed for detection by TaqMan qRT-PCR targeting the hemagglutinin HA gene (SuperScript III Platinum one-step qRT-PCR system, Invitrogen) according to the recommendations of the Pasteur Institute (Van der Werf S. & Enouf V., SOP/FluA/130509). Confirmed pH1N1/2009v infection was defined as a positive qRT-PCR detection of the HA gene in at least one nasal swab.

Hemagglutination inhibition assay (HIA). A standard hemagglutination inhibition technique was adapted to detect and quantify pH1N1/2009v antibodies [17] . The antigen was prepared by diluting a non-inactivated cell culture supernatant producing a pdm H1N1v strain (strain OPYFLU-1 isolated from a young patient returning from Mexico in early May 2009) [18] . Briefly, the virus was propagated onto MDCK cells under standard conditions. The last passage (used for antigen preparation) was performed in the absence of trypsin

and ht-FBS. The supernatant was collected at day seven p.i. clarified by centrifugation at 8006 g for 10 min at room temperature, aliquoted and conserved at 280uC. The hemagglutinating titer of the non inactivated viral antigen was immediately determined under the HIA format described below. The dilution providing 5.33 hemagglutinating units in a volume of 25 mL was used for subsequent HIA. Sera were heat-inactivated at 56uC for 30 min prior to use. Sequential twofold dilutions in PBS (1/10 to 1/1280) in volumes of 25 mL were performed and distributed in V-bottom 96 well microplates. Human red blood cells (RBC) were used for hemagglutination experiments. Detection and quantification of antibody to pH1N1/2009v was performed as follows: 25 mL of virus suspension was added to the serum dilution (25 mL) and incubated for 1 hour at room temperature. Each well was then filled with 25 mL of a 1% RBC suspension in PBS (v/v: 0.33%), followed by another 30 min incubation at room temperature. The HIA titer was determined as the last dilution providing clear inhibition of hemagglutination. All experiments were performed in the presence of the same negative and positive controls, the latter including sera with 1/40, 1/80, 1/160 and 1/320 antibody titers.

The results reported in this study were based only on serological analysis of paired sera. For the sake of analysis, four successive phases were identified throughout the pandemic wave: phase A (weeks 30-31) corresponded to early epidemic time, phase B (W32-39) to the epidemic unfolding, phase C (W40-44) to the immediate post-epidemic stage and phase D (W45-52) to the late post-epidemic stage. Seropositivity was defined as a HIA titer of 1/ 40 or more. The baseline-proxy seroprevalence rate was estimated on serum samples collected in phase A. The cumulative incidence rate of infection measured the raise between the raw seroprevalence rate at any given time during the epidemic phases (S_{2pi}) and the age-specific baseline-proxy seroprevalence rate (S_{1pA}) ($s_{2pi} - s_{1pA}$). Seroconversion was defined as a shift from seronegative at inclusion (sample 1: HIA ,1/40) to seropositive on follow-up (sample 2: HIA \$1/40), or for sera tested seropositive on inclusion as a four-fold increase of HIA titers between sample 1 and sample 2 paired sera. We also calculated the proportion of sera that tested seropositive in sample 1 for which the HIA titer decreased fourfold and passed under the cut-off value of 1/40 in sample 2. We considered this proportion as a "seronegation" rate.

The sample size was calculated for identifying risk factors in the prospective cohort study. Considering on average three individuals per household, an intra-household correlation of 0.3, a power greater than 80% could be obtained with a sample size of 840 comprising 2500 individuals, assuming exposure levels ranging from 10% to 90% and a relative risk

greater than 1.3. With 2,500 subjects, the study allowed 1-2% absolute precision around the estimated values for seroconversion rates. Data entry used EpiData version 3.1 (The Epidata Association, Odense, Denmark). SAS version 9.1 (SAS Inc., Cary, NC, USA) was used for statistical analysis. The characteristics of the study cohort were compared to those of the population of Reunion Island and a Chi2 test (or Fisher's exact test when non applicable) was used to analyse differences in age, sex and geographic location. Cumulative incidence rates of infection (i.e. seroincidence) and seroconversion rates were standardized according to the age structure of the community (French National Institute for Statistics and Economical Studies (INSEE) source).

Baseline-proxy seroprevalence, cumulative incidence rates of infection, as well as seroconversion and seronegation rates, were expressed as percentages. Cumulative reverse distribution curves were used to show the distribution of antibody titers. In all tests, a P value, 0.05 was considered significant.

We estimated 95% confidence intervals (CIs) of proportions by using a cluster bootstrap technique with 1000 re-samples [19]. After bootstrapping, we used an ANOVA model to compare mean cumulative incidence proportions between pandemic phases, within each age group. We used an alternating logistic regression model (ALR) with an exchangeable log Odds Ratio (OR) to test the intra-household correlation-adjusted association between factors and the seroconversion outcome.

Data were analysed with respect to subject age. Initially, four age groups were considered: the children and adolescents (<20 yrs), young adults (20-39 yrs), middle-age adults (40-59 yrs), and elderly adults (≥ 60 yrs). As the cumulative incidence of infection of the second and third groups were very close, both groups were merged into one adults group (20-59 yrs). Therefore we refer further in our study to three age groups: children and adolescents (<20 yrs), adults (20-59 yrs), elderly (≥ 60 yrs).

A total of 2,164 individuals from 772 households were enrolled between weeks 30 and 44 in the CoPanFlu-RUN cohort, allowing the collection of 1,932 sera at inclusion (sample 1). During this period, 136 households (17.7% of households) containing 464 individuals (21.4% of individuals) reported at least one case of ILI. Sixty subjects among the 464 individuals (12.9%, belonging to 33 households [24.3%]) were qRT-PCR positive, which

documented the pH1N1/2009v infection. No positive qRT-PCR could be detected after week 37 and no ILI was reported after week 40, the end of the epidemic wave. The second follow up serum sample (sample 2) was obtained for 1,759 subjects at least five weeks after the end of the epidemic wave (weeks 45-52) which allowed the constitution of a serobank of 1,687 paired-sera. The profile of the cohort and the major outcomes are displayed in Figure 1 . Details on inclusions and serum sample timing with respect to the circulation of pH1N1/2009v over the island are provided in figure 2 . The socio-demographic and space-time characteristics of the cohort are detailed in Table 1 . Compared to the community of Reunion Island, the sample of 1,687 individuals for whom paired sera were available, was older (,20 yrs: 27% vs 35%, and ≥60 yrs: 17,9% vs 11,3%) and composed of a slight excess of females (54.1% vs 51.5%). The imbalance was due to a deficit in subjects aged under 40 years, reflecting men at work and the fact that parents declined the second serum for children younger than five.

Baseline-proxy (,pre-epidemic) HIA titers to the pH1N1/ 2009v were measured on sample 1 (Table 2) , obtained from 249 subjects (103 households) recruited at the very beginning of the investigation during weeks 30 and 31 (phase A, Figure 2), when the epidemic activity in the cohort was still very low. Age distribution in this group was similar to that of the whole cohort (data not shown). The overall, the baseline-proxy seroprevalence rate (HIA ≥1/40), over all ages, was 43.4% (95%CI: 37.4%-49.6%). However the majority of positive sera had low antibody titers, at the cut off value for confirmation (i.e. = 1/40). The proportions of sera with HIA titer ≥1/40 were 0%, 3.0% and 24.6% in the young, middle-aged and older age groups respectively. These results indicate that pre-epidemic baseline antibody cross reactivity was stronger in the elderly (≥60 yrs) and weaker in children and adolescents (<20 yrs) and adults (20-59 yrs), with highly significant differences between age groups (P<0.0001).

The reverse cumulative distribution curves of HIA titers are displayed for each age group and for the whole cohort on Figure 3 . The proportion of seropositive sera (HI ≥1/40) steadily increased during the epidemic unfolding (phase B, W32-39) and in immediate post epidemic period (phase C, W40-44) when it reached its maximum level, then declined in the late post epidemic period (phase D, W45-52). This decline was significant enough to return the reverse cumulative distribution curve to baseline levels in the elderly. The cumulative incidence rates, obtained after subtraction of the age-specific baseline-proxy seroprevalence from the raw seroprevalence at each phase of the epidemic are shown in Table 2 (note that the cumulative incidence rates of infection represented for the group "all

ages" were standardized according to age structure of the community). The cumulative incidence rates were much higher in children and adolescents (<20 yrs), indicating very active transmission of infection within this age group. As mentioned earlier, cumulative incidence rates peaked in phase C (W40-44), and then declined indicating some lability of the humoral immune response against the pH1N1/2009v. The age-related difference observed in the incidence rates was highly statistically significant ($P=0.0001$).

To estimate more appropriately the decline of antibody titers occurring after the peak of the humoral response to the pH1N1/2009v, we considered paired-sera from the group of 264 subjects for whom the first serum sample (sample 1) was obtained just after the epidemic wave (phase C, W40-44), and the corresponding second sample was collected at the end of the survey (phase D, W45-52). Seronegation rates were 27.0% (61/226) for all age groups, 17.4% (12/69) in children and adolescents (<20 yrs), 32.3% (41/127) in adults (20-59 yrs) and 26.7% (8/30) in the elderly (≥ 60 yrs). Differences between the seronegation rates according to age were statistically weakly significant ($P = 0.0671$).

We then considered the 1687 individuals for whom paired sera were available and we measured the seroconversion rates according to age and to the time of first serum sample collection (phase A, B or C). Criteria of seroconversion were defined in the method section. As shown in table 3, there was a sharp decline in seroconversion rates across all the age groups, depending on whether participants were enrolled during phase A, phase B, or phase C ($P=0.0001$). To interpret these data, one should remember that antibodies at seroprotective levels (HIA $\geq 1/40$), in serum samples 1 collected during the pre epidemic phase B or early post epidemic phase C could represent either base line cross reactive antibodies or rising pH1N1/2009 specific antibodies due to a recent or ongoing infection. This ambiguity could lead to underestimation of the seroconversion rate for subjects enrolled in phases B and C. In order to solve this ambiguity, we specifically considered the group of 249 subjects in whom cross reactive antibodies were detected at the time of phase A (W30-31). The seroconversion rate of this group is the most indicative of the exposure of individuals to the whole epidemic wave. It was the highest (63.2%, $P=0.0001$) in children and adolescents (<20 yrs), and still significantly high in adults (39.4%, $P=0.0001$).

We then tested in this particular group, the impact of (baseline) pre-epidemic cross reactive antibodies on the rate of seroconversion to pH1N1/2009 (Table 4). No subject with HIA titer superior to 1/40 had evidence of seroconversion to pH1N1/2009. The

seroconversion rate in individuals with a HIA titer equal to 1/40 was linked with age, being more important in children and adolescents (<20 yrs). The highest seroconversion rate (.56%) was registered in subjects with HIA titers inferior to 1/40, particularly for the under 20 years where it reached 85%. Hence, the risk of seroconversion decreased when pre-epidemic HIA titer was high after controlling for age ($P=0.0001$) (Figure 4). The multivariate adjusted odds ratio for seroconversion were 0.15 (95%CI: 0.06-0.37, $P=0.0001$) per two-fold increase in baseline titer, 1.79 (95%CI: 1.23-2.59, $P=0.003$) per other household members who seroconverted, 5.33 (95%CI: 1.56-19.27, $P=0.008$) Figure 1. The cohort profile and major outcomes. Figure 1 details the three phases of the protocol: i) inclusion (weeks 30-44) and serum samples S1 collection; ii) follow up for detection of ILI in households, qRT-PCR on nasal swabs and estimation of cumulative seroincidence rates; iii) end of the study (weeks 45-52) and samples S2 collection. HIA on paired sera (S1+S2) allowed estimating seroconversion rates. doi:10.1371/journal.pone.0025738.g001 Bp (baseline-proxy) seroprevalence rates were estimated on weeks 30-31 in each age group. b Cumulative incidence rates measured the raise between raw seroprevalence rates and age-specific baseline-proxy seroprevalence rate. In the group "All ages", cumulative incidence rates were standardized according to age structure of the community. doi:10.1371/journal.pone.0025738.t002 Data are numbers, percentages (95% confidence intervals) and ALR parameter test P value for comparison of seroconversion proportions according to time of first sample (S1) collection at inclusion, in each age group, after controlling for household selection. In the group "All ages", rates of seroconversion were standardized according to age structure of the community. NA: not assessed. Seroconversion was defined as a shift from seronegative at inclusion (i.e. HIA titer, 1/40) to seropositive on follow-up sample, or as a 4-fold increase of reciprocal HIA titer between first and second paired samples for sera tested seropositive on inclusion (i.e. HIA titer $\geq 1/40$). for age <20 years (vs age ≥ 60 years) and 11.35 (95%CI: 0.41-4.47, $P=0.62$) for age 20-60 years (vs age ≥ 60 years). The observed and predicted seroconversion rates according to age and baseline HIA titer are displayed Figure 4.

Finally, we considered the 46 subjects who had been infected by the pandemic virus over the course of the study, verified by a positive qRT-PCR nasal swab, and for whom paired sera were available. Initial HIA antibody titers in this group were 1/40,

The CoPanFlu-RUN cohort was set up to conduct a prospective population-based study investigating the herd immunity induced by the 2009 pandemic influenza virus and identifying risk factors for pH1N1/2009v infection from paired sera collected in an entire

community. Most works published to date have used either extensive cross-sectional serosurveys on pre-and post-epidemic independent serum samples, the baseline immunity being assessed from stored frozen samples [5, 7, 8] , or non representative adult cohorts (military, health care workers, long-stay patients). Antibody titers were measured by HIA using a cut-off value set at 1/40 as classically recommended. This HIA titer at 1/40 is considered protective, i.e. conferring 50% protection against a viral challenge [20] . Our assay has introduced some changes in the experimental protocol compared to the classic one. The use of a non-inactivated viral antigen, i.e. a native virus, with nondenatured epitopes probably allows detection of antibodies to epitopes of the hemagglutinin not detected in the classic HIA test. This can induce slight differences in the sensitivity of detection of cross-reacting antibodies, but this does not modify the kinetics of Ab and the epidemiological evolution of seroprevalence and does not jeopardize the global comparability of serological results. This is confirmed by the fact that our HI assay detected seroprotective antibody titers in 93.5% and gave evidence seroconversion in 73.9% of qRT-PCR confirmed pH1N1/2009 influenza, all figures close to those reported in the literature [5, 21] .

We considered that titers of .1/40, in sera collected from individuals enrolled during weeks 30 and 31 were cross reactive antibodies and not de novo antibodies triggered by the pandemic virus and hence used them as a proxy for baseline pre epidemic immunity. Several arguments support this assumption: i) the first case indicating autochthonous transmission in Reunion Island was reported by the epidemiological surveillance department of La Réunion on 21st July (week 30), i.e. the same day when inclusion started in our study cohort; ii) 7 to 15 days are required to develop an antibody response after viral infection; iii) On weeks 30 and 31, the epidemic activity due to the pandemic virus was very low in our study cohort and it became significant only after week 32. Hence, during weeks 30-31, 103 households were recruited and only 2 households reported ILI cases. Nasal swabs collected from these 2 individuals were tested qRT-PCR negative to the pandemic virus whereas one had evidence of coronavirus and rhinovirus using a multiplex RT-PCR to respiratory viruses (H. Pascalis, manuscript in preparation). In contrast, during weeks 32 to 39, 199 individuals belonging to 99 households reported ILI, among whom 60 individuals had documented infection by the pandemic virus.

Our study shows that a substantial proportion of Reunion Island's population had pre-existing immunity to 2009 pandemic influenza virus with the highest baseline-proxy seroprevalence rate observed among adults aged of 60 years or more. Other studies from

all continents had also reported high pre-epidemic seropositivity rates among the elderly [5, 6, 8, [22] [23] [24] [25] [26] , though large variations do exist between countries [10, 11, 23, 27, 28] . These cross reactive antibodies have been interpreted as being the residual signature of the remote exposure of these individuals to H1N1 viruses circulating before 1957 [24, 25, 29, 30] . Baseline seropositivity rates that we report in children and in younger adults (i.e. 30%-35%) were notably higher than those reported from other parts of the world [6, 8, 22, 23, [31] [32] [33] . However one should note that these baseline antibodies were of low titer, just at the level of the HIA threshold (i.e. 1/40). Several factors could have contributed to this comparatively high baseline rates found in our study: i) It may reflect the fact that the HI test used in our study was marginally more sensitive than the classic one [17] ; ii) Some individuals may have already been infected with pH1N1/ 2009 virus at weeks 30 and 31 and may have triggered an antibody response to the virus. This hypothesis seems unlikely in view of the arguments presented above and of a similar high proportion of sera titering HIA = 1/40 among 122 sera from adult patients sent for diagnostic purposes to the Regional Hospital microbiology laboratory, during the first half of 2009 (i.e. before the 2009 pandemic) (data not shown). However we cannot formally exclude this hypothesis in view of a recently reported study from Taiwan [11] that showed evidence of subclinical community transmission with proved seroconversion several weeks before report of the first documented case in the island. A similar conclusion was also drawn from Australia [34] ; iii) our serological test might detect cross-reactive antibodies triggered by recent vaccination with trivalent seasonal influenza vaccine as reported [4, [35] [36] [37] [38] [39] . However, seasonal influenza vaccines were of rather limited use in Reunion Island, especially in children and young adults; iv) Finally the high baseline titers may reflect the infectious history of the individuals to seasonal influenza viruses cross antigenic with pH1N1/2009 virus as recently suggested for seasonal 2007 H1N1 infection [40] . This serosurvey indicates that a large fraction of the Reunion Island population was infected with the pandemic virus. Younger people, have paid the main tribute to the epidemic as almost two thirds show evidence of seroconversion, confirming earlier clinical reports from the island [12] and accumulating reports from other countries [17, 32, 41, 42] and suggesting that school children have likely played the central role in the epidemic diffusion of the pandemic virus. Lower infection rates were found in adults and the lowest rates were recorded in the elderly.

Based on clinical cases reported to the epidemiological surveillance services [12] , it was estimated that 66,915 persons in Reunion Island who consulted a physician were infected by the pH1N1/2009 virus during the 9 weeks of the epidemic, giving a cumulative attack rate of 8.26%. Taking into account those who did not consult a physician, the number of

symptomatic infected persons was estimated to 104,067 (attack rate: 12.85%). In fact, the attack rate of pH1N1/2009 infection in our serosurvey was about 42%-44% at the peak of the antibody response (i.e., weeks 40-44), a figure which is at least 3 to 4 times higher than rates of infection based on clinical cases. The wide gap between the two estimates indicates that a large fraction (almost two thirds) of those who got infected by pH1N1/2009 virus escaped medical detection, probably because they developed mild disease or asymptomatic infection, a further indication of the benign nature of the virus, at least at the community level. In England, Baguelin et al. [43] estimated that the cumulative incidence rates of infection by the pandemic virus in children were 20 to 40 times higher than that estimated from clinical surveillance.

Our study, as others [6], indicates that pre-existing cross reactive antibodies to pH1N1/2009 at titers $\geq 1/40$ prevented seroconversion in response to the pandemic virus. This level of pre-existing cross reactive immunity likely confers true protection against infection as about two thirds and one third of documented infection (qRT-PCR positive) in our series have occurred in individuals with baseline HIA titers $\geq 1/40$ and $< 1/40$ respectively and less than 5% of documented infections occurred in individuals with baseline titers $< 1/40$. The protection was effective not only in older adults but also in younger persons. This indicates that protection was conferred not only by baseline cross reactive antibodies triggered by close pH1N1/2009 viruses that circulated before 1957 (as in the elderly), but also by antibodies likely resulting from recent exposure to seasonal influenza epidemics (as shown in younger persons) [40]. The observed seroconversion rates depend on age, after adjusting for baseline pH1N1/2009 titers. The protective role of increasing age might be explained by a stronger cross-immunity in adults and elderly or by a higher exposure of young subjects to the virus during the 2009 epidemic (due to social contacts and mixing patterns). It may also indicate that immune mechanisms other than cross reactive antibodies detected by HIA (i.e. immunity to neuraminidase and conserved T cells epitopes [44]) might develop throughout life, providing additional protection from infection or severe disease, especially in the elderly. Interestingly, evidence is seen for a decline in antibody titers, which occurred soon after the passage of the epidemic wave. In paired sera, this decline was significant enough to bring, within a few weeks, almost 27% of sera that tested positive (i.e. HI titers $\geq 1/40$) in the immediate post epidemic phase to levels under the cut-off value in the second serum sample. This decay accounts for the observation that older adults (≥ 60 yrs) in the study cohort were apparently almost completely spared by the epidemic if one only considers cumulative incidence rates derived from IHA titration on samples 2 (weeks 45-52). In fact, the cumulative incidence rate in older adults measured just after the epidemic peak (i.e. weeks 40-44) was 20.4%.

Similar results of early antibody decay were recently reported [10, 45] . More generally, these data show that serosurveys conducted months after passage of the epidemic, likely underestimate the real extent of pH1N1/2009 infection, compared to antibody titration performed earlier, when humoral responses are at their highest level. Whether the decline in antibody titers has functional immunologic consequence to individuals or within the communities warrants further investigation. However, one should note that there was no second epidemic wave in Reunion Island during the subsequent austral winter seasons in 2010 and 2011.

Influenza during the 2010 winter was at a level not higher than the usual passages of seasonal flu, though almost two thirds of documented cases in 2010 were also due to pH1N1/2009v [46] . In addition many fewer pandemic virus isolates were noted during the ongoing 2011 austral winter, strongly suggesting that the first epidemic wave had conferred a solid herd immunity, at the community level.

Our study has some limitations. The fact that the epidemic progression coincided with the implementation of the prospective study, we were not able to collect, strictly speaking, pre-epidemic sera from the cohort members. Therefore we used as proxy base line seroprevalence data from individuals recruited at the very beginning of the investigation when the epidemic activity in the cohort was very low. This may overestimate the base line immunity if subclinical community transmission had occurred before the first cases of pH1N1/2009 influenza were reported. Antibodies to the pandemic virus were detected by HIA, a test that has a good specificity but a rather low sensitivity [46] . Hence, the threshold of 1/40 may underestimate the number of infected individuals. However, rates of seroconversion, the serologic gold standard test based on paired sera, likely gave the most accurate picture of the pandemic in at the community level in Reunion Island.

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High Burden of Non-Influenza Viruses in Influenza-Like Illness in the Early Weeks of H1N1v Epidemic in France

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Abstract: **BACKGROUND:** Influenza-like illness (ILI) may be caused by a variety of pathogens. Clinical observations are of little help to recognise myxovirus infection and implement appropriate prevention measures. The limited use of molecular tools underestimates the role of other common pathogens. **OBJECTIVES:** During the early weeks of the 2009–2010 flu pandemic, a clinical and virological survey was conducted in adult and paediatric patients with ILI referred to two French University hospitals in Paris and Tours. Aims were to investigate the different pathogens involved in ILI and describe the associated symptoms. **METHODS:** H1N1v pandemic influenza diagnosis was performed with real time RT-PCR assay. Other viral aetiologies were investigated by the molecular multiplex assay RespiFinder19®. Clinical data were collected prospectively by physicians using a standard questionnaire. **RESULTS:** From week 35 to 44, endonasal swabs were collected in 413 patients. Overall, 68 samples (16.5%) were positive for H1N1v. In 13 of them, other respiratory pathogens were also detected. Among H1N1v negative samples, 213 (61.9%) were positive for various respiratory agents, 190 in single infections and 23 in mixed infections. The most prevalent viruses in H1N1v negative single infections were rhinovirus (62.6%), followed by parainfluenza viruses (24.2%) and adenovirus (5.3%). 70.6% of H1N1v cases were identified in patients under 40 years and none after 65 years. There was no difference between clinical symptoms observed in patients infected with H1N1v or with other pathogens. **CONCLUSION:** Our results highlight the high frequency of non-influenza viruses involved in ILI during the pre-epidemic period of a flu alert and the lack of specific clinical signs associated with influenza infections. Rapid diagnostic screening of a large panel of respiratory pathogens may be critical to define and survey the epidemic situation and to provide critical information for patient management.

Text: In order to monitor the spread of influenza and alert health handlers, several epidemiological tools have been developed. In France, a network of 1300 general practitioners, "Réseau Sentinelles", working throughout the country, provides real-time clinical data used to evaluate regional and national influenza spreading [1, 2] . The criteria used by this network to define clinical influenza-like illness (ILI) are the occurrence of a sudden fever above 39°C with myalgia and respiratory signs. In general no formal viral diagnosis is carried out. The Groupes Régionaux d'Observation de la Grippe (GROG) is a second French network that surveys the emergence and the spread of the influenza viruses [3, 4] . This network is based on clinical surveillance of acute respiratory infections and laboratory analysis of nasal specimens collected from adults and children by volunteer general practitioners and pediatricians.

According to the sentinel network's criteria, French health authorities proclaimed that flu epidemic level was reached during the second week of September 2009 (week 37) [5, 6] . On the contrary, data provided by the GROG showed only sporadic H1N1v activity until the last week of October (week 44) [6, 7] . Thus, it became rapidly obvious that a variety of viruses were circulating in the community and that an overestimation of myxovirus infection was at stake [8, 9, 10, 11] .

As a better knowledge of the epidemic status was a key feature for national healthcare organization, hospital preparedness, patient management and disease control, unambiguous viral diagnosis appeared critical. In France, data on viral aetiologies associated with ILI were at best sporadic and correlations with clinical symptoms were often lacking. Extensive molecular assays to screening for respiratory viruses were not available countrywide for routine diagnosis. Therefore the epidemiological pattern of respiratory pathogens with overlapping seasonality was poorly known.

The aim of the present study was to investigate respiratory pathogens involved in ILI during the early weeks of the 2009-2010 H1N1v diffusion in France (weeks 35 through 44) and describe the associated symptoms in paediatric and adult populations.

This study was a non-interventional study with no addition to usual procedures. Biological material and clinical data were obtained only for standard viral diagnostic following

physicians' prescriptions (no specific sampling, no modification of the sampling protocol, no supplementary question in the national standardized questionnaire). Data analyses were carried out using an anonymized database. According to the French Health Public Law (CSP Art L 1121-1.1), such protocol does not require approval of an ethics committee and is exempted from informed consent application.

In the two academic hospitals, Saint-Louis hospital (SLS) in Paris and Tours hospital (TRS), influenza-like illness (ILI) was defined as a patient suffering from at least one general symptom (fever above 38°C, asthenia, myalgia, shivers or headache) and one respiratory symptom (cough, dyspnoea, rhinitis or pharyngitis), in agreement with the guidelines from the French Institut de Veille Sanitaire (InVS), a governmental institution responsible for surveillance and alert in all domains of public health [12]. Criteria for severe clinical presentation were temperature below 35°C or above 39°C despite antipyretic, cardiac frequency above 120/min, respiratory frequency above 30/min, respiratory distress, systolic arterial pressure below 90 mmHg or altered consciousness. Predisposing factors of critical illness were children younger than one year old, pregnant women, diabetes, chronic pre-existing disease (such as respiratory, cardiovascular, neurologic, renal, hepatic or hematologic diseases) and immunosuppression (associated with HIV infection, organ or hematopoietic stem cells transplantation, receipt of chemotherapy or corticosteroids) [13, 14]. A cluster of suspected influenza infections was defined as at least three possible cases in a week in a closed community (household, school,...) [15].

In the two institutions, the prescription of H1N1v molecular testing was recommended for patients with ILI and with either a severe clinical presentation, an underlying risk factor of complications or a condition which was not improving under antiviral treatment. Investigation of grouped suspected cases was also recommended. From week 35 (last week of August) to 44 (last week of October), 413 endonasal swabs were collected in 3 ml of Universal Transport Medium (Copan Diagnostics Inc, Murrieta, CA) from adults and children seen in emergency rooms for suspected ILI (Table 1) and sent to SLS and TRS laboratories for H1N1v detection. The two microbiology laboratories participated in the reference laboratories network for the detection of pandemic influenza H1N1v.

Clinical data were collected at the time of medical attention and reported by clinicians on a national standardized questionnaire provided by InVS [1, 12]. This questionnaire included the presence or absence of the main general and respiratory symptoms associated with ILI

(fever, asthenia, myalgia, shivers, headache, cough, rhinitis, pharyngitis, sudden onset) [12] .

Total nucleic acid was extracted from 400 mL of Universal Transport Medium using the EasyMag System (Biomérieux, Marcy l'Etoile, France) in SLS or the EZ1 Advanced XL (Qiagen, Courtaboeuf, France) in TRS, according to the manufacturers' instructions (elution volume: 100 mL in SLS or 90 mL in TRS). Before extraction, 5 mL of an Internal Amplification Control (IAC) which contained an encephalomyocarditis virus (EMC) RNA transcript was added into the sample.

Pandemic H1N1v infection was diagnosed by real-time reverse transcription-PCR (RT-PCR) assay on a 7500 Real Time PCR System (Applied Biosystems, Foster City, CA) according to the protocol of the Centers for Disease Control (CDC) [16] . Other respiratory infections were investigated by a multiplex molecular assay based on the Multiplex Ligation-dependent Probe-Amplification (MLPA) technology (RespiFinder19H, Pathofinder, Maastricht, The Netherlands) that allows the detection and differentiation of 14 respiratory viruses, including influenza virus A (InfA), influenza virus B (InfB), rhinovirus (RHV), parainfluenza viruses 1 to 4 (PIV-1 to PIV-4), human metapneumovirus (hMPV), adenovirus (ADV), respiratory syncytial virus A (RSVA), respiratory syncytial virus B (RSVB) and human coronaviruses 229E, OC43 and NL63 (Cor-229E, Cor-OC43, Cor-NL63) [17] . The test allows also the detection of H5N1 influenza A virus and of four bacteria: *Chlamydomphila pneumoniae* (CP), *Mycoplasma pneumoniae* (MP), *Legionella pneumophila* (LP) and *Bordetella pertussis* (BP). The amplified MLPA products were analyzed on an ABI 3100 genetic analyzer (Applied Biosystems, Foster City, CA). Fragment sizing analysis was performed with the GeneMarker software (SoftGenetics, LLC, State College, PA).

Further testing for H1N1v was carried out with Simplexa TM Influenza A H1N1 (2009) (Focus Diagnostics, Cypress, California) when the CDC real time RT-PCR assay was negative for H1N1 and the RespiFinder19H assay was positive for Influenza A. If this latter assay was negative, H3N2 typing was performed as previously described [18] .

Data from our study are summarized as frequencies and percentages for categorical variables. Quantitative variables are presented as medians, 25th and 75th percentiles. To compare those variables according to the viral infection status, Fisher tests

By using CDC reference assay, H1N1v was detected in 66 samples out of 413 (16.6%), more frequently in SLS (38 samples) than in TRS (28 samples) (p, 10⁻²⁴). Overall, weekly percentage of H1N1v positive endonasal swabs remained under 10% until week 41 and increase significantly after (P Trend, 0.0001) (Figure 1). Rate of H1N1v detection reached 30% in SLS at week 42 and in TRS at week 44. Overall, this rate was in agreement with results provided by the GROG network, showing an earlier start of H1N1v epidemic in Paris area [7, 19] .

All 413 nucleic acid extracts were analyzed using the RespiFinder19H assay (Figure 2). Sixty six patients tested H1N1v positive with CDC real time RT-PCR assay were confirmed with the multiplex assay. Thirteen were also co-infected by one or two other respiratory pathogens (multiple infections) (Figure 2). Three of the 347 H1N1v negative samples could not be studied with the multiplex assay because they contained RT-PCR inhibitors (no amplification of the internal control). Two hundred and fifteen (62.5%) of the remaining 344 H1N1v negative samples were found positive for at least one respiratory pathogen (Figure 2). Two hundred and twelve were positive for non influenza pathogens (189 single infections and 23 mixed infections with two, three or four viruses) and three additional single infections by influenza A were identified in SLS, including two by pandemic H1N1v and one by seasonal H3N2, as determined after molecular typing (data not shown).

Overall, 68 patients (16.5%) were then positive for H1N1v, one for H3N2 and 212 for non influenza pathogens. There were 245 single infections (55 with H1N1v and 190 with other respiratory pathogens) and 36 mixed infections (13 with H1N1v and 23 without H1N1v) (Figure 2).

Among H1N1v negative single infections, the most prevalent viruses were rhinovirus (62.6%, 119 patients), followed by parainfluenza viruses 1 to 4 (24.2%, 46 patients), adenovirus (5.3%, 10 patients), human coronavirus 229E, OC43 and NL63 (3.2%, 6 patients) and respiratory syncytial virus A and B (2.6%, 5 patients) (Figure 2). In addition, RespiFinder19H assay identified three patients with bacterial infection, two with *Mycoplasma pneumoniae* (one 25 years old female in SLS and one 39 years old female in TRS) and one with *Bordetella pertussis* (one 60 years old male in SLS). No single infection by influenza B, hMPV, *Chlamydia pneumoniae* or *Legionella pneumophila* was identified (Figure 2). To analyze if viral co-infections occurred more frequently for some

viruses, we carried out a two by two comparisons, that showed a higher proportion of co-infection only for ADV ($p = 0.05$).

Non-influenza respiratory viruses presented a different epidemic profile compared to H1N1v. Overall, in both hospitals, weekly rate of non-H1N1v respiratory viruses whether alone or involved in co-infection increased between week 37 and 39 (from 51.4% to 81.3%) and then consistently decreased (Figure 3). RHV infections that represented nearly half of non-H1N1v viral infections (141 out of 213, 66.2%) were a significant contributing factor. In both hospitals, emergence of H1N1v cases was associated with a rapid decline of RHV rate of infection from 50-60% down to less than 20% with a one to two weeks gap between SLS and TRS.

Data on age (In both institutions, 85.5% (106/124) children younger than 15 years of age were infected by at least one respiratory pathogen (Table 2). H1N1v infected patients were not significantly younger than H1N1v non infected patients (27 years old vs. 25 years old, $p = 0.80$) (Figure 4) . However, 70.6% (48/68) of H1N1v cases were identified in patients under 40 years old (22 in SLS and 26 in TRS) and no case was observed in patients older than 65 years (Table 2). PIV infection occurred in very young patients (median (Figure 4) . Consequently, PIV and ADV were more frequently detected in the younger population of TRS versus SLS ($p, 10/24$ and $p, 10/23$ respectively). In contrast, although individuals with RHV infection were slightly younger than individuals without (median age = 24 vs. 29 for patients without RHV, $p = 0.05$) (Figure 4) , influenza-like illness associated with RHV was more frequent in SLS than in TRS ($p = 0.012$). Finally, patients with viral multiple infection were significantly younger than those with single infection (median, IDR: 4, 2-18.5 vs. 25, 6-43) and rates of mixed infection At the time of medical attention, 383 (92.7%) standardized clinical questionnaires were collected out of 413 patients. Four of them could not be exploited because they were too incomplete. A review of the 379 workable questionnaires showed that 90.8% (344/379) of the patients included in this study fulfilled the criteria of ILI as defined above, and 52.5% had either a severe clinical presentation or an underlying risk factor of complications (45.9%, 174/379), or were in a suspected cluster of grouped cases (6.6%, 25/379).

Overall, most patients have fever (93.9%) and cough (86.1%) (Table 3). Other classical clinical signs associated with ILI such as asthenia, myalgia, shivers, headache, rhinitis or pharyngitis were less frequent. A sudden onset was also described in 59.2% of cases. Only

32.5% of the patients had a temperature above 39°C; the age of these patients ranged from zero to 86 years, with a median age of 32 years and a mean age of 34 years (data not shown).

In H1N1v infected patients (including single and multiple infections), the main symptoms were also fever (98.2%) and cough (89.5%) (We then compared clinical characteristics between patients positive for H1N1v, patients positive for other respiratory pathogens and negative for H1N1v and patients without any detection of respiratory pathogens (as detected with RespiFin-der19H) (Table 3). There was no difference between the three groups except for fever, cough, pharyngitis. However for these latter symptoms, the comparison between patients positive for H1N1v and those positive for other respiratory pathogens or between patients positive for H1N1v and those without any detection of respiratory pathogens, showed no difference except for pharyngitis, which was less frequent in patients positive for H1N1v than in patients positive for other respiratory pathogens (Table 3) .

As RHV was the most frequent aetiology in ILI, we also compared clinical symptoms observed in patients with a single infection by RHV or by H1N1v (data not shown). There was no difference except that rhinitis and pharyngitis were significantly more frequent in RHV infection (62.7% vs. 34.1% [$p = 0.006$] and 39.0% vs. 10.0% [$p = 0.001$], respectively). Viral multiple infection (including samples with H1N1v) was not associated with a different clinical presentation. Fever and cough were observed in over 90% of the patients (90.6% and 90.3%, respectively), but only 33.3% of these patients had a temperature above 39°C, which was not different from patients with single viral infection (28.6%).

Our results highlight the high frequency of non-influenza viruses involved in acute respiratory infections during the epidemic period of a flu alert as defined by the Réseau Sentinelles according to ILI definition (a sudden fever above 39°C accompanied by myalgia and respiratory signs). These data extend previous observations in Europe reporting high prevalence of RHV infections before seasonal influenza [4, 20] or in 2009, before H1N1v pandemic influenza [1, 8, 9, 11, 21] . We confirm that RHV represent the most frequent aetiology of acute respiratory Table 2 . Age of patients with respiratory samples positive for H1N1v, positive for other respiratory pathogens or negative. infections both in adult and paediatric populations and may represent more than 50% of cases. We show that other viral infections than influenza and RHV may represent up to 30% of aetiologies. We

observed differences between the two hospitals, with a higher frequency of parainfluenza and ADV infections in Tours in contrast with a higher frequency of RHV in Paris, likely explained by the higher proportion of paediatric samples collected in Tours. However, despite the distance between the two institutions (about 250 km) and differences between the two populations, both presented similar patterns of high frequency of non-influenza viruses in acute respiratory infections before the flu epidemic wave and a decline when influenza reached epidemic levels.

In the two cities, high frequencies of RHV were seen at the same level with a likely different evolution speed, with sudden increase and decrease in SLS and more progressive variation in TRS. In both institutions, there was a decrease in the proportion and number of RHV diagnoses roughly in parallel with the increase of influenza diagnoses. Indeed, H1N1v exceeds 20% of positive detection's rate only when RHV dropped under 40%. These data are thus consistent with negative interaction of the two epidemics at the population level. It was previously hypothesised that RHV epidemic could interfere with the spread of pandemic influenza [20, 21, 22] . Few in vitro data support this hypothesis. It has been reported that interferon and other cytokines production by RHV infected cells induced a refractory state to virus infection. These data include the three patients whose respiratory samples could not be studied with the multiplex assay because of RT-PCR inhibitors. of neighbouring cells [23] . Further work is needed to confirm in vitro and in vivo such negative interactions and if viral interference are really translated to a population level. Analysis of rhinovirus and influenza epidemics in previous years should also help to determine if similar interferences were observed with seasonal influenza and to elaborate modelling and prediction of the spread of influenza according to respiratory viruses' circulation. Systematic extensive screening of respiratory viruses at a national level should be implemented for this purpose.

Very few RSV infections were observed in contrast to usual epidemiology which was characterized the last four past years by a start of epidemics in weeks 44-45 [1] . It has been confirmed by other laboratories and the French InVS that the 2009-10 RSV epidemic was delayed and had a lower impact compared with the previous winter season [1, 24] . Delayed and reduced RSV spread may be due to viral interference between RSV and influenza. Another possible explanation is better prevention behaviour about respiratory infections as recommended by a national campaign including recommendations for hands washing after sneezing and the use of mask [1] .

Influenza infections were mainly detected in patient under 40 years old and no case was found in patients older than 65. These results corroborate previous data suggesting that past seasonal H1N1 infections or vaccination may give partial crossed protection [10, 13, 25] . We have previously shown that the neutralizing titers against pandemic H1N1v virus correlate significantly with neutralizing titers against a seasonal H1N1 virus, and that the H1N1v pandemic influenza virus neutralizing titer was significantly higher in subjects who had recently been inoculated by a seasonal trivalent influenza vaccine [26] .

Viral co-infections were predominantly seen in paediatric patients, as previously described [4, 27, 28, 29] , both in influenza and non-influenza cases at a similar rate. No evidence of more pronounced respiratory impact was seen in these patients.

Our results showed the lack of specific clinical signs associated with proven H1N1v infections. Clinical characteristics did not differ between influenza infections or other viral infections. In particular, the proportion of patients with fever above 39°C was not higher in H1N1v positive patients. In addition, the patients without any evidence of respiratory viral infections did not have different symptoms. These patients may have been infected with other virus not included in the multiplex assay (human Bocavirus, coronavirus HKU1) [9, 10, 11] or were seen too late at the time of viral shedding was cleared [30] . However, to determine how specific the symptoms are for influenza would require to assess also the distribution of respiratory pathogens (H1N1v and other respiratory viruses) and related symptoms in patients presented at the emergency departments in SLS and TRS with respiratory syndromes, but not tested for H1N1v. In addition, despite some underlying conditions that were associated with complications not previously observed in seasonal influenza, most illnesses caused by the H1N1v virus were acute and self-limited [13, 31] . The higher proportion of non influenza viruses reported in ILI in 2009 was thus most likely a consequence of more frequent visits to a doctor for respiratory tract infections than usually observed for fear of the flu pandemic. The general lack of difference in symptoms in the particular context of H1N1v pandemic has therefore to be considered with caution and does not rule out that more significant differences may arise in future influenza epidemics with other influenza viruses. Our data confirm that it may be virtually impossible to recognize symptoms heralding H1N1v infections and virological data should be helpful along with clinical reports to monitor influenza epidemic [10] .

Molecular multiplex detection has recently emerged as a potent diagnostic tool to determine acute respiratory infections' aetiologies [11, 32, 33] . These data show that sensitive molecular multiplex detection of respiratory viruses is feasible and efficient for the detection of virus involved in acute respiratory infections and provides insights into their epidemic profile. Our results confirm the performance of RespiFinder19H assay to detecting respiratory viruses in the general population as recently shown in transplant patients with ILI [34] . RespiFinder19H confirmed all H1N1 infections detected by the CDC reference assay and was able to identify two additional H1N1 cases suggesting a high sensitivity of this multiplex assay to detect influenza A infections.

In conclusion, our results highlight that successive and mixed outbreaks of respiratory viral infections may affect influenza epidemiology and can lead to misinterpret the early development of a flu epidemic. Rapid diagnostic screening of a large panel of respiratory pathogens may be critical to define and survey the epidemic situation and to provide critical information for patient management.

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Livestock Drugs and Disease: The Fatal Combination behind Breeding Failure in Endangered Bearded Vultures

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Abstract: There is increasing concern about the impact of veterinary drugs and livestock pathogens as factors damaging wildlife health, especially of threatened avian scavengers feeding upon medicated livestock carcasses. We conducted a comprehensive study of failed eggs and dead nestlings in bearded vultures (*Gypaetus barbatus*) to attempt to elucidate the proximate causes of breeding failure behind the recent decline in productivity in the Spanish Pyrenees. We found high concentrations of multiple veterinary drugs, primarily fluoroquinolones, in most failed eggs and nestlings, associated with multiple internal organ damage and livestock pathogens causing disease, especially septicaemia by swine pathogens and infectious bursal disease. The combined impact of drugs and disease as stochastic factors may result in potentially devastating effects exacerbating an already high risk of extinction and should be considered in current conservation programs for bearded vultures and other scavenger species, especially in regards to dangerous veterinary drugs and highly pathogenic poultry viruses.

Text: Environmental pollutants are increasingly documented as a driver of wildlife endangerment due to their roles in organ damage, hormonal disruption and alteration of the immune system [1, 2] . Disease may also facilitate endangerment and extinction at global and local scales, especially when pathogens interact with other drivers such as pollutants [3] . There is increasing concern about the impact of veterinary drugs and livestock pathogens as factors damaging wildlife health [4] [5] [6] , and even causing declines approaching extinction [7] . These threats may be especially detrimental to wildlife as they increasingly concur and interact as a consequence of the elimination of livestock residues containing veterinary pharmaceuticals and resistant pathogens due to growing intensive livestock operations worldwide [6, 8, 9] . In particular, the ingestion of antimicrobials, primarily fluoroquinolones, has been recently related to immunodepression-mediated acquisition of opportunistic pathogens and disease, as well as to organ damage in nestling vultures [6, 10, 11] . Fluoroquinolone residues have also been found in avian scavenger eggs and are associated with severe alterations in the development of embryo cartilage and bones that could preclude embryo movement and subsequently normal development, pre-hatch position and successful hatching [12] . Therefore, antimicrobials and other drugs may negatively affect embryo and nestling health with potentially devastating consequences on breeding success and conservation of vultures and other threatened avian scavengers.

The bearded vulture (*Gypaetus barbatus*) is one of the most endangered birds in Europe, with a main stronghold in the Pyrenees. Increasing declines in productivity (average number of fledglings raised per territorial pair) have recently been reported in the Spanish Pyrenees associated with habitat saturation processes [13, 14]. Given that bearded vultures may raise only one fledgling per breeding attempt, this productivity decline should be linked to increasing breeding failure when the proportion of territorial pairs that are breeding does not greatly vary with time [15]. The proximate mechanisms by which density can affect productivity have been investigated, including habitat heterogeneity, with progressively poorer territories being used, territory shrinkage and interference with breeders and floaters [13]. However, the proximate causes of breeding failure are poorly known despite the long-term interests in the conservation of this species [16]. To evaluate these causes, the examination of failed eggs and dead nestlings is imperative, including the study of the presence and impact of injury, developmental problems, poor nutritional condition, pollutants, organ damage, pathogens causing disease, etc. in order to determine the most likely cause of breeding failure.

Here, we conducted a comprehensive study of failed eggs and dead nestling bearded vultures collected during recent years in the Pyrenees. Both the productivity and survival rates of adults and young birds have reached the lowest values since the bovine spongiform encephalopathy (BSE) crisis [13, 14, 17]. This temporal decline could be related to illegal poisoning [17] and recent changes in the abundance, distribution and quality of carrion available to avian scavengers as a consequence of EU regulations derived from the BSE crisis [6, [18] [19] [20]. In particular, the BSE crisis caused the lack or scarcity of unstabled livestock available to scavengers and their subsequent increase in the consumption of carrion from stabled livestock, which is intensively medicated [21]. Therefore, we specifically focused on determining whether breeding failure in bearded vultures is related to the ingestion of veterinary drugs from stabled livestock carrion, as documented in other avian scavenger species [12]. We also assessed the potential effects of veterinary drugs on embryo damage and immunodepression increasing the probability of acquisition and proliferation of pathogens causing fatal disease [6, [10] [11] [12] 21]. Because veterinary drugs should be exclusively acquired from the ingestion of carrion from livestock medicated to combat disease, we predict that their presence should be associated with that of pathogens acquired from the same livestock, especially poultry pathogens more likely transmitted between avian species [22]. Alternatively, if the temporal decline in productivity was primarily associated with breeding failure due to the effects of habitat saturation processes [13, 17], we should expect egg and nestling mortality to be directly related to developmental and nutritional problems indicating

progressively lower quality territories (e.g. embryo emaciation, nestling starvation) and interference by both conspecifics and heterospecifics (e.g. incubation failure, injury due to predation attempts or disturbance).

Failed eggs (n = 5) and dead nestlings (n = 4) were collected from bearded vulture nests located in the Spanish Pyrenees between 2005 and 2008. The study of this material did not require of the approval of an ethics committee because it was collected after breeding failure (egg or nestling death) was confirmed in the field. Three of the specimens (two nestlings and one egg) were collected in 2005, 2007 and 2008 from a particular territory. Eggs and nestlings were collected after breeding failure and frozen. Necropsies were performed on all specimens according to standard protocols [12] . The age of embryos and nestlings were estimated according to size and development. Samples of liver, kidney, spleen, large and small intestines, lungs, brain, lymphoid organs (thymus, bursa of Fabricius, Peyer's patches) and knee joints were fixed in 10% buffered formalin, sectioned at 4 mm and stained for histopathological analysis [10, 12] .

Liver (dead nestlings and failed embryos) and yolk (failed embryos) were used for the determination of the presence of veterinary drugs, including fluoroquinolones (enrofloxacin and ciprofloxacin), other antimicrobials (amoxicillin and oxytetracycline), non-steroidal anti-inflammatories (NSAIDs) such as diclofenac, flunixin meglumine, ketoprofen, ibuprofen, meloxicam, sodium salicylate, acetaminophen, and antiparasitics (metronidazole, diclazuril, fenbendazole, ivermectin) as described previously [12] . The limits of quantification, percentage recoveries, and interand intra-assay reproducibility were adequate [10, 12] .

Other contaminants potentially affecting eggs and embryos were determined in liver, including heavy metals (Cd, Zn, Pb and Hg), following Blanco et al. [23] , dithiocarbamate thiram, disulfuram, polybrominated diphenyl ethers, organochlorines and brominated flame retardants, following Lemus et al. [12] and carbamate and organophosphate pesticides (carbofuran, aldicarb and fenthion) following Elliot et al. [24] . We measured brain cholinesterase activity to assess early exposure to anticholinesterase pesticides [25] . Potential contamination was assessed by comparison with levels from apparently normal wild birds of other species [26] in the absence of basal levels for bearded vultures.

Determination of bacterial and fungal pathogens were conducted by sampling oropharynx, lung, liver, kidney, spleen, and intestine with sterile swabs and cultured using standard microbiology protocols [10, 12, 27, 28] . *Salmonella* serotypes and phage types were determined in the Spanish Reference Laboratory (Laboratorio Central Veterinario, Algete, Madrid). For confirmation of the identification of the alpha hemolytic *Streptococcus pneumoniae* we used a specific identification test (Accuprobe, Salem, MA) based on the detection of specific ribosomal RNA sequences. Samples of lesions found in internal organs and tissues during necropsies were taken with sterile swabs and cultured using the same standard microbiology protocols. In addition, we determined the presence of selected avian pathogens, including bacterial, viral, fungal, and protozoan pathogens by means of PCR-based methods (see Table S1 for details). The presence of *Chlamyophila psittaci* and *Mycoplasma* sp. in blood was determined as described previously [29, 30] . The presence of poxvirus, the paramyxovirus causing Newcastle disease, the serotypes H5, H7 and H9 of avian influenza, falcon adenovirus, circovirus, herpesvirus, polyomavirus, reovirus and West Nile virus were determined following the PCR-based methods available in the literature [31] [32] [33] [34] [35] [36] [37] [38] [39] . We also searched for helminths and protozoans in the gastrointestinal tract by macroscopic and microscopic observations using standard protocols [40] .

Specific immunocytochemical procedures were used for detection of mielodepressive virus, including the alphaherpesvirus causing Marek disease [41] in kidney and bursa of Fabricius, the gyrovirus causing infectious chicken anaemia [42] in thymus and bone marrow, the birnavirus causing infectious bursal disease (IBD, [43]) in bursa of Fabricius, and the coronavirus causing chicken infectious bronchitis in kidney [44] . In addition, we conducted a specific immunocytochemical procedure for West Nile virus antigen detection [45] in brain, spinal medulla, thymus and thyroid. All immunohistochemistry analyses were conducted at the Department of Veterinary Anatomy, Veterinary Faculty, Universidad Complutense de Madrid, Spain and at the Pathology Department of the Veterinary Faculty, University of Utrecht, The Netherlands. The presence of these viruses was also determined by PCR-based methods [43, [46] [47] [48] .

All dead nestlings and three of five unhatched embryos showed two to six different veterinary drugs in liver (nestlings) and egg yolk (embryos). In addition, the two embryos with fluoroquinolones in the yolk also had them in the liver (Table 1) . Fluoroquinolones were the most prevalent drugs and showed the highest concentrations (Table 1) . Other drugs such as NSAIDs and antiparasitics were found in most nestlings at variable

concentrations, but in no eggs (Table 1) . Other toxic compounds were detected in lower prevalence and concentrations (see Table 1 for those more relevant values; all insecticides were found at concentrations ,0.001 ppb), which was further supported by basal levels of brain cholinesterase (Table 1) .

Dead embryos and nestlings showed a moderate to good nutritional state. Major histopathological lesions were primarily located in the kidney, including glomerulonephritis and/or glomerulonephrosis present in all individuals with fluoroquinolones, but not in those without drugs (Table 1) . All individuals with fluoroquinolones also showed joint lesions, including arthritis and/ or arthrosis of the long bone articulations, as well as massive osseous stroma of the spongy bones.

The fungi *Candida albicans* was isolated from the oral cavity of five individuals. All individuals showed non-specific mixed bacterial flora. Enterotoxigenic *Escherichia coli* and *Salmonella* spp. were isolated in four cases (Table 1 (cont.) *Samples from the same territory in different years. 1 Veterinary drugs. EN: enrofloxacin (mg/g), CI: ciprofloxacin (mg/g), OX: oxytetracycline (mg/g), FL: flunixin meglumine (mg/g), AS: sodium salicylate (mg/g), IV: Ivermectin (mg/g). 2 Other toxicants. OR: organochlorines (ng/g), Pb: lead (ng/g). 12. z29 (three cases, see Table 1) . One individual showed infection by *Salmonella enterica* enteritidis (see above) and enterotoxigenic *Escherichia coli* O86 in all examined organs (septicaemia) except brain, which rejected the possibility of post-mortem contamination. *Pasteurella multocida* was isolated in a single individual that also showed enterotoxigenic *Escherichia coli* O86 (Table 1) ; all of these individuals contained fluoroquinolones. One of the failed embryos without veterinary drugs showed suppurative myocarditis, multiple microabscesses in head muscles, suppurative leptomeningitis, as well as lower jaw gangrenous inflammation with loss of the osseous stroma due to a mixed infection with *Streptococcus suis* and *Streptococcus pneumoniae* in brain, meninges and neck muscles; this embryo also showed infection by chicken infectious bronchitis (Table 1) . Both immunocytochemistry for the detection of poultry viruses and PCR pathogen survey were positive to IBDV in six individuals with fluoroquinolones (Table 1) . Immunocytochemical procedures failed to detect West Nile virus antigens in individuals in which PCR for this virus had been positive. Parasitology was negative for all helminths, helminth eggs and protozoans.

We found multiple veterinary drugs, primarily fluoroquinolones, in most failed eggs and dead nestling bearded vultures from the Pyrenees. They also showed multiple internal organ damage and pathogens potentially acquired from medicated livestock carrion, especially viruses often infecting poultry. Recorded drug concentrations were among the highest reported in avian scavengers [6, [10] [11] [12] 21] . NSAIDs and antiparasitics were found in lower prevalence than fluoroquinolones, but at higher concentrations than those found in other avian scavengers, especially for flunixin meglumine and sodium salicylate [6, 12, 21] . On the contrary, we found no sterile eggs, poor nutritional conditions or injury in any failed embryo or nestling. Other pollutants were found in low prevalence and concentrations posing low risk to embryo and nestling health.

Fluoroquinolones may cause generalized direct developmental damage precluding embryo hatching, physiological alterations due to their impact on liver and kidney and immunodepression reducing resistance to opportunistic pathogens [6, [10] [11] [12] 21] . These pathogens may be acquired at the same time that drugs used to treat diseased livestock are ingested, as indicated by their high prevalence in embryos and nestlings. Therefore, despite the relatively small sample size resulting from low abundance, endangerment and logistic difficulties in reaching nests in this species, the results provide evidence of a combined impact of veterinary drugs and livestock disease as the primary cause of breeding failure in the sampled individuals.

The presence of West Nile virus is not likely to be associated with nestling disease or mortality because the lack of lesions in target tissues and viral antigen particles in the immunohistochemistry study. Fatal septicaemia caused by *Streptococcus suis*, one of the most important swine pathogens worldwide [49] , in combination with septicaemia from *Streptococcus pneumoniae* and infection by chicken infectious bronchitis virus were found in a single embryo. This concentration of livestock pathogens has not been reported before and, to our knowledge, this is the first report of the three pathogens causing disease in a wild bird. Other pathogens recorded in embryos and nestlings, including *Salmonella* serotypes and phages typical of livestock [50] , and enterotoxigenic *Escherichia coli* O86 causing septicaemia, were potentially transmitted by consumption of carcasses of infected poultry and other livestock [22, 27, 28] . In addition, we found that the IBD virus infected most individuals alone or together with other pathogens also potentially acquired from livestock carrion. This virus causes a highly contagious immunosuppressive bursal disease in poultry [51] and may be transmitted to wildlife in contact with poultry waste or by ingestion of carcasses [22, 52] . Nestlings are especially susceptible to IBD because of

the primary role of bursa of Fabricius in immune function development at this age. In fact, immunosuppression due to IBD was indicated by the inflammation, necrosis and loss of lymphocytes in the bursa of Fabricius together with the presence of viral antigens recorded by means of immunocytochemical procedures. The potential impact of highly pathogenic and contagious poultry viruses has been previously recognized as a threat to wildlife health due to the increasing contact of wildlife with livestock operations in general, and poultry farms and their residues in particular, in natural areas worldwide [52] [53] [54] [55] . However, damage from IBD virus on the bursa of Fabricius represents, to our knowledge, the first evidence of clinical disease compatible with death caused by this poultry virus in wildlife. The presence of IBD has been not previously recorded in embryos of wild birds, probably because vertical transmission has been ruled out in poultry and, as consequence, it has probably not been evaluated in other species until now. This striking and concerning result could be related to the longer egg development and incubation periods of bearded vultures compared with poultry, and/or due to contrasting environmental conditions during incubation between bearded vultures and poultry. Thus, embryo infection with IBD may occurs via the female or during incubation as a consequence of egg contact between the egg and poultry remains in the nests of bearded vultures, which requires more research.

Despite their potential effects on population dynamics and conservation through a reduction of productivity and changes in mating behaviour [13, 14] , habitat saturation processes were apparently not directly related to particular proximate causes of egg and nestling failure in this study or in these sampled individuals. As an alternative non-mutually exclusive explanation, we suggest that the recent decline in productivity could also be linked to the increasing ingestion of veterinary drugs and acquisition of pathogens from medicated stabled livestock carcasses due to decreasing availability of unstabled livestock carcasses -the traditional primary food of bearded vultures [16] since the BSE crisis [21] , accompanied by a possible increasing use of antibiotics in stabled livestock operations. In this sense, it is remarkable that bearded vultures primarily feed upon livestock bones, which are one of the major target tissues of fluoroquinolones in medicated animals [56] , therefore, rendering this species especially sensitive to the consequences of an increase in the consumption of stabled intensively medicated livestock. The presence of veterinary drugs in eggs implies their previous presence at least in breeding females [12] , but also probably in breeding males and non-breeders frequently using artificial feeding sites and livestock carcass dumps [17] , where veterinary drugs may be ingested from medicated livestock carcasses [10, 21] . Therefore, further research is required to determine the impact of veterinary drugs and livestock disease on fitness of full-grown

individuals, including the potentially subtle, sublethal or indirect effects of these factors on population dynamics.

The link between veterinary drugs and livestock disease should be further investigated in scavenger species, because both threats may concur in food and because the immunodepressive effects and other physiological alterations caused by drugs may facilitate the acquisition and proliferation of pathogens [6, 11, 21] . Given that both threats acting together may greatly contribute to breeding failure decreasing productivity, their potential as stochastic factors with potentially devastating effects increasing the risk of extinction should be not overlooked in current conservation programs of bearded vultures and other scavenger species, especially regarding dangerous veterinary drugs and highly pathogenic viruses frequently infecting poultry. In addition, restricted geographic distribution and low genetic variability [57] common to many threatened species may favour pathogen transmission and reduce the ability of a naïve immune system to fight against novel pathogens [3, 28, 58] , making them especially vulnerable to the potential cross-species transmission of highly virulent virus strains able to cause important outbreaks, as reported in poultry [59] [60] [61] .

The association of pollution and disease may further increase extinction risk if it interacts with the effects of habitat saturation processes [13, 14, 17] . These processes may facilitate conspecific contact and interactions also likely to increase intra-and interspecific pathogen transmission rates in breeding and feeding areas, especially of highly contagious poultry diseases [22] . This could be further enhanced by the artificially high numbers of bearded vultures and other scavengers attracted to feeding points and carcass refuse dumps, both as a result of management and due to the scarcity of unstabled livestock carcasses since the BSE crisis [17, 21] . Whatever the potential contribution of underlying ultimate mechanisms reducing productivity, our findings highlight the need to determine the proximate causes of breeding failure and mortality in wildlife populations in order to understand the processes regulating demography from an ecological framework perspective.

Table S1

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A Schiff Base-Derived Copper (II) Complex Is a Potent Inducer of Apoptosis in Colon Cancer Cells by Activating the Intrinsic Pathway

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Abstract: Metal-based drugs with extensive clinical applications hold great promise for the development of cancer chemotherapeutic agents. In the last few decades, Schiff bases and their complexes have become well known for their extensive biological potential. In the present study, we examined the antiproliferative effect of a copper (II) complex on HT-29 colon cancer cells. The Cu(BrHAP)(2) Schiff base compound demonstrated a potent antiproliferative effect in HT-29 cells, with an IC(50)value of 2.87 µg/ml after 72 h of treatment. HT-29 cells treated with Cu (II) complexes underwent apoptosis death, as exhibited by a progressive elevation in the proportion of the G(1) cell population. At a concentration of 6.25 µg/ml, the Cu(BrHAP)(2) compound caused significant elevation in ROS production following perturbation of mitochondrial membrane potential and cytochrome c release, as assessed by the measurement of fluorescence intensity in stained cells. Furthermore, the activation of caspases 3/7 and 9 was part of the Cu (II)

complex-induced apoptosis, which confirmed the involvement of mitochondrial-mediated apoptosis. Meanwhile, there was no significant activation of caspase-8. Taken together, these results imply that the Cu(BrHAP)(2) compound is a potential candidate for further in vivo and clinical colon cancer studies to develop novel chemotherapeutic agents derived from metal-based agents.

Text: Cancer is a debilitating disease that afflicts a substantial portion of the world population in all generations and is a major health problem of global concern [1] . Among the various types of cancer, colorectal cancer is the second and third most prevalent cancer among males and females in the United States, respectively. In spite of all the considerable progress in protective methods and recent improvements in screening techniques and chemotherapy, the 1-year and 5-year relative survival rates for patients suffering from colorectal cancer are 83.2% and 64.3%, respectively [2] . In addition, due to bitter controversy over optimal methods for early detection, full compliance of patients with screening recommendations remains a major hindrance for diagnosis at the early stages of cancer development. Development of resistance to chemotherapy also represents a critical issue for which simultaneous treatment with various classes of therapeutics to reduce the resistance has yielded some success [3] . Moreover, the numerous side effects of chemotherapeutic drugs on cancer patients, including hair loss, diarrhea, bleeding, and immunosuppression, have made the process 2

The Scientific World Journal of treatment more complicated [4] . The highly regulated programmed cell death process of apoptosis is a matter of great interest in oncology and cancer therapy and represents a common molecular pathway for drug resistance and carcinogenesis [5] .

Maintenance of a constant cell number in the colonic mucosa is highly regulated through the balance between apoptosis and cell proliferation. The perturbation in this balance leads to an escape from normal cell number homeostasis and is associated with the progression of cancer cells [6, 7] . Thus, suppression of proliferation and elevation of apoptosis in these aberrant cells are suggested to be the essential mechanism for the inhibition of colon cancer. Furthermore, apoptosis and the factors involved in its mechanism of action also present a window that can be exploited for the improvement of potential therapeutic agents with high effectiveness and less adverse side effects [8] . Hence, screening for novel compounds capable of inducing apoptosis in colon cancer

cells that can be used alone or in combination with other chemotherapeutic drugs is a significant need and represents a critical challenge in medicinal chemistry.

Metal complexes have been extensively utilized in clinics for centuries and have attracted numerous inorganic chemists to analyze them, with the main focus being medical applications [9, 10]. Copper, an essential trace element with an oxidative nature and bioessential activity in human metabolism, does not exist in an ionic form in biological systems. Thus, measurement of copper in the body is evaluated in the form of complexes with organic compounds [11]. Schiff bases are a critical class of compounds in medical chemistry that have demonstrated significant chemotherapeutic and antibacterial application [12, 13]. Schiff base Cu(II) complexes revealed great potential for antiproliferative, antibacterial, and gastroprotective activity [14] [15] [16] [17] [18]. This study evaluated the anticancer potential of a copper (II) complex derived from N,N - dimethyl ethylene diamine and 2-hydroxyacetophenone Schiff base ligand, Cu(BrHAP) 2. Furthermore, the possible apoptotic mechanism underlying this activity was also examined. Dulbecco's Modified Eagle Medium (DMEM, Life Technologies, Inc., Rockville, MD) containing 10% fetal bovine serum, 100 g/mL streptomycin, and 100 U/mL penicillin G at 37 ° C in a humidified atmosphere of 5% CO₂ /95% air. The cells were plated at a fitting density in tissue culture flasks (Corning, USA) according to each experimental scale. Cell viability was measured by a conventional MTT [3-(4,5-dimethylthiazol-2yl)-2,5-diphenyltetrazolium bromide] reduction assay. After 48 h exposure to six concentrations of Cu(BrHAP) 2, cells were treated with MTT solution (2 mg/mL) for 2 h. The dark formazan crystals formed in intact cells were dissolved in DMSO, and the absorbance was measured at 570 nm and 650 nm as a background using a microplate reader (Hidex, Turku, Finland). The IC₅₀ value was determined as the concentration of Cu(BrHAP) 2 required to reduce the absorbance of treated cells to 50% of the DMSO-treated control cells. All samples were prepared in triplicates.

Assay. Measurement of lactate dehydrogenase (LDH) release is a biomarker for determining the cytotoxicity of a compound. Briefly, HT-29 cells were treated with different concentrations of Cu(BrHAP) 2 and Triton X-100 (positive control) for 48 h, and the supernatants of the untreated and treated cells were transferred to a new 96-well plate for LDH activity analysis. Next, 100 L of LDH reaction solution was added to each well, the plate was incubated at room temperature for 30 min, and the absorbance was read at 490 nm using a Tecan Infinite 200 Pro (Tecan, Männedorf, Switzerland) microplate reader. The amount of formazan salt and intensity of red color in treated and untreated samples were

represented as the LDH activity of cells. The LDH release level in cells treated with Cu(BrHAP) 2 was expressed as a percentage of the positive control.

A propidium iodide (PI) and acridine orange (AO) double staining assay were carried out for detection of apoptosis in the treated cells using a fluorescent microscope (Leica attached with Q-Fluoro software) according to a standard procedure. HT-29 cells (5×10^4 cells/mL in a 25 mL culture flask) were plated, treated with Cu(BrHAP) 2 at the IC 50 concentration, and incubated for 24, 48, and 72 h. After harvesting the cells, they were stained with fluorescent dyes and observed under a UV-fluorescent microscope (Olympus BX51) within 30 min.

In brief, HT-29 cells (1×10^4 cells/well in 96-well plate) were supplemented with Cu(BrHAP) 2 (2 g/mL) or DMSO (negative control) for 24 h. The live cells were then incubated with BrdU and Phospho-Histone H3 dyes for 30 min. After the cells were fixed and stained as described by the manufacturer's instructions, they were visualized and analyzed using the Cellomics ArrayScan HCS reader (Thermo Scientific). The fluorescence intensities of the dyes were measured using a target activation bioapplication module.

To confirm the result of the fluorescence cell cycle analysis, HT-29 cells (5×10^4 cells/mL) were treated with Cu(BrHAP) 2 for 24, 48, and 72 h for flow cytometry analysis. After incubation, HT-29 cells were spun down at 1800 rpm for 5 min. Next, fixation of a cell population for flow cytometry analysis was carried out to restore integrity. In brief, the cell pellets were fixed by mixing them with 700 L of cold ethanol (90%) and were then kept at 4 °C overnight. Treated HT-29 cells were spun down, and the ethanol was discarded. After washing and suspending the cells in PBS, 25 L of RNase A (10 mg/mL) and 50 L of propidium iodide (PI) (1 mg/mL) were added to the fixed cells for 1 h at 37 °C. The added RNase A limited the ability of PI to bind to only DNA molecules. At the end, the DNA content of the cells was analyzed by a flow cytometer (BD FACSCanto II).

The oxygen radical antioxidant capacity (ORAC) assay was carried out based on the protocols described in detail previously [19]. In brief, Cu(BrHAP) 2 at the concentration of 100 g/mL was used for this assay in a total reaction volume of 200 L. The experiment was performed in a black 96-well microplate with 25 L of compound, blank (solvent/PBS), standard (trolox), or positive control (quercetin). The plate was then supplemented with the

working fluorescein solution (150 L), followed by a 5 min incubation at 37 °. The total volume of 200 L was made up by adding 25 L of AAPH working solution. Fluorescence intensity was measured at an excitation wavelength of 485 nm and an emission wavelength of 538 nm every 2 min for 2 h. The result was quantified by calculating the differences of area under the fluorescence decay curve (AUC) of samples and blank. The values were Trolox equivalents (TE).

In brief, HT-29 cells (1×10^4 cells/mL) were seeded in 96-well plates and treated with different concentrations of Cu(BrHAP) 2 and DMSO (negative control) for 24 h. After 30 min treatment with dihydroethidium (DHE) dye, cells were fixed and washed with wash buffer as described by the manufacturer's instructions. In the presence of superoxides, DHE dye is oxidized to ethidium. The fluorescence intensity was determined by a fluorescent plate reader at an extension wavelength of 520 nm and an emission wavelength of 620 nm.

The critical factors for monitoring the cell health, namely, cell loss, changes in cell permeability, cytochrome release, mitochondrial membrane potential changes, nuclear size, and morphological changes, were studied using a Cellomics Multiparameter Cytotoxicity 3 Kit as described in detail previously [20]. Plates with stained cells were analyzed using the ArrayScan HCS system (Cellomics, PA, USA).

Caspases 3/7, -8, and 9 activities were determined using the commercial caspase-Glo 3/7, 8, and 9 assay kit (Promega, Madison, WI). HT-29 cells (1.0×10^4 cells/well) were seeded overnight in white-walled 96-well plates and treated with different concentrations of Cu(BrHAP) 2 for 24 h. According to the manufacturer's protocol, the treated cells were supplemented with caspase-Glo reagent (100 L) and incubated at room temperature for 30 min. The active caspases from apoptotic cells caused the cleavage of aminoluciferin-labeled synthetic tetrapeptide, leading to the release of substrate for the luciferase enzyme. Caspase activities were analyzed using a Tecan Infinite 200 Pro (Tecan, Männedorf, Switzerland) microplate reader.

In brief, HT-29 cells (1.0×10^4 cells/well in a 96-well plate) were treated with different concentrations of Cu(BrHAP) 2 for 3 h, followed by stimulation with TNF-(1 ng/mL) for 30 min. After discarding the medium, cells were fixed and stained using a Cellomics nucleus factor-B (NF-B) activation kit (Thermo Scientific) according to the manufacturer's

instructions. Next, an Array Scan HCS Reader was used for evaluation of the plate. Cytoplasmic and nuclear NF- κ B intensity ratios were calculated using Cytoplasm to Nucleus Translocation Bioapplication software. The average intensity of 200 cells/well was determined. The ratios for untreated, treated, and TNF-stimulated cells were compared.

All the experiments were performed at least three times independently. The results were presented as the mean $\pm$ standard deviation (SD) of the number of experiments shown in the legends. An analysis of variance (ANOVA) was carried out using the prism statistical package (GraphPad Software, USA). < 0.05 was considered statistically significant.

Cells of the Colon. Initially, the cytotoxicity of Cu(BrHAP) 2 was tested on HT-29 and CCD 841 cell lines. The IC₅₀ values of the Schiff base compound were determined based on the result collected from three independent MTT experiments. As indicated in Table 1, Cu(BrHAP) 2 elicited a significant cytotoxicity and cell inhibitory effect after 24, 48, and 72 h of treatment on HT-29 cell. **2-Induced LDH Release.** Lactate dehydrogenase (LDH) release in the medium is a marker that shows the loss of membrane integrity, apoptosis, or necrosis. The cytotoxicity of the Cu(BrHAP) 2 compound, as determined by the LDH release assay, was quantified on HT-29 cells treated with various concentrations of the Schiff base compound for 48 h. Cu(BrHAP) 2 induced a significant elevation in LDH release, demonstrating cytotoxicity at the 6.25 and 12.5 μ g/mL concentrations compared to the control cells (Figure 2).

Microscopy and AO/PI Double Staining. Morphological changes in HT-29 cells treated with Cu(BrHAP) 2 compound were observed under a fluorescent microscope at 24, 48, and 72 h. The cells were scored under a fluorescent microscope to analyze viable cells, early apoptosis, and late apoptosis. Early apoptosis, defined as intervening AO within the fragmented DNA, was observed under bright green fluorescence. At the same time, control cells were visualized with a green intact nuclear structure. After 24 and 48 h of treatment with Cu(BrHAP) 2, moderate apoptosis was observed in the form of blebbing and nuclear chromatin condensation. Furthermore, in the late stage of apoptosis, changes, such as the presence of a reddish-orange color due to binding of PI to denatured DNA, were observed after 72 h of treatment (Figure 3). The results showed that the Cu(BrHAP) 2 compound induced morphological features of apoptosis in a time-dependent manner. Figure 4, demonstrated that there is no cell cycle arrest in the S/M phases. The lack of cell cycle arrest in the S/M phases suggested possible cell cycle arrest in the G₁ /G₂ phases. To

determine the exact arrested phase, treated HT-29 cells were analyzed for cell cycle progression using flow cytometry. As expected, there was no significant arrest in the S/M phases. Meanwhile, significant cell cycle arrest in the G 1 phase was observed for HT-29 cells after 24 and 48 h of treatment (Figure 5).

Assay. Antioxidant capacity was measured by ORAC assay, which is the only assay that involves the use of peroxy radical as a prooxidant and quantifies activity via the area under the curve (AUC) technique. In our experiment, quercetin was used as a positive control. The result demonstrated that Cu(BrHAP) 2 exhibited low to moderate antioxidant activity compared to quercetin (Table 2) .

Formation. HT-29 cells were treated with different concentrations of Cu(BrHAP) 2 for 24 h and stained with DHE dye to determine the influence of the Schiff base compound on ROS production. The fluorescence intensities of DHE oxidation by ROS were quantified using a fluorescence microplate reader. As depicted in Figure 6 , exposure to the Schiff base compound caused a significant elevation in the ROS levels of treated HT-29 cells at the 6.25 g/mL concentration.

To investigate the induction of apoptosis by Cu(BrHAP) 2 , nuclear morphological changes in HT-29 cells were analyzed by detection of nuclear condensation. As shown in Figure 7 , Hoechst 33342 staining demonstrated that nuclear condensation, which is directly related to apoptotic chromatin changes, emerged in some cells after treatment with Cu(BrHAP) 2 . Meanwhile, the permeability of treated cells was also elevated. Mitochondria are the main source for the production of ROS and adenosine triphosphate (ATP) and are critical in controlling the death and survival of cells. The reduction in fluorescence intensity depicted in Figure 6 Cu(BrHAP) 2 triggered the translocation of cytochrome from mitochondria into the cytosol during apoptosis in HT-29 cells.

Activation. The elevation in ROS production associated with a collapse in MMP may lead to the activation of the caspase cascade. To investigate caspase activation, the bioluminescent intensities representing caspases 3/7, 8, and 9 activities were quantified in HT-29 cells treated with different concentrations of Cu(BrHAP) 2 for 24 h. As shown in Figure 8 , significant elevation in the activity of caspase-3/7 at the 6.25 g/mL concentration and caspase-9 at the 6.25 and 12.5 g/mL concentrations was observed in Cu(BrHAP)

2-treated cells, while no significant change in the activity of caspase-8 was detected between treated and untreated HT-29 cells. Thus, the apoptosis induced by the Schiff base compound in HT-29 cells is possibly mediated via the intrinsic pathway, but not the extrinsic pathway.

is a transcription factor that has a critical role in cytokine gene expression. NF- κ B activation and translocation to the nucleus to enable DNA-binding activity and facilitate target gene expression are mediated by inflammatory cytokines such as tumor necrosis factor-(TNF-). The Cu(BrHAP) 2 Schiff base compound did not exhibit any inhibitory effect on translocation of TNF--stimulated NF- κ B in HT-29 treated cells, and TNF--stimulation led to NF- κ B translocation from the cytoplasm to the nucleus (Figure 9).

Carcinogenesis is a multistage process in which unregulated cell proliferation as well as a reduction in apoptosis incidence serves as initial characterizations for its progression [21]. One of the defense procedures in multicellular organisms is the destruction of undesirable cell development, which is defined as programmed cell death. Apoptosis is the most noticed programmed cell death mechanism and is characterized by distinct morphological changes such as membrane permeability, cell shrinkage, disruption of the mitochondrial membrane, and chromatin condensation [22, 23]. The disruption of cellular homeostasis between cell death and cell proliferation leads to cancer incidence [24], and agents that can induce apoptosis are known to have potential anticancer effects [25, 26]. Apoptosis pathways are effective targets for cancer therapy as well as chemoprevention. Numerous chemopreventive drugs have been determined to regulate key events or molecules in apoptosis-inducing signal transduction pathways [27]. In the present study, the Cu(BrHAP) 2 Schiff base compound was evaluated for its ability to inhibit the growth of HT-29 cells using an MTT assay. HT-29 cells have recently been characterized as a suitable model for colon cancer studies [28] [29] [30]. human colon cancer cells in a time- and dose-dependent manner. Meanwhile, the nontumorigenic colon cell line (CCD 841) showed no cytotoxicity after treatment with the compound. The cytotoxic effect of the Cu(II) compound was also confirmed by measuring the level of LDH release from treated cells. Considerably elevated LDH release showed that the cytotoxicity of the Cu(BrHAP) 2 compound possibly occurred via the loss of membrane integrity, whether through activation of apoptosis or the necrosis pathway [31]. The observation of early apoptosis and late apoptosis by fluorescent microscopy analysis and AO/PI double staining following treatment of HT-29 cells with the compound included some signs of apoptosis, namely, cytoplasmic shrinkage, membrane blebbing, and DNA fragmentation [32, 33]. We found

that the number of cells with early apoptosis features was higher at earlier stages of treatment. However, when treatment time increased to 72 h, late apoptosis or necrosis characterizations were dominant among treated HT-29 cells. Concurrent detection of late apoptosis or necrosis is scientifically possible because treated HT-29 cells undergoing apoptosis may have progressed into necrosis due to the prolonged incubation with the Schiff base compound.

To elucidate the mechanisms underlying the observed antiproliferative effect of the Cu(II) complex on cancer cells, cell cycle distribution was analyzed using BrdU and Phospho-Histone H3 staining along with flow cytometry [34] [35] [36]. BrdU dye can attach to the synthesized DNA of replicating cells during the S phase of the cell cycle, while Phospho-Histone H3 dye stains cells in different mitotic stages. The cell cycle results from the BrdU and Phospho-Histone H3 double staining assay indicated that there were no significant changes in the number of cells in the S/M phases after the exposure of HT-29 cells to the Schiff base compound. This result suggests the possibility that the cells were arrested in the G₁ or G₂ phase of the cell cycle. Thus, the flow cytometry analysis of the cell cycle was performed to determine the exact arrested phase, and the results demonstrated significant cell cycle arrest at G₁ after 24 and 48 h of treatment, suggesting proliferative suppression via induction of apoptosis [37, 38].

Perturbation of mitochondrial membrane potential is one of the earliest intracellular events that occur following the induction of apoptosis [39]. As the main source of cellular ROS and adenosine triphosphate (ATP), mitochondria are the key regulators of mechanisms controlling the survival or death of cells. After confirming that the Cu(BrHAP)₂ Schiff base compound did not have significant antioxidant capacity in HT-29 cancer cells using the ORAC assay, the induction of ROS production in treated cells was analyzed. According to our study, after exposing the Cu(II) compound to HT-29 cells and analyzing the levels of ROS, it was demonstrated that the level of ROS in treated HT-29 cells was significantly elevated at a compound concentration of 6.25 g/mL.

In metal-induced apoptosis, the mitochondria have the crucial role in mediating apoptosis through metal-induced ROS [40]. The intrinsic or mitochondrial-dependent signaling pathway involves different factors of nonreceptor-mediated stimuli that induce intracellular signals. These signals, mainly through the p53 protein, act on the mitochondrial-initiated events. Excessive ROS production is a negative signal that can result

in the failure of suppression of antiapoptotic factors, thereby triggering apoptosis. Therefore, we used mitochondrial membrane potential (MMP) fluorescent probes to examine the effect of elevated ROS production on the function of mitochondria in treated HT-29 cells. As shown in Figure 7, changes in MMP after treatment with the Cu(BrHAP) 2 Schiff base compound leading to the membrane depolarization of the mitochondria were demonstrated by Rhodamine 123 release to the cytoplasm from the mitochondria matrix. The result implies that the induction of apoptosis by Cu(II) Schiff base complexes may be associated with the mitochondrial pathway [26, 41, 42]. One of the important signals to initiate the procedure of apoptosis is cytosolic cytochrome c. The release of cytochrome c into the cytosol and reduction of its levels in the mitochondria have been shown to occur as a result of changes in MMP [30]. As the result illustrated, the synthetic Schiff base compound also led to an increase in the level of cytochrome c in the cytosol compared to the control.

The excessive production of ROS from mitochondria and the collapse of MMP may activate the downstream caspase molecules and consequently lead to apoptotic cell death. After the binding of cytochrome c to apoptotic activating factor-1, caspase-9 is activated via apoptosome formation, which leads to active caspase-3/7, the most effective caspase with many cellular targets [43]. In the extrinsic pathway, apoptosis is mediated by death receptors. As an example, FAS ligand interacts with the FAS receptor, leading to the activation of caspase-8 [44]. Caspase-8 activation cleaves and activates downstream executioner caspases such as caspase-3/7 [45, 46]. In our study, the Cu(BrHAP) 2 Schiff base compound induced significant elevation in the caspases 3/7 and 9 activities compared to the control. Meanwhile, there was no activation of caspase-8, suggesting that the apoptosis induced in HT-29 cells was mediated via the intrinsic mitochondrial pathway but not the extrinsic, death receptor-linked caspase-8 pathway.

The supporting evidence of LDH release, ROS production, MMP suppression, elevation in the level of cytochrome c, and activation of caspases 3/7 and 9 demonstrated the promising anticancer activity of the Cu(BrHAP) 2 Schiff base compound against the HT-29 colon cancer cell line via the intrinsic mitochondrial pathway.

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--- START OF ARTICLE 63 ---

Multivalent HA DNA Vaccination Protects against Highly Pathogenic H5N1 Avian Influenza Infection in Chickens and Mice

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Abstract: **BACKGROUND:** Sustained outbreaks of highly pathogenic avian influenza (HPAI) H5N1 in avian species increase the risk of reassortment and adaptation to humans. The ability to contain its spread in chickens would reduce this threat and help maintain the capacity for egg-based vaccine production. While vaccines offer the potential to control avian disease, a major concern of current vaccines is their potency and inability to protect against evolving avian influenza viruses. **METHODOLOGY / PRINCIPAL FINDINGS:** The ability of DNA vaccines encoding hemagglutinin (HA) proteins from different HPAI H5N1 serotypes was evaluated for its ability to elicit neutralizing antibodies and to protect against homologous and heterologous HPAI H5N1 strain challenge in mice and chickens after DNA immunization by needle and syringe or with a pressure injection device. These vaccines elicited antibodies that neutralized multiple strains of HPAI H5N1 when given in combinations containing up to 10 HAs. The response was dose-dependent, and breadth was determined by the choice of the influenza virus HA in the vaccine. Monovalent and trivalent HA vaccines were tested first in mice and conferred protection against lethal H5N1 A/Vietnam/1203/2004 challenge 68 weeks after vaccination. In chickens, protection was observed against heterologous strains of HPAI H5N1 after vaccination with a trivalent H5 serotype DNA vaccine with doses as low as 5 µg DNA given twice either by

intramuscular needle injection or with a needle-free device.

CONCLUSIONS/SIGNIFICANCE: DNA vaccines offer a generic approach to influenza virus immunization applicable to multiple animal species. In addition, the ability to substitute plasmids encoding different strains enables rapid adaptation of the vaccine to newly evolving field isolates.

Text: The highly pathogenic H5N1 influenza virus causes lethal multi-organ disease in poultry, resulting in significant economic losses and a public health concern in many parts of the world. The greatest threats posed by this virus are its ability to cause mortality in humans, its potential to compromise food supplies, and its possible economic impacts. Viral maintenance in poultry potentiates the risk of human-to-human transmission and the emergence of a pandemic strain through reassortment. An effective, safe poultry vaccine that elicits broadly protective immune responses to evolving flu strains would provide a countermeasure to reduce the likelihood of transmission of this virus from domestic birds to humans and simultaneously would protect commercial poultry operations and subsistence farmers.

DNA vaccines have been shown to elicit robust immune responses in various animal species, from mice to nonhuman primates [1] [2] [3] [4] [5] [6] [7] [8] [9] [10] [11] . In human trials, these vaccines elicit cellular and humoral immune responses against various infectious agents, including influenza, SARS, SIV and HIV. In addition to their ability to elicit antibody responses, they also stimulate antigenspecific and sustained T cell responses [1] [2] [3] 6, 12, 13] . DNA vaccination has been used experimentally against various infectious agents in a variety of mammals, including cattle (against infectious bovine rhinotracheitis/bovine diarrhea virus, leptospirosis and mycobacteriosis) [14, 15] , pigs (against classical swine fever virus and mycoplasmosis) [16] , and horses (against West Nile virus and rabies) [17] . In addition, DNA vaccines have been tested against avian plasmodium infection in penguins [18] and against influenza and infectious bursal disease in chickens [7, 8, 19] , duck hepatitis B virus in ducks [6] , and avian metapneumovirus and Chlamydia psittaci in turkeys [20, 21] (reviewed in ref. [22]). While they have been used in chickens to generate antisera to specific influenza viruses and confer protection against the low pathogenicity H5N2 strain [23] , there is only one previous report of a monovalent DNA vaccine effective against H5N1 (and that only against a matched H5N1 isolate) [24] ; no protection with multivalent DNA vaccines against heterologous strains has been reported.

Development and characterization of a DNA vaccine modality for use in poultry offers a potential countermeasure against HPAI H5N1 avian influenza outbreaks. The virus can infect humans, typically from animal sources, including commercial and wild avian species, livestock, and possibly other non-domesticated animal species [25] [26] [27] . While there is marked diversity in the host range of type A influenza viruses, many experts have speculated that a pandemic strain of type A influenza could evolve in avian species or avian influenza viruses could contribute virulent genes to a pandemic strain through reassortment [28, 29] . Thus, there is reason to consider vaccination of poultry that would stimulate potent and broad protective immune responses [7, 30, 31] . In undertaking such efforts, it is important that there be a differentiation of infected from vaccinated animals [32] so that animals can be protected and permit monitoring of new infections using proven and sensitive methodologies.

In this study, we used an automated high capacity needle-free injection device, Agro-JetH (Medical International Technology, Inc., Denver, CO) to explore the feasibility of DNA vaccination of poultry. After optimization of injection conditions, alternative multivalent DNA vaccine regimens were analyzed and compared for magnitude and breadth of neutralizing antibodies, as well as protective efficacy after challenge in mouse and chicken models of HPAI H5N1 infection. The findings suggest that it is possible to develop a multivalent DNA vaccine for poultry that can protect against multiple HPAI H5N1 strains and that could keep pace with the continued evolution of avian influenza viruses.

Immunogenicity and neutralizing antibody specificity of alternative HA DNA vaccines in mice To evaluate the efficacy of multivalent DNA vaccines, initial studies were performed in mice. Expression vectors encoding HAs from ten phylogenetically diverse strains of influenza viruses [33] were generated by synthesis of cDNAs (see Materials and Methods) in plasmid expression vectors, pCMV/R or pCMV/R 8kB, which mediates high level expression and immunogenicity in vivo [34, 35, 36] . Animals were immunized with each expression vector intramuscularly (IM) at three week intervals, and the antisera were evaluated on day 14 after the third immunization for their ability to neutralize HPAI H5N1 pseudotyped lentiviral vectors as previously described [35, 36] . We have previously shown that lentiviral assay inhibition (LAI) yields similar results to microneutralization and HAI analyses with higher sensitivity in mice [35, 36]

To determine whether immunization with multiple HAs simultaneously could expand the breadth of the neutralizing antibody response without significant loss of magnitude, a combination of 10 HA DNA vaccine immunogens was administered IM at proportionally lower concentration (1.5 mg per immunogen) into groups of 10 mice (see Materials and Methods). Remarkably, despite a log lower DNA concentration of each component, significant neutralizing antibody titers were generated to each of the 10 immunogens, with .80% neutralization against 6 out of 12 H5 HA pseudoviruses at dilutions of up to 1:400 (Fig. 2A) .

To evaluate whether similar breadth of immunity could be generated with fewer immunogens, two different combinations of 5 immunogens were selected, based on the phylogenetic diversity of HA among the avian influenza viruses [33] and the crossreactivity of the neutralizing antibody responses of select individual immunogens (Fig. 1). As expected, there were substantial differences in the breadth of neutralization between these two sets of 5 immunogen multivalent vaccines (Fig. 2 , B vs. C). In one set, while neutralization of homologous strains was comparable to the monovalent and the 10 immunogen multivalent immune response, fewer cross-reactive antibodies were detected, directed most prominently against A/Iraq Protection of DNA-vaccinated mice against challenge with heterologous H5N1 A/Vietnam/1203/2004 influenza virus Mice immunized as described above were challenged with a heterologous H5N1 virus 68 weeks after the final DNA vaccination. Animals were then challenged with 10 LD 50 of the highly pathogenic A/Vietnam/1203/2004 virus intranasally, and morbidity and mortality were monitored for 21 days after the viral challenge. The control animals, injected with the plasmid expression vector with no insert, died within 10 days of infection. Complete survival was observed in the groups immunized with the 10 component and set 2 of the 5 component multivalent DNA vaccines (Fig. 3) . Immunization with HA derived from the A/ Indonesia/05/2005 strain or set 1 of the 5 component multivalent DNA vaccine showed a survival rate approaching 90%. In contrast, animals injected with HA plasmid DNA derived from A/ Anhui/1/2005, which has diverged more from A/Vietnam/ 1203/2004, showed a lower percent survival (70%) after lethal viral challenge. Survival differences between groups were assessed using a log-rank test and the Gehan-Wilcoxon test on the survival curves for pairs of groups. A test was deemed significant if the pvalue was ,0.01. Mice injected IM with different HAs, A/ Indonesia/5/05, A/Anhui/1/05, 10HA, 5 HA (Set 1), or 5 HA (Set 2) showed a significant difference compared to control (all p values,0.001). Among the HA-immunized groups, there was no significant difference between any two groups (p.0.08 for all comparisons). For example, no significant difference was observed between the A/Anhui/1/05 group, which had the least survival among the HA immunized groups (7 out of

10), and other HA groups: A/Indonesia/5/05 ($p = 0.377$), 10 HA ($p = 0.082$), 5 HA (Set 1) ($p = 0.101$), or 5 HA (Set 2) ($p = .411$). Therefore, we cannot exclude the possibility that the 3 deaths in the A/Anhui/1/05 group may have been due to random chance.

Since it is desirable to confer protective immunity in poultry and HA DNA vaccination was effective in mice, we next examined the breadth and potency of single or multiple HA plasmid immunization in chickens. The ability of chickens to generate specific antibodies was assessed with three strains that showed broad cross protection in mouse studies (A/Vietnam/1203/2004, A/Anhui/ 1/2005 and A/Indonesia/05/2005), administered individually or in combination, by different injection methods. In addition to needle injection, a needle-free repetitive injection device, Agro-JetH (Medical International Technology, Inc., Denver, CO), was analyzed. This device disperses the 0.1 to 5 ml injection doses into the dermal, subcutaneous, or intramuscular tissue depending upon the pressure adjustments, powered by a CO₂ gas pressure plunger [39]. The injection conditions were determined by histologic analysis of tissues that received injections of India ink; a pressure of 48 psi was chosen since it enabled consistent delivery into intradermal and subcutaneous tissues (Fig. S1).

Immunization of chickens with the control plasmid (CMV/R) without an HA gene insert elicited minimal neutralizing antibody titers compared to HA-immunized animals 1 week after 3 DNA immunizations. Nearly all chickens immunized with either monovalent or multivalent HA DNA vaccines generated significant neutralization titers (Fig. 4 and Table S1). In general, there was a progressive increase in the amount of neutralization after each successive DNA vaccination (data not shown) with maximal response at 1 week after the 3rd DNA immunization, with highest and most consistent levels in the trivalent vaccine group delivered with the Agro-JetH device. Neutralization of the Indonesia HA strain was the most robust, with neutralization nearing 100% at titers greater than 1:3200. Both the monovalent and multivalent vaccines elicited robust homologous (vaccine (Fig. 4). Even though one chicken (238) in the multivalent vaccine group produced almost the same degree of neutralization at each time point and was protected, it did not produce a high neutralizing antibody titer for reasons that were uncertain but possibly related to a non-specific inhibitor in the sera.

To determine whether chickens immunized with single or multiple DNA vaccines were protected from a lethal challenge of a heterologous HPAI H5N1 virus, vaccinated chickens

were In panels B and C, mice (n = 10) were immunized with 15 mg of plasmid (3 mg each) three times at 3 week intervals. Serum pools from the immunized animals were collected 14 days after the third immunization. The antisera were tested against the 12 indicated pseudotyped lentiviral vectors at varying dilutions. Error bars at each point indicate the standard deviation; each sample was evaluated in triplicate. In general, the immunized serum neutralized all tested pseudotyped lentiviruses at low dilutions while differences were often observed at high dilution. doi:10.1371/journal.pone.0002432.g002 inoculated with 20 LD 50 of highly pathogenic A/Vietnam/1203/ 2004 heterologous virus intranasally using standard methods [25, 40] and monitored for morbidity, mortality, viral shedding and serum antibodies. While all the control animals died within 2 days of infection, 100% survival was noted in the rest of the chickens (Fig. 5A). The animals that were healthy, showing no signs of clinical disease or malaise, were euthanized on day 14. There was no evidence for viral shedding monitored via tracheal and cloacal swabs of infected chickens 2-14 days after challenge as determined by embryonal inoculation (data not shown: egg infectious dose 50 (EID 50) limit of detection ,100 virus particles).

To compare the relative efficacy of DNA vaccines delivered IM by needle and syringe versus the needle-free Agro-JetH device injection, a dose-response study was performed with amounts of DNA vaccine ranging from 500 to 0.5 mg with two inoculations. In these experiments, the HA derived from A/chicken/Nigeria/641/ 2006 was substituted for A/Vietnam/1203/2004 since it represented a more contemporary isolate. The observed rate of protection was higher among the animals receiving 5 mg by Agro-Jet (8/8) than by IM injection (6/8) (Fig. 5, B vs. C). Both modes provided complete protection for all animals at doses higher than this, and 25% protection for the animals receiving 0.5 mg doses (Fig. 5B, C) . Survival differences between consecutive doses were assessed using a log-rank test on the survival curves for pairs of groups. A test was deemed significant if the p-value was $\leq .01$, and marginally significant if the p-value was $\leq .05$ but $> .01$. Chickens injected IM showed a marginally significant difference between 0.5 and 5 mg ($p = .047$). In the same group there was a significant difference between control and 5, 50 and 500 mg ($p < .001$ for all comparisons) and the difference between control and 0.5 mg was marginally significant ($p = .016$). Chickens that were injected using Agro-JetH showed a significant difference between 0.5 and 5 mg ($p = .004$) and between control and 5, 50, and 500 mg ($p < .001$ for all comparisons). There were no differences between control and 0.5 mg or between 5, 50, and 500 mg. Lastly, the survival differences between Agro-JetH and IM for each dose group were not significant. The neutralizing antibody response to homologous and heterologous HAs corresponded with protection and correlated with dose, with higher titers elicited by injection with Agro-JetH compared to needle (Table S2) . We assessed viable viral shedding

after inoculation by chick embryo inoculation three days after virus challenge (Week 8). While we noted some embryonic lethality at the 0.5 mg dose, there was no embryonic lethality at 5, 50 or 500 mg groups (data not shown).

Since the HPAI H5N1 virus first appeared ten years ago, this highly pathogenic avian influenza virus has shown increasing diversification and dissemination in Asia, Africa, and Europe [28, [41] [42] [43] [44] . In addition to its effects on human health by crossspecies transmission [28, 45, 46] and ability to compromise food sources, it poses a continuing threat to public health as it evolves and adapts in different species. The pandemic potential of this virus, especially as it relates to the poultry industry and for reservoir avian hosts, underscores the need for a vaccine that offers broad spectrum immunity and protection against lethal viral challenge. While the virus remains restricted in its ability to infect humans and undergo efficient human-to-human transmission [28, 47] , its persistence and spread in poultry increases the risk of the emergence of a pandemic strain. One approach to pandemic risk reduction is to limit the propagation of the virus in poultry and other relevant avian species.

We have previously reported that DNA vaccines encoding HA can confer protection against a highly lethal human pandemic influenza virus, the 1918 H1N1 virus, in mice [36] . DNA vaccines offer several advantages, including the ability to express diverse antigens, tolerability in various hosts, ease of delivery, and stability for storage and distribution without the necessity of maintaining a cold chain; they have been shown to be safe and efficacious in a variety of animal models [2, 4, 12, 22, 48] . Because they do not contain other viral proteins used to screen for infection, they also address the need to differentiate vaccinated from infected animals. There is evidence that DNA vaccination elicits cell-mediated immunity against influenza HA in addition to inducing an antibody response [36] , an effect that could significantly contribute to protective immunity as viruses show genetic drift and reduced susceptibility to neutralization.

Ideally, a highly effective influenza vaccine should not only be able to let the host develop a protective immune response against a matching live virus challenge but also elicit robust protective immune responses against a broad range of homologous and heterologous H5 influenza strains. A multivalent H5 vaccine containing diverse serotypes could expand the antigenic breadth sufficiently to provide protection against heterologous challenge and may preclude the emergence of vaccine-resistant strains that may arise due to

evolutionary vaccine pressure on the virus. Due to the antigenic drift and shift of the influenza virus genome, it has been very difficult to predict the next dominant strain of an avian endemic outbreak. DNA vaccines can be synthesized in a relatively short period of time, and the targeted mutations can be tailored to specific viral serotypes. The mutations promote a focused and enhanced immune response [3, 49, 50] that may be particularly important in the event of an outbreak where specificity is the key to epidemic control. The use of modified codons ensures maximal expression in the host and eliminates the possibility of recombination with influenza viruses that might potentially generate new strains.

A more broadly protective murine vaccine was developed here by including more HAs from varying strains in the multivalent vaccine (Figs. 2 and 3) . However, it is less practical to include large numbers of different HAs in one vaccine due to the cost and complexity of manufacturing such a vaccine. Therefore, we simplified the vaccine regimen based on cross-neutralization studies and phylogenetic relationships. A trivalent vaccine was subsequently identified for further studies. Due [51] .

While three DNA immunizations were used initially to demonstrate protective immunity and have been used previously to elicit protection in mice [36] , we found that effective protective immunity could be induced with two DNA vaccinations and as little as 5 mg trivalent DNA immunization using the ID/SC route with the Agro-JetH device. In addition, based on the chick embryo inoculation data, we believe that there is effective neutralization of the virus and lack of infectious viral shedding in chicken vaccinated with as little as 5 mg of DNA. The device's capacity for rapid repetitive injection and the lower quantity and stability of DNA enhance the practicality and utility of this approach for vaccination of endangered species in captivity or administration to poultry or other animals.

A/Vietnam/1203/2004 (H5N1) (A/VN/1203/04) was obtained from the repository at the Centers for Disease Control and Prevention (CDC), Atlanta, Georgia. The virus was propagated in 10-day old embryonated chicken eggs at 35°C and stored at -270°C until use. The virus was titrated by the Reed and Muench method to determine EID₅₀ [52] . GenBank ABD28180) were synthesized using human-preferred codons (GeneArt, Regensburg, Germany) [36] . HA cDNAs from diverse strains of influenza viruses were then inserted into plasmid expression vectors, pCMV/R or pCMV/R 8kB, which mediates high

level expression and immunogenicity in vivo [34, 35, 36] . For initial trivalent immunizations in chickens, the A/Vietnam/1203/ 2004, A/Anhui/1/2005 and A/Indonesia/05/2005 strains were used and in the dose response study, the Vietnam strain was replaced with A/chicken/Nigeria/641/2006. The immunogens used in DNA vaccination contained a cleavage site mutation (PQRERRRKKRG to PQRETRG) as previously described [35, 36] . This mutation was generated by site-directed mutagenesis using a QuickChange kit (Stratagene, La Jolla, CA).

DNA immunization of mice [6] [7] [8] week old female BALB/c mice were purchased from The Jackson Laboratory and maintained in the AAALAC-accredited Vaccine Research Center Animal Care Facility (Bethesda, MD) under specific pathogen-free conditions. All experiments were approved by the Vaccine Research Center Animal Care and Use Committee. The mice were immunized as previously described [5] . Briefly, mice (10 animals for all test groups, 20 animals for the

The study was carried out in the AAALAC-accredited animal facility at the University of Maryland School of Medicine. Six groups of 8 one-day-old male and female SPAFAS White Leghorn Chickens, *Gallus domesticus*, were obtained from Charles River Laboratories (Connecticut). The animals were housed in brooder and grower cages (McMurray Hatcheries, Iowa). Feed (Teklad Japanese Quail Diet -3050, Harlan-Teklad, WI) and water were provided to the animals ad libitum. The study was performed in strict accordance with the "Guide" after approvals from the Animal Care and Use Committees of the Vaccine Research Center, NIH and the University of Maryland. DNA immunizations were performed as described at 0, 3 and 6 weeks. A total dose of 500 mg of one or a combination of the following DNA plasmids in a volume of 250 ml was administered to each animal: pCMV/ R, pCMV/R-HA

Agro-JetH is a needle-free device used for mass delivery of vaccines and drugs in livestock and poultry. The device is semiautomatic and requires a small CO₂ tank or compressed air for low pressure delivery. Upon trigger activation, CO₂ disperses the injectate at a precise dose into the muscle, dermis or subcutaneous tissue depending on the setting that was standardized for our use. We used an effective volume of 0.1 ml in our injectate [39] . In this study we were able to effectively deliver 0.1 ml of injectate into the animal's dermis/subcutaneous tissue at a pressure of 48-55 psi.

Sixty-eight weeks after the last immunization, female BALB/c mice were lightly anesthetized with Ketamine/Xylazine and inoculated intranasally with 10 LD 50 of A/Vietnam/1203/2004 virus diluted in phosphate-buffered saline in a 50 μ l volume. Mice were monitored daily for morbidity and measured for weight loss and mortality for 21 days post infection. Any mouse that had lost more than 25% of its body weight was euthanized. All experiments involving the HPAI virus were conducted in an AAALAC accredited facility (BioQual Inc., Gaithersburg, MD) under BSL 3 conditions that included enhancements required by the USDA and the Select Agent Program.

White Leghorn chickens were challenged one week after the last immunization with 20 lethal dose 50 (LD 50) of A/Vietnam/1203/04 (H5N1) influenza A virus, equivalent to 2610 4 EID 50 based on previous challenges [53] . Chickens were infected with 200 μ l virus intranasally. Tracheal and cloacal swabs were collected days 3 and 5 post-challenge and stored in glass vials containing BHI medium (BBL TM Brain Heart Infusion, Becton Dickinson) at 280uC. Blood was collected 14 days post-challenge and serum was titered by microneutralization assay. Chickens were observed and scored daily for clinical signs of infection, morbidity and mortality. Chickens that survived the study were bled and humanely euthanized at day 14 post-challenge. Lungs, heart, intestine and kidney were collected and samples were stored in formalin for histopathology. Experiments were carried out under BSL3+ conditions with investigators wearing appropriate protective equipment and compliant with all Institutional Animal Care and Use Committee-approved protocols and under Animal Welfare Act regulations at the University of Maryland, College Park, Maryland.

Representative tracheal and cloacal swabs were chosen to run an EID 50 assay for comparison and virus titers were determine by the method of Reed and Meunch [52] . Briefly, swabs were used to infect 10 day-old embryonated chicken eggs in 10-fold dilutions. Three eggs were inoculated per dilution and incubated for 48 hours before titration.

Neutralizing antibodies were titrated from serum samples collected week 5 and 7 post-vaccination and day 14 post-challenge. The microneutralization assay was performed using a 96-well plate format. Serum was treated with receptor-destroying enzyme (Denka Seiken Co.) and treated at 37uC per the manufacturer's instructions. After an overnight incubation and subsequent inactivation samples were brought to a final dilution of 1:10

using PBS and each sample was serially diluted and virus, diluted to 100 TCID₅₀, was added to each well. The plates were then incubated at 37°C, 5% CO₂ for 1-2 hours. Following incubation, supernatants were used to infect a second 96-well plate of MDCK cells. Microplates were incubated at 4°C for 15 minutes and then 37°C, 5% CO₂ for 45 minutes. Supernatants of serum and virus were then discarded and 200 µl of OptiMEM (containing 1X antibiotics/antimycotics, 1 mg/ml TPCK-trypsin) was added and incubated at 37°C, 5% CO₂ for 3 days. After 3 days, 50 µl of the supernatant from each well was transferred into a new 96-well microplate, and an HA assay was performed to calculate the antibody titers. Virus and cell controls were included in the assay.

Two-fold dilutions of heat-inactivated sera were tested in a microneutralization assay as previously described [54] for the presence of antibodies that neutralized the infectivity of 100 TCID₅₀ (50% tissue culture infectious dose) of the A/Vietnam/1203/2004 H5N1 virus on MDCK cell monolayers by using two wells per dilution on a 96-well plate.

The recombinant lentiviral vectors expressing a luciferase reporter gene were produced as previously described [35, 36]. For the neutralization assay, antisera from immunized animals were heat-inactivated at 55°C for 30 minutes and mixed with 50 µl of pseudovirus at various dilutions. The sera/virus mixture was then added to 293A cells in 96-well B&W TC Isoplates (Wallac, Turku, Finland; 12,000 cells/well). Two hours later, the plates were washed and fresh medium was added. Cells were lysed in mammalian cell lysis buffer (Promega, Madison, WI) 24 hrs after infection and luciferase activity was measured using the Luciferase Assay System (Promega, Madison, WI).

The following strains were used for the production of pseudotyped viruses: for HA we used A/Thailand/1(KAN-

The HA/HI titers were determined as previously described [54]. Briefly, HA titers were calculated using 50 µl of 0.5% chicken red blood cell suspension in PBS added to 50 µl of twofold dilutions of virus in PBS. This mix was incubated at room temperature for 30 minutes. The HA titers were calculated as the reciprocal value of the highest dilution that caused complete hemagglutination. HI titers were calculated by titrating 50 µl of antiserum treated with receptor-destroying enzyme and an equivalent amount of A/Vietnam/1203/2004 virus (four hemagglutinating doses) was added to each well. Wells

were incubated at room temperature for 30 minutes and 50 ml of a 0.5% suspension of chicken red blood cells was added. HI titers were calculated after 30 minutes as the reciprocal of the serum dilution that inhibited hemagglutination.

Table S1 Hemagglutination inhibition (HI), microneutralization titer (NT), and LAI of sera from individual chickens immunized with different vaccines. Sera from immunized animals were obtained at week 5 or 7, a week before or after the final boost, and neutralization was assessed by HI, microneutralization (NT) and LAI (shown as IC 50). Individual animal serum of each group is shown and was analyzed as described in the Materials and Methods section. Figure S1 Characterization of needle-free (Agro-JetH) DNA immunization in chickens. To evaluate the distribution of fluid into superficial or deep layers of subcutaneous tissues after delivery by AgroJetH, 4 or 7 week old chickens were injected with a solution containing India ink with this needle-free device at various pressures, ranging from 45 to 55 mm Hg. Three sites (thigh, wing and breast) were used, and biopsies were taken for routine hematoxylin and eosin staining. Representative sections of thigh injections are shown from 7 week old chickens and were similar at 4 weeks (data not shown). While the 48 mm Hg pressure deposited the injectate into the dermis/subcutaneous region (left), the higher pressure injections, 52 and 58 mm Hg, deposited the injectate into the subcutaneous and muscle layers (middle, right). 48 mm Hg consistently provided an optimal pressure to deposit the injectate into the dermis and subcutaneous tissue and was chosen for all AgroJetH immunizations. Found at: doi:10.1371/journal.pone.0002432.s003 (10.74 MB DOC)

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Metabolic engineering of *Escherichia coli* into a versatile glycosylation platform:
production of bio-active quercetin glycosides

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Abstract: **BACKGROUND:** Flavonoids are bio-active specialized plant metabolites which mainly occur as different glycosides. Due to the increasing market demand, various biotechnological approaches have been developed which use *Escherichia coli* as a microbial catalyst for the stereospecific glycosylation of flavonoids. Despite these efforts, most processes still display low production rates and titers, which render them unsuitable for large-scale applications. **RESULTS:** In this contribution, we expanded a previously developed *in vivo* glucosylation platform in *E. coli* W, into an efficient system for selective galactosylation and rhamnosylation. The rationale of the novel metabolic engineering strategy constitutes of the introduction of an alternative sucrose metabolism in the form of a sucrose phosphorylase, which cleaves sucrose into fructose and glucose 1-phosphate as precursor for UDP-glucose. To preserve these intermediates for glycosylation purposes, metabolization reactions were knocked-out. Due to the pivotal role of UDP-glucose, overexpression of the interconverting enzymes *galE* and *MUM4* ensured the formation of both UDP-galactose and UDP-rhamnose, respectively. By additionally supplying exogenously fed quercetin and overexpressing a flavonol galactosyltransferase (*F3GT*) or a rhamnosyltransferase (*RhaGT*), 0.94 g/L hyperoside (quercetin 3-O-galactoside) and 1.12 g/L quercitrin (quercetin 3-O-rhamnoside) could be produced, respectively. In addition, both strains showed activity towards other promising dietary flavonols like kaempferol, fisetin, morin and myricetin. **CONCLUSIONS:** Two *E. coli* W mutants were engineered that could effectively produce the bio-active flavonol glycosides hyperoside and quercitrin starting from the cheap substrates sucrose and quercetin. This novel fermentation-based glycosylation strategy will allow the economically viable production of various glycosides. **ELECTRONIC SUPPLEMENTARY MATERIAL:** The online version of this article (doi:10.1186/s12934-015-0326-1) contains supplementary material, which is available to authorized users.

Text: Flavonoids are a class of plant secondary metabolites, which are chemically characterized by a 15-carbon backbone that consists of two phenyl rings and a heterocyclic ring. To date, over 10,000 flavonoids have been characterized from various plants, which are classified according to their chemical structure, i.e., the number and presence of hydroxyl groups and further functional group modifications into various subgroups, such as anthoxanthins, flavanones, and flavanonols [1, 2] .

In recent years flavonoids have garnered much attention from various application domains because of the various beneficial effects on human health that have been attributed to them, such as anticancer [3] and antioxidant [4] to anti-inflammatory [5] , antimicrobial [6] and antiviral [6, 7] effects. As final step in their biosynthesis, flavonoids are often glycosylated which has a profound effect on their solubility, stability and bio-activity [8, 9] . For example, the best studied flavonol quercetin, which makes up to 75 % of our daily flavonoid intake, predominantly occurs as different glycosides. Over 350 different quercetin glycoforms have been reported to date with varying pharmacological properties [10, 11] . In this context, hyperoside (quercetin 3-O-galactoside) and quercitrin (quercetin 3-O-rhamnoside) (Fig. 1) have gained a lot of attention as valuable products for the pharmaceutical industry e.g., as powerful antioxidants with cytoprotective effects [12] [13] [14] [15] and as promising antiviral agents that block replication of the influenza virus [16] or inhibit the viruses hepatitis B [17] and SARS [18] . Furthermore, they have been attributed with anti-inflammatory [19, 20] , antidepressant [21, 22] , apoptotic [23] and antifungal [24] activities, rendering them interesting therapeutics resulting in a steadily increasing market demand.

To date, the majority of quercetin and its glycosides are extracted from plant material, which is generally a laborious and low-yielding process requiring many purification steps [25] . In vitro plant cell cultures or engineered plants can be used to overcome the low yields and improve production [26] [27] [28] , however since metabolic engineering of plants is both very controversial and still in its infancy [29] , this approach is often restricted to small-scale production. Although chemical synthesis of quercetin (glycosides) has proven to be feasible [30] [31] [32] , stereoselective formation of glycosidic linkages is often hampered by the presence of various reactive groups [33] , which requires many protecting and deprotecting steps [34] . In addition, the generation of toxic waste and a low atom efficiency [35] render these production processes neither sustainable nor economically viable.

As a result, in the last two decades enormous efforts have been invested in the development of alternative production methods for these specialized (secondary) plant metabolites [36]. Advances in the fields of protein engineering, systems and synthetic biology have accelerated these efforts to transform model organisms like *Escherichia coli* and *Saccharomyces cerevisiae* in real microbial cell factories for the sustainable production of flavonoids [37] [38] [39]. Subsequently, strategies for the *in vivo* glycosylation of flavonoids have also been developed. These are typically based on both the overexpression of specific glycosyltransferases, which transfer a sugar residue from an activated nucleotide sugar to an aglycon in a stereo- and regioselective way, and the engineering or introduction of the targeted nucleotide sugar pathway. In this way, Fig. 1 Transformation of *E. coli* W into a sucrose-based galactosylation and rhamnosylation platform. The metabolic engineering strategy applied makes use of several gene deletions (indicated in red) and overexpressions of genes (indicated in green). The rationale of a split metabolism is applied, whereby sucrose is divided by sucrose phosphorylase (BaSP) in fructose to be used for growth and a glucose 1-phosphate as activated precursor for UDP-glucose. The latter is a universal pivot molecule for the formation of UDP-galactose and UDP-rhamnose, interconversions catalyzed by the enzymes GalE and MUM4, respectively. To ensure growth-coupled production, various genes, involved in the metabolism of these UDPsugars and their precursors, were knocked out (shown in red). The production of the bioactive quercetin glycosides hyperoside and quercitrin was chosen to evaluate the versatility of the engineered production platform. Finally, the introduction of either the glycosyltransferase F3GT or RhaGT ensures efficient galactosylation or rhamnosylation, respectively various quercetin glycosides have already been produced in *E. coli* such as the naturally occurring 3-O-glucoside [40], 3-O-xyloside [41] and 3,7-O-bisrhamnoside [42], or the new-to-nature quercetin 3-O-(6-deoxytalose) [43]. However, despite these engineering efforts, the reported product rates and titers are still in the milligram range, rendering these microbial production hosts unsuitable for industrial applications. The developed production processes are typically biphasic bioconversion processes using resting cells, which makes it difficult to improve production rates [44]. Furthermore, such systems often entail expensive growth media or the addition of enzyme inducers, making the overall process very costly.

To tackle these problems, we previously developed an efficient platform for the glucosylation of small molecules in *E. coli* W [45]. Through metabolic engineering, a mutant was created which couples the production of glucosides to growth, using sucrose

as a cheap and sustainable carbon source. By introducing the sucrose phosphorylase from *Bifidobacterium adolescentis* (BaSP) sucrose can be split into fructose to be used for growth purposes and glucose 1-phosphate (glc1P) to be used as precursor for UDP-glucose (UDP-glc) formation (Fig. 1) . To impede the conversion of glc1P into biomass precursors, several endogenous genes involved in its metabolism such as phosphoglucomutase (pgm) and glucose-1-phosphatase (agp) were knocked out. Subsequently, glc1P can efficiently be channeled towards UDP-glc by overexpressing the uridylyltransferase from *Bifidobacterium bifidum* (ugpA). Metabolization of UDP-glc is prevented by knocking out the UDP-sugar hydrolase (ushA) and the galactose operon (galETKM). However, in view of the pivotal role of UDP-glc in the production of a large variety of UDP-sugars, this glucosylation system can easily be extended towards other UDP-sugars, such as UDP-galactose (UDP-gal), UDP-rhamnose (UDP-rha) and UDP-glucuronate.

In the present contribution, this previously developed *E. coli* W-based glucosylation platform is transformed into a platform for galactosylation and rhamnosylation (Fig. 1) , whose potential is demonstrated using the galactosylation and rhamnosylation of exogenously fed quercetin yielding hyperoside and quercitrin, respectively, as case study.

Escherichia coli W is a fast-growing non-pathogenic strain which tolerates osmotic stress, acidic conditions, and can be cultured to high cell densities, making it an attractive host for industrial fermentations [46] . Moreover, *E. coli* W is able to grow on sucrose as sole carbon source [46] , which is an emerging feedstock for the production of bio-products. Hence, *E. coli* W was selected as host for sucrose-based *in vivo* glycosylation. Prior to the production of the glycosides hyperoside and quercitrin in *E. coli* W, the toxicity of their aglycon quercetin was investigated. To this end, the wild type (WT) strain was grown on minimal sucrose medium containing different concentrations of quercetin (0, 0.15 and 1.5 g/L). The specific growth rates (h^{-1}) (0.96 ± 0.06 , 0.92 ± 0.05 and 0.87 ± 0.06 , respectively) were not significantly different (p ANOVA = 0.12) nor from the one previously determined for the WT [45] ($p = 0.69$, $p = 0.98$ and $p = 0.68$, respectively). On the other hand, the optical density at 600 nm after 24 h incubation (6.36 ± 0.12 , 5.18 ± 0.16 and 4.77 ± 0.20 , respectively) was lower (about 20 %) when quercetin was added ($p = 0.0002$ and $p = 0.0001$). No significant difference in optical density could be observed between 0.15 and 1.5 g/L quercetin ($p = 0.14$). In view of the above, it was opted to add 1.5 g/L quercetin to evaluate the potential of the developed glycosylation platform.

To evaluate the *in vivo* glycosylation potential, strains sGAL1 and sRHA1, which constitutively express the flavonol 3-O-galactosyltransferase from *Petunia hybrida* and the flavonol 3-O-rhamnosyltransferase from *A. thaliana*, respectively, were cultured in minimal medium with 1.5 g/L of quercetin for 16 h. TLC analysis of the supernatants of both cultures yielded two new yellow product spots. The TLC spot obtained from the sGAL1 culture, which had the same retention time as the hyperoside standard ($R_f = 0.5$), was subsequently purified and analyzed. Both NMR and MS analysis confirmed the production of quercetin 3-O-galactoside. However, the product spot obtained from the sRHA1 culture had a different retention factor ($R_f = 0.55$) than the quercitrin standard ($R_f = 0.74$), and was identified as isoquercitrin (quercetin 3-O-glucoside). As opposed to other reports on wild type *E. coli* strains expressing RhaGT, which simultaneously produced quercitrin (quercetin 3-O-rhamnoside) and isoquercitrin [47, 48], no rhamnoside could be detected. Examination of the *E. coli* W genome revealed that the gene cluster responsible for the endogenous production of dTDP-rhamnose, which functions as an alternative rhamnosyldonor for RhaGT in *E. coli* B and K12 derivatives [47], was not present [46, 49].

In a follow-up experiment, sGAL1 and sRHA1 were grown on minimal medium with two different concentrations (0.15 and 1.5 g/L) of quercetin. Growth and glycoside formation were monitored during 30 h. The final titers (C_p) and specific productivities (q_p) are shown in Fig. 2. Remarkably, an increase in quercetin concentration resulted in a two to threefold increase in productivity and titer, indicating that quercetin supply is rate-limiting and crucial for efficient *in vivo* glycosylation. However, while sGAL1 continuously produced hyperoside during the exponential phase, which is also reflected in the relatively high specific productivity, sRHA1 only started to accumulate significant amounts of isoquercitrin at the end of the exponential phase. This production start coincides with a reduction in specific growth rate, which dropped from 0.35 ± 0.04 to $0.06 \pm 0.01 \text{ h}^{-1}$.

As described in detail in the Background section, we previously metabolically engineered *E. coli* W to create a platform for *in vivo* glucosylation of small molecules [45]. In the original base glucosylation strain, sucrose phosphorylase encoded by BaSP was located on a mediumcopy plasmid and transcribed from a medium-strong constitutive promoter (P22) [50]. For reasons of comparison and flexibility, it was opted to integrate BaSP in the genome of *E. coli* W. In addition, chromosomal integration is advantageous because of a significant increase in gene stability. Since the level of gene expression can considerably be impacted by the genome integration site [51] due to structural differences such as supercoiling DNA regions, two different DNA sites were assessed for BaSP integration, i.e.,

melA and glgC, which encode an α -galactosidase and a glucose-1-phosphate adenylyltransferase, respectively. To this end, an adapted knockin-knockout procedure for large DNA fragments was applied, which is schematically shown in Additional file 1: Figure S2. BaSP under control of promoter P22 was knocked in at the two different loci in *E. coli* W Δ cscAR, which resulted in the *E. coli* W strains Δ cscAR Δ melA::L4-P22-BaSP-L5 and Δ cscAR Δ glgC::L4-P22-BaSP-L5. Their maximal specific growth rate (μ_{max}) on minimal sucrose medium, which is shown in Fig. 3, was compared to the original strain Δ cscAR + pBaSP. The influence of the knockin locus on the maximal specific growth rate is clear. Interestingly, integration at the melA locus resulted in a strain with a μ_{max} which was not significantly different from the reference strain Δ cscAR + pBaSP. In view of the latter and considering the aimed growth-coupled production, it was opted to integrate BaSP at the melA locus leading to the final production base strain *E. coli* W Δ cscAR Δ pgm Δ agp Δ ushA Δ lacZYA::P22-lacY Δ galETKM Δ melA::L4-P22-BaSP-L5 (sGLYC) as shown in Table 1.

In nature, UDP-glc serves as a pivot molecule in the formation of a variety of UDP-sugars [44]. For example, using the interconverting enzymes UDP-glucose 4-epimerase (GalE) and UDP-rhamnose synthase (MUM4) UDP-glc can be converted to UDP-gal and UDP-rha, respectively. Though GalE is natively present in *E. coli* W an alternative homologous epimerase (GalE2) from *B. bifidum* was also selected and cloned due to the Fig. 2 Comparison of the specific glycoside productivities (q_p) and glycoside titers (C_p) for strains sGAL1, which produces the 3-O-galactoside, and sRHA1, which produces the 3-O-glucoside, when grown for 30 h on minimal medium containing 0.15 or 1.5 g/L of quercetin. Error bars represent standard deviations tight and complex regulation of GalE expression in *E. coli* W. On the other hand, UDP-rhamnose synthesis is restricted to plants. Due to lack of the rfb cluster [46] *E. coli* W is even unable to form endogenous dTDP-rhamnose as alternative rhamnosyl donor. Hence, the MUM4 gene from *A. thaliana* was expressed from plasmid pMUM4 to achieve UDP-rhamnose formation in *E. coli* (Fig. 1).

The constructed galactosylation (sGAL) and rhamnosylation (sRHA) strains were grown on minimal medium with two levels (0.15 and 1.5 g/L) of quercetin. Growth and production were monitored to determine the specific productivities, as shown in Fig. 4. Again, higher extracellular quercetin concentrations resulted in a fivefold increase in q_p . However, no significant difference in productivity was observed between sGAL2 and sGAL3 at 1.5 g/L quercetin, indicating that UDP-galactose formation is as efficient with both GalE homologs and not likely the rate limiting step. With sGAL3, the highest hyperoside productivity (68.7

mg/g CDW/h) and titer (0.94 g/L) were obtained, the latter being 3.5-fold higher compared to sGAL1.

In contrast to sRHA1, TLC analysis of the supernatant of the cultures of sRHA2 and sRHA3 resulted in a product spot with a retention factor that corresponds to quercitrin, which was confirmed by MS analysis, thus showing *in vivo* activity of MUM4. A quercitrin titer of 1.18 g/L and specific productivity of 47.8 mg/g CDW/h were obtained after 30 h incubation of sRHA3 when 1.5 g/L quercetin was added to the medium, which corresponded to a 53 % conversion. Also 51 mg/L of isoquercitrin was produced extracellularly which corresponds with a quercitrin:isoquercitrin production ratio of 24:1. This suggests the preference of RhaGT for UDP-rhamnose when different UDP-sugar donors are present. Possible explanations for the significantly lower specific productivity (fivefold decrease) of sRHA2 as compared to sRHA3 are either a higher metabolic burden [52] caused by the two plasmid system or a too limited activity of the native GalU, which could be insufficient for adequate UDP-glc formation [45] .

To demonstrate the scalability of the developed bioprocess, strain sGAL3 was cultured in a 1-L bioreactor, which also ensures a constant pH set at 6.80 and avoid oxygen limitation. A detailed overview of the consumption of sucrose, growth and hyperoside production is given in Fig. 5 . After a lag-phase, the strain displayed a growth rate of $0.32 \pm 0.02 \text{ h}^{-1}$ while simultaneously producing hyperoside. The observed specific productivity ($65.9 \pm 2.6 \text{ mg/g CDW/h}$) was comparable to the one obtained on shake flask scale. When nearly all quercetin was converted, hyperoside formation slowed down, which can be explained either by the observed correlation between quercetin concentration and q_p , or by the reported reversibility of F3GT [53] . It is likely that further improvements in titer and productivity can be realized by optimizing the supply of quercetin using a fed-batch system.

To the best of our knowledge, the results obtained in this study with the engineered sGAL and sRHA strains for the production of hyperoside and quercitrin are the highest reported to date both in terms of titer and production rate. The maximal production rate obtained in this contribution was 6 to 50-fold higher compared to the maximal production rates ($r_{p,max}$) of processes reported in the literature [47, 54] as is illustrated in Fig. 6 .

The increased performance, in terms of titer and productivity, obtained with the developed platform can be attributed to the use of a split metabolism in combination with optimally rerouting the flux from glucose 1-phosphate towards UDP-galactose and UDP-rhamnose. The undesired conversion of the activated sugars into biomass Fig. 3 Effect of the chromosomal integration locus of the knockin of BaSP on the growth rate. Strains were grown in shake flasks and the resulting maximal growth rates ($\mu_{\max}$) were compared with *E. coli* W Δ cscAR with plasmid-based BaSP expression (+pBaSP). Error bars represent standard deviations. Biomass formation is impeded by gene deletions, which guarantees a high product yield. In addition, since biomass formation, which is fueled by the fructose moiety of sucrose, and glycoside synthesis go hand in hand and subsequently are performed at the same time at a high rate, a high productivity is equally guaranteed (one-step fermentation process).

Besides quercetin also other flavonols such as kaempferol, fisetin, morin and myricetin significantly contribute to our daily flavonoid intake, which also have extremely diverse beneficial effects [55, 56]. As the sugar moiety is a major determinant of the intestinal absorption of dietary flavonoids and their subsequent bioactivity [57, 58], to this end, strains sGAL3 and sRHA3 were grown in tubes with 5 mL minimal medium, each containing 1.5 g/L of either kaempferol, myricetin, morin or fisetin. Growth and production were monitored over 48 h and various spots were observed on TLC with similar retention factors as hyperoside and quercitrin. Mass spectrometry was used to identify the compounds produced, which confirmed the *in vivo* galactosylation of myricetin, kaempferol, morin and fisetin (Table 2). All compounds occurred with an m/z of $[M + 114]$, due to complexation with trifluoroacetic acid from the mobile phase. The galactoside of morin was produced at a slow rate, which is in accordance to the very low *in vitro* activity of F3GT towards this flavonol [53]. A possible explanation for this limited activity may be the presence of an unusual hydroxyl group at the 2' position, which may sterically hinder deprotonation and consequent galactosylation of morin at hydroxyl group 3 [59].

Incubation of sRHA3 with the different flavonols investigated showed two distinct glycoside spots on TLC, which corresponded to the 3-O-rhamnoside and 3-O-glucoside. Kaempferol proved to be the best substrate for RhaGT and was predominantly rhamnosylated (8:1 ratio), with a titer exceeding 400 mg/L, which is twofold higher than previously reported [47]. Fisetin on the other hand was efficiently glucosylated, yet the formation of its rhamnoside was not as efficient, with a titer below 5 mg/L. A similar preference towards glucoside formation was also observed with myricetin and morin, which indicates that the positioning of the hydroxyl groups is the determining factor for glycosylation with RhaGT. The

production of the desired rhamnosides, galactosides or glucosides may be improved considerably by using UGTs that are more specific towards certain flavonols and UDP-sugars. Transformation of the corresponding UGTs in the developed in vivo glycosylation strains presents a promising alternative for the large-scale production of various flavonol glycoforms, which are to date mainly extracted from plant material.

On the other hand, due to the pivotal role of UDP-glc, various other UDP-sugars can be formed in vivo (e.g. UDP-glucuronate, UDP-xylose, UDP-arabinose). In combination with the modularity of the developed glycosylation platform, which permits rapid introduction of any UGT or UDP-sugar pathway, virtually any glycoside can be produced. Hence, this demonstrates that the proposed microbial platform is a robust, versatile and efficient microbial cell factory for the glycosylation (e.g. glucosylation, rhamnosylation, galactosylation) of small molecules. Although obtained productivities are the highest reported today and compete with the current production processes, further improvement can be limited due to solubility issues of the aglycon or of the glycoside. To this end follow-up research can focus on further metabolic engineering (e.g. introduction of the aglycon pathway allowing in vivo gradual production of the aglycon) or on process optimization [e.g. 2-phase (bilayer) fermentation which enables in situ recovery] to improve these issues.

In this contribution, a biotechnological platform was developed for the galactosylation and rhamnosylation of small molecules, such as secondary metabolite natural products, starting from a previously created glucosylation host. To this end, the routes to convert UDP-glucose into UDP-galactose and UDP-rhamnose were introduced by expressing a UDP-glucose epimerase (galE) and a UDP-rhamnose synthase (MUM4), respectively. As a proof of concept, the bio-active flavonol quercetin was selected for galactosylation and rhamnosylation, yielding hyperoside (quercetin 3-O-galactoside) and quercitrin (quercetin 3-O-rhamnoside), respectively. Next, the flavonol 3-O-galactosyltransferase (F3GT) from *Petunia hybrida* and the flavonol 3-O-rhamnosyltransferase from *Arabidopsis thaliana* (RhaGT) were overexpressed in the metabolically engineered *E. coli* W mutants. The strains created were able to produce 940 mg/L of hyperoside and 1176 mg/L of quercitrin at specific production rates of 68.7 mg/g CDW/h and 47.8 mg/g CDW/h, respectively, which are the highest reported to date. Interestingly, both GTs showed in vivo activity towards other dietary flavonols, whereby for example over 400 mg/L of kaempferol 3-O-rhamnoside could be formed extracellularly.

All plasmids used were constructed using Gibson assembly [60] or CLIVA [61] . All PCR fragments were amplified using Q5 polymerase from New England Biolabs (Ipswich, Massachusetts). Oligonucleotides were purchased from IDT (Leuven, Belgium). The plasmids and bacterial strains used in this study are listed in Table 1 . A list of primers for the creation of gene knockouts/knockins and for the cloning of the expression plasmids is given in Additional file 2: Table S1 . E. coli DH5 α was used for plasmid cloning and propagation, while E. coli W was used for expression of the production plasmids and the creation of gene knockouts and knockins. Hyperoside, quercitrin, isoquercitrin, kaempferol and myricetin were purchased from Carbosynth (Berkshire, UK). All other chemicals used were purchased from Sigma Aldrich (Germany) unless otherwise indicated.

The expression plasmids for the production of hyperoside and quercitrin were constructed as depicted in Additional file 3: Figure S1A Figure S1D). The galE [Genbank: JW0742] and galE2 [Genbank: KJ543703] sequences were amplified from the genomic DNA of E. coli and Bifidobacterium bifidum, respectively. CLIVA assembly resulted in the intermediary plasmid pBaSP/F3GT/UgpA (Figure S1A), which was subsequently used for the amplification of the F3GT/ UgpA backbone. Gibson assembly of the GalE or GalE2 inserts with this backbone resulted in the final galactosylation plasmids pGalE/F3GT/UgpA and pGalE2/F3GT/ UgpA, respectively (Figure S1B). Similarly, MUM4 and RhaGT were introduced using a 3-pieces Gibson assembly (Figure S1C), resulting in the final rhamnosylation plasmid pMUM4/RhaGT/UgpA.

The overall E. coli W knockout mutants were created using the one step deletion system of Datsenko and Wanner [62] . The strategy for chromosomal integration of BaSP under control of the constitutive promoter P22 flanked by L4 and L5 at the melA and glgC loci is depicted and explained in Additional file 1: Figure S2 . Transformants were plated on minimal sucrose medium agar plates and grown overnight for screening. The in-house strain E. coli W Δ cscAR Δ pgm Δ agp Δ ushA Δ lacZYA::P22-lacY Δ galETKM [45] was used for the chromosomal integration of L4-P22-BaSP-L5 at the melA site, yielding the base strain sGLYC (Table 1) . This strain and the E. coli W wild type were transformed with the production plasmids described above, resulting in the galactosylation (sGAL) and rhamnosylation (sRHA) strains given in Table 1 .

Composition of LB and minimal sucrose medium was described previously [45] . Minimal medium agar plates with sucrose (50 g/L) had the same composition as minimal sucrose medium, but contained additionally 15 g/L of agarose. The agarose and salts were autoclaved separately at 121 °C for 21 min. Sucrose was filter sterilized through a 0.22 µm corning filter (Fisher, Belgium) and heated for 1 min in a microwave oven at 800 W prior to mixing it with the warm agarose and salt solutions. 1 mL/L of mineral solution [45] was sterilely added prior to pouring the plates.

Escherichia coli W mutant precultures were grown in 5 mL LB medium with the antibiotics (50 µg/mL kanamycin or carbenicillin) required for maintenance and selection of the plasmids. The cultures were grown for 16 h at 37 °C and 200 rpm and used for the 2 % inoculation of 100 mL minimal sucrose medium in 500 mL shake flasks. For the production of hyperoside and quercitrin, quercetin was added to the minimal medium at a concentration of 0.15 or 1.5 g/L. Growth conditions were the same as previously described [45] . Samples were taken at regular intervals from the broth and, after centrifugation, the supernatant was used for the analysis and quantification of sugars. For the analysis of quercetin and its glycosides, 200 µL of the culture was collected and extracted with 800 µL ethyl acetate. The organic layer was collected, evaporated in a SpeedVac™ vacuum concentrator (Thermo Fisher, USA) and dissolved in 200 µL of DMSO for HPLC quantification.

The bioreactor set-up and fermentation conditions used are the same as previously described [45] . Production experiments were performed on minimal sucrose medium without MOPS buffer and with the addition of quercetin as acceptor.

Culture samples were primarily analyzed by TLC on Silica gel 60 F 254 precoated plates (Merck, Germany). All plates were run in a closed TLC chamber and developed using standard visualization techniques and agents: UV fluorescence (254 nm) or by staining with 10 % (v/v) H₂SO₄ and subsequent charring. The mobile phase for detecting the various flavonols and corresponding glycosides consisted of an ethyl acetate:acetic acid:formic acid:water (100:11:11:27 v/v) mixture [63] . Product spot intensities of other flavonol glycosides were processed and quantified using ImageJ [64] . HPLC quantification of sucrose, fructose and glucose was performed using an X-bridge Amide column (35 µm, Waters, USA) as described previously [45] . Quercetin, hyperoside, quercitrin and isoquercitrin were detected with the method described by Pandey et al. [41] using a Varian

HPLC system (Agilent technologies, California). Mass spectrometry for determination of the various flavonol glycosides was performed with a Micromass Quattro LC (McKinley Scientific, USA). Detection was performed in negative mode ESI-224 MS with a capillary voltage of 2.53 kV, a cone voltage of 20 V, cone and desolvation gas flows of 93 and 420 L/h, and source and cone temperatures of 150 and 350 °C, respectively.

Quercetin glycosides were extracted from the broth with an equal volume of ethyl acetate after which the organic layer was evaporated to dryness. The remaining product was dissolved in the solvent system described above and run on a preparative TLC plate. The band containing hyperoside (R_f 0.53) or quercitrin (R_f 0.75) was scraped off, extracted with ethyl acetate and evaporated to yield a bright yellow powder. Products were confirmed by NMR. Spectra were reported elsewhere [47, 65] .

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Clara Cell 10 kDa Protein Alleviates Murine Hepatitis Virus Strain 3-Induced Fulminant Hepatitis by Inhibiting Fibrinogen-Like Protein 2 Expression

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Abstract: Background: Fulminant hepatitis (FH) is a serious threat to human life, accompanied by massive and rapid necroinflammation. Kupffer cells, the major immune cell population involved in innate immune responses, are considered to be central for FH. Fibrinogen-like protein 2 (Fgl2) is a pro-coagulant protein that is substantially induced in macrophages upon viral infection, and Fgl2 depletion represses murine hepatitis virus strain 3 (MHV-3) infection. Clara cell 10 kDa (CC10) protein is a secretory protein with anti-inflammatory properties in allergic rhinitis and asthma. However, its mechanisms of action and pathogenic roles in other disease are still unclear. In this study, we aimed to determine the role of CC10 in FH and the regulation of Fgl2 by CC10. Methods: A mouse FH model was established by peritoneal injection of MHV-3. The mice received CC10 protein through tail vein injection before viral infection. Survival rate, liver function, liver histology, fibrin deposition, and necrosis were examined. The regulatory effect of CC10 on Fgl2 expression was investigated using THP-1 cells and mouse peritoneal macrophages in vitro. Results: In the mouse FH model induced by MHV-3, the survival rate increased from 0 to 12.5% in the CC10 group compared to that in the saline-only control group. Meanwhile, the levels of ALT and AST in serum were significantly decreased and liver damage was reduced. Furthermore, hepatic Fgl2, TNF- α , and IL-1 β expression was obviously downregulated together with fibrin deposition, and hepatocyte apoptosis was reduced after administration of CC10 protein. In vitro, CC10 was found to significantly inhibit the expression of Fgl2 in IFN- γ -treated THP-1 cells and MHV-3-infected mouse peritoneal macrophages by western blot and real-time PCR. However, there was no direct interaction between CC10 and Fgl2 as shown by co-immunoprecipitation. Microarray investigations suggested that HMG-box transcription factor 1 (HBP1) was significantly low in CC10-treated and IFN- γ -primed THP-1 cells. HBP1-siRNA treatment abrogated the inhibitory effect of CC10 on Fgl2 expression in Human Umbilical Vein Endothelial cells (HUVECs). Conclusion: CC10 protects against MHV-3-induced FH via suppression of Fgl2 expression in macrophages. Such effects may be mediated by the transcription factor HBP1.

Text: Fulminant hepatitis (FH) is a serious life-threatening disease characterized by massive hepatocyte necrosis, severe liver damage, and high mortality. The underlying mechanisms and the pathogenesis of FH are not clear. However, accumulating evidence suggests that, regardless of the pathogenesis of FH, the host's inflammatory responses contribute to liver microcirculatory disorders and injuries. Accordingly, It has been shown that immune cell activation and inflammatory cytokines play an important role in FH (1) . In recent years, our laboratory has conducted extensive research on the pathogenesis of FH and found that immune cells play a key role in it. Kupffer cells, natural killer (NK) cells (2, 3) , cytotoxic T-

lymphocytes (CTLs), and double negative T-cells (DNT) (4) (5) (6) in liver and the cytokines that are produced by these cells cause liver damage.

Prothrombinase Fgl2 belongs to the fibrinogen superfamily and is produced by activated macrophages or endothelial cells, transforming prothrombin directly into thrombin, so as to quickly initiate the process of coagulation. This promotes the conversion of fibrinogen into fibrin, resulting in thrombosis (7) (8) (9) (10) (11) (12) . Our study found that Fgl2 was highly expressed in peripheral blood mononuclear cells (PBMCs) and in liver tissue of humans or mice with severe viral hepatitis, and was positively related to the severity of the disease (13, 14) . Gene therapy targeting Fgl2 silencing showed that the survival rate of fulminant hepatitis mice increased from 0 to 33.3% (15) . Thus far, the discovery and related research involving Fgl2 have provided new insights into the molecular mechanism of hepatocyte necrosis in FH. In view of the important role of Fgl2 in severe viral hepatitis, investigations concerning the regulation of Fgl2 will be beneficial in the search for new strategies for treatment of severe hepatitis.

Clara cell 10 kDa protein (CC10), also considered to be uteroglobin, Clara cell secretory protein, is one of members of secretoglobin superfamily. Expressed in mucosal epithelial cells of organs (including lungs and nose) that communicated with the outside world (16) . CC10 has immunomodulatory and anti-inflammatory effects. Compared to wild-type mice, CC10-knockout mice exhibited excessive airway inflammation Abbreviations: FH, fulminant hepatitis; MHV-3, murine hepatitis virus strain 3; Fgl2, Fibrinogen-like protein 2; CC10, Clara cell 10 KDa protein; ALF, acute liver failure; PFU, plaque-forming units; PBS, phosphate-buffered saline; ALT, alanine aminotransferase; AST, aspartate aminotransferase; PCA, pro-coagulant activity; HRP, horseradish peroxidase; TUNEL, terminal deoxynucleotidyl transferase dUTP nick end labeling. caused by allergic reaction and bacterial and viral infections (17) . Reduced levels of CC10 are associated with inflammatory and allergic airway diseases, including sinusitis, asthma and allergic rhinitis (18) (19) (20) (21) .

Previous studies and published articles show that CC10 protein can not only inhibit Th17 cell responses by inhibiting expression of related molecules of dendritic cells and cytokines in mice with allergic rhinitis, but also can inhibit chitosan-3 like protein 1 (22, 23) . Moreover, CC10 inhibits the expression of an important immune regulator, osteopontin (OPN), in models of allergic rhinitis (21) .

In this study, we investigated the role of CC10 in hepatitis virus strain 3 (MHV-3)-induced FH in mice and explored whether CC10 protein could regulate Fgl2 in the disease process.

Female BALB/cJ mice (Shanghai Silaike Animal Seed Center, Shanghai, China), 6-8 weeks of age, with a body weight of 18.0-20.0 g, were kept in Tongji Hospital with food and water. Mice were divided into two groups: CC10 group (experimental group) and phosphate-buffered saline (PBS) group (control group). This study was carried out in accordance with the recommendations of the guidelines of the National Institutes of Health and the Animal Experiment Committee of Tongji hospital. This study was reviewed and approved by the Animal Experiment Committee of Tongji hospital.

The human monocyte cell line THP-1 was purchased from the Cell Institute of the Chinese Academy of Sciences (Shanghai, China). Human Umbilical Vein Endothelial Cells (HUVECs) were obtained from the Biology Treasure Center of Wuhan University, China. The Chinese hamster ovary (CHO) cell line was acquired from the typical culture preservation commission cell bank, the Chinese Academy of Sciences (Shanghai, China). Human Umbilical Vein Endothelial Cells (HUVECs) and CHO cells were cultured in Dulbecco's modified Eagle's medium (DMEM), and THP-1 cells were maintained in RPMI 1,640 containing 10% heat inactivated fetal bovine serum (FBS, Gibco Life Technologies, USA), 100 U/mL penicillin, and 100 mg/mL streptomycin and cultured at 37 °C, 50 mL/L CO₂ and 95% humidity.

Peritoneal exudative macrophages (PEMs) were obtained from BALB/cJ mice. Cells were resuspended in RPMI 1,640 supplemented with 10% FBS at $1-2 \times 10^6$ cells/mL in a 6-well plate and incubated for 4 h. They were then washed with RPMI 1640 medium and non-adherent cells discarded. The adherent cells were macrophages and were incubated for a further 12 h. Peritoneal exudative macrophages (PEMs) were divided into two groups. One group was supplemented with CC10 protein (150 ng/mL) and in the other group, PBS was added. After 2 h of stimulation, 1,000 plaque forming units (PFUs) of MHV-3 was added to the cells, which were then cultured for 4 h. Peritoneal exudative macrophages (PEMs) were harvested and lysed for real-time PCR and western blotting analysis.

Cell apoptosis was detected by the terminal deoxynucleotidyl transferase dUTP nick end labeling (TUNEL) method with a TUNEL apoptosis detection kit (Roche, Switzerland). Briefly, 5 µm sections were deparaffinized, dehydrated through an alcohol series and incubated with proteinase K for 30 min at 37 • C. After stopping the proteinase K digestion reaction with PBS, the samples were incubated with terminal deoxynucleotidyl transferase end-labeling cocktail (a mixture of terminal deoxynucleotidyl transferase and dUTP at a ratio of 2:29, respectively), for 2 h at 37 • C in an immunohistochemistry wet box. Following washing and blocking, each section was supplemented with reagent (converter-POD) to cover the tissues and incubated for 30 min at 37 • C in a wet box. Then, the liver tissue sections were washed with PBS, and colored with diaminobenzidine (DAB) subsequently. Hepatocytes with nucleus stained brownish yellow were considered to be apoptotic cells.

The expression of Fgl2 on THP-1 cells was measured by flow cytometry (BD FACS Canto II, USA). Briefly, cells (2×10^5 per tube) were incubated with Human TruStrain FcX (Fc Receptor Blocking solution, BioLegend, USA) for 10 min at room temperature and then incubated in the dark with mouse anti-Fgl2 antibody (1:100, Abnova,) or normal goat serum (an isotype control) at 4 • C for 40 min. Cells were washed with PBS and incubated in the dark with PE-conjugated goat anti-mouse IgG antibody (1:50, BioLegend, USA) at 4 • C for 30 min. Cells were then washed with PBS and resuspended in 300 µL PBS for study.

Liver slices were fixed in 4% paraformaldehyde and then embedded in paraffin. Immunohistochemistry of liver tissues was performed using SP-9001 SPlink Detection Kits (Biotin-Streptavidin HRP Detection Systems) (ZSGB-BIO, Beijing, China) according to the manufacturer's instructions. For immunohistochemistry staining, the expression of Fgl2, fibrinogen, Fas and TNF-receptor 1 in mouse liver tissues was detected with polyclonal rabbit anti-mouse Fgl2 antibody (1:100, Proteintech, USA), polyclonal rabbit anti-mouse fibrinogen antibody (1:1,000, Abcam, EngLand), polyclonal rabbit antimouse Fas antibody (1:50, Abcam, EngLand), and polyclonal rabbit anti-mouse TNF-receptor 1 antibody (1:500, Abcam, EngLand), respectively. After incubation with an horseradish peroxidase (HRP)-labeled goat IgG fraction to rabbit IgG Fc, the target protein was detected using a DAB kit (ZSGB-BIO, Beijing, China). The slides were then counterstained with hematoxylin and visualized under a microscope (Olympus, Tokyo, Japan).

Liver tissue and cells were homogenized in RIPA lysis buffer with phenyl methane sulfonyl fluoride (PMSF) protease inhibitor. Protein lysates were separated by SDS-PAGE, and

western blotting was performed using a monoclonal mouse antihuman/mouse Fgl2 (1:750, Abnova), a monoclonal mouse antihuman HBP1 (1:100, Santa Cruz, USA), and a monoclonal rabbit anti-human/mouse β -actin (1:1,000, Cell Signaling Technology, USA).

Liver tissues were collected from MHV-3-infected BALB/cJ mice at 72 h, and total RNA was extracted using Trizol Reagent (Invitrogen, USA) and then reverse transcribed into cDNA by using ReverTra Ace qPCR RT kit (TOYOBO, Japan). The cDNA was then amplified by RT-PCR by using Dream Taq Green PCR Master Mix (2 $\times$) (Thermo Scientific, USA). Realtime quantitative PCR (qPCR) with SYBR Green Real-time PCR Master Mix (TOYOBO, Japan) was performed using a CFX96 real-time PCR detection system (Bio-Rad, USA) and mRNA levels were normalized with reference to those of the house keeping gene GAPDH. Primer sequences for qPCR amplification were as follows: mTNF- α forward, 5' -TTT GAG ATC CAT GCC GTT GG-3' ; mTNF- α reverse, 5' -GCCA CCA CGC TCT TCT GT-3' ; mL-1 β forward, 5' -TGT AAT GAA AGA CGG CAC ACC-3' ; mL-1 β reverse, 5' -TCT TCT TTG GGT ATT GCT TGG-3' . mFgl2 forward, 5' -GCC AAA TGT GAG TCC CTG GAA-3' ; mFgl2 reverse, 5' -TTC CAC CCA AGA GCA CGT TTA AG-3' ; hFgl2 forward 5' -ACA GTT CAG GCT GGT GGT-3' ; hFgl2 reverse, 5' -GGC TTA AAG TGC TTG GGT-3' ; HBP1 forward, 5' -TGA AGC AGA AGC TGG GAGT-3' ; HBP1 reverse,

THP-1 cells were treated with 100 ng/ml phorbol 12-myristate 13-acetate (PMA) (Sigma, USA) for 48 h to induce differentiation toward adherent macrophage-like cells as reported previously (24) . The CC10 group was supplemented with CC10 protein (150 ng/ml). After 2 h of stimulation, IFN- γ (10 ng/ml) was added to these cells, which were then cultured for 12 h before they were collected for western blotting and real-time PCR studies.

The Chinese hamster ovary (CHO) cells were cultured in 10 cm cell culture dishes with DMEM supplemented with 10% FBS until 80-90% confluence. Next, 12 μ g pcDNA3.1-hFgl2 (constructed in our lab) was mixed with 12 μ g pcDNA3.1-hCC10 in serumfree DMEM. The mixture was then combined with Lipofectamine 2,000 (Invitrogen, USA) and mixed gently. After incubation at 27 $^{\circ}$ C for 20 min, the solution was added to CHO cells and incubated at 37 $^{\circ}$ C in 5% CO₂ . Four to Six hour after transfection, the medium was removed and fresh medium containing 10% FBS was added. At 48 h after transfection, the cells were collected for co-immunoprecipitation analysis to evaluate the interaction of CC10 with Fgl2.

Both HUVEC and THP-1 cells express fgl2. However, in the transfection experiments, it is difficult to transfect the THP-1 cells with siRNA, so we use HUVEC instead of THP-1. Human Umbilical Vein Endothelial Cells (HUVECs) were cultured in FIGURE 1 | CC10 protein increased survival rate and reduced liver damage in mice. (A) The survival rate of CC10 group is higher than the control group comprised of MHV-3-infected BALB/cJ mice treated with saline. CC10 protein (2 µg) or saline were injected into mice by tail vein. BALB/cJ mice then received 100 PFU of MHV-3 intraperitoneally 24 h later to develop fulminant viral hepatitis. Then, CC10 protein (2 µg) or saline were injected into mice by tail vein following MHV-3 infection 24 h later. The survival rate was observed for 10 days (n = 24/group). Representative data from three independent experiments are shown. The survival curve was analyzed by using the Log-Rank Test. ***P < 0.001 compared with saline group. (B) Histopathology of liver tissues (H&E staining; original magnification, ×400, n = 5/group) at 72 h post-MHV-3 infection was evaluated in the two groups of MHV-3-infected BALB/cJ mice. Livers were collected from saline-treated (a) and CC10-treated (b) BALB/cJ mice at 72 h after MHV-3 infection. Arrows point to inflammatory cell infiltration areas or necrotic regions with inflammation. (C) Effect of CC10 on serum ALT and AST levels (n = 6-8/group). Values represent means and standard error of three independent experiments performed in triplicate. **P < 0.01 compared with the saline group.

six-well plates with DMEM supplemented with 10% FBS until 70-80% confluence. 50 pmol HBP1-siRNA was mixed with 125 µl serum-free DMEM. Two microliter Lipofectamine 2,000 was gently mixed with serum-free DMEM. After incubation at 27 °C for 5 min, the solution was added to HUVECs and incubated at 37 °C. Four hour after transfection, the medium was removed and fresh medium containing 10% FBS was added. At 48 h after transfection, cells were collected for real-time PCR and western blot analysis to evaluate the effects of HBP1 on Fgl2. At 24 h after transfection, the CC10 group was supplemented with the CC10 protein (150 ng/mL). After 4 h of stimulation, IFN-γ (10 ng/mL) was added to these cells. These cells were then cultured for 24 h before they were harvested for real-time PCR studies to evaluate the effects of CC10 on Fgl2 by HBP1. Negative control was used as a control.

To detect whether there was a potential interaction between CC10 protein and Fgl2, CHO cells were transfected with pcDNA3.1-hCC10 and pcDNA3.1-hFgl2 for 48 h. Cells transfected with empty plasmid pcDNA3.1 (mock) were used as negative controls for CC10 gene transfection. Immunoprecipitation and immunoblotting were performed by using Pierce Co-Immunoprecipitation Kit (Pierce, USA). Total cell proteins were extracted as

previously described (25) . The proteins were immunoprecipitated by mouse anti-human Fgl2 antibody (1:500, Abnova). For co-immunoprecipitation experiments, western blotting was performed using both rat anti-human uteroglobin/SCGB1A1 Antibody (1:750, R&D, USA) *Frontiers in Immunology* | www.frontiersin.org and mouse anti-human Fgl2 antibody (1:500, Abnova). Control isotype rat IgG1 was used as a negative control for primary antibodies.

The human CC10 coding region gene, including a 389 bp sequence, was amplified from homogenized human turbinate tissue by RT-PCR. In this study, the sequences of PCR primers for CC10 were as follows: hCC10-forward, 5' -CCC TCC ACC ATG AAA CTCG-3' ; hCC10-reverse, 5' -TGA GAT GCT TGT GGT TTA TTG AAG-3' . The PCR products were cloned into pEASY-T1 cloning vector (TransGEN, Beijing, China) and then subcloned into HindIII/XbaI site of pcDNA3.1 vector (Invitrogen, USA) to form eukaryotic expression plasmids pcDNA3.1-hCC10.

Microarray analysis was used to screen changes in genome-wide gene expression patterns in THP-1 cells with or without CC10 protein. The changes in over 47,000 human gene expression patterns were assessed using Affymetrix gene microarrays (Human Genome U133 Plus 2.0) (CapitalBio Co.,Ltd., Beijing, China). Three replicates were used for microarrays analysis.

Data obtained from the experiments are expressed as means $\pm$ SEM. Survival curve comparisons were performed with the Log Rank test. Multiple group analyses for data were evaluated by one-way analyses of variance. Analyses of two group results were performed using Student's t-test to evaluate the statistical significance of differences. Values of $P < 0.05$ indicated significance.

To establish an animal model of mouse FH, MHV-3 was injected intraperitoneally to BALB/cJ mice (24 mice/group). To further study the role of CC10 in FH, recombinant mouse CC10 protein (2 μ g/mouse) or saline was administrated into the tail vein 24 h prior to MHV-3 infection. The same dose of CC10 protein or saline was then administered 24 h later. The survival rate of the CC10 and saline groups was observed for 10 days. The results showed that mice in the two groups began to die at 48 h after injection of MHV-3 and exhibited symptoms of horripilation, slow activity, and reduced food consumption. In the CC10

group 24 mice were alive on day 3 after infection, 4 mice alive on day 4, and 3 of 24 (12.5%) mice recovered from fulminant viral hepatitis. At the same time, in saline treated group, there were 5 mice alive on day 3, 1 mice alive on day 4 after infection, and no mice survived to day 5. That is to say, the mice in the saline group died within 3 or 4 days. Three of 24 (12.5%) mice of the CC10 group recovered from fulminant viral hepatitis (Figure 1A) .

To better understand the mechanisms underlying the biological effects of the CC10 protein, liver function (ALT and AST levels in serum) and liver histology in mice of MHV-3-infected was performed. Liver tissues were harvested 72 h following MHV-3 infection, and liver histology was detected by H&E staining. These results showed that there was substantial inflammatory cell infiltration and widespread necrosis of hepatocytes in the liver tissue of the saline group mice (Figure 1Ba). There were rare or no infiltrating inflammatory cells, and few or no hepatocyte necrosis in the livers of mice in the CC10 group 72 h after MHV-3 infection (Figure 1Bb) . Serum ALT and AST levels in mice were observed 72 h after MHV-3 infection. The results showed that serum ALT and AST levels in the saline group reached a peak 72 h after MHV-3 infection, but there was no significant increase in the CC10 group compared to the levels in the control group ($P < 0.01$, Figure 1C) . These results suggested that CC10 protein has a role in protection against MHV-3-induced liver injury in mice.

To further elucidate the mechanisms of reduced liver injury following CC10 protein injection, we investigated the cytokines TNF- α and IL-1 β expression. Because these two cytokines play a crucial role in the liver damage of FH. They are characterized by an increase in apoptosis. Levels of TNF- α and IL-1 β in liver tissues were markedly reduced in the CC10 group (as shown in Figure 2A) . Hepatic apoptosis (Figure 2B) was significantly reduced in the CC10 group.

We and collaborators have a long standing interest in studying the role of fgl2 in viral hepatitis. Fgl2 has been verified to play an essential role in the progression of fulminant viral hepatitis as we appreciate from previous reports. We have provided liver pathology figures and liver function for MHV-3 infected mice with a fgl2 gene knockout as shown in Supplementary Figure 1 .

The data was comparable with previous reports from our center and collaborators. From this current study we shown that CC10 plays a protective role in liver damage.To study the related molecules of CC10 in MHV-3-induced FH mice, we evaluated whether there was crosstalk between Fgl2 and CC10. We found that the expression of Fgl2 in the liver of mice was reduced 72 h after MHV-3 infection and treatment with CC10 protein (Figures 3A,B) . Furthermore, fibrin deposition, an indicator of liver injury associated with Fgl2 expression in FH, was also decreased in the livers of CC10-treated mice compared to that in controls (Figure 3C). This indicates that CC10 treatment reduced liver injury after viral infection by inhibiting Fgl2 expression.

We examined the effect of increasing doses of CC10 protein (0, 50, 150, and 300 ng/mL) on IFN- γ -induced Fgl2 expression in THP-1 cells. CC10 treatment showed a 10.1% decrease in THP-1 cells compared to that in control after stimulation with 10 ng/mL IFN- γ for 12 h. CC10 protein inhibited Fgl2 expression between doses of 0 ng/mL and 300 ng/mL (Figure 4A). In particular, 150 ng/mL CC10 protein had the strongest inhibitory effect on Fgl2 expression among the doses, and we chose this dose for the following experiments. We explored the effect of different time points of stimulation with a concentration of 150 ng/mL CC10 protein. After stimulation with CC10 protein for 6, 12, and 24 h compared to the PBS control, the strongest inhibitory effect on Fgl2 expression was noted at 12 h; hence, we chose this time point for the following studies (Figure 4B). An increasing number of studies suggest that macrophages are the primary source of Fgl2. In order to ascertain that CC10 has a direct effect on macrophages, we treated THP-1 cells with recombinant CC10 and assessed the expression of Fgl2. Unlike in controls, IFN- γ induced a significant increase in Fgl2 expression. This effect was attenuated when cells were treated with CC10 protein (Figures 4C,D) , revealing that CC10 directly reduces the levels of Fgl2 in macrophages. To further explore the possibility that CC10 protein directly acts on macrophages, we infected murine PEMs with MHV-3 in the presence of recombinant CC10 and determined Fgl2 expression. Compared to levels in the controls, MHV-3infected macrophages exhibited a significant increase in Fgl2 production, and this effect was abolished by using CC10 protein (Figures 5A,B) , indicating that CC10 directly modulates Fgl2 production in macrophages.

In order to determine genes that were downregulated after stimulation by CC10 protein, we used DNA microarray analysis to screen for differentially expressed genes. THP-1 cells were cultured and PMA was added to induce differentiation into macrophages. The production of Fgl2 was stimulated by IFN γ . The experimental group was treated with CC10 protein for

microarray detection of differentially expressed genes. The results showed that the most obviously downregulated genes were UBE2W, HECTD1, MIR612, ATRX, SOX4, HBP1, and Fgl2 (Supplementary Table 1) . And then these genes were tested by qPCR. However, UBE2W, HECTD1, MIR612, ATRX, and SOX4 was not differentially expressed by qPCR, while HBP1 and fgl2 were still down-regulated genes. DNA microarray analysis identified HBP1 as a down-regulated gene involved in the pathological processes of the regulation of CC10.

Recently, very limited studies have explored the role of HBP1 in FH. Nevertheless, the mechanistic functions of HBP1 in FH remain largely unexplored. Therefore, we selected this gene for further study. qPCR analysis confirmed that mRNA levels of HBP1 were significantly decreased in THP-1 cells after CC10

protein stimulation compared to that in the PBS control group (Figure 6A) .

We knocked down HBP1 using HBP1-siRNA. Then, transfection of HBP1-SiRNA into HUVECs was detected by qPCR and western-blotting methods. As expected, HBP1 knockdown led to significantly decreased expression of HBP1 (Figures 6B,C) . Furthermore, HBP1 knockdown impaired expression of Fgl2 (Figure 6D), suggesting that HBP1 was able to activate Fgl2.

HBP1-SiRNA was used to transfect HUVECs. Then, IFN- γ was added to induce the expression of Fgl2 followed by stimulation with CC10 protein (150 ng/ml) after 2 h. Finally, we explored the expression of Fgl2 by qPCR. The results showed that HBP1-SiRNA treatment abrogated the inhibitory effect of CC10 on Fgl2 expression in HUVECs (Figure 7) . That is to say, CC10 could suppress Fgl2 expression in macrophages. Such an effect may be mediated by the transcription factor HBP1.

It is well-known that CC10 protein can suppress the immune response. In animal models of allergic diseases of the respiratory tract, most of evidences confirm this inhibition (26) . Its function in FH has not been investigated yet. Here, we used a murine FH model established by MHV-3 infection to explore the effects of CC10 in this disease process. To determine the role of CC10 in the pathogenesis of FH, CC10 protein was injected into a mouse FH model established by MHV-3 infection. MHV-3-induced liver injury in CC10-treated mice occurred

rarely and the areas of lesions were much fewer than those in saline-treated control mice. In summary, these results suggested that CC10 could reduce pathological liver damage in this FH model together with lower mortality rates followed by MHV-3 infection.

MHV-3 induced fulminant viral hepatitis progresses rapidly and infected mice die within 3-5 days. Previous studies suggested fgl2 played a vital role in this process with a 15-40% increase of survival when fgl2 was deleted (12, 15, 27, 28) . Multiple inflammatory factors or mediators including TNF- α and IFN- γ , IL-1 β and C5aR have been demonstrated to promote FH progression with significant discrepancies between liver damage and survival rate (29) (30) (31) (32) , which is accordant with our observation that CC10 substantially alleviated liver injury though survival rate improved mildly. The survival rate based on hours may be more accurate to examine the effect of CC10 on FH.

It is speculated that fgl2 can mediate lethality in MHV-3-induced FH. This is due to the fact that fgl2 induces the deposition of fibrinogen, which leads to activation of the coagulation cascade and induction of procoagulant activity (15) . To determine whether the tissue necrosis was mediated by Fgl2 in CC10-treated mice following infection, Fgl2 expression was observed. Results suggested that the expression of Fgl2 was significantly increased in MHV-3-induced FH mice and CC10 treatment significantly reduced the production of Fgl2 in the infected liver and serum. In addition, decreased fibrinogen deposition was also observed in the livers of CC10-treated mice. Therefore, our research results strongly clarify that the lower mortality of CC10-treated mice after MHV-3 infection is due to the lower levels of Fgl2 and decreased fibrinogen deposition.

Indeed, it has been reported that Fgl2 is expressed on macrophages, and the expression of Fgl2 is believed to be induced by IFN- γ and TNF- α (22) . Cultured THP-1 cells activated by IFN- γ or IL-2 have been demonstrated, with induction of Fgl2 expression and enhanced activation of human prothrombin (23) . Therefore, in this study, we explored this cell line to investigate the modulation of CC10 on Fgl2. Surprisingly, we found that CC10 directly inhibited IFN- γ -induced Fgl2 expression in THP-1 cells. As we know, IFN- γ has proved to be the main cytokine that leads to the development and progression of FH. Also, it was shown that IFN- γ might exert its own proinflammatory biological function through enhancing Fgl2 expression. Therefore, in our study, CC10 might counter the effect of IFN- γ in the setting of FH, which substantiates its role in FH. These results demonstrated that CC10 regulates the expression of Fgl2 in macrophages.

In the current study, we used co-immunoprecipitation to analyze binding between CC10 and Fgl2. In this study, we investigated possible protein-protein interactions between CC10 and Fgl2 in vitro. The Chinese hamster ovary (CHO) cells transfected with pcDNA3.1-hCC10 and pcDNA3.1-hFgl2. Cellular proteins were immunoprecipitated with anti-CC10 antibody or anti-Fgl2 antibody. Immunoblotting was performed with anti-Fgl2 and anti-CC10 antibodies. Immunoprecipitation of protein extracts from pcDNA 3.1-CC10 and pcDNA3.1-Fgl2 co-transfected CHO cells with anti-Fgl2 or anti-CC10 antibody followed by western blotting with Fgl2 and CC10 antibodies indicated that CC10 did not co-immunoprecipitate with Fgl2, showing that there is no direct relationship between CC10 and Fgl2 (data not shown). The results showed that CC10 has no direct interaction with Fgl2. From our previous study the gene of fgl2 contributed profoundly in MHV-3 induced fulminant hepatitis and is extensively expressed in macrophages and endothelium (12, 33). Our microarray indicated a CC10 down-regulated fgl2 expression and this is further confirmed by qPCR and Western blotting in vivo (peritoneal macrophages) and in vitro (THP-1, macrophage cell line). Therefore, it is reasonable to focus on macrophages to display the effect of CC10 on fgl2 expression and eventually mice survival. We entirely agree there may be other possibilities for a protective effect of CC10 to contribute to the disease process. This is worth further studies. The potential receptor of CC10 has not been revealed yet. Our previous study have demonstrated that CC10 have effect of dendritic cells in allergic rhinitis (34). In this research, we evaluated the effect of CC10 on macrophages functions and found Fgl2 was substantially down-regulated upon CC10 treatment, therefore, we speculate that potential CC10 receptor may be also expressed on macrophages. The potential target of CC10 on other immune cells cannot be excluded.

DNA microarray analysis is one of the most powerful approaches for the potential identification of unexpected genes involved in pathogenic processes. By using this approach, HMGbox transcription factor 1 (HBP1) was found to be one of the most downregulated genes after CC10 treatment of THP-1 cells. HBP1 is a well-described transcriptional repressor that modulates expression of genes involved in cell cycle progression. In a recent study, it was found that HBP1 is a direct target of miR-21 and confirmed that HBP1 modulates the inhibitory function of miR-21-ASO in hepatosteatosis and carcinogenesis simultaneously (23). HBP1 is an endogenous inhibitor of the Wnt signaling pathway in both normal and cancer cells. The tumor suppressor role of HBP1 has been reported in some malignancies, such as oral cancer and glioma (35). However, an association between HBP1 and Fgl2 has not been investigated yet. The current study

clearly demonstrated that CC10 protects against MHV-3 induced FH via suppression of Fgl2 expression. Such effects might be mediated by HBP1. However, the functional status of HBP1 in the CC10 pathway requires further research, and such studies are conducting in our laboratory.

In conclusion, we demonstrated that CC10 could limit the immunopathological damage in MHV-3-induced FH mice. Our results suggest that enhancing CC10 expression by an immunotherapeutic approach might be an effective treatment for FH.

HY performed all the described experiments and wrote the manuscript. YL assisted with some experiments, analyzed experimental results, and edited the manuscript. HW analyzed experimental results. XW reviewed and edited the manuscript. JH, WY, DX, XL, GS, and QN provided experimental help and design.

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Evidence for the Convergence Model: The Emergence of Highly Pathogenic Avian Influenza (H5N1) in Viet Nam

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Abstract: Building on a series of ground breaking reviews that first defined and drew attention to emerging infectious diseases (EID), the ‘convergence model’ was proposed to explain the multifactorial causality of disease emergence. The model broadly hypothesizes disease emergence is driven by the co-incidence of genetic, physical environmental, ecological, and social factors. We developed and tested a model of the emergence of highly pathogenic avian influenza (HPAI) H5N1 based on suspected convergence factors that are mainly associated with land-use change. Building on previous geospatial statistical studies that identified natural and human risk factors associated with urbanization, we added new factors to test whether causal mechanisms and pathogenic landscapes could be more specifically identified. Our findings suggest that urbanization spatially combines risk factors to produce particular types of peri-urban landscapes with significantly higher HPAI H5N1 emergence risk. The work highlights that peri-urban areas of Viet Nam have higher levels of chicken densities, duck and geese flock size diversities, and fraction of land under rice or aquaculture than rural and urban areas. We also found that land-use diversity, a surrogate measure for potential mixing of host populations and other factors that likely influence viral transmission, significantly improves the model’s predictability. Similarly, landscapes where intensive and extensive forms of poultry production overlap were found at greater risk. These results support the convergence hypothesis in general and demonstrate the potential to improve EID prevention and control by combining geospatial monitoring of these factors along with pathogen surveillance programs.

Text: Two decades after the Institute of Medicine's seminal report [1] recognized novel and reemerging diseases as a new category of microbial threats, the perpetual and unexpected nature of the emergence of infectious diseases remains a challenge in spite of significant clinical and biomedical research advances [2] . Highly Pathogenic Avian Influenza (HPAI) (subtype H5N1) is the most significant newly emerging pandemic disease since HIV/AIDS. Its eruption in Southeast Asia in 2003-4 and subsequent spread globally to more than 60 countries fits the complex systems definition of "surprise" [3] . In this same year that IOM had published its final report on microbial threats which highlighted H5N1's successful containment in Hong Kong in 1997 [4] , massive outbreaks occurred in Southeast Asia where it remains endemic, along with Egypt's Nile Delta. Since 2003, HPAI H5N1 has killed millions of poultry in countries throughout Asia, Europe, and Africa, and 402 humans have died from it in sixteen countries according to WHO data as of January 2015. The threat of a pandemic resulting in millions of human cases worldwide remains a possibility [5] .

Lederberg et al. [1] first pointed to the multiplicity of factors driving disease emergence, which later were elaborated and described in terms of 'the convergence model' [6]. The model proposes emergence events are precipitated by the intensifying of biological, environmental, ecological, and socioeconomic drivers. Microbial "adaptation and change," along with "changing ecosystems" and "economic development and land use" form major themes. Joshua Lederberg, the major intellectual force behind the studies summed-up saying "Ecological instabilities arise from the ways we alter the physical and biological environment, the microbial and animal tenants (humans included) of these environments, and our interactions (including hygienic and therapeutic interventions) with the parasites" [6].

Combining such disparate factors and associated concepts from biomedicine, ecology, and social sciences in a single framework remains elusive. One approach suggested has been to employ social-ecological systems theory that attempts to capture the behavior of so-called 'coupled natural-human systems', including the inevitable unexpected appearance of new diseases, themselves one of the "emerging properties" of complex adaptive systems (CAS) [7, 8]. The convergence model can be so adapted by incorporating the dynamics of urban, agricultural, and natural ecosystem transformations proposed with this framework. These associated multifaceted interactions including feedbacks that affect ecological communities, hosts and pathogen populations, are the proximate drivers of disease emergence.

The initial HPAI H5N1 outbreaks in Vietnam represent an ideal opportunity to adapt and test a CAS-convergence model. Emergence risk should be highest in the most rapidly transforming urban areas, peri-urban zones where mixes of urban-rural, modern-traditional land uses and poultry husbandry coincide most intensely. Specifically we hypothesized a positive association between the presence of HPAI outbreaks in poultry at the commune level and: 1) peri-urban areas, as defined by Saksena et al. [9], 2) land-use diversity, and 3) co-location of intensive and extensive systems of poultry.

We used the presence or absence at the commune level of HPAI H5N1 outbreaks in poultry as the dependent variable. Vietnam experienced its first HPAI H5N1 outbreak in late 2003, since then, there have been five waves and sporadic outbreaks recorded over the years [10, 11]. We chose to study the first wave (Wave 1) that ended in February 2004 and the second wave (Wave 2) that occurred between December 2004 and April 2005. We used data from

the Viet Nam 2006 Agricultural Census to develop an urbanicity classification that used data collected at a single point in time (2006) but across space (10,820 communes) to infer processes of change (urbanization, land-use diversification, and poultry intensification) [9]. The 58 provinces in Vietnam (not counting the 5 urban provinces that are governed centrally) are divided into rural districts, provincial towns, and provincial cities. Rural districts are further divided into communes (rural areas) and towns, and provincial towns and cities are divided into wards (urban subdistricts) and communes. A commune in Viet Nam is thus the third level administrative subdivision, consisting of villages/hamlets. For the purpose of simplicity we will henceforth use the term "commune" to refer to the smallest administrative unit whether it is a commune, town, or ward. We included risk factors documented in previous work. We also aimed to understand the differences, if any, in risk dynamics at different scales; comparing risks at the national scale to those at two sub-national agro-ecological zones. For this purpose we chose to study the Red River and Mekong River deltas, well known hot spots of the disease. Hence we conducted two sets of analyses (waves 1 and 2) for three places (nation, Red River Delta, and Mekong Delta) producing a total of 6 wave-place analyses. Data on outbreaks were obtained from the publicly available database of Viet Nam's Department of Animal Health. Given the highly complex dynamics of the epidemics and in keeping with recent methodological trends, we used multiple modeling approaches-parametric and non-parametric-with a focus on spatial analysis. We used both 'place' oriented models that can take into account variations in factors such as policies and administration as well as 'space' oriented models that recognize the importance of physical proximity in natural phenomenon [12].

Very few empirical studies have attempted to determine whether urbanization is related to EID outbreaks or whether urbanization is associated primarily with other factors related to EID outbreaks. One immediate problem researchers face is defining what is rural, urban, and transitional (i.e., peri-urban). Some studies have used official administrative definitions of urban and rural areas, but this approach is limited in its bluntness [13]. Other studies prioritized human population density as a satisfactory surrogate [11, [14] [15] [16] [17] [18] [19] [20], but this approach ignores the important fact that density is not a risk factor if it is accompanied by sufficient infrastructure to handle the population. Spencer [21] examined urbanization as a non-linear characteristic, using household-level variables such as water and sanitation services. He found evidence that increased diversity in water supply sources and sanitation infrastructure were associated with higher incidences of HPAI. These studies employed a limited definition of urbanization that lacked a well-defined characterization of peri-urbanization.

Still other studies have mapped the relative urban nature of a place, a broad concept that is often referred to as 'urbanicity' [22] [23] [24] [25] . While these studies show differences in the rural/ urban nature of communities across space and time, they have been limited to small-to medium-scale observational studies; and they have failed to distinguish between different levels of "ruralness". Perhaps the best known model of peri-urbanization is McGee's concept of *desakota* (Indonesian for "village-town") [26] . McGee identified six characteristics of *desakota* regions: 1) a large population of smallholder cultivators; 2) an increase in non-agricultural activities; 3) extreme fluidity and mobility of population; 4) a mixture of land uses, agriculture, cottage industries, suburban development; 5) increased participation of the female labor force; and 6) "grey-zones", where informal and illegal activities group [26] . Saksena et al. [9] built on McGee's *desakota* concepts and data from the 2006 Viet Nam Agricultural Census to establish an urbanicity classification. That study identified and mapped the 10,820 communes, the smallest administrative unit for which data are collected, as being rural, peri-urban, urban, or urban core. This project used the Saksena classification to assess associations between urbanicity classes, other risks factors, and HPAI outbreaks.

Researchers have estimated that almost 75% of zoonotic diseases are associated with landcover and land-use changes (LCLUC) [27, 28] . LCLUC such as peri-urbanization and agricultural diversification frequently result in more diverse and fragmented landscapes (number of land covers or land uses per unit of land). The importance of landscape pattern, including diversity and associated processes, which equate to host species' habitat size and distribution, and thus pathogen transmission dynamics is axiomatic though the specific mechanisms depend on the disease [29, 30] . Landscape fragmentation produces ecotones, defined as abrupt edges or transitions zones between different ecological systems, thought to facilitate disease emergence by increasing the intensity and frequency of contact between host species [31] Furthermore, fragmentation of natural habitat tends to interrupt and degrade natural processes, including interspecies interactions that regulate densities of otherwise opportunistic species that may serve as competent hosts [32] , although it is not clear if reduced species diversity necessarily increases pathogen transmission [33] . Rarely has research connected land-use diversification to final health endpoints in humans or livestock; this study attempts to link land-use diversity with HPAI H5N1 outbreaks.

Human populations in the rapidly urbanizing cities of the developing world require access to vegetables, fruits, meat, etc. typically produced elsewhere. As theorized by von Thünen in 1826 [34], much of this demand is met by farms near cities [35], many in areas undergoing processes of peri-urbanization [26]. Due to the globalization of poultry trade, large-scale chicken farms raising thousands of birds have expanded rapidly in Southeast Asia and compete with existing small backyard farmers [36]. Large, enterprise-scale (15,000-100,000 birds) operations are still rare in Viet Nam (only 33 communes have such a facility). On the other hand, domestic and multinational companies frequently contract farmers to raise between 2,000 and 15,000 birds.

Recent studies have examined the relative role of extensive (backyard) systems and intensive systems [15, [17] [18] [19] 37]. In much of Asia there is often a mix of commercial and backyard farming at any one location [36]. Experts have suggested that from a biosecurity perspective the co-location of extensive and intensive systems is a potential risk factor [38]. Intensive systems allow for virus evolution (e.g. Low Pathogenic Avian Influenza to HPAI) and transformation, while extensive systems allow for environmental persistence and circulation [39]. Previous studies of chicken populations as a risk factor have distinguished between production systems-native chickens, backyard chickens; flock density; commercial chickens, broilers and layers density, etc. [15, [17] [18] [19] 37]. In isolation, however, none of these number and/or density based poultry metrics adequately measures the extent of co-location of intensive and extensive systems in any given place. Intensive and extensive systems in Viet Nam have their own fairly well defined flock sizes. A diversity index of the relative number of intensive and extensive systems of poultry-raising can better estimate the effect of such co-location; this study attempts to link a livestock diversity index with the presence or absence of HPAI H5N1 outbreaks at the commune level.

This study investigated for the 10,820 communes of Viet Nam a wide suite of socio-economic, agricultural, climatic and ecological variables relevant to poultry management and the transmission and persistence of the HPAI virus. Many of these variables were identified based on earlier studies of HPAI (as reviewed in Gilbert and Pfeiffer [40]). Three novel variables were included based on hypotheses generated by this project. All variables were measured or aggregated to the commune level. The novel variables were:

- Degree of urbanization: We used the urbanicity classification developed by Saksena et al. [9] to define the urban character of each commune. The classification framework is based on four characteristics: 1) percentage of households whose main income is from agriculture, aquaculture and forestry, 2) percentage of households with modern forms of toilets, 3) percentage of land under agriculture, aquaculture and forestry and 4) the Normalized Differentiated Vegetation Index (NDVI). The three-way classification enabled testing for non-linear and non-monotonous responses.

- Land-use diversity: We measured land-use diversity using the Gini-Simpson Diversity Index [41]. The Gini-Simpson Diversity Index is given by $1-\lambda$, where λ equals the probability that two entities taken at random from the dataset of interest represent the same type. In situations with only one class (complete homogeneity) the Gini-Simpson index would have a value equal to zero. Such diversity indices have been used to measure land-use diversity [42]. We used the following five land-use classes: annual crops, perennial crops, forests, aquaculture and built-up land (including miscellaneous uses) for which data were collected in the 2006 Agricultural Census. The area under the last class was calculated as the difference between the total area and the sum of the first four classes.

The following variables are listed according to their role in disease introduction, transmission and persistence, though some of these factors may have multiple roles.

- Human population related transmission.

Human population density [11, 14-16, 18, 19, 44, 45].

- Poultry trade and market.

Towns and cities were assumed to be active trading places [10, 18, 37, 44, 46]. So, the distance to the nearest town/city was used as indicator of poultry trade.

Trade is facilitated by access to transportation infrastructure [37, 47, 48] . So, the distance to the nearest a) national highway and b) provincial highway was used as indicator of transportation infrastructure.

- Disease introduction and amplification.

The densities of chicken were calculated based on commune area [15, 19, 37, 49] .

- Intermediate hosts.

Duck and geese densities were calculated using total commune area [11, 19, 49] .

As previous studies have shown a link between scavenging in rice fields by ducks and outbreaks, we also calculated duck density using only the area under rice.

- Agro-ecological and environmental risk factors.

Previous studies have shown that the extent of rice cultivation is a risk factor, mainly due its association with free ranging ducks acting as scavengers [10] . We used percentage of land under rice cultivation as a measure of extent.

Rice cropping intensity is also a known risk factor [11, 17, 37] . We used the mean number of rice crops per year as a measure of intensity.

The extent of aquaculture is a known risk factor [10] , possibly because water bodies offer routes for transmission and persistence of the virus. The percentage of land under aquaculture was used as a metric.

Proximity to water bodies increases the risk of outbreaks [47, [50] [51] [52] , possibly by increasing the chance of contact between wild water birds and domestic poultry. We measured the distance between the commune and the nearest: a) lake and b) river.

Climatic variables-annual mean temperature and annual precipitation-have been associated with significant changes in risk [48, 53] .

Elevation, which is associated with types of land cover and agriculture, has been shown to be a significant risk factor in Vietnam [10] .

Compound Topographical Index (CTI, also known as Topographical Wetness Index) is a measure of the tendency for water to pool. Studies in Thailand and elsewhere [54] have shown that the extent of surface water is a strong risk factor, possibly due to the role of water in long-range transmission and persistence of the virus. In the absence of reliable and inexpensive data on the extent of surface water we used CTI as a proxy. CTI has been used in Ecological Niche Models (ENM) of HPAI H5N1 [55, 56] . However, given the nature of ENM studies, the effect of CTI as a risk factor has been unknown so far. CTI has been used as a risk factor in the study of other infectious and non-infectious diseases [57] . Some studies have shown that at local scales, the slope of the terrain (a component of CTI) was significantly correlated with reservoir species dominance [58] . CTI is a function of both the slope and the upstream contributing area per unit width orthogonal to the flow direction. CTI is computed as follows: $CTI = \ln(A_s / (\tan(\beta)))$ where; A_s = Area Value calculated as $((\text{flow accumulation} + 1) \times (\text{pixel area in m}^2))$ and β is the slope expressed in radians [59] .

Though previous studies have indicated that Normalized Difference Vegetation Index (NDVI) is a risk factor [10, 20, 55, 60, 61], we did not include it explicitly in our models, as the urban classification index we used included NDVI [9] .

We obtained commune level data on HPAI H5N1 outbreaks from the publicly available database of the Department of Animal Health [10] . Viet Nam experienced its first major epidemic waves between December 2003 and February 2006 [10] . We chose to study the first wave (Wave 1) that ended in February 2004 and the second wave (Wave 2) that

occurred between December 2004 and April 2005. In Wave 1, 21% of the communes and in Wave 2, 6% of the communes experienced outbreaks. We used data from the 1999 Population Census of Viet Nam to estimate human population per commune. We relied on data from two Agriculture Censuses of Viet Nam. This survey is conducted every five years covering all rural households and those peri-urban households that own farms. Thus about three-fourths of all of the country's households are included. The contents of the survey include number of households in major production activities, population, labor classified by sex, age, qualification, employment and major income source; agriculture, forestry and aquaculture land used by households classified by source, type, cultivation area for by crop type; and farming equipment by purpose. Commune level surveys include information on rural infrastructure, namely electricity, transportation, medical stations, schools; fresh water source, communication, markets, etc. Detailed economic data are collected for large farms. We used the 2006 Agriculture Census for most variables because the first three epidemic waves occurred between the Agricultural Censuses of 2001 and 2006 but were closer in time to the 2006 census [10] . However, for data on poultry numbers we used the 2001 Agriculture Census data set because between 1991 and 2003 the poultry population grew at an average rate of 7% annually. However, in 2004, after the first wave of the H5N1 epidemic, the poultry population fell 15%. Only by mid-2008 did the poultry population return close to pre-epidemic levels. Thus, we considered the poultry population data from the 2001 census to be more representative. We aggregated census household data to the commune level. A three-way classification of the rural-to-urban transition was based on a related study [9] .

Raster data on annual mean temperature and precipitation were obtained from the World-Clim database and converted to commune level data. The bioclimatic variables were compiled from the monthly temperature and precipitation values and interpolated to surfaces at 90m spatial resolution [62] . This public database provides data on the average climatic conditions of the period 1950-2000.

Elevation was generated from SRTM 90 meter Digital Elevation Models (DEM) acquired from the Consortium for Spatial Information (CGIAR-CSI). Compound Topographical Index (CTI) data were generated using the Geomorphometry and Gradient Metrics Toolbox for Arc-GIS 10.1.

Prior to risk factor analysis we cleaned the data by identifying illogical values for all variables and then either assigning a missing value to them or adjusting the values. Illogical values occurred mainly (less than 1% of the cases) for land-related variables such as percentage of commune land under a particular type of land use. Next we tested each variable for normality using the BestFit software (Palisade Corporation). Most of the variables were found to follow a log-normal distribution and a log-transform was used on them. We then examined the bi-variate correlations between all the risk factors (or their log-transform, as the case may be). Correlations were analyzed separately for each place. Certain risk factors were then eliminated from consideration when $|r| \leq 0.5$ (r is the Pearson correlation coefficient). When two risk factors were highly correlated, we chose to include the one which had not been adequately studied explicitly in previously published risk models. Notably, we excluded a) elevation (correlated with human population density, chicken density, duck density, percentage land under paddy, annual temperature and compound topographical index), b) human population density (correlated with elevation and CTI), c) chicken density (only at national level, correlated with CTI), d) duck and goose density (correlated with elevation, chicken density, percentage land under paddy, land use diversity index and CTI), e) annual temperature (correlated with elevation and CTI) and f) cropping intensity (correlated with percentage land under paddy).

Considering the importance of spatial autocorrelation in such epidemics, we used two modeling approaches: 1) multi-level Generalized Linear Mixed Model (GLMM) and 2) Boosted Regression trees (BRT) [63, 64] with an autoregressive term [65]. GLMM is a 'place' oriented approach that is well suited to analyzing the effect of administrative groupings, while BRT is a 'space' oriented approach that accounts for the effects of physical proximity. We began by deriving an autoregressive term by averaging the presence/absence among a set of neighbors defined by the limit of autocorrelation, weighted by the inverse of the Euclidean distance [65].

The limit of the autocorrelation of the response variable was obtained from the range of the spatial correlogram $\rho(h)$ [66]. To determine which predictor variables to include in the two models, we conducted logistic regression modeling separately for each of them one by one but included the autoregressive term each time. We finally included only those variables whose coefficient had a significance value $p \leq 0.2$ (in at least one wave-place combination) and we noted the sign of the coefficient. This choice of p value for screening risk factors is common in similar studies [15, 18, 45, 67]. We used a two-level GLMM (communes nested under districts) to take account of random effects for an area influenced by its neighbors,

and thus, we studied the effect of spatial autocorrelation. We used robust standard errors for tests of fixed effects. Boosted regression trees, also known as stochastic gradient boosting, was performed to predict the probability of HPAI H5N1 occurrence and determine the relative influence of each risk factor to the HPAI H5N1 occurrence. This method was developed recently and applied widely for distribution prediction in various fields of ecology [63, 64] . It is widely used for species distribution modeling where only the sites of occurrence of the species are known [68] . The method has been applied in numerous studies for predicting the distribution of HPAI H5N1 disease [16, 51, [69] [70] [71] . BRT utilizes regression trees and boosting algorithms to fit several models and combines them for improving prediction by performing iterative loop throughout the model [63, 64] .

The advantage of BRT is that it applies stochastic processes that include probabilistic components to improve predictive performance. We used regression trees to select relevant predictor variables and boosting to improve accuracy in a single tree. The sequential process allows trees to be fitted iteratively through a forward stage-wise procedure in the boosting model. Two important parameters specified in the BRT model are learning rate (lr) and tree complexity (tc) to determine the number of trees for optimal prediction [63, 64] . In our model we used 10 sets of training and test points for cross-validation, a tree complexity of 5, a learning rate of 0.01, and a bag fraction of 0.5. Other advantages of BRT include its insensitivity to co-linearity and non-linear responses. However, for the sake of consistency with the GLMM method, we chose to eliminate predictors that were highly correlated with other predictors and to make log-transforms where needed. In the GLMM models we used $p < 0.05$ to identify significant risk factors.

The predictive performances of the models were assessed by the area under the curve (AUC) of the receiver operation characteristic (ROC) curve. AUC is a measure of the overall fit of the model that varies from 0.5 (chance event) to 1.0 (perfect fit) [72] . A comparison of AUC with other accuracy metrics concluded that it is the most robust measure of model performance because it remained constant over a wide range of prevalence rates [73] . We used the corrected Akaike Information Criteria (AICc) to compare each GLMM model with and without its respective suite of fixed predictors.

We used SPSS version 21 (IBM Corp., New York, 2012) for GLMM and R version 3.1.0 (The R Foundation for Statistical Computing, 2014) for the BRT. For calculating the spatial correlogram we used the spdep package of R.

The fourteen predictor variables we modeled (see tables) were all found to be significantly associated with HPAI H5N1 outbreaks ($p < 0.05$) in at least one wave-place combination based on univariate analysis (but including the autoregressive term) (Table 1). Land-use diversity, chicken density, poultry flock size diversity and distance to national highway were found to have significant associations across five of the six wave-place combinations.

The predictive power of the GLMM models, as measured by the AUC, is very good with AUC values ranging from 0.802 to 0.952 (Tables 2-7). The predictive power of the national models was higher than that of the delta models. The predictive power of the BRT models is good, with AUCs ranging from 0.737 to 0.914. The BRT models also had a better predictive power at the national level than at the delta level. These values are higher than those reported for Wave 1 (AUC = 0.69) and Wave 2 (AUC = 0.77) by Gilbert et al. [11]. Both Gilbert et al. [11] and this study found that at the national level the predictive performance for Wave 2 was higher than that for Wave 1. Wave 2 mainly affected the Mekong River Delta. Previous studies indicated the duck density was an important predictor [11]; our results, however, indicated that the diversity of duck flock size was a more important predictor than duck density.

Both the GLMM and BRT models found annual precipitation to be a significant factor. The GLMM model indicated a negative association; similar to what was found by studies in China [51] and in the Red River Delta [53]. A global study of human cases also found occurrence to be higher under drier conditions [74]. Generally, the role of precipitation was found to be far more significant in the deltas than for the country as a whole.

The unadjusted Relative Risk (RR) of peri-urban areas in comparison with non-peri-urban areas was 1.41 and 1.60 for Waves 1 and 2, respectively. In terms of urbanicity, we found that chicken density, percentage of land under rice, percentage of land under aquaculture, flock size diversity for duck and geese, and the Compound Topographical Index (CTI) to be highest in peri-urban areas (Fig 1a-1e). We also found that land-use diversity was higher in rural areas, but peri-urban areas had diversity levels only marginally lower (Fig 1f). The urbanicity variable alone, however, was not found to be significantly associated with HPAI

H5N1 in any place according to the GLMM model except for the urban level in Red River Delta for Wave 2 and in the Mekong River Delta for Wave 1. The BRT model ranked urbanicity as one of the least influential variables. Land-use diversity was found to be significantly associated with HPAI H5N1 in both waves for Viet Nam according to the GLMM model, but at the delta level the association was significant only for Wave 2 in the Mekong River Delta. The BRT model indicated that land-use diversity highly influenced HPAI H5N1 at the national level in Wave 2. For the remaining waveplace combinations land-use diversity had middle to below-middle rank of influence.

Both the GLMM and BRT models indicated that the diversity of chicken flock-size had a strong association with HPAI H5N1 for both waves at the national level. This was generally found to be true at the delta levels with some exceptions. The diversity of duck and goose flock size was also significantly associated with HPAI H5N1 in all places, but the associations were much stronger in Wave 2 than in Wave 1.

The GLMM model indicated that the CTI had a very strong association with HPAI H5N1 at the national level in both waves although this was not true in the two deltas. The CTI is a steady state wetness index commonly used to quantify topographic control on hydrological processes. Accumulation numbers in flat areas, like deltas, are very large; hence the CTI was not a relevant variable in the GLMM model in these areas. The BRT model however indicated that CTI had middle to low influence in all waves and places. We found very high spatial clustering effects as indicated by the fact that in all waves and places the BRT model found the spatial autocorrelation term to have the highest rank of influence. As expected, the relative influence of the autocorrelation term at the national level was higher (60-78%) than at the delta levels (14-35%). In the GLMM models we found the Akaike Information Criterion (AIC) using the entire set of 14 variables to be much lower than the AICs of a GLMM model without fixed effects. This indicated that though clustering effects were significant, our theory driven predictor variables improved model performance.

A limitation of using surveillance methods for the dependent variable (poultry outbreaks) is that the data may have reporting/detection biases [11]. Under-reporting/detection in rural areas as compared to peri-urban areas is possible. We believe that the urbanicity and the shortest distance to nearest town risk factors serve as rough proxies for reporting/detection efficiency. Previous studies have tended to use human population density as a proxy for this purpose. In our study we found a strong association between human population

density and urbanicity. But we acknowledge that a categorical variable such as urbanicity may provide less sensitivity than a continuous variable such as human population density in this specific context.

This study explored the validity of a general model for disease emergence that combined the IOM 'convergence model' [6] and the social-ecological systems model [7, 8], for investigating the specific case of HPAI in Vietnam. We sought to test the hypotheses that measures of urbanization, land-use diversification, and poultry intensification are correlated with outbreaks in poultry. Our results generally support the hypothesis that social-ecological system transformations are associated with H5NI outbreaks in poultry.

The results presented here highlight three main findings: 1) when relevant risk factors are taken into account, urbanization is generally not a significant independent risk factor; but in peri-urban landscapes emergence factors converge, including higher levels of chicken densities, duck and geese flock size diversities, and fraction of land under rice or aquaculture; 2) high land-use diversity landscapes, a variable not previously considered in spatial studies of HPAI H5N1, are at significantly greater risk for HPAI H5N1 outbreaks; as are 3) landscapes where intensive and extensive forms of poultry production are co-located.

Only one other study has explicitly examined urbanicity in the context of HPAI H5N1. Loth et al. [17] found peri-urban areas in Indonesia were significantly associated with HPAI H5N1 cases, even based on multivariate models. Our study, however, attempted both to associate HPAI H5N1 with degree of urbanicity and to determine the features of peri-urban areas that place them at risk. When those features (i.e., chicken densities, duck and geese flock size diversities, and the fraction of land under rice or aquaculture) are included in multivariate models, the role of the urbanization variable per se diminishes. We found in the main river deltas in Viet Nam (Red River and Mekong), urbanization had no significant association with HPAI H5N1. This may be due to the fact that the deltas are more homogenous, in terms of urbanization, than the country as a whole. This is the first study to examine land-use diversity as a risk factor for HPAI H5N1. Measured by the Gini-Simpson Diversity Index of the five land-use classes on which data were collected in the 2006 Viet Nam Agricultural Census, and the presence or absence of HPAI outbreaks at the commune level, our results indicate a strong association between land-use diversity and HPAI H5N1 at the national level and in the Mekong River Delta. This metric captures both the variety of

habitats and of the complexity of geospatial patterning likely associated with transmission intensity. Our results are similar to what has been observed by studies of other EIDs using fragmentation metrics (e.g. [75] [76] [77]). This is one of the few studies, however, to link landscape fragmentation to an EID disease in poultry and not just to the vector and/or hosts of the EID.

Previous studies have focused on poultry production factors such as type of species, size of flocks, and extent of commercialization (e.g. [15, [17] [18] [19]). This study expands on those findings by providing evidence that when intensive and extensive systems of chicken and/or duck and geese production co-exist in the same commune, the commune experiences higher risk of disease outbreak. Future studies need to examine the biological causal mechanisms in this context.

We suggest that national census data (particularly agricultural censuses) compiled at local levels of administration provide valuable information that are not available from remotely sensed data (such as poultry densities) or require a large amount of labor to map at national to larger scales (land-use diversity). Mapping land-use classes at the national scale for local administrative units (i.e., the 10,820 communes in Viet Nam) is not an insignificant task. Future studies, however, could examine the correlation between a census-based metric with metrics derived from remote sensing used to measure proportional abundance of each landcover type within a landscape [78] . Vietnam is relatively advanced in making digital national population and agricultural census data available in a format that can be linked to administrative boundaries. While other nations are beginning to develop similar capacities, in the short term the application of this method to other countries may be limited. Ultimately, both census and remotely sensed data can be used independently to map the urban transition and diversity of land use; these tools, however, may provide their greatest insights when used together.

Another important contribution of this study was the discovery of the importance of CTI. So far CTI had been used only in ecological niche modeling studies of HPAI H5N1; the specific role and direction of influence of CTI had so far been unknown. Our study, the first to use CTI as a risk factor, found it had a large positive influence on HPAI H5N1 risk at the national level. Previous studies have highlighted the role of surface water extent in the persistence and transmission of the HPAI H5N1 virus. These studies measured surface water extent as area covered by water, magnitude of seasonal flooding, distance to the

nearest body of water, or other variables that are often difficult to map using remotely sensed data, especially for large area studies. CTI on the other hand has the potential to serve as an excellent surrogate which can easily be measured in a GIS database. The national and regional (delta) models differed quite considerably, both in terms of performance and significant risk factors. In the deltas we commonly found only chicken density, duck flock size diversity and annual precipitation to be significant. This suggests dynamics of risk at the commune level are strongly dependent on the spatial range of analysis, consistent with another study in the Mekong Delta [61] . Though that study's model initially included three dozen commonly known risk factors, the significant risk factors were limited to poultry flock density, proportion households with electricity, re-scaled NDVI median May-October, buffalo density and sweet potato yield. Another study in the Red River Delta [79] found that in addition to the typical poultry density metrics, only the presence of poultry traders was significant. We speculate that for smaller regions, especially for known hot-spots, the relevant risk factors are those that reflect short-range, short-term driving forces such as poultry trading, presence of live bird markets and wet markets etc. Improving model performance for smaller regions would require highly refined and nuanced metrics for poultry trading, road infrastructure, water bodies, etc.-data that are typically not available through census surveys. The differences between the national and regional models suggest that our results can inform planners making decisions at different hierarchical levels of jurisdiction: national, region and local.

Our study has the potential to inform the design of future research related to the epidemiology of other EIDs in Viet Nam and elsewhere. For example, we speculate that in Southeast Asia, Japanese encephalitis, the transmission of which is associated with rice cultivation and flood irrigation [80] , may also show a strong association with peri-urbanization. In some areas of Asia these ecological conditions occur near, or occasionally within, urban centers. Likewise, Hantaan virus, the cause of Korean hemorrhagic fever, is associated with the field mouse *Apodemus agrarius* and rice harvesting in fields where the rodents are present [80] . Our work has demonstrated that the percentage of land under rice in peri-urban areas and rural areas is similar. Hence diseases associated with rice production are likely to peak in peri-urban areas given other risk factors such as land-use diversity, CTI, and distance to infrastructure. Our poultry flock-size diversity findings may also be relevant to understanding the dynamics of other poultry related infections such as Newcastle disease. Finally, these results suggest the validity of a general model of zoonotic disease emergence that integrates IOM's convergence model with the subsequently proposed social-ecological systems and EID framework. Thus, convergence represents the coalescence in time and space of processes associated with land-cover and land-use

changes. Project results question whether the urban/rural land-use dichotomy is useful when large areas and parts of the population are caught between the two. Planners need better tools for mapping the rural-urban transition, and for understanding how the specific nature of peri-urban environments creates elevated health risk that require adaptation of existing planning, land use, and development practices.

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Hydroxychloroquine, a less toxic derivative of chloroquine, is effective in inhibiting SARS-CoV-2 infection in vitro

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Abstract: nan

Text: Hydroxychloroquine, a less toxic derivative of chloroquine, is effective in inhibiting SARS-CoV-2 infection in vitro Jia Liu 1 , Ruiyuan Cao 2 , Mingyue Xu 1,3 , Xi Wang 1 , Huanyu Zhang 1,3 , Hengrui Hu 1,3 , Yufeng Li 1,3 , Zhihong Hu 1 , Wu Zhong 2 and Manli Wang 1
Dear Editor, The outbreak of coronavirus disease 2019 (COVID-19) caused by the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2/2019-nCoV) poses a serious threat

to global public health and local economies. As of March 3, 2020, over 80,000 cases have been confirmed in China, including 2946 deaths as well as over 10,566 confirmed cases in 72 other countries. Such huge numbers of infected and dead people call for an urgent demand of effective, available, and affordable drugs to control and diminish the epidemic.

We have recently reported that two drugs, remdesivir (GS-5734) and chloroquine (CQ) phosphate, efficiently inhibited SARS-CoV-2 infection in vitro ¹. Remdesivir is a nucleoside analog prodrug developed by Gilead Sciences (USA). A recent case report showed that treatment with remdesivir improved the clinical condition of the first patient infected by SARS-CoV-2 in the United States ², and a phase III clinical trial of remdesivir against SARS-CoV-2 was launched in Wuhan on February 4, 2020. However, as an experimental drug, remdesivir is not expected to be largely available for treating a very large number of patients in a timely manner. Therefore, of the two potential drugs, CQ appears to be the drug of choice for large-scale use due to its availability, proven safety record, and a relatively low cost. In light of the preliminary clinical data, CQ has been added to the list of trial drugs in the Guidelines for the Diagnosis and Treatment of COVID-19 (sixth edition) published by National Health Commission of the People's Republic of China. CQ (N4-(7-Chloro-4-quinolinyl)-N1,N1-diethyl-1,4pentanediamine) has long been used to treat malaria and amebiasis. However, *Plasmodium falciparum* developed widespread resistance to it, and with the development of new antimalarials, it has become a choice for the prophylaxis of malaria. In addition, an overdose of CQ can cause acute poisoning and death ³. In the past years, due to infrequent utilization of CQ in clinical practice, its production and market supply was greatly reduced, at least in China. Hydroxychloroquine (HCQ) sulfate, a derivative of CQ, was first synthesized in 1946 by introducing a hydroxyl group into CQ and was demonstrated to be much less (~40%) toxic than CQ in animals ⁴. More importantly, HCQ is still widely available to treat autoimmune diseases, such as systemic lupus erythematosus and rheumatoid arthritis. Since CQ and HCQ share similar chemical structures and mechanisms of acting as a weak base and immunomodulator, it is easy to conjure up the idea that HCQ may be a potent candidate to treat infection by SARS-CoV-2. Actually, as of February 23, 2020, seven clinical trial registries were found in Chinese Clinical Trial Registry (<http://www.chictr.org.cn>) for using HCQ to treat COVID-19. Whether HCQ is as efficacious as CQ in treating SARS-CoV-2 infection still lacks the experimental evidence.

To this end, we evaluated the antiviral effect of HCQ against SARS-CoV-2 infection in comparison to CQ in vitro. First, the cytotoxicity of HCQ and CQ in African green monkey

kidney VeroE6 cells (ATCC-1586) was measured by standard CCK8 assay, and the result showed © The Author(s) 2020

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this license, visit <http://creativecommons.org/licenses/by/4.0/>. (Fig. 1a) . To better compare the antiviral activity of CQ versus HCQ, the dose-response curves of the two compounds against SARS-CoV-2 were determined at four different multiplicities of infection (MOIs) by quantification of viral RNA copy numbers in the cell supernatant at 48 h post infection (p.i.). The data summarized in Fig. 1a and Supplementary Table S1 show that, at all MOIs (0.01, 0.02, 0.2, and 0.8), the 50% maximal effective concentration (EC₅₀) for CQ (2.71, 3.81, 7.14, and 7.36 μ M) was lower than that of HCQ (4.51, 4.06, 17.31, and 12.96 μ M). The differences in EC₅₀ values were statistically significant at an MOI of 0.01 ($P < 0.05$) and MOI of 0.2 ($P < 0.001$) (Supplementary Table S1) . It is worth noting that the EC₅₀ values of CQ seemed to be a little higher than that in our previous report (1.13 μ M at an MOI of 0.05) ¹ , which is likely due to the adaptation of the virus in cell culture that significantly increased viral infectivity upon continuous passaging. Consequently, the selectivity index (SI = CC₅₀ /EC₅₀) of CQ (100.81, 71.71, 38.26, and 37.12) was higher than that of HCQ (55.32, 61.45, 14.41, 19.25) at MOIs of 0.01, 0.02, 0.2, and 0.8, respectively. These results were corroborated by immunofluorescence microscopy as evidenced by different expression levels of virus nucleoprotein (NP) at the indicated drug concentrations at 48 h p.i. (Supplementary Fig. S1) . Taken together, the data suggest that the anti-SARS-CoV-2 activity of HCQ seems to be less potent compared to CQ, at least at certain MOIs.

Both CQ and HCQ are weak bases that are known to elevate the pH of acidic intracellular organelles, such as endosomes/lysosomes, essential for membrane fusion ⁵ . In addition, CQ could inhibit SARS-CoV entry through changing the glycosylation of ACE2 receptor and spike protein ⁶ . Time-of-addition experiment confirmed that HCQ effectively inhibited the entry step, as well as the post-entry stages of SARS-CoV-2, which was also found upon CQ

treatment (Supplementary Fig. S2). To further explore the detailed mechanism of action of CQ and HCQ in inhibiting virus entry, co-localization of virions with early endosomes (EEs) or endolysosomes (ELs) was analyzed by immunofluorescence analysis (IFA) and confocal microscopy. Quantification analysis showed that, at 90 min p.i. in untreated cells, 16.2% of internalized virions (anti-NP, red) were observed in early endosome antigen 1 (EEA1)-positive EEs (green), while more virions (34.3%) were transported into the late endosomal-lysosomal protein LAMP1 + ELs (green) ($n > 30$ cells for each group). By contrast, in the presence of CQ or HCQ, significantly more virions (35.3% for CQ and 29.2% for HCQ; $P < 0.001$) were detected in the EEs, while only very few virions (2.4% for CQ and 0.03% for HCQ; $P < 0.001$) were found to be co-localized with LAMP1 + ELs ($n > 30$ cells) (Fig. 1b, c) . This suggested that both CQ and HCQ blocked the transport of SARS-CoV-2 from EEs to ELs, which appears to be a requirement to release the viral genome as in the case of SARS-CoV 7 .

Interestingly, we found that CQ and HCQ treatment caused noticeable changes in the number and size/morphology of EEs and ELs (Fig. 1c) . In the untreated cells, most EEs were much smaller than ELs (Fig. 1c) . In CQ and HCQ-treated cells, abnormally enlarged EE vesicles were observed (Fig. 1c , arrows in the upper panels), many of which are even larger than ELs in the untreated cells. This is in agreement with previous report that treatment with CQ induced the formation of expanded cytoplasmic vesicles 8 . Within the EE vesicles, virions (red) were localized around the membrane (green) of the vesicle. CQ treatment did not cause obvious changes in the number and size of ELs; however, the regular vesicle structure seemed to be disrupted, at least partially. By contrast, in HCQ-treated cells, the size and number of ELs increased significantly (Fig. 1c , arrows in the lower panels).

Since acidification is crucial for endosome maturation and function, we surmise that endosome maturation might be blocked at intermediate stages of endocytosis, resulting in failure of further transport of virions to the ultimate releasing site. CQ was reported to elevate the pH (see figure on previous page) Fig. 1 Comparative antiviral efficacy and mechanism of action of CQ and HCQ against SARS-CoV-2 infection in vitro. a Cytotoxicity and antiviral activities of CQ and HCQ. The cytotoxicity of the two drugs in Vero E6 cells was determined by CCK-8 assays. Vero E6 cells were treated with different doses of either compound or with PBS in the controls for 1 h and then infected with SARS-CoV-2 at MOIs of 0.01, 0.02, 0.2, and 0.8. The virus yield in the cell supernatant was quantified by qRT-PCR at 48 h p.i. Y-axis represents the mean of percent inhibition normalized to the PBS group. The

experiments were repeated twice. b, c Mechanism of CQ and HCQ in inhibiting virus entry. Vero E6 cells were treated with CQ or HCQ (50 μ M) for 1 h, followed by virus binding (MOI = 10) at 4°C for 1 h. Then the unbound virions were removed, and the cells were further supplemented with fresh drug-containing medium at 37°C for 90 min before being fixed and stained with IFA using anti-NP antibody for virions (red) and antibodies against EEA1 for EEs (green) or LAMP1 for ELs (green). The nuclei (blue) were stained with Hoechst dye. The portion of virions that co-localized with EEs or ELs in each group (n > 30 cells) was quantified and is shown in b. Representative confocal microscopic images of viral particles (red), EEA1 + EEs (green), or LAMP1 + ELs (green) in each group are displayed in c. The enlarged images in the boxes indicate a single vesicle-containing virion. The arrows indicated the abnormally enlarged vesicles. Bars, 5 μ m. Statistical analysis was performed using a one-way analysis of variance (ANOVA) with GraphPad Prism (F = 102.8, df = 5,182, ***P < 0.001).

of lysosome from about 4.5 to 6.5 at 100 μ M⁹. To our knowledge, there is a lack of studies on the impact of HCQ on the morphology and pH values of endosomes/ lysosomes. Our observations suggested that the mode of actions of CQ and HCQ appear to be distinct in certain aspects.

It has been reported that oral absorption of CQ and HCQ in humans is very efficient. In animals, both drugs share similar tissue distribution patterns, with high concentrations in the liver, spleen, kidney, and lung reaching levels of 200-700 times higher than those in the plasma¹⁰. It was reported that safe dosage (6-6.5 mg/kg per day) of HCQ sulfate could generate serum levels of 1.4-1.5 μ M in humans¹¹. Therefore, with a safe dosage, HCQ concentration in the above tissues is likely to be achieved to inhibit SARS-CoV-2 infection.

Clinical investigation found that high concentration of cytokines were detected in the plasma of critically ill patients infected with SARS-CoV-2, suggesting that cytokine storm was associated with disease severity¹². Other than its direct antiviral activity, HCQ is a safe and successful anti-inflammatory agent that has been used extensively in autoimmune diseases and can significantly decrease the production of cytokines and, in particular, pro-inflammatory factors. Therefore, in COVID-19 patients, HCQ may also contribute to attenuating the inflammatory response. In conclusion, our results show that HCQ can efficiently inhibit SARS-CoV-2 infection in vitro. In combination with its anti-inflammatory function, we predict that the drug has a good potential to combat the

disease. This possibility awaits confirmation by clinical trials. We need to point out, although HCQ is less toxic than CQ, prolonged and overdose usage can still cause poisoning. And the relatively low SI of HCQ requires careful designing and conducting of clinical trials to achieve efficient and safe control of the SARS-CoV-2 infection.

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Factors Associated With Mental Health Outcomes Among Health Care Workers Exposed to Coronavirus Disease 2019

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Abstract: **IMPORTANCE:** Health care workers exposed to coronavirus disease 2019 (COVID-19) could be psychologically stressed. **OBJECTIVE:** To assess the magnitude of mental health outcomes and associated factors among health care workers treating patients exposed to COVID-19 in China. **DESIGN, SETTINGS, AND PARTICIPANTS:** This cross-sectional, survey-based, region-stratified study collected demographic data and mental health measurements from 1257 health care workers in 34 hospitals from January 29, 2020, to February 3, 2020, in China. Health care workers in hospitals equipped with fever clinics or wards for patients with COVID-19 were eligible. **MAIN OUTCOMES AND**

MEASURES: The degree of symptoms of depression, anxiety, insomnia, and distress was assessed by the Chinese versions of the 9-item Patient Health Questionnaire, the 7-item Generalized Anxiety Disorder scale, the 7-item Insomnia Severity Index, and the 22-item Impact of Event Scale–Revised, respectively. Multivariable logistic regression analysis was performed to identify factors associated with mental health outcomes. **RESULTS:** A total of 1257 of 1830 contacted individuals completed the survey, with a participation rate of 68.7%. A total of 813 (64.7%) were aged 26 to 40 years, and 964 (76.7%) were women. Of all participants, 764 (60.8%) were nurses, and 493 (39.2%) were physicians; 760 (60.5%) worked in hospitals in Wuhan, and 522 (41.5%) were frontline health care workers. A considerable proportion of participants reported symptoms of depression (634 [50.4%]), anxiety (560 [44.6%]), insomnia (427 [34.0%]), and distress (899 [71.5%]). Nurses, women, frontline health care workers, and those working in Wuhan, China, reported more severe degrees of all measurements of mental health symptoms than other health care workers (eg, median [IQR] Patient Health Questionnaire scores among physicians vs nurses: 4.0 [1.0-7.0] vs 5.0 [2.0-8.0]; $P = .007$; median [interquartile range {IQR}] Generalized Anxiety Disorder scale scores among men vs women: 2.0 [0-6.0] vs 4.0 [1.0-7.0]; $P < .001$; median [IQR] Insomnia Severity Index scores among frontline vs second-line workers: 6.0 [2.0-11.0] vs 4.0 [1.0-8.0]; $P < .001$; median [IQR] Impact of Event Scale–Revised scores among those in Wuhan vs those in Hubei outside Wuhan and those outside Hubei: 21.0 [8.5-34.5] vs 18.0 [6.0-28.0] in Hubei outside Wuhan and 15.0 [4.0-26.0] outside Hubei; $P < .001$). Multivariable logistic regression analysis showed participants from outside Hubei province were associated with lower risk of experiencing symptoms of distress compared with those in Wuhan (odds ratio [OR], 0.62; 95% CI, 0.43-0.88; $P = .008$). Frontline health care workers engaged in direct diagnosis, treatment, and care of patients with COVID-19 were associated with a higher risk of symptoms of depression (OR, 1.52; 95% CI, 1.11-2.09; $P = .01$), anxiety (OR, 1.57; 95% CI, 1.22-2.02; $P < .001$), insomnia (OR, 2.97; 95% CI, 1.92-4.60; $P < .001$), and distress (OR, 1.60; 95% CI, 1.25-2.04; $P < .001$). **CONCLUSIONS AND RELEVANCE:** In this survey of health care workers in hospitals equipped with fever clinics or wards for patients with COVID-19 in Wuhan and other regions in China, participants reported experiencing psychological burden, especially nurses, women, those in Wuhan, and frontline health care workers directly engaged in the diagnosis, treatment, and care for patients with COVID-19.

Text: Abbreviation: PHQ-9, 9-item Patient Health Questionnaire; GAD-7, 7-item Generalized Anxiety Disorder; ISI, 7-item Insomnia Severity Index; IES-R, 22-item Impact of Event Scale–Revised; IQR, interquartile range. Hyperarousal, median (IQR) 6.0(2.0, 10.0) 6.0(2.0, 9.0) .29

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A missense mutation in *Katnal1* underlies behavioural, neurological and ciliary anomalies

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Abstract: Microtubule severing enzymes implement a diverse range of tissue-specific molecular functions throughout development and into adulthood. Although microtubule severing is fundamental to many dynamic neural processes, little is known regarding the role of the family member Katanin p60 subunit A-like 1, *KATNAL1*, in central nervous system (CNS) function. Recent studies reporting that microdeletions incorporating the *KATNAL1* locus in humans result in intellectual disability and microcephaly suggest that *KATNAL1* may play a prominent role in the CNS; however, such associations lack the functional data required to highlight potential mechanisms which link the gene to disease symptoms. Here we identify and characterise a mouse line carrying a loss of function allele in *Katnal1*. We show that mutants express behavioural deficits including in circadian rhythms, sleep, anxiety and learning/memory. Furthermore, in the brains of *Katnal1* mutant mice we reveal numerous morphological abnormalities and defects in neuronal migration and morphology. Furthermore we demonstrate defects in the motile cilia of the ventricular

ependymal cells of mutants, suggesting a role for *Katnal1* in the development of ciliary function. We believe the data we present here are the first to associate *KATNAL1* with such phenotypes, demonstrating that the protein plays key roles in a number of processes integral to the development of neuronal function and behaviour.

Text: Microtubule severing enzymes are a family of AAA-ATPase proteins that participate in fundamental cellular processes such as mitosis, ciliary biogenesis and growth cone motility. In neurons this family is known to control such processes as axonal elongation¹ and synaptic development.² In addition, mutations in microtubule severing enzyme genes *SPG4*, *KATNB1* and *KATNAL2* are associated with hereditary spastic paraplegia, cerebral malformations and autism, respectively, [3] [4] [5] [6] and mutations in *Fign* cause a range of phenotypes in mice.⁷ Currently the microtubule severing enzyme *KATNAL1* is poorly characterised and it is not yet understood how the enzyme functions in the nervous system. Recent evidence from genetic characterisation of human patients suggests that haploinsufficiency of *KATNAL1* is linked with a number of symptoms including intellectual disability (ID) and craniofacial dysmorphologies.^{8, 9} It is also notable that a very rare *KATNAL1* mutation has been associated with schizophrenia¹⁰ (<http://atgu.mgh.harvard.edu/~spurcell/genebook/genebook.cgi?user=guest&cmd=verb-gene&tbox=KATNAL1>) and that Peters syndrome and autism have both been associated with the chromosomal region containing the *KATNAL1* locus.^{11, 12} Although such association studies strongly suggest that *KATNAL1* plays a fundamental role in the central nervous system (CNS), additional studies using cellular or animal models are required to understand how the gene may be causative for disease. Here we present the first study describing neural and behavioural deficits associated with a loss of function allele of *Katnal1* in the mouse. This mutant mouse line was independently identified in two parallel phenotyping screens, which demonstrated that mutant mice showed both male sterility and circadian phenotypes. Subsequent behavioural investigations demonstrated that this mutation is associated with anxiety and memory deficits. Underlying these behavioural phenotypes, we identified histopathological abnormalities in the brains of *Katnal1* 1H/1H mutants, including disordered cellular layers in the hippocampus and cortex and substantially larger ventricles. Further investigations demonstrated that *Katnal1* 1H/1H mice show neuronal migration and ciliary function deficits suggesting *KATNAL1* plays an essential role in these processes. These findings are the first to our knowledge to conclusively show that mutations in *Katnal1* lead to behavioural and neuronal disturbances and provide insight regarding the clinical associations that have been linked to the gene. performed on mouse cohorts that were partially or completely congenic on the C57BL/6 J background.

Circadian wheel running was performed as previously described. 14 Sleep assessment by electroencephalography and electromyography Electroencephalography and electromyography was performed as previously described. 15 Behavioural phenotyping Spontaneous alternation. Mice were placed in a walled T-maze (black polyvinyl chloride, lined with sawdust; stem = 88 × 13 cm; arms = 32 × 13 cm) and allowed to enter a goal arm of their choice. The mouse was confined in the goal arm for 30 s, before being allowed a second free choice of goal arm. An alternation was recorded if the second choice differed from that of the first. One trial was performed per day for 10 days.

Open field behaviour. Mice were placed into a walled arena (grey polyvinyl chloride; 45 × 45 cm) and allowed to explore for 20 min. Animals were monitored by EthoVision XT analysis software (Noldus, Wageningen, Netherlands).

Video tracking in the home cage. Activity in the home cage was recorded by video tracking as previously described. 16 Morris water maze and ultrasonic vocalisation. These tests were performed as previously described. 17 Brain histology and immunofluorescence Brains were mounted in OCT (VWR) and 12 µm coronal sections taken. Sections were stained with hematoxylin and eosin, or immunolabelled following standard protocols.

In vivo neuronal migration assessment was performed as previously described 18 using embryos at either E13 or E15 (three mothers per age group) and pups at P9. Cell counts were performed using ImageJ (NIH, Bethesda, MD, USA).

In vitro neuronal migration assessment was performed using a Boyden chamber migration protocol as previously described. 19 Micro-computed tomography scanning Micro-computed tomography was performed using a Skyscan 1172 at 90 kV, 112 µA using an aluminium and copper filter, a rotation step of 0.250 degrees and a pixel size of 4.96 µm.

Segmentation, volume calculation and 3D modelling was performed using ITK-SNAP version 3.0.0 (ref. 20) and 3DSlicer. 21 Golgi-Cox staining of neurons Golgi-Cox neuronal staining was performed using the FD Rapid GolgiStain Kit (FD NeuroTechnologies, Columbia, MD, USA). Neurons were analysed using ImageJ.

Brains from P2 mice were dissected, and the dorsal cerebral half was sectioned (250 μ m) through the floor of the lateral and 3rd ventricle using a vibratome. Ciliary beat frequency and pattern was analysed as previously described. 22 Electron microscopy For Scanning Electron Microscopy the ependymal lining of the lateral ventricle was fixed in 2.5% glutaraldehyde, 2% paraformaldehyde in 0.1 M phosphate buffer, incubated in 2% osmium tetroxide, and dehydrated through ethanol solutions. Samples were critical point dried using an Emitech K850 (Quorum Technologies, East Sussex, UK), coated with platinum using a Quorum Q150R S sputter coater (Quorum Technologies). and visualised using a JEOL LSM-6010 scanning electron microscope (Jeol, Herts, UK).

Transmission electron microscopy was performed as previously described. 22 Statistical analysis Data was analysed using two-tailed students T test or AVOVA using SPSS (IBM) or GraphPad Prism 5.0 (GraphPad Software, La Jolla, CA, USA). Significance level for all analysis was set at $P < 0.05$. All graphs are presented showing mean $\pm$ s.e.m.

Additional and more detailed methods can be found in supplementary information.

Identification and cloning of the *Katnal1* 1H mutation To identify novel gene mutations affecting circadian behaviour we undertook a circadian running wheel screen of pedigrees of N-ethyl-N-nitrosourea mutagenised mice. 13 In one pedigree 17.65% of animals showed a short circadian period in constant darkness (o 23 h observed in 12 out of 68 animals screened). An outcross using an affected female produced no affected animals (33 animals screened). In subsequent intercross screens 15.5% of animals were affected (53 out of 342 animals screened), suggesting that the pedigree carries a mutation causing a recessive circadian phenotype which is 60% penetrant. We found no gender bias in affected animals (proportion of affected animals: male = 47.2%; female = 52.8%).

Concurrently a male sterility phenotype was identified within the same pedigree. 23 Genome-wide SNP linkage analysis mapped the circadian and sterility phenotypes to the same region on chromosome 5 and subsequent sequencing identified the causative mutation as a T to G single point mutation within exon seven of the *Katnal1* gene. For full details of mapping and identification of the mutation see reference 23. This mutant allele was designated *Katnal1* 1H , and results in a leucine to valine substitution at residue 286 of

the protein. In vitro functional analysis demonstrated that the mutation is a recessive loss-of-function allele. 23 3D modelling of the protein suggests that this loss of function is due to hydrophobic changes in the AAA domain of the enzyme (Supplementary Figure S1). Genotyping confirmed that the mutation was homozygous in affected circadian animals and wild type or heterozygous in unaffected animals, confirming that *Katnal1* 1H was causative for the circadian phenotype.

Circadian and sleep anomalies in *Katnal1* 1H/1H mice More extensive circadian phenotyping conducted on *Katnal1* homozygotes (*Katnal1* 1H/1H) and wild-type littermates (*Katnal1* +/+) confirmed that *Katnal1* 1H/1H mice had a shorter free-running circadian period (Figures 1a-c) and furthermore revealed that *Katnal1* 1H/1H animals were more active in the light phase of the light/dark cycle (Figure 1d), showed increased anticipation of light to dark transitions and greater shift in activity onset when released from light/dark cycles to constant darkness (Figure 1e). Data and cohort details are given in Supplementary Table S1 . Bioluminescence recordings performed using PER2::LUCIFERASE reporter mice carrying the *Katnal1* 1H mutation revealed that these circadian changes were not due to changes to the core molecular clock of the suprachiasmatic nucleus (the site of the master circadian clock in the brain; Supplementary Figure S2).

Circadian disruptions are often associated with deficits in sleep homeostasis. Therefore to complement our circadian studies we conducted wireless electroencephalography recordings over a baseline period of 24 h and following a 6 h period of sleep deprivation. A detailed summary of electroencephalography analysis is given in Supplementary Table S1. Compared to wildtype littermates, the non-REM delta power of *Katnal1* 1H/1H mice was higher in the dark phase of baseline sleep (mixed ANOVA, interaction factors 'genotype X time, $F(1,88) = 8.91$, $P = 0.0175$) (Figure 1f) and in both the light and dark phases of recovery sleep (mixed ANOVA, interaction factors 'genotype X time', $F(1,136) = 11.93$, $P = 0.0086$; Figure 1g). All other sleep parameters were unaffected in *Katnal1* 1H/1H animals.

Katnal1 1H/1H mice display a spectrum of behavioural deficits Human patients carrying a heterozygous deletion incorporating the *Katnal1* locus show a number of cognitive deficits including ID and a delay in language acquisition. 8, 9 We therefore investigated whether these deficits were modelled in *Katnal1* 1H/1H mice by subjecting animal cohorts to a battery of behavioural tests. Data and cohort details are given in Supplementary Table S2 .

Both working memory and spatial memory were significantly poorer in *Katnal1* 1H/1H mice, as evidenced by reduced spontaneous alternations in a T-maze (Figure 2a) and in the Morris water maze where mutants take longer to find the platform in acquisition trials (Figure 2b). Compared to wild-type littermates, *Katnal1* 1H/1H animals have a shorter period (c), are more active in the light phase of the light/dark cycle (d) and show an earlier onset of activity in light/dark transitions and in the transition from light/dark cycles to constant darkness (e). In EEG recordings during sleep, *Katnal1* 1H/1H mice show increased non-REM delta power in the dark phase of the light/dark cycle (f) and following sleep deprivation (g). * $P \leq 0.05$; ** $P \leq 0.01$; *** $P \leq 0.001$. EEG, electroencephalography; DD, constant darkness; LD, light/dark cycle. type = 164 ± 12 m, *Katnal1* 1H/1H = 243 ± 20 m, $P = 0.02$; distance travelled in periphery of open field: wild type = 4.3 ± 0.2 m, *Katnal1* 1H/1H = 6 ± 0.3 m, $P = 0.004$). Conversely when mouse activity was recorded in the home cage, we found no difference between genotypes (distance travelled over 24 h: wild type = 399 ± 77 m, *Katnal1* 1H/1H = 418 ± 41 m, $P = 0.833$) suggesting that the former activity differences were due to the novel environment of the open field rather than generalised hyperactivity in *Katnal1* 1H/1H animals. Finally, in certain conditions (such as maternal separation) mice emit ultrasonic vocalisations (USVs). To test whether *Katnal1* 1H/1H animals vocalised differently to wild types, we separated pups at postnatal days 7-8 (the age at which mice show peak of USV emission 24) and recorded their USVs. In these tests, compared to wild types, *Katnal1* 1H/1H pups produced fewer (Figure 2g) and shorter (Figure 2h) vocalisations, containing fewer phrases (Figure 2i).

Gross brain morphological abnormalities in *Katnal1* 1H/1H mice Since we observed a number of behavioural phenotypes in *Katnal1* 1H/1H mice, we performed histological analysis to ascertain whether differences in brain histology underlied these behaviours. Data and cohort details are given in Supplementary Table S3 . Analysis of hematoxylin and eosin stained brain sections revealed that, compared to wildtype littermates, *Katnal1* 1H/1H animals had less tightly packed pyramidal cell layers in the hippocampus (Figures 3a and b) and a narrower cortical layer 1 and wider cortical layer 6 (Figures 3c-e) . To confirm these cortical layer differences, immunofluorescence was performed using the (Figures 3l and m) .

Quantification of fluorescence intensity demonstrated that in *Katnal1* 1H/1H cortex both calbindin and CUX1 labelling was more intense closer to the cortical surface, which is consistent with the reduction in the size of layer 1 (two-way analysis of variance (ANOVA), interaction factors 'genotype X distance of fluorescence from cortical surface', calbindin: $F(75,988) = 16.8$, $P < 0.0005$; CUX1: $F(93,372) = 2.17$, $P = 0.001$; Figures 3h and k) . Similar quantification revealed that FOXP2 labelling extended further from layer 6b (as labelled by CTGF) in the *Katnal1* 1H/1H cortex, which is consistent with an increase in the size of layer

6 (two-way ANOVA, interaction factors 'genotype X distance of fluorescence from CTGF labelling:' $F(93,372) = 1.32$, $P = 0.038$; Figure 3n). Finally, three dimensional models of the ventricular system were constructed from brain micro-computed tomography scans (Figures 3o and p) . Volumetric analysis revealed that *Katnal1* 1H/1H mice had substantially larger ventricles than wild types (Figure 3q).

Neuronal migration and morphology defects in *Katnal1* 1H/1H brains The histological phenotypes of *Katnal1* 1H/1H mouse brains described above are suggestive of neuronal migration defects. 18 We therefore investigated whether *Katnal1* 1H/1H mice showed abnormal neuronal migration using BrdU labelling of E13 and E15 embryos and quantified labelled cells in the cortex of P9 pups (described in reference 18). At both ages *Katnal1* 1H/1H animals had greater numbers of labelled neurons in bins close to the cortical surface neurons positioned closer to the cortical surface compared to wild type. To confirm these results we used a Boyden chamber 19 and performed in vitro neuronal migration analysis in E13.5 primary cortical neuronal cultures. Here we found that a greater proportion of *Katnal1* 1H/1H cortical neurons migrated to the base of the cell culture insert compared to wildtype controls (Supplementary Figure S3) .

Since in both BrdU labelling and the Boyden assay neurons from *Katnal1* 1H/1H animals migrated further than those of wild-type littermates, these results suggest that *Katnal1* 1H/1H cortical neurons show defects in the termination of cortical neuronal migration. Given its role in cytoskeletal organisation, we also hypothesised that neuronal morphology is modulated by *Katnal1*. Analysis of golgi stained neurons from layers 2-3 of the cortex (Figures 4g and i) demonstrated that, compared to wild-type littermates, *Katnal1* 1H/1H neurons had larger soma (Figure 4k) , and shorter and thinner axons (Figures 4l and m) (data and cohort details are given in Supplementary Table S3). Furthermore, analysis at higher magnification (Figures 4h and j) , demonstrated that the number of synaptic spines on *Katnal1* 1H/1H neurons was significantly reduced compared to wild type (Figure 4n) .

Recent studies have demonstrated that mutations in some microtubule severing enzymes can cause defects in cilia. 5 Since such ciliary defects could underlie the phenotypes described above we studied the motile cilia of the ependymal lining of the lateral ventricle in sections of postnatal day 2 mouse brains from both *Katnal1* 1H/1H ($n = 4$) and wild-type animals ($n = 3$). We found that the ciliary beat frequency (CBF) of *Katnal1* 1H/1H animals was significantly attenuated compared to wild-type (CBF: wildtype = 22.39 ± 0.94 Hz,

Katnal1 1H/1H = 14.25 ± 0.92 Hz, $P = 0.0001$; Figure 5a , Supplementary Movies S1). This reduction in CBF in Katnal1 1H/1H animals was also associated with an increased proportion of cilia with an abnormal beat pattern (ciliary dyskinesia) (proportion of dyskinetic cilia: wild type = 17%, Katnal1 1H/1H = 75%) (Figure 5b and Supplementary Movies S1). Visual inspection of the cilia identified a number of ciliary abnormalities such as a swollen ciliary tip (Supplementary Movie S3) or extremely long cilia (Supplementary Movie S4) scattered throughout the field of cilia in Katnal1 1H/1H ventricles. These abnormalities were observed in approximately 25% of Katnal1 1H/1H brain slices. The abnormal cilia always showed a dyskinetic beat pattern and lower beat frequency. To further investigate ciliary morphology we performed scanning electron microscopy upon the ependymal lining of the lateral ventricles of both Katnal1 1H/1H ($n = 3$) and wild-type animals ($n = 3$; Figures 5c and d) . Cilia measurements showed no significant differences in average cilia length between genotypes (average cilia length: wild type = 6.22 ± 0.86 μm , Katnal1 1H/1H = 6.54 ± 0.94 μm , $P = 0.303$). However in Katnal1 1H/1H samples we noted the presence of both long and short cilia (Figures 5e and f ; defined as two standard deviations longer or shorter than the average cilia length) that were not present in wild-type samples. In addition, inspection of Katnal1 1H/1H cilia identified ciliary abnormalities including bifurcated cilia (Figure 5g) , abnormal kinks and bends in the cilia (Figure 5h) and swellings along the length of the cilia (Figure 5i). Transmission electron microscopy of ependymal cilia found that vesicular aggregates were present within the ciliary swellings described above (Figure 5j). Although these abnormalities were present in only a small proportion (<1%) of Katnal1 1H/1H cilia, they were notably absent from wild-type cilia.

Microtubule severing enzymes play diverse roles in the nervous system. However, at present the microtubule severing enzyme Katnal1 is poorly defined in the context of CNS development and function. Here we present a detailed phenotypic analysis of Katnal1 1H and show that the mutation is associated with changes in circadian rhythms, sleep and behaviour. Furthermore we demonstrate that defects in brain histopathology, neuronal migration and neuronal morphology underlie these phenotypes. Finally we also demonstrate that Katnal1 1H causes a range of defects in the motile cilia of ventricular ependymal cells. The data we present here are the first to associate KATNAL1 with such dysfunctions with important implications for clinical association studies.

The Katnal1 1H mutation was initially identified with a circadian deficit including a short free-running period and advanced activity onset. However subsequent ex vivo experiments

using SCN slices of animals carrying the PER2::LUC reporter gene demonstrated no defects in SCN cellular rhythms, suggesting that the core circadian clock was unperturbed by the mutation. Phenotypes in circadian running wheel rhythms that are not associated with changes to the core clock mechanism have also been reported in mouse models of schizophrenia. 25 Here it has been suggested that the wheel running changes observed are the result in defects in output pathways from the SCN circadian clock. Similarly, in *Katnal1* 1H/1H mice we hypothesise that the defects we demonstrate in neuronal anatomy and neuronal morphology may disrupt output signals from the SCN. Alternatively given that various neuropeptides such as oxytocin are secreted in a circadian manner from ependymal cells lining the third ventricle of the brain, 26 altered ventricular morphology and ciliary function in *Katnal1* 1H/1H mice may disrupt the circulation of factors secreted by the ciliated ventricular ependymal cells and contribute to the disruption of the behavioural rhythms observed.

The behavioural consequences of microtubule severing enzyme dysfunction in mouse models have been poorly characterised. Currently the phenotypes described are limited to motor dysfunction in mice lacking the *Spg4* gene 27 and head shaking and circling in the *Fign* mutant. 7, 28, 29 In contrast, here we demonstrate that loss of function of *Katnal1* is associated with a range of behavioural phenotypes, including changes in circadian activity, poor learning and memory, hyperactivity in a novel environment (the open field) and deficits in USVs. Notably the learning and memory, anxiety and vocalisation phenotypes reprise the clinical symptoms of ID, increased anxiety in novel situations and delays in language acquisition reported in human patients who carry microdeletions incorporating haploinsufficiency of *KATNAL1*. 8, 9 While it is also worth noting that mutant mice spend more time the centre of the open field than wild types (implying that *Katnal1* 1H/1H animals show reduced anxiety), we suggest that this result is confounded by the hyperactivity in novel environments phenotype we also describe in mutant mice. This observation is backed up by the fact that mutant animals showed increased activity in all regions of the open field rather than just the anxiolytic periphery. Here we also highlight defects in *Katnal1* 1H/1H mice such as compromised neuronal migration and morphology which may underpin such phenotypes. In *Drosophila*, the homologue of *Katnal1* (*kat-60L1*) has been demonstrated to play a critical role in neuronal morphology during development, 30 however the data that we present here is the first to demonstrate a similar phenotype in mammals and furthermore suggests how subtle perturbations to *KATNAL1* function may contribute to specific neural and behavioural conditions. For example, defects in neuronal migration, synaptic spines and neuronal morphology such as those we have demonstrated here, have been suggested to underpin ID in conditions such as lissencephaly, 18 Down's

syndrome 31 and Rett syndrome. 32 While we are not suggesting that *Katnal1* is causative for these conditions, similarities in symptoms and neuronal phenotypes between these conditions and those linked to *Katnal1* dysfunction should be appreciated. Furthermore a rare mutation in *KATNAL1* has been associated with schizophrenia 10 (<http://atgu.mgh.harvard.edu/~spurcell/genebook/genebook.cgi?user=guest&cmd=verb-gene&tbox=KATNAL1>) and *KATNAL1* has been shown to interact with the schizophrenia associated gene *DISC1*. 33 In line with these observations we note that increases in ventricular volume and reductions in synaptic spines have been reported in schizophrenic patients 34, 35 and our data demonstrates the same phenotypes in *Katnal1* 1H/1H mice. Thus the range of phenotypes associated with defects in the function of *Katnal1* strongly suggests that the gene should be considered in the pathology of disorders such as ID and schizophrenia.

We do note one key genetic difference between the human patients and *Katnal1* 1H/1H animals. While the human patients were all heterozygous for the *Katnal1* deletion, we found no phenotype in heterozygous mutant mice (data not shown) suggesting that while haploinsufficiency is causative for phenotypes in humans, mice require complete loss of *KATNAL1* function to show similar effects. A similar discrepancy between humans and mice has also been noted for the intellectual disability candidate gene *CTNNB1*. 17 While heterozygous loss of function mutations in *CTNNB1* are causative for intellectual disability in humans, conditional knock outs for *CTNNB1* have no reported behavioural or craniofacial phenotypes. 36, 37 These differences demonstrate that while mouse models of intellectual disability are of great use in our understanding of the causative mechanisms which underlie the condition, there are still genetic and neurodevelopmental differences between species which also must be taken into account. We also note that while the *Katnal1* 1H mutation shows a loss of catalytic function in both HEK293 cells and Sertoli cells, 23 this loss of function has not been verified in neuronal cells. However, given that our data demonstrates that the *Katnal1* 1H mutation lies in an essential catalytic domain and that we show neuronal phenotypes in *Katnal1* 1H/1H mice, we would expect to see the same loss of catalytic function in neurons.

The data we present here also demonstrate defects in motile cilia in *Katnal1* 1H/1H mice. Ciliary disruptions in humans (ciliopathies) include Bardet-Biedl and Joubert syndrome. 38 While there is currently limited data available regarding the behavioural phenotypes of mouse models of ciliopathies, we note that ciliary dysfunction in mice has been linked with learning and memory 39 and vocalisation phenotypes, 40 both of which were disturbed in

the *Katnal1* 1H/1H mice described here. It is also notable that the neuronal migration and enlarged ventricle phenotypes that we describe in *Katnal1* 1H/1H mice recapitulate features associated with known ciliopathy gene mutations. [41] [42] [43] [44] Furthermore in Bardet-Biedl syndrome mouse models ciliary defects such as reduced CBF 45 and structural defects such as abnormal lengthening and swellings along their length 41 have been described, that are similar to those we describe in *Katnal1* 1H/1H mice. There is strong evidence that ciliopathy associated genes play a number of roles in neuronal development by affecting processes such as progenitor proliferation or maintenance of the radial glia scaffold. 43 However it is also clear that defects in microtubule organisation also affect synaptic structure. 2 At present it is difficult to disentangle the relative contributions of defects in microtubule severing and ciliary abnormalities to the overall phenotypes we observe in *Katnal1* 1H/1H mice. Further investigations are required to clarify the impacts of these two processes. However it is notable that while defects in cilia structure may contribute to the phenotypes we describe in *Katnal1* 1H/1H mice, they are far less prominent in *Katnal1* 1H/1H mice than in other mouse ciliopathy models, 41 suggesting that the ciliary component of KATNAL1 dysfunction may be mild compared to other ciliopathies. Similarly while hydrocephalus has been suggested to be a component of some ciliopathy mouse models, 46 *Katnal1* 1H/1H mice showed only increased ventricle size rather than an increased incidence of hydrocephalus, further suggesting the ciliary defects in these animals are mild compared to other ciliopathies.

In summary the data presented here clearly demonstrate that KATNAL1 plays an important role in a variety of neuronal processes including neuronal migration, neuronal morphology and ependymal ciliary function. The downstream effect of these defects leads in turn to a number of behavioural changes including in learning and memory, reaction to anxiogenic situations and circadian rhythms. These data therefore highlight how perturbations in KATNAL1 may play a role in neuronal dysfunction and demonstrates that the enzyme is a novel candidate in the study of behavioural and neurodevelopmental disorders.

The authors declare no conflict of interest.

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Screening of FDA-Approved Drugs for Inhibitors of Japanese Encephalitis Virus Infection

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Abstract: Japanese encephalitis virus (JEV), an arthropod-borne flavivirus, is a major cause of acute viral encephalitis in humans. No approved drug is available for the specific treatment of JEV infections, and the available vaccines are not effective against all clinical JEV isolates. In the study described here, a high-throughput screening of an FDA-approved drug library for inhibitors of JEV was performed. Five hit drugs that inhibited JEV infection with a selective index of >10 were identified. The antiviral activities of these five hit drugs against other flavivirus, including Zika virus, were also validated. As three of the five hit drugs were calcium inhibitors, additional types of calcium inhibitors that confirmed that calcium is essential for JEV infection, most likely during viral replication, were utilized. Adaptive mutant analysis uncovered that replacement of Q130, located in transmembrane domain 3 of the nonstructural NS4B protein, which is relatively conserved in flaviviruses, with R or K conferred JEV resistance to manidipine, a voltage-gated Ca(2+) channel (VGCC) inhibitor, without an apparent loss of the viral growth profile. Furthermore, manidipine was indicated to protect mice against JEV-induced lethality by decreasing the viral load in the brain, while it abrogated the histopathological changes associated with JEV infection. This study provides five ant flavivirus candidates and identifies cytoplasmic calcium to be a novel antiviral target for the treatment of JEV infection. The findings reported here provide therapeutic possibilities for combating infections caused by flaviviruses. **IMPORTANCE** No approved therapy for the treatment of Japanese encephalitis virus infection is currently available. Repurposing of approved drugs would accelerate the development of a therapeutic stratagem. In this study, we screened a library of FDA-approved drugs and

identified five hit drugs, especially calcium inhibitors, exerting ant flavivirus activity that blocked viral replication. The in vivo efficacy and toxicity of manidipine were investigated with a mouse model of JEV infection, and the viral target was identified by generating an adaptive mutant.

Text: Flaviviruses are taxonomically classified in the genus *Flavivirus* and family *Flaviviridae*.

These viruses comprise over 70 different pathogens, such as Japanese encephalitis virus (JEV), Zika virus (ZIKV), dengue virus (DENV), West Nile virus (WNV), and yellow fever virus (YFV). Most flaviviruses are arthropod borne and cause public health problems worldwide (1). The development and usage of vaccines against some flaviviruses, such as JEV, YFV, and tick-borne encephalitis virus (TBEV), have decreased the rates of morbidity and mortality from infections caused by these viruses (2); however, flavivirus-induced diseases are still pandemic, and few therapies beyond intensive supportive care are currently available.

Flaviviruses have an approximately 11-kb positive-stranded RNA genome containing a single open reading frame (ORF) flanked by untranslated regions (UTRs) at both termini. The ORF encodes three structural proteins, including the capsid (C), membrane (premembrane [prM] and membrane [M]), and envelope (E), and seven nonstructural proteins (NS1, NS2A, NS2B, NS3, NS4A, NS4B, and NS5) (3). These seven nonstructural proteins participate in viral replication, virion assembly, and virus escape from immune surveillance.

To date, no specific antivirals with activity against flaviviruses are available. To address this, we conducted a screen of a library of 1,018 FDA-approved drugs. Since flaviviruses are similar in structure and pathogenesis, we first utilized JEV as the prototype to screen the drug library and subsequently validated the antiviral activities with ZIKV, WNV, and DENV type 2 (DENV-2). The hit drugs identified in this study offer potential new therapies for the treatment of flavivirus infection and disease.

Screening of an FDA-approved drug library for inhibitors of JEV infection. Recombinant viral particles (RVPs) with the luciferase-reporting replicon enveloped by the JEV structural

proteins were used to select inhibitors, with a focus on those that inhibit virus entry and replication, by a high-throughput screening (HTS) assay (4, 5) . The number of genomic RNA copies of RVP was determined to be 8.4×10^6 copies/ml by using a standard curve generated with plasmids carrying the infectious clone. The HTS assay conditions, including the seeding cell density and RVP dose, were optimized to be 10,000 cells per 96-well plate and 20 l (16 copies/cell) RVP for the infective dose, respectively. Under the optimized conditions, the signal-to-basal (S/B) ratio, coefficient of variation (CV), and Z= factor were 38,374, 2.8%, and 0.89, respectively, which demonstrated that the assay was robust and suitable for the large-scale screening of compounds.

A schematic of the HTS assay is depicted in Fig. 1B . After three rounds of screening, five hits with a selective index (SI; which is equal to the 50% cytotoxic concentration [CC 50] /50% inhibitory concentration [IC 50]) of C10 were selected. The CC 50 values of the hit drugs exhibited in Fig. 1B were similar to those previously published for diverse cell systems but determined using different toxicity assays (6) (7) (8) (9) (10) (11) (12) (13) . Three of the hit drugs, manidipine, cilnidipine, and benidipine hydrochloride, were dihydropyridine (DHP) voltage-gated Ca 2 α channel (VGCC) antagonists, while pimecrolimus is an inhibitor of inflammatory cytokine secretion and nelfinavir mesylate is an HIV-1 protease blocker. All five drugs exhibited a dose-dependent inhibition of JEV RVP infection (Fig. 1C) . To validate the antiviral effect, hit drugs were purchased from other commercial sources and tested. In the reconfirmation screen, all hit drugs showed antiviral and cytotoxic effects similar to those found in the primary screen.

Validation of hit drugs. To verify the results obtained by the luciferase reporter assays, we also investigated the antiviral effect of the five hit drugs on wild-type JEV strain AT31. As expected from the HTS assay, all five drugs robustly inhibited virus production, with a reduction of approximately 4 to 5 log units at the highest concentration and an approximately 1-log-unit decrease with 2.5 M the drugs (Fig. 2B) . A sharp decrease in JEV RNA levels was also detected (Fig. 2C) . The attenuated RNA levels in the high-dose, middle-dose, and low-dose groups were all above 40%. In particular, in the manidipine-treated group, the inhibitory effect was at least 80% compared to that for the control, which showed a strong inhibition of viral replication. Consistent with the inhibition of virus replication and production, expression of the viral structural protein prM was hardly detectable following treatment with the drugs at the high concentration (Fig. 2D) . Overall, the results in Fig. 2 confirmed that the five hit drugs inhibited JEV infection in a dose-dependent manner in vitro.

Drugs inhibit JEV infection during viral RNA synthesis. Because RVPs, which have a natural virus-like envelope on the outside and a replicon on the inside, permitted the quantification of JEV productive entry and replication, a time-of-addition experiment was performed to investigate whether the hit drugs blocked the entry step or the replication step. As shown in Fig. 3B, no suppression of luciferase activity by any of the hit drugs was observed when they were used as treatments before infection or during infection or as a virucide, suggesting that these drugs do not inhibit JEV infection either by inactivating the virus directly or by blocking JEV entry. However, these drugs exerted fully inhibitory effects when they were added at 1 h postinfection, suggesting that viral replication was the stage at which these drugs showed inhibitory activity.

To confirm this suggestion, we investigated the inhibitory effects of these drugs on the JEV replicon. The highest concentration of manidipine and nelfinavir mesylate tested in baby hamster kidney (BHK-21) cells was adjusted to 5 M and 10 M, respectively. It was shown that all five drugs inhibited JEV RNA synthesis in a dose-dependent manner, while neither drug inhibited the initial translation of replicon RNA (5, 14) (Fig. 3C), confirming that these drugs inhibited JEV infection at the stage of replication.

Hit drugs exhibit broad-spectrum ant flavivirus activity. In order to determine whether the antiviral activity of the five hit drugs extended to other flaviviruses, we explored their antiviral effect against ZIKV. Similar to the findings for JEV, the ZIKV titer was decreased by multiple log units when ZIKV was treated with a high concentration of each of the drugs (Fig. 4A). Moreover, ZIKV exhibited a higher sensitivity to the two calcium channels inhibitors manidipine and cilnidipine than JEV, with no plaque formation being observed at 10 M. Consistent with this result, sharp decreases in the level of replication of ZIKV RNA and the level of expression of viral protein were also detected (Fig. 4A). Notably, treatment with 5 M manidipine produced a 95% inhibition of viral replication, translation, and viral yields. Taken together, these results indicate that the hit drugs could effectively inhibit ZIKV infection.

Since these drugs exhibited their anti-JEV effects at the stage of viral replication, we further tested the effects against WNV and DENV-2 by using WNV and DENV-2 replicons. Similar to the results for JEV, a dose-dependent reduction in the level of WNV replication was observed with the drug treatments. The same phenotype was observed for DENV-2 for all

drugs except nelfinavir mesylate, which showed no effect at the concentrations tested (Fig. 4B and C). Together, these results indicate that the five hit drugs are excellent candidates for broad-spectrum ant Flavivirus treatment. Antiviral effect of calcium inhibitors. Since three hit drugs, manidipine, cilnidipine, and benidipine hydrochloride, were DHP VGCC inhibitors, we asked whether other calcium antagonists could block JEV infection. To address this question, we employed four different classes of inhibitors. Verapamil, a prototype phenylalkylamine (PAA) VGCC inhibitor (15) , exhibited a dose-dependent inhibition of JEV on both African Green monkey kidney (Vero) and human hepatocellular carcinoma (Huh-7) cells (Fig. 5) , which was consistent with the inhibitory effects of the DHP inhibitors, suggesting that calcium channels play an important role in JEV infection. Cyclosporine and 2-aminobiphenyl borate (2-APB), which inhibit the efflux of Ca^{2+} from the mitochondrial and endoplasmic reticulum (ER) pool, respectively (16) (17) (18) (19) , were also found to block JEV infection effectively. Similarly, treatment with the cell-permeant Ca^{2+} chelator 1,2-bis-(o-aminophenoxy)-ethane-N,N,N',N'-tetraacetic acid, tetraacetoxymethyl ester (BAPTA-AM), could also suppress JEV infection. Taken together, we concluded that intracellular Ca^{2+} is essential for JEV infection and cytoplasmic calcium is a potent target for ant Flavivirus treatment.

Selection and characterization of manidipine-resistant JEV. To identify the viral target of the calcium channel inhibitor, we selected a manidipine-resistant virus by serially passaging JEV in the presence of manidipine. Viruses from passage 20 (P20) showed robust resistance compared with the wild type (WT) (Fig. 6A). When JEV from P20 was treated with 5 M or 10 M manidipine, the viral titer was about 10-and 100-fold higher than that of the WT, respectively. Individual virus clones were isolated, and two isolates were randomly selected and amplified. An amino acid substitution was observed in two isolated clones, resulting in a glutamine (Q)-to-arginine (R) switch at amino acid position 130 in transmembrane domain 3 (TMD3) of NS4B, i.e., position 2401 of the translated polyprotein in the JEV infectious cDNA clone (Fig. 6B). Sequence alignment of NS4B indicated that Q130 was conserved in all flaviviruses except YFV, which possessed a lysine at that position (Fig. 6B) . The conserved Q130 of NS4B may account for the sensitivity of JEV, ZIKV, WNV, and DENV-2 to manidipine, as described above (Fig. 4) , while YFV showed resistance to the drug (data not shown).

To confirm that the Q130R mutation did confer manidipine resistance and to investigate the role of Q130 in NS4B function, we produced JEV clones with the Q130R, Q130K, Q130E, or Q130A mutation by introducing the desired mutations into the infectious cDNA clone and

rescuing the mutant viruses. To investigate the biological properties of the mutant viruses, we first examined the growth kinetics of the rescued viruses. As shown in Fig. 6C , all mutant viruses had an accumulation of infectious virions and reached the highest titer at 60 h postinfection. Infection of the Q130R and Q130K mutant viruses resulted in growth curves similar to the growth curve for the WT (Fig. 6C) , while the Q130E and Q130A mutants produced smaller amounts of viruses between 24 and 60 h. Analysis of the plaque morphology revealed that the plaques of the Q130R, Q130K, and Q130E mutants were similar to the plaques of the WT, whereas the plaques of the Q130A mutant were smaller than those of the WT.

We next investigated the sensitivity of the four mutant viruses to manidipine. As shown in Fig. 6D , the Q130R and Q130K mutant viruses were resistant to manidipine. At a 10 M concentration, manidipine efficiently inhibited WT JEV infection and reduced the viral yields by approximately 4 log units, while the Q130R and Q130K mutant viruses were resistant to manidipine and the viral titer decreased less than 2 log units. The Q130A mutant virus demonstrated moderate resistance and a slightly higher Taken together, it could be concluded that Q130 not only is critical for conferring manidipine sensitivity but also is important for JEV replication. The replacement of glutamine with basic amino acids conferred resistance to manidipine without an apparent loss of growth.

In vivo efficacy of manidipine. As manidipine exhibited the strongest inhibitory activities on JEV replication as well as ZIKV infection when its activities were compared with those of the five hit drugs (Fig. 2 and 4A) , we further examined the protective effect of manidipine against JEV-induced lethality in a mouse model. As anticipated, mice in the JEV-infected vehicle-treated group started to show symptoms, including limb paralysis, restriction of movement, piloerection, body stiffening, and whole-body tremor, from day 5 postinfection. Within 21 days postinfection, most mice in the JEV-infected group succumbed to the infection, with the mortality rate being 73% (4 out of 15 animals survived). Manidipine treatment following JEV infection reduced the mortality rate to 20% (12 out of 15 animals survived) (Fig. 7A). Mice treated with manidipine alone or treated with manidipine and infected with JEV showed little abnormal behavior, similar to the findings for the mice in the vehicle-treated group. These results suggest that manidipine provided effective protection against JEV-induced mortality.

To further relate these protective effects to the viral load and histopathological changes in the mouse brains, the viral titer was determined and mouse brain sections were collected and assayed at day 5 and day 21 postinfection, since mice started to show symptoms of JEV infection from day 5 postinfection and most of the surviving mice had recovered at day 21. The results indicated that, during the progression of the disease, manidipine treatment significantly reduced the viral load in infected mice compared to that in infected mice not receiving treatment, while no plaques formed in either the manidipine- or vehicle-treated group, and viral loads were undetectable in each group on day 21 postinfection (Fig. 7B). As JEV was rapidly cleared from the blood after inoculation and was present in the lymphatic system during the preclinical phase, the effects of manidipine on infection of serum and the spleen were evaluated at earlier time points to detect whether the drug reduced the peripheral viral loads (20, 21). As shown in Fig. 7C, manidipine had little effect on peripheral JEV infection, which indicated that manidipine protected the mice against JEV-induced lethality by decreasing the viral load in the brain. Similarly, apparent damage in the brain, including meningitis, perivascular cuffing, vacuolar degeneration, and glial nodules, was observed in the JEV-infected and vehicle-treated group on day 5 postinfection, while manidipine treatment remarkably alleviated these phenomena (Fig. 7D). These results indicate that the alleviation of histopathological changes was accompanied by a reduction in the viral load as well as a reduction in the rate of mortality, further confirming the curative effects of manidipine on viral encephalitis.

Among the five hit drugs, manidipine, cilnidipine, and benidipine hydrochloride were VGCC inhibitors. It has been well documented in the literature that Ca²⁺ inhibitors serve to inhibit virus infection at the stage of either entry (15, 22) or replication (18) and even at the stage of budding (23). To this end, we first reviewed all 21 calcium inhibitors included in the current library of FDA-approved drugs and found that, in addition to the four DHP VGCC inhibitors listed in Fig. 1B, two other calcium inhibitors, i.e., flunarizine dihydrochloride and lomerizine hydrochloride, were also identified to be primary candidates with levels of inhibition of >90%. Similarly, three calcium channel antagonists, nisoldipine, felodipine, and nifedipine hydrochloride, showed levels of inhibition of 75%, 72%, and 66%, respectively, in the primary screen. Together, 9 of the 21 calcium inhibitors in the library, accounting for nearly half of the calcium inhibitors, exhibited levels of flavivirus inhibition of greater than 50%, suggesting that calcium, especially the calcium channel, is a potential antiviral target. To address this, another type of VGCC inhibitor, verapamil, an FDA-approved drug not yet included in the drug library used in this study, was investigated. Likewise, a Ca²⁺ chelator, BAPTA-AM, as well as the Ca²⁺ inhibitors 2-APB and cyclosporine, targeting ER and the mitochondrial Ca²⁺ channel, respectively, were

employed to investigate the response of JEV infection to the decrease in intracellular Ca²⁺ levels. In line with the activities of the three hit DHP VGCC inhibitor drugs, the additional Ca²⁺ inhibitors exerted anti-JEV activity, which indicated that Ca²⁺ is indispensable for JEV infection. Thus, Ca²⁺ inhibitors might be utilized as effective treatments for flavivirus infection.

As the hit drugs exerted full inhibitory activity when they were added posttreatment, we believe that Ca²⁺ is important for flavivirus genome replication. Furthermore, selection and genetic analysis of drug-resistant viruses revealed that NS4B is the viral target of manidipine. NS4B is part of the viral replication complex and is supposed to anchor the viral replicase to the ER membrane (24). Meanwhile, the N-terminal 125-amino-acid domain of DENV NS4B was indicated to be responsible for inhibition of the immune response (25). Notably, several structurally distinct compounds have been identified to inhibit flavivirus replication by intensively targeting the TMD of NS4B (26) (27) (28) (29) (30) (31) (32). It is thus conceivable that inhibitors targeting TMD of NS4B would perturb its function, leading to the suppression of viral RNA replication. In this study, the replacement of Q130 of NS4B with a basic amino acid conferred the resistance effect without suppressing JEV replication, suggesting that position 130 could tolerate a basic amino acid and that the basic amino acid might be involved in the interplay of NS4B with host proteins rather than viral proteins.

Moreover, the efficacy and toxicity of manidipine were monitored *in vivo*, with manidipine demonstrating effective antiviral activity with favorable biocompatibility. However, the dose used in this study was higher than the dose typically used clinically, representing one of the scenarios most commonly encountered in drug repurposing (33, 34). As manidipine was approved for use for the long-term treatment of hypertension (35, 36), pulse-dose treatment with manidipine over the shorter period of time required for the treatment of virus infection might be relatively safe. Moreover, use of a combination of manidipine with other Ca²⁺ inhibitors might improve its therapeutic efficacy, reduce its toxicity, and reduce the risk of resistance development (37) (38) (39).

Besides the three VGCC inhibitors, two hit drugs, pimecrolimus and nelfinavir mesylate, showed equivalent inhibitory activities on the replication of JEV, ZIKV, WNV, and DENV-2. Although there has been no report on the use of pimecrolimus for the treatment of infectious diseases, we showed that it had a robust effect against JEV with an SI of C32. The

maximum plasma concentration (C max) of nelfinavir mesylate achieved with an adult dose was 3 to 4 g/ml (40) , which was comparable to the IC 50 reported here. Notably, nelfinavir mesylate was confirmed to inhibit herpes simplex virus 1 (HSV-1) and the replication of several other herpesviruses by interfering directly or indirectly with the later steps of virus formation, such as glycoprotein maturation or virion release, other than functioning in herpesviruses protease (41, 42) . Whether nelfinavir mesylate inhibits flavivirus by interference with the virus protease or by other off-target effects is unknown. Understanding of the mechanism of the anti-flavivirus effects of these drugs might uncover novel targets of the drugs, providing further insight into the pathogenesis of flaviviruses.

Above all, the findings reported here provide novel insights into the molecular mechanisms underlying flavivirus infection and offer new and promising therapeutic possibilities for combating infections caused by flaviviruses.

Cells and viruses. BHK-21, SH-SY5Y (human neuroblastoma), Vero, and Huh-7 cells were cultured in Dulbecco modified Eagle medium (HyClone, Logan, UT, USA) supplemented with 10% fetal bovine serum (Gibco, Grand Island, NY, USA). JEV strain AT31, the WNV replicon, and the DENV-2 replicon expressing Renilla luciferase (Rluc) were kindly provided by Bo Zhang, Wuhan Institute of Virology, Chinese Academy of Sciences (CAS), China. JEV replicon recombinant viral particles (RVPs) were generated as previously described (4, 5) . ZIKV strain H/PF/2013, kindly provided by the European Virus Archive Goes Global, was propagated and titrated in Vero cells.

Optimization of HTS assay conditions. The cell density and RVP dose were optimized for the HTS assay. Vero cells at different densities (2,500 to 12,500 cells per well) were infected with from 1.25 to 20 l RVPs (1 to 16 copies per well). The appropriate cell density as well as the RVP dose was selected by comparing the S/B ratio, CV, and Z= values under different conditions as previously described (43) . Methyl- β -cyclodextrin and dimethyl sulfoxide (DMSO) were used as positive and negative controls, respectively.

HTS assay of an FDA-approved compound library. A library of 1,018 FDA-approved drugs was purchased from Selleck Chemicals (Houston, TX, USA). The compounds were stored as 10 mM stock solutions in DMSO at 4°C until use. The first round of the HTS assay was carried out as shown in Fig. 1A . The criteria used to identify the primary candidates were

no apparent cytotoxicity and an average level of inhibition of C90% in duplicate wells. The criteria of dose-dependent inhibition and cell viability of C80% were applied for the reconfirmation screen. Furthermore, the CC 50 of each compound was calculated, and those compounds displaying SIs over 10 were considered hits in this study.

Identification of antiviral effects of five hit drugs. The antiviral effects of the drugs were evaluated by quantitative reverse transcription-PCR (qRT-PCR), immunofluorescence assay (IFA), and plaque assay as previously reported (44) (45) (46) (47) . The experimental timeline is depicted in Fig. 2A .

To ensure the effectiveness of the hit drugs in flavivirus replication, BHK-21 cells transfected with the JEV, WNV, or DENV-2 replicon were incubated with each drug at the concentrations indicated above, and the luciferase activities were determined 24 h, 48 h, or 72 h later, respectively.

Time-of-addition experiment. To evaluate which stage of the JEV life cycle was inhibited by each hit, a time-of-addition experiment was performed as previously described (43) . Vero cells were infected with 20 I RVPs for 1 h (0 to 1 h). The test compounds were incubated with the cells for 1 h before infection (Δ 1 to 0 h), during infection (0 to 1 h), and for 23 h postinfection (1 to 24 h) (Fig. 3A) . To exclude a possible direct inactivating effect of the drugs, RVPs were incubated with each drug at 37°C for 1 h, and the mixtures were diluted 25-fold to infect Vero cells. Twenty-four hours later, the luciferase activities were determined as described above (Fig. 3A) .

Manidipine-resistant virus. Manidipine-resistant virus was generated by passaging of JEV on Vero cells in the presence of manidipine. Passages 1 to 10 used 5 M manidipine, and passages 11 to 20 used 10 M manidipine. As a control, WT virus was passaged in the presence of 2% DMSO in parallel. Passaging was terminated at passage 20, when no further improvement in resistance was detected. Two manidipine-resistant virus isolates were plaque purified and amplified in the presence of manidipine. Viral RNA was extracted, amplified, and purified for sequencing. An infectious cDNA clone of JEV, strain AT31 (pMWJEAT), kindly provided by T. Wakita, Tokyo Metropolitan Institute for Neuroscience, was used to recover WT and mutant viruses as described previously (4) . Virus titers and manidipine sensitivities were determined by plaque assay in Vero cells.

Manidipine administration to JEV-infected mice. Adult female BALB/c mice (age, 4 weeks) were kept in the Laboratory Animal Center of Wuhan Institute of Virology, CAS (Wuhan, China). The mice were randomly divided into four groups (30 mice per group): a JEV-infected and vehicle (2% Tween 80 plus 5% DMSO in phosphate-buffered saline [PBS])-treated group, a manidipine-treated group, a JEV-infected and manidipine-treated group, and a vehicle-treated group. For infection, mice were infected intraperitoneally with 5×10^6 PFU of JEV strain AT31. For the manidipine and vehicle treatments, mice were injected intraperitoneally with 25 mg/kg of body weight manidipine or PBS with 2% Tween 80 and 5% DMSO, respectively. Treatments were administered twice a day for the first 2 days and then consecutively administered once a day for up to 21 days. Five mice from each group were sacrificed on days 1, 3, and 5 postinfection. Serum, spleen tissue, and brain tissue samples were collected for viral titer determination and histopathology investigation. Fifteen mice were monitored daily for morbidity and mortality. The mice that showed neurological signs of disease were euthanized according to the Regulations for the Administration of Affairs Concerning Experimental Animals in China. The protocols were reviewed and approved by the Laboratory Animal Care and Use Committee at the Wuhan Institute of Virology, CAS (Wuhan, China).

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Rotavirus A in wild and domestic animals from areas with environmental degradation in the Brazilian Amazon

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Abstract: Acute gastroenteritis is one of the main causes of mortality in humans and young animals. Domestic and mainly wild animals such as bats, small rodents and birds are highly diversified animals in relation to their habitats and ecological niches and are widely distributed geographically in environments of forest fragmentation in some areas of the Amazon, being considered important sources for viruses that affect humans and other animals. Due to the anthropical activities, these animals changed their natural habitat and adapted to urbanized environments, thus representing risks to human and animal health. Although the knowledge of the global diversity of enteric viruses is scarce, there are reports demonstrating the detection of rotavirus in domestic animals and animals of productive systems, such as bovines and pigs. The present study investigated the prevalence of Rotavirus A in 648 fecal samples of different animal species from the northeastern mesoregion of the state of Pará, Brazil, which is characterized as an urbanized area with forest fragments. The fecal specimens were collected from October 2014 to April 2016 and subjected to a Qualitative Real-Time Polymerase Chain Reaction (RT-qPCR), using the NSP3 gene as a target. It was observed that 27.5% (178/648) of the samples presented positive results for RVA, with 178 samples distributed in birds (23.6%), canines (21.35%), chiropterans (17.98%), bovines (14.6%), horses (8.43%), small rodents (6.74%), pigs (3.93%) and felines (3.37%), demonstrating the circulation of RVA in domestic animals and suggesting that such proximity could cause transmissions between different species and the occurrence of rearrangements in the genome of RVA as already described in the literature, associated to the traces of environmental degradation in the studied areas.

Text: Emerging and reemerging infectious diseases are increasing each year in several countries, with an impact both on human populations and on domestic and wild animals living in areas with considerable forest remnants [1] . Most of these diseases are of viral origin, suggesting the emergence and reemergence of viruses that are triggered by human activities that modify the environment [2] .

The populations of wild animals that inhabit forest fragments are strategic groups for studies of public health and the transmission of zoonosis, given that they act as indicators in the assistance and intervention in the human populations, aiming at the prevention of outbreaks and epidemics [3] .

Acute gastroenteritis can be caused by infection in the gastrointestinal tract, caused by different infectious or parasite agents [4] [5] [6] [7] . They represent one of the main causes of mortality in humans, and in young animals, counting for about 25% of mortality [8] . Rotavirus is widely distributed in animals, which act as sources of rotavirus emergent strains, with these animals acting in the transmission between species and through reassortment leading to the emergence of new strains which have been reported in human infections [9] [10] [11] [12] .

The rotavirus (RV) belongs to the Reoviridae family and comprises nine species known as Rotavirus group A to I, with a recent proposal of the J species [13, 14] . Rotavirus A (RVA) is widespread worldwide and predominantly infects humans, bovines and other mammal species, as well as birds [15] . They have a double-stranded ribonucleic acid (dsRNA) genome, divided into 11 segments coding for structural proteins (VP1-VP4, VP6 and VP7) and nonstructural (NSP1-NSP5/NSP6) proteins [16, 17] .

There are records of a close relationship between Amazonian wildlife and human populations [18] , and this interaction is the effect of anthropogenic urbanization activities that result in the deforestation of forest areas, causing the degradation of previously isolated sites such as caves and small caves, a continuous and nature progressive process that has led not only to changes in wildlife habitats but also to a greater relationship with human populations in rural and urban environments, contributing to the occurrence and emergence of diseases different from what normally occurs in endemic regions [19] [20] [21] [22] .

Although the results of RVA have already been described globally [12, [23] [24] [25] [26] [27] [28] [29] [30] , in Brazil, the occurrence, diversity and role of rotavirus in these animals are still poorly studied, considering the large number of present species [4, [31] [32] [33] [34] .

In the Brazilian Amazon, especially in the state of Pará, the city of Belém and Northeast metropolitan mesoregions are some of the areas with the highest indexes of environmental changes [35], which are concentrated, along with the fact that the knowledge of the global diversity of enteric virus in animals is scarce [36].

Therefore, it is important to monitor the health of domestic and wild animals in their natural habitat, especially in areas with anthropic alterations that have an interface with rural communities and enterprises, in order to investigate the occurrence of RVA in this population. These communities are ecologically complex, because they have multiple hosts and endless pathogens that may eventually circulate in contiguous urban centers, in addition to the fact that it should also be considered that there is still a lack of studies showing the significance of these viruses infecting this population, as in the context of epidemiological surveillance, these animals become important, since they can be considered as natural sources, with the possibility of transmission to humans [37] [38] [39].

The qualitative real-time polymerase chain reaction (qRT-PCR) used the NSP3 gene and the TaqMan probe from a highly conserved region of the rotavirus non-structural protein 3 (NSP3), which was previously used in samples from human origin and with low viral loads

Precipitation data were obtained from The Brazilian National Institute of Meteorology (Inmet <http://www.inmet.gov.br/>) for the years of capture in the Expedito Ribeiro Settlement (2014) and Açailândia (2015) of the Data Collection Platforms (PCDs) of Belém, located 50 km from Santa Bárbara do Pará, and Tracuateua, located 50 km from Peixe-Boi and 100 km from Viseu. Garmin GPSMap 64s Global Positioning System (GPS) coordinates were collected in the field. The municipal boundaries were obtained on the website of the Brazilian Institute of Geography and Statistics (IBGE) (<http://www.ibge.gov.br/>) and data on deforestation and land use were obtained from the PRODES [43] and TerraClass [44] Projects. PRODES has annual data in digital format since 2000 and TerraClass presents biannual data since 2004.

The satellite image was generated using the sensor Sentinel 2 of the European Space Agency (ESA) (<https://sentinel.esa.int/web/sentinel/user-guides/sentinel-2-msi>) with Open Access CC-BY License (<http://open.esa.int/>) from the years of 2017 and 2018.

All the data obtained was stored in a Geographic Database (BDG). The BDG was imported/ stored in a GIS for the editing of the graphic elements, establishment of topological relations between the elements and their respective attributes, spatial analysis and visualization of the result through thematic maps.

For the present study, forest fragments of similar size, shape and Phyto physiology were chosen, considering an open peri urban matrix with similar soil use. The selected fragments were distributed within the mesoregions studied, and in each selected fragment fecal samples were randomly collected from domestic and wild animals [45] .

Soil use classes were obtained from the TerraClass data mosaic from 2004 to 2016, because the study sites were in an area with a high cloud presence, which prevented observation (the area was not observed).

The data processing, interpretation, visualization and spatial analysis were performed in ArcGIS software (<http://www.arcgis.com/>). For the analysis of data related to the determination of the richness, composition and abundance of the fauna of the animals studied in the study area, considering the collection methods adopted and the species available in each city, each sample was considered as an independent sample.

The richness of wild fauna and domestic animals was determined by the total number of species including all collection methods, and the similarity of species was made by the chi-square analysis between the samples of the different treatments with the aid of the EstimateS 8.0 software [46] .

For the calculation of the Test T, the Statistica software was used, and the indices of infected animals in the two environments (forest fragment and peridomicile) were calculated for each treatment sampled by collection area, using the software Past 1.92. Aiming at comparing the values of the diversity indexes through the paired test, as well as the descriptive analysis of the anthropic effects [47] .

The data obtained for the occurrence of RVA and the questionnaires was inserted into a database for a descriptive analysis of the epidemiological profile of the animal population in the three forest ecosystems studied. In this analysis, descriptive statistical treatments were carried out, using customized "row-columns" type charts, referring to the data, in order to characterize the sample and quantify the results using absolute frequency values using the chi-square test and the Test T.

Population study, collection of clinical specimens and laboratory methodology. The flying animals (wild birds and chiroptera) were captured using mist nets which were opened at dawn (4:00 a.m.) and closed in the morning (9:00 a.m.) and were inspected every one hour until the closing, with a sampling effort of 15 days. This research was approved by National All procedures with animals were performed by veterinarians, being birds and bats identified and released at the same capture site. The fecal specimens were collected by stimulation of the rectal ampulla with the use of a "Zaragatoa", packed in cryogenic vials, identified, stored in liquid nitrogen, and later sent to the Laboratory.

Wild animals (small non-flying mammals) were trapped within live-containment traps of the Tomahawk cage (size 45x16x16cm) and Sherman type aluminum (size 30x9x8cm). In each sample plot, 61 traps were distributed, 20 Shermans and 41 Tomahawks being baited with a mixture made with peanut paste, sardines, cod liver oil and corn meal, as well as fruit like banana, apple and pineapple. All the traps used were inspected daily in the morning, the baits being exchanged when necessary and later after the capture in bags of cloth and at least five specimens of each species were chosen for the collection of biological material. The wild animals were sedated with a combination of ketamine 20mg/kg and xylazine 2mg/kg intramuscularly and subsequently, euthanized with anesthetic overdose of 2% lidocaine in the foramen magnum, according to the recommendation of the National Council for the Control of Animal Experimentation (CONCEA).

From October 2014 to April 2016, 1,282 fecal samples were collected from wild and domestic animals. Among these, 648 (50.5%) samples were randomly selected for RVA research and handled in Level Three Biosafety Laboratory (NB3).

The viral genome was extracted using the TRIZOL LS REAGENT protocol (INVITRO-GEN, USA/KIT QIAGEN), following the manufacturer's recommendations, with minor adaptation according to the protocol described in the supplemental data.

The qRT-PCR was conducted according to Zeng et al. [40] for the detection of RVA using the NSP3 segment of RVA as the target gene sequence. The assay was conducted in a mixture containing: RNase-free H₂O, TaqMan RT-PCR Mix (2x), TaqMan RT Enzyme Mix (40x), primers for the NSP3 gene, Primer NSP3 Forward (20mM), Primer NSP3 Reverse (20mM), probe NSP3 S (10nm), Template (RNA) 3μL, having a total reaction volume of 17μL and reverse transcription cycling of 50 °C, 30 minutes, denaturation of 95 °C, 10 minutes, annealing of 45 cycles of 95 °C, 15 seconds and extension of 60 °C, 1 minute.

The analyzes were considered positive when presenting the cycle threshold (CT) 40. In order to guarantee a reliable test result, the measurements of contamination control were performed with the use of positive animal control (SA11 prototype) and a negative control (ultrapure water).

All RVA-positive samples were subjected to reverse transcription-polymerase chain reaction (RT-PCR) according to Mijatovic et al [41] to genotyping low viral loads samples. First round was performed with consensus primers N-VP4F1/N-VP4R1 and the Nested-PCR was conducted with N-VP4F2/N-VP4R2 primers to amplify VP4 gene. Amplicons were purified and sequencing for VP4 gene using the same primers of Nested-PCR. The sequences were collected from an automated ABI Prism 3130xl DNA sequencer (Applied Biosystems). The sequence fragments were assembled and edited using the Geneious Bioinformatics software platform v.8.1.7. Posteriorly, the data were compared with other sequences from the National Center for Biotechnology Information GenBank database using BLAST alignment tool to elucidate the RVA genotype of the samples.

From October 2014 to April 2016, a total of 648 fecal samples of wild and domestic animals belonging to three forest fragments areas were tested for the NSP3 gene by qualitative qPCR, and 178 (27.5%) were positive for RVA, distributed among the species: birds (23.6%), canines (21.35%), bats (17.98%), cattle (14.6%), horses (8.43%), small rodents (6.74%), swine (3.93%) and felines (3.37%). The CT interval ranged from 28. 47 It

was possible to detect viral strains in all genders of animals studied and in the harvesting period none of the animals showed signs of acute infection and / or diarrhea.

Rotavirus A (RVA) detected in the present study of wild and domestic animals belonging to the three areas of forest fragment, according to Fig 2. In relation to the evaluated bovines, only in the city of Viseu, these species were studied because they were created extensively. In addition, most of the animals were young with ages varying from 1 day to 8 yearsold, history of deficient vaccination, lack of technical assistance and raised in the form of subsistence. The animals showed no symptoms of diarrhea, only low weight performance and poor sanitary management status. In relation to chiroptera, 32 (17.98%) positive samples for RVA were distributed among *Carollia perspicillata* species, with 12 (37.5%) being all adults, 9 (28.12%) *Desmodus rotundus* samples (4 young and 5 adults), 5 (15.6%) of *Uroderma bilobata* (15.62%), 3 (9.37%) of *Artibeus lituratus* and the species *Artibeus Planirostus*, *Diaemus iyoug* and *Glossophagine* with 1 (3.12%) each. These animals came from areas of forest fragments located near bovine and equine farms, in addition to inhabiting small chicken farms. Fig 3 shows the results obtained for all the species of animals investigated in the forest fragment as well as in the peridomicillus area.

The anthropic variables were analyzed for the three cities studied, as well as the use of the soil within the range of the animals, obeying the domicile, the peridomicile and the forest fragment where the traps of small rodents, birds and various species of animals were captured (Fig 4 and Fig 5) .

Considering the factors related to the anthropic activities in the three studied areas within the three cities of the present study, it was observed that the city of Santa Bárbara is the one that has a better area of preserved forest and the city of Viseu a smaller area. However, in the city of Santa Bárbara, a greater concentration of occupations was observed around the area of forest fragment. It was observed in this chosen area of the city, the presence of different families living in a rural settlement, surviving from the exploitation of forest resources and the creation of small animals for subsistence, such as poultry and fish farming, as well as family farming products.

The breeding of animals in native pastures was only observed in the cities of Peixe Boi and Viseu. Extensive livestock farming was practiced with beef cattle, equines for work and

small animals (swine and goats). In relation to the most preserved pasture area, the city of Peixe Boi had the largest area, according to the data shown in Fig 5, however, in the city of Viseu, a higher regeneration was observed in the pastures during the period of the study, with significant secondary vegetation.

When comparing the climates of the three areas it was observed that the predominant climate is megathermal and humid with average annual temperature around 27 °C. The months of October, November and December are the hottest, with temperatures between 32 °C and 34 °C and absolute maximums around 41 °C. Annual rainfall is quite high, generally around 2,350 mm, but strongly concentrated from January to June (80%). From September to December, on the contrary, rainfall is rare, about 7%, with a short dry season, of moderate water deficit in those months. The relative humidity of the average air oscillates around 85%, as shown in Fig 6 [48] .

The description of the accumulated precipitation in the year of capture of the fecal specimens compared to the Climatological Normals (CLINO) for the period from 1961-1990 of the PCDs closest to the locations of the Expedito Ribeiro / Santa Bárbara settlement (Belém PCD), Vila Ananim / Peixe-Boi and Açateua / Viseu (Tracauateua PCD) show the frequency of rainfall in the regions, which facilitates the renewal of the pastures and the regeneration of the impacted forests, being an important indicator of the reduction of the damages caused by deforestation in the region.

The average deforestation index in the three study areas was calculated from data obtained from INPE information systems. It was observed that in the years of 2013 to 2014 there were no changes in these regions; in the period from 2014 to 2015 about 4.1% of the city of Viseu was changed and 1.6% of the city of Peixe Boi. In relation to the period of 2016, great changes were observed in Peixe Boi (79%) and in the city of Viseu (70%), thus demonstrating that changes in the natural ecosystem may be associated with the frequencies for RVA in the studied areas, according to Fig 7. When assessing the infected animals in relation to the uninfected animals in both the forest fragment and the peridomicile, considering as animals of the forest fragment the birds, the chiroptera and the small rodents and as animals of the peridomicile the canines, bovines, pigs, felines and horses, a percentage of 37.07% infected peri domestic animals (86/232) and 22.12% infected forest fragment animals (92/416) were obtained. Applying the selected statistical analysis, a Pearson χ^2 Chi-square value was obtained: 16.7159, df = 1 and $p < 0.001$,

meaning that the hypothesis was corroborated, that is, the greater the degradation of the environment, the more likely it will be the search for food by wild animals in adjacent areas, or in the edge of the forest or even in the peri domiciliary region. In this sense, the possibility of contagion with other species of animals, even humans, should be considered because of the capacity of the rotavirus to be transmitted via the fecal / oral route or through direct contact with the environment. It is important to point out that the animals detected in this study are important sources of viral strains.

A total of 80 stool samples were selected, reextracted and analyzed using PCR for the VP4 gene. Eight strains (10%) were positive for VP4 gene, being 2 strains bellowed to P [6] genotype and 6 to P[4]-type, according to

In the present study, RVA was detected circulating in 27.5% of the animals; 36% in domestic animals and 64% in wild animals, providing a unique dataset with qRT-PCR detecting a low viral load of RVA in different species, which further correlates with the deforestation index. These data are important because there is a lack of tests for RVA diagnosis in animals, since the current methods of RVA detection does not always detect in these populations [8] . With the advent of real-time PCR (qPCR), there was an exponential growth, compared to conventional PCR essays, since its superior accuracy, sensitivity and specificity is remarkable, and it is Rotavirus A in wild and domestic animals possible to detect RVA in a variety of animal species using NSP3 gene [49] . The sensitivity of RT-qPCR significantly improved the rate of RVA detection in clinical samples from animals and in this context, the present study proposed an interesting study metrics using virus spreading in the wild animals which inhabit forest fragments to indicate human population interventions, with the goal of preventing the virus outbreaks leveraged on the unique geographic characteristics of Brazil and its large number of species in Amazon.

Currently, no data have been described in the literature regarding the RVA detection using real-time qPCR technique in a wide variety of wild animal species. However, a study by Soltan et al. [50] conducted with horses and cattle detected RVA by RT-PCR, commercial RT-PCR and RT-qPCR in 36.7%, 51.4% and 56.9% respectively, differently from the present study that showed higher positivity for chiropterans (17.98%), canines (21.35%), birds (23.6%) and cattle (14.6%).

The first description of RVA in chiroptera was recorded in feces of *Eidolon helvum* caught in Vihiga, Kenya [51] . Afterwards, several strains of RVA were detected by different molecular techniques involving chiroptera, in several countries, including Kenya (*E. helvum*), China (*Rhinolophus hipposideros* and *Aselliscus stoliczkanus*), France (*Myotis mystacinus*), Cameroon (*E. helvum*) and Brazil [31, [51] [52] [53] [54] [55] . The present study shows the occurrence of RVA in 17.98% of the chiroptera, being among the species *Carollia perspicillata* (37.5%), *Desmodus rotundus* (28.12%), *Uroderma bilobata* (15.6%), *Artibeus lituratus* (9.37%), *Artibeus Planirostus* (3.12%), *Diaemus iyoug* and *Glossophagine* (3.12%).

Barquez et al. [56] reported that *Desmodus rotundus* is one of the three hematophagous species of the Phyllostomidae family, found throughout South America, Central America and Mexico. Of the positive chiroptera for RVA in the present study, a prevalence of 28.12% was of *Desmodus rotundus*. This species feeds on birds, can feed on mammals, mostly medium or large, facilitating the dissemination of viral spores among the community within the habitat, as observed in the present study. These findings show the importance of epidemiological data on the studied species due to the lack of studies involving species of neotropical chiroptera, and it is not possible to establish comparative parameters for these animals.

Regarding the circulation of RVA in canines and birds, the prevalence was 53% and 29%, respectively. Although in the Amazon region there are records of RVA, RVD, RVF and RVG that infect birds [57] [58] and RVD in migratory birds [59] , all were detected by RT-PCR assays differently from the present study which detected the RVA by RT-qPCR involving a variety of animal species.

On the other hand, the prevalence in felines (16%) and pigs (22%) was lower, probably because there are few animals of these species in the region, as well as few creations.

The study detected the presence of RVA in different species of animals both in areas near the home and in areas located in fragments of forest, characterized as forest remnants, since they were located in cities that suffered high environmental impacts due to vegetal extractivism, pasture formation for cattle breeding, exploitation of natural resources, and direct reflexes on the habitats of wild animals that can serve as virus sources, thus facilitating the dispersion of RVA among communities of coexisting animals.

It is worth emphasizing that these animals have a greater contact with the human populations of the studied areas since they cohabit with the humans in the region, besides having a high flow of movement between the forest extracts and environments chosen for the present study. However, it is noteworthy that only in the communities of Santa Bárbara and Viseu were collected fecal specimens from asymptomatic humans for diarrhea and tested for RVA, but all were negative. It is notorious yet, the existence of different levels of degradation in the studied environments, considering the presented data. The fragmentation of the forest generates many consequences on the Amazonian biota, being able to alter the diversity and the composition of the animal communities in the fragments and even to interfere in the ecological processes, without considering that the fragments of forest in the Amazon are influenced by the climate, possibly facilitating the dispersion of pathogens by the environment, since the wild animals detected in the present study are asymptomatic and have low viral load for RVA.

The occurrence of RVA in this population of animals may explain the possibility of dispersion of viral strains, since there is a proximity to the human population, besides the biological characteristics of these species that may represent important sources for gastroenteric viruses, along with the fact that all animals were asymptomatic for diarrhea.

Wild birds have unlimited flight capacity, were captured in an interface region between the peridomicile and forest fragments and it is believed that this region has not been influenced by anthropic activities such as those observed in the area of the present study. On the other hand, the breeding method for poultry and canines close to homes and the forest ecosystem, as they are created in the communities surveyed, probably facilitates direct contact with possible sources of contamination, since in the areas the use of septic tanks is deficient and sometimes non-existent, which may facilitate or even increase the risk of viral dispersion throughout the environment.

The high rates of increase and the analysis of land use in the researched areas may be important indicators of how these animals interact, since with deforestation, the populations of wild animals seek refuge in nearby communities facilitating the dispersal of infectious agents and the possible occurrence of carrier animals by direct contact or contamination of the local environment.

To our knowledge, this is the first study in which a real-time PCR assay was applied for the detection of RVA involving a wide variety of domestic and wild animals, facilitating practical utility in epidemiological and molecular studies and assisting in a perspective in the elaboration of sanitary control and monitoring, preventing possible outbreaks in the studied communities. The detection of positive animals was useful to monitor the infection of the agent in the animal population and to provide an early warning signal to predict an impending epidemic and a favorable risk for the human population, given the evidence of RVA circulation in the different forest fragments.

In addition, the RT-qPCR assay may be a useful alternative for the differential diagnosis of RV in possible coexisting mixed infections clinically indistinguishable such as those caused by other viral strains that cause gastroenteritis such as: astrovirus, coronavirus, picobirnavirus, calicivirus, among others as observed in the studies of Jing et al. [60] and Waruhiu et al. [61] .

Diarrhea associated with RVA infections in pigs is an important cause of increased mortality and economic losses in Europe. The most prevalent genotypes isolated from feces of Belgian diarrheal and non-diarrheal piglets in 2012 [62] demonstrate a wide range of combinations of genotypes G / P including; G3P [6] , G4P [6] , G5P [6] , G4P [7] , G5P [7] , G9P [7] , G9P [13] and G9P [23] . On the other hand, in the present study it was possible to detect only P [6] genotype, since majority of samples was asymptomatic for diarrhea.

Finding shows that different P genotypes of RVA strains interact with distinct blood group histological antigens (HBGA, ABOH, Lewis) and sialic acids via VP4 providing insight into the regional prevalence and increased zoonotic potential of some RVA of origin swine [63] . The genotype P [6] was identified in piglets in Brazil [64] and in Italy and Japan resembling genotype P [6] human [65, 66] .

In the population of animals studied the zoonotic transmission can be frequent, since the animals live in contact with humans and in precarious sanitary conditions. In Brazil, this genotype was described in animal and human populations in studies of Luchs et al. [32] ; Honma et al. [67] ; Araújo et al. [68] ; Mascarenhas et al. [69] and Lorenzetti et al. [70] such studies corroborate the importance of continuing to monitor genotypes to verify if

uncommon strains or new strains are emerging and can infect animal populations or inter-species transmissions.

Regarding the genotype P [4], it was most detected in our samples in bats, dogs, swine and feline. This genotype is not common in animals, being more detected in human and environmental samples in various parts of the world and included our region [71]. It is important to emphasize that the indicators of environmental contamination in Brazil are significant and contribute to the possibility of human-animal transmission [71]. Such data need further investigation in later work to better characterize the interspecies transmission, since the occurrence of enteric viruses in different matrices demonstrates the anthropogenic impact of the exposed population around and points to the potential risk of infection by the possible exposure of individuals susceptible. Our findings may be useful for tracking fecal contamination in the environment using animals as possible sources thus minimizing the risk of infection by exposure to susceptible individuals, in this case different animal species or even human populations.

RVA were detected in wild and domestic animals using a RT-qPCR assay that analyzed samples that had low viral load for RVA. Although the samples are asymptomatic for diarrhea, it is necessary to conduct strategies for the monitoring and control of the animals in the areas studied in the human population as well as in other species of animals, as well as the implementation of preventive measures aimed at future outbreaks in communities animals in the resident population in these impacted areas. Therefore, the present study is unprecedented in the region and in the country in relation to the research of RVA in wild animals. It is noteworthy that, although the quality of the analyzed samples is characterized as low detectable viral load, the technique presented a good analytical response in the detection of the source animals for RVA, facilitating the selection of the samples for future genetic characterization tests.

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Regulation and Maintenance of an Adoptive T-Cell Dependent Memory B Cell Pool

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Abstract: We investigated the ability of monoclonal B cells to restore primary and secondary T-cell dependent antibody responses in adoptive immune-deficient hosts. Priming induced B cell activation and expansion, AID expression, antibody production and the generation of IgM(+)IgG(-) and IgM(-)IgG(+) antigen-experienced B-cell subsets that persisted in the lymphopenic environment by cell division. Upon secondary transfer and recall the IgM(-)IgG(+) cells responded by the production of antigen-specific IgG while the IgM(+) memory cells secreted mainly IgM and little IgG, but generated new B cells expressing germinal center markers. The recall responses were more efficient if the antigenic boost was delayed suggesting that a period of adaptation is necessary before the transferred cells are able to respond. Overall these findings indicate that reconstitution of a functional and complete memory pool requires transfer of all different antigen-experienced B cell subsets. We also found that the size of the memory B cell pool did not rely on the number of the responding naïve B cells, suggesting autonomous homeostatic controls for naïve and memory B cells. By reconstituting a stable memory B cell pool in immune-deficient hosts using a monoclonal high-affinity B cell population we demonstrate the potential value of B cell adoptive immunotherapy.

Text: Immune responses to infectious agents have different out-comes that can either protect or fail to control disease. Protection from re-infection relies on the establishment of efficient secondary immune responses that require the generation of antigen-specific "memory" B and T lymphocytes. The generation and selection of T-cell dependent "memory" B cells involves distinct molecular mechanisms: immunoglobulin isotype recombination and somatic hyper mutation, both dependent on the expression of AID [1] .

Therefore, a long-standing paradigm defined memory B cells as IgM⁻IgG⁺ isotype switched cells [2] . Different lines of evidence indicate that this is not always the case. In humans, it has been shown that some IgM⁺ B cells bear the phenotype of other memory cells, being CD27⁺ , and carry frequent point mutations in the V region of the Ig genes, suggesting that they must represent highly selected B cell populations [3] . In mice, populations of CD19⁺ IgM⁺ able to mount secondary responses have been identified [4] [5] [6] [7] . Overall these findings suggest that the T-cell dependent memory B cell pool comprises distinct subsets of memory B cells with different properties and effector functions [4] [5] [6] .

The biological properties that ensure the long-term persistence of memory and efficient secondary antibody responses have not been yet completely established. While initial studies proposed that after transfer memory B cells faded rapidly [8, 9] suggesting that long-lasting memory required the continuous recruitment of new cells [8] and/or antigen persistence [9, 10] , others suggested that memory B cells were able of extended survival without cell division [11] in the absence of antigen [2] . Long-term persistence of antibody responses has also been attributed to populations of long-lived plasma cells mainly resident in the bone marrow following immunization [12, 13] . The demonstration of the compartmentalization of "antibody memory" into different cellular layers suggested that the separate subsets of memory B cells behave differently. Accordingly, it has been reported that IgG⁺ cells that could rapidly respond upon challenge did not persist long, while IgM⁺ cells could generate a second wave of germinal center responses allowing persistence of memory [4] [5] [6] [14] .

Currently, immunotherapy approaches using passive antibody transfer [15, 16]) is limited by the short half-life of immunoglobulin. Therefore new therapy strategies may require the adoptive transfer of high-affinity memory B cells, ready to respond and able to persist. The development of these new strategies requires a profound understanding of the mechanisms that regulate memory B cell numbers and ensure long persistence upon adoptive transfer. Moreover, knowledge of the mechanisms that determine the size of the memory B cell pool may be also critical to devise new reconstitution strategies. So far, studies comparing populations of naïve and memory B cells have been hindered both by the vast clonal heterogeneity of the cells involved and by our inability to generate significant numbers of antigen specific memory B cells. Indeed in a normal laboratory mouse the population of B cells bearing a "memory IgG⁺ phenotype" represent a small fraction of the total B cell pool (<0.5%) and upon immunization the number of the clonal diverse antigen-specific memory B cells generated is generally very limited (<10³) [1, 6] .

To circumvent these limits, we decided to compare the properties of homogeneous populations of naïve and memory B cells of known antigen specificity, belonging to the same clone. We used SW HEL transgenic mice where B cells bear a high-affinity BCR specific for HEL and are capable of class switch recombination and somatic hypermutation (SHM) [17, 18] . To identify "memory B cells" the SW HEL mice were crossed with mice where AID transcription provokes the permanent expression of an YFP reporter in post-germinal center lymphocytes [19] . These mice were in a Rag2-deficient background and therefore contain a pure population of monoclonal HEL-specific B cells. To generate memory cells, purified naïve B cells from the SW HEL .AID/YFP.Rag2 -/mice were transferred into adoptive hosts together with monoclonal OVA-specific CD4 + T cells from OTII.Rag2 -/-TCR transgenic mice. Upon immunization with OVA-HEL complexes, we obtained a significant number of persisting HEL-specific IgM + Ig-G -YFP + and IgM -IgG + YFP + memory B cells, number that did not correlate to the number of precursor naïve cells initially injected suggesting that the memory B cell pool is regulated independently. We characterized the functional capacity of these two memory cell types in immune deficient hosts.

Mice B6 and B6.Rag2-/- [20] mice were kept at the Centre Des Techniques Avancées (CDTA), Centre National de la Recherche Scientifique (CNRS), Orleans, France; SWHEL.AID/YFP.Rag-/-mice, obtained by crossing SWHEL (18)(a gift of Dr. Robert Brink) and AID/YFP [19] (a gift of Dr. Rafael Casellas) with B6.Rag2-/-mice. OTII.Rag-/-mice were kept in our animal facilities at the Pasteur Institute. Experiments were performed according to Pasteur Institute Safety Committee in accordance with French and European guidelines and the ethics Committee of Paris 1 (permits 2010-0002, -0003 and -0004). Euthanasia of the mice was performed by cervical dislocation. This specific study was approved by the European Research Council (ERC) committee related to the grant AdG09 249740-QSIS. The general status of the mice was controlled daily by monitoring the appearance of obvious pain, distress or suffering (prostration, respiratory issues, loss of weight). The end-point of the experiment was determined by a loss of more than 20% of the weight or as soon as the distress signs appeared. In this case, experiment was stopped and the animals were euthanized.

Single-cell suspensions of B cells from spleens and lymph nodes of SW HEL .AID/YFP.Rag -/mice together with CD4 + T cells from spleens and lymph nodes of OTII.Rag -/mice were

transferred intravenously into the retro-orbital sinus of B6.Ly5 a IgH a or B6.Rag2 -/recipient mice. Mice received 10^6 HEL + B cells and 10^6 CD4 + T cells unless stated otherwise. Mice were immunized 24H later with 1 mg of Ovalbumin coupled to Hen Egg Lysozyme (OVA-HEL) in 50µg of Alu-S-Gel (Serva) we determined as the optimal dose of Ag (data not shown).

Naive cells from SW HEL .AID/YFP.Rag -/mice and memory B cells subsets from immunized B6.Rag -/hosts mice were purified from spleens and lymph nodes by flow cytometry sorting. Single-cell suspensions containing 5×10^4 B cells and 10^6 T cells were transferred intravenously into B6.Rag2 -/recipient hosts. The purity of sorted cells was above 98%. 24 h after transfer, mice were immunized with 1 mg of OVA-HEL.

Spleen, bone marrow, inguinal and mesenteric lymph nodes single-cell suspensions were stained for cell surface or intracellular proteins with appropriate combinations of the following monoclonal antibodies conjugated to pacific blue, Qdot-655, Brilliant Violet 605, allophycocyanin, peridinin chlorophyll protein-cyanine 5.5, phycoerythrin, phycoerythrin-cyanine7: anti-CD19 (6D5), anti-IgM (R6-60.2), anti-IgG1 (X56), anti-CD138 (281-2), anti-GL7 (GL7), anti-CD95 (Jo2), anti-CD62L (MEL-14), anti-CD69 (H1-2F3), anti-BAFFR (7H22-E16), anti-CXCR5 (L138D7), anti-IA b (AF6-120.1), anti-CD80 (16-10A1), anti-CD73 (TY-11-8) and anti-PDL2 (TY25) and anti-Ki-67 (mm1) purchased from Becton Dickinson Pharmingen, Biolegend, Invitrogen and eBioscience. Cells were also stained with HEL (Sigma) coupled with AF594 using Alexa Fluor1 594 Protein Labeling Kit from Life technologies. Before staining, cells were treated with Fc-Block (CD16/CD32, Becton Dickinson Pharmingen). Dead cells were excluded during analysis according to their light-scattering characteristics. For intracellular stainings, cells were first stained with antibodies specific for cell surface antigens. Then, cells were fixed and permeabilized according the manufacturer's recommendations (BD Biosciences). For proliferation assay, mice were injected i.p. with 50 mg/kg of BrdU (Sigma-Aldrich) and were killed 24 or 72 hours later. Incorporated BrdU was detected intracellularly using anti-BrdU APC-conjugated antibodies according to the manufacturer's recommendations (BD Biosciences). All data acquisitions and analyses were performed with LSRFortessa (Becton Dickinson) interfaced with BD FACSDiva (Becton Dickinson) and FlowJo (Tree Star) software. Subsets of memory B cells were sorted as CD19 + HEL + YFP + IgM + or IgG + and naive cells as CD19 + HEL + YFP -IgM + using a FACSriaIII flow cytometer. The purity of the sorted populations varied from 90-95%.

Sera HEL-specific Ig concentrations were quantified by ELISA. Plates were coated with HEL and saturated with PBS-5% Milk. Dilutions of sera were added. After incubation (2 hours, 37 ° C) and washing, HRP-labeled anti-mouse IgM or IgG antibodies were added. After incubation and washing, bound antibodies were revealed with the substrate O-phenylenediamine and H₂O₂. The reaction was stopped after 10 min. by addition of 10% SDS and the absorbance read at 492nm in a multiscan spectrometer. Ig concentrations were determined by comparing the displacement of the dilution curves in the linear interval between standards at a concentration of 1 mg/ml and the serum samples.

The quantification of IgG or IgM secreting cells was assayed by ELISpot technique. Briefly, plates were coated with HEL. After saturating, the cells were distributed into the micro wells in RPMI1640-2%FCS. The plates were incubated for 12 h at 37 ° C, 5% CO₂ atmosphere. After extensive wash, plates were incubated with goat anti-mouse IgM or anti-IgG labeled with alkaline phosphatase. After washing, the revealing substrate was added (2,3 mM 5-bromo-4-chloro-3-indolyl phosphate diluted in 2-amino-2-methyl-1-proprenolol buffer).

Spleens from 14 day-immunized mice were initially fixed with paraformaldehyde and embedded in 4% low-gelling-temperature agarose (type VII-A; Sigma-Aldrich) prepared in PBS. 150µm slices were cut with a vibratome (VT 1000S; Leica) in a bath of ice-cold PBS. For immunolabeling, samples were saturated with PBS supplemented with 10% of fetal calf serum, then were labeled with primary antibodies anti-B220-APC (clone RA3-6B2) and anti-IgD-PE (clone 11-26c.2a) and analyzed with a spinning disk confocal microscope equipped with a CoolSnap HQ2 camera (Photometrics) and a 20x objective. Images were acquired and analyzed with MetaMorph 7 imaging software (Molecular Devices).

Sample means were compared using the Student's t test. Sample means were considered significantly different at $p < 0.05$.

During an immune response the complexity of determinants expressed by immunizing antigen and the degeneracy of antigen-specific recognition results in a vast heterogeneity of responding cells rendering impossible the direct comparison of the properties of naïve and memory B cells belonging to the same clone. We have devised an experimental system that permits the comparison between naïve and memory B cells expressing the same

antigen receptor and allows marking permanently memory B cells. For that purpose we used SW HEL transgenic mice in a Rag2-deficient background holding a single population of monoclonal B cells, all bearing a high-affinity BCR specific for HEL and capable of class switch recombination and somatic hypermutation (SHM) [17, 18] . To identify antigen-experienced B cells the SW HEL . Rag2 ^{-/-} mice were crossed with mice where AID transcription induces the permanent expression of an YFP reporter in post-germinal center lymphocytes [19] . Since in intact Tg mice immune responses were not traceable, probably because of the presence of low level pre-existing anti-HEL antibodies that neutralize the immunizing protein, we used an adoptive cell transfer strategy to study the ability of the high affinity monoclonal B cell to reconstitute response in immune-deficient hosts and generate antibody memory. Purified naïve B cells from the SW HEL .AID/YFP.Rag2 ^{-/-} mice were transferred into Rag2-deficient mice together with monoclonal OVA-specific CD4 + T helper cells from OTII.Rag2 ^{-/-} TCR transgenic mice. The day after, host mice were immunized with OVA-HEL complexes (Fig 1A) . In these conditions, antigenic challenge resulted in B cell activation and the development of significant numbers of CD19 + HEL + AID/YFP + B cells, which were not detected in non-immunized mice or in mice immunized in absence of helper T cells (Fig 1B) . We followed the early kinetics of this response. The number of HEL-specific B cells increased from the initial 2x10⁶ transferred to about 15x10⁶ at day 14 (Fig 1C left) the B cells expressing AID/YFP being the dominant population (Fig 1C right) . A fraction of the HEL-specific B cells underwent class switch recombination and at day 14 we recovered both IgM + IgG -AID/YFP + and IgM -IgG + AID/YFP + cell populations (Fig 1B) . B cell expansion and phenotypic changes were accompanied by the production of IgM and IgG HEL-specific antibodies (Fig 1D) . Two weeks after antigenic challenge we observed the formation of germinal centers in the spleen of the host mice (Fig 1E) . Coherently we found that while upon adoptive transfer all B cells expressed CD95, only after antigenic challenge most YFP + B cells expressed the germinal center specific marker GL7 (Fig 1F) .

In conclusion, the adoptive cell transfer strategy allowed the development of a primary immune response with B cell activation and expansion, induction of AID expression, class switch recombination, antigen-specific IgM and IgG antibody production and germinal center formation.

We studied the evolution of the B cell response. From two weeks onwards the total number of B cells contracted and at four weeks we recovered about 2-4x10⁶ cells, number that remained stable up to week 20 (Fig 2A) . High titers of HEL-specific IgG were kept from

week 3 to 8, declined thereafter, but were still significantly elevated 20 weeks later (Fig 2B) . A population of cells secreting HEL-specific Igs was present in the spleen (Fig 2C) , but not in the BM (not shown) even at the late time points. About 60% of the recovered cells exhibited the phenotype of antigen-experienced ("memory") CD19 + HEL + AID/YFP + expressing either IgM or IgG (Fig 2D and 2E) . We compared the phenotype of the two AID/YFP + IgM + and AID/YFP + IgM -IgG + memory cell populations recovered with that of the naïve B cells (Fig 2F) . We found that antigen-experience and naïve B cells expressed similar levels of CD62L, CD69 and BAFFR (not shown). Antigen-experienced cells presented sustained expression of CD95 and increased levels of PNA, but the vast majority lost expression of the germinal center marker GL7 present at earlier times post-immunization (Fig 2F compare to Fig 1F) . Compared to naïve B cells, AID/ YFP + cells expressed higher levels of CD80 and MHC class II and down-regulated expression of CXCR5 (Fig 2F) . These findings indicate that the post-germinal center AID/YFP + B cells express an activated phenotype [5, 21] , have increased antigen-presenting capacity [22] , but may lose the ability to re-enter primary follicles being CXCR5 low [23] . We have also compared the patterns of gene expression (RNAseq) by naïve, activated (YFPcells of immunized mice) and both populations of YFP + memory cells. The data shows a clear discrimination of naïve and activated/memory cells while indicating only minor differences between both subsets of YFP + memory cells (Fig 3) .

mRNA was isolated from sort-purified Naïve (CD19 + HEL + YFP -IgM +) IgM + IgG or IgM - IgG + HEL + CD19 + YFP + memory B cells from spleen of different recipient mice. Total RNA was isolated and sequenced using the Illumina HiSeq 2500 (Illumina, San Diego, CA) as recommended by the manufacturer. The validated libraries were then subjected to DNA sequencing. The analysis is performed using the R software, Bioconductor packages including DESeq2 and the PF2tools package (version 1.2.9) developed at PF2 (Institut Pasteur). Normalization and differential analysis are carried out according to the DESeq2 model and package (version 1.8.1). Fig 3A shows a representative heat map of the different cells populations. Fig 3B shows

Late in the immune response persistent B cell numbers were kept by active cell division as a significant fraction of the cells were Ki67 + (Fig 2G left) and incorporated BrdU (Fig 2G middle) . The frequency of BrdU + cells was higher among the AID/YFPcells (15%) than in the major AID/YFP + memory population (3%) and similar between the IgM + and IgM -AID/YFP + populations (Fig 2G middle and not shown) . Three days after BrdU pulse populations were clear of BrdU + cells (Fig 2G right) attesting their high division rate. In spite of their increased proliferation rate, memory cells numbers were stable indicating

that proliferation may be compensated by cell death as suggested by the frequency of caspase3 + cells (Fig 2H) . The frequency of Caspase3 + cells was higher among the AID/YFP + cells suggesting that a fraction of these cells may represent cells undergoing terminal differentiation. Importantly, these findings demonstrate that the transfer strategy allowed the generation of significant numbers of persisting antigen-experienced YFP + cells.

It is not yet known whether the number of antigen-experienced memory B cells correlated to the number of naïve B cells or if it is controlled independently of the initial number of antigen-specific B cells present. To approach this question we transferred different numbers of mature naïve B cells from SW HEL .AID/YFP.Rag2⁻/donors (ranging from 10⁵ to 5.10⁶) into Rag2-deficient mice together with an excess of CD4 + T helper cells (10⁶) and immunize the hosts the day after cell transfer with OVA-HEL in optimal non-limiting quantities. To directly compare the results obtained after the transfer of different all numbers we allowed the responses to reach steady-state eight weeks after antigenic challenge. We studied the amplitude of the immune response by measuring the serum titers of HEL-specific IgG antibodies and enumerating the number of HEL-specific B cells recovered. We found that in the presence of excess T cell help, the levels of the HEL-specific IgGs (Fig 4C) , and both the total number of HEL-specific (Fig 4A) and of memory YFP + B cells recovered (Fig 4B) , did not correlate to the number of antigen specific naïve B cells initially transferred.

Memory B cells are defined functionally by their ability to induce secondary IgG antibody responses upon secondary antigenic challenge. We investigated whether the subsets of AID/ YFP + IgM + and AID/YFP + IgM -IgG + antigen-experienced (memory) B cells persisting at late time points could mount secondary IgG responses and persist after secondary transfer. For this purpose we followed two different experimental strategies. In the first, 5x10⁴ cells of either IgM + or IgM -IgG + memory B cells, were transferred with an excess helper OTII CD4 + T cells into secondary Rag-deficient hosts that were boosted with OVA-HEL the day after cell transfer. In the absence of immunization antibody levels were undetectable (not shown) and three weeks after transfer recovery of both memory B cell subsets was about 10-20% of the initial cell input, exceeding naïve B cell recovery (Fig 5A) , supporting the notion that memory B cells may not require specific ligand recognition to survive (2). One cannot exclude, however, that cross-reactivity of the BCR transgene with environmental antigens may allow signaling sufficient to maintain naïve and memory cell survival in the absence of HEL [24] . Following immunization, the secondarily transferred

AID/YFP + IgM -IgG + cells responded promptly with the exclusive production of significant levels HEL-specific IgG thus confirming their memory statute (11) . The AID/YFP + IgM + B cells in response to antigenic boost produced only limited amounts of IgM antibodies (Fig 4B) , little IgG antibodies, but did generate GL7 + B cells more efficiently than the IgG + memory B cell population (Fig 5D) . Thus the IgM + subset may contain precursors able to generate a secondary germinal center reaction and a new progeny of IgG + effectors (4). With time antibody levels decayed rapidly suggesting that the number of transferred memory B cells declined in the secondary hosts after antigenic boost. Indeed, IgM + and IgG + memory B cells failed to expand and 3 weeks after immunization cell recovery was similar to the retrieval observed in the non-immunized hosts (compare Fig 5E and 5A) . In similar experimental conditions, naïve B cells following immunization expanded, acquired AID/ YFP expression and their numbers more than doubled the number initially injected (Figs 5F and 2A). These data suggest that a significant fraction of the memory B cells generated have a reduced expansion capacity being programmed for rapid differentiation for effector functions.

Besides long-term survival memory B cells must maintain functional activity in the absence of nominal antigen to be fully effective. To test this we used an alternative approach where memory cells were parked in secondary Rag-deficient hosts for 30 days before re-immunization. We found that under these conditions antigenic challenge resulted in the production of HEL-specific IgG antibodies and in a 100 fold increase in the number of cells recovered, expansion that largely exceed that observed after immediate challenge (Fig 5G) .

The aim of this study was to characterize the fate of activated B cells and the generation of memory B cells. To do this, we adoptively transferred monoclonal B cells into immune deficient hosts followed by immunization in presence of T cell help. This strategy resulted in the development of different B cell memory subsets, namely IgM + and IgG + , as described for in situ generated memory cells [4, 6, 14] . These findings indicate that distinct memory B cell subsets are not the result of the heterogeneity of initially responding naïve cells, but originate from the differentiation of a single B cell clone.

While studying the respective rate of proliferation of both types of memory B cells, we found the same high rate of proliferation for IgM + and IgG + memory B cells. These results contrast with previous published data. First it was been reported that "in situ" memory B

cells persist as resting non-dividing cells [11, 25] . However, we have shown that upon adoptive transfer and in absence of competing cells, B cells increase their division rate to occupy the available empty niche [26] , which may explain the higher division rate observed here using this adoptive cell transfer strategy. Secondly, comparing life spans among heterogeneous memory B cell populations it was previously reported a lower division rate among the IgM + subset compared to the IgG + polyclonal subset [6] . Differences in BCR affinity between IgM + and IgG + memory clones may explain the higher division rate previously observed among the IgG + cells [6] . In contrast we compared memory B cell subsets belonging to the same clone bearing the same high affinity BCR. Overall these observations support the notion that lymphocyte division rates and life spans are not an intrinsic cell property, but rather determined by the environment and the presence of competing populations [27] . They demonstrate that upon the correct conditions memory B cells can persist by cell division.

An important question was whether the number of memory B cells depends on the number of initial naïve B cells. We found that, in the presence of an excess of T cell help, that was not the case. However, it was previously reported during polyclonal responses that serum titers of anti-HSA was proportional to the number of cells transferred into irradiated mice [28] . It is possible that limited antigen-specific T-B cell encounters may constraint the number of responding B cells and thus determine linear precursor-progeny between naïve and memory B cells. Our findings indicate that within a single clone the number of precursor naïve B cells present in the peripheral B cell pool does determine neither the intensity nor the final number of memory B cells in response to an optimal dose of antigen. They suggest that the size of memory B cell pool may be controlled independently of the number of naïve B cell precursors and that in the absence of clonal competition the memory niche can be filled with a single monoclonal population. Considering diverse polyclonal populations, the limited niche for memory cells will imply strong competition among clones resulting in the selection of best fit (high affinity) cells: rare mutated clones being able to out compete more frequent but less avid clones. In our settings, the transgenic memory B cells are likely to counter select any new mutant clones since they express a very high affinity BCR selected in the course of a secondary immune response [29] . Thus, notwithstanding the expression of AID and proliferation we did not detect any BCR VH and VL Ig-chain nucleotide mutations among the recovered memory B cells (not shown). These findings may have implication for vaccination protocols as they indicate that each new antigenic exposure or unrelated immunization would add extra competing clones supporting the need for repeated antigenic boosts to prevent memory B cell attrition. They

also demonstrate that the memory B cell pool can be reconstituted from a relatively small number of antigen-specific cells.

It is likely that the relatively poor memory B cell expansion observed after immediate boost after adoptive transfer could be due to the lack in Rag-deficient hosts of the appropriate environment required for memory B cell survival and function. It should be pointed out that B cell transfer into transgenic ML5 Rag-deficient hosts expressing low levels of HEL [29] resulted in rapid cell loss and recovery suggesting that in these hosts, B cells are trapped by antigen in locations where they are unable to survive (not shown). Nevertheless, it has been shown that B cells can drive the maturation of follicular dendritic cells and the organization of lymphoid follicles [30]. Similarly, transferred helper cells may also modify their immediate environment. Thus, by allowing lymphocytes to adapt and modify their immediate environment we improved their response and more important, we recovered the memory B cell pool size present in the original donor mice.

In this study we show that it is possible to fully reconstitute a primary response and the establishment of antibody memory in immune deficient mice after adoptive transfer of antigen-specific monoclonal B cells together with a population of monoclonal helper T cells. Indeed, it is generally believed that in immune deficiencies, B cell therapy has restricted application due to intrinsic defects of host's lymphoid organs structure that may prevent development of immune responses, germinal center formation, establishment of antibody memory and limit cell survival. In contrast we showed that after adoptive transfer in immune deficient hosts antigen immunization induced B cell activation and expansion, induction of AID expression, class switch recombination, antigen-specific IgM and IgG antibody production, germinal center formation and the generation of two subsets of AID/YFP + IgM + IgG and AID/YFP + IgM - IgG + antigen-experienced B cell subsets able to persist in a lymphopenic environment by cell division mimicking responses obtained in intact non-Tg mice [4]. Upon challenge the AID/YFP + IgM - IgG + cells responded promptly with the production of HEL-specific IgG while the AID/YFP + IgM + B cells secreted only limited amounts of IgM antibodies and fail to produce IgG. In contrast the AID/YFP + IgM + B cells could give rise to new GL7 + B cells, suggesting that full reconstitution of the memory B cell pool may require transfer of the different antigen-experienced B cell subsets. Importantly, we found that the recall responses were more efficient if the transferred memory cells were given the required time to adapt to their new environment, suggesting that a period of accommodation is necessary before the transferred cells are fully capable to respond. Our findings also show that different processes can modify the survival

conditions of memory B cells. Finally, we found that the generation of the memory B cell pool in response to an optimal dose of Ag did not rely on the number of the initially responding B cells, suggesting autonomous homeostatic controls for naïve and memory B cells a property that may allow reconstitution of the memory pool in immune-deficient hosts using a limited number of precursor naïve B cells. An autonomous control of the memory B cell pool where each antigenic exposure adds new competing clones supports the notion of vaccination strategies using antigenic boosting to prevent memory B cell attrition. Overall the findings reported demonstrate that it is possible to reconstitute the memory B cell pool of an immune deficient host with an artificially induced population of monoclonal high affinity memory B cells.

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Optimization Method for Forecasting Confirmed Cases of COVID-19 in China

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Abstract: In December 2019, a novel coronavirus, called COVID-19, was discovered in Wuhan, China, and has spread to different cities in China as well as to 24 other countries. The number of confirmed cases is increasing daily and reached 34,598 on 8 February 2020. In the current study, we present a new forecasting model to estimate and forecast the

number of confirmed cases of COVID-19 in the upcoming ten days based on the previously confirmed cases recorded in China. The proposed model is an improved adaptive neuro-fuzzy inference system (ANFIS) using an enhanced flower pollination algorithm (FPA) by using the salp swarm algorithm (SSA). In general, SSA is employed to improve FPA to avoid its drawbacks (i.e., getting trapped at the local optima). The main idea of the proposed model, called FPASSA-ANFIS, is to improve the performance of ANFIS by determining the parameters of ANFIS using FPASSA. The FPASSA-ANFIS model is evaluated using the World Health Organization (WHO) official data of the outbreak of the COVID-19 to forecast the confirmed cases of the upcoming ten days. More so, the FPASSA-ANFIS model is compared to several existing models, and it showed better performance in terms of Mean Absolute Percentage Error (MAPE), Root Mean Squared Relative Error (RMSRE), Root Mean Squared Relative Error (RMSRE), coefficient of determination (R^2), and computing time. Furthermore, we tested the proposed model using two different datasets of weekly influenza confirmed cases in two countries, namely the USA and China. The outcomes also showed good performances.

Text: A large family of viruses, called coronaviruses, are severe pathogens for human beings, which infect respiratory, hepatic, gastrointestinal, and neurologic diseases. They are distributed among humans, birds, livestock, mice, bats, and other wild animals [1] [2] [3]. The outbreaks of two previous coronaviruses, SARS-CoV and MERS-CoV in 2003 and 2012, respectively, have approved the transmission from animal to animal, and human to human [4]. In December 2019, the World Health Organization (WHO) received notifications from China for many cases of respiratory illness that were linked to some people who had visited a seafood market in Wuhan [5]. Currently, Wuhan city suffers from the spreading of a novel coronavirus, called COVID-19 (previously, it was called 2019-nCoV). In [6], the authors concluded that COVID-19 likely originated in bats, because it is more similar to two bat-derived coronavirus strains. However, the source of the COVID-19 is not confirmed yet, and its communities, Hong Kong and Toronto, were 1.2 and 1.32, respectively. Ong et al. [20] proposed a monitoring and forecasting model for influenza A (H1N1-2009). Furthermore, Nah et al. [21] proposed a probability-based model to predict the spread of the MERS.

The Adaptive Neuro-Fuzzy Inference System (ANFIS) [22] is widely applied in time series prediction and forecasting problems, and it showed good performance in many existing applications. It offers flexibility in determining nonlinearity in the time series data, as well as combining the properties of both artificial neural networks (ANN) and fuzzy logic

systems. It has been applied in various forecasting applications, for example, in [23] , a stock price forecasting model was proposed using ANFIS and empirical mode decomposition. Chen et al. [24] proposed a TAIEX time series forecasting model based on a hybrid of ANFIS and ordered weighted averaging (OWA). In [25] , another time series forecasting method was presented for electricity prices based on ANFIS. Svalina et al. [26] proposed an ANFIS based forecasting model for close price indices for a stock market for five days. Ekici and Aksoy [27] presented an ANFIS based building energy consumption forecasting model. More so, ANFIS is also applied to forecast electricity loads [28] . Kumar et al. [29] proposed an ANFIS based model to forecast return products. Ho and Tsai [30] applied ANFIS to forecast product development performance. However, estimating ANFIS parameters is a challenge that needs to be improved. Therefore, in previous studies, some individual swarm intelligence (SI) methods have been applied to the ANFIS parameters to enhance time series forecasting because these parameters have a significant effect on the performance of ANFIS. The SI methods include the particle swarm optimization (PSO) [31, 32] , social-spider optimization [33] , sine-cosine algorithm (SCA) [34] , and multi-verse optimizer (MVO) [35] . For example, in [34] SCA algorithm was applied to improve the ANFIS model to forecast oil consumption in three countries, namely, Canada, Germany, and Japan. In the same context, in [35] , The MVO algorithm was used to enhance the ANFIS model to forecast oil consumption in two countries. In addition, in [36] the PSO was used with ANFIS to predict biochar yield. However, individual SI algorithms may stock at local optima. Therefore, one solution is to apply hybrid SI algorithms to avoid this problem. In [37] , a hybrid of two SI algorithms, namely GA and SSA, was presented to improve the ANFIS model. The proposed new model called GA-SSA-ANFIS was applied to forecast crude oil prices for long-term time series data. However, the previously mentioned methods suffer from some limitations that can affect the performance of the forecasting output such as slow convergence and the ability to balance between exploration and exploitation phases can influence the quality of the final output. This motivated us to propose an alternative forecasting method dependent on the hybridization concept. This concept avoids the limitations of traditional SI techniques by combining the strengths of different techniques, and this produces new SI techniques that are better than traditional ones.

In the current study, we propose an improved ANFIS model based on a modified flower pollination algorithm (FPA) using the salp swarm algorithm (SSA). The FPA is an optimization algorithm proposed by Yang [38] , which was inspired by the flow pollination process of the flowering plants. The FPA was employed in various optimization applications, for example to estimate solar PV parameter [39, 40] , solving sudoku puzzles

[41] , feature selection [42] , antenna design [43] , and other applications [44] [45] [46] [47] . Moreover, SSA is also an optimization algorithm proposed by Mirjalili et al. [48] inspired by the behavior of salp chains. In recent years, the SSA was utilized to solve different optimization problems, such as feature selection [49, 50] , data classification [51] , image segmentation [52] , and others [53, 54] .

The proposed method called FPASSA is a hybrid of FPA and SSA, in which the SSA is applied as a local search method for FPA. The proposed FPASSA starts by receiving the historical COVID-19 dataset. Then a set of solutions is generated where each of them represents the value for the parameters of the ANFIS model. Then the quality of each solution is calculated using the fitness value, and the solution that has the best fitness value is chosen to represent the best solution. Then the probability of each solution is computed. Then the current solution will be updated, either using global or local strategy in FPA. However, in the case of local strategy, the operators of SSA or FPA will be used according to the probability of the fitness value for each solution. The process of updating the solutions is repeated until reaching the stop condition, and the best parameter configurations are used to forecast the number of confirmed cases of COVID-19.

The main contribution points of the current study are as follows:

- 1.

We propose an efficient forecasting model to forecast the confirmed cases of the COVID-19 in China for the upcoming ten days based on previously confirmed cases.

An improved ANFIS model is proposed using a modified FPA algorithm, using SSA.

We compare the proposed model with the original ANFIS and existing modified ANFIS models, such as PSO, GA, ABC, and FPA.

The rest of this study is organized as follows. The preliminaries of ANFIS, FPA, and SSA are described in Section 2. Section 3 presents the proposed FPASSA, and Section 4 presents the experimental setup and results. We conclude this study in Section 5.

The principles of the ANFIS are given in this section. The ANFIS model links the fuzzy logic and neural networks [22]. It generates a mapping between the input and output by applying IF-THEN rules (it is also called Takagi-Sugeno inference model). Figure 1 illustrates the ANFIS model where, y and x define the inputs to Layer 1 whereas, O_{1i} is its output of node i that is computed as follows:

where μ denotes the generalized Gaussian membership functions. A_i and B_i define the membership values of μ . α_i and β_i denote the premise parameters set. The output of Layer 2 (it is also known as the firing strength of a rule) is calculated as follows:

Meanwhile, the output of Layer 3 (it is also known as the normalized firing strength) is calculated as follows:

The output of Layer 4 (it is also known as an adaptive node) is calculated as follows:

where r_i , q_i , and p_i define the consequent parameters of the node i . Layer 5 contains only one node; its output is computed as:

Flower Pollination Algorithm is an optimization method proposed by Yang [38]. It simulates the transfer of flowers' pollen by pollinators in nature. This algorithm utilizes the two types of pollination (i.e., self-pollination and cross-pollination). In self-pollination, the pollination occurs with no pollinators, whereas, in cross-pollination, the pollens are moved between different plants. In more detail, the self-pollination can be represented as a local pollination while the cross-pollination can be called global pollination.

The global pollination or cross-pollination can be mathematically formed as follows:

where $x_{t,i}$ defines the pollen i at iteration t . L denotes the pollination's strength or the step size. F^* is the target position or best solution. In some cases, insects can fly with different distance steps for a long space; therefore, Levy fly distribution is applied to simulate this movement.

where $\lambda = 1.5$. $\Gamma(\lambda)$ denotes the gamma function. This distribution is available for large steps $s > 0$. The self-pollination or local pollination can be mathematically formed as follows:

where $x_{t,i}$ and $x_{k,i}$ represent pollens from different flower in the same plant. in the range $[0,1]$ The process of pollination can be done using cross-pollination or self-pollination. Therefore, the random variable p , in the range $[0, 1]$, is used to determine this process.

SSA is an optimization technique introduced by [48] . It simulates the Salps' behavior in nature. This behavior is called salp chain. The mathematical model of SSA begins by splinting its population into a leader group and followers group. The leader is the front salp, whereas, the followers are the other salps. The search space is determined in n -dimensions with n variables. Equation (10) works to update the salps' positions.

where $x_{1,j}$ denotes the leader's position in j -th dimension. F_j is the target position. ub_j and lb_j represent the max and min bounds, respectively. c_2 and c_3 denote random numbers in $[0, 1]$. c_1 is an important parameter; it balances between the exploration and exploitation phases. It is computed as follows:

where the current loop number is t and the max loop' number is t_{max} . Then, the followers' position is updated as follows:

where $x_{i,j}$ defines the i -th position of the follower in j -th dimension. $i > 1$.

This section explains the proposed FPASSA-ANFIS method. It is a time series method for forecasting the confirmed cases of the COVID-19, as given in Figure 2 . The FPASSA-ANFIS utilizes the improved FPA to train the ANFIS model by optimizing its parameters. The

FPASSA-ANFIS contains five layers as the classic ANFIS model. Layer 1 contains the input variables (the historical COVID-19 confirmed cases). Whereas Layer 5 produces the forecasted values. In the learning phase, the FPASSA is used to select the best weights between Layer 4 and Layer 5.

The FPASSA-ANFIS starts by formatting the input data in a time series form. In our case, the autocorrelation function (ACF) was considered. ACF is one of the methods applied to find patterns in the data; it presents information about the correlation between points separated by various time lags. Therefore, in this paper, the variables with ACF greater than 0.2 are considered i.e., 5-lags.

Besides, the training data contains 75% of the dataset, whereas the testing data contains 25% of them. The number of clusters is defined by the fuzzy c-mean (FCM) method to construct the ANFIS model.

The parameters of the ANFIS model are prepared by the FPASSA algorithm. In the training phase, the calculation error (as in Equation (13)) between the real data and the predicted data is used to evaluate the parameters' quality.

where T is the real data, and P is the predicted data. N is the sample length. The smaller values of the objective function indicate good ANFIS's parameter.

On the other hand, the updating phase of the followers' positions in the SSA algorithm is applied to improve the global pollination phase in the FPA algorithm. In this improvement, there is a random variable (r) used to switch between both phases. If $r > 0.5$, then the operators of the SSA is used; otherwise, the operators of the FPA are used. In general, The FPASSA starts by constructing the population (X); afterward, the objective function is calculated for each solution. The solution with the lowest error value is saved to the next iteration. This sequence is repeated until meeting the stop condition, which in this paper, is the maximum number of iterations. Then the best solution is passed to train the parameters of the ANFIS model.

After finishing the training phase, the testing phase is started with the best solution to compute the final output. The performance of the proposed method is evaluated by comparing the real data with the predicted data using the performance measures. Finally, the FPASSA produces a foretasted value for confirmed cases of COVID-19 in China in the next day. The steps of the proposed FPASSA are presented in Algorithm 1.

Input: Historical COVID-19 dataset, size of population N , total number of iterations t_{max} .

Divide the data into training and testing sets.

Using Fuzzy c-mean method to determine the number of membership functions.

Constructing the ANFIS network.

Set the initial value for N solutions (X). Return the best solution that represents the best configuration for ANFIS.

Apply the testing set to the best ANFIS model.

Forecasting the COVID-19 for the next ten days.

This section presents the description of the used dataset, the performance measures, the parameter setting for all methods, the experiment results, and discussions.

The main dataset of this study is COVID-19 dataset. It was collected from the WHO website (<https://www.who.int/emergencies/diseases/novel-coronavirus-2019/situation-reports/>). It contains the daily confirmed cases in China from 21 January 2020 to 18 February 2020, as shown in Table 1. We used 75% from the dataset to train the model while the rest is used to test it.

Moreover, we evaluated the performance of the proposed method using two datasets of weekly influenza confirmed cases. The first one is called DS1; it was collected from the Centers for Disease Control and Prevention (CDC) (<https://www.cdc.gov/flu/weekly/>). It starts from week number 40 in 2015 and continues until week number 6 in 2020. Whereas, the second one is called DS2. It was collected from the WHO website (<https://www.who.int/influenza>). It contains the data of weekly influenza confirmed cases in China from week number 1 in 2016 to week number 8 in 2020.

The quality of the proposed method is evaluated using a set of performance metrics as follows:

- Root Mean Square Error (RMSE):

where Y_p and Y are the predicted and original values, respectively. • Mean Absolute Error (MAE):

- Mean Absolute Percentage Error (MAPE):

- Root Mean Squared Relative Error (RMSRE):

N represents the sample size of the data. • Coefficient of Determination (R^2):

where $\bar{Y}$ represents the average of Y .

The lowest value of RMSE, MAE, MAPE, and RMSRE refers to the best method. The higher value of R^2 indicates better correlation for the method.

This paper aims to assess the ability of the FPASSA to forecast the COVID-19 by comparing its performance with other methods, namely the ANFIS and the trained ANFIS models using PSO, GA, ABC, FPA, and FPASSA. The parameters' setting for these models is listed in Table 2 .

The common parameters, such as population size, are set to 25 and 100 iterations are applied. Besides, each algorithm is performed for 30 independent runs to fair comparisons. The selected parameters are chosen because they produced good behavior in previous experiments, such as [34, 35, 55, 56] . Table 2 . Parameters' setting.

Parameters Setting

Max. epochs = 100, Error goal = 0, Initial step = 0.01, Decrease rate = 0.9, Increase rate = 1.

In this section, the performance of the proposed FPASSA to predict the DS1 and DS2 is discussed. It can be concluded from Table 3 that the performance of FPASSA outperformed the compared methods in all measures, whereas the FPA is ranked second. The results of DS2 indicate that the FPASSA is ranked first in terms of RMSE, MAPE, R^2 , and the CPU time. Whereas, the PSO is ranked second, followed by the FPA, GA, then ABC. These results denote that the proposed method can optimize the parameters of the ANFIS model effectively and produce good results in terms of the performance measures. Comparison results between the proposed FPASSA and other models to forecast COVID-19 are given in Table 4 . It can be concluded that the FPASSA outperforms other models. For example, by analyzing the results of RMSE, MAE, MAPE, RMSRE, and CPU time(s) it can be observed that the FPASSA achieves the smallest value among the comparison algorithms, and this indicates the high quality of the FPASSA. Meanwhile, the FPA allocates the second rank, which provides better results than the rest of the methods.

Moreover, the value of R^2 refers to the high correlation between the prediction obtained by the proposed FPASSA method and the original COVID-19, which has nearly 0.97. This can also be noticed from Figure 3 , which depicts the training of the algorithms using the historical data of the COVID-19 as well as their forecasting values for ten days. Table 5 depicts the forecasting value for the confirmed cases of the COVID-19 in China from

19/2/2020 to 28/2/2020. From these results, it can be noticed that the outbreak will reach its highest level on the day 28/2/2020. The average percentage of the increase over the forecasted period is 10%, the highest percentage is 12% on 28/2/2020, and the lowest percentage is 8.7% on 19/2/2020. From the previous results, it can be concluded that the proposed FPASSA-ANFIS has a high ability to forecast the COVID-19 dataset. These results avoid the limitations of traditional ANFIS because of the combination with the modified FPA method. Moreover, the operators of SSA are combined with the local strategy of FPA to enhance their exploitation ability. However, the time computational of the proposed FPASSA method still requires more improvements.

This paper proposed a modified version for the flower pollination algorithm (FPA) using the salp swarm algorithm (SSA). This modified version, called FPASSA, is applied to improve the performance of the ANFIS through determining the optimal value for its parameters. The developed FPASSA-ANFIS model is applied as a forecasting technique for a novel coronavirus, called COVID-19, that was discovered in Wuhan, China at the end of last year and January of the current year. The proposed FPASSA-ANFIS model has a high ability to predict the number of confirmed cases within ten days. Besides, FPASSA-ANFIS outperforms other forecasting models in terms of RMSE, MAE, MAPE, RMSRE, and R^2 . Furthermore, two datasets of weekly influenza confirmed cases in the USA and China were used to evaluate the proposed method, and the evaluation outcomes showed its good performance. According to the promising results obtained by the proposed FPASSA-ANFIS, it can be applied in different forecasting applications.

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The vacuolar-type ATPase inhibitor archazolid increases tumor cell adhesion to endothelial cells by accumulating extracellular collagen

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Abstract: The vacuolar-type H⁽⁺⁾-ATPase (v-ATPase) is the major proton pump that acidifies intracellular compartments of eukaryotic cells. Since the inhibition of v-ATPase resulted in anti-tumor and anti-metastatic effects in different tumor models, this enzyme has emerged as promising strategy against cancer. Here, we used the well-established v-ATPase inhibitor archazolid, a natural product first isolated from the myxobacterium *Archangium gephyra*, to study the consequences of v-ATPase inhibition in endothelial cells (ECs), in particular on the interaction between ECs and cancer cells, which has been neglected so far. Human endothelial cells treated with archazolid showed an increased adhesion of tumor cells, whereas the transendothelial migration of tumor cells was reduced. The adhesion process was independent from the EC adhesion molecules ICAM-1, VCAM-1, E-selectin and N-cadherin. Instead, the adhesion was mediated by β 1-integrins expressed on tumor cells, as blocking of the integrin β 1 subunit reversed this process. Tumor cells preferentially adhered to the β 1-integrin ligand collagen and archazolid led to an increase in the amount of collagen on the surface of ECs. The accumulation of collagen was accompanied by a strong decrease of the expression and activity of the protease cathepsin B. Overexpression of cathepsin B in ECs prevented the capability of archazolid to increase the adhesion of tumor cells onto ECs. Our study demonstrates that the inhibition of v-ATPase by archazolid induces a pro-adhesive phenotype in endothelial cells that promotes their interaction with cancer cells, whereas the transmigration of tumor cells was reduced. These findings further support archazolid as a promising anti-metastatic compound.

Text: The vacuolar-type H⁺ -ATPase (v-ATPase) is the major proton pump responsible for acidification of intracellular compartments in eukaryotic cells [1] . The enzyme consists of two multi-subunit complexes, the soluble V₁ transmembrane V_o subcomplex required for the proton transport across membranes [1, 2] . In most cell types v-ATPases are only expressed in the endomembrane system to regulate and maintain the acidic pH of intracellular compartments such as lysosomes, endosomes, the Golgi apparatus,

secretory granules and coated vesicles [3] . The function of v-ATPases is essential for cellular processes such as vesicular trafficking, receptor-mediated endocytosis and protein degradation and processing. In specialized cell types including osteoclasts and renal epithelial cells, v-ATPases can also be expressed on the plasma membrane, where they pump protons into the extracellular space [2] [3] [4] . In cancer cells v-ATPases are expressed on the plasma membrane in order to eliminate toxic cytosolic H^+ . Most importantly, v-ATPases contribute to the acidic tumor microenvironment, which leads to the activation of proteases, thus facilitating tumor cell migration, invasion and angiogenesis [5] [6] [7] . Since the inhibition of v-ATPase was shown to reduce the invasiveness of cancer cells and metastasis formation [8, 9] , this enzyme has emerged as a promising drug target in the recent years. Archazolid A and B are highly potent and specific inhibitors of v-ATPases [10] . They were first isolated from the myxobacterium *Archangium gephyra* [11] . These compounds inhibit v-ATPase at low nanomolar concentrations [10, 12] by binding to the subunit c of the V_o complex. As their biological activity is comparable to the v-ATPase inhibitors bafilomycin and concanamycin [10, 11] , archazolids are natural compounds of high interest that can be used both as a tool to study the consequences of v-ATPase inhibition and as a lead for drug development. Archazolids can be either produced by fermentation [11] or by total synthesis [13, 14] .

In the field of cancer research several studies reported on interesting pharmacological effects of archazolid: It reduced the migration of different invasive tumor cells in vitro and cancer cell metastasis in vivo in a breast tumor mouse model [15] . Furthermore, archazolid activated pathways of cellular stress response and apoptosis in highly invasive tumor cells [16] . In classically activated macrophages, archazolid selectively induced the generation of tumor necrosis factor α (TNF α), which may indirectly promote tumor suppression [17] .

Up to now, the role of v-ATPases in endothelial cells has only rarely been investigated. The endothelium plays a crucial role in the pathogenesis and progression of cancer: The metastatic cascade includes local angiogenesis at the site of the primary tumor and adhesion of tumor cells at the site of metastasis [18] . Angiogenesis, the development of new blood vessels out of existing ones, depends on the proliferation, migration and differentiation of endothelial cells [19] . This process ensures the nutrient supply of the tumor and its growth [20] . Circulating cancer cells can adhere to the endothelium at distant sites. This adhesive interaction is mediated by receptors and corresponding ligands expressed on tumor and endothelial cells [18, 21] . V-ATPases have been reported to

regulate intracellular pH and cell migration in microvascular endothelial cells [22, 23] . A recent study showed that the inhibition of v-ATPase by concanamycin prevented proliferation, reduced migration and impaired angiogenesis-related signaling in endothelial cells [24] . So far, there are no investigations on the role of endothelial v-ATPases for the process of tumor cell adhesion onto the endothelium. Thus, we were interested in the consequences of the inhibition of endothelial v-ATPase by archazolid on the interaction between endothelial and cancer cells. Various cell adhesion molecules on the endothelium, such as intercellular adhesion molecule 1 (ICAM-1), vascular cell adhesion protein (VCAM-1), E-selectin or N-cadherin [21] as well as integrins expressed on cancer cells have been reported to mediate cell adhesion of cancer cells onto endothelial cells [25] [26] [27] . Accordingly, we focused on these cell adhesion molecules and integrins. For the first time, our study revealed a link between the function of v-ATPases and the adhesion and transmigration properties of endothelial cells.

CellTiter-Blue Cell Viability Assay (Promega, Mannheim, Germany) was performed according to the manufacturer's protocol for determining the cell viability of cells after treatment with archazolid. This assay is based on the ability of metabolically active cells to reduce resazurin which results in fluorescent resorufin. The CellTiter-Blue Reagent was added to the cells 4 h before the endpoint of treatment. Fluorescence was measured with an Infinite F200 pro microplate reader (Tecan, Männedorf, Switzerland) at 560 nm (excitation) and 590 nm (emission).

CytoTox 96 Non-Radioactive Cytotoxicity Assay (Promega) was performed according to the manufacturer's instructions for determining the lactate dehydrogenase (LDH) release after treatment with archazolid. Lysis buffer was added to untreated cells 45 min before the end of treatment to induce the release of this enzyme. LDH is a cytosolic enzyme that is released by leaky cells. Released LDH catalyzes the enzymatic conversion of lactate to pyruvate which provides NADH for the conversion of iodonitrotetrazolium violet into a red formazan product in the presence of diaphorase. The absorbance was measured with a Varioskan Flash microplate reader (Thermo Fisher Scientific) at 490 nm.

LysoTracker Red DND-99 (Life Technologies, Thermo Fisher Scientific) is a dye to measure pH values in viable cells. HUVECs were cultured to confluence on collagen G-coated μ -slides (80826, ibidi, Martinsried, Germany) before they were treated with archazolid for 24 h. 1 μ g/ ml Hoechst 33342 (Sigma-Aldrich, Munich, Germany) was used to visualize the

nuclei and 50 nM LysoTracker Red DND-99 was used to visualize the acidic compartments which correspond to the lysosomes. Both dyes were incubated for 10 min at 37 °C before acquisition of single images by a Leica DMI6000 B fluorescence microscope (Leica Microsystems, Wetzlar, Germany).

HUVECs were seeded in collagen G-coated 24-well plates and grown to confluence for two days before treatment. The cells were incubated with indicated concentrations of archazolid for 24 h. Untreated MDA-MB-231 or PC-3 cells were labeled with CellTracker Green CMFDA Dye (5 µM in serum-free DMEM, 37 °C) for 30 min before 100,000 cells per well were added to HUVECs and were allowed to adhere for various time points at 37 °C. Non-adherent tumor cells were washed off three times with PBS containing Ca²⁺ and Mg²⁺. Tumor cell adhesion was determined by fluorescence measurements with an Infinite F200 pro microplate reader (Tecan) at 485 nm (excitation) and 535 nm (emission).

For blocking the integrin β1 subunit on MDA-MB-231 or PC-3 cells, CellTracker Greenlabeled MDA-MB-231 or PC-3 cells were incubated with an anti-integrin β1 antibody (P5D2, ab24693, Abcam, Cambridge, United Kingdom) at a concentration of 1 µg antibody per one million cells in 1 ml DMEM. Before adding to archazolid-treated HUVECs, MDA-MB-231 or PC-3 cells were washed once with DMEM. For blocking the integrin β1 subunit on HUVECs, the cells were incubated with the anti-integrin β1 antibody (0.1 µg/well in ECGM). HUVECs were washed once with ECGM before untreated MDA-MB-231 or PC-3 cells were added to HUVECs.

For the adhesion of MDA-MB-231 or PC-3 cells onto extracellular matrix (ECM) components 24-well plates were coated with collagen G (10 µg/ml in PBS), human plasma fibronectin (10 µg/ml PBS) or laminin-411 (10 µg/ml in Dulbecco's PBS [DPBS] containing Ca²⁺ and Mg²⁺) at 4 °C overnight. The adhesion of MDA-MB-231 and PC-3 cells onto these three most prominent ECM components was carried out as described above (10 min adhesion at 37 °C).

HUVECs were grown on a porous filter membrane (Transwell insert, polycarbonate membrane, 8 µm pores; Corning, New York, USA) for 48 h and were treated as indicated. Untreated MDA-MB-231 cells were labeled with CellTracker Green CMFDA Dye (as described in the section cell adhesion assay) and resuspended in medium 199 (PAN-

Biotech) containing 0.1% BSA. HUVECs were washed twice with medium 199 containing 0.1% BSA before MDA-MB-231 cells were allowed to transmigrate through the endothelial monolayer for 24 h. Medium 199 containing 0.1% BSA was used as negative control and medium 199 containing 20% FCS was used as chemoattractant for transmigration in the lower compartment. Non-migrated cells remaining in the upper compartment were carefully removed using a cotton swab. Transmigrated cells were lysed in radioimmunoprecipitation assay (RIPA) buffer and transmigration was quantified by measuring the fluorescence signal at 485 nm (excitation) and 535 nm (emission).

HUVECs were grown to confluence on 6-well plates before they were treated with archazolid for 12 h. The cells were induced to upregulate the gene expression of cell adhesion molecules by TNF α . RNA was isolated using the RNeasy Mini Kit from Qiagen (Hilden, Germany) according to the manufacturer's protocol. On-column DNase digestion was performed to remove genomic DNA. RNA was transcribed into cDNA by Superscript II (Life Technologies, Thermo Fisher Scientific). qPCR experiments were performed using a StepOnePlus System (Applied Biosystems, Thermo Fisher Scientific) and data was analyzed by the StepOne and StepOnePlus Software v2.3. Power SYBR Green PCR Master Mix (Life Technologies) and the comparative C_T quantitation method ($2^{-\Delta\Delta C_T}$) were used.

HUVECs were grown to confluence on 12-well plates before they were treated with archazolid for 24 h. Cells were treated with TNF α for 24 h to induce the expression of cell adhesion molecules. Subsequently, the cells were detached with HyClone HyQTase (GE Healthcare, Freiburg, Germany). In the case of ICAM-1 the detached cells were fixed with 4% formaldehyde (Polysciences, Hirschberg an der Bergstraße, Germany) in PBS for 10 min and washed once with PBS before incubating with the fluorescein isothiocyanate (FITC)-labeled anti-human CD54 (mouse, ICAM-1) antibody (MCA1615F, Biozol, Eching, Germany) at room temperature for 45 min. For all other proteins, the cells were not fixed and washed once with PBS before incubating with the antibodies phycoerythrin (PE)-labeled anti-human CD106 (mouse, VCAM-1), PE-labeled anti-human CD62E (mouse, E-selectin) and PE-labeled anti-human CD325 (mouse, N-cadherin) from Becton Dickinson on ice for 45 min. These antibodies were diluted in PBS containing 0.2% BSA. The surface expression of cell adhesion molecules was measured by flow cytometry (FACSVerse, Becton Dickinson, Heidelberg, Germany).

To stain the surface collagen on HUVECs, cells were incubated with an anti-human collagen antibody (rabbit, 1:40, ab36064, Abcam) on ice for 30 min. The staining procedure was performed on ice to ensure that surface proteins or antibodies are not endocytosed. The cells were washed once with PBS containing Ca^{2+} and Mg^{2+} before they were fixed with Roti-Histofix (Carl Roth). Alexa Fluor 488-conjugated anti-rabbit antibody (goat, 1:400, A11008, Life Technologies) was used as secondary antibody and Hoechst 33342 (1 $\mu\text{g}/\text{ml}$, Sigma-Aldrich) was used to visualize nuclei.

Confluent HUVECs in 6-well plates were treated as indicated. Cells were washed with ice-cold PBS and lysed with RIPA buffer supplemented with protease inhibitors (Complete Mini EDTA-free; Roche, Mannheim, Germany), sodium orthovanadate, sodium fluoride, phenylmethylsulphonyl fluoride, β -glycerophosphate, sodium pyrophosphate and H_2O_2 . Protein determination was performed using the Pierce BCA Protein Assay Kit (Thermo Fisher Scientific). Equal amounts of proteins (10-20 μg) were separated by sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE; Bio-Rad Laboratories, Munich, Germany). Separated proteins were transferred onto polyvinylidene difluoride membranes by tank blotting (Bio-Rad Laboratories) for immunodetection. Membranes were blocked with 5% bovine grade milk powder (Carl Roth) in TBS containing 0.1% Tween 20 (Sigma-Aldrich). The following antibodies were used: mouse anti-human cathepsin B antibody (IM27L, Merck) (1:500), mouse anti- β -actin- peroxidase antibody (A3854, Sigma-Aldrich) (1:100,000) and antimouse IgG horse radish peroxidase (HRP)-linked antibody (7076, Cell Signaling, Frankfurt, Germany) (1:5,000). ImageJ version 1.49m was used for densitometric analysis.

Cathepsin B activity assay was performed as described in the publication by Kubisch et al. [28]. Confluent HUVECs or HMEC-1 seeded in 6-well plates were treated as indicated. Cells were washed with PBS and lysed with the non-denaturing M-PER mammalian protein extraction reagent (78501, Thermo Fisher Scientific) supplemented with protease inhibitors (Complete Mini EDTA-free, Roche), sodium orthovanadate, sodium fluoride, phenylmethylsulphonyl fluoride. The fluorogenic cathepsin B substrate Z-Arg-Arg-7-amido-4-methylcoumarin hydrochloride (C5429, Sigma-Aldrich) was added to 30 μg of the cell lysate diluted in assay buffer supplemented with 2 mM L-cysteine (C7880, Sigma-Aldrich) and incubated for 30 min at 40 °C. Fluorescence was measured at 348 nm (excitation) and 440 nm (emission) with a microplate reader (Varioskan Flash, Thermo Fisher Scientific). The intensity of the fluorescence signal corresponded to the cathepsin B enzyme activity.

For background subtraction the cathepsin B inhibitor CA-074Me (Enzo Life Sciences, Lörrach, Germany) was added to an additional reaction.

The HUVEC Nucleofector Kit (Lonza, Cologne, Germany) was used to transfect HUVECs. The transfection was performed according to the manufacturer's protocol using 2.5 µg plasmid DNA for 500,000 cells (Nucleofector 2b Device, Lonza). 48 h after transfection the cells were treated for further experiments. The addgene plasmid #11249 hCathepsin B was kindly provided by Hyeryun Choe [29]. hCathepsin B was digested with PmeI and XbaI and the linear DNA fragment not corresponding to the human CTSB gene was religated to generate the empty pcDNA3.1 (-) delta MCS plasmid that was used for control transfections. The original backbone of hCathepsin B is the pcDNA3.1 (-) from Thermo Fisher Scientific. The control vector pcDNA3.1 (-) delta MCS used for our transfections was cloned on the basis of hCathepsin B and is therefore lacking almost the whole part of the multiple cloning site of the pcDNA3.1 (-).

Statistical analyses were performed using GraphPad Prism 5.0 (San Diego, USA). One-way ANOVA followed by Tukey's post-hoc test or unpaired t-test was used for the evaluation of a minimum of three independent experiments. The numbers of independently performed experiments (n) are stated in the corresponding figure legends. p 0.05 was considered as statistically significant. Data are expressed as mean ± standard error of the mean (SEM).

Since the v-ATPase inhibitor archazolid has never been used for studies in endothelial cells, we first performed cytotoxicity assays. We treated confluent HUVECs with up to 1 nM archazolid for 24 and 48 h and observed that this treatment has neither an influence on the metabolic activity nor on the release of LDH after 24 h (Fig 1A and 1B, left panels). The metabolic activity and the release of LDH were only slightly affected by the highest concentration of archazolid after 48 h (Fig 1A and 1B, right panels). Consequently, the following experiments were all carried out after 24 h (or less) of archazolid treatment in order to exclude any cytotoxic effects of archazolid within our experimental settings.

Microscopic analysis revealed that also the integrity of the endothelial monolayer was not affected by archazolid, but the cells showed a slightly different morphology (Fig 2A): Archazolid-treated cells were swollen compared to control cells, which was not unexpected, as vacuolation of the endoplasmic reticulum (ER) has been described for

other cell types and is typical for v-ATPase inhibitors [11, 16, 24, 30] . This effect was obvious both in subconfluent and in confluent cells (Fig 2A) . Inhibition of v-ATPase prevents the acidification of lysosomes [1, 31] . Using the cell-permeable dye LysoTracker Red DND-99, it is possible to label the acidic lysosomes in living cells. Thus, this dye can serve as an indicator of v-ATPase inhibition. To proof that archazolid is also functionally active as a v-ATPase inhibitor in HUVECs, cells were treated with 1 nM archazolid before they were incubated with LysoTracker Red DND-99 and Hoechst 33342. As shown in Fig 2B, the red vesicular staining corresponding to acidified lysosomes in control cells disappeared completely after treatment with archazolid. In summary, archazolid treatment for 24 h was not cytotoxic to quiescent HUVECs, but inhibited the functionality of the v-ATPase.

We analyzed the adhesion of MDA-MB-231 cells onto HUVECs. Confluent HUVECs were treated with up to 1 nM archazolid for 24 h. Untreated MDA-MB-231 cells were labeled with Cell-Tracker Green CMFDA Dye. Interestingly, v-ATPase inhibition strongly increased the attachment of the metastatic breast carcinoma cell line MDA-MB-231 onto HUVECs after 10 and 120 min of adhesion (Fig 3A and 3B) . We also investigated the influence of archazolid on the transendothelial migration of MDA-MB-231 cells. HUVECs seeded in a Boyden chamber were treated with 1 nM archazolid for 24 h. CellTracker Green-labeled MDA-MB-231 cells (not treated with archazolid) were allowed to transmigrate through the endothelial monolayer for 24 h. As shown in Fig 3C, archazolid significantly decreased the transendothelial migration of MDA-MB-231 cells.

The influence of archazolid on tumor cell adhesion was not only studied in HUVECs, which represent macrovascular endothelial cells, but also in microvascular HMEC-1 cells. Moreover, besides the breast cancer cell line MDA-MB-231, also PC-3 prostate cancer cells were used as a second metastatic cancer cell line. Archazolid treatment of endothelial cells increased the attachment of MDA-MB-231 cells onto the HMEC-1 monolayer after 120 min of adhesion (Fig 4A) and increased the attachment of PC-3 cells onto the HUVEC monolayer after 30 and 60 min of adhesion (Fig 4B) . Of note, the adhesion of non-metastatic Jurkat cells, an acute T cell leukemia cell line, remained unaffected after treatment of HUVECs with archazolid (S1A Fig). Taken together, archazolid treatment augmented the adhesive properties of both micro-and macrovascular endothelial cells for metastatic tumor cells, but not for non-metastatic ones. Of note, cancer cell adhesion onto archazolid-activated endothelial cells increased with the time of adhesion.

The adhesion of tumor cells onto the endothelium is in principle similar to that of leukocytes, but slightly differs in the molecules that mediate the adhesion process. Ligands for the endothelial cell adhesion molecules ICAM-1, VCAM-1, E-selectin and N-cadherin were found to be expressed on tumor cells and to mediate tumor-endothelial cell interaction [21]. Inhibition of the v-ATPase might affect the expression of endothelial cell adhesion molecules on mRNA or protein levels. To determine the mRNA expression of ICAM-1, VCAM-1, E-selectin and Ncadherin, HUVECs were treated with archazolid for 12 h. TNF α is known to upregulate the expression of ICAM-1, VCAM-1 and E-selectin [32] and, thus, served as positive control. Quantitative real-time PCR showed that v-ATPase inhibition in HUVECs did not alter the mRNA levels of ICAM-1, VCAM-1, E-selectin and N-cadherin (Fig 5A). The protein expression of these adhesion molecules on the surface of endothelial cells was analyzed by flow cytometry. Archazolid (1 nM, 24 h) did not affect the cell surface expression of ICAM-1, VCAM-1, E-selectin and N-cadherin (Fig 5B).

Besides ICAM-1, VCAM-1, E-selectin and N-cadherin, also integrins are able to mediate the process of cell adhesion [33] [34] [35]. Since none of the cell adhesion molecules expressed on HUVECs were regulated upon archazolid treatment, we considered integrins as potential interaction partners. Within this protein family β 1-integrins have been reported to mediate tumor cell adhesion onto quiescent endothelial cells [25]. In order to elucidate the role of β 1-integrins for the archazolid-induced tumor cell adhesion, the integrin β 1-subunit was blocked either on MDA-MB-231 cells, PC-3 cells or on HUVECs. (Of note, as in all experiments throughout this study, only endothelial cells were treated with archazolid.) After blocking β 1-integrins on MDA-MB-231 or PC-3 cells, the archazolid-induced tumor cell adhesion was reduced almost to control level (Fig 6A and 6B, left panels), whereas blocking of β 1-integrins on HUVECs had no significant effect on the increase of tumor cell adhesion by v-ATPase inhibition (Fig 6A and 6B, right panels).

Depending on their α subunit, β 1-integrins have a variety of ligands including extracellular matrix (ECM) components such as collagen, fibronectin and laminin [35]. Therefore, we hypothesized that archazolid treatment of endothelial cells might lead to an upregulation of these components. MDA-MB-231 and PC-3 cells were allowed to adhere onto plastic that was coated with these ECM components. This cell adhesion assay revealed that MDA-MB-231 as well as PC-3 cells favor the interaction with the ECM component collagen, as the adhesion onto collagen is much higher than onto the uncoated plastic control (Fig 7A).

MDA-MB-231 and PC-3 cells also adhered to fibronectin-coated plastic, but to a much lesser extent compared to the collagen coating. Therefore, we focused on the interaction between these two tumor cell lines and collagen. Blocking of the integrin $\beta 1$ subunit on MDA-MB-231 and PC-3 cells clearly abolished the interaction with collagen (Fig 7B) , indicating that the attachment of these tumor cells to collagen is mediated by $\beta 1$ -integrins.

Since collagen is the major ECM component MDA-MB-231 and PC-3 cells interact with, the next step was to prove whether v-ATPase inhibition influences the amount of collagen expressed by HUVECs as extracellular matrix. To detect collagen on the endothelial surface, archazolid-treated HUVECs were labeled with an antibody against collagen type I-IV on ice to prevent endocytosis and to ensure that the antibody does not bind to intracellular collagen. Interestingly, archazolid increased the amount of surface collagen on HUVECs by about 50% (Fig 7C) . Control stainings were performed using an antibody against the cytosolic p65 subunit of the transcription factor nuclear factor κB (NF κB) to show that intracellular proteins were not detected by this staining method (S2 Fig) .

It was reported that v-ATPase inhibition by archazolid impairs the activity of cathepsin B [28, 36] , a lysosomal enzyme that degrades extracellular matrix components including collagen [37] [38] [39] [40] [41] . As collagen is degraded by cathepsin B and the activation of cathepsin B depends on v-ATPase activity [28, [36] [37] [38] 42] , we suggested that an accumulation of collagen on the surface of endothelial cells might be a consequence of an impaired functionality of cathepsin B. Therefore, an enzyme activity assay based on the proteolysis of a fluorogenic cathepsin B substrate was performed. In archazolid-treated HUVECs and HMEC-1 the activity of cathepsin B was induced both the mRNA (1 ng/ml TNF) and the cell surface expression (10 ng/ml TNF) of ICAM-1, VCAM-1, E-selectin and Ncadherin.

<https://doi.org/10.1371/journal.pone.0203053.g005> Inhibition of endothelial vATPase increases tumor cell adhesion to endothelial cells strongly decreased by approximately 50% compared to control cells at an archazolid concentration of 1 nM (Fig 8A) . In line with this result, western blot analysis showed that archazolid (1 nM) reduces the protein expression of the mature, active form of cathepsin B to less than 40% of the control in HUVECs (Fig 8B) . To proof whether the archazolid-induced tumor cell adhesion is a consequence of the decreased amount of cathepsin B, HUVECs were transfected with a plasmid coding for human cathepsin B or with the empty vector as control. After 48 h, the

transfected cells were treated with 1 nM archazolid. The level of cathepsin B after transfection and treatment was assessed by western blot analysis (Fig 9A) . Overexpression of cathepsin B strongly diminished both the basal and the archazolid-induced adhesion of MDA-MB-231 cells (Fig 9B) .

Targeting the proton pump v-ATPase for cancer therapy has gained great interest since its inhibition was reported to reduce the invasiveness of cancer cells and, most importantly, also metastasis [8, 9] . Thus, intensive research related to v-ATPases was done in cancer cells, whereas there are only few studies investigating v-ATPases in endothelial cells indicating a role in migration, proliferation and possibly angiogenesis [22] [23] [24] . In the present study we used the myxobacterial natural product archazolid to investigate the consequences of v-ATPase inhibition in the endothelium on tumor-endothelial cell interactions.

For the first time, we were able to show a link between v-ATPase and the adhesion and transmigration properties of the endothelium. Inhibition of the v-ATPase in endothelial cells by archazolid significantly increased the adhesion of metastatic cancer cells and decreased the transendothelial migration of cancer cells which was attributed to augmented collagen levels on the surface on archazolid-treated endothelial cells. Of note, adhesion of the non-metastatic Jurkat cell line onto archazolid-treated endothelial cells remained unaffected. The archazolid-induced adhesion of tumor cells was independent from the endothelial cell adhesion molecules ICAM-1, VCAM-1, E-selectin and N-cadherin, as their expression was not regulated by the compound. However, we found that the archazolid-induced tumor cell adhesion was mediated by $\beta 1$ -integrins expressed on MDA-MB-231 breast cancer and PC-3 prostate cancer cells as blocking of the integrin $\beta 1$ subunit on these tumor cells reversed the pro-adhesive effect of archazolid. In adhesion experiments on plastic coated with extracellular matrix components, we could show that MDA-MB-231 and PC-3 cells clearly favored the interaction with collagen, whereas the adhesion of non-metastatic Jurkat cells was largely independent from extracellular matrix proteins (S1B Fig). The different adhesion properties of metastatic cancer cells and Jurkat cells might be a result of the distinct integrin expression pattern of each cell line. MDA-MB-231 and PC-3 cells express $\alpha 2\beta 1$ - and $\alpha 3\beta 1$ -integrins, which represent collagen receptors [43, 44] , while Jurkat cells express $\alpha 4\beta 1$ -integrins but lack $\alpha 2\beta 1$ -, $\alpha 3\beta 1$ -integrins [44] . $\alpha 4\beta 1$ integrins are receptors for VCAM-1 and fibronectin [35] and it has been shown that Jurkat cells interact with human endothelial cells that express VCAM-1 after cytokine treatment or cells transfected with VCAM-1 [45] . Our results are in line with previous

studies showing that $\alpha 2\beta 1$ - and $\alpha 3\beta 1$ -integrin expressing MDA-MB-231 and PC-3 cells were able to rapidly attach to collagen in the cortical bone matrix. In contrast, Jurkat cells were not able to adhere [44] and might preferentially interact with cell adhesion molecules rather than with ECM proteins. $\alpha 2\beta 1$ - and $\alpha 3\beta 1$ -integrins can additionally act as laminin receptors [46] and at least $\alpha 3\beta 1$ integrins recognize fibronectin [46, 47]. Though expressing receptors for fibronectin and laminin, MDA-MB-231 and PC-3 cells adhered to fibronectin to a much lesser extent and did not adhere to laminin, probably due to lower affinities to these extracellular matrix components.

Importantly, v-ATPase inhibition by archazolid increased the surface levels of the extracellular matrix component collagen, which might explain that the increase of MDA-MB-231 and PC-3 cells onto archazolid-treated HUVECs is independent of endothelial cell adhesion molecules. By performing a live cell proteolysis assay, Cavallo-Medved et al. demonstrated ECM degradation, in particular of gelatin and collagen IV, in association with active cathepsin B in caveolae of endothelial cells during tube formation [40]. In addition, recent studies reported that v-ATPase inhibition impairs the activity of cathepsin B in cancer cells [28, 36]. Therefore, we suggested that the accumulation of collagen on the endothelial surface might be a consequence of impaired cathepsin B activity or expression in endothelial cells. In fact, we confirmed the impairment of cathepsin B activity by archazolid as the expression levels of the mature active form of this enzyme was strongly reduced. Cathepsin B is synthesized as preprocathepsin B on membrane-bound ribosomes. Following transport to the Golgi apparatus, the preprocathepsin B is glycosylated with mannose-containing oligosaccharides. The targeting of procathepsin B to lysosomes is mannose-6-phosphate receptor-dependent and its dissociation from the receptor as well as its proteolytic processing into mature cathepsin B requires acidification of the compartment [48]. In cancer cells v-ATPase inhibition by archazolid impaired the mannose-6-phosphate receptor-mediated trafficking from the trans-Golgi network to prelysosomal compartments resulting in a decrease of active lysosomal proteases like cathepsin B [28]. We assumed that the archazolid-induced decrease in cathepsin B activity and expression was based on the same mechanism. Interestingly, overexpression of cathepsin B attenuated the archazolid-induced adhesion of breast cancer cells onto endothelial cells, indicating that the adhesion negatively correlates with the expression of cathepsin B.

As cathepsin B can also degrade other extracellular matrix components such as fibronectin and laminin [38, 49], v-ATPase inhibition could lead to an accumulation of these proteins

and an increased adhesion of cells expressing fibronectin or laminin receptors. However, we did not focus on these ECM components since they were not relevant for the adhesion of MDA-MB-231 and PC-3 cells. These cells predominantly adhered to collagen, while the adhesion of Jurkat cells is mostly independent from the ECM proteins collagen, fibronectin or laminin (S1B Fig). Interestingly [50] . In hepatic cancer cells, archazolid reduces Ras/Raf/MEK/ERK signaling by altering the membrane composition and fluidity [51] . We assume that archazolid affects endothelial cells in a similar way leading to inhibition of Ras signaling and, therefore, reduced transendothelial migration of MDA-MB-231 cells.

Taken together, our study shows that archazolid reduces the activity and expression of cathepsin B in endothelial cells. As a result, the amount of collagen on the surface of endothelial cells was significantly upregulated, which finally resulted in an increased adhesion of the β 1-integrin-expressing metastatic cancer cell lines MDA-MB-231 and PC-3 onto archazolid-treated endothelial cells, whereas the adhesion of non-metastatic Jurkat cells was unaffected. This study shows that the v-ATPase plays an important role in regulating the adhesion of cells expressing receptors for extracellular matrix components. Archazolid represents a promising tool to elucidate the role of v-ATPase in endothelial cells. Moreover, we for the first time linked the function of v-ATPase to the adhesion and transmigration of tumor cells onto endothelial cells as well as to the remodeling of the extracellular matrix on the surface of endothelial cells. The fact that the adhesion of metastatic tumor cells onto endothelial cells is increased while their transendothelial migration is reduced upon inhibition of endothelial v-ATPase by archazolid further supports the view of archazolid as a potential anti-metastatic compound.

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Gastroenteritis and respiratory infection outbreaks in French nursing homes from 2007 to 2018: Morbidity and all-cause lethality according to the individual characteristics of residents

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Abstract: **BACKGROUND:** Gastroenteritis (GE) and respiratory tract infection (RTI) outbreaks are a significant issue in nursing homes. This study aimed to describe GE and RTI outbreaks with infection and all-cause lethality rates according to the individual characteristics of nursing home residents. **METHODS:** Clinical and virological surveillance were conducted (2007 to 2018). Virus stratifications for the analysis were: outbreaks with positive norovirus or influenza identifications (respectively NoV+ or Flu+), episodes with no NoV or influenza identification or testing (respectively NoV- or Flu-). Associations between individual variables (sex, age, length of stay (LOS), autonomy status) and infection and lethality rates were tested with univariate and Mantel-Haenszel (MH) methods. **RESULTS:** 61 GE outbreaks and 76 RTI outbreaks (total 137 outbreaks) were recorded involving respectively 4309 and 5862 residents. In univariate analysis, higher infection rates and age were associated in NoV+, NoV-, and Flu+ contexts, and lower infection rates were associated with longer stays (NoV+ and NoV-). In MH stratified analysis (virus, sex (female/male)) adjusted for LOS (<4 or ≥4 years), the odds of being infected remained significant among older residents (≥86 years): NoV+/male (Odds ratio (OR(MH)): 1.64, 95% confidence interval (CI): 1.16–2.30) and Flu+/female and male (respectively OR(MH): 1.50, CI: 1.27–1.79 and 1.73, CI: 1.28–2.33). In univariate analysis, lower autonomy status (NoV+, Flu+ and Flu-) and increased age (Flu+) were associated with higher lethality. In MH adjusted analysis, significant OR(age) adjusted for autonomy was: Flu+/ ≥86 years compared with <86 years, 1.97 (1.19–3.25) and OR(autonomy) adjusted for age for the more autonomous group (compared with the less autonomous group) was: Flu+, 0.41 (0.24–0.69); Flu-, 0.42 (0.20, 0.90). **CONCLUSION:** The residents of nursing homes are increasingly elderly and dependent. The specific infection and lethality risks according to these two factors indicate that surveillance and infection control measures are essential and of high priority.

Text: Introduction Gastroenteritis (GE) and respiratory tract infection (RTI) outbreaks represent a significant burden of illness in nursing homes. Viruses cause the majority of these outbreaks, and noroviruses and influenza viruses are the most common pathogens [1, 2] .

Previous studies have suggested that viral respiratory infections and norovirus outbreaks are a common cause of hospitalization or death, particularly among elderly individuals [3] [4] [5] .

The impact of outbreaks has been described in terms of both frequency and epidemiology, but little is known about infection rates and all-cause lethality in GE and RTI nursing home outbreaks in relation to the individual characteristics of the residents [6] . The residents of these institutions are increasingly elderly and dependent, and the impact of this trend on the seasonal outbreak burden requires in-depth investigation. The results of studies focused on this issue could yield valuable information for nursing homes, allowing them to adapt their infection control strategies, in particular for improved assessment of infection risk.

Our objective was to describe GE and RTI infection and all-cause lethality rates according to the individual characteristics of nursing home residents (sex, age, length of stay, autonomy status), and to identify specific susceptibility patterns related to these types of viral outbreaks in these facilities.

The present study explored outbreaks in 14 sites (28 units with geriatric nursing home activities for a total of 1121 beds) caring for dependent people in southern Alsace (an area in northeastern France). Data were collected between September 2007 and August 2018 [7, 8] .

Each site was geographically independent and autonomous for social and care management. Units were located within the larger sites and were defined as a place having a dedicated team at one location.

During outbreaks at one site, only the residents in the units with confirmed cases were included. Outbreak inclusion depended on institutional alert to the hygiene team. Surveillance was done in each unit independently, and the members of staff had to inform a physician or charge nurse when two or more potential related cases of pneumonia or GE were observed within four days and when three or more cases were observed for other RTI. Units also had to inform the hygiene team when these threshold values were exceeded. For influenza, the first suspected case led to a local alert and the hygiene team was contacted. A practitioner from the hygiene team collected the information and evaluated the clinical signs, the virology information and the epidemiological context with the physician in the affected unit. The detected cluster was only put under surveillance if the hygiene team considered that there was a potential outbreak phenomenon. The duration of 4 days was in relation with the national protocol with alert to the authorities when 5 cases occurred within 4 days [9, 10] . On a local level and in addition to the clusters reported to the authorities, clusters with at least 3 cases within a period of seven days in one unit could be recorded if they were reported to the hygiene team. Because several outbreaks could potentially occur in the same unit during the surveillance period, a resident could be included repeatedly in different clusters. As a result, the observed patterns reflected the characteristics of an institutional population with longitudinal and pluriannual exposures.

Personal information and clinical information was collected by a practitioner from the hygiene team directly from the residents' health care records. Personal information was collected for all those present the first day of the outbreak. The collected information included: month and year of birth, sex, date of arrival at the nursing home and autonomy status. The autonomy status of residents in French nursing homes is assessed using the AGGIR scale (Autonomy Gerontology Groups Iso-Resources), which is the legal instrument for evaluating dependency in the elderly and whose primary purpose is the allocation of means and resources [11] .

With the AGGIR scale, autonomy is classified into 6 Iso-Resource Groups (GIR): GIR 1 (bedridden or armchair-bound persons, mental functions seriously altered and requiring continuous presence), GIR 2 (bedridden or armchair-bound persons, mental functions not totally altered and requiring assistance in most activities of daily living, or mental functions altered with preserved ability to get around), GIR 3 (preserved mental autonomy with partially preserved motor autonomy and assistance several times a day for physical

autonomy), GIR 4 (moves around the home and sometimes assistance for washing, dressing, physical activities or eating), GIR 5 (only occasional assistance for washing, meal preparation, and housework), GIR 6 (autonomy for essential tasks of daily living). In outbreaks where the influenza virus was identified, influenza vaccination status and oseltamivir prescriptions were recorded as well. A file is transmitted in the Supporting Information with all previous data (S1 Data).

GE was defined as the sudden onset of vomiting and/or diarrhea over a 24 h period: (i) diarrhea 3 episodes, (ii) and/or vomiting 3 episodes, (iii) or diarrhea or vomiting <3 episodes with two or more other symptoms (diarrhea, vomiting, stomach ache, abdominal cramps, nausea, fever, mucus in stools) [1] .

RTI presentation in older adults may be atypical, like for other acute illnesses in this age group [12] . We used the recommended definitions for RTI surveillance in geriatric units, divided in 3 subcategories: (i) common cold syndromes or pharyngitis (at least two of the following criteria: runny nose or sneezing, stuffy nose (i.e. congestion), sore throat or hoarseness or difficulty swallowing, dry cough, swollen or tender glands in the neck (cervical lymphadenopathy)), (ii) influenza-like illness (both the following criteria must be met: fever AND at least three other symptoms (chills, new headache or eye pain, myalgia or body aches, malaise or loss of appetite, sore throat, new or increased dry cough)) and (iii) lower respiratory tract infection (both of the following criteria must be met: at least two respiratory signs or symptoms (new or increased cough, new/increased sputum production, O₂ saturation <94% or reduced >3% from baseline, abnormal lung examination (new or changed), pleuritic chest pain, respiratory rate ≥ 25 breaths/min AND one or more constitutional signs/symptoms (fever, leukocytosis, confusion, acute functional decline)) [13] [14] [15] [16] . Infection corresponding to one of these three subcategories was included in this study and classified as RTI.

For both infection types, the practitioner from the hygiene team obtained clinical information from the patient's health care records, and members of the health care team were consulted if necessary to complete any missing information. At the end of the episode (within seven days after the last identified case), case inclusion as exposed and not infected (ENI) or exposed and infected (EI) was determined with a resident physician.

In order to study the lethality, the presence of each infected resident was evaluated once at least 56 days after the last case of each outbreak. Each resident was followed up retrospectively during eighth 7-day interval (I n, n = 1 to 8, total 56 days, between the date of onset of symptoms and the fifty-sixth day of the studied period) with three different possibilities: present (alive and officially residing in the institution), lost to follow-up (alive at the date of departure but no longer residing in the institution (return home, transfer to another institution)) or death (death recorded in the health care record). The dates of death and lost to follow-up were recorded.

Testing for the virus was not systematic and was decided by the physicians in each institution in the presence of clinical signs.

For GE, stool samples were sent to the National Reference Centre for Gastroenteritis Viruses in Dijon for laboratory testing, as previously described [8] . For RTI surveillance, rapid tests were used to identify the influenza virus. The rapid immunoassay diagnosis tests used for influenza detection were: Clearview1 Exact Influenza A and B (Inverness Medical, Cologne, Germany) from 2007 to 2014 and InfluenzaTop1 (Alldiag, Strasbourg, France) from 2014 to 2018. Given the low sensitivity of influenza rapid tests, they were no longer used once the control measures had been implemented and the influenza outbreak was under control. Samples were also occasionally sent to hospital laboratories or to the National Reference Centre for Influenza Viruses to detect viruses with real-time RT-PCR [17] . Most testing targeted the norovirus (NoV) and influenza virus, but other tests were occasionally performed by the National Reference Centre for Influenza Viruses (rhinovirus, respiratory syncytial virus, human metapneumovirus, parainfluenza 1, 2, 3 and 4, and coronavirus) and the National Reference Centre for Enteric Viruses (rotavirus, astrovirus, and adenovirus).

Because testing for the viruses was variable (from one institution/physician to another, not used in some outbreaks, types of virus sought) and considering the poor sensitivity of the rapid influenza tests, these two sources of data were used to define the epidemiological context of confirmed outbreaks. Consequently, individual cases were included consistently in all episodes according to clinical signs and medical evaluation. When virus testing was negative, the clinical signs were recorded and medical evaluation was used as previously to classify the included residents as infected or not infected.

The epidemiological context of each outbreak was defined according to whether the virus had been identified or not. One or more positive samples led to the qualification of a NoV (NoV+) or flu (Flu+) context. The other episodes were qualified as flu or NoV outbreaks with no specific identification or testing (NoV-and Flu-).

Flu and NoV contexts did not eliminate other potential enteric or respiratory pathogens.

Sex, age, length of stay (LOS, in years) and autonomy status were described for all exposed residents. Influenza vaccination and oseltamivir administration rates were calculated for confirmed influenza outbreaks. Dichotomous or categorical variables were expressed as percentages. In univariate analysis, the categories were specific in order to obtain a precise description of the age and LOS variables. Class intervals were 5 years for age and one year for LOS. The residents classified as GIR 4 to 6 (sometimes, occasional and no assistance) were grouped together because they were few.

For the multi-level analysis with 2x2 tables, a median value was used to define the two-level age categories. For LOS, assessing the longest stays was necessary to identify the effect of longer exposure in a nursing home. Consequently, a four-year cutoff was chosen to create the two categories. For autonomy, the two most dependent categories (GIR 2) were grouped together and compared with the more autonomous categories (GIR 3).

The outbreak epidemiological contexts were used with the four categories: NoV+, Flu+, NoV-and flu-. Other GE or RTI viruses were occasionally identified, but the number of results was too limited to develop separate analyses. However, all the results are available in the tables about the virus investigations along with NoV and influenza identifications.

For the different categories, infection rate (EI/Exposed Residents (ER), in percentage) was calculated according to sex, age group, LOS and autonomy status. To investigate all-cause lethality and define the appropriate period for the 56-day monitoring (D 1 to 56), the all-cause lethality rate per 7-day interval ($LR_n = I_n / n$, $n = 1$ to 8) was calculated: (number of deaths during interval I n / (EI alive the first day of nth studied interval minus lost to follow-up EI during the interval I n) $\times 100$).

Seeing as successive clusters could occur within the same site, potentially influencing all-cause lethality, the serial interval in days (SI_d) was calculated. The SI_d was the time period between the onset of symptoms of the last case in initial outbreak (N) and the onset of symptoms of the first case in the following outbreak (N+1). Investigations were performed when SI_d was shorter or equal to the length of the previous D₁ to 56 and the following parameters were evaluated for these specific situations: number of episodes, residents infected in both outbreaks, and death among the identified individuals.

Finally, according to the death rate and the impact of successive outbreaks, the number of 7-day intervals (N₇) to take into account was defined, and the all-cause lethality rate was analyzed during these periods (I₁ to N_{th} I_n or D₁ to 7_N).

All-cause lethality rates were calculated with the following formula: (number of deaths from D₁ to 7_N / (EI number at D₁ minus lost to follow-up among EI during the period D₁ to 7_N) * 100). Estimation of the turnover rate per 7-day interval among the infected residents was calculated on the base of the LOS (median in years) with the following formula: (proportion of discharged residents: 50.0% in the case of the median) / [(median LOS - 365) / 7]. The average rate of residents discharged per 7-day period was calculated: [(number of lost to follow up during the period D₁ to 7_N / number of exposed and infected residents at D₁) / N_{7-day interval}] * 100.

As some residents were included in several outbreaks during the surveillance, the observations were not completely independent; non-parametric tests were used as a result. In univariate analysis, Chi-square or Fisher exact tests (expected number of frequencies fewer than 5) were used to compare infection and lethality rates according to the studied parameters and the odds ratio was calculated by median-unbiased estimation. The Kruskal-Wallis test was used to compare median values. Confidence intervals for medians were calculated with bootstrap methods.

Covariate adjusted analyses were performed with two tables (2x2). The respective impact of each individual factor was tested with Mantel-Haenszel chi-squared tests. The equality of the stratum odds ratios was tested with the Woolf test of homogeneity. Finally, for each virus context, multiple tables (2x2) were generated and tested with confounding variables, effect modifiers or covariables. Statistical analyses were done using R for Mac OS X version

R 3.4.1 software with RStudio version 1.0.153. A file is transmitted in the Supporting Information with all R codes and the packages used (S1 R Codes). Differences were considered significant at $p = 0.05$.

The French Data Protection Authority approved data collection and analysis (DE-2013-074) and the local ethics committee (Espace Local de Réflexion Ethique, Centre Hospitalier de Rouffach) approved the study protocol (ERLE-32). According to the French law for biomedical research and human experimentation, individual written consent was not required from the patients or their relatives for data collection. Each year, the referring local practitioner of the study coordinated with the doctors working in the nursing home. At the beginning of the surveillance period, information regarding participation in the study was displayed in the family visiting area, including a document about their right to access and rectify personal data. After collection, data were rendered anonymous. No specific authorization was needed to retrospectively analyze anonymous data collected during routine care in the context of routine surveillance.

A total of 137 outbreaks were recorded in the 14 sites. RTI outbreaks were more frequent than GE outbreaks (76 outbreaks and 5862 exposed residents vs. 61 outbreaks and 4309 exposed residents, respectively). Overall, 7643 of the exposed residents were women and 2528 were men. The median age was 86.7 years old (interquartile range: 81.1-91.0 years).

Virus investigations (respectively 389 samples for RTI and 143 for GE with all the detailed results in S1-S4 Tables) confirmed a considerable number of norovirus-related GE outbreaks (34/61) and influenza-related RTI outbreaks (46/76). For GE outbreaks, 2524 residents were in a NoV+ context versus 1785 in a NoV-context, and for RTI outbreaks, 3479 residents were in a Flu+ context versus 2383 in a Flu-context.

For GE surveillance in the NoV+ context, there were 1093 EI residents versus 1431 ENI residents, whereas in the NoV-context, there were 583 EI residents versus 1202 ENI residents. Therefore, the infection rate was higher in the NoV+ context (43.3%) than in the NoV-context (32.7%, (odds ratio (OR): 0.63, 95% confidence interval (CI): 0.56-0.72), $p < 0.001$).

For RTI surveillance, the rates of infection were similar with and without confirmed influenza: 31.5% (N = 1095 EI residents /3479 exposed residents) vs. 30.5% (N = 728 EI residents/ 2383 exposed residents, OR: 0.96, CI: 0.85-1.07, p = 0.47). Moreover, infection rate in the NoV + context was higher than the three other contexts: NoV-(OR: 0.63, CI: 0.56-0.72), Flu+ (OR: 0.60, CI:0.54-0.67) and Flu-(OR: 0.58, CI: 0.51-0.65).

In univariate analysis, certain individual characteristics were associated with significant variations in the infection rate (S5 Table) . The infection rate increased with age (except in the Flucontext) and, decreased with LOS during GE outbreaks. The covariate adjusted analysis revealed specific significant effect modification according to sex (NoV+) and LOS (NoV-) (S6 Table) . In analyses stratified according to virus and sex, age adjusted for LOS remained significant for Flu+ and NoV+ outbreaks (males). In NoV-context, the effect modification of LOS remained significant (Table 1) . Finally, when autonomy was included and adjusted for age (virus, sex, LOS stratification), the less autonomous residents (female/LOS<4 years/age<86/ GIR 1-2) were affected more severely by Flu+ outbreaks with specific effect modification according to age (S7 Table) . The study of lethality rates in infected residents over the 56 days after onset indicated that there were significant variations for RTI but no change for GE (Table 2) . Significant differences appeared after 28 days in the context of Flu+ outbreaks and other RTI outbreaks.

The analysis of successive or simultaneous clusters in the same institutions was performed when the time period between the onset of symptoms of the last case in outbreak N and the onset of symptoms of the first case in outbreak N+1 was 56 days (S8 Table) . 44 of the 137 outbreaks (32.12%) were identified, and 194 of the 3499 exposed and infected residents contracted multiple infections. The percentage of exposed and infected residents implicated in more than one virus stratification was 11.09% ((194 - 2)/3499). Moreover, two deceased residents were included in the NoV-Na and Flu lethality analyses because death occurred within 56 days for both infections. The analysis of virus stratification of the 44 outbreaks showed the absence of successive clusters for the same category. The same analysis for the first four 7-day intervals (Days 1 to 28) showed the respective values: 26 outbreaks (18.98%), 117 residents ((6.69% ((117 - 2)/3499)), one dead resident.

Finally, according to the higher lethality impact during the first four 7-day intervals and to limit the impact of successive clusters in the same site, all cause lethality rates were

studied according to individual parameters for the four 7 days intervals with the respective number of According to the surveillance type (GE or RTI), the lethality rates differed significantly: 1.6% versus 3.4% (respectively NoV+ and NoV-contexts, OR: 2.24, CI: 1.16-4.39, $p = 0.02$) and 8.3% versus 5.6% (respectively Flu+ and Flu-, OR: 0.67, CI: 0.45-0.97, $p = 0.04$).

In univariate analysis (S9 Table) , low autonomy status in the NoV+, Flu+ and Flu-contexts was most significantly associated with increased all-cause lethality, and age was associated with higher lethality in the Flu+ context. In the adjusted analysis, no significant statistical differences were identified in GE outbreaks. For RTI episodes, the adjusted analysis showed that autonomy had a significant impact when adjusted for sex, age or LOS (Flu+ and Flu-NA) and that age had a significant impact when adjusted for sex, autonomy or LOS (Flu+) (S10 Table) .

In Table 3 , the specific effects of age or autonomy were tested. Significant OR age adjusted for autonomy were: Flu+/age ≥ 86 years (compared with the <86 group), 1.97 (1.19-3.25). OR autonomy adjusted for age were for GIR 3-6 (compared with GIR 1-2): Flu+, 0.41 (0.24-0.69); Flu-, 0.42 (0.20, 0.90).

Finally, despite the low number of residents and deaths per category, and consequently the limited robustness of the results, autonomy adjusted for age with stratification according to virus, sex and LOS showed that the effects were higher among subgroups of less autonomous residents (female or male/LOS <4 years/GIR 1-2) in Flu+ outbreaks, and there was also higher mortality in the small subgroup of autonomous men with LOS ≥ 4 years (higher mortality) (S11 Table) .

In the Flu+ context, data regarding vaccination status and oseltamivir prescriptions were available but not used in this study.

In the present study, surveillance data obtained during GE and RTI outbreaks in nursing homes were used to construct stratified analyses and to identify specific infection and all-cause lethality rates according to the residents' individual characteristics.

The infection rates observed here were similar to those found in previous studies of NoV and Influenza outbreaks (odds of being infected during a Flu+ outbreak were around 40% less than during a NoV+ outbreak). Reported infection rates were close to 30.0% in influenza outbreaks and 40.0% in NoV outbreaks [18] [19] [20] .

Older age appeared to increase the likelihood of GE and influenza infection, with increasing rates among older residents. Age is a well-known factor for influenza and norovirus severity in the elderly and in nursing homes [21, 22] . For NoV, the highest incidence estimates (5-year age strata) was found in the 85 year-category (approximately 800 men and for 1,400 women per 100,000 inhabitants). In our study, univariate analysis (NoV+) showed that the odds of being infected were 1.5 to 1.6 times higher if a resident was older than 85. Moreover, an adjusted analysis of GE outbreaks highlighted different effects among subgroups of residents according to sex and LOS. Indeed, multiple and/or repeated exposure to GE viruses while institutionalized may lead to susceptibility or possible increased immunity in some residents [23] . For the sex variable, two factors could explain the effect: a possible selection bias with men reporting mild infections less than women (particularly in the <86 years subgroup) or that male susceptibility was different (age, LOS, immunity, . . .). A German study from 2013 also reported a greater impact in women [21] . Moreover, when age analysis was stratified by sex and LOS, no Epidemic impacts in nursing homes significant impact was observed in women; the only significant differences were fewer infections in men in the <86-subgroup (except in the NoV-with LOS <4 years).

For the RTI outbreaks, sex and LOS variables did not have a significant effect. In residents older than 86, the odds of being infected in Flu+ context were 1.5 times higher for women and 1.7 for men. In univariate analysis, contrary to the other virus contexts where odds ratios were rarely above 2, residents over 95 years old had increased odds of infection of 2.8 compared with the 70-year-old category, and for the 100 year-old group the odds were approximately 3.8. In the Flu+ context, autonomy adjusted for age (virus, sex and LOS stratification) revealed a possible increase in infection rates among less autonomous residents. A previous study found that when elderly residents were exposed to the A(H3N2) virus, there were higher rates of infection and reinfection, and more significant effects on the institution than with other influenza types/subtypes. In the community, the relative illness ratio (RIR) in the [22, 24] .

The incidence of influenza infection and the associated risks were well described by age group, but the specific impact according to age was not studied. The results of this work highlighted the specific age distribution of influenza illnesses among the nursing home residents and the more significant impact among the older residents. This specific susceptibility could be a critical factor in the institutional exposure and dissemination of influenza and could partly explain the high infection impact in the elderly institutional population.

In this work, autonomy status was not the main factor associated with infection (no significant impact in GE and in Flu-contexts). However, in Flu+ outbreaks, a high level of dependency was associated with a higher risk of falling ill. This observation implies that staff could play a role in the spread of infection (highly dependent and less mobile residents are less likely to contaminate themselves) or that the more active residents may be less fragile and/or have a greater involvement in the recommended infection control measures. Finally, improving compliance with personal hygiene measures both for nursing staff and residents might be expected to have a beneficial effect on infection rates. Previous studies identified higher NoV infection rates in highly dependent individuals, but the results were not adjusted for age and LOS to take into account the potential correlation with the autonomy status [20] .

Lethality is difficult to assess in nursing homes because death is frequent. Our GE and RTI episodes occurred during the winter seasons, and there are possible interactions between outbreaks and increased mortality at this time of the year [25] . The all-cause lethality rate of the infected residents in our study reflected global mortality including GE and RTI outbreaks and the global epidemiological context. Not surprisingly, a higher all-cause lethality rate was observed in the influenza contexts, as reported in previous studies [25] [26] [27] . Age and autonomy had similar effects in the different contexts, but in GE and to a lesser degree in Flu-outbreaks, the relatively small number of deaths could have limited the power of the statistical tests. In nursing homes, residents are generally discharged due to death. The number of residents lost to follow up was low (0.14% in the first 28 days), so the 7-day interval turnover rate calculated on the base of the median LOS provides a good indication of the average case fatality rate. The lethality rate for NoV+ outbreaks was similar to the estimated 7-day interval turnover rate (1.6%) indicating that this context had a limited impact on the death rate.

The all-cause lethality rate was most affected by age and autonomy. Both individual characteristics were significant in the Flu+ outbreaks, and autonomy adjusted for age was significant in the Flu-episodes. The influence of age on mortality in a context of influenza has already been described: a very high mortality rate (831/100,000 inhabitants) was reported in persons 90 years of age and older compared with those aged 65-69 years (23/100,000 inhabitants) [28] . In our univariate analysis, the higher risk was observed in the 90 group whose risk of death was at least 2.6 higher than the <70 group. When adjusted for autonomy, the impact of age was not significant in more autonomous residents in the Flu+ context and not at all in the Flucontext. The opposite analysis (autonomy adjusted for age) showed higher global impact in the less autonomous group (Flu-) or only in the 86 age group (Flu+). Age and autonomy are a reflection of resident's level of frailty. Clinical frailty scores were not used in this study, but in a previous study of patients with critical illness, they were associated with greater mortality, regardless of age [29] . This suggests that in addition to age, autonomy can be a valuable indicator for the assessment of outbreak impact in outbreak surveillance. Other studies have suggested that age and certain comorbidities are independent risk factors for the influenza mortality rate or that mortality increase according to the number of risk factors [28, 30] . Comorbidities and underlying diseases of various severities could reflect overall frailty and consequently the risk of death. In nursing homes, information about autonomy and age are easier to collect and interpret than data on comorbidities. These various approaches should be evaluated and compared in the goal of optimizing risk assessment among nursing home residents.

The present work has two main limitations. First, the virus information was incomplete (limited identification, mainly influenza rapid tests for the RTI). Consequently, some episodes in the levels with no available identification may also have been associated with influenza or norovirus, and multiple contaminations could have been underestimated or not taken into account. Moreover, vaccination and oseltamivir prescriptions were recorded but not included because the influenza genotype was not determined and identification was limited.

Secondly, the deaths of uninfected residents were not recorded in this protocol even though such data would have provided valuable information about the global epidemiological context.

In conclusion, specific susceptibility patterns were observed among exposed residents. In this cohort of nursing homes, infection rates varied according to virus, sex, length of stay and age, and there were major differences in lethality depending on virus, age and autonomy score. The collected data were easy to record and could be used to improve the characterization of seasonal outbreaks in nursing homes, whose residents are particularly vulnerable. Finally, as the average age and dependency level of residents continues to increase, subsequently increasing the risk of infection and death, health care staff will have to be increasingly vigilant during seasonal outbreaks and targeted interventions should be implemented.

Supporting information S1 Data. (CSV) S1 R Codes. (R) S1 Table. (XLSX) S10 Table. (XLSX) S11 Table. (XLSX)

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Critical care response to a hospital outbreak of the 2019-nCoV infection in Shenzhen, China

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Abstract: nan

Text: The main challenge may include (1) early identification of outbreak, (2) rapid expansion of patients, (3) high risk of nosocomial transmission, (4) unpredictability of size impacted, and (5) lack of backup resource. These challenges have caused severe shortage of healthcare workers, medical materials, and beds with isolation. The Spring Festival holiday has greatly aggravated the shortage of human resources and heavy traffic flow due to the vacation of healthy workers and factory workers, which further magnified the risk of transmission. The key point is to discriminate the infectious disease outbreak from regular clustering cases of flu-like diseases at early stage. There is a trade-off between false alarm causing population panic and delayed identification leading to social crisis.

Early identification of 2019-nCoV infection presents a major challenge for the frontline clinicians. Its clinical symptoms largely overlap with those of common acute respiratory illnesses, including fever (98%), cough (76%), and diarrhea (3%), often more severe in older adults with pre-existing chronic comorbidities [1] . Usually, the laboratory abnormalities include lymphocytopenia and hypoxemia [1] . The initial chest radiographs may vary from minimal abnormality to bilateral ground-glass opacity or subsegmental areas of consolidation [1] . In addition, asymptomatic cases and lack of diagnosis kits result in delayed or even missed diagnosis inevitable and makes many other patients, visitors, and healthcare workers exposed to the 2019-nCoV infection.

Critical care response to the outbreak of coronavirus should happen not only at the level of hospital, but also at the level of the city which is dominated by the government. At the early stage, the size of the patients' population is not beyond the capability of local infectious diseases hospital (IDH). The general hospital is responsible for fever triage, identifying suspected cases, and transferring to the local IDH. Such a plan is mandatory for every hospital. Shenzhen city has established a preexisting Infectious Disease Epidemic Plan (IDEP), which has facilitated managing and containing local outbreak of the 2019-nCoV. In case the patient load exceeds the hospital capability of the IDH, new IDHs should be considered either by building a temporary new IDH or reconstructing an existing hospital. Wuhan, the epicenter of the outbreak, is racing against time to build two specialized hospitals for nCoV patients, namely Huoshenshan and Leishenshan hospital, whereas a different strategy has been undertaken in Shenzhen city by reconstructing an existing hospital to become an IDH with capability of 800 beds.

2019-nCoV patients should be admitted to singlebedded, negative pressure rooms in isolated units with intensive care and monitoring [2] . Clinical engineering should have plans to reconstruct standard rooms [2] . Retrofitting the rooms with externally exhausted HEPA filters may be an expedient solution. Also, the general hospital may consider procedures such as suspending elective surgeries, canceling ambulatory clinics and outpatient diagnostic procedures, transferring patients to other institutions, and restricting hospital visitors [2] . More importantly, because the hospitals' ability to respond to the outbreak largely depends on their available ICU beds, the plan to increase ICU bed capacity needs to be determined.

Caring for 2019-nCoV patients represents a substantial exposure risk for ICU staff because of the following reasons: highly contagious with multiple transmission route, high exposure dose, long daily contact hours, and ICU stay. The basic reproductive number was estimated to be 2.2 (95% CI, 1.4 to 3.9) [3] , or as high as between 3.6 and 4.0 [4] . The 2019-nCoV is proved to be transmitted by respiratory droplets, contact, and fecal-oral, even transmission through the eye is possible [5, 6] . The higher viral load and aerosol-generating procedures, such as noninvasive ventilation, magnify the exposure and transmission risk [2, 7, 8] . Moreover, virus shedding can be prolonged and last for > 3 weeks according to some literature and our unpublished data [2] . Healthcare providers and those in contact with infected patients should utilize contact, droplet, and airborne precautions with N95 respirator. Strict infection prevention and control practices have been implemented and audited in our units following the infection prevention and control plan published by China's National Health Committee (CNHC). In addition, well-equipped fever clinic as triage station with trained staff knowing 2019-nCoV case definitions is established. For suspected 2019-nCoV infection, several key points are crucial procedures: recording a detailed history, standardizing pneumonia workup, obtaining lower respiratory tract specimens [2, 8] , and implementing droplet isolation to break the transmission chain in the healthcare setting [2] .

The risk of 2019-nCoV exposure may cause significant psychosocial stress on healthcare workers [2] . The death of a retired ENT physician from a 2019-nCoV infection has added to fears in January 2020. Psychotherapists have also been invited to join medical teams to evaluate and deal with potential stress and depression for the safety of the healthcare workers.

Critical management 2019-nCoV management was largely supportive, including intubation, early prone positioning, neuromuscular blockade, and extracorporeal membrane oxygenation (ECMO) according to the recommendations updated by CNHC. Low-dose systematic corticosteroids, lopinavir/ritonavir, and atomization inhalation of interferon were encouraged. These critical managements have worked well so far, as our 2019-nCoV patients had zero mortality. On the contrary, the previously reported mortality of 2019-nCoV patients in Wuhan ranged from 11 to 15% [1, 9] .

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Statistics-Based Predictions of Coronavirus Epidemic Spreading in Mainland China

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Abstract: Information about the open-access article 'Statistics-Based Predictions of Coronavirus Epidemic Spreading in Mainland China' in DOAJ. DOAJ is an online directory that indexes and provides access to quality open access, peer-reviewed journals.

Text: Here, we consider the development of an epidemic outbreak caused by coronavirus COVID-19 (the previous name was 2019-nCoV) (see e.g., [1] [2] [3]). Since long-term data are available only for mainland China, we will try to predict the number of coronavirus

victims V (number of persons who caught the infection and got sick) only in this area. The first estimations of $V(t)$ exponential growth versus time t , typical for the initial stages of every epidemic (see e.g., [4]) have been done in [3] . For long-time predictions, more complicated mathematical models are necessary. For example, a susceptible-exposed-infectious-recovered (SEIR) model was used in [2] . Nevertheless, complicated models need more effort for unknown parameters identification. This procedure may be especially difficult if reliable data are limited.

In this study, we use the known SIR model for the dynamics of an epidemic [4] [5] [6] [7] [8] . For the parameter identification, we will use the exact solution of the SIR set of linear equations and statistical approach developed in [4] (tested also in [9]). These methods were applied for investigation of the children disease, which occurred in Chernivtsi (Ukraine) in 1988-1989. We will estimate some of the epidemic characteristics and present the dependencies for victim numbers, infected and removed persons versus time.

We shall analyze the daily data for the number of confirmed cases in mainland China, which origins from the National Health Commission of the People's Republic of China [1] . A part of the official diagram (its version, presented on February 15, 2020) is shown in Fig. 1 . For calculations, we have used the data for the period of time from January 16 to February 9, 2020. The numbers shown after February 9 were used for verification of predictions.

On February 12, 2020, the National Health Commission of the People's Republic of China has added 12289 new cases (not previously included in official counts) as "clinically diagnosed cases". The cases, reported by this official organization before, have the name of "tested confirmed cases" [1] . To avoid confusion, we will denote "tested confirmed cases" as W_j ; j corresponds to the different time moments t_j (see the Table) . Let us denote the "clinically diagnosed cases" as Q_j . The sum of W_j and Q_j is shown in the last column in Fig. 1 and in the Table. The Table shows that the precise time of the epidemic beginning t_0 is unknown. Therefore, the optimization procedures have to determine the optimal value of this parameter as well as for other parameters of SIR model. The sum of "tested confirmed cases" and "clinically diagnosed cases" $W_j + Q_j$

16	0	45	1	16	14380	Unknown
17	1	62	2	17	17205	Unknown
18	2	121	3	18	20440	Unknown
19	3	198	4	19	24324	Unknown
20	4	291	5	20	28018	Unknown
21	5	440	6	21	31161	Unknown
22	6	571	7	22	34568	Unknown
23	7	830	8	23	37198	Unknown
24	8	1287	9	24	40171	Unknown
25	9	1975	10	25	42638	Unknown
26	10	2744	11	26	44653	Unknown
27	11	4515	12	27	46472	

58761 28 12 5974 13 28 48467 63851 29 13 7711 14 29 49970 66492 30 14 9692 ---
 -31 15 11791 ----

The SIR model for an infectious disease can be written as follows [6, 7] :

,

The number of susceptible persons is S , infected (persons who are sick and spread the infection) I , removed (persons who do not spread the infection anymore, this number is the sum of isolated, recovered and dead people) R ; the infection and immunization rates are β and γ respectively.

It follows from (1) and (2) that

Integration of (5) with the initial conditions (4) yields:

Function I has a maximum at $S = S^*$ and tends to zero at infinity, see [6, 7] . In comparison, the number of susceptible persons at infinity 0, S^* and can be calculated with the use of (6) from a non-linear equation

yields:

Thus, for every set of parameters $N, \beta, \gamma, \alpha, \delta, \epsilon, \theta$ and a fixed value of V the integral (10) can be calculated and the corresponding moment of time can be determined from (9) . Then I can be calculated from (6) by putting $S = N - V$ and function R from

Statistical approach for parameter identification. Linear regression As in paper [4] , we shall use the fact that the random function $1 - (I/N)^{\alpha}$ has a linear distribution (see (9)). Then we can apply the linear regression (see [10]) for every pair of parameters N and α and

calculate the corresponding values of 0 t and τ . The optimal (the most reliable) values of N and τ correspond to the maximum value of the correlation coefficient r (see [4, 9]).

Since we did not know and still don't know the values of Q_j before February 12, 2020, we supposed that $V_j = W_j$ and have done the calculations with the use of data for the time period from January 16 to February 9, 2020. The optimal values of the parameters are:

The corresponding correlation coefficient is very high $r = 0.997966487046645$. The solution of (7) yields the value 45579.

The corresponding number of infected I , susceptible S and removed R persons versus time (starting from January 16, 2020) were calculated and shown in Fig. 2. The blue line represents the number of victims $V = I + R$ and is in good agreement with "tested confirmed cases" W_j , reported by the National Health Commission of the People's Republic of China [1] (blue markers).

Unfortunately, many cases have not been included in the official counts and have appeared in the official Table from [1] only on February 12 as "clinically diagnosed cases" Q_j (see Fig. 1). Since the National Health Commission of the the People's Republic of China has proposed two different ways of registration of the same disease [1], V_j must be the sum of W_j and Q_j , i.e. $V_j = W_j + Q_j$ (provided that no new methods of registering the same disease would appear). Values W_j after February 9 are shown in Fig. 3 by "stars". "Crosses" represent the sum $W_j + Q_j$.

Since the optimal curve was obtained only with the use of W_j and the difference between W_j and V_j is very big (e.g., it was 12 289 persons on February 12, 2020), the predictions shown in Fig. 2 and reported in [11] are no longer relevant. To have better predictions, it is necessary to have exact Q_j data for the period before February 12. Blue markers show the "tested confirmed cases" W_j , reported by the National Health Commission of the People's Republic of China [1]. The "circles" correspond to the points used for calculations (it was supposed that $V_j = W_j$); "stars" - to the points used only for verification. "Circles" show the "tested confirmed cases" W_j for the period from January 16 to February 9, 2020, [2]. These points were used to calculate the prediction curve. "Stars" correspond to the "tested

confirmed cases" W_j for the period from February 10 to February 14, 2020, [1] . "Crosses" represent the sum $W_j + Q_j$ from [1]

The simple mathematical model was used to predict the characteristics of the epidemic caused by coronavirus in mainland China. The numbers of infected, susceptible, and removed persons versus time were predicted and compared with the new data obtained after February 10, 2020, when the calculations were completed. Unfortunately, many cases have not been included in the official counts and have appeared on February 12 only. It makes the predictions reported on February 10, 2020, no longer relevant. Further research should focus on updating the predictions with the use of corrected data and more complicated mathematical models.

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Vesicular stomatitis virus with the rabies virus glycoprotein directs retrograde transsynaptic transport among neurons in vivo

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Abstract: Defining the connections among neurons is critical to our understanding of the structure and function of the nervous system. Recombinant viruses engineered to transmit across synapses provide a powerful approach for the dissection of neuronal circuitry in vivo. We recently demonstrated that recombinant vesicular stomatitis virus (VSV) can be endowed with anterograde or retrograde transsynaptic tracing ability by providing the virus with different glycoproteins. Here we extend the characterization of the transmission and gene expression of recombinant VSV (rVSV) with the rabies virus glycoprotein (RABV-G), and provide examples of its activity relative to the anterograde transsynaptic tracer form of rVSV. rVSV with RABV-G was found to drive strong expression of transgenes and to spread rapidly from neuron to neuron in only a retrograde manner. Depending upon how the RABV-G was delivered, VSV served as a polysynaptic or monosynaptic tracer, or was able to define projections through axonal uptake and retrograde transport. In animals co-infected with rVSV in its anterograde form, rVSV with RABV-G could be used to begin to characterize the similarities and differences in connections to different areas. rVSV with RABV-G provides a flexible, rapid, and versatile tracing tool that complements the previously described VSV-based anterograde transsynaptic tracer.

Text: Mapping neuronal connectivity in the central nervous system (CNS) of even simple organisms is a difficult task. Recombinant viruses engineered to trace synaptic connections and express transgenes promise to enable higher-throughput mapping of connections among neurons than other methods, e.g., serial reconstruction from electron micrographs (Bock et al., 2011; Briggman et al., 2011) . The Pseudorabies (PRV) and Rabies viruses (RABV) have been the best characterized and most utilized circuit tracing viruses to date (Ugolini et al., 1989; Kelly and Strick, 2000) . RABV was recently modified by Wickersham and colleagues such that it can travel across only one synapse, allowing for a straightforward definition of monosynaptic connections (Wickersham et al., 2007b) . This strategy permitted the first unambiguous identification of retrogradely connected cells from an initially infected cell ("starter cell"), without the need for electrophysiology. Moreover, the starter cell could be defined through the expression of a specific viral receptor that limited the initial infection.

Recently, we created an anterograde monosynaptic virus that complements the previously available retrograde viral tracers (Beier et al., 2011) . Vesicular stomatitis virus (VSV), a virus related to RABV, with its own glycoprotein (G) gene (VSV-G), or with a G from the unrelated lymphocytic choriomeningitis virus (LCMV-G), spreads in the anterograde direction across synapses. VSV can be used as a polysynaptic tracer that spreads across

many synapses, owing to the fact that the normal, replication-competent form of the virus does not cause serious diseases in humans (Brandly and Hanson, 1957; Johnson et al., 1966; Brody et al., 1967) . Whether the virus is a monosynaptic or polysynaptic tracer is determined by the method of delivery of the G gene (Figure 1A) . Advantages of VSV are that it is well-characterized, is relatively simple in comparison to PRV, and it rapidly grows to high titer in tissue culture cells. It is also being developed as a vaccine vector, often using a G of another virus as the immunogen, as well as being developed as a cytotoxic agent that will target tumor cells in humans (Balachandran and Barber, 2000; Stojdl et al., 2000 Stojdl et al., , 2003 .

Previous studies of the anatomical patterns of transmission, as well as physiological recordings, have shown that the transmission of VSV and RABV among neurons is via synapses (Kelly and Strick, 2000; Wickersham et al., 2007b; Beier et al., 2011) . In addition, it has been shown that RABV, as well as lentiviruses with RABV-G in their envelope, travel retrogradely from an injection site (Mazarakis et al., 2001; Wickersham et al., 2007a) . We hypothesized that providing a recombinant VSV (rVSV) with the RABV-G would create a retrograde polysynaptic transsynaptic tracer without the biosafety concerns inherent to RABV. Our initial characterization of rVSV with RABV-G showed that indeed

FIGURE 1 | Synaptic tracing strategies using VSV. (A) Schematic illustrating the strategies for polysynaptic or monosynaptic retrograde or anterograde transsynaptic transmission of rVSV encoding GFP. The initially infected cell is indicated by an asterisk. VSV encoding a glycoprotein (G) within its genome can spread polysynaptically. The direction of the spread depends on the identity of the glycoprotein. Infected neurons are shown in green. In some cases, the initially infected starter cell can be defined by the expression of an avian receptor, TVA (tagged with a red fluorescent protein). The TVA-expressing neurons can then be specifically infected by rVSV G with the EnvA/RABV-G (A/RG) glycoprotein (Wickersham et al., 2007b) on the virion surface [rVSV G(A/RG)]. These starter cells are then yellow, due to viral GFP and mCherry from TVA-mCherry expression. For monosynaptic tracing, the G protein is expressed in trans in the TVA-expressing cell, and thus complements rVSV G to allow transmission in a specific direction. (B) Genomic diagrams of rVSV vectors. All VSVs contain four essential proteins: N, P, M, and L. Some viruses encode a G gene in their genome, which allows them to spread polysynaptically. rVSV vectors typically encode a transgene in the first position, while others carry an additional transgene in the G position. (C) Morphological characterization of rVSV-infected neurons in several locations within the mouse brain. (i,ii) Caudate-putamen (CP) neurons at 4 dpi from an injection of the CP with rVSV(VSV-G) viruses encoding (i) CFP or (ii) Korange. (iii) Labeled neurons of the CA1 region of the hippocampus are shown at 5 dpi following injection into the hippocampus of

rVSV(VSV-G) encoding Venus. (iv,v) Cortical pyramidal neurons are shown following injection into the CP of rVSV(RABV-G) expressing (iv) GFP at 24 hpi, or (v) mCherry at 48 hpi. Inset in (iv) is a high magnification of the neuron in panel (iv), highlighting labeling of dendritic spines. (vi) Multiple viruses can be co-injected into the same animal. Here, individual rVSV G(VSV-G) viruses encoding CFP, GFP , Venus, Korange, and mCherry were used to infect the cortex. Scale bars = 50 μ m.

www.frontiersin.org February 2013 | Volume 7 | Article 11 | 2 it could be taken up as a retrograde tracer (Beier et al., 2011) .

To determine if it could transmit among neurons following its replication in neurons, and to further analyze the transmission patterns of both the monosynaptic and polysynaptic forms of rVSV with RABV-G, we made injections into several CNS and peripheral locations. In addition, we performed co-infections of rVSV with RABV-G and the anterograde form of rVSV in order to exploit the differences in the directionality of transmission of these two viruses in mapping circuits.

Schematics of viruses created and used throughout this study are shown in Figure 1 . We created rVSV vector plasmids carrying different transgenes in either the first or fifth genomic positions (Figure 1B) . After rescuing each virus, we tested the ability of each to express transgenes in different brain regions through intracranial injections (Figure 1C) . All rVSV vectors drove robust fluorophore expression 1 or 2 days post-infection (hpi) (Figure 1C) (van den Pol et al., 2009) . In fact, by 12 hpi, labeling was sufficiently bright to image fine morphological details, such as dendritic spines (Figure 1C ,iv).

To characterize the physiological properties of cells infected with rVSV, we tested a replication-competent rVSV encoding GFP, with RABV-G in the genome in place of VSV-G [hereafter designated rVSV(RABV-G)]. van den Pol et al. reported that hippocampal neurons infected with replication-incompetent (G-deleted or " G") rVSV were physiologically healthy at 12-14 hpi, but were less so by 1 day post-infection (dpi) (van den Pol et al., 2009) . Given the known toxicity of both VSV and RABV-G (Coulon et al., 1982) , we tested the physiology of cortical pyramidal neurons in the motor cortex (M1) infected with rVSV(RABV-G). Between 12 and 18 hpi, the membrane capacitance, input resistance, resting membrane potential, and current-to-action potential firing relationship were indistinguishable

between infected and uninfected neurons (Figure 2) . However, by 2 dpi, electrophysiological properties were so abnormal in the infected cortical pyramidal cells that physiological measurements could not be made.

The speed and strength of the expression of transgenes encoded by VSV depends upon the gene's genomic position (van den Pol et al., 2009; Beier et al., 2011) . Genes in the first position are expressed the most highly, with a decrease in the level of expression in positions more 3 within the viral plus strand. When GFP was inserted into the first position of VSV, GFP fluorescence was first detectable at approximately 1 hpi in cultured cells (van den Pol et al., 2009) . In order to quantify the relative expression of a fluorescent protein in the first genomic position in neurons, rat hippocampal slices were infected with a replication-incompetent rVSV that expresses mCherry (rVSV G, Figures 1A,B) . This was a G virus which had the RABV-G supplied in trans during the preparation of the virus stock [referred to as rVSV G(RABV-G)].

Average fluorescence intensity of the infected cells was measured every hour over the course of 18 h. By 4 hpi at 37 • C, red fluorescence was clearly visible, and reached maximal levels by approximately 14 hpi (N = 3, Figure 3) . Similar results were obtained with a virus encoding GFP in the first genomic position rather than mCherry (i.e., Figure 1B) (N = 3).

We previously demonstrated that rVSV(RABV-G) could be taken up retrogradely by neurons (Beier et al., 2011) , but these experiments did not distinguish between direct axonal uptake of the initial inoculum vs. retrograde transsynaptic transmission following viral replication. To distinguish between these two mechanisms and to extend the previous analyses, we conducted further experiments in the mammalian visual system (Figures 4A-G) . As visual cortex area 1 (V1) does not receive direct projections from retinal ganglion cells (RGCs), but rather receives secondary input from RGCs via the lateral geniculate nucleus (LGN), infection of RGCs from injection of V1 would demonstrate retrograde transmission from cells which supported at least one round of viral replication. Following a V1 injection with rVSV(RABV-G), GFP-positive RGCs were observed in the retina by 3 dpi (N = 3; Figure 4G) . Importantly, viral labeling in the brain was restricted to primary and secondary projection areas, even at 7 dpi. These included the LGN (Figure 4D) and the hypothalamus (Figure 4E) , two areas known to project directly to V1 (Kandel, 2000) . Selective labeling was observed in other areas, such as cortical areas surrounding V1

(Figure 4C) , which project directly to V1, and also in the superior colliculus (SC) stratum griseum centrale, which projects to the LGN (Figure 4F) . Labeling was also observed in the nucleus basalis, which projects to the cortex, as well as many components of the basal ganglia circuit, which provide input to the thalamus [such as the caudate-putamen (CP), globus pallidus (GP), and the subthalamic nucleus (STn)]. The amygdala, which projects to the hypothalamus, was also labeled. Consistent with a lack of widespread viral transmission, animals did not exhibit signs of disease at 7 dpi.

These data show that rVSV(RABV-G) can spread in a retrograde direction from the injection site, but do not address whether the virus can spread exclusively in the retrograde direction. Directional transsynaptic specificity can only be definitively addressed using a unidirectional circuit. We therefore turned to the primary motor cortex (M1) to CP connection, in which neurons project from the cortex to the CP, but not in the other direction (Figure 4H) (Beier et al., 2011) . Injections of rVSV(RABV-G) into M1 should not label neurons in the CP if the virus can only label cells across synapses in the retrograde direction. Indeed, at 2 dpi, areas directly projecting to the injection site, including the contralateral cortex, were labeled (Figure 4I). Only axons from cortical cells were observed in the CP, with no GFP-labeled cell bodies present in the CP (Figure 4J) , consistent with lack of anterograde transsynaptic spread. By 3 dpi, a small number of medium spiny neurons (MSNs) in the CP were observed, likely via secondary spread from initially infected thalamic or GP neurons (data not shown).

A particular advantage of retrograde viral tracers is the ability to label CNS neurons projecting to peripheral sites. This has been a powerful application of both RABV and PRV (Ugolini et al., 1989; Standish et al., 1994) . To test if rVSV(RABV-G) could also perform this function, we examined the innervation of the dura surface by neurons of the trigeminal ganglion, a neuronal circuit thought to be involved in migraine headaches (Penfield and McNaughton, 1940; Mayberg et al., 1984) . These neurons have axons, but not canonical dendrites, and send projections into the spinal cord and brainstem. Therefore, the only way trigeminal neurons could become labeled from viral application to the dura is through retrograde uptake of the virus. We applied rVSV(RABV-G) to the intact dura mater and analyzed the dura, trigeminal ganglion, and CNS for labeling (Figure 4K). At the earliest time point examined, 3 dpi, we observed axons traveling along the dura, but little other evidence of infection (Figure 4L) . No labeled neuronal cell bodies on the dura were observed, consistent with the lack of neurons on this surface. In contrast, we did find labeled cell bodies in the trigeminal ganglion (Figure 4M). No infection was seen in the

CNS, even at 4 dpi, consistent with the lack of inputs from the brain into the trigeminal ganglion (N = 4 animals).

To further characterize patterns and kinetics of viral transmission and directional specificity of transsynaptic spread, injections of rVSV(RABV-G) were made into the CP (Figure 5A). In order to determine which cells were labeled by direct uptake of virus in the inoculum, a separate set of animals were injected into the CP with the replication-incompetent rVSV G(RABV-G) (N = 3 animals, analyzed 3 dpi). Cells labeled by rVSV G(RABV-G) were observed in the CP, GP, substantia nigra (SN), thalamus, and layers 3 and 5 of the cortex, consistent with infection at the axon terminal and retrograde labeling of cell bodies of neurons known to project directly to the CP (Figure 5C) (Albin et al., 1995) . Areas labeled by CP injection are indicated in Figure 5B .

The patterns of spread for the replication-competent rVSV(RABV-G) were characterized over the course of 1-5 dpi (Figures 5D-H) . During this interval, progressively more cells in infected regions were labeled by rVSV(RABV-G), including within the CP, nucleus basalis, cortex, and GP (listed in Figure 5B) . In addition, more cortical cells were labeled in clusters near cortical pyramidal neurons, both ipsilateral and contralateral to the injected side, including neurogliaform cells (data not shown). These data are in contrast to those observed following infection with an anterograde transsynaptic tracing virus, such as rVSV with its own G gene, rVSV(VSV-G) (Figure 5B) . At 3 dpi following rVSV(VSV-G) injection into the CP, the cerebral cortex was not labeled, but regions receiving projections from the CP, such as the STn, GP, and SN, were labeled (Beier et al., 2011) .

In order to investigate other areas for evidence of cell-to-cell retrograde transsynaptic spread, the nucleus basalis was examined following infection of the CP with replication-competent rVSV(RABV-G). The nucleus basalis was labeled by 2 dpi (Figures 5E-H) , consistent with at least a single transsynaptic jump, as this area does not directly project to the CP. The virus appeared to travel transsynaptically at the rate of roughly 1 synapse per day, as evidenced by the lack of labeled neurogliaform cells in the cortex, and lack of neurons in the nucleus basalis at 1 dpi, and label appearing in these cell types/areas at 2 dpi, as previously observed (Beier et al., 2011) . Labeling remained well-restricted to the expected corticostriatal circuits at 5 dpi, suggesting that viral spread becomes less efficient after crossing one or two connections, consistent with injections into V1 (Figure 4) . While glial cells can be infected and were observed near the injection site (van den Pol et

al., 2002; Chauhan et al., 2010) , infected glial cells away from the injection site generally were not observed.

One advantage of having both anterograde and retrograde forms of the same virus is that they can be used in parallel, or in tandem, to trace circuitry to and from a single or multiple sites of injection, with each virus having similar kinetics of spread and gene expression. In fact, if different fluorophores are used in different viruses, e.g., rVSV(VSV-G) and rVSV(RABV-G), then the viruses can be co-injected into the same site and their transmission can be traced independently (Figure 6A). This is most straightforward if there are no cells at the injection site that are initially infected by both viruses. Co-infected cells can be easily detected, as they would express both fluorescent proteins shortly after injection.

In order to determine whether two viruses would allow simultaneous anterograde and retrograde transsynaptic tracing from a single injection site, a rVSV(VSV-G) expressing Venus and a rVSV(RABV-G) expressing mCherry were injected individually (Figures 6B-D) or co-injected (Figures 6E-G) into the motor cortex, and brains were examined 3 dpi. The pattern of labeling from the co-injected brains was equivalent to the patterns observed when each virus was injected individually: rVSV(VSV-G) was observed to infect neurons in the cortex, CP, and downstream nuclei, whereas the rVSV(RABV-G) was not observed to infect neurons in the CP, but rather in the thalamus and nucleus basalis (N = 4). The initial co-infection rate is dependent upon The presence or absence of labeling is indicated by (+) and (-), respectively. The extent of labeling is indicated by the number of (+). Some animals were infected with G viruses to determine which areas were labeled by direct uptake of the virions, rather than by replication and transmission. These were sacrificed at 3 dpi. (C) Parasagittal section of a brain infected with VSV[Δ g](RABV-G). The injection site is marked by a red arrow. Several areas that project directly to the CP were labeled due to direct uptake of the virions, including the cortex, thalamus, and GP (arrowheads), 3 dpi. the dose of the initial inocula. When injecting 3×10^3 focus forming units (ffu) rVSV(VSV-G) and 3×10^4 ffu rVSV(RABV-G), no co-infection was observed at the injection site. Thus, co-infection of the same brain region, without co-infection of the same cells,

does not alter the spreading behavior of either rVSV(VSV-G) or rVSV(RABV-G). One example of how this dual retrograde and anterograde transsynaptic tracing system can be used is to determine if three Frontiers in Neural Circuits www.frontiersin.org February 2013 | Volume

7 | Article 11 | 8 distinct regions are connected and the directionality of any connections. For example, the anterograde transsynaptic virus can be injected into one region, the retrograde into another, and a third region can then be examined for evidence of labeling by either or both viruses (e.g., Figure 6H). To test this possibility, rVSV(VSV-G) was injected into the motor cortex, rVSV(RABV-G) was injected into the substantia nigra pars reticulara (SNr), and animals were sacrificed at 3 dpi. We observed that cells were singly labeled, either with Venus [rVSV(VSV-G)] or with mCherry [rVSV(RABV-G)], and were located largely in different regions of the CP (Figures 6I,J) (N = 3) . These results suggest that the anterograde connections from the cells infected with rVSV(VSV-G) in the M1 were with CP MSNs that did not project to the region of the SNr injected with rVSV(RABV-G) (N = 3 animals).

In addition to polysynaptic tracing, VSV can be modified to trace circuits monosynaptically (Beier et al., 2011) . With RABV, this was achieved in vivo by first infecting with an adeno-associated virus (AAV) expressing TVA, a receptor for an avian retrovirus, and RABV-G (Wall et al., 2010) . This was followed 3 weeks later by infection with a G RABV with an EnvA/RABV-G chimeric glycoprotein on the virion surface (Wickersham et al., 2007b) , which allowed infection specifically of the cells expressing TVA. A similar strategy was used to test rVSV's ability to monosynaptically trace retrogradely connected neurons in vivo. Inputs to choline acetyltransferase (ChAT)-expressing neurons in the striatum were used for this test. These neurons primarily receive input from the cortex and the thalamus (Thomas et al., 2000; Bloomfield et al., 2007) (Figure 7A) . In order to mark this population, we crossed ChAT-Cre mice to Ai9 mice, which express tdTomato in cells with a Cre expression history (Madisen et al., 2010) . Six-week-old mice from this cross were injected in the CP with two AAV vectors: one expressing a Cre-conditional ("floxed") TVA-mCherry fusion protein, and another expressing a floxed RABV-G. Two weeks later, the mice were injected in the same coordinates with rVSV G with the EnvA/RABV-G chimeric glycoprotein on the virion surface [rVSV G(A/RG)] (Beier et al., 2011) . Cells successfully infected with these two AAV vectors could host infection by a rVSV and should be able to produce rVSV virions with RABV-G on the surface. Such starter cells should also express tdTomato and GFP. If rVSV were to be produced, and if it were to transmit across the synapse retrogradely, cortical and thalamic neurons should be labeled by GFP. Mice injected with these AAV and rVSV viruses were sacrificed 5 days after rVSV infection, and brains analyzed for fluorescence. As expected for starter cells, some neurons in the CP expressed both tdTomato and GFP (Figure 7B) . Outside of the CP, small numbers of GFP+ neurons that were not mCherry+ were observed in the cortex (Figures 7C,D) and thalamus (Figure 7E) , consistent with retrograde spread. Control animals not expressing Cre, or not injected with

AAV encoding RABV-G, did not label cells in the cortex or thalamus (N = 3 for both controls and experimental condition).

Here, we report on the use of rVSV as a retrograde transsynaptic tracer for CNS circuitry. VSV can be modified to encode the RABV-G protein in the viral genome, allowing the virus to replicate and transmit across multiple synaptically connected cells, i.e., as a polysynaptic tracer. Alternatively, if the virus has the G gene deleted from its genome and RABV-G is provided in trans, it behaves as a monosynaptic tracer (Beier et al., 2011) . Although it has been known for many years that RABV travels retrogradely among neurons (Astic et al., 1993; Ugolini, 1995; Kelly and Strick, 2003) , and pseudotyping lentiviruses with RABV-G is sufficient for axonal transport (Mazarakis et al., 2001) , the retrograde transmission specificity among neurons had not been clearly shown to be a property of the G protein itself, as it might have been due to other viral proteins in addition to, or instead of, the viral G protein. Since native VSV does not have these retrograde transsynaptic properties (van den Pol et al., 2002; Beier et al., 2011) , and the only alteration to the VSV genome was the substitution of the VSV G gene with the G gene of RABV, it is clear that the RABV glycoprotein is responsible for retrograde direction of viral transmission across synapses, at least in the case of rVSV.

The early onset of gene expression from VSV relative to RABV (one hour vs. multiple hours) makes it beneficial in experimental paradigms in which the experiment needs to be done within a narrow window of time, such as tissue slices and explants. In addition, more than one transgene can be encoded in the viral genome without the need of a 2A or IRES element. The use of the first position of the genome enhances the expression level of the transgene inserted at that location, since VSV (and RABV) express genes in a transcriptional gradient; therefore, the first gene is the most highly transcribed (Knipe, 2007) . This leads to rational predictions of expression levels so that one can choose the position of insertion of a transgene, or transgenes, according to this gradient and the desired level of expression. The size of the viral capsid is apparently not rigid, allowing for the inclusion of genomes that are substantially larger than the native genome, unlike the rigid capacity for some other viral vectors, such as AAV Yan et al., 2000) .

The fact that VSV can be made to spread anterogradely (Beier et al., 2011) or retrogradely across synapses with the change of a single gene affords several advantages over viral tracers that heretofore have not shown such flexibility in the directionality of tracing. In

addition to the obvious application of tracing anterograde connections, combinations can be made to exploit the different forms of the virus. One example that employs the simultaneous infection with an anterograde and retrograde form of VSV is demonstrated in Figure 6 . This experiment was designed to address whether the anterograde projections from the cortex to the CP would label the same brain regions as were labeled by a retrograde virus injected into the SN. Although a block of superinfection by the virus may preclude infection of the same cell with multiple rVSVs, adjacent cells could still become labeled by different viruses (Whitaker-Dowling et al., 1983) . The observed results could be due to a preferential labeling by the anterograde transsynaptic virus of indirect pathway MSNs in this experiment, which then synapse onto the GP, thereby reflecting a viral bias. Alternatively, it could indicate that the cortical neurons in the injected region largely do not label the MSNs that project to the area of the SN injected with the retrograde virus. One further possibility is that too little virus was used to observe co-labeling of a given region. However, given the density of infection (i.e., Figures 6I,J) , the latter possibility seems unlikely. Additionally, the spread of the polysynaptic rVSV(RABV-G) appears to attenuate with increasing numbers of synapses crossed, permitting an analysis of more restricted viral spread. This is quite fortuitous, as if spread were to continue, it would lead to widespread infection and lethality. In addition, reconstruction of connectivity would be more difficult. This reduced efficiency appears to also hold for the monosynaptic form of VSV complemented with RABV-G, as the efficiency of transmission appeared lower than the comparable experiment with RABV (Watabe-Uchida et al., 2012) . This is likely due to viral attenuation when VSV-G is replaced with RABV-G.

We were attracted to the use of VSV as a viral tracer due to its long track record as a safe, replication-competent laboratory agent. Laboratory workers using VSV have not contracted any diseases, and natural VSV infections among human populations in Central America and the southwestern United States (Rodríguez, 2002) occur without evident pathology (Johnson et al., 1966; Brody et al., 1967) . VSV was thus an attractive candidate for its use as a polysynaptic tracer for CNS studies, which requires an ability to replicate through multiple transmission cycles. Both replicationcompetent and incompetent forms of VSV are in use under Biosafety Level 2 containment. Replication-competent RABV is Biosafety Level 3, due to the fact that infection with replicationcompetent RABV is almost always fatal to humans and in mice when infected intracerebrally (Smith, 1981; Knipe, 2007) . Differences in pathogenicity between VSV and RABV are likely due to the ability of RABV to evade the innate immune system, particularly interferon (Hangartner et al., 2006; Junt et al., 2007; Lyles and Rupprecht, 2007; Rieder and Conzelmann, 2009; Iannacone et al., 2010) . VSV infection efficiently triggers an interferon response, and it has not evolved a

method of escape from this response, unlike RABV (Brzózka et al., 2006) . In fact, VSV is being pursued as a vaccine for other viruses, including RABV (Lichty et al., 2004; Publicover et al., 2004; Kapadia et al., 2005; Schwartz et al., 2007; Iyer et al., 2009; Geisbert and Feldmann, 2011) . VSV does not typically spread beyond the initially infected site in the periphery (Kramer et al., 1983; Vogel and Fertsch, 1987) . This likely is the cause of the minor or absent symptoms in humans and animals infected in nature. Polysynaptic VSV vectors are thus predicted to be much safer than polysynaptic RABV vectors. We have tested this prediction by injecting a series of mice in the footpads and hind leg muscles with rVSV(RABV-G), with the result that no injected animals showed any evidence of morbidity or mortality (Beier, Goz et al., in preparation) .

While safer for laboratory workers than RABV, the main drawback to using VSV is its rapid cellular toxicity (van den Pol et al., 2009; Beier et al., 2011) . Toxicity is due to suppression of cellular transcription and a block in the export of cellular RNAs from the nucleus to the cytoplasm (Black and Lyles, 1992; Her et al., 1997; Ahmed and Lyles, 1998; Petersen et al., 2000; von Kobbe et al., 2000) , as well as inhibition of the translation of cellular mRNAs (Francoeur et al., 1987; Jayakar et al., 2000; Kopecky et al., *Frontiers in Neural Circuits* www.frontiersin.org February 2013 | Volume 7 | Article 11 | 10 2001). VSV is much quicker to enact its gene expression program than is RABV, such that cells suffer the toxic effects more quickly than after RABV infection. One aspect of VSV that can be exploited in the future to ameliorate the speed of toxicity is the use of VSV mutants and variants. One such mutant is the M51R, which permitted us to conduct physiological analyses of pre- and post-synaptic cells (Beier et al., 2011) . We are in the process of examining the transmission properties of this mutant *in vivo*, as well as the effects of other mutations or viral variants on prolonging the health of neurons after infection.

rVSV vectors can be used to study the connectivity of neuronal circuitry. In addition to combinations of replication-competent forms of VSV, the replication-incompetent, monosynaptic forms of the virus can be easily combined, without the need to change viruses (Beier et al., 2011) . This allows a straightforward way to study both the projections into, and out from, a genetically defined cell population. This can be done with the same viral genome, with the only change needed being the glycoprotein, for the selection of the direction of transmission. This flexibility of VSV makes it a powerful, multi-application vector for studying connectivity in the CNS.

All rVSV clones were cloned from the rVSV G backbone (Chandran et al., 2005) . mCherry, Kusabira orange, Venus, and CFP were cloned into the first (GFP) position using XhoI and MscI sites, and VSV-G (a gift from Richard Mulligan, Harvard Medical School, Boston, MA) and RABV-G (a gift from Ed Callaway, Salk Institute, San Diego, CA) were cloned into the fifth (G) position using the MluI and NotI restriction sites. Genes for fluorescent proteins were obtained from Clontech. Viruses were rescued as previously described (Whelan et al., 1995) . At 95% confluency, eight 10 cm plates of BSR cells were infected at an MOI of 0.01. Viral supernatants were collected at 24-h time intervals and ultracentrifuged at 21,000 RPM using a SW28 rotor and resuspended in 0.2% of the original volume. For titering, concentrated viral stocks were applied in a dilution series to 100% confluent BSR cells and plates were examined at 12 hpi. Viral stocks were stored at -80°C .

For G viruses, 293T cells were transfected with PEI (Ehrhardt et al., 2006) at 70% confluency on 10 cm dishes with 5 μg of pCAG-RABV-G. Twenty-four hours post-infection, the cells were infected at an MOI of 0.01 with rVSV G expressing either GFP or mCherry. Viral supernatants were collected for the subsequent 4 days at 24 h intervals.

Virus preparations are now available from the Salk GT3 viral core (<http://vectorcore.salk.edu/>). All plasmids are available from Addgene (<http://www.addgene.org/>).

AAV-FLEX-RABV-G and AAV-FLEX-TVA-mCherry plasmids originated from the Lab of Naoshige Uchida (Watabe-Uchida et al., 2012) , and virus stocks were generous gifts from Brad Lowell, Harvard Medical School.

ChAT-Cre (B6;129S6-Chat^{tm1}(cre)Lowl /J) and Ai9 (B6.Cg-Gt(ROSA)26Sor^{<tm9}(CAG-tdTomato)Hze>/J) mice were obtained from the Jackson Laboratory (Madison et al., 2010) .

Eight-week-old CD-1 mice were injected using pulled capillary microdispensers (Drummond Scientific, Cat. No: 5-000-2005) , using coordinates from The Mouse Brain in Stereotaxic Coordinates (Franklin and Paxinos, 1997) . Injection coordinates (in mm) used were: For multi-color analysis (Figures 1C,D) , 3×10^9 ffu/mL rVSV was injected into various regions. For CP injections, 100 nL of rVSV(RABV-G) or rVSV(VSV-G) at 3×10^7

ffu/mL was injected at a rate of 100 nL/min. For the replication-incompetent viruses, 100 nL of 1×10^7 ffu/mL rVSV G(RABV-G) or rVSV G (VSV-G) was injected. In the motor cortex, 100 nL of 1×10^7 ffu/mL rVSV(RABV-G) was injected, and mice harvested 2 dpi. For V1 injections, 100 nL of 3×10^{10} ffu/mL rVSV(RABV-G) was injected, and mice were examined 3 or 7 dpi.

For infections of the dura mater, 1 μ L of 3×10^{10} ffu/mL rVSV(RABV-G) was applied to the surface of the dura. The virus was allowed to absorb, and the surface was subsequently covered in bone wax, and the wound sutured.

For co-injections of virus into the same animal, 100 nL of a combination of 3×10^7 ffu/mL rVSV(VSV-G) and 3×10^8 ffu/mL rVSV(RABV-G) were co-injected into the motor cortex, and brains examined 3 dpi. For injections of the viruses into different regions, 100 nL of 3×10^7 ffu/mL rVSV(VSV-G) was injected into M1, and 100 nL of 3×10^8 ffu/mL rVSV(RABV-G) into the SNr, and brains examined 3 dpi. A lower titer of rVSV(VSV-G) was used, as rVSV(RABV-G) is attenuated.

All mouse work was conducted in biosafety containment level 2 conditions and was approved by the Longwood Medical Area Institutional Animal Care and Use Committee.

Recordings were made from cortical pyramidal neurons in slices taken from postnatal day 12-18 mice, inoculated in the CP 12-18 h prior with rVSV(RABV-G). Coronal slices (300 μ m thick) were cut in ice-cold external solution containing (in mM): 110 choline, 25 NaHCO₃, 1.25 NaH₂PO₄, 2.5 KCl, 7 MgCl₂, 0.5 CaCl₂, 25 glucose, 11.6 Na-ascorbate, and 3.1 Na-pyruvate, bubbled with 95% O₂ and 5% CO₂. Slices were then transferred to artificial cerebrospinal fluid (ACSF) containing (in mM): 127 NaCl, 25 NaHCO₃, 1.25 NaH₂PO₄, 2.5 KCl, 1 MgCl₂, 2 CaCl₂, and 25 glucose, bubbled with 95% O₂ and 5% CO₂. After an incubation period *Frontiers in Neural Circuits* www.frontiersin.org February 2013 | Volume 7 | Article 11 | 11 of 30-40 min at 34 °C, slices were stored at room temperature. All experiments were conducted at room temperature (25 °C). In all experiments, 50 μ M picrotoxin, 10 μ M 2,3-Dioxo-6-nitro-1, 2, 3, 4 -tetrahydrobenzo [f]quinoxaline -7-sulfonamide (NBQX), and 10 μ M 3-((R)-2-Carboxypiperazin-4-yl)-propyl-1-phosphonic acid (CPP) were present in the ACSF to block GABAA/C, AMPA, and NMDA receptor-mediated transmission, respectively. All chemicals were from Sigma or Tocris.

Whole-cell recordings were obtained from infected and uninfected deep layer cortical pyramidal neurons identified with video-IR/DIC and GFP fluorescence was detected using epifluorescence illumination. With the deep layers of the cortex, 2-photon laser scanning microscopy (2PLSM) was used to confirm the cell types based on morphology. Deep layer pyramidal neurons had large cell bodies, classic pyramidal shape and dendritic spines. Glass electrodes (2-4 M Ω) were filled with internal solution containing (in mM): 135 KMeSO₄, 5 KCl, 5 HEPES, 4 MgATP, 0.3 NaGTP, 10 Na₂HPO₄, 1 EGTA, and 0.01 Alexa Fluor-594 (to image neuronal morphology) adjusted to pH 7.4 with KOH. Current and voltage recordings were made at room temperature using a AxoPatch 200B or a Multiclamp 700B amplifier. Data was filtered at 5 kHz and digitized at 10 kHz.

Imaging and physiology data were acquired and analyzed as described previously (Carter and Sabatini, 2004). Resting membrane potential was determined by the average of three 5-s sweeps with no injected current. Passive properties of the cell, membrane (R_m) and series resistance (R_s) and capacitance (C_m), were measured while clamping cells at -65 mV and applying voltage steps from -55 to -75 mV. The current-firing relationship was determined in current clamp with 1-s periods of injected current from 100 to 500 pA.

The time course of viral gene expression experiments were carried out in organotypic hippocampal slice cultures prepared from postnatal day 5-7 Sprague-Dawley rats as described previously (Stoppini et al., 1991). Slices were infected after 7 days in vitro, and images were acquired on a two-photon microscope.

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The Intranasal Application of Zanamivir and Carrageenan Is Synergistically Active against Influenza A Virus in the Murine Model

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4459876/>

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Abstract: **BACKGROUND:** Carrageenan is a clinically proven and marketed compound for the treatment of viral upper respiratory tract infections. As infections caused by influenza virus are often accompanied by infections with other respiratory viruses the combination of a specific anti-influenza compound with the broadly active antiviral polymer has huge potential for the treatment of respiratory infections. Thus, the combination of the specific anti-influenza drug Zanamivir together with carrageenan in a formulation suitable for intranasal application was evaluated in-vitro and in-vivo. **PRINCIPAL FINDINGS:** We show in-vitro that carrageenan and Zanamivir act synergistically against several influenza A virus strains (H1N1(09)pdm, H3N2, H5N1, H7N7). Moreover, we demonstrate in a lethal influenza model with a low pathogenic H7N7 virus (HA closely related to the avian influenza A(H7N9) virus) and a H1N1(09)pdm influenza virus in C57BL/6 mice that the combined use of both compounds significantly increases survival of infected animals in comparison with both mono-therapies or placebo. Remarkably, this benefit is maintained even when the treatment starts up to 72 hours post infection. **CONCLUSION:** A nasal spray containing carrageenan and Zanamivir should therefore be tested for prevention and treatment of uncomplicated influenza in clinical trials.

Text: The periodic appearance of new influenza variants poses a worldwide pandemic threat. Since the emergence of the new A(H7N9) virus, more than 400 human cases were reported to the WHO with a mortality rate of more than 35%. Most patients with A(H7N9) infections had contact with poultry or visited live animal markets. However, some sporadic cases seemed to be a result of human to human transmissions [1, 2] . In contrast to pandemic viruses which fulminantly enter the human population and cause high mortality rates, seasonal influenza viruses generally cause uncomplicated and transient infections in humans, with virus replication localized to the upper respiratory tract [3, 4] . However, in its

fully developed form influenza is an acute respiratory disease resulting in hospitalizations and deaths mainly among high-risk groups. Worldwide, annual epidemics result in about three to five million cases of severe illness, and about 250,000 to 500,000 deaths [5] . For this reason WHO [6] and CDC [7] recommend antiviral treatment for any patient with suspected influenza who is at risk for influenza complications without previous laboratory confirmation.

It is known that influenza virus infections are often accompanied by other viral pathogens [8] . Depending on the detection method (qRT-PCR or immunofluorescence) different ratios of co-infections have been found. Analysis by qRT-PCR revealed that 54.5-83.3% of influenza A or B positive patients were found to have at least one concomitant respiratory viral infection [9] [10] [11] [12] . The detection frequency with immunofluorescence was found to be even higher (90-100%) [13, 14] . Potential concomitant viral pathogens of influenza virus infections include human rhinovirus (hRV), respiratory syncytial virus, adenovirus, human coronavirus, human metapneumovirus and parainfluenza virus [14, 15] .

As a result of the multiple infections, a specific anti-influenza mono-therapy treats the influenza virus infection only, but not the infection with the concomitant viral pathogen. Hence, the therapy often fails to sufficiently resolve symptoms. This is also reflected by the fact that neuraminidase inhibitors (NI) are highly efficacious in animal models investigating influenza mono-infections [16, 17] but show lower efficacy against influenza symptoms in clinical trials in adults with natural infections [18] . Therefore, there is a high medical need for a broadly acting antiviral therapy in combination with a specific anti-influenza therapy for treatment of patients suffering from upper respiratory tract symptoms. Ideally, the substances present in the combination complement each other by different modes of action, leading to a treatment that provides full protection against a broad range of different respiratory viruses as well as different influenza strains with a low probability to induce escape mutations.

One approach for a broad antiviral therapy is the creation of a protective physical barrier in the nasal cavity using carrageenan. Carrageenan is a high molecular weight sulfated polymer derived from red seaweed (Rhodophyceae) that has been extensively used in food, cosmetic and pharmaceutical industry and is generally recognized as safe by the FDA (GRAS) (reviewed in [19]). Three main forms of carrageenans are commercially used:

kappa, iota and lambda. They differ from each other in the degree of sulfation, solubility and gelling properties [20] . The antiviral mechanism of carrageenan is based on the interference with viral attachment; as a consequence, viral entry is inhibited [21, 22] . Its antiviral activity is dependent on the type of polymer as well as the virus and the host cells [23] [24] [25] [26] [27] [28] [29] [30] [31] [32] and has been reviewed in [33] [34] [35] . We published that iota-carrageenan is a potent inhibitor of hRV [36] and influenza A [37] replication and demonstrated the antiviral efficacy of iota-carrageenan against common cold viruses by intranasal application in several randomized, double-blind, parallel group, placebo-controlled clinical trials [38] [39] [40] . The pooled analysis of two studies conducted in 153 children and 203 adults revealed that patients infected with any respiratory virus, who were intranasally treated with iota-carrageenan showed a 1.9 day faster recovery from common cold symptoms than placebo treated patients in the intention-to-treat population [41, 42] . The anti-influenza activity was shown by subgroup analysis of 49 influenza infected patients who benefited from a 3.3 days faster recovery from symptoms. The use of carrageenan nasal spray was associated with a significant reduction of the influenza viral load in nasal fluids and a significant increase in the number of virus free patients within the treatment period of 7 days. In good accordance Prieschl-Grassauer are co-founders of Marinomed Biotechnologie GmbH. Marinomed Biotechnologie GmbH had a role in study design, data collection and analysis, decision to publish, preparation of the manuscript and is financing the processing charge of the manuscript. with the literature [9] [10] [11] [12] [13] [14] we observed that the majority of influenza virus infected patients suffered from a concomitant respiratory viral infection (66%) as determined by real-time PCR. Carrageenan containing nasal sprays are already marketed for the treatment of respiratory viral infections under different brand names in 18 countries.

At present the only available effective drugs for treatment and post exposure prevention of influenza are the NI (Oseltamivir and Zanamivir worldwide; Peramivir in Japan and South Korea). Since the large-scale use of M2 blockers for prophylaxis and treatment in humans [43] and farming [44] , the currently circulating influenza viruses already lack sensitivity to this drug group [45] .

We have already shown an additive therapeutic effect of a combination therapy with intranasally applied iota-carrageenan and orally administered Oseltamivir in lethally H1N1 A/PR/ 8/34 infected mice and a treatment start 48 hours post infection (hpi) [37] .

Due to these very promising results we further developed the concept of combining carrageenan with an NI therapy. In contrast to Oseltamivir, which needs to be activated by metabolic conversion, Zanamivir is directly applied as active drug and can also be administered intranasally [46] [47] [48] [49] [50] [51] [52] . The potential of an intranasal administration of Zanamivir was investigated by GlaxoSmithKline. In seven clinical challenge trials 66 volunteers were infected with influenza B/Yamagata/16/88 and 213 with influenza A/Texas/36/91 (H1N1). 156 of these participants got intranasally applied Zanamivir at different doses (daily dose levels from 6.4 mg to 96 mg) for prophylaxis or therapy [46, 47, 53, 54] . These challenge trials showed that treatment starting before and up to 36 hours post virus inoculation was associated with prevention of laboratory confirmed influenza and febrile illness as well as a reduction in viral titers, duration of shedding and symptoms. In total, safety data from 1092 patients after intranasal application of Zanamivir were published and no evidence for Zanamivir induced adverse events or increased frequencies of local nasal intolerance in comparison to placebo groups was found [46, 49, 52] .

Taken together, the combination of a carrageenan nasal spray that provides broad antiviral activity against upper respiratory infections-including influenza-with Zanamivir, a specific anti-influenza drug, meets the existing medical need to treat multiple viral infections. In the present work we investigate the therapeutic effect of a combination of carrageenan and Zanamivir in-vitro and in an animal model.

Kappa-carrageenan and iota-carrageenan were purchased from FMC Biopolymers (Philadelphia, PA). The identity, purity (>95%) of carrageenan subtypes and the molecular weight (>100,000) was confirmed by NMR analysis as described elsewhere [55] and the presence of lambda-carrageenan was below the detection limit of 3%. The dry polymer powders were dissolved in aqua bidest (Fresenius Kabi, Austria) to a final concentration of 2.4 mg/ml iota-and 0.8 mg/ml kappa-carrageenan. This 2x stock solution was sterile filtered through a 0.22 µm filter (PAA, Switzerland) and stored at room temperature until use. For further testing the stock solution was diluted to a mixture containing 1.2 mg/ml iota-carrageenan and 0.4 mg/ml kappa-carrageenan (hereinafter referred to as "carrageenan").

Zanamivir was purchased as powder (Haosun Pharma, China) and the identity and purity was confirmed by NMR analysis. Zanamivir was either dissolved in carrageenan or placebo solutions, followed by sterile filtration through a 0.22 µm filter (Sarstedt, Germany). For in-vivo studies all Zanamivir containing solutions were freshly prepared.

Madin-Darby canine kidney (MDCK) cells were obtained from the American Type Culture Collection (ATCC, Manassas, VA) and cultivated in a 37°C incubator (Sanyo, Japan; CO₂ : 5%, relative humidity: >95%). MDCK cells were grown in Dulbecco's minimal essential (DMEM) high glucose medium (PAA, Austria) supplemented with 10% fetal bovine serum (FBS; PAA, Austria; heat inactivated).

Influenza virus A/Hansa Hamburg/01/09 (H1N1(09)pdm) was kindly provided by Peter Staeheli Department of Virology, University of Freiburg, Germany and previously described in [56] ; A/Teal/Germany/Wv632/05 (H5N1) previously published in [57] (accession numbers CY061882-9) and A/Turkey/Germany/R11/01 (H7N7) (taxonomy ID 278191, accession number AEZ68716) were supplied by courtesy of Martin Beer, Institute of Diagnostic Virology, Friedrich-Loeffler-Institute, Riems, Germany; A/Aichi/2/68 (H3N2) was purchased from the ATCC. All influenza viruses were propagated in MDCK cells at 37°C and 5% CO₂ in influenza medium [Opti-Pro serum free medium (Gibco, Austria) supplemented with 4 mM L-glutamine (PAA, Austria), 1% antibiotic-antimycotic mix (PAA, Austria) and 5 µg/ml trypsin (Sigma Aldrich, Austria)].

To determine the 50% inhibitory concentration (IC₅₀) and the combination effect of carrageenan and Zanamivir, a semi-liquid plaque assay was developed. Into 96 well tissue culture plates 1.7x10⁴ MDCK cells/well were seeded and infected at 90% confluence (24-28 hours later). Serial dilutions of carrageenan and Zanamivir were prepared in assay medium (influenza medium without trypsin). For infection, viruses were diluted to an MOI of 0.003 (H1N1(09)pdm and H3N2 Aichi), 0.015 (H5N1) or 0.004 (H7N7), respectively, in assay medium and incubated at room temperature (RT) for 10 min with the serial dilutions of carrageenan and/or Zanamivir, respectively. For evaluation of the combination effect of carrageenan and Zanamivir, viruses were diluted in assay medium containing constant concentrations of either carrageenan or Zanamivir. The other substance was serially diluted and used for virus incubation. Cells were infected in 6 replicates/compound dilution, respectively, and incubated at RT for 45 min before inoculum removal. Cells were further incubated with the respective concentration of the investigated substances present

in the overlay [influenza medium with 2.25% Carboxymethylcellulose (CMC, Fluka, Austria)] for 30-42 hours at 37°C. Evolving plaques were evaluated after methanol/acetone cell fixation by immune staining with antibodies either directed against the influenza A nucleoprotein (AbD Serotec, Germany) (for H1N1(09)pdm, H5N1 and H7N7) or the hemagglutinin (AbD Serotec, Germany) (for H3N2). Analysis was done with a HRP labeled detection antibody (Thermo Scientific, Germany) using TMB (Biolegend, Germany) as substrate and a microplate reader at 450 nm. The reduction of detected signal represents a reduction in the number and size of plaques and indicates suppression of viral replication during infection and cultivation.

After the immunostaining cells were stained with 0.005% crystal violet solution to assess the condition of the cell layer and the toxicity of the compounds. IC 50 values and standard deviations were calculated for a sigmoidal dose response model using XLfit Excel add-in version 5.3.1.3.

All animal experiments were carried out according to the guidelines of the "European Convention for the Protection of Vertebrate Animals used for Experimental and other Scientific Purposes" and the Austrian law for animal experiments. All animal experiments were approved by the Veterinary University of Vienna institutional ethics committee and performed under the Austrian Federal Ministry of Science and Research experimental animal license numbers BMWF-68.205/0262-II/3b/2011 and BMWF-68.205/0142-II/3b2012. C57BL/6 mice were purchased from Janvier Labs, France and maintained under standard laboratory conditions in the animal facilities of the Veterinary University of Vienna. For euthanasia and anesthesia asphyxiation through CO₂ was used and all efforts were made to minimize suffering.

For infection experiments, 3-5 weeks old female mice were intranasally inoculated with 50 µl influenza virus solution (25 µl/nostril) containing 2.27×10^3 or 1.65×10^3 plaque-forming unit of H1N1(09)pdm or H7N7, respectively. Subsequently, treatment started 24, 48 or 72 hpi, as indicated for the different experiments. Treatment was performed intranasally either with 50 µl therapeutic solution or placebo twice per day for 5 days. As therapy either carrageenan (containing 1.2 mg/ml iota-carrageenan and 0.4 mg/ml kappa-carrageenan to provide a daily dose of 12 mg/kg body weight (BW)), Zanamivir (containing either 130 µg/ml or 390 µg/ml Zanamivir, to provide a daily dose of 1 or 3 mg/kg BW, respectively) or a combination of carrageenan and Zanamivir were used. Carrageenan and Zanamivir are

used at non-toxic concentrations as shown by [58] and [59] . Mice were monitored twice daily for 15 days for survival and weight loss. Mortality also includes mice that were sacrificed for ethical considerations when they had lost more than 25% of their initial body weight. We confirm the viral infection in these animals by necropsy and scoring of the lung inflammation.

As the mechanisms underlying the antiviral activity of NI and carrageenans are fundamentally distinct, they are likely to exhibit different activities towards the individual influenza virus strains. As a result, in combination they could complement each other to provide protection against a broader spectrum of influenza virus strains than the individual compounds.

To test this hypothesis, we investigated the sensitivity of various influenza virus strains to Zanamivir and carrageenan in an adapted plaque reduction assay with semi-liquid overlay in MDCK cells [60, 61] . Using this method, we determined the IC₅₀ of Zanamivir and carrageenan against influenza A viruses of human and animal origin, namely H1N1(09)pdm (A/Hansa Hamburg/01/09), H3N2 (A/Aichi/2/68), low pathogenic (LP) H5N1 (A/Teal/Germany/ Wv632/05) and LP H7N7 (A/Turkey/Germany/R11/01) (Table 1) . Both substances were nontoxic at the highest tested concentration (400 μ M Zanamivir and 533 μ g/ml carrageenan), neither was their combination. Furthermore, CMC in the overlay did not show any virus inhibitory effect (data not shown). Inhibition of viral replication of all tested influenza strains was achieved with both substances. However, the IC₅₀ values varied widely depending on the influenza virus strain. The IC₅₀ values of Zanamivir ranged between 0.18 μ M for H5N1 and 22.97 μ M for H7N7 and that of carrageenan from 0.39 μ g/ml to 118.48 μ g/ml for H1N1(09)pdm and H7N7, respectively (see Table 1). These results demonstrate that carrageenan and Zanamivir target individual influenza strains to different extents so that they may complement each other to provide broader anti-influenza activity.

The type of compound interaction was characterized by employing isobolograms (Fig 1) . As described in [62] , isobolograms graphically compare the doses of two compounds needed to reach 50% inhibition to the predicted doses calculated based on a model of drug additivity. A curve linearity of ~ 1 is expected for an additive compound interaction whereas a curve progression <1 argue for synergistic and >1 for an antagonistic compound interaction.

Two virus strains were selected for those experiments, one being the most sensitive to carrageenan (H1N1(09)pdm) and one being the least sensitive (H7N7). In both cases the isobolograms show a synergistic interaction of carrageenan and Zanamivir (Fig 1) . Thus, it was shown that Zanamivir and carrageenan target individual influenza viruses with different efficiencies, most probably due to their different antiviral strategies. As a result, the combination provides synergistic activity with higher protection against a broader spectrum of influenza virus strains than the individual compounds.

In the influenza animal model, C57Bl/6 mice are challenged with a lethal dose of the respective virus and treated with different regimens in comparison to a vehicle control (placebo). Infection and treatment (twice a day for 5 days) are done intranasally without anesthesia. We investigated whether the combination of Zanamivir and carrageenan is more efficacious in reducing mortality than the corresponding mono-therapies.

First, we determined the minimal effective dose of a Zanamivir mono-therapy that significantly improved survival time of H1N1 and H7N7 infected mice. For the H7N7 lethal infection the minimal effective dose of Zanamivir as mono-therapy ranged between 1 and 3 mg/kg BW/ day (data not shown). Next, we compared the antiviral activity of carrageenan (12 mg/kg BW/ day) and Zanamivir (1 and 3 mg/kg BW/day) mono-therapies with the respective combination versus placebo treatment. Survival rates of mice with treatment starting 24 hpi are shown in Fig 2A. All placebo treated mice died between day 7 and 9 and also in all mono-therapy groups 100% lethality was observed until day 15. In contrast, the combination therapies led to 50% and 90% survival, depending on the Zanamivir concentration. Statistical analysis showed that the Zanamivir mono-therapy 1 mg/kg BW/day did not show a significant benefit ($p = 0.1810$), whereas the mono-therapy with 3 mg/kg BW/day significantly increased the survival rate compared with placebo treated mice ($p = 0.0016$). Both Zanamivir concentrations experienced significant benefit in survival by the combination with carrageenan ($p < 0.0001$). Similarly, the combination therapies resulted in remarkably increased survival ($p = 0.0421$ for 1 mg and $p < 0.0001$ for 3 mg/kg BW/day) when compared to the carrageenan mono-therapy. No statistically significant difference was observed between the combination containing 3 mg/kg BW/day Zanamivir and that containing 1 mg/kg BW/day ($p = 0.0525$). However, a trend for an increased survival rate with the higher Zanamivir concentration was evident. Therefore, for further investigation the combination therapy containing 3 mg/kg BW/day Zanamivir was evaluated in lethally H7N7 infected mice.

Next, the therapeutic potential of the combination with a delayed therapy start 48 or 72 hpi versus placebo treatment was explored. The survival rates of mice are shown in Fig 2B. All placebo treated mice died until day 10 and also in the group with the treatment start 72 hpi 100% lethality was found. In contrast, the combination therapy starting 48 hpi provided a statistically significant enhanced survival rate in comparison to placebo-treated mice ($p = 0.0010$).

In summary, the combination of two effective, established mono-therapies resulted in a significantly enhanced survival in lethally H7N7 infected mice. Additionally, the combination therapy was highly efficient in comparison to placebo treatment even after a treatment onset up to 48 hpi.

Intranasal therapy with carrageenan and Zanamivir starting 72 hpi significantly protects lethally influenza H1N1(09)pdm infected mice. Next, the minimal effective dose of Zanamivir used as mono-therapy was evaluated in a lethal H1N1(09)pdm mouse model, following the same scheme as described in the H7N7 experiments. The lowest effective dose of Zanamivir after a treatment start 24 hpi was 1 mg/kg BW/ day and its combination with carrageenan was highly effective (data not shown). In the following experiment the therapeutic potential of the combination with a therapy start 48 or 72 hpi was investigated in comparison with the respective placebo treatment.

As shown in Fig 3, the survival rates of mice treated with the combination therapy were highly significantly increased in comparison to the placebo group ($p < 0.0001$). There was no difference in survival between the two therapy starting points, 48 or 72 hpi, which both resulted

We investigated the antiviral effect of a combination of carrageenan with the NI Zanamivir in cell culture studies and in mouse influenza infection models. We have previously shown that a combined therapy of iota-carrageenan with the NI Oseltamivir led to significantly enhanced survival in mice infected with H1N1 PR/8/34 in comparison with the respective mono-therapies [37]. However, Oseltamivir is an orally administered prodrug, which has to be converted into its active form by metabolic processing. Therefore, a further development of a combination nasal spray was not possible with Oseltamivir. Instead

Zanamivir-a NI that is applied as active drug-was chosen for the development of a compound combination.

During the evaluation process we found that the binding efficiency of different carrageenan subtypes on different influenza strains varies. The combined use of iota-and kappa-carrageenan for the treatment of lethally influenza infected C57Bl/6 mice revealed a better therapeutic effect than the use of iota-carrageenan alone (S1 Fig). Thus, to provide a broader spectrum of activity against different influenza virus strains, a mixture of iota-and kappa-carrageenan (designated as carrageenan) was used for further evaluation.

For investigation of the effect of a compound combination of carrageenan and Zanamivir, we examined their inhibition efficiency, individually and in combination, against influenza viruses in an adapted plaque reduction assay with semi-liquid overlay in MDCK cells. The combination showed a synergistic inhibition of virus replication in in-vitro assays with all tested influenza viruses (Fig 1) . This indicates that the physical interaction of the polymer with the virus does not disturb the inhibition of the neuraminidase by Zanamivir. This was confirmed in in-vitro tests examining a potential influence of the polymer on the neuraminidase inhibiting activity of Zanamivir (data not shown). Hence, the observed synergistic effect is based on the combination of two distinct underlying mechanisms. As a result, in the proposed combination both mechanisms would complement each other to provide more efficient protection against a broader spectrum of influenza virus strains than the individual compounds.

The synergistic effect was also shown in lethal mice models (Fig 2 and Fig 3) . The pathogenicity of influenza viruses in mice varies and is dependent on the strain and its adaptation to the host. Depending on virus dose and strain, influenza viruses can induce lethal infections in certain mouse strains usually within two weeks [37, 63] . In our model, C57Bl/6 mice are challenged intranasally with a lethal dose of the respective virus and treated with different regimens in comparison to a vehicle control (placebo). In such a model, early virus replication takes place in the upper respiratory tract. From there, virus spreads to the lung and causes lethal pneumonia. The effect of the treatment on mortality is assessed in comparison to placebo-treated control mice. Of all in-vitro tested influenza strains the H1N1(09)pdm and the LP H7N7 are particularly interesting for two reasons. First, they are highly relevant pathogens, as placebo or with the mono-therapies consisting of carrageenan (12 mg/kg BW/day) or Zanamivir (1 and 3 mg/ kg BW/day) or a combination

thereof. Treatment started 24 hpi and continued for 5 days. (B) Mice (n = 20 per group) were lethally intranasally infected without anesthesia on day 0 and intranasally treated twice per day either with placebo or a combination of carrageenan with Zanamivir (3 mg/kg BW/day). Treatment started either 48 hpi or 72 hpi and continued for 5 days. On the y-axis the survival of mice [%] and on the x-axis the time post infection [days] is given. Placebo treated uninfected control mice showed 100% survival in both experiments (data not shown). Statistical analyses were conducted using log rank test and are shown beneath the graphs. Values of $p < 0.05$ were considered statistically significant; non-significance (n.s.) was obtained with p -values > 0.05 . both are involved in recent influenza outbreaks. The H1N1(09)pdm is associated with more than 18,400 deaths in the season 2009/2010 while the LP H7N7 carries an HA closely related to that of the avian influenza H7N9 virus which has caused more than 175 deaths until October 2014 [64] . Second, they are of special interest for the carrageenan/Zanamivir combination approach. They have shown to differ in in-vitro susceptibility to carrageenan, Zanamivir (Table 1) and the combination thereof (Fig 1) . While H1N1(09)pdm was highly sensitive to inhibition by both substances alone, H7N7 required much higher concentrations of carrageenan and Zanamivir, respectively, to achieve similar inhibition efficiencies. Therefore, both virus strains were chosen to further explore the efficiency of the combination therapy in a mouse model.

We established lethal mouse models with both viruses that resulted in 6.8 and 8.5 mean survival days for LP H7N7 and H1N1(09)pdm, respectively. These results are in good accordance to similar already published lethal influenza models [65] [66] [67] . In our models the lowest effective dose for Zanamivir at a treatment start 24 hpi was found to be between 1 to 3 mg/kg BW/day for both viruses. This concentration range is relatively high in comparison to other published studies. However, these studies were done under anesthesia with different viruses and a prophylactic therapy start [65, 66] . The fact that a higher dose of NI is needed for an effective treatment when the therapy starts 24 hpi is already known for Oseltamivir [68] . Nonetheless, also data with much higher effective concentrations (10 mg/kg BW/day [69]) and with similar concentrations of Zanamivir (2.5 mg/kg BW/day [67]) were published as well.

We found that the combination of carrageenan with 3 mg/kg BW/day Zanamivir used for treatment of H7N7 infected mice resulted in significantly enhanced survival of mice in comparison to both mono-therapies (Fig 2) . The significantly enhanced survival compared to the placebo treated group was also found after a delayed treatment start 48 hpi. Furthermore, in the H1N1(09)pdm model the combination of carrageenan with 1 mg/kg

BW/day Zanamivir showed statistically significant enhanced survival in comparison to placebo treatment even after a treatment start 72 hpi. This is a remarkable finding since NIs are normally not effective when applied 72 hpi.

The finding supports the development of the Zanamivir and carrageenan combination approach. As the intranasal treatment regime is incapable to effectively treat virus infections of the lung, the primary target of such a product is the prophylaxis and therapy of uncomplicated influenza. Since the majority of influenza infections causes uncomplicated illnesses and practically all cases of influenza start with an infection of the nasal cavity or the upper respiratory tract, the therapeutic potential is huge. However, clinical studies are required to elucidate and demonstrate the potential of the proposed combination therapy.

Combination of antiviral strategies has led to impressive achievements in the combat against other viral disease like HIV. In particular the problem of antiviral resistance could be addressed with this strategy. In the last decade concerns have been raised about the increased emergence of Oseltamivir resistant influenza viruses. The augmented appearance of viruses carrying the mutation H275Y in the neuraminidase of H1N1(09)pdm viruses that confers resistance to Oseltamivir left Zanamivir as only treatment option for symptomatic patients infected with an Oseltamivir resistant influenza strain [70] . In contrast to Oseltamivir, resistance to Zanamivir is less frequent. To date, Zanamivir resistant influenza has been detected only once, in an immunocompromised patient [71, 72] . However, lessons should be learned from previous anti-influenza interventions which resulted in occurrence of resistance against currently approved drugs [73] . Therefore, concerns are comprehensible that an increased Zanamivir use may also lead to the rapid emergence of resistances [74] . To overcome this threat, a combination of antivirals which inhibits virus replication by distinct mechanisms is a valid strategy. We checked for the possibility of generating double compound escape mutant viruses while passaging viruses in the presence of increasing concentrations of compound combinations. After 10 passages in MDCK cells no resistance to the compound combination for any tested influenza virus could be found (data not shown). However, this finding does not guarantee that emergence of Zanamivir escape mutants can be completely halted.

In summary, we demonstrated that the anti-influenza mechanisms of both single compounds complement each other. The combination provides synergistically better protection against a broader spectrum of influenza viruses than the individual compounds.

A nasal spray containing carrageenan together with Zanamivir provides an easy to apply treatment of upper respiratory tract infections in patients under suspicion to be influenza infected. Patients would benefit from the fast and efficient treatment of uncomplicated influenza in the upper respiratory tract. Due to the faster influenza virus clearance from the upper respiratory tract and the independent antiviral mechanism of carrageenan and Zanamivir the likelihood to develop escape mutations against Zanamivir will be reduced. Both individual compounds are able to reduce severity and/or duration of the influenza illness and a combination is expected to work similarly. Additionally, due to the broad antiviral effectiveness of carrageenan, patients will receive in parallel a treatment of concomitant viral infections. Therefore, patients will benefit from a decreased probability to develop complications. In consideration of the complications known to accompany an influenza virus illness this combinational therapy meets an urgent medical need.

A second scope of this combination is the protection against newly emerging pandemic viruses during the time until identification of the virus followed by manufacturing and distribution of vaccines [43] . Even if, due to new reverse genetic techniques, less time for production of vaccines is needed, it still takes months before large quantities of vaccine are available [75] . During this time the human population should be protected to decelerate viral spread. At the moment the only available opportunities for personal protection are hygiene measures and the use of Tamiflu (brand name of Oseltamivir).

Novel protection and treatment options for influenza are desperately needed. Based on our encouraging results in mice we suggest testing a nasal spray containing carrageenan in combination with the neuraminidase inhibitor Zanamivir in clinical trials for prevention or treatment of uncomplicated influenza infections.

Supporting Information S1 Fig. Therapeutic efficacy of iota-carrageenan solely or together with kappa-carrageenan in influenza H7N7 lethal infected mice. Mice (n = 20 per group) were lethally intranasally infected without anesthesia on day 0 and accordingly intranasally treated twice per day either with placebo or with iota-carrageenan or with a mixture of iota- and kappa-carrageenan. Treatment started 24 hpi and continued for 5 days. On the y-axis the survival of mice [%] and on the x-axis the time post infection [days] is given. Placebo treated, uninfected control mice showed 100% survival (data not shown). Statistical analyses were conducted using log rank test and are shown beneath the graphs. Values of

p<0.05 were considered statistically significant; non-significance (n.s.) was obtained with p-values >0.05. (TIFF)

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Etiology of Influenza-Like Illnesses from Sentinel Network Practitioners in Réunion Island, 2011-2012

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5031398/>

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Abstract: In Réunion Island, despite an influenza surveillance established since 1996 by the sentinel general practitioner's network, little is known about the etiology of Influenza like-illness (ILI) that differs from influenza viruses in a tropical area. We set up a retrospective study using nasal swabs collected by sentinel GPs from ILI patients in 2011 and 2012. A total of 250 swabs were randomly selected and analyzed by multiplex reverse transcriptase polymerase chain reaction (RT-PCR) including research of 18 viruses and 4 bacteria. We detected respiratory viruses in 169/222 (76.1%) samples, mostly rhinovirus (23.4%), influenza A virus (21.2%), influenza B virus (12.6%), coronavirus (4.9%) and Human metapneumovirus (3.6%). Nine swabs (5.3% of positive swabs) revealed co-infections with two viruses identified, among which six concerned co-infections with influenza viruses. We observed important seasonal differences, with circulation of Human

Metapneumoviruses, RSV A and B and coronavirus only during summer; whereas parainfluenza viruses were identified only during winter. In conclusion, this study highlights a substantial circulation of multiple respiratory pathogens in Réunion Island throughout the year. It shows that ILI are not only attributable to influenza and underlines the need for biological surveillance. As the use of multiplex RT-PCR showed its efficacy, it is now used routinely in the surveillance of ILI.

Text: Influenza like-illness (ILI) or acute respiratory infections can be caused by several types of respiratory viruses or bacteria in humans [1] . Influenza viruses, Respiratory Syncytial viruses (RSV) and Parainfluenza viruses are identified as major viruses mostly responsible for ILI and pneumonia in several studies [2] . However practitioners cannot diagnose the infection without a biological test confirmation. Unfortunately, these infections causes are identified in less than 50% [3] .

Réunion Island, a French overseas territory with 850,000 inhabitants, is located in the southern hemisphere between Madagascar and Mauritius in the Indian Ocean (Latitude: 21°05.2920 S Longitude: 55°36.4380 E.). The island benefits from a healthcare system similar to mainland France and epidemiological surveillance has been developed by the regional office of the French Institute for Public Health Surveillance (Cire OI), based on the surveillance system of mainland France [4] . Influenza activity generally increases during austral winter, corresponding to summer in Europe [5] . Since 2011, influenza vaccination campaign in Reunion Island starts in April and the vaccine used corresponds to World Health Organization recommendations for the southern hemisphere.

Since 1996, clinical and biological influenza surveillance has been based on a sentinel practitioner's network [6] . In 2014, this network was composed of 58 general practitioners (GPs) spread over the island and represented around 7% of all Réunion Island GPs. Nasal swabs are randomly collected all along the year and are tested by RT-PCR for influenza viruses. Among these surveillance samples, 40 to 50% are tested positive for influenza A virus, A(H1N1)pdm09 or B virus by the virological laboratory of the University Hospital Center of Réunion. Thus ILI samples tested negative for influenza are of unknown etiology.

Several biological tools allow identifying respiratory pathogens from nasal swab. In recent years, multiplex reverse transcriptase polymerase chain reaction (RT-PCR) has been

developed to identify several viruses simultaneously [7] [8] [9] [10] . We therefore used this new method to set up a retrospective study using swabs collected by sentinel GPs from 2011 to 2012.

The main objective of our study was to characterize respiratory pathogens responsible for ILI consultations in sentinel GPs in 2011 and 2012. Secondary objectives were to highlight seasonal trends on respiratory pathogens circulation and to describe occurrence of co-infections, especially during the flu season.

ILI was defined as a sudden onset of fever more than 38 degrees Celsius and cough, associated or not with other symptoms such as breathing difficulty, headache, etc. Every week, all GPs of the sentinel network were encouraged to collect a nasal swab from the first two patients who presented ILI since less than three days. After being tested for influenza viruses, the 994 swabs collected in 2011 and 2012 are frozen at -80°C at the university hospital center (CHU) laboratory.

Based on the budget, a season-stratified sample of 250 swabs was randomly selected in order to describe circulating viruses including outside flu season. Random sampling was performed with Excel 1 using the anonymized surveillance database of the Cire OI. The sampling frame contained identification number of swab assigned by Cire OI, laboratory identification number, sex, age, date of onset of symptoms, date of swab collection and result of influenza RT-PCR.

We used Respifinder 1 Smart 22 kits a multiplex RT-PCR (PathoFinder, Maastricht, The Netherlands) which can detect 22 respiratory pathogens. This assay is based on the multiplex ligation-dependent probe amplification (MLPA) technology. The reverse transcription and preamplification steps were performed on the eppgradient Mastercycler 1 (Eppendorf) and the hybridization, ligation and detection steps on the LightCycler 1 480 system (Roche Applied Science). This method was chosen because of its high specificity, compared to other same methods (78% versus 33%) [3, 11] . Multiplex analysis allows for rapid production of diagnostic results. It thus allows highlighted the possible presence of eighteen respiratory viruses and four bacteria in one reaction by melt curve analysis:

Influenza A not (H1N1

Statistical analyses were performed with Stata 1 and Excel 1 . Two seasons were defined to identify possible seasonal trends in circulation of the viruses: winter season during weeks 23 to 39 between June and September and summer season during the rest of the year.

Data and swabs result from a surveillance system that received regulatory approvals, including the CNIL (National Commission for Information Technology and Civil Liberties Number 1592205) approval in July 2012. All the patients have received oral information and gave their consent for swab and data collection. Data were collected for surveillance purpose and are totally anonymous.

Among the 250 randomly-selected swabs, 26 were not available anymore as they were sent to Influenza Reference Center for confirmation and characterization of the pathogenic agent. According to the sensitivity of the assay two samples could be discordant results between Influenza PCR initially realized and Multiplex PCR. Thus they were deleted from the analysis: one is positive for Influenza in singleplex and negative for all tested pathogens in multiplex and one is positive for Influenza in singleplex and positive for PIV2 in multiplex. In total, 222 analyses were considered. Moreover, 53 samples were negative for all analyzed respiratory pathogens (23.9%) and 169 samples had at least one detected pathogen (76.1%), finally a total of 178 pathogens was identified.

During the study period, a minority of the weeks (21 i.e. 20%) did not include any sampled swab, mainly outside flu season.

Patients' sex-ratio was 0.63 (86 men and 136 women) and mean age was 28.4 years [min 0; max 81]. Ten percent had less than 5 years, 24% 5-15 years, 63% 15-65 years and only 3% were 65 and older.

The respiratory pathogens most frequently identified in ILI swabs were rhinovirus (23.4%), influenza A not H1N1 (21.2%) and influenza B (12.6%) (Table 1) .

Among the 22 respiratory pathogens tested by the multiplex, only three were not found in any analyzed sample: Parainfluenza3, Legionella pneumophila and Bordetella pertussis.

Regarding co-infections, nine swabs revealed the presence of two viruses, among which 6 involved influenza viruses (Table 2) .

Analyses showed that some viruses are possibly seasonal and were circulating during a specific period of the year. They are detected only in summer for Human Metapneumovirus, RSV A and B, and influenza A(H1N1)pdm09. For the latter, it is specific to the studied period since the influenza A(H1N1)pdm09 virus reappeared in Réunion Island in October 2012 and was no longer circulating since late 2010. On the opposite, Parainfluenza 1,2 and 4 viruses were identified only in winter. For other pathogens, no specific period of detection was observed.

A weekly description of samples was realized to study the distribution of respiratory pathogens in 2011 and 2012 (Fig 1) . Results of biological analyses were compared with data of ILI consultations declared by sentinel GPs in 2011 and 2012. We observed in 2011, after a first wave in June mainly due to influenza A not H1N1 virus, a second wave of ILI consultations with mainly identification of Parainfluenza viruses and not influenza viruses. In 2012, the second epidemic wave at the end of austral winter coincided with Influenza viruses and Rhinovirus circulation.

Regarding negative swabs (Fig 2) , we observed no seasonality during the study period with a similar proportion whatever the season.

This retrospective study based on a sentinel GPs network showed that not only influenza viruses are responsible for ILI consultations. Indeed, an important circulation of multiple pathogens was observed throughout the year, with 12 different types of pathogens identified in 2011 and 2012. Respiratory viral pathogens were present in 76.1% of samples, which is largely above results from annual influenza surveillance [12] . After influenza viruses, Rhinovirus and Coronavirus were the most common respiratory viruses in Réunion Island. Although samples were not taken every week, sample was representative of ILI activity and consistent with flu season. Nevertheless, according to the low number of samples, it is difficult to conclude about seasonality. However in our study, RSV was circulating in summer season which is hot and rainy, which is confirmed by other studies in tropical region [13] .

This study also highlighted several co-infections, showing that concomitant the multiple etiology of ILI. Co-circulation was already observed in Réunion Island during the A(H1N1) pdm09 pandemic in addition to influenza virus, with identification of other respiratory viruses such as Rhinovirus or Coronavirus [14] . In mainland France, during this pandemic, circulation of major respiratory viruses was found, such as Rhinovirus, Parainfluenza, Coronavirus, Human Metapneumovirus, like in our publication [15] [16] . In our study, only 5.3% of positive swabs were co-infections whereas in two studies in Madagascar co-infections represented 27.3% and 29.4% [17] [18] .

Despite the distance of 9,300 km between Réunion and France, the island is directly connected to Europe with four daily flights to France. These exchanges can impact respiratory pathogens circulation in southern and northern hemisphere. Results of this study can therefore be of interest to both Indian Ocean and Europe countries.

Among the 148 swabs initially negative for influenza because not previously tested for any other viruses, the study found an etiology for 95 swabs. In total, only 53 swabs, representing 24% of the sample, remained without etiology with negative multiplex PCR results all along the year. Multiple hypotheses can explain this result: a poor quality of swabs, preventing from identifying a pathogen, noninfectious causes or other pathogens not included in the multiplex PCR. However, we couldn't test the negative swabs for RNase P, a marker of human cells, which could provide a modicum of assurance that the swab contained human cells.

Concerning the two samples divergent for influenza identification between the multiplex and singleplex PCR, we discarded them for the analysis; one was positive in Influenza with singleplex and positive in PIV with multiplex. It could be a false positive result from singleplex. Indeed, as the multiplex PCR assay has a good sensitivity and is considered as a gold-standard, we decided to keep seven negative results for Influenza in singleplex and positive in Influenza in multiplex [7] [8] [9] [10] .

No case of *Bordetella pertussis* which causes whooping cough and *Legionella pneumophila* which causes Legionnaires' disease was identified in this study. However,

these diseases are rare in Réunion Island, around three cases of Legionnaires' disease are declared each year.

A limit of the study is that no clinical data were available in the virological surveillance system of influenza in Réunion Island. It was impossible to compare clinical symptoms according to each pathogen and to know if there are different pathogens which cause for instance rhinitis, laryngitis or bronchitis (diseases included in ILI). A specific prospective study including clinical data might provide useful elements in the semiotics of diseases.

In conclusion, this study highlighted an important circulation of multiple pathogens in Réunion Island throughout the year. It shows that ILI is not specific to influenza and so it is essential to have biological results in order to establish the differential diagnosis and thus explain the etiology of symptoms. For a better understanding of respiratory pathogens circulating in Réunion Island, information from this study may also be useful to practitioners who see many patients in consultation with ILI. As the use of multiplex RT-PCR showed its efficacy in the ILI surveillance and allowed to highlight the circulation of other viruses and bacterial causes of respiratory infections, it is now used routinely in the surveillance of ILI. Moreover, it would be interesting to repeat this study every 3 or 5 years adding clinical data to monitor the evolution of respiratory pathogens in Réunion Island over time.

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No credible evidence supporting claims of the laboratory engineering of SARS-CoV-2

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7054935/>

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Abstract: nan

Text: The emergence and outbreak of a newly discovered acute respiratory disease in Wuhan, China, has affected greater than 40,000 people, and killed more than 1,000 as of Feb. 10, 2020. A new human coronavirus, SARS-CoV-2, was quickly identified, and the associated disease is now referred to as coronavirus disease discovered in 2019 (COVID-19) (<https://globalbiodefense.com/novel-coronavirus-covid-19-portal/>).

According to what has been reported [1] [2] [3] , COVID-2019 seems to have similar clinical manifestations to that of the severe acute respiratory syndrome (SARS) caused by SARS-CoV. The SARS-CoV-2 genome sequence also has ~80% identity with SARS-CoV, but it is most similar to some bat beta-coronaviruses, with the highest being >96% identity [4, 5] .

Currently, there are speculations, rumours and conspiracy theories that SARS-CoV-2 is of laboratory origin. Some people have alleged that the human SARS-CoV-2 was leaked directly from a laboratory in Wuhan where a bat CoV (RaTG13) was recently reported, which shared ~96% homology with the SARS-CoV-2 [4] . However, as we know, the human SARS-CoV and intermediate host palm civet SARSlike CoV shared 99.8% homology, with a total of 202 single-nucleotide (nt) variations (SNVs) identified across the genome [6] . Given that there are greater than 1,100 nt differences between the human SARS-CoV-2 and the bat RaTG13-CoV [4] , which are distributed throughout the genome in a naturally occurring pattern following the evolutionary characteristics typical of CoVs, it is highly unlikely that RaTG13 CoV is the immediate source of SARS-CoV-2. The absence of a logical targeted pattern in the new viral sequences and a close relative in a wildlife species (bats) are the most revealing signs that SARS-CoV-2 evolved by natural evolution. A search for an intermediate animal host between bats and humans is needed to identify animal CoVs more closely related to human SARS-CoV-2. There is speculation that pangolins might carry CoVs closely related to SARS-CoV-2, but the data to substantiate this is not yet published ([https:// www.nature.com/articles/d41586-020-00364-2](https://www.nature.com/articles/d41586-020-00364-2)).

Another claim in Chinese social media points to a Nature Medicine paper published in 2015 [7] , which reports the construction of a chimeric CoV with a bat CoV S gene (SHC014) in the backbone of a SARS CoV that has adapted to infect mice (MA15) and is capable of infecting human cells [8] . However, this claim lacks any scientific basis and must be discounted because of significant divergence in the genetic sequence of this construct with the new SARS-CoV-2 (>5,000 nucleotides).

The mouse-adapted SARS virus (MA15) [9] was generated by serial passage of an infectious wildtype SARS CoV clone in the respiratory tract of BALB/c mice. After 15 passages in mice, the SARS-CoV gained elevated replication and lung pathogenesis in aged mice (hence M15), due to six coding genetic mutations associated with mouse adaptation. It is likely that MA15 is highly attenuated to replicate in human cells or patients due to the mouse adaptation.

It was proposed that the S gene from bat-derived CoV, unlike that from human patients-or civets-derived viruses, was unable to use human ACE2 as a receptor for entry into human cells [10, 11] . Civets were proposed to be an intermediate host of the bat-CoVs, capable of spreading SARS CoV to humans [6, 12] . However, in 2013 several novel bat coronaviruses were isolated from Chinese horseshoe bats and the bat SARS-like or SL-CoV-WIV1 was able to use ACE2 from humans, civets and Chinese horseshoe bats for entry [8] . Combined with evolutionary evidence that the bat ACE2 gene has been positively selected at the same contact sites as the human ACE2 gene for interacting with SARS CoV [13] , it was proposed that an intermediate host may not be necessary and that some bat SL-CoVs may be able to directly infect human hosts. To directly address this possibility, the exact S gene from bat coronavirus SL-SHC014 was synthesized and used to generate a chimeric virus in the mouse adapted MA15 SARS-CoV backbone. The resultant SL-SHC014-MA15 virus could indeed efficiently use human ACE2 and replicate in primary human airway cells to similar titres as epidemic strains of SARS-CoV. While SL-SHC014-MA15 can replicate efficiently in young and aged mouse lungs, infection was attenuated, and less virus antigen was present in the airway epithelium as compared to SARS MA15, which causes lethal outcomes in aged mice [7] .

Due to the elevated pathogenic activity of the SHC014-MA15 chimeric virus relative to MA15 chimeric virus with the original human SARS S gene in mice, such experiments with

SL-SHC014-MA15 chimeric virus were later restricted as gain of function (GOF) studies under the US government-mandated pause policy (<https://www.nih.gov/about-nih/who-we-are/nih-director/statements/nih-lifts-funding-pause-gain-function-research>). The current COVID-2019 epidemic has restarted the debate over the risks of constructing such viruses that could have pandemic potential, irrespective of the finding that these bat CoVs already exist in nature. Regardless, upon careful phylogenetic analyses by multiple international groups [5, 14], the SARS-CoV-2 is undoubtedly distinct from SL-SHC014-MA15, with >6,000 nucleotide differences across the whole genome. Therefore, once again there is no credible evidence to support the claim that the SARS-CoV-2 is derived from the chimeric SL-SHC014-MA15 virus.

There are also rumours that the SARS-CoV-2 was artificially, or intentionally, made by humans in the lab, and this is highlighted in one manuscript submitted to BioRxiv (a manuscript sharing site prior to any peer review), claiming that SARS-CoV-2 has HIV sequence in it and was thus likely generated in the laboratory. In a rebuttal paper led by an HIV-1 virologist Dr. Feng Gao, they used careful bioinformatics analyses to demonstrate that the original claim of multiple HIV insertions into the SARS-CoV-2 is not HIV-1 specific but random [15]. Because of the many concerns raised by the international community, the authors who made the initial claim have already withdrawn this report.

Evolution is stepwise and accrues mutations gradually over time, whereas synthetic constructs would typically use a known backbone and introduce logical or targeted changes instead of the randomly occurring mutations that are present in naturally isolated viruses such as bat CoV RaTG13. In our view, there is currently no credible evidence to support the claim that SARS-CoV-2 originated from a laboratory-engineered CoV. It is more likely that SARS-CoV-2 is a recombinant CoV generated in nature between a bat CoV and another coronavirus in an intermediate animal host. More studies are needed to explore this possibility and resolve the natural origin of SARS-CoV-2. We should emphasize that, although SARS-CoV-2 shows no evidence of laboratory origin, viruses with such great public health threats must be handled properly in the laboratory and also properly regulated by the scientific community and governments.

No potential conflict of interest was reported by the author(s).

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Effectiveness of zinc supplementation on diarrhea and average daily gain in pre-weaned dairy calves: A double-blind, block-randomized, placebo-controlled clinical trial

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6619766/>

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Abstract: The objective of this clinical trial was to evaluate the effectiveness of zinc supplementation on diarrhea and average daily weight gain (ADG) in pre-weaned dairy calves. A total of 1,482 healthy Holstein heifer and bull calves from a large California dairy were enrolled at 24 to 48 hours of age until hutch exit at approximately 90 days of age. Calves were block-randomized by time to one of three treatments: 1) placebo, 2) zinc methionine (ZM), or 3) zinc sulfate (ZS) administered in milk once daily for 14 days. Serum total protein at enrollment and body weight at birth, treatment end, and hutch exit were measured. Fecal consistency was assessed daily for 28 days post-enrollment. For a random sample of 127 calves, serum zinc concentrations before and after treatment and a fecal antigen ELISA at diarrhea start and resolution for *Escherichia coli* K99, rotavirus, coronavirus, and *Cryptosporidium parvum* were performed. Linear regression showed that ZM-treated bull calves had 22 g increased ADG compared to placebo-treated bulls ($P =$

0.042). ZM-treated heifers had 9 g decreased ADG compared to placebo-treated heifers ($P = 0.037$), after adjusting for average birth weight. Sex-stratified models showed that high birth weight heifers treated with ZM gained more than placebo-treated heifers of the same birth weight, which suggests a dose-response effect rather than a true sex-specific effect of ZM on ADG. Cox regression showed that ZM and ZS-treated calves had a 14.7% ($P = 0.015$) and 13.9% ($P = 0.022$) reduced hazard of diarrhea, respectively, compared to placebo-treated calves. Calves supplemented for at least the first five days of diarrhea with ZM and ZS had a 21.4% ($P = 0.027$) and 13.0% ($P = 0.040$) increased hazard of cure from diarrhea, respectively, compared to placebo-treated calves. Logistic regression showed that the odds of microbiological cure at diarrhea resolution for rotavirus, *C. parvum*, or any single fecal pathogen was not different between treatment groups. Zinc supplementation delayed diarrhea and expedited diarrhea recovery in pre-weaned calves. Additionally, zinc improved weight gain differentially in bulls compared to heifers, indicating a research need for sex-specific dosing.

Text: Introduction Diarrhea is the leading cause of morbidity and mortality and the most common reason for antimicrobial drug treatments in pre-weaned dairy heifers [1, 2]. A USDA survey of preweaned dairy heifers reported that 24% experienced diarrhea and 18% received antimicrobial treatment for it [1]. Diarrhea is also a leading cause of morbidity and the second foremost cause of mortality in children with over 1 billion cases and a half a million deaths annually [3, 4]. Zinc supplementation in children decreases the incidence, duration, and severity of diarrhea, increases recovery rates, decreases the use of antibiotics and antidiarrheal medications, and reduces mortality [5] [6] [7] [8] [9] [10]. In a clinical trial that established a non-toxic zinc dose and investigated its therapeutic use for diarrhea in neonatal dairy calves, zinc-treated calves had numerically quicker clinical recovery, increased weight gain, and higher odds of fecal clearance of *Cryptosporidium parvum* between diarrhea onset and recovery compared with placebotreated calves [11]. As a result, zinc supplementation may be beneficial for prevention of diarrhea in dairy calves and, thus, minimize antimicrobial use. However, studies investigating zinc's potential effectiveness are lacking.

In children, both organic (zinc acetate, zinc gluconate, and zinc methionine) and inorganic (zinc sulfate and zinc oxide) zinc formulations are beneficial in the prevention and treatment of childhood diarrhea [12] [13] [14] [15]. However, differing bioavailability was observed in several animal studies [16] [17] [18] [19]. In addition, the underlying mechanism of action of oral zinc is unknown [6]. Hence, contrasting the effect, if any, of

organic compared to inorganic zinc formulations in pre-weaned calves may help identify differences in mode of action.

The objective of this clinical trial was to compare average daily weight gain (ADG) and the incidence and duration of diarrhea in pre-weaned dairy calves randomly assigned to receive either organic zinc methionine (ZM), inorganic zinc sulfate (ZS), or a placebo in milk once daily for 14 days. By elucidating the potential role of zinc supplementation in prevention of diarrhea in preweaned dairy calves, calf morbidity, mortality, and antimicrobial usage may be mitigated.

A double-blind, block randomized, placebo-controlled clinical trial was conducted between December 14, 2015 and June 15, 2016 on a large dairy in California's San Joaquin Valley. The dairy was selected based on the owner and calf manager's willingness to participate in the study. The dairy herd was composed of 5,500 lactating cows, predominantly Holsteins, and housed approximately 1,600 pre-weaned calves. Approximately 75% of the calves were born on the participating dairy and 25% were born on an affiliated dairy located approximately 10 miles away. Calves enrolled in the trial included healthy Holstein heifer or bull calves 24 to 48 hours of age. Calves were determined to be healthy via visual examination by a veterinarian (HF) or a trained researcher. Calves were excluded if they had obvious morbidities or congenital defects, were non-Holstein, born on the affiliated dairy, younger than 24 hours of age or older than 48 hours of age at the time of enrollment. Calves from the affiliated dairy were excluded due to differences in physical location and management practices of pre-partum cows. All procedures were approved by the University of California Davis Institutional Animal Care and Use Committee (protocol number 18067 Approved: March 6, 2014) .

107 g between treatment groups (Stata, College Station, TX). After allowing for 15% attrition and assuming 50% incidence of diarrhea based on study authors expert opinion, and a difference in ADG of 107g [11] , a sample size of 500 calves per treatment group (n = 1500 total) was deemed required.

Newborn calves were removed from the dam within an hour of birth and placed in a strawbedded, group calf pen where their navels were dipped in an iodine-based solution. Each calf received 4 liters of colostrum within 1 hour of birth and a second colostrum

feeding (2 liters) 6-10 hours after birth. Colostrum was refrigerated for < 48 hours and heated in a hot water bath prior to feeding using an esophageal tube feeder. Within 18 hours of birth, pre-weaned calves were transported to individual metal hutches initially bedded with almond shells. Straw hay was later added to wet and muddy hutches throughout the pre-weaning period.

For the first 14 days of life, pre-weaned calves were bottle-fed 1.9 liters of milk twice daily and 1.9 liters of a commercial oral electrolyte solution (Calva Lyte; Calva Products, Inc., Acampo, CA) once daily between milk feedings. Milk consisted of a combination of pasteurized waste milk, rehydrated commercial milk replacer powder (Strauss Feeds LLC, Watertown, WI), tetracycline and neomycin powder, and additional supplements (S1 Table). The proportion of pasteurized waste milk to milk replacer varied with each feeding, as the volume of waste milk varied with changes in the number and production of cows contributing to the waste milk tank. After 14 days of age, calves were bottle-fed 2.8 liters of milk twice daily. Calves with clinical diarrhea received 1.9 liters of a commercial oral electrolyte solution (NuLife; Genex Cooperative, Inc., Shawano, WI) once daily between milk feedings. A list of ingredients that make up the two oral electrolyte solutions can be found in S2 Table. All pre-weaned calves had free choice access to water and a calf starter grain mix. Calves were gradually weaned over a 10-day period, starting at approximately 60 days of age after which calves received a grower grain mix until 90 days of age when they were moved to group pens.

Each calf received 1 mL of a selenium supplement (MU-SE; Merck Animal Health, Boxmeer, Netherlands) intramuscularly within 24 hours of age and an intranasal vaccine (Inforce 3, Zoetis, Inc., Florham Park, NJ) within 48 hours of age and again near the time of weaning. Approximately 8.3% (n = 126) of all enrolled calves were also vaccinated using an autogenous *Moraxella bovis/bovoculi* bacterin vaccine (Newport Laboratories, Inc., Worthington, MN) for the prevention of pinkeye at 5 and 7 weeks of age. All pre-weaned calves were evaluated daily by dairy personnel for calfhood diseases and treated according to standard on-farm treatment protocols. With regard to diarrhea therapy, calves less than two weeks of age with clinical diarrhea received an oral mixture of 118.5 mL (2.08 g) bismuth subsalicylate (Bismusal Suspension, Durvet, Inc., Blue Springs, MO) and 31.5 mL (1575 mg) spectinomycin (SpectoGard, Bimeda, Inc., Le Sueur, MN) once daily for two days. Calves older than two weeks of age with clinical diarrhea received oral sulfamethoxazole (1600 mg)/trimethoprim (320 mg) (Amneal Pharmaceuticals of NY,

Hauppauge, NY) once daily for 2 to 3 days. Repeated treatment was at the discretion of the calf manager.

(inductively coupled plasma mass spectrometry). Financial limitations restricted the ability to test more than two samples of each dietary component. Due to variation of zinc content in duplicate samples, the maximum concentration was used to estimate the daily zinc intake during the first 14 days of life (S3 Table) .

In blocks of 36, enrolled calves were randomized using a random number generator (Microsoft Corporation, Redmond, WA) to one of three treatment groups: 1) placebo, 2) ZM, or 3) ZS to be administered in the morning milk feeding once daily for 14 days, starting on the day after enrollment. During the 14-day zinc treatment period, study calves that did not drink the entire milk bottle were tube-fed the remaining milk by trained technicians using an esophageal feeder disinfected between uses. The ZM treatment group received 0.45 g zinc methionine complex (equivalent to 80 mg of elemental zinc) as the product Zinpro180 (Zinpro Corporation, Eden Prairie, MN) combined with 0.44 g milk replacer powder. The ZS treatment group received 0.22 g zinc sulfate monohydrate (equivalent to 80 mg of elemental zinc) (Sigma-Aldrich Company, St. Louis, MO) combined with 0.44 g milk replacer powder. The placebo treatment group received approximately 0.44 g fresh milk replacer powder. Zinc supplementation was based on a previously published clinical trial, toxicological studies, and nutritional guidelines [11, [22] [23] [24] [25] . The milk replacer powder used in treatment preparation was the same product used in the pre-weaned calf milk ration. Treatments were weighed (GX-2000 precision scale; A&D Co Ltd., San Jose, CA) at the Dairy Epidemiology Laboratory at the University of California, Davis Veterinary Medicine Teaching and Research Center (VMTRC; Aly Lab) in Tulare, CA and placed in 2.0 mL microcentrifuge tubes with polypropylene snap caps (Fisher Scientific, Pittsburgh, PA). Prior to study commencement, a color was randomly and permanently assigned to each treatment group, after which treatment tubes, calf milk bottles and calf hutches were marked with either pink, orange, or yellow ink. The study investigators and technicians responsible for treatment preparation, allocation, administration and data collection were blinded to the color assignment until completion of the trial.

For each calf, the study period started at enrollment (24 to 48 hours of age) and ended when the calf exited the hutch (approximately 90 days of age) and from here onwards will be referred to as the "pre-weaning period." Calf enrollment and study procedures were

performed daily at the time of morning feeding. At enrollment, calf characteristics, including sex, birth date, time of first colostrum feeding, and treatment color were recorded. Attitude and feces were assessed daily until 28 days post-enrollment using previously published methods [11] by two study investigators, a veterinarian and a trained researcher. Attitude scoring was based on a three-point scale. A calf with an attitude score of 1 was bright, alert, and readily stood with stimulation; a calf with a score of 2 was quiet, alert, and stood only with moderate stimulation; a calf with a score of 3 exhibited a dull mentation and remained recumbent in response to stimulation. Fecal scoring was performed only on fresh feces and was also based on a three-point scale, as 1 (solid), 2 (semi-formed/loose), or 3 (watery). If no fresh feces were observed in the hutch, "none seen" (NS) was recorded. Body weight was measured using a digital scale at birth, end of treatment, and hutch exit by farm employees with the exception of end of treatment weights, which were recorded by study investigators. Treatments for farm-diagnosed illnesses were performed and recorded on hutch cards by the calf manager. Study investigators regularly recorded this information from cards in addition to extracting treatment event reports from DairyComp 305 (Valley Agricultural Software, Tulare, CA). Though daily diagnosis and treatment of study calves was performed and recorded by the calf manager, a veterinarian was responsible for examining and determining whether study calves met specific criteria for euthanasia. A calf was euthanized if morbidity was severe enough to significantly depress appetite, hydration status, attitude, mentation, and/or ambulatory capability and the calf showed limited to no immediate response to therapy or supportive care. Study calves were euthanized by the calf manager using an on-farm captive bolt protocol established by the herd veterinarian within 3 hours of the decision to euthanize. Enrolled calves that died prior to hutch exit were necropsied within 24 hours of death by a veterinarian. All calves were monitored throughout the study period for evidence of zinc toxicity. At the end of the study period, calves were cared for at the dairy in accordance with standard commercial operations.

Using the same random number generator, a random sample of 127 calves was selected for additional biologic sampling. Approximately 8 to 10% of the study population was selected due to the financial constraints of additional laboratory testing. Serum zinc concentration at baseline and in response to treatment were evaluated for the three treatment groups. Feces collected on the first day of diarrhea and at diarrhea resolution were evaluated for four fecal pathogens (*Escherichia coli* K99, bovine rotavirus and coronavirus, and *Cryptosporidium parvum* oocysts).

Serum total protein. At enrollment, blood from each calf was collected from the jugular vein using a 20 gauge 1-inch multi-sample needle (Exelint International Co., Redondo Beach, CA) and placed into a 10 mL red top serum tube (BD Vacutainer, Franklin Lakes, NJ) for determination of total protein. Samples remained at room temperature for up to 12 hours until clotting and were then centrifuged (International Equipment Company, CRU-5000, Needham Heights, MA) for 15 minutes. Total protein (g/dL) was measured by a single investigator (HF) on decanted serum using a handheld refractometer (Sper Scientific, Model 300005, Scottsdale, AZ).

Serum zinc. For the 127 randomly-sampled calves, additional blood was collected at enrollment and on the last day of treatment, as described above, and placed into 6.0 mL trace element tubes (BD Vacutainer, Franklin Lakes, NJ). Serum was extracted, as described above, and placed in a 2.0 mL microcentrifuge tube with a polypropylene snap cap (Fisher Scientific, Pittsburgh, PA) and stored at -20°C until analysis. Using the same random number generator, 36 of the 127 sampled calves were randomly selected for analysis due to limited financial resources. The pre-and post-zinc supplementation serum samples from each calf ($n = 72$) were analyzed for zinc concentration (ppm) by ICP-OES (inductively coupled plasma-optical emission spectroscopy) at the CAHFS Laboratory. Quality control samples, including method blanks, laboratory control spikes, and reference Sigma serum, were run with each set of study samples.

Fecal analysis. For 127 randomly-sampled calves at the first diarrhea episode, fecal samples were collected at two time points, the first day of diarrhea (fecal score > 1) and the day diarrhea resolved (second day of fecal score = 1). Using new gloves and sterile lubricant, fresh feces was collected by digital rectal stimulation into 20 mL polypropylene twist-top jars (The Cary Company, Addison, IL) and stored at -20°C until analysis. Fecal samples were tested at the Dairy Epidemiology Laboratory, VMTRC by a veterinarian for *E. coli* K99, bovine rotavirus and coronavirus, and *C. parvum* oocysts using a commercial kit (Pathasure Enteritis 4; Biovet, Quebec, Canada) that is highly specific ($> 90\%$) and sensitive (*E. coli* K99, 93%; rotavirus, 100%; coronavirus, 77%) [26, 27]. For calves with a first-day diarrhea sample on or before 7 days of age, both fecal samples were tested for all four pathogens. For calves with a first-day diarrhea sample after 7 days of age, both fecal samples were tested for three pathogens (*C. parvum*, bovine rotavirus and coronavirus). Samples from calves older than 7 days of age were not tested for *E. coli* K99 based on calves' susceptibility [28]. Testing was performed according to kit manufacturer guidelines, and a low-temperature incubator (Fisher Scientific, Model 146, Pittsburgh, PA) was used

during incubation periods. Test results were recorded as positive or negative using control wells for color comparison. If the color change was darker than the negative control, the sample was considered positive.

Milk zinc. For each of the 107 study days (December 15, 2015 to March 31, 2016) of zinc supplementation, approximately 1.5 mL of treated milk from two bottles of each treatment group were randomly collected into 2.0 mL microcentrifuge tubes with polypropylene snap caps (Fisher Scientific, Pittsburgh, PA), and stored at -20°C until analysis. At the time of analysis, milk samples were thawed at 4°C , vortexed, pooled by week and treatment group, and analyzed for zinc concentration (ppm) by ICP-MS (inductively coupled plasma mass spectrometry) at the CAHFS Laboratory. Quality control samples, including method blanks, laboratory control spikes, National Institute of Standards and Technology (NIST) reference materials (NIST 1640), and a spiked milk sample, were run with each set of study samples.

Data analysis was performed using R Statistic Software version 3.3.1 and Stata IC 14.2 (College Station, TX). Statistical differences were determined at the 5% level of significance using per protocol analysis. An ANOVA was used to compare calves in each treatment group at enrollment with respect to birth weight (kg), serum total protein (g/dL), attitude score, and fecal score. Oral zinc dose at the start and end of treatment was calculated as the zinc supplementation dose (80 mg) divided by calf body weight (kg) at birth and on the last day of treatment, respectively. An ANOVA was used to compare oral zinc dose (mg/kg) at treatment start and end as well as mean body weight (kg) at birth, end of treatment, and hutch exit between treatment groups and between bulls and heifers. A Chi-Square test of Independence was used to compare the proportions of calves by sex as well as mortality between treatment groups. For all analyses, Tukey's Honest Significant Difference method was used to generate pairwise comparisons to further characterize significant differences identified by ANOVA. Residual diagnostics, including Residuals vs. Fitted, Scale-Location, Normal Q-Q, and Cook's distances plots, were used to validate all ANOVA model assumptions. The non-parametric Kruskal-Wallis Rank Sum test was used when assumptions were violated.

For the randomly-sampled calves, Fisher exact tests were used to compare fecal pathogen prevalence on the first day of diarrhea and at diarrhea resolution between treatment groups. Pairwise comparisons with Bonferroni adjustment were used to identify specific

differences in pathogen prevalence. An ANOVA was used to compare serum zinc concentration before and after treatment between treatment groups and between bulls and heifers. A Kruskal-Wallis Rank Sum test was used to compare rank sums of zinc concentrations in pooled milk samples from different treatment groups. Post-hoc Nemenyi-tests for pairwise multiple comparisons of ranked data were used to identify specific differences in zinc concentrations between groups.

For all regression models in this study, univariate regression was first used to evaluate associations between individual predictor variables and outcomes. All variables with statistical and/or biological significance were initially included in multivariate regression models. The final models were built using a manual backwards elimination procedure, with a significance level of $P > 0.05$ as the removal criterion. Confounding was assessed using the method of change of estimates, where a 10% or greater change in the estimate of the treatment group regression coefficient between the models with and without the confounder variable was used as evidence of confounding [29] [30] [31] [32]. Variables identified as confounders were included in the final model. All possible interactions between treatment group and predictor variables were explored and retained in the final model if statistically significant.

Microbiological cure. For the randomly-sampled calves, logistic regression was used to evaluate associations between microbiological cure and treatment group. Other predictor variables of interest included sex, serum total protein, and age on the first day of diarrhea. Microbiological cure was defined as a negative fecal ELISA test at resolution of clinical diarrhea for calves with a positive ELISA test on the first day of diarrhea for at least one of the four fecal pathogens (*E. coli* K99, bovine rotavirus and coronavirus, and *C. parvum*). Models were generated for each fecal pathogen individually and an overall model, which evaluated microbiological cure at clinical diarrhea resolution for calves that tested positive for any single pathogen on the first day of diarrhea. Serum total protein and calf age at first diarrhea were included in all final models to control for potential confounding by passive transfer status and age.

Mean daily weight change. Linear regression was used to evaluate associations between ADG (kg) and treatment group during the treatment and pre-weaning periods separately. Other predictor variables of interest included sex, birth weight (kg), serum total protein, number of days having diarrhea, age, and volume (L) of milk, Calva, and NuLife electrolytes

fed at either end of treatment or hutch exit. For each calf, ADG during the treatment and pre-weaning period was calculated as the difference between birth and end treatment or hutch exit weight, respectively, divided by the number of days between these time points. To explore the possibility of an interaction between treatment group, sex, and birth weight, the final linear regression model for ADG during the pre-weaning period was stratified by sex. Age at the end of treatment or hutch exit and number of days with diarrhea were dropped from all final models in favor of improved Akaike information criterion (AIC).

Onset of diarrhea and clinical cure. For all survival analyses, diarrhea was defined as a fecal score greater than 1 while diarrhea cure was defined as the second consecutive day of normal feces (fecal score of 1) following the first diarrhea episode. Subsequent episodes of diarrhea were not included in the analysis. Calves that died or did not experience diarrhea or cure from diarrhea were censored. If fresh feces were not observed on daily calf hutch assessment, a fecal score was not recorded for that day and not included in the analyses. Kaplan-Meier analysis was used to determine median days to first diarrhea event and, for those calves that developed diarrhea during the assessment period, median days to clinical diarrhea cure. A Log Rank test of equality was used to compare survivor functions between treatments.

Cox Proportional Hazards regression analysis was used to estimate and compare the hazard of diarrhea and diarrhea cure between treatment groups. Sex, age, serum total protein at enrollment, birth weight (kg), antimicrobial therapy, and application of fresh straw to the hutches were evaluated as predictor variables and potential confounders. When modeling the hazard of diarrhea cure, a binary variable termed therapeutic supplementation indicating whether calves were treated with either ZM, ZS or placebo for all or at least the first 5 days of diarrhea was evaluated as an additional covariate. A five-day period was selected by the authors based on clinical experience, as five days represents a reasonable duration over which most therapeutic treatments for calf diarrhea should be applied and be expected to alleviate disease. The proportional hazards assumption that the hazard of diarrhea is independent of time was assessed using analysis of Schoenfeld residuals and testing whether the log hazard-ratio function is constant over time. Any variable found to violate the proportional hazards assumption was included in the final regression model as a time varying covariate.

A total of 1,513 calves were enrolled in the trial. However, due to failure to immediately recognize exclusion criteria, 23 calves were excluded shortly after enrollment. In addition, 8 calves were excluded due to treatment errors. Therefore, a total of 1,482 calves (placebo = 500, ZM = 491, ZS = 491) were included in the final analyses. A total of 242 calves (16.3%) had minimal fecal output at the time of enrollment while 125 calves (8.4%) had abnormal fecal scores of 2 or 3 that were described as meconium. All enrolled calves appeared healthy on visual assessment and hence were assumed to have a normal fecal score at the time of enrollment. The three treatment groups at enrollment did not differ significantly in mean birth weight (kg) ($P = 0.244$), mean serum total protein (g/dL) ($P = 0.541$), mean attitude score ($P = 0.845$), mean fecal score ($P = 0.522$), as shown in Table 1, or distribution of calf sex ($P = 0.472$). Of the 1,482 study calves, 21 (1.4%) died during the trial: 5 calves in the placebo group (1.0%), 11 calves in the ZM group (2.2%), and 5 calves in the ZS group (1.0%). Of these 21 calves that died, 14 (66.7%) were bulls and 7 (33.3%) were heifers. Eighteen of the 21 calves (85.7%) were found dead, rather than euthanized, due to acute and spontaneous death without previous obvious clinical signs of disease. The remaining 3 calves were euthanized prior to death due to severe and/or prolonged morbidity. Characteristics and causes of death based on field necropsy of these calves can be found in S4 Table. There was no significant difference in the proportion of calves that died between treatment groups ($P = 0.168$), though mortality was significantly higher in bulls compared to heifer calves ($P = 0.049$). Birth weight data were available for all 1,482 calves. Due to calf mortality between enrollment and completion of treatment ($n = 4$), end treatment weight data were available for 1,478 calves. Similarly, due to calf mortality ($n = 21$) or missing data for body weight at hutch exit from the dairy's records ($n = 40$), hutch exit weight data were available for 1,421 calves. A summary of body weight data stratified by treatment group and sex is presented in Table 2. Within each treatment group, bull calves showed consistently higher birth weight ($P < 0.001$), end treatment weight ($P < 0.001$), and exit hutch weight ($P < 0.001$) compared to heifer calves. However, at birth, end of treatment, or hutch exit there were no differences in body weight between treatment groups for bulls ($P > 0.1$) or heifers ($P > 0.1$). The mean attitude scores during the study period were 1.2 across the three treatment groups ($P = 0.208$). Of 1,482 calves included in the final analysis,

A total of 629 treated milk samples were obtained throughout the 107-day study period and pooled by treatment group and week, yielding a total of 16 pooled samples per treatment group. Zinc concentrations (ppm) were significantly higher in pooled milk samples treated with ZM ($P < 0.001$) and ZS ($P < 0.001$) compared to placebo-treated samples, and there were no significant differences between ZM and ZS-treated samples ($P = 1.000$), as shown

in S5 Table. Within zinc treatment groups, oral zinc dose at the start and end of treatment is summarized by sex in S6 Table. For both ZM- and ZS-treated calves, oral zinc dose at the start ($P < 0.001$) and end ($P < 0.001$) of treatment was significantly higher in heifer versus bull calves. Serum zinc concentrations before and after treatment were obtained from 36 calves ($n = 12$ for each treatment group) and are summarized in S7 Table. Overall, there were no significant differences in mean pre-treatment serum zinc concentrations between treatment groups ($P = 0.233$). Mean post-treatment serum zinc concentrations were significantly higher in calves treated with ZM ($P < 0.001$) and ZS ($P = 0.002$) compared to placebo-treated calves, and there were no significant differences among calves treated with ZM and ZS ($P = 0.406$). Stratification of serum zinc data by treatment group and sex demonstrated that for ZM-treated calves, heifers had a numerically higher post-treatment serum zinc concentration compared to bulls, though the difference was not statistically significant ($P = 0.199$). In contrast, in ZS-treated calves, heifers had a numerically lower post-treatment serum zinc concentration compared to bulls, though the difference was also not statistically significant ($P = 0.538$).

Fecal analysis data were analyzed for 92 of the 127 randomly-selected calves. The remaining 35 calves were not included in the analysis due to not acquiring diarrhea during the assessment period ($n = 10$), death prior to final sampling ($n = 1$), exclusion due to improper treatment regimen ($n = 1$), or incorrect sampling day ($n = 23$). The 92 calves had a mean age at onset of diarrhea of 13.3, 11.0 and 11.3 days for ZM, ZS and placebo-treated groups, respectively; and a mean age at resolution of diarrhea of 18.8, 16.1 and 15.7 days for ZM, ZS and placebo-treated groups, respectively. There were no significant differences in the prevalence of *E. coli* K99 ($P = 0.694$), rotavirus ($P = 0.331$), coronavirus ($P = 0.819$), or *C. parvum* ($P = 0.719$) fecal shedding on the first day of diarrhea between treatment groups (S8 Table). There were no significant differences in the prevalence of *E. coli* K99 ($P = 0.256$), rotavirus ($P = 0.344$), or coronavirus ($P = 1.000$) fecal shedding at resolution of diarrhea between treatment groups, though there was a difference in *C. parvum* ($P = 0.006$) fecal shedding between treatment groups (S9 Table). The prevalence of *C. parvum* fecal shedding at resolution of diarrhea was significantly higher in calves treated with ZM ($P = 0.009$) and ZS ($P = 0.023$) compared to placebo-treated calves, and there were no significant differences among calves treated with ZM and ZS ($P = 1.000$).

Results of logistic regression models for overall and pathogen-specific microbiological cure are presented in Tables 3-5. For pathogen-specific cure, all calves that tested positive on the first day of diarrhea for coronavirus ($n = 4$) tested negative at clinical diarrhea

resolution. For calves that tested positive on the first day of diarrhea for *E. coli* K99 ($n = 9$), all placebo-treated calves ($n = 2$) tested negative at clinical diarrhea resolution while half ($n = 2$) of all ZS-treated calves tested either positive or negative at clinical diarrhea resolution, resulting in omission of ZS treatment variable from the model due to collinearity. Hence, logistic regression analyses for microbiological cure in calves that tested positive for coronavirus and *E. coli* K99 were not possible. For 59 calves that tested positive for rotavirus on the first day of diarrhea (Table 3), calves treated with ZM had a 50% increased odds of testing negative at diarrhea resolution compared to placebo-treated calves, though this difference was not significant ($P = 0.549$). Likewise, calves treated with ZS had 100% increased odds (2 times the odds) of testing negative for rotavirus at diarrhea resolution compared to placebo-treated calves, though this difference was also not significant ($P = 0.314$). However, this model demonstrated a significant main effect of serum total protein, such that for every 1 unit (g/dL) increase in serum total protein at enrollment, the odds of microbiological cure of rotavirus decreased by 79% ($P = 0.026$). For 40 calves that tested positive for *Cryptosporidium parvum* on the first day of diarrhea (Table 4), calves treated with ZM had an 87% reduced odds of testing negative at diarrhea resolution. Effect of zinc on diarrhea in calves compared to placebo-treated calves, though this difference was not significant ($P = 0.119$). Likewise, calves treated with ZS had a 74% reduced odds of testing negative for *Cryptosporidium parvum* at diarrhea resolution compared to placebo-treated calves, though this difference was also not significant ($P = 0.183$). For 55 calves that tested positive for any one of the four fecal pathogens (*E. coli* K99, rotavirus, coronavirus, *Cryptosporidium parvum*) on the first day of diarrhea (Table 5), calves treated with ZM had a 25% increased odds of testing negative at diarrhea resolution compared to placebo-treated calves, though this difference was not significant ($P = 0.769$). Likewise, calves treated with ZS had a 52% increased odds of testing negative for *Cryptosporidium parvum* at diarrhea resolution compared to placebo-treated calves, though this difference was also not significant ($P = 0.633$). However, this model demonstrated that heifer calves had 71% lower odds of microbiological cure of any single fecal pathogen compared to bull calves ($P = 0.076$).

A total of 1,482 calves were included in the linear regression model results for ADG during the treatment period, which are presented in Table 6. There was no significant difference in ADG for ZM- or ZS-treated calves compared to placebo-treated calves, though there were significant main effects of sex, birth weight, and milk volume. Specifically, heifer calves gained 70 g bodyweight per day less compared to bull calves ($P < 0.001$). For every 1 kg increase in birth weight, calves gained 16 g per day less than their herd mates ($P < 0.001$). For every 1 L increase in milk volume per day during the treatment period, calves gained an

additional 13 g per day ($P < 0.001$). Table 7 summarizes the linear regression analysis of ADG during the pre-weaning period for 1,421 calves which showed a significant difference in ADG for ZM-treated calves compared to placebo-treated calves and in bull versus heifer calves. Milk volume had a significant effect on ADG, such that for every 1 L increase in milk volume per day during the treatment period, calves gained an additional 2 g bodyweight per day ($P = 0.001$). Results of the final model showed a significant main effect for ZM-treated bull calves and a significant interaction term for ZM treatment by sex. After controlling for milk volume received during the treatment Table 6 calves ($n = 1,482$) Effect of zinc on diarrhea in calves period, ZM-treated bulls gained 22 g body weight per day on average more than placebotreated bull calves ($P = 0.042$) and ZM-treated heifers gained 12 g less body weight per day on average compared to placebo-treated heifers ($P = 0.019$). When considering the model coefficients for treatment group, sex, and their interaction, bull calves treated with ZM gained 454 g per day ($0.432 + 0.022$) while female calves treated with ZM gained 0.404 g per day ($0.432 + 0.022 - 0.016 - 0.034$), hence 50 g per day more gain in male calves compared to heifers treated with ZM ($P = 0.019$). For ZS-treated calves, there was a numerical decrease in weight gain of 5 g per day in bulls and 11 g per day in heifers compared to placebo-treated calves, though the differences were not significant ($P = 0.673$ bulls; $P = 0.681$ heifers). Linear regression models of ADG during the pre-weaning period were stratified by sex in order to avoid interpreting a three-way interaction between treatment group, sex, and birth weight. In the heifer model (S10 Table), the interaction between ZM treatment and birth weight was significant which implied that birth weight modified the effect of ZM treatment on ADG. At a 29 kg birth weight (two standard deviations below the mean), ZM-treated heifers gained 49 g body weight per day on average less than placebo-treated heifers ($P = 0.037$). However, at a 49 kg birth weight (two standard deviations above the mean), ZM-treated heifers gained 30 g body weight per day on average more than placebo-treated heifers ($P = 0.037$). In the bull calves model (S11 Table) there was no significant interaction between treatment group and birth weight.

A total of 1,482 calves were included in the Kaplan-Meier survival analysis of time to first diarrhea event (Fig 1). There were no significant differences in median age at onset of diarrhea, specifically, 8, 8 and 7 days for the ZM, ZS and placebo-treated calves, respectively ($P = 0.402$). Cox proportional hazard regression model for diarrhea hazard are presented in Table 8. After controlling for age, calves treated with ZM had a 14.7% reduced hazard of diarrhea compared to placebo-treated calves ($P = 0.015$). Calves treated with ZS had 13.9% reduced hazard of diarrhea compared to placebo-treated calves ($P = 0.022$). calves ($n = 1,421$) A total of 1,394 calves were included in the Kaplan-Meier survival analysis of time to clinical diarrhea cure (Fig 2), as 88 calves failed to acquire diarrhea

during the assessment period. There were no significant differences in the median days to diarrhea cure which was 7 days across all 3 treatment groups ($P = 0.264$). Cox proportional hazard regression model for diarrhea cure hazard are presented in Table 9 . Results of the final model showed a significant interaction term between treatment and therapeutic supplementation as well as the need for age as a time varying covariate. When considering calves that did not receive supplementation, respective to each of the 3 groups, for at least the first five days of diarrhea there was no significant difference between either ZM-and ZS-treated calves compared to placebo-treated calves ($P = 0.223$ ZM, $P = 0.134$ ZS). However, when considering calves that were supplemented for at least the first five days of diarrhea, ZM-treated calves experienced a 21.4% higher hazard of cure from diarrhea compared to placebo-treated calves ($P = 0.027$). Likewise, ZS-treated calves experienced a 13.0% higher hazard of cure from diarrhea compared to placebo-treated calves ($P = 0.040$).

The current trial demonstrated evidence for the beneficial effect of ZM on ADG and neonatal diarrhea as well as an effect of ZS on diarrhea in dairy calves during the pre-weaning period. It is important to consider these results in the context of the entire pre-weaning and hutch period. On average, after 90 days from birth to hutch exit, placebo-treated bull calves gained 38.88 kg body weight while ZM-treated bull calves gained an additional 1.98 kg (40.86 kg). In contrast, the effect of zinc on weight gain in treated heifers depended on birth weight. Low birth weight heifers treated with ZM gained on average less than a placebo-treated heifer of the same birth weight. In contrast, high birth weight heifers treated with ZM gained more than placebo-treated heifers of the same birth weight. The switch in direction of the association between ZM treatment and ADG in heifer calves depending on birth weight suggests a doseresponse effect rather than a true sex-specific effect of ZM on ADG. Hence, low birth weight calves (including heifers) may require a lower dose of ZM to mitigate any negative effect of what is otherwise a suitable dose for higher birth weight calves. These findings are in agreement with a previous randomized clinical trial testing the effect of daily oral zinc in diarrheic neonatal Holstein calves which, showed that ZM-treated calves had a numerically, though not significantly increased ADG compared to calves treated with zinc oxide or placebo due to small sample size [11] . In general, our trial findings are in agreement with the large body of human literature supporting the use of oral zinc for the prevention and treatment of diarrhea and impaired growth in children [5, 10, 33] .

Zinc supplementation is widely accepted by global health organizations as a vital component of therapy for childhood diarrhea [3, 4] , however, recent reviews of the

literature demonstrated heterogeneity in study results on the basis of age, baseline zinc status, geographic location, and supplementation regimen [10, 34] . Similar to our findings, a sex-specific response to zinc supplementation has been demonstrated in several human studies. Zinc gluconate administered for diarrhea prevention reduced the incidence of dysentery in treated boys but not girls [35] ; when given therapeutically, it reduced diarrhea duration and frequency more dramatically in boys compared to girls [36] . Similarly, zinc sulfate was shown to improve Effect of zinc on diarrhea in calves diarrhea outcomes in boys but improved growth rates in girls [13] . Broadly, these differences between male and female responses to zinc supplementation are not understood, though theories regarding differences in immune function and response [13, 35] , diarrhea etiology [13] , and nutrient requirements [35] have been proposed. In the current study, ZM-treated bulls demonstrated increased ADG compared to placebo-treated bulls while ZM-treated heifers demonstrated decreased ADG compared to placebo-treated heifers. However, due to a significant interaction between ZM treatment and birth weight, this reduction in ADG in ZMtreated heifers was overcome with increasing birth weight, such that ZM-treated heifers with birth weights above 42 kg experienced increased ADG during the pre-weaning period, compared to placebo-treated heifers with birth weights above 42 kg. Differences in the growth response to ZM supplementation between bull and heifer calves may have been related to its effect on feed intake. Previous research on the effects of feeding various doses of oral zinc oxide to pre-ruminant dairy calves demonstrated that high levels of oral zinc supplementation resulted in reduced feed intake [23] . In the current trial, oral ZM dose was estimated to be significantly higher in heifers compared to bulls due to the significantly lower birth weight of heifers. Additionally, serum zinc concentrations in ZM-treated heifers were numerically higher than that of bulls, though this difference was not significant, likely due to the small sample size. Perhaps the higher zinc dose in heifers was associated with reduced feed intake, leading to reduced growth, and that this effect was more pronounced for ZM compared to ZS. The fact that ZM-treated heifers with birth weights approaching those of average bull calves (and, therefore, a similar zinc dose to that in bulls) experienced an increase in ADG over placebo-treated heifers similar to that of bull calves partially supports this theory. Although management practices on the study dairy were designed to be identical for both bulls and heifers, it is possible that subtle, unrecognized differences in nutritional and health management may also have contributed to sex-specific differences in weight gain. Nevertheless, future trials are warranted to investigate the potential differences in the dose-response to zinc supplementation between bulls and heifers.

We hypothesized that ADG would be increased in zinc-supplemented calves compared to placebo-supplemented calves due to the potential preventive and therapeutic effects of

zinc supplementation on neonatal diarrhea. In other words, calf diarrhea is mitigated by zinc supplementation and, therefore, on the causal pathway between zinc and ADG. However, considering the similarly-reduced hazard of diarrhea and increased hazard of cure from diarrhea in both ZM and ZS treatment groups but a lack of effect of ZS on ADG, it is likely that the effect of ZM on ADG is not solely mediated through its effects on diarrhea. Differences in effectiveness between organic and inorganic formulations also may exist. In fact, the underlying mechanism of action of oral zinc remains unknown [6] . Several theories of the mechanisms of action of zinc in the prevention and treatment of childhood diarrhea exist, including a mucosal-protective role, a diarrhea-induced zinc deficiency, an essential element in cell-mediated immunity, and a modifier of intra-luminal electrolyte secretion and absorption [6, [37] [38] [39] .

The clinical and practical implications of effects of ZM supplementation on ADG and diarrhea must be considered. Pre-weaned calf diarrhea remains an ongoing issue for the dairy industry. The deleterious effects on calf health and performance and the resulting economic burden create a strong incentive to treat and prevent diarrhea in pre-weaned calves. On large dairy operations like those in California's Central Valley, small changes in disease incidence and duration as well as animal growth and performance can have profound economic consequences. As a non-antimicrobial product, zinc may become increasingly attractive as antimicrobials in livestock feed are under increased scrutiny and regulation due to concerns about antimicrobial resistance [2, 40] . Prevalence of *C. parvum* fecal shedding in a random sample of 92 study calves at onset and resolution of diarrhea was significantly higher in calves treated with zinc compared to Placebo-treated calves. In contrast, a previous study where calves that tested positive for *C. parvum* at the start of diarrhea and were treated with ZM had 16 times higher odds of being fecal ELISA negative at exit compared to the Placebo group ($P = 0.08$; power = 72.3%) [11] . The difference in findings may be due to the differences in the timing of diarrhea across treatment groups. For the current study's random sample of calves that acquired, survived, and were sampled on the correct days, the mean age of calves on both onset and resolution of diarrhea was higher for ZM and ZS calves compared to placebo-treated calves. Although *C. parvum* oocyst shedding in infected calves can occur as early as 3 days of age, peak shedding occurs at about 14 days of age [41] . It is possible that the increase in prevalence of *C. parvum* shedding in ZM and ZS treated groups was due to the increased age of zinc-treated calves compared to placebo-treated calves at resolution of diarrhea. The latter explanation is also supported by our findings that the odds of microbiological cure from *C. parvum* significantly decreased in older calves, with no significant differences in the odds

of cure between treatment groups. In addition, the current testing did not estimate the concentration of *C. parvum* shedding which may still differ between treatment groups.

Despite the large sample size, the current trial was limited to a single California dairy, which may represent other large dairies but does not reflect all the dairy management systems in California or elsewhere. Additionally, our results show that calves respond to zinc supplementation for diarrhea prevention differently depending on chemical formulation and calf sex. The latter could be due to differences in body weight between bulls and heifers and may point towards the need for sex-specific dosing. Furthermore, the current research did not evaluate the potential economic utility of zinc supplementation. Future studies on more accurate dosing of zinc by calf sex, the practical feasibility of weight-based dosing, and the expected cost-effectiveness of zinc administration as part of the management of pre-weaned dairy calves are warranted. Finally, our clinical trial was performed on a single, large, predominately Holstein, California dairy over a six-month period, which precluded our ability to evaluate differences due to season or breed. Hence, future studies to assess any modifying effect of breed and seasonal differences on the effect of zinc on calf health and weight gain are also needed.

The current double blind, block-randomized placebo controlled clinical trial tested the effect of a prophylactic daily oral zinc supplementation in neonatal Holstein calves. Bull calves treated with ZM had a significantly increased ADG (22 g per day) during the pre-weaning period compared to placebo-treated bulls. In comparison, ZM-treated heifers had significantly lower average daily gain (9 g per day) compared to placebo-treated heifers, although higher ZM doses in low birthweight heifers may explain the lower ADG. Calves treated with either ZM or ZS had significantly lower risks of diarrhea and significantly higher risk of cure from diarrhea over the first 30 days of life compared to placebo-treated calves and hence the current trial demonstrated that zinc supplementation delayed diarrhea and expedited diarrhea recovery in pre-weaned calves. Additionally, zinc improved weight gain differentially in bulls compared to heifers, indicating the need for further research to investigate zinc dosing in calves.

Supporting information S1 (DOCX) S1 Dataset. Raw data collected from trial, organized as separate excel sheets for enrollment, daily assessment, serum total protein, birth weight, exit treatment weight, exit trial weight, serum zinc testing, fecal samples, fecal testing, milk testing, and dead calves. (XLSX)

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Hantaviruses in the Americas and Their Role as Emerging Pathogens

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Abstract: The continued emergence and re-emergence of pathogens represent an ongoing, sometimes major, threat to populations. Hantaviruses (family Bunyaviridae) and their associated human diseases were considered to be confined to Eurasia, but the occurrence of an outbreak in 1993–94 in the southwestern United States led to a great increase in their study among virologists worldwide. Well over 40 hantaviral genotypes have been described, the large majority since 1993, and nearly half of them pathogenic for humans. Hantaviruses cause persistent infections in their reservoir hosts, and in the Americas, human disease is manifest as a cardiopulmonary compromise, hantavirus cardiopulmonary syndrome (HCPS), with case-fatality ratios, for the most common viral serotypes, between 30% and 40%. Habitat disturbance and larger-scale ecological disturbances, perhaps including climate change, are among the factors that may have increased the human caseload of HCPS between 1993 and the present. We consider here the features that influence the structure of host population dynamics that may lead to viral outbreaks, as well as the macromolecular determinants of hantaviruses that have been regarded as having potential contribution to pathogenicity.

Text: Emerging pathogens cause new or previously unrecognized diseases, and among them, emerging zoonotic diseases are a major concern among scientists studying infectious diseases at different spatial and temporal scales [1, 2] . Changes in biotic and abiotic conditions may alter population disease dynamics and lead to the emergence of zoonotic infections [3] [4] [5] [6] . During the last decades, several outbreaks of emerging and re-emerging viral pathogens have occurred, affecting both purely-local and worldwide/pandemic involvement of human populations. Among the conspicuous examples are influenza A, Ebola virus, hepatitis C virus, severe adult respiratory distress (SARS), coronavirus, and human immunodeficiency virus, which challenge prevention and control measures of public health systems [7] . In the Americas, the recent outbreak of pandemic influenza A subtype H1N1 became a major target for control due to its rapid spread, and uncertainties in virulence and transmissibility, yet vaccine availability was limited when significant activity occurred in advance of the traditional influenza season [8] . However, in the last century outbreaks of several viral-related diseases have emerged or re-emerged involving arenaviruses and dengue viruses, and more recently, hantaviruses, and the expansion of the geographic range of West Nile virus. Among zoonotic diseases, small mammals are hosts of several pathogenic RNA viruses, especially Arenaviridae and Bunyaviridae: Hantavirus [9] [10] [11] .

Hantavirus infections became a concern in the Americas after the description of an outbreak of acute respiratory distress occurred in the Four Corners area in 1993 [12] . The newly recognized disease, hantavirus cardiopulmonary syndrome, HCPS (or hantavirus pulmonary syndrome), was linked to infection by the newly-discovered Sin Nombre virus (SNV), and the rodent *Peromyscus maniculatus* (deer mouse) was identified as the reservoir [13] . However, hantavirus infections have a much longer history. A review of ancient Chinese writings, dating back to approximately 960 AD, revealed descriptions closely resembling hemorrhagic fever with renal syndrome (HFRS), the syndrome caused by Old World hantaviruses [14] . During the twentieth century, cases of acute febrile disease with renal compromise were described from several Eurasian countries and Japan, often in association with military engagements [15] . HFRS as a distinct syndrome, however, was first brought to the attention of western medicine in association with an outbreak that occurred among United Nations troops during the Korean conflict between 1951 and 1954, where more than 3,200 soldiers were afflicted [16] . It took more than two decades until the etiologic agent, Hantaan virus (HTNV), was isolated from the striped field mouse *Apodemus agrarius*, detected in part by the binding of antibodies from patient

serum samples to the lung tissues of healthy, wild-caught field mice [17, 18] . The virus was later found to represent the type species of a new genus Hantavirus of the family Bunyaviridae, although it was later apparent that the first hantavirus to be isolated was the shrew-borne Thottapalayam virus [19] . The categorization of hantaviruses as belonging to the family Bunyaviridae is due in part to the consistent presence of three RNA genomes that are circularized in vivo as a result of the presence of terminal complementary nucleotides that help fold the genome into a -hairpin morphology, first described for the Uukuniemi phlebovirus [19, 20] . Table 1 is a list of the predominant, serologically distinct pathogenic hantaviruses. Many other named genotypes are described, but such other pathogenic forms are generally closely related to Andes or, in some cases, Sin Nombre virus.

During virus maturation, the precursor form GPC is processed using a membrane -bound protease into Gn and Gc, a cleavage that occurs, and appears to be signaled, after the conserved peptide signal WAASA at the C-terminal of Gn [24] . Although the two proteins can be expressed independently through transfection, they can be retained in the wrong cellular compartment (ER or aggresome); they thus must be co-expressed to allow them stability so that the two can be assembled correctly in the Golgi [25, [27] [28] [29] .

A number of activities and properties have been identified for the hantavirus envelope glycoproteins, including some features that are suspected to be involved in the pathogenicity of the disease-causing serotypes, a possibility that has engendered experimental attention. The glycoproteins are the known or presumed ligands for at least two distinct cellular receptors, the α 3 integrin chain and decay accelerating factor, or DAF [30, 31] ; with gC1qR/p32 also identified as another potential entry receptor [32] . Comparisons with the tick-borne encephalitis virus E protein, led Tischler et al. to consider the Gc glycoprotein as a potential class II fusion protein, perhaps imparting fusion activity to the virion, and this hypothesis has gained support in other studies [33, 34] .

Additional activities have been identified with, or claimed to be related to, Gn. For many of these studies, an underlying premise has held that there are differences between the glycoproteins of -pathogenic hantaviruses relative to viruses in the genus that are dubbed to be -non-pathogenic. While it is true that it has not yet been possible to link Prospect Hill virus (PHV) to human disease, the absence of evidence for its pathogenicity should perhaps not be equated with the evidence of its absence. One might only consider that the

level of disease (e.g., lethargy, fever, proteinuria, and azotemia) associated with infection of nonhuman primates by PHV is not significantly different from that recorded for nonhuman primate models using the known-pathogen Puumala virus (PUUV) [35, 36] . For the purpose of this discussion we will presume that apathogenic hantaviruses are indeed apathogenic.

While some studies have suggested that Gn glycoproteins are directed more rapidly into the ubiquitin-proteasome pathway than are apathogenic forms, others have interpreted differences in the handling of Gn glycoproteins across hantavirus species by the ubiquitin-proteasomal system as independent of pathogenicity [37] [38] [39] . Some investigators have directed their efforts toward identifying a differential capacity, either kinetic or in absolute magnitude, in the ability of pathogenic and apathogenic hantaviruses to elicit an interferon response in cells. One premise that emerges is that apathogenic forms would tend to induce an earlier innate response that would render it more likely that the virus would be quickly cleared or rendered less competent in its replication so as to blunt any pathological response in the host [40] [41] [42] . The anti-hantavirus innate response can in some cases be attributed to viral interaction as a ligand of TLR-3, but not in others, and in endothelial cells, it appears not to require more than the viral particle itself, even when introduced in replication-incompetent form [43, 44] . Proteins and mRNAs prominently induced by hantaviruses include MxA and IFIT-1 (ISG-56) and others including some with known or suspected anti-viral activity. Those hantaviruses, often highly pathogenic strains, that fail to induce a potent antiviral response, are suspected or presumed to have a (more) potent interferon-pathway antagonism mechanism relative to other viruses, a mechanism that acts positively to prevent an effective innate response from forming, at least early in infection [42, 45] . Yet some instances are reported wherein highly pathogenic hantaviruses, such as SNV, are also able to induce expression of interferon-stimulated gene mRNAs, even very early in infection, with ISG proteins, as expected, taking longer to appear in the cell [44] . Anti-interferon activities have also been attributed to the NSs protein that may be elaborated in cells infected by serotypes that encode this protein [46] . Other investigators have examined the activities of hantavirus glycoproteins and other proteins that might themselves directly affect some aspects of the pathogenic progression associated with hantavirus infection of humans, such as vascular permeability changes. While early attempts to directly cause increases in permeability of endothelial monolayers with viral particles or viral infection were largely disappointing, hantaviruses have been identified as adversely affecting endothelial migration over substrata and in potentiating VEG-F-induced endothelial permeability [47, 48] .

The shorter (250-kD) nucleocapsid or N protein is a structural component of the viral nucleocapsid, along with the genomic viral RNA segments. As an RNA-binding protein that engages the hairpin termini of the genomic segments with high affinity [49, 50], it limits the access of the RNA to host nucleases and helps to render viral replication a closed process within the cytoplasm. It also acts as a peripheral membrane protein, as does the L protein [51], an activity that could play a role in its presumed, but not yet demonstrated function as matrix [52]. Until recently, it had not been appreciated that N has a wide variety of other activities, some of which can be linked, not only to fundamental requirements of replication, but also to the interference with an array of the intracellular processes of the normal cell. Thus, an interaction between the amino terminus of the hantavirus N protein and the cellular protein Daxx has been proposed, with the suggestion of potential pro-apoptotic consequences [51]. N is also reported to interact with actin microfilaments, and the SUMO-1 protein [53, 54]. Using reporter-gene based assays, Connie Schmaljohn and her colleagues have reported that Hantaan virus' nucleocapsid protein has an inhibitory role in inflammatory responses mediated by NF kappa B (NF- κ B). The effects on NF- κ B expression appeared to be confined to prevention of its nuclear translocation after its attempted activation with lipopolysaccharide, LPS [55]. In the cytoplasm of infected cells, N protein can be found in cellular P bodies where it sequesters and protects 5' caps. It may locate the caps through its interaction with DCP1, a key constituent of P bodies. During hantavirus infection, the viral RNAs become concentrated in P bodies, through their interaction with N and DCP1. The N protein demonstrates preferential protection of mRNAs engineered to prematurely terminate their encoded protein in comparison to native mRNAs [56]. N protein has been increasingly linked to viral replication and translation, sometimes in previously unanticipated ways. It is among a growing family of diverse viral proteins that can serve as a nonspecific -RNA chaperone, an activity that should facilitate the L polymerase's access to vRNA for transcription and replication, in that it can transiently dissociate misfolded RNA structures [57]. Some of N protein's effects on translation might not immediately be recognized to be adaptive in nature. It can replace the entire EIF4F translational initiation complex, simultaneously presenting the ribosome with a replacement for the cap-binding activity of eIF 4E, binding to the 43S pre-initiation complex as does eIF 4G, while replacing the helicase activity of eIF 4A, which is presumed to be needed to dissociate higher-order RNA structure [56, 58]. These three factors normally work together to achieve translational initiation. In P bodies, N protein's ability to bind at high affinity to capped native cellular oligoribonucleotides, along with its activity in protecting capped RNAs from degradation likely facilitates the access of capped oligonucleotides for use in transcriptional initiation by L polymerase (-cap snatching)).

Trafficking of N for viral assembly: Classically, N protein in infected cells appears to be clustered or particulate in nature, with a heavy concentration at a single perinuclear location, widely considered to be the Golgi [27] . The N proteins of hantaviruses are found in association with particulate fractions, and confocal microscopy and biochemical-inhibitor studies have shown that N tracks along microtubules but not with actin filaments [52] . The ultimate destination for N, for its assembly into viral particles is the Golgi, and it traffics there via the endoplasmic reticulum-Golgi intermediate complex (ERGIC), also known as vesicular-tubular cluster [52] . A dominant negative inhibitor, dynamin, associated with dynein-mediated transport, reduced N's accumulation in the Golgi. Later studies suggested that the specific dependence on microtubular transport is specific to Old World hantaviruses such as HTNV, but that the New World hantavirus ANDV is instead associated with actin filaments [59] . However, recent data indicates that microtubular transport is indeed utilized for the New World hantavirus SNV [60] .

Hantavirus diseases of man have long been suspected of having an immunopathogenic basis in part because of their relatively long incubation period of 2-3 weeks and the observed temporal association between immunologic derangements and the first appearance of signs and symptoms of hantavirus illness. HFRS and HCPS share many clinical features, leading many investigators to consider them to be, in essence, different manifestations of a similar pathogenic process, differing mainly in the primary target organs of disease expression (Table 2). The pathogenesis of hantavirus infections is the topic of a continuously-updated review in the series UpToDate [61] .

By the time symptoms appear in HCPS, both strong antiviral responses, and, for the more virulent viral genotypes, viral RNA can be detected in blood plasma or nucleated blood cells respectively [63, 64] . At least three studies have correlated plasma viral RNA with disease severity for HCPS and HFRS, suggesting that the replication of the virus plays an ongoing and real-time role in viral pathogenesis [65] [66] [67] . Several hallmark pathologic changes have been identified that occur in both HFRS and HCPS. A critical feature of both is a transient (~ 1-5 days) capillary leak involving the kidney and retroperitoneal space in HFRS and the lungs in HCPS. The resulting leakage is exudative in character, with chemical composition high in protein and resembling plasma.

The continued experience indicating the strong tissue tropism for endothelial cells, specifically, is among the several factors that make $\beta 3$ integrin an especially attractive

candidate as an important in vivo receptor for hantaviruses. It is likely that hantaviruses arrive at their target tissues through uptake by regional lymph nodes, perhaps with or within an escorting lung histiocyte. The virus seeds local endothelium, where the first few infected cells give rise, ultimately, to a primary viremia, a process that appears to take a long time for hantavirus infections [62, 63] . By the time that secondary viremia emerges, the agents of the more severe forms of HFRS and HCPS have begun to achieve sufficient mass as to induce, through PAMP-PRR interactions and other means, the expression of proinflammatory cytokines [64] . For HCPS, that expression favors the pulmonary bed and lymphoid organs, yet, for unknown reasons, spares the retroperitoneum and, in general, the kidney. In HFRS the situation is reversed, and yet it is often not appreciated that the expected preferential tissue tropism of HFRS-associated viruses and their HCPS-associated counterparts for the renal and pulmonary beds, respectively, is not as one would predict through the manifestations of the two diseases.

Local elaboration of inflammatory and chemotactic mediators is considered to be a requirement for the development of systemic disease symptoms, with those abnormalities sometimes culminating in shock and death. Yet it is not hypoxemia, due to the prominent pulmonary edema, that leads to death in most fatal cases of HCPS, but rather intoxication of the heart by as-yet-undefined mediators that leads to the low cardiac output state and the associated shock syndrome [64, 65] . It is tempting to speculate that mediators produced in the lung in connection with the inflammatory infiltrate can percolate through the coronary circulation with minimal dilution in HCPS, a disadvantageous consequence of the close anatomic juxtaposition of the two organs. Thus, at least three classes of potential mechanisms, some overlapping and all certainly nonexclusive of the others, could be presumed to underlie the pathogenesis of HCPS. These include:

(1) Innate immune mechanisms. The nature of interactions between hantavirus pathogen-associated molecular patterns (PAMP) with the pattern recognition receptors (PRR) of susceptible endothelial cells are beginning to be clarified. The prototypical HTNV appears to be recognized by TLR-3 [43] . Such an infection has consequences such as increased expression of HLA-DR in dendritic cells [66] and differentiation of monocytes toward dendritic cells [67] .

(2) Direct viral effects. The observed correlation between viral load and disease severity leaves the possibility open that hantavirus particles or RNA can themselves have toxic

effects on cells or on signaling. Some investigators have favored direct viral toxicity, acting through the inhibition of endothelial cell barrier function, as an explanation for much of the capillary leak, although there is widespread agreement that multiple mechanisms that mediate pathogenesis likely operate simultaneously in the affected patient [68] . A potentially important clue toward the mechanism by which hantavirus infections deplete blood platelets and, in some cases cause hemorrhagic manifestations, was advanced by the recent discovery that pathogenic hantaviruses are able to recruit platelets to adhere to endothelial cell surfaces, with $\beta 3$ integrin used as a critical binding element [69] .

(3) Pathogenic effects caused by the activities of specific viral macromolecules. We have reviewed some of the activities associated with the Gn, Gc and N, virally-encoded polypeptides in previous sections.

Testing models of pathogenesis can be done more effectively when there is an animal model that mimics key aspects of the disease. There is no such model that closely mimics HFRS, but animal models exist for both the asymptomatic carriage of PUUV and SNV by their native carrier rodents, the bank vole *Myodes glareolus* and the deer mouse *P. maniculatus*; as well as a Syrian hamster model using ANDV or the related Maporal virus from Venezuela, for which an HCPS-mimetic disease is observed [70] [71] [72] [73] .

The ANDV-Syrian hamster model has a number of features in common with the human disease, as well as some differences. Unlike the neurologic diseases that have been possible to elicit with HTNV, the hamster model for HCPS appears to be caused by capillary leak that results in pulmonary edema and the production of a pleural effusion with exudative characteristics. Typically the hamsters die between 11 and 14-d post-inoculation, reflecting a slightly accelerated incubation period in comparison to human infections. As with human HCPS, the microscopic examination of the lung reveals abundant fibrin deposition, thickened alveolar septa, and viral antigen expressed abundantly in the microvascular endothelium. ANDV-infected hamsters fitted with physiologic monitoring devices exhibited diminished pulse pressures, tachycardia, and hypotension that appear to closely mimic the shock that is believed to be the proximate cause of demise in patients who succumb to HCPS [65, 74] .

Compared to the human disease, ANDV-infected hamsters exhibit exceptionally high titers of live ANDV in their tissues, with much of the viral replication occurring in hepatocytes, which are spared in the human disease. Titers of live ANDV in some cases exceed 10^8 /g, whereas hantavirus isolates from human tissues have been notoriously difficult to obtain. Despite the universal occurrence of mildly-elevated hepatic enzymes in patients with HCPS, hepatic enzymes do not appear to be present at elevated levels in the blood of diseased hamsters even immediately before death [75] .

The protracted incubation period associated with hantavirus disease gives the host considerable time to mount a mature immune response against the virus. Thus, in contradistinction to infections of comparable severity and related symptomatology associated with arenaviruses and filoviruses, hantavirus infections of humans are associated with antibody responses of significant titer by the time symptoms commence. Despite this observation, it appears to be possible that natural variation in individual neutralizing antibody responses among patients with SNV infections can be linked to disease severity, suggesting that administration of antiviral antibodies could prove effective therapeutically [76] . In the case of ANDV infection, new evidence has emerged indicating that the apparent clearance of the virus from the blood does not result in the complete removal of antigenic stimulus by the virus, suggesting that the virus may persist, perhaps in some as-yet undetermined immunologically privileged site [77] .

A role for T cell-mediated pathological responses in HFRS and HCPS has been the source of speculation for a variety of reasons. The severity of SNV-associated HCPS may have made it more apparent that the onset of pulmonary edema, tachycardia and hypertension seemed to be all but universally temporally associated with the appearance of a spectrum of highly-activated cells of the lymphoid lineage in the peripheral blood. Cells with a close morphologic similarity to these -immunoblasts were detected in the congested, heavy lungs of patients who came to autopsy, as well as in lymphoid organs and in the portal triads [63, [78] [79] [80] . These observations led to speculation that some component of hantavirus pathogenesis could be linked to the appearance of antiviral T cells that could stimulate or contribute to the appearance of a -storm of mediators and the associated capillary leak phenotype. Subsequent studies have borne out the expectation that a significant fraction of the immunoblast population in patients with HCPS are T cells with specificity for specific class I HLA-presented epitopes of viral antigens, including Gn, Gc and N [77, [81] [82] [83] . Presumably, the antiviral activities of such cells, manifested in

part through their elaboration of mediators in the affected interstitium, can contribute to the endothelial/capillary leak that lies at the heart of hantavirus pathogenesis.

Because early cases of HCPS often came to autopsy, it became possible to examine necropsied tissues for expression of cytokines. The study by Mori et al. (1999) revealed high relative expression of proinflammatory cytokines including TNF α , IL-1 α , IL-6, providing evidence in favor of a -cytokine storm model for pathogenesis [64] . The authors believed, based on the morphology of cytokine-secreting cells, that both monocytes and lymphocytes were contributing to the production of cytokines. That proinflammatory mediators are found in elevated levels in the plasma as well as the renal interstitium of patients with acute hantaviral illness has been recognized for some time as well [84, 85] .

While diagnosis of HCPS as well as HFRS is best accomplished with IgM serology, in the acute stage of SNV infection, RT-PCR can also be used if blood cells or blood clot are used instead of plasma or serum, where sensitivity even using nested PCR primers drops to about 70% [86] [87] [88] . In a facility at which many cases of HCPS are treated, the University of New Mexico medical center in Albuquerque, a diagnostic service has long been offered in which the patient's hematologic findings are analyzed to establish the probability that a patient has HCPS. The combination of thrombocytopenia, elevated abundance of -immunoblast lymphocytes, left-shifted polymorphonuclear cell population without strong morphologic evidence for their activation, and elevated hemoglobin or hematocrit values is highly specific for HCPS and allows clinicians the ability to put presumptive-HCPS patients on extracorporeal membrane oxygenation (ECMO), which is believed to have saved many patients from a lethal outcome [89] .

Human infection by hantaviruses is thought to follow contact with secretions or excretions produced by infected rodents. In the United States, 538 human infections by hantavirus were reported through late December 2009 [90] , with New Mexico, Arizona and Colorado exhibiting the highest case-loads. While the prototypical central American hantavirus in central America was Rio Segundo virus of *Reithrodontomys mexicanus* from Costa Rica, the first human disease appeared some years later in Panama, where Choclo virus (CHOV) arose as the etiologic agent and is believed to be responsible for all known cases of HCPS. The fulvous pygmy rice rat *Oligoryzomys fulvescens* has been identified as the rodent reservoir [91] . In Panama, the first cases of HCPS, albeit with little or no evident cardiac involvement, were reported in 1999, and since then, 106 human infections have occurred

with a 26% mortality rate [92] . Serosurveys of mammals in Mexico and Costa Rica have found anti-hantavirus antibodies [93] [94] [95] [96] , and seroprevalences ranging between 0.6 to 1.6% in human populations were reported despite the absence of known HCPS cases [97] . In South America, HCPS cases have been indentified in Argentina, Bolivia, Brazil, Chile, Paraguay and Uruguay, and evidence for human exposure to hantaviruses have also been reported in Venezuela [98] and Perú [99] . In southern South America, ANDV is the main etiologic agent with cases in Chile and Argentina reported since 1995. In Chile, 671 cases of HCPS due to ANDV have occurred during the period 2001-2009 [100] . Since 1995, more than 1,000 HCPS cases have been reported in Argentina [101] ; in Brazil, approximately 1,100 HCPS cases have been identified between 1993 and 2008 [102] . Case-fatality ratios in those three countries have been similar, ranging from 30% (Argentina), 36% (Chile) and 39% (Brazil).

Hantavirus infections occur more frequently in men than women, although the male/female ratio is highly variable. For example, Panamanian communities showed a ratio of 55 men to 45 women [103] , while in Chile the ratio is more biased to males (71%) [104] . In the Paraguayan Chaco the male-female ratio approaches 50% [105] . In North America, by December 2009 63% of case-patients were males [90] . All ethnic and racial groups seem to be susceptible to hantavirus infections, and the differences between certain groups (as indigenous and non-indigenous) are more likely correlated with the type habitat where the population resides (e.g., rural versus urban areas). In fact, rural communities account for the highest hantavirus incidences overall and are therefore at higher risk [92, [105] [106] [107] [108] [109] [110] [111] , although the importance of peridomestic settings as a major area of exposure has also been emphasized [112, 113] .

The main mechanism by which humans acquire hantavirus infection is by exposure to aerosols of contaminated rodent feces, urine, and saliva [114, 115] . This can occur when humans reside in areas in close proximity to those that rodents inhabit, live in areas infested with rodents, or when rodents invade human settings, which are more frequent in rural habitats. There is a long history of human co-existence with rodents, raising questions about the apparent recent increases in hantavirus-related illnesses, especially HCPS. Other than an apparent association with El Niño southern oscillation (ENSO) events in some regions [116, 117] , the recent increases in incidence of HCPS do not seem to follow a readily-defined temporal or spatial pattern. However, some landscape features such as habitat fragmentation or human-disturbed areas may influence rodent population dynamics and impact viral incidence [118] [119] [120] [121] . Despite the stochasticity

associated with contraction of hantavirus infection, certain scenarios have been recognized as posing higher risk. Human activities in poorly ventilated buildings that aerosolize particulates that are then inhaled (i.e., cleaning, shaking rugs, dusting) are frequently identified among patients admitted for HCPS [11, 122] . Outdoor activities are thought to convey lower risk due to lability of hantaviruses to UV radiation and the presumed tendency to be dispersed in wind, although certain environmental conditions seem to maintain the virus for longer periods outside its natural host allowing for indirect transmission [123] . An alternative but uncommon route of virus transmission is by rodent bites [124] [125] [126] . Field workers handling mammals are potentially at higher risk of exposure with hantavirus infections, although when quantified through serosurveys the absolute risk appears rather slight [127] . A new study in Colorado suggests the possibility that a rodent bite may have been the proximate vehicle for outdoor transmission of SNV [128] , which re-emphasizes the use of personal protective equipment during field work activities [129] . As a particular case within hantaviruses, person-to-person transmission has exclusively been documented for the South American Andes virus [130] [131] [132] [133] [134] [135] . The identification of this transmission route has been made using both molecular tools and epidemiological surveys, but the mechanism of interpersonal transmission is not well established. Recent findings show that family clusters and specifically sexual partners share the greater risk of interpersonal transmission, although sexual transmission per se can be neither inferred nor refuted presently [130, 135] . Interestingly, ANDV may also be shed by humans through other biological fluids such as urine [136] , illustrating the particular properties that differentiate this virus from other hantaviruses. Although interpersonal transmission seems to be unique for ANDV, viral RNA of PUUV has been detected in saliva of patients with HFRS, and some patients with SNV-HCPS have viral RNA in tracheal secretions [88, 137] .

Hantaviruses in the Americas are naturally hosted by rodents (Muridae and Cricetidae) as well as shrews (Soricidae) and moles (Talpidae) (Figure 1) . Three shrew and one mole species have been reported to host hantaviruses and their pathogenicity for humans remains unknown [22, 138, 139] . At least 15 rodent species have been identified as carriers of different pathogenic hantaviruses, with some South American genotypes such as Castelo do Sonhos (CDSV) or Hu39694 only identified after human infections (Figure 1) . Hantaviruses typically show high species-specificity and no intermediate host [140] . However, some hantavirus genotypes have been described in the same rodent species. Such is the case of Playa de Oro (OROV) and Catacamas (CATV) identified in *Oryzomys couesi* [141, 142] , or Maporal (MAPV) and Choclo (CHOV) hosted by *O. fulvescens* [91, 143] . In North America both Muleshoe and Black Creek Canal hantaviruses have been

detected in geographically-distant *Sigmodon hispidus* [144, 145] . Also, one hantavirus genotype (e.g., Juquitiba-like virus) may be carried by more than one rodent species (*O. nigripes*, *Oxymycterus judex*, *Akodon montesis*). Another example is Laguna Negra virus (LANV) which after being identified in *Calomys laucha* [146] has also been reported in *C. callosus* [147] . The rapid increase in the discovery of new hantaviruses and the identification of their hosts does not seem likely to end soon as new small mammal species are screened [95] . This subject is complicated by continued controversy in the criteria for the classification of distinct hantaviruses [148, 149] , which is also tied to host taxonomic classification and taxonomic rearrangements.

Cross-species transmission is a major process during spread, emergence, and evolution of RNA viruses [6, 150] . Particularly within hantaviruses, spillover to secondary hosts are increasingly identified as more extensive studies are performed [151] [152] [153] [154] [155] [156] . For example, ANDV is the predominant etiologic agent of HCPS in South America, and *O. longicaudatus* the main rodent reservoir. Spillover in at least four other rodent species that co-occur with the reservoir have been identified, with *Abrothrix longipilis* showing the second higher prevalence to ANDV-antibodies, and there is presently no question that the virus is extremely similar genetically between the two host rodents [157, 158] . In North America, spillover of Bayou virus (BAYV) may have occurred from the main reservoir *O. palustris* to *S. hispidus*, *R. fulvescens*, *P. leucopus*, and *B. taylori* [159] [160] [161] . Hantavirus spillover is more likely to occur with host populations inhabiting sympatric or syntopic regions [151, 162] , and cross-species transmission would presumably have greater chances of success if the host species are closely related [163] . An interesting exception is found between Oxbow virus (OXBV) and Asama virus (ASAV) in which a host-switch process seemed to have occurred between mammals belonging to two families (Talpidae and Soricidae), likely as a result of alternating and recurrent co-divergence of certain taxa through evolutionary time [138] .

Hantaviruses are horizontally transmitted between rodents and are not transmitted by arthropods (unlike other viruses of the family Bunyaviridae). Spillover infection to nonhuman mammals usually results in no onward (or -dead-end) transmission, but if humans are infected may result in high morbidity and mortality [122, 164] . During the spring of 1993, an outbreak of patients with HCPS due to SNV occurred in the Four Corners states resulting in more than 60% case-fatality among the initial cases, many involving members of the Navajo tribe [12, 121] . In Panama, an outbreak was reported during 1999-2000 in Los Santos, and 12 cases were identified with three fatalities [165, 166] . This

represented the first report of human hantavirus infections in Central America. In South America, the first largest identified outbreak occurred in the Chaco region in northwestern Paraguay during 1995-1996. Seventeen individuals were identified with SNV antibody (ELISA) or were antigen (IHC) positive out of 52 suspected cases [167] . Major outbreaks due to ANDV occurred in 1996 in southern Argentina [131, 134] ; in southern Chile clusters of patients presented with hantavirus illness in 1997 [158] . In Brazil, the first outbreak was identified in the Brazilian Amazon (Maranhão State) in 2000, and involved small villages that resulted in a 13.3% prevalence of those tested (398 total residents) [168] .

The factors that trigger hantavirus outbreaks are still poorly understood, probably because they result from several interacting biotic and abiotic features whose key parameters are difficult to model. However, the use of new modeling approaches that involve geographical and environmental features seem to be promising in predicting potential hantavirus outbreaks and/or areas of higher risk [169] [170] [171] [172] . Because hantaviruses are known to be directly transmitted from infected to susceptible hosts, the first natural approach is to relate outbreaks to the ecology of the viral hosts. Hantavirus transmission and persistence in rodent populations depends on several factors that interact to affect ecological dynamics of the host, which in turn is strongly influenced by the behavioral characteristics of individual rodent species, to landscape structure, and environmental features [173, 174] . Viral transmission depends on contact rates among susceptible hosts, and despite the prevailing notion that a higher density increases encounters and hence secondary infected hosts, contrasting patterns relating rodent population size and virus prevalence can be found [175] . In addition, it has been shown that SNV transmission follows a contact heterogeneity pattern, where individuals in the population have different probability of transmitting the infection [176] . The understanding of viral transmission proves to be far more complex when species other than the main reservoir host are incorporated in the model. In fact, recent studies have shown that higher hosts species diversity is correlated with lower infection prevalence in North America for *P. maniculatus* [177] , in Central America for *O. fulvescens* (reservoir of Choclo virus) and *Zygodontomys brevicauda* (reservoir of Calabazo virus) [178] , and in South America for *Akodon montensis* (reservoir of Jabora virus) [162] . Contact rates vary according to the spatial distribution of populations and seem to be strongly influenced by landscape structure. For example, SNV prevalence in *P. maniculatus* was higher in landscapes with a higher level of fragmentation of the preferred habitat [179] . In addition, certain properties of the landscape such as elevation, slope, and land cover seem to be useful in detecting areas with persistent SNV infections, and therefore thought to be refugial areas where the virus can be maintained for years [169] . Changes in the natural environment of reservoir species, such as forest

fragmentation and habitat loss, may alter population abundance and distribution and lead to hantavirus outbreaks, as observed in the Azurero Peninsula of Panama [118, 119] . Also, differences in the microhabitat, including overstory cover, may lead to differences in the ecological dynamics within populations and affect the rate of exposure to the virus [180] . Differences in hantavirus infections through contrasting landscapes in the latitudinal span have been found in rodent populations of *O. longicaudatus* in Chile, suggesting that humans are differentially exposed to the virus [107, 181] .

Rodent population dynamics are affected by seasonal changes of weather and climate [182, 183] . In the case of the ENSO-associated outbreaks, a complex cascade of events triggered by highly unusual rains in the precedent year have been postulated to result in an increase of primary production and rodent densities, also increasing the likelihood of transmission of the virus to humans, but it has proved difficult to precisely demonstrate the suggested intermediate events such as increased rodent densities in the increased caseload [116, 121, 184] . In South America, effects of climate change and hantavirus outbreaks have not been well studied, despite the knowledge that several rodents species that are reservoirs of emerging diseases have dramatically been affected by events like El Niño [185] . Changes in host population dynamics are also affected by seasonality, which may lead to disease outbreaks when processes that equilibrate rodent populations from season to season are interrupted [186] .

Viral emergence may continue to be promoted as human-introduced changes continue to increase in the environment at different geographical scales. Human incursions into previously uncultivated environments may lead to new contacts between rodent reservoirs and humans, increasing the likelihood of contracting infections [187] . These changes may also alter rodent's population structure and dynamics and interspecies interactions creating conditions that may lead to viral outbreaks, viral establishment in new hosts, and emergence of HCPS [102, 162] , even with seemingly slight ecological disturbance to the virus-host system [188] .

Certain pathophysiologic characteristics, including thrombocytopenia and shock, of hantavirus diseases of humans, bear substantial similarity to the hemorrhagic fevers induced by other viruses such as arenaviruses, filoviruses and flaviviruses, despite sharing essentially no sequence similarities therewith. Such observations raise questions about

whether such commonalities in pathogenesis are chance similarities of phenotype, or instead report the presence of common molecular mechanisms among the viruses.

In this review we discuss the general properties, discoveries and epidemiology/ecology of the New World forms of pathogenic hantaviruses, and also seek to identify some of the characteristics of the viral macromolecules and immunologic mechanisms that have been proposed as potential direct mediators of the pathogenic events that characterize the human disease HCPS. While it is unlikely that expression of any particular viral protein or RNAs in isolation can be relied upon to replicate key phenotypes of infection by the complete virus, some of the findings have been sufficiently consistent with what is known of the pathogenesis in vivo that they offer plausible first-pass leads in the search for therapeutic targets. We look forward to the mechanistic revelations that will follow the inevitably expanded usage of powerful methods such as deep sequencing, ever-more advanced imaging, and microscopic methods, and animal models that can at last be said to be close mimics of human hantavirus disease.

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Pre-existing immunity against vaccine vectors – friend or foe?

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3542731/>

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Abstract: Over the last century, the successful attenuation of multiple bacterial and viral pathogens has led to an effective, robust and safe form of vaccination. Recently, these vaccines have been evaluated as delivery vectors for heterologous antigens, as a means of simultaneous vaccination against two pathogens. The general consensus from published studies is that these vaccine vectors have the potential to be both safe and efficacious. However, some of the commonly employed vectors, for example *Salmonella* and adenovirus, often have pre-existing immune responses in the host and this has the potential to modify the subsequent immune response to a vectored antigen. This review examines the literature on this topic, and concludes that for bacterial vectors there can in fact, in some cases, be an enhancement in immunogenicity, typically humoral, while for viral vectors pre-existing immunity is a hindrance for subsequent induction of cell-mediated responses.

Text: In the fields of medicine and veterinary medicine, there are numerous live, attenuated bacterial and viral vaccines in use today worldwide. The safety and efficacy of such vaccines is well established and allows further development as vector systems to deliver antigen originating from other pathogens. Various attenuated bacteria, including *Escherichia coli*, *Vibrio cholerae*, lactic acid bacteria (LAB), specifically *Lactococcus lactis*, *Mycobacterium*, *Listeria*, *Shigella* and *Salmonella*, have been tested for the targeted delivery of heterologous antigens of bacterial, viral and parasitic origin into a variety of animal hosts (Bahey-El-Din et al., 2010; Innocentin et al., 2009; Johnson et al., 2011; Tobias et al., 2008 Tobias et al., , 2010 Tobias & Svennerholm, 2012) . Bacteria such as *E. coli* and lactic acid bacteria have recently gained favour, as *E. coli* is a commensal and lactic acid bacteria are present in most fermented food items and are therefore naturally present in the host. They are also a much safer option than traditional attenuated vaccines in children and immunocompromised people. As this review discusses the effects of pre-existing immune responses to attenuated vaccines, further discussion of LAB and *E. coli* as potential vectors will not be undertaken; however, the reader is directed to several interesting reviews (Bermúdez-Humarán et al., 2011; Wells & Mercenier, 2008) . Intracellular bacteria from the genera *Mycobacterium* (Guleria et al., 1996) , *Listeria* (Gentschev et al., 2001) , *Shigella* (Levine et al., 1997) and *Salmonella* (Dogan et al., 1987) are considered to be suitable candidates for the delivery of vaccine antigens due to their capability to induce robust T cell immune responses (Alderton et al., 1991; Lo et al., 1999; Mastroeni et al., 2001; Mittrücker & Kaufmann, 2000; Nauciel, 1990) . *Salmonella* is one genus that has been well examined as a vector, building on the extensive research available

on the micro-organism's physiology and pathogenesis (Basso et al., 2000; Killeen & DiRita, 2000; Sirard et al., 1999; Ward et al., 1999) . There exist several commercial vaccines that are used as anti-Salmonella vaccines in humans and animals (e.g. Ty21a for typhoid fever in humans, several Salmonella serovars against salmonellosis in chickens and other animals). The general strategy for vectoring heterologous antigen is depicted in Fig. 1 . The first clinical trial of a recombinant, which was conducted over 20 years ago using an attenuated Salmonella as a delivery vector, led to the widespread testing of this bacterium as a mucosal delivery system for antigens from non-Salmonella pathogens (Dougan et al., 1987) . These studies have demonstrated the utility of live bacteria to deliver expressed antigens and DNA vaccines to the host immune system (Atkins et al., 2006; Hussein & Hensel, 2008; Jiang et al., 2004; Kirby et al., 2004) . Since then several other intracellular bacterial vectors have been successfully tested for their capability to deliver a variety of antigens from various pathogens, as well as vaccination against cancer. One genus which has been widely tested as vector is Listeria. Listeria species are Gram-positive intracellular food-borne pathogens. The advantages of Listeria are that it can invade a variety of cells, including antigen presenting cells (APCs). After invading the host cell, Listeria resides inside the phagosome; however, it can escape the phagosome with the help of listeriolysin O (LLO; Hly) and reside in the cytoplasm of the cells, thereby efficiently presenting antigen to both CD8 and CD4 T cells (Cossart & Mengaud, 1989; Kaufmann, 1993; Pamer et al., 1997) . Several studies have demonstrated the effectiveness and ease of using Listeria monocytogenes to deliver heterologous vaccine antigens and DNA vaccines Jensen et al., 1997; Johnson et al., 2011; Peters et al., 2003; Shen et al., 1995; Yin et al., 2011) .

Similarly, various viral vectors have been successfully tested for their capability to deliver heterologous vaccine antigens, and this generally results in the induction of strong CTL immune responses. In the veterinary field, there are numerous viral vector vaccines that are currently licensed for use in livestock and domesticated animals. These recombinant vaccines are based on both DNA viruses (such as fowlpox virus-based vaccines which target avian influenza virus and fowlpox virus, or vaccinia virusbased vectors against the rabies virus in wildlife) and RNA viruses [such as Newcastle disease virus-based vaccines to be used in poultry or yellow fever virus (YFV)-based vaccines to be used in horses against West Nile virus] (Draper & Heeney, 2010) . Based on the safety record in the veterinary field, many viruses have been studied for human use as a vector in vaccine development (Beukema et al., 2006; Esteban, 2009; Schirrmacher & Fournier, 2009; Stoyanov et al., 2010; Weli & Tryland, 2011) . Amongst them, YFV (YF-17D strain) was the first to be licensed for use in humans, where the cDNAs encoding the envelope proteins of YFV were replaced with the corresponding genes of an attenuated Japanese encephalitis

virus strain, SA14-14-2 (Appaiahgari & Vрати, 2010; Rollier et al., 2011) . Poxviruses are also studied extensively as candidate vectors for human use, among which attenuated derivatives of vaccinia virus [such as modified vaccinia virus Ankara (MVA) and New York attenuated vaccinia virus NYVAC strains] are the most promising vectors (Esteban, 2009; Gó mez et al., 2008; Rimmelzwaan & Sutter, 2009). They are ideal candidate vectors due to their large DNA-packing capacity and their thermal and genetic stability (Minke et al., 2004) . The NYVAC vector has been shown to induce CD4 + T cell-dominant responses, and MVA induces both CD4 + and CD8 + T cell responses (Mooij et al., 2008) . The adenovirus (Ad) vector is another of the most widely evaluated vectors to date to express heterologous antigens, due to ease of production, safety profile, genetic stability, the ease of DNA genome manipulation, and the ability to stimulate both innate and adaptive immune responses and induce both T and B cell responses (Alexander et al., 2012; Fitzgerald et al., 2003; Gabitzsch & Jones, 2011; Lasaro & Ertl, 2009; Vemula & Mittal, 2010; Weyer et al., 2009) . They have been extensively examined as a delivery vector in several preclinical and clinical studies for infectious diseases such as anthrax, hepatitis B, human immunodeficiency virus (HIV)-1, influenza, measles, severe acute respiratory syndrome (SARS), malaria and tuberculosis M. Saxena and others (Chengalvala et al., 1994; Gao et al., 2006; Hashimoto et al., 2005; Hsu et al., 1992; Limbach & Richie, 2009; Radosevic et al., 2007; Shiver et al., 2002) .

However, before vectored vaccines can be used in the human population they need to satisfy several important criteria. Safety is a major concern, as even a low level of toxicity is unacceptable (of course the minor discomfort that accompanies many vaccinations is normal). Secondly, a vaccine should be inexpensive, so that it can be administered to a large population at minimal cost, and this is particularly important in resource-poor countries (Killeen & DiRita, 2000) . Similar constraints apply to veterinary vaccines, with cost often an even more important consideration. Finally, long-lasting cellular and (where appropriate) humoral immune responses to the vectored antigen must be induced following administration of these vaccines, preferably with a single dose (Atkins et al., 2006) .

As some of the vectors in use will have been seen by the host immune system prior to vaccination, whether the presence of pre-existing immune responses is detrimental for the further development of a vector-based vaccine scheme, or can augment responses to the vectored antigen, needs to be considered in detail. This is the subject of this review. In discussing the possible effects on pre-existing immunity, the natural immunity to the vector

needs to be considered. Therefore, considering a vector such as *Salmonella*, if a host has previously been infected there will exist robust B and T memory responses, and as such, when a vaccination is delivered, an anamnestic response to the *Salmonella* antigens will be induced (while the response to the vectored antigen will be a primary response). This will theoretically reduce the exposure of the heterologous antigen to the immune system, as the vector is rapidly cleared. Surprisingly, as will be seen in some of the examples given below, this can have results that differ depending on the magnitude of the response to the vectored antigen. Similarly, for virally vectored antigens, the existence of pre-existing immunity to the vector (particularly neutralizing antibody) will restrict delivery of the virus into cells, thereby effectively reducing the dose of the vectored antigen. Again, this might be expected to result in a reduction in the antigenicity of the vectored antigen.

In the case of bacterial vectors, the effect of pre-existing immune responses has only been tested using *Salmonella* serovars and *Listeria* spp. Concern that prior immunological experience of the host with either the homologous *Salmonella* vector strain or a related strain might compromise its ability to deliver heterologous vaccine antigen was first raised in 1987 (Dougan et al., 1987) . Bao and Clements subsequently reported experimental evidence of the consequences of prior exposure of animals to the vector strain (Bao & Clements, 1991) . This work showed that both serum and mucosal antibody responses against the foreign antigen were in fact upregulated in animals with prior exposure to the vector strain. Whittle & Verma (1997) reported similar findings. Mice immunized via the intra-peritoneal route with a *Salmonella* dublin aroA mutant expressing heterologous antigen after being exposed to the same vector showed a higher immune response to the vectored antigen in comparison to mice without any immunological memory against the vector.

Subsequently, several studies have been conducted to examine the effect of pre-existing immunity in the host against *Salmonella*. These results are summarized in Table 1 .

The various reports are contradictory in their findings and seem to paint a rather confusing picture. Some studies concluded that pre-existing immunity against the *Salmonella* vector leads to stronger immune responses against the delivered antigen (Bao & Clements, 1991; Jespersgaard et al., 2001; Kohler et al., 2000a, b; Metzger et al., 2004; Saxena et al., 2009; Sevil Domènech et al., 2008; Whittle & Verma, 1997) , with others considering pre-existing immunity to be a limiting factor in the long-term use of *Salmonella* as an efficient vector for

antigen delivery (Attridge et al., 1997; Gahan et al., 2008; Roberts et al., 1999; Sevil Domènech et al., 2007; Vindurampulle & Attridge, 2003a, b) .

A slight majority of the studies listed in Table 1 (10 versus eight) indicate the upregulation of immune responses after animals have been exposed to either homologous or related strains before the delivery of heterologous antigen using a Salmonella vector. A study by Metzger and co-workers on human volunteers using Salmonella Typhi as a vector suggested that there was no change in the T cell immune response against the heterologous antigen in human volunteers who were exposed to empty vector in comparison with volunteers who were immunologically naive of the vector strain (Metzger et al., 2004) . In these subjects, humoral responses were moderately elevated in preexposed individuals. Similarly, Saxena et al. (2009) indicated higher humoral and T cell responses in mice pre-exposed to homologous or heterologous Salmonella strains. The interleukin 4 (IL4) response was significantly higher when the animal host was exposed to the homologous strain, whereas pre-exposure to a related species did not have such an impact on IL4 responses. Conversely interferon (IFN)- γ responses were higher, irrespective of the strain to which mice were pre-exposed. This study also indicated that the presence of homologous or heterologous opsonizing antibodies leads to a higher uptake of Salmonella by macrophages in vitro, which may explain the higher immune responses in exposed mice. As may be expected, uptake was higher when homologous sera were used as the opsonin rather than heterologous sera. This is depicted in Fig. 2 .

Conversely, there are reports that indicate that pre-existing immunity against the bacterial vector downregulates immune responses against the delivered heterologous antigen using similar or related vectors. Attridge and coworkers reported that the presence of immunity against the bacterial vector prior to the delivery of vectored antigenic

Microbiology 159 protein can downregulate immune responses in mice against the delivered antigen (Attridge et al., 1997) . Similar results were reported by Roberts et al. (1999) and Vindurampulle & Attridge (2003a, b) . However, the latter authors found that the hypo-responsiveness could be largely eliminated by exposing animals to the foreign antigen prior to vectorpriming (Vindurampulle & Attridge, 2003b) . Unfortunately, this would appear to be impractical for an immunization regimen! A study presented by Gahan et al. (2008) immunized mice with *S. Typhimurium* expressing C fragment of tetanus toxin antigen from an expression plasmid or as a DNA vaccine. Vaccinated mice developed

humoral responses to LPS and tetC (for the plasmid-bearing vaccines). Animals from all groups (including a previously unvaccinated group) were immunized on day 182 with *Salmonella* expressing tetC. At this time, the anti-LPS and tetC titres were beginning to wane. Fourteen days after the second immunization, the colonization of various mouse organs was assessed. The ability to colonize was found to be significantly reduced in groups that had been previously vaccinated with *Salmonella*. In view of this finding, it was perhaps not surprising that at day 210 the LPS titres were not significantly different between groups receiving one or two vaccinations. More interestingly, mice that had been primed with *Salmonella* alone, and then boosted with *Salmonella* expressing tetC, induced much lower anti-tetC responses than mice that had not been primed. This argues strongly that prior immunological immunity to the vector can seriously dampen subsequent antigen-specific humoral responses. Whether the same is true for cellular responses was not evaluated.

Other studies have evaluated cellular responses. A study by Sevil Domènech and colleagues reported that pre-existing anti-vector immunity seriously compromises CD8 + responses in mice when exposed to a similar strain used as vector (Sevil Domènech et al., 2007) . In contrast, another study by the same authors reported that animals exposed to related vectors induce much higher CD8 + responses when compared with animals which do not have any pre-existing *Salmonella* immunity (Sevil Domènech et al., 2008) . The difference between these two studies was that in the first, the prime and boost were with identical serovars, while in the second study, different serovars were used. This may point to a way of avoiding downregulation of CD8 responses by pre-existing immunity. This is important, as one of the advantages of using *Salmonella* (an intracellular pathogen) is that strong cellular immune responses can be induced.

It must be noted that in the case of *Salmonella* vaccines, effects other than strictly immunological responses (particularly adaptive responses) should be considered. In the context of innate immunity, it was shown that administration of non-virulent *Salmonella* to gnotobiotic pigs eliminated disease following challenge with a virulent strain (Foster et al., 2003) . Interestingly, protection was not by competitive exclusion, as the virulent strain was in high numbers in the gut but did not distribute systemically. The protection was proposed to be mediated by the infiltration of a large number of polymorphonuclear leukocytes into the gut, and although perhaps impractical as a general prophylactic (as the time between vaccination and infection is short), this may be an option for short-term or perhaps therapeutic vaccination (as reviewed by Foster et al., 2012) .

Chickens (*Gallus gallus*) are a natural animal reservoir for *Salmonella*, which makes them an important source of *Salmonella*-associated gastroenteritis in humans. The ability to use oral *Salmonella* vaccines to immunize against heterologous pathogens would be of enormous benefit to Uptake of STM-1 by J774 macrophages, relative to the highest uptake percentage. X, Opsonized with naive sera; m, opsonized with serum from mice exposed to *Salmonella enteritidis*; &, opsonized with serum from mice exposed to STM-1.

Pre-existing immunity against vaccine vectors the poultry industry in both broiler and layer flocks. Both vertical and horizontal transmission is associated with *Salmonella* in chickens (Liljebjelke et al., 2005) . Vertical transmission via in ovo transmission is particularly important, because if there is prior exposure to the vaccine strain, subsequent vaccination using an oral *Salmonella* vector could be severely compromised. A considerable number of studies on cross-protective immunity and competitive exclusion have been undertaken in chickens. Protective cross-reactive immunity against *Salmonella* strains has been demonstrated against both homologous and heterologous challenges (Beal et al., 2006) , although cross-serogroup protection was not strong. Furthermore, a recent study reported that pretreatment of newly hatched chickens with different *Salmonella* strains could produce a complete invasioninhibition effect on any subsequent exposure to both homologous and heterologous strains (Methner et al., 2010) . Pre-exposure with a highly invasive form of *Salmonella Enteritidis* caused a large influx of heterophils to the caecal mucosa in 1-day-old chicks, and subsequent heterologous caecal colonization was inhibited for a period of 48 h (Methner et al., 2010) . The implications of this kind of colonization-inhibition study on the immunological status of the affected chickens are yet to be fully elucidated. It should be noted that the studies listed in Tables 1 and 2 are controlled laboratory studies, with the possibility of a competitive exclusion component to immunity not discussed.

Similarly studies of *L. monocytogenes* and the effects of preexisting immune responses indicate conflicting results. A study by Bouwer et al. (1999) indicates that pre-existing immune responses against the *Listeria* vector do not diminish immune responses against the delivered heterologous antigen, and a similar study by Starks et al. (2004) also concluded that prior exposure of mice to the empty *Listeria* vector did not influence anti-cancer immune responses when a similar mutant was used as a carrier of a melanoma cancer antigen. Similar findings were reported by Whitney et al. (2011) in rhesus macaques

in which *L. monocytogenes* was used as a carrier of gag-HIV antigen. Conversely, studies by Stevens et al. (2005) in which *L. monocytogenes* was used to deliver feline immunodeficiency virus (FIV) gag protein and as a carrier of DNA vaccines to vaccinate cats against FIV envelope protein indicated lower immune responses against the delivered antigen in cats exposed to empty *Listeria* vector in comparison with naive animals (Stevens et al., 2005). Similar findings have been reported by Tvinnereim et al. (2002) and Leong et al. (2009). However, taken together, these studies conclude that prior exposure of host animals to empty vector does not abrogate immune responses to the vectored antigen, but only reduces them somewhat. Only the study by Vijh et al. (1999) indicated that exposure to the empty vector may completely abrogate immune responses against the delivered antigens (Vijh et al., 1999). However, these studies also indicate that downregulation of antigen-specific immune responses is highly dependent on dose and time. Leong et al. (2009) also demonstrated that the negative impact of vector-specific immune responses can also be countered by repeated immunization with the same vaccine and dose; this in effect leads to higher priming of naive T cells against the delivered antigen. Of course, such repeated vaccination may not be practicable in real-world situations.

Despite the many advantages which viral vectoring can offer, pre-existing immunity is a major obstacle of many viral-vectored vaccines, such as Ad serotype 5 or herpes simplex virus type 1 (HSV-1), where the rate of seroprevalence to these viruses is very high [40-45 % and 70 % (or more) of the US population, respectively] (Hocknell et al., 2002; Pichla-Gollon et al., 2009). Vector-specific antibodies may impede the induction of immune responses to the vaccine-encoded antigens, as they may reduce the dose and time of exposure of the target cells to the vaccinated antigens (Pichla-Gollon et al., 2009; Pine et al., 2011). In a large-scale clinical trial (STEP) of an Ad serotype 5 (AdHu5)-based HIV-1 vaccine, the vaccines showed a lack of efficacy and tended to increase the risk of HIV-1 infection in vaccine recipients who had pre-existing neutralizing antibodies to AdHu5 (Buchbinder et al., 2008). For an HSV-1-based vector vaccine, it has been demonstrated that pre-existing anti-HSV-1 immunity reduced, but did not abolish, humoral and cellular immune responses against the vaccine-encoded antigen (Hocknell et al., 2002; Lauterbach et al., 2005). However, Brockman and Knipe found that the induction of durable antibody responses and cellular proliferative responses to HSV-encoded antigen were not affected by prior HSV immunity (Brockman & Knipe, 2002). Similarly, pre-existing immunity to poliovirus has little effect on vaccine efficacy in a poliovirus-vectored vaccine (Mandl et al., 2001). Different effects of pre-existing immunity on the efficacy of recombinant viral vaccine vectors are summarized in Table 2.

There are several approaches to avoiding pre-existing vector immunity, such as the use of vectors derived from nonhuman sources, using human viruses of rare serotypes (Kahl et al., 2010; Lasaro & Ertl, 2009) , heterologous prime-boost approaches (Liu et al., 2008) , homologous reimmunization (Steffensen et al., 2012) and removing key neutralizing epitopes on the surface of viral capsid proteins (Gabitzsch & Jones, 2011; Roberts et al., 2006) . The inhibitory effect of pre-existing immunity can also be avoided by masking the Ad vector inside dendritic cells (DCs) (Steffensen et al., 2012) . In addition, mucosal vaccination or administration of higher vaccine doses can overcome pre-existing immunity problems (Alexander et al., 2012; Belyakov et al., 1999; Priddy et al., 2008; Xiang et al., 2003) .

As we search for new vaccine approaches for the array of pathogens for which none is yet available, revisiting proven vaccines and developing these further has gained M. Saxena and others momentum. Hence, attenuated bacteria and viruses which have a long history of efficacy and safety are being brought into use. While very attractive, a common theme in these experimental approaches has been the limitations that preexisting immunity to the vector may pose. However, as this examination of the relevant literature shows, there is a rather confusing picture, with some studies in fact indicating that pre-existing immunity may be a friend, rather than foe.

Few studies using viral vectors have reported on the influence of pre-existing immunity on humoral responses. Generally speaking, for bacterial-delivered antigens, the humoral responses were influenced by pre-existing immunity, with slightly more studies finding augmentation rather than diminution. Why is there variation? This may be due to several factors, including the type of Salmonella used and its invasiveness. Dunstan and colleagues tested the ability of six isogenic Salmonella serovar Typhimurium strains harbouring different mutations for their ability to induce immune responses against the C fragment of tetanus toxin and concluded that the strain which had the least ability to colonize Peyer's patches induced the lowest immune responses (Dunstan et al., 1998) .

Similarly, the boosting time and nature of the antigen used might be important. Attridge and colleagues indicated the importance of boosting time. In one experiment, boosting mice at 10 weeks led to complete inhibition of antibody responses against the delivered heterologous antigen; however, when the mice were boosted at 4 weeks, the

downregulation of antibody responses was not so prominent (Attridge et al., 1997) . A similar study conducted by Kohlers and colleagues shows that boosting at 7 weeks after pre-exposing animals to empty vector leads to lower antigen-specific IgG and secretory IgA responses; however, boosting at 14 weeks leads to higher IgG and secretory IgA responses (Kohler et al., 2000b) . This is in conflict with the above result, although it should be mentioned that they used different Salmonella species. Vindurampulle and Attridge also examined the impact of the Salmonella strain and the nature of the antigens used. In their study, they used S. Dublin and Salmonella Stanley aroA mutants to deliver E. coli K88 and LT-B antigens, and concluded that the effect of pre-existing immunity depends on both the strain used and the type of antigen delivered (Vindurampulle & Attridge, 2003b) .

All these studies on the effect of pre-existing immunity discuss the impact on humoral responses. Sevil Domenech and colleagues reported that pre-exposing animals to the homologous Salmonella vector leads to a significant reduction in CD8 + responses; however, exposure of animals to a heterologous strain leads to significantly higher CD8 + responses (Sevil Domènech et al., 2007 , 2008 . Saxena and colleagues also reported that antigenspecific T cell responses were either similar or significantly higher, with no downregulation in T cell responses observed after pre-exposing mice to either homologous or heterologous strains (Saxena et al., 2009) .

For viral vectors, the impact of cell-mediated immunity was more pronounced, and as depicted in Table 2 , almost always resulted in a reduction in the subsequent immune response. Presumably this is because viruses will induce neutralizing antibody on the first dose, and in subsequent doses this antibody will limit the number of transduced cells, therefore limiting the responses. This is particularly a problem with a common viral vector such as Ad, where a large proportion of the population will have immunological memory against common serotypes (Lasaro & Ertl, 2009) . As these authors conclude, it will be possible to utilize such vectors only by developing vaccines from alternative serotypes. It may be that a vector such as Pre-existing immunity against vaccine vectors attenuated influenza virus, with the ability to easily develop reassortants, will be useful in this context.

In addition, immunological memory in the form of opsonizing antibody certainly plays an important role in the early uptake of Salmonella by macrophages and DC. This may be beneficial, as the live bacterial vector used for delivery purposes harbours mutations in genes encoding proteins responsible for their survival in the animal host. This not only

encumbers their ability to cause disease, making them safe live vectors, but also limits the number of replications. The presence of opsonizing antibodies should mean a higher level of bacterial uptake, leading to higher presentation to the immune system and therefore a better immune response. We have previously shown that this is indeed the case (Saxena et al., 2009) (depicted in Fig. 2). It would be of great benefit to address these issues not only in mice but also in other organisms such as chickens, which are the most likely host to be targeted for the use of live Salmonella vectors, specifically where the vaccines are developed for use in livestock and poultry.

To summarize, bacterial vectors such as Salmonella and viral vectors such as Ad show great promise as delivery vehicles for heterologous antigens; however, prior exposure to the vector must be considered. By judicious selection of the strain/serotype it will be possible to avoid the negative effects and it may indeed be possible to positively influence the response, particularly for humoral immunity.

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Mucosal immune responses induced by oral administration recombinant *Bacillus subtilis* expressing the COE antigen of PEDV in newborn piglets

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Abstract: Porcine epidemic diarrhea (PED) is a highly contagious disease in newborn piglets and causes substantial economic losses in the world. PED virus (PEDV) spreads by fecal–oral contact and can be prevented by oral immunization. Therefore, it is necessary to develop an effective oral vaccine against PEDV infection. Currently, *Bacillus subtilis* as recombinant vaccine carrier has been used for antigen delivery and proved well in immune effect and safety. The present study evaluated the immunogenicity of recombinant *Bacillus subtilis* (B. subtilis-RC) in piglets via oral administration. After oral immunization in piglets, B. subtilis-RC significantly increased the local mucosal immune responses. Oral administration with B. subtilis-RC significantly improved the level of specific mucosal immunoglobulin A (IgA) antibodies against PEDV infection, through enlarging the area of Peyer's patches (PPs) and increasing the number of ileum IgA(+) secreting (SIgA) cells. In the meantime, B. subtilis-RC remarkably increased the number of intraepithelial lymphocytes (IELs). We also observed that oral administration of B. subtilis-RC significantly increased CD3(+)T lymphocytes' numbers and up-regulated the ratio of CD4(+)/CD8(+) T cells. Furthermore, high titers of specific serum immunoglobulin G (IgG) revealed satisfactory systemic immune response against PEDV infection. In summary, our study demonstrated that oral administration of B. subtilis-RC could trigger a high level of local and systemic immune responses and would be a promising candidate vaccine against PEDV infection in piglets.

Text: Porcine epidemic diarrhea (PED) characterized by highly fatal acute diarrhea in piglets, results in enormous losses in the worldwide pig industry [1] . The causative agent PED virus (PEDV) belongs to the porcine coronaviruses (CoVs). PEDV infection mainly spreads through the digestive tract [2] , and damages the host intestine mucosal surfaces by infecting the intestine epithelial cells [3] . Therefore enhancing intestinal mucosal immunity can elicit effective mucosal immune responses against PEDV infection [4] . Currently, traditional vaccines (intramuscular route or subcutaneous injection) have been developed and applied widely in the market [5] . These vaccines administered parenterally cannot effectively induce high titers of maternal antibodies and virus-specific IgA antibodies, resulting in inadequate mucosal protection to against PEDV infection [6] . Furthermore, these maternal antibodies in the milk were always degraded by gastric acid and pepsin before entering the intestinal tract. Effective PEDV vaccines must provide adequate mucosal protection in the intestinal tract. However, the effective vaccines are currently lacking [7] .

As a superior way of mucosal immunization, oral administration can protect the gut and stimulate the common mucosal immune system [8] . Besides, oral immunization has several attractive features which include safety, and a straightforward, inexpensive, and needle-free approach [9] . Therefore, oral immunization often delivers large amounts of antigens to prevent the diarrheal diseases [10] . Nevertheless, there are several challenges by oral immunization, which consist of physical, chemical, and biological barriers when delivering antigens to the gastrointestinal (GI) tract (such as gastric acids, pepsin, and trypsin in the GI tract) [11] .

It is a substantial problem that digestive acids and proteases can degrade antigen proteins for nutrient absorption [12] . Therefore, the vaccine delivery system has been applied to solve the problem. The system can protect antigens from the severe environment of the GI tract and deliver antigens to intestinal mucosa [13] . Currently, *Bacillus subtilis* (*B. subtilis*) is widely used as a vaccine delivery system for its unique characteristics.

As a nonpathogenic Gram-positive bacterium, *B. subtilis* has been regarded as a novel probiotic and food additive in humans and animals [14] . The *B. subtilis* has adjuvant activity and can deliver heterologous antigens to the GI tract, providing additional immunity stimulation [15] . Besides, research had shown that orally administered *B. subtilis* could also enhance immune regulation and gut health in pigs [16] . Moreover, oral administration of *B. subtilis* could elicit humoral and cellular immune responses to the maintenance of gut homeostasis by dendritic cells (DCs) [17] . DCs are the most important professional antigen-presenting cells and can effectively regulate antibody titers [18] . DCs naturally exist in the gut-associated lymphoid tissue (GALT), including Peyer's patches (PPs), isolated lymphoid follicles (ILFs), mesenteric lymph nodes (MLNs), and scatter throughout the subepithelial lamina propria (LP) of the small intestine and colon [19] . Furthermore, *B. subtilis* is convenient for genetic manipulation and has developed a large variety of genetic tools [20] . Therefore, *B. subtilis* is widely used as an effective vaccine delivery system to induce mucosal immune responses and shows unique effect on the immune system.

In the present report, we explored the immune effect of a recombinant *B. subtilis* (*B. subtilis*-RC) which had been successfully constructed with expressing PEDV COE protein in piglets. Our research indicated that *B. subtilis*-RC was beneficial to the mucosal immune system development, and could effectively generate specific antibodies against PEDV infection, suggesting a potential approach for preventing PEDV infection.

The *B. subtilis* WB800 was kindly provided by Dr. Xuewen Gao (from the department of plant pathology, Nanjing Agricultural University) [21]. *B. subtilis*-RC previously constructed in our laboratory was able to express the gene COE (499-638 amino acids in S protein). Prior to oral administration, the recombinant strain was grown in LB broth at 37 °C for 12 h, and then washed twice with PBS, and suspended in PBS to reach a final concentration of 1×10^{10} CFU/ml. The PEDV Zhejiang08 strain was provided by the Veterinary Medicine Research Centre of the Beijing Dabeinong Technology Group Co., Ltd. [22]. The virus was cultured in African green monkey kidney cells (Vero cells) and purified by using a discontinuous sucrose density gradient. The virus was UV-inactivated at UV dose of 4 J/cm² for 24 h to achieve a complete loss of infectivity [23]. The purified virus concentration was measured using the BCA protein assay kit (Thermo Fisher, MA, U.S.A.). ELISA: Rabbit anti-pig IgG (horseradish peroxidase (HRP)), Goat Anti-Pig IgA (HRP) were purchased from Abcam. Second antibody: DyLight 649-conjugated goat anti-mouse IgG antibody, DyLight 488-conjugated goat anti-rabbit IgG antibody, DyLight 594-conjugated goat anti-rabbit IgG antibody were purchased from Multi-science, Hangzhou, China. ABC-based system (biotinylated goat anti-rabbit IgG antibody) was used as the secondary antibody with DAB as a chromogen was purchased from Boster, Wuhan, China.

Specific pathogen-free (SPF) DLY piglets (Duroc and Landrace and Yorkshire) were kindly provided by Jiangsu Academy of Agricultural Sciences (Nanjing, China). The animal experiments had been approved by the Institutional Animal Care and Use Committee of Nanjing Agricultural University and followed the National Institutes of Health's guidelines for the performance of animal experiments. Twelve newborn piglets were randomly divided into three groups (four piglets in each group), and housed under similar conditions in different stables in order to avoid probiotic cross-contamination. The piglets were orally dosed with 100 µl of *B. subtilis*-RC. The control groups of piglets were orally administered with inactivated PEDV (100 µg/dose) and equal volume of PBS. The immunization protocol was performed on the piglets that were 5 days old (Figure 1C), and signed as 0 day. Then booster immunizations were administered on 5 days.

Specimen collection was then performed every 7 days post boost immunization (Figure 1C). Blood samples were collected weekly from all piglets after the boost immunization and allowed to clot overnight at room temperature to collect serum. Blood samples were separated by centrifugation and stored at -20 °C in order to detect the levels of specific

IgG and IgA. Three swabs were collected every week lasting for 1 month, including nasal, oral, and feces swabs for the ELISA. The piglets were sacrificed in 33 days. The same location of the small intestine and ileum tissues from each piglet were fixed with Bonn's liquid and 4% paraformaldehyde.

The small intestine tissues in same location were fixed with Bouin Fixative Solution for 24 h, embedded in paraffin, and sectioned at 4- μ m thickness. The sections were placed on glass slides. Hematoxylin-eosin staining was applied to the paraffin sections, then observing and taking photographs under optical microscope (OLYMPUS CX23). The number of intraepithelial lymphocytes (IELs) were counted in every 100 epithelial cells under the same multiple light microscope amongst ten pictures from each group [24] .

The immunohistochemistry detection was performed with the SABC kit (Boster Bioscience). Hydrogen peroxide was used to deactivate intrinsic peroxidase. Antigen retrieval was performed in a water bath using citrate-EDTA buffer (10 mM citric acid, 2 mM EDTA, 0.05% Tween 20, pH 6.2). Sections were incubated with diluted anti-IgA antibody (1:100; Abcam) overnight at 4 • C. As negative controls, immunostaining performed by incubating samples with control antiserum instead of primary antibody. The addition of biotin-labeled secondary antibody to the slides was followed by adding HRP-labeled streptavidin. After staining with DAB, the slides were recorded using a digital camera (Leica-DM4000B) [25] .

The isolated intestines with PPs were transferred to ice-cold PBS. Then, remaining fat and connective tissue was removed and washed thoroughly with ice-cold PBS. Next, the intestine was cut longitudinally into 0.5-cm fragments. The fragments were incubated with 5 ml of 30 mM EDTA and placed in 5 ml digestion solution containing 4% FBS, 0.5 mg/ml each of Collagenase D (Roche) and DNase I (Sigma), and 50 U/ml Dispase (Fisher). The fragments were incubated with Dulbecco's PBS (DPBS) for 20 min at 37 • C by slow rotation (100 rpm). After incubating, the epithelial cells layer which contained the IELs were separated by intensive vortex and passed through a 70- μ m cell strainer. Single cell suspension was collected and washed twice by DPBS, the solution was vortexed intensely and passed through a 40- μ m cell strainer. Supernatants was washed by precooled RPMI medium 1640 (Thermo Fisher Scientific) and suspended by 10 ml of the 40% fraction of a 40:80 Percoll gradient, overlaid on 5 ml of the 80% fraction in a 15-ml Falcon tube. Percoll gradient separation was performed by centrifuging for 20 min at 2500 rpm. LP lymphocytes

(LPLs) were collected at the interphase of the Percoll gradient, then washed and suspended in FACS buffer or T cell medium. In the meantime, flow cytometry analysis was performed on BD FacsCalibur (BD Biosciences) instruments and analyzed by FlowJo software. All antibodies were purchased from BD Pharmingen or eBiosciences. Isolated single-cell suspensions were stained with anti-CD3-APC, anti-CD4-FITC, anti-CD8-PE, all at 1:100 dilution for 30 min on ice, and washed with PBS twice, and analyzed by FACS [26] .

Cytokines interleukin (IL) 10 (IL-10) and IL-1 β (Abcam) were measured by ELISA according to the manufacturer's instructions. Data were acquired on an automated ELISA plate reader at OD 450 nm immediately.

PEDV neutralizing antibodies were measured in intestine washing liquid by plaque reduction neutralization test (PRNT). The test was performed as previously described with minor modifications [27] . A total of 450 μ l of intestine washing liquid was two-fold serially diluted and mixed with 50 μ l viral suspension containing 10³ TCID₅₀ PEDV virus for 1 h at 37 • C in 12-well flat bottomed tissue culture plates. The mixture was then inoculated for 1 h at 37 • C and 5% CO₂ . Then, the mixture was inoculated with Vero cells suspension (approximately 1.0 $\times$ 10⁶ ml⁻¹) for another 3-4 days. After staining with Crystal Violet, the plates were observed under a microscope for cytopathic effect.

Data were obtained as the means + – S.E.M. of three replicates per test in a single experiment. GraphPad Prism V6.0 (San Diego, CA, U.S.A.) used to perform statistical analyses. Tukey's multiple comparison tests and one-way ANOVA were used to analyze the significance of the difference between means. P-values less than 0.05 (P<0.05) were considered significant and P-values less than 0.01 (P<0.01) as highly significant.

PPs are a concentrate of lymphoid tissue and the primary site for immunoglobulin A (IgA) production which is crucial to regulate the homeostatic balance of intestine [28] . The area of PPs is a key immunity indicator. Oral administration with *B. subtilis*-RC significantly (P<0.01) increased the area of PPs compared with two control groups as shown in Figure 1A . In addition, the villi length of ileum got longer by oral administration with *B. subtilis*-RC (P<0.01) than the other two groups (Figure 1B) . These primarily confirmed that *B. subtilis*-RC was beneficial to maintain the structure of intestine.

Intestinal IELs are a large and diverse population of lymphoid cells residing within the intestinal epithelial cells (IECs), and forming the intestinal mucosal barrier [29] . IELs are important part of the gut mucosal immune system.

The level of specific anti-PEDV ileum IgA + secreting (SIgA) antibody in piglets was measured by ELISA in the mouth and feces. As shown in Figure 3A ,B, antigen-specific mucosal SIgA in the above sites was clearly higher than inactivated PEDV group ($P<0.05$ or $P<0.01$). As expected, the mouth had higher levels of SIgA than other sites. After oral immunization, the level of serum anti-PEDV IgG antibody in piglets immunized with B. subtilis-RC, inactivated PEDV or PBS were determined by ELISA, as shown in Figure 3C . The results indicated that although the titers dropped during sampling period, the IgG level of B. subtilis-RC still significantly increased from 0 to 33 days than inactivated PEDV group ($P<0.05$ or $P<0.01$).

CD3 + T lymphocytes are the fundamental cell surface markers of T lymphocytes, therefore, the number of CD3 + T lymphocytes could represent the quantity of T lymphocytes. Consequently, we analyzed the number of CD3 + T lymphocytes in ileum. The data indicated that both B. subtilis-RC and inactivated PEDV could dramatically ($P<0.05$) increase CD3 + T lymphocytes compared with PBS group (Figure 4A). These changes showed confident evidence that oral administration with B. subtilis-RC had a good influence on intestinal mucosal immunity in piglets.

SIgA is the main immunoglobulin isotype in animals, largely secreted across the intestinal mucosal surface especially in the small intestine [30] . SIgA plays an important role in intestinal mucosal immunity and reflects on the intestinal mucosal immunity. After oral administration with B. subtilis-RC, the number of IgA secreting cells had quickly risen compared with the other two groups ($P<0.05$) (Figure 4B) . These results showed that oral administration with B. subtilis-RC was conducive to intestinal mucosal immunity and could increase the number of IgA secreting cells to produce positive effects on against PEDV infection.

A great deal of immune cells are scattered in the epithelial cells. IECs indirectly or directly interact with innate and adaptive immune cells by presenting antigens to lymphocytes [31] . Consequently, learning about how the lymphocytes are distributed in the small intestinal

mucosa is very meaningful for mucosal immunology. Previous data had shown that CD3 + T lymphocytes significantly ($P < 0.05$) increased (Figure 4A), so we further analyzed the immunological classification of CD3 + T lymphocytes. The lymphocyte of the ileum with PPs junction was isolated and the lymphocytes of CD3, CD4, and CD8 were analyzed by three colors flow cytometry (Figure 5A). These results showed that CD3 + CD4 + T cells have obviously ($P < 0.01$) increased (Figure 5B), nevertheless the CD3 + CD8 + T cells remarkably ($P < 0.05$) declined (Figure 5C). After calculation, the ratio of CD4 + /CD8 + T cells increased (Figure 5D). This ratio could also further measure the immunity levels of piglets.

Cytokine IL-1 β and IL-10 levels were determined to evaluate cellular immune responses induced by B. subtilis-RC as shown in Figure 6A ,B. As we can see from the diagram, significantly ($P < 0.01$) higher IL-1 β and IL-10 were produced after oral administration with B. subtilis-RC than the other two groups. These all revealed that B. subtilis-RC could stimulate cytokines release to mediate communication with and between cells of the immune system, improving the mucosal immune response to PEDV infection.

The PEDV neutralizing antibodies were detected by PRNT assay. Oral administration with B. subtilis-RC could effectively reduce the plaque-forming ability of PEDV ($P < 0.01$) compared with other two groups in Figure 7 . This revealed that B. subtilis-RC could stimulate high level of PEDV neutralizing antibodies against PEDV infection.

Amidst the PEDV outbreak, various vaccines have been developed to control diseases and the effects are unsatisfactory. Oral vaccines can induce more robust mucosal immunity than injectable counterparts [32] . Therefore, oral immunization has appeared as an effective strategy for controlling PEDV outbreak [33] .

It is now clear that effective mucosal immune response requires serum IgG and mucosal SIgA [34] . SIgA is the basis of the mucosal immune system, playing an important role in maintaining the immune homeostasis, and neutralizing the invasive pathogens. Serum IgG represents systemic immune responses. During PEDV infections, oral immunization elicits not only mucosal but also systemic immune responses very well [35] . Our data showed a strong and long-lasting anti-PEDV IgG response were detected by oral administration with B. subtilis-RC in piglets. Although as time went on, the antibody titers declined a little, it

still stayed on overhead compared with control groups and with accordance to the changeable tendency of antibodies. The change of specific IgA showed similar results in mouth and feces mucosa. All these changes had contributed to fight PEDV infection. As the extra immunity boost, *B. subtilis*-RC reduced the ability of pathogens to cross the intestinal mucosa and the systemic spread of invasive pathogens [36]. The mucosal immune system generates immune responses through immune cells that reside in mucosal compartments. T lymphocytes residing in the mucosa play important roles in mucosal immunity [37]. We further explored the species, amounts, and distribution of T lymphocytes in the intestine mucosa. CD3 is a fundamental cell surface marker of T lymphocytes [38]. The result showed that the number of CD3 + T lymphocytes significantly increased, and these revealed that *B. subtilis*-RC could stimulate T-cell maturation. According to the molecules expressed on the cell surface, T lymphocytes can further divide into T helper cells (CD4 + T cells) and cytotoxic T cells (CD8 + T cells) [39]. Furthermore, we observed that the ratio of CD4 + /CD8 + T cells increased by oral administration. The CD4/CD8 ratio measures the ratio of T helper cells to cytotoxic T cells. Therefore, we could see that oral administration *B. subtilis*-RC could strengthen Th1 immune response by raising the ratio of CD4 + /CD8 + T cells.

Small intestine morphology can directly reflect the intestinal health and plays an important role in maintaining the intestine immune system [40]. The early stage of PEDV infection is frequently accompanied by necrosis and exfoliation of infected villous epithelial cells, ultimately resulting in acute, severe villous atrophy [41]. Therefore, the effective work of maintaining intestine morphology is a good indicator for assessing the efficacy of vaccines. After oral administration with *B. subtilis*-RC, we found the area of PPs expanded significantly. PPs are small masses of lymphatic tissue and form an important part of the immune system by recruiting and inducing the T cells to prevent the growth of pathogens in the intestines. Furthermore, an increase in the number of IELs demonstrated the effectiveness of *B. subtilis*-RC. Moreover, the villi length of ileum showed some encouraging results that a well-formed intestine morphology came into being by *B. subtilis*-RC. The satisfactory intestine morphology was the first step on the road against PEDV infection. Several morphology results proved that *B. subtilis*-RC could remarkably maintain the intestine morphology and form comprehensive protection.

As previously mentioned, oral administration with *B. subtilis*-RC could stimulate T-cell proliferation and differentiation and modulate the immune response. Moreover, cytokines are small-molecule proteins with wide biological activity, synthesized and secreted by

immune cells and some non-immune cells [42] . As a cell signaling molecule, it mainly acts to regulate immune responses, participating in the differentiation and development of immune cells, mediating inflammatory responses, stimulating hematopoiesis, and participating in tissue repair. Previous studies had demonstrated that PEDV inhibited both NF- κ B and pro-inflammatory cytokines [43] . Therefore, cytokines are a key indicator for evaluating the ability of a vaccine to stimulate immune responses. In this study, we had observed that IL-1 β and IL-10 increased ($P < 0.01$) remarkably. IL-1 β as one of the earliest pro-inflammatory cytokines and is centrally involved in the initiation and regulation of inflammatory and innate immune responses. Research had shown that IL-1 β could significantly up-regulate the local and systemic immune tissues post microbial infection [44] . In addition, IL-10 is a potent anti-inflammatory cytokine that plays an essential role in preventing inflammatory and autoimmune pathologies [45] . In summary, both data showed that oral administration with *B. subtilis*-RC regulated and enhanced immunity by up-regulating cytokines IL-1 β and IL-10.

In conclusion, the present results demonstrated that oral immunization with *B. subtilis*-RC could effectively induce local mucosal and systematic immune responses against PEDV infection, while enhancing and regulating the immune function by raising the ratio of CD4 + /CD8 + T cells and cytokines IL-1 β and IL-10, thus pointing to a promising oral vaccine candidate for PEDV infection in piglets.

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SARS to novel coronavirus – old lessons and new lessons

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Abstract: The response to the novel coronavirus outbreak in China suggests that many of the lessons from the 2003 SARS epidemic have been implemented and the response improved as a consequence. Nevertheless some questions remain and not all lessons have been successful. The national and international response demonstrates the complex link between public health, science and politics when an outbreak threatens to impact on global economies and reputations. The unprecedented measures implemented in China are a bold attempt to control the outbreak – we need to understand their effectiveness to balance costs and benefits for similar events in the future.

Text: On 29 December 2019 clinicians in a hospital in Wuhan City, China noticed a clustering of cases of unusual pneumonia (with the first case identified at that time on 12 December) with an apparent link to a market that sells live fish, poultry and animals to the public. This event was reported to the World Health Organisation (WHO) on 31 December [1]. Within 4 weeks, by 26 January 2020, the causative organism had been identified as a novel coronavirus, the genome of the virus had been sequenced and published, reverse transcription polymerase chain reaction tests had been developed, the WHO R&D Blueprint had been activated to accelerate diagnostics, therapeutics and vaccine development and a candidate vaccine was ready for initial laboratory testing. Currently Chinese health authorities are building a 1000 bed hospital in Wuhan in 10 days.

By 26 January also, almost 50 million people in Wuhan and neighbouring cities had effectively been placed in quarantine while the WHO had determined that the event should not yet be declared as a Public Health Emergency of International Concern (PHEIC) [2] and had recommended no specific travel restrictions. The WHO have emphasised the importance of exit screening at ports in countries showing transmission of the novel coronavirus and have provided guidance for countries implementing entry screening at airports while acknowledging that evidence for the effectiveness of entry screening is equivocal.

This response is one of the swiftest, coordinated global responses to an emerging infectious disease the world has seen in modern times, but is it the appropriate response, will it be effective and is it sustainable?

According to the situation report published by the WHO on 28 January 2020 [3], a total of 2798 confirmed 2019-nCoV cases have been reported globally; of these, 2761 cases were from China, including Hong Kong (8 cases), Macau (5) and Taipei (4). Thirty-seven confirmed cases have been reported outside of China in eleven countries in Europe, North America, Australia and Asia; of these 37 exported cases, 36 had a travel history from China or an epidemiological link to a case from China. Of the confirmed cases in China, 461 have been reported as severely ill, with 80 deaths to date.

This outbreak and the response to it illustrate some key issues about how global preparedness and response capacity for outbreaks have evolved over almost two decades since the severe acute respiratory syndrome (SARS) epidemic of 2002/3 and what lessons have, or have not, been learned. It also raises questions about the impact these lessons have had on the way agencies and governments respond to these events and about the role of the WHO and the International Health Regulations (IHR).

One of the critical lessons from the SARS experience was the absolute necessity to be able to coordinate the international resources that are available in an outbreak and to get them focussed on identifying priorities and solving problems. The WHO established the means to do this for SARS and it has since been further developed and integrated into global preparedness, especially after the West Africa Ebola epidemic. Organisations such as the Global Outbreak Alert and Response Network (GOARN), the Coalition for Epidemic Preparedness Innovations (CEPI), the Global Research Collaboration For Infectious Disease Preparedness (GloPID-R) and the Global Initiative on Sharing All Influenza Data (GISAID) have been supported by the WHO Research Blueprint and its Global Coordinating Mechanism to provide a forum where those with the expertise and capacity to contribute to managing new threats can come together both between and during outbreaks to develop innovative solutions to emerging problems. This global coordination has been active in the novel coronavirus outbreak. WHO's response system includes three virtual groups based on those developed for SARS to collate real time information to inform real time guidelines, and a first candidate vaccine is ready for laboratory testing within 4 weeks of the virus being identified.

Another key factor in successfully preventing and managing emerging threats is the rapid and transparent sharing of information between countries and agencies. There was extensive criticism of China for its perceived failure to share information about the emerging SARS infection early enough in the outbreak to allow countries to prepare and respond. There were similar concerns about information sharing as Middle East Respiratory Syndrome (MERS) emerged and evolved in the Middle East in 2012, particularly in Saudi Arabia, and about the emergence of Ebola in West Africa in 2014.

On this occasion information sharing seems to have been rapid and effective (while recognising that the information available in the early stages of an outbreak is always less than the global community would like). The WHO was notified of the original clustering within days and the full genomic sequence of the new virus was published less than 2 weeks after the cluster was first detected. The WHO has expressed its satisfaction with the actions of the Chinese authorities in sharing information with the WHO.

Working with journalists and the media to help them understand the science and epidemiology, particularly in a fast moving event, will improve risk communication to the public and reduce inappropriate concerns and panic.

While reporting of this outbreak shows signs of the efforts of epidemiologists, infectious disease experts, national and international public health agencies and others engaging with journalists, there are also signs that this is not yet achieving its goal. For example, the public perception is that the increase in case numbers reported daily by the Chinese authorities represents a daily escalation in the epidemic while the reality is that these numbers are also the result of active, aggressive, case finding in China and some of these cases are 'old' cases newly recognised as being due to the novel coronavirus. Similarly the virus is usually described by the media as 'deadly' and although this is true in the sense that it has caused deaths, the nuances of uncertain case fatality rates in the early stages of an outbreak are not being communicated. The current estimated case fatality rate seems to be around 3% which is significant but not comparable to the 10% rate for SARS or 34% reported for MERS. These misperceptions are still driving public anxiety.

To supplement formal reporting mechanisms between countries and with WHO (including the IHR), the use of informal mechanisms such as media and social media reports was advocated in the light of the SARS experience. There are now globally several systems that provide collated information from informal reporting including networks of experts and scanning of media and social media. These contribute to, and amplify, epidemic intelligence and are being integrated with national and international surveillance systems.

The value, and the challenges, of this additional source of information has been evident in the current outbreak. The value comes from ensuring that early indications of cases beyond the initial outbreak city have been detected and can supplement the global risk assessment and monitoring of the evolution of the outbreak. The challenges lie in the volume and diversity of the information available and the relative lack of verification mechanisms, such that one of these systems (ProMed) has commented that it was becoming increasingly difficult to assimilate the information being supplied [4] and to make meaningful interpretations.

Early in the outbreak it was reported that health workers had not been infected. This was reassuring because it is health workers who many times, and inadvertently, amplify transmission. Failure to wash hands between patients, for example, can result not only in autoinfection, but also in infection of patients hospitalised for other causes when they provide care. Autoinfection is not only a risk for the health worker, but also for their families and the communities in which they live, depending on the transmissibility and means of transmission. More recently infection, and at least one death, in health workers has been confirmed. Although not unexpected this does add to the epidemiological risk.

A characteristic of the SARS outbreak was the variability of transmissibility between cases and the occurrence of 'superspreading events' where a case infected significantly more contacts than the average. This was also seen with MERS in the outbreak in the Republic of Korea (RoK). In this current novel coronavirus outbreak, such superspreading events have not been documented but the epidemiology is still not clear. Confirming whether or not this is happening must be an urgent task for the Chinese investigation. Modellers have suggested reproductive rates (R_0) of 3.8 (95% confidence interval, 3.6-4.0) [5] and 2.6 (1.5-3.5) [6]; R_0 for SARS was estimated at around 3 in the absence of control measures [7].

The economic impact of major outbreaks can be substantial for the affected country. This was seen clearly in SARS, MERS in RoK and Ebola in West Africa. One analyst estimates that the current coronavirus outbreak's likely impact will range from a 0.8% cut to real GDP if the epidemic is controlled within 3 months, to a 1.9% cost to GDP if the epidemic lasts 9 months [8] . This may increase substantially in the light of the extended restrictions on movement, and therefore trade and commerce, within China.

The emergence of a significant respiratory illness linked to a novel coronavirus represents a test of the global capacity to detect and manage emerging disease threats. Its emergence in China adds an additional dimension in the light of previous experience with SARS. The timing of the outbreak immediately before the Chinese Lunar New Year with its attendant population movements adds extra risk and urgency to the response.

The rapid sharing of information in this outbreak and the speed of the coordinated response both in the country and internationally suggest that lessons have been learned from SARS that improve global capacity. The international networks and forums that now exist have facilitated the bringing together of expertise from around the world to focus research and development efforts and maximise the impact.

At this early stage in the outbreak information remains incomplete and key clinical and epidemiological questions have not yet been answered, but the deficit seems to be due more to the constraints of investigating an emerging disease than to any unwillingness to engage and share information with partners.

There are some indications of areas where further improvement is necessary. The global media response to the unfolding events has been relatively balanced and informed but the nuances of the evolving situation have not been critically examined in partnership with the media and as a result the public perception of the risk may be exaggerated although it of course remains possible that the outbreak will develop in a way that matches up to the perceived risk. The lack of appreciation of the uncertainties in determining a meaningful case fatality rate and the significance of ascertainment bias at the beginning of an outbreak, along with the impact of aggressive case finding on case numbers, are examples of where understanding could be improved. This is always a challenging process when balancing the resources focussed on analysing the situation on the ground with resources

directed at interpreting the information for journalists but in SARS, the R 0 was seen to decrease in response to information reaching the public and the public then adopting risk reduction actions [6] ; so accurate public risk communication is critical to success. It would be helpful to find a forum where this can be explored with the media community after the event.

The increase in access to early information from diverse sources including media and social media adds an important dimension to identifying and tracking new events globally and is a key part of the overall epidemic intelligence system. However, it is also a potential source of disinformation. When, as has been seen in this outbreak, the volume of information coming in exceeds any capacity to collate and analyse it and to attempt to cross-reference and verify separate items, there is a risk that the information fuels speculation and media and public concern. Again there is a fine balance between information that encourages appropriate risk avoidance actions and information that encourages inappropriate actions; however the public health is usually better served by more information rather than less.

The role of a declaration of a PHEIC in managing a serious outbreak has been questioned in the light of Ebola in West Africa and in the Democratic Republic of Congo [9] and has been challenged again with this outbreak. The binary nature of a PHEIC declaration (either an event is a PHEIC or it isn't there are no intermediate options) and the specificity of the three defined criteria for a PHEIC have caused difficulty for Emergency Committees in considering whether a given event should be a PHEIC. The lack of a clear understanding of what a PHEIC declaration is meant to achieve adds to the Emergency Committee's difficulties, as does the relative paucity of clinical and epidemiological answers at this stage of the investigation. In this instance the Emergency Committee were divided in coming to a conclusion but decided on balance that the current situation, although an emergency, should not as yet be declared a PHEIC [2]. As with Ebola in the DRC, there has been criticism of the WHO for this decision but, as with Ebola, it is not immediately clear what would be different in the response if a PHEIC was declared.

The WHO is working on improving the way in which Emergency Committees develop their advice for the Director General but, as recommended by this Emergency Committee and the post-Ebola IHR Review Committee in 2015, the development of an intermediate alert alongside WHO's risk assessment process may be helpful.

A key function of a PHEIC declaration is that it is the (only) gateway to the WHO Temporary Recommendations on possible travel and trade restrictions to limit international spread of a disease. In this case several countries globally had already implemented entry screening at airports and China had begun closing down international travel from Wuhan before the Emergency Committee had finished their deliberations. While the WHO would not, and could not, interfere with the sovereign decisions of member states, the lack of influence on travel and trade decisions could prove problematic.

Alongside the speed of the response in this outbreak, we have seen dramatic changes in the scale of the response. The imposition of very extensive quarantine measures on millions of people as an attempt to break the transmission of the virus is unprecedented. We do not know whether they will be effective; indeed we do not know how we will determine if they have been effective what end point can we measure that will provide an answer to that question? If recent suggestions that people infected with this coronavirus may be infectious while incubating or asymptomatic, and the reports that up to 5 m people left Wuhan before the travel restrictions were imposed, are confirmed, the efficacy of these control measures will be more challenged.

Given the likely impact on at least the Chinese economy and probably the global economy, it will be important to understand the role and the effectiveness of public health measures on this scale for the future.

However, the imposition of these dramatic measures does also raise a wider question: if there is an impact from these measures, what other countries would (or could) implement such measures? Would other countries accept the self-imposed economic damage that China has accepted to try and contain this outbreak? Is it reasonable to consider that national governments would close down public transport into and out of London, New York or Paris in the week before Christmas even if it were shown to be an effective control measure?

These decisions and questions cross the interface between public health, science and politics. The response to this outbreak in

China was inevitably influenced by the historical reaction to the country's response to SARS and the world's suspicion of China's lack of cooperation at that time. The current response is therefore framed within a context of not wanting to be seen to be behaving in the same way with this event.

This may indicate another impact of the SARS (and MERS and Ebola) experience on the response to subsequent outbreaks a tendency to look at worst case scenarios and respond accordingly and a fear of 'getting it wrong'. This can deter leaders at all levels, from outbreak teams to national governments, from making judgements when all the information they would like is not available in case those judgments turn out to be wrong when the full information becomes available.

In emergency response it is generally better to over-react and then scale back if necessary rather than under-react and then act too late. Response should be on a 'no regrets' basis make the best decisions possible on the basis of the best information and science available at the time but do not judge or criticise if later information suggests a different course of action. The early response must recognise what is known and what is not known and look at what of the unknowns can reasonably be estimated by reference to previous outbreaks, similar pathogens, early reporting and modelling, etc. The risk assessment and response can then be modified and refined as information on the unknowns evolves.

Key to that approach, however, is confidence that decisions will not be criticised based on information that was not available at the time. It is also important to be ready to change decisions when the available information changes something that both scientists and politicians can find difficult.

In that context, China should not be judged for implementing what might appear to be extreme measures but China should also be prepared to discontinue the measures quickly if evidence suggests they are not the best way to solve the problem. By closing airports the international spread from Wuhan may be decreased, but success will depend on how effective the measures really are at stopping people moving out of the affected area as well as on the behaviour of the virus. As always, only time will tell but time is scarce.

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Deep sequencing of primary human lung epithelial cells challenged with H5N1 influenza virus reveals a proviral role for CEACAM1

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Abstract: Current prophylactic and therapeutic strategies targeting human influenza viruses include vaccines and antivirals. Given variable rates of vaccine efficacy and antiviral resistance, alternative strategies are urgently required to improve disease outcomes. Here we describe the use of HiSeq deep sequencing to analyze host gene expression in primary human alveolar epithelial type II cells infected with highly pathogenic avian influenza H5N1 virus. At 24 hours post-infection, 623 host genes were significantly upregulated, including the cell adhesion molecule CEACAM1. H5N1 virus infection stimulated significantly higher CEACAM1 protein expression when compared to influenza A PR8 (H1N1) virus, suggesting a key role for CEACAM1 in influenza virus pathogenicity. Furthermore, silencing of endogenous CEACAM1 resulted in reduced levels of proinflammatory cytokine/chemokine production, as well as reduced levels of virus replication following H5N1 infection. Our study provides evidence for the involvement of CEACAM1 in a clinically relevant model of H5N1 infection and may assist in the development of host-oriented antiviral strategies.

Text: Influenza viruses cause acute and highly contagious seasonal respiratory disease in all age groups. Between 3-5 million cases of severe influenza-related illness and over 250 000 deaths are reported every year. In addition to constant seasonal outbreaks, highly pathogenic avian influenza (HPAI) strains, such as H5N1, remain an ongoing pandemic threat with recent WHO figures showing 454 confirmed laboratory infections and a mortality rate of 53%. It is important to note that humans have very little pre-existing immunity towards avian influenza virus strains. Moreover, there is no commercially available human H5N1 vaccine. Given the potential for H5N1 viruses to trigger a pandemic 1,2 , there is an urgent need to develop novel therapeutic interventions to combat known deficiencies in our ability to control outbreaks. Current seasonal influenza virus prophylactic and therapeutic strategies involve the use of vaccination and antivirals. Vaccine efficacy is highly variable as evidenced by a particularly severe 2017/18 epidemic, and frequent re-formulation of the vaccine is required to combat ongoing mutations in the influenza virus genome. In addition, antiviral resistance has been reported for many circulating strains, including the avian influenza H7N9 virus that emerged in 2013 3, 4 . Influenza A viruses have also been shown to target and hijack multiple host cellular pathways to promote survival and replication 5, 6 . As such, there is increasing evidence to suggest that targeting host pathways will influence virus replication, inflammation, immunity and pathology 5, 7 . Alternative intervention strategies based on modulation of the host response could be used to supplement the current prophylactic and therapeutic protocols.

While the impact of influenza virus infection has been relatively well studied in animal models 8, 9 , human cellular responses are poorly defined due to the lack of available human autopsy material, especially from HPAI virus-infected patients. In the present study, we characterized influenza virus infection of primary human alveolar epithelial type II (ATII) cells isolated from normal human lung tissue donated by patients undergoing lung resection. ATII cells are a physiologically relevant infection model as they are a main target for influenza A viruses when entering the respiratory tract 10 . Human host gene expression following HPAI H5N1 virus (A/Chicken/ Vietnam/0008/04) infection of primary ATII cells was analyzed using Illumina HiSeq deep sequencing. In order to gain a better understanding of the mechanisms underlying modulation of host immunity in an anti-inflammatory environment, we also analyzed changes in gene expression following HPAI H5N1 infection in the presence of the reactive oxygen species (ROS) inhibitor, apocynin, a compound known to interfere with NADPH oxidase subunit assembly 5, 6 .

The HiSeq analysis described herein has focused on differentially regulated genes following H5N1 infection. Several criteria were considered when choosing a "hit" for further study. These included: (1) Novelty; has this gene been studied before in the context of influenza virus infection/pathogenesis? (2) Immunoregulation; does this gene have a regulatory role in host immune responses so that it has the potential to be manipulated to improve immunity? (3) Therapeutic reagents; are there any existing commercially available therapeutic reagents, such as specific inhibitors or inhibitory antibodies that can be utilized for in vitro and in vivo study in order to optimize therapeutic strategies? (4) Animal models; is there a knock-out mouse model available for in vivo influenza infection studies? Based on these criteria, carcinoembryonic-antigen (CEA)-related cell adhesion molecule 1 (CEACAM1) was chosen as a key gene of interest. CEACAM1 (also known as BGP or CD66) is expressed on epithelial and endothelial cells 11, as well as B cells, T cells, neutrophils, NK cells, macrophages and dendritic cells (DCs) [12] [13] [14]. Human CEACAM1 has been shown to act as a receptor for several human bacterial and fungal pathogens, including *Haemophilus influenza*, *Escherichia coli*, *Salmonella typhi* and *Candida albicans*, but has not as yet been implicated in virus entry [15] [16] [17]. There is however emerging evidence to suggest that CEACAM1 is involved in host immunity as enhanced expression in lymphocytes was detected in pregnant women infected with cytomegalovirus 18 and in cervical tissue isolated from patients with papillomavirus infection 19.

Eleven CEACAM1 splice variants have been reported in humans 20. CEACAM1 isoforms (Uniprot P13688-1 to -11) can differ in the number of immunoglobulin-like domains present, in the presence or absence of a transmembrane domain and/or the length of their cytoplasmic tail (i.e. L, long or S, short). The full-length human CEACAM1 protein (CEACAM1-4L) consists of four extracellular domains (one extracellular immunoglobulin variable-region-like (IgV-like) domain and three immunoglobulin constant region 2-like (IgC2-like) domains), a transmembrane domain, and a long (L) cytoplasmic tail. The long cytoplasmic tail contains two immunoreceptor tyrosine-based inhibitory motifs (ITIMs) that are absent in the short form 20. The most common isoforms expressed by human immune cells are CEACAM1-4L and CEACAM1-3L 21. CEACAM1 interacts homophilically with itself 22 or heterophilically with CEACAM5 (a related CEACAM family member) 23. The dimeric state allows recruitment of signaling molecules such as SRC-family kinases, including the tyrosine phosphatase SRC homology 2 (SH2)-domain containing protein tyrosine phosphatase 1 (SHP1) and SHP2 members to phosphorylate ITIMs 24. As such, the presence or absence of ITIMs in CEACAM1 isoforms influences signaling properties and

downstream cellular function. CEACAM1 homophilic or heterophilic interactions and ITIM phosphorylation are critical for many biological processes, including regulation of lymphocyte function, immunosurveillance, cell growth and differentiation 25, 26 and neutrophil activation and adhesion to target cells during inflammatory responses 27 . It should be noted that CEACAM1 expression has been modulated in vivo using an anti-CEACAM1 antibody (MRG1) to inhibit CEACAM1-positive melanoma xenograft growth in SCID/NOD mice 28 . MRG1 blocked CEACAM1 homophilic interactions that inhibit T cell effector function, enhancing the killing of CEACAM1+ melanoma cells by T cells 28 . This highlights a potential intervention pathway that can be exploited in other disease processes, including virus infection. In addition, Ceacam1-knockout mice are available for further in vivo infection studies.

Our results show that CEACAM1 mRNA and protein expression levels were highly elevated following HPAI H5N1 infection. Furthermore, small interfering RNA (siRNA)-mediated inhibition of CEACAM1 reduced inflammatory cytokine and chemokine production, and more importantly, inhibited H5N1 virus replication in primary human AII cells and in the continuous human type II respiratory epithelial A549 cell line. Taken together, these observations suggest that CEACAM1 is an attractive candidate for modulating influenza-specific immunity. In summary, our study has identified a novel target that may influence HPAI H5N1 immunity and serves to highlight the importance of manipulating host responses as a way of improving disease outcomes in the context of virus infection.

Three experimental groups were included in the HiSeq analysis of H5N1 infection in the presence or absence of the ROS inhibitor, apocynin: (i) uninfected cells treated with 1% DMSO (vehicle control) (ND), (ii) H5N1-infected cells treated with 1% DMSO (HD) and (iii) H5N1-infected cells treated with 1 mM apocynin dissolved in DMSO (HA). These three groups were assessed using pairwise comparisons: ND vs. HD, ND vs. HA, and HD vs. HA. H5N1 infection and apocynin treatment induce differential expression of host genes. AII cells isolated from human patients 29, 30 were infected with H5N1 on the apical side at a multiplicity of infection (MOI) of 2 for 24 hours and RNA extracted. HiSeq was performed on samples and reads mapped to the human genome where they were then assembled into transcriptomes for differential expression analysis. A total of 13,649 genes were identified with FPKM (fragments per kilobase of exon per million fragments mapped) > 1 in at least one of the three experimental groups. A total of 623 genes were significantly upregulated and 239 genes were significantly downregulated (q value < 0.05, ≥ 2 -fold change) following H5N1 infection (ND vs. HD) (Fig. 1A ; Table S1). HPAI H5N1 infection of AII cells activated

an antiviral state as evidenced by the upregulation of numerous interferon-induced genes, genes associated with pathogen defense, cell proliferation, apoptosis, and metabolism (Table 1; Table S2). In addition, Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway mapping showed that many of the upregulated genes in the HD group were mapped to TNF signaling (hsa04668), Toll-like receptor signaling (hsa04620), cytokine-cytokine receptor interaction (hsa04060) and RIG-I-like receptor signaling (hsa04622) (In the H5N1-infected and apocynin-treated (HA) group, a large number of genes were also significantly upregulated (509 genes) or downregulated (782 genes) (Fig. 1B ; Table S1) relative to the ND control group. Whilst a subset of genes was differentially expressed in both the HD and HA groups, either being upregulated (247 genes, Fig. 1D) or downregulated (146 genes, Fig. 1E), a majority of genes did not in fact overlap between the HD and HA groups (Fig. 1D , E). This suggests that apocynin treatment can affect gene expression independent of H5N1 infection. Gene Ontology (GO) enrichment analysis of genes upregulated by apocynin showed the involvement of the type I interferon signaling pathway (GO:0060337), the defense response to virus (GO:0009615), negative regulation of viral processes (GO:48525) and the response to stress (GO:0006950) (Table S2 , "ND vs. HA Up"). Genes downregulated by apocynin include those that are involved in cell adhesion (GO:0007155), regulation of cell migration (GO:0030334), regulation of cell proliferation (GO:0042127), signal transduction (GO:0007165) and oxidation-reduction processes (GO:0055114) (Table S2 , "ND vs. HA Down").

A total of 623 genes were upregulated following H5N1 infection ("ND vs. HD Up", Fig. 1F). By overlapping the two lists of genes from "ND vs. HD Up" and "HD vs. HA Down", 245 genes were shown to be downregulated in the presence of apocynin (Fig. 1F). By overlapping three lists of genes from "ND vs. HD Up", "HD vs. HA Down" and "ND vs. HA Up", 55 genes out of the 245 genes (190 plus 55 genes) were present in all three lists (Fig. 1G) , indicating that these 55 genes were significantly inhibited by apocynin but to a level that was still significantly higher than that in uninfected cells. The 55 genes include those involved in influenza A immunity (hsa05164; DDX58, IFIH1, IFNB1, MYD88, PML, STAT2), Jak-STAT signaling (hsa04630; IFNB1, IL15RA, IL22RA1, STAT2), RIG-I-like receptor signaling (hsa04622; DDX58, IFIH1, IFNB1) and Antigen processing and presentation (hsa04612; TAP2, TAP1, HLA-DOB) (Tables S3 and S4) . Therefore, critical immune responses induced following H5N1 infection were not dampened following apocynin treatment. The remaining 190 of 245 genes were not present in the "ND vs. HA Up" list, suggesting that those genes were significantly inhibited by apocynin to a level that was similar to uninfected control cells (Fig. 1G). The 190 genes include those involved in TNF signaling (hsa04668; CASP10, CCL2, CCL5, CFLAR, CXCL5, END1, IL6, TRAF1, VEGFC), cytokine-cytokine receptor

interaction (hsa04060; VEGFC, IL6, CCL2, CXCL5, CXCL16, IL2RG, CD40, CCL5, CCL7, IL1A), NF-kappa B signaling pathway (hsa04064: TRAF1, CFLAR, CARD11, TNFSF13B, TICAM1, CD40) and PI3K-Akt signaling (hsa04151; CCND1, GNB4, IL2RG, IL6, ITGA2, JAK2, LAMA1, MYC, IPK3AP1, TLR2, VEGFC) (Tables S3 and S4). This is consistent with the role of apocynin in reducing inflammation 31 . By overlapping the three lists of genes from "ND vs. HD Up", "HD vs. HA Down" and "ND vs. HA Down", 11 genes were found in all three comparisons (Fig. 1H). This suggests that these 11 genes are upregulated following H5N1 infection and are significantly reduced by apocynin treatment to a level lower than that observed in uninfected control cells (Fig. 1H). Among these were inflammatory cytokines/chemokines genes, including CXCL5, IL1A, AXL (a member of the TAM receptor family of receptor tyrosine kinases) and TMEM173/STING (Stimulator of IFN Genes) (Table S4) .

Our previous study demonstrated that H5N1 infection of A549 cells in the presence of apocynin enhanced expression of negative regulators of cytokine signaling (SOCS), SOCS1 and SOCS3 6 . This, in turn, resulted in a reduction of H5N1-stimulated cytokine and chemokine production (IL6, IFNB1, CXCL10 and CCL5 in A549 cells), which was not attributed to lower virus replication as virus titers were not affected by apocynin treatment 6 . We performed a qRT-PCR analysis on the same RNA samples submitted for HiSeq analysis to validate HiSeq results. IL6 (Fig. 2A) , IFNB1 (Fig. 2B) , CXCL10 (Fig. 2C) , and CCL5 (Fig. 2D) gene expression was significantly elevated in A549 cells following infection and was reduced by the addition of apocynin (except for IFNB1). Consistent with previous findings in A549 cells 6 , H5N1 infection alone induced the expression of SOCS1 as shown by HiSeq and qRT-PCR analysis (Fig. 2E). Apocynin treatment further increased SOCS1 mRNA expression (Fig. 2E). Although HiSeq analysis did not detect a statistically significant increase of SOCS1 following apocynin treatment, the Log2 fold-changes in SOCS1 gene expression were similar between the HD and HA groups (4.8-fold vs 4.0-fold) (Fig. 2E). HiSeq analysis of SOCS3 transcription showed significant increase following H5N1 infection and apocynin treatment (Fig. 2F). qRT-PCR analysis showed that although SOCS3 mRNA was only slightly increased following H5N1 infection, it was further significantly upregulated in the presence of apocynin (Table 2). Representatives of over-represented KEGG pathways with a maximum P-value of 0.05 and the number of genes contributing to each pathway that is significantly upregulated following H5N1 infection ("ND vs. HD Up"). The full list of KEGG pathways is presented in Table S3 . of apocynin (Fig. 2F) . Therefore, apocynin also contributes to the reduction of H5N1-stimulated cytokine and chemokine production in A549 cells. Apocynin, a compound that inhibits production of ROS, has been shown to influence influenza-specific responses in vitro 6 and in vivo 5 . Although virus

titers are not affected by apocynin treatment in vitro⁶, some anti-viral activity is observed in vivo when mice have been infected with a low pathogenic A/HongKong/X31 H3N2 virus⁶. HiSeq analysis of HPAI H5N1 virus gene transcription showed that although there was a trend for increased influenza virus gene expression following apocynin treatment, only influenza non-structural (NS) gene expression was significantly increased (Fig. 2G). The reduced cytokine and chemokine production in H5N1-infected ATII cells (Fig. 2A-F) is unlikely to be associated with lower virus replication.

GO enrichment analysis was performed on genes that were significantly upregulated following HPAI H5N1 infection in ATII cells in the presence or absence of apocynin to identify over-presented GO terms. Many of the H5N1-upregulated genes were broadly involved in defense response (GO:0006952), response to external biotic stimulus (GO:0043207), immune system processes (GO:0002376), cytokine-mediated signaling pathway (GO:0019221) and type I interferon signaling pathway (GO:0060337) (Table 1; Table S2). In addition, many of the H5N1-upregulated genes mapped to metabolic pathways (hsa01100), cytokine-cytokine receptor interaction (hsa04060), Influenza A (hsa05164), TNF signaling (hsa04668) or Jak-STAT signaling (hsa04630) (Table S3). However, not all the H5N1-upregulated genes in these pathways were inhibited by apocynin treatment as mentioned above (Fig. 1F; Table S3). Fold-changes following qRT-PCR analysis were calculated using $2^{-\Delta\Delta C_t}$ method (right Y axis) normalized to β -actin and compared with the ND group. Data from HiSeq was calculated as Log₂ fold-change (left Y axis) compared with the ND group. IFNB1 transcription was not detected in ND, therefore HiSeq IFNB1 data from HD and HA groups was expressed as FPKM. * $p < 0.05$ and ** $p < 0.01$, *** $p < 0.001$ compared with ND; # $p < 0.05$, ## $p < 0.01$, compared with HD. (G) HiSeq analysis of H5N1 influenza virus gene expression profiles with or without apocynin treatment in primary human ATII cells. # $p < 0.05$, compared with HD. Upregulation of the cell adhesion molecule CEACAM1 in H5N1-infected ATII cells. The cell adhesion molecule CEACAM1 has been shown to be critical for the regulation of immune responses during infection, inflammation and cancer²⁰. The CEACAM1 transcript was significantly upregulated following H5N1 infection (Fig. 3A). In contrast, a related member of the CEACAM family, CEACAM5, was not affected by H5N1 infection (Fig. 3B). It is also worth noting that more reads were obtained for CEACAM5 (>1000 FPKM) (Fig. 3B) than CEACAM1 (~7 FPKM) (Fig. 3A) in uninfected ATII cells, which is consistent with their normal expression patterns in human lung tissue³². Therefore, although CEACAM1 forms heterodimers with CEACAM5²³, the higher basal expression of CEACAM5 in ATII cells may explain why its expression was not enhanced by H5N1 infection. Endogenous CEACAM1 protein expression was also analyzed in uninfected or influenza virus-infected A549 (Fig. 3C) and

ATII cells (Fig. 3D). CEACAM1 protein expression was slightly, but not significantly, increased in A549 cells infected with A/Puerto Rico/8/1934 H1N1 (PR8) virus for 24 or 48 hours when compared to uninfected cells (Fig. 3C). No significant difference in CEACAM1 protein levels were observed at various MOIs (2, 5 or 10) or between the 24 and 48 hpi timepoints (Fig. 3C).

After examining CEACAM1 protein expression following infection with PR8 virus in A549 cells, CEACAM1 protein expression was then examined in primary human ATII cells infected with HPAI H5N1 and compared to PR8 virus infection (Fig. 3D). ATII cells were infected with PR8 virus at a MOI of 2, a dose that induced upregulation of cytokines and influenza Matrix (M) gene analyzed by qRT-PCR (data not shown). Lower MOIs of 0.5, 1 and 2 of HPAI H5N1 were tested due to the strong cytopathogenic effect H5N1 causes at higher MOIs. Endogenous CEACAM1 protein levels were significantly and similarly elevated in H5N1-infected ATII cells at the three MOIs tested. CEACAM1 protein expression in ATII cells infected with H5N1 at MOIs of 0.5 were higher at 48 hpi than those observed at 24 hpi (Fig. 3D). HPAI H5N1 virus infection at MOIs of 0.5, 1 and 2 stimulated higher endogenous levels of CEACAM1 protein expression when compared to PR8 virus infection at a MOI of 2 at the corresponding time point (a maximum ~9-fold increase induced by H5N1 at MOIs of 0.5 and 1 at 48 hpi when compared to PR8 at MOI of 2), suggesting a possible role for CEACAM1 in influenza virus pathogenicity (Fig. 3D).

In order to understand the role of CEACAM1 in influenza pathogenesis, A549 and ATII cells were transfected with siCEACAM1 to knockdown endogenous CEACAM1 protein expression. ATII and A549 cells were transfected with siCEACAM1 or siNeg negative control. The expression of four main CEACAM1 variants, CEACAM1-4L, -4S, -3L and -3S, and CEACAM1 protein were analyzed using SYBR Green qRT-PCR and Western blotting, respectively. SYBR Green qRT-PCR analysis showed that ATII cells transfected with 15 pmol of siCEACAM1 significantly reduced the expression of CEACAM1-4L and -4S when compared to siNeg control, while the expression of CEACAM1-3L and -3S was not altered (Fig. 4A). CEACAM1 protein expression was reduced by approximately 50% in both ATII and A549 cells following siCEACAM1 transfection when compared with siNeg-transfected cells (Fig. 4B). Increasing doses of siCEACAM1 (10, 15 and 20 pmol) did not further downregulate CEACAM1 protein expression in A549 cells (Fig. 4B). As such, 15 pmol of siCEACAM1 was chosen for subsequent knockdown studies in both ATII and A549 cells. It is important to note that the anti-CEACAM1 antibody only detects L isoforms based on

epitope information provided by Abcam. Therefore, observed reductions in CEACAM1 protein expression can be attributed mainly to the abolishment of CEACAM1-4L.

The functional consequences of CEACAM1 knockdown were then examined in A549 and ATII cells following H5N1 infection. IL6, IFNB1, CXCL10, CCL5 and TNF production was analyzed in H5N1-infected ATII and A549 cells using qRT-PCR. ATII (Fig. 5A) and A549 cells (Fig. 5B) transfected with siCEACAM1 showed significantly lower expression of IL6, CXCL10 and CCL5 when compared with siNeg-transfected cells. However, the expression of the anti-viral cytokine, IFNB1, was not affected in both cell types. In addition, TNF expression, which can be induced by type I IFNs 33, was significantly lower in siCEACAM1-transfected A549 cells (Fig. 5B), but was not affected in siCEACAM1-transfected ATII cells (Fig. 5A). Hypercytokinemia or "cytokine storm" in H5N1 and H7N9 virus-infected patients is thought to contribute to inflammatory tissue damage 34, 35. Downregulation of CEACAM1 in the context of severe viral infection may reduce inflammation caused by H5N1 infection without dampening the antiviral response. Furthermore, virus replication was significantly reduced by 5.2-fold in ATII (Figs. 5C) and 4.8-fold in A549 cells (Fig. 5D) transfected with siCEACAM1 when compared with siNeg-transfected cells. Virus titers in siNeg-transfected control cells were not significantly different from those observed in mock-transfected control cells (Fig. 5C,D).

Influenza viruses utilize host cellular machinery to manipulate normal cell processes in order to promote replication and evade host immune responses. Studies in the field are increasingly focused on understanding and modifying key host factors in order to ameliorate disease. Examples include modulation of ROS to reduce inflammation 5 and inhibition of NFkB and mitogenic Raf/MEK/ERK kinase cascade activation to suppress viral replication 36, 37. These host targeting strategies will offer an alternative to current interventions that are focused on targeting the virus. In the present study, we analyzed human host gene expression profiles following HPAI H5N1 infection and treatment with the antioxidant, apocynin. As expected, genes that were significantly upregulated following H5N1 infection were involved in biological processes, including cytokine signaling, immunity and apoptosis. In addition, H5N1-upregulated genes were also involved in regulation of protein phosphorylation, cellular metabolism and cell proliferation, which are thought to be exploited by viruses for replication 38. Apocynin treatment had both anti-viral (Tables S2-S4) 5 and pro-viral impact (Fig. 2G), which is not surprising as ROS are potent microbicidal agents, as well as important immune signaling molecules at different concentrations 39. In our hands, apocynin treatment reduced H5N1-induced

inflammation, but also impacted the cellular defense response, cytokine production and cytokine-mediated signaling. Importantly, critical antiviral responses were not compromised, i.e. expression of pattern recognition receptors (e.g. DDX58 (RIG-I), TLRs, IFIH1 (MDA5)) was not downregulated (Table S1). Given the significant interference of influenza viruses on host immunity, we focused our attention on key regulators of the immune response. Through HiSeq analysis, we identified the cell adhesion molecule CEACAM1 as a critical regulator of immunity. Knockdown of endogenous CEACAM1 inhibited H5N1 virus replication and reduced H5N1-stimulated inflammatory cytokine/chemokine production. H5N1 infection resulted in significant upregulation of a number of inflammatory cytokines/chemokines genes, including AXL and STING, which were significantly reduced by apocynin treatment to a level lower than that observed in uninfected cells (Table S4) . It has been previously demonstrated that anti-AXL antibody treatment of PR8-infected mice significantly reduced lung inflammation and virus titers 40 . STING has been shown to be important for promoting anti-viral responses, as STING-knockout THP-1 cells produce less type I IFN following influenza A virus infection 41 . Reduction of STING gene expression or other anti-viral factors (e.g. IFNB1, MX1, ISG15; Table S1) by apocynin, may in part, explain the slight increase in influenza gene transcription following apocynin treatment (Fig. 2G) . These results also suggest that apocynin treatment may reduce H5N1-induced inflammation and apoptosis. Indeed, the anti-inflammatory and anti-apoptotic effects of apocynin have been shown previously in a number of disease models, including diabetes mellitus 42 , myocardial infarction 43 , neuroinflammation 44 and influenza virus infection 6 .

Recognition of intracellular viral RNA by pattern recognition receptors (PRRs) triggers the release of pro-inflammatory cytokines/chemokines that recruit innate immune cells, such as neutrophils and NK cells, to the site of infection to assist in viral clearance 45 . Neutrophils exert their cytotoxic function by first attaching to influenza-infected epithelial cells via adhesion molecules, such as CEACAM1 46 . Moreover, studies have indicated that influenza virus infection promotes neutrophil apoptosis 47 , delaying virus elimination 48 . Phosphorylation of CEACAM1 ITIM motifs and activation of caspase-3 is critical for mediating anti-apoptotic events and for promoting survival of neutrophils 27 . This suggests that CEACAM1-mediated anti-apoptotic events may be important for the resolution of influenza virus infection in vivo, which can be further investigated through infection studies with Ceacam1-knockout mice.

NK cells play a critical role in innate defense against influenza viruses by recognizing and killing infected cells. Influenza viruses, however, employ several strategies to escape NK effector functions, including modification of influenza hemagglutinin (HA) glycosylation to avoid NK activating receptor binding 49 . Homo- or heterophilic CEACAM1 interactions have been shown to inhibit NK-killing 25, 26 , and are thought to contribute to tumor cell immune evasion 50 . Given these findings, one could suggest the possibility that upregulation of CEACAM1 (to inhibit NK activity) may be a novel and uncharacterized immune evasion strategy employed by influenza viruses. Our laboratory is now investigating the role of CEACAM1 in NK cell function. Small-molecule inhibitors of protein kinases or protein phosphatases (e.g. inhibitors for Src, JAK, SHP2) have been developed as therapies for cancer, inflammation, immune and metabolic diseases 51 . Modulation of CEACAM1 phosphorylation, dimerization and the downstream function with small-molecule inhibitors may assist in dissecting the contribution of CEACAM1 to NK cell activity.

The molecular mechanism of CEACAM1 action following infection has also been explored in A549 cells using PR8 virus 52 . Vitenshtein et al. demonstrated that CEACAM1 was upregulated following recognition of viral RNA by RIG-I, and that this upregulation was interferon regulatory factor 3 (IRF3)-dependent. In addition, phosphorylation of CEACAM1 by SHP2 inhibited viral replication by reducing phosphorylation of mammalian target of rapamycin (mTOR) to suppress global cellular protein production. In the present study, we used a more physiologically relevant infection model, primary human ATII cells, to study the role of CEACAM1. Further studies will be required to investigate/confirm the molecular mechanisms of CEACAM1 upregulation following influenza virus infection, especially in vivo. As upregulation of CEACAM1 has been observed in other virus infections, such as cytomegalovirus 18 and papillomavirus 19 , it will be important to determine whether a common mechanism of action can be attributed to CEACAM1 in order to determine its functional significance. If this can be established, CEACAM1 could be used as a target for the development of a pan-antiviral agent.

In summary, molecules on the cell surface such as CEACAM1 are particularly attractive candidates for therapeutic development, as drugs do not need to cross the cell membrane in order to be effective. Targeting of host-encoded genes in combination with current antivirals and vaccines may be a way of reducing morbidity and mortality associated with influenza virus infection. Our study clearly demonstrates that increased CEACAM1 expression is observed in primary human ATII cells infected with HPAI H5N1 influenza virus. Importantly, knockdown of CEACAM1 expression resulted in a reduction in influenza virus

replication and suggests targeting of this molecule may assist in improving disease outcomes.

Isolation and culture of primary human ATII cells. Human non-tumor lung tissue samples were donated by anonymous patients undergoing lung resection at University Hospital, Geelong, Australia. The research protocols and human ethics were approved by the Human Ethics Committees of Deakin University, Barwon Health and the Commonwealth Scientific and Industrial Research Organisation (CSIRO). Informed consent was obtained from all tissue donors. All research was performed in accordance with the guidelines stated in the National Statement on Ethical Conduct in Human Research (2007) . The sampling of normal lung tissue was confirmed by the Victorian Cancer Biobank, Australia. Lung specimens were preserved in Hartmann's solution (Baxter) for 4-8 hours or O/N at 4 °C to maintain cellular integrity and viability before cells are isolated. Human alveolar epithelial type II (ATII) cells were isolated and cultured using a previously described method 30, 53 with minor modifications. Briefly, lung tissue with visible bronchi was removed and perfused with abundant PBS and submerged in 0.5% Trypsin-EDTA (Gibco) twice for 15 min at 37 °C. The partially digested tissue was sliced into sections and further digested in Hank's Balanced Salt Solution (HBSS) containing elastase (12.9 units/mL; Roche Diagnostics) and DNase I (0.5 mg/mL; Roche Diagnostics) for 60 min at 37 °C. Single cell suspensions were obtained by filtration through a 40 µm cell strainer and cells (including macrophages and fibroblasts) were allowed to attach to tissue-culture treated Petri dishes in a 1:1 mixture of DMEM/F12 medium (Gibco) and small airway growth medium (SAGM) medium (Lonza) containing 5% fetal calf serum (FCS) and 0.5 mg/mL DNase I for 2 hours at 37 °C. Non-adherent cells, including ATII cells, were collected and subjected to centrifugation at 300 g for 20 min on a discontinuous Percoll density gradient (1.089 and 1.040 g/mL). Purified ATII cells from the interface of two density gradients was collected, washed in HBSS, and re-suspended in SAGM medium supplemented with 1% charcoal-filtered FCS (Gibco) and 100 units/mL penicillin and 100 µg/mL streptomycin (Gibco). ATII cells were plated on polyester Transwell inserts (0.4 µm pore; Corning) coated with type IV human placenta collagen (0.05 mg/mL; Sigma) at 300,000 cells/cm² and cultured under liquid-covered conditions in a humidified incubator (5% CO₂ , 37 °C). Growth medium was changed every 48 hours. These culture conditions suppressed fibroblasts expansion within the freshly isolated ATII cells and encouraged ATII cells to form confluent monolayers with a typical large and somewhat square morphology 54

Cell culture and media. A549 carcinomic human alveolar basal epithelial type II-like cells and Madin-Darby canine kidney (MDCK) cells were provided by the tissue culture facility of Australian Animal Health Laboratory (AAHL), CSIRO. A549 and MDCK cells were maintained in Ham's F12K medium

(GIBCO) and RPMI-1640 medium (Invitrogen), respectively, supplemented with 10% FCS, 100 U/mL penicillin and 100 µg/mL streptomycin (GIBCO) and maintained at 37 °C, 5% CO₂.

Virus and viral infection. HPAI A/chicken/Vietnam/0008/2004 H5N1 (H5N1) was obtained from AAHL, CSIRO. Viral stocks of A/Puerto Rico/8/1934 H1N1 (PR8) were obtained from the University of Melbourne. Virus stocks were prepared using standard inoculation of 10-day-old embryonated eggs. A single stock of virus was prepared for use in all assays. All H5N1 experiments were performed within biosafety level 3 laboratories (BSL3) at AAHL, CSIRO.

Cells were infected with influenza A viruses as previously described 6, 29. Briefly, culture media was removed and cells were washed with warm PBS three times followed by inoculation with virus for 1 hour. Virus was then removed and cells were washed with warm PBS three times, and incubated in the appropriate fresh serum-free culture media containing 0.3% BSA at 37 °C. Uninfected and infected cells were processed identically. For HiSeq analysis, ATII cells from three donors were infected on the apical side with H5N1 at a MOI of 2 for 24 hours in serum-free SAGM medium supplemented with 0.3% bovine serum albumin (BSA) containing 1 mM apocynin dissolved in DMSO or 1% DMSO vehicle control. Uninfected ATII cells incubated in media containing 1% DMSO were used as a negative control. For other subsequent virus infection studies, ATII cells from a different set of three donors (different from those used in HiSeq analysis) or A549 cells from at least three different passages were infected with influenza A viruses at various MOIs as indicated in the text. For H5N1 studies following transfection with siRNA, the infectious dose was optimized to a MOI of 0.01, a dose at which significantly higher CEACAM1 protein expression was induced with minimal cell death at 24 hpi. For PR8 infection studies, a final concentration of 0.5 µg/mL L-1-Tosylamide-2-phenylethyl chloromethyl ketone (TPCK)-treated trypsin (Worthington) was included in media post-inoculation to assist replication. Virus titers were determined using standard plaque assays in MDCK cells as previously described 55.

RNA extraction, quality control (QC) and HiSeq analysis. ATII cells from three donors were used for HiSeq analysis. Total RNA was extracted from cells using a RNeasy Mini kit (Qiagen). Influenza-infected cells were washed with PBS three times and cells lysed with RLT buffer supplemented with β-mercaptoethanol (10 µL/mL; Gibco). Cell lysates were

homogenized with QIAshredder columns followed by on-column DNA digestion with the RNase-Free DNase Set (Qiagen), and RNA extracted according to manufacturer's instructions. Initial QC was conducted to ensure that the quantity and quality of RNA samples for HiSeq analysis met the following criteria; 1) RNA samples had OD_{260/280} ratios between 1.8 and 2.0 as measured with NanoDrop™ Spectrophotometer (Thermo Scientific); 2) Sample concentrations were at a minimum of 100 ng/μl; 3) RNA was analyzed by agarose gel electrophoresis. RNA integrity and quality were validated by the presence of sharp clear bands of 28S and 18S ribosomal RNA, with a 28S:18S ratio of 2:1, along with the absence of genomic DNA and degraded RNA. As part of the initial QC and as an indication of consistent H5N1 infection, parallel quantitative real-time reverse transcriptase PCR (qRT-PCR) using the same RNA samples used for HiSeq analysis was performed in duplicate as previously described 6 to measure mRNA expression of IL6, IFNB1, CXCL10, CCL5, TNF, SOCS1 and SOCS3, all of which are known to be upregulated following HPAI H5N1 infection of A549 cells 6 Sequencing analysis and annotation. After confirming checksums and assessing raw data quality of the FASTQ files with FASTQC, RNA-Seq reads were processed according to standard Tuxedo pipeline protocols 56 , using the annotated human genome (GRCh37, downloaded from Illumina iGenomes) as a reference. Briefly, raw reads for each sample were mapped to the human genome using TopHat2, sorted and converted to SAM format using Samtools and then assembled into transcriptomes using Cufflinks. Cuffmerge was used to combine transcript annotations from individual samples into a single reference transcriptome, and Cuffquant was used to obtain per-sample read counts. Cuffdiff was then used to conduct differential expression analysis. All programs were run using recommended parameters. It is important to note that the reference gtf file provided to cuffmerge was first edited using a custom python script to exclude lines containing features other than exon/cds, and contigs other than chromosomes 1-22, X, Y. GO term and KEGG enrichment. Official gene IDs for transcripts that were differentially modulated following HPAI H5N1 infection with or without apocynin treatment were compiled into six target lists from pairwise comparisons ("ND vs. HD Up", "ND vs. HD Down", "ND vs. HA Up", "ND vs. HA Down", "HD vs. HA Up", "HD vs. HA Down"). Statistically significant differentially expressed transcripts were defined as having ≥2-fold change with a Benjamini-Hochberg adjusted P value < 0.01. A background list of genes was compiled by retrieving all gene IDs identified from the present HiSeq analysis with FPKM > 1. Biological process GO enrichment was performed using Gorilla, comparing unranked background and target lists 57 . Redundant GO terms were removed using REVIGO 58 . Target lists were also subjected to KEGG pathway analysis using a basic KEGG pathway mapper 59 and DAVID Bioinformatics Resources Functional Annotation Tool 60,61 .

Quantitative real-time reverse transcriptase polymerase chain reaction (qRT-PCR). mRNA concentrations of genes of interest were assessed and analyzed using qRT-PCR performed in duplicate as previously described ⁶ . Briefly, after total RNA extraction from influenza-infected cells, cDNA was prepared using SuperScript™ III First-Strand Synthesis SuperMix (Invitrogen). Gene expression of various cytokines was assessed using TaqMan Gene Expression Assays (Applied Biosystems) with commercial TaqMan primers and probes, with the exception of the influenza Matrix (M) gene (forward primer 5'-CTTCTAACCGAGGTCGAAACGTA-3'; reverse primer 5'-GGTGACAGGATTGGTCTTGTCTTTA-3'; probe 5'-FAM-TCAGGCCCCCTCAAAGCCGAG-NFQ-3') ⁶² . Specific primers ⁶³ (Table S5) were designed to estimate the expression of CEACAM1-4L, -4S, -3L and -3S in A/TII and A549 cells using iTaq Universal SYBR Green Supermix (Bio-Rad) according to manufacturer's instruction. The absence of nonspecific amplification was confirmed by agarose gel electrophoresis of qRT-PCR products (15 µL) (data not shown). Gene expression was normalized to β-actin mRNA using the 2^{-ΔΔCT} method where expression levels were determined relative to uninfected cell controls. All assays were performed in duplicate using an Applied Biosystems® StepOnePlus™ Real-Time PCR System. Western blot analysis. Protein expression of CEACAM1 was determined using Western blot analysis as previously described ⁶ . Protein concentrations in cell lysates were determined using EZQ® Protein Quantitation Kit (Molecular Probes™, Invitrogen). Equal amounts of protein were loaded on NuPAGE 4-12% Bis-Tris gels (Invitrogen), resolved by SDS/PAGE and transferred to PVDF membranes (Bio-Rad). Membranes were probed with rabbit anti-human CEACAM1 monoclonal antibody EPR4049 (ab108397, Abcam) followed by goat anti-rabbit HRP-conjugated secondary antibody (Invitrogen). Proteins were visualized by incubating membranes with Pierce enhanced chemiluminescence (ECL) Plus Western Blotting Substrate (Thermo Scientific) followed by detection on a Bio-Rad ChemiDoc™ MP Imaging System or on Amersham™ Hyperfilm™ ECL (GE Healthcare).

To use β-actin as a loading control, the same membrane was stripped in stripping buffer (1.5% (w/v) glycine, 0.1% (w/v) SDS, 1% (v/v) Tween-20, pH 2.2) and re-probed with a HRP-conjugated rabbit anti-β-actin monoclonal antibody (Cell Signaling). In some cases, two SDS/PAGE were performed simultaneously with equal amounts of protein loaded onto each gel for analysis of CEACAM1 and β-actin protein expression in each sample, respectively. Protein band density was quantified using Fiji software (version 1.49J10) ⁶⁴ . CEACAM1 protein band density was normalized against that of β-actin and expressed as fold changes compared to controls.

Knockdown of endogenous CEACAM1. A772 and A549 cells were grown to 80% confluency in 6-well plates then transfected with small interfering RNA (siRNA) targeting the human CEACAM1 gene (siCEACAM1; s1976, Silencer[®] Select Pre-designed siRNA, Ambion[®]) or siRNA control (siNeg; Silencer[®] Select Negative Control No. 1 siRNA, Ambion[®]) using Lipofetamine 3000 (ThermoFisher Scientific) according to manufacturer's instructions. Transfection and silencing efficiency were evaluated after 48 hours by Western blot analysis of CEACAM1 protein expression and by qRT-PCR analysis of CEACAM1 variants. In parallel experiments, virus replication and cytokine/chemokine production was analyzed in siCEACAM1-or siNeg-transfected cells infected with H5N1 virus (MOI = 0.01) at 24 hpi. Statistical analysis. Differences between two experimental groups were evaluated using a Student's unpaired, two-tailed t test. Fold-change differences of mRNA expression (qRT-PCR) between three experimental groups was evaluated using one-way analysis of variance (ANOVA) followed by a Bonferroni multiple-comparison test. Differences were considered significant with a p value of <0.05. The data are shown as means ± standard error of the mean (SEM) from three or four individual experiments. Statistical analyses were performed using GraphPad Prism for Windows (v5.02).

All data generated or analyzed during this study are included in this published article or the supplementary information file. The raw and processed HiSeq data has been deposited to GEO (GSE119767; <https://www.ncbi.nlm.nih.gov/geo/>).

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Improved Pharmacological and Structural Properties of HIV Fusion Inhibitor AP3 over Enfuvirtide: Highlighting Advantages of Artificial Peptide Strategy

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Abstract: Enfuvirtide (T20), is the first HIV fusion inhibitor approved for treatment of HIV/AIDS patients who fail to respond to the current antiretroviral drugs. However, its clinical application is limited because of short half-life, drug resistance and cross-reactivity with the preexisting antibodies in HIV-infected patients. Using an artificial peptide strategy, we designed a peptide with non-native protein sequence, AP3, which exhibited potent antiviral activity against a broad spectrum of HIV-1 strains, including those resistant to T20, and had remarkably longer in vivo half-life than T20. While the preexisting antibodies in HIV-infected patients significantly suppressed T20's antiviral activity, these antibodies neither recognized AP3, nor attenuated its anti-HIV-1 activity. Structurally different from T20, AP3 could fold into single-helix and interact with gp41 NHR. The two residues, Met and Thr, at the N-terminus of AP3 form a hook-like structure to stabilize interaction between AP3 and NHR helices. Therefore, AP3 has potential for further development as a new HIV fusion inhibitor with improved antiviral efficacy, resistance profile and pharmacological properties over enfuvirtide. Meanwhile, this study highlighted the advantages of artificially designed peptides, and confirmed that this strategy could be used in developing artificial peptide-based viral fusion inhibitors against HIV and other enveloped viruses.

Text: The sequences of gp41 NHR-or CHR-derived peptides. The residues corresponding to the NHR pocket region are marked in red. The residues for the PBD are marked in blue, and the MT-hook residues adjacent to the N terminus of PBD are marked in green. 5HRu peptide consists of 5 copies of artificial sequence template (AEELAKK) underlined. The mutant residues in PBD of AP2 and AP3 were highlighted in pink. (b) The inhibitory activity of AP1, AP2, AP3 and T20 on infection by HIV-1 IIIB (subtype B, X4) in MT-2 cells (left panel) by HIV-1 Bal (subtype B, R5) in M7 cells (right panel). Each sample was tested in triplicate and the experiment was repeated twice. The data are presented as means $\pm$ SD.

To address these obstacles, many efforts have been made to optimize T20 and gp41 CHR-derived peptides. Some of these peptides have better inhibitory activities against T20-resistant strains and/or longer half-life than T20. However, they still have the problem to cross-react with the preexisting antibodies in the sera of HIV-infected patients because they contain some native CHR sequences. Based on the universal artificial peptide template of 5HRu, we previously designed the artificial peptides of AP1 (PBD-m4HR) and AP2 (PBDtrp-m4HR), and have made preliminary research on their inhibitory activity against HIV-1 Env-mediated cell-cell fusion 16 . In the present study, we designed a new artificial peptide, AP3 (Fig. 1a) , aiming to apply the "M-T hook" structure to stabilize the interaction of the artificial peptide with the hydrophobic pocket on the gp41 NHR trimer 17, 18 . After comprehensively studying its antiviral activity, biochemical property, crystal structure, functional mechanism, in vivo half-life and, for the first time, the effect of preexisting antibodies in the sera of HIV-infected patients, we found that the newly designed artificial peptide, AP3, exhibited improved antiviral activity, drug resistance profile and pharmacological properties over T20. Particularly, the preexisting antibodies in the sera of HIV-infected patients did not suppress, but enhanced the anti-HIV-1 activity of AP3. These results suggest that AP3 has potential for development as a new anti-HIV drug and confirm that this strategy can be used for designing artificial antiviral peptides against other enveloped viruses, such as SARS-CoV 19 , MERS-CoV 20 , and paramyxovirus 21 .

AP3 inhibited HIV-1 infection with higher potency than T20. Our previously designed artificial peptides AP1 and AP2 could inhibit HIV-1 Env-mediated cell-cell membrane fusion 16 . He and colleagues reported that adding two amino acids of Met and Thr to the N-terminus of a CHR-peptide could enhance their anti-HIV-1 activity 17, 18 . Here we designed a new artificial peptide, AP3, by adding Met and Thr to the N-terminus of AP2 (Fig. 1a) . We then compared AP3 with AP1, AP2 and T20 for their anti-HIV-1 activity against divergent HIV-1 strains, including the laboratory-adapted viruses, IIB (subtype B, X4) and Bal (subtype B, R5), and a series of primary HIV-1 isolates, as well as the T20-resistant strains. As shown in Fig. 1b , AP3 exhibited higher inhibitory activities on infection by HIV-1 IIB and HIV-1 Bal strains (IC₅₀ : 3.06 and 15.09 nM, respectively) than AP1 (IC₅₀ : 86.25 and 396.14 nM, respectively), AP2 (IC₅₀ : 23.05 and 49.95 nM, respectively), and T20 (IC₅₀ : 13.63 and 30.21 nM, respectively). The inhibitory activity of AP3 on infection by divergent primary HIV-1 isolates with distinct genotypes (subtypes A -E and group O) and phenotypes (R5 and X4) was also higher than that of AP2 and T20 (Table 1) . While T20 was not effective against T20-resistant HIV-1 strains at the concentration as high as 2,000 nM, AP3 could

effectively inhibit infection of these strains with IC₅₀ in the range of 13 ~ 90 nM, which was about 2-to 4-fold more effective than AP2 (Table 1). These results indicate that the artificial peptide AP3 has remarkably improved anti-HIV-1 activity against a broad spectrum of HIV-1 strains, including T20-resistant variants, over T20 and the artificial peptides AP1 and AP2. The preexisting antibodies in HIV-1-infected patients neither recognized AP3, nor attenuated its anti-HIV-1 activity. Previous studies have shown that the preexisting antibodies in HIV-1-infected patients, including those cross-reacting with T20 and those specific for the binding sites of T20 in gp120 (e.g., the C1 and V3 loop regions) and gp41 (e.g., the NHR domain), could significantly block the fusion inhibitory activity of T20 14, 15 . Here we investigated the influence of preexisting antibodies against AP3 peptide. As shown in Fig. 2a , both T20 and C46 reacted with the antibodies in sera from five HIV-1-infected patients; however, none of the three artificial peptides AP1, AP2 and AP3 was recognized by the preexisting antibodies. The inhibitory activity of T20 on HIV-1 IIIB infection was reduced about 1.9-fold to > 3.6-fold in the presence of the sera from HIV-1-infected patients (Fig. 2b and Supplementary Table S1), confirming that the preexisting antibodies in sera of HIV/AIDS patients can attenuate the anti-HIV-1 activity of T20 14, 15 . However, none of the artificial peptides in the present study showed significant decrease of anti-HIV-1 activity in the presence of patients' sera. Instead, the antiviral activity of AP3 increased in the presence of antisera from HIV-1-infected patients (Fig. 2b and Supplementary Table S1), suggesting that anti-HIV-1 antibodies actually enhanced the anti-HIV-1 activity of AP3, possibly because the binding of the antibodies to some sites in gp120 or gp41 promote the interaction of AP3 with viral gp41 NHR region.

AP3 had longer half-life than T20. Although T20 has shown efficacy in inhibiting HIV-1 infection, its major weakness lies in its short half-life in plasma (about 2 h) [22] [23] [24] . As a result, T20 has to be administered subcutaneously twice daily at 90 mg per dose, often causing serious injection-site reactions 25, 26 . Here, we performed pharmacokinetic studies by intravenous administration of AP3, AP2, and T20, respectively, to SD rat at a dose of 1 mg/kg, in order to compare their in vivo circulation time. As expected, T20 exhibited a shorter half-life and lower AUC (0-t) from systemic circulation, while AP3 and AP2 demonstrated much higher concentration and longer circulation time (Table 2). The pharmacokinetic profiles of AP3 and AP2 fit a non-compartment model. The pharmacokinetic parameters were calculated with PK Solver. The in vivo elimination half-life of AP3 ($t_{1/2} = 6.02$ h) was about 2.8-fold longer than that of T20 ($t_{1/2} = 1.57$ h). This result provided the theoretical basis for reducing the injection frequency and dose of the fusion inhibitor, in conjugation with the improved antiviral potency of AP3. Therefore, replacement of T20 with AP3 may significantly reduce injection-site reactions and the drug

cost, which would promote the clinical applications of the HIV fusion inhibitor in resource-poor regions or countries.

AP3 was much more resistant than T20 to proteolytic degradation by proteinase K and rat liver homogenate. We compared the stability of T20 and AP3 in the presence of proteinase K (a broad-spectrum serine proteinase) and rat liver homogenate. After treatment with 20 ng/mL of proteinase K for 2 h at 37 °C, only 29% of the parental T20 peptide remained, as detected by LC-MS analysis. Under the same condition, AP3 retained 100% of its prototype (Fig. 3a). In addition, AP3 showed a significantly enhanced in vitro metabolic stability over T20 in the presence of liver homogenate (Fig. 3b) .

These results indicate that the artificial peptide AP3 is much more resistant to proteolytic degradation than the natural peptide T20, which may contribute to its significant longer in vivo half-life than T20 as described above.

AP3 formed stable α -helical complex and block gp41 6-HB formation. To investigate the antiviral mechanism of AP3, the thermal stability of AP3/N36 complex was compared with that of AP1/N36, AP2/N36, T20/N36, and C34/N36 complexes by circular-dichroism (CD) spectroscopy 27 . Because T20 lacks the pocket-binding domain (PBD), the T20/N36 complex did not show a typical α -helical conformation, in consistence with our previous studies 8, 9 . Similar to the α -helicity of C34/N36 complex 3 , the AP1/N36, AP2/N36 and AP3/N36 complexes all formed a saddle-shaped negative peak at 208 nm and 222 nm, indicating their α -helical structures (Fig. 4a) Fig. 4b) , indicating that the α -helical complex formed by AP3 and N36 is the most stable among the four complexes.

Then we compared the inhibitory activity of AP3 with that of AP1 and AP2 on 6-HB formation between C34 and N36. Since T20 cannot block 6-HB formation 8, 9 , we used a small-molecule HIV-1 fusion inhibitor, ADS-J1 28, 29 , to replace T20 as a control of 6-HB inhibition. As expected, ADS-J1 could effectively inhibit 6-HB formation with IC 50 of 2.75 μ M 8, 9, [27] [28] [29] . AP3 was highly effective against 6-HB formation in a dose-dependent manner with an IC 50 value of 0.24 μ M, about 30- and 15-fold more potent than AP1 and AP2, respectively (Fig. 4c) , confirming that AP3 can potently block gp41 6-HB fusion core formation, thus inhibiting HIV-1 fusion with the target cell membrane.

Structural basis for the potent fusion inhibitory activity of the artificial peptide AP3. To elucidate the molecular determinants of these artificial peptides, we successfully solved all three complex structures of AP1/AP2/AP3 peptides binding with gp41 NHR. For AP1 and AP2, an optimized linker Each sample was tested in triplicate and the experiment was repeated twice. The data are presented as means $\pm$ SD. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. "SGGRGG" was used to assemble the NHR and the artificial peptide into a single recombinant protein (N36-L6-AP1 or N36-L6-AP2). However, a similar strategy failed on the crystallization of AP3; therefore, we decided to cocrystallize the synthetic peptide N45 and AP3 peptide, and eventually the complex crystals were obtained. Interestingly, the crystals of three different inhibitors belong to three distinctive space groups: P2₁ for N36-L6-AP1, R32 for N36-L6-AP2, and P6₃ for N45/AP3. As expected, the NHR portions in three structures all form a trimeric core, while the AP1, AP2 or AP3 portion folds into a single-helix conformation and binds to NHR-trimer to form a typical 6-HB, similar to that of the HIV-1 gp41 core structure formed by the native CHR peptide C34 and N36 (Fig. 5a). Also, the conserved hydrophobic residues, such as W43, W46 and I50, in the artificial peptides were deeply buried into the hydrophobic Table 2. Pharmacokinetic parameters of AP2, AP3 and T20 following intravenous administration at 1 mg/kg in male SD rats ($n = 2$). Figure 3. Sensitivity of AP3 and T20 to proteolytic degradation by proteinase K and rat liver homogenate. (a) After digestion by proteinase K at pH 7.2 and (b) rat liver homogenate, the residual amount of AP3 and T20 was detected by LC-MS analysis. The experiment was performed in triplicate and the data are presented as means $\pm$ SD. The inhibition of AP1, AP2, AP3, T20 and ADS-J1 against 6-HB formation between N36 and C34 was detected by ELISA using the 6-HB-specific mAb NC-1. Each sample was tested in triplicate, and the data are presented as means $\pm$ SD. grooves formed between each pair of NHR helices, similar to the corresponding residues of W628, W631 and I635 in the native gp41 CHR (Fig. 5b). AP peptides exhibited better affinity against gp41 natural CHR. In C34, which contains the natural CHR sequence from W628 to L661, no strong interaction between I642 and Q565 in the viral gp41 NHR-CHR complex was found (Fig. 5c). However, in the corresponding sequence (from W43 to K76) of AP1 and AP2, a hydrogen bond was established between S57 (corresponding to I642 in CHR) and Q18 (corresponding to Q565 in NHR) in N36-L6-AP1, N36-L6-AP2 and N45/AP3. Thus, S57 in AP1/AP2/AP3 plays a role in stabilizing the interactions between the artificial peptide inhibitor and its NHR target, resulting in their stronger binding affinity. Moreover, in NHR-CHR, L567 and L568 on two adjacent NHRs form a hydrophobic groove, in which T639 is buried (Supplementary Fig. S1a). However, in N36-L6-AP1, N36-L6-AP2 and N45/AP3, I54 (corresponding to T639 in CHR) can strongly bind to L20 and L21 through fully hydrophobic side chain interactions. Similarly, the interaction of I64 (corresponding to S659 in CHR) with L10 and L11

(corresponding to L567 and L568 in NHR, respectively) in N36-L6-AP1, N36-L6-AP2 and N45/AP3 has been significantly enhanced (Supplementary Fig. S1b).

Like the gp41 CHR helix, the helices of AP1, AP2 and AP3 also have two different sides, a hydrophobic side facing toward the NHR and a hydrophilic one facing outward. It is expected that the enhancement of the hydrophilicity of the exposed side of the inhibitors can increase their antiviral activity and solubility. To achieve this goal, the amino acid residues with hydrophobicity, or low hydrophilicity, like N637, S640, L641 and S644 in CHR, were changed to the amino acid residues with high hydrophilicity, like E52, K55, K56 and E59 in AP1, AP2 and AP3, respectively. Moreover, the hydrophobic residue M629 in CHR was replaced with a hydrophilic residue E44 in AP2 and AP3 (Supplementary Fig. S2). These hydrophilic residues, such as glutamic acid and lysine, can increase the solubility of whole peptide and, hence, stabilize the complex formed by the inhibitor and its target. It has been proved that the EE-KK double salt bridge can stabilize helix conformation 30 . We have identified this kind of interaction between i and $i + 3$ or $i + 4$ positions on the three complex structures. In N36-L6-AP1, R48 interacts with E45 and E52 to form a salt bridge network. In N36-L6-AP2, E45 interacts with K48, and E52 binds to K56, while in N45/AP3, K69 binds to E66 (Supplementary Fig. S2). These strong salt bridges formed by the oppositely charged residues stabilize AP peptide conformation, bringing its inhibitory effect into full play.

As previously reported, addition of the "M-T hook" to the CHR peptides C34 and sifuvirtide could dramatically improve the anti-HIV-1 activity 17, 18 . As expected, the N-terminal Met and Thr of AP3 forms a hook-like structure (Fig. 5d) . The hydrophobic methionine side chain of M41 accommodates the groove between AP3 and NHR helices, capping the hydrophobic pocket. This interaction leads to a series of conformational changes. The main chain of AP3 at W43 moves 1.91 Å closer to NHR compared to AP2 (Supplementary Fig. S3). The side chain of W43 in AP3 flips around 90 degrees and is buried deeper than that of AP2. The side chain of E44 turns back to interact to D47, but the E45 side chain turns back from K48 and interacts with T42. Therefore, this M-T hook structure could further stabilize the binding between AP3 and NHR target.

Enfuvirtide, also known as T20, was approved by the U.S. FDA as the first HIV entry inhibitor-based antiviral drug for use with other anti-HIV medicines to treat HIV-1 infected adults and children at ages 6-16 years 23,31,32 (<http://www.fuzeon.com>). Although T20 is

an indispensable anti-HIV drug for HIV/ AIDS patients who have failed to respond to the current antiretroviral therapeutics, its shortcomings have limited its clinical application. T20 has lower anti-HIV activity and shorter half-life than other CHR peptides containing PBD, such as C34 and C38 8, 9, 33 . In addition, T20-resistant HIV-1 variants emerged shortly (e.g., 14 days) after its use in patients 34 . Most of the T20-resistant viruses carried mutations in the GIV motif (residues 36-45: GIVQQNNLL) in the gp41 NHR domain 10, [34] [35] [36] [37] [38] . The lack of PBD contributes to the major weaknesses of T20 described above. Since the conserved hydrophobic pocket in the gp41 NHR-trimer plays a critical role in stabilizing the interaction between the gp41 NHR and CHR and formation of the fusogenic 6-HB core 1, 39, 40 , the PBD-containing CHR-peptide, like C34, can bind to viral gp41 trimer more strongly and stably, thus possessing more potent anti-HIV activity than T20, a CHR peptide without PBD 8, 9 . In the absence of PBD, T20 mainly interacts with the middle region of the NHR domain containing the GIV motif. Therefore, a virus with mutations in this motif is generally resistant to T20 10, [34] [35] [36] [37] [38] . Compared with other anti-HIV drugs, another weakness of T20 is its cross-reactivity with the preexisting antibodies in HIV-1-infected patients. Besides gp41, T20 could also bind to some regions in gp120. The preexisting antibodies specific for the T20's binding sites in gp120 and gp41 may indirectly suppress the anti-HIV activity of T20 14, 15 .

Addition of PBD to the N-terminus of T20, such as T-1249, could significantly improve the anti-HIV-1 potency, half-life and drug-resistance profile 33, [41] [42] [43] . Addition of M-T hook structure to the N-terminus of a PBD-containing CHR-peptides, such as MT-C34 or MT-SFT, could further increase the anti-HIV-1 activity of the corresponding CHR-peptides 17, 18 . Deletion of the GIV-motif-binding domain from a CHR-peptide, such as CP621-652 and CP32M, is another effective approach to increase the genetic barrier to drug resistance 44, 45 . However, none of the above approaches is effective in preventing the cross-reaction of T20 with the preexisting anti-gp41 antibodies in HIV/AIDS patients, since the above-modified peptides mainly contain the native sequences of the HIV-1 gp41 CHR domain. Our previous studies have shown that AP1 and AP2, artificial peptides with non-native protein sequences, could form coiled-coil structure to interact with gp41 NHR and inhibit HIV-1 Env-mediated cell-cell fusion 16 . In the present study, we designed a new artificial peptide, AP3, by adding M-T hook structure to the N-terminus of AP2 (Fig. 1a) , followed by investigating the influence of preexisting anti-gp41 antibodies in HIV-infected patients on AP3, using AP1, AP2 and T20 as controls. We demonstrated that sera of HIV-infected patients could bind to T20 and significantly reduce its potency against HIV-1. However, these same serum samples did not interact with the three artificial peptides and hardly impaired their antiviral activity. Surprisingly, the antibodies in the sera could even

enhance AP3's anti-HIV-1 activity (Fig. 2a,b and Supplementary Table S1). These results confirmed, for the first time, that replacement of the native viral sequence in T20 with an artificial sequence is an effective approach to overcome a key shortcoming of T20 whereby its anti-HIV activity could be attenuated by preexisting anti-gp41 antibodies in HIV/AIDS patients. It is worthwhile to explore why the antibodies in the sera is able to enhance the anti-HIV-1 activity of AP3. Our recent study has demonstrated that T20's anti-HIV-1 activity is enhanced by a non-neutralizing antibody directed against the NHR domain of the HIV-1 gp41 46 . We thus hypothesize that some of the anti-gp41 antibodies in HIV/AIDS patients may bind to a site in NHR domain adjacent to the AP3's binding region, resulting in increased interaction between AP3 and NHR-trimer and enhanced antiviral activity of AP3.

We then compared the inhibitory activity of AP3 with M-T hook and T20/AP2 without M-T hook on infection by divergent HIV-1 strains. AP3 was more effective than either AP2 or T20 in inhibiting infection by the laboratory-adapted strains and the primary isolates of HIV-1, including those resistant to T20 (Fig. 1b, Table 1). One may question whether AP3 can also induce drug-resistant viruses in patients if it is used in clinics to treat HIV-infected patients. We believe that AP3 is expected to have much higher genetic barrier to resistance than T20 because AP3 contains PBD, while T20 lacks PBD. Dwyer et al. 33 used T2544, a PBD-containing CHR-peptide, to carry out a passaging experiment, using T20 as a control. They demonstrated that T20 could induce a mutant virus with high resistance (81-fold) to T20 in about 1 month, while T2544 failed to induce a resistant strain in more than 2 months in culture. After extending the passaging experiment for almost 8 months, they identified one strain with a weak resistance (8.3-fold) to T-2544, and the related mutation sites were not in the gp41 pocket region, suggesting that the PBD-containing CHR-peptides, including AP3, may have difficulty to induce drug-resistance.

AP3 also had longer half-life than T20 (Table 2) , possibly because the artificial peptide AP3 is less sensitive to the proteolytic enzymes than T20 with native viral protein sequence. Removal of the proteolytic enzymes' cleavage sites in AP3 peptide is expected to further extend its half-life. These results confirmed that replacement of native protein sequence with artificial sequence and addition of the M-T hook to the PBD-containing peptide is a sound strategy for designing HIV fusion inhibitory peptides with improved antiviral activity and pharmacological properties when compared to T20.

Since the three-dimensional structures of AP peptides had not been investigated before the present study, the optimization of these artificial peptide inhibitors could not be performed rationally. Our structural studies of the artificial peptides AP1/AP2/AP3 in complex with NHR showed that AP peptides, just like the CHR peptide C34, could bind to gp41 NHR to form a canonical 6-HB structure (Fig. 5a) . It is well known that a deep hydrophobic pocket exists in each groove on the surface of the viral gp41 NHR trimer. The hydrophobic residues I635, W631 and W628 in the gp41 CHR bind with the hydrophobic residues in the wall of this pocket, resulting in the formation of stable 6-HB by the strong interaction between CHR and NHR. This important feature has been well preserved in the AP1/AP2/AP3 6-HB structures (Fig. 5b) , which may account for the potent HIV-1 fusion inhibitory activities of these artificial peptides.

A new hydrogen bond, which was established between S57 and Q18 in AP1/AP2/AP3 complexes, does not exist in the viral gp41 CHR-NHR complex, suggesting that S57 may play an important role in stabilizing the interactions between the peptide and NHR, resulting in binding affinities of AP1/AP2/AP3 that are stronger than those of HIV-1 gp41 CHR to NHR. Furthermore, the EE-KK double salt bridge formed between the *i* and *i* + 4 positions in the AP1/AP2/AP3 structures could stabilize helix conformation and increase the inhibitory effect of these peptides. Compared with AP1, triple-site mutations were introduced in AP2 and AP3, i.e. M44E, R48K and E49K. Those substitutions not only increase solubility of the peptide, but also trigger a series of rearrangements of certain intrahelical salt bridges to improve the stability of CHR helix structure and HIV-1 fusion inhibitory activity.

M-T hook was previously demonstrated to be an effective step toward increasing the stable interaction between a CHR-peptide and the HIV-1 gp41 pocket 17, 18 . Therefore, AP2 was further optimized by incorporating Met and Thr at its N-terminus. CD spectroscopy and thermal denaturation results both indicate that the incorporation of M-T hook contribute to the formation of a more stable 6-HB core structure between AP3 (M-T hook-optimized AP2) and N36. In addition, the EE-KK double salt bridge formed between *i* and *i* + 4 positions in the N36-L6-AP3 structure contributed to increased CHR helix and 6-HB stability, resulting in improved potency of AP3, as has been noted in studies of CHR-peptides with EE-KK double mutations 30, 33, 47, 48 . Also, the HIV-1 fusion activity and half-life of AP2 may have been strengthened and extended, respectively, by the addition of M-T hook in the design of AP3.

In conclusion, AP3, an artificial peptide with both PBD and M-T hook structures, exhibited improved anti-HIV-1 activity and drug-resistance profile, as well as prolonged half-life. Moreover, it did not react with the preexisting antibodies in the sera of HIV/AIDS patients. Consequently, its antiviral activity Scientific Reports | 5:13028 | DOI: 10.1038/srep13028 was not significantly affected by these antibodies. Therefore, AP3 shows promise as a candidate for further development as a new HIV fusion inhibitor for clinical use. This study also provides important structure and activity information for the rational design of novel artificially peptide inhibitors. Besides, our results highlighted the advantages of artificially designed peptides and confirmed that this strategy could be widely used in development of artificial peptide-based virus fusion inhibitors against HIV-1 and other enveloped viruses with class I membrane fusion proteins, such as SARS-CoV 19 , MERS-CoV 20 , and paramyxovirus 49 .

Ethics statement. This study did not involve human experimentation; the only human materials used were serum samples obtained from HIV-1-infected individuals with the approval by the Ethics Committee of the Shanghai Public Health Clinical Center, Fudan University (Protocol No. SPHCC-125-2). The methods were carried out in accordance with the approved guidelines. All of these sera samples came from adults; no minor was involved in this study. Written informed consent for the use of the clinical specimens was obtained from all patients involved in this study.

Peptide synthesis. A panel of peptides (Fig. 1a) , including T20, C34, C46, AP1, AP2, AP3, as well as NHR-derived N-peptides, N36 and N45, were synthesized with a standard solid-phase Fmoc method, as described previously 8, 50 . All peptides were acetylated at the N terminus and amidated at the C terminus. The peptides were found to be about 95% pure by HPLC and were identified by mass spectrometry (Perseptive Biosystems, Framingham, MA, USA). Concentrations of the peptides were determined by UV absorbance and a theoretically calculated molar-extinction coefficient based on tryptophan and tyrosine residues.

Qualification assay. Chromatographic analyses were performed using an ODS-C8 column (5 μ m, 100 mm $\times$ 2.0 mm ID) kept at ambient temperature. The mobile phase was composed of acetonitrile-water-formic acid in the ratio of 50:50:0.1 (v/v/v) at a flow rate of 0.3 mL/min. The sample injection volume was 10 μ L. Acetonitrile was HPLC grade, and

other chemical reagents and solvents were analytical grade. A Thermo TSQ Quantum Discovery MAX triple-quadrupole tandem mass spectrometer equipped with ESI source (San Jose, CA) and Surveyor LC pump were used for LC-MS analysis. Data acquisition and data processing were performed by using Xcalibur software and LCQuan 2.0 data analysis program (Thermo Finnigan), respectively. Optimized MS parameters were as below: 4800 V spray voltage, 40.0 psi sheath gas pressure, 1.0 psi auxiliary valve flow, and 300 °C of capillary temperature. When running collision-induced dissociation (CID), the pressure was set to 1.5 mTorr. The selected reaction monitoring (SRM) mode was used for AP3 while the selected ion monitoring (SIM) mode was preformed for T20. The following transitions were recorded: m/z 670.5 for AP3, m/z 1498.6 for T20. The masses of synthetic peptides T20, AP1, AP2 and AP3 were determined by MALDI-TOF-MS (Supplementary Fig. S4 and S5).

Expression and purification of fusion protein N36-L6-AP1 and N36-L6-AP2. Using overlapping PCR, the DNA fragment encoding AP1 or AP2 peptide was attached to the 3'-end of the cDNA of gp41 NHR ("N36", 546-581), with a short linker ("L6", SGGRGG) between them. Then, the whole sequence was subcloned into the pET-28a vector (Novagen, USA) with an artificial SUMO-tag between the N-terminal His-tag and the target protein. The pET-28a-SUMO-N36-L6-AP1-or pET-28a-SUMO-N36-L6-AP2-transformed E. coli cells were induced by adding 1 mM IPTG and incubating overnight at 16 °C. Fusion protein was purified by Ni-NTA affinity resin (Qiagen, Valencia, CA, USA), and the His-SUMO-tag was cleaved off by Ulp1 enzyme treatment at 4 °C for 2 h. The purified N36-L6-AP1 or N36-L6-AP2 was applied onto a Superdex-75 gel filtration column (GE Healthcare, Piscataway, NJ, USA). Fractions containing N36-L6-AP1 or N36-L6-AP2 trimer were collected and concentrated to different concentrations by ultrafiltration.

Crystallization, data collection, and structure determination. The fusion protein N36-L6-AP1 was crystallized at 16 °C using the hanging drop, vapor-diffusion method. The drops were set on a siliconized cover clip by equilibrating a mixture containing 1 µ l protein solution (25 mg/ml N36-L6-AP1 trimer in 20 mM Tris-HCl pH 8.0 and 150 mM NaCl) and 1 µ l reservoir solution (0.1 M Tris-HCl pH 8.5, 32% (w/v) PEG3350, and 0.2 M MgCl₂) against a 400 µ l reservoir solution. After one week, single crystals formed and were flash frozen by liquid nitrogen for future data collection. Fusion protein N36-L6-AP2 was crystallized in a similar way with a different reservoir solution (0.1 M Tris-HCl pH 8.0, 34% (w/v) PEG3350, and 0.2 M MgCl₂). To obtain the complex crystal of AP3 and NHR, synthesized AP3 was first mixed with peptide N45 at 1:1 molar ratio and then applied onto a Superdex-75 gel

filtration column (GE Healthcare, Piscataway, NJ, USA) to isolate the formed 6-HB. Fractions containing N45/AP3 trimer were collected and concentrated to 30 mg/ml, then crystallized at 16 °C using the hanging drop, vapor-diffusion method. The drops were set on a siliconized cover clip by equilibrating a mixture containing 1 μ l protein solution (20 mM Tris-HCl pH 8.0 and 150 mM NaCl) and 1 μ l reservoir solution (0.2 M Ammonium Sulfate, 0.1 M Bis-Tris pH 6.5, and 25% w/v PEG 3350) against a 400 μ l reservoir solution. After 3 days, single crystals formed and were flash frozen by liquid nitrogen for future data collection. The datasets of N36-L6-AP1 were collected at 100 K at beamline 19-ID of the Advanced Photon Source (Argonne National Laboratory, USA). The datasets of N36-L6-AP2 were collected on an in-house x-ray source (MicroMax 007 x-ray generator, Rigaku, Japan) at the Institute of Biophysics, Chinese Academy of Sciences. The datasets of AP3/N45 complex crystals were collected at beamline BL-19U1 of the Shanghai Synchrotron Radiation Facility, China. X-ray diffraction data were integrated and scaled using the HKL2000 program 51. The phasing problem of all three structures was solved by the molecular replacement method using PHENIX.phaser 52 with a crystal structure of HIV gp41 NHR-CHR (PDB entry: 1SZT) as a search model. The final models were manually adjusted in COOT 53 and refined with PHENIX.refine 54. All coordinates were deposited in the Protein Data Bank (N36-L6-AP1: 5CMU; N36-L6-AP2: 5CN0; and N45/AP3: 5CMZ). The statistics of data collection and structure refinement are given in Supplementary Table S2.

Determination of the cross-reactivity of the native and artificial peptides with the preexisting antibodies in HIV-1-infected patients by sandwich ELISA. A sandwich ELISA was conducted to determine the cross-reactivity of the peptides with the preexisting antibodies in HIV-1-infected patients. T20, C46, AP1, AP2 and AP3 were coated onto the wells of 96-well polystyrene plates (Costar, Corning Inc., Corning, NY) at 10 μ g/ml. The wells were then blocked with 1% gelatin, followed by addition of 50 μ l of serially diluted sera from HIV-1-infected patients and incubation at 37 °C for 1 h. Then, HRP-labeled goat-anti-human IgG (Abcam, UK) and TMB were added sequentially. A450 was determined with an ELISA reader (Ultra 384, Tecan).

patients. Inhibition of peptides on HIV-1 IIIB (subtype B, X4) infection in the presence of HIV-1-infected patients' sera was determined as previously described 55. Briefly, each peptide was mixed with serially diluted serum from an HIV-1-infected patient at room temperature for 30 min. Next, the mixture of peptide/serum and HIV-1 (100 TCID₅₀) were added to MT-2 cells (1 $\times$ 10⁵/ml) in RPMI 1640 medium containing 10% FBS. After incubation at 37 °C overnight, the culture supernatants were replaced with fresh culture medium. On the

fourth day post-infection, culture supernatants were collected for detection of p24 antigen by ELISA.

CD Spectroscopy and Thermal Midpoint Analysis. The secondary structure of AP1, AP2 or AP3 peptides mixed with N36 was analyzed by CD spectroscopy as previously described 56 . Briefly, each peptide or peptide mixture was dissolved in phosphate-buffered saline (PBS: 50 mM sodium phosphate and 150 mM NaCl, pH 7.2) at the final concentration of 10 μ M and incubated at 37 °C for 30 min before cooling down to 4 °C. The CD spectra of each sample were acquired on a Jasco spectropolarimeter (Model J-815, Jasco Inc., Japan) at 4 °C using a 5 nm bandwidth, 0.1 nm resolution, 0.1 cm path length, and an average time of 5.0 sec. Spectra were corrected by the subtraction of a blank corresponding to the solvent composition of each sample. Thermal midpoint analysis was used to determine the temperature at which 50% of the 6-HB formed by the CHR and NHR would decompose. It was monitored at 222 nm from 4 °C to 98 °C by applying a thermal gradient of 5 °C/min. The melting curve was smoothed, and the midpoint of the thermal unfolding transition (T_m) values was calculated using Jasco software utilities as described above.

Inhibition of gp41 six-helix bundle formation by sandwich ELISA. Inhibition of gp41 six-helix bundle formation by a testing peptide was determined with a sandwich ELISA described previously 57 . Briefly, a testing peptide (ADS-J1 as a control) at graded concentrations was preincubated with peptide N36 (1 μ M) at 37 °C for 30 min, followed by the addition of peptide C34 (1 μ M) and incubation at 37 °C for another 30 min. The mixture was added to a 96-well polystyrene plate (Costar, Corning Inc., Corning, NY) precoated with anti-N36/C34 antibodies (2 μ g/ml) purified from mouse antisera specifically against the gp41 six-helix bundle 58 . Then, mAb NC-1, HRP-labeled rabbit-anti-mouse IgG (Sigma), and TMB were added in order. A450 was determined by an ELISA reader (Ultra 384, Tecan).

Inhibition activities of AP1, AP2, and AP3 on HIV-1 infection were determined as previously described 57 . For inhibition of HIV-1 IIIB (subtype B, X4) infection, 100 TCID₅₀ of the virus was added to 1 $\times$ 10⁵ /ml MT-2 cells in RPMI 1640 medium containing 10% FBS in the presence or absence of the test peptide overnight. Then, the culture supernatants were changed to fresh media. On the fourth day post-infection, culture supernatants were collected for detection of p24 antigen by ELISA. For inhibition of infection by the HIV-1 strain Bal (subtype B, R5), M7 cells (1 $\times$ 10⁵ /ml) were precultured overnight and infected with Bal at 100 TCID₅₀ in the presence or absence of the test peptide or protein overnight.

Then, the culture supernatants were changed to fresh media. On the fourth day post-infection, the culture supernatants were discarded, and fresh media were complemented again. The supernatants were collected on the seventh day post-infection and tested for p24 antigen by ELISA as previously described 55 . The percent inhibition of p24 production was calculated. Analysis of the half-life of peptide inhibitors. Four male SD rats weighing approximately 200 g each were obtained from the Shanghai Medical School Animal Center and were used for the half-life assay. Animals were treated in accordance with the Animal Welfare Act and the "Guide for the Care and Use of Laboratory Animals" (NIH Publication 86-23, revised 1985). Either AP2 or AP3 was intravenously injected at the concentration of 1 mg/ml. After injection, blood samples were acquired from rat orbit at several time points (8 and 30 min and 1.5, 3, 6, 9, 12, and 24 h after peptide injection) and placed in clean tubes. To study the pharmacokinetics of AP2 and AP3 in rats and provide experimental evidence for the possible pharmacokinetics in human, a double-antibody sandwich ELISA method was established for rapid determination of AP2 and AP3 in rat plasma. Briefly, 96-well polystyrene plates (Costar, Corning Inc., Corning, NY) were precoated with antibody against AP2 or AP3 (5 μ g/ml) purified from rabbit anti-sera 59 . They were then preincubated with serum samples diluted 20 times at 37 °C for 1 h, followed by the addition of anti-AP2 or anti-AP3 antibody (1:1000) purified from mouse antisera specifically against AP2 or AP3 59 at 37 °C for another 1 h. Then, HRP-labeled rabbit-anti-mouse IgG (Sigma, USA) and TMB were added in order. Absorbance at 450 nm was determined by an ELISA reader (Ultra 384, Tecan). The standard peptide parameters were obtained first. Then, the plasma peptide concentrations were determined as a function of time, and the half-life was calculated by using PK Solver for Microsoft Excel to obtain pharmacokinetic parameters.

Assessment of sensitivity of peptides to proteolytic digestion by proteinase K and proteolytic enzymes in liver homogenate. The peptides (10 μ g/mL) were prepared in PBS pH 7.2 containing 20 ng/ml proteinase K. The resulting mixture were incubated at 37 °C in a water bath and taken out at different time intervals (0, 5, 15, 30, 60, 120 minutes), followed by quenching the samples with ethyl alcohol and quantitating the peptides by LC-MS analysis as described above.

To test the sensitivity of peptides to the proteolytic enzymes in liver homogenate, 3 male SD rats (250 $\pm$ 20 g) were sacrificed under anesthesia. The whole liver was quickly removed from each rat, washed in ice-cold PBS (50 mM, pH 7.2), weighed and cut into small pieces, which were resuspended in PBS to 100 mg wet liver tissue/2.5 ml PBS. The samples were

pooled and homogenized, followed by centrifugation at 9,000 g for 20 min at 4 °C. The supernatants were collected. The test peptides were added to the liver homogenate at a final concentration of 10 µg/ml. The resulting mixture was incubated 37 °C in a water bath, and the residue peptides in the mixture were quantitated as described above.

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Safe patient transport for COVID-19

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Abstract: nan

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Dear Editor, Although COVID-19 has not been officially labelled as a pandemic yet, the global burden of disease is significant and continues to rise. The virus has a high human-to-human transmissibility via airborne, droplet and contact routes [1] . Patient numbers can surge, and hospitals should be ready not just with the infrastructure, but also staff to be familiar with workflows. Kain and Fowler [2] have eloquently detailed influenza pandemic

preparations for hospitals and intensive care units, and we feel the principles described in the article are relevant to COVID-19. Staff must consider patient transfers in between wards, as COVID-19 patients are admitted in isolation facilities to contain infected cases and to avoid nosocomial spread [1] .

Infectious cases may be intentionally brought out of isolation rooms for various reasons. Intra-hospital transfer may be required from emergency departments to the wards, from the general floor to the intensive care unit and from the wards to radiology suites. Inter-hospital transfer may be required for extracorporeal membrane oxygenation (ECMO) if patients with COVID-19 develop severe acute respiratory distress syndrome within hospitals with only basic ventilation facilities. During episodes of patient transport outside of isolation, potential breaches of infection control can occur. At the same time, when COVID-19 patients turn ill during transport, their management is exceptionally challenging as accompanying staff would be wearing cumbersome personal protective equipment (PPE) [3] .

Mitigating the spread of COVID-19 is a national priority in Singapore [4] , and part of this effort involves planning and conducting safe patient transport for suspected or confirmed cases. HCWs who handle the transport of COVID-19 patients must consider the following principles (see Table 1): firstly, early recognition of the deteriorating patient; secondly, HCW safety; thirdly, bystander safety; fourthly, contingency plans for medical emergencies during transport; fifthly, post-transport decontamination. Specific action steps require designated zones for transport [5] , sufficient supplies of PPE, staff training and support personnel like security officers and cleaning crews. Powered air-purifying respirators add a layer of safety on top of N95 respirators [3] and should be used if possible for high-risk cases, such as those requiring ambulance transport to ECMO centres.

Given the continued global spread of COVID-19, we expect that more hospitals will need to deal with this disease. Haphazard transport of infected cases leading to nosocomial spread can stymie efforts to break the chains of transmission. We hope that our suggestions can aid others in ensuring safe patient transport for COVID-19 and reduce nosocomial spread.

Not applicable.

Availability of data and materials Not applicable.

Ethics approval and consent to participate Not applicable.

Not applicable.

The authors declare that they have no competing interests. prior to embarking on the same ambulance • Staff to doff PPE in the nearest clinical area, for example ambulance bay, upon arrival • Terminal cleaning of ambulance upon arrival when back at primary hospital BVM bag-valve-mask, CO₂ carbon dioxide, ECMO extracorporeal membrane oxygenation, EMD emergency, GW general ward, HEPA high-efficiency particulate air, ICU intensive care unit, PAPR powered air-purifying respirator, PPE personal protective equipment

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Viruses Causing Gastroenteritis: The Known, The New and Those Beyond

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Abstract: The list of recently discovered gastrointestinal viruses is expanding rapidly. Whether these agents are actually involved in a disease such as diarrhea is the essential question, yet difficult to answer. In this review a summary of all viruses found in diarrhea is presented, together with the current knowledge about their connection to disease.

Text: The gastrointestinal tract is a vulnerable organ for infections as there is constant contact with the outside, mainly via the oral route. Inflammation of the stomach and the intestines (gastroenteritis) can cause nausea, vomiting and diarrhea. Gastroenteritis is responsible for two to three million deaths each year, making it one of the most common causes of mortality [1] . Mainly children in developing countries, but also immuno-compromised individuals in developed countries, suffer from diarrhea. While bacterial and parasitic gastrointestinal infections are declining as a result of proper disposal of sewage and safe drinking water, viral gastroenteritis is not declining in developing countries [2] . In the developed world, viruses are already the most common pathogens causing diarrhea [3] .

Although viruses infecting humans had already been described since 1901 [4] and viruses were suspected to play a role in diarrhea, it lasted until 1972, when the first virus causing gastroenteritis (norovirus) was identified in an outbreak of diarrhea in Norwalk (California, United States) [5] . Shortly after the discovery of norovirus several other viruses causing gastroenteritis were discovered: rotavirus in epithelial cells of children with gastroenteritis [6] , astrovirus in infantile diarrhea cases [7] , enteric adenoviruses in the feces of children with acute diarrhea [8] , and sapovirus during an outbreak of gastroenteritis in an orphanage in Sapporo, Japan [9] . All these viruses spread via the fecal-oral route through person-to-person transmission and are described in more detail below.

Noroviruses are part of the family Caliciviridae and outbreaks of norovirus gastroenteritis have been reported in cruise ships, health care settings, schools, and in the military, but norovirus is also responsible for around 60% of all sporadic diarrhea cases (diarrhea cases where an enteropathogen could be found), reviewed in the literature [10, 11] . The pathogenesis of norovirus infection has been tested in vivo. Filtrated norovirus was given to healthy volunteers after which most of them developed diarrhea [12] . Culturing of the virus, however, has been a problem since its discovery, yet one study has recently described the cultivation of norovirus in B cells, and has revealed that co-factors, such as

histo-blood antigen expressing enteric bacteria, are probably needed before enteric viruses can be cultured in vitro [13]. Sapoviruses are also members of the Caliciviridae. There are five human genogroups of sapovirus described [14] which account for 2.2%-12.7% of all gastroenteritis cases around the globe [14, 15]. Sapovirus outbreaks occur throughout the year and can be foodborne [16]. For sapoviruses it has been described that the virus was not found before onset of an outbreak, and that it was found in 95% of the patients during an outbreak, while it declined to 50% after an outbreak, indicating that the virus introduces disease in a naturally infected host [17].

Rotavirus infection is the most common cause of viral gastroenteritis among children; however, parents of infected children also often become ill and as a result rotavirus is the second most common cause of gastroenteritis in adults [18]. Studies in human volunteers have shown that infection with rotavirus causes diarrhea, results in shedding of the virus and a rise in antibody anti-virus titer after infection [19]. Additionally, astroviruses infections are common, accounting for about 10% of all sporadic diarrhea cases [20]. Astrovirus has been isolated from diseased people, filtrated and administered to healthy individuals after which in some of the volunteers diarrheal disease was observed and astrovirus was shed in their stools [21]. The virus can replicate in human embryonic kidney cells and was detected by electron microscopy (EM) [21]. Adenoviruses are responsible for around 1.5%-5.4% of the diarrhea cases in children under the age of 2 years, reviewed in the literature [22]. Of the 57 identified adenovirus types [23], only adenoviruses type 40 and 41 are associated with diarrhea [24]. Next to these two types, adenovirus type 52 can also cause gastroenteritis [25], although it has been argued whether type 52 is actually a separate type since there is not sufficient distance to adenovirus type 41 [26]. Adenoviruses can generally be propagated in cell lines; however, enteric adenovirus 40/41 are difficult to culture, reviewed in the literature [27].

In the 1980s and 1990s some viral agents were identified for which the direct association with disease is less clear. Aichi viruses are members of the Picornaviridae identified in fecal samples of patients with gastroenteritis [28]. Aichi virus infection has been shown to elicit an immune response [29]. Since their discovery, two case-control studies were performed, but, although both studies only found Aichi virus in stools of diarrheic patients, the prevalence of Aichi virus (0.5% and 1.8%) was too low to find a significant association with diarrhea [30, 31]. In immuno-compromised hosts the virus is found in higher quantities and is not associated with diarrhea [32]. Toroviruses, part of the Coronaviridae, were first identified in 1984 in stools of children and adults with gastroenteritis [33]. Torovirus

infection is associated with diarrhea [34] and is more frequently observed in immuno-compromised patients and in nosocomial infected individuals [34] . Retrospective analysis of nosocomial viral gastroenteritis in a pediatric hospital revealed that in 67% of the cases torovirus could be detected [35] . However, only a limited number of studies report the detection of torovirus and therefore the true pathogenesis and prevalence of this virus remains elusive. Picobirnaviruses belong to the Picobirnaviridae and were first detected in the feces of children with gastroenteritis [36] . Since the initial discovery, the virus has been detected in fecal samples of several animal species, and it has been shown that the viruses are genetically highly diverse without a clear species clustering, reviewed in the literature [37] . This high sequence diversity has also been observed within particular outbreaks of gastroenteritis [38, 39] , limiting the likelihood that picobirnaviruses are actually causing outbreaks, as no distinct single source of infection can be identified.

In 1907 the first tissue culture system was developed which was regarded as the golden standard for virus detection for a long time, reviewed in the literature [40] . In the 1930's serology and electron microscopy were introduced which boosted the discovery of new viruses. During these years, these methods developed fruitfully but viruses infecting the gastrointestinal tract were especially difficult to culture. Throughout the last several decades, several DNA-based techniques have been developed for virus discovery that boosted the identification of novel viruses in stool samples. The four most used methods are: 1. Universal primer-PCR [41] ; 2. Random priming-based PCR [42] ; 3. Virus Discovery cDNA, Amplified Fragment Length Polymorphism (VIDISCA) [43] ; and 4. Sequence-Independent Single Primer Amplification (SISPA) [44] . Universal primer-PCR is a virus discovery technique that uses universal primers designed on conserved parts of a specific viral family, which can be used to detect novel variants of this viral family. Random priming-based PCR is a technique that randomly amplifies all nucleic acids present in samples, after which the resulting PCR products can be cloned and sequenced. SISPA and VIDISCA are virus discovery techniques that are based on digestion with restriction enzymes, after which adaptors can be ligated. These methods have been successful in the discovery of novel viruses, but there are some limitations. Universal primers are useful for discovering novel viruses of a chosen family, but the primers, based on our present knowledge of the viral family, may not fit on all unknown variants. Random priming PCR, SISPA and VIDISCA are sequence independent amplification techniques. The disadvantage of random priming PCR, SISPA and VIDISCA is that the virus needs to be present at a high concentration, while the host background DNA and/or RNA should be minimal and preferably not complex.

In recent years, sequence independent amplification techniques improved considerably by coupling these techniques to next-generation sequencing platforms and as a result several novel viruses have been described in gastroenteritis cases, such as cosavirus [45] , Saffold virus [46] , klassevirus/salivirus [47, 48] , polyomavirus [49] , bufavirus [50] , tusavirus [51] , and recovirus [52] . Although these viruses are found in individuals with diarrhea, for most of them the degree of circulation (prevalence) and the ability to cause morbid conditions or disease (pathogenesis) remains to be determined, as described below (also see Table 1). Only found in low prevalence; **: Only limited data is available about this virus; ***: Antibodies against astrovirus HMO-C were observed whereas no antibodies against astrovirus HMO-A were found (HMO = human-mink-ovine-like astrovirus); -No published data available; ^ Picobirnavirus, tusavirus and recovirus were identified in the gastrointestinal tract after next-generation sequencing, but no information regarding antibody response or association with diarrhea is available.

In the last decade, two novel clades of astroviruses have been discovered in stool samples from patients with diarrhea that are genetically far distinct from the classical astroviruses. The first clade consists of the VA-1, VA-2, VA-3, VA-4, and VA-5 astroviruses, which are genetically related to feline and porcine astroviruses, while the second clade consists of the MLB1, MLB2 and MLB3 astroviruses and form a separate cluster [55, 57, [74] [75] [76] [77] [78] . For these novel clades the pathogenesis remains to be determined since the viruses have been identified in patients with and without diarrhea, and in some studies the viruses were associated with diarrhea whilst in others no association could be found [55] [56] [57] . In addition an antibody response was observed against some but not all novel astrovirus types [54, 58] . Recently, astrovirus MLB2 has also been detected in blood plasma of a febrile child [79] and astrovirus VA1 in a frontal cortex biopsy specimen from a patient with encephalitis [80] , suggesting that astrovirus infection may not be limited to the gastrointestinal tract.

In 2008, Saffold virus was detected in a stool sample from a pediatric patient with fever of unknown origin [46] . Although Saffold virus type 3 was cultured on a human epithelial cervical carcinoma (HeLa) cell line, cytopathic effects were observed and neutralizing antibodies have been found in serum samples [59] , subsequent case-control studies showed that the virus was not significantly associated with diarrhea [53, 60, 61] . Additionally, in 2008 cosavirus was identified in a patient with diarrhea [45] . However, a case-control study showed that this virus was also detected in a substantial amount of individuals without diarrhea and is not associated with diarrhea [32, 62, 63] .

Klassevirus/salivirus was identified in 2009 in two fecal samples from infants with gastrointestinal disorders [47, 48] . In two studies the detection of this virus was associated with diarrhea [48, 53] , while in another study no association with disease was found [65] . Serological evidence of human klassevirus infection was obtained, suggesting that the virus infects human cells [64] .

With the use of next-generation sequencing techniques, three novel polyomaviruses were also identified in human fecal samples. MW polyomavirus was identified in the stool of a healthy child from Malawi in 2012 [49] , and in the same year MX polyomavirus was found in stool samples of patients with and without diarrhea from Mexico, United States and Chili [68] . One year later, STL polyomavirus was found in the stool of a healthy child from Malawi [71] . An antibody response against MX polyomavirus [66] and MW polyomavirus [69] was observed, although MW polyomavirus [67] and STL polyomavirus [70] were not significantly associated with diarrhea in two independent case-control studies.

Bufavirus is a member of the Parvoviridae and was first described in 2012 [50] . Two case-controls in Thailand and in Turkey showed that the virus was only found in patients with diarrhea and not in controls [72, 73] ; however, because of the low prevalence (respectively 0.3% in Thailand and 1.4% in Turkey), no significant association with disease was found. Tusavirus, another recently described member of the Parvoviridae, was identified in the feces of a child from Tunisia with unexplained diarrhea [51] , and thus far this is the only study describing this virus. Recovirus is a novel member of the Caliciviridae and was found in diarrhea samples from Bangladesh [52] . Similar to tusavirus, this is the only study describing this virus thus far.

The identification of the above-mentioned novel viruses certainly increased our knowledge about viruses that can be found in the gastrointestinal tract of humans, yet it is unknown how many of these novel viruses are actually enteropathogens. Human stool contains a wide variety of viruses which can be derived from different hosts: Besides genuine human viruses, plant dietary viruses [32, 81] and animal dietary viruses [82] can also be found in human stool, as well as bacteriophages and viruses infecting protozoa [32] . Even viruses derived from other parts of the body can be found in fecal samples, such as the John Cunningham Polyoma virus originating from the kidney ending up in feces via urine [83] , and rhinoviruses [84] , bocaviruses [85] and coronaviruses [86] originating from the respiratory tract and probably swallowed. Furthermore, viruses infecting blood cells such

as human immunodeficiency virus (HIV)-1 can also be detected in fecal samples [87] . Therefore, once a novel virus has been identified in human stool samples it does not indicate that this virus is replicating in human intestinal cells.

Koch recognized as early as 1891 that associating the presence of a certain agent with a certain disease is complex, and he therefore postulated guidelines that should be followed before an agent can be classified as a pathogen [88] . His postulates can be summarized in three points: (1) The microbe occurs in every case of the disease in question and under circumstances which can account for the pathological changes and clinical course of the disease; (2) the microbe occurs in no other disease as a fortuitous and nonpathogenic parasite; and (3), after being fully isolated from the body and repeatedly grown in pure culture, the microbe can induce the disease anew. If a microbe has fulfilled these three postulates it can be stated that "the occurrence of the microbe in the disease can no longer be accidental, but in this case no other relation between it and the disease except that the microbe is the cause of the disease can be considered". For enteric viruses, however, these postulates are not applicable. Firstly, the enteric viruses are not easily cultured [89] [90] [91] , and, secondly, prolonged shedding of viral agents and asymptomatic infection have been described [92] , reviewed in the literature [93] . Although attempts have been made to adjust the Koch's postulates specifically for viruses and the current methodologies deployed [94] [95] [96] , fulfilling these postulates is still not feasible on most occasions due to the lack of an efficient cell culture system, difficulties in antigen synthesis and high levels of viral genetic diversity within viral groups, reviewed in the literature [97] .

Several approaches have been made to develop a methodology that adds more significance to the discovery of a novel virus. One approach is based on the enrichment of immunogenic viruses before next-generation sequencing by making use of autologous antibody capture prior to sequencing. This method was tested and validated on several fecal samples containing adenovirus, sapovirus and norovirus, and has shown to enrich immunogenic viruses, while plant viruses and bacteriophages were not enriched after antibody capture [98] . Another method to enrich for relevant viruses prior to next-generation sequencing is the so-called virome capture sequencing platform for vertebrate viruses (VirCapSeq-VERT) which uses ~2 million probes which cover the genomes of all members of the viral taxa known to infect vertebrates [99] . However, both methods have limitations: For the antibody capture method, viruses need to be present in high viral loads, and convalescent blood, serum or plasma needs to be available. A disadvantage of the

VirCapSeq-VERT technique is that completely novel viruses, e.g., viruses from a novel virus family, will not be identified.

The most straightforward method to demonstrate association with disease is using case-control studies. In order to perform such studies, matched stool samples have to be collected in case and control groups from the same geographical locations in the same period of the year. Additionally, whereas in recent years case-control studies have been performed using conventional real-time PCRs (RT-PCR), in the future, sequence independent next-generation sequencing techniques can be used for such case-control studies. Since it allows detection of virtually all nucleic acids, next-generation sequencing has several advantages compared to specific RT-PCRs. Next-generation sequencing prevents the necessity to perform numerous RT-PCRs to screen for all viruses suspected to be associated with disease, and novel variants of currently known viral families or novel virus species can be detected which can be particularly beneficial if only few reference genomes are available. The major benefit of such a database is that in the immediate future the most important question can be answered if a novel virus is identified in diarrhea cases: Is the virus likely to cause disease?

In conclusion, the long list of viruses identified in the gastrointestinal tract is most probably not final yet. It is to be expected that several novel viruses will be described in the near future, since detection of these agents using the current next-generation sequence technologies is no longer a difficulty. Therefore, adding relevance to the discovery of novel viruses should be the main goal for future studies.

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Neutralization Interfering Antibodies: A “Novel” Example of Humoral Immune Dysfunction Facilitating Viral Escape?

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Abstract: The immune response against some viral pathogens, in particular those causing chronic infections, is often ineffective notwithstanding a robust humoral neutralizing response. Several evasion mechanisms capable of subverting the activity of neutralizing antibodies (nAbs) have been described. Among them, the elicitation of non-neutralizing and interfering Abs has been hypothesized. Recently, this evasion mechanism has acquired an increasing interest given its possible impact on novel nAb-based antiviral therapeutic and prophylactic approaches. In this review, we illustrate the mechanisms of Ab-mediated interference and the viral pathogens described in literature as able to adopt this “novel” evasion strategy.

Text: Hypervariable viruses adopt several mechanisms to cope with the host humoral immune response. The most studied mechanism is the accumulation of point mutations on immunodominant regions of surface proteins, making them no longer recognizable by previously generated neutralizing antibodies (nAbs) [1] [2] [3] [4] . Other escape mechanisms involving surface proteins include glycosylation of functionally pivotal residues (the glycan shield) or their association with host serum components (e.g., lipoproteins) in order to mask them from the immune system [5] [6] [7] [8] [9] (Figure 1A). Other known escape mechanisms are (i) a sort of protected route of virus spreading, such as cell-to-cell transmission [10, 11] ; (ii) the molecular mimicry between viral proteins and host self-antigens or (iii) the viral-induced stimulation of subfamily-restricted antibodies (Abs), both with obvious implications in viral-induced autoimmune diseases such as cryoglobulinemia for HCV [12] [13] [14] . The possible interfering effect of non-neutralizing Abs (non-nAbs) was originally proposed by Dulbecco et al. in 1956 [15] , to explain the apparent inhibition of virus neutralization exerted by some serum samples. Recently, this proposed immune escape mechanism has re-acquired a relevant interest, especially considering the potential clinical use of neutralizing anti-infectious nAbs or the design of

epitope-based vaccinal approaches [16] . To date, two main mechanisms have been proposed for the interfering effects of non-nAbs: (i) direct binding interference by steric hindrance, (ii) inhibition of binding following conformational changes of the viral antigen bound by interfering non-nAbs. Moreover, it has been speculated that, even when not directly interfering with nAbs binding, non-nAbs may also lead to the enhancement of viral infection through interaction with Fc receptors or complement receptors [17] .

Overall, possibly elicited non-nAbs in infected or vaccinated individuals may interfere with the neutralizing potential of nAbs. In more detail, these interfering Abs are able to bind viral proteins at the level of immunodominant but functionally irrelevant regions of viral proteins, decreasing or blocking the binding of nAbs to crucial viral epitopes (e.g., receptor-binding domains) (Figure 1B) [18] . A candidate antiviral monoclonal antibody (mAb) or polyclonal preparation should not be subjected to this mechanism of interference, or to the other escape mechanisms previously mentioned. Similarly, novel vaccinal approaches should avoid the elicitation of interfering Abs that could even worsen the disease in case of a real infection.

In the following paragraphs we discuss these mechanisms with specific examples of their role in the course of the viral infections where they have been described.

Hepatitis C virus (HCV) is a positive-sense single stranded RNA enveloped virus causing chronic hepatitis in most untreated patients (about 80%), with the consequent risk of developing cirrhosis and hepatocellular carcinoma. More than 170 million people (2%-3% of the world population) are infected worldwide, and a protective vaccine is not yet available, whereas therapeutic options are still limited and not completely effective [19] . For these reasons chronic HCV infection represents the major indication for liver transplantation in Europe and United States. Moreover, transplanted recipients are subject to high risk of graft re-infection and to a more severe and rapid progression of the liver disease [20] .

Schematic representation of viral escape mechanisms from humoral immune response against surface viral proteins: point mutations on immunodominant regions, glycosylation of functionally pivotal residues (glycan shield) of the viral surface proteins and virus association with host serum components (e.g., lipoproteins) (B) Mechanisms of

interference on nAb-mediated virus neutralization by the binding of interfering non-nAbs: non-neutralizing/interfering Abs might interfere with the binding of nAbs by steric hindrance following a spatial occupancy of their epitope or a competition for the binding; otherwise the binding of non-neutralizing/interfering Abs may induce conformational changes on the viral protein, thus affecting nAb binding to the antigen. Non-neutralizing/interfering Abs are depicted in black while nAbs in yellow.

The HCV genome encodes a single polyprotein of about 3,000 aminoacids that is processed by host and viral proteases into at least 3 structural (core, E1 and E2) and 7 non-structural (p7, NS2, NS3, NS4A, NS4B, NS5A and NS5B) proteins [21, 22] . In particular, the envelope type I membrane glycoproteins E1 and E2 form non-covalent heterodimers on the surface of the HCV envelope and allow clathrin-mediated virus endocytosis interacting consecutively with several entry cellular factors such as glycosaminoglycans [23] [24] [25] , low-density lipoprotein receptor [26, 27] , scavenger receptor class B type I [28] , the tetraspanin CD81 [29] , the tight-junction proteins claudin-1 and occludin, and the recently described Niemann-Pick C1-like 1 cholesterol absorption receptor [30] [31] [32] [33] [34] . The development of effective prophylactic and therapeutic approaches against this virus has been hindered mainly by its high mutation rate that gives rise to highly diversified viral variants, even within a single patient (quasispecies) [35] . Indeed, seven major genotypes, varying by up to 30% in nucleotide sequence, and several subtypes are recognized, each characterized by different clinical features such as different evolutionary rates to chronic liver diseases or different response to available antiviral therapies [21, 36, 37] .

The development and use of anti-HCV mAbs capable of targeting structurally and functionally conserved regions of the highly variable viral particles are being considered as novel therapeutic tools [38] [39] [40] [41] [42] [43] . In particular, the production of potent nAbs in acute infections has been shown to correlate with viral clearance in a single-source outbreak cohort [44] . Moreover, in vaccinated chimpanzees, a sustained Ab response to envelope glycoproteins E1 and E2 correlates with reduced viremia [45] , while the passive administration of neutralizing mAbs in a uPA-SCID chimeric mouse model of infection was able to protect against challenge with a HCV quasispecies inoculum [46] . Broadly cross-neutralizing human mAbs directed against the surface E2 glycoprotein of HCV (HCV/E2) are typically directed against functionally important regions within the CD81 binding site [47] [48] [49] [50] [51] [52] [53] [54] , as well as against other critical residues highly conserved among different genotypes [55, 56] . This aspect is crucial for the possible therapeutic in vivo use of such mAbs, but it may not be sufficient since it has been recently supposed that

other non-nAb populations may interfere with their neutralizing activity [39, [57] [58] [59] [60] [61] . In fact, in persistently infected individuals anti-HCV/E2 cross-nAbs are generally elicited at low titer and in a late stage of the infection, leading to a poor control of viremia, whereas quasispecies-specific neutralizing or high titer non-nAbs are elicited earlier [53, [58] [59] [60] [61] [62] . Moreover, the in vivo use of anti-HCV polyclonal immunoglobulin preparations in both chimpanzees and humans has been disappointing, and clinical studies have shown that these preparations fail to prevent recurrent infections in patients after liver transplantation [63] .

At this regard, a recent paper has suggested that the effect of some of these nAbs, directed against functionally important residues involved in the viral binding to CD81 (within epitope I, encompassing aminoacid residues 412-426), could be hindered by the presence of non-nAbs binding residues within epitope II on HCV/E2 (aminoacid residues 434-446) [58] . In particular, blocking of these interfering epitope II-specific Abs not only raised the neutralizing titer of serum containing both epitope I-and epitope II-specific Abs, but also uncovered a broader cross-genotype neutralizing response [58] . However, the role (and the existence itself) of these interfering Abs in influencing HCV infection is still controversial. Some authors recently corroborated the data of Zhang et al. by in vitro neutralization assays using serum-derived HCV of genotype 4a and polyclonal Abs derived from immunized goats with different conserved peptides spanning aminoacid residues 412-419, 430-447 and 517-531 of HCV/E2 glycoprotein [64] . In particular, this group found an interfering activity exerted by the weakly neutralizing 430-447-elicited Abs on the neutralizing activity of both the 412-419 and the 517-531-elicited Abs [64] . Interestingly, according to the putative model for E2 folding, all the three aforementioned regions would lie next to each other on the glycoprotein [48] . Therefore, this structural prediction possibly supports the interfering effect of epitope II-directed Abs. However, while this predicted structure is currently the best model available, these conclusions cannot be absolutely ascertained. For this purpose, the availability of E1-E2 crystal will certainly accelerate the fine elucidation of the spatial proximities of neutralizing and interfering mAbs on the E1-E2 structure and, consequently, structure-based vaccine progress.

Moreover, it is noteworthy that individuals with Abs that target the region of E2 encompassing epitope I frequently harbor Abs that recognize the region containing epitope II, thus confirming the co-immunogenicity of these epitopes [58] . Finally, it has been shown both a low prevalence (less than 2.5%) and a low titer of epitope I-reactive Abs in sera from both chronic and acute resolved infections thus supporting the hypothesis of a

conformational masking by adjacent regions such as that containing epitope II [65] . In fact, Zhang et al. originally put forward the idea that once epitope II is bound to an Ab, the site of epitope I becomes masked and can no longer be recognized by specific nAbs. Indeed, depletion of Abs to epitope II in plasma from a chronically infected HCV patient and vaccinated chimpanzees recovered an otherwise undetectable cross-genotype neutralizing activity [58] . Another possibility is that the initial binding of interfering Abs to the region containing epitope II may induce conformational changes on E2 that inhibit the binding by epitope I-directed Abs, as recently suggested by Lapierre et al. for other anti-HCV/E2 Abs [66] .

Conversely, these conclusions were not supported in a recent study by Tarr et al. using murine (AP33) and rat (2/69a) mAbs, as well as human immunoglobulin fractions affinity-purified on linear peptides representing distinct HCV/E2 domains clustering within the regions 412-426 and 434-446 [67] . Although confirming the previously reported co-immunogenicity of these two regions, the authors failed to demonstrate any inhibition between these two groups of Abs. Considering their results, the authors indeed suggested that interference by non-nAbs, at least to the region encompassing residues 434-446, is not a possible mechanism for HCV persistence in chronically infected individuals, as it had been originally proposed by Zhang et al. In accordance with the findings of Tarr and colleagues, Keck et al. described anti-HCV/E2 human mAbs binding conformation-sensitive epitopes encompassing also some residues within the 434-446 interfering region [56] . These mAbs are broadly neutralizing and do not lead to viral escape mutants, demonstrating the functional importance of their epitopes. The authors conclude that not all Abs directed against epitope II are interfering, but they also speculate that the interfering activity could be limited to Abs recognizing linear epitopes within it [56] .

Recently, we have partly confirmed the observations of Zhang et al. using a panel of anti-HCV/E2 mAbs: the well characterized mouse anti-HCV/E2 mAb AP33, whose epitope encompasses epitope I (aminoacid residues 412-423), and a weakly neutralizing human anti-HCV/E2 mAb (named e509), whose epitope encompasses epitope II [68] . In particular, we found that e509 is able to interfere with the neutralizing activity of AP33 on genotype 1a virus (strain H77). Instead, we found that e509 does not minimally interfere with the activity of two other broadly cross-neutralizing human anti-HCV/E2 mAbs, named e20 and e137 [49, 69] . Interestingly, we found that both e20 and e137 bind also residues within epitope II, at a higher affinity compared to e509, thus displacing it from the interfering epitope and, therefore, keeping unaltered their neutralizing activity. Thus, in our opinion, the described

divergent observations reported above may depend on the different Ab specificities present in the polyclonal preparations used and, probably, also on the different HCV genotypes infecting the studied patients [68] . Moreover, the different strategies adopted in isolating epitope I-and epitope II-directed Abs followed in the studies above could explain the different data obtained. In fact, immunoglobulins purified on peptides representing distinct HCV/E2 regions [67] are obviously directed against linear epitopes; these preparations are certainly different from mAbs cloned using a full-length HCV/E2 glycoprotein, which are more probably directed against conformational epitopes including also residues outside the investigated linear regions [54] .

To summarize, in the HCV field several works support the existence of interfering Ab populations and hypothesize their possible role in HCV persistence, as demonstrated using human plasma-derived immunoglobulin preparations, human mAbs, and sera of animals vaccinated with recombinant HCV/E2 peptides. The possible mechanism leading to the interference is still controversial, but both direct steric hindrance and induced antigen conformational changes have been hypothesized. On the other hand, other papers do not confirm these findings, suggesting that the putative interfering epitope II may be targeted by Abs endowed with a broadly neutralizing activity. Our recent paper, using well characterized mAbs [68] , shows that the interfering Abs do exist but that their overall effect may be biased by the presence of nAbs with different binding features and by the infecting HCV genotype. Future works investigating the in vivo role of these interfering Ab subpopulations in HCV persistence will certainly be very useful.

The influenza viruses circulate worldwide in animal reservoirs, especially water fowl, potentially affecting humans of any age group. Influenza viruses are classified into types A, B or C based on antigenic differences of their nucleoprotein and matrix protein. The most clinically relevant and variable type is influenza A which is divided in several subtypes, according to the antigenic characteristic of the two envelope glycoproteins, and causes epidemic and pandemic infections [70] . The yearly recurring influenza epidemics are associated with significant morbidity and mortality, particularly among risk groups (such as elderly people or those with chronic medical conditions, pregnant women and children) [71] ; the global spread of pandemic influenza viruses can cause millions of deaths [72] .

Within the enveloped influenza virion eight segments of negative single-stranded RNA are protected by the nucleocapsid protein, forming the ribonucleoprotein (RNP). The first six

RNA segments each code for a single protein: PB2, PB1, and PA (all constituting the RNA-dependent RNA polymerase), the hemagglutinin (HA), the nucleoprotein (NP), the neuraminidase (NA). The last two segments each code for two different proteins: the matrix proteins (M1 and M2) and the non-structural proteins (NS1 and NS2). Three different proteins (HA, NA and M2) are present on the viral envelope. The HA glycoprotein is the most abundant and it is the major target of the humoral immune response. Together with the NA transmembrane glycoprotein, HA is capable of eliciting a subtype-specific immune responses which is fully protective within, but only partially protective across different subtypes [73] . HA is synthesized as inactive precursor that transits into its active form upon cleavage by host cell proteases, and which is present on the viral membrane as homotrimers. HA trimers bind to 2,6-linked sialic acid molecules on cell membrane proteins or lipids through domains located in the globular head of each monomer. Subsequently, the viral envelope fuses by clathrin-dependent and -independent mechanisms with the endocytic vesicle membrane through the HA fusion peptide located in the stem region of each monomer. As a consequence, viral components are released into the host cell and can subvert the synthetic capabilities of the host cell for production and release of progeny particles [74] .

The humoral immunity plays an important role in the host defense against influenza virus infection as most of Abs neutralize influenza viruses and, hence, limit infection [75] [76] [77] [78] . In fact, a large body of experimental works suggests that occlusion of the receptor-binding site on HA by Abs is the main mechanism of influenza viral neutralization. Less common, but more broadly nAbs may neutralize influenza virus by inhibiting fusion of the viral envelope with the endocytic-vesicle membrane [50, [79] [80] [81] [82] [83] . Aminoacid changes on HA, more frequent on the immunodominant globular head, have complex effects on viral neutralization by Abs, usually allowing the mutated variants to escape from previously generated nAbs [84] . Classical studies using neutralizing mouse mAbs identified five distinct antigenic sites (A-E) on the HA1 globular head region in the three-dimensional structure of the H3 HA molecule (A/Hong Kong/1/68) [85] [86] [87] as well as in H1 [88] and H2 subtypes [89] .

During the first few days of an infection, the nAb titer is often low, while the titer of non-nAbs is higher and may play a role in the outcome of an infection, as recently observed for influenza A/2009 H1N1 pandemic virus infected patients by To et al. [90] . In particular, this group found that the amount, as well as the avidity, of non-nAbs were higher for patients with severe disease than for those with mild disease. The authors concluded that an

exaggerated non-nAb response during the early stage of infection was associated with severe disease [90] . Moreover, the authors speculated that non-nAbs present in patients' sera during the early stage of infection were likely to be either preexisting or the result of a secondary heterosubtypic humoral immune response against more conserved epitopes on several influenza proteins [91] . This early humoral response can be elicited within a few days after infection, because of immune priming by previous exposure to shared viral epitopes. In fact, the matrix proteins and nucleoprotein have conserved aminoacid sequences, and therefore Abs against these proteins from previous seasonal influenza virus infection or vaccination could be induced [92] . Indeed, upon infection with influenza virus, memory B-cells can proliferate rapidly and generate a large amount of these high avidity non-nAbs, especially in patients with severe disease. This is consistent with the observation that the number of peripheral blood B-cells is higher in patients with severe disease than in those with mild disease during the early stage of infection.

The mechanism of Ab neutralization interference has been indirectly speculated also by Ndifon et al., who observed that some aminoacid changes on HA actually increase the efficiency of neutralization of escape variants by previously generated Abs, even if not directly influencing their binding [93] . In detail, this group suggested that the increase in neutralizing activity after HA mutation could be the resultant of a lesser steric interference between Abs. Specifically, if there is a steric competition for binding to HA by Abs with different neutralization efficiency, then a mutation that reduces the binding of Abs with low neutralizing activity could increase the overall viral neutralization. Indeed, similarly to what has been speculated for HCV, Abs that bind to HA epitopes located at a distance from the receptor-binding site may therefore fail to occupy this site efficiently, thereby leading to a decreased viral neutralization. Moreover, it has been shown that Abs that bind to a certain HA epitope can prevent further binding of Abs to other epitopes of the same HA protein, and even to epitopes found on adjacent HA proteins. The above observations suggest that Abs that bind to low-neutralization efficiency epitopes of HA might interfere with the binding of nAbs to close high-neutralization efficiency epitopes, thereby impeding the neutralization of influenza viruses. Considering the HA structure, the binding of the interfering Abs would lie at the level of epitope C and E located far from the receptor-binding site on the globular head of the HA. However, the binding of these Abs may influence the binding of nAbs to epitopes A, B and D, located closer to the receptorbinding site [93] . At this regard changes to epitope A, B and D could be highly favored by natural selection, whereas changes to epitopes C and E could be disadvantageous to influenza viruses [93] .

Similarly to HCV, but with a sounder confirm due to the availability of the crystal structure, these speculations raise the intriguing possibility that the influenza viruses may have evolved by favoring the preferential elicitation of Abs recognizing epitopes with a low-neutralization profile. Indeed, steric hindrance by Abs that bind these epitopes could greatly reduce the extent of mutation required for a virus to evade neutralization by host Abs. Consequently, a decrease in the affinity of Abs for epitopes with low-neutralization efficiency could lead to an increase in viral neutralization. This suggests a possible approach to design "low-interference" vaccines that could greatly diminish the impact of Ab interference. These immunogens are genetically modified from viral target only at the level of low-neutralization efficiency epitopes. Indeed, vaccine-induced Abs only recognize high-neutralization efficiency epitopes of the target and Abs induced by low-interference vaccine strain have low affinity for low-neutralization efficiency epitopes of the target circulating virus strain. Therefore, they do not interfere with Abs to high-neutralization efficiency epitopes, implying an improved neutralization. Consequently, limiting Ab-mediated interference, the target virus cannot escape from vaccine-induced Abs through small epitope changes. Alternatively, vaccines could be designed to include only those regions that correspond to epitopes with high-neutralization efficiency.

Furthermore, antiviral drugs could be designed to include viral proteins carrying modifications at the level of high-neutralization efficiency epitopes; these "decoy" proteins would compete with virus for binding to low-neutralization efficiency Abs in a manner similar to that played by neuraminidase inhibitors.

In synthesis, the availability of HA crystal structure has helped to confirm the existence and to explain the mechanisms of interference by non-or weakly-neutralizing anti-HA Abs. The recent work by To et al. [90] evidencing that a non-nAb response during the early stage of infection is associated with a severe disease, may be the first proof of the role of these interfering Abs in the course of a natural infection.

The [94] . More than 8,000 cases, including almost 800 deaths, were reported during the outbreak period and increasing age and comorbidity were risk factors for severe disease and death [95] . Since 2003, only sporadic cases have been reported; however, the possibility that SARS outbreaks could reemerge naturally or be deliberately released is a public health concern. Like influenza viruses, SARS-CoV circulates in animal reservoirs,

with bats that are thought to transmit the virus to small mammals with exposure to these small animals as the source of human infections [96] . The clinical disease is similar to other severe acute respiratory infections, including influenza, and the SARS case definition includes clinical, epidemiologic, and laboratory criteria [97, 98] .

The basic genome organization and replicative cycle is similar for all CoVs. Gene 1 encodes all predicted replicase/transcriptase proteins, which are translated from input genomic RNA, while genes 2-9 encode structural and accessory proteins, including the envelope spike (S) protein, which are translated from separate subgenomic mRNAs. CoVs use a unique discontinuous mechanism to transcribe a series of progressively larger subgenomic mRNAs, and each contains a leader RNA sequence that is derived from the 5' end of the genome [99] .

The S protein of CoVs is inserted in the envelope of the virion mediating binding and fusion events necessary for infection, and it is the major target of the humoral protective immunity [100] . Although the S protein of SARS-CoV (SARS-S) shares little aminoacid identity (approximately 20%-27%), it shares common structural features with S proteins of the other members of the Coronaviridae family. SARS-S protein is a type I transmembrane glycoprotein of approximately 1,255 amino acids in length and divided into two functional domains: S1 (aminoacid residues 15-680) and S2 (aminoacid residues 681-1,255) [101] . In many CoVs, the S protein is cleaved during biogenesis and these two functional domains are held together non-covalently; however, as in the case of human CoV 229E, the S protein is not cleaved in SARS-CoV [102] . The S1 domain forms a globular structure that mediates interaction of the S protein with its main receptor, angiotensin-converting enzyme 2 (ACE2), while the S2 domain mediates fusion and contains the putative fusion peptide and two conserved helical regions (HR1 and HR2) that upon cleavage by the endosomal protease cathepsin L form the six helix bundle fusion core [103] .

Vaccine strategies aiming at blocking/limiting infection by SARS-CoV mainly focus on targeting the SARS-S viral glycoprotein [100] . Nonetheless, such a strategy poses a singular dilemma for CoVs, as previous vaccination protocols have highlighted the possibility of immune-mediated enhancement of the disease [104] .

At this regard, the group of Zhong et al. investigated the role of non-neutralizing interfering Abs also in the case of SARS-CoV infection [105]. In particular, they found that two mAbs directed against the region encompassing aminoacid residues 491-510 of SARS-S (341C and 540C) act synergistically to inhibit SARS-CoV infection in vitro, while a non-neutralizing mAb (240C) whose epitope encompasses the above mentioned region, disrupted the neutralizing activity of both 341C and 540C [105, 106]. By analyzing the crystal structure of the SARS-S protein, the authors proposed a possible explanation to what observed, evidencing that the epitopes of all the mAbs are closely packed and proximal to each other but distal from the ACE2 receptor binding site [105]. Moreover, the epitope of the non-neutralizing mAb 240C partially overlaps by at least 2 aminoacids (P507 and A508) with that of the neutralizing mAb 341C. As a consequence, mAb 240C could inhibit mAb 341C binding in an equilibrium-related manner. On the other hand, the authors found that the 240C mAb could sterically interfere with the binding of the 540C mAb through the proposed mechanism of spatial occupancy (Figure 1B). In fact, the accessibility of mAb 540C to its epitope may be blocked by the mAb 240C binding that masks the surface area containing it. In fact, as speculated by Davies and Cohen, the buried area of an Ab can range from 500 Å² to more than 800 Å² corresponding to 21-32 aminoacids, although only 9-20 aminoacid residues (the real epitope) make direct contacts with the Ab [107]. In fact, as previously observed for HCV, influenza and other human and animal viruses [108], one of the possible mechanisms is that the steric block by non-nAbs reduces the binding of nAbs on the SARS-S protein disabling neutralization. Conversely, notwithstanding the epitopes of mAbs 341C and 540C are located on a single loop; they are spatially separated thereby providing distinct interfaces for independent Ab binding.

To conclude, SARS-CoV can elicit potentially interfering non-nAbs by presenting on its surface closely packed regions with different biological features. On the other hand, the host can mount a vigorous neutralizing humoral response by producing Abs that recognize distinct epitopes and act synergistically. In particular, these results suggest that a cocktail of neutralizing human mAb that can bind to unique epitopes and have different mechanisms of action might be of clinical utility against SARS-CoV infection, and indicate that a similar approach may be applied to treat other viral infections [109].

The human immunodeficiency virus (HIV) is a positive single-stranded RNA retrovirus, causing substantial morbidity and mortality across the globe, particularly in developing countries. Human immunodeficiency viruses type 1 and 2 (HIV-1 and HIV-2) are the results of multi-interspecies transmissions from simian virus to humans. HIV-2 prevalence is low

and there is an higher proportion of HIV-2 infected individuals that do not progress to acquired immunodeficiency disease syndrome (AIDS) compared with those infected with HIV-1 [110] . HIV-1 viruses are very divergent and are classified in four groups: M, N, O and P. In particular, the group M is subdivided in nine subtypes and numerous circulating recombinant forms [111] .

The genome of all retroviruses encode the Gag, Pol and Env structural proteins. Among the HIV structural proteins, gp120 and gp41 surface envelope glycoproteins form heterodimers that are organized as trimers on the surface of the viral membrane. HIV-1 entry into target cells is initiated by the interaction of these surface envelope glycoproteins with CD4 and a co-receptor (typically CCR5 or CXCR4) on target cells [112] . The gp120 portion binds the target cell receptors, while gp41 promotes fusion of viral and cellular membranes [113] . Upon binding to the CD4 receptor, gp120 undergoes a conformational change, resulting in the exposure of epitopes that can be bound by co-receptor molecules and in the eventual formation of the transient pre-hairpin intermediate conformation [114] [115] [116] . In the pre-hairpin intermediate, the gp41 molecules reorganize so that the N-terminal peptides form a trimer of helices that expose the fusion peptide to the target cell, while the C-terminal helices remain anchored to the viral membrane [113] . This stage is vulnerable to a number of nAbs and peptides capable of binding either the N-or C-terminal peptides [117, 118] . Upon fusion with the target cell membrane, further gp41 reorganization results in the association of N-and C-terminal peptides to create a six-helix post-fusion bundle [119] . After fusion and delivery of the viral capsid in the cytoplasm, uncoating leads to the release of viral enzymes, proteins, and genomic RNA inside the cell.

Reverse transcription of the viral genomic single-stranded positive RNA is then initiated to yield a double-stranded proviral DNA to be imported in the nucleus and integrated into host chromosome. Active transcription from the integrated proviral DNA occurs in the presence of NF- κ B and viral Tat. Splicing of viral mRNA yields early accessory proteins like Tat, Rev, and Nef, which help in transcription, splicing, and modification of the cellular machinery, respectively. Accumulation of Rev protects the viral mRNA from splicing, thus yielding increasingly longer mRNAs able to code for structural and envelope proteins, and finally viral genomic RNAs is ready to be encapsidated [111] . Antiretroviral drug therapy for HIV is highly effective in controlling the infection; however, the eradication of this virus is currently not practicable and the treatment is therefore lifelong and burdened by considerable toxicity and drug resistance. A vaccine is widely viewed as being crucial for the control of the epidemic but several advanced efforts to develop an effective prophylaxis resulted

unsuccessful [120, 121] . One of the greatest challenges in developing a vaccine against HIV is to overcome its ability to constantly mutate and escape anti-HIV immune responses [122] . This high mutation rate is a direct result of the presence of the virus' low fidelity RNA polymerase as well as the high levels of recombination it undergoes and the constantly evolving glycan shield of the envelope glycoproteins [123] [124] [125] . At this regard, both cytotoxic T lymphocytes and nAbs have long been reported to select for immune escape variants during the course of HIV-1 infection [126] [127] [128] .

A candidate passive immunotherapy could consist, as previously suggested for SARS-CoV infection, in the administration of a cocktail of broadly neutralizing mAbs, that could minimize the onset of viral escape mutants [129] . Various combinations of human mAbs have been studied over the past several years which have shown additive, synergistic, or antagonistic effects on the neutralization of HIV-1 [130] [131] [132] [133] [134] [135] . Antagonistic effect in HIV-1 neutralization has been previously reported with a pair of anti-gp120 mAbs directed against the V3-loop and the CD4 binding site, respectively [136] . The molecular mechanisms determining the antagonism have not been further studied in details.

The only study describing for the first time at the molecular level a possible mechanism of interference also for HIV was performed using pair combinations of anti-gp41 mAbs [137] . More in details, the authors noted an antagonistic effect when the anti-gp41 neutralizing mAbs 2F5 or 50-69 were combined with the non-neutralizing anti-gp41 mAb 98-6 [137] . In particular mAbs 50-69 and 98-6 recognize different gp41 epitopes located within cluster I (aminoacid residues 579-613) and cluster II (aminoacid residues 644-667), respectively. On the other hand mAb 2F5 recognize a different epitope from mAb 98-6, within the gp41 membrane-proximal external region (MPER), in a portion adjacent to the cluster II region of gp41. Moreover, there is some overlap between cluster II epitopes and the epitope recognized by mAb 2F5 [138] , explaining the inhibition of mAb 2F5 binding by mAb 98-6 [137, 139] .

Thus, in the case of the antagonism between mAbs 2F5 and 98-6 the author hypothesized a mechanism of steric hindrance between the two mAbs as they could bind peptides and peptide complexes representing the pre-fusogenic and fusogenic forms of gp41 [140] . In particular, mAb 98-6 had a higher affinity for the peptide complexes representing the fusogenic form, than did 2F5. Thus, the binding of 98-6, which fails to neutralize the HIV-1

isolate 89.6 (HIV-1 89.6), could interfere with the binding of 2F5, leading to the neutralization antagonism. In contrast, mAbs 50-69 and 2F5 recognize distinct epitopes on gp41, and display independent (additive) reactivity against HIV-1 89.6 in combination with most of the other anti-gp41 and anti-gp120 mAbs tested [137] .

To conclude, anti-gp120 and anti-gp41 Abs are induced in HIV-1-infected individuals but are predominantly non-neutralizing, since the functionally important regions of HIV surface proteins are almost completely hidden to the immune system [139] . An intriguing hypothesis is that, together with other HIV escape mechanisms, the effect of the extremely rare anti-gp41 and anti-gp120 nAbs may be also hindered by the overwhelming amount of interfering non-nAbs. To date, the existence of interfering non-nAbs has been clearly evidenced only using anti-gp41 mAbs with different biological features, whereas no data have been generated using anti-gp120 mAbs. The possible role of non-nAbmediated interference in facilitating HIV escape in the course of the natural infection certainly deserves future studies.

Immunoprophylactic or immunotherapeutic approaches with mAbs are still considered a possible supporting tool in the management of infectious diseases. In particular, the availability of broadly neutralizing mAbs directed against viral pathogens, whose actual prophylactic and therapeutic approaches are far from effective, has led to many ongoing clinical trials. However, the evidence reported in this review suggest that candidate mAbs to be possibly used in antiviral passive immunization approaches, or to be elicited by future vaccine strategies, have not only to be highly cross-neutralizing molecules [141, 142] , but also tailored molecules whose activity is not influenced by possible interfering Abs produced in the course of infection. To this end, they must either be directed against highly neutralizing epitopes not subjected to the mechanism of interference, or must feature high affinity for the antigen in order to displace the binding of possible interfering Abs [51, 68] .

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The Evolutionary Dynamics of the Lion *Panthera leo* Revealed by Host and Viral Population Genomics

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Abstract: The lion *Panthera leo* is one of the world's most charismatic carnivores and is one of Africa's key predators. Here, we used a large dataset from 357 lions comprehending 1.13 megabases of sequence data and genotypes from 22 microsatellite loci to characterize its recent evolutionary history. Patterns of molecular genetic variation in multiple maternal (mtDNA), paternal (Y-chromosome), and biparental nuclear (nDNA) genetic markers were compared with patterns of sequence and subtype variation of the lion feline immunodeficiency virus (FIV(Ple)), a lentivirus analogous to human immunodeficiency virus (HIV). In spite of the ability of lions to disperse long distances, patterns of lion genetic diversity suggest substantial population subdivision (mtDNA $\Phi(ST) = 0.92$; nDNA $F(ST) = 0.18$), and reduced gene flow, which, along with large differences in sero-prevalence of six distinct FIV(Ple) subtypes among lion populations, refute the hypothesis that African lions consist of a single panmictic population. Our results suggest that extant lion populations derive from several Pleistocene refugia in East and Southern Africa (~324,000–169,000 years ago), which expanded during the Late Pleistocene (~100,000 years ago) into Central and North Africa and into Asia. During the Pleistocene/Holocene transition (~14,000–7,000 years), another expansion occurred from southern refugia northwards towards East Africa, causing population interbreeding. In particular, lion and FIV(Ple) variation affirms that the large, well-studied lion population occupying the greater Serengeti Ecosystem is derived from three distinct populations that admixed recently.

Text: Lion fossils trace to the Late Pliocene in Eastern Africa and the Early Pleistocene in Eastern and Southern Africa coincident with the flourishing of grasslands ,2-1.5 million years ago [1, 2] . By Mid Pleistocene (,500,000 years ago), lions occupied Europe and by the Late Pleistocene (,130,000-10,000 years ago) lions had the greatest intercontinental distribution for a large land mammal (excluding man), ranging from Africa into Eurasia and the Americas [3] . Lions were extirpated from Europe 2,000 years ago and within the last 150 years from the Middle East and North Africa. Today, there are less than 50,000 free-ranging lions [4] that occur only in sub-Saharan Africa and the Gir Forest, India (Figure 1A) .

Understanding the broader aspects of lion evolutionary history has been hindered by a lack of comprehensive sampling and appropriately informative genetic markers [5] [6] [7] [8] [9] , which in species of modern felids requires large, multigenic data sets due to its generally rapid and very recent speciation [10, 11] . Nevertheless, the unique social ecology of lions [12] [13] [14] and the fact that lions have experienced well-documented infectious disease outbreaks, including canine distemper virus, feline parvovirus, calicivirus, coronavirus, and lion feline immunodeficiency virus (FIV Ple) [15] [16] [17] [18] provide a good opportunity to study lion evolutionary history using both host and virus genetic information. Indeed, population genetics of transmitted pathogens can accurately reflect the demographic history of their hosts [19, 20] . Unlike other of the 36 cat species, lions have a cooperative social system (prides of 2-18 adult females and 1-9 males) and their populations can have high frequencies of FIV Ple , a lentivirus analogous to human immunodeficiency virus (HIV), which causes AIDS-like immunodeficiency disease in domestic cats. FIV Ple is a retrovirus that integrates into the host genome and is transmitted by cell-to-cell contact, which in felids occurs during mating, fighting and mother-to-offspring interactions. Thus, viral dissemination is a function in part of the frequency of contact between infected and naïve lions within and among populations. The virus is quite genetically diverse in lions [15, 18] , offering a unique marker for assessing ongoing lion demographic processes.

To unravel lion population demographic history we used a large multigenic dataset. Distinct sets of markers may not necessarily

The lion *Panthera leo*, a formidable carnivore with a complex cooperative social system, has fascinated humanity since pre-historical times, inspiring hundreds of religious and cultural allusions. Here, we use a comprehensive sample of 357 individuals from most of

the major lion populations in Africa and Asia. We assayed appropriately informative autosomal, Y-chromosome, and mitochondrial genetic markers, and assessed the prevalence and genetic variation of the lion-specific feline immunodeficiency virus (FIV Ple), a lentivirus analogous to human immunodeficiency virus (HIV) that causes AIDS-like immunodeficiency disease in domestic cats. We compare the large multigenic dataset from lions with patterns of genetic variation of the FIV Ple to characterize the population-genomic legacy of lions. We refute the hypothesis that African lions consist of a single panmictic population, highlighting the importance of preserving populations in decline rather than prioritizing larger-scale conservation efforts. Interestingly, lion and FIV Ple variation revealed evidence of unsuspected genetic diversity even in the well-studied lion population of the Serengeti Ecosystem, which consists of recently admixed animals derived from three distinct genetic groups. Historical and current geographic distribution of lion, *Panthera leo*. A three-letter code pointing to a white dotted circle represents the geographic location of the 11 lion populations determined by Bayesian analyses [22] and factorial correspondence analyses [23] of the genetic distinctiveness of 357 lion samples (see text): GIR, Gir Forest, India; UGA, Uganda (Queen Elizabeth National Park); KEN, Kenya (Laikipia), SER, Serengeti National Park, Tanzania; NGC, Ngorongoro Crater, Tanzania; KRU, Kruger National Park, South Africa; BOT-I, southern Botswana and Kalahari, South Africa; BOT-II, northern Botswana; and NAM, Namibia. Green squares represent captive individual samples to explore the relationship of lions from more isolated/ endangered/depleted areas: ATL, Morocco Atlas lions (n = 4); ANG, Angola (n = 2); and ZBW, Zimbabwe (n = 1). Deduced historical expansions (M1 and M2) are represented by red arrows (see text). (B) Haplotype frequencies observed in the 11 lion populations for nDNA (ADA and TF), and mtDNA (12S-16S) genes, paralleled with the FIV Ple serum-prevalence frequencies (black - sero-positive; gray -indeterminate; white -sero-negative). Population sample sizes are indicated within parenthesis. (C) Statistical parsimony networks of lion ADA, TF, and 12S-16S haplotypes. Circle size is proportional to the haplotype frequency and crossbars represent the number of step mutations connecting haplotypes. The mtDNA haplotypes H5 and H6 are shaded gray as they were detected only in the individual samples from ANG, ATL, and ZBW, which do not group in unique population clusters (see text).

doi:10.1371/journal.pgen.1000251.g001 yield similar inferences of population history, as coalescent times vary as a function of their pattern of inheritance [21]. There is also a large variance in coalescent times across loci sharing a common pattern of inheritance especially in complex demographical histories (Table 1). However, the accurate interpretation of the differences among loci can provide a more resolved and coherent population history, affording more-nuanced insights on past demographic processes, levels of admixture, taxonomic issues, and on the most appropriate steps for effective conservation and management of remaining populations.

The goal of this study was to assess the evolutionary history of lion by (1) characterizing lion population structure relative to patterns of FIV Ple genetic variation, (2) detect signatures of migration using both host and viral population genomics, and (3) reconstruct lion demographic history and discuss its implication for lion conservation. We assess genetic variation from 357 lions from most of its current distribution, including mitochondrial (mtDNA; 12S-16S, 1,882 bp), nuclear (nDNA) Y-chromosome (SRY, 1,322 bp) and autosomal (ADA, 427 bp; TF, 169 bp) sequences, and 22 microsatellites markers. We further document patterns of FIV Ple variation in lions (FIV Ple pol-RT gene, up to 520 bp).

Genetic analyses of 357 lions from throughout the extant species range showed that paternally inherited nDNA (SRY) and maternally inherited (mtDNA) sequence variation was generally low (only one paternal SRY-haplotype and 12 mtDNA haplotypes; $p = 0.0066$) (Figure 1 ; Figure S1 ; Tables S1 and S2). The most common mtDNA haplotype H11 was ubiquitous in Uganda/Tanzania and parts of Botswana/South Africa, H1 was common in Southern Africa, and H7 and H8 were unique to Asian lions. The autosomal nDNA sequences showed fairly distinct patterns of variation (Figure 1 ; Figure S1). Of the five ADA haplotypes, A2 was the most common and most-widely distributed. The other four haplotypes, which are derived and much less common, included one (A5) that was fixed in Asian lions. The three TF haplotypes were more widely and evenly distributed.

Levels of population subdivision among lions were assessed using microsatellite and sequencing data. Eleven groups were identified using Bayesian analyses [22] and three-dimensional factorial correspondence analyses [23] (Figure 2 ; Table S3). Most clusters represented geographically circumscribed populations: Namibia (NAM), Kruger National Park (KRU), Ngorongoro Crater (NGC), Kenya (KEN), Uganda (UGA), and Gir (GIR). Two distinct clusters were found in Botswana, BOT-I that included lions from southern Botswana and Kalahari (South Africa) ($F_k = 0.24$) and BOT-II found exclusively in northern Botswana ($F_k = 0.18$). Surprisingly, three distinct clusters were found in a single geographical locale (approximately 60640 km square) in the large panmyctic population of the Serengeti National Park (SER-I/SER-II/SER-III) ($F_k = 0.18, 0.21, \text{ and } 0.15$, respectively).

Two captive lions from Angola (ANG), one from Zimbabwe (ZBW) and four Morocco Zoo Atlas lions (ATL; presently extinct from the wild) (Figure 1A) were included in the analyses to explore the relationship of lions from more isolated, endangered, or depleted areas.

ANG and ZBW lions were assigned to BOT-II ($q = 0.90$ and 0.87 ; 90%CI: 0.47-1.00) and KRU ($q = 0.85$; 90%CI: 0.52-1.00) (Bayesian analyses [22]) populations, respectively, as expected based on their geographical proximity. However, these lions differed from BOT-II and KRU by up to 8 mtDNA mutations, sharing haplotypes with the ATL lions (H5 in ANG and H6 in ZBW) (Figure 1B and 1C). The ATL lions did not group in a unique cluster.

Both nDNA and mtDNA pairwise genetic distances among the 11 lion populations showed a significant relationship with geographic distance ($R^2 = 0.75$; Mantel's test, $P = 0.0097$; and $R^2 = 0.15$; Mantel's test, $P = 0.0369$; respectively) (Figure 3). The significant positive and monotonic correlation across all the scatterplot pairwise comparisons for the nDNA markers (biparental) was consistent with isolation-by-distance across the sampled region. However, the correlation between nDNA F_{ST} and geographic distance considerably decreased when the Asian GIR population was removed ($R^2 = 0.19$; Mantel's test, $P = 0.0065$) suggesting that caution should be taken in interpreting the pattern of isolation-by-distance in lions. We further compared linearized F_{ST} estimates [24] plotted both against the geographic distance (model assuming habitat to be arrayed in an infinite onedimensional lattice) and the log geographic distance (model assuming an infinite two-dimensional lattice). The broad distribution of lions might suggest a priori that a two-dimensional isolationby-distance model would provide the best fit for the nDNA data ($R^2 = 0.25$; Mantel's test, $P = 0.0022$), but instead the onedimensional isolation-by-distance model performed better ($R^2 = 0.71$; Mantel's test, $P = 0.0476$) (Figure S2). The pattern observed for the mtDNA (maternal) was more complex. While there was a significant relationship between mtDNA F_{ST} and geographic distance, there was an inconsistent pattern across broader geographic distances (Figure 3). This is partly due to the fixation or near fixation of haplotype H11 in six populations and the fixation of a very divergent haplotype H4 in KEN population (Figure 1B and 1C). The removal of the KEN population considerably increased the correlation between mtDNA F_{ST} and geographic distance ($R^2 = 0.27$; Mantel's test, $P = 0.0035$). Thus, the null hypothesis of regional equilibrium for mtDNA across the entire sampled region is rejected despite the possibility that isolation-by-distance may occur regionally.

These contrasting nDNA and mtDNA results may be indicative of differences in dispersal patterns between males and females, which would be consistent with evidence that females are more phylopatric than males. Alternatively, selection for matrilineally transmitted traits upon which neutral mtDNA alleles hitchhike is possible, given the low values of nucleotide diversity of the mtDNA ($p = 0.0066$). A similar process has been

suggested in whales ($p = 0.0007$) [25] and African savannah elephants ($p = 0.0200$) [26], where both species have female philopatry and like lions, a matriarchal social structure. However, genetic drift tends to overwhelm selection in small isolated populations, predominantly affecting haploid elements due to its lower effective population size (Table 1). Therefore, we suggest that the contrasting results obtained for nDNA and mtDNA are more likely further evidence that lion populations underwent severe bottlenecks. The highly structured lion matrilineages comprise four monophyletic mtDNA haplo-groups (Figure 4A; Figure S3). Lineage I consisted of a divergent haplotype H4 from KEN, lineage II was observed in most Southern Africa populations, lineage III was widely distributed from Central and Northern Africa to Asia, and lineage IV occurred in Southern and Eastern Africa.

Seroprevalence studies indicate that FIV Ple is endemic in eight of the 11 populations but absent from the Asian GIR lions in India and in Namibia and southern Botswana/Kalahari regions (NAM/ BOT-I) in Southwest Africa (Figure 1B). Phylogenetic analysis of the conserved pol-RT region in 117 FIV-infected lions depicted monophyletic lineages [15, 18] that affirm six distinct subtypes (A-F) that are distributed geographically in three distinct patterns (Figure 4B; Figure S4). First, multiple subtypes may circulate within the same population as exemplified by subtypes A, B and C all ubiquitous within the three Serengeti lion populations (SER-I, SER-II and SER-III) and subtypes A and E within lions of Botswana (BOT-II) (Figure 4B and 4C and Figure S4). Second, unique FIV Ple subtypes may be restricted to one location as subtype F in Kenya, subtype E in Botswana (BOT-II), subtype C in Serengeti, and subtype D in Krugar Park (Figure 4B and 4C and Figure S4). Third, intra-subtype strains cluster geographically, as shown by distinct clades within subtype A that were restricted to lions within Krugar Park, Botswana and Serengeti and within subtype B that corresponded to Uganda, Serengeti and Ngorongoro Crater lions (Figure 4B and 4C and Figure S4).

Not unexpectedly, FIV Ple pairwise genetic distances, represented as population F_{ST} among the eight lion FIV-infected populations, were not significantly correlated with geographic distance ($R^2 = 0.08$; Mantel's test, $P = 0.165$) (Figure 3), and affirms that patterns of viral dissemination do not conform to a strict isolation-by-distance model. Rather, the two distinct clusters observed (Figure 3) reflect the complex distribution of FIV Ple among African lions. Indeed, despite the low geographic distance within East-African lion populations, the FIV Ple genetic divergence showed a broader range in F_{ST} (0.03 to 0.79 for most of first cluster; Figure 3). By contrast, approximately half of the range in F_{ST}

(0.26 to 0.69 for the second cluster; Figure 3) was observed among East and Southern Africa in spite of its large geographic separation. In contrast with the patterns observed in lions, linearized F_{ST} estimates [24] for FIV Ple were better correlated with log geographic distance (two-dimensional lattice model) ($R^2 = 0.15$) than with geographic distance (one-dimensional model) ($R^2 = 0.02$), although in both cases the Mantel's test was not significant ($P=0.2474$) (Figure S2).

The mtDNA coalescence dating suggested that the East African lineage I (KEN haplotype H4) had an old origin of ,324,000 years (95% CI: 145,000-502,000). Extant East African populations (KEN/NGC/SER-I/SER-II/SER-III) also showed a slightly significant higher nDNA allelic richness and genetic diversity (Table S4) . Moreover, the FIV Ple subtype diversity was higher in East African clades (exhibiting four out of the six known viral-strains), including the most divergent FIV Ple subtype C (Figure 4B and 4C). These genetic data from lions and FIV Ple is consistent with the older origin of extant East African lions, which is further supported by the oldest lion fossils discovered in East Africa [1] .

Relative to East Africa, Southern lions have a slightly more recent mtDNA coalescence. Lineage II, found in NAM, BOT-II and KRU has an estimated coalescence of 169,000 years (95% CI: 34,000-304,000) and the more widespread lineage IV found in the Southern populations of BOT-I, BOT-II and KRU as well as the Eastern populations of SER (I, II, and III), NGC and UGA, coalesces ,101,000 years ago (95% CI: 11,000-191,000). However, the similar levels of nDNA genetic diversity, the occurrence of an exclusively Southern mtDNA lineage II and highly divergent FIV Ple subtypes, FIV Ple subtype D found only in KRU and subtype E exclusive to BOT-II, suggests that both East and Southern Africa were important refugia for lions during the Pleistocene. Therefore, the co-occurrence of divergent mtDNA haplotypes (6 to 10 mutations; Figure 1B and 1C) in southern populations may be the consequence of further isolation within refugia during colder climatic periods. Contemporary fragmentation of lion populations could further explain the results of nested-clade phylogeographical analysis (NCPA [27]) (Figure S5), which inferred restricted gene flow with isolation-by-distance between mtDNA haplotypes H9 (BOT-II) and H10 (KRU) ($\chi^2 = 10.00$, $P = 0.0200$), between haplotypes H1 (BOT-II/NAM) and H2 (KRU) ($\chi^2 = 71.00$, $P=0.0001$), and between haplotypes H9-H10 (BOT-II/KRU) and haplotypes H11-H12 (BOT-I/KRU/SER/NGC/UGA) ($\chi^2 = 187.83$, $P=0.0001$).

Further isolation within refugia (sub-refugia) may also have occurred in East Africa. This is suggested by the distinctive mtDNA haplotype H4 and the unique FIV Ple subtype F found in the Kenya population, which may have resulted from reduced gene flow across the Rift valley, a scenario that has been suggested for several bovid and carnivore populations (see [28] and references therein).

The best example of concordance between host genome markers and viral transmission patterns is observed in the Serengeti National Park in Tanzania. Our previous findings described markedly high levels of FIV Ple subtype A, B and C circulating within the Serengeti lion population to such an extent that 43% of the lions sampled were multiply-infected with two or three subtypes [15, 18] and were hypothesized to represent recent admixture of three formerly separated populations. Such result is confirmed here by lion genomic markers (Figure 2). Further, although lions within the Serengeti can be assigned to one of three populations (SER-I, SER-II or SER-III) by host genomic markers, FIV Ple subtypes are distributed ubiquitously in all three, characteristic of rapid horizontal retroviral transmission subsequent to host population admixture. The possible isolating mechanism remains to be elucidated as there is no apparent barrier to gene flow in this ecosystem.

Based on patterns of genetic diversity and phylogenetic analysis of lion nDNA/mtDNA and FIV Ple markers, we propose a scenario of a period of refugia/isolation in the Late Pleistocene followed by two major lion expansions across Africa and Asia. The first expansion, supported by the mtDNA NCPA [27] ($\chi^2 = 690.00$, $P = 0.0001$; Figure S5), was a long-distance colonization of mtDNA lineage-III (GIR/ATL/ANG/ZBW) around 118,000 years ago (95% CI: 28,000-208,000), with subsequent fragmentation of haplotypes H5-H6 into Central and North Africa and haplotypes H7-H8 into West Asia (M1- Figure 1A) . Support for this initial expansion is also found in nDNA. The ADA haplotype A5 fixed in GIR is also present in KEN, SER-II, and SER-III, suggesting that lions likely colonized West Asia from the East Africa refugia (Figure 1B) . Such an expansion may have been favored by the start of a warmer and less arid period in Africa 130,000-70,000 years ago [29] . This "out-of-Africa event" would have occurred much later than the initial lion expansion through Eurasia based on fossils (,500,000 years ago) [3] . It is likely that multiple lion expansions occurred in the Pleistocene, as occurred with humans [21] .

A second, more recent lion expansion probably occurred at the Pleistocene/Holocene transition, this one from Southern Africa toward East Africa (M2- Figure 1A, Figure 3). This is reflected in the mtDNA lineage IV, where haplotypes present in Southern lions are basal (older) to those found in the East. Overall, mtDNA population nucleotide diversity decreases from Southern to East Africa (Figure 1B and 1C) , a finding supported by pairwise mismatch analysis [30] (raggedness, $r = 0.086$; $P, 0.001$). The fixation of mtDNA haplotype H11 in BOT-I (otherwise fixed only in East Africa populations) suggests that the colonizing lions expanded northwards from the Kalahari Desert, which included bush, woodland and savannah habitats during the climatic fluctuations of the Pleistocene [31] . This expansion would have occurred relatively recently as the single rare tip mtDNA haplotype H12, found only in SER-I, is derived from the interior widespread haplotype H11 (,14,000-7,000 years; given one mtDNA substitution every 7,000 years; Table 1). This expansion is also supported by FIV Ple subtype A where haplotypes present in Southern lions (KRU and BOT-I) are basal to those found in the East (SER-I, SER-II and SER-III) and a decrease of nucleotide diversity of this FIV Ple subtype is observed from Southern ($p = 0.15$) to Eastern Africa ($p = 0.03$) (Figure 3B). Interestingly, a similar northward colonization process from Southern Africa has been suggested for some of the lion preys, namely the impala, greater kudu, and wildebeest [32, 33] .

If we had restricted our inferences to mtDNA, we might have concluded that East African lion populations, which are fixed or nearly fixed for haplotype H11, went extinct during the Pleistocene/Holocene transition (similar to the well known mega-fauna extinctions of the Late Pleistocene [34]) and were then colonized by Southern populations. However, our population genomics data better fit a scenario of lion population expansion and interbreeding rather than simple replacement. First, genetic diversity and allelic richness at nDNA are slightly higher in East Africa populations relatively to those in Southern Africa. This is contrary to the expected pattern of population expansion in which there is usually a progressive decline in genetic diversity and allelic richness. Second, SER lions carry two diverse FIV Ple subtypes found only in East Africa (B-C), and not only FIV Ple subtype A, which was presumably introduced in East Africa coincidently with the mtDNA expansion event northwards from South. Third, the East African FIV Ple subtype B found in UGA/SER-I/SER-II/SER-III/NGC showed evidence of a population expansion (raggedness, $r = 0.004$; $P, 0.01$; $F_s = 220.37$; $P, 0.00001$) and the highest nucleotide diversity observed within FIV Ple subtypes ($p = 0.09$). Four, the FIV Ple subtype diversity is higher in East African clades (four out of the six viral strains).

The utility of FIV Ple pol-RT as a marker of lion population structure and natural history is that it can be informative on a contemporaneous time scale, though it may be less useful at capturing more ancient demographic events. The extreme divergence among FIV Ple subtypes, considered with high sero-prevalence in eight of the 11 lion populations, and combined with patterns of geographic concordance, support the hypothesis that FIV Ple is not a recent emergence within modern lions [35]. Populations that harbor one private FIV Ple subtype such as KEN (subtype F), BOT-II (subtype E), and KRU (subtype D) must have been sufficiently isolated for enough time for the virus to evolve into unique subtypes, a result corroborated by the high nDNA and mtDNA genetic structure present in these lion populations (Figure 4). Thus, it is possible that the initial emergence of FIV Ple pre-dates the Late-Pleistocene expansions of contemporary lion populations [36], but present day distributions are more useful indicators of very recent host population dynamics, a result also observed with FIV Pco in a panmictic population of pumas in western North America [19].

Accurate interpretation of past and contemporary population demographic scenarios is a primary goal for the effective conservation of endangered species. In this study, we found substantial population subdivision, reduced gene flow, and large differences in FIV Ple sequence and sero-prevalence among lion populations, as well as evidence of historic secondary contact between populations (Figure 3C; Table S4 to S9). The very low population level of mtDNA nucleotide diversity, the number of haplotypes private to a single population (Figure 1), and probably also the lack of SRY genetic variation across all male lions (haplotype S1, $n = 183$) suggests that lion numbers diminished considerably following the Late Pleistocene. The last century reduction in lion distribution further eroded its genomic diversity, and microsatellite variation suggested recent population bottlenecks in seven out of the 11 populations (standardized differences test, $P < 0.05$; Table S5) [37].

Although we did not explicitly try to address the adequacy of lion subspecies designations (currently only one African subspecies is widely recognized) [38, 39], we provided strong evidence that there is no evidence of substantial genetic exchange of matrilineages among existing populations as the AMOVA [40] withinpopulation component was uniformly high in all distinct subdivision scenarios ($W_{ST} < 0.920$; $P < 0.0001$; three-six groups; Table S6). Similarly, significant population structure was detected from nDNA ($F_{ST} = 0.18$), with low levels of admixture evident from Bayesian analysis [22] ($a = 0.033$). Therefore, employing a bottom-up perspective that prioritizes populations, rather than large-scale units (e.g. all

African lions), might preserve and maintain lion diversity and evolutionary processes most efficiently [41] .

A total of 357 individuals were obtained across most of the lion range in Africa and Asia (Figure 1A ; Table S1). Genetic variation among lion specimens was assessed using maternal (12S and 16S genes), paternal (SRY gene) and bi-parental autosomal (22 microsatellite loci, and the ADA and TF genes) markers (GenBank accession numbers: FJ151632-FJ151652). Analyses of mtDNA in *Panthera* species are complicated by the presence of a 12.5 kb mtDNA integration into chromosome F2 [42] . Accordingly, mtDNA specific primers were designed for the 12S and 16S genes (Table S2) and we used long-range PCR amplification. We designed primers to amplify segments of the ADA (exon 10 and intron 10) and the TF (intron 3) genes (Table S2) , two of the most variable protein loci in lion populations [5] , localized on the domestic cat *Felis catus* chromosome A3p and C2q, respectively. The Y-chromosome SRY-39UTR gene was also amplified [43] .

PCR products were amplified from 50 ng of genomic DNA in a 25 μ L reaction system containing 1.5 mM MgCl₂ , 1.0 mM dNTPs, 0.25 units of AmpliTaq Gold DNA polymerase (Applied Biosystems), and 16 PCR buffer II; the amplification protocol was: denaturation 10 min at 95°C, a touch-down cycle of 95°C for 30 s, 52°C for 60 s decreased by 1°C in the next cycle for 10 cycles, 72°C for 120 s, then 35 amplification cycles of 95°C for 30 s, 52°C for 60 s, and 72°C for 120 s, followed by an extension of 10 min at 72°C. PCR products were sequenced on an ABI 377. Sequences were aligned and cleaned using SEQUENCHER (Gene Codes).

Twenty two polymorphic microsatellite loci (20 dinucleotide repeats: FCA006, FCA008, FCA014, FCA069, FCA077, FCA085, FCA091, FCA098, FCA105, FCA126, FCA129, FCA139, FCA205, FCA208, FCA211, FCA224, FCA229, FCA230, FCA247, and FCA281; and two tetranucleotide repeats: FCA391 and FCA441) were amplified [44] . Microsatellites were scored in an ABI 377 and analyzed using GENESCAN 2.1 and GENOTYPER 2.5. These loci are located on 11 of the 19 *F. catus* chromosomes, occurring in different linkage groups or at least 12 centimorgans apart [44, 45] .

Western blots using domestic cat and lion FIV as antigen were performed as previously described [46, 47] . The supernatant from virus-infected cells was centrifuged at 200 g for

10 min at 5uC. The resultant supernatant was centrifuged at 150,000 g at 4uC for 2 hours. Pelleted viral proteins were resuspended in 1/20 th of the original volume and total protein content was assayed using the Biorad Protein Assay. Twenty mg of viral protein were run on 4-20% Tris-Glycine gels and transferred to PDVF membranes (BioRad). Membrane strips were exposed 2-12 h to a 1:25 or 1:200 dilution of serum or plasma. After washing, samples were labeled with goat anti-cat HRP or phosphate conjugated antibody (KPL laboratories) at a 1:2000 dilution, washed, and incubated in ECL Western Blotting detection reagents (Amersham Biosciences) for 2 min, then exposed to Lumi-Film Chemiluminescent Detection Film (Boehringer Mannheim) or incubated in BCIP/ NBP phosphatase substrate (KPL laboratories) for 15 min [46] [47] [48] . Results were visualized and scored manually based on the presence and intensity of antibody binding to the p24 gag capsid protein.

Nested PCR amplification of partial FIV Ple pol-RT was performed [18, 46] . Briefly, first round PCR reactions used 100 ng of genomic DNA, 2.5 mM MgCl₂ and an annealing temperature of 52uC. Second round PCR reactions used identical conditions with 1-5 ml of first-round product as template. All PCR products were sequenced as described above for lion genetic analyses (GenBank accession numbers: AY549217-AY552683; AY878208-AY878235; FJ225347-FJ225382).

We used the GENETIX 4.02 [49] , GENEPOP 3.3 [50] , MICROSAT [51] , and DNASP 4.10 [52] to calculate the following descriptive statistics: (i) percentage of polymorphic loci (P_{95}), number of alleles per locus (A), observed and expected heterozygosity (H_E and H_O), and number of unique alleles (A_U); (ii) assess deviations from HWE; (iii) estimate the coefficient of differentiation (F_{ST}), and (iv) nucleotide (p) and haplotype (h) diversity.

We tested the hypothesis that all loci are evolving under neutrality for both the lion and the FIV Ple data. For frequency data, we used the method described by Beaumont and Nichols [53] and implemented in FDIST (<http://www.rubic.rdg.ac.uk/mab/software.html>). The F_{ST} values estimated from microsatellite loci plotted against heterozygosity showed that all values fall within the expected 95% confidence limit and consequently no outlier locus were identified. For sequence data (lion nDNA/ mtDNA and FIV Ple pol-RT), we ruled out any significant evidence for genetic hitchhiking and background selection by assessing Fu and Li's D^* and F^* tests [54] and Fu's F_S statistics [55] .

A Bayesian clustering method implemented in the program STRUCTURE [22] was used to infer number of populations and assign individual lions to populations based on multilocus genotype (microsatellites) and sequence data (ADA, TF, and mtDNA genes) and without incorporating sample origin. For haploid mtDNA data, each observed haplotype was coded with a unique integer (e.g. 100, 110) for the first allele and missing data for the second (STRUCTURE [22] analyses with or without the mtDNA data were essentially identical). For K population clusters, the program estimates the probability of the data, $\Pr(X|K)$, and the probability of individual membership in each cluster using a Markov chain Monte Carlo method under the assumption of Hardy-Weinberg equilibrium (HWE) within each cluster. Initial testing of the HWE in each of the populations defined by the geographic origin of sampling revealed no significant deviation from HW expectations with the exception of SER and BOT population (later subdivided by STRUCTURE [22] in 3 and 2 clusters, respectively; such deviations from HW expectations were interpreted as evidence of further population structuring). We conducted six independent runs with $K = 1-20$ to guide an empirical estimate of the number of identifiable populations, assuming an admixture model with correlated allele frequencies and with burn-in and replication values set at 30,000 and 10 6 , respectively. STRUCTURE also estimates allele frequencies of a hypothetical ancestral population and an alpha value that measures admixed individuals in the data set. The assignment of admixed individuals to populations using STRUCTURE [22] has been considered in subsequent population analyses. For each population cluster k, the program estimates F_k , a quantity analogous to Wright's F_{ST} , but describing the degree of genetic differentiation of population k from the ancestral population.

Patterns of gene flow and divergence among populations were described using a variety of tests. First, to visualize subtle relationships among individual autosomal genotypes, threedimensional factorial correspondence analyses [23] (FCA) were performed in GENETIX [49], which graphically projects the individuals on the factor space defined by the similarity of their allelic states. Second, neighbor-joining (NJ) analyses implementing the Cavalli-Sforza & Edwards' chord genetic distance [56] (D_{CE}) were estimated in PHYLIP 3.6 [57], and the tree topology support was assessed by 100 bootstraps. Third, the difference in average H_O and A was compared among population groups using a twosided test in FSTAT 2.9.3.2 [58], which allows to assess the significance of the statistic OS_x using 1,000 randomizations. Four, the equilibrium between drift and gene flow was tested using a regression of pairwise F_{ST} on geographic distance matrix among all populations for host nDNA(microsatellites)/mtDNA and FIV Ple data. A Mantel test [59] was used to estimate the 95% upper probability for each matrix correlation. Assuming a stepping stone model of migration where gene flow is more likely between adjacent populations, one can reject the

null hypothesis that populations in a region are at equilibrium if (1) there is a nonsignificant association between genetic and geographic distances, and/or (2) a scatterplot of the genetic and geographic distances fails to reveal a positive and monotonic relationship over all distance values of a region [60]. We also evaluated linearized F_{ST} [i.e. $F_{ST}/(12F_{ST})$] [24] among populations. We tested two competing models of isolation-by-distance, one assuming the habitat to be arrayed in an infinite one-dimensional lattice and another assuming an infinite two-dimensional lattice. Both models showed that genetic differentiation increased with raw and logtransformed Euclidean distances, respectively [24]. We determined the confidence interval value of the slope of the regression for the nDNA data using a non parametric ABC bootstrap [61] in GENEPOP 4.0 [62].

The demographic history of populations was compared using a variety of estimators based on the coalescence theory. First, signatures of old demographic population expansion were investigated for mtDNA and FIV Pol-RT haplotypes using pairwise mismatch distributions [63] in DNASP [52]. The goodness-of-fit of the observed data to a simulated model of expansion was tested with the raggedness (r) index [64].

Second, the occurrence of recent bottlenecks was evaluated for microsatellite data using the method of Cornuet & Luikart [37] in BOTTLENECK [65] and using 10,000 iterations. This approach, which exploits the fact that rare alleles are generally lost first through genetic drift after reduction in population size, employs the standardized differences test, which is the most appropriate and powerful when using 20 or more polymorphic loci [37]. Tests were carried out using the stepwise mutation model (SMM), which is a conservative mutation model for the detection of bottleneck signatures with microsatellites [66].

Third, to discriminate between recurrent gene flow and historical events we used the nested-clade phylogeographical analysis [27, 67] (NCPA) for the mtDNA data. When the null hypothesis of no correlation between genealogy and geography is rejected, biological inferences are drawn using a priori criteria. The NCPA started with the estimation of a 95% statistical parsimony [68] mtDNA network in TCS 1.20 [69]. Tree ambiguities were further resolved using a coalescence criteria [70]. The network was converted into a series of nested branches (clades) [71, 72], which were then tested against their geographical locations through a permutational contingency analysis in GEODIS 2.2 [73]. The inferences obtained were also corroborated with the automated implementation of the NCPA in ANECA [74]. To address potential weaknesses in some aspects of the NCPA analysis [75,

76] , we further validated the NCPA inferences with independent methods for detecting restricted gene-flow/isolation-by-distance (using matrix correlation of pairwise F_{ST} and geographic distance) and population expansion (using pairwise mismatch distributions).

Four, to test the significance of the total mtDNA genetic variance, we conducted hierarchical analyses of molecular variance [40] (AMOVA) using ARLEQUIN 2.0 [77] . Total genetic variation was partitioned to compare the amount of difference among population groups, among populations within each groups, and within populations.

Phylogenetic relationships among mtDNA and FIV Ple pol-RT sequences were assessed using Minimum evolution (ME), Maximum parsimony (MP), and Maximum likelihood (ML) approaches implemented in PAUP [78] . The ME analysis for mtDNA consisted of NJ trees constructed from Kimura two-parameter distances followed by a branch-swapping procedure and for FIV Ple data employed the same parameter estimates as were used in the ML analysis. The MP analysis was conducted using a heuristic search, with random additions of taxa and tree-bisection-reconnection branch swapping. The ML analysis was done after selecting the best evolutionary model fitting the data using MODELTEST 3.7 [79] . Tree topologies reliability was assessed by 100 bootstraps. For the FIV Ple data, the reliability of the tree topology was further assessed through additional analyses using 520 bp of FIV Ple pol-RT sequences in a representative subset of individuals.

The time to the most recent common ancestor (TMRCA) for the ADA and TF haplotypes was estimated following Takahata et al. [80] , where we calculate the ratio of the average nucleotide differences within the lion sample to one-half the average nucleotide difference between leopards (*P. pardus*) and lions and multiplying the ratio by an estimate of the divergence time between lions and leopards (2 million years based on undisputed lion fossils in Africa) [81, 82] . The mtDNA TMRCA was estimated with a linearized tree method in LINTREE [83] and using the equation $H = 2mT$, where H was the branch height (correlated to the average pairwise distance among haplotypes), m the substitution rate, and T the divergence time. Leopard and snow leopard (*P. uncia*) sequences were used as outgroups. Inference of the TMRCA for microsatellite loci followed Driscoll et al. [6] where the estimate of microsatellite variance in average allele repeat-size was used as a surrogate for evolutionary time based on the rate of allele range reconstitution subsequent to a severe founder effect. Microsatellite allele variance has been shown to be a reliable estimator for microsatellite evolution and demographic inference in felid species [6] . [24] for lion (nDNA

and mtDNA) and FIV Ple (pol-RT) genetic data plotted both against the geographic distance (model assuming habitat to be arrayed in an infinite one-dimensional lattice; one-dimension isolation-by-distance [IBD]) and the log geographic distance (model assuming an infinite two-dimensional lattice; twodimension isolation-by-distance) on geographic distance. Found at: doi:10.1371/journal.pgen.1000251.s002 (0.13 MB PDF) Figure S3 Phylogenetic relationships of the 12S-16S mtDNA lion haplotypes. Neighbour-joining tree of the 1,882 bp 12S-16S mtDNA sequences. Bootstrap values are placed at each branchpoint for the minimum evolution/maximum parsimony/maximum likelihood analyses, respectively (ME/MP/ML). Outgroups: Ppaleopard, Panthera pardus; Pun -snow-leopard, Panthera uncia. The symbol (N) represents nodes with bootstrap support ,50 or an inferred polytomy in the bootstrap 50% majority-rule consensus tree.

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Beyond phage display: non-traditional applications of the filamentous bacteriophage as a vaccine carrier, therapeutic biologic, and bioconjugation scaffold

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Abstract: For the past 25 years, phage display technology has been an invaluable tool for studies of protein–protein interactions. However, the inherent biological, biochemical, and biophysical properties of filamentous bacteriophage, as well as the ease of its genetic

manipulation, also make it an attractive platform outside the traditional phage display canon. This review will focus on the unique properties of the filamentous bacteriophage and highlight its diverse applications in current research. Particular emphases are placed on: (i) the advantages of the phage as a vaccine carrier, including its high immunogenicity, relative antigenic simplicity and ability to activate a range of immune responses, (ii) the phage's potential as a prophylactic and therapeutic agent for infectious and chronic diseases, (iii) the regularity of the virion major coat protein lattice, which enables a variety of bioconjugation and surface chemistry applications, particularly in nanomaterials, and (iv) the phage's large population sizes and fast generation times, which make it an excellent model system for directed protein evolution. Despite their ubiquity in the biosphere, metagenomics work is just beginning to explore the ecology of filamentous and non-filamentous phage, and their role in the evolution of bacterial populations. Thus, the filamentous phage represents a robust, inexpensive, and versatile microorganism whose bioengineering applications continue to expand in new directions, although its limitations in some spheres impose obstacles to its widespread adoption and use.

Text: The filamentous bacteriophage (genera Inovirus and Plectrovirus) are non-enveloped, rod-shaped viruses of *Escherichia coli* whose long helical capsids encapsulate a single-stranded circular DNA genome. Subsequent to the independent discovery of bacteriophage by Twort (1915) and d'Hérelle (1917), the first filamentous phage, f1, was isolated in Loeb (1960) and later characterized as a member of a larger group of phage (Ff, including f1, M13, and fd phage) specific for the *E. coli* conjugative F pilus (Hofschneider and Mueller-Jensen, 1963; Marvin and Hoffmann-Berling, 1963; Zinder et al., 1963; Salivar et al., 1964). Soon thereafter, filamentous phage were discovered that do not use F-pili for entry (If and Ike; Meynell and Lawn, 1968; Khatoon et al., 1972), and over time the list of known filamentous phage has expanded to over 60 members (Fauquet et al., 2005), including temperate and Gram-positivetropic species. Work by multiple groups over the past 50 years has contributed to a relatively sophisticated understanding of filamentous phage structure, biology and life cycle (reviewed in Marvin, 1998; Rakonjac et al., 2011; Rakonjac, 2012).

In the mid-1980s, the principle of modifying the filamentous phage genome to display polypeptides as fusions to coat proteins on the virion surface was invented by Smith and colleagues (Smith, 1985; Parmley and Smith, 1988). Based on the ideas described in Parmley and Smith (1988), groups in California, Germany, and the UK developed phage-display platforms to create and screen libraries of peptide and folded-protein variants

(Bass et al., 1990; Devlin et al., 1990; McCafferty et al., 1990; Scott and Smith, 1990; Breitling et al., 1991; Kang et al., 1991) . This technology allowed, for the first time, the ability to seamlessly connect genetic information with protein function for a large number of protein variants simultaneously, and has been widely and productively exploited in studies of protein-protein interactions. Many excellent reviews are available on phage-display libraries and their applications (Kehoe and Kay, 2005; Bratkovic, 2010; Pande et al., 2010) . However, the phage also has a number of unique structural and biological properties that make it highly useful in areas of research that have received far less attention.

Thus, the purpose of this review is to highlight recent and current work using filamentous phage in novel and nontraditional applications. Specifically, we refer to projects that rely on the filamentous phage as a key element, but whose primary purpose is not the generation or screening of phage-displayed libraries to obtain binding polypeptide ligands. These tend to fall into four major categories of use: (i) filamentous phage as a vaccine carrier; (ii) engineered filamentous phage as a therapeutic biologic agent in infectious and chronic diseases; (iii) filamentous phage as a scaffold for bioconjugation and surface chemistry; and (iv) filamentous phage as an engine for evolving variants of displayed proteins with novel functions. A final section is dedicated to recent developments in filamentous phage ecology and phage-host interactions. Common themes shared amongst all these applications include the unique biological, immunological, and physicochemical properties of the phage, its ability to display a variety of biomolecules in modular fashion, and its relative simplicity and ease of manipulation.

Nearly all applications of the filamentous phage depend on its ability to display polypeptides on the virion's surface as fusions to phage coat proteins (Table 1) . The display mode determines the maximum tolerated size of the fused polypeptide, its copy number on the phage, and potentially, the structure of the displayed polypeptide. Display may be achieved by fusing DNA encoding a polypeptide of interest directly to the gene encoding a coat protein within the phage genome (type 8 display on pVIII, type 3 display on pIII, etc.), resulting in fully recombinant phage. Much more commonly, however, only one copy of the coat protein is modified in the presence of a second, wild-type copy (e.g., type 8+8 display if both recombinant and wild-type pVIII genes are on the phage genome, type 8+3 display if the Parmley and Smith (1988), McConnell et al. (1994) , Rondot et al. (2001) Hybrid (type 3+3 and 3+8 systems) Type 3+3 system <1 2 Smith and Scott (1993) , Smith and Petrenko (1997) pVI Hybrid (type 6+6 system) Yes <1 2 >25 kDa Hufton et al. (1999) pVII

Fully recombinant (type 7 system) No ~5 >25 kDa Kwasnikowski et al. (2005) Hybrid (type 7+7 system) Yes <1 2 Gao et al. (1999) pVIII Fully recombinant (landscape phage; type 8 system)

No 2700 3 ~5-8 residues Kishchenko et al. (1994) , Petrenko et al. (1996) Hybrid (type 88 and 8+8 systems) Type 8+8 system ~1-300 2 >50 kDa Scott and Smith (1990) , Greenwood et al. (1991) , Smith and Fernandez (2004) pIX Fully recombinant (type 9+9 * system) Yes ~5 >25 kDa Gao et al. (2002) Hybrid (type 9+9 system) No <1 2 Gao et al. (1999) , Shi et al. (2010) , Tornetta et al. (2010) 1 Asterisks indicate non-functional copies of the coat protein are present in the genome of the helper phage used to rescue a phagemid whose coat protein has been fused to a recombinant polypeptide. 2 The copy number depends on polypeptide size; typically <1 copy per phage particle but for pVIII peptide display can be up to ~15% of pVIII molecules in hybrid virions. 3 The total number of pVIII molecules depends on the phage genome size; one pVIII molecule is added for every 2.3 nucleotides in the viral genome. recombinant gene 8 is on a plasmid with a phage origin of replication) resulting in a hybrid virion bearing two different types of a given coat protein. Multivalent display on some coat proteins can also be enforced using helper phage bearing nonfunctional copies of the relevant coat protein gene (e.g., type 3 * +3 display). By far the most commonly used coat proteins for display are the major coat protein, pVIII, and the minor coat protein, pIII, with the major advantage of the former being higher copy number display (up to ~15% of recombinant pVIII molecules in a hybrid virion, at least for short peptide fusions), and of the latter being the ability to display some folded proteins at an appreciable copy number (1-5 per phage particle). While pVIII display of folded proteins on hybrid phage is possible, it typically results in a copy number of much less than 1 per virion (Sidhu et al., 2000) . For the purposes of this review, we use the term "phage display" to refer to a recombinant filamentous phage displaying a single polypeptide sequence on its surface (or more rarely, bispecific display achieved via fusion of polypeptides to two different capsid proteins), and the term "phage-displayed library" to refer to a diverse pool of recombinant filamentous phage displaying an array of polypeptide variants (e.g., antibody fragments; peptides). Such libraries are typically screened by iterative cycles of panning against an immobilized protein of interest (e.g., antigen for phage-displayed antibody libraries; antibody for phage-displayed peptide libraries) followed by amplification of the bound phage in E. coli cells.

Early work with anti-phage antisera generated for species classification purposes demonstrated that the filamentous phage virion is highly immunogenic in the absence of adjuvants (Meynell and Lawn, 1968) and that only the major coat protein, pVIII, and the

minor coat protein, pIII, are targeted by antibodies (Pratt et al., 1969; Woolford et al., 1977). Thus, the idea of using the phage as carrier to elicit antibodies against poorly immunogenic haptens or polypeptide was a natural extension of the ability to display recombinant exogenous sequences on its surface, which was first demonstrated by de la Cruz et al. (1988). The phage particle's low cost of production, high stability and potential for high valency display of foreign antigen (via pVIII display) also made it attractive as a vaccine carrier, especially during the early stages of development of recombinant protein technology.

Building upon existing peptide-carrier technology, the first filamentous phage-based vaccine immunogens displayed short amino acid sequences derived directly from proteins of interest as recombinant fusions to pVIII or pIII (de la Cruz et al., 1988). As library technology was developed and refined, phage-based antigens displaying peptide ligands of monoclonal antibodies (selected from random peptide libraries using the antibody, thus simulating with varying degrees of success the antibody's folded epitope on its cognate antigen; Geysen et al., 1986; Knittelfelder et al., 2009) were also generated for immunization purposes, with the goal of eliciting anti-peptide antibodies that also recognize the native protein. Some of the pioneering work in this area used peptides derived from infectious disease antigens (or peptide ligands of antibodies against these antigens; Table 2), including malaria and human immunodeficiency virus type 1 (HIV-1). When displayed on phage, peptides encoding the repeat regions of the malarial circumsporozoite protein and merozoite surface protein 1 were immunogenic in mice and rabbits (de la Cruz et al., 1988; Greenwood et al., 1991; Willis et al., 1993; Demangel et al., 1996), and antibodies raised against the latter cross-reacted with the full-length protein. Various peptide determinants (or mimics thereof) of HIV-1 gp120, gp41, gag, and reverse transcriptase were immunogenic when displayed on or conjugated to phage coat proteins (Minenkova et al., 1993; di Marzo Veronese et al., 1994; De Berardinis et al., 1999; Scala et al., 1999; Chen et al., 2001; van Houten et al., 2006; van Houten et al., 2010), and in some cases elicited antibodies that were able to weakly neutralize lab-adapted viruses (di Marzo Veronese et al., 1994; Scala et al., 1999). The list of animal and human infections for which phage-displayed peptide immunogens have been developed as vaccine leads continues to expand and includes bacterial, fungal, viral, and parasitic pathogens (Table 2). While in some cases the results of these studies have been promising, antibody epitope-based peptide vaccines are no longer an area of active research for several reasons: (i) in many cases, peptides incompletely or inadequately mimic epitopes on folded proteins (Irving et al., 2010; see below); (ii) antibodies against a single epitope may be of limited utility, especially for highly variable pathogens (Van Regenmortel, 2012); and (iii) for pathogens for

which protective immune responses are generated efficiently during natural infection, peptide vaccines offer few advantages over recombinant subunit and live vector vaccines, which have become easier to produce over time.

More recently, peptide-displaying phage have been used in attempts to generate therapeutic antibody responses for chronic diseases, cancer, immunotherapy, and immunocontraception. Immunization with phage displaying Alzheimer's disease β -amyloid fibril peptides elicited anti-aggregating antibodies in mice and guinea pigs (Frenkel et al., 2000 (Frenkel et al., , 2003 Esposito et al., 2008; Tanaka et al., 2011) , possibly reduced amyloid plaque formation in mice (Frenkel et al., 2003; Solomon, 2005; Esposito et al., 2008) , and may have helped maintain cognitive abilities in a transgenic mouse model of Alzheimer's disease (Lavie et al., 2004) ; however, it remains unclear how such antibodies are proposed to cross the blood-brain barrier. Yip et al. (2001) found that antibodies raised in mice against an ERBB2/HER2 peptide could inhibit breast-cancer cell proliferation. Phage displaying peptide ligands of an anti-IgE antibody elicited antibodies that bound purified IgE molecules (Rudolf et al., 1998) , which may be useful in allergy immunotherapy. Several strategies for phage-based contraceptive vaccines have been proposed for control of animal populations. For example, immunization with phage displaying follicle-stimulating hormone peptides on pVIII elicited antibodies that impaired the fertility of mice and ewes (Abdennebi et al., 1999) . Phage displaying or chemically Rubinichik and Chow (2000) conjugated to sperm antigen peptides or peptide mimics (Samoylova et al., 2012a,b) and gonadotropin-releasing hormone (Samoylov et al., 2012) are also in development.

For the most part, peptides displayed on phage elicit antibodies in experimental animals (Table 2) , although this depends on characteristics of the peptide and the method of its display: pIII fusions tend toward lower immunogenicity than pVIII fusions (Greenwood et al., 1991) possibly due to copy number differences (pIII: 1-5 copies vs. pVIII: estimated at several hundred copies; Malik et al., 1996) . In fact, the phage is at least as immunogenic as traditional carrier proteins such as bovine serum albumin (BSA) and keyhole limpet hemocyanin (KLH; Melzer et al., 2003; Su et al., 2007) , and has comparatively few endogenous B-cell epitopes to divert the antibody response from its intended target (Henry et al., 2011) . Excepting small epitopes that can be accurately represented by a contiguous short amino acid sequence, however, it has been extremely difficult to elicit antibody responses that cross-react with native protein epitopes using peptides. The overall picture is considerably bleaker than that painted by Table 2 , since in several studies either: (i)

peptide ligands selected from phage-displayed libraries were classified by the authors as mimics of discontinuous epitopes if they bore no obvious sequence homology to the native protein, which is weak evidence of non-linearity, or (ii) the evidence for cross-reactivity of antibodies elicited by immunization with phage-displayed peptides with native protein was unconvincing. Irving et al. (2010) describe at least one reason for this lack of success: it seems that peptide antigens elicit a set of topologically restricted antibodies that are largely unable to recognize discontinuous or complex epitopes on larger biomolecules. While the peptide may mimic the chemistry of a given epitope on a folded protein (allowing it to crossreact with a targeted antibody), being a smaller molecule, it cannot mimic the topology of that antibody's full epitope.

Despite this, the filamentous phage remains highly useful as a carrier for peptides with relatively simple secondary structures, which may be stabilized via anchoring to the coat proteins (Henry et al., 2011) . This may be especially true of peptides with poor inherent immunogenicity, which may be increased by high-valency display and phage-associated adjuvanticity (see Immunological Mechanisms of Vaccination with Filamentous Phage below).

The filamentous phage has been used to a lesser extent as a carrier for T-cell peptide epitopes, primarily as fusion proteins with pVIII (Table 3) . Early work, showing that immunization with phage elicited T-cell help (Kölsch et al., 1971; Willis et al., 1993) , was confirmed by several subsequent studies (De Berardinis et al., 1999; Ulivieri et al., 2008) . From the perspective of vaccination against infectious disease, De Berardinis et al. (2000) showed that a cytotoxic T-cell (CTL) epitope from HIV-1 reverse transcriptase could elicit antigen-specific CTLs in vitro and in vivo without addition of exogenous helper T-cell epitopes, presumably since these are already present in the phage coat proteins (Mascolo et al., 2007) . Similarly, efficient priming of CTLs was observed against phage-displayed T-cell epitopes from Hepatitis B virus (Wan et al., 2001) and *Candida albicans* (Yang et al., 2005a; Wang et al., 2006 Wang et al., , 2014d , which, together with other types of immune responses, protected mice against systemic candidiasis. Vaccination with a combination of phage-displayed peptides elicited antigen-specific CTLs that proved effective in reducing porcine cysticercosis in a randomized controlled trial (Manoutcharian et al., 2004; Morales et al., 2008) .

While the correlates of vaccine-induced immune protection for infectious diseases, where they are known, are almost exclusively serum or mucosal antibodies (Plotkin, 2010) ,

In certain vaccine applications, the filamentous phage has been used as a carrier for larger molecules that would be immunogenic even in isolation. Initially, the major advantages to phage display of such antigens were speed, ease of purification and low cost of production (Gram et al., 1993) . *E. coli* F17a-G adhesin (Van Gerven et al., 2008) , hepatitis B core antigen (Bahadir et al., 2011) , and hepatitis B surface antigen (Balcioglu et al., 2014) all elicited antibody responses when displayed on pIII, although none of these studies compared the immunogenicity of the phage-displayed proteins with that of the purified protein alone. Phage displaying *Schistosoma mansoni* glutathione S-transferase on pIII elicited an antibody response that was both higher in titer and of different isotypes compared to immunization with the protein alone (Rao et al., 2003) . Two studies of anti-idiotypic vaccines have used the phage as a carrier for antibody fragments bearing immunogenic idiotypes. Immunization with phage displaying the 1E10 idiotype scFv (mimicking a *Vibrio anguillarum* surface epitope) elicited antibodies that protected flounder fish from *Vibrio anguillarum* challenge (Xia et al., 2005) . A chemically linked phage-BCL1 tumor-specific idiotype vaccine was weakly immunogenic in mice but extended survival time in a B-cell lymphoma model (Roehnsch et al., 2013) , and was well-tolerated and immunogenic in patients with multiple myeloma (Roehnsch et al., 2014) . One study of DNA vaccination with an anti-laminarin scFv found that DNA encoding a pIII-scFv fusion protein elicited stronger humoral and cell-mediated immune responses than DNA encoding the scFv alone (Cuesta et al., 2006) , suggesting that under some circumstances, endogenous phage T-cell epitopes can enhance the immunogenicity of associated proteins. Taken together, the results of these studies show that as a particulate virus-like particle, the filamentous phage likely triggers different types of immune responses than recombinant protein antigens, and provide additional T-cell help to displayed or conjugated proteins. However, the low copy number of pIII-displayed proteins, as well as potentially unwanted phage-associated adjuvanticity, can make display of recombinant proteins by phage a suboptimal vaccine choice.

Although our understanding of the immune response against the filamentous phage pales in comparison to classical model antigens such as ovalbumin, recent work has begun to shed light on the immune mechanisms activated in response to phage vaccination (Figure 1) . The phage particle is immunogenic without adjuvant in all species tested to date, including mice (Willis et al., 1993) , rats (Dente et al., 1994) , rabbits (de la Cruz et al., 1988)

, guinea pigs (Frenkel et al., 2000; Kim et al., 2004), fish (Coull et al., 1996; Xia et al., 2005), non-human primates (Chen et al., 2001), and humans (Roehnsch et al., 2014). Various routes of immunization have been employed, including oral administration (Delmastro et al., 1997) as well as subcutaneous (Grabowska et al., 2000), intraperitoneal (van Houten et al., 2006), intramuscular (Samoylova et al., 2012a), intravenous (Vaks and Benhar, 2011), and intradermal injection (Roehnsch et al., 2013); no published study has directly compared the effect of administration route on filamentous phage immunogenicity. Antibodies are generated against only three major sites on the virion: (i) the surface-exposed N-terminal ~12 residues of the pVIII monomer lattice (Terry et al., 1997; Kneissel et al., 1999); (ii) the N-terminal N1 and N2 domains of pIII (van Houten et al., 2010); and (iii) bacterial lipopolysaccharide (LPS) embedded in the phage coat (Henry et al., 2011). In mice, serum antibody titers against the phage typically reach $1:10^5$ – $1:10^6$ after 2–3 immunizations, and are maintained for at least 1 year postimmunization (Frenkel et al., 2000). Primary antibody responses against the phage appear to be composed of a mixture of IgM and IgG2b isotypes in C57BL/6 mice, while secondary antibody responses are composed primarily of IgG1 and IgG2b isotypes, with a lesser contribution of IgG2c and IgG3 isotypes (Hashiguchi et al., 2010). Deletion of the surface-exposed N1 and N2 domains of pIII produces a truncated form of this protein that does not elicit antibodies, but also results in a non-infective phage particle with lower overall immunogenicity (van Houten et al., 2010).

FIGURE 1 | Types of immune responses elicited in response to immunization with filamentous bacteriophage. As a virus-like particle, the filamentous phage engages multiple arms of the immune system, beginning with cellular effectors of innate immunity (macrophages, neutrophils, and possibly natural killer cells), which are recruited to tumor sites by phage displaying tumor-targeting moieties. The phage likely

activates T-cell independent antibody responses, either via phage-associated TLR ligands or cross-linking by the pVIII lattice. After processing by antigen-presenting cells, phage-derived peptides are presented on MHC class II and cross-presented on MHC class I, resulting in activation of short-lived CTLs and an array of helper T-cell types, which help prime memory CTL and high-affinity B-cell responses.

Although serum anti-phage antibody titers appear to be at least partially T-cell dependent (Kölsch et al., 1971; Willis et al., 1993; De Berardinis et al., 1999; van Houten et al., 2010), many circulating pVIII-specific B cells in the blood are devoid of somatic mutation even after repeated biweekly immunizations, suggesting that under these conditions, the phage activates T-cell-independent B-cell responses in addition to high-affinity T-cell-dependent responses (Murira, 2014). Filamentous phage particles can be processed by antigen-presenting cells and presented on MHC class II molecules (Gaubin et al., 2003; Ulivieri et al., 2008) and can activate TH1, TH2, and TH17 helper T cells (Yang et al., 2005a; Wang et al., 2014d). Anti-phage TH2 responses were enhanced through display of CTLA-4 peptides fused to pIII (Kajihara et al., 2000). Phage proteins can also be cross-presented on MHC class I molecules (Wan et al., 2005) and can prime two waves of CTL responses, consisting first of short-lived CTLs and later of long-lived memory CTLs that require CD4+ T-cell help (Del Pozzo et al., 2010). The latter CTLs mediate a delayed-type hypersensitivity reaction (Fang et al., 2005; Del Pozzo et al., 2010).

The phage particle is self-adjuvanting through multiple mechanisms. Host cell wall-derived LPS enhances the virion's immunogenicity, and its removal by polymyxin B chromatography reduces antibody titers against phage coat proteins (Grabowska et al., 2000). The phage's singlestranded DNA genome contains CpG motifs and may also have an adjuvant effect. The antibody response against the phage is entirely dependent on MyD88 signaling and is modulated by stimulation of several Toll-like receptors (Hashiguchi et al., 2010), indicating that innate immunity plays an important but largely uncharacterized role in the activation of anti-phage adaptive immune responses. Biodistribution studies of the phage after intravenous injection show that it is cleared from the blood within hours through the reticuloendothelial system (Molenaar et al., 2002), particularly of the liver and spleen, where it is retained for days (Zou et al., 2004), potentially activating marginal-zone B-cell responses. Thus, the filamentous phage is not only a highly immunogenic carrier, but by virtue of activating a range of innate and adaptive immune responses, serves as an excellent model virus-like particle antigen.

Long before the identification of filamentous phage, other types of bacteriophage were already being used for antibacterial therapy in the former Soviet Union and Eastern Europe (reviewed in Sulakvelidze et al., 2001). The filamentous phage, with its nonlytic life cycle, has less obvious clinical uses, despite the fact that the host specificity of Inovirus and Plectrovirus includes many pathogens of medical importance, including *Salmonella*, *E. coli*, *Shigella*, *Pseudomonas*, *Clostridium*, and *Mycoplasma* species.

In an effort to enhance their bactericidal activity, genetically modified filamentous phage have been used as a "Trojan horse" to introduce various antibacterial agents into cells. M13 and Pf3 phage engineered to express either BglII restriction endonuclease (Hagens and Blasi, 2003; Hagens et al., 2004) , lambda phage S holin (Hagens and Blasi, 2003) or a lethal catabolite gene activator protein (Moradpour et al., 2009) effectively killed *E. coli* and *Pseudomonas aeruginosa* cells, respectively, with no concomitant release of LPS (Hagens and Blasi, 2003; Hagens et al., 2004) . Unfortunately, the rapid emergence of resistant bacteria with modified F pili represents a major and possibly insurmountable obstacle to this approach. However, there are some indications that filamentous phage can exert useful but more subtle effects upon their bacterial hosts that may not result in the development of resistance to infection. Several studies have reported increased antibiotic sensitivity in bacterial populations simultaneously infected with either wild type filamentous phage (Hagens et al., 2006) or phage engineered to repress the cellular SOS response (Lu and Collins, 2009) . Filamentous phage f1 infection inhibited early stage, but not mature, biofilm formation in *E. coli* (May et al., 2011) . Thus, unmodified filamentous phage may be of future interest as elements of combination therapeutics against certain drug-resistant infections.

More advanced therapeutic applications of the filamentous phage emerge when it is modified to express a targeting moiety specific for pathogenic cells and/or proteins for the treatment of infectious diseases, cancer and autoimmunity (Figure 2) . The first work in this area showed as proof-of-concept that phage encoding a GFP expression cassette and displaying a HER2specific scFv on all copies of pIII were internalized into breast tumor cells, resulting in GFP expression (Poul and Marks, 1999) . M13 or fd phage displaying either a targeting peptide or antibody fragment and tethered to chloramphenicol by a labile crosslinker were more potent inhibitors of *Staphylococcus aureus* growth than high-concentration free chloramphenicol (Yacoby et al., 2006; Vaks and Benhar, 2011) . M13 phage loaded with doxorubicin and displaying a targeting peptide on pIII specifically killed prostate cancer cells in vitro (Ghosh et al., 2012a) . Tumorspecific peptide:pVIII fusion proteins selected from "landscape" phage (Romanov et al., 2001; Abbineni et al., 2010; Fagbohun et al., 2012 Fagbohun et al., , 2013 Lang et al., 2014; Wang et al., 2014a) were able to target and deliver siRNA-, paclitaxel-, and doxorubicincontaining liposomes to tumor cells (Jayanna et al., 2010a; Wang et al., 2010a Wang et al., ,b,c, 2014b Bedi et al., 2011 Bedi et al., , 2013 Bedi et al., , 2014 ; they were non-toxic and increased tumor remission rates in mouse models (Jayanna et al., 2010b; Wang et al., 2014b,c) . Using the

B16-OVA tumor model, Eriksson et al. (2007) showed that phage displaying peptides and/or Fabs specific for tumor antigens delayed tumor growth and improved survival, owing in large part to activation of tumor-associated macrophages and recruitment of neutrophils to the tumor site (Eriksson et al., 2009) . Phage displaying an scFv against β -amyloid fibrils showed promise as a diagnostic (Frenkel and Solomon, 2002) and therapeutic (Solomon, 2008) reagent for Alzheimer's disease and Parkinson's disease due to the unanticipated ability of the phage to penetrate into brain tissue (Ksendzovsky et al., 2012) . Similarly, phage displaying an immunodominant peptide epitope derived from myelin oligodendrocyte glycoprotein depleted pathogenic demyelinating antibodies in brain tissue in the murine experimental autoimmune encephalomyelitis model of multiple sclerosis (Rakover et al., 2010) . The advantages of the filamentous phage in this context over traditional antibody-drug or protein-peptide conjugates are (i) its ability to carry very high amounts of drug or peptide, and (ii) its ability to access anatomical compartments that cannot generally be reached by systemic administration of a protein.

Unlike most therapeutic biologics, the filamentous phage's production in bacteria complicates its use in humans in several ways. First and foremost, crude preparations of filamentous phage typically contain very high levels of contaminating LPS, in the range of $\sim 10^2$ - 10^4 endotoxin units (EU)/mL (Boratynski et al., 2004; Branston et al., 2015) , which have the potential to cause severe adverse reactions. LPS is not completely removed by polyethylene glycol precipitation or cesium chloride density gradient centrifugation (Smith and Gingrich, 2005; Branston et al., 2015) , but its levels can be reduced dramatically using additional purification steps such as size exclusion chromatography (Boratynski et al., 2004; Zakharova et al., 2005) , polymyxin B chromatography (Grabowska et al., 2000) , and treatment with detergents such as Triton X-100 or Triton X-114 (Roehnisch et al., 2014; Branston et al., 2015) . These strategies routinely achieve endotoxin levels of <1 EU/mL as measured by the limulus amoebocyte lysate (LAL) assay, well below the FDA limit for parenteral administration of 5 EU/kg body weight/dose, although concerns remain regarding the presence of residual virion-associated LPS which may be undetectable. A second and perhaps unavoidable consequence of the filamentous phage's bacterial production is inherent heterogeneity of particle size and the spectrum of host cell-derived virion-associated and soluble contaminants, which may be cause for safety concerns and restrict its use to high-risk groups.

Many types of bacteriophage and engineered phage variants, including filamentous phage, have been proposed for prophylactic use *ex vivo* in food safety, either in the production

pipeline (reviewed in Dalmasso et al., 2014) or for detection of foodborne pathogens post-production (reviewed in Schmelcher and Loessner, 2014) . Filamentous phage displaying a tetracysteine tag on pIII were used to detect *E. coli* cells through staining with biarsenical dye . M13 phage functionalized with metallic silver were highly bactericidal against *E. coli* and *Staphylococcus epidermidis* . Biosensors based on surface plasmon resonance (Nanduri et al., 2007) , piezoelectric transducers (Olsen et al., 2006) , linear dichroism (Pacheco-Gomez et al., 2012) , and magnetoelastic sensor technology (Lakshmanan et al., 2007; Huang et al., 2009) were devised using filamentous phage displaying scFv or conjugated to whole IgG against *E. coli*, *Listeria monocytogenes*, *Salmonella typhimurium*, and *Bacillus anthracis* with limits of detection on the order of 10^2 - 10^6 bacterial cells/mL. Proof of concept has been demonstrated for use of such phage-based biosensors to detect bacterial contamination of live produce (Li et al., 2010b) and eggs (Chai et al., 2012) .

The filamentous phage particle is enclosed by a rod-like protein capsid, ~1000 nm long and 5 nm wide, made up almost entirely of overlapping pVIII monomers, each of which lies ~27 angstroms from its nearest neighbor and exposes two amine groups as well as at least three carboxyl groups (Henry et al., 2011) . The regularity of the phage pVIII lattice and its diversity of chemically addressable groups make it an ideal scaffold for bioconjugation (Figure 3) . The most commonly used approach is functionalization of amine groups with NHS esters (van Houten et al., 2006 (van Houten et al., , 2010 Yacoby et al., 2006) , although this can result in unwanted acylation of pIII and any displayed biomolecules. Carboxyl groups and tyrosine residues can also be functionalized using carbodiimide coupling and diazonium coupling, respectively (Li et al., 2010a) . Carrico et al. (2012) developed methods to specifically label pVIII N-termini without modification of exposed lysine residues through a two-step transamination-oxime formation reaction. Specific modification of phage coat proteins is even more easily accomplished using genetically modified phage displaying peptides (Ng et al., 2012) or enzymes (Chen et al., 2007; Hess et al., 2012) , but this can be cumbersome and is less general in application.

For more than a decade, interest in the filamentous phage as a building block for nanomaterials has been growing because of its unique physicochemical properties, with emerging applications in magnetics, optics, and electronics. It has long been known that above a certain concentration threshold, phage can form ordered crystalline suspensions (Welsh et al., 1996) . Lee et al. (2002) engineered M13 phage to display a ZnS-binding peptide on pIII and showed that, in the presence of ZnS nanoparticles, they self-assemble into highly ordered film biomaterials that can be aligned using magnetic fields. Taking

advantage of the ability to display substrate-specific peptides at known locations on the phage filament Hess et al., 2012) , this pioneering **FIGURE 3 | Chemically addressable groups of the filamentous bacteriophage major coat protein lattice.** The filamentous phage virion is made up of ~2,500-4,000 overlapping copies of the 50-residue major coat protein, pVIII, arranged in a shingle-type lattice. Each monomer has an array of chemically addressable groups available for bioorthogonal conjugation, including two primary amine groups (shown in red), three carboxyl groups (show in blue) and two hydroxyl groups (show in green). The 12 N-terminal residues generally exposed to the immune system for antibody binding are in bold underline. Figure adapted from structural data of Marvin, 1990 , freely available in PDB and SCOPe databases.

work became the basis for construction of two- and three-dimensional nanomaterials with more advanced architectures, including semiconducting nanowires (Mao et al., 2003 (Mao et al., , 2004 , nanoparticles , and nanocomposites (Oh et al., 2012; Chen et al., 2014) . Using hybrid M13 phage displaying Co₃O₄- and gold-binding peptides on pVIII as a scaffold to assemble nanowires on polyelectrolyte multilayers, Nam et al. (2006) produced a thin, flexible lithium ion battery, which could be stamped onto platinum microband current collectors (Nam et al., 2008) . The electrochemical properties of such batteries were further improved through pIII-display of single-walled carbon nanotube-binding peptides (Lee et al., 2009) , offering an approach for sustainable production of nanostructured electrodes from poorly conductive starting materials. Phage-based nanomaterials have found applications in cancer imaging (Ghosh et al., 2012b; Yi et al., 2012) , photocatalytic water splitting (Nam et al., 2010a; Neltner et al., 2010) , light harvesting (Nam et al., 2010b; Chen et al., 2013) , photoresponsive technologies (Murugesan et al., 2013) , neural electrodes (Kim et al., 2014) , and piezoelectric energy generation (Murugesan et al., 2013) .

Thus, the unique physicochemical properties of the phage, in combination with modular display of peptides and proteins with known binding specificity, have spawned wholly novel materials with diverse applications. It is worth noting that the unusual biophysical properties of the filamentous phage can also be exploited in the study of structures of other macromolecules. Magnetic alignment of high-concentration filamentous phage in solution can partially order DNA, RNA, proteins, and other biomolecules for measurement of dipolar coupling interactions (Hansen et al., 1998 (Hansen et al., , 2000 Dahlke Ojennus et al., 1999) in NMR spectroscopy.

Because of their large population sizes, short generation times, small genome sizes and ease of manipulation, various filamentous and non-filamentous bacteriophages have been used as models of experimental evolution (reviewed in Husimi, 1989; Wichman and Brown, 2010; Kawecki et al., 2012; Hall et al., 2013) . The filamentous phage has additional practical uses in protein engineering and directed protein evolution, due to its unique tolerance of genetic modifications that allow biomolecules to be displayed on the virion surface. First and foremost among these applications is in vitro affinity maturation of antibody fragments displayed on pIII. Libraries of variant Fabs and single chain antibodies can be generated via random or sitedirected mutagenesis and selected on the basis of improved or altered binding, roughly mimicking the somatic evolution strategy of the immune system (Marks et al., 1992; Bradbury et al., 2011) . However, other in vitro display systems, such as yeast display, have important advantages over the filamentous phage for affinity maturation (although each display technology has complementary strengths; Koide and Koide, 2012) , and regardless of the display method, selection of "improved" variants can be slow and cumbersome. Iterative methods have been developed to combine computationally designed mutations (Lippow et al., 2007) and circumvent the screening of combinatorial libraries, but these have had limited success to date.

Recently, Esvelt et al. (2011) developed a novel strategy for directed evolution of filamentous phage-displayed proteins, called phage-assisted continuous evolution (PACE), which allows multiple rounds of evolution per day with little experimental intervention. The authors engineered M13 phage to encode an exogenous protein (the subject for directed evolution), whose functional activity triggers gene III expression from an accessory plasmid; variants of the exogenous protein arise by random mutagenesis during phage replication, the rate of which can be increased by inducible expression of error-prone DNA polymerases. By supplying limiting amounts of receptive E. coli cells to the engineered phage variants, Esvelt et al. (2011) elegantly linked phage infectivity and production of offspring with the presence of a desired protein phenotype. Carlson et al. (2014) later showed that PACE selection stringency could be modulated by providing small amounts of pIII independently of protein phenotype, and undesirable protein functions negatively selected by linking them to expression of a truncated pIII variant that impairs infectivity in a dominant negative fashion. PACE is currently limited to protein functions that can be linked in some way to the expression of a gene III reporter, such as protein-protein interaction, recombination, DNA or RNA binding, and enzymatic catalysis (Meyer and Ellington, 2011) .

This approach represents a promising avenue for both basic research in molecular evolution (Dickinson et al., 2013) and synthetic biology, including antibody engineering.

Filamentous bacteriophage have been recovered from diverse environmental sources, including soil (Murugaiyan et al., 2011) , coastal fresh water (Xue et al., 2012) , alpine lakes (Hofer and Sommaruga, 2001) and deep sea bacteria (Jian et al., 2012) , but not, perhaps surprisingly, the human gut (Kim et al., 2011) . The environmental "phageome" in soil and water represent the largest source of replicating DNA on the planet, and is estimated to contain upward of 10³⁰ viral particles (Ashelford et al., 2003; Chibani-Chennoufi et al., 2004; Suttle, 2005) . The few studies attempting to investigate filamentous phage environmental ecology using classical environmental microbiology techniques (typically direct observation by electron microscopy) found that filamentous phage made up anywhere from 0 to 100% of all viral particles (Demuth et al., 1993; Pina et al., 1998; Hofer and Sommaruga, 2001) . There was some evidence of seasonal fluctuation of filamentous phage populations in tandem with the relative abundance of free-living heterotrophic bacteria (Hofer and Sommaruga, 2001) . Environmental metagenomics efforts are just beginning to unravel the composition of viral ecosystems. The existing data suggest that filamentous phage comprise minor constituents of viral communities in freshwater (Roux et al., 2012) and reclaimed and potable water (Rosario et al., 2009) but have much higher frequencies in wastewater and sewage (Cantalupo et al., 2011; Alhamlan et al., 2013) , with the caveat that biases inherent to the methodologies for ascertaining these data (purification of viral particles, sequencing biases) have not been well validated. There are no data describing the population dynamics of filamentous phage and their host species in the natural environment.

At the individual virus-bacterium level, it is clear that filamentous phage can modulate host phenotype, including the virulence of important human and crop pathogens. This can occur either through direct effects of phage replication on cell growth and physiology, or, more typically, by horizontal transfer of genetic material contained within episomes and/or chromosomally integrated prophage. Temperate filamentous phage may also play a role in genome evolution (reviewed in Canchaya et al., 2003) . Perhaps the best-studied example of virulence modulation by filamentous phage is that of *Vibrio cholerae*, whose full virulence requires lysogenic conversion by the cholera toxin-encoding CTX ϕ phage (Waldor and Mekalanos, 1996) . Integration of CTX ϕ phage occurs at specific sites in the genome; these sequences are introduced through the combined action of another filamentous phage, fs2 ϕ , and a satellite filamentous phage, TLC-Kn ϕ 1 (Hassan et al., 2010) . Thus,

filamentous phage species interact and coevolve with each other in addition to their hosts. Infection by filamentous phage has been implicated in the virulence of *Yersinia pestis* (Derbise et al., 2007) , *Neisseria meningitidis* (Bille et al., 2005 (Bille et al., , 2008 , *Vibrio parahaemolyticus* (Iida et al., 2001) , *E. coli* 018:K1:H7 (Gonzalez et al., 2002) , *Xanthomonas campestris* (Kamiunten and Wakimoto, 1982) , and *P. aeruginosa* (Webb et al., 2004) , although in most of these cases, the specific mechanisms modulating virulence are unclear. Phage infection can both enhance or repress virulence depending on the characteristics of the phage, the host bacterium, and the environmental milieu, as is the case for the bacterial wilt pathogen *Ralstonia solanacearum* (Yamada, 2013) . Since infection results in downregulation of the pili used for viral entry, filamentous phage treatment has been proposed as a hypothetical means of inhibiting bacterial conjugation and horizontal gene transfer, so as to prevent the spread of antibiotic resistance genes (Lin et al., 2011) .

Finally, the filamentous phage may also play a future role in the preservation of biodiversity of other organisms in at-risk ecosystems. Engineered phage have been proposed for use in bioremediation, either displaying antibody fragments of desired specificity for filtration of toxins and environmental contaminants (Petrenko and Makowski, 1993) , or as biodegradable polymers displaying peptides selected for their ability to aggregate pollutants, such as oil sands tailings (Curtis et al., 2011 (Curtis et al., , 2013 . Engineered phage displaying peptides that specifically bind inorganic materials have also been proposed for use in more advanced and less intrusive mineral separation technologies (Curtis et al., 2009) .

The filamentous phage represents a highly versatile organism whose uses extend far beyond traditional phage display and affinity selection of antibodies and polypeptides of desired specificity. Its high immunogenicity and ability to display a variety of surface antigens make the phage an excellent particulate vaccine carrier, although its bacterial production and preparation heterogeneity likely limits its applications in human vaccines at present, despite being apparently safe and well-tolerated in animals and people. Unanticipated characteristics of the phage particle, such as crossing of the blood-brain barrier and formation of highly ordered liquid crystalline phases, have opened up entirely new avenues of research in therapeutics for chronic disease and the design of nanomaterials. Our comparatively detailed understanding of the interactions of model filamentous phage with their bacterial hosts has allowed researchers to harness the phage life cycle to direct protein evolution in the lab. Hopefully, deeper knowledge of phage-host

interactions at an ecological level may produce novel strategies to control bacterial pathogenesis. While novel applications of the filamentous phage continue to be developed, the phage is likely to retain its position as a workhorse for therapeutic antibody discovery for many years to come, even with the advent of competing technologies.

KH and JS conceived and wrote the manuscript. MA-G read the manuscript and commented on the text.

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Nanopore sequencing as a revolutionary diagnostic tool for porcine viral enteric disease complexes identifies porcine kobuvirus as an important enteric virus

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Abstract: Enteric diseases in swine are often caused by different pathogens and thus metagenomics are a useful tool for diagnostics. The capacities of nanopore sequencing for viral diagnostics were investigated here. First, cell culture-grown porcine epidemic diarrhea virus and rotavirus A were pooled and sequenced on a MinION. Reads were already detected at 7 seconds after start of sequencing, resulting in high sequencing depths (19.2

to 103.5X) after 3 h. Next, diarrheic feces of a one-week-old piglet was analyzed. Almost all reads (99%) belonged to bacteriophages, which may have reshaped the piglet's microbiome. Contigs matched Bacteroides, Escherichia and Enterococcus phages. Moreover, porcine kobuvirus was discovered in the feces for the first time in Belgium. Suckling piglets shed kobuvirus from one week of age, but an association between peak of viral shedding (10(6.42)–10(7.01) copies/swab) and diarrheic signs was not observed during a follow-up study. Retrospective analysis showed the widespread (n = 25, 56.8% positive) of genetically moderately related kobuviruses among Belgian diarrheic piglets. MinION enables rapid detection of enteric viruses. Such new methodologies will change diagnostics, but more extensive validations should be conducted. The true enteric pathogenicity of porcine kobuvirus should be questioned, while its subclinical importance cannot be excluded.

Text: metagenomics is a valuable asset for diagnostics in pigs, leading to discovery of novel viruses and identification of porcine viral enteric disease complexes. Although standardized procedures have been developed to study viral metagenomes in fecal samples, they still require an extensive sample preparation, including random or targeted pre-amplification of viral genomes present in the sample ¹³. Most sequencing platforms still require capital investments and high sample turnover rates to be cost-effective. Performing the necessary analyses often results in long time periods between sample arrival and diagnostic reporting, since results can only be processed after finishing the sequencing run. Third-generation sequencing using MinION (Oxford Nanopore Technologies, ONT) might be a useful and affordable diagnostic tool for swine veterinary medicine as it allows rapid sample preparation and real-time sequence analysis. The flowcells used for sequencing consist of a membrane containing multiple CsgG nanopore proteins from Escherichia coli ¹⁴. An ion current is established through this pore resulting in typical current changes upon passage of specific nucleotides. This signal is converted into a nucleotide sequence by computational algorithms (basecalling). Since the release of MinION technology, major advances have been made in terms of the number and the quality of reads generated ¹⁵. In the field of virology, the technology has mainly been applied in human medicine. Using nanopore sequencing, it was possible to distinguish three poxviruses with 98% nucleotide similarity at strain level ¹⁶. MinION has also been used as a diagnostic tool during recent Ebolavirus outbreaks in West Africa, allowing fast on-site characterization of circulating strains ^{17, 18}. Coupled to a laptop-based bioinformatics workflow, MinION was able to detect Chikungunya virus, Ebola virus and hepatitis C virus in less than 6 hours using earlier versions of the technology ¹⁹. A multiplex PCR method for complete on-site Zikavirus genome sequencing in samples with

low viral loads has recently been developed by Quick and coworkers 20 . Partial dengue virus genomes were isothermally amplified followed by sequencing, allowing classification of strains in serotypes 21 . In veterinary virology, the use of nanopore sequencing is growing. A novel species of papillomavirus was identified in warts from giraffes, using rolling-circle amplification and nanopore sequencing 22 . The entire genome of a parapoxvirus isolated from a seal was obtained by combining data from Illumina next-generation sequencing with nanopore sequencing data 23 . One study has reported the detection of Venezuelan equine encephalitis virus from unamplified cDNA created from poly-A tailed RNA using cell culture grown viruses 24 . To the author's knowledge, the present study is the first using MinION as an aid in porcine health management. This study was aimed to explore the possibilities of MinION as a rapid and easy-to-use diagnostic tool in pig health management for diagnosis of viral enteric disease complexes. The ability to detect high loads of cell culture-grown rotavirus and coronavirus, mimicking shedding quantities observed in diarrheic piglets, was evaluated. In a second case, the ability to detect (novel) viruses in diarrheic feces of a one-week-old piglet with diarrhea was investigated. No gene-specific or random pre-amplification of viral nucleic acids was conducted to challenge the MinION's sensitivity. A porcine kobuvirus was discovered in the latter case and a longitudinal field study was conducted hereafter to elucidate the shedding patterns of this virus. Moreover, archival (2014) fecal samples from diarrheic suckling piglets less than two weeks old were investigated for the presence of kobuviruses, to study their epidemiology in Belgium.

be performed for 243,313 reads with a mean length of 740 nucleotides. Reads with a q-score lower than 7 were filtered out, resulting in 179,015 remaining sequences (mean length 816 nt) for use in downstream analyses. Results of the sequencing run, including taxonomical classification and mapping of reads against PEDV and rotavirus A (RVA) reference genomes are shown in Fig. 1A . After 24 hours of sequencing, a total of 15,232 reads were classified as viral by sensitive tBLASTx comparison against a complete viral database. Of these, 39.3% ($n = 5,985$) and 10.3% ($n = 1,564$) were assigned to viral families comprising Porcine epidemic diarrhea virus (family Coronaviridae) and Rotavirus (family Reoviridae, subfamily Sedoreovirinae), respectively. A fraction of the reads (29.3%, $n = 4,468$) were assigned to order Caudovirales. These reads originated from the lambda phage DNA used in a previous control run on the same flowcell. At 7.5 and 24.2 seconds after the start of sequencing, respectively, the first reads matching PEDV and RVA were translocated through a nanopore. Most reads were generated in the first twelve hours of sequencing and read accumulation was most exponential in the first three hours of sequencing (Fig. 1B) . PEDV and RVA sequences were extracted from the dataset and mapped against viral

reference genes to calculate sequencing depths over time (Fig. 1C) . After one hour, sequencing depths were higher for PEDV (43.0X) than for RVA (4.9 to 22.1X). High sequencing depths were acquired after three hours of sequencing for PEDV (103.5X) and for most RVA gene segments (19.2 to 48.2X).

De novo assembly was executed on the quality-filtered reads prior to identification (tBLASTx) to recover viral genomes. This resulted in the recovery of the almost complete PEDV genome and RVA gene segments with identities varying between 95 and 99% compared to the reference genes (Table 1) . Higher assembly accuracies (97 to 99%) were obtained when only the reads matching against rotavirus and PEDV were included for de novo assembly (Table 1) . However, execution of de novo assembly prior to taxonomical classification (tBLASTx) reduced the time to identify entire viral genomes in the dataset.

Virome composition of a young diarrheic piglet using nanopore sequencing. A total of 30,088 reads were generated by sequencing the diarrheic fecal sample for three hours. Of these, 25,466 reads (q-score >7, mean read length 653 nt) were used for further analyses. Different methods were used to compare the reads against a viral database using the HPC cluster of Ghent University and results are shown in Fig. 2 . Comparison against a complete viral database resulted in the detection of 6,781 to 8,677 potential viral reads, depending on the BLAST settings. BLASTn resulted in rapid taxonomical identification of reads at almost similar sensitivity compared to tBLASTx. However, there was a very high difference between wall times on the HPC cluster, with only 26 seconds of analysis time for BLASTn, versus almost 24 hours for tBLASTx. The majority of sequences were assigned to bacteriophages within the order Caudovirales and families Siphoviridae (n = 3,213 to 4,163 reads), Podoviridae (n = 2,506 to 3,002 reads) and Myoviridae (n = 912 to 1,202 reads). A de novo assembly was executed on the basecalled, quality filtered reads and the resulting contigs were used as input material for VirSorter analysis. Nineteen contigs were classified as sure (n = 4; category 1), somewhat sure (=14; category 2) and not so sure (=1; category 3) to be phage-like contigs (Fig. 2B) . Comparison of these contigs against the GenBank database using BLAST allowed classification into four different groups. Ten contigs showed moderate to high nucleotide similarities to the Bacteroides phage B124-14, suggesting that they all belonged to one phage genome. This was also supported by the fact that all these contigs mapped nicely distributed across the reference genome of Bacteroides phage B124-14 (data not shown). The longest contig with a size of 39,069 nucleotides, together with four other contigs showed similarities (95% nt identity) to different Escherichia phages. As they also mapped nicely distributed across the reference genome of

Escherichia phage vB_EcoP_PhAPEC7, it seems that they must also belong to one phage genome (data not shown). Two contigs showed poor similarity to both the Enterococcus phage vB_EfaS_IME_196, isolated from hospital sewage in China from an Enterococcus faecalis strain, and the Enterococcus hirae bacterial genome. The latter might be a prophage inserted in the bacterial genome. Interestingly, three contigs were identified for which no similarities were found with existing viruses in GenBank, but contig 0105 mapped to the reference genome of the Enterococcus phage vB_EfaS_IME196 (data not shown). These might be novel phages or divergent variants from existing phages present in GenBank.

Three eukaryotic porcine viruses, porcine kobuvirus (n = 18 to 22 reads), enterovirus G (n = 5 to 9 reads) and astrovirus (n = 4 reads) were found at much lower abundancies. The genera Kobuvirus and Enterovirus belong to the family Picornaviridae, whereas the genus Mamastrovirus belongs to the family Astroviridae. Kobuvirus reads were mapped against a European reference strain S-1/HUN/2007/Hungary, as shown in Fig. 2C . However, full-genome coverage at high sequencing depth was not obtained.

Shedding of porcine kobuvirus and rotaviruses in suckling piglets. The shedding of porcine kobuvirus, RVA and rotavirus C (RVC) was quantitatively investigated in 5 suckling pigs of the same farm from which the diarrheic feces originated. The fecal shedding patterns of the different viruses and presence of diarrheic signs are shown in Fig. 3A . All piglets started shedding porcine kobuvirus at the end of the first week after parturition. In two piglets (A and D) the shedding was sustained and lasted for at least 2 weeks (above the limit of quantification). Peak shedding titers of the porcine kobuvirus varied between 6.42 and 7.01 log₁₀ copies/swab, which is generally lower than peak shedding observed for typical enteric viruses such as rotavirus and PEDV. Moreover, the peak of shedding was not related to diarrheic episodes, questioning the role of this virus in the pathogenesis of diarrhea on the farm. Diarrheic signs were only noticed in two piglets (A and B). In piglet B, an association between high RVC shedding and diarrheic episodes was observed. In contrast, there was no direct association between peak shedding of kobuvirus and diarrheic episodes. Interestingly, a peak in kobuvirus shedding was observed in piglet C at day 11 post-farrowing. This animal died shortly hereafter, but it was unclear if this can be attributable to the kobuvirus infection. Acute RVA shedding was observed at the end of the suckling period in three of five piglets, even though all sows were vaccinated before farrowing using a bovine inactivated rotavirus vaccine.

Retrospective analysis of porcine kobuviruses shedding in Belgian diarrheic suckling pigs and phylogenetic analysis. A total of 44 diarrheic fecal samples collected in 2014 were screened for the presence of kobuvirus using the new RT-qPCR. Of these, 25 samples (56.8%) tested positive and 18 samples showed quantifiable viral loads (4.31 to 6.83 log₁₀ copies/swab). Seven samples were positive, but viral loads were too low to allow accurate quantification. The presence of RVA and RVC had been quantitatively assessed in these samples in a previous study and the occurrence of co-infections between rotaviruses and kobuvirus is shown in Fig. 3B 25 . Kobuvirus was found in equal ratios in rotavirus-negative and -positive samples. Twelve samples contained a single rotavirus infection with a high RVA load and in four of these, a high kobuvirus load (5.16 to 5.42 log₁₀ copies/g) was observed. A single RVC infection was found in seven samples and in four of these tested positive for kobuvirus at high loads (4.31 to 5.59 log₁₀ copies/g). A dual RVA/RVC infection was seen in two samples, but neither contained quantifiable kobuvirus loads. Many (n = 10) of the rotavirus-negative samples contained high kobuvirus loads.

Strain 17V079 showed high similarity to other Belgian porcine kobuvirus isolates from 2014 (92.1 to 94.0% nucleotide sequence identity) and the Hungarian reference strain S-1/Hun/2017 (93.4%). Furthermore, there was a high level of genetic variability between the 2014 Belgian porcine kobuvirus isolates, with nucleotide sequence identities ranging between 90.1 and 97.2%. A phylogenetic analysis, using the 3D gene of 17V079 and twelve Belgian isolates from 2014 (Fig. 3C) , shows the Belgian strains clustering between strains from different geographical locations.

Prevention and treatment of enteric disease problems in young piglets is frequently hampered by a lack of diagnostic tools. Veterinarians are restricted to a short list of known viruses, bacteria, parasites and management factors to define a differential diagnosis. Only the most likely cause(s) of the disease will be diagnostically investigated, often leading to negative, inconclusive or incomplete results. However, metagenomics studies have indicated the existence of viral enteric disease complexes, potentially involving multiple known and novel viruses 3, 4, 6, 7, 9, 10 . Detection of nucleic acids from pathogens using NGS-based metagenomics approaches is a partial solution to diagnostic testing problems and can provide a complete readout of viruses and other pathogens present in a sample. However, most NGS platforms require large investments and processing of the reads can only start at the end of the sequencing run. Viral metagenomics also requires extensive

laboratory preparations, including centrifugation, filtration and nuclease treatment to discard bacterial and host nucleic acids that make up to the bulk of all nucleic acids present 13 . Furthermore, the amount of viral nucleic acids in a sample is very low, requiring targeted or random amplification of these genomes before NGS analysis. Amplification may induce bias and hampers the development of a fast diagnostic pipelines due to considerable time loss. All these factors lead to a long turnover time between sample collection and diagnostic reporting. The third-generation sequencing device MinION (ONT), holds promise as a diagnostic platform, as it allows real-time sequencing and analyses of all DNA/RNA in a sample, theoretically without needing pre-amplification of viral nucleic acids. It was the aim of the present study to evaluate this technology for use as a rapid tool for porcine viral enteric disease complex identification, without the conduction of viral nucleic acid amplification. In a first experiment, cell culture-grown PEDV and RVA, known to induce diarrhea in young pigs, were pooled at high loads mimicking shedding quantities in diarrheic piglets. Sequencing of this pooled sample with the MinION resulted in rapid identification of both viruses. Real-time analysis of the sequencing reads was not conducted, but is achievable as previously demonstrated by Greninger and colleagues using the SURPI analysis pipeline for rapid identification of human viruses from different clinical matrices 19 . Interestingly, the first reads matching PEDV and RVA were generated respectively after 7 and 24 seconds of sequencing. High sequencing depths (43.0X) were acquired within one hour of sequencing for PEDV and within three hours for most of the eleven RVA gene segments (19.2-48.2X). Overall, higher sequencing depths were generated for PEDV that could indicate that sequencing of longer viral genomes is favored over smaller gene segments, as PEDV has a genome size of approximately 28 kb, and RVA gene segments are shorter (0.6 to 3.3 kb). This bias might have been introduced during the ligation of the sequencing adapters to the viral nucleic acids. It can be hypothesized that adapters are more easily attached to longer DNA fragments, and bias should be avoided by standardization of viral nucleic acid input length. Rapid read generation allows flexible use of the sequencing platform and sequences can be read until enough genome information of the viruses of interest is available. While the technology can be useful for giving fast readouts of viruses (<3 hours) present in a sample, thorough validation, using well-defined virus stocks, spiking experiments in matrices (e.g. feces) and real clinical samples is necessary to make sure that all members of the porcine viral enteric disease complex are accurately being diagnosed. Furthermore, the accuracy of the technology needs further improvement, as error rates of contigs from de novo assemblies still ranged between 1 and 5%, hindering the precise analysis of subtle but important mutations in the viral genome.

After the successful identification of the cell culture-grown viruses, the performance of the MinION was further explored by analyzing a diarrheic fecal sample of a one-week-old suckling piglet. Real-time PCR analyses were conducted for RVA, RVC, PEDV and TGEV. *Enterococcus hirae* was isolated at a private diagnostic laboratory, but this bacterial species is not considered a typical cause of diarrheic disease in pigs [26]. Viral metagenomics was conducted on this sample using the MinION and two different BLAST search algorithms were used to taxonomically identify the reads by comparing them against a complete viral database. Overall, tBLASTx with an e-value of 10^{-3} was able to identify the most viral reads compared to other search options conducted. However, BLASTn search options also reached high sensitivity, but at much lower time cost: 26 seconds instead of almost 24 hours. For rapid read analysis and searching for closely related non-divergent viral sequences, BLASTn or another fast methodology should thus be preferentially used. However, tBLASTx might pick up more divergent or novel viruses, improving overall sensitivity.

Three porcine viruses, including porcine kobuvirus, porcine mamastrovirus and enterovirus G, were identified in sample 17V079. Astro- and enteroviruses have been detected earlier in both diarrheic and non-diarrheic feces of Belgian pigs and in feces from pigs around the globe [9,10,27]. In a recent study from Thailand, the difference in prevalence of astrovirus in diarrheic (8.4%) versus non-diarrheic (4.6%) piglets less than 4-weeks-old was not statistically significant. Also other studies have shown that the role of porcine astrovirus in the pathogenesis of pig diarrhea is not completely clear [28]. In contrast, associations between diarrhea and human astrovirus infections have been made [29]. A recent study in 5 European countries (Hungary, Spain, Germany, Austria and Sweden) have indicated the widespread of porcine astroviruses in the swine population. A one hundred percent prevalence of astrovirus was found in diarrheic and non-diarrheic pigs from Austria and Spain. Porcine astroviruses have recently also been linked to outbreaks of neurological disorders in weaned piglets from Hungary, and in 5-week-old pigs and sows in the United States [30, 31]. The gut might be a hypothetical entry port for such neurological astrovirus infections.

Enteroviruses have been more generally linked to neurological disorders in pigs, although they are commonly found in feces as well [11, 12, [32] [33] [34]]. In a study from Vietnam, no significant correlation was found between diarrhea status and presence of enterovirus G in feces [35]. The involvement of both astro- and enteroviruses in the pathogenesis of enteric disorders might be questioned here, but cannot be completely ruled out. Furthermore,

while sensitive tBLASTx searches were used here, there is still a possibility that a completely novel virus might be present in the dark matter of the sequencing reads. However, reporting of a porcine kobuvirus in Belgian piglets with MinION is unique. In Belgium, kobuviruses had previously only been found in diarrheic samples of calves and young cattle in Belgium 36 . In the present study, a novel RT-qPCR assay, targeting the conserved 3D gene encoding the RNA-dependent-RNA-polymerase, was developed and used to assess, for the first time, longitudinal quantitative shedding kinetics of porcine kobuvirus in pigs under field conditions. Similar kinetics were also analyzed for porcine rotavirus A and C. While suckling piglets started shedding porcine kobuvirus from one week of age, an association between peak viral shedding (6.42 to 7.01 log 10 copies/swab) and diarrheic signs was not observed. In one pig, an association was made between diarrheic episodes and the peak of rotavirus C shedding, a well-known enteric pathogen 37, 38 . Very interestingly, kobuvirus fecal loads were typically lower than those reported of well-described enteric viruses of which the pathogenicity has been proven using piglet infection models, such as PEDV and rotavirus [39] [40] [41] . Similar viral loads for porcine kobuvirus were also found in case (4.60 ± 1.76 copies/qPCR reaction) and control pigs (4.79 ± 1.72 copies/qPCR reaction) during a recent Danish study to evaluate the role of viruses in the pathogenesis of the new neonatal porcine diarrhea syndrome. The study demonstrated that kobuvirus, astrovirus, rotavirus A, porcine teschovirus, porcine norovirus and porcine coronaviruses were not involved in the pathogenesis of the syndrome 42 . The finding of low kobuvirus loads in feces casts doubt over the true enteric pathogenic tropism of the virus. Hypothetically, its replication is likely not distributed across the whole villus but limited to either enterocytes at the villus' tips or to immune cells present in the gut. The presence of kobuvirus RNA in serum has also been demonstrated in Hungarian pigs, but it was not known if the virus is also replicating in other organs 43 . Both the oro-fecal route and the feeding of milk to sucklings pigs could be involved in virus transmission. Highest rates of infection were observed in suckling piglets, compared to older pigs, in other countries [44] [45] [46] . In our study, relatively long shedding of porcine kobuvirus was observed in three out of five animals, which may indicate that this virus may induce persistent infections. A 2011 Brazilian study demonstrated the presence of kobuvirus RNA in serum from 3-day-old piglets, which had disappeared by day 21, indicating viral clearance from the blood and excluding systemic persistence 45 . A complete lack of pathogenicity cannot be excluded, as porcine kobuviruses might play a role as a subclinically important virus. Such subclinical, yet immunosuppressive, properties have been attributed to the economically important swine pathogen porcine circovirus 47 . Of interest, one of the piglets died at the peak of kobuvirus shedding, although it was not clear if there was any causality between virus replication and the piglet's death. In vivo animal experiments in a model of neonatal, conventional kobuvirus-negative piglets should be conducted to elucidate the

pathogenesis of porcine kobuviruses. Attempts were made to isolate the virus in different cell lines (MA104, ST and SK), and peripheral blood mononuclear cells. There was no evidence of cytopathogenic effect after several days of incubation. Antibodies to visualize antigen expression were not available and therefore the possibility of replication without SClenTIFIC REPORTS | (2018) 8:9830 | DOI:10.1038/s41598-018-28180-9 evident cytopathogenic effect cannot be ruled out. Efforts will be made to isolate the virus in porcine primary enterocyte cultures, once available.

To assess more broadly the prevalence of kobuvirus in the Belgian swine industry, a retrospective analysis of diarrheic samples from suckling piglets less than two weeks old was conducted. A high proportion (40.9%) of the samples (n = 44) contained quantifiable viral loads ranging between 4.31 to 6.83 log₁₀ copies/g feces. Viral loads found were thus comparable to the loads excreted by piglets in the longitudinal analysis and the above-mentioned study from Denmark, demonstrating the endemic presence of the virus in the Belgian swine population 42 .

In the present study, non-diarrheic piglets were not included and therefore no association between kobuvirus prevalence and disease can be made. However, the prevalence of kobuvirus has been widely described in pigs from several European countries (The Netherlands (16.7%), Slovakia (63.4%), Hungary (81.0%), Czech Republic (87.3%), Austria (46.2%), Italy (52.4%), Germany (54.5%) and Sweden (45.0%)), American countries (The United States (21.9%) and Brazil (53.0%)), African countries (Kenya (14.9%) and Uganda (15.5%)) and Asian countries (Thailand (99%), South Korea (52.1%) and Vietnam (29.3%)) [44] [45] [46] [48] [49] [50] [51] [52] [53] . In a small proportion of these studies, statistically significant associations between prevalence of kobuvirus and diarrhea in pigs were demonstrated, such as in Hungary (54.5% prevalence in healthy pigs vs 92.3% prevalence in diarrheic pigs), Spain (47.5% healthy vs 74.4% diarrheic), Brazil (41% vs 78.4%), Thailand (19.3% vs 84.5%) and Vietnam (27.6% to 40.9%) 35, 45, 46, 52 . Indeed, it is difficult to make correlations between prevalence of the virus and diarrhea, as the pathogenicity of the virus could be largely influenced by other factors such as co-infections with other enteric viruses, microbiota and management factors.

Belgian isolates showed genetic moderate to high genetic variability, with nucleotide identities between 90.1 and 97.2%. Furthermore, they clustered diffusely between strains

from different countries around the world, indicating that strains are not distinguishable based on their geographical origin.

Because most (99%) of the reads generated during sequencing of the fecal sample 17V079 matched bacteriophages upon analysis with BLAST, bacteriophages may have played an important role in the pathogenesis of the diarrheic disease. De novo assembled contigs were analyzed using VirSorter, a software package for mining viral signals from microbial genomic data. Such tools allow maximizing the possibility of detecting dsDNA phages 54 . Several contigs showed high similarities to the *Bacteroides* phage B124-14, found in municipal wastewater and human fecal samples. It was shown to be absent in 30 samples collected from different animal species, including pigs, and is therefore considered a human-specific phage 55, 56 . The finding of several contigs, genetically similar to phage B124 and likely belonging to one phage genome, indicates that this phage found in the pig fecal sample may also replicate in the microbiome of the young pig gut and not solely in humans. However, it is possible that the phage's replication ability in the pig's gut is age-dependent and that very young age groups were not sampled in previous studies. Interestingly, several of the contigs found also showed similarities to *Escherichia* phages. Two of the contigs were similar to *Escherichia* phages PhAPEC5 and PhAPEC7, isolated from Belgian rivers in the neighborhood of poultry houses and known to cause lytic infections in avian pathogenic *Escherichia coli*. Electron microscopic images of the phages PhAPEC5 and PhAPEC7 indicated that they belonged to the family Podoviridae 57 . Two other contigs were similar to two closely related *Escherichia* phages, St11Ph5 and G7C, found in sewage and horse feces, respectively 58 . Finally, one contig showed limited similarity to an *Enterococcus* phage, isolated from hospital sewage in China, while a last contig showed moderate similarities to the bacterial *Enterococcus hirae* genome. This region may be a prophage, inserted in the bacterial genome.

The phages found in this piglet may have reshaped the gut microbiota, allowing opportunistic bacteria such as *Enterococcus hirae* to proliferate and to start secreting toxins. It is also possible that a phage infection of bacteria in the pig's gut led to a stress status for these bacteria, prompting the secretion of toxins. The new neonatal diarrhea syndrome described above shows high similarities to the disease described in the case 17V079 and it may be that bacteriophages are involved in the pathogenesis of this syndrome. So far, the role of phages has not been considered in the pathogenesis of several enteric disorders, but given the high abundance here, it should be in future studies.

It is clear that new technologies will change the way diagnostics are performed in the near future. Pricing might currently be an aspect hampering high-throughput analysis of samples in swine veterinary medicine, but as the technology evolves fast, this might become very soon less relevant. Complete overviews of all viruses and other pathogens in a sample will be given in a single readout instead of requiring different diagnostic assays. However, care should be given to the interpretation of such results, as they should only be analyzed by trained veterinarians.

Viruses. Porcine rotavirus A (RVA) strain RVA/Pig-tc/BEL/12R046/2012/G9P [23] was isolated from a diarrheic piglet and grown for three successive passages in MA104 cells to an infectious virus titer of $10^{7.8}$ CCID₅₀ / ml. The nucleotide sequences of the 11 gene segments of this strain were resolved earlier using Sanger sequencing (GenBank accession numbers: KM82070 (VP1), KM820707 (VP2), KM827014 (VP3), KM820720 (VP4), KM820728 (VP6), KM820735 (VP7), KM820742 (NSP1), KM820672 (NSP2), KM820679 (NSP3), KM820686 (NSP4) and KM820693 (NSP5)) 59 . A porcine epidemic diarrhea virus strain (PEDV, CV777) isolated in Belgium in the 1970s was adapted for growth in Vero cells in the 1980s 60 . In our Laboratory, the virus was grown to an infectious virus titer of $10^{6.0}$ CCID₅₀ /ml (GenBank accession number: AF353511).

Origin of a fecal sample from diarrheic suckling piglets. A diarrheic fecal sample was collected from a Belgian pig on a farm housing a total of 620 sows and using a 2-week batch-production system, with a weaning age of 23 days. Topigs Norsvin sows were crossed with Piétrain boars, producing 32. toxins (Suiseng, Hipra). Rotavirus A vaccination was done off-label with an inactivated bovine rotavirus A vaccine (Lactovac, Zoetis). Until recently, diarrheic problems were rarely present in suckling piglets and also very low mortality percentages (6.2-7.1%) were observed. Since the spring of 2017, enteric disease started causing more severe problems accompanied with mortality on this farm, mainly in 7-days-old suckling piglets. A diarrheic fecal sample of such a piglet was investigated at a private diagnostic laboratory (Dialab, Belsele, Belgium) and labeled 17V079. No virological cause was found to explain the diarrheic problems on the farm. The only isolated bacterium was *Enterococcus hirae*. This bacterium was thereon added to the sow vaccination schedule (inactivated autovaccine). No other pathogens were found in this sample. As the clinical picture hinted at a viral cause for the disease, the sample was sent to the Laboratory of Virology at the Faculty of Veterinary Medicine (Ghent University) for

further analysis. The sample tested negative for RVA, RVC, PEDV and TGEV using in-house RT-qPCR assays 25, 61, 62 . Therefore, it was decided to perform a metagenomics analysis with MinION described in this study.

Purification of viral nucleic acids. First, viral enrichment was done based on the NetoVIR protocol to obtain pure viral nucleic acids for sequencing library preparation 13 . MinION analyses of cell culture grown viruses RVA and PEDV were conducted at the Laboratory of Clinical Virology (Rega Institute, KU Leuven), whereas the diarrheic fecal sample was analyzed at the Laboratory of Virology (Faculty of Veterinary Medicine, Ghent University). RVA and PEDV stocks were centrifuged at $17,000 \times g$ for 3 min. The supernatant of both suspensions was diluted to $6 \log_{10}$ CCID₅₀ /ml and 500 μ l of each suspension was mixed to reach an equal concentration of both viruses. This mixture was filtered using a 0.8 μ m polyethersulphone filter for 1 min at $17,000 \times g$, followed by a nuclease treatment for 2 hours at 37 °C to digest free nucleic acids in the suspension: 250 μ l of the sample was added to 14 μ l of home-made buffer (1 M Tris, 100 mM CaCl₂ and 30 mM MgCl₂ , pH 8), 4 μ l of Benzonase Nuclease (Millipore) and 2 μ l Micrococcal Nuclease (NEB) as described earlier 13 . Fourteen microliters of EDTA were added to stop the reaction, followed by extraction of nucleic acids from the viral particles using the QIAamp Viral RNA Mini Kit (Qiagen). The manufacturer's instructions were followed but no carrier RNA was added and elution was done in 30 μ l of AVE to concentrate the viral nucleic acid extract.

The diarrheic fecal sample 17V079 was processed similarly as the cell culture grown viruses, with some minor modifications. A 10% w/v suspension of the diarrhea was made in Minimum Essential Medium and centrifuged. The supernatant was filtered through a 0.45 μ m syringe filter (Sarstedt) and treated with Benzonase Nuclease for 1 hour to speed up the diagnostic pipeline. Viral nucleic acids were extracted using the QIAamp Cadore Pathogen Mini Kit according to the manufacturer's instructions without addition of carrier RNA. Elution was done in a volume of 50 μ l. cDNA and second strand synthesis for nanopore sequencing. Nucleic acids were heated at 95 °C for 2 min and chilled on ice to resolve secondary RNA structures and to denature double-stranded RNA. Superscript IV Reverse Transcriptase (ThermoScientific) was used to generate cDNA. Ten microliters of template nucleic acids were mixed with 0.5 μ l random hexamer primers (Random Primer 6, New England Biolabs), 1 μ l dNTP mix (NEB) and 2.5 μ l nuclease-free water. Primer annealing was conducted at 65 °C for 5 min, after which 4 μ l Superscript IV Reaction Buffer (ThermoScientific), 1 μ l dithiothreitol (ThermoScientific) and 1 μ l SuperScript IV Reverse Transcriptase (ThermoScientific) were added in a total reaction volume of 20 μ l. The

reaction conditions were as follows: 23 °C for 10 min, 50 °C for 10 min, 80 °C for 10 min and an infinite hold step at 10 °C.

A second strand of DNA was generated from single stranded (c)DNA molecules using the NEBNext Second Strand Synthesis Kit (NEB). Twenty microliters cDNA reaction mixture were added to 10 µl NEBNext Second Strand Synthesis Reaction Buffer, 5 µl NEBNext Second Strand Synthesis Enzyme Mix and 45 µl nuclease-free water (80 µl total reaction volume). Isothermal amplification was done at 16 °C for 1 h and double-stranded nucleic acids were purified using 144 µl of magnetic AMPure XP Beads (Beckman Coulter). Two washing steps with freshly prepared 70% ethanol were conducted before eluting in 52 µl nuclease-free water.

Nanopore sequencing library preparation. A deoxyadenosine was ligated to the 3'-end of double-stranded nucleic acids to allow binding of complimentary sequencing adapters. Fifty microliters of (un) amplified DNA were mixed with 7 µl Ultra II End-Prep Reaction Buffer (New England Biolabs) and 3 µl Ultra II End-prep enzyme mix (New England Biolabs), and incubated at 20 °C for 5 min and 65 °C for 5 min. Next, nucleic acids were purified using 60 µl AMPure XP Beads and eluted in 31 µl nuclease-free water. Sequencing adapters, provided with the Ligation Sequencing Kit 1D (R9.4) (SQK-LSK108, ONT), were ligated to the dA-tailed nucleic acids. End-prepped DNA (30 µl) was mixed with 20 µl adapter mix (AMX, ONT) and 50 µl Blunt/TA Ligation Master Mix (New England Biolabs) in a total reaction volume of 100 µl and incubated at room temperature for 10 min. The sequencing library, containing double-stranded DNA with adapters ligated to the 3' ends, was then purified using 40 µl AMPure XP beads. Two washing steps were conducted using 140 µl Adapter Bead Binding Buffer (ABB, ONT) before eluting in 15 µl of Elution Buffer (ELB, ONT). (EXP-LLB001, ONT), 12 µl adapted and tethered library and 12.5 µl nuclease-free water. Sequencing was done using the software programme MinKNOW software (ONT).

Bio-informatics analyses. Raw reads were produced by MinKNOW. Live basecalling was enabled for the first experiment using MinKNOW version 1.5.5. In the second experiment, basecalling was done after the sequencing run using Albacore (version 1.2.5., ONT). Quality scores and read lengths were visualized using NanoPlot, followed by quality filtering with NanoFilt 63 . Reads with a q-score lower than 7 were omitted. Sequences were then analyzed using different BLAST methods including BLASTn and tBLASTx (BLAST version 2.6.0; e-value cut-off $1e^{-3}$ - $1e^{-10}$) to compare sensitivity and run-times to detect

viral sequences among the reads. A complete viral database was composed of all virus sequences in GenBank (taxonomy ID 10239, containing sequences up to 17th of September 2017). The best hit (lowest e-value) was visualized using KronaTools 64 . Reads matching viruses were extracted using Seqtk (<https://github.com/lh3/seqtk>) and used in downstream analyses. GraphMap (version 0.5.2) and Samtools (version 1.6) were used for mapping of reads against reference sequences, while Canu 1.6 was used for de novo assembly of viral genomes [65] [66] [67] [68] . VirSorter was run using the 'Viromes' database to look for phages, with the Virome Decontamination Mode on to identify phage contigs 54 . Bio-informatics analyses were executed on a local computer cluster and the high-performance computing facilities of Ghent University. The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request. Sanger sequencing of porcine kobuvirus polymerase gene. A porcine kobuvirus was discovered in the sample 17V079 using the MinION. The sequence of the 3D gene of porcine kobuvirus encodes the polymerase and is considered to be most conserved among different strains. The exact nucleotide sequence of this virus was verified using reverse transcription polymerase chain reaction followed by Sanger sequencing, as low coverage was obtained with MinION. RT-PCR was executed using the OneStep RT-PCR Kit (Qiagen) with the newly designed primers Kobu_6049Fw and Kobu_7524Rv (IDT DNA Technologies) (Table 2). The RT-PCR reaction contained 5 µl 5 × Qiagen OneStep RT-PCR Buffer, 1 µl dNTPs, 3 µl of each primer (10 µM), 7 µl nuclease-free water, 1 µl OneStep RT-PCR enzyme mix and 5 µl template RNA or water (total reaction volume of 25 µl). RT-PCR conditions were as follows: 50 °C for 30 min, 95 °C for 15 min, followed by 30 cycles of amplification (94 °C for 30 s, 50 °C for 30 s and 72° for 90 s) and a final extension step at 72 °C for 1 min. Reactions were held at 10 °C prior to loading 5 µl PCR product with 1 µl of loading dye in a 1.5% agarose gel. Electrophoresis was conducted for 30 min at 100 V and PCR product was visualized by ethidium bromide staining and UV light. The amplicon was sent to GATC (Constance, Germany) for Sanger sequencing using an ABI 3730xl DNA Analyzer system. Quality control of the raw chromatograms was done using 4Peaks (Nucleobytes BV, The Netherlands) and BLASTn (NCBI, United States).

Specific RT-qPCR primers (Table 2) for the porcine kobuvirus polymerase-encoding gene were designed using Primerquest and Oligoanalyzer (IDT DNA Technologies) to allow exact quantification in feces of piglets. Each RT-qPCR reaction consisted of 10 µl PrecisionPlus OneStep qRT-PCR Mastermix containing SYBR Green, ROX and an inert blue pipetting dye (Primerdesign, Southampton, United Kingdom), 0.4 µl of each primer (200 nM) and 6.2 µl nuclease-free water. Three microliters of template RNA or water were added to each tube containing 17 µl mastermix. A synthetic RNA positive control (175nt) was generated by RT-

PCR using the primers Kobu3D_qPCR +T7_Fw and Kobu3D_qPCR_Rv, followed by in vitro transcription of this PCR product using a T7 RNA polymerase. The positive control was measured using Nanodrop and used to setup a standard curve over a linear dynamic range (LDR) from six to one log 10 copies/reaction. Reaction conditions were as follows: 55 °C for 10 min and 95 °C for 2 min, followed by 40 cycles of denaturation (95 °C for 10 s) and annealing (58 °C for 60 s). Detection of SYBR Green fluorescence was done at the end of each annealing phase. A melt curve analysis was executed to assess specificity of the amplicons generated. Each dilution point in the standard curve and each sample was tested in duplicates. Amplicons were analyzed once on an agarose gel to assess the correct length of the amplicon and Sanger sequencing was conducted to confirm the amplification of the partial porcine kobuvirus polymerase gene. Assays were valid if the efficiency over the LDR was between 90 and 110%, and R² of the standard curve replicates was >0.99. Quantification of the viral loads was possible if the C_q-values of two qPCR replicates fell within the LDR of the assay. Both replicates had to be positive for a sample to be considered as positive. If the C_q-values of specific amplicons have fallen behind the lowest point of the standard curve, the sample was considered positive but not quantifiable. Longitudinal investigation of kobuvirus and rotavirus shedding in suckling piglets. Upon characterization of the virome with the MinION, a longitudinal follow-up study was setup between August and September 2017. To warrant the health status of the pig stock, entrance to the farm was strictly regulated. Sampling was performed by the farmer. Detailed instructions and sampling materials were provided to the farmer. Sample collection in the longitudinal field study was done in agreement with the European legislation on animal experiments. Sample collection was approved by and done in accordance to the requirements of the Local Ethical Committee of the Faculty of Veterinary Medicine and Bioscience Engineering of Ghent University. One day after parturition of the sows, five litters were selected at random. Within each litter, one piglet was identified for longitudinal follow-up during the entire suckling period. A dry cotton rectal swab (Copan) was collected from each individual piglet at days 1, 5, 8, 11, 14, 17, 20 and 22 after birth. The swab was placed immediately in 2 ml of viral transport medium (phosphate buffered saline containing 1000 U/ml penicillin (Continental Pharma, Puurs, Belgium), 1 mg/ml streptomycin (Certa, Braine l'Alleud, Belgium), 1 mg/ml gentamicin (Life Technologies) and 0.01% v/v Fungizone (Bristol-Myers Squibb, Braine l'Alleud, Belgium)) in a sterile 15 ml falcon tube (Sarstedt) and stored at -20 °C. Every week, samples were collected from the farm and transported to the Laboratory of Virology. The farmer was asked to mark the tube of each sample for presence or absence of diarrheic signs. Upon arrival in the Laboratory of Virology, the samples were thawed and placed on a shaker for 30 min at 4 °C to release viral particles in the transport medium. Samples were extracted using the QIAamp Cadore Pathogen Mini Kit according to the manufacturer's instructions and purified nucleic acids

were eluted in 100 µl of AVE and stored at –70 °C until RT-qPCR analysis. RT-qPCR analysis was conducted, as described above, to quantify porcine kobuvirus genome copies per swab. Furthermore, RVA and RVC shedding was assessed using previously described in-house RT-qPCR assays 25, 61 .

Belgian suckling pigs. Fecal samples (n = 44) of diarrheic suckling piglets less than 2 weeks old were sent to a private laboratory by veterinarians (Dialab, Belsele, Belgium) for etiological diagnosis, as described earlier.

These samples were collected in 2014 and stored at –70 °C in the laboratory. They had previously been evaluated for the presence of rotaviruses using RT-qPCR 25 . RNA extraction was conducted using the QIAamp Cadore Pathogen Mini Kit (Qiagen) as described above and RT-qPCR was done to quantify the load of kobuvirus RNA copies. Samples with a quantifiable viral load were subjected to RT-PCR to amplify the 3D polymerase gene, after which Sanger sequencing was performed. The sequences encoding the polymerase of 11 Belgian porcine kobuvirus isolates were deposited into GenBank with accession numbers MH184664-MH184674. The sequences were used to conduct a multiple sequence alignment together with other porcine kobuvirus strains in MEGA 7 using the ClustalW plug-in 69 . A maximum-likelihood phylogenetic tree was constructed with RAxML using a general time reversible model with gamma distribution (20 cats, alpha: 0.121, LogLK = 14938.461) and heuristic branch swapping 70 . Tree editing was done using Affinity Designer (Serif). Pairwise distances were calculated using the p-distance model in Mega with bootstrap values set at 500 replicates.

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C. difficile 630Δerm Spo0A Regulates Sporulation, but Does Not Contribute to Toxin Production, by Direct High-Affinity Binding to Target DNA

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3485338/>

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Abstract: *Clostridium difficile* is a Gram positive, anaerobic bacterium that can form highly resistant endospores. The bacterium is the causative agent of *C. difficile* infection (CDI), for which the symptoms can range from a mild diarrhea to potentially fatal pseudomembranous colitis and toxic megacolon. Endospore formation in Firmicutes, including *C. difficile*, is governed by the key regulator for sporulation, Spo0A. In *Bacillus subtilis*, this transcription factor is also directly or indirectly involved in various other cellular processes. Here, we report that *C. difficile* Spo0A shows a high degree of similarity to the well characterized *B. subtilis* protein and recognizes a similar binding sequence. We find that the laboratory strain *C. difficile* 630 Δ erm contains an 18bp-duplication near the DNA-binding domain compared to its ancestral strain 630. In vitro binding assays using purified C-terminal DNA binding domain of the *C. difficile* Spo0A protein demonstrate direct binding to DNA upstream of *spo0A* and *sigH*, early sporulation genes and several other putative targets. In vitro binding assays suggest that the gene encoding the major clostridial toxin TcdB may be a direct target of Spo0A, but supernatant derived from a *spo0A* negative strain was no less toxic towards Vero cells than that obtained from a wild type strain, in contrast to previous reports. These results identify for the first time direct (putative) targets of the Spo0A protein in *C. difficile* and make a positive effect of Spo0A on production of the large clostridial toxins unlikely.

Text: Sporulation is an adaptive strategy that enables bacteria to survive harsh environmental conditions for prolonged periods of time, and is an integral part of the transmission of sporulating pathogens and their tolerance and resistance towards antimicrobial compounds.

Spo0A is the key regulator for sporulation [1, 2] . Most of our knowledge about the protein is based on work in Bacilli. Spo0A is a response regulator that demonstrates phosphorylation

dependent binding to DNA [3] [4] [5] . Phosphorylation occurs through the concerted action of several proteins that together form a so called phosphorelay [6] . The signaling cascade allows for the integration of environmental signals into the regulation of Spo0A dependent processes, including sporulation. The two functional domains, the N-terminal phosphorylation and dimerization domain (receiver domain), and the C-terminal DNA binding (effector) domain are separated by a hinge region that is relatively poorly conserved [7] . Phosphorylation is believed to result in a structural rearrangement that facilitates dimerization [8, 9] , resulting in the disruption of transcription-inhibitory contacts between the receiver and effector domains. The isolated DNA binding domain can bind legitimate targets of the Spo0A protein due to the absence of the transcription inhibitory contacts, thereby bypassing the need for phosphorylation [10] . Extensive characterization of Spo0A targets has revealed a motif that represents a high affinity Spo0A binding site, the 0A box [10, 11] . The crystal structure of the DNA binding domain confirms specific and non-specific contacts between the protein and the consensus sequence [12, 13] . It is noteworthy that Spo0A regulates many other processes than sporulation, such as competence for genetic transformation, DNA replication, and biofilm formation in *B. subtilis* [14] [15] [16] , virulence factors and stress responses in for instance *B. anthracis* and *B. thuringiensis* [17] [18] [19] [20] [21] , and solvent production in *Clostridium acetobutylicum* [22, 23] .

C. difficile is a Gram positive, anaerobic bacterium that is the causative agent of *C. difficile* infection (CDI) (for recent reviews see [24, 25]). Though many people are asymptotically colonized by *C. difficile*, the bacterium can cause serious health problems, such as pseudomembranous colitis and toxic megacolon, under the influence of risk factors such as age and antibiotic use. As a result, CDI was long regarded a nosocomial infection. Recently, however, an increase in the cases of community acquired CDI can be observed [26] . Outbreaks of CDI have been linked to so called hypervirulent strains, such as PCR ribotypes 027 (BI/ NAP1) and 078 [27, 28] . Its main virulence factors are the major clostridial toxins A and B [29, 30] . In addition, certain strains of *C. difficile*, including ribotypes 027 and 078, additionally encode a binary toxin [31, 32] . *C. difficile* is transmitted via the fecal-oral route. It is believed that spores are crucial to successfully infect new hosts, as they are able to withstand the harsh environment of the stomach, and survive antibiotic treatments that alter the endogenous flora, after which *C. difficile* can overgrow [24, 25] .

There is limited knowledge about the regulation of sporulation in *C. difficile*. It has been reported that *spo0A*, as expected, is required for the formation of spores [33] and the gene is required for persistence and transmission in mice [34]. Though the pathways downstream of *Spo0A* seem to a large extent conserved between *B. subtilis* and *Clostridia*, this is less so for the pathways leading to activation of *Spo0A* [2]. It has been suggested that the orphan histidine kinase CD2492 is involved in the activation of *Spo0A* [35]. Similarly, it was reported that multiple orphan histidine kinases can phosphorylate *Spo0A* in *C. acetobutylicum* [36]. Recently, it was reported that *spo0A* can be transcribed from a SigH-dependent promoter [37]. It is unknown which genes are regulated by direct binding of *Spo0A* to their upstream regions.

Here, we establish an in vitro binding assay for *C. difficile* *Spo0A* and demonstrate for the first time direct binding of this transcription factor to DNA upstream of several putative target genes.

Escherichia coli strains were routinely grown in Luria-Bertani broth or plates, supplemented with appropriate antibiotics. Chloramphenicol was used at a final concentration of 20 mg/mL for agar plates and 10 mg/mL for liquid cultures. Ampicillin was used at a final concentration of 100 mg/mL. Kanamycin was used at a final concentration of 20 mg/mL. Cloning was carried out using *E. coli* DH5a, overexpression was performed in *E. coli* Rosetta(DE3) pLysS (Novagen). *C. difficile* strains were grown in a glucose-free trypton-yeast based medium (TTY; 3% w/v bactotrypton (BD), 2% yeast extract (Fluka), 0.1% w/v thioglycollate (Sigma) pH 7.4), supplemented with 20 mg/mL of lincomycin when appropriate, or on CLO or TSS plates (Biomerieux).

All plasmids are listed in Table 1. Primers (obtained from Sigma Aldrich) are listed in Text S1 and specific cycling conditions are available on request. Unless noted otherwise, PCR reactions were carried out using Pfu polymerase (Fermentas) according to the instructions of the manufacturer.

Plasmid pWKS1251, for the overproduction of *Spo0A*-DBD carrying a C-terminal 66His-tag, was constructed as follows. A sequence corresponding to the DNA binding domain of *Spo0A* was amplified using primers oWKS-1123a and oWKS-1124 using chromosomal DNA from *C. difficile* strain 630Derm as a template. The resulting fragment was cloned into

pCR2.1-TOPO (Invitrogen), yielding pWKS1247. This plasmid was digested with NdeI and XhoI, separated on a 1% agarose/0.56 TAE (20 mM Tris Acetate, 0.5 mM EDTA) gel, the fragment corresponding to the DNA binding domain was recovered by gel-isolation (using a GeneJET Gel Extraction kit, Fermentas) and cloned into similarly digested pMF14 [10] that had been gel-isolated in the same manner. The construct was verified by PCR, restriction analyses and DNA sequencing using primers oWKS-135 and oWKS-136 (see below).

Plasmid pWKS1245, for the production of full length Spo0A carrying a C-terminal 6xHis-tag, was constructed in a similar manner using chromosomal DNA from *C. difficile* 630Derm as a template, but using the PCR product of primers oWKS-1122 and oWKS-1123a.

Plasmids used as PCR templates for generating EMSA probes were constructed by cloning the PCR products into pCR2.1-TOPO. The inserts, and in the case of the mutated PabrB promoters the presence of the desired point mutations in the consensus 0A box, were verified by DNA sequencing using primers oWKS-24 and oWKS-25 (see below).

Sequence grade plasmids were isolated using a Nucleospin Plasmid QuickPure kit (Macherey Nagel) according to the manufacturer's instructions, except that two lysis reactions were combined onto a single filter and eluted with 65uC prewarmed AE buffer. All constructs were sequenced using BigDye Terminator chemistry (Invitrogen) on an ABI3130 sequencer (Perkin Elmer), according to the instructions of the manufacturers. In short, 200 ng of plasmid was mixed with 3.2 pmol of primer, 1 mL Terminator Ready Reaction Mix (Invitrogen) in a final volume of 20 mL. After thermocycling, DNA was precipitated and washed with 65% isopropanol, and dissolved in 12 mL HiDi formamid (Invitrogen) at 96uC for 2 mins and stored in the dark at 4uC until the sequencing run. Sequence analyses were performed in CloneManager Professional Suite 7 (SciEd) and Geneious version 5.6.2 (Biomatters Ltd).

Plasmids pWKS1245 and pWKS1251 were transformed into *E. coli* Rosetta(DE3) pLysS (Novagen). Transformants were used to inoculate 25 mL of LB with appropriate antibiotics. After overnight incubation, the cells were 1:100 diluted in 500 mL fresh medium containing appropriate antibiotics. Protein production was induced with 1 mM IPTG at an OD600 of 0.7 and growth was continued for another three hours before harvesting. Cells were washed with ice cold PBS and stored at 280uC for later use. Purification of the proteins was

essentially done as described [10] . In short, cells were disrupted in 4 mL lysis buffer (2 mM PMSF, 10 mM imidazole, 5 mM beta-mercaptoethanol, 300 mM NaCl, 50 mM NaH₂PO₄, pH 7.9). Cleared cell lysates were incubated with 2 mL pre-equilibrated 50% TALON slurry (Clontech) in a final volume of 15 mL lysis buffer for 1 hr. The resin was allowed to settle on a Poly-Prep column (BioRad) and washed with 2 mL wash buffer (20 mM imidazole, 300 mM NaCl, 50 mM NaH₂PO₄, pH 7.9). The protein was stepwise eluted in 1 mL fractions after applying 2 mL elution buffer to the column (identical to wash buffer but with 50, 100, 250 or 500 mM imidazole). The whole procedure was carried out at 4°C. Fractions were assayed for purity and yield and suitable fractions were dialysed against 26 L dialysis buffer (50 mM Tris-HCl pH 8, 1 mM EDTA, 0.5 mM DTT) using Slide-A-Lyzer cassettes with a molecular weight cut-off of 3.5 kDa (Pierce).

Proteins were stored at -80°C in storage buffer (identical to dialysis buffer but containing 20% glycerol). Protein concentrations were determined using Bradford reagent (BioRad), according to the manufacturer's instructions.

DNA fragments for use in EMSA experiment were generated by PCR using GoTaq polymerase (Promega) and chromosomal DNA from *B. subtilis* JH642 (Bacillus Genetic Stock Center 1A96; <http://www.bgsc.org>), plasmids listed in Table 1 , or chromosomal DNA from *C. difficile* 630Derm [38] as a template. Primers and specific cycling conditions for generation of the EMSA probes are listed in Text S1. DNA fragments of the expected size were isolated from a 16TAE/8% native polyacrylamide gel using diffusion buffer (0.5 M ammonium acetate, 10 mM magnesium acetate, 1 mM EDTA pH 8, 0.1% SDS) and a QIAexII kit (Qiagen), according to the manufacturer's instructions. Recovered DNA was end-labeled with ³²P- γ -ATP using FR buffer and T4 kinase (Invitrogen) according to the instructions of the manufacturer. Specific activity was determined on a LS6000 scintillation counter (Beckman).

EMSA conditions were based on previous studies [10] . In short, binding reactions were carried out in binding buffer (10 mM Tris-HCl pH 7.6, 1 mM EDTA, 50 mM NaCl, 1 mM DTT, 5% glycerol) in the presence of 200 mg/mL bovine serum albumin (NEB) and 200 cpm/mL radiolabeled DNA fragment. Reactions were incubated for 20 minutes at 30°C prior to loading on a 16TAE/8% non-denaturing polyacrylamide gel that was prerun for 20 minutes at 50 V in 16 TAE buffer. Electrophoresis was carried out for 120 min at 85 V. After vacuum

drying the gels onto filter paper, they were imaged after overnight exposure on Phosphorimager screens on a Typhoon instrument (GE Healthcare).

The toxic effects of *C. difficile* culture supernatants on Vero cells (a kind gift of Eric Snijder [39]) were determined as follows. Supernatant from a bacterial culture was harvested by centrifuging cells for 3 minutes at 14000g and filtered on a 0.45 µm cellulose acetate filter using a syringe. Supernatants were 2-fold serially diluted in cell culture medium (Dulbecco modified Eagle medium (Lonza) supplemented with 100 mg/mL penicillin, 100 U/mL streptomycin, 10% fetal calf serum), before applying them to a monolayer of Vero cells, and incubation was continued for another hour. As a positive control, 50 µL 1:10 diluted purified toxin (Techlab) was added to the cells. To determine if observed cytotoxic effects were specific for the large clostridial toxins, commercially available anti-toxin against TcdA and TcdB (Techlab) was added to 10-fold diluted bacterial supernatant for 60 min prior to incubation on the Vero cells. Toxin end-point titres were defined as the lowest dilution at which no cytopathological effects (cell rounding) were observed.

Statistical significance was evaluated with an independent sample t-test.

Immunization of mice with full length *C. difficile* Spo0A-6xHis was kindly performed at the Wellcome Trust Sanger Institute (Hinxton, UK). Cells from 1 mL of *C. difficile* culture were collected by centrifugation for 1 min at 14000 rpm in a table top centrifuge and resuspended in 200 µL resuspension buffer (10 mM Tris HCl pH 8, 10 mM EDTA, 0.5 mg/mL lysozyme, 1 mM Pefabloc SC (Roche)). After incubation for 30 mins at 37°C, 50 µL of 56 SDS sample buffer (0.1 M DTT, 2% SDS, 50 mM Tris HCl pH 6.8, 10% glycerol, 0.0025% BPB) was added, and samples were heated to 96°C for 5 mins. Total cell lysates (amounts corrected for OD 600) were separated on a 12% SDS-PAGE gel prior to semi-dry blotting for 1 h at 10 V to a polyvinylidene fluoride (PVDF) membrane. Membranes were blocked in PBST buffer (phosphate buffered saline with 0.1% v/v Tween-20) containing 5% membrane blocking reagent (Amersham Biosciences). To visualize Spo0A protein cleared polyclonal serum from a single mouse at a 1:3000 dilution was used, followed by either a goat-anti-mouse HRP-conjugated secondary antibody followed by ECL+ detection (Amersham Bioscience), or a goat anti-mouse-biotin-conjugated secondary antibody (Dako) followed by a tertiary mouse-anti-biotin Cy3-conjugated antibody (Jackson). Detection was done using on a Typhoon instrument (GE Healthcare). Background corrected peak volumes were quantified using ImageQuant TL (Amersham Biosciences).

Alignments of *B. subtilis* and *C. difficile* spo0A were made using ClustalW2 (<http://www.ebi.ac.uk/Tools/msa/clustalw2/>) on the basis of the published genome sequences, Genbank accession numbers AL009126 and AM180355, respectively, and the 630Derm spo0A sequence as determined in this study. The sequence for spo0A of *C. difficile* strain 630Derm was deposited in Genbank (accession no JX050222). Consensus Spo0A boxes were identified using a Single string Search command in Genome2D [40], allowing 0 mismatches. The box positions were linked to up and downstream genes using the "Add nearest gene to List of DNA Motifs" feature and Microsoft Excel. The results were manually inspected for those boxes within 500 bp upstream of a gene on the same strand. Figures for publication were prepared using ImageQuant TL (Amersham Biosciences), Adobe Photoshop CS3 (Adobe Systems Inc) and Corel Graphics Suite X5 (Corel Corporation).

In order to characterize *C. difficile* Spo0A, the full length protein and its DNA binding domain (DBD) were expressed as a Cterminally 66His-tagged protein in the heterologous host *Escherichia coli* (Fig. 1A) and purified to near homogeneity using metal affinity chromatography (Fig. 1A; lanes P). Full length protein was used to raise antibodies to detect Spo0A in total lysates of *C. difficile* strains, and the purified DNA binding domain was used in subsequent in vitro binding assays (see below).

We determined the expression of *C. difficile* Spo0A throughout growth. We found that the protein is present in lysates from exponential to stationary growth phase cells. We performed immunoblotting using polyclonal antibodies against *C. difficile* Spo0A on total lysates of wild type and spo0A mutant cells grown in a trypton-yeast based medium (TTY). We found a clear signal of the size expected for full length Spo0A (31 kDa) as early as 3 hours post inoculation (exponential growth phase), through transition phase (8 h) as well as 24 and 48 hours post inoculation (stationary growth phase) (Figure 1B; 630Derm). The signals were specific for *C. difficile* Spo0A as they were absent from lysates from the *C. difficile* spo0A mutant (Fig. 1B, CT::spo0A). We obtained similar results in other media, such as the commonly used supplemented brain heart infusion broth (BHIS; data not shown).

To determine relative levels of Spo0A throughout growth, we performed an immunoblot experiment using fluorescent antibodies, which gives more quantitative information

compared to the use of horseradish peroxidase conjugated antibodies in our hands. We found that the levels of Spo0A increases approximately 20-fold from 6 hours post inoculation and remains at similar levels from 8 to 48 hours post inoculation (Figure 1C).

Though it should be noted that the Western blots do not provide information on the phosphorylation state of the protein, we conclude that the protein in active or inactive form is present throughout growth and is more abundant in stationary growth phase.

Spo0A of *C. difficile* Strain 630Derm Contains a 6aminoacid Duplication BLAST homology searches readily identify a homolog of the well-characterized *B. subtilis* Spo0A protein in *C. difficile* 630 (CD1214) and previous work demonstrated that a *spo0A* mutant (an insertional inactivation of *cd1214*) -as expected -no longer forms spores [41] . In silico analyses suggest a similar secondary structure for both proteins (Fig. 2A) , with a conserved dimerization and DNA binding domain, separated by a poorly conserved hinge region [7, 12] .

We compared the sequence of CD1214 obtained from our lab strain 630Derm [38] to that of the published *C. difficile* 630 genome [42] . Strain 630Derm is a spontaneous erythromycin sensitive strain, which is commonly used in mutagenesis studies and was obtained by serial passaging of strain 630 [33, 38] . The 630Derm *spo0A* sequence (Genbank accession no JX050222) was derived from the expression plasmids constructed for this study, and confirmed in a whole genome sequence of strain 630Derm generated in our lab (data not shown). We found that 630Derm *spo0A* contains an 18 base pair direct repeat, resulting in a 6 amino acid (NVGNIE) duplication compared to the published reference sequence. The duplication maps to a region of the protein with relatively low sequence conservation (hinge), flanking the highly conserved DNA binding domain (Fig. 2A and B) . We verified the absence of this duplication in strain 630 by PCR (Fig. 2C) as well as sequencing from the chromosomal DNA of *C. difficile* 630 (data not shown), to rule out an error in the original genome sequence and to demonstrate that the difference in size of the PCR product was specific to the 18 bp insertion. In addition, we checked several other strains of PCR ribotypes 12 (to which 630 and 630Derm belong) by PCR, but the duplication was found to be unique to 630Derm among the isolates tested (data not shown).

C. difficile Spo0A-DBD Shows Similar Specificity as *B. subtilis* Spo0A-DBD Next, we examined the conservation of the DNA binding domain of Spo0A (Spo0A-DBD) between *B. subtilis* and *C. difficile*. In *B. subtilis* amino acid residues contacting the backbone of the DNA and interacting with specific residues of the Spo0A binding sequence have been defined [13]. We found that all these residues were conserved in the *C. difficile* protein sequence (Fig. 2B), indicating that the protein likely recognizes a similar motif.

DNA binding by full length Spo0A in *B. subtilis* requires phosphorylation dependent dimerization [8, 9]. However, it was shown that the isolated DBD is capable of binding to legitimate targets of the full length protein [10]. Analogously, we purified the *C. difficile* Spo0A-DBD for use in in vitro binding assays. As no direct targets for the *C. difficile* protein have been reported so far, we used the upstream region of the *abrB* gene (*PabrB*) of *B. subtilis*. *PabrB* is commonly used as a high-affinity control in binding assays with the *B. subtilis* Spo0A or Spo0A-DBD protein [43, 44]. It is noteworthy that we failed to identify a homolog of *abrB* in *C. difficile* using BLAST, indicating that potential indirect regulation by Spo0A cannot occur through *abrB* in *C. difficile* as it does in *B. subtilis*. We found that *C. difficile* Spo0A-DBD bound with high affinity to *PabrB* (Fig. 2D and E). We performed electrophoretic mobility shift assays (EMSAs) using radiolabeled *PabrB* and increasing amounts of purified *C. difficile* Spo0A-DBD that was purified using a C-terminal 66His-tag. The addition of protein leads to a dose-dependent retardation of the DNA fragment with an apparent K_D of 50 nM. In the same range of protein concentrations, no binding was observed for a negative control (a DNA fragment of *B. subtilis* *citG* [45]) (Fig. 2E), suggesting that binding was specific for the *abrB* promoter region.

B. subtilis Spo0A recognizes a distinct sequence (0A box), that is characterized by a 7 bp core motif (TGTCGAA) [10, 11]. Structural studies have revealed that the protein makes specific contacts with the G at position 2 (G2), and the C at position 4 (C4) and 5 (G5) of this motif [13]. We introduced G2A, C4A, G5A, G2A/C4A and C4A/G5A mutations in the perfect consensus core 0A-box present in *PabrB*. We found that the affinity of *C. difficile* Spo0A for these mutated *PabrB* fragments was highly reduced (Fig. 2E). We performed EMSAs using radiolabeled *PabrB* containing the mutated core sequence. For the single point mutations in the DNA, the affinity decreased 10-fold. There did not seem to be an additive effect of a second point mutation for the two combinations tested. None of the mutations abolished binding of *C. difficile* Spo0A completely, most likely as the result of binding of Spo0A to other (non-consensus) 0A boxes in the *abrB* promoter [44].

Taken together, we conclude that the guanine and cytosine residues in the core TGTCGAA motif of PabrB are important for specific binding of this fragment by *C. difficile* Spo0A-DBD.

Value for Binding by *C. difficile* Spo0A-DBD Above, we have established that the Spo0A-DBD of *C. difficile* is highly homologous to that of the *B. subtilis* Spo0A protein, and that the proteins recognize a similar consensus sequence (Fig. 2). Based on this information, we identified the several genes as putative direct targets of *C. difficile* Spo0A.

We queried the *C. difficile* 630 genome sequence for perfect matches to the core 0A box using Genome2D [40] . Such an analysis revealed the presence of 102 matching motifs, of which 45 were located within 500 bp of the initiating ATG of an open reading frame on the same strand (see Table S1). Our attention was drawn to *spo0A* and *sigH*, as these two genes were previously found to be regulated by Spo0A in *B. subtilis* and/or play important roles in sporulation [3, [46] [47] [48] . We found that *C. difficile* Spo0A bound to DNA sequences upstream of *spo0A* and *sigH*.

We performed EMSAs with DNA encompassing 220-281 bp upstream of the initiating ATG codon of the *spo0A*, *sigH* and *spoVG* open reading frames. We found that the addition of Spo0A-DBD to the reactions caused retardation of the *spo0A* and *sigH* DNA fragments (Fig. 3A) , but not of a *spoVG* fragment which did not contain a consensus 0A box (Fig. 3B) . It should be noted that the affinity of Spo0A-DBD for the region upstream of *spo0A* was the highest we have observed so far for any *C. difficile* DNA. Moreover, the presence of multiple shifted species could indicate the presence of more than one strong binding site. These results establish that *spo0A* and *sigH* are likely legitimate targets of Spo0A in *C. difficile*, and confirm that *spoVG* is not, in line with results obtained in *B. subtilis* [10] .

We were interested to see if Spo0A in *C. difficile* could potentially regulate genes that have no documented function in sporulation. Our in silico analysis identified several genes with no obvious link to sporulation that had a consensus 0A box within 100 bp upstream of their start codon. This positioning is similar to that observed for *spo0A* (275) and *sigH* (278). We confirmed in vitro binding of the *C. difficile* Spo0A-DBD to the promoter regions of *lplA* and *ssuA*.

We carried out EMSA experiments using probes that included the perfect consensus site and purified Spo0A-DBD protein. We observed binding of the protein to fragments upstream of the *lplA* gene (CD1654; box at 267) and the *ssuA* gene (CD1484; box at 282) (Fig. 3A). The *lplA* gene encodes a predicted lipoate-protein ligase, and *ssuA* is annotated as an aliphatic sulfonates ABC transporter; to our knowledge, neither of these have been directly implicated in sporulation or have found to be targets for Spo0A in other organisms.

Together our results establish the potential for binding of Spo0A to DNA upstream of *spo0A* and *sigH*, two genes that are important for sporulation, and indicate that Spo0A may have functions that go beyond the regulation of sporulation in *C. difficile*.

It has been established that a *spo0A* mutant of *C. difficile* does not produce any spores, consistent with a crucial role in the sporulation pathway [33]. However, the *in silico* identification of upstream regions with a consensus Spo0A binding site did not point to any of the early sporulation genes (downstream of *spo0A* itself) as direct targets of Spo0A. This is likely the result of variations in the 0A-box in these promoters that were disregarded in the box search. In support of this, many well-characterized legitimate direct targets of *B. subtilis* Spo0A (such as *spolIAA* and *spolIE*) do not contain a 100% match to the core motif, but rather one or more near-consensus boxes [5, 49]. We found that Spo0A- We performed EMSA experiments using increasing amounts of purified Spo0A-DBD from *C. difficile* 630Derm and the DNA fragments indicated above (Fig. 3B). For *spolIAA* (encoding an antianti sigma-factor) and *spolIE* (encoding a serine phosphatase), we observed a low intensity shifted species at concentrations as low as 150 nM. For *spolIGA* (encoding a sporulation specific protease) we observed the shifted species only at higher concentrations of protein (.200 nM). The negative control (*spoVG*) did not demonstrate binding of Spo0A-DBD at these concentrations. Moreover, the shift we observed was reversible using unlabeled DNA containing a high affinity binding site, but not using unlabeled DNA that lacked such a site (Figure S1B-D). Therefore, we consider the binding to *spolIAA*, *spolIE* and *spolIGA* genes to be specific, despite the fact that increasing the amount of protein did not seem to cause a significant increase in the amount of DNA in the complex.

Together, these results suggest that Spo0A in *C. difficile* might regulate the transcription of at least a subset of early sporulation genes by direct binding to their promoter regions.

C. difficile Spo0A-DBD Binds to DNA Upstream of tcdB It has previously been reported that the deletion of Spo0A in *C. difficile* results in a significantly lower toxin production and a ,1000-fold reduction in the toxicity of culture supernatant derived from spo0A negative cells towards Vero cells [35] . Considering the absence of a homolog of the abrB repressor, direct binding of Spo0A and concomitant activation of toxin gene transcription is a likely mechanism through which this could occur. We found evidence for direct binding of Spo0A-DBD to the region upstream of tcdB, encoding one of the major clostridial toxin genes, and possibly tcdC, but this did not seem to result in lower toxin levels in our hands.

We performed EMSAs using DNA upstream of tcdR (encoding a sigma factor responsible for the activation of toxin gene transcription), tcdB (encoding toxin B), tcdA (encoding toxin A). In order to test regions upstream of all open reading frames in the PaLoc, we also tested binding of Spo0A to DNA upstream of tcdE (encoding a holin-like protein [50, 51]) and tcdC (encoding a putative negative regulator of toxin production [52] [53] [54]), even though this regulator does not have a significant effect on toxin levels under the conditions we used [55, 56] . Of the regions tested, we only observed a clear shifted species, indicative of Spo0A binding, for tcdB (Figure 4A); the shifted species in our EMSA assay was reversed by the addition of unlabeled DNA containing a high affinity binding site, but not by DNA lacking such a site (Figure S1E). For tcdC, some smearing was observed at all concentrations of proteins tested (Figure 4A), and there did not seem to be a clear effect of the addition of unlabeled DNA fragments (Figure S1F). The probes for tcdA, tcdE and tcdR were indistinguishable from those obtained with our negative control, spoVG.

We wanted to determine if toxin levels in culture supernatants were directly or indirectly affected by Spo0A, as was previously suggested. We found no lower toxicity towards Vero cells of culture supernatants derived from spo0A mutant cells compared to wild type.

We grew three independent biological replicates of a wild type (630Derm) or Clostron-generated spo0A mutant (CT::spo0A -a kind gift of the Minton lab) in glucose-free TTY medium. We harvested culture supernatant at late-exponential phase (approximately 7 hours post inoculation), the transition phase between exponential and stationary growth phase (approximately 9 hours post inoculation), as well as two time points in stationary phase (24 and 48 hours post inoculation) and determined the toxin endpoint titres (see Materials and Methods). In contrast to previous findings, we observed a small (#4-fold)

increase in the toxicity of supernatants derived from spo0A mutant cells compared to wild type, but in all cases this difference was not statistically significant (p.0.05, independent sample t-test). In other medium (BHIS), we observed no differences at all (data not shown).

We conclude that Spo0A does not positively affect toxin production in *C. difficile* 630Derm and the in vivo relevance of the binding to regions upstream of tcdB and/or tcdC is therefore limited under our experimental conditions.

The Spo0A-box of *C. difficile* In *B. subtilis*, the binding site of Spo0A on target DNA has been well-characterized, through a combination of in vitro binding assays, determination of in vivo binding profiles and mutagenesis of regulated promoter sequences. This work has led to the identification of a conserved core motif, TGTCGAA, or Spo0A box [5, 10, 11, 45] . Depending on the analysis, this motif is flanked by one or more adenine or thymine residues [10, 11] . Interestingly, many target genes do not harbor a perfect match to this consensus sequence, but rather contain one or more degenerate motifs. The differences in these motifs may reflect different promoter architectures (e.g. AT content), modes of action (e.g. activation or repression) or levels of regulation. Spo0A genes in *B. subtilis* can be divided in different classes that respond to different levels of phosphorylated Spo0A [43, 57] .

For *C. difficile*, we conclude that the Spo0A protein likely recognizes a motif that is similar to the *B. subtilis* Spo0A box on the basis of four lines of evidence; 1. All DNA binding/contacting residues are conserved (Fig. 2B) , 2. *C. difficile* Spo0A can bind with high affinity to a target of *B. subtilis* Spo0A (Fig. 2D) , 3. Mutagenesis of key residues in the *B. subtilis* Spo0A box reduces affinity of *C. difficile* Spo0A for DNA (Fig. 2E) and 4. A *B. subtilis* Spo0A box has predictive value for DNA binding by *C. difficile* Spo0A (Fig. 3A) . It is conceivable that our model system, using the purified DNA binding domain, does not accurately reflect binding to all target sites, if target site selectivity is determined in part by other parts by of the full length protein. It is likely that differences do exist between the preferred binding sites for both proteins that will be evident when a comprehensive analysis is performed of in vivo DNA binding of *C. difficile* Spo0A; based on the limited data set of this study, a MEME analysis [58] already suggests possible differences in the extended Spo0A motif (W.K. Smits, unpublished observations). These differences may relate to the much higher AT content of *C. difficile* compared to *B. subtilis* (71 vs. 56.5%, respectively), or phosphorylation dependent dimerization, for instance.

The initiation of sporulation in *B. subtilis* is subject to complex regulation (for review see ref [1, 59]). The activation of Spo0A is controlled by a multi-component phosphorelay that can integrate environmental cues [60] and ensures a gradual increase in the level of phosphorylated Spo0A in the cell [57]. In addition, the transcription of the *spo0A* gene is controlled by multiple feedback loops. For instance, Spo0A regulates its own transcription by binding to the *spo0A* promoter [46], as well as by indirectly stimulating the transcription of *sigH*, encoding a sigma factor that recognizes the *spo0A* promoter [48].

In *C. difficile*, there are some interesting differences and similarities in the regulatory pathways. Most notably, there seems to be no phosphorelay [2] and the phosphorylation state of Spo0A is supposedly controlled by orphan histidine kinases [35]. The transcription of *spo0A* in *C. difficile* is under control of the transition state sigma factor Sigma H [37], as it is in *B. subtilis* [61]. Our data indicate that both *spo0A* and *sigH* could be targets for direct regulation by Spo0A in *C. difficile* (Fig. 3A), raising the possibility of auto-regulation of *spo0A*. The putative direct regulation of *sigH* by Spo0A may reflect that the *C. difficile* genome does not harbor a homolog of the pleiotropic regulator AbrB, which is responsible for the Spo0A-dependent regulation of *sigH* in *B. subtilis* [48]. Consistent with a model in which *spo0A* is positively autoregulated, we noted a sharp increase in the levels of Spo0A as cells approach the stationary growth phase (Figure 1C).

Downstream of Spo0A, we found binding of Spo0A to DNA upstream of several early sporulation genes, such as *spolI*AA, *spolI*E, and *spolI*GA (Fig. 3B). All these observations are consistent with direct regulation of these genes by Spo0A in other organisms [5, 45, 49, 62], and the conservation of the sporulation pathway [2].

Though Spo0A is the key regulator for sporulation in Firmicutes, it regulates numerous other processes in various bacteria. In the non-pathogenic *B. subtilis*, for instance, the protein also affects competence development, biofilm formation, the production of and resistance to antimicrobial compounds, chromosome dynamics and aspects of phage biology [10, [14] [15] [16]. Importantly, several of these processes are indirectly regulated, through the Spo0A-dependent repression of *abrB*. Additionally, transcription of *abrB* responds already to low levels of Spo0A,P [43]. As a result these effects are detectable in late-exponential and early stationary phase, as some Spo0A is present throughout growth in *B. subtilis* cells.

Though *abrB* is absent from *C. difficile*, this does not exclude the possibility of indirect transcriptional regulation through Spo0A-dependent effects on other regulators. Alternatively, Spo0A may exert a direct effect. In *Clostridium acetobutylicum* and *C. beijerinckii*, Spo0A is a direct regulator of solvent formation, as well as sporulation [22, 23]. It seems therefore conceivable that Spo0A in *C. difficile* also affects aspects of metabolism. In this respect, it is important to note that also in *C. difficile* Spo0A is detectable from early exponential growth phase on (Figure 1B).

We observed direct binding of *C. difficile* Spo0A to the promoter region of *sigH* (Fig. 3A). This gene encodes the key sigma factor for the transition phase, and regulates processes outside sporulation as well [37]. Moreover, we found significant levels of Spo0A from early stationary phase on (Fig. 1B and unpublished observations), indicating the regulatory actions of Spo0A need not be limited to stationary phase in *C. difficile*. In line with this idea, we found a potential regulatory link between Spo0A and two genes that to our knowledge are not related to the sporulation process, the lipotease *lplA* and the aliphatic sulfonates transporter *ssuA* (Fig. 3A). The presence of a putative Spo0A binding site upstream of these genes, as well as the spacing compared to the start codon, is conserved in the problematic Stoke-Mandeville strain (R20291), a member of PCR ribotype 27. This could indicate that these aspects of regulation by Spo0A are conserved in multiple strains of *C. difficile*.

It should be noted that our work so far has been limited to an in vitro analysis of Spo0A binding, and therefore does not indicate whether activation or repression of the putative target genes occurs in vivo. To answer this question, detailed transcriptome and/or proteome studies have to be performed. In order to distinguish direct from indirect effects, in vivo binding profiles of Spo0A should be performed. The antibodies generated for this study should prove to be useful for this type of experiments.

Amongst the pathogenic Firmicutes, Spo0A has been reported to affect toxin production in multiple species. In *B. anthracis* a *spo0A* mutation results in elevated levels of *AbrB*, and concomitantly lower levels of the toxin genes *pagA*, *cya* and *lef* that are under *AbrB* control [17]. Similarly, the production of the emetic toxin cereulide in *B. cereus* is greatly repressed in a *spo0A* mutant, in an *AbrB*-dependent manner [63]. In contrast, Spo0A directly represses the expression of the *cry* toxin genes in *B. thuringiensis* and a *spo0A* mutant is

therefore a hyper-producer of the insecticidal crystal protein [18, 21] . In *Clostridium perfringens* TpeL, a member of the large clostridial toxins just like TcdA and TcdB, is directly dependent on Spo0A [64] and also the production of enterotoxin in this organism seems to be (indirectly) dependent on sporulation [65, 66] .

In *C. difficile* an insertional *spo0A* mutant generated using Clostron technology was reported to have ,10-fold reduced levels of toxin A (TcdA), both intracellularly and extracellularly as well as ,1000-fold reduced toxicity towards Vero cells, which are primarily sensitive towards toxin B (TcdB) [35] . Our in vitro binding data indicate a potential binding site for Spo0A upstream of *tcdB* and possibly *tcdC* (Fig. 4A) . However, the in vivo relevance of this binding seems limited as in our hands an independently derived but otherwise identical mutant (a kind gift of the Minton lab; [33]) did not demonstrate a reduced toxicity towards Vero cells. In contrast, we found that in TTY medium toxin levels were slightly elevated in *spo0A* mutant cells compared to wild type (#2fold in exponential phase cells up to 4-fold in late-stationary phase cells). The small, and not significant, differences in toxin levels in our experiments might be attributed to differences in the susceptibility of cells for lysis rather than the production of toxin, but could also indicate a negative regulatory effect of Spo0A on toxin production. In support of the latter hypothesis, it was recently reported that a *spo0A* mutant of *C. difficile* strain R20291 (a PCR ribotypes 027/BI/NAP1 epidemic strain) demonstrates ,10fold higher toxin levels than its isogenic wild type 30 h post inoculation, and is significantly more virulent in a mouse model of disease [34] .

The differences between Underwood et al [35] on the one hand and our study as well as the study of Deakin and coworkers [34] on the other hand may be explained by differences in experimental conditions, such as the medium used. However, we observed no difference in cytotoxicity between supernatant derived from wild type or *spo0A* mutant cells when they were grown in BHIS, a medium nearly identical to that used previously (data not shown). Alternatively, the differences could indicate integration of the group II intron at more than one location in the chromosome in the strain used in Underwood et al [35] . In the absence of a complementation experiment and/or Southern blot data, this remains to be established.

In summary, our data are consistent with a model in which the regulation of the major clostridial toxins in *C. difficile* is not positively affected by Spo0A, in contrast to previous

findings and other pathogenic Clostridia. Whether Spo0A is truly a negative regulator of toxin production remains to be confirmed using in vitro and in vivo transcription assays.

In the present study we have for the first time demonstrated direct binding of the DNA binding domain of *C. difficile* Spo0A to putative target DNA. This work has revealed that aspects of Spo0A binding are conserved between *Bacillus* and *C. difficile* (0A box, possible auto-regulation and binding to early sporulation promoters), whereas others are not (the absence of *abrB* as a direct target in *C. difficile*, binding to DNA upstream of *lplA*, *ssuA*). The effects of Spo0A on toxin production may be similar to those observed for *B. thuringiensis* [18, 21]. Future work will be aimed at determining the effect of Spo0A on the transcription of the putative target genes, and carry out a comprehensive analysis of Spo0A binding in vivo. The identification of genes affected by Spo0A in *C. difficile* may shed light on the role of the protein in virulence and pathogenesis of this organism.

Figure S1 Specificity controls for binding by Spo0A-DBD-his6. Arrows indicate the position of shifted species (DNA:protein complexes). Titrations with PCR fragments of *PabrB* (containing a high affinity binding site) and *PtcdA* (lacking such a site) correspond to approximately 0.1 nM/mL -0.03 nM/mL. A. Comparison of binding of Spo0A-DBD-his6, Spo0A-his6 and CD2195-his6 binding to the upstream region of *spolIAA*. B. Binding of Spo0A-DBD-his6 to the upstream region of *spolIAA* is reversed by the addition of *PabrB*, but not by the addition of *PtcdA*. C. Binding of Spo0A-DBD-his6 to the upstream region of *spolIE* is reversed by the addition of *PabrB*, but not by the addition of *PtcdA*. D. Binding of Spo0A-DBD-his6 to the upstream region of *spolIGA* is reversed by the addition of *PabrB*, but not by the addition of *PtcdA*. E. Binding of Spo0A-DBD-his6 to the upstream region of *tcdB* is reversed by the addition of *PabrB*, but not by the addition of *PtcdA*. F. Binding of Spo0A-DBD-his6 to the upstream region of *tcdC* is not or moderately affected by the addition of *PabrB* and/or *PtcdA*. (TIF) Text S1 Oligonucleotides used in this study and PCR cycling conditions for the EMSA probes. (PDF)

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Venezuelan Equine Encephalitis Virus Induces Apoptosis through the Unfolded Protein Response Activation of EGR1

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Abstract: Venezuelan equine encephalitis virus (VEEV) is a previously weaponized arthropod-borne virus responsible for causing acute and fatal encephalitis in animal and human hosts. The increased circulation and spread in the Americas of VEEV and other encephalitic arboviruses, such as eastern equine encephalitis virus and West Nile virus, underscore the need for research aimed at characterizing the pathogenesis of viral encephalomyelitis for the development of novel medical countermeasures. The host-pathogen dynamics of VEEV Trinidad donkey-infected human astrocytoma U87MG cells were determined by carrying out RNA sequencing (RNA-Seq) of poly(A) and mRNAs. To identify the critical alterations that take place in the host transcriptome following VEEV infection, samples were collected at 4, 8, and 16 h postinfection and RNA-Seq data were acquired using an Ion Torrent PGM platform. Differential expression of interferon response, stress response factors, and components of the unfolded protein response (UPR) was observed. The protein kinase RNA-like endoplasmic reticulum kinase (PERK) arm of the UPR was activated, as the expression of both activating transcription factor 4 (ATF4) and CHOP (DDIT3), critical regulators of the pathway, was altered after infection. Expression of the transcription factor early growth response 1 (EGR1) was induced in a PERK-dependent manner. EGR1(−/−) mouse embryonic fibroblasts (MEFs) demonstrated lower susceptibility to VEEV-induced cell death than isogenic wild-type MEFs, indicating that EGR1 modulates proapoptotic pathways following VEEV infection. The influence of EGR1 is of great importance, as neuronal damage can lead to long-term sequelae in individuals who have survived VEEV infection. **IMPORTANCE** Alphaviruses represent a group of clinically relevant viruses transmitted by mosquitoes to humans. In severe cases, viral spread targets

neuronal tissue, resulting in significant and life-threatening inflammation dependent on a combination of virus-host interactions. Currently there are no therapeutics for infections caused by encephalitic alphaviruses due to an incomplete understanding of their molecular pathogenesis. Venezuelan equine encephalitis virus (VEEV) is an alphavirus that is prevalent in the Americas and that is capable of infecting horses and humans. Here we utilized next-generation RNA sequencing to identify differential alterations in VEEV-infected astrocytes. Our results indicated that the abundance of transcripts associated with the interferon and the unfolded protein response pathways was altered following infection and demonstrated that early growth response 1 (EGR1) contributed to VEEV-induced cell death.

Text: Venezuelan equine encephalitis virus (VEEV) is a New World alphavirus in the family *Togaviridae* that is endemic to the Americas. VEEV is a positive-strand RNA virus that is transmitted by mosquitoes and that is naturally present in rodent reservoirs (1). There are six subtypes that are categorized by their geographic range and pathology in equines and humans. The two epizootic strains, IA/B and IC, arose from mutations among the enzootic strains (2). The IA/B and IC strains are of particular concern due to increased rates of morbidity and mortality and the risks associated with viral amplification and potential species spillover (2). In humans, VEEV causes a febrile illness typified by fever, malaise, and vomiting. In some cases, infection progresses to the central nervous system (CNS) and neurological symptoms, such as confusion, ataxia, and seizures, manifest. The mortality rate among cases with neurological symptoms can be as high as 35% in children and 10% in adults, with long-term neurological deficits often being seen in survivors (2). In 1995, an outbreak of VEEV in Colombia and Venezuela resulted in over 100,000 human cases (3). In addition to natural outbreaks, VEEV is also a concern from a bioterrorism perspective, as it can be grown to high titers, requires a low infectious dose, and contains multiple serotypes. Both the former Soviet Union and the United States previously weaponized the virus, producing large quantities for their now defunct offensive bioweapons programs (4). Currently, vaccine strain TC83 is used in horses and for high-risk personnel; however, due to the low rate of seroconversion achieved with this vaccine (5) and its reliance on two single attenuating mutations (6), it is considered unfit for mass distribution (7). To date there are no FDA-approved therapeutics for VEEV infection, and further studies are required for clarification of the mechanisms associated with the underlying pathogenesis of VEEV.

Viral and host transcriptomic studies can provide a wealth of information on the underlying pathogenic mechanisms and interactions following the course of an infection. The use of

highthroughput next-generation sequencing has led to the discovery of previously uncharacterized viruses and the establishment of numerous novel experimental systems redefining virus-host interactions. To date a number of studies have examined the alterations in the host transcriptome following VEEV infection. A comparative microarray analysis between cells persistently infected with VEEV and cells able to clear VEEV resulted in the identification of PARP12L as an antiviral factor (8) . A molecular comparison utilizing microarrays of host-based responses to the TC83 strain was able to identify biomarkers differentiating between vaccine responder and vaccine nonresponder groups, as well as the involvement of interferon (IFN), interferon-induced pathways, Toll-like receptor (TLR), and interleukin 12 (IL-12) related pathways (9) . A study examining the role of adhesion and inflammatory factors in VEEV-infected CD-1 mice found viral modulation of the expression of extracellular matrix and adhesion genes, such as integrins (Itg_X, Itg2, 3, and 7), cadherins 1 and 2, vascular cell adhesion molecule 1, and intracellular adhesion molecule 1 (ICAM-1), in the brains of VEEV-infected mice (10) . Follow-up experiments utilizing ICAM-1-knockout mice demonstrated reduced inflammation in the brain and a subsequent delay in the onset of neurological sequelae (10) . A study by Sharma et al. utilized microarrays to analyze gene expression changes in the brain tissue of VEEV-infected mice over the course of an infection, discovering numerous immune pathways involved in antigen presentation, inflammation, apoptosis, and the traditional antiviral response (Cxcl10, Cxcl11, Ccl5, lfr7, lfi27, Oas1b, Fcerg1, Mif, clusterin, and major histocompatibility complex [MHC] class II) (11) . A second study by the same group identified the regulation of microRNAs (miRNAs) in the brains of VEEV-infected mice, which enabled the correlation of the miRNA changes with earlier mRNA expression data (11, 12) . These analyses suggest that VEEV may be utilizing cellular miRNAs in order to regulate downstream mRNA, which may correspond with the VEEV-induced histological changes to the nervous system (11, 12) .

In the current study, next-generation RNA sequencing (RNA-Seq) was used to identify clinically relevant alterations in the mRNA transcriptome of human astrocytes infected with wildtype (WT) VEEV strain Trinidad donkey (TrD). The analysis of host mRNAs by RNA-Seq provides novel insight into how a host responds to a viral infection through the identification of a wide and dynamic range of transcripts in an unbiased manner. Selective sequencing of mRNAs, specifically, polyadenylated [poly(A)] transcripts, which account for ~1% of the entire transcriptome, enhances the detection of the most relevant and low-abundance transcripts (13) . As VEEV has been shown to productively infect astrocytes both in vitro and in vivo (14, 15) , we chose astrocytes as our model of interest. Astrocytes are the most abundant cell in the brain, outnumbering neurons by at least 5-fold (16) ,

providing an abundant resource for viral replication within the brain. In addition to their well-described structural role in neuronal tissue, astrocytes play critical roles in other processes, including the regulation of blood flow and of the blood-brain barrier, synapse transmission, and the response to infection (16). VEEV-infected astrocytes have been shown to produce multiple cytokines, including IL-8, IL-17, interferon gamma (IFN- γ), and gamma interferon-induced protein 10, all of which were found to be associated with viral attenuation (14).

In order to obtain a dynamic view of the virus-host interactome, RNA-Seq was used to monitor changes in gene expression in VEEV TrD-infected astrocytes at 4, 8, and 16 h postinfection (hpi). By viewing the alterations at multiple early time points using triplicate biological replicates, a robust and dynamic range of information is generated, and this information provides an increase in both the power and the accuracy of detection of differentially expressed transcripts in a highly relevant clinical model (17). Among VEEV-infected cells, an increase in interferon-regulated genes, including IFIT1, IFIT2, IFIT3, and OASL, was observed. The increased expression of genes involved in the stress-induced unfolded protein response (UPR) pathway was also noted. Interestingly, VEEV infection resulted in an increase in early growth response protein 1 (EGR1), which may serve as a link between the two pathways. The identification of host mRNAs whose expression is altered following VEEV replication, specifically, EGR1 and its interactors up- and downstream, may provide novel host-based therapeutic targets critical for VEEV replication and a greater understanding of the underlying mechanisms underpinning alphavirus replication.

Viral infections and plaque assays. VEEV TrD was obtained from BEI Resources. All experiments with VEEV TrD were performed under biosafety level 3 (BSL-3) conditions. All work involving select agents is registered with the Centers for Disease Control and Prevention and was conducted at George Mason University's Biomedical Research Laboratory, which is registered in accordance with federal select agent regulations. For infections, VEEV was added to supplemented Dulbecco modified Eagle medium (DMEM) to achieve a multiplicity of infection (MOI) of 0.05, 0.5, or 5. Cells were infected for 1 h at 37°C and rotated every 15 min to ensure adequate coverage. The cells were then washed with phosphate-buffered saline (PBS), and complete growth medium was added back to the cells. Viral supernatants and cells were collected at various times postinfection for further analysis. Plaque assays were performed as previously described (18). mRNA isolation and poly(A) library preparation. RNA from U87MG cells was purified from both VEEV TrD-infected (biosafety level 3) and mock-infected U87MG cells at 4, 8, and 16 hpi utilizing a

mirVana isolation kit (Life Technologies). Quality control of purified RNA was then performed using an Agilent 2100 bioanalyzer, and an RNA integrity number (RIN) cutoff of 8 was utilized for all samples. An External RNA Controls Consortium (ERCC) RNA spike-in control mix was then added to the total RNA inputs (10 g RNA) before poly(A) selection using a Life Technologies Dynabeads mRNA Direct kit. Preparation of a whole-transcriptome RNA library from purified mRNA was then performed using an Ion Total RNA-Seq kit (v2; Life Technologies). Quality control of the cDNA libraries was then performed using the Agilent 2100 bioanalyzer along with sterility testing for removal of libraries for sequencing from a BSL-3 to BSL-2 laboratory.

RNA sequencing. Library template preparation was performed on a One Touch 2 platform (Life Technologies). Next-generation RNA sequencing was performed on an Ion Torrent PGM platform and was carried out for each sample to assess the differential gene expression of infected versus uninfected cells over time.

Data filtering and RNA-Seq analysis pipeline. A total of 1119 million sequencing reads and an average of 6.6 million reads per sample were used as the input into our analysis pipeline. Unless otherwise noted, downstream RNA-Seq analysis was carried out using the CLC bio Genomics Workbench (v7). Raw RNA-Seq reads were trimmed to remove any residual sequencing adapter fragments that remained on the 5' or 3' ends after sequencing. In addition, end trimming of reads was done using the modified Mott algorithm with a Q20 quality score, and any reads of less than 15 bp were discarded. Following read trimming, the reads were mapped to human genome hg19 with the following RNA-Seq parameters: a 10-hit limit for multiple mapped positions, a similarity fraction of 0.8, a length fraction of 0.8, a mismatch cost of 2, and an indel cost of 3. The expression level of individual genes and transcripts was calculated using the number of reads per kilobase of the exon model per million mapped reads (RPKM) method of Mortazavi et al. (19). In addition, unmapped reads were also mapped to the ERCC92 synthetic RNA sequence set (20), as well as to the VEEV reference genome (GenBank accession number L01442). In all samples, the correlation coefficient (R^2) between the expected and the mapped number of reads for the ERCC92 spike-in controls was above 0.90. A summary of the overall sequencing results is shown in Table 1.

Postmapping filtering of all RNA-Seq data was carried out next to include only genes with at least one uniquely mapped read (26,230 genes remained across all data sets) and only

those with a nonzero interquartile range across the entire experiment. Principal component analysis of the resulting filtered data set (13,906 genes in total) was carried out using raw counts of uniquely mapped reads (see Fig. 2A). The remaining RPKM expression values for each gene included in the filtered data set were subjected to quantile normalization with a 5% cutoff. A box plot of log₂transformed RPKM values for each sample before normalization is shown in Fig. 2B . The R² value for pairwise sample-to-sample variation within each biological replicate set was observed to range from 0.89 to 0.99, indicating that our biological replicates were consistent and showed no strong bias (data not shown).

Differential gene expression analysis. Differentially expressed genes (DEGs) were identified using two approaches. First, the empirical analysis of differential gene expression algorithm, part of the edgeR Bioconductor package (21) , was applied to the integrated data set of all 18 experiments using the default parameters and a false discovery rate-corrected P value. At each time point, infected and mock-infected samples were compared, and genes whose expression differed by more than 2-fold with a significance with a P value of ≤ 0.05 were provisionally considered to be differentially expressed.

In addition to the method described above, an orthogonal statistical test of differential expression was applied to the data using a statistical test developed by Baggerly et al. (22) to count the number of expressed sequence tags associated with individual genes, a common feature of both serial analysis of gene expression (SAGE) data and RNA-Seq data. When infected and mock-infected samples were compared, individual genes were provisionally considered differentially expressed when their expression differed by more than 2-fold with a significance with a P value of ≤ 0.05 . Differentially expressed genes found to be in the intersection of the sets of genes identified by both of the methods outlined above were considered high-quality candidates and used as the starting point for further investigation.

Clustering and GSEA. Filtered, normalized expression data were subjected to k-means clustering using a Euclidian distance metric where genes were grouped by means of normalized gene expression (RPKM) values for each experimental condition. Clustering was fitted to 20 distinct clustering groups, and the individual gene expression profiles clustered were further tested for enrichment of gene ontology (GO) terms associated with individual genes. Gene annotations were obtained from Reactome, a database of biological pathway and gene functional annotations (23) . Enrichment analysis was

performed using two approaches. First, a hypergeometric test on GO annotations was carried out using an implementation of the GOSTats package on each of the individual clusters obtained from k-means clustering (24) . In addition, gene set enrichment analysis (GSEA) was carried out on the entire filtered data set using 100,000 permutations, while duplicates were removed and an analysis of variance was applied. A total of 1,419 categories passed a minimum feature size of 10 and were used for further investigation.

Cohorts of genes with shared patterns of expression over time were identified by k-means clustering. Those found to be enriched for DEGs were subsequently subjected to pathway analysis using the GeneMania system (25) . Using an ad hoc manual approach, relevant pathways and the connections between them were identified on the basis of existing data in the literature coupled with the temporal gene expression data obtained from this study.

qRT-PCR analysis. Purified mRNA was converted to cDNA using a high-capacity RNA-to-cDNA kit (Life Technologies) according to the manufacturer's instructions. Analysis of the viral copy numbers was performed by quantitative reverse transcription-PCR (qRT-PCR) as previously described (26) . Host expression of the following genes was assayed with TaqMan assays (indicated in parentheses): activating transcription factor 3 (ATF3; Hs00231069_m1), ATF4 (Hs00909569_g1), CEBPB (Hs00270923_s1), CEBPD (Hs00270931_s1), DDIT3 (Hs00358796_g1), FOS (Hs04194186_s1), JUN (Hs01103582_s1), EGR1 (Hs00152928_m1), IFI6 (Hs00242571_m1), IFIT1 (Hs01911452_s1), IFIT2 (Hs01922738_s1), IFIT3 (Hs01922738_s1), ISG15 (Hs01921425_s1), ISG20 (Hs00158122_m1), OASL (Hs00984387_m1), BIRC5 (Mm00599749_m1), and XIAP (Mm01311594_mH). Assays for 18S rRNA (Hs99999901_s1 or Mm04277571_s1) were used for normalization. Assays were performed according to the manufacturer's instructions using an ABI StepOne Plus instrument.

Treatment with PERKi and collection for Western blot analysis. U87MG cells were pretreated for 2 h with 10 μ M the protein kinase RNAlike endoplasmic reticulum (ER) kinase (PERK) inhibitor (PERKi) GSK2606414 (catalog number 516535; EMD Millipore) or dimethyl sulfoxide (DMSO) in DMEM prior to infection with VEEV TrD (MOI, 5). After 1 h, the viral inoculum was removed and cells were washed with sterile PBS (1 $\times$). The medium was replaced with medium containing the inhibitor or DMSO. At 16 hpi, the medium was removed, and the cells were washed with PBS and then collected for Western blot analysis.

Knockdown of EGR1 with siRNA. U87MG cells seeded at 6.7×10^4 cells per well in a 12-well plate were transfected with 50 nM siGenome Protein lysate preparation and Western blot analysis. Protein lysate preparation and Western blot analysis were performed as previously described (27). Primary antibodies to the following were used: EGR1 (antibody 44D5; catalog number 4154; Cell Signaling), polyclonal anti-Venezuelan equine encephalitis virus TC83 (subtype IA/B) capsid protein (BEI Resources), CHOP (antibody L63F7; catalog number 2895; Cell Signaling), phosphorylated α subunit of eukaryotic initiation factor 2 (p-eIF2 α ; Ser51; antibody D9G8; catalog number 3398; Cell Signaling), ATF4 (antibody D4B8; catalog number 11815; Cell Signaling), activated caspase 3 (antibody Asp175; catalog number 9661; Cell Signaling), and horseradish peroxidase-conjugated β -actin (catalog number ab49900-100; Abcam).

Immunofluorescence analysis. U87MG cells were grown on coverslips in a 6-well plate, infected with VEEV TrD as described above, washed with PBS (without Ca and Mg), and then fixed with 4% formaldehyde. Cells were permeabilized with 0.5% Triton X-100 in PBS for 20 min and then washed twice with PBS. The cells were blocked for 10 min at room temperature in 3% bovine serum albumin in PBS. Primary antibodies consisting of a VEEV capsid protein (catalog number NR-9403; BEI Resources) diluted 1:600 and an EGR1 antibody (antibody 44D5; catalog number 4154; Cell Signaling) diluted 1:400 were incubated in fresh blocking buffer at 37°C for 1 h and washed 3 times for 3 min each time in 300 mM NaCl with 0.1% Triton X-100. Alexa Fluor 568 donkey anti-goat secondary antibody (catalog number A11057; Invitrogen) and Alexa Fluor 488 donkey anti-mouse secondary antibody (catalog number A21202; Invitrogen) diluted 1:400 were used as secondary antibodies and treated in the same manner as the primary antibodies. DAPI (4',6-diamidino-2-phenylindole) diluted 1:1,000 was used to visualize the nuclei. Coverslips were mounted onto glass slides using 10 μ l of Fluoromount G mounting medium (catalog number 0100-01; Southern Biotech). A Nikon Eclipse TE2000-U fluorescence microscope was used for fluorescence microscopy. Images were viewed using a 60 $\times$ objective oil immersion lens. Five images of each sample were obtained, and a representative image of each sample is shown below. All images were subjected to fourline averaging. The images were processed through Nikon NIS-Elements AR Analysis (v3.2) software.

CellTiter Glo and Caspase 3/7 Glo assays. Wild-type and EGR1 Δ/Δ mouse embryonic fibroblasts (MEFs) were infected with TrD at various MOIs for an hour and then washed with PBS, and the medium was replaced. Cell viability was measured at 24 h postinfection using a Promega CellTiter luminescent cell viability assay (catalog number G7571) according to

the manufacturer's protocol. Luminescence was read using a Beckman Coulter DTX 880 multimode detector with an integration time of 100 ms per well. Similarly, caspase activation in infected wildtype and EGR1 Δ/Δ MEFs was measured at 24 h postinfection using a Promega Caspase 3/7 Glo assay (catalog number G8090) according to the manufacturer's protocol. Luminescence was read using the DTX 880 multimode detector with an integration time of 100 ms per well.

Nucleotide sequence accession numbers. The raw sequencing data for all RNA-Seq runs included in this work are publically available in the NCBI BioProject database under accession number PRJNA300864 (<http://www.ncbi.nlm.nih.gov/bioproject/PRJNA300864>).

VEEV replication kinetics in U87MG astrocytes. VEEV replicates in vivo in monocytes, macrophages, neurons, and astrocytes (14). Common cell lines used to study VEEV infection include Vero and BHK cells; in this study, U87MG astrocytes were chosen as an in vitro model due to their physiological relevance and greater clinical significance. Initial experiments were performed to characterize viral replication in U87MG cells. VEEV replication kinetics in U87MG cells were measured using plaque assays and by monitoring viral protein and RNA expression levels and the cytopathic effect (CPE) on the infected cells (Fig. 1). Viral release was observed as early as 4 hpi, with 10^4 log units of virus being observed, followed by a consistent increase in replication at 8 and 16 hpi (Fig. 1A). Viral replication peaked at 16 hpi, and no additional increase in viral titers was observed at 24 hpi. Viral capsid expression followed a similar pattern, with protein being detected at 8 hpi and expression plateauing at 16 hpi (Fig. 1B). Among infected U87MG cells, a significant CPE was observed by microscopy at 24 hpi, with little to no CPE being detected at 16 hpi (data not shown). Consistent with these observations, increased caspase 3/7 activity was observed only at 24 hpi (Fig. 1C). On the basis of these data, times of 4, 8, and 16 hpi, reflecting the early, middle, and late stages of the viral life cycle, respectively, were selected for RNA-Seq analysis in order to provide a dynamic view of the host-pathogen transcriptome profile.

RNA sequencing analysis of VEEV-infected astrocytes. mRNA from triplicate sets of mock- and VEEV-infected U87MG cell cultures was isolated, purified at 4, 8, and 16 hpi, and used to prepare cDNA libraries for downstream RNA-Seq (see Materials and Methods). A high-level summary of the RNA-Seq results is shown in Table 1. VEEV RNA samples were assayed by quantitative RT-PCR at each time point as a control to demonstrate the

increasing viral RNA load over time (Fig. 1D) , consistent with the increasing number of RNA-Seq reads mapped to the VEEV genome at later time points (Table 1) .

For RNA-Seq analysis, individual genes were expressed as the number of reads per kilobase of the exon model per million mapped reads (RPKM) (19) . Log 2 -normalized RPKM expression values for each experimental sample are shown in Fig. 2A and can be found in Data Set S1 in the supplemental material. Minimal sample-to-sample variation in expression values within biological replicates was consistently detected ($R^2 \geq 0.89$ for all replicates; data not shown). In addition, intersample variation was also found to be minimal when it was tested pairwise across the entire experiment by using RPKM values for ERCC97 synthetic spike-in control RNAs ($R^2 \geq 0.90$ for all comparisons; data not shown).

As anticipated, two-component principal component analysis of the RNA-Seq data for mock-infected cells versus VEEV-infected cells showed a clear separation of the samples at 16 hpi from the samples at earlier time points (Fig. 2B) . However, the clustering of VEEV-infected samples with mock-infected samples at earlier time points suggested that the response to viral infection was limited to a narrow subset of early response genes, thus placing a higher burden of proof on identifying differentially expressed genes (DEGs) during the first few hours of infection. Along these lines, two orthogonal methods were used to identify DEGs suitable for further characterization: the edgeR method (21) and the method developed by Baggerly et al. (22) . Genes identified by one method were provisionally considered DEGs, and those identified by both methods were candidate DEGs to be confirmed by qRT-PCR. In addition to comparing individual gene expression values for mock-infected cells and VEEV-infected cells at each time point, gene expression values were also compared serially within each time series of VEEV-infected cells for genes that did not show any statistically significant changes in expression in mock-infected cells. A schematic of the comparative analysis is shown in Fig. 2C . The number of statistically significant DEGs identified by each of these comparisons is shown in Fig. 2D . Furthermore, k-means clustering (against normalized RPKM values) was employed to identify gross changes in gene expression over time for cohorts of genes potentially sharing the same pathway or regulatory triggers (Fig. 3 ; see also Data Set S2 in the supplemental material). Gene set enrichment analysis (GSEA; see Material and Methods and Data Set S3 in the supplemental material) was carried out on each kmeans cluster. In particular, cluster 20 (Table 2) was significantly enriched for genes involved in translational control, the type I interferon-mediated signaling pathway, and the unfolded protein response (UPR) pathway (GSEA P value ≤ 0.01). Although there is a well-established connection between

translational control and UPR, a novel connection between UPR and the type I interferon-mediated response in response to viral replication was suggested by pathway analysis (see Materials and Methods), implicating early growth response 1 (EGR1) as a potential bridge between these two pathways (Fig. 4). EGR1 belongs to cluster 20 and is strongly induced during VEEV infection, and several other genes associated with the interferon response belong to the same cluster: IRF1, IFIT1, IFIT2, ISG15, and ILF3. EGR1 has been associated with increases in the expression of activating transcription factor 3 (ATF3) (28), which is a key component of the UPR and which also belongs to cluster 20. This connection represented a potential biological process annotations obtained from Reactome for cluster 20. Reactome annotation identifiers are indicated for each annotation. Only traceable author submission (TAS)-classified annotations are considered. TAP, transporter associated with antigen processing; SRP, signal recognition particle. b Full set, the total number of genes in the genome with an annotated biological process; subset, total number of differentially expressed genes with an annotated biological process.

Network of type I interferon response- and UPR-related genes. Large circles, differentially expressed genes; small circles, genes with no significant change in expression; red circles, type I interferon response factors; yellow circles, genes regulating DNA transcription; blue circles, unfolded protein response genes; red lines, genes involved in physical protein-protein interactions; blue lines, genes involved in a common pathway. This network was seeded with k-means clusters 18 and 20, and many ribosomal protein genes were removed.

bridge between the UPR pathway and the interferon response pathway, with EGR1 being one of the potential key transcription factors driving this connection. Consequently, 15 genes from this analysis were selected for further characterization by qRT-PCR (see below): ATF3, activating transcription factor 4 (ATF4), CEBPB, CEBPD, DDIT3/CHOP, EGR1, FOS, IFI6, IFIT1, IFIT2, IFIT3, ISG15, ISG20, JUN, and OASL. The expression values of these genes, as measured by RNA-Seq, are shown in Fig. 5A and B. Confirmatory qRT-PCR analysis indicated concordant gene expression (Fig. 5C and D). The interferon response genes induced are in agreement with those detected in previously published studies (11, 29, 30), and these genes served as an internal positive control. Moreover, the link between EGR1 and the interferon pathway has been demonstrated; EGR1 is induced by IFN- γ in mouse fibroblasts and by IFN- α , IFN- β , and IFN- γ in human fibroblasts (31, 32). EGR1 and the UPR pathway were selected for further analysis, as their role in VEEV infection has not been elucidated.

The RNA-Seq and pathway analysis data indicated that UPR and stress response genes were induced after VEEV infection. During an infection, host cells respond to cellular stresses resulting from increased viral protein translation and secretion by triggering the onset of the UPR pathway. The UPR pathway is an adaptive cellular response activated by endoplasmic reticulum (ER) stress due to protein misfolding. In order to regulate cellular homeostasis during protein folding and secretion, the UPR pathway has developed three classes of sensors to ensure proper cellular regulation: inositolrequiring enzyme 1 (IRE1), protein kinase RNA-like ER kinase (PERK), and activating transcription factor 6 (ATF6) (33, 34) . During VEEV infection, the PERK arm of the UPR appeared to be altered, as two critical regulators of this pathway were differentially expressed: ATF4 and CHOP (DDIT3) (35) . To determine if DEGs altered subsequent protein expression, Western blot analysis was performed for CHOP, ATF4, and phosphorylated eIF2_α (p-eIF2_α). Tunicamycin, a glycosylation inhibitor and inducer of UPR (36) , was included as a positive control. A time course analysis of U87MG cells treated with 1 M tunicamycin indicated that 8 h of treatment provided the most robust induction of UPR proteins (data not shown). VEEV-infected but not mock-infected or UV-inactivated VEEV (UV-VEEV)-infected cells displayed a dramatic increase in p-eIF2_α expression and a modest but consistent increase in CHOP and ATF4 expression at 16 hpi (Fig. 6A) . No change in protein expression was observed at 4 hpi (data not shown). Confocal microscopy confirmed CHOP and ATF4 up- regulation, demonstrating a more robust and nuclear staining pattern in VEEV-infected cells than in mock-infected cells (Fig. 6C to E). While ATF4 protein expression levels increased, ATF4 mRNA abundances decreased following VEEV infection (Fig. 5B and D). These results are consistent with the observation that ATF4 expression is regulated at the translational level upon UPR induction (37) . As eIF2_α can be phosphorylated by multiple kinases (PERK, protein kinase double-stranded RNA dependent [PKR], general control nonderepressible-2 [GCN2], and hemeregulated inhibitor [HRI]) (38) , the PERK inhibitor (PERKi) GSK2606414 was used to determine if the observed phosphorylation was PERK dependent. Treatment of VEEV-infected cells with PERKi resulted in a marked decrease in eIF2_α phosphorylation (Fig. 6B) . These results indicate that PERK contributes to eIF2_α phosphorylation but that there is likely an additional kinase contributing to the phosphorylation event. Collectively, these findings indicate that the PERK arm of the UPR pathway is induced at later time points following VEEV infection.

EGR1 is upregulated in infected cells and localizes to the nucleus. EGR1 is a transcription factor that can be induced by numerous signals, including oxidative stress, hypoxemia, and

growth factors (39, 40) . It can also be activated upon infection by both DNA and RNA viruses, including Epstein-Barr virus, mouse hepatitis virus, murine coronavirus, and Japanese encephalitis virus (41) (42) (43) . Treatment of MEFs with the UPR activator thapsigargin has been shown to induce EGR1 expression in a PERK-dependent manner (44) . Given the link between EGR1 and UPR and the robust induction of EGR1 mRNA expression following VEEV infection (Fig. 4 and 5) , EGR1 was chosen for further study. EGR1 protein expression after VEEV infection was analyzed by Western blot analysis. As previous studies have indicated that EGR1 can be activated by mouse hepatitis virus independently of virus replication (likely due to cellular membrane disruption following entry) (41), a UV-inactivated virus control (UV-VEEV) was included. EGR1 protein levels were increased following VEEV infection compared to those in mock-infected cells and UV-VEEV-infected cells (Fig. 7A; compare lanes 3, 6, and 9) . The most dramatic upregulation of EGR1 occurred at 16 hpi; this correlates with the highest levels of VEEV capsid production (Fig. 1B) . Following induction, EGR1 has been shown to translocate to the nucleus to induce gene expression through binding to the Egr binding sequence (EBS) [GCG(G/T)GGCG] (40, 45) . Confocal microscopy revealed high levels of EGR1 in the nuclei of infected cells, whereas only low levels of both nuclear and cytoplasmic EGR1 were detected in mock-infected cells (Fig. 7B) . PERKi treatment of VEEV-infected cells resulted in a complete loss of EGR1 induction (Fig. 7C) , indicating that EGR1 was induced in a PERK-dependent fashion. These results demonstrate that EGR1 protein levels and nuclear localization are increased following VEEV infection and that the induction of EGR1 is dependent on PERK.

The loss of EGR1 inhibits VEEV-induced apoptosis but does not alter VEEV replication kinetics. As EGR1 influences cell survival and apoptosis (46) , the impact of EGR1 on VEEV-induced cell death was assessed. Caspase 3 cleavage was observed in WT MEFs at 24 hpi when they were infected at an MOI of 0.5 and started as early as 16 hpi when they were infected at an MOI of 5 (Fig. 8A) . In contrast, EGR1 Δ/Δ cells showed little to no detectable caspase cleavage following infection with VEEV. Two sets of experiments were performed to quantitatively confirm these results: CellTiter Glo assays to measure total cell viability (ATP production) and Caspase 3/7 Glo assays to measure caspase 3/7 activity. Both WT and EGR1 Δ/Δ MEFs displayed dose-dependent decreases in cell viability following VEEV infection, with EGR1 Δ/Δ cells having significantly more viable cells at each MOI examined (Fig. 8B) . Concordantly, a dose-dependent increase in caspase 3/7 activity was observed following VEEV infection, with EGR1 Δ/Δ cells demonstrating reduced caspase 3 activity at MOIs of 0.5 and 5 (Fig. 8C) . These results were replicated in U87MG cells transfected with siRNA targeting EGR1 (Fig. 8D) . EGR1 has been shown to negatively regulate the transcription of BIRC5 (survivin), an inhibitor of apoptosis (IAP) family member (47) . RNA-

Seq data indicated that BIRC5 gene expression was decreased following VEEV infection: log₂-transformed fold change values of normalized gene expression were $\Delta 1.16$, $\Delta 1.18$, and $\Delta 1.50$ at 4, 8, and 16 hpi, respectively (see Table S1 in the supplemental material and NCBI BioProject accession number PRJNA300864). WT and EGR1 Δ/Δ MEFs were used to determine if EGR1 influenced BIRC5 gene expression following VEEV infection. BIRC5 expression was significantly decreased at 16 hpi in VEEV-infected WT MEFs, but this reduction was not observed in VEEV-infected EGR1 Δ/Δ MEFs (Fig. 8E). Expression of the gene for the X-linked inhibitor of apoptosis (XIAP), another IAP family member, was not significantly differentially altered after infection (data not shown). Collectively, these results demonstrate that EGR1 contributes to VEEV-induced apoptosis.

VEEV replication kinetics were determined for both EGR1 Δ/Δ and WT MEFs to determine the relevance of EGR1 in viral replication. Cells were infected at two different MOIs (0.5 and 5), and viral supernatants were collected at 4, 8, 16, and 24 hpi and analyzed by plaque assay. The replication kinetics were similar between EGR1 Δ/Δ and WT MEFs at both MOIs, with titers peaking at 16 hpi (Fig. 9A). A lack of EGR1 expression was confirmed by Western blotting (Fig. 9B). These results were replicated in U87MG cells transfected with siRNA targeting EGR1. Transfection of siRNA targeting EGR1 resulted in a ~90% decrease in EGR1 protein expression (Fig. 9D) without any significant effect on viral replication (Fig. 9C). These results suggest that the decrease in apoptosis observed in EGR1 Δ/Δ MEFs was not due to altered VEEV replication kinetics.

Despite being recognized as an emerging threat, relatively little is known about the virulence mechanisms of alphaviruses, largely due to a knowledge gap in the host-pathogen interactome. VEEV infection often results in fatal encephalitis and is known to inhibit both cellular transcription and translation in order to downregulate the innate immune response (1, 48). In contrast, in the CNS VEEV has been shown to upregulate numerous genes in both the inflammatory response and apoptotic pathways (1, 48). Specifically, numerous proinflammatory cytokines, including interleukin-1 β (IL-1 β), IL-6, IL-12, glycogen synthase kinase 3 β , inducible nitric oxide synthase, and tumor necrosis factor alpha (TNF- α), have all been shown to play a role in VEEV pathogenesis (49) (50) (51) (52) (53). The use of high-throughput next-generation sequencing technologies, such as RNA-Seq, allows an in-depth and unbiased look into the virus-host transcriptome, thus enabling changes in the expression of specific mRNAs to be connected with phenotypic outcomes. To this end, identification of critical differentially expressed transcripts among

clinically relevant infected cells will help lead to a greater understanding of viral pathogenesis and may prove beneficial for the identification of therapeutic targets.

In this study, network analysis/RNA-Seq data and the results of protein expression studies revealed that VEEV infection resulted in activation of the PERK arm of the UPR pathway, including the activation of ATF4, CHOP, and eIF2 γ phosphorylation. Several alphaviruses have previously been reported to hijack key components of the UPR pathway in order to promote viral replication, as the reliance of enveloped viruses on the ER for the synthesis of viral envelope-associated glycoproteins and their transport to the plasma membrane often stresses the ER due to rapid viral protein production (54, 55). Modulation of the UPR is not unique to alphaviruses; rather, it is a shared trait of many positive-sense RNA viruses.

Dengue virus has been shown to suppress PERK by inhibiting continued eIF2 γ phosphorylation in order to inhibit immediate apoptosis, increasing viral protein translation and extending the length of productive viral replication (34). Studies with hepatitis E virus (HEV) have demonstrated that expression of HEV capsid protein open reading frame 2 (ORF2) activates the expression of CHOP and ATF4 (56). In HEV, ORF2 was shown to stimulate CHOP through both ER stressors and amino acid response elements (AARE) through interaction with ATF4 (56).

The results shown here indicate that during VEEV infection, initiation of the UPR pathway and subsequent activation of EGR1 play a role in the outcome of virus-induced apoptosis. During the initial detection of ER stress, PERK is able to identify misfolded proteins in the lumen of the ER and phosphorylates eIF2 γ in order to initiate prosurvival pathways in the UPR through the general At 24 hpi caspase 3/7 activity was analyzed using the Caspase 3/7 Glo assay. The fold change values for mock-infected cells were set to a value of 1. **, $P < 0.001$. (E) EGR1 Δ/Δ and WT MEFs were mock or VEEV infected (MOI, 5). RNA was prepared, and gene expression was determined by qRT-PCR using a TaqMan assays for BIRC5 (survivin). The data shown are the values of the fold change of normalized gene expression determined by the $\Delta\Delta C_T$ threshold cycle (C_T) method. *, $P < 0.005$ (comparison of VEEV-infected WT and EGR1 Δ/Δ cells). inhibition of protein synthesis (33, 34). VEEV appears to induce the UPR and promote increased eIF2 γ phosphorylation, which results in the translational inhibition of most mRNAs, while UPR selectively increases the translation of ATF4. ATF4 is responsible for the expression of genes that encode proteins involved in apoptosis, redox processes, amino acid metabolism, and ER chaperone recruitment and is a well-known mediator of the PERK pathway and CHOP (33, 34). CHOP activation

facilitates the increased expression of cellular chaperones in order to counteract the buildup of misfolded proteins (57) . Failure to suppress protein misfolding in persistently stressed cells, such as during a viral infection, can then result in activation of the proapoptotic transcription factor CHOP, leading to suppression of the antiapoptotic protein B cell lymphoma-2 (Bcl-2). CHOP can also function as a prosurvival transcription factor by dephosphorylating eIF2 α through activation of the DNA damage-inducible protein (GADD34) in a self-regulating feedback loop (33, 34) . However, the data presented here support a model whereby VEEV infection leads CHOP to function in its proapoptotic role, as no change in GADD34 gene expression was detected by RNA-Seq analysis.

While the UPR was induced following VEEV infection, robust activation was not observed until later time points after infection. This is somewhat surprising, as VEEV infection is expected to induce significant ER stress due to the massive production of viral proteins during the course of an acute robust infection. The structural proteins of VEEV are translated from the viral subgenomic RNA into polyproteins on the rough ER. The E1 and pE2 precursor glycoproteins are then assembled as heterodimers in the ER, undergoing conformational changes requiring numerous chaperones (1, 58) . It is possible that VEEV has developed mechanisms to subvert the induction of the UPR. In order to counteract the UPR, the nonstructural proteins (nsPs) of Chikungunya virus (CHIKV) have been shown to inhibit expression of ATF4 and other known UPR target genes, including GRP78/BiP, GRP94, and CHOP (59) . Through nsP activity, CHIKV has developed a means of suppressing the UPR activity resulting from viral glycoprotein-induced ER stress, thus preventing immediate autophagy and apoptotic activation. The VEEV capsid is responsible for interfering with nucleocytoplasmic trafficking and inhibiting rRNA and mRNA transcription and has been implicated in the regulation of type I IFN signaling and the antiviral response through the regulation of both viral RNA and protein production (1, 48, 60) . Therefore, we hypothesize that the ability of the VEEV capsid to inhibit cellular transcription and block nucleocytoplasmic trafficking results in delayed induction of the UPR.

The results of a detailed network analysis based on existing data in the literature, coupled with the temporal gene expression profiles obtained from this study, point toward EGR1 being an important node in the novel link between VEEV activation of the type I interferon response and UPR. EGR1 is known to form a DNA binding complex with C/EBPB, a critical dimerization partner of CHOP (61) . Previous studies have demonstrated that the nuclear localization of CHOP may act as an inducer of EGR1 and that CHOP may act as a transcriptional cofactor for regulation of C/EBPB-EGR1 target genes (61) . The results of the

Western blot and microscopy analysis presented in this study support this model, as VEEV infection was found to increase both the overall levels and the nuclear distribution of CHOP along with those of EGR1. Previous studies demonstrated EGR1 mRNA induction by IFN- γ in mouse fibroblasts and by TNF- α , TNF- β , IL-1, IFN- α , IFN- β , and IFN- γ in human fibroblasts (31, 32). EGR1, also known as Zif268 and NGF1-A, is a zinc finger protein and mammalian transcription factor. It has been implicated in cellular proliferation and differentiation, but it may also have proapoptotic functions, depending on the cell type and stimulus (62). Of particular interest, EGR1 directly controls proliferation when activated by the mitogen-activated protein kinase/extracellular signal-regulated kinase pathway in mitogen-stimulated astrocytes (63). Virus-induced changes in EGR1 expression have been observed in several in vitro systems. In HIV-1-infected astrocytes, EGR1 upregulation was found to be induced by Tat through transactivation of the EGR1 promoter, leading to cellular dysfunction and Tat-induced neurotoxicity (64). Increased amounts of EGR1 mRNA have also been demonstrated to act in a region-specific manner, corresponding temporally with viral RNA production in the brain tissues of rats infected with either rabies virus or Borna disease virus (65).

In summary, the current study demonstrates a potential link between UPR activation and EGR1. EGR1 Δ/Δ MEFs demonstrated lower levels of susceptibility to VEEV-induced cell death than wild-type MEFs, indicating that EGR1 modulates proapoptotic pathways following infection. Studies are under way to determine if alteration of the UPR through small molecule inhibitors or siRNA interference influences VEEV replication and/or cell death. To date the mechanisms underlying VEEV pathogenesis and subsequent neuronal degeneration have been only partially elucidated. Therefore, determining the role of EGR1 and UPR may play a significant role in the development of a novel therapeutic target resulting in decreased neuronal death and the subsequent neuronal sequelae that result from infection.

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--- START OF ARTICLE 97 ---

Host resilience to emerging coronaviruses

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Abstract: Recently, two coronaviruses, severe acute respiratory syndrome coronavirus and Middle East respiratory syndrome coronavirus, have emerged to cause unusually severe respiratory disease in humans. Currently, there is a lack of effective antiviral treatment options or vaccine available. Given the severity of these outbreaks, and the possibility of additional zoonotic coronaviruses emerging in the near future, the exploration of different treatment strategies is necessary. Disease resilience is the ability of a given host to tolerate an infection, and to return to a state of health. This review focuses on exploring various host resilience mechanisms that could be exploited for treatment of severe acute respiratory syndrome coronavirus, Middle East respiratory syndrome coronavirus and other respiratory viruses that cause acute lung injury and acute respiratory distress syndrome.

Text: The 21st century was heralded with the emergence of two novel coronaviruses (CoV) that have unusually high pathogenicity and mortality [1] [2] [3] [4] [5] . Severe acute respiratory syndrome coronavirus (SARS-CoV) was first identified in 2003 [6] [7] [8] [9] . While there was initially great concern about SARS-CoV, once no new cases emerged, funding and research decreased. However, a decade later Middle East respiratory syndrome coronavirus (MERS-CoV), also known as HCoV-EMC, emerged initially in Saudi Arabia [3, 10] . SARS-CoV infected about 8000 people, and resulted in the deaths of approximately 10% of those infected [11] . While MERS-CoV is not as widespread as SARS-CoV, it appears to have an even higher mortality rate, with 35-50% of diagnosed infections resulting in death [3, [12] [13] . These deadly betacoronavirus viruses existed in animal reservoirs [4] [5] 9, [14] [15] . Recently, other CoVs have been detected in animal populations raising the possibility that we will see a repeat of these types of outbreaks in the near future [11, [16] [17] [18] [19] [20] . Both these zoonotic viruses cause a much more

severe disease than what is typically seen for CoVs, making them a global health concern. Both SARS-CoV and MERS-CoV result in severe lung pathology. Many infected patients have acute lung injury (ALI), a condition that is diagnosed based on the presence of pulmonary edema and respiratory failure without a cardiac cause. In some patients there is a progression to the more severe form of ALI, acute respiratory distress syndrome (ARDS) [21] [22] [23] .

In order to survive a given infection, a successful host must not only be able to clear the pathogen, but tolerate damage caused by the pathogen itself and also by the host's immune response [24] [25] [26] . We refer to resilience as the ability of a host to tolerate the effects of pathogens and the immune response to pathogens. A resilient host is able to return to a state of health after responding to an infection [24, [27] [28] . Most currently available treatment options for infectious diseases are antimicrobials, For reprint orders, please contact: reprints@futuremedicine.com REviEW Jamieson future science group and thus target the pathogen itself. Given the damage that pathogens can cause this focus on rapid pathogen clearance is understandable. However, an equally important medical intervention is to increase the ability of the host to tolerate the direct and indirect effects of the pathogen, and this is an area that is just beginning to be explored [29] . Damage to the lung epithelium by respiratory pathogens is a common cause of decreased resilience [30] [31] [32] . This review explores some of the probable host resilience pathways to viral infections, with a particular focus on the emerging coronaviruses. We will also examine factors that make some patients disease tolerant and other patients less tolerant to the viral infection. These factors can serve as a guide to new potential therapies for improved patient care.

Both SARS-CoV and MERS-CoV are typified by a rapid progression to ARDS, however, there are some distinct differences in the infectivity and pathogenicity. The two viruses have different receptors leading to different cellular tropism, and SARS-CoV is more ubiquitous in the cell type and species it can infect. SARS-CoV uses the ACE2 receptor to gain entry to cells, while MERS-CoV uses the ectopeptidase DPP4 [33] [34] [35] [36] . Unlike SARS-CoV infection, which causes primarily a severe respiratory syndrome, MERS-CoV infection can also lead to kidney failure [37, 38] . SARS-CoV also spreads more rapidly between hosts, while MERS-CoV has been more easily contained, but it is unclear if this is due to the affected patient populations and regions [3] [4] 39]. Since MERS-CoV is a very recently discovered virus, [40, 41] more research has been done on SARS-CoV. However, given the

similarities it is hoped that some of these findings can also be applied to MERS-CoV, and other potential emerging zoonotic coronaviruses.

Both viral infections elicit a very strong inflammatory response, and are also able to circumvent the immune response. There appears to be several ways that these viruses evade and otherwise redirect the immune response [1, [42] [43] [44] [45] . The pathways that lead to the induction of the antiviral type I interferon (IFN) response are common targets of many viruses, and coronaviruses are no exception. SARS-CoV and MERS-CoV are contained in double membrane vesicles (DMVs), that prevents sensing of its genome [1, 46] . As with most coronaviruses several viral proteins suppress the type I IFN response, and other aspects of innate antiviral immunity [47] . These alterations of the type I IFN response appear to play a role in immunopathology in more than one way. In patients with high initial viral titers there is a poor prognosis [39, 48] . This indicates that reduction of the antiviral response may lead to direct viral-induced pathology. There is also evidence that the delayed type I IFN response can lead to misregulation of the immune response that can cause immunopathology. In a mouse model of SARS-CoV infection, the type I IFN response is delayed [49] . The delay of this potent antiviral response leads to decreased viral clearance, at the same time there is an increase in inflammatory cells of the immune system that cause excessive immunopathology [49] . In this case, the delayed antiviral response not only causes immunopathology, it also fails to properly control the viral replication. While more research is needed, it appears that MERS has a similar effect on the innate immune response [5, 50] .

The current treatment and prevention options for SARS-CoV and MERS-CoV are limited. So far there are no licensed vaccines for SAR-CoV or MERS-CoV, although several strategies have been tried in animal models [51, 52] . There are also no antiviral strategies that are clearly effective in controlled trials. During outbreaks several antiviral strategies were empirically tried, but these uncontrolled studies gave mixed results [5, 39] . The main antivirals used were ribavirin, lopinavir and ritonavir [38, 53] . These were often used in combination with IFN therapy [54] . However, retrospective analysis of these data has not led to clear conclusions of the efficacy of these treatment options. Research in this area is still ongoing and it is hoped that we will soon have effective strategies to treat novel CoV [3,36,38,40, [55] [56] [57] [58] [59] [60] [61] [62] [63] [64] .

The lack of effective antivirals makes it necessary to examine other potential treatments for SARS-CoV and MERS-CoV. Even if there were effective strategies to decrease viral burden, for these viruses, the potential for new emerging zoonotic CoVs presents additional complications. Vaccines cannot be produced in time to stop the spread of an emerging virus. In addition, as was demonstrated during SARS-CoV and MERS-CoV outbreaks, there is always a challenge during a crisis situation to know which Host resilience to emerging coronaviruses REviEW future science group www.futuremedicine.com antiviral will work on a given virus. One method of addressing this is to develop broad-spectrum antivirals that target conserved features of a given class of virus [65] . However, given the fast mutation rates of viruses there are several challenges to this strategy. Another method is to increase the ability of a given patient to tolerate the disease, i.e., target host resilience mechanisms. So far this has largely been in the form of supportive care, which relies on mechanical ventilation and oxygenation [29, 39, 66] .

Since SARS-CoV and MERS-CoV were discovered relatively recently there is a lack of both patient and experimental data. However, many other viruses cause ALI and ARDS, including influenza A virus (IAV). By looking at data from other high pathology viruses we can extrapolate various pathways that could be targeted during infection with these emerging CoVs. This can add to our understanding of disease resilience mechanisms that we have learned from direct studies of SARS-CoV and MERS-CoV. Increased understanding of host resilience mechanisms can lead to future host-based therapies that could increase patient survival [29] .

One common theme that emerges in many respiratory viruses including SARS-CoV and MERS-CoV is that much of the pathology is due to an excessive inflammatory response. A study from Josset et al. examines the cell host response to both MERS-CoV and SARS-CoV, and discovered that MERS-CoV dysregulates the host transcriptome to a much greater extent than SARS-CoV [67] . It demonstrates that glucocorticoids may be a potential way of altering the changes in the host transcriptome at late time points after infection. If host gene responses are maintained this may increase disease resilience. Given the severe disease that manifested during the SARS-CoV outbreak, many different treatment options were empirically tried on human patients. One immunomodulatory treatment that was tried during the SARS-CoV outbreak was systemic corticosteroids. This was tried with and without the use of type I IFNs and other therapies that could directly target the virus [68] . Retrospective analysis revealed that, when given at the correct time and to the appropriate patients, corticosteroid use could decrease mortality and also length of hospital stays [68]

. In addition, there is some evidence that simultaneous treatment with IFNs could increase the potential benefits [69] . Although these treatments are not without complications, and there has been a lack of a randomized controlled trial [5, 39] .

Corticosteroids are broadly immunosuppressive and have many physiological effects [5, 39] . Several recent studies have suggested that other compounds could be useful in increasing host resilience to viral lung infections. A recent paper demonstrates that topoisomerase I can protect against inflammation-induced death from a variety of viral infections including IAV [70] . Blockade of C5a complement signaling has also been suggested as a possible option in decreasing inflammation during IAV infection [71] . Other immunomodulators include celecoxib, mesalazine and eritoran [72, 73] . Another class of drugs that have been suggested are statins. They act to stabilize the activation of aspects of the innate immune response and prevent excessive inflammation [74] . However, decreasing immunopathology by immunomodulation is problematic because it can lead to increased pathogen burden, and thus increase virus-induced pathology [75, 76] . Another potential treatment option is increasing tissue repair pathways to increase host resilience to disease. This has been shown by bioinformatics [77] , as well as in several animal models [30-31, 78-79]. These therapies have been shown in cell culture model systems or animal models to be effective, but have not been demonstrated in human patients. The correct timing of the treatments is essential. Early intervention has been shown to be the most effective in some cases, but other therapies work better when given slightly later during the course of the infection. As the onset of symptoms varies slightly from patient to patient the need for precise timing will be a challenge.

Examination of potential treatment options for SARS-CoV and MERS-CoV should include consideration of host resilience [29] . In addition to the viral effects, and the pathology caused by the immune response, there are various comorbidities associated with SARS-CoV and MERS-CoV that lead to adverse outcomes. Interestingly, these additional risk factors that lead to a more severe disease are different between the two viruses. It is unclear if these differences are due to distinct populations affected by the viruses, because of properties of the virus themselves, or both. Understanding these factors could be a key to increasing host resilience to the infections. MERS-CoV patients had increased morbidity and mortality if they were obese, immunocompromised, diabetic or had cardiac disease [4, 12] .

REVIEW Jamieson future science group Risk factors for SARS-CoV patients included an older age and male [39] . Immune factors that increased mortality for SARS-CoV were a higher neutrophil count and low T-cell counts [5, 39, 77] . One factor that increased disease for patients infected with SARS-CoV and MERS-CoV was infection with other viruses or bacteria [5, 39] . This is similar to what is seen with many other respiratory infections. A recent study looking at malaria infections in animal models and human patients demonstrated that resilient hosts can be predicted [28] . Clinical studies have started to correlate specific biomarkers with disease outcomes in ARDS patients [80] . By understanding risk factors for disease severity we can perhaps predict if a host may be nonresilient and tailor the treatment options appropriately.

A clear advantage of targeting host resilience pathways is that these therapies can be used to treat a variety of different infections. In addition, there is no need to develop a vaccine or understand the antiviral susceptibility of a new virus. Toward this end, understanding why some patients or patient populations have increased susceptibility is of paramount importance. In addition, a need for good model systems to study responses to these new emerging coronaviruses is essential. Research into both these subjects will lead us toward improved treatment of emerging viruses that cause ALI, such as SARS-CoV and MERS-CoV.

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- Severe acute respiratory syndrome coronavirus and Middle East respiratory syndrome coronavirus are zoonotic coronaviruses that cause acute lung injury and acute respiratory distress syndrome.
- Antivirals have limited effects on the course of the infection with these coronaviruses.

- There is currently no vaccine for either severe acute respiratory syndrome coronavirus or Middle East respiratory syndrome coronavirus.
- Host resilience is the ability of a host to tolerate the effects of an infection and return to a state of health.
- Several pathways, including control of inflammation, metabolism and tissue repair may be targeted to increase host resilience.
- The future challenge is to target host resilience pathways in such a way that there are limited effects on pathogen clearance pathways. Future studies should determine the safety of these types of treatments for human patients.

Papers of special note have been highlighted as:

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Potential Rapid Diagnostics, Vaccine and Therapeutics for 2019 Novel Coronavirus (2019-nCoV): A Systematic Review

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Abstract: Rapid diagnostics, vaccines and therapeutics are important interventions for the management of the 2019 novel coronavirus (2019-nCoV) outbreak. It is timely to systematically review the potential of these interventions, including those for Middle East respiratory syndrome-Coronavirus (MERS-CoV) and severe acute respiratory syndrome (SARS)-CoV, to guide policymakers globally on their prioritization of resources for research and development. A systematic search was carried out in three major electronic databases (PubMed, Embase and Cochrane Library) to identify published studies in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines. Supplementary strategies through Google Search and personal communications were used. A total of 27 studies fulfilled the criteria for review. Several laboratory protocols for confirmation of suspected 2019-nCoV cases using real-time reverse transcription polymerase chain reaction (RT-PCR) have been published. A commercial RT-PCR kit developed by the Beijing Genomic Institute is currently widely used in China and likely in Asia. However, serological assays as well as point-of-care testing kits have not been developed but are likely in the near future. Several vaccine candidates are in the pipeline. The likely earliest Phase 1 vaccine trial is a synthetic DNA-based candidate. A number of novel compounds as well as therapeutics licensed for other conditions appear to have in vitro efficacy against the 2019-nCoV. Some are being tested in clinical trials against MERS-CoV and SARS-CoV, while others have been listed for clinical trials against 2019-nCoV. However, there are currently no effective specific antivirals or drug combinations supported by high-level evidence.

Text: Since mid-December 2019 and as of early February 2020, the 2019 novel coronavirus (2019-nCoV) originating from Wuhan (Hubei Province, China) has infected over 25,000 laboratory-confirmed cases across 28 countries with about 500 deaths (a case-fatality rate of about 2%). More than 90% of the cases and deaths were in China [1] . Based on the initial reported surge of cases in Wuhan, the majority were males with a median age of 55 years and linked to the Huanan Seafood Wholesale Market [2] . Most of the reported cases had similar symptoms at the onset of illness such as fever, cough, and myalgia or fatigue. Most cases developed pneumonia and some severe and even fatal respiratory diseases such as acute respiratory distress syndrome [3] .

The 2019 novel coronavirus (2019-nCoV), a betacoronavirus, forms a clade within the subgenus sarbecovirus of the Orthocoronavirinae subfamily [4] . The severe acute respiratory syndrome coronavirus (SARS-CoV) and Middle East respiratory syndrome coronavirus (MERS-CoV) are also betacoronaviruses that are zoonotic in origin and have been linked to potential fatal illness during the outbreaks in 2003 and 2012, respectively [5, 6] . Based on current evidence, pathogenicity for 2019-nCoV is about 3%, which is significantly lower than SARS-CoV (10%) and MERS-CoV (40%) [7] . However, 2019-nCoV has potentially higher transmissibility (R_0 : 1.4-5.5) than both SARS-CoV (R_0 : [2] [3] [4] [5] and MERS-CoV (R_0 : <1) [7] .

With the possible expansion of 2019-nCoV globally [8] and the declaration of the 2019-nCoV outbreak as a Public Health Emergency of International Concern by the World Health Organization, there is an urgent need for rapid diagnostics, vaccines and therapeutics to detect, prevent and contain 2019-nCoV promptly. There is however currently a lack of understanding of what is available in the early phase of 2019-nCoV outbreak. The systematic review describes and assesses the potential rapid diagnostics, vaccines and therapeutics for 2019-nCoV, based in part on the developments for MERS-CoV and SARS-CoV.

A systematic search was carried out in three major electronic databases (PubMed, Embase and Cochrane Library) to identify published studies examining the diagnosis, therapeutic drugs and vaccines for Severe Acute Respiratory Syndrome (SARS), Middle East Respiratory Syndrome (MERS) and the 2019 novel coronavirus (2019-nCoV), in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.

There were two independent reviewers each focusing on SARS, MERS, and 2019-nCoV, respectively. A third independent reviewer was engaged to resolve any conflicting article of interest. We used the key words "SARS", "coronavirus", "MERS", "2019 Novel coronavirus", "Wuhan virus" to identify the diseases in the search strategy. The systematic searches for diagnosis, therapeutic drugs and vaccines were carried out independently and the key words "drug", "therapy", "vaccine", "diagnosis", "point of care testing" and "rapid diagnostic test" were used in conjunction with the disease key words for the respective searches.

Examples of search strings can be found in Table S1 . We searched for randomized controlled trials (RCTs) and validation trials (for diagnostics test) published in English, that measured (a) the sensitivity and/or specificity of a rapid diagnostic test or a point-of-care testing kit, (b) the impact of drug therapy or (c) vaccine efficacy against either of these diseases with no date restriction applied. For the 2019-nCoV, we searched for all in vitro, animal, or human studies published in English between 1 December 2019 and 6 February 2020, on the same outcomes of interest. In addition, we reviewed the references of retrieved articles in order to identify additional studies or reports not retrieved by the initial searches. Studies that examined the mechanisms of diagnostic tests, drug therapy or vaccine efficacy against SARS, MERS and 2019-nCoV were excluded. A Google search for 2019-nCoV diagnostics (as of 6 February 2020; Table S2) yielded five webpage links from government and international bodies with official information and guidelines (WHO, Europe CDC, US CDC, US FDA), three webpage links on diagnostic protocols and scientific commentaries, and five webpage links on market news and press releases. Six protocols for diagnostics using reverse transcriptase polymerase chain reaction (RT-PCR) from six countries were published on WHO's website [9] . Google search for 2019-nCoV vaccines yielded 19 relevant articles.

With the emergence of 2019-nCoV, real time RT-PCR remains the primary means for diagnosing the new virus strain among the many diagnostic platforms available ([10] [11] [12] [13] [14] [15] [16] [17] [18] [19] ; Table S3). Among the 16 diagnostics studies selected, one study discussed the use of RT-PCR in diagnosing patients with 2019-nCoV [11] (Table 1). The period and type of specimen collected for RT-PCR play an important role in the diagnosis of 2019-nCoV. It was found that the respiratory specimens were positive for the virus while serum was negative in the early period. It has also suggested that in the early days of illness, patients have high levels of virus despite the mild symptoms.

Apart from the commonly used RT-PCR in diagnosing MERS-CoV, four studies identified various diagnostic methods such as reverse transcription loop-mediated isothermal amplification (RT-LAMP), RT-insulated isothermal PCR (RT-iiPCR) and a one-step rRT-PCR assay based on specific TaqMan probes. RT-LAMP has similar sensitivity as real time RT-PCR. It is also highly specific and is used to detect MERS-CoV. It is comparable to the usual diagnostic tests and is rapid, simple and convenient. Likewise, RT-iiPCR and a one-step rRT-PCR assay have also shown similar sensitivity and high specificity for MER-CoV. Lastly, one study focused on the validation of the six commercial real RT-PCR kits, with high

accuracy. Although real time RT-PCR is a primary method for diagnosing MERS-CoV, high levels of PCR inhibition may hinder PCR sensitivity (Table 1) .

There are eleven studies that focus on SARS-CoV diagnostic testing (Table 1) . These papers described diagnostic methods to detect the virus with the majority of them using molecular testing for diagnosis. Comparison between the molecular test (i.e RT-PCR) and serological test (i.e., ELISA) showed that the molecular test has better sensitivity and specificity. Hence, enhancements to the current molecular test were conducted to improve the diagnosis. Studies looked at using nested PCR to include a pre-amplification step or incorporating N gene as an additional sensitive molecular marker to improve on the sensitivity (Table 1).

In addition, there are seven potential rapid diagnostic kits (as of 24 January 2020; Table 2) available on the market for 2019-nCoV. Six of these are only for research purposes. Only one kit from Beijing Genome Institute (BGI) is approved for use in the clinical setting for rapid diagnosis. Most of the kits are for RT-PCR. There were two kits (BGI, China and Veredus, Singapore) with the capability to detect multiple pathogens using sequencing and microarray technologies, respectively. The limit of detection of the enhanced realtime PCR method was 10² -fold higher than the standard real-time PCR assay and 10⁷ fold higher than conventional PCR methods In the clinical aspect, the enhanced realtime PCR method was able to detect 6 cases of SARS-CoV positive samples that were not confirmed by any other assay [25] • The real time PCR has a threshold sensitivity of 10 genome equivalents per reaction and it has a good reproducibility with the inter-assay coefficients of variation of 1.73 to 2.72%. • 13 specimens from 6 patients were positive with viral load range from 362 to 36,240,000 genome equivalents/mL. The real-time RT-PCR reaction was more sensitive than the nested PCR reaction, as the detection limit for the nested PCR reaction was about 10³ genome equivalents in the standard cDNA control. [34] Real-time reverse-transcription PCR (rRT-PCR); RNA-dependent RNA polymerase (RdRp); open reading frame 1a (ORF1a); Loop-mediated isothermal amplification (LAMP); enzyme-linked immunosorbent assay (ELISA); immunofluorescent assay (IFA); immunochromatographic test (ICT); nasopharyngeal aspirate (NPA).

With the emergence of 2019-nCoV, there are about 15 potential vaccine candidates in the pipeline globally (Table 3), in which a wide range of technology (such as messenger RNA, DNA-based, nanoparticle, synthetic and modified virus-like particle) was applied. It will

likely take about a year for most candidates to start phase 1 clinical trials except for those funded by Coalition for Epidemic Preparedness Innovations (CEPI). However, the kit developed by the BGI have passed emergency approval procedure of the National Medical Products Administration, and are currently used in clinical and surveillance centers of China [40] .

Of the total of 570 unique studies on 2019-nCoV, SARS CoV or MERS-CoV vaccines screened, only four were eventually included in the review. Most studies on SARS and MERS vaccines were excluded as they were performed in cell or animal models (Figure 1). The four studies included in this review were Phase I clinical trials on SARS or MERS vaccines (Table 4) [44] [45] [46] [47] . There were no studies of any population type (cell, animal, human) on the 2019-nCoV at the point of screening. The published clinical trials were mostly done in United States except for one on the SARS vaccine done in China [44] . All vaccine candidates for SARS and MERS were reported to be safe, well-tolerated and able to trigger the relevant and appropriate immune responses in the participants. In addition, we highlight six ongoing Phase I clinical trials identified in the ClinicalTrials.gov register ([48, 49]); Table S4) [50] [51] [52] . These trials are all testing the safety and immunogenicity of their respective MERS-CoV vaccine candidates but were excluded as there are no results published yet. The trials are projected to complete in December 2020 (two studies in Russia [50, 51]) and December 2021 (in Germany [52]).

Existing literature search did not return any results on completed 2019-nCoV trials at the time of writing. Among 23 trials found from the systematic review (Table 5) , there are nine clinical trials registered under the clinical trials registry (ClinicalTrials.gov) for 2019-nCoV therapeutics [53] [54] [55] [56] [57] [58] [59] [60] [61] . Of which five studies on hydroxychloroquine, lopinavir plus ritonavir and arbidol, mesenchymal stem cells, traditional Chinese medicine and glucocorticoid therapy usage have commenced recruitment. The remaining four studies encompass investigation of antivirals, interferon atomization, darunavir and cobicistat, arbidol, and remdesivir usage for 2019-nCoV patients (Table 5) . Seroconversion measured by S1-ELISA occurred in 86% and 94% participants after 2 and 3 doses, respectively, and was maintained in 79% participants up to study end at week 60. Neutralising antibodies were detected in 50% participants at one or more time points during the study, but only 3% maintained neutralisation activity to end of study. T-cell responses were detected in 71% and 76% participants after 2 and 3 doses, respectively. There were no differences in immune responses between dose groups after 6

weeks and vaccine-induced humoral and cellular responses were respectively detected in 77% and 64% participants at week 60.

[47] Molecules developed by the university scientists inhibit two coronavirus enzymes and prevent its replication. The discovered drug targets are said to be more than 95% similar to enzyme targets found on the SARS virus. Researchers note that identified drugs may not be available to address the ongoing outbreak but they hope to make it accessible for future outbreaks.

[85] Besides the six completed randomized controlled trials (RCT) selected from the systematic review (Table 6) , there is only one ongoing randomized controlled trial targeted at SARS therapeutics [92] . The studies found from ClinicalTrials.gov have not been updated since 2013. While many prospective and retrospective cohort studies conducted during the epidemic centered on usage of ribavirin with lopinavir/ritonavir or ribavirin only, there has yet to be well-designed clinical trials investigating their usage. Three completed randomized controlled trials were conducted during the SARS epidemic-3 in China, 1 in Taiwan and 2 in Hong Kong [93] [94] [95] [96] [97] . The studies respectively investigated antibiotic usage involving 190 participants, combination of western and Chinese treatment vs. Chinese treatment in 123 participants, integrative Chinese and Western treatment in 49 patients, usage of a specific Chinese medicine in four participants and early use of corticosteroid in 16 participants. Another notable study was an open non-randomized study investigating ribavirin/lopinavir/ritonavir usage in 152 participants [98] . One randomized controlled trial investigating integrative western and Chinese treatment during the SARS epidemic was excluded as it was a Chinese article [94] .

There is only one ongoing randomized controlled trial targeted at MERS therapeutics [99] . It investigates the usage of Lopinavir/Ritonavir and Interferon Beta 1B. Likewise, many prospective and retrospective cohort studies conducted during the epidemic centered on usage of ribavirin with lopinavir/ritonavir/ribavirin, interferon, and convalescent plasma usage. To date, only one trial has been completed. One phase 1 clinical trial investigating the safety and tolerability of a fully human polyclonal IgG immunoglobulin (SAB-301) was found in available literature [46] . The trial conducted in the United States in 2017 demonstrated SAB-301 to be safe and well-tolerated at single doses. Another trial on MERS therapeutics was found on ClinicalTrials.gov-a phase 2/3 trial in the United States

evaluating the safety, tolerability, pharmacokinetics (PK), and immunogenicity on coadministered MERS-CoV antibodies REGN3048 & REGN3051 [100].

Rapid diagnostics plays an important role in disease and outbreak management. The fast and accurate diagnosis of a specific viral infection enables prompt and accurate public health surveillance, prevention and control measures. Local transmission and clusters can be prevented or delayed by isolation of laboratory-confirmed cases and their close contacts quarantined and monitored at home. Rapid diagnostic also facilitates other specific public health interventions such as closure of high-risk facilities and areas associated with the confirmed cases for prompt infection control and environmental decontamination [11, 101] .

Laboratory diagnosis can be performed by: (a) detecting the genetic material of the virus, (b) detecting the antibodies that neutralize the viral particles of interest, (c) detecting the viral epitopes of interest with antibodies (serological testing), or (d) culture and isolation of viable virus particles.

The key limitations of genetic material detection are the lack of knowledge of the presence of viable virus, the potential cross-reactivity with non-specific genetic regions and the short timeframe for accurate detection during the acute infection phase. The key limitations of serological testing is the need to collect paired serum samples (in the acute and convalescent phases) from cases under investigation for confirmation to eliminate potential cross-reactivity from non-specific antibodies from past exposure and/or infection by other coronaviruses. The limitation of virus culture and isolation is the long duration and the highly specialized skills required of the technicians to process the samples. All patients recovered.

Significantly shorted time from the disease onset to the symptom improvement in treatment (5.10 ± 2.83 days) compared to control group (7.62 ± 2.27 days) ($p < 0.05$) No significant difference in blood routine improvement, pulmonary chest shadow in chest film improvement and corticosteroid usgae between the 2 groups. However, particularly in the respect of improving clinical symptoms, elevating quality of life, promoting immune function recovery, promoting absorption of pulmonary inflammation, reducing the dosage of cortisteroid and shortening the therapeutic course, treatment with integrative chinese

and western medicine treatment had obvious superiority compared with using control treatment alone. Single infusions of SAB-301 up to 50 mg/kg appear to be safe and well-tolerated in healthy participants. [46] Where the biological samples are taken from also play a role in the sensitivity of these tests. For SARS-CoV and MERS-CoV, specimens collected from the lower respiratory tract such as sputum and tracheal aspirates have higher and more prolonged levels of viral RNA because of the tropism of the virus. MERS-CoV viral loads are also higher for severe cases and have longer viral shedding compared to mild cases. Although upper respiratory tract specimens such as nasopharyngeal or oropharyngeal swabs can be used, they have potentially lower viral loads and may have higher risk of false-negatives among the mild MERS and SARS cases [102, 103] , and likely among the 2019-nCoV cases.

The existing practices in detecting genetic material of coronaviruses such as SARS-CoV and MERS-CoV include (a) reverse transcription-polymerase chain reaction (RT-PCR), (b) real-time RT-PCR (rRT-PCR), (c) reverse transcription loop-mediated isothermal amplification (RT-LAMP) and (d) real-time RT-LAMP [104] . Nucleic amplification tests (NAAT) are usually preferred as in the case of MERS-CoV diagnosis as it has the highest sensitivity at the earliest time point in the acute phase of infection [102] . Chinese health authorities have recently posted the full genome of 2019-nCoV in the GenBank and in GISAID portal to facilitate in the detection of the virus [11] . Several laboratory assays have been developed to detect the novel coronavirus in Wuhan, as highlighted in WHO's interim guidance on nCoV laboratory testing of suspected cases. These include protocols from other countries such as Thailand, Japan and China [105] .

The first validated diagnostic test was designed in Germany. Corman et al. had initially designed a candidate diagnostic RT-PCR assay based on the SARS or SARS-related coronavirus as it was suggested that circulating virus was SARS-like. Upon the release of the sequence, assays were selected based on the match against 2019-nCoV upon inspection of the sequence alignment. Two assays were used for the RNA dependent RNA polymerase (RdRP) gene and E gene where E gene assay acts as the first-line screening tool and RdRp gene assay as the confirmatory testing. All assays were highly sensitive and specific in that they did not cross-react with other coronavirus and also human clinical samples that contained respiratory viruses [11] .

The Hong Kong University used two monoplex assays which were reactive with coronaviruses under the subgenus Sarbecovirus (consisting of 2019-nCoV, SARS-CoV and SARS-like coronavirus). Viral RNA extracted from SARS-CoV can be used as the positive control for the suggested protocol assuming that SARS has been eradicated. It is proposed that the N gene RT-PCR can be used as a screening assay while the Orf1b assay acts as a confirmatory test. However, this protocol has only been evaluated with a panel of controls with the only positive control SARS-CoV RNA. Synthetic oligonucleotide positive control or 2019-nCoV have yet to be tested [106] .

The US CDC shared the protocol on the real time RT-PCR assay for the detection of the 2019-nCoV with the primers and probes designed for the universal detection of SARS-like coronavirus and the specific detection of 2019-nCoV. However, the protocol has not been validated on other platforms or chemistries apart from the protocol described. There are some limitations for the assay. Analysts engaged have to be trained and familiar with the testing procedure and result interpretation. False negative results may occur due to insufficient organisms in the specimen resulting from improper collection, transportation or handling. Also, RNA viruses may show substantial genetic variability. This could result in mismatch between the primer and probes with the target sequence which can diminish the assay performance or result in false negative results [107] . Point-of-care test kit can potentially minimize these limitations, which should be highly prioritized for research and development in the next few months.

Serological testing such as ELISA, IIFT and neutralization tests are effective in determining the extent of infection, including estimating asymptomatic and attack rate. Compared to the detection of viral genome through molecular methods, serological testing detects antibodies and antigens. There would be a lag period as antibodies specifically targeting the virus would normally appear between 14 and 28 days after the illness onset [108] . Furthermore, studies suggest that low antibody titers in the second week or delayed antibody production could be associated with mortality with a high viral load. Hence, serological diagnoses are likely used when nucleic amplification tests (NAAT) are not available or accessible [102] .

Vaccines can prevent and protect against infection and disease occurrence when exposed to the specific pathogen of interest, especially in vulnerable populations who are more prone to severe outcomes. In the context of the current 2019-nCoV outbreak, vaccines will

help control and reduce disease transmission by creating herd immunity in addition to protecting healthy individuals from infection. This decreases the effective R_0 value of the disease. Nonetheless, there are social, clinical and economic hurdles for vaccine and vaccination programmes, including (a) the willingness of the public to undergo vaccination with a novel vaccine, (b) the side effects and severe adverse reactions of vaccination, (c) the potential difference and/or low efficacy of the vaccine in populations different from the clinical trials' populations and (d) the accessibility of the vaccines to a given population (including the cost and availability of the vaccine).

Vaccines against the 2019-nCoV are currently in development and none are in testing (at the time of writing). On 23 January 2020, the Coalition for Epidemic Preparedness Innovations (CEPI) announced that they will fund vaccine development programmes with Inovio, The University of Queensland and Moderna, Inc respectively, with the aim to test the experimental vaccines clinically in 16 weeks (By June 2020). The vaccine candidates will be developed by the DNA, recombinant and mRNA vaccine platforms from these organizations [109] .

Based on the most recent MERS-CoV outbreak, there are already a number of vaccine candidates being developed but most are still in the preclinical testing stage. The vaccines in development include viral vector-based vaccine, DNA vaccine, subunit vaccine, virus-like particles (VLPs)-based vaccine, inactivated whole-virus (IWW) vaccine and live attenuated vaccine. The latest findings for these vaccines are based on the review by Yong et al. (2019) in August 2019 [110] . As of the date of reporting, there is only one published clinical study on the MERS-CoV vaccine by GeneOne Life Science & Inovio Pharmaceuticals [47] . There was one SARS vaccine trial conducted by the US National Institute of Allergy and Infectious Diseases. Both Phase I clinical trials reported positive results, but only one has announced plans to proceed to Phase 2 trial [111] .

Due to the close genetic relatedness of SARS-CoV (79%) with 2019-nCoV [112] , there may be potential cross-protective effect of using a safe SARS-CoV vaccine while awaiting the 2019-nCoV vaccine. However, this would require small scale phase-by-phase implementation and close monitoring of vaccinees before any large scale implementation.

Apart from the timely diagnosis of cases, the achievement of favorable clinical outcomes depends on the timely treatment administered. ACE2 has been reported to be the same cell entry receptor used by 2019-nCoV to infect humans as SARS-CoV [113] . Hence, clinical similarity between the two viruses is expected, particularly in severe cases. In addition, most of those who have died from MERS-CoV, SARS-CoV and 2019-nCoV were advance in age and had underlying health conditions such as hypertension, diabetes or cardiovascular disease that compromised their immune systems [114] . Coronaviruses have error-prone RNA-dependent RNA polymerases (RdRP), which result in frequent mutations and recombination events. This results in quasispecies diversity that is closely associated with adaptive evolution and the capacity to enhance viral-cell entry to cause disease over time in a specific population at-risk [115] . Since ACE2 is abundantly present in humans in the epithelia of the lung and small intestine, coronaviruses are likely to infect the upper respiratory and gastrointestinal tract and this may influence the type of therapeutics against 2019-nCoV, similarly to SAR-CoV.

However, in the years following two major coronavirus outbreaks SARS-CoV in 2003 and MERS-CoV in 2012, there remains no consensus on the optimal therapy for either disease [116, 117] . Well-designed clinical trials that provide the gold standard for assessing the therapeutic measures are scarce. No coronavirus protease inhibitors have successfully completed a preclinical development program despite large efforts exploring SARS-CoV inhibitors. The bulk of potential therapeutic strategies remain in the experimental phase, with only a handful crossing the in vitro hurdle. Stronger efforts are required in the research for treatment options for major coronaviruses given their pandemic potential. Effective treatment options are essential to maximize the restoration of affected populations to good health following infections. Clinical trials have commenced in China to identify effective treatments for 2019-nCoV based on the treatment evidence from SARS and MERS. There is currently no effective specific antiviral with high-level evidence; any specific antiviral therapy should be provided in the context of a clinical study/trial. Few treatments have shown real curative action against SARS and MERS and the literature generally describes isolated cases or small case series.

Many interferons from the three classes have been tested for their antiviral activities against SARS-CoV both in vitro and in animal models. Interferon β has consistently been shown to be the most active, followed by interferon α . The use of corticosteroids with interferon alfacon-1 (synthetic interferon α) appeared to have improved oxygenation and faster resolution of chest radiograph abnormalities in observational studies with untreated

controls. Interferon has been used in multiple observational studies to treat SARS-CoV and MERS-CoV patients [116, 117]. Interferons, with or without ribavirin, and lopinavir/ritonavir are most likely to be beneficial and are being trialed in China for 2019-nCoV. This drug treatment appears to be the most advanced. Timing of treatment is likely an important factor in effectiveness. A combination of ribavirin and lopinavir/ritonavir was used as a post-exposure prophylaxis in health care workers and may have reduced the risk of infection. Ribavirin alone is unlikely to have substantial antiviral activities at clinically used dosages. Hence, ribavirin with or without corticosteroids and with lopinavir and ritonavir are among the combinations employed. This was the most common agent reported in the available literature. Its efficacy has been assessed in observational studies, retrospective case series, retrospective cohort study, a prospective observational study, a prospective cohort study and randomized controlled trial ranging from seven to 229 participants [117]. Lopinavir/ritonavir (Kaletra) was the earliest protease inhibitor combination introduced for the treatment of SARS-CoV. Its efficacy was documented in several studies, causing notably lower incidence of adverse outcomes than with ribavirin alone. Combined usage with ribavirin was also associated with lower incidence of acute respiratory distress syndrome, nosocomial infection and death, amongst other favorable outcomes. Recent in vitro studies have shown another HIV protease inhibitor, nelfinavir, to have antiviral capacity against SARS-CoV, although it has yet to show favorable outcomes in animal studies [118]. Remdesivir (Gilead Sciences, GS-5734) nucleoside analogue in vitro and in vivo data support GS-5734 development as a potential pan-coronavirus antiviral based on results against several coronaviruses (CoVs), including highly pathogenic CoVs and potentially emergent BatCoVs. The use of remdesivir may be a good candidate as an investigational treatment.

Improved mortality following receipt of convalescent plasma in various doses was consistently reported in several observational studies involving cases with severe acute respiratory infections (SARIs) of viral etiology. A significant reduction in the pooled odds of mortality following treatment of 0.25 compared to placebo or no therapy was observed [119]. Studies were however at moderate to high risk of bias given their small sample sizes, allocation of treatment based on the physician's discretion, and the availability of plasma. Factors like concomitant treatment may have also confounded the results. Associations between convalescent plasma and hospital length of stay, viral antibody levels, and viral load respectively were similarly inconsistent across available literature. Convalescent plasma, while promising, is likely not yet feasible, given the limited pool of potential donors and issues of scalability. Monoclonal antibody treatment is progressing. SARS-CoV enters host cells through the binding of their spike (S) protein to angiotensin converting enzyme 2

(ACE2) and CD209L [118] . Human monoclonal antibodies to the S protein have been shown to significantly reduce the severity of lung pathology in non-human primates following MERS-CoV infection [120] . Such neutralizing antibodies can be elicited by active or passive immunization using vaccines or convalescent plasma respectively. While such neutralizing antibodies can theoretically be harvested from individuals immunized with vaccines, there is uncertainty over the achievement of therapeutic levels of antibodies.

Other therapeutic agents have also been reported. A known antimalarial agent, chloroquine, elicits antiviral effects against multiple viruses including HIV type 1, hepatitis B and HCoV-229E. Chloroquine is also immunomodulatory, capable of suppressing the production and release of factors which mediate the inflammatory complications of viral diseases (tumor necrosis factor and interleukin 6) [121] . It is postulated that chloroquine works by altering ACE2 glycosylation and endosomal pH. Its anti-inflammatory properties may be beneficial for the treatment of SARS. Niclosamide as a known drug used in antihelminthic treatment. The efficacy of niclosamide as an inhibitor of virus replication was proven in several assays. In both immunoblot analysis and immunofluorescence assays, niclosamide treatment was observed to completely inhibit viral antigen synthesis. Reduction of virus yield in infected cells was dose dependent. Niclosamide likely does not interfere in the early stages of virus attachment and entry into cells, nor does it function as a protease inhibitor. Mechanisms of niclosamide activity warrant further investigation [122] . Glycyrrhizin also reportedly inhibits virus adsorption and penetration in the early steps of virus replication. Glycyrrhizin was a significantly potent inhibitor with a low selectivity index when tested against several pathogenic flaviviruses. While preliminary results suggest production of nitrous oxide (which inhibits virus replication) through induction of nitrous oxide synthase, the mechanism of Glycyrrhizin against SARS-CoV remains unclear. The compound also has relatively lower toxicity compared to protease inhibitors like ribavirin [123] . Inhibitory activity was also detected in baicalin [124] , extracted from another herb used in the treatment of SARS in China and Hong Kong. Findings on these compounds are limited to in vitro studies [121] [122] [123] [124] .

Due to the rapidly evolving situation of the 2019-nCoV, there will be potential limitations to the systematic review. The systematic review is likely to have publication bias as some developments have yet to be reported while for other developments there is no intention to report publicly (or in scientific platforms) due to confidentiality concerns. However, this may be limited to only a few developments for review as publicity does help in branding to some extent for the company and/or the funder. Furthermore, due to the rapid need to

share the status of these developments, there may be reporting bias in some details provided by authors of the scientific articles or commentary articles in traditional media. Lastly, while it is not viable for any form of quality assessment and metaanalysis of the selected articles due to the limited data provided and the heterogeneous style of reporting by different articles, this paper has provided a comprehensive overview of the potential developments of these pharmaceutical interventions during the early phase of the outbreak. This systematic review would be useful for cross-check when the quality assessment and meta-analysis of these developments are performed as a follow-up study.

Rapid diagnostics, vaccines and therapeutics are key pharmaceutical interventions to limit transmission of respiratory infectious diseases. Many potential developments on these pharmaceutical interventions for 2019-nCoV are ongoing in the containment phase of this outbreak, potentially due to better pandemic preparedness than before. However, lessons from MERS-CoV and SARS-CoV have shown that the journeys for these developments can still be challenging moving ahead.

Supplementary Materials: The following are available online at www.mdpi.com/xxx/s1, Table S1 : Example of full search strategy in Pubmed, Table S2 : Google Search: 2019-nCoV diagnostics, Table S3 : Summary of diagnostic assays developed for 2019-nCoV, Table S4

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Port d'Entrée for Respiratory Infections – Does the Influenza A Virus Pave the Way for Bacteria?

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5742597/>

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Abstract: Bacterial and viral co-infections of the respiratory tract are life-threatening and present a global burden to the global community. *Staphylococcus aureus*, *Streptococcus pneumoniae*, and *Streptococcus pyogenes* are frequent colonizers of the upper respiratory tract. Imbalances through acquisition of seasonal viruses, e.g., Influenza A virus, can lead to bacterial dissemination to the lower respiratory tract, which in turn can result in severe pneumonia. In this review, we summarize the current knowledge about bacterial and viral co-infections of the respiratory tract and focus on potential experimental models suitable for mimicking this disease. Transmission of IAV and pneumonia is mainly modeled by mouse infection. Few studies utilizing ferrets, rats, guinea pigs, rabbits, and non-human primates are also available. The knowledge gained from these studies led to important discoveries and advances in understanding these infectious diseases. Nevertheless, mouse and other infection models have limitations, especially in translation of the discoveries to humans. Here, we suggest the use of human engineered lung tissue, human ex vivo lung tissue, and porcine models to study respiratory co-infections, which might contribute to a greater translation of the results to humans and improve both, animal and human health.

Text: In recent years the human microbiota is more and more recognized to play a crucial role in pathogenesis of many diseases (Weinstock, 2012) . The upper respiratory tract is a natural niche for potentially pathogenic bacteria embedded in commensal communities forming the nasopharyngeal microbiome. In particular, the microbial communities of the nasopharynx (Hilty et al., 2012) are associated with respiratory diseases, i.e., severe pneumonia, which are responsible for substantial mortality and morbidity in humans worldwide (Prina et al., 2016) . The composition of the nasopharyngeal microbiome is highly dynamic (Biesbroek et al., 2014a,b,c) and many factors, including environmental and host factors, can affect microbial colonization (Koppen et al., 2015) . Recent studies on neonates have shown that the respiratory microbiota develops from initially maternally transmitted mixed flora with predominance of *Streptococcus viridans* species to niche-specific bacterial profiles containing mostly *Staphylococcus aureus* at around 1 week of

age (Bosch et al., 2016a) . Between 2 weeks and 6 months after birth, the staphylococcal predominance declines and colonization with *Streptococcus pneumoniae* (pneumococci) as a predominant pathobiont emerges (Miller et al., 2011; Bosch et al., 2016a,b) . The dynamic microbiome composition is guaranteed through the interplay between bacterial species, other microbes, and changing environmental conditions, as well as host-bacteria interactions (Blaser and Falkow, 2009) . Most of the time, the microbiome and its interplay with the human host are believed to be beneficial for both (Pettigrew et al., 2008; Murphy et al., 2009) . However, imbalances in microbial composition can lead to acquisition of new viral or bacterial species and invasion of potential pathogens, which in turn can become detrimental, especially in elderly people and children with an exhausted or immature immune system (Pettigrew et al., 2008; Blaser and Falkow, 2009; Murphy et al., 2009) .

One particular example showing imbalances introduced by single dosage of antibiotics was demonstrated by Ichinohe and colleagues (Ichinohe et al., 2011) . While commensal respiratory microbiota facilitated immune-support against Influenza A virus infection (IAV), oral treatment with antibiotics resulted not only in a shift of bacterial composition, but also in impaired CD4 T-, CD8 T-, and B-cell immunity following infection with IAV in mice (Ichinohe et al., 2011) . Analyses of human oropharyngeal microbiomes during the 2009 H1N1 IAV pandemic revealed that at the phylum level, the abundance of Firmicutes and Proteobacteria was augmented in pneumonia patients as compared to healthy controls (Leung et al., 2013) . However, another study published in the same year contradicted these results (Chaban et al., 2013) . Chaban and colleagues analyzed microbiomes of 65 patients from H1N1 IAV outbreak in 2009. Although the phylogenetic composition of pneumonia patients was dominated by Firmicutes, Proteobacteria, and Actinobacteria, no significant differences between the patients and healthy controls or any other variables tested, including age and gender, were observed (Chaban et al., 2013) .

In this review we discuss secondary bacterial infections of the respiratory tract after primary infection by IAV with a focus on mechanisms by which these interactions are potentially mediated, and we will provide insight into the host contribution and immunological consequences. We further focus on potential animal models suitable for mimicking asymptomatic bacterial colonization and disease progression and thus, enabling to study adaptation strategies, viral-bacterial interactions, and immune responses in these highly lethal co-infections.

Influenza A viruses belong to the family of Orthomyxoviridae and based on the antigenicity of their haemagglutinin (HA) and neuraminidase (NA) they are classified into 16 classical HA and 9 classical NA subtypes (Neumann et al., 2009) . The 8-segmented genomes of influenza A viruses are characterized by a significant plasticity. Due to point mutations and re-assortment events new variants or strains with epidemic or pandemic potential emerge (Neumann et al., 2009) . In addition, influenza can be transmitted between animals, including swine, birds, horses, and humans, making it a zoonotic disease (van der Meer et al., 2010) . Seasonal influenza usually resolves without consequences in healthy individuals. However, it is estimated that seasonal influenza effects 5-10% of the world's population resulting in about 250,000 to 500,000 deaths annually (Tjon-Kon-Fat et al., 2016) . At greater risk to develop secondary bacterial pneumonia are individuals with comorbidities, elderly people (age > 65), pregnant women, and children under the age of one (Rothberg et al., 2008) .

For a long time it was considered that the H1N1 strain, an avian-like H1N1 virus, directly caused most of the fatalities during the 1918-1919 pandemic (Spanish Flu), often from a hemorrhagic pneumonitis rapidly progressing to acute respiratory distress syndrome and death (Osterholm, 2005; Gerberding, 2006; Oxford et al., 2006) . The pandemic killed around 50 million people worldwide and remains unique in its severity compared to other big outbreaks. However, many of the findings have been reinterpreted in recent years (Brundage and Shanks, 2007; Chien et al., 2009) . It is estimated that around 95% of all severe cases and deaths were attributed to secondary infections with bacterial pathogens, most predominantly by *Streptococcus pneumoniae* (Morens et al., 2008) . Individual studies limited to certain regions identified also other pathogens commonly colonizing the respiratory tract, including *Staphylococcus aureus*, group A streptococcus (GAS) and *Haemophilus influenzae* (Brundage and Shanks, 2008) . During the next two pandemics (H2N2 Asian Flu 1957 and H3N2 Hong Kong Flu 1968 –1969 bacterial co-infections were less likely the cause of death compared to the Spanish Flu (Giles and Shuttleworth, 1957; Trotter et al., 1959) . Still, pneumonia accounted for about 44% of deaths during the Asian Flu (Giles and Shuttleworth, 1957) . Most fatalities resulting from pneumonia occurred in individuals with chronic conditions, i.e., chronic lung diseases, rheumatic carditis, and hypertension (Giles and Shuttleworth, 1957) . In 1957-1958, *S. aureus* was predominantly isolated from fatal pneumonia cases (Hers et al., 1957 (Hers et al., , 1958 Robertson et al., 1958; Martin et al., 1959) , whereas *S. pneumoniae* returned as predominant cause of severe pneumonia during the Hong Kong Flu (Sharrar, 1969; Bisno et al., 1971; Burk et al., 1971; Schwarzmann et al., 1971) . Forty years later in 2009, a novel H1N1 virus of swine origin emerged and caused again a pandemic (Dawood et al., 2009 (Dawood et al., , 2012 .

In contrast to Asian and Hong Kong Flu, mortality rates were rather low, but most deaths occurred in healthy young individuals with no underlying conditions (Reichert et al., 2010; Monsalvo et al., 2011; Dawood et al., 2012) . About 25-50% of severe or fatal cases were linked to complications due to bacterial pneumonia (Dominguez-Cherit et al., 2009; Estenssoro et al., 2010; Mauad et al., 2010; Shieh et al., 2010) . Although regional variations occurred, pneumococci and *S. aureus* were the most frequently isolated bacterial species (Mauad et al., 2010; Shieh et al., 2010; Rice et al., 2012) . Group A streptococcus was absent in many local pneumonia outbreaks associated with viruses, but was predominant in others (Brundage and Shanks, 2008; Ampofo et al., 2010) . When it does appear, it is typically third in incidence (Chaussee et al., 2011) . Overall, data on pandemic outbreaks suggest that disease severity and mortality can be linked to secondary bacterial pathogens with variations depending on regions and state of immunity of the population (Brundage and Shanks, 2008; Shanks et al., 2010 Shanks et al., , 2011 McCullers, 2013) .

There is increasing evidence that the nasopharyngeal microbiota plays an important role in the pathogenesis of acute viral respiratory infections (Teo et al., 2015; de Steenhuijsen Piter et al., 2016; Rosas-Salazar et al., 2016a,b) . Respiratory viruses, including IAV, have been shown to alter bacterial adherence and colonization leading to an increased risk of secondary bacterial infections (Tregoning and Schwarze, 2010) . Pneumococci, *S. aureus*, and GAS are important human Gram-positive pathogens. All of them are frequent colonizers of the human nasopharynx and they share many features including pathogenic mechanisms and clinical aspects (Figure 1) . However, they also have unique properties.

Staphylococcus aureus colonizes persistently about 30% of the human population and typical niches include nares, axillae, and skin (Peacock et al., 2001; von Eiff et al., 2001; van Belkum et al., 2009) . They cause a variety of clinical manifestations ranging from mild skin infections to fatal necrotizing pneumonia. In the last decades, the pathogen became resistant to an increasing number of antibiotics and methicillin-resistant *S. aureus* (MRSA) is now a major cause of hospital acquired infections (Hartman and Tomasz, 1984; Ubukata et al., 1989; Zetola et al., 2005) . Also the rise of community-acquired *S. aureus* strains is of special concern, because certain clones are associated with very severe infections (Rasigade et al., 2010) . Recent prospective studies demonstrated an increase in proportion of communityacquired methicillin-sensitive *S. aureus* in severe pneumonia cases (McCaskill et al., 2007; Sicot et al., 2013) .

The pneumococcus is a typical colonizer of the human nasopharynx. About 20-50% of healthy children and 8-30% of healthy adults are asymptotically colonized (McCullers, 2006) . Pneumococci cause diseases ranging from mild, i.e., sinusitis, conjunctivitis, and otitis media, to more severe and potentially life-threatening infections, including communityacquired pneumonia, bacteraemia, and meningitis (Bogaert et al., 2004; Valles et al., 2016) . This bacterium is associated with high morbidity and mortality rates in risk groups such as immunocompromised individuals, children, and elderly (Black et al., 2010; Valles et al., 2016) .

Group A streptococci colonize the mouth and upper respiratory tract in about 2-5% of world's population (Okumura and Nizet, 2014) . The most common, non-invasive and mild infections caused by GAS are tonsillitis and pharyngitis with estimated 600 million cases per year (Carapetis et al., 2005) .

Listed as number nine in the list of global killers with around 500,000 deaths annually (Carapetis et al., 2005) , it is obvious that this pathogen can cause severe invasive infections, including pneumonia, sepsis, streptococcal toxic shock syndrome, and necrotizing skin infections (Cunningham, 2000; Carapetis et al., 2005) .

Although all three pathogens are able to cause highly lethal diseases, the most fatal remains the pneumococcus, estimated to cause ca. 10% of all deaths in children below 5 years of age (O'Brien et al., 2009) , in the elderly (Marrie et al., 2017) , and in immuno-compromised individuals (Baxter et al., 2016) .

Influenza A virus binds via HA to either $\alpha 2,3$ or $\alpha 2,6$ -linked sialic acid at the surface of epithelial cells of the upper and lower respiratory tract (Webster et al., 1992) . Seasonal strains show usually affinity to $\alpha 2,6$ -linked sialic acids that are expressed in the human trachea, whereas avian-like viruses preferentially bind to $\alpha 2,3$ -linked sialic acids of alveolar type II cells (Shinya et al., 2006; van Riel et al., 2007 van Riel et al., , 2010 . The release of viral genomic RNA into the cytosol activates different immune response pathways. Binding of viral RNA to retinoic acid inducible gene 1 induces the expression of type I and III interferons and activates transcription factor NF- κ B, which in turn activates the release of pro-inflammatory cytokines (Durbin et al., 2013; Iwasaki and Pillai, 2014) . In addition, inflammasome activation leads to the release of IL-1 β and IL-18 (Pothlichet et al., 2013;

Iwasaki and Pillai, 2014) . All these responses are supposed to promote viral clearance. However, the presence of viral proteins during infection induces also direct activation of the intrinsic or indirectly the activation of the extrinsic apoptotic pathway via production of inflammatory cytokines, resulting in apoptosis or even necrosis of the epithelium (Korteweg and Gu, 2008) . Furthermore, aberrant coagulation induced by virus infection causes a hyper-inflammatory response (Yang and Tang, 2016) . All these events contribute to lung tissue injury (Imai et al., 2008; Davidson et al., 2014) . The epithelial damage due to viral replication provides a beneficial environment for initial bacterial attachment (Plotkowski et al., 1993) . On the other hand, already colonized bacteria might enhance influenza virus virulence either by directly secreting proteases that cleave and activate HA (Figure 2) (Bottcher-Friebertshauser et al., 2013) or, indirectly, by activating host proteases such as plasminogen, which increases replication rates and infectivity of the virus (Scheiblaue et al., 1992; Tse and Whittaker, 2015) .

Potentially pathogenic bacteria, including the three species mentioned above, express an arsenal of virulence factors responsible for attachment to human host structures. Microbial surface components recognizing adhesive matrix molecules FIGURE 1 | Potential models to study bacterial and viral co-infections of the respiratory tract. *S. pneumoniae*, *S. aureus*, *S. pyogenes*, and *S. suis* are frequent colonizers of the upper respiratory tract. Seasonal IAV infection can lead to an increased risk of secondary bacterial infections, i.e., pneumonia. Several experimental models can be used for studying these severe infections. Patient samples, including ex vivo lung tissue are materials of choice, but they are rare due to ethical considerations. Tissue engineering approaches closely resemble the 3D architecture, cellular composition, and matrix complexity of the respective organ and were proven as useful tool to study infectious diseases. In vivo bacterial and viral co-infections are mainly performed in mice, which does not necessarily resemble the human physiology and immune system. Thus, we suggest using the porcine model, which nearly resembles over 80% of the human immune system.

(MSCRAMMs), such as PspC, PspA, and PsaA in pneumococci (Hammerschmidt, 2006) , SPA, FnbA, ClfA, and ClfB in *S. aureus* (Bartlett and Hulten, 2010; Otto, 2010) , and M-protein, PrtF1, and PrtF2 in GAS (Cunningham, 2000) , respectively, and so-called moonlighting proteins expressed by all three species, e.g., GAPDH, enolase or PGK (Fulde et al., 2013) , enable the bacteria to attach to damaged cells or molecules of the extracellular matrix, including fibronectin, fibrin, fibrinogen, and collagens, or fibrinolytic proteins like plasminogen (McCullers and Rehg, 2002; Bergmann and Hammerschmidt, 2007; Linke et

al., 2012; Siemens et al., 2012; Voss et al., 2012) . Once the initial attachment occurs, bacterial cytotoxins including pneumolysin of pneumococci (Garcia-Suarez Mdel et al., 2007; Zahlten et al., 2015) , α -hemolysin and leukocidins of *S. aureus* (Mairpady Shambat et al., 2015) , and Streptolysins S and O and Streptococcal pyrogenic exotoxin B of *S. pyogenes* (Tsai et al., 1998; Gurel et al., 2013; Siemens et al., 2015 Siemens et al., , 2016 , can synergize with viral counterparts to further increase lung tissue pathology. Additional potential mechanisms by which the initial colonization of the lower respiratory tract and lung tissue damage might occur include potentiation of the development of pneumonia by IAV neuraminidase through enzymatic removal of sialic acid from the lung, thus exposing host receptors for pneumococcal adherence (McCullers and Bartmess, 2003) . The host inflammatory state in response to viral infection can alter presentation of receptors on the surface, thus allowing bacterial invasion (Cundell and Tuomanen, 1994) . As the patient begins to recover from viral infection, secondary bacterial infections might occur (Louria et al., 1959) due to the incomplete wound healing and exposure of host membrane components, including laminin, collagens type I and IV to classical bacterial MSCRAMMs (Louria et al., 1959; Puchelle et al., 2006) .

Epithelial cells are the first responders to infections in the lung, followed by the tissue resident alveolar macrophages. They promote viral clearance via phagocytosis, efferocytosis, and release of cytokines and chemokines to promote immune responses (Hashimoto et al., 2007; Kumagai et al., 2007; Wang et al., 2012; Hillaire et al., 2013) . Respiratory viruses like IAV are able to induce suppression and killing of the resident alveolar macrophages (Figure 2) (Ghoneim et al., 2013) . These cells are usually replaced by differentiation of recruited blood derived monocytes into macrophages of different polarization patterns. This in turn creates a delay in pathogen clearance and opens a window for host susceptibility to secondary bacterial infections, colloquially named superinfections (Ghoneim et al., 2013) . In addition, induction of interferons as a response to viral infection compromises the immune sensing of Gram-positive bacteria by neutrophils and macrophages, which would normally clear the bacteria from the lungs (Figure 2) (Sun and Metzger, 2008; Tian et al., 2012) . The exact mechanism underlying this phenomenon is still not understood. Several studies suggested that viral RNA activates Toll-like receptors (TLR) 2 and TLR4 and, consequently, the production of type I interferons to promote an antiviral state (Shahangian et al., 2009) . The subsequent infection with Gram-positive bacteria, e.g., pneumococci, enhances the type I interferon expression, which in turn suppresses production of the CCL2 chemokine and recruitment of macrophages (Nakamura Frontiers in Microbiology | www.frontiersin.org FIGURE 2 | The interplay between IAV, bacteria, and the human host. The epithelial damage due to viral

replication provides a beneficial environment for bacterial (Bact.) attachment. IAV is able to induce suppression and killing of resident alveolar macrophages (AM), which in turn delays viral clearance. The release of viral RNA activates different immune response pathways resulting in cytokine storm. Type I and III interferons compromise the immune recognition of Gram-positive bacteria by neutrophils and macrophages. In addition, they might suppress natural killer cell function (NK), including release of TNF, which activates alveolar macrophages. After initial inflammation, the situation might worsen due to cellular infiltration of the lungs by neutrophils (PMN), leading to an increased degranulation and tissue damage by effector molecules, including heparin-binding protein (HBP). et al., 2011). Another study by Shahangian et al. (2009) revealed that the antiviral state leads to impaired production of neutrophil chemoattractants CXCL1 and CXCL2, which in turn promotes less effective immune responses due to attenuated neutrophil functions during the early phase of pneumococcal invasion. Other studies found that IAV exposed lungs had impaired natural killer (NK) cell responses in the airway to subsequent *S. aureus* infection (Small et al., 2010) . Reduced TNF α production by NK cells was identified as a crucial upstream mechanism of depressed antimicrobial activities by alveolar macrophages (Figure 2) (Small et al., 2010) . It seems likely that IAV NA is also able to activate host cell receptors in a TGF- β dependent manner, which in turn promotes GAS invasion and subsequent lung pathology (Li et al., 2015) . In vitro studies on the interplay between IAV-pneumococci and human dendritic cells revealed TLR3 as a crucial sensor of viral and bacterial RNA leading to enhanced IL-12p70 production, which in turn might promote an anti-viral state by upregulation of interferons (Yamamoto et al., 2004; Spelmink et al., 2016) . However, it should be noted that depending on the bacterial species the disease manifestation and underlying innate immune responses might vary (Sharma-Chawla et al., 2016) .

A lot of the experimental studies on disease mechanisms and immune responses are based on a subsequent bacterial infection within hours or a few days post IAV infection. However, bacterial infiltrations of the lungs might occur much later, i.e., during the onset of wound healing after partial clearance of IAV, which has been reported in most studies performed in recent years (Snelgrove et al., 2008; Hussell and Cavanagh, 2009) . These processes are characterized by a general anti-inflammatory state and suppression of mechanisms involved in pathogen clearance due to increased interleukin-10 production (van der Sluijs et al., 2004; Metzger and Sun, 2013) . The anti-inflammatory state suppresses the expression of pattern recognition receptors (PRR) on professional phagocytes leading to impaired phagocytosis and killing of microbes. These events might

allow bacterial overgrowth in the lungs and tissue pathology (Sun and Metzger, 2008; Goulding et al., 2011) .

Like other severe infectious diseases caused by single agents, pneumonia is characterized by hyper-inflammatory conditions of the lungs at the onset of infection followed by a hypoinflammatory state with immune paralysis (Morton et al., 2014) . In co-infections, after initial inflammation in response to viral infection the situation might worsen due to bacterial invasion and enhanced cellular infiltration of the lungs by neutrophils, leading to an increased tissue damage and cytokine storm (Figure 2) (Conenello et al., 2007; McAuley et al., 2007 McAuley et al., , 2010 Porto and Stein, 2016) . Furthermore, the coagulation system becomes activated and contributes to the pathophysiological response to infection (van der Poll and Herwald, 2014). Bacteria like pneumococci, *S. aureus*, and GAS can activate and modulate the coagulation system, leading to extensive expression of tissue factor and increasing the risk of severe coagulopathy Shannon et al., 2013; Walters et al., 2016) .

Bacterial pathogens also express a variety of cytolytic toxins that can contribute to inflammation and tissue pathology. Pneumolysin, a pneumococcal pore-forming toxin with low affinity to lung epithelial cells, can damage neutrophils by utilizing P2X7 receptor (Domon et al., 2016) . Staphylococcal cytotoxins (α -toxin and leukocidins, including Pantan-Valentine leucocidin, PVL) are associated with severe tissue pathology, strong upregulation of chemokines, and increased neutrophil influx of the lungs (Mairpady Shambat et al., 2015) . GAS toxins, including SLO and SpeB, are capable of directly causing tissue damage and promoting pro-inflammatory states through neutrophil lysis (Snall et al., 2016; Uhlmann et al., 2016) . The cytolytic effects caused by bacterial toxins might synergize with the outcome of IAV cytotoxic accessory protein, PB1-F2, mediated tissue pathology leading to enhanced cytokine production (Ramos and Fernandez-Sesma, 2012) . Taken together, most likely synergistic effects of the pathways that are involved in bacterial and viral inflammation lead to enhanced immune activation and higher morbidity and mortality (Joyce et al., 2009; Koppe et al., 2012; Ramos and Fernandez-Sesma, 2012; Bucasas et al., 2013; Kuri et al., 2013) . Figure 2 summarizes the interplay between virus, bacteria, and host.

Experimental animal models are a useful tool to study in vivo effects of different infectious agents and they represent approximately 3% of all pneumonia research published in

peerreview journals (Hraiech et al., 2015) . However, the constant increase of animal studies in the last decades is in contrast to their reproducibility in humans (Hackam and Redelmeier, 2006) . Hackam and colleagues identified 2,000 articles published between 1980 and 2006 in seven leading scientific journals that regularly publish animal studies (Hackam and Redelmeier, 2006) . Seventy-six out of 2,000 were highly cited with a median citation count of 889. Out of these 76 studies 28 were replicated in human randomized trials, 14 were contradicted, and 34 remained untested (Hackam and Redelmeier, 2006) . Only 1.4% of the animal studies published in high-impact journals were translated in human randomized trials (Hackam and Redelmeier, 2006) , whereas about 44% replication rate was reported for highly cited human studies (Ioannidis, 2005) . In pneumonia models, mammals are mostly used because of their anatomical and physiological proximity to humans (Hraiech et al., 2015) . To monitor extensive physiological studies, larger mammalian species, including ferrets, dogs, rabbits, pigs, and baboons are the models of choice (Mizgerd and Skerrett, 2008) . However, rodents and in particular mice are used more frequently as a pneumonia model organisms. Rapid reproductive rate, small size, less complicated handling, the ability to reproduce and compare results with already published bacterial and viral mono-infections, detailed knowledge of genetics and immune responses, and a plethora of available reagents to study infections in mice are reasons for the use of these animals. To avoid variations in responses due to genetic diversity inbred mice strains are useful tools for studies aiming to elucidate molecular mechanisms of diseases. In addition, genetic engineering allowed to generate a wide variety of mouse variants with gain-of-function, loss-of-function or reporter genes (Mizgerd and Skerrett, 2008) .

As outlined above, many in vivo mice studies on bacterial and viral co-infections provided useful insights into severe pneumonia, including (i) the fact that viral infection primes the host for bacterial susceptibility leading to severe secondary infection (Hashimoto et al., 2007; Shahangian et al., 2009; Chaussee et al., 2011; Nakamura et al., 2011) , (ii) pathogen synergism (Tsai et al., 1998; McCullers and Rehg, 2002; Garcia-Suarez Mdel et al., 2007; Gurel et al., 2013; Mairpady Shambat et al., 2015; Zahlten et al., 2015) , (iii) enhanced inflammatory response at the onset of infection (Korteweg and Gu, 2008; Durbin et al., 2013; Pothlichet et al., 2013; Iwasaki and Pillai, 2014) leading to increased alveolar damage followed by immune paralysis with defective clearance of microorganisms (Shinya et al., 2006; van Riel et al., 2007; van Riel et al., 2010) , and (iv) host receptor availability for sustained bacterial infection (Louria et al., 1959; Plotkowski et al., 1993; Cundell and Tuomanen, 1994; Puchelle et al., 2006; Korteweg and Gu, 2008) . However, mouse models for bacterial and/or viral infections have several limitations. Most of the bacterial and viral

species under study are human pathogens. In recent years it was also shown that host genetic variations and sex differences have an impact on predisposition, severity, and outcome of infection (Chella Krishnan et al., 2015). While C57BL/6 and BALB/c mice are characterized by a higher resistance, DBA/2 strains are more susceptible and permissive to bacterial and viral strains (Alymova et al., 2011; Chella Krishnan et al., 2015). In addition, transmission of IAV and bacteria is inefficient in adult mice, thus requiring alternative animal models, including neonatal mice or ferrets (Diavatopoulos et al., 2010; McCullers et al., 2010). IAV was shown to be essential for pneumococcal transmission from colonized mice to their naive littermates and the transmission occurred only when all mice were infected with IAV (Diavatopoulos et al., 2010). Ferrets are naturally susceptible to IAV isolated from different species, including humans, birds, and swine (Thangavel and Bouvier, 2014). The infection of ferrets with human seasonal IAV isolates results in an upper respiratory tract infection similar to human influenza infection (Tripp and Tompkins, 2009). In contrast to mice, non-adapted human IAV can be used for the infection. Unfortunately, there are only few reports on bacterial and IAV co-infections in this model organism. A report by Sanford and Ramsay showed enhanced staphylococcal colonization of the upper respiratory tract in IAV infected animals as compared to non-infected, while no difference between both groups was observed in group B streptococcal infection (Sanford and Ramsay, 1987). In contrast, Smith and McCullers reported lack of establishment of staphylococcal infection even when ferrets were pre-infected with IAV (Smith and McCullers, 2014). The biggest advantages of using ferrets as a model include (i) their susceptibility to nonadapted human pathogens, (ii) efficiency in transmitting IAV and bacteria from one individual to another, and (iii) presentation of the clinical signs of disease manifestation akin to human influenza infection. Unfortunately, their limited availability, complex husbandry, and limited accessibility to ferret-specific reagents makes this research difficult to perform (Bouvier and Lowen, 2010).

In recent years, the guinea pig (*Cavia porcellus*) was also used in pneumonia research. The physiology and anatomy of the guinea pig lung resembles to a certain extent the human lung and this model organism is often used in non-infectious lung diseases, including asthma and chronic obstructive pulmonary disease (Canning and Chou, 2008). In addition, its commercial availability, ease of husbandry, the ability to work with nonadapted pathogens and the efficiency of transmission are reasons for using this in vivo model (Bouvier and Lowen, 2010). Guinea pigs are susceptible to human, avian, and swine influenza viruses. Although viral replication can be readily detected upon intranasal inoculation in the upper respiratory tract and the lungs, guinea pigs exhibit only minor clinical symptoms (Lowen et al., 2006; Gabbard et al., 2014). However, the lung pathology

of human IAV infected guinea pigs correlates with the clinical severity of human infection (Gabbard et al., 2014) . Transmission of pneumococci in guinea pigs is promoted by co-infection with Sendai virus (Saito et al., 1988) . Guinea pigs infected with pneumococci alone and cage-mated with non-treated contact animals transmitted the bacteria only in 7% of cases, while Sendai-virus infected, co-housed guinea pigs acquired pneumococcal infection in 83% of contacts (Saito et al., 1988) . Another study evaluated antibiotic efficacy in invasive pulmonary infection caused by penicillin resistant pneumococcus (Ponte et al., 1996) . Intratracheal instillation of 3×10^9 CFU of *S. pneumoniae* induced a fatal pneumonia and bacteremia in 85% of untreated animals within 46 h (Ponte et al., 1996) . As with ferrets, there is a paucity of data describing immune responses to pulmonary infectious agents. This is in parts due to the lack of species specific reagents, which is a disadvantage in using this model organism.

Recently, the cotton rat (*Sigmodon hispidus*) was reported to be susceptible to IAV. Nasal and pulmonary infection in adult inbred cotton rats did not require viral adaptation (Ottolini et al., 2005) . The infection led to increased breathing rates accompanied by weight loss and decreased body temperature. Replication of IAV was more extensive in nasal tissues than the lung, and persisted for six consecutive days. Tissue pathology included damage of bronchiolar epithelium and the animals developed pneumonia which persisted for nearly 3 weeks (Ottolini et al., 2005) . In bacteriological studies rats are more frequently used. There are numerous rat models investigating the impact of diabetes (Oliveira et al., 2016) , metabolic syndromes (Feng et al., 2015) , cirrhosis , pharmaco-kinetics and dynamics (Antonopoulou et al., 2015; Hoover et al., 2015) , intoxication (Davis et al., 1991) , immunization (Iinuma and Okinaga, 1989) , and general bacterial virulence factors (Shanley et al., 1996) on development of pneumococcal, streptococcal, and staphylococcal pneumonia and lung pathology. Unfortunately, there are only few studies on bacterial and viral co-infections in rats. The first was performed by Harford et al., 1946 (Harford et al., 1946) . The authors concluded that the secondary bacterial pneumonia does not convert the sub-lethal viral infection to a lethal outcome (Harford et al., 1946) . Another study on human respiratory syncytial virus and *S. pneumoniae* revealed that rats were easily colonized with pneumococci, but viral replication after subsequent infection was strain dependent. In addition, neither pneumococci nor the virus spread from the upper to the lower respiratory tract, and neither pathogen was transmitted to naive cage mates . Although rats share a lot of immune features with humans, including nitric oxide production by macrophages (Carsillo et al., 2009) , the biggest disadvantages are low animal availability, aggressiveness of the species, and the lack of specific reagents.

Rabbits (*Oryctolagus cuniculus*) are well known for their use in studying cardiovascular diseases, antibody production, and eye research. Rabbits were also employed to study pneumonia, although only a few models are available. Typical read-out parameters include survival, leukocyte infiltration of the lungs, lung pathology, and assessment of drug concentration in serum. One of the first studies on pneumococcal pneumonia in rabbits was performed in Kline and Winternitz (1913) . This study revealed that rabbits possess an active immunity if they have recovered from one attack of experimental pneumonia and they may subsequently resist repeated intra-tracheal dosages of pneumococci (Kline and Winternitz, 1913) . In 1926 an infection by inhalation of Type I pneumococci was established in rabbits (Stillman and Branch, 1926) . The bacteria infiltrated easily the lower respiratory tract and pneumococci which reached the lungs usually disappeared within hours and fatal septicemia appeared in some of the animals (Stillman and Branch, 1926) . Most recent rabbit models of pneumococcal and staphylococcal pneumonia are based on intra-bronchial or intra-pulmonary infections which make them useful for pathogenesis (Diep et al., 2010 (Diep et al., , 2017 , as well as drug efficiency and efficacy studies (Cabellos et al., 1992; Croisier-Bertin et al., 2011) . However, this infection route requires surgery and species-specific reagents are scarce. In IAV research rabbits are frequently used for antibody production and for studies on antibody kinetics following single or multiple IAV administrations (Loza-Tulimowska et al., 1977) . Also, rabbits are used for safety investigations of vaccines (e.g., CoVaccine HT or Aflunov) (Heldens et al., 2010; Gasparini et al., 2012) . In recent years the shedding of avian IAV by cottontails (*Sylvilagus* spp.) was investigated revealing that nasally and orally inoculated cottontails shed relatively large quantities of viral RNA (Root et al., 2014) . Notably, low viral titers were found to be sufficient to initiate viral replication in cottontails (Root et al., 2017) . However, despite their susceptibility to IAV infection, rabbits are only rarely used as model for IAV pathogenesis since they offer no improvement over other established infection models.

Macaques represent the major non-human primate for studying infectious diseases. They are omnivorous and adaptable. The species most commonly used are rhesus macaques (*Macaca mulatta*) and cynomolgus macaques (*Macaca fascicularis*).

Although it was shown early that macaques were susceptible to IAV (Saslaw et al., 1946) , the animal models of choice remained ferrets and mice. Recently, macaques have been used to compare the pathogenesis of highly virulent 1918 pandemic IAV and the

pathogenic bird flu strain (H5N1) with a conventional H1N1 strain (Rimmelzwaan et al., 2001) . Cynomolgus macaques infected with highly pathogenic H5N1 developed acute respiratory distress syndrome, fever, and necrotizing pneumonia (Rimmelzwaan et al., 2001) . The 1918 IAV strain induced dysregulation of the antiviral response leading to insufficient protection of the host, which in turn resulted in acute respiratory distress and a fatal outcome (Kobasa et al., 2007) . The 2009 pandemic H1N1 US isolate caused severe pathological lesions in the lungs of the macaques (Itoh et al., 2009) . The three studies mentioned above used combined intratracheal delivery of high doses of virus. A recent study by Marriott et al. analyzed the outcome of challenge routes, including inhaled aerosol and intra-nasal instillation with low to moderate doses of H1N1 in cynomolgus macaques (Marriott et al., 2016) . Virus replication was detected in all challenge groups, although the disease remained sub-clinical.

In bacteriological studies non-human primates are rarely used. For group A streptococcal infection longitudinal transcriptome analyses were performed in experimental pharyngitis (Virtaneva et al., 2005) and lower respiratory tract infection in cynomolgus macaques (Olsen et al., 2010a) . The lower respiratory tract disease observed in macaques after GAS infection mimicked the clinical and pathological features of severe bronchopneumonia in humans (Olsen et al., 2010a) . Another study by Olsen and colleagues analyzed the contribution of PVL of a highly virulent USA300 *S. aureus* strain in respiratory infection (Olsen et al., 2010b) . Although the lower respiratory tract disease observed in monkey mimicked the clinical and pathological features of early mild to moderate pneumonia in humans, no involvement of PVL in lung pathology or immune cell influx of the lungs could be detected (Olsen et al., 2010b) . The same research group has developed a non-lethal IAV (H3N2)-*S. aureus* co-infection model in cynomolgus macaques (Kobayashi et al., 2013) . Pneumonia progression was monitored by clinical parameters assessment, blood chemistry, nasal swabs, and pathology of the lungs. Seasonal IAV infection in healthy cynomolgus macaques caused mild pneumonia, but did not predispose the animals to subsequent severe infection with the USA300 clone (Kobayashi et al., 2013) .

Although macaques are frequently used for evaluation of pneumococcal vaccine efficacy, including testing the impact of 13-valent pneumococcal conjugate vaccine and 23-valent pneumococcal polysaccharide vaccine on antigen-specific memory B cell repertoires (Jia et al., 2017) , only two studies on pneumococcal carriage and pneumonia were conducted in the last decade. In 2013, Philipp and colleagues analyzed the carriage rate of pneumococcus in 158 colony animals. None of the surveyed rhesus macaques carried *S.*

pneumoniae in the nasopharynx (Philipp et al., 2012) . The authors concluded that rhesus macaque is probably not a natural host of pneumococci. But, when infants were colonized with 19F strain via nasopharyngeal instillation, the colonization was induced in eight of eight infants, lasted for 2 weeks in all animals and for 7 weeks in more than 60% (Philipp et al., 2012) . The same group tested detoxified pneumolysin (dPly) and pneumococcal histidine triad protein D (PhtD) as potential vaccine candidates to prevent pneumonia (Denoel et al., 2011) . After immunization the rhesus macaques were challenged with a 19F pneumococcal strain. AS02-adjuvanted PhtD-dPly vaccine protected the animals against *S. pneumoniae*-induced pneumonia, which was linked to the capacity (i) to greatly reduce bacterial load within the first week post-challenge and (ii) the levels of PhtD-and Ply-specific antibodies (Denoel et al., 2011) . Although only a few macaque studies on pneumonia exist, due to the close proximity to humans in terms of physiology and immunity, these animals can be a good model in the context of translational studies evaluating therapeutics and prophylaxis.

Despite the wide use of different animal models, the optimal in vivo model for human pneumonia remains to be identified. Small mammals including rodents are well known from a biological, genetic, and immunological point of view and are easy to maintain. The choice of these particular animals for infectious disease studies is often a result of a compromise between technical and financial options. However, they are also far from humans' anatomy, physiology, immunology, and susceptibility to exclusively human pathogens. The experimental animal model should be chosen based on responses comparable to humans. Primates are usually legally reserved to specific topics. In this case, pigs could be an appropriate model system for studying infectious diseases including pneumonia (Figure 1) . The composition and size of the porcine genome is comparable to that of humans (Hart et al., 2007) . In addition, human and porcine organs have many common features and functions (Swindle et al., 2012) . The upper respiratory tract of humans and pigs, including the lymphoid tissue in the nasopharynx, is anatomically similar. Furthermore, like humans, pigs possess tonsils, which are absent in mice (Horter et al., 2003) . A major advantage of studying infectious diseases by utilizing pigs as a host organism is that pigs have a full set of innate and adaptive immune effectors. According to whole genome sequencing results the porcine immune system resembles over 80% of the human immune system, whereas mice share less than 10% with humans (Dawson et al., 2016) . Most of the immune cell compartments identified in humans are also present in pigs (Piriou-Guzylack and Salmon, 2008; Fairbairn et al., 2011) . In contrast to mice and similar to humans, pigs have 50-70% of circulating polymorph nuclear cells (Fairbairn et al., 2011) . In addition, all functional cytokines or orthologs involved in Th1, Th2, Th17, and

Treg paradigm and corresponding immune cells have been described in pigs (Murtaugh et al., 2009; Kaser et al., 2011; Kiros et al., 2011) . Especially the very prominent human pro-inflammatory chemo-attractant, CXCL8, is present as an ortholog in pigs, whereas there is no homologue in mice (Fairbairn et al., 2011) . In contrast to human monocytes, which can be divided in three subclasses (classical CD14 + CD16 – , nonclassical CD14 + CD16 + , and intermediate CD14 ++ CD16 +), porcine monocytes consist of four subclasses (Chamorro et al., 2005; Fairbairn et al., 2013) . Like human monocytes they express adhesion molecules, such as VLA-4 and LFA-1 and costimulatory molecules, including CD80 and CD86 (Chamorro et al., 2005) .

The pig has previously been used to mimic a number of human infectious diseases. Examples for *S. aureus* infections with this model organism are wound infections Svedman et al., 1989) , osteomyelitis (Jensen et al., 2010) , and sepsis (Nielsen et al., 2009) . Intravenous inoculation of piglets with pneumococci led to bacteremia during a 5 days period and was associated with fever and septic arthritis. Intranasal inoculation of piglets led to colonization for at least six consecutive days without causing clinical signs (De Greeff et al., 2016) . In addition, research on respiratory infections of pigs by human pathogens including *S. aureus* (Luna et al., 2009) , *Mycobacterium tuberculosis* (Gil et al., 2010) , *Bordetella pertussis* (Elahi et al., 2007) , *Pseudomonas aeruginosa* (Luna et al., 2009) , and IAV (Khatri et al., 2010) , was performed in recent years. The fact that pigs and humans are infected with identical subtypes of IAV (H1N1, H3N2), and show similar clinical presentation and pathogenesis, makes pigs an ideal model organism for studies on respiratory co-infections (Van Reeth et al., 1998) . Especially IAV infections are already well established in swine (Van Reeth et al., 1998 , 2002a Jung et al., 2007; Khatri et al., 2010; Barbe et al., 2011) .

In addition to the limited number of publications on pigs and human pathogens, a lot can be translated and learned from studies on the porcine zoonotic pathogen *Streptococcus suis*. *S. suis* usually inhabits mucosal surfaces of tonsils, nares, genital and alimentary tract of piglets. Once the microbial balance is disturbed, the bacteria can cause meningitis, septicemia, arthritis, and pneumonia in pigs (Staats et al., 1997) . Some *S. suis* strains are considered to be hyper-virulent and others hypo-or avirulent. In general, serotype 2 is most frequently isolated from diseased pigs (Staats et al., 1997) . *S. suis* can also cause severe diseases in humans including septicemia, meningitis, arthritis, and streptococcal toxic shock syndrome (Tang et al., 2006; Yu et al., 2006; Gottschalk et al., 2007) . Although many in vivo studies on *S. suis* have been performed by utilizing mice as a

model organism (Seitz et al., 2012; Auger et al., 2016) , several other studies have shown the advantage of using swine as a natural host for *S. suis* (Bi et al., 2014; Ferrando et al., 2015) . A recent publication by Lin and colleagues on H1N1 and *S. suis* co-infected piglets demonstrated the synergistic effects of both pathogens (Lin et al., 2015) . Co-infected piglets had more severe clinical presentation and pathological changes in the lung, as compared to animals infected with single pathogens (Lin et al., 2015) . In addition, genes associated with immune responses, inflammatory cytokine production, and apoptotic pathways were highly overexpressed in the coinfecting group (Lin et al., 2015) . Although the porcine model seems to be ideal to mimic human infectious diseases, there are also disadvantages, including, e.g., requirement for specialized experimental animal facilities, time consuming management, high maintenance costs, and limited availability of transgenic animals.

Although the use of animals contributes greatly to our understanding of infectious diseases, human 3D-organotypic tissue models and ex vivo organ tissues should be considered, as they are most valuable tools to study host-pathogen interactions in a more complex setting (Figure 1) . Tissue engineering approaches were originally focused on regenerative medicine (Langer and Vacanti, 1993) . In contrast to standard monolayer cell cultures, tissue models much more closely resemble the 3D architecture, cellular composition, and matrix complexity of the respective organ. In recent years tissue engineering was also successfully employed in a number of studies in infectious diseases, including Zika virus infections of cerebral organoids (Lancaster et al., 2013; Dang et al., 2016) , *Helicobacter pylori* infections of gastric epithelial organoids (McCracken et al., 2014; Schlaermann et al., 2016) , *Escherichia coli* and Rotavirus infections of gastrointestinal and small intestinal enteroids (Saxena et al., 2015; VanDussen et al., 2015) , *Entamoeba histolytica* or Hepatitis B virus infections of hepatic sinusoid tissue (Petropolis et al., 2014 (Petropolis et al., , 2016 , group A and G streptococcal or staphylococcal infections of skin tissue models Mairpady Shambat et al., 2016) , and staphylococcal and Andes hantavirus infections of human lung tissue (Mairpady Shambat et al., 2015; Sundstrom et al., 2016) . The adaptability of these tissue-engineered models to multiple pathogens suggests a great potential for studies of infectious diseases. For instance, the lung tissue model relevant for pneumonia consists of lung fibroblasts embedded in a collagen matrix with a stratified epithelial layer on top (Nguyen Hoang et al., 2012) . The engineered tissue is suitable for implanting and studying immune cells, including dendritic cells, monocytes, macrophages, and even peripheral blood mononuclear cells (Nguyen Hoang et al., 2012; Mairpady Shambat et al., 2015) . A recent publication demonstrated a two-hit-event of lung pathology in staphylococcal necrotizing pneumonia (Mairpady

Shambat et al., 2015) . While the α -toxin had direct damaging effect on the lung epithelium, PVL induced lung pathology indirectly through the lysis of neutrophils (Mairpady Shambat et al., 2015) . All the studies mentioned above highlight a significant progress in the field of infectious diseases not only from a scientific point of view but also by contributing to the three R principle of animal experimentation (Russell, 1995) .

On these terms, the use of cultured ex vivo human organ biopsies, which are rare due to ethical considerations, is an additional option to study host-pathogen interactions. This ex vivo system may overcome even the limitations of the engineered tissue. In recent years human ex vivo lung tissue infections with various microorganisms, including pneumococci (Szymanski et al., 2012; Fatykhova et al., 2015) , *Bacillus anthracis* (Chakrabarty et al., 2007) , *Haemophilus influenzae* (Zhang et al., 2016) , and IAV (Nicholls et al., 2007; Chan et al., 2009) , were performed. In the human setting, most of the work focused on tropism, severity of infections, release of inflammatory mediators, and replication rates of the microorganisms. In addition, recently also experiments on swine influenza virus (SIV) and *S. suis* co-infections of the porcine ex vivo lung slices were reported. Meng and colleagues showed that SIV promotes subsequent bacterial infections in a two-step process of which the first initial step was dependent on capsule expression, whereas the second step of bacterial invasion into deeper layers was capsule-independent and required virus-mediated damage (Meng et al., 2015) . However, this is just a beginning and more investigations are needed to unravel the complexity underlying these highly invasive infections.

In summary, bacterial and viral co-infections of the respiratory tract are highly lethal and present a dramatic burden for the global health system. The synergy between bacterial and viral infectious agents is related to a variety of factors, including epithelial barrier damage, exaggerated innate immune response, and cytokine storm. Despite many advances in recent years, more knowledge on mechanisms and immunology of disease progression is needed. The synergistic mechanisms between viruses and bacteria leading to enhanced morbidity and mortality are poorly understood. In vivo characterizations of these severe infections are mainly performed in mice which poorly resemble the human physiology and immune system. Several efforts have been made to establish other models, including ferrets, guinea pigs, rabbits, rats, and non-human primates. However, all have limitations. Here, we suggest using the porcine model, which provides obvious advantages in studies of human infectious diseases and should be considered much more frequent for future studies on severe infectious diseases, including pneumonia.

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Architectural Insight into Inovirus-Associated Vectors (IAVs) and Development of IAV-Based Vaccines Inducing Humoral and Cellular Responses: Implications in HIV-1 Vaccines

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Abstract: Inovirus-associated vectors (IAVs) are engineered, non-lytic, filamentous bacteriophages that are assembled primarily from thousands of copies of the major coat protein gp8 and just five copies of each of the four minor coat proteins gp3, gp6, gp7 and gp9. Inovirus display studies have shown that the architecture of inoviruses makes all coat proteins of the inoviral particle accessible to the outside. This particular feature of IAVs allows foreign antigenic peptides to be displayed on the outer surface of the virion fused to its coat proteins and for more than two decades has been exploited in many applications including antibody or peptide display libraries, drug design, and vaccine development against infectious and non-infectious diseases. As vaccine carriers, IAVs have been shown to elicit both a cellular and humoral response against various pathogens through the display of antibody epitopes on their coat proteins. Despite their high immunogenicity, the goal of developing an effective vaccine against HIV-1 has not yet materialized. One possible limitation of previous efforts was the use of broadly neutralizing antibodies, which exhibited autoreactivity properties. In the past five years, however, new, more potent broadly neutralizing antibodies that do not exhibit autoreactivity properties have been

isolated from HIV-1 infected individuals, suggesting that vaccination strategies aimed at producing such broadly neutralizing antibodies may confer protection against infection. The utilization of these new, broadly neutralizing antibodies in combination with the architectural traits of IAVs have driven the current developments in the design of an inovirus-based vaccine against HIV-1. This article reviews the applications of IAVs in vaccine development, with particular emphasis on the design of inoviral-based vaccines against HIV-1.

Text: Filamentous bacterial viruses are a group of thread-like viruses containing single-stranded DNA genomes. Collectively, they constitute the genus Inovirus in the family Inoviridae, the terms deriving from the Greek word 'Iva for filament [1] [2] [3] , and they are commonly called filamentous bacteriophages. There are over 50 different known individual species of filamentous viruses; the majority of them capable of infecting Gram-negative bacteria. The complex interaction between filamentous phages and their bacterial hosts is specified by receptor organelles that are usually encoded by transmissible plasmids [1, 4] . One of the most intriguing features of inoviruses is that they are assembled at the host membrane, where the major capsid protein subunits replace the single-stranded DNA binding protein, and progeny virions are continuously extruded into the medium without killing the infected cell, giving rise to titers of up to 10^{13} virions per milliliter of liquid culture [5, 6] . The high virus production is associated only with a mild retardation of the host's growth, which gives rise to the formation of opaque plaques on bacterial lawns. In this sense, filamentous viruses bear a resemblance to symbiotic non-pathogenic animal viruses rather than phages, the term coming from the Greek word φάγος for destroyer. Inovirus virions are flexible and slender cylindrical filaments [2, 3] less than 10 nm in diameter and in the order of 1000 nm in length (see details in Figure 1). Each virion has several thousand identical major capsid or coat protein subunits packaging a circular single-stranded DNA molecule. Each virion also has a few specific minor proteins at each end, those at one end (proximal end) for attachment during infection, and those at the other end (distal end) for nucleation and initiation of the assembly process at the host's membrane.

The number of species of inovirus isolated and characterized in different parts of the world is rather large [1] . Among *E. coli* inoviruses, the best-studied and most-exploited is Ff, a group of closely related viruses (fd, f1 and M13) (for review see [4, 6]) that infect male (F +) strains of *E. coli*. All Ff have almost identical DNA and protein sequences, gene organization, and most other structural parameters [7] [8] [9] . Fd, M13 and f1 differ in their

genomes at only about 100 positions out of 6408 nucleotides for f1 and fd or 6407 nucleotides for M13. The genetics and life cycle of three viruses f1, fd and M13 have been studied extensively and we know a great deal about them. The Ff genome contains ten tightly arranged genes and a non-coding intergenic region, which contains the packaging signal [10] [11] [12], the (+) and (–) origins of DNA replication, and the major rho-dependent transcriptional terminator (for recent review see [4]). Five of the ten viral genes encode proteins found in the virion (g3, g6, g8, g7 and g9). Genes 3 and 6 are found on the proximal end of the virion and are essential for infectivity and stabilization, whereas, gene 7 and gene 9 proteins, gp7 and gp9, respectively, are located at the distal end of the virion and are responsible for the initiation assembly [13] [14] [15]. In the end-to-end model illustrated in Figure 1, both the proximal (gp7 and gp9) and distal (gp3 and gp6) minor coat protein subunits maintain the fivefold axial symmetry of the major coat protein gp8 subunits along the virion.

The life cycle of Ff filamentous viruses starts with the adsorption of the virus to the tip of the F⁺ specific pilus of *E. coli*. Attachment takes place by means of an adsorption structure composed of five copies of gp3, located at the proximal end of the virion (see Figure 1), mainly through sequential binding of the gp3 N2 domain with the tip of the F pilus and the N1 domain with the periplasmic domain III of TolA (for recent review see reference 4). After adsorption, the virus is drawn to the surface of the cell where the major coat protein gp8 subunits become associated with the inner membrane of the cell [16] [17] [18] and the circular single-stranded DNA (cssDNA) is released into the cytoplasm. Inside the cytoplasm, the cssDNA is converted into a parental double-stranded replicative form (RF). Ff inoviruses replicate their genome by a rolling-circle mechanism. Consecutive transcription from the RF DNA, and translation as well as an asymmetric single strand DNA synthesis lead to an intracellular pool of viral precursor complexes, which contain viral single-stranded DNA molecules bound with gp5 protein subunits, except a small hairpin loop that serves as the packaging signal for virion assembly [10] [11] [12]. The major coat protein gp8 subunits are synthesized with a signal peptide sequence that facilitates their transport to and insertion into the bacterial membrane, where they are cleaved by signal peptidase. After cleavage, the mature coat protein is left spanning the membrane with its C terminus in the cytoplasm and the N terminus outside the cytoplasm in the periplasm [19] [20] [21]. The assembly of filamentous viruses takes place at the membrane where the gp5 subunits are replaced by gp8 subunits. The first step of the assembly sequence is the binding of the packaging signal, a site of about 30 nucleotides that forms a hairpin loop, with presumably five subunits of each of the minor coat proteins gp7 and gp9 [22] [23] [24] [25] [26]. Virus assembly proceeds as single-stranded DNA passes through the membrane

and more mature coat protein gp8 subunits are added until the entire DNA molecule is packaged. At the distal end of the virion, five copies of each of gp6 and gp3 are added and the complete virion is released into the medium [23, 27] . The assembly of inoviruses on the bacterial inner membrane is a harmonized sequential process that involves a variety of interactions between viral-encoded proteins (gp1, gp4 and gp11) and host-encoded proteins, without killing the host bacterium (reviewed by [4]).

The structure of the Ff virus has been extensively studied by a number of laboratories in the last five decades. However, despite all of the efforts, the structure has not been completely determined and some critical questions remain unanswered. The major difficulty is that these viruses cannot be crystallized. They can be oriented in fibers suitable for X-ray fiber diffraction studies, but these are not crystals. Interpretations of the fiber diffraction patterns together with a number of physicochemical measurements, have shown that the major coat protein gp8 subunits have a five-start helical symmetry (5-fold rotation axis) and are referred to as Class I [28] [29] [30] strictly based on the fundamental symmetry of the protein subunits helices as determined by fiber diffraction (reviewed by [2]). On the other hand, the structure of the packaged ssDNA molecule in the virion, including its helical symmetry and the interactions with the protein sheath, is one of the least understood aspects. The structure of the DNA inside these viruses cannot be determined by conventional X-ray fiber diffraction techniques, partly because of the low DNA content in the virions. Theoretical studies have demonstrated that the single-stranded DNA molecule is uniquely packaged inside the Ff virion by predominant electrostatic interactions [3, 31] .

A 3D scale schematic model of an end-to-end Ff (fd) inoviral virion. The model is based on published physical data including the determined helical parameters of the major coat protein gp8 and the X-ray structure of the N1-N2 domains of the minor coat protein gp3. The model shows the relative location of the circular single-stranded DNA (cssDNA) genome (6408 nucleotides long, illustrated as blue ribbons), some structural details of the outer virion capsid (major coat protein gp8) and the four minor coat proteins (gp6, gp3, gp7 and gp9) present at the ends of the virion. On top, a digital scanning transmission electron micrograph (STEM) of unstained fd virus, prepared by the wet-film technique according to previously established procedures of the Brookhaven STEM facility [32] [33] [34] . The ends of one complete virion are designated by arrows. The data were collected in collaboration with L.A. Day and J. S. Wall at the Brookhaven National Laboratory, Upton New York. Under these STEM conditions fd virions are about 8800 Å long and about 65 Å in diameter [3] . In the middle, a proposed end-to-end scale 3D diagram of fd virion is presented. The entire fd

virion is composed of about 2700 subunits of gp8 with the exception of its two ends. Architectural details of an axial slab 176 Å long (about 1/50 of the virion length) consisting entirely of subunits of major coat protein gp8. The structure of the 50-amino-acid-long and extended α -helical gp8, shown below in both surface and ribbon images, is presented in the virion model as a cylindrical stack of 25 gray disks about 70 Å long and 10 Å in diameter. The images of gp8 were derived from coordinates of RCSB PDB database accession number 2cOW [35] using PyMOL [36]. The gp8 subunits are arranged with a helical symmetry that includes a two-fold screw axis and a five-fold rotation axis, consisting of two pentamers of pg8 [28] [29] [30] [35]. The two pentamers are architecturally related to each other by a translation of about 16 Å along the virion axis and a rotation of 36° about the axis [28]. The proximal end of the virion, shown on the left, is composed of five copies of each of the minor coat proteins gp6 and gp3 (for a recent review see [4]). The proximal end is modeled based on partial information known about the structures of gp6 and gp3. Specifically, the N-terminal portion of gp6 was modeled following the helical parameters of gp8, based on protein sequence homology between the two [23] [24] [25]. Five copies of gp3 subunits were modeled based on structural information of the N1-N2 domains. The images of the N1-N2 domains of gp3 are shown below and were derived from coordinates of RCSB PDB database accession number 1g3p [36] using PyMOL. The domain organization of gp3 is also shown. The distal end of the virion (right) consists of five copies of each minor coat proteins gp7 and gp9, modeled following the helical parameters of gp8 according to a previously published model [25].

The easy genetic manipulation of inoviruses and the possibility of inserting random oligonucleotides into their genome set the foundation for inovirus display (phage display) technology [37, 38]. Expression of these genetically modified inoviruses results in the presentation of oligopeptides as fusion proteins on the surface of the virion and are herein termed IAVs for inovirus-associated vectors. IAVs can be modified to express an oligopeptide on either all or on some copies of a particular capsid protein. One possibility is to insert an oligonucleotide sequence of interest in the viral genome to create a fusion with capsid protein gp3, gp7, gp8 or gp9, so that the oligopeptide is displayed on every copy of the capsid protein. Alternatively, a phagemid vector can be used, which carries an extra copy of a capsid protein to which the oligonucleotide is fused. Coinfection of bacterial hosts with the phagemid vector and a replication deficient helper phage, that carries the wild type capsid protein, would result in mosaic inovirus particles. That is, they will contain copies of both the wild type coat protein and the recombinant protein that contains the oligopeptide of interest [39]. Non-mosaic and mosaic IAVs that display a peptide on gp3 or gp8 have been recently reviewed [4] while IAVs that display a peptide on

gp7 or gp9 have been reviewed elsewhere [40] [41] [42] . In contrast to the other four-capsid proteins, gp6 capsid protein has only been utilized for the production of mosaic virions [43] [44] [45] . Figure 2 illustrates the display of an antigen on each of the five capsid proteins of an IAV as indicated in published literature. It also introduces a new terminology to denote the gene to which the oligopeptides are fused to and whether the virion is a mosaic. To display many copies of an oligopeptide on an IAV, the ideal capsid protein to utilize is gp8. The resulting non-mosaic IAV can display a peptide on each of the approximately 2700 copies of gp8 on its capsid surface. The tradeoff, however, is a significant limitation in the size of the peptide: only peptides up to 6 amino acids may be displayed without distorting the assembly of the virus (see Figures 2 and 3). This size restriction of the displayed peptide may be overcome by generating a mosaic IAV that displays the foreign peptide in only a minority of gp8 on the viral surface [38] . With regards to the display of peptides on gp3, it is possible through mosaic IAVs to present even a whole protein on the viral surface [46] . Although in such a case the protein is expected to be present in up to five copies per virion, studies show that virions carry none, or just one copy of the protein on their surface [39] .

Random Peptide Libraries (RPL) is one common application of IAVs, where random oligopeptides are displayed on different clones of an inovirus particle. The vast diversity of an RPL depends on the size of the oligopeptide where the complexity of an RPL increases exponentially as the size of the oligopeptide increases. RPLs can subsequently be used in many applications including the identification of peptide ligands by receptors, the mapping of substrate sites for enzymes, and the creation of antibody peptide libraries. These applications are reviewed elsewhere [38, 46, 47] . Inovirus display technology has also been used for epitope mapping and vaccine design purposes. RPLs can be used to characterize the epitope of an antibody of interest through biopanning with a monoclonal antibody of interest, which can result in the isolation of mimotopes; these are oligopeptides that mimic the native antibody epitope. The selected recombinant inoviruses that carry mimotopes can then be isolated in order to determine the DNA and amino acid sequence of the displayed oligopeptides. DNA sequencing of such inserts as well as structure prediction analysis can potentially identify the previously unknown target of an antibody. Besides antibody characterization, inoculation of recombinant inoviruses isolated through this approach can potentially be used as vaccine carriers. For example, if a neutralizing antibody against a pathogen is used to screen an RPL, the selected peptides (fused to inoviruses) would mimic the original antibody epitope. Vaccination of animals with these inoviruses could ultimately induce the production of similar antibodies by the vaccinated individual, offering protection from infection against the pathogen. Inovirus

display technology has been successfully applied in the development of vaccines against various pathogens (Tables 1 and 2). The potential for inovirus display technology to facilitate the mapping of antibody epitopes is of great importance, especially in the case of HIV-1. Epitopes of broadly neutralizing monoclonal anti-HIV-1 antibodies could be rare, vulnerable spots on the surface of a frequently mutating virus such as HIV-1 and therefore, their identification and further study could lead to new drug therapies or vaccine targets. This review focuses particularly on the applications of inovirus display technology that utilizes capsid proteins gp3 and gp8, as those have been used in vaccine development.

Schematic representations of antigen display on the surface of Ff inovirus-associated vectors (IAVs). Foreign antigens are shown as red spheres. The designation on the left denotes the inoviral gene, which can be genetically modified to express an antigen on the outer architecture of the virion. IAVs that contain both the wild type and antigen display capsid proteins are designated by "m" which indicates that the virion is a mosaic. Each Ff virion contains about 2700 copies of major capsid protein gp8, and five copies of each of the minor capsid proteins, gp3, gp6, gp7 and gp9. inovirus-associated vector (bottom) showing the major coat protein gp8 subunits arranged with a combined five-fold rotation axis and an approximate two-fold screw axis [28] . Right, the corresponding surface lattices, identical to those previously published [30] . The lattice diagrams show the relative position of each gp8 subunit on the outer virion surface. The five gp8 subunits of each of the two interlocking pentamers constituting the helical symmetry of the virion are indicated by blue and green dots respectively. The relative virion surface area (about $1400 \text{ \AA}^2$) associated with each gp8 subunit is marked in yellow. The virion perimetrical (azimuthal) distance is calculated based on a virion diameter of about $65 \text{ \AA}$. The displayed antigens, represented by red spheres, are arranged on the surface of the Ff.g8 inovirus-associated vector according to helical symmetry of the virion outer architecture (bottom).

IAVs are effective vaccine carriers and, as shown in Tables 1 and 2 , they have been used successfully in numerous vaccine development studies. They have been utilized in the development of vaccines against a wide variety of viral, protozoan and worm parasites and also against non-infectious diseases like Alzheimer and various types of cancer. All these attempts can be divided in two main sub-categories. The first sub-category, inovirus display technology was used to screen RPLs with monoclonal antibodies and then to select the immunogenic peptides of interest. The selected peptide was used as a vaccine in its soluble form or in conjugation with carrier proteins [55,66,70,81-84, 86-88,90-92,98] . In the second sub-category, similar to the first sub-category, inovirus display technology

has been used for epitope mapping and isolation, but in this case, inoviruses were also used as the vaccine carriers for the immunogenic peptides [48] [49] [50] [51] [52] [53] [54] [56] [57] [59] [60] [61] [62] [63] [64] [65] [67] [68] [69] [71] [72] [73] [74] [75] [76] [77] [78] [79] [80] . In contrast to IAVs, soluble peptides have the disadvantage of being less stable than the same peptides fused to inoviral particles. Soluble peptides have a flexible 3D structure and thus, do not always retain the 3D structure of the desirable epitope. As a result, soluble peptides, unlike inovirus-bound peptides, are less capable of inducing the production of the desirable antibodies [99] . Additionally, the inoviral vectors displaying peptides are highly immunogenic. Their high immunogenicity is reinforced by the ability of inoviruses to display multiple copies of peptides on their surface. Additionally, inoviruses are known for their structural simplicity, which allows the immune system to focus selectively on the displayed peptides and not on the viral carrier [100] . Furthermore, since inoviruses can replicate in *E. coli* cultures, the cost of vaccine production is low. In summary, IAVs can be used as efficient and cost effective vaccine carriers.

A large number of research studies have focused on the application of inovirus-based vaccines against infectious diseases. A common approach in many of these efforts was to vaccinate animals with inoviruses and to then challenge them with specific pathogens, in order to assess the level of protection against the pathogens. Three of these studies focused on immunization against viral parasites. In 1997, Bastien et al. fused a 15-mer linear epitope of Human Respiratory Syncytial Virus (RSV) glycoprotein G on inovirus gp3 and used the recombinant inovirus to vaccinate mice [69] . This resulted in a specific humoral response, with the vaccinated animals having complete protection from challenge with the RSV virus [69] . In 2000, Grabowska et al. used monoclonal antibodies against Herpes Simplex Virus type 2 (HSV-2) glycoprotein G2 to screen 15-mer RPLs [65] . The selected recombinant inoviruses were used to vaccinate mice, resulting in a specific humoral response. A high survival rate of vaccinated mice after challenge with a lethal dose of the virus was observed, and the level of protection was proportional to the dose of the inoviral vaccine [65] . Additionally, in 2000, Yu et al. used monoclonal antibodies against the surface glycoprotein of Neurotropic Murine Coronavirus to screen various RPLs [67] . A selected clone displaying a 13-mer peptide induced a humoral immune response but no cellular response in mice. Even so, after lethal virus challenge, three out of six mice survived. [67] . Besides viral infections, inoviral vaccine research has also been applied for systemic candidiasis, a fungal infection caused by *Candida albicans*. In two separate studies, a specific six amino acid peptide epitope of the fungal heat shock protein 90 was displayed as a fusion with the inoviral coat protein gp8 [73, 74] . After infection with the parasite, mice immunized with the recombinant inoviruses had fewer colony forming units

of *Candida albicans* in the kidney and a longer lifespan [73] . The use of inovirus as a vaccine carrier was particularly successful against *Taenia solium*, a parasitic worm that uses pigs as intermediate hosts and causes neurocysticercosis, a parasitic disease of the central nervous system, in humans [79] . In 2004, Manoutsarian et al. fused four antigenic peptides (GK1, KETc1, KETc7, KETc12) to inoviruses and the cocktail of recombinant inoviruses was used to vaccinate pigs; as a result, a specific cellular response was induced. Vaccination of pigs protected them against challenge with the pathogen: 1/3 of pigs were totally protected and 2/3 had reduced number of cysticerci [79] . Based on these results, large-scale vaccination of 1047 rural pigs in 16 villages of central Mexico was conducted in 2008. The immunization was successful since the vaccine conferred significant protection against the parasite. Furthermore, this inovirus-based vaccine was more economical compared with to another vaccine made of synthetic peptides. The study was particularly important, not only due to the large scale vaccination attempt with inoviruses, but also due to its effectiveness in significantly reducing the number of cysticerci in the vaccinated animals [101] . Efforts were also made to develop vaccines against the worm *Schistosoma japonicum*. In 2004, Tang et al. screened a 12-mer RPL with polyclonal serum from infected mice [76] . The selected recombinant inoviruses induced a specific humoral response, which conferred partial protection from parasite challenge in the vaccinated mice [76] . Following this study, in 2006, Wang et al. used the monoclonal antibody SSj14, which targets the parasite to screen a 12-mer RPL [77] . The recombinant inoviruses induced both humoral and cellular responses that significantly protected the vaccinated mice against the worm infection [77] . Additionally, in 2006, Wu et al. used polyclonal serum from infected rabbits to screen a 12-mer RPL [78] . A humoral response was induced in the vaccinated mice, which conferred partial protection from the parasite [78] . In 2008, Villa-Mancera et al., produced a vaccine against *Fasciola hepatica*, by screening a previously constructed 12-mer inovirus RLP with an anti-cathepsin L monoclonal antibody [75] . Although immunization of sheep with the selected recombinant inoviruses induced a weak, specific humoral response after challenge with the parasite, the vaccinated animals had a remarkable reduction in worm burden compared to controls [75] . An inovirus-based vaccine has been constructed against the parasitic worm *Trichinella spiralis*. In this case, Gu et al. used a monoclonal antibody against rTs87 antigen to screen a 12-mer RPL [80] . As a result, a humoral response was induced and the vaccinated mice gained partial protection after challenge against the parasite [80] . In summary, the results of the above studies show that the construction of an efficient inovirus-based vaccine that confers protection to the vaccinated animals against the infectious pathogen is achievable through the induction of humoral or cellular immune response or both.

As previously mentioned, inovirus display technology has also been used for the design of vaccines against non-infectious diseases and in some cases, the capability of the vaccine to limit the progression of the disease was evaluated. In 2005, Fang et al. displayed an epitope of the Melanoma Antigen A1 (MAGE A1) in the surface of inovirus fused on gp8 [85]. This resulted in an induction of cellular immune response against the melanoma tumor and in the significant inhibition of tumor growth in the vaccinated mice. In addition, there was an important increase in the survival rate of vaccinated animals [85]. Similar results were obtained in 2002 by Wu et al. in an effort to design a vaccine against murine mastocytoma P815 [89]. An epitope of the P1A tumor antigen was fused to the inoviral surface and the vaccinated mice gained significant protection against tumor growth. Moreover, there was a significant increase in survival rate due to an anti-tumor cellular response that was induced [89]. Some important efforts have also been made for the design of an inovirus-based vaccine against Alzheimer's disease. The main target of these vaccines was the induction of antibodies against β -amyloid plaques. In these studies, the antigenic epitope that was displayed in the inoviral surface was the epitope EGFR, which consists of the four amino acids E, G, F and R and it is part of the β -amyloid peptide. In mice immunized with recombinant inoviruses, the researchers observed a reduction in β -amyloid plaque burden and a specific humoral response [93] [94] [95] [96]. Collectively, these studies show that it is possible to protect vaccinated animals against disease progression, thus alluding to the promising use of such vaccines against non-infectious diseases in humans.

In summary, the utilization of inoviral vectors for vaccine development against infectious and non-infectious non-HIV-1 diseases has produced significant and promising results. First, in the majority of cases, the vaccine was successful since it induced specific humoral or cellular response, or both. Furthermore, in many cases, there was an attempt to evaluate the efficacy of the vaccine after challenge against the pathogen in vaccinated animals. In all cases, the vaccine could provide partial, significant or even complete protection against the pathogen. This established the effectiveness of the use of inoviruses as vaccine vectors.

Our knowledge of HIV-1 neutralization epitopes initiated from the isolation of several neutralizing monoclonal antibodies (2F5, 4E10, b12 and 2G12) that were described between 1993 and 1994 [102] [103] [104]. Thus far, neutralizing monoclonal antibodies

have been found to target four major epitopes on the HIV-1 envelope gp41 and gp120 glycoproteins [105] [106] [107] . These monoclonal antibodies target the MPER epitope of gp41 (monoclonal antibodies 2F5, 4E10, M66.6, CAP206-CH12 and 10e8) [107] [108] [109] [110] [111] [112] [113] [114] [115] ; the V1V2-glycan of gp120 (PG9, PG16, CH01-04 and PGT 141-145) [116] [117] [118] [119] ; the glycan dependent site of the gp120 V3 loop (PGT121-123, PGT125-131 and PGT135-137) [119] ; and the CD4-binding site (b12, HJ16, CH103-106, VRC01-03, VRC-PG04, VRC-PG04b, VRC-CH30-34, 3BNC117, 3BNC60, NIH45-46, 12A12, 12A21, 8ANC131, 8ANC134, INC9 and IB2530 [102, 114, [120] [121] [122] [123] [124] [125] [126] [127] [128] . Monoclonal antibody 2G12 targets the surface glycans on the outer domain of gp120 that is distinct from the four major epitope target sites described above [102, 104, 129] .

The inovirus display technology has also been applied in vaccination strategies against HIV-1 [130] . However, unlike the successful development of vaccines against non-HIV-1 parasites, these efforts failed. In all published studies (see Table 1), the HIV-1 inovirus-display vaccines were made utilizing the broadly neutralizing antibodies 2F5, 2G12 and b12 [50, 52, 54, 55, 59, 62] . The first study for the construction of a vaccine against HIV-1 using inovirus display technology was performed in 1993 by Keller et al., using the 447-52D monoclonal antibody to screen a 15-mer RPL [50] . Vaccination of selected recombinant inoviruses in rabbits resulted in the induction of type-specific neutralizing antibodies [50] . A few years later, in 2001, Zwick et al. used b12 monoclonal antibody to screen a variety of linear and constrained RPLs [52] . However, the vaccination in mice and rabbits with the selected recombinant inoviruses did not result in the production of b12-like antibodies at detectable levels [52] . The same antibody was used in 2005 by Dorgham et al. , in order to screen a 15-mer RPL, and the selected mimotopes were fused to the capsid of inoviruses which were then used to vaccinate mice [54] . The induced antibodies could bind gp160 but they did not have neutralizing potency [54] . In another study in 2007, Wilkinson et al. screened a 9-mer and a constrained 10-mer library with antibody 5145A [55] . This was the only case in which the selected mimotopes were fused in to small heat shock protein (HSP) of the archaeon *Methanococcus jannaschii* as a carrier protein. Following vaccination of HSP-mimotopes in rabbits, anti-gp120 antibodies without neutralizing potency were produced [55] . The 2G12 antibody was used for the first time in 2008 by Menendez et al. for the screening of a variety of linear and constrained RPLs [59] . Nevertheless, vaccination of selected inoviruses in rabbits induced the production of antibodies that could not bind to gp120 [59] . More recently, in 2011, Rodriguez et al. used the 2F5 antibody to screen a 12-mer and a constrained 7-mer RPL [62] . Vaccination in mice and rabbits led to the production of non-neutralizing antibodies [62] . In most of these studies, the use of inoviral

vectors resulted in the induction of a specific humoral response. However, the sera of the vaccinated animals did not have broadly neutralizing ability.

Besides monoclonal antibodies, polyclonal sera from HIV-1-infected patients were also used for the screening RPLs (Table 1) [51, 53, [56] [57] [58] 63] . This approach carries a degree of uncertainty, since it is based on the premise that the polyclonal sera will contain at least one broadly neutralizing monoclonal anti-HIV-1 antibody, meaning that a new neutralizing epitope against it can be isolated from the RPL. It is suggested that long-term non-progressors (HIV-1-infected patients who remain asymptomatic for a long time) are more likely to produce neutralizing antibodies in comparison to AIDS patients, and it is suggested that these neutralizing antibodies in the serum of long-term non-progressors may contribute to the control of viral load [131] [132] [133] . However, this hypothesis has been questioned [134] . Polyclonal serum for the screening of RPLs was used for the first time in 1999 by Scala et al., who screened both linear and constrained 9-mer RPLs [51] . After that, vaccination of selected inoviruses in mice led to the production of neutralizing antibodies. A few years later, in 2007, Rodriguez et al. used polyclonal serum to screen a 7-mer, a 12-mer and a constrained 7-mer RPL [56] . Vaccination in mice induced the production of antibodies that could bind to gp41, but no information was provided about their neutralization potency [56] . Additionally, in 2007, Humbert et al. used polyclonal serum to screen a 7-mer, a 12-mer and a constrained 7-mer RPL [57] . Vaccination of selected recombinant inoviruses in mice induced the production of neutralizing antibodies [57] . The screening of the same RPLs using polyclonal serum from a monkey infected with a Simian-Human Immunodeficiency Virus (SHIV) was performed by the same group. In this study, vaccination in mice was performed using the prime-boost strategy: DNA vaccine as prime (encoding gp160) and a cocktail of recombinant inoviruses as boost. The result was the induction of neutralizing antibodies [58] . In 2013, Gazarian et al. screened a linear 12-mer and constrained 7-mer RPLs using sera from HIV-infected individuals [63] . Vaccination in rabbits resulted in the production of antibodies that bind gp160 [63] . In 2001, non-human primates were used by Chen et al., as an animal model for vaccination with recombinant inoviruses [53] . This group performed screening of a 9-mer and a constrained 9-mer RPL with polyclonal serum isolated from an infected donor. After that, the vaccination of selected inoviruses was performed in rhesus macaques. As a result, sera from the vaccinated macaques exhibited neutralizing activity. Furthermore, the vaccinated macaques were not protected from infection, but four out of five animals were able to control the viral load after challenge against the pathogenic SHIV89.6PD virus. This was a very important result, which underlines the potential of the method. Additionally, no

specific CTL response was detected, implying that the control of viral load was an exclusive result of humoral response [53] .

Since the first attempt to induce the production of anti-HIV-1 neutralizing antibodies using inovirus-based vaccines, all efforts to date have not led to the production of broadly neutralizing antibodies in vaccinated animals. This failure could be to a certain extent explained by the usage of the "old generation" monoclonal antibodies (2F5, 2G12 and b12) which were shown to demonstrate autoreactivity properties in vitro studies [135] [136] [137] [138] [139] . In 2010, Verkoczy et al. demonstrated in an in vivo study that Pre-B cells expressing 2F5-like antibodies were unable to mature in mice, suggesting a triggering of immunological tolerance due to the autoreactive properties of the 2F5-like antibody [140] . Collectively, these studies implied that the screening RPLs using 2F5, 2G12 and b12 could result in inoviral-based vaccines that could trigger immunological tolerance in vaccinated animals. It is important to note, however, that the lack of autoreactivity properties for several of the "next generation" broadly neutralizing antibodies (10e8, PG9, PG16, VRC01-03, VRC-PG04, VRC-PG04b, and VRC-CH30-34) could solve the autoreactivity problems encountered by 2F5, 2G12, and b12.

Despite the fact that the recent isolation of new broadly neutralizing antibodies against HIV-1 has focused the attention of HIV-1 vaccine development on the induction of a humoral anti-HIV-1 response, recent results underline the importance of a cellular anti-HIV-1 response as well [141, 142] . The experimental results concerning inovirus-based vaccines in non-HIV-1 diseases prove that inoviruses can also induce a strong specific cellular response (Tables 1 and 2); this property makes them great candidates as vectors for HIV-1 vaccine design. The ability of inoviruses to induce a cellular immune response was first demonstrated in 2000 by DeBerardinis et al. [143] . In this work, recombinant inoviruses carrying the RT2 epitope and the pep23 epitope of HIV-1 reverse transcriptase in gp8 could induce specific cellular responses in human cell lines in vitro and in mice in vivo against the RT2 peptide. It is interesting that without the pep23 epitope, the cellular response was undetectable. This indicates that the pep23 is a CTL epitope that is necessary for cellular response, possibly because it enables internalization of the recombinant inovirus into the APCs [143] . Therefore, a question arises of whether an epitope fused to the surface of the inovirus can induce a cellular or a humoral response or both. It is suggested that the type of immune response caused by an epitope fused on the surface of the inovirus is dependent on the length and sequence of the peptide [143] . In 2003, Gaubin et al. demonstrated that FITC-labeled fd virions can be internalized in human

EBV-B cell lines and the fd virions are successively degraded and targeted both to MHC class I and class II antigen-processing pathways [144] . This was confirmed after endocellular localization of the labeled virions with confocal microscopy. This experiment showed that the inoviruses could be internalized in APCs even without carrying a CTL epitope, but in very low rate. For in vivo experiments however, it is possible that the requirement of a CTL epitope is critical for the induction of a cellular response [144] . In 2011, Sartorius et al. showed that a hybrid fd virion with the anti-DEC-205 scFv antibody fragment fused on gp3 and the OVA257-264 antigenic epitope fused on gp8 can be internalized in human dendritic cells through a specific interaction between the anti-DEC205 scFv and the DEC-205 receptor [145] . In addition, inoculation of mice with the hybrid virions induced a specific cellular response against the OVA257-264 epitope [145] . This important characteristic of inoviral vectors to induce a cellular response was reported in only two studies aimed at developing a HIV-1 inoviral vaccine, possibly due to the complexity of detecting a cellular response and also because it is a labor intensive and time-consuming process. In 2001, Chen et al. attempted to detect a cellular response, but such a response was not induced in that experiment [53] . A more recent research study related to HIV-1, which clearly demonstrates the ability of inoviruses to induce strong cellular response, was performed in 2009 by Pedroza-Roldan et al. [60] . In this effort, an immunodominant CTL epitope of the V3 loop of gp120 (residues 311-320) was expressed as fusion to gp8 of an M13 inovirus. A random peptide library was created by inserting mutations in certain positions of this epitope. A cocktail of inoviruses carrying the V3 loop epitope or variations of this epitope were used for the vaccination of mice and, as a result, a CTL response was induced. The most important result of this study was that the immunization induced long-lasting memory T-cell responses, which were detected seven months after a single immunization [60] . The same vaccination also induced a strong humoral response, since the sera of vaccinated mice could neutralize five out of ten pseudoviruses from a panel [61] . All the above experiments clearly show that the inoviruses are capable of inducing a specific cellular immune response: the ability of the inoviral vectors to induce both arms of adaptive immunity is unique and it could prove to be valuable in the development of a successful HIV-1 vaccine.

In the general field of HIV-1 vaccine design, all studies for the production of anti-HIV-1 broadly neutralizing antibodies through vaccination with either soluble peptides or viral vectors have been unsuccessful. For this reason, some efforts have been directed to the induction of cellular immune response [146, 147] . In recent years, in HIV-1 vaccine phase I and phase II clinical trials, adenoviral vectors have been used in order induce a cellular immune response in HIV-1 vaccinated individuals [148] [149] [150] [151] . However, there

are concerns about the safety of these viruses. In 2007, the large-scale phase IIB Merck trial was abruptly terminated because there was evidence that the individuals vaccinated with adenovirus rAd5 vector (expressing gag, pol and nef) became more vulnerable to HIV-1 infection in comparison to controls. It was suggested that the group with the increased risk of being infected with HIV-1 consisted of individuals who were Ad5 seropositive [152] . Other eukaryotic viruses that infect other species and do not replicate in human cells were also tested as candidates HIV vaccine carriers and in theory are safer. For example, the canarypox vector was used in the RV144 phase III clinical trial, the only clinical trial that had positive results to date, offering partial protection (31%) to vaccinated individuals in comparison with control [152, 153] . Apart from safety reasons, the use of adenovirus-based vectors has not been protective. The recent HVTV 505 phase IIB trial that used adenovirus rAd5 as a boost and DNA as prime for vaccination of 2504 human volunteers was abandoned as futile [150] . Recently, a rhesus macaque cytomegalovirus (RhCMV) vector successfully induced a persisting CTL response in rhesus macaques that strongly protected the vaccinated animals from challenge against the pathogenic SIVmac239 strain. Importantly, this study, 50% of the vaccinated animals reduced the viral load to undetectable levels [141, 142] . However, the design of a human version of this CMV vector could impose safety risks, since the human CMV is a persistent and pathogenic human virus [146] . Therefore, various types of eukaryotic viral vectors are currently under investigation. These are reviewed elsewhere [154, 155] . However, the use of a eukaryotic virus is accompanied by serious safety concerns. As an alternative, the use of prokaryotic viruses such as inoviruses, may be utilized which have a decisively lower safety risk to humans. Even if inoviruses could infect a eukaryotic cell, the assembly of the new prokaryotic virions cannot take place without the specific conditions that exist in the inter-membrane area of the E. coli and without the presence of the specific E. coli enzyme leader peptidase that does not exist in human cells [156, 157] . Furthermore, there was a phase I case study in 2006 where fd inoviruses were intravenously infused in humans (for purposes unrelated to vaccination), causing no side effects or even allergic reactions in any of the eight volunteers. To our knowledge, this is the only case where inoviruses were infused in humans [158] . Therefore, in contrast to other viral vectors, the use of inoviruses does not impose a major safety risk to humans. This characteristic of inoviral vectors along with their capability to trigger both cellular and humoral immune responses makes them an attractive option as vaccine vectors.

During the last two decades, inoviral vectors have been used in the development of vaccines against various infectious parasites and against non-infectious diseases like cancer and Alzheimer's with promising results. While the applications of inovirus display

technology in vaccine design against non-HIV-1 diseases have been mostly successful, the design of a HIV-1 vaccine development has so far been disappointing. A major obstacle has been the use of neutralizing monoclonal antibodies plagued with autoreactivity properties. Screening of RPLs with these antibodies resulted in the isolation of peptides that, as vaccine antigens, were unsuccessful in inducing a specific humoral response that would produce neutralizing antibodies. The recent isolation of antibodies such as VRC01 and 10E8 with more neutralizing breadth and potency without autoreactivity properties than the previously utilized antibodies may overcome this obstacle [115, 124] . Particularly, the induction of VRC01-like and 10E8-like antibodies could be a feasible target, since these antibodies also seem to be produced from a significant percentage of the HIV-1-infected population [115, 126, 127] . While humoral responses have been well documented, cellular responses have not been assessed in most studies for the design of a HIV-1 vaccine using inoviruses. The only study in which a cellular anti-HIV-1 response was detected also reported a successful induction of a long-lasting memory CTL response seven months after a single vaccination in mice with inoviral particles [60] , demonstrating that the induction of cellular immunity against HIV-1 using inoviruses is feasible. IAVs are advantageous in that they can induce both arms of adaptive immunity. This finding could therefore be of importance in future efforts for the design of a HIV-1 vaccine.

Moreover, IAVs have unique characteristics compared to other viral vectors: they are stable, they can display a peptide in multiple (from few to thousands) copies on their surface and such constructs are very immunogenic without the use of an adjuvant. In addition, IAVs allow the immune system to focus on a specific epitope of interest instead of the whole protein, which is of great importance, since an important aspect for successful HIV-1 vaccine design is to focus on the induction of neutralizing antibodies against the specific neutralizing epitopes while at the same time avoiding the induction of ineffective antibodies against the numerous non-neutralizing epitopes of the HIV-1 glycoproteins, which act as decoys for the immune system. Ideally, an effective HIV-1 vaccine should be able to stimulate both humoral and cellular immune responses. Recently, adenoviral vectors were tested in clinical trials as HIV-1 vaccine carriers in order to induce cellular immunity, but they were shown to impose serious health risks for humans. On the other hand, IAVs are able to induce cellular immunity and at the same time they have been demonstrated to be safe for administration in animals and humans. These characteristics of IAVs, conferred by their structural and biological properties, make them effective antigen display vectors that can induce strong and specific humoral and cellular immune responses against the displayed antigen. These properties of IAVs along with newly

discovered broadly neutralizing anti-HIV-1 antibodies, pave the way for the development of an effective HIV-1 vaccine.

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--- START OF ARTICLE 101 ---

Mucosal Vaccination with Recombinant *Lactobacillus casei*-Displayed CTA1-Conjugated Consensus Matrix Protein-2 (sM2) Induces Broad Protection against Divergent Influenza Subtypes in BALB/c Mice

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Abstract: To develop a safe and effective mucosal vaccine against pathogenic influenza viruses, we constructed recombinant *Lactobacillus casei* strains that express conserved matrix protein 2 with (pgsA-CTA1-sM2/*L. casei*) or without (pgsA-sM2/*L. casei*) cholera toxin subunit A1 (CTA1) on the surface. The surface localization of the fusion protein was verified by cellular fractionation analyses, flow cytometry and immunofluorescence microscopy. Oral and nasal inoculations of recombinant *L. casei* into mice resulted in high levels of serum immunoglobulin G (IgG) and mucosal IgA. However, the conjugation of cholera toxin subunit A1 induced more potent mucosal, humoral and cell-mediated immune responses. In a challenge test with 10 MLD(50) of A/EM/Korea/W149/06(H5N1), A/Puerto

Rico/8/34(H1N1), A/Aquatic bird /Korea/W81/2005(H5N2), A/Aquatic bird/Korea/W44/2005(H7N3), and A/Chicken/Korea/116/2004(H9N2) viruses, the recombinant pgsA-CTA1-sM2/L. casei provided better protection against lethal challenges than pgsA-sM2/L. casei, pgsA/L. casei and PBS in mice. These results indicate that mucosal immunization with recombinant L. casei expressing CTA1-conjugated sM2 protein on its surface is an effective means of eliciting protective immune responses against diverse influenza subtypes.

Text: Vaccination remains most economical and effective means against respiratory diseases caused by influenza viruses [1] . Based on the circulating viruses in the population, trivalent vaccine strains have been developed and are used for the influenza virus protection [2] . The most acceptable current available strategy is the intramuscular administration of inactivated vaccines produced by egg-based manufacturing systems which while effective, are hampered by limited capacity and flexibility [3] . However, vaccine strains must be frequently adapted to match the circulating viruses throughout the world [4] . In addition, the levels of antibody induced by the inactivated vaccine have been observed to decrease by 75% over an 8-month period [2, 5] . Therefore, alternative strategies for developing broadly cross-protective, safe and effective vaccines against influenza viral infections are of prominent importance.

Matrix protein 2 (M2) is highly conserved among influenza A virus strains, indicating that M2 is an attractive target for developing a universal vaccine [6] . In previous studies, various constructs of the M2 vaccine have been developed and tested, including recombinant Escherichia coli (E. coli) expressing M2 fusion protein, adenoviral vectors expressing the M2 protein, plasmid DNA encoding M2 [7] [8] [9] and peptides encoding M2e [11] , each of which was able to elicit protective immune responses in mice. However, the drawback of these M2-based vaccines is their low immunogenicity; additionally, most of them would require intramuscular injections. Therefore, many strategies have been applied focusing on increasing the immunogenicity of M2-based vaccines, for example, fusion of M2 with different carrier molecules like human papilloma virus L protein [12] , keyhole limpet hemocyanin [10] and flagellin [13] . Furthermore, vaccinations with different adjuvants and routes of administration have been applied to evaluate their protection against divergent strains of influenza viruses. Mice immunized mucosally with an M2 or virus like particles (VLPs) adjuvanted with cholera toxin (CT) demonstrated better protection compared to mice subjected to parenteral immunization [14, 15] . However, due to the adverse effects of CT in humans, investigators have attempted to identify nontoxic subunits with adjuvanticity

by removing either subunit A or subunit B [16] . *E. coli* expressing cholera toxin subunit A1 (CTA1) fused with the D-fragment of *Staphylococcus aureus* showed the adjuvant effects without any reactogenicity of the A1 subunit in the mucosal vaccine [6] . Although, chemical or genetic conjugation of M2 may not present M2 in its native tetrameric form, extracellularly accessible antigens expressed on the surfaces of bacteria are better recognized by the immune system than those that are intracellular [17] . Thus, choice of delivery vehicle is also an important concern for potential mucosal vaccines.

Recently, lactic acid bacteria (LAB) presenting influenza virus antigens have been studied [3, 18, 19] . For mucosal immunization, LAB is a more attractive delivery system than other live vaccine vectors, such as *Shigella*, *Salmonella*, and *Listeria* [20, 21] . It is considered safe and exhibits an adjuvant-like effect on mucosal and systemic immunity [18, 22, 23] . Anchoring of the target protein to the cell surfaces of LAB is primarily intended to use in mucosal vaccines. The transmembrane protein *pgsA* is one of the poly-cglutamate synthetase complexes of *Bacillus subtilis* [17, 24, 25] , which is a well-studied anchor protein is able to fuse the target protein to its C terminus and stabilize the complex by anchoring it in the cell membrane. Since sM2 is a highly conserved and promising target for a universal vaccine and CTA1 is strong mucosal adjuvant, in this study, we developed constructs using a consensus sM2 gene reconstituted from the analysis of H1N1, H5N1 and H9N2 influenza viruses (no trans-membrane domain) with or without the fusion of CTA1. To achieve this, we used a novel expression vector that can express a *pgsA* gene product as an anchoring matrix. Our target antigens, sM2 and CTA1, were displayed on the surface of *Lactobacillus casei*, and the oral or intranasal administration of recombinant *L. casei* induced systemic and mucosal immune responses that have the potential to protect against the lethal challenges of divergent influenza subtypes.

A total of 672 female BALB/c mice (5 weeks old) were purchased from Samtako (Seoul, Korea) and housed in ventilated cages. The mice were managed with pelleted feed and tap water ad libitum, maintained in a specific-pathogen-free environment and all efforts were made to minimize suffering following approval from the Institutional Animal Care and Use Committee of Bioleaders Corporation, Daejeon, South Korea, protocol number: BSL-ABLS-13-002. Immunizations of animal were conducted in biosafety level (BSL)-2 laboratory facilities. Mice were divided into 6 experimental sets, each consisting of 2 subsets: 1 for oral and 1 for intranasal administration which contained 4 groups each. Out of 6, 4 sets had 14 mice per group. One sets had 17 (3 mice for lung histopathology and

immunohistochemistry), and the last contained 11 mice per group (3 mice for CTL response).

Concentrations of recombinant *L. casei* were determined by colony forming units (CFU). In each subset, 2 groups received 10¹⁰ CFU of pgsA-sM2/*L. casei* or pgsA-CTA1-sM2/*L. casei*, and the remaining two groups received the same concentration of pKV-pgsA/*L. casei* or PBS in 100 ml orally via intragastric lavage at days 0 to 3, 7 to 9 and 21 to 23. Similarly, 10⁹ CFU of recombinant cells were administered in 20 ml suspensions into the nostrils of lightly anesthetized mice on days 0 to 3, 7 to 9 and 21. Blood samples were collected from the retro-orbital plexus at days 21, 14 and 28; sera were separated by centrifugation for 5 minutes at 12,000g and stored at 220uC until analysis. At day 28, 3 mice in each group were randomly sacrificed to collect IgA sample from lungs and intestine and stored at 270uC until analysis. Spleens were collected aseptically at day 28 for the analysis of the CTL response randomly from 3 mice of one set. The rest of the mice from the same set were maintained for 6 months from the date of the last boosting to measure the long-lasting immune responses and protection efficacy.

The avian influenza viruses A/EM/Korea/W149/06(H5N1), A/Puerto Rico/8/34(H1N1), A/Aquatic bird/Korea/W81/2005 (H5N2), A/Aquatic bird/Korea/W44/2005(H7N3), and A/Chicken/Korea/116/2004(H9N2) used in this study were kindly provided by Dr. Young-Ki Choi (College of Medicine and Medical Research Institute, Chungbuk National University, Cheongju, Republic of Korea). All viruses were propagated in the allantoic fluid of 10-day-old chicken embryos, and 50% mouse lethal doses (MLD₅₀) were determined in 8-week-old naive BALB/c mice. Ether narcosis-anesthetized mice were intranasally infected with 10 times the MLD₅₀ of challenge viruses in 20 ml of PBS. Six mice in each group were sacrificed on 3 and 5 dpi to check virus titer in lungs and other 5 mice remained in each group have been used for survival. Mice were monitored every alternate day at fixed time point for measuring the weight loss and survival. Mice were euthanized if moribund, i.e. weight loss, ruffled fur, shivering, tachypnea, respiratory distress, hypothermia and poorly responsive to external stimuli, remaining were considered as survival number. After final monitoring, all the survived mice were humanely euthanized using CO₂ inhalation for 5 minutes.

At 180 days after the final vaccination, mice from one set were challenged with H5N2 for measuring the long lasting immune responses. All challenge tests were conducted inside

an approved BSL-3+ facility under appropriate conditions. Bacterial Strains and Cloning for the Construction of Recombinant Plasmid PgsA-sM2/L. casei and PgsA-CTA1-sM2/L. casei In this study, *E. coli* JM83 was used for cloning and *L. casei* L525 was used for surface expression of the target protein. These bacteria were grown in LB and MRS media, respectively. The plasmid pKV-Pald-PgsA, harboring the pgsA genes of *Bacillus subtilis*, was used to construct the surface display plasmid, which was a kind gift from the Bioleaders Corporation (Daejeon, South Korea). A gene encoding the consensus sequence of M2 spanning the residues of the extracellular and cytoplasmic domains without the transmembrane domain of influenza virus was generated. The consensus sequences were created based on the most common amino acids in each position of the alignment of H1N1, H5N1 and H9N2; then, they were synthesized and used as templates for the construction of the plasmids pgsA-sM2/L. casei and pgsA-CTA1-sM2/L. casei by cloning, as described previously [26, 27]. The sM2 gene was modified by adding a Kpn I site at the 59 terminal and Sal I at the 39 terminal for cloning. The polymerase chain reaction (PCR) was performed to amplify the gene using the primer pair 59-GGGGTACCTCATTATTAACA-39, and 59-ACGTCGACT-CATTATTCAAGTTCAATAATG AC-39. Similarly, a BamH I site at the 59 terminal and a Kpn I site at the 39 terminal end were added to the CTA1 gene using primers 59-CGGGATCCAAT-GATGATAAGTTATAT-39 and 59-GGGT ACCCGAT-GATCTTGGAGC ATT-39. The modified genes were ligated into the T Easy Vector (Invitrogen, Seoul, Korea). Genes were then digested with Kpn I-Sal I for sM2 and BamH I-Kpn I for CTA1. The digested sM2 was ligated to the plasmid vector pKV-pgsA for the construction of pKV-pgsA-sM2. Similarly, CTA1 was ligated for the construction of pKV-pgsA-CTA1-sM2. The ligated products were transformed into *E. coli* JM83 competent cells, as previously described, using an electroporation method [17]. The profiles of the recombinant plasmids were confirmed by restriction endonuclease digestion and DNA sequencing (Solgent, Seoul, Korea). After confirmation, the plasmids were transformed into *L. casei* L525 by electroporation and named pgsA-sM2/L. casei and pgsA-CTA1-sM2/L. casei.

The recombinant *L. casei* containing pgsA, pgsA-sM2 and pgsA-CTA1-sM2 genes were grown at 30°C for 48 hours. Cells were harvested by centrifugation at 6,000g for 10 minutes at 4°C, followed by washing two times with sterile phosphate-buffered saline (PBS). Bacterial lyses were performed by sonication and centrifuged at 12,000g for 20 minutes at 4°C. Cell wall and cytoplasmic fractions were separated by centrifugation at 25,000g at 4°C for 2 hours. Pellets (cell wall) were resuspended in 100 ml of 1% sarcosol containing 1 mM phenylmethylsulfonyl fluoride (PMSF, Sigma-Aldrich, St. Louis, USA) as a protease inhibitor. Fractions were analyzed by western blotting, as described previously. For the immune detection of fusion proteins, the membranes were probed with rabbit anti-

cholera toxin (1:2000, Abcam, UK), rabbit anti-pgsA (1:1000) and rabbit anti-M2 (1:1000) antibodies. The rabbit anti-pgsA and rabbit anti-M2 antibodies used in this experiment were generated by the i.m. inoculation of KLH-conjugated pgsA or M2 peptide in rabbit, respectively, two times at 2 weeks-interval. The membranes were reacted with a 1:10,000 dilution of anti-rabbit immunoglobulin G conjugated with horseradish peroxidase (IgG HRP). Finally, the target proteins were detected using the WEST-ZOL plus Western Blot Detection System (iNtRON Biotechnology, Gyeonggi-do, Korea) and visualized by enhanced chemiluminescence (ECL) [17, 26, 28] .

To investigate the expression of sM2 or CTA1-sM2 on the surface of *L. casei*, recombinant *L. casei* were grown in 30uC for 48 hours in the MRS broth. Bacteria were harvested by centrifugation at 5,000g for 10 minutes at 4uC, washed three times with sterile phosphate-buffered saline containing 0.01% Tween-20 (PBST) and probed with polyclonal rabbit anti-M2 or rabbit anti-CT antibody overnight. Following another washing, the cells were treated with fluorescein isothiocyanate (FITC)conjugated anti-rabbit IgG antibodies (Burlingame, CA, USA) for 2 hours. Finally, 10,000 cells were analyzed by flow cytometry (Becton Dickinson, Oxnard, CA, USA). For the immunofluorescence, cells were prepared under the same condition described for the flow cytometry. The pgsA/*L. casei* was used as a negative control and Immunofluorescence analysis was examined using a Carl Zeiss Axioskop 2 fluorescence microscope.

ELISA Antibody titers were measured by enzyme-linked immunosorbent assay (ELISA) using serum or mucosal samples from vaccinated mice. First, 96-well immunosorbent plates (Nunc) were incubated with 300 ng/well purified sM2 or CTA1 proteins at 4uC overnight. The recombinant sM2 and CTA1 proteins used in this study were purified from *E. coli*. Next, the wells were blocked with 10% skim milk for 2 hours in RT, washed five times with PBST, treated with diluted serum samples (1:200) in triplicate for detecting IgG and undiluted tissue homogenized supernatant for detecting local IgA and incubated for 2 hours at 37uC. After washing three times, goat anti-mouse IgG HRP (1:1000, sigma) or anti-mouse IgA was added to each well and incubated for an additional 2 hours at 37uC. Following another round of washing, the plates were reacted with the substrate solution containing tetramethylbenzidine and H₂O₂ and allowed to precede the reaction for 10 minutes. After adding the stop solution 2N-H₂SO₄, the optical density (OD) was measured at 450 nm using an ELISA autoreader (Molecular devices).

The development and counting of cytokines were performed by ELISPOTs, as described previously [31, 32] . Briefly, the day before the isolation of splenocytes, ELISPOT 96-well plates were coated with monoclonal anti-mouse IFN- γ and IL-4 capture antibodies (5 mg/ml) in PBS and incubated at 4°C overnight. The plates were washed with PBS, and 200 μ l/well of blocking solution containing complete RPMI 1640 medium and 10% fetal bovine serum, was added (Invitrogen, Carlsbad, CA, USA) and incubated for 2 hours in RT. Splens from the vaccinated mice were isolated aseptically and added at 5 $\times$ 10⁴ cells/well in media containing sM2 protein, M2 peptide (SLLTEVETPTRNGWECKCSD) (1 mg/well), only medium (negative control), or 5 mg/ml phytohemagglutinin (positive control, Invitrogen, Carlsbad, CA, USA). After adding cells and stimulators, the plates were incubated for 24 hours at 37°C with 5% CO₂ . The plates were sequentially treated with biotinylated anti-mouse IFN- γ and IL-4 antibodies, streptavidin horseradish peroxidase, and substrate solution. Finally, the spots were counted using an ImmunoScan Entry analyzer (Cellular Technology, Shaker Heights, USA).

The lungs were collected aseptically, and virus titers were determined by 50% tissue culture infectious dose (TCID₅₀), as described previously [33] . Briefly, lung tissues were homogenized in 500 μ l of PBS containing antibiotics (penicillin, and streptomycin) and antimycotics (Fungizone) compounds (Gibco, Grand Island, NY, USA). Mechanically homogenized lung samples were centrifuged (15 minutes, 12,000g and 4°C) to remove the cellular debris before their storage at 280°C. MDCK cells were inoculated with a 10-fold serially diluted sample and incubated at 37°C in a humid atmosphere of 5% CO₂ for an hour. After absorption, the media was removed, and overlay medium containing L-1-tosylamido-2-phenylethyl chloromethyl ketone (TPCK) trypsin (Thermo Fisher Scientific, Rockford, USA) was added to the infected cells and incubated for 72 hours. Viral cytopathic effects were observed daily, and the titers were determined by the HA test. The viral titer of each sample was expressed as 50% tissue infected doses using the Reed-Muench method [34] .

For histopathology, lung tissues were collected at 5 dpi from ether narcosis-anesthetized mice. Tissues were immediately fixed in 10% formalin containing neutral buffer, embedded in paraffin wax, sectioned at 4-6 μ m thickness using a microtome machine, mounted onto slides, and stained with eosin stain. Histopathological changes were examined by light microscopy, as previously described [29, 30, 35] . Furthermore, slides were stained using an immunoperoxidase method with an antibody (rabbit anti-M2, 1:500) directed against the matrix protein-2 of influenza A virus. A Goat-anti-rabbit IgG HRP (1:2000, Sigma-Aldrich, St.

Louis, USA) was used as the secondary antibody for the detection of virus infected cells in respective tissues [57] .

Data are presented as the means $\pm$ standard deviations (S.D.) and are representative of at least three independent experiments. Differences between groups were analyzed by analysis of variance (ANOVA), and means were compared by Student's t-test. P-values less than 0.05 were regarded as significant. Results for percent initial body weight were also compared by using Student's t test.

Comparison of survival was done by log-rank test using GraphPad Prism 6 version.

The pgsA-expressing vector was used to construct plasmids containing the highly conserved consensus sM2 gene, with (pgsA-CTA1-sM2) or without (pgsA-sM2) the cholera toxin subunit A1 (CTA1, Fig. 1A). Plasmids were transformed into *L. casei* cells. The expression levels of pgsA-sM2 and pgsA-CTA1-sM2 were monitored by immunoblotting using anti-pgsA, anti-M2 or anti-CT polyclonal antibodies (data not shown).

To determine the cellular localization of the sM2 and CTA1 proteins expressed on the surface of *L. casei* via the cell wall anchor protein pgsA, membrane and cytoplasmic fractions were subjected to western blot analysis. As expected, both pgsA-sM2 and pgsA-CTA1-sM2 fusion proteins were detected by anti-pgsA, anti-M2 or anti-CT polyclonal antibodies in the membrane, not in cytoplasmic fractions (Fig. 1B, lane 2, 3 and 4) . Immunoreactions were performed with anti-pgsA, and bands representing the size of the fused proteins pgsA-sM2 and pgsA-CTA1-sM2 were detected, while during the reactions with anti-M2 or anti-CT antibodies, no other bands were detected (Fig. 1B, lane 3 and 4) . This finding may have resulted from the degradation that occurs during the membrane fractionation procedure.

Fluorescence-activated cell sorting (FACS) and immunofluorescence labeling of the cells were used to verify the localization of the fusion pgsA-sM2 and pgsA-CTA1-sM2 protein on the surface of *L. casei*. Flow cytometric analysis using rabbit anti-M2 and anti-CT antibodies revealed increase level of fluorescence intensity of pgsA-sM2/*L. casei* or pgsA-CTA1-sM2/*L. casei* cells, compared to that of control *L. casei* cells (Fig. 1C) .

Immunofluorescence microscopy also showed recombinant bacteria harboring pgsA-sM2 or pgsA-CTA1-sM2 that immunostained positive for sM2 and CTA1, but this was not found in control cells. These results demonstrated that recombinant *L. casei* could efficiently display the sM2 and CTA1-sM2 fusion proteins on the surface, using pgsA as a membrane anchor protein.

Immune Responses Induced by Mucosal Immunization with *L. casei* Surface Displayed sM2 and CTA1-sM2

Preliminary experiment was conducted to determine the doses and schedule of pgsA-CTA1-sM2/*L. casei* vaccine candidate on influenza virus protection (data not shown). To characterize the immunogenicity of the *L. casei* surface-displayed sM2 and CTA1-conjugated sM2, BALB/c mice were immunized nasally (10^9 cells/20 ml dose) or orally (10^{10} cells/100 ml dose) with recombinant live pgsA-sM2/*L. casei* and pgsA-CTA1-sM2/*L. casei* bacteria. As a negative control, mice were immunized with *L. casei* harboring the parental plasmid pKV-pgsA (pgsA/*L. casei*) and PBS. Serum samples were collected at 0, 14 and 28 days and analyzed by ELISA, using sM2 and CTA1 proteins (purified from *E. coli*) as a coating antigen. After the first series of immunization, comparatively low levels of serum IgG were detected both in the i.n. and orally immunized group. However, high antibody levels were detected shortly after the second series of immunization, and the CTA1-conjugated sM2 group induced serum IgG at significant level, compared to sM2-only group and negative controls (Fig. 2A and B) . Although the conjugation of CTA1 with sM2 was expected to have an adjuvant function only, a significant level of anti-CTA1 antibodies was detected in both the nasal and oral vaccinations (Fig. 2A and B right panel) . In comparison with the oral group, the nasally immunized group showed higher levels of serum IgG specific to both sM2 and CTA1.

To assess the mucosal immune responses, the local IgA levels were determined by ELISA. Lung and intestinal tissues were collected at day 28 of immunization and examined using sM2 protein as a coating antigen. In both routes of vaccination, pgsA-CTA1-sM2/*L. casei* induced significantly increased levels of sM2-specific mucosal IgA compared to the pgsA-sM2/*L. casei* and control groups. However, as expected, higher levels of antibody titers were detected at the site of inoculation than at the remote site. A similar pattern of antibody responses was observed for both routes of immunization, in which the pgsA-CTA1-sM2/*L. casei* groups dominated (Fig. 2C and D) . These data demonstrated that

cholera toxin subunit A1-conjugated sM2 resulted in significant enhancements to the sM2-specific IgG and mucosal IgA levels compared with sM2 alone or with controls immunized with pgsA/ *L. casei* or PBS.

Mucosal Immunization with *L. casei* Surface-displayed sM2 and CTA1-sM2 Stimulated M2-specific Cellular Immune Response

To determine whether mucosal vaccination with *L. casei* surfacedisplayed sM2 and CTA1-conjugated sM2 could induce cellular immunity, IFN- γ and IL-4 ELISPOT were performed. Splenocytes from vaccinated mice were stimulated with 10 mg/ml of recombinant sM2 protein or M2 peptide, and the cytokine ELISPOTs were developed. The spots were counted to measure the differences in the CTL responses between the groups. Cells from the mice immunized i.n. with pgsA-CTA1-sM2/*L. casei* showed significant levels of IFN- γ in response to stimulation with sM2 protein and M2 peptide (Fig. 3A) . Similarly, we observed that i.n. administered groups both for pgsA-sM2/*L. casei* and pgsA-CTA1-sM2/*L. casei* showed detectable levels of IL-4 secreting splenocytes following stimulation with either sM2 protein or M2 peptide (Fig. 3B) . IFN- γ and IL-4 secreting cells were also observed in mice immunized orally with pgsA-sM2/*L. casei* and pgsA-CTA1-sM2/*L. casei* (Fig. 3C) although their levels were lower than i.n. group and were not significant. Control group immunized with pgsA/*L. casei* showed background spot level for both in intranasal and oral groups. These findings indicate that highly conserved sM2 can induce M2-specific IFN- γ and IL-4 secreting T cell responses, while mucosal delivery through *L. casei* and CTA1 conjugation with sM2 enhanced the cell mediated immunity, which may contribute to broadening the protective immunity.

M2 is known as a potential target for the development of broad spectrum influenza vaccine with minimum variability [36, 37] . To confirm the variability of sM2 sequences of the challenged viruses used in this study, we compared the sM2 of influenza subtypes available from U.S. National Center for Biotechnology Information (NCBI) with our consensus sM2 sequence particularly the whole conserved ecto and some portion of cytoplasmic domain (CD) although entire CD was included in vaccine construct (Table 1) . We found that, viruses used in this study contain 0-8 mismatched amino acids among the amino acids of sM2 compared in this study. To evaluate the efficacy of the sM2 vaccine, week after the final immunization, mice were challenged i.n. with the 10 MLD 50 of A/Aquatic bird/Korea/W81/2005 (H5N2) influenza virus subtypes that was homologous to

the consensus sM2 sequence. Mice immunized orally with pgsA-sM2/ *L. casei* and pgsA-CTA1-sM2/*L. casei* showed 40 and 60% protection respectively. Similarly, i.n. immunization groups conferred 40 and 80%, against the lethal infection with highly virulent H5N2 virus. In contrast, none of the unimmunized mice survived after lethal infection (Fig. 4A and B, right panel) . Morbidity was increased in the mice immunized via oral route, whereas mice that received i.n. immunization with pgsA-CTA1-sM2/*L. casei* lost ,20% of their initial body weight and started recovering by 9 day post infection (dpi) and had completely recovered by day 13 (Fig. 4A and B, left panel) .

We next evaluated the protection efficiency of sM2 vaccine candidate against A/Puerto Rico/8/34(H1N1), which contains 8 mismatched amino acids relative to the sM2 consensus sequence. Sets of vaccinated mice were challenged with 10 MLD 50 of the H1N1 virus. As shown in figure 4C and D, mice immunized by the The mice were grouped as mentioned in materials and methods and received oral or nasal administrations, according to the schedule. Arrows indicated the immunization routes and periods of pgsA/*L. casei*, pgsA-sM2/*L. casei* or pgsA-CTA1-sM2/*L. casei* cells. Sera were collected at days 0, 14 and 28; samples from the lungs and intestines were collected at day 28 after immunization. A week after the final immunization, spleens were excised from 3 mice in each group, with one set for CTL analysis. Two or 24 weeks after the last immunization, all mice were challenged with a lethal dose of influenza subtypes through intranasal route and monitored for 13 days. On days 3 and 5 post infection, the lungs were excised from 3 mice in each group to determine the virus titer. On 5 dpi, the mice from one set were sacrificed for lung histopathology and immunohistochemistry. doi:10.1371/journal.pone.0094051.g001 CTA1-sM2 Induces Protective Immunity to Pathogenic Influenza A Viruses PLOS ONE | www.plosone.org i.n route exhibited a higher level of protection than the orally immunized groups, and mice immunized with pgsA-CTA1-sM2/ *L. casei* showed a significantly higher level of protection compared to mice immunized with pgsA-sM2/*L. casei* (Fig. 4C and D, right panel) . Unimmunized mice lost up to 40% of their body weight and died by 9 dpi. Mice immunized with pgsA-CTA1-sM2/*L. casei* lost approximately 10% of their body weight, whereas mice immunized with pgsA-sM2/*L. casei* lost .20% of their initial body weight by 9 dpi and recovered more slowly than mice immunized with pgsA-CTA1-sM2/*L. casei* (Fig. 4C and D, left panel) .

Another set of vaccinated mice were infected with A/Chicken/ Korea/116/2004(H9N2) to check the range of protection ability of sM2 vaccine induced immune responses. The sM2 sequence of H9N2 contains 2 mismatched relative to the sM2 consensus sequence. The

mice immunized with pgsA-CTA1-sM2/L. casei showed negligible body weight losses and gradual recovery compared to those of mice immunized with pgsA-sM2/L. casei and the unimmunized mice for both the i.n and oral routes (Fig. 4E and F left panel) . None of the unimmunized mice survived, whereas 100% and 80% of the mice immunized with pgsA-CTA1-sM2/L. casei via the i.n. and oral routes survived, respectively. The survival rates of mice immunized with pgsA-sM2/L. casei were 80% and 60% for the i.n. and oral routes, respectively (Fig. 4E and F, right panel) .

The breadth of protection of the sM2 vaccine against divergent influenza subtypes was also evaluated. Set of immunized mice were challenged with high pathogenic avian influenza (HPAI) A/ EM/Korea/W149/06(H5N1), which contains 2 amino acid mismatches relative to the sM2 consensus sequence. Mice immunized via the i.n. and oral routes with pgsA-CTA1-sM2/L. casei showed higher protection efficacies, 80% and 60%, respectively, compared with mice immunized with pgsA-sM2/L. casei, for which the rates were 60% and 20%, respectively (Fig. 4G and H, right panel) . Regarding morbidity, mice immunized with pgsA-CTA1-sM2/L. casei showed lower morbidity than mice immunized with pgsA-sM2/L. casei (Fig. 4G and H, left panel) . One more set of vaccinated mice were challenged with the A/Aquatic bird/ Korea/W44/2005 (H7N3) virus, which contains 1 mismatch relative to the consensus sM2 sequence, and the body weight and survival were observed for 13 dpi. As shown in figure 4I and J, unimmunized mice lost as much as 30% of their body weight than mice immunized with pgsA-sM2/L. casei and pgsA-CTA1-sM2/L. casei (Fig. 4I and J, left panel) . Mice immunized with pgsA-CTA1-sM2/L. casei through the i.n route showed significantly higher level of protection against the H7N3 influenza virus than the other groups (Fig. 4I and J, right panel) . Taken together, the results indicate that i.n. immunization with pgsA-CTA1-sM2/L. casei induced immune responses that conferred significant levels of protection against divergent subtypes of influenza viruses containing mismatched amino acids ranging from 0 to 8 of the consensus sM2, regardless of whether it was complete or partial.

Virus titers in the lungs of challenged mice were measured to estimate replication at 3 and 5 dpi. Mice were immunized via the i.n and oral routes with pgsA-sM2/L. casei and pgsA-CTA1-sM2/ L. casei and challenged with the H5N2, H1N1, H9N2, H5N1 or H7N3 influenza subtypes. On 3 and 5 dpi, 3 mice were sacrificed randomly from each group, and their lung virus titers were measured using the TCID 50 method. Mice immunized with pgsA-CTA1-sM2/L. casei had lower titers at 3 dpi and had significantly reduced viral replication at 5 dpi compared to mice immunized with pgsA-sM2/L. casei or the control groups at the same

time (Fig. 5A-J) . Reduced viral titers in the lungs were observed in groups of mice immunized via the i.n route relative to the mice immunized via the oral route, particularly at day 3 post infections (Fig. 5) . These reduced titers may be due to routes of vaccination and challenge being the same, and the titers correlated with the survival results for lethal infections with H5N2, H1N1, H9N2, H5N1 and H7N3. Taken together, these results demonstrate that the consensus sM2 protein fused with CTA1 afforded better protection than sM2, and the i.n route was more potent than the oral route of immunization with regard to protection against a lethal challenge of divergent influenza subtypes.

Histopathology and immunohistochemistry were performed to corroborate the lung virus titer findings. At 5 dpi, lungs were randomly collected from each group of one set, fixed and stained with eosin before being examined under a light microscope. As shown in figure 5K , clear signs of profound pulmonary inflammation were observed in the lungs of mice treated with PBS or pgsA/L. casei for both the oral and i.n routes of administration, whereas the lungs of the mice immunized with pgsA-CTA1-sM2/L. casei showed no remarkable pulmonary inflammation compare to the pgsA-sM2/L. casei-treated mice (Fig. 5K, middle and left panel) . For immunohistochemistry, immunoperoxidase method with an antibody directed against the matrix protein-2 of influenza A virus was used for the detection of virus infected cells in the respective tissues. Virus antigen in epithelial cells appears as brown coloration of the nucleus and cytoplasm. As shown in figure 5K, at 5 days p.i., numerous virusinfected cells were detected in control or pgsA-sM2/L. casei vaccinated mice, whereas highly reduced number of antigen positive cells were found in the mice vaccinated with pgsA-CTA1-sM2/L. casei, both in i.n. and orally immunized group (Fig. 5K right panel) . These results indicate that mice immunized with pgsA-CTA1-sM2/L. casei developed immune responses that are strong enough to inhibit virus replication, which promotes the survival of mice after a lethal infection by influenza A.

The PgsA-CTA1-sM2/L. casei Vaccination Induced Longlasting Cross Protection

The duration of protection is an important criterion for a potential vaccine. Thus, the longevity of the immunity induced by sM2 and CTA1-conjugated sM2 were investigated by detecting serum IgG and mucosal IgA by ELISA. Significantly increase levels of sM2-specific serum IgG as well as lung and intestinal IgA were observed 180 days after vaccination (Fig. 6A and C) compare to PBS and pgsA/L. casei groups. Mice were challenged with A/ Aquatic bird/Korea/W81/2005(H5N2), and the body weight changes and

survival were monitored until 13 dpi. The unimmunized mice showed .30% body weight loss (Fig. 6B and D left panel) and died by day 9 post infection in both the oral and i.n. groups. In contrast, the mice immunized with pgsA-CTA1-sM2/L. casei showed negligible body weight loss, which was recovered by 13 dpi; 80% survived in the i.n. immunized group (Fig. 6B right panel) , and 60% survived in the orally immunized group (Fig. 6D right panel) . This result indicates that the CTA1 conjugated sM2 mucosal vaccine conferred protection against a lethal infection 6 months after the final immunization.

The mucosal immune system is the first immunological barrier against the pathogens that invade the body via the mucosal surface. Thus, the induction of mucosal immunity is necessary to ensure protection against multiple subtypes of influenza A virus. A respiratory virus, influenza A is responsible for annual seasonal epidemics worldwide and, occasionally, pandemics, which are caused by emerging novel subtypes/strains derived through reassortment with avian or porcine viruses. Current influenza vaccines provide strain-specific protection only. Thus, it is crucial to establish a broadly cross-protective influenza vaccine. Antigens that are well conserved among influenza A viruses are considered promising targets for the induction of cross-protection against these different subtypes. However, the goal should be the development of a first line of defense by effectively eliminating pathogens at the mucosal surface. Influenza matrix protein-2 (M2) is relatively well conserved among the influenza subtypes and can be considered a promising influenza vaccine antigen [30] . It consists of the following three structural domains: a 24-amino-acid extracellular domain, a 19-amino-acid transmembrane domain, and a 54-amino-acid cytoplasmic tail domain [39, 40] . The extracellular and cytoplasmic domains, which are well conserved among influenza viruses and play an important role in viral assembly and morphogenesis, were used in this study. Here, we developed sM2 consensus derived from the analysis of sequences of H5N1, H1N1 and H9N2 subtypes in the database. Considering the previous findings that extracellular domain particularly (aa, 1-13) is highly conserved among the influenza virus subtypes and recognized as epitope for the induction of monoclonal antibodies, which could protect influenza virus infection [56] , sM2 backbone sequence from the H5N1 virus were used. For the possible homology among other subtypes we changed at the position of 14 (E-G) and 18 (R-K) and kept unchanged the conserved epitope (aa, 1-13). As shown in sequence alignment, sM2 of consensus sequence has 0-8 mismatches among the subtypes used in this study (Table 1)

.

Moreover, the incorporation of an adjuvant is considered essential to boost the interaction of the vaccine with the mucosal immune system [41] . Various adjuvants, such as liposomes, nanoparticles, and immunostimulating complexes (ISCOMs), have been studied and were found to improve the immune response [42] , but their efficacies were not optimal. Despite its potential as a mucosal adjuvant [43] , the use of cholera toxin (CT) in vaccines is limited by its innate toxicity. Thus, the toxicity of CT would have to be separated from its adjuvant activity before it could be used as a vaccine adjuvant. Studies have shown that constructs consisting of M2e fused with cholera toxin subunit A1 along with a strong ADP-ribosylating agent and a dimer of the D-fragment of *Staphylococcus aureus* protein A vaccine elicited complete protection and reduced morbidity [6, 44] . CTA1 retains the adjuvant function of CT without its toxic side effects, such as reactogenicity at the site of its administration and binding to or accumulation in the nervous tissues [45] . Based on previous findings, it has been hypothesized that the consensus sM2 fragment, when fused with the potent mucosal adjuvant CTA1, may induce broad protective immunity against divergent subtypes of influenza virus. In this study, we used the whole 22-kDa CTA1 protein (an ADP-ribosyltransferase), which consists of three distinct subdomains: CTA11 (residues 1 to 132), CTA12 (residues 133 to 161), and CTA13 (residues 162 to 192). It has been reported that CTA1 lacking CTB has strong adjuvant activities without any toxicity. CTA1 enhances the IgA and IgG antibody responses, as well as CTL activity [47] .

For the development of a universal mucosal influenza vaccine with a conserved sM2 peptide and potent adjuvant CTA1, recombinant *L. casei* displaying sM2 fused with or without CTA1

The lungs of the mice vaccinated with pgsA-CTA1-sM2/*L. casei* showed clear alveoli without inflammatory cell infiltration, in contrast to the lungs of mice vaccinated with pgsA-sM2/*L. casei* or control mice, both of which revealed features of severe pneumonitis (middle and left panel). Reduced number of viral antigen were detected in lungs of the mice vaccinated with pgsA-CTA1-sM2/*L. casei*, in contrast to the lungs of mice vaccinated with pgsA-sM2/*L. casei* or control revealed features of severe pneumonitis with increase virus antigen (right panel). Micrographs are representative for each treatment group at a magnification of 200X. Virus antigen in epithelial cells appears as brown coloration of the nucleus and cytoplasm. In lung titers, bars denote mean $\pm$ S.D. The asterisk indicates a significant difference between pgsA-CTA1-sM2/*L. casei* and other groups (* P ,0.05). doi:10.1371/journal.pone.0094051.g005 were constructed for mucosal delivery by the widely used live vaccine vehicle LAB [38] . The pgsA gene used in this study is an anchor for

display on the surface of LAB which is derived from the pgsBCA enzyme complex of *Bacillus subtilis* and consists of transmembrane domain near its N-terminus with the domain located on the outside of the cell membrane. Thus, pgsA is able to cross the cell wall and display the heterologous protein fused to its C-terminus [17] .

The developed vaccines were tested through two major routes. We found that vaccination with pgsA-CTA1-sM2/L. casei was able to induce a significantly higher level of sM2-specific serum IgG (Fig. 2A and B) and mucosal IgA (Fig. 2C and D) compared to pgsA-sM2/L. casei, and conferring protection against divergent influenza subtypes of both phylogenetic group 1 (H1, H5, H9) and group 2 (H7) [46] (Fig. 4) . This study also revealed that i.n. administration was superior to the oral route of vaccination, which is consistent with other observations [48] . There may be two possible reasons to explain this phenomenon. First, the challenge route is the same as that of the vaccination; specific mucosal IgA can prevent viral colonization in the respiratory tract. Second, the volume of the inocula was 5 times lower than that for oral inoculation, which may have allowed the concentrated form of the antigen to be presented to immune cells. Because greater levels of serum IgG and mucosal IgA were detected in intranasally immunized mice than in those immunized orally (Fig. 2) , an alternative explanation could be that the antigens are processed and/or presented differently to immune cells in the two mucosal compartments. Importantly, our study demonstrated for the first time that mucosal immunization with the LAB surface-displayed CTA1-conjugated sM2-based vaccine candidate induced broad protection against challenge with divergent influenza subtypes.

However, the mechanism by which Abs against sM2 mediated this broad protection is not fully understood. Previous studies have demonstrated that Abs to the N-terminus of M2e, particularly positions 1-10, inhibited the replication of the influenza A virus [49, 50] . Other studies revealed that anti-M2e IgG-mediated cellular cytotoxicity or phagocytosis can induce the removal of infected cells before progeny virus budding and spread [54, 55] which is supporting our findings of lung virus titer and immunohistochemistry data detected at 5 dpi in our challenge experiments. Therefore, in this study, combination of those responses and Abs to the N-terminus of the sM2 sequence which is conserved among the challenge viruses (Table 1) may protect the divergent influenza subtypes after mucosal immunization with the recombinant LAB CTA1-conjugated sM2-based vaccine candidate. Moreover, the cellular immune response plays an important role in controlling viral replication. We examined the Th1-type (IFN- γ) and Th2-type (IL-4) cytokine responses by the ELISPOT assay. Significantly higher levels of IFN- γ were detected in response to

stimulation with both the sM2 protein and M2 peptide in mice immunized with pgsA-CTA1-sM2/L. casei compared to the levels in mice in the pgsA-sM2/L. casei and control groups (Fig. 3A and C) . Similarly, substantially high levels of IL-4 were observed in mice immunized with pgsA-CTA1-sM2/ L. casei upon stimulation with the sM2 protein and M2 peptide (Fig. 3B and D) . These results further support the findings that the antibodies and cell-mediated cytotoxicity were specific to the M2 antigen and that their anti-viral activities were induced by monomeric M2, three copies of M2 fused with ASP-1 [34, 51, 52] . Together, these results indicate that sM2 adjuvanted with fused CTA1 induced immune responses in mice, which protected them from divergent influenza subtypes. In this regard, our results have significance for the use of CTA1, which has adjuvant function, in vaccine candidates.

As clinical protection is not the only parameter by which vaccine performance is assessed, we evaluated the immunogenicity of the recombinant LAB vaccine on the basis of other parameters, such as the reduction of pathological lesions and virus shedding. In this study, low titers of the challenge virus were titrated from the lungs after vaccination with pgsA-CTA1-sM2/L. casei, whereas challenge virus could be detected at higher titers in the mock mice and those vaccinated with pgsA-sM2/L. casei (Fig. 5A-J) . Reduced gross and histopathological lesions consistent with viral infection are the primary parameters indicative of influenza vaccine efficacy. Here, we demonstrated that vaccination with pgsA-CTA1-sM2/L. casei remarkably limited the severity of the damage by inhibiting viral replication and the accumulation of inflammatory cells and virus antigen in the lung alveolar tissues, relative to the severity in the unimmunized mice and the mice vaccinated with pgsA-sM2/L. casei (Fig. 5K) .

Our study further demonstrated, for the first time, that recombinant L. casei expressing CTA1-sM2 induced long-lasting immunity and conferred protection against lethal infections by influenza, even at 6 months after the final vaccination (Fig. 6) , which is important for any successful vaccine. Similar results were observed in previous studies, in which M2 VLP conferred longterm immunity and cross protection and the antibodies in the sera and mucosal sites were long lived [53, 54] .

In conclusion, our findings revealed that the mucosal immunization of mice with recombinant L. casei expressing CTA1 conjugated sM2 can induce systemic and local, as well as cell mediated, immune responses against divergent influenza virus subtypes. Thus,

the recombinant *L. casei* expressing CTA1 conjugated consensus sM2 mucosal vaccine may be a promising vaccine candidate for influenza pandemic preparedness.

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--- START OF ARTICLE 102 ---

Severe Acute Respiratory Syndrome Coronavirus Viroporin 3a Activates the NLRP3 Inflammasome

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6361828/>

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Abstract: Nod-like receptor family, pyrin domain-containing 3 (NLRP3) regulates the secretion of proinflammatory cytokines interleukin 1 beta (IL-1 β) and IL-18. We previously showed that influenza virus M2 or encephalomyocarditis virus (EMCV) 2B proteins stimulate IL-1 β secretion following activation of the NLRP3 inflammasome. However, the mechanism by which severe acute respiratory syndrome coronavirus (SARS-CoV) activates the NLRP3 inflammasome remains unknown. Here, we provide direct evidence that SARS-CoV 3a protein activates the NLRP3 inflammasome in lipopolysaccharide-primed macrophages. SARS-CoV 3a was sufficient to cause the NLRP3 inflammasome activation. The ion channel activity of the 3a protein was essential for 3a-mediated IL-1 β secretion. While cells uninfected or infected with a lentivirus expressing a 3a protein defective in ion channel activity expressed NLRP3 uniformly throughout the cytoplasm, NLRP3 was redistributed to the perinuclear space in cells infected with a lentivirus expressing the 3a

protein. K(+) efflux and mitochondrial reactive oxygen species were important for SARS-CoV 3a-induced NLRP3 inflammasome activation. These results highlight the importance of viroporins, transmembrane pore-forming viral proteins, in virus-induced NLRP3 inflammasome activation.

Text: Severe acute respiratory syndrome coronavirus (SARS-CoV), a member of the genus Betacoronavirus within the family Coronaviridae, is an enveloped virus with a single-stranded positive-sense RNA genome of approximately 30 kb in length. The 5 two-thirds of the genome encodes large polyprotein precursors, open reading frame (ORF) 1 and ORF1b, which are proteolytically cleaved to generate 16 non-structural proteins (Tan et al., 2005) . The 3 one-third of the genome encodes four structural proteins, spike (S), envelope (E), matrix (M) and nucleocapsid (N), and non-structural proteins, along with a set of accessory proteins (3a, 3b, 6, 7a, 7b, 8a, 8b, and 9b) (Perlman and Dandekar, 2005; Tan et al., 2005) . SARS-CoV is the etiological agent of SARS (Drosten et al., 2003; Fouchier et al., 2003; Ksiazek et al., 2003; Kuiken et al., 2003; Peiris et al., 2003) . At least 8,098 laboratory-confirmed cases of human infection, with a fatality rate of 9.6%, were reported to the World Health Organization from November 2002 to July 2003. High levels of proinflammatory cytokines, including tumor necrosis factor (TNF)- α , interleukin (IL)-1 β , and IL-6, were detected in autopsy tissues from SARS patients (He et al., 2006) . Although dysregulation of inflammatory cytokines may be involved in lung injury and the pathogenesis of SARS-CoV, the underlying molecular mechanisms are not fully understood.

The innate immune systems utilizes pattern recognition receptors (PRRs) to detect pathogen-associated molecular patterns (Medzhitov, 2001; Kawai and Akira, 2010) . Recognition of virus infection plays an important role in limiting virus replication at the early stages of infection. Nod-like receptor family, pyrin domain-containing 3 (NLRP3) is activated by a wide variety of stimuli, including virus infection (Bauernfeind et al., 2011) . Four models describing activation of the NLRP3 inflammasome have been proposed thus far (Hornung and Latz, 2010; Schroder et al., 2010; Tschopp and Schroder, 2010) . First, the disturbances in intracellular ionic concentrations, including K⁺ efflux and Ca²⁺ influx, play an important role (Fernandes-Alnemri et al., 2007; Petrilli et al., 2007; Arlehamn et al., 2010; Ichinohe et al., 2010; Ito et al., 2012; Murakami et al., 2012; Munoz-Planillo et al., 2013) . Second, cathepsin B and L, which are specific lysosomal cysteine proteases, are though to play a role after phagocytosis of cholesterol crystals (Duewell et al., 2010) , fibrillar peptide amyloid-beta , silica crystals, and aluminum salts . Third is the release of reactive oxygen species (ROS) or mitochondrial DNA from damaged mitochondria (Zhou et

al., , 2011 Nakahira et al., 2011; Shimada et al., 2012) . Finally, viral RNA or RNA cleavage products generated by RNase L activate the NLRP3 inflammasome via the DExD/H-box helicase, DHX33 (Allen et al., 2009; Mitoma et al., 2013; Chen et al., 2014; Chakrabarti et al., 2015) . Upon activation, the NLRP3 is recruited to the mitochondria via association with mitochondrial antiviral signaling (MAVS) or mitofusin 2 expressed on the outer mitochondrial membrane Subramanian et al., 2013) ; these molecules then recruit the apoptosis-associated speck-like protein containing a caspase recruitment domain (ASC) and pro-caspase-1 to form the NLRP3 inflammasome. This event activates the downstream molecule, caspase-1, which catalyzes the proteolytic processing of pro-IL-1 β and pro-IL-18 into their active forms and stimulates their secretion (Kayagaki et al., 2015; Shi et al., 2015) .

It is increasingly evident that NLRP3 detects RNA viruses by sensing the cellular damage or distress induced by viroporins (Ichinohe et al., 2010; Ito et al., 2012; Triantafilou et al., 2013; Nieto-Torres et al., 2015) , transmembrane pore-forming proteins, encoded by certain RNA viruses; these proteins alter membrane permeability to ions by forming membrane channels (Tan et al., 2005; Chen and Ichinohe, 2015) . A recent study shows that the SARS-CoV E protein, which comprise only 76 amino acids, forms Ca²⁺ - permeable ion channels and activates the NLRP3 inflammasome (Nieto-Torres et al., 2015) . Although the E and 3a proteins of SARS-CoV, which comprise 274 amino acids and contain three transmembrane domains (Zeng et al., 2004; Lu et al., 2006) , are thought to act as Na⁺ /K⁺ and K⁺ channels, respectively (Wilson et al., 2004; Lu et al., 2006; Torres et al., 2007; Parthasarathy et al., 2008; Pervushin et al., 2009; Wang et al., 2011) , the role of the 3a protein in activating the NLRP3 inflammasome remains unknown. Here, we examined the role of the 3a protein in activating the NLRP3 inflammasome.

Six-week-old female C57BL/6 mice were purchased from The Jackson Laboratory. All animal experiments were approved by the Animal Committees of the Institute of Medical Science (The University of Tokyo).

Bone marrow-derived macrophages (BMMs) were prepared as described previously (Ichinohe et al., 2009) . In brief, bone marrow was obtained from the tibia and femur by flushing with Dulbecco's modified Eagle's medium (DMEM; Nacalai Tesque). Bone marrow cells were cultured for 5 days in DMEM supplemented with 30% L929 cell supernatant containing macrophage colony-stimulating factor, 10% heat-inactivated fetal bovine serum

(FBS), and L-glutamine (2 mM) at 37 °C/5% CO₂. HEK293FT cells (a human embryonic kidney cell line) and HeLa cells (a human epithelial carcinoma cell line) were maintained in DMEM supplemented with 10% FBS, penicillin (100 units/ml), and streptomycin (100 µg/ml) (Nacalai Tesque). MDCK cells (Madin-Darby canine kidney cells) and HT-1080 cells (a human fibrosarcoma cell line) were grown in Eagle's minimal essential medium (E-MEM; Nacalai Tesque) supplemented with 10% FBS, penicillin (100 units/ml), and streptomycin (100 µg/ml) (Nacalai Tesque).

Influenza A virus strain A/PR8 (H1N1) was grown at 35 °C for 2 days in the allantoic cavities of 10-day-old fertile chicken eggs (Ichinohe et al., 2009). The viral titer was quantified in a standard plaque assay using MDCK cells (Pang et al., 2013).

Plasmids cDNAs encoding the E and M proteins of SARS-CoV Frankfurt 1 strain (Matsuyama et al., 2005) were obtained by reverse transcription and PCR of total RNA extracted from SARS-CoV-infected Vero cells, followed by PCR amplification using specific primers. pcDNA3.1D-3a-V5His was provided by Ming-Fu Chang (National Taiwan University College of Medicine, Taipei, Taiwan). To generate the plasmids pLenti6-E-V5His, pLenti6-3a-V5His, and pLenti-M-V5His, cDNA fragments of E, 3a, and M were amplified from pcDNA3.1D-E-V5His, pcDNA3.1D-3a-V5His, and pcDNA3.1D-M-V5His using specific primer sets and then ligated into pLenti6-TOPO vectors (Invitrogen). To generate plasmids pCA7-flag-E, pCA7-flag-3a, and pCA7-flag-M, pCA7-HA-E, pCA7-HA-3a, and pCA7-HA-M, cDNA fragments of E, 3a, and M were amplified from pcDNA3.1D-E-V5His, pcDNA3.1D-3a-V5His, and pcDNA3.1D-M-V5His using specific primer sets, digested with EcoR I and Not I, and subcloned into the EcoR I-Not I sites of the pCA7-flag-ASC plasmid or pCA7-HA-M2 plasmid, respectively (Ito et al., 2012). To construct plasmids expressing the E mutant V25F, the mutated E fragments were amplified by inverse PCR with wildtype E-containing plasmids and specific primer sets. The PCR products were cleaved by Dpn I, ligated in a ligase- and T4 kinase-containing reaction and then transformed into DH5α competent cells (TOYOBO). To construct plasmids expressing the 3a mutant 3a-CS, fragments were amplified from wildtype 3a-containing plasmids using 3a-specific primer sets and transformed as described above.

HEK293FT cells were seeded in 24-well cluster plates and transfected with 1 µg pLenti6-E/3a/M-V5His, pLenti-GFP (green fluorescent protein), or pLenti-M2 using polyethylenimine (PEI) Max. At 24 h post-transfection, the cells were lysed with RIPA buffer (50 mM Tris-HCl,

1% NP-40, 0.05% sodium dodecyl sulfate (SDS), 150 mM NaCl and 1 mM EDTA). And the lysates were subjected to SDS-polyacrylamide gel electrophoresis (PAGE) followed by electroblotting onto polyvinylidene difluoride (PVDF) membranes. The membranes were incubated over night with mouse anti-V5-tag (R960-25, Invitrogen), mouse anti-influenza A virus M2 (14C2, Abcam), mouse anti-GFP (GF200, Nacalai Tesque), or rabbit antitubulin (DM1A, Santa Cruz) antibodies, followed by horseradish peroxidase-conjugated anti-mouse IgG (Jackson Immuno Research Laboratories) or anti-rabbit IgG (Invitrogen). After washing 3 times with washing buffer (0.05% Tween-20/PBS), the membranes were exposed using Chemi-Lumi One Super (Nacalai Tesque), and the chemiluminescent signals were captured by an ImageQuant LAS-4000 mini apparatus (GE Healthcare).

To generate lentiviruses expressing V5-tagged SARS-CoV E, 3a, and M proteins, the full-length cDNA encoding each viral protein was cloned into the pLenti6.3/V5-TOPO vector (Invitrogen) using the following primers: SARS-CoV E forward, 5'-caccatgtactcattcggttcgga-3', and reverse, 5'-gaccagaagatcaggaactc-3'; SARS-CoV 3a forward, 5caccatggattgtttatgagatt-3', and reverse, 5'-caaaggcacgctagtagtcg-3'; SARS-CoV M forward, 5'-caccatggcagacaacggtactat-3', and reverse, 5'-ctgtactagcaaagcaatat-3'. Sub-confluent monolayers of HEK293FT cells seeded in a collagen-coated dish (10 cm in diameter) were transfected with 3 µg of pLenti6.3/V5-TOPO vector expressing each viral protein or EGFP together with ViraPower Packaging Mix (Invitrogen) using Lipofectamine 2000 (Invitrogen). The supernatants containing lentiviruses were harvested and filtered through a 0.45 µm filter (Millipore) at 72-96 h post-transfection (Ito et al., 2012). The lentiviral titer was then quantified using HT-1080 cells as described previously.

Bone marrow-derived macrophages were plated at a density of 8×10^5 in 24-well plate and infected with A/PR8 influenza virus or lentivirus at a multiplicity of infection (MOI) of 5 or 0.2 for 1 h, respectively. Then, BMMs were stimulated with 1 µg/ml of LPS and cultured for additional 23 h in complete media. Supernatants were collected at 24 h post-infection and centrifuged to remove cell debris. The amount of IL-1β in the supernatants was measured in an enzyme-linked immunosorbent assay (ELISA) using paired antibodies (eBioscience) (Ichinohe et al., 2010).

To clarify the cellular localization of the wild-type and mutant 3a proteins of SARS-CoV, HeLa cells were cultured on coverslips and transfected with 1 µg of pCA7-flag-3a or pCD7-flag-3a-CS together with 0.5 µg of ER-mCherry or DsRed-Golgi (Ito et al., 2012). At 24 h

post-transfection, cells were fixed with 4% paraformaldehyde and permeabilized with 1% Triton X-100/PBS. After washing with PBS and blocking with 4% BSA/PBS, the cells were incubated with a mouse anti-flag antibody (M2, Sigma) followed by incubation with Alexa Fluor 488-conjugated goat anti-mouse IgG (H+L) (Life Technologies).

To observe the cellular distribution of NLRP3 in the E-or 3a-expressing cells, HeLa cells were cultured on coverslips and transfected with 1 µg of pCA7-HA-E, pCA7-HA-EV25F, pCA7-HA-3a, pCA7-HA-3a-CS, or pCA7 control vector together with 0.5 µg of pCA7-NLRP3. At 24 h post-transfection, cells were fixed and permeabilized with 4% paraformaldehyde and 1% Triton X-100/PBS. After washing and blocking, the cells were incubated with rabbit anti-HA (561, MBL) and mouse anti-NLRP3 (Cryo-2; AdipoGen) antibodies, followed by Alexa Fluor 488-conjugated goat anti-rabbit IgG (H+L) and Alexa Fluor 568-conjugated goat anti-mouse IgG (H+L) (Life Technologies). Fluorescent signals were observed by confocal microscopy (A1R + , Nikon).

Statistical significance was tested using a two-tailed Student's t-test. P-values < 0.05 were considered statistically significant.

We previously demonstrated that the influenza virus M2 protein (a proton-selective ion channel), its H37G mutant (which has lost its proton selectivity and enables the transport of other cations such as Na⁺ and K⁺), and the EMCV 2B protein (a Ca²⁺ channel) stimulates NLRP3 inflammasome-mediated IL-1β secretion (Ichinohe et al., 2010; Ito et al., 2012). In addition, the SARS-CoV E protein acts as a Ca²⁺-permeable ion channels that activates the NLRP3 inflammasome (Nieto-Torres et al., 2015). The fact that 3a protein of SARS-CoV acts as viroporin prompted us to examine whether it also triggers inflammasome activation. Thus, we first generated lentivirus plasmids expressing V5-tagged proteins and confirmed their expression in HEK293FT cells by immunoblot analysis (Figures 1A-C). We next transduced lipopolysaccharide (LPS)-primed BMMs with the lentiviruses expressing the SARS-CoV E, 3a, M, influenza virus M2, or EMCV 2B proteins. Consistent with previous reports (Ichinohe et al., Figure 1D). Similarly, the lentiviruses expressing the SARS-CoV E or 3a proteins stimulated IL-1β release from LPS-primed BMMs (Figure 1D). Furthermore, IL-1β secretion from LPS-primed BMMs co-infected with E- and 3a-expressing lentiviruses was significantly higher than that from SARS-CoV E-expressing lentivirus-infected cells (Figure 1E). These data indicated that the expression of SARS-CoV viroporin 3a is sufficient to stimulate IL-1β secretion by LPS-primed BMMs.

Previous studies demonstrated that the N-terminal 40 amino acids of the SARS-CoV E protein are important for ion channel formation, and that mutations N15A and V25F [located in the transmembrane domain (from amino acid residues 7-38)] prevent ion conductivity (Wilson et al., 2004; Torres et al., 2007; Verdía-Baguena et al., 2012). In addition, the SARS-CoV 3a protein contains a cysteine-rich domain (amino acid residues 127-133) that is involved in the formation of a homodimer to generate the ion channel (Lu et al., 2006; Chan et al., 2009). Thus, mutation of the cysteine-rich domain blocks the ion conductivity by the 3a protein (Chan et al., 2009). To this end, we substituted amino acids Cys-127, Cys-130, and Cys-133 within the cysteine-rich domain of the SARS-CoV 3a protein with serine to generate a lentivirus expressing the ion channel activity-loss mutant, 3a-CS (Chan et al., 2009; Figure 2A). To test whether the ion channel activity of the SARS-CoV 3a protein is required to stimulate secretion of IL-1 β , we transduced LPS-primed BMMs with lentiviruses expressing the SARS-CoV E, V25F, 3a, 3a-CS, or M proteins. Consistent with a previous report (Nieto-Torres et al., 2015), we found that the V25F mutant lentivirus failed to stimulate IL-1 β release from BMMs (Figure 2B). Notably, the 3a-CS mutant completely abrogated IL-1 β secretion (Figure 2B), suggesting that the ion channel activity of the 3a protein is required for SARS-CoV 3a-induced IL-1 β secretion.

FIGURE 4 | NLRP3 inflammasome activation by SARS-CoV 3a. HeLa cells were transfected with the expression plasmid encoding NLRP3 and that encoding HA-tagged SARS-CoV 3a, 3a-CS, E, or V25F, and by with a confocal microscope. Scale bars, 10 μ m. Data are representative of at least three independent experiments.

Next, we determined the subcellular localization of the SARS-CoV 3a protein using confocal microscopy. When the SARS-CoV Cell-free supernatants were collected at 24 h (lentiviruses) or 6 h (ATP) post-infection or stimulation, and analyzed for IL-1 β by ELISA. Data are representative of at least three independent experiments, and indicate the mean $\pm$ SD; ** $P < 0.01$ and *** $P < 0.001$.

3a protein was expressed in HeLa cells, we observed two main distribution patterns. Consistent with previous reports (Yu et al., 2004; Yuan et al., 2005), the 3a protein localized to the Golgi apparatus (Figure 3A). In addition, the 3a proteins concentrated in spot structures, which mainly localized to the endoplasmic reticulum (ER) (Figure 3B). By contrast, the 3a-CS mutant was concentrated in the Golgi apparatus rather than in the ER

and did not form spot structures (Figures 3A,B) . We next examined the intracellular localization of NLRP3. Activation of the NLRP3 inflammasome led to a redistribution from the cytosol to the perinuclear space, a process considered as a hallmark of NLRP3 activation (Zhou et al., 2011; Ito et al., 2012; Johnson et al., 2013; Moriyama et al., 2016) . Although cells expressing the ion channel activity-loss mutants 3a-CS or V25F uniformly expressed NLRP3 throughout the cytoplasm, it was redistributed to the perinuclear region in SARS-CoV 3a-or E-expressing cells (Figure 4) . Together, these data provide evidence that the ion channel activity of the SARS-CoV 3a protein is essential for triggering the NLRP3 inflammasome.

Both K⁺ Efflux and ROS Production Are Involved in the IL-1 β Release Induced by the SARS-CoV 3a Protein

Finally, we investigated the mechanism by which SARS-CoV 3a triggers NLRP3 inflammasome activation. A previous study showed that the 3a protein of SARS-CoV acts as a K⁺ channel (Lu et al., 2006) . In addition, K⁺ efflux is a well-known activator of the NLRP3 inflammasome (Mariathasan et al., 2006; Petrilli et al., 2007) . These observations prompted us to examine whether K⁺ efflux is required for 3a-mediated IL-1 β secretion. To this end, BMMs in K⁺-rich medium were infected with influenza A virus or lentiviruses expressing the SARS-CoV E or 3a proteins. In agreement with a previous result (Ichinohe et al., 2010) , we found that IL-1 β secretion caused by influenza virus was completely blocked when the extracellular K⁺ concentration was increased to 130 mM (Figure 5A) . The inhibitory effect of the K⁺-rich medium was also observed when cells were stimulated with lentiviruses expressing the SARS-CoV E or 3a proteins (Figure 5B) . Since mitochondrial ROS are important for NLRP3 inflammasome activation (Nakahira et al., 2011; Zhou et al., 2011) , we next stimulated BMMs with extracellular ATP or lentiviruses expressing the SARS-CoV E or 3a proteins in the presence or absence of the antioxidant, Mito-TEMPO, a scavenger that is specific for mitochondrial ROS (Trnka et al., 2009) . As reported previously (Nakahira et al., 2011; Ito et al., 2012) , treatment of BMMs with Mito-TEMPO completely blocked IL-1 β secretion in response to ATP (Figure 6A) . Similarly, IL-1 β release induced by the SARS-CoV E and 3a proteins was significantly inhibited by Mito-TEMPO (Figure 6B) . These observations indicate that the SARS-CoV 3a protein disrupts intracellular ionic concentrations and causes mitochondrial damages, thereby activating the NLRP3 inflammasome.

In summary, we found that the ion channel activity of SARS-CoV 3a protein is essential for activation of the NLRP3 inflammasome. In addition, both K⁺ efflux and mitochondrial ROS production are required for SARS-CoV 3a-mediated IL-1 β secretion.

Thus far, several models have been proposed to explain NLRP3 inflammasome activation by RNA viruses. First, viral RNA or RNA cleavage products generated by RNase L activate the NLRP3 inflammasome via the DExD/H-box helicase, DHX33 (Allen et al., 2009; Mitoma et al., 2013; Chen et al., 2014; Chakrabarti et al., 2015). Second, viroporins encoded by RNA viruses activate the NLRP3 inflammasome (Ichinohe et al., 2010; Ito et al., 2012; Triantafilou et al., 2013; Nieto-Torres et al., 2015). In the case of influenza virus, the proton-selective M2 ion channel in the acidic trans-Golgi network activates the NLRP3 inflammasome (Ichinohe et al., 2010). Interestingly, an M2 mutant in which histidine was substituted with glycine at position 37 (H37G), causing loss of proton selectivity, enables transport of other cations (i.e., Na⁺ and K⁺), thereby leading to enhanced secretion of IL-1 β from LPS-primed BMMs and dendritic cells when compared with the wild-type M2 protein.

In addition, the 2B proteins of EMCV, poliovirus, enterovirus 71 (EV71), and human rhinovirus (a member of the Picornaviridae family) triggers NLRP3 inflammasome activation by inducing Ca²⁺ flux from the ER and Golgi compartments (Ito et al., 2012; Triantafilou et al., 2013). Furthermore, hepatitis C virus stimulates NLRP3 inflammasome-mediated IL-1 β production through its p7 viroporin (Negash et al., 2013; Farag et al., 2017). Third, a recent study has demonstrated that the 3D protein of EV71 directly interacts with NLRP3 to facilitate the assembly of NLRP3 inflammasome complex (Wang et al., 2017).

In the case of SARS-CoV, the viroporin E forms Ca²⁺-permeable ion channels and activates the NLRP3 inflammasome (Nieto-Torres et al., 2015). In addition, another viroporin 3a was found to induce NLRP3 inflammasome activation (Yue et al., 2018). Although alanine substitution at Cys-133, which is required for dimer or tetramer formation (Lu et al., 2006), still allows activation of the NLRP3 inflammasome by interacting with caspase-1 (Yue et al., 2018), the ion channel activity-loss mutant 3a-CS (Cys-to-Ser substitution at positions Cys-127, Cys-130, and Cys-133) (Chan et al., 2009) completely abrogated IL-1 β secretion from LPS-primed BMMs, suggesting that the 3a protein of SARS-CoV has the ability to induce the NLRP3 inflammasome activation by multiple mechanisms. Previous studies show that the 3a protein of SARS-CoV is localized to the

plasma membrane (Minakshi and Padhan, 2014) and acts as a K⁺ channel (Lu et al., 2006), thereby (presumably) stimulating the K⁺ efflux at the plasma membrane. Indeed, we found that IL-1 β secretion caused by the 3a protein was significantly inhibited when the extracellular K⁺ concentration increased to 130 mM. Although it remains unclear whether another viroporin 8a of SARS-CoV (Castano-Rodriguez et al., 2018) activates the NLRP3 inflammasome, these data highlights the importance of viroporins in SARS-CoV-induced NLRP3 inflammasome activation. A better understanding of the mechanism that governs the NLRP3 inflammasome will facilitate the development of more effective interventions for the treatment of infectious diseases and increase our understanding of viral pathogenesis.

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A novel anti-mycobacterial function of mitogen-activated protein kinase phosphatase-1

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Abstract: **BACKGROUND:** Mycobacterium tuberculosis (MTB) is a major cause of morbidity and mortality in the world. To combat against this pathogen, immune cells release cytokines including tumor necrosis factor- α (TNF- α), which is pivotal in the development of protective granulomas. Our previous results showed that Bacillus Calmette Guerin (BCG), a mycobacterium used as a model to investigate the immune response against MTB,

stimulates the induction of TNF- α via mitogen-activated protein kinase (MAPK) in human blood monocytes. Since MAPK phosphatase-1 (MKP-1) is known to regulate MAPK activities, we examined whether MKP-1 plays a role in BCG-induced MAPK activation and cytokine expression. RESULTS: Primary human blood monocytes were treated with BCG and assayed for MKP-1 expression. Our results demonstrated that following exposure to BCG, there was an increase in the expression of MKP-1. Additionally, the induction of MKP-1 was regulated by p38 MAPK and extracellular signal-regulated kinase 1 and 2 (ERK1/2). Surprisingly, when MKP-1 expression was blocked by its specific siRNA, there was a significant decrease in the levels of phospho-MAPK (p38 MAPK and ERK1/2) and TNF- α inducible by BCG. CONCLUSIONS: Since TNF- α is pivotal in granuloma formation, the results indicated an unexpected positive function of MKP-1 against mycobacterial infection as opposed to its usual phosphatase activity.

Text: Tuberculosis (TB) remains a major cause of morbidity and mortality in the world, especially in the developing countries [1] . The disease is caused by *Mycobacterium tuberculosis* (MTB) and approximately one third of the world's population has been infected by this pathogen. In a recent report, World Health Organization (WHO) estimated that there are 9.2 million new TB cases around the world in 2006 [1] .

In response to MTB infection, induction of cytokines by immune cells is an important defense mechanism. The infected macrophages secrete intercellular signaling factors, proinflammatory cytokines, to mediate the inflammatory response leading to the formation of granuloma and induction of T-cell mediated immunity [2] . In order to understand TB pathogenesis, signaling pathways induced by mycobacteria have long been a subject of interest. Mitogen activated protein kinases (MAPKs) including extracellular signal-regulated kinase 1 and 2 (ERK1/2), p38 MAPK, and c-Jun N-terminal kinase (JNK) have been implicated as important cellular signaling molecules activated by mycobacteria [3] . Previous reports have shown that p38 MAPK and ERK1/2 are required in the induction of TNF- α expression in human monocytes infected with *M. tuberculosis* H37Rv [4] . We have further revealed the significant role of MAPKs in the signal transduction events of mycobacterial activation of primary human blood monocytes (PBMo) leading to cytokine expressions via the interaction with PKR [5] . However, the subsequent events as to how MAPK is regulated and how such regulation affects cytokine production in response to mycobacteria remain to be elucidated.

Since MAPKs are activated by phosphorylation, dephosphorylation of MAPKs seems to be an efficient process to inactivate their activities. It can be achieved by specific protein kinase phosphatases which can remove the phosphate group from MAPKs. Examples of these phosphatases include tyrosine phosphatases, serine/threonine phosphatases, and dual-specificity phosphatases (DUSPs). Some DUSPs are also known as MAPK phosphatases (MKPs) [6] [7] [8] . Currently, there are at least 10 MKPs identified, while MKP-1 is the most studied member of the family. The regulatory role of MKP-1 on cytokine induction is best demonstrated by MKP-1 knockout (KO) macrophages in response to lipopolysaccharide (LPS), a cell wall component of Gram-negative bacteria. MKP-1 KO macrophages showed prolonged phosphorylation of p38 MAPK and JNK as well as increased production of TNF- α in response to LPS treatment [9] . Consistent with these results, another group further revealed that LPS-treated MKP-1 KO bone marrow-derived macrophages show increased AP-1 DNA-binding activity [10] . Also, they showed that LPS-induced MKP-1 expression is dependent on myeloid differentiation factor 88 (MyD88) and TIR domain-containing adaptor inducing IFN- β (TRIF) [10] , thus demonstrating the role of MKP-1 in signal transduction.

Not only LPS, other TLR inducers including CpG, peptidoglycan, poly IC, and Pam 3 Cys can regulate cytokine expressions including TNF- α , IL-10 via MKP-1 activities [10, 11] . In these processes, MKP-1 serves to mitigate the undesirable effects of septic shock and maintain organ functions by restraining the inflammatory responses following bacterial infection. Another example of MKP-1 function is the immune response to *Staphylococcus aureus* (*S. aureus*), a Gram positive bacteria. There are higher levels of cytokine production including TNF- α , IL-6, and MIP-1 α in MKP-1 KO mice infected with *S. aureus* [12] . Also, the mice would have a rapid development of multiorgan dysfunction as well as faster mortality rate upon challenge with heat-killed *S. aureus* [12] . Taken together, these results suggest that MKP-1 protects the host from overactivation of the immune system in response to Gram negative or Gram positive bacteria.

In the past, it was believed that different MKP/DUSP family members have overlapping functions. However, the emergence of DUSP2 turned the concept up side down [13] . It was shown that DUSP2 behaves differently and is opposite to the function as stated above. In DUSP2 KO cells, they produced less inflammatory mediators, implying that DUSP2 may play a role in mediating instead of limiting inflammation. For instances, when DUSP2 KO macrophages were treated with LPS, there were less TNF, IL-6, nitric oxide, IL-12-producing cells when compared to those of the wild type counterparts [13] . When the DUSP2 KO

bone marrow-derived mast cells were first sensitized with immunoglobulin E (IgE) receptor (FcεRI) and then stimulated with dinitrophenol-heat stable antigen, they produced lower TNF mRNA levels, diminished IL-6 production, less phosphorylation of ERK1/2, p38 MAPK, and less transcriptional activities by Elk1 and NFAT-AP-1 [13] .

These unexpected positive regulations of immune cell functions by DUSP2 have been hypothesized to be due to crosstalks between MAPKs [13] . Stimulation of KO mast cells and macrophages showed increases in phosphorylation of JNK. Moreover, inhibition of JNK by small molecule inhibitors showed increases in phosphorylation of ERK [13] . The authors also showed that there were physical interactions of DUSP2 with ERK2, DUSP2 with JNK2, as well as DUSP2 and p38 MAPK after stimulation of the cells with dinitrophenol-heat stable antigen. Nevertheless, the details of the crosstalks between MAPKs and phosphatases need further investigation. Thus, the MKP family plays a critical role in the regulation of immune responses.

Innate immune response protects the host from MTB infection by secretion of cytokines including TNF-α in immune cells. Meanwhile, MAPK is one of the critical proteins in the regulation of immunity and cytokine expression. Since MAPK is regulated by MKP-1 in response to LPS and the activation of MAPK is important in BCG-induced cytokine expression, we hypothesize that MKP-1 plays a critical role in the immune regulation of BCG in human monocytes. We examined the involvement of MKP-1 in BCG-induced MAPK activation and its consequent cytokine expression. Here, we present evidences that MKP-1 plays an unexpected role in the regulation of cytokine induction by BCG through its control of MAPK phosphorylation.

It has been reported that many inducers including growth factors, LPS, peptidoglycan, and dexamethasone can stimulate the expression of MKP-1 in human macrophages, microglia, mast cells or fibroblasts [6] . To investigate the role of different TLR inducers in MKP-1 induction process in human blood monocytes, the level of MKP-1 mRNA was measured by quantitative polymerase chain reaction (QPCR) method. PBMo were isolated from primary human blood mononuclear cells and stimulated with Pam 3 Cys (TLR2 agonist), poly IC (TLR3 agonist), or LPS (TLR4 agonist) for 1 and 3 hours. Following exposure to Pam 3 Cys or LPS, there were significant inductions of MKP-1 mRNA levels within 1 hour of treatment (Figure 1A). These effects on MKP-1 induction continued for 3 hours post-treatment with

Pam 3 Cys (Figure 1A). In contrast, poly IC did not induce MKP-1 (Figure 1A). The results indicate that different inducers showed differential up-regulation of MKP-1 expression.

LPS has been extensively used to demonstrate the role of MKP-1 in immune response both in vivo and in vitro [9, 12] . To establish a foundation for interpretation of subsequent experimental results, LPS was used as a positive control for the induction of MKP-1 expression. To determine the levels of MKP-1 in response to LPS, kinetics of MKP-1 transcription were determined by QPCR. There was a significant induction of MKP-1 mRNA, which peaked as early as 1 hour upon LPS stimulation, and the levels gradually decreased over a course of 6 hours. These results showed that LPS induced MKP-1 expression (Figure 1B) .

Next, to demonstrate the induction of specific phosphatases by BCG, kinetics of MKP-1 expression in PBMo was studied by using QPCR during BCG treatment. Similar to the results produced by LPS, upon the addition of BCG (MOI = 1 CFU/cell), there was a significant induction of MKP-1 mRNA within 1 hour of BCG treatment as determined by Taqman probe specific for MKP-1 (Figure 2A). The effects lasted for at least 6 hours (Figure 2A).

To examine whether the changes of protein production were in parallel to that of the mRNA levels, the protein levels of MKP-1 were measured by Western blotting. In response to BCG, PBMo produced the MKP-1 protein as early as 30 minutes after treatment. The protein levels were maintained for 2 hours and dropped to basal levels at 3 hours (Figure 2B). The results demonstrated that there was MKP-1 induction in response to BCG activation in human monocytes.

It has been shown that inhibition of p38 MAPK either by specific inhibitor or siRNA reduced the expression of MKP-1 in LPS-or peptidoglycan-treated macrophages [14] . To determine the mechanisms involved in the BCG-induced MKP-1 expression, PBMo were pretreated with several inhibitors including PD98059 (inhibitor for MAP kinase kinase [MEK] or ERK1/2), SB203580 (inhibitor for p38 MAPK), SP600125 (inhibitor for JNK), and CAPE (inhibitor for NF- κ B) for 1 hour. A range of concentrations of each inhibitor was used to test their optimal concentrations and effects on cell viability and kinase inhibitions. BCG was added afterwards and total RNA was harvested. The results demonstrated that, with the

inhibition of ERK1/2 and p38 MAPK activities by their corresponding relatively specific inhibitors, MKP-1 expressions were significantly reduced (Figure 3). In addition, using higher dose of SB203580, we showed that the inhibition is increased further (data not shown). On the contrary, pretreatment of the cells with CAPE and SP600125 did not affect the induction of MKP-1 by BCG (Figure 3). These results suggest that BCG-induced MKP-1 expression is dependent on both p38 MAPK and ERK1/2.

Throughout the above experiments, the primary goal was to examine the induction of MKP-1 by BCG in human monocytes. Thus, to further examine the role of MKP-1 in BCG-induced signaling, transfection of siRNA into PBMo was used to knockdown the activity of MKP-1. To demonstrate that the MKP-1 siRNA can indeed knockdown the target gene, PBMo were first transfected with control or MKP-1 siRNA and then treated with BCG for 3 hours. Levels of MKP-1 mRNA were measured by RT-PCR method.

In Figure 4A , BCG stimulated MKP-1 expression (lanes 1 and 2). In MKP-1 siRNA transfected monocytes, induction of MKP-1 by BCG was significantly decreased (lanes 2 and 4). The results showed that the siRNA does abrogate the levels of MKP-1 mRNA.

To further determine whether MKP-1 siRNA affects BCGinduced MKP-1 at protein levels, PBMo were treated as above and MKP-1 proteins were measured by Western blotting. The results showed that BCG could induce MKP-1 proteins as usual for cells transfected with control siRNA (Figure 4B , lanes 1-3). However, the levels of BCGinduced MKP-1 protein expression were reduced in cells transfected with MKP-1 siRNA (Figure 4B , lanes 4-6). Together, the results suggest that MKP-1 siRNA not only reduced the MKP-1 mRNA in BCG treatment but also abrogated the BCG-induced MKP-1 protein.

As stated in the literature [9] , MKP-1 KO mice showed increased TNF- α production in response to LPS. On the basis of the above MKP-1 siRNA results, LPS was then used as a control to demonstrate the effects of this MKP-1 siRNA system. cytokine expression induced by LPS in MKP-1 siRNA transfected cells suggest that the siRNA system is effective in knocking down the MKP-1 expression and MKP-1 acts as a negative regulator in LPS-induced TNF- α expression.

To investigate the effect of MKP-1 siRNA on BCG-induced cytokine expression, the levels of TNF- α , IL-6 and IL-10 mRNA were measured by QPCR method. PBMo were transfected with either control or MKP-1 siRNA. Following exposure to BCG with control siRNA, there were significant inductions of TNF- α , IL-6 and IL-10 mRNA levels for 3 hours after treatment as previously reported ([5] and data not shown). Next, the effects of MKP-1 siRNA were examined on the cytokine expression induced by BCG. Surprisingly, there was a significant abrogation of BCG-induced TNF- α expression by MKP-1 siRNA (Figure 4D). With the knockdown of MKP-1, the level of BCG-induced TNF- α was only 60% compared to that of the control cells, while BCG-induced IL-6 and IL-10 were unchanged in MKP-1 siRNA transfected cells. The results revealed that MKP-1 plays a role in the induction of TNF- α expression upon BCG stimulation, which may be different from that of its conventional functions in which MKP-1 acts as a negative regulator in LPS-induced signaling pathways [7] .

The unexpected observations in cytokine expression lead to the investigation on the effects of MKP-1 siRNA on BCG-induced MAPK activation. MKP-1 was found to have a preferential substrate binding to p38 MAPK and JNK than ERK1/2 [7] . The phosphorylation status of MAPKs was assessed in control or MKP-1 siRNA transfected PBMo. Western blotting results demonstrated that BCG-induced both p38 MAPK and ERK1/2 phosphorylation in 15 minutes (data not shown) and peaked at 30 minutes, and then returned to basal levels in cells treated with the control siRNA (Figure 5). Similar to the results of cytokine expression, phosphorylation of both p38 MAPK and ERK1/2 in response to BCG was decreased in monocytes transfected with MKP-1 siRNA instead of the expected increase in phosphorylation (Figure 5). The results suggest that MKP-1 knockdown would result in reduced MAPK phosphorylation by BCG, implying that the reduced level of TNF- α production in BCG stimulated monocytes is due to reduced phosphorylation of MAPKs by MKP-1 siRNA.

This report presented evidences that a novel function of MKP-1 is uncovered in cytokine regulation in response to mycobacterial infection. BCG induces MKP-1 as a rapid response (Figure 2) . The induction mechanism of MKP-1 by BCG is dependent on both ERK1/2 and p38 MAPK (Figure 3). Using siRNA approach, the functions of MKP-1 can be examined in primary human monocytes. The results showed that the BCG-induced MAPKs activation as well as cytokine expression are downstream of MKP-1 (Figures 4D and 5) . Thus, MKP-1 is a critical signaling molecule that is involved in BCG-induced cytokine expression.

Previous reports have shown that MKP-1 induced by LPS or peptidoglycan is dependent on p38 MAPK [14] . Accordingly, BCG-induced MKP-1 can be inhibited by both p38 MAPK and ERK1/2 inhibitors. Interestingly, it has been shown that degradation of MKP-1 is reduced after ERK1/2 phosphorylation [15] . It can be hypothesized that BCG-induced MKP-1 proteins can be stabilized by ERK1/2 and the detailed mechanisms involved require more exploration. Also, since the inhibition of MKP-1 expression by both inhibitors (for p38 MAPK and ERK1/2) was not complete, it is believed that other proteins may be involved in the BCG-induced MKP-1 expression.

On the basis of the literature results on LPS effects (Figure 6), the original expectation for this project is that MKP-1 acts as a negative regulator. LPS-stimulated MKP-1 KO peritoneal macrophages showed prolonged phosphorylation of p38 MAPK and JNK as well as increased production of TNF- α [9] . In doing so, LPS-induced MKP-1 could BCG-induced MAPK phosphorylation is decreased by MKP-1 siRNA prevent prolonged TNF- α production as in sepsis which may lead to severe damage to the host. It was expected that BCG induces MKP-1 and its induction would correlate with the dephosphorylation of MAPKs including p38 MAPK. By blocking the MKP-1 using siRNA, it was expected to have increased p38 MAPK phosphorylation and prolonged TNF- α production in response to BCG. Nevertheless, our results shown here are diametrically opposite. One possibility for the unexpected results may be due to non-specific effects of transfection or siRNA. However, this was not the case since there was a prolonged and increased TNF- α expression after the MKP-1 siRNA-transfected monocytes were treated with LPS (Figure 4C).

There is now a new hypothesis to explain such paradoxical effects of MKP-1 in TNF- α regulation in which the phosphatase plays a role in positive regulation of TNF- α production in response to BCG as in the case of DUSP2 [13] . The structures of MKP-1 and DUSP2 are similar, with which they both contain a MAPK-interacting domain and a phosphatase catalytic site. By contrast, other DUSP may have extra domains, e.g., PEST [6] . Here, we postulate that the function of MKP-1 in BCG-induced signaling is similar to that of the DUSP2/PAC1.

Actually, the discovery of DUSP2 has initially created some paradoxical questions. As described, DUSP2 behaves differently from other MKP family members [13] . In DUSP2 KO macrophages treated with LPS, they produced less inflammatory mediators including less

TNF, IL-6, nitric oxide, and IL-12-producing cells, when compared to that of the wild type counterparts [13] . Indeed, the results of these published studies on DUSP2 studies are quite similar to that of our reported results here.

It is plausible that these unexpected positive regulations of immune cell functions by DUSP2 were due to crosstalks between MAPKs [13] . It was shown that there are interactions between JNK and ERK1/2 pathways [16] .

Here, we showed that the sustained activation of JNK blocks ERK activation (Figure 6). In the DUSP2 situation, stimulation of KO mast cells and macrophages shows increased phosphorylation of JNK, and inhibition of JNK by its own specific inhibitor restores phosphorylation of ERK1/2 [13] .

In the BCG-MKP-1 situation, there is an early phosphorylation of p38 MAPK and ERK1/2. Therefore, it is possible that JNK may play a role in the crosstalk interaction of MAPK. However, our preliminary data suggest that the level of phosphorylated JNK was not increased in PBMo MKP-1 plays a critical role in the regulation of cytokine expression upon mycobacterial infection Figure 6 MKP-1 plays a critical role in the regulation of cytokine expression upon mycobacterial infection. LPS model was provided according to literature findings (Left). In this scenario, LPS activates MKP-1, which in turn dephosphorylates and deactivates phospho-p38 MAPK, resulting in less TNF- α induction. However, the situation in DHP-HSA activation of DUSP2 is more complicated (Middle), since the phosphatase activity causes subsequent inhibition of phospho-JNK which leads to the derepression of phospho-p38 MAPK. Consequently, the combined effects of this cascade results in more TNF- α expression. The unexpected antimycobacterial role of MKP-1 (Right) may be explained by events similar to the DUSP2 effects. In this case (Right), there was an inhibition of unknown pathways or kinases downstream of MKP-1, and the unknown factor in turn inhibits MAPKs activation leading to more TNF- α induction. The details and kinase targets are yet to be identified. transfected with MKP-1 siRNA (data not shown). Thus, the details of the crosstalk between MAPKs need further investigation. Here, we present a model to summarize the results and to hypothesize the existence of an as yet unidentified intermediary factor or factors in the pathways downstream of MKP-1 effects in the BCG-induced signaling cascade. The unexpected antimycobacterial role of MKP-1 (Figure 6) may be explained by events similar to the DUSP2 effects. In this case, BCG induces MKP-1 expression while also activates MAPKs including p38 MAPK and ERK1/2. Downstream of

MKP-1, there is an inhibition of unknown pathways or kinases. The unknown factor in turn inhibits MAPKs activation, which ultimately leads to more TNF- α induction (Figure 6).

In summary, MKP-1 plays a critical role in the regulation of cytokine expression upon mycobacterial infection. Inhibition of unknown pathways or kinases downstream of MKP-1, which in turn inhibits MAPKs activation, may be used to explain the novel function of MKP-1 in enhancing MAPK activity and consequent TNF- α expression following BCG treatment (Figure 6). Taken together, the role of MAPK crosstalks need further exploration. (3) TNF- α , 30 cycles (TM = 56°C), upstream, 5'-GGCTCCAGGCGGTGCTTGTTTC-3', downstream, 5'-AGACGGCGATGCGGCTGATG-3'. PCR products were analyzed on a 1% agarose gel with ethidium bromide and visualized under ultraviolet light. In order to check the size of the PCR products, 1 kb Plus DNA Lad-der™ (Invitrogen, USA) was run along with the PCR products.

To perform QPCR, the levels of MKP-1, and TNF- α mRNA as well as the reference gene GAPDH (as internal control) were assayed by the gene-specific Assays-on-Demand reagent kits (Applied Biosystems, USA). All samples were run in duplicates or triplicates and with no template controls on an ABI Prism 7700 Sequence Detector. The analysis method of QPCR was the comparative cycle number to threshold (C_T) method as described in user bulletin no. 2 of the ABI Prism 7700 Sequence Detection System. The number of C_T of the targeted genes was normalized to that of GAPDH in each sample (Δ C_T). The C_T value of the treated cells was compared with that of the untreated or mock-treated cells ($\Delta\Delta$ C_T). The relative gene expression of the targeted genes (fold induction) was calculated as $2^{-\Delta\Delta C_T}$.

Total cellular proteins were extracted by lysing cells in lysis buffer containing 1% Triton X-100, 0.5% NP-40, 150 mM NaCl, 10 mM Tris-HCl (pH 7.4), 1 mM EDTA, 1 mM EGTA (pH 8.0), 1% SDS, 0.2 mg/ml PMSF, 1 μ g/ml aprotinin, 1 mM sodium orthovanadate, 2 μ g/ml pepstatin, 2 μ g/ml leupeptin, and 50 mM sodium fluoride for 5 minutes. The homogenate was then boiled for 10 minutes and stored at -70°C until use. The concentrations of total protein in cell extracts were determined by BCA™ Protein Assay Kit (Pierce, IL, USA).

Western blot was done as described [20]. Equal amounts of protein were separated by 10% SDS-PAGE, electroblotted onto nitrocellulose membranes (Schleicher & Schuell), and

followed by probing with specific antibodies for Actin, MKP-1 (Santa Cruz Biotech., USA), phospho-p38 MAPK, phospho-ERK1/2 (Cell Signaling, USA). After three washes, the membranes were incubated with the corresponding secondary antibodies. The bands were detected using the Enhanced Chemiluminescence System (Amersham Pharmacia Biotech) as per the manufacturer's instructions.

Transfection of siRNA into human monocytes was done as described [21]. MKP-1 siRNA included (i) MKP1-HSS102982, AAACGCUUCGUAUCCUCCUUUGAGG; (ii) MKP1-HSS102983, UUAUGCCCAAGGCAUCCAG-CAUGUC; and (iii) MKP1-HSS102984, UGAUG-GAGUCUAUGAAGUCAUGGC. MKP-1 knockdown in PBMo was conducted by using MKP1-HSS102983 only or a pool of the above three different MKP-1 Stealth™ Select RNAi (ratio = 1:1:1, 200 nM, Invitrogen, USA). Stealth™ RNAi Negative Control Duplex (200 nM) was used as a control for sequence independent effects for the siRNA transfection. Transfection of monocytes was done by using jetPEI™ DNA transfection reagent (Polyplus Transfection, USA) according to the manufacturer's instructions. After transfecting the cells for 24 h, the transfectants were treated with different inducers as described above.

Statistical analysis was performed by Student's t test. Differences were considered statistically significant when p values were less than 0.05.

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Virus-Vectored Influenza Virus Vaccines

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Abstract: Despite the availability of an inactivated vaccine that has been licensed for >50 years, the influenza virus continues to cause morbidity and mortality worldwide. Constant evolution of circulating influenza virus strains and the emergence of new strains diminishes the effectiveness of annual vaccines that rely on a match with circulating influenza strains. Thus, there is a continued need for new, efficacious vaccines conferring cross-clade protection to avoid the need for biannual reformulation of seasonal influenza vaccines. Recombinant virus-vectored vaccines are an appealing alternative to classical inactivated vaccines because virus vectors enable native expression of influenza antigens, even from virulent influenza viruses, while expressed in the context of the vector that can improve immunogenicity. In addition, a vectored vaccine often enables delivery of the vaccine to sites of inductive immunity such as the respiratory tract enabling protection from influenza virus infection. Moreover, the ability to readily manipulate virus vectors to produce novel influenza vaccines may provide the quickest path toward a universal vaccine protecting against all influenza viruses. This review will discuss experimental virus-vectored vaccines for use in humans, comparing them to licensed vaccines and the hurdles faced for licensure of these next-generation influenza virus vaccines.

Text: Seasonal influenza is a worldwide health problem causing high morbidity and substantial mortality [1] [2] [3] [4] . Moreover, influenza infection often worsens preexisting medical conditions [5] [6] [7] . Vaccines against circulating influenza strains are available and updated annually, but many issues are still present, including low efficacy in the populations at greatest risk of complications from influenza virus infection, i.e., the young and elderly [8, 9] . Despite increasing vaccination rates, influenza-related hospitalizations are increasing [8, 10] , and substantial drug resistance has developed to two of the four currently approved anti-viral drugs [11, 12] . While adjuvants have the potential to improve efficacy and availability of current inactivated vaccines, live-attenuated and virus-vectored vaccines are still considered one of the best options for the induction of broad and efficacious immunity to the influenza virus [13] .

The general types of influenza vaccines available in the United States are trivalent inactivated influenza vaccine (TIV), quadrivalent influenza vaccine (QIV), and live attenuated influenza vaccine (LAIV; in trivalent and quadrivalent forms). There are three types of inactivated vaccines that include whole virus inactivated, split virus inactivated, and subunit vaccines. In split virus vaccines, the virus is disrupted by a detergent. In subunit vaccines, HA and NA have been further purified by removal of other viral components. TIV is administered intramuscularly and contains three or four inactivated viruses, i.e., two type A strains (H1 and H3) and one or two type B strains. TIV efficacy is measured by induction of humoral responses to the hemagglutinin (HA) protein, the major surface and attachment glycoprotein on influenza. Serum antibody responses to HA are measured by the hemagglutination-inhibition (HI) assay, and the strain-specific HI titer is considered the gold-standard correlate of immunity to influenza where a four-fold increase in titer post-vaccination, or a HI titer of $\geq 1:40$ is considered protective [4, 14] . Protection against clinical disease is mainly conferred by serum antibodies; however, mucosal IgA antibodies also may contribute to resistance against infection. Split virus inactivated vaccines can induce neuraminidase (NA)-specific antibody responses [15] [16] [17] , and anti-NA antibodies have been associated with protection from infection in humans [18] [19] [20] [21] [22] . Currently, NA-specific antibody responses are not considered a correlate of protection [14] . LAIV is administered as a nasal spray and contains the same three or four influenza virus strains as inactivated vaccines but on an attenuated vaccine backbone [4] . LAIV are temperature-sensitive and cold-adapted so they do not replicate effectively at core body temperature, but replicate in the mucosa of the nasopharynx [23] . LAIV immunization induces serum antibody responses, mucosal antibody responses (IgA), and T cell responses. While robust serum antibody and nasal wash (mucosal) antibody responses are associated with protection from infection, other immune responses, such as CD8 + cytotoxic lymphocyte (CTL) responses may contribute to protection and there is not a clear correlate of immunity for LAIV [4, 14, 24] .

Currently licensed influenza virus vaccines suffer from a number of issues. The inactivated vaccines rely on specific antibody responses to the HA, and to a lesser extent NA proteins for protection. The immunodominant portions of the HA and NA molecules undergo a constant process of antigenic drift, a natural accumulation of mutations, enabling virus evasion from immunity [9, 25] . Thus, the circulating influenza A and B strains are reviewed annually for antigenic match with current vaccines, Replacement of vaccine strains may occur regularly, and annual vaccination is recommended to assure protection [4, 26, 27] . For the northern hemisphere, vaccine strain selection occurs in February and then manufacturers begin production, taking at least six months to produce the millions of

vaccine doses required for the fall [27] . If the prediction is imperfect, or if manufacturers have issues with vaccine production, vaccine efficacy or availability can be compromised [28] . LAIV is not recommended for all populations; however, it is generally considered to be as effective as inactivated vaccines and may be more efficacious in children [4, 9, 24] . While LAIV relies on antigenic match and the HA and NA antigens are replaced on the same schedule as the TIV [4, 9] , there is some suggestion that LAIV may induce broader protection than TIV due to the diversity of the immune response consistent with inducing virus-neutralizing serum and mucosal antibodies, as well as broadly reactive T cell responses [9, 23, 29] . While overall both TIV and LAIV are considered safe and effective, there is a recognized need for improved seasonal influenza vaccines [26] . Moreover, improved understanding of immunity to conserved influenza virus antigens has raised the possibility of a universal vaccine, and these universal antigens will likely require novel vaccines for effective delivery [30] [31] [32] .

Virus-vectored vaccines share many of the advantages of LAIV, as well as those unique to the vectors. Recombinant DNA systems exist that allow ready manipulation and modification of the vector genome. This in turn enables modification of the vectors to attenuate the virus or enhance immunogenicity, in addition to adding and manipulating the influenza virus antigens. Many of these vectors have been extensively studied or used as vaccines against wild type forms of the virus. Finally, each of these vaccine vectors is either replication-defective or causes a self-limiting infection, although like LAIV, safety in immunocompromised individuals still remains a concern [4, 13, [33] [34] [35] . Table 1 summarizes the benefits and concerns of each of the virus-vectored vaccines discussed here.

There are 53 serotypes of adenovirus, many of which have been explored as vaccine vectors. A live adenovirus vaccine containing serotypes 4 and 7 has been in use by the military for decades, suggesting adenoviruses may be safe for widespread vaccine use [36] . However, safety concerns have led to the majority of adenovirus-based vaccine development to focus on replication-defective vectors. Adenovirus 5 (Ad5) is the most-studied serotype, having been tested for gene delivery and anti-cancer agents, as well as for infectious disease vaccines.

Adenovirus vectors are attractive as vaccine vectors because their genome is very stable and there are a variety of recombinant systems available which can accommodate up to 10

kb of recombinant genetic material [37] . Adenovirus is a non-enveloped virus which is relatively stable and can be formulated for long-term storage at 4 °C, or even storage up to six months at room temperature [33] . Adenovirus vaccines can be grown to high titers, exceeding 10¹⁰ plaque forming units (PFU) per mL when cultured on 293 or PER.C6 cells [38] , and the virus can be purified by simple methods [39] . Adenovirus vaccines can also be delivered via multiple routes, including intramuscular injection, subcutaneous injection, intradermal injection, oral delivery using a protective capsule, and by intranasal delivery. Importantly, the latter two delivery methods induce robust mucosal immune responses and may bypass preexisting vector immunity [33] . Even replication-defective adenovirus vectors are naturally immunostimulatory and effective adjuvants to the recombinant antigen being delivered. Adenovirus has been extensively studied as a vaccine vector for human disease. The first report using adenovirus as a vaccine vector for influenza demonstrated immunogenicity of recombinant adenovirus 5 (rAd5) expressing the HA of a swine influenza virus, A/Swine/Iowa/1999 (H3N2). Intramuscular immunization of mice with this construct induced robust neutralizing antibody responses and protected mice from challenge with a heterologous virus, A/Hong Kong/1/1968 (H3N2) [40] . Replication defective rAd5 vaccines expressing influenza HA have also been tested in humans. A rAd5-HA expressing the HA from A/Puerto Rico/8/1934 (H1N1; PR8) was delivered to humans epicutaneously or intranasally and assayed for safety and immunogenicity. The vaccine was well tolerated and induced seroconversion with the intranasal administration had a higher conversion rate and higher geometric mean HI titers [41] . While clinical trials with rAd vectors have overall been successful, demonstrating safety and some level of efficacy, rAd5 as a vector has been negatively overshadowed by two clinical trial failures. The first trial was a gene therapy examination where high-dose intravenous delivery of an Ad vector resulted in the death of an 18-year-old male [42, 43] . The second clinical failure was using an Ad5-vectored HIV vaccine being tested as a part of a Step Study, a phase 2B clinical trial. In this study, individuals were vaccinated with the Ad5 vaccine vector expressing HIV-1 gag, pol, and nef genes. The vaccine induced HIV-specific T cell responses; however, the study was stopped after interim analysis suggested the vaccine did not achieve efficacy and individuals with high preexisting Ad5 antibody titers might have an increased risk of acquiring HIV-1 [44] [45] [46] . Subsequently, the rAd5 vaccine-associated risk was confirmed [47] . While these two instances do not suggest Ad-vector vaccines are unsafe or inefficacious, the umbra cast by the clinical trials notes has affected interest for all adenovirus vaccines, but interest still remains.

Immunization with adenovirus vectors induces potent cellular and humoral immune responses that are initiated through toll-like receptor-dependent and independent

pathways which induce robust pro-inflammatory cytokine responses. Recombinant Ad vaccines expressing HA antigens from pandemic H1N1 (pH1N1), H5 and H7 highly pathogenic avian influenza (HPAI) virus (HPAIV), and H9 avian influenza viruses have been tested for efficacy in a number of animal models, including chickens, mice, and ferrets, and been shown to be efficacious and provide protection from challenge [48, 49] . Several rAd5 vectors have been explored for delivery of non-HA antigens, influenza nucleoprotein (NP) and matrix 2 (M2) protein [29, [50] [51] [52] . The efficacy of non-HA antigens has led to their inclusion with HA-based vaccines to improve immunogenicity and broaden breadth of both humoral and cellular immunity [53, 54] . However, as both CD8 + T cell and neutralizing antibody responses are generated by the vector and vaccine antigens, immunological memory to these components can reduce efficacy and limit repeated use [48] .

One drawback of an Ad5 vector is the potential for preexisting immunity, so alternative adenovirus serotypes have been explored as vectors, particularly non-human and uncommon human serotypes. Non-human adenovirus vectors include those from non-human primates (NHP), dogs, sheep, pigs, cows, birds and others [48, 55] . These vectors can infect a variety of cell types, but are generally attenuated in humans avoiding concerns of preexisting immunity. Swine, NHP and bovine adenoviruses expressing H5 HA antigens have been shown to induce immunity comparable to human rAd5-H5 vaccines [33, 56] . Recombinant, replication-defective adenoviruses from low-prevalence serotypes have also been shown to be efficacious. Low prevalence serotypes such as adenovirus types 3, 7, 11, and 35 can evade anti-Ad5 immune responses while maintaining effective antigen delivery and immunogenicity [48, 57] . Prime-boost strategies, using DNA or protein immunization in conjunction with an adenovirus vaccine booster immunization have also been explored as a means to avoid preexisting immunity [52] .

Adeno-associated viruses (AAV) were first explored as gene therapy vectors. Like rAd vectors, rAAV have broad tropism infecting a variety of hosts, tissues, and proliferating and non-proliferating cell types [58] . AAVs had been generally not considered as vaccine vectors because they were widely considered to be poorly immunogenic. A seminal study using AAV-2 to express a HSV-2 glycoprotein showed this virus vaccine vector effectively induced potent CD8 + T cell and serum antibody responses, thereby opening the door to other rAAV vaccine-associated studies [59, 60] .

AAV vector systems have a number of engaging properties. The wild type viruses are non-pathogenic and replication incompetent in humans and the recombinant AAV vector systems are even further attenuated [61] . As members of the parvovirus family, AAVs are small non-enveloped viruses that are stable and amenable to long-term storage without a cold chain. While there is limited preexisting immunity, availability of non-human strains as vaccine candidates eliminates these concerns. Modifications to the vector have increased immunogenicity, as well [60] .

There are limited studies using AAVs as vaccine vectors for influenza. An AAV expressing an HA antigen was first shown to induce protective in 2001 [62] . Later, a hybrid AAV derived from two non-human primate isolates (AAVrh32.33) was used to express influenza NP and protect against PR8 challenge in mice [63] . Most recently, following the 2009 H1N1 influenza virus pandemic, rAAV vectors were generated expressing the HA, NP and matrix 1 (M1) proteins of A/Mexico/4603/2009 (pH1N1), and in murine immunization and challenge studies, the rAAV-HA and rAAV-NP were shown to be protective; however, mice vaccinated with rAAV-HA + NP + M1 had the most robust protection. Also, mice vaccinated with rAAV-HA + rAAV-NP + rAAV-M1 were also partially protected against heterologous (PR8, H1N1) challenge [63] . Most recently, an AAV vector was used to deliver passive immunity to influenza [64, 65] . In these studies, AAV (AAV8 and AAV9) was used to deliver an antibody transgene encoding a broadly cross-protective anti-influenza monoclonal antibody for in vivo expression. Both intramuscular and intranasal delivery of the AAVs was shown to protect against a number of influenza virus challenges in mice and ferrets, including H1N1 and H5N1 viruses [64, 65] . These studies suggest that rAAV vectors are promising vaccine and immunoprophylaxis vectors. To this point, while approximately 80 phase I, I/II, II, or III rAAV clinical trials are open, completed, or being reviewed, these have focused upon gene transfer studies and so there is as yet limited safety data for use of rAAV as vaccines [66] .

Alphaviruses are positive-sense, single-stranded RNA viruses of the *Togaviridae* family. A variety of alphaviruses have been developed as vaccine vectors, including Semliki Forest virus (SFV), Sindbis (SIN) virus, Venezuelan equine encephalitis (VEE) virus, as well as chimeric viruses incorporating portions of SIN and VEE viruses. The replication defective vaccines or replicons do not encode viral structural proteins, having these portions of the genome replaced with transgenic material.

The structural proteins are provided in cell culture production systems. One important feature of the replicon systems is the self-replicating nature of the RNA. Despite the partial viral genome, the RNAs are self-replicating and can express transgenes at very high levels [67] .

SIN, SFV, and VEE have all been tested for efficacy as vaccine vectors for influenza virus [68] [69] [70] [71] . A VEE-based replicon system encoding the HA from PR8 was demonstrated to induce potent HA-specific immune response and protected from challenge in a murine model, despite repeated immunization with the vector expressing a control antigen, suggesting preexisting immunity may not be an issue for the replicon vaccine [68] . A separate study developed a VEE replicon system expressing the HA from A/Hong Kong/156/1997 (H5N1) and demonstrated varying efficacy after in ovo vaccination or vaccination of 1-day-old chicks [70] . A recombinant SIN virus was used as a vaccine vector to deliver a CD8 + T cell epitope only. The well-characterized NP epitope was transgenically expressed in the SIN system and shown to be immunogenic in mice, priming a robust CD8 + T cell response and reducing influenza virus titer after challenge [69] . More recently, a VEE replicon system expressing the HA protein of PR8 was shown to protect young adult (8-week-old) and aged (12-month-old) mice from lethal homologous challenge [72] .

The VEE replicon systems are particularly appealing as the VEE targets antigen-presenting cells in the lymphatic tissues, priming rapid and robust immune responses [73] . VEE replicon systems can induce robust mucosal immune responses through intranasal or subcutaneous immunization [72] [73] [74] , and subcutaneous immunization with virus-like replicon particles (VRP) expressing HA-induced antigen-specific systemic IgG and fecal IgA antibodies [74] . VRPs derived from VEE virus have been developed as candidate vaccines for cytomegalovirus (CMV). A phase I clinical trial with the CMV VRP showed the vaccine was immunogenic, inducing CMV-neutralizing antibody responses and potent T cell responses. Moreover, the vaccine was well tolerated and considered safe [75] . A separate clinical trial assessed efficacy of repeated immunization with a VRP expressing a tumor antigen. The vaccine was safe and despite high vector-specific immunity after initial immunization, continued to boost transgene-specific immune responses upon boost [76] . While additional clinical data is needed, these reports suggest alphavirus replicon systems or VRPs may be safe and efficacious, even in the face of preexisting immunity.

Baculovirus has been extensively used to produce recombinant proteins. Recently, a baculovirus-derived recombinant HA vaccine was approved for human use and was first available for use in the United States for the 2013-2014 influenza season [4] .

Baculoviruses have also been explored as vaccine vectors. Baculoviruses have a number of advantages as vaccine vectors. The viruses have been extensively studied for protein expression and for pesticide use and so are readily manipulated. The vectors can accommodate large gene insertions, show limited cytopathic effect in mammalian cells, and have been shown to infect and express genes of interest in a spectrum of mammalian cells [77] . While the insect promoters are not effective for mammalian gene expression, appropriate promoters can be cloned into the baculovirus vaccine vectors.

Baculovirus vectors have been tested as influenza vaccines, with the first reported vaccine using *Autographa californica* nuclear polyhedrosis virus (AcNPV) expressing the HA of PR8 under control of the CAG promoter (AcCAG-HA) [77] . Intramuscular, intranasal, intradermal, and intraperitoneal immunization of mice with AcCAG-HA elicited HA-specific antibody responses, however only intranasal immunization provided protection from lethal challenge. Interestingly, intranasal immunization with the wild type AcNPV also resulted in protection from PR8 challenge. The robust innate immune response to the baculovirus provided non-specific protection from subsequent influenza virus infection [78] . While these studies did not demonstrate specific protection, there were antigen-specific immune responses and potential adjuvant effects by the innate response.

Baculovirus pseudotype viruses have also been explored. The G protein of vesicular stomatitis virus controlled by the insect polyhedron promoter and the HA of A/Chicken/Hubei/327/2004 (H5N1) HPAIV controlled by a CMV promoter were used to generate the BV-G-HA. Intramuscular immunization of mice or chickens with BV-G-HA elicited strong HI and VN serum antibody responses, IFN- γ responses, and protected from H5N1 challenge [79] . A separate study demonstrated efficacy using a bivalent pseudotyped baculovirus vector [80] .

Baculovirus has also been used to generate an inactivated particle vaccine. The HA of A/Indonesia/CDC669/2006(H5N1) was incorporated into a commercial baculovirus vector controlled by the e1 promoter from White Spot Syndrome Virus. The resulting recombinant virus was propagated in insect (Sf9) cells and inactivated as a particle vaccine [81, 82] . Intranasal delivery with cholera toxin B as an adjuvant elicited robust HI titers and

protected from lethal challenge [81] . Oral delivery of this encapsulated vaccine induced robust serum HI titers and mucosal IgA titers in mice, and protected from H5N1 HPAIV challenge. More recently, co-formulations of inactivated baculovirus vectors have also been shown to be effective in mice [83] .

While there is growing data on the potential use of baculovirus or pseudotyped baculovirus as a vaccine vector, efficacy data in mammalian animal models other than mice is lacking. There is also no data on the safety in humans, reducing enthusiasm for baculovirus as a vaccine vector for influenza at this time.

Newcastle disease virus (NDV) is a single-stranded, negative-sense RNA virus that causes disease in poultry. NDV has a number of appealing qualities as a vaccine vector. As an avian virus, there is little or no preexisting immunity to NDV in humans and NDV propagates to high titers in both chicken eggs and cell culture. As a paramyxovirus, there is no DNA phase in the virus lifecycle reducing concerns of integration events, and the levels of gene expression are driven by the proximity to the leader sequence at the 3' end of the viral genome. This gradient of gene expression enables attenuation through rearrangement of the genome, or by insertion of transgenes within the genome. Finally, pathogenicity of NDV is largely determined by features of the fusion protein enabling ready attenuation of the vaccine vector [84] .

Reverse genetics, a method that allows NDV to be rescued from plasmids expressing the viral RNA polymerase and nucleocapsid proteins, was first reported in 1999 [85, 86] . This process has enabled manipulation of the NDV genome as well as incorporation of transgenes and the development of NDV vectors. Influenza was the first infectious disease targeted with a recombinant NDV (rNDV) vector. The HA protein of A/WSN/1933 (H1N1) was inserted into the Hitchner B1 vaccine strain. The HA protein was expressed on infected cells and was incorporated into infectious virions. While the virus was attenuated compared to the parental vaccine strain, it induced a robust serum antibody response and protected against homologous influenza virus challenge in a murine model of infection [87] . Subsequently, rNDV was tested as a vaccine vector for HPAIV having varying efficacy against H5 and H7 influenza virus infections in poultry [88] [89] [90] [91] [92] [93] [94] . These vaccines have the added benefit of potentially providing protection against both the influenza virus and NDV infection.

NDV has also been explored as a vaccine vector for humans. Two NHP studies assessed the immunogenicity and efficacy of an rNDV expressing the HA or NA of A/Vietnam/1203/2004 (H5N1; VN1203) [95, 96]. Intranasal and intratracheal delivery of the rNDV-HA or rNDV-NA vaccines induced both serum and mucosal antibody responses and protected from HPAIV challenge [95, 96]. NDV has limited clinical data; however, phase I and phase I/II clinical trials have shown that the NDV vector is well-tolerated, even at high doses delivered intravenously [44, 97]. While these results are promising, additional studies are needed to advance NDV as a human vaccine vector for influenza.

Parainfluenza virus type 5 (PIV5) is a paramyxovirus vaccine vector being explored for delivery of influenza and other infectious disease vaccine antigens. PIV5 has only recently been described as a vaccine vector [98]. Similar to other RNA viruses, PIV5 has a number of features that make it an attractive vaccine vector. For example, PIV5 has a stable RNA genome and no DNA phase in virus replication cycle reducing concerns of host genome integration or modification. PIV5 can be grown to very high titers in mammalian vaccine cell culture substrates and is not cytopathic allowing for extended culture and harvest of vaccine virus [98, 99]. Like NDV, PIV5 has a 3'-to 5' gradient of gene expression and insertion of transgenes at different locations in the genome can variably attenuate the virus and alter transgene expression [100]. PIV5 has broad tropism, infecting many cell types, tissues, and species without causing clinical disease, although PIV5 has been associated with -kennel coughll in dogs [99]. A reverse genetics system for PIV5 was first used to insert the HA gene from A/Udorn/307/72 (H3N2) into the PIV5 genome between the hemagglutinin-neuraminidase (HN) gene and the large (L) polymerase gene. Similar to NDV, the HA was expressed at high levels in infected cells and replicated similarly to the wild type virus, and importantly, was not pathogenic in immunodeficient mice [98]. Additionally, a single intranasal immunization in a murine model of influenza infection was shown to induce neutralizing antibody responses and protect against a virus expressing homologous HA protein [98]. PIV5 has also been explored as a vaccine against HPAIV. Recombinant PIV5 vaccines expressing the HA or NP from VN1203 were tested for efficacy in a murine challenge model. Mice intranasally vaccinated with a single dose of PIV5-H5 vaccine had robust serum and mucosal antibody responses, and were protected from lethal challenge. Notably, although cellular immune responses appeared to contribute to protection, serum antibody was sufficient for protection from challenge [100, 101]. Intramuscular immunization with PIV5-H5 was also shown to be effective at inducing neutralizing antibody responses and protecting against lethal influenza virus challenge [101]. PIV5 expressing the NP protein of HPAIV was also efficacious in the murine immunization and challenge model, where a single intranasal immunization induced

robust CD8 + T cell responses and protected against homologous (H5N1) and heterosubtypic (H1N1) virus challenge [102] .

Currently there is no clinical safety data for use of PIV5 in humans. However, live PIV5 has been a component of veterinary vaccines for -kennel coughll for >30 years, and veterinarians and dog owners are exposed to live PIV5 without reported disease [99] . This combined with preclinical data from a variety of animal models suggests that PIV5 as a vector is likely to be safe in humans. As preexisting immunity is a concern for all virus-vectored vaccines, it should be noted that there is no data on the levels of preexisting immunity to PIV5 in humans. However, a study evaluating the efficacy of a PIV5-H3 vaccine in canines previously vaccinated against PIV5 (kennel cough) showed induction of robust anti-H3 serum antibody responses as well as high serum antibody levels to the PIV5 vaccine, suggesting preexisting immunity to the PIV5 vector may not affect immunogenicity of vaccines even with repeated use [99] .

Poxvirus vaccines have a long history and the notable hallmark of being responsible for eradication of smallpox. The termination of the smallpox virus vaccination program has resulted in a large population of poxvirus-naï ve individuals that provides the opportunity for the use of poxviruses as vectors without preexisting immunity concerns [103] . Poxvirus-vectored vaccines were first proposed for use in 1982 with two reports of recombinant vaccinia viruses encoding and expressing functional thymidine kinase gene from herpes virus [104, 105] . Within a year, a vaccinia virus encoding the HA of an H2N2 virus was shown to express a functional HA protein (cleaved in the HA1 and HA2 subunits) and be immunogenic in rabbits and hamsters [106] . Subsequently, all ten of the primary influenza proteins have been expressed in vaccine virus [107] .

Early work with intact vaccinia virus vectors raised safety concerns, as there was substantial reactogenicity that hindered recombinant vaccine development [108] . Two vaccinia vectors were developed to address these safety concerns. The modified vaccinia virus Ankara (MVA) strain was attenuated by passage 530 times in chick embryo fibroblasts cultures. The second, New York vaccinia virus (NYVAC) was a plaque-purified clone of the Copenhagen vaccine strain rationally attenuated by deletion of 18 open reading frames [109] [110] [111] .

Modified vaccinia virus Ankara (MVA) was developed prior to smallpox eradication to reduce or prevent adverse effects of other smallpox vaccines [109] . Serial tissue culture passage of MVA resulted in loss of 15% of the genome, and established a growth restriction for avian cells. The defects affected late stages in virus assembly in non-avian cells, a feature enabling use of the vector as single-round expression vector in non-permissive hosts. Interestingly, over two decades ago, recombinant MVA expressing the HA and NP of influenza virus was shown to be effective against lethal influenza virus challenge in a murine model [112] . Subsequently, MVA expressing various antigens from seasonal, pandemic (A/California/04/2009, pH1N1), equine (A/Equine/Kentucky/1/81 H3N8), and HPAI (VN1203) viruses have been shown to be efficacious in murine, ferret, NHP, and equine challenge models [113] . MVA vaccines are very effective stimulators of both cellular and humoral immunity. For example, abortive infection provides native expression of the influenza antigens enabling robust antibody responses to native surface viral antigens. Concurrently, the intracellular influenza peptides expressed by the pox vector enter the class I MHC antigen processing and presentation pathway enabling induction of CD8 + T cell antiviral responses. MVA also induces CD4 + T cell responses further contributing to the magnitude of the antigen-specific effector functions [107, [112] [113] [114] [115] . MVA is also a potent activator of early innate immune responses further enhancing adaptive immune responses [116] . Between early smallpox vaccine development and more recent vaccine vector development, MVA has undergone extensive safety testing and shown to be attenuated in severely immunocompromised animals and safe for use in children, adults, elderly, and immunocompromised persons. With extensive pre-clinical data, recombinant MVA vaccines expressing influenza antigens have been tested in clinical trials and been shown to be safe and immunogenic in humans [117] [118] [119] . These results combined with data from other (non-influenza) clinical and pre-clinical studies support MVA as a leading viral-vectored candidate vaccine.

The NYVAC vector is a highly attenuated vaccinia virus strain. NYVAC is replication-restricted; however, it grows in chick embryo fibroblasts and Vero cells enabling vaccine-scale production. In non-permissive cells, critical late structural proteins are not produced stopping replication at the immature virion stage [120] . NYVAC is very attenuated and considered safe for use in humans of all ages; however, it predominantly induces a CD4 + T cell response which is different compared to MVA [114] . Both MVA and NYVAC provoke robust humoral responses, and can be delivered mucosally to induce mucosal antibody responses [121] . There has been only limited exploration of NYVAC as a vaccine vector for influenza virus; however, a vaccine expressing the HA from A/chicken/Indonesia/7/2003

(H5N1) was shown to induce potent neutralizing antibody responses and protect against challenge in swine [122] .

While there is strong safety and efficacy data for use of NYVAC or MVA-vectored influenza vaccines, preexisting immunity remains a concern. Although the smallpox vaccination campaign has resulted in a population of poxvirus-naï ve people, the initiation of an MVA or NYVAC vaccination program for HIV, influenza or other pathogens will rapidly reduce this susceptible population. While there is significant interest in development of pox-vectored influenza virus vaccines, current influenza vaccination strategies rely upon regular immunization with vaccines matched to circulating strains. This would likely limit the use and/or efficacy of poxvirus-vectored influenza virus vaccines for regular and seasonal use [13] . Intriguingly, NYVAC may have an advantage for use as an influenza vaccine vector, because immunization with this vector induces weaker vaccine-specific immune responses compared to other poxvirus vaccines, a feature that may address the concerns surrounding preexisting immunity [123] .

While poxvirus-vectored vaccines have not yet been approved for use in humans, there is a growing list of licensed poxvirus for veterinary use that include fowlpox-and canarypox-vectored vaccines for avian and equine influenza viruses, respectively [124, 125] . The fowlpox-vectored vaccine expressing the avian influenza virus HA antigen has the added benefit of providing protection against fowlpox infection. Currently, at least ten poxvirus-vectored vaccines have been licensed for veterinary use [126] . These poxvirus vectors have the potential for use as vaccine vectors in humans, similar to the first use of cowpox for vaccination against smallpox [127] . The availability of these non-human poxvirus vectors with extensive animal safety and efficacy data may address the issues with preexisting immunity to the human vaccine strains, although the cross-reactivity originally described with cowpox could also limit use.

Influenza vaccines utilizing vesicular stomatitis virus (VSV), a rhabdovirus, as a vaccine vector have a number of advantages shared with other RNA virus vaccine vectors. Both live and replication-defective VSV vaccine vectors have been shown to be immunogenic [128, 129] , and like Paramyxoviridae, the Rhabdoviridae genome has a 3'-to-5' gradient of gene expression enabling attention by selective vaccine gene insertion or genome rearrangement [130] . VSV has a number of other advantages including broad tissue tropism, and the potential for intramuscular or intranasal immunization. The latter delivery

method enables induction of mucosal immunity and elimination of needles required for vaccination. Also, there is little evidence of VSV seropositivity in humans eliminating concerns of preexisting immunity, although repeated use may be a concern. Also, VSV vaccine can be produced using existing mammalian vaccine manufacturing cell lines.

Influenza antigens were first expressed in a VSV vector in 1997. Both the HA and NA were shown to be expressed as functional proteins and incorporated into the recombinant VSV particles [131]. Subsequently, VSV-HA, expressing the HA protein from A/WSN/1933 (H1N1) was shown to be immunogenic and protect mice from lethal influenza virus challenge [129]. To reduce safety concerns, attenuated VSV vectors were developed. One candidate vaccine had a truncated VSV G protein, while a second candidate was deficient in G protein expression and relied on G protein expressed by a helper vaccine cell line to provide the virus receptor. Both vectors were found to be attenuated in mice, but maintained immunogenicity [128]. More recently, single-cycle replicating VSV vaccines have been tested for efficacy against H5N1 HPAIV. VSV vectors expressing the HA from A/Hong Kong/156/97 (H5N1) were shown to be immunogenic and induce cross-reactive antibody responses and protect against challenge with heterologous H5N1 challenge in murine and NHP models [132] [133] [134].

VSV vectors are not without potential concerns. VSV can cause disease in a number of species, including humans [135]. The virus is also potentially neuroinvasive in some species [136], although NHP studies suggest this is not a concern in humans [137]. Also, while the incorporation of the influenza antigen into the virion may provide some benefit in immunogenicity, changes in tropism or attenuation could arise from incorporation of different influenza glycoproteins. There is no evidence for this, however [134]. Currently, there is no human safety data for VSV-vectored vaccines. While experimental data is promising, additional work is needed before consideration for human influenza vaccination.

Current influenza vaccines rely on matching the HA antigen of the vaccine with circulating strains to provide strain-specific neutralizing antibody responses [4, 14, 24]. There is significant interest in developing universal influenza vaccines that would not require annual reformulation to provide protective robust and durable immunity. These vaccines rely on generating focused immune responses to highly conserved portions of the virus that are refractory to mutation [30] [31] [32]. Traditional vaccines may not be suitable for these

vaccination strategies; however, vectored vaccines that have the ability to be readily modified and to express transgenes are compatible for these applications.

The NP and M2 proteins have been explored as universal vaccine antigens for decades. Early work with recombinant viral vectors demonstrated that immunization with vaccines expressing influenza antigens induced potent CD8 + T cell responses [107, [138] [139] [140] [141] . These responses, even to the HA antigen, could be cross-protective [138] . A number of studies have shown that immunization with NP expressed by AAV, rAd5, alphavirus vectors, MVA, or other vector systems induces potent CD8 + T cell responses and protects against influenza virus challenge [52, 63, 69, 102, 139, 142] . As the NP protein is highly conserved across influenza A viruses, NP-specific T cells can protect against heterologous and even heterosubtypic virus challenges [30] .

The M2 protein is also highly conserved and expressed on the surface of infected cells, although to a lesser extent on the surface of virus particles [30] . Much of the vaccine work in this area has focused on virus-like or subunit particles expressing the M2 ectodomain; however, studies utilizing a DNA-prime, rAd-boost strategies to vaccinate against the entire M2 protein have shown the antigen to be immunogenic and protective [50] . In these studies, antibodies to the M2 protein protected against homologous and heterosubtypic challenge, including a H5N1 HPAIV challenge. More recently, NP and M2 have been combined to induce broadly cross-reactive CD8 + T cell and antibody responses, and rAd5 vaccines expressing these antigens have been shown to protect against pH1N1 and H5N1 challenges [29, 51] .

Historically, the HA has not been widely considered as a universal vaccine antigen. However, the recent identification of virus neutralizing monoclonal antibodies that cross-react with many subtypes of influenza virus [143] has presented the opportunity to design vaccine antigens to prime focused antibody responses to the highly conserved regions recognized by these monoclonal antibodies. The majority of these broadly cross-reactive antibodies recognize regions on the stalk of the HA protein [143] . The HA stalk is generally less immunogenic compared to the globular head of the HA protein so most approaches have utilized -headless HA proteins as immunogens. HA stalk vaccines have been designed using DNA and virus-like particles [144] and MVA [142] ; however, these approaches are amenable to expression in any of the viruses vectors described here.

The goal of any vaccine is to protect against infection and disease, while inducing population-based immunity to reduce or eliminate virus transmission within the population. It is clear that currently licensed influenza vaccines have not fully met these goals, nor those specific to inducing long-term, robust immunity. There are a number of vaccine-related issues that must be addressed before population-based influenza vaccination strategies are optimized. The concept of a -one size fits all vaccine needs to be updated, given the recent ability to probe the virus-host interface through RNA interference approaches that facilitate the identification of host genes affecting virus replication, immunity, and disease. There is also a need for revision of the current influenza virus vaccine strategies for at-risk populations, particularly those at either end of the age spectrum. An example of an improved vaccine regime might include the use of a vectored influenza virus vaccine that expresses the HA, NA and M and/or NP proteins for the two currently circulating influenza A subtypes and both influenza B strains so that vaccine take and vaccine antigen levels are not an issue in inducing protective immunity. Recombinant live-attenuated or replication-deficient influenza viruses may offer an advantage for this and other approaches.

Vectored vaccines can be constructed to express full-length influenza virus proteins, as well as generate conformationally restricted epitopes, features critical in generating appropriate humoral protection. Inclusion of internal influenza antigens in a vectored vaccine can also induce high levels of protective cellular immunity. To generate sustained immunity, it is an advantage to induce immunity at sites of inductive immunity to natural infection, in this case the respiratory tract. Several vectored vaccines target the respiratory tract. Typically, vectored vaccines generate antigen for weeks after immunization, in contrast to subunit vaccination. This increased presence and level of vaccine antigen contributes to and helps sustain a durable memory immune response, even augmenting the selection of higher affinity antibody secreting cells. The enhanced memory response is in part linked to the intrinsic augmentation of immunity induced by the vector. Thus, for weaker antigens typical of HA, vectored vaccines have the capacity to overcome real limitations in achieving robust and durable protection.

Meeting the mandates of seasonal influenza vaccine development is difficult, and to respond to a pandemic strain is even more challenging. Issues with influenza vaccine strain selection based on recently circulating viruses often reflect recommendations by the World Health Organization (WHO)-a process that is cumbersome. The strains of influenza A viruses to be used in vaccine manufacture are not wild-type viruses but rather reassortants

that are hybrid viruses containing at least the HA and NA gene segments from the target strains and other gene segments from the master strain, PR8, which has properties of high growth in fertilized hen's eggs. This additional process requires more time and quality control, and specifically for HPAI viruses, it is a process that may fail because of the nature of those viruses. In contrast, viral-vectored vaccines are relatively easy to manipulate and produce, and have well-established safety profiles. There are several viral-based vectors currently employed as antigen delivery systems, including poxviruses, adenoviruses, baculovirus, paramyxovirus, rhabdovirus, and others; however, the majority of human clinical trials assessing viral-vectored influenza vaccines use poxvirus and adenovirus vectors. While each of these vector approaches has unique features and is in different stages of development, the combined successes of these approaches supports the virus-vectored vaccine approach as a whole. Issues such as preexisting immunity and cold chain requirements, and lingering safety concerns will have to be overcome; however, each approach is making progress in addressing these issues, and all of the approaches are still viable. Virus-vectored vaccines hold particular promise for vaccination with universal or focused antigens where traditional vaccination methods are not suited to efficacious delivery of these antigens. The most promising approaches currently in development are arguably those targeting conserved HA stalk region epitopes. Given the findings to date, virus-vectored vaccines hold great promise and may overcome the current limitations of influenza vaccines.

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Nucleolar Protein Trafficking in Response to HIV-1 Tat: Rewiring the Nucleolus

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Abstract: The trans-activator Tat protein is a viral regulatory protein essential for HIV-1 replication. Tat trafficks to the nucleoplasm and the nucleolus. The nucleolus, a highly dynamic and structured membrane-less sub-nuclear compartment, is the site of rRNA and ribosome biogenesis and is involved in numerous cellular functions including transcriptional regulation, cell cycle control and viral infection. Importantly, transient nucleolar trafficking of both Tat and HIV-1 viral transcripts are critical in HIV-1 replication, however, the role(s) of the nucleolus in HIV-1 replication remains unclear. To better understand how the interaction of Tat with the nucleolar machinery contributes to HIV-1 pathogenesis, we investigated the quantitative changes in the composition of the nucleolar proteome of Jurkat T-cells stably expressing HIV-1 Tat fused to a TAP tag. Using an organellar proteomic approach based on mass spectrometry, coupled with Stable Isotope Labelling in Cell culture (SILAC), we quantified 520 proteins, including 49 proteins showing significant changes in abundance in Jurkat T-cell nucleolus upon Tat expression. Numerous proteins exhibiting a fold change were well characterised Tat interactors and/or known to be critical for HIV-1 replication. This suggests that the spatial control and subcellular compartmentalisation of these cellular cofactors by Tat provide an additional layer of control for regulating cellular machinery involved in HIV-1 pathogenesis. Pathway analysis and network reconstruction revealed that Tat expression specifically resulted in the nucleolar enrichment of proteins collectively participating in ribosomal biogenesis, protein homeostasis, metabolic pathways including glycolytic, pentose phosphate, nucleotides and amino acids biosynthetic pathways, stress response, T-cell signaling pathways and genome integrity. We present here the first differential profiling of the nucleolar proteome of T-cells expressing HIV-1 Tat. We discuss how these proteins collectively participate in interconnected networks converging to adapt the nucleolus dynamic activities, which favor host biosynthetic activities and may contribute to create a cellular environment supporting robust HIV-1 production.

Text: The nucleolus is a highly ordered subnuclear compartment organised around genetic loci called nucleolar-organising regions (NORs) formed by clusters of hundreds of rDNA gene repeats organised in tandem head-to-tail repeat [1, 2]. A membrane-less organelle originally described as the "Ribosome Factory", the nucleolus is dedicated to RNA-

polymerase-I-directed rDNA transcription, rRNA processing mediated by small nucleolar ribonucleoproteins (sRNPs) and ribosome assembly. Ribosome biogenesis is essential for protein synthesis and cell viability [2] and ultimately results in the separate large (60S) and small (40S) ribosomal subunits, which are subsequently exported to the cytoplasm. This fundamental cellular process, to which the cell dedicates most of its energy resources, is tightly regulated to match dynamic changes in cell proliferation, growth rate and metabolic activities [3] .

The nucleolus is the site of additional RNA processing, including mRNA export and degradation, the maturation of uridine-rich small nuclear RNPs (U snRNPs), which form the core of the spliceosome, biogenesis of t-RNA and microRNAs (miRNAs) [4] . The nucleolus is also involved in other cellular processes including cell cycle control, oncogenic processes, cellular stress responses and translation [4] .

The concept of a multifunctional and highly dynamic nucleolus has been substantiated by several studies combining organellar proteomic approaches and quantitative mass spectrometry, and describing thousands of proteins transiting through the nucleolus in response to various metabolic conditions, stress and cellular environments [5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16] . Collectively, the aforementioned studies represent landmarks in understanding the functional complexity of the nucleolus, and demonstrated that nucleolar proteins are in continuous exchange with other nuclear and cellular compartments in response to specific cellular conditions.

Of importance, the nucleolus is also the target of viruses including HIV-1, hCMV, HSV and KSHV, as part of their replication strategy [2, 17] . Proteomics studies analysing the nucleoli of cells infected with Human respiratory syncytial virus (HRSV), influenza A virus, avian coronavirus infectious bronchitis virus (IBV) or adenovirus highlighted how viruses can distinctively disrupt the distribution of nucleolar proteins [2, 17, 18, 19, 20, 21, 22, 23, 24] . Interestingly, both HIV-1 regulatory proteins Tat and Rev localise to the nucleoplasm and nucleolus. Both their sequences encompass a nucleolar localisation signal (NoLS) overlapping with their nuclear localisation signal (NLS), which governs their nucleolar localisation [25, 26, 27, 28, 29, 30, 31] . Furthermore, Tat and Rev interact with the nucleolar antigen B23, which is essential for their nucleolar localisation [25, 26, 27, 28, 29, 30] . Nevertheless, a recent study described that in contrast to Jurkat T-cells and other transformed cell lines where Tat is associated with the nucleus and nucleolus, in primary T-

cells Tat primarily accumulates at the plasma membrane, while trafficking via the nucleus where it functions [32] . While the regulation of their active nuclear import and/or export, as mediated by the karyopherin/importin family have been well described, the mechanisms distributing Tat and Rev between the cytoplasm, nucleoplasm and the nucleolus remains elusive [33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48] . Importantly, two major studies by Machienzi et al. have revealed important functional links between HIV-1 replication and the nucleolus [49, 50] . First, they could inhibit HIV-1 replication and Tat transactivation function employing a TAR decoy specifically directed to the nucleolus. Furthermore, using a similar approach, with an anti-HIV-1 hammerhead ribozyme fused to the U16 small nucleolar RNA and therefore targeted to the nucleolus, they could dramatically suppress HIV-1 replication. Collectively, these findings strongly suggest that HIV-1 transcripts and Tat nucleolar trafficking are critical for HIV-1 replication. However the nature of these contributions remains to be elucidated.

In this report, we systematically analysed the nucleolar proteome perturbations occurring in Jurkat T-cells constitutively expressing HIV-1 Tat, using a quantitative mass spectrometry approach. Following the detailed annotation of the quantitative abundance changes in the nucleolar protein composition upon Tat expression, we focussed on the Tat-affected cellular complexes and signalling pathways associated with ribosome biogenesis, spliceosome, molecular chaperones, DNA replication and repair and metabolism and discuss their potential involvement in HIV-1 pathogenesis.

In this study, we investigated the quantitative changes in the nucleolar proteome of Jurkat T cells constitutively expressing HIV-1 Tat (86aa) versus their Tat-negative counterpart, using stable isotope labelling with amino acids in cell culture (SILAC) technology, followed by ESI tandem mass spectrometry and implemented the experimental approach described in Figure 1A . First, using retroviral gene delivery, we transduced HIV-1 Tat fused to a tandem affinity purification (TAP) tag (consisting of two protein G and a streptavidin binding peptide) or TAP tag alone (control vector) in Jurkat leukemia T cell clone E6-1 and sorted the transduced cells (GFP positive) by FACS. This resulted in a highly enriched population of polyclonal transduced cells presenting different expression levels of the transgene (Figure 1B) . The functionality of TAP-Tat was confirmed by transfecting Jurkat TAP-Tat and TAP cells with a luciferase reporter gene vector under the control of the HIV-1 LTR (pGL3-LTR) [36] . TAP-Tat up regulated gene expression from the HIV-1 LTR by up to 28 fold compared to control (Figure 1C) . To further address the functionality of Tat fused to TAP, we compared Jurkat TAP-Tat with Jurkat-tat, a cell line stably expressing untagged Tat [51] . Both cell line

exhibited comparable HIV-1 LTR activity following transfection with pGL3-LTR (Figure S1). Next, Tat expression and subcellular localization was verified by subcellular fractionation followed by WB analysis (Figure 1E). TAP-Tat displayed a prominent nuclear/nucleolar localization but could also be detected in the cytoplasm. These observations were further validated by immunofluorescence microscopy (Figure 1E). Of note, Jurkat-tat presented similar patterns for Tat subcellular distribution as shown by immunofluorescence microscopy and subcellular fractionation followed by WB analysis (Figure S2 and S3). We next compared the growth rate and proliferation of the Jurkat TAP and TAP-Tat cell lines (Materials and Methods S1), which were equivalent (Figure S4A). Similarly, FACS analysis confirmed that the relative populations in G1, S, and G2/M were similar for Jurkat TAP-Tat and TAP cells (Figure S4B).

We labeled Jurkat TAP-Tat and Jurkat TAP cells with light (R0K0) and heavy (R6K6) isotope containing arginine and lysine, respectively. Following five passages in their respective SILAC medium, 85 million cells from each culture were harvested, pooled and their nucleoli were isolated as previously described (Figure 1A) [52] . Each step of the procedure was closely monitored by microscopic examination. To assess the quality of our fractionation procedure, specific enrichment of known nucleolar antigens was investigated by Western Blot analysis (Figure 1D). Nucleolin (110 kDa) and Fibrillarin (FBL) (34 kDa), two major nucleolar proteins known to localise to the granular component of the nucleolus, were found to be highly enriched in the mixed nucleolar fraction. Of note, nucleolin was equally distributed between the nuclear and cytoplasmic fractions. This distribution pattern for nucleolin appears to be specific for Jurkat T-cells as show previously [52, 53] . The nuclear protein PARP-1 (Poly ADPribose polymerase 1) (113 kDa) was present in the nuclear and nucleoplasmic fraction but was depleted in the nucleolar fraction. Alpha-tubulin (50 kDa) was highly abundant in the cytoplasmic fraction and weakly detected in the nuclear fractions. Collectively, these results confirmed that our methods produced a highly enriched nucleolar fraction without significant cross contamination.

Subsequently, the nucleolar protein mixture was trypsin digested and the resulting peptides were analysed by mass spectrometry. Comparative quantitative proteomic analysis was performed using MaxQuant to analyse the ratios in isotopes for each peptide identified. A total of 2427 peptides were quantified, representing 520 quantified nucleolar proteins. The fully annotated list of the quantified nucleolar proteins is available in Table S1 and the raw data from the mass spectrometry analysis was deposited in the Tranche repository

database ([https:// proteomecommons.org/tranche/](https://proteomecommons.org/tranche/)), which can be accessed using the hash keys described in materials and methods. We annotated the quantified proteins using the ToppGene Suite tools [54] and extracted Gene Ontology (GO) and InterPro annotations [55]. The analysis of GO biological processes (Figure 1F) revealed that the best-represented biological processes included transcription (24%), RNA processing (23%), cell cycle process (13%) and chromosome organisation (15%), which reflects nucleolar associated functions and is comparable to our previous characterisation of Jurkat T-cell nucleolar proteome [52]. Subcellular distribution analysis (Figure 1F) revealed that our dataset contained proteins known to localise in the nucleolus (49%), in the nucleus (24%) while 15% of proteins were previously described to reside exclusively in the cytoplasm. The subcellular distribution was similar to our previous analysis of the Jurkat T-cell nucleolar proteome [52]. Table S1. The distribution of protein ratios are represented in Figure 1G as log₂ (abundance change). The SILAC ratios indicate changes in protein abundance in the nucleolar fraction of Jurkat TAP-Tat cells in comparison with Jurkat TAP cells. The distribution of the quantified proteins followed a Gaussian distribution (Figure 1G). A total of 49 nucleolar proteins exhibited a 1.5 fold or greater significant change ($p < 0.05$) upon Tat expression (Table 1). Of these, 30 proteins were enriched, whereas 19 proteins were depleted. Cells displayed no changes in the steady state content of some of the major and abundant constituents of the nucleolus, including nucleophosmin (NPM1/ B23), C23, FBL, nucleolar protein P120 (NOL1), and nucleolar protein 5A (NOL5A). The distinct ratios of protein changes upon Tat expression could reflect specific nucleolar reorganization and altered activities of the nucleolus.

We performed WB analysis to validate the SILAC-based results obtained by our quantitative proteomic approach (Figure 2). 15 selected proteins displayed differential intensity in the nucleolar fractions upon Tat expression, including 9 enriched (HSP90b, STAT3, pRb, CK2a, CK2a', HSP90a, Transportin, ZAP70, DDX3), and 3 depleted (ILF3, BOP1, and SSRP1) proteins. In addition, we also tested by WB analysis, protein abundance not affected by Tat expression (Importin beta, FBL, B23, C23). These results highlight the concordance in the trend of the corresponding SILAC ratios, despite some differences in the quantitative ranges. Of note, using WB, we could observe a change of intensity for protein with a SILAC fold change as low as 1.25-fold.

Of note, the question remains as to which fold change magnitude might constitute a biologically relevant consequence. On the one hand, the threshold of protein abundance changes can be determined statistically and would then highlight the larger abundance

changes as illustrated in Table 1 . Alternatively, the coordinated enrichment or depletion of a majority of proteins belonging to a distinct cellular complex or pathway would allow the definition of a group of proteins of interest and potential significance. Therefore, we next focused on both enriched or depleted individual proteins with activities associated with HIV-1 or Tat molecular pathogenesis, and on clustered modifications affecting entire cellular signaling pathways and macromolecular complexes.

We initially focused on signaling proteins interacting with Tat and/or associated HIV-1 molecular pathogenesis and whose abundance in the nucleolus was modulated by Tat expression.

Phospho-protein phosphatases. Phospho-protein phosphatase PP1 and PP2A are essential serine/threonine phosphatases [56, 57] . Importantly, PP1 accounts for 80% of the Ser/Thr phosphatase activity within the nucleolus. In our study, PP1 was found to be potentially enriched by 1.52-fold in the nucleolus of Jurkat cells expressing Tat, which supports previous studies describing the nuclear and nucleolar targeting of PP1a by HIV-1 Tat and how PP1 upregulates HIV-1 transcription [58, 59, 60, 61, 62] . PP1 c was also identified as part of the in vitro nuclear interactome [63] . Similarly, PPP2CA, the PP2A catalytic subunit (1.29-fold) and its regulatory subunit PP2R1A (1.27-fold) were similarly enriched upon Tat expression. Interestingly, Tat association with the PP2A subunit promoters results in the overexpression and up regulation of PP2A activity in lymphocytes [64, 65] . Furthermore, PP2A contributes to the regulation of HIV-1 transcription and replication [61, 66] .

Retinoblastoma Protein. The tumour suppressor gene pRb protein displayed a 1.4-fold change in the nucleolus upon Tat expression [67] . Furthermore, WB analysis confirmed the distinct translocation of pRb from the nucleoplasm to the nucleolus by Tat (Figure 2). Depending on the cell type, pRb can be hyperphosphorylated or hypophosphorylated upon Tat expression and can negatively or positively regulate Tat-mediated transcription respectively [68, 69, 70] . Interestingly, the hyperphosphorylation of pRB triggers in its translocation into the nucleolus [71] . Phosphorylation of pRB is also associated with an increase in ribosomal biogenesis and cell growth [72] .

STAT3. The transcription factor signal transducer and activator of transcription 3 (STAT3) was significantly enriched (1.86-fold) in the nucleolar fraction by Tat constitutive expression. Furthermore, WB analysis indicated that Tat expression could promote the relocalisation of STAT3 from the cytoplasm to the nucleus, with a distinct enrichment in the nucleolus (Figure 2). Interestingly, previous studies have demonstrated Tat-mediated activation of STAT3 signaling, as shown by its phosphorylation status [73] . Interestingly, STAT3 phosphorylation induced dimerisation of the protein followed its translocation to the nucleus [74] .

YBX1. YBX1, the DNA/RNA binding multifunctional protein was enriched by 1.38-fold in the nucleolus of Jurkat cells upon Tat expression. Interestingly, YBX1 interacts with Tat and TAR and modulates HIV-1 gene expression [63, 75] .

ZAP70. The protein tyrosine kinase ZAP70 (Zeta-chainassociated protein kinase 70) was enriched by 1.24-fold in the nucleolus of Jurkat cells expressing Tat [76] . Furthermore, WB analysis revealed that Tat expression could promote the relocalisation of ZAP70 from the cytoplasm to the nucleus, with a distinct enrichment in the nucleolus (Figure 2). Of note, ZAP70 is part of the in vitro nuclear Tat interactome [63] .

Matrin 3. The inner nuclear matrix protein, Matrin 3 (MATR3), presented a 1.39-fold change in the nucleolus of Jurkat cells expressing Tat. It localizes in the nucleolasm with a diffuse pattern excluded from the nucleoli [77] . Matrin 3 has been identified as part of the in vitro HIV-1 Tat nuclear interactome [63] . Two recent studies have described Matrin 3 as part of ribonucleoprotein complexes also including HIV-1 Rev and (Rev Response Element) RRE-containing HIV-1 RNA, and promoting HIV-1 post-transcriptional regulation [78, 79, 80] .

CASP10. The pro-apototic signaling molecule, Caspase 10 (CASP10), was significantly depleted from the nucleolus of Jurkat-Tat cells (0.82-fold) [81] . Importantly, Tat expression downregulates CASP10 expression and activity in Jurkat cells [82] .

ADAR1. Adenosine deaminase acting on RNA (ADAR1), which converts adenosines to inosines in double-stranded RNA, was significantly depleted from the nucleolus of Jurkat-Tat cells (0.78-fold). Interestingly, ADAR1 over-expression up-regulates HIV-1 replication via

an RNA editing mechanism [83, 84, 85, 86, 87, 88] . Furthermore, ADAR1 belongs to the in vitro HIV-1 Tat nuclear interactome [63] .

To underline the structural and functional relationships of the nucleolar proteins affected by HIV-1 Tat, we constructed a network representation of our dataset. We employed Cytoscape version 2.6.3 [89] and using the MiMI plugin [90] to map previously characterised interactions, extracted from protein interaction databases (BIND, DIP, HPRD, CCSB, Reactome, IntAct and MINT). This resulted in a highly dense and connected network comprising 416 proteins (nodes) out of the 536 proteins, linked by 5060 undirected interactions (edges) (Figure 3A). Centrality analysis revealed a threshold of 23.7 interactions per protein. Topology analysis using the CentiScaPe plugin [91] showed that the node degree distribution follows a power law (Figure S5), characteristic of a scale-free network. Importantly, when we analysed the clustering coefficient distribution (Figure S6) we found that the network is organised in a hierarchical architecture [92] , where connected nodes are part of highly clustered areas maintained by few hubs organised around HIV-1 Tat. Furthermore, node degree connection analysis of our network identified HIV-1 Tat as the most connected protein (Figure S6). Specifically, the topology analysis indicated that the values for Tat centralities were the highest (Node degree, stress, radiality, closeness, betweenness and centroid), characterising Tat as the main hub protein of the nucleolar network. Indeed, a total of 146 proteins have been previously described to interact with Tat (Figure 3B , Table S2). These proteins are involved in a wide range of cellular processes including chromosomal organization, DNA and RNA processing and cell cycle control. Importantly, over the third of these proteins exhibit an increase in fold ratio change (59 proteins with a ratio ≥ 1.2 fold).

In parallel, we characterised the magnitude of the related protein abundance changes observed in distinct cellular pathways (Figure 4) .

Ribosomal biogenesis. We initially focused on ribosome biogenesis, the primary function of the nucleolus. We could observe a general and coordinated increase in the abundance of ribosomal proteins in the nucleolus by Tat expression (Figure 4). While some ribosomal proteins remained unaffected, Tat caused the nucleolar accumulation of several distinct large and small ribosomal proteins, except RPL35A, for which Tat expression caused a marked decrease at the nucleolar level (0.29-fold). Similarly, several proteins involved in rRNA processing exhibited an overall increase in nucleolar accumulation upon Tat

expression. These include human canonical members of the L7ae family together with members participating in Box C/D, H/ACA and U3 snoRNPs (Figure 4) . Conversely, BOP1, a component of the PeBoW (Pescadillo Bop1 WDR12) complex essential for maturation of the large ribosomal subunit, was significantly depleted from the nucleolus of Jurkat TAP-Tat cells (0.81-fold) and this was confirmed by WB analysis (Figure 2) [93] . Nevertheless, the other PeBoW complex components, Pes1 (0.94-fold) and WDR12 (1.1fold), were not affected by Tat expression. Of note, we did not detect change in the abundance of protein participating in rDNA transcription such as RNAPOLI, UBF.

Spliceosome. We identified and quantified in our dataset 55 proteins out of the 108 known spliceosomal proteins [94] . These proteins include the small nuclear ribonucleoproteins U1, U2 and U5, Sm D1, D2, D3, F and B, and the heterogeneous nuclear ribonucleoproteins. Our data suggested a distinct increase in the abundance of specific spliceosome complex proteins upon expression of HIV-1 Tat in Jurkat T-cells (Figure 3 and 4) . The only three proteins that were significantly depleted from the nucleolus upon expression of HIV-1 Tat were RBMX (0.89-fold), HNRNPA2B1 (0.84-fold) and SNRPA (0.81-fold). Several investigations showed expression alteration in cellular splicing factors in HIV-1 infected cells [95, 96] .

Molecular chaperones. We have identified several molecular chaperones, co-chaperones and other factors involved into proteostasis to be highly enriched in the nucleolus of T-cells upon Tat expression (Figure 3 and 4) , many of which were previously characterised as part of the Tat nuclear interactome [63] . Several heat-shock proteins including DNAJs, specific HSP90, HSP70 and HSP40 isoforms and their co-factors were distinctively enriched in the nucleolar fraction of Jurkat cells expressing Tat (Figure 4) . As shown by WB, while HSP90a and b are mostly cytoplasmic, Tat expression triggers their relocalisation to the nucleus and nucleolus, corroborating our proteomic quantitative approach (Figure 2) . Similarly, heat-shock can cause the HSP90 and HSP70 to relocalise to the nucleolus [97, 98, 99, 100, 101] . In a recent study, Fassati's group has shown that HSP90 is present at the HIV-1 promoter and may directly regulate viral gene expression [102] . We also observed the coordinated increased abundance of class I (GroEL and GroES) and class II (chaperonin containing TCP-1 (CTT)) chaperonin molecules (Figure 3 and 4) upon Tat expression.

Ubiquitin-proteasome pathway. The ubiquitin-proteasome pathway is the major proteolytic system of eukaryotic cells [103] . Importantly, the nuclear ubiquitin-proteasome pathway

controls the supply of ribosomal proteins and is important to ribosome biogenesis [104, 105] . The 26S proteasome is composed of the 20S core particle (CP) and the 19S regulatory particle (RP). Alternatively, CP can associate with the 11S RP to form the immunoproteasome. All the quantified proteins in our study are part of the 19S regulatory complex and include PSMD2 (1.5-fold), PSMD3 (1.32-fold), PSMD11 (1.25-fold) and PSMD13 (0.72-fold), the only proteasome component significantly depleted from the nucleolus in the presence of Tat (Figure 4) . Interestingly, Tat interacts with distinct subunits of the proteasome system, including the 19S, 20S and 11S subunits. The consequences of these interactions include the competition of Tat with 11S RP or 19S RP for binding to the 20S CP, which resulted in the inhibition of the 20S peptidase activity [106, 107, 108, 109, 110, 111] . Furthermore, Tat was shown to modify the proteasome composition and activity, which affects the generation of peptide antigens recognized by cytotoxic T-lymphocytes [112] . Importantly, a recent study demonstrated that in the absence of Tat, proteasome components are associated to the HIV-1 promoter and proteasome activity limits transcription [113] . Addition of Tat promoted the dissociation of the 19S subunit from the 20S proteasome, followed by the distinct enrichment of the 19S-like complex in nuclear extracts together with the Tat-mediated recruitment of the 19S subunits to the HIV-1 promoter, which facilitated its transcriptional elongation [113] . We also quantified UBA1 (1.36-fold), the E3 ubiquitin-protein ligase UHRF1 (1.13-fold), UBC (1-fold) and two Ubiquitinspecific-peptidases, USP30 (1.28-fold) and USP20 (0.06-fold) (Figure 4) .

DNA replication and repair. Upon HIV-1 Tat expression, we observed the coordinated nucleolar enrichment of several cellular factors associated with DNA replication and repairs pathways (Figure 4) .

Tat induced the coordinated enrichment of the miniature chromosome maintenance MCM2-7 complex (from 1.23-to 3.30fold, respectively) [114] . MCM7, 6 and 3 were identified as part of the in vitro nuclear interactome of HIV-1 Tat [63] . The structural maintenance of chromosomes 2, SMC2, was enriched (1.35-fold) in the nucleolar fraction by Tat expression. SMC2 was identified as part of the in vitro nuclear interactome of HIV-1 Tat [63] . While replication factor C1 (RFC1) and RFC2 (1.31-and 1.28-fold respectively) displayed an increased fold change and RFC5/3 were not affected, RFC4 was severely depleted (0.69-fold) from the nucleolar fraction upon Tat expression [115] . RFC1 and RFC2 were identified as part of the in vitro nuclear interactome of HIV-1 Tat [63] . Tat induced the enrichment of XRCC6 (1.27-fold) and XRCC5 (1.36-fold) in the nucleolus, which are involved in the repair of non-homologous DNA end joining (NHEJ) [116] . XRCC6 associates

with viral preintegration complexes containing HIV-1 Integrase and also interact with Tat and TAR [117, 118, 119] . Furthermore, in a ribozyme-based screen, XRCC5 (Ku80) knockdown decreased both retroviral integration and Tat-mediated transcription [120] . As part of the base excision repair (BER), we have identified a major apurinic/apyrimidinic endonuclease 1 (APEX1) (1.29-fold) . Importantly, in a siRNA screen targeting DNA repair factors, APEX1 knockdown was found to inhibit HIV-1 infection by more 60% [121] . The high mobility group (HMG) protein, HMGA1 (1.30-fold), was enriched in the nucleolus following Tat expression [122] . HMGA1 interact with HIV-1 Integrase and is part of the HIV-1 pre-integration complex [123, 124] . Importantly, HMGA1 has been identified in a proteomic screen, as a cellular cofactor interacting with the HIV-1 59leader [125] .

Metabolism. Our proteomic data suggest that Tat induces perturbations in glycolysis, the pentose phosphate pathway, and nucleotide and amino acid biosynthesis (Figure 4 and Figure S7). Notably, in T cells expressing Tat, we detected co-ordinated changes in the abundance of proteins not previously known to be associated with Tat pathogenesis, which revealed unexpected connections with glycolysis and the pentose phosphate pathway, including the following glycolytic enzymes, lactate dehydrogenase B (LDHB) (1.37-fold), glyceraldehyde 3-phosphate dehydrogenase (GAPDH) (1.17-fold) and phosphoglyceric acid mutase (PGAM1) (0.89-fold) (Figure 4 and Figure S7). Briefly, GPI catalyzes the reversible isomerization of glucose-6-phosphate in fructose-6-phosphate. Subsequently, PFKP catalyzes the irreversible conversion of fructose-6-phosphate to fructose-1,6-bisphosphate and is a key regulatory enzyme in glycolysis. At the end of the glycolytic pathway, PKM2, in its tetrameric form, is known to generate ATP and pyruvate, while LDHB diverts the majority of the pyruvate to lactate production and regeneration of NAD⁺ in support to continued glycolysis, a phenomenon described for proliferative T cells [126] . Of note, in highly proliferating cells, PKM2 can be found in its dimeric form and its activity is altered. This upregulates the availability of glucose intermediates, which are rerouted to the pentose phosphate and serine biosynthesis pathways for the production of biosynthetic precursors of nucleotides, phospholipids and amino acids. As part of the pentose phosphate pathway, we have characterised the significant enrichment of glucose-6-phosphate dehydrogenase (G6PD) (2.11-fold), which branches of the glycolysis pathway to generate NADPH, ribose-5-phosphate an important precursor for the synthesis of nucleotides. Consistent with this, we detected the coordinated increase in the abundance of enzymes which plays a central role in the synthesis of purines and pyrimidines. More specifically, IMPDH2 (1.66-fold), a rate-limiting enzyme at the branch point of purine nucleotide biosynthesis, leading to the generation of guanine nucleotides, phosphoribosyl pyrophosphate synthetase 2 (PRPS2) (1.41-fold), cytidine-5-prime-triphosphate synthetase

(CTPS) (1.74-fold) which catalyses the conversion of UTP to CTP and the ribonucleotide reductase large subunit (RRM1) (1.56-fold). In parallel, we noted the increased abundance of the phosphoserine aminotransferase PSAT1 (1.90-fold), an enzyme implicated in serine biosynthesis, which has been linked with cell proliferation *in vitro*.

The host-virus interface is a fundamental aspect in defining the molecular pathogenesis of HIV-1 [127, 128, 129, 130, 131, 132, 133]. Indeed, with its limited repertoire of viral proteins, HIV-1 relies extensively on the host cell machinery for its replication. Several recent studies have capitalized on the recent advances in the "OMICS" technologies, and have revealed important insights into this finely tuned molecular dialogue [132, 134]. HIV-1 Tat is essential for viral replication and orchestrates HIV-1 gene expression. The viral regulatory protein is known to interact with an extensive array of cellular proteins and to modulate cellular gene expression and signaling pathway [135, 136]. We and others have employed system-level approaches to investigate Tat interplay with the host cell machinery, which have characterised HIV-1 Tat as a critical mediator of the host-viral interface [137, 138, 139, 140, 141, 142, 143, 144, 145, 146, 147, 148, 149]. Here, we have investigated the nucleolar proteins trafficking in response to HIV-1 Tat expression in T-cells, with the view to provide unique and novel insights on the role of proteins compartmentalisation by Tat in the fine-tuning of protein availability and function.

We have developed for this study, a cellular model using Jurkat T-cells stably expressing Tat fused in its N-terminal to TAP-tag.

Jurkat T-cells are robust and present the advantage to grow without stimulations and are easily transduced using retroviral gene delivery. Importantly, they have been widely employed to evaluate Tat-mediated pathogenesis using system-wide approaches and to analyse T-cell key cellular signaling pathways and functions [144, 150, 151, 152]. Indeed, we have found them particularly suited for prolonged *in vitro* culture in SILAC medium and subsequent isolation of their nucleolus followed by MS analysis, which requires up to 85 millions of cells. We fused Tat to the TAP tag to enable future downstream applications such as Tandem affinity purification or Chromatin IP analysis. Importantly, we have confirmed that N-terminal TAP-tag did not interfere with Tat function nor its localisation in Jurkat cells, when compared to untagged-Tat. Of note, Tat subcellular distribution can vary according to the cell type employed. While Tat is known to accumulate in the nucleus and nucleolus in Jurkat cells and other transformed cell lines, in primary T-cells, Tat was described to

primarily accumulate at the plasma membrane, while trafficking via the nucleus where it functions [32] . These differences remain to be characterised but could be related to different expression levels of transport factors in transformed cell lines versus primary cells, as recently described by Kuusisto et al. [39] . Furthermore, Stauber and Pavlakis have suggested that Tat nucleolar localisation could be the results of Tat overexpression [31] . Here, we have selected and employed a polyclonal population of Jurkat T-cells expressing Tat at different levels. We propose that this heterogeneity in Tat expression levels might reflect Tat stochastic expression described during viral replication [153] .

Using a quantitative proteomic strategy based on an organellar approach, we quantified over 520 nucleolar proteins, including 49 proteins exhibiting a significant fold change. The extent to which the induced variations in the abundance of nucleolar proteins are biologically relevant and can affect cellular and/or viral processes remains to be determined. Nevertheless, the biological nature of the pathways and macromolecular complexes affected enable us to discuss their potential associations with HIV-1 pathogenesis.

HIV-1 Tat is expressed early following HIV-1 genome integration and mediates the shift to the viral production phase, associated with robust proviral gene expression, viral proteins assembly and ultimately, virions budding and release. In this context and based on our results, we propose that Tat could participate in shaping the intracellular environment and metabolic profile of T cells to favor host biosynthetic activities supporting robust virions production. Indeed, we observed the distinct nucleolar enrichment of ribosomal proteins and enzymes associated with ribosomal biogenesis, which could be indicative of an increase in protein synthesis. With the notable exception of RPL35A nucleolar depletion, ribosomal proteins and enzymes associated with ribosomal biogenesis were in the top 20 most enriched nucleolar proteins (NHP2L1, RLP14, RPL17, RPL27, RPS2, RPL13). Furthermore, this effect appears to be specific to HIV-1 Tat since transcription inhibition by Actinomycin D resulted in the overall depletion of ribosomal proteins in the nucleolus [9] . Moreover, quantitative proteomics analysis of the nucleolus in adenovirus-infected cells showed a mild decrease in ribosomal proteins [24] . Whether this reflect a shift in ribosome biogenesis and/or a change in the composition of the ribosomal subunits remains to be determined. Nevertheless, the adapted need for elevated ribosome production is intuitive for a system that needs to support the increased demand for new viral proteins synthesis. In parallel, we observed the concordant modulation of pathways regulating protein homeostasis. We noted the significant nucleolar accumulation of multiple molecular

chaperones including the HSPs, the TCP-1 complex, and CANX/CALR molecules and the disrupted nucleolar abundance of proteins belonging to the ubiquitin-proteasome pathway, which controls the supply of ribosomal proteins [104, 105] . These observations further support previous studies describing the modulation of the proteasomal activity by Tat, which affect the expression, assembly, and localization of specific subunits of the proteasomal complexes [106, 107, 108, 109, 110, 111, 113] . We also observed the concomitant depletion of CASP10 in the nucleolus of Jurkat TAP-Tat. It has been suggested that CASP10 could be targeted to the nucleolus to inhibit protein synthesis [154] . Interestingly, the presence and potential roles of molecular chaperones in the nucleolus have been highlighted by Banski et al, who elaborate on how the chaperone network could regulate ribosome biogenesis, cell signaling, and stress response [97, 155] . As viral production progresses into the late phase and cellular stress increases, nucleolar enrichment of molecular chaperones by Tat could not only enable adequate folding of newly synthesised viral proteins but could also promote tolerance of infected cells to stress and maintain cell viability.

Coincidentally, we observed the marked nucleolar enrichment of enzymes belonging to metabolic pathways including glycolysis, pentose phosphate, nucleotide and amino acid biosynthetic pathways. Similarly, these pathways are elevated in proliferative T-cells or in cancer cells following a metabolic shift to aerobic glycolysis, also known as the Warburg effect [156, 157, 158, 159] . There, glucose intermediates from the glycolysis pathway are not only committed to energy production and broke-down into pyruvate for the TCA cycle, but are redirected to alternative pathways, including the pentose phosphate pathway, and used as metabolic precursors to produce nucleotides, amino acids, acetyl CoA and NADPH for redox homeostasis. Consistently, we also noted the concomitant nucleolar enrichment of enzymes belonging to the nucleotide synthesis pathway, including IMPDH2, a rate limiting enzyme known to control the pool of GTP. Similarly, we noted the nucleolar enrichment of PSAT1, an enzyme involved in serine and threonine metabolism, which is associated with cellular proliferation [160] . Collectively, we propose that by controlling protein homeostasis and metabolic pathways, Tat could meet both the energetic and biosynthetic demand of HIV-1 productive infection.

Of note, while nucleotide metabolism enzymes are associated with the nucleus, glycolysis takes place in the cytoplasm. Nevertheless, glycolytic enzymes have been detected in both the nuclear and nucleolar fractions by proteomic analyses [8, 161] . Furthermore glycolytic enzymes, such as PKM2, LDH, phosphoglycerate kinase, GAPDH, and aldolase, also have

been reported to display nuclear localization and bind to DNA [162] . More specifically, PKM2 is known to associate with promoter and participate in the regulation of gene expression as a transcriptional coactivator [163] .

HIV-1 Tat has previously been described as an immunoregulator and more specifically, has been reported both to inhibit or to promote TCR signaling [164] . We have observed the nucleolar enrichment by Tat of key proximal or downstream components of T-cell signaling pathways, including ZAP70, ILF3 and STAT3, which play crucial roles in T-cell development and activation. We had previously identified them as T-cell specific components of the nucleolus, and IF studies suggested that their association with the nucleolus could be regulated by specific conditions [165] . Our results further support that Tat could contribute to the dysregulation of TCR-derived signals and that the nucleolus could represent an important spatial link for TCR signaling molecules.

We observed the coordinated nucleolar enrichment of key components of the DNA replication, recombination and repair pathways by Tat. These include XRCC5 and XRCC6, HMGA1, APEX1, MCM2-7, SMC2, RFC1 and RFC2, while RFC4 was found to be significantly depleted. Interestingly, these cofactors have been associated with the efficiency of retroviral DNA integration into the host DNA or the integrity of integrated provirus [166] . Whether the increased abundance of these factors within the nucleolus could be associated with their potential participation in the integration and maintenance of provirus gene integrity, remains to be determined.

The mechanisms of Tat-mediated segregation and compartmentalisation of proteins in or out of the nucleolus may depend on factor(s) inherent for each protein and the nature of their relationship with Tat, since subcellular fractionation combined with WB analysis showed that the pattern and extent of subcellular redistribution between proteins varied. We could observe cases where Tat upregulated the expression of proteins which resulted in a general increase of these proteins throughout the cellular compartments including the nucleolus (DDX3, TNPO1). Alternatively, Tat could trigger the nucleolar translocation of proteins directly from the cytoplasm or the nucleoplasm (pRb). Additionally, we observed cytoplasmic proteins redistributed to both the nucleoplasm and nucleolus upon Tat expression (STAT3, ZAP70 and HSP90). Finally, we also noted protein depletion in the nucleolar fraction accompanied by an increase in the nucleoplasm (SSRP1). It remains difficult at this stage, to appreciate whether the accumulation of specific proteins would

result in their activation or inhibition by sequestering them away from their site of action. Conversely, the depletion of a protein from the nucleolus could either result in the down-regulation of its activity in this location or could be the result of its mobilization from its storage site, the nucleolus, to the nucleoplasm or cytoplasm where it can perform its function. Remarkably, we identified several known HIV-1 Tat partners involved in HIV-1 pathogenesis, which suggests that Tat could physically modulate their nucleolar targeting or their recruitment to specific site in the nucleoplasm or cytoplasm. Tat could also promote post-translational modifications, which could mediate the targeting of specific proteins to the nucleolus. This is exemplified by the following enriched proteins, pRb, PP1 and STAT3, for which phosphorylation is induced by Tat. Importantly, their phosphorylation status determines their subcellular distribution, thus providing a potential mechanism for their redistribution by Tat. Moreover, our data indicates that serine/threonine kinases (CK2 a') and phosphatases (PP1) were significantly enriched in the nucleolar fractions of Jurkat TAP-Tat. These enzymes account for the majority of the phosphorylation/dephosphorylation activity in the nucleolus and can act as regulators of nucleolar protein trafficking. In addition, Tat significantly decreased the levels of SUMO-2 in the nucleolus. Similarly, SUMO-mediated post-translational modifications are known to modulate nucleolar protein localization [104] . Given the potential importance of post-translational modifications, including phosphorylation in the Tat-mediated change of abundance of nucleolar proteins, a more targeted proteomic approach such as the enrichment for phosphopeptides, would extend the resolution of our screening approach.

The control of protein turnover is also an important mean to modulate the abundance of nucleolar proteins. Ribosomal proteins are degraded by the Ubiquitin-Proteasome pathway to ensure their abundance matches up with rRNA transcription levels. Conversely, heat shock proteins HSP90s protect them from degradation. Interestingly, our data showing that Tat modulation the abundance proteins associated with the Ubiquitin-proteasome and heat-shock pathway. This could contribute to the observed enrichment of ribosomal proteins by Tat. Nevertheless, we cannot exclude that the increased abundance of ribosomal proteins in the nucleolus could be the result of Tat-mediated prevention of their export to the cytoplasm. Interestingly, using a different cellular system, a drosophila melanogaster Tat transgenic strain, Ponti et al, analysed the effects of Tat on ribosome biogenesis, following 3 days heat shock treatment to induce Tat expression under the control of the hsp70 promoter [167] . Following Tat expression, they observed a defect in pre-rRNA processing associated with a decrease in the level of 80S ribosomes [167] . Nevertheless, the different cellular system employed combined with the 3 days heatshock induction make their results difficult to compare with ours.

While previous system-level studies have monitored the effects of HIV-1 Tat expression on T cells, to our knowledge, we have presented here the first proteomic analysis of dynamic composition of the nucleolus in response to HIV-1 Tat expression. Using quantitative proteomics, we have underlined the changes in abundance of specific nucleolar proteins and have highlighted the extensive and coordinated nucleolar reorganization in response to Tat constitutive expression. Our findings underscore that Tat expressing T-cells exhibit a unique nucleolar proteomic profile, which may reflect a viral strategy to facilitate the progression to robust viral production. Importantly, we noted the functional relationship of nucleolar proteins of our dataset with HIV-1 pathogenesis and HIV-1 Tat in particular. This further increases our confidence in our experimental strategy and suggests a role for Tat in the spatial control and subcellular compartmentalization of these cellular cofactors. Ultimately, our study provides new insights on the importance of Tat in the cross talk between nucleolar functions and viral pathogenesis. Importantly, we have also identified changes in nucleolar protein abundance that were not previously associated with HIV-1 pathogenesis, including proteins associated with metabolic pathways, which provide new potential targets and cellular pathways for therapeutic intervention.

Jurkat T-cells, clone E6.1 (ATCC), Jurkat NTAP-Tat and Jurkat NTAP were maintained in RPMI-1640 medium supplemented with 10% (v/v) foetal bovine serum (Gibco, EU approved), and antibiotics. Phoenix-GP cells (G.P. Nolan; www.stanford.edu/group/nolan/), were maintained in DMEM medium supplemented with 10% (v/v) foetal bovine serum (GIBCO, EU approved). Cells were counted using Scepter TM 2.0 Cell Counter (Millipore).

The sequence of HIV-1 Tat (HIV-1 HXB2, 86 amino acids) was sub-cloned into pENTR 2B vector (Invitrogen, A10463). Using the Gateway technology (Invitrogen), we introduced the HIV-1 Tat sequence into the plasmid pCeMM-NTAP(GS)-Gw [168] . Phoenix cells (G.P. Nolan; www.stanford.edu/group/nolan/), were transfected using Eugene 6 (Roche) with 5 mg of the plasmid NTAP-Tat or NTAP and 3 mg of the pMDG-VSVG. Viral supernatants were collected after 48 h, filtered and used to transduce the Jurkat cell lines. The construct is termed NTAP-Tat, the empty vector was termed NTAP. Using retroviral gene delivery, we stably transduced Jurkat cells (clone E6.1 (ATCC)). The positive clones named Jurkat NTAP-Tat and Jurkat NTAP were sorted to enrich the population of cells expressing GFP using the BC MoFlo XDP cell sorter (Beckman Coulter).

Sub-cellular fractions (10 mg) were resolved by SDS-PAGE and transferred onto BioTrace PVDF membranes (Pall corporation). The following primary antibodies were used: α -Tubulin (Sc 5286), C23 (Sc 6013), and Fibrillarin (Sc 25397) were from Santa Cruz Biotechnology, and PARP (AM30) from Calbiochem, mouse anti-ZAP 70 (05-253, Millipore), rabbit anti-STAT3 (06-596, Millipore), rabbit anti-ILF3 (ab92355, Abcam), rabbit anti-HSP90 beta (ab32568, Abcam), mouse anti-ADAR1 (ab88574, Abcam), rabbit anti-HDAC1 (ab19845, Abcam), rabbit anti-SSRP1 (ab21584, Abcam) rabbit anti-BOP1 (ab86982, Abcam), mouse anti-KpNB1 (ab10303, Abcam), rabbit anti-HIV-1 Tat (ab43014, Abcam), rabbit anti-CK2A (ab10466, Abcam), rabbit anti-DDX3X (ab37160, Abcam), mouse anti-TNPO1 (ab2811, Abcam), mouse anti-HSP90A (CA1023, MERCK), and rabbit-anti RB1 (sc-102, Santa Cruz). The following secondary antibodies were used ECL: Anti-mouse IgG and ECL Anti-rabbit IgG (GE Healthcare), and Donkey anti-goat IgG (Sc 2020) (Santa Cruz Biotechnology).

For SILAC analysis SILAC-RPMI R0K0 and SILAC-RPMI R6K6 (Dundee cells) media supplemented with 10% dialyzed FBS (GIBCO, 26400-036) were used. The Jurkat cells expressing NTAP-Tat and NTAP were serially passaged and grown for five doublings to ensure full incorporation of the labelled amino acids. Cells viability was checked with Trypan Blue (0.4% solution, SIGMA) and further confirmed using PI staining and FACS analysis. Cells were mixed to the ratio 1:1 to obtain 140610 6 cells. Nucleoli were isolated from the mixed cell population as previously described in Jarboui et al., [165] .

Nucleolar extracts (100 mg) were resuspended in 50 mM ammonium bicarbonate and in solution trypsin digested as previously described in Jarboui et al. [165] . Sample was run on a Thermo Scientific LTQ ORBITRAP XL mass spectrometer connected to an Eksigent NANO LC.1DPLUS chromatography system incorporating an auto-sampler. Sample was loaded onto a Biobasic C18 PicofritTM column (100 mm length, 75 mm ID) and was separated by an increasing acetonitrile gradient, using a 142 min reverse phase gradient (0-40% acetonitrile for 110 min) at a flow rate of 300 nL min⁻¹. The mass spectrometer was operated in positive ion mode with a capillary temperature of 200°C, a capillary voltage of 46V, a tube lens voltage of 140V and with a potential of 1800 V applied to the frit. All data was acquired with the mass spectrometer operating in automatic data dependent switching mode. A high resolution MS scan was performed using the Orbitrap to select the 5 most intense ions prior to MS/MS analysis using the Ion trap.

The incorporation efficiency of labelled amino-acids was determined by analysing the peptides identified in isolated nucleoli from cell population maintained in "Heavy" medium as described in [169] . Our analysis showed that we had an incorporation efficiency .95% (data not shown).

The MS/MS spectra were searched for peptides identification and quantification using the MaxQuant software [170] (version 1.1.1.36), the Human IPI Database (version 3.83) and the Andromeda search engine associated to MaxQuant [171] . Standard settings were used for MaxQuant with the Acetyl (Protein N-term) as variable modification and Carbamidomethyl (Cys) as fixed modification, 2 missed cleavage were allowed, except that the filtering of labelled amino acids was prohibited. Initial mass deviation of precursor ion and fragment ions were 7 ppm and 0.5 Da, respectively. Each protein ratio was calculated as the intensity-weighted average of the individual peptides ratios. Proteins were identified with the minimum of one peptide with a false discovery rate less than 1%.

Gene ontology, KEGG pathway and Pfam terms were extracted from UNIPROT entries using Perseus, a software from the MaxQuant Data analysis package (<http://www.maxquant.org>), and the ToppGene suite tools [54] .

The Jurkat NTAP-Tat and Jurkat NTAP were transfected using the Amaxa electroporation system (Amaxa biosystem) with the pGL3 (pGL3-LTR) (Promega) as recommended by Amaxa Biosystem. Dual-luciferase assays (Promega) were performed according to the manufacturer's instructions. Luciferase activity was measured and normalized against the total amount of proteins as quantified by the BCA protein quantification kit (Pierce, Thermo Scientific).

To preserve their original shape, we performed immunostaining of Jurkat cells in suspension. Cells were fixed in 2% PFA for 10 min at RT, permeabilised in 0.5% Triton X-100 for 15 min at RT and blocked with 5% FCS. Cells were incubated with the rabbit HIV-1 Tat antibody (ab43014, Abcam) followed by the secondary antibody anti-Rabbit alexa fluor 647 (A-21246, Invitrogen). Cells were allowed to attach to Cell-Tak (BD) coated Silanised Slides (DaoCytomation), and stained with DAPI. Images were captured with a Carl Zeiss Confocal Microscope equipped with a Plan-Apochromat 63X/1.4 oil DIC objective.

The proteomics RAW Data file from the mass spectrometry analysis was deposited to the Tranche repository([https:// proteomecommons.org/tranche/](https://proteomecommons.org/tranche/)) [172] . The file can be accessed and downloaded using the following hash key:

(R3O5SV5Z6HvWqrBNDhp21tXFetluDWYxvwMIfU-h6e1kMgarauCSq4dINcxeUvFOHDEzLeDcg4X5Y8reSb6-MUA6wM1kIAAAAAAAB/w = =).
Materials and Methods S1 Description of the methods employed to examine cell cycle, cell viability and cell proliferation analysis.

(DOCX)

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Interactome analysis of the lymphocytic choriomeningitis virus nucleoprotein in infected cells reveals ATPase Na(+)/K(+) transporting subunit Alpha 1 and prohibitin as host-cell factors involved in the life cycle of mammarenaviruses

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5834214/>

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Abstract: Several mammalian arenaviruses (mammarenaviruses) cause hemorrhagic fevers in humans and pose serious public health concerns in their endemic regions. Additionally, mounting evidence indicates that the worldwide-distributed, prototypic mammarenavirus, lymphocytic choriomeningitis virus (LCMV), is a neglected human pathogen of clinical significance. Concerns about human-pathogenic mammarenaviruses are exacerbated by the lack of licensed vaccines, and current anti-mammarenavirus therapy is limited to off-label use of ribavirin that is only partially effective. Detailed understanding of virus/host-cell interactions may facilitate the development of novel anti-mammarenavirus strategies by targeting components of the host-cell machinery that are required for efficient virus multiplication. Here we document the generation of a recombinant LCMV encoding a nucleoprotein (NP) containing an affinity tag (rLCMV/Strep-NP) and its use to capture the NP-interactome in infected cells. Our proteomic approach combined with genetics and pharmacological validation assays identified ATPase Na(+)/K(+) transporting subunit alpha 1 (ATP1A1) and prohibitin (PHB) as pro-viral factors. Cell-based assays revealed that ATP1A1 and PHB are involved in different steps of the virus life cycle. Accordingly, we observed a synergistic inhibitory effect on LCMV multiplication with a combination of ATP1A1 and PHB inhibitors. We show that ATP1A1 inhibitors suppress multiplication of Lassa virus and Candid#1, a live-attenuated vaccine strain of Junín virus, suggesting that the requirement of ATP1A1 in virus multiplication is conserved among genetically distantly related mammarenaviruses. Our findings suggest that clinically approved inhibitors of ATP1A1, like digoxin, could be repurposed to treat infections by mammarenaviruses pathogenic for humans.

Text: Introduction Mammarenaviruses (Arenaviridae: Mammarenavirus) cause chronic infections of rodents worldwide [1] . Invasion of human dwellings by infected rodents can result in human infections through mucosal exposure to aerosols or by direct contact of abraded skin with infectious material. Several mammarenaviruses cause viral hemorrhagic fevers (VHFs) in humans and pose important public health problems in their endemic areas [2] [3] [4] [5] [6] . Mammarenaviruses are classified into two main groups, Old World (OW) and New World (NW) [1] . The OW Lassa virus (LASV), causative agent of Lassa fever (LF), is the most significant OW mammarenaviral pathogen. LASV is estimated to infect several hundred thousand individuals annually in Western Africa, resulting in a high number of LF cases associated with high morbidity and lethality. Moreover, LASV endemic regions are expanding [7] , and the association of the recently identified mammarenavirus Lujo virus with a VHF outbreak in Southern Africa [8, 9] has raised concerns about the emergence of novel VHF-causing mammarenaviruses. The most significant NW mammarenavirus is Junín virus (JUNV), which causes Argentinian hemorrhagic fever [10] . The worldwide-distributed

OW mammarenavirus lymphocytic choriomeningitis virus (LCMV) is a neglected human pathogen of clinical significance especially in congenital infections [11] [12] [13] [14] [15] . Moreover, LCMV poses a particular threat to immunocompromised individuals, as has been illustrated by fatal cases of LCMV infection associated with organ transplants [16, 17] . However, LCMV research can be safely performed at BSL-2 containment, rather than the BSL-4 containment necessary for live LASV or JUNV research [18] .

No US Food and Drug Administration (FDA)-licensed vaccines are available for the treatment of arenavirus infections, although a live attenuated vaccine strain of JUNV, Candid#1, is licensed in Argentina. Likewise, current anti-mammarenavirus therapy is limited to an offlabel use of the nucleoside analogue ribavirin that is only partially effective and can cause significant side effects [19] [20] [21] . Development of effective anti-mammarenavirus drugs has been hampered by the lack of detailed understanding of virus/host-cell interactions required for mammarenavirus multiplication that could represent amenable targets for antiviral therapy. the potential problem that overexpression of a single viral gene product may potentiate PPI interactions that are not relevant during the course of a natural virus infection. To overcome this issue, we designed a recombinant LCMV (rLCMV) encoding a tandem [WSHPQFEK (GGGS) 3 WSHPQFEK] Strep-tag fused to the amino-terminus of NP (rLCMV/Strep-NP) (Fig 1A and 1B) . To facilitate the identification of specific PPI between NP and host cell proteins, we used our mammarenavirus tri-segmented (r3) platform [30] to design an r3LCMV expressing a C-terminus Strep-tag version of enhanced green fluorescent protein (r3LCMV/eGFP-Strep) that we used as a negative control (Fig 1A and 1B) . We rescued rLCMV/Strep-NP and r3LCMV/eGFP-Strep and confirmed the expression of strep-tagged NP and eGFP in rLCMV/ Strep-NP-and r3LCMV/eGFP-Strep-infected cells, respectively (Fig 1C) . Next, we examined the growth properties of rLCMV/Strep-NP and r3LCMV/eGFP-Strep in three different cells lines from hamsters, humans, and nonhuman primates (BHK-21, A549, and Vero E6 cells, respectively) (Fig 1D) . The fitness of rLCMV/Strep-NP and r3LCMV/eGFP-Strep was modestly decreased compared to that observed with wild-type (WT) Armstrong (rLCMV ARM) and Clone 13 (rLCMV Cl-13) strains of LCMV. However, both rLCMV/Strep-NP and r3LCMV/eGFP-Strep had WT-like growth kinetics and reached high titers. As with WT LCMV, infection with rLCMV/Strep-NP prevented production of bioactive IFN-I by cells in response to Sendai virus (SeV) infection as determined using an IFN bioassay based on protection against the cytopathic effect (CPE) induced by infection with vesicular stomatitis virus (VSV) (Fig 1E) . Vero cells treated for 16 h with tissue cultured supernatants (TCS) from A549 cells infected first with WT LCMV or rLCMV/Strep, followed by 24 h infection with SeV, remained fully susceptible to VSV-induced CPE. In contrast, Vero cells treated with TCS

from A549 cells infected with rLCMV/NP(D382A), a mutant unable to prevent induction of IFN-I [30] , and subsequently with SeV, were protected against VSV induced CPE.

We selected the human A549 cell line because lung epithelial cells are one of the initial cell targets of humans following inhalation of mammarenavirions. We infected A549 cells (multiplicity of infection [MOI] = 0.1) with either rLCMV/Strep-NP or r3LCMV/eGFP-Strep (Fig 2A) . At 48 h post-inoculation (pi), we prepared total cell lysates for pull-down (PD) assays using a sepharose resin coated with strep-tactin. Aliquots of the protein complexes present in the PD samples were fractionated by sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) (Fig 2B) followed by SYPRO Ruby protein gel staining. We compared the pattern of stained protein bands detected between rLCMV/Strep-NP-and r3LCMV/eGFP-Strepinfected samples and confirmed the presence of Strep-NP and eGFP-Strep in pertinent samples (Fig 2B) . Protein complexes in the rest of eluates from the PD samples were concentrated by trichloroacetic acid (TCA) precipitation and subjected to trypsin digestion (Fig 2A) . Digested peptides were subjected to liquid chromatography-tandem mass spectrometry (LC-MS/MS) analysis using a hybrid mass spectrometer consisting of linear quadrupole ion dual cell trap (LTQ) Velos and an Orbitrap analyser.

We classified host-cell proteins identified by LC-MS/MS analysis from two independent biological replicates into two groups: 1) proteins found only in Strep-NP PD samples with at least five spectral counts (Table 1) , and 2) proteins found in both Strep-NP and eGFP-Strep PD samples with five or higher spectral counts in Strep-NP samples and at least 2-fold higher spectral counts in Strep-NP PD compared to eGFP PD samples (Table 2) . Filtering the data using these criteria resulted in identification of 139 candidate host-cell proteins as NP-interacting partners. Among 53 proteins found present in both NP-and eGFP-PD samples, 36 had spectral counts in the NP-PD sample that were less than two-fold higher than their corresponding spectral counts in the eGFP-PD sample (Fig 2C and S1 Table) . Based on the criteria we described above, we considered these proteins to be highly unlikely specific NPinteractors with an involvement in the LCMV life cycle, and therefore we did not consider these hits for further analysis. However, we acknowledge that we cannot formally rule out that some of these hits could still play a role on the life cycle of LCMV.

The protein annotation through evolutionary relationship (PANTHER) protein family classification (Biological Process) of the NP-interacting protein candidates revealed that a large number of proteins were involved in metabolic and cellular processes (Fig 2D) . We

also analyzed the molecular functions of the NP-interacting protein candidates according to the PAN-THER protein profile classification (Fig 2E) , which revealed diversified biochemical functions enriched for nucleic acid-binding proteins and chaperones.

To initially assess pro-or anti-viral roles of NP-interacting host-cell proteins identified by LC-MS/MS, we examined the effect of small interfering RNA (siRNA)-mediated knockdown (kd) of each of the corresponding genes on multiplication of rLCMV expressing reporter gene ZsGreen (ZsG) in A549 cells (Fig 3A) . The siRNAs we used were from the genome-wide ON TARGET-Plus (OTP) Human siRNA library (18,301 genes, 4 siRNAs/gene). OTPs are latest generation of siRNAs and offer significant advantages over previous generations. Off-target effects are primarily driven by antisense strand microRNA (miR)-like seed activity. In OTPs, the sense strand is modified to favor antisense strand uptake whereas the antisense strand seed region is modified to drastically reduce seed-related off-targeting [33] . In addition, OTPs are designed on the foundation of the SMARTselection algorithm (Dharmacon), widely considered to be the best algorithm for rational siRNA design strategy. Numerous host-cell factors showed an anti-LCMV effect (increased ZsG expression by kd of the genes), including microtubule-associated protein 1B (MAP1B) [[38] , dengue virus 2 (DENV-2) [39] , and chikungunya virus (CHIKV) [40] .

To begin assessing the potential biological implications of the identified NP-host cell protein interactions, we selected ATP1A1 and PHB given the availability of reagents, existing knowledge about their roles in cell physiology, and evidence of participation in multiplication of other viruses. We confirmed that cells transfected with siRNA specific to ATP1A1 and PHB exhibited the predicted reduced levels in ATP1A1 and PHB protein expression (Fig 3B) . Likewise, we examined whether siRNA-mediated reduced expression levels of ZsGreen correlated with reduced LCMV progeny titers. For this, we transfected A549 cells with siRNA targeting Expression of Strep-tagged proteins. A549 cells seeded (5.0×10^5 cells/well) in a 6-well plate and cultured overnight were infected (MOI = 0.1) with the indicated rLCMV. At 48 h pi, total cell lysates were prepared, and protein expression was analyzed by western blotting. (D) Growth properties of rLCMV expressing Strep-tagged proteins. BHK-21 (1.75×10^5 cells/well), A549 (1.25×10^5 cells/well), or Vero E6 (1.25×10^5 cells/well) cells seeded in 24-well plates and cultured overnight were infected (MOI = 0.01) with the indicated rLCMV. At the indicated times pi, TCSs were collected and virus titers determined by IFFA. Results represent means $\pm$ SD of the results of three independent experiments. (E) Lack of induction of IFN-I in cells infected with rLCMV/Strep-NP. A549 cells were infected (MOI = 0.1) with the indicated rLCMV or mock-infected, and 36

h later infected with SeV (MOI = 1). At 24 h pi with SeV, TCS were collected and used, following virus inactivation by U.V., to treat Vero E6 cells for 16 h, followed by infection with rVSV (MOI = 0.1) [rVSV(+)] or mockinfection [rVSV (-)]. At 24 h pi with rVSV, cells were fixed with 4% PFA and stained with crystal violet to assess rVSV-induced cytopathic effect. We used as control rLCMV/NP(D382A) that contains mutation D382A within NP, which has been shown to abrogate the NP's ability to counteract the induction of IFN-I production.

<https://doi.org/10.1371/journal.ppat.1006892.g001> ATP1A1 or with NC siRNA 72 h prior to infection with rLCMV/ZsG. We found that siRNA-mediated kd of ATP1A1 dramatically inhibited ZsGreen expression (Fig 3Ci) , which was associated with a significant reduction of infectious LCMV progeny (Fig 3Cii) . Our attempts to see interaction between NP and ATP1A1 or NP and PHB by co-immunoprecipitation were unsuccessful. Several possibilities could account for this, including interactions of low affinity or high on/off rate or both. Another consideration is that only a minor fraction of NP might be engaged in the interaction with a given host cell protein, and therefore, detection of these interactions would require highly sensitive methods such as LC-MS/MS. To overcome this problem we used confocal microscopy to examine the co-localization of NP with ATP1A1 and PHB in LCMV-infected cells. Weighted co-localization coefficients (CC), determined by taking into consideration the brightness of each channel signal, were significantly higher than non-weighted CC, indicating the presence of brighter pixels in the co-localized regions compared to the non-co-localized regions (Fig 4) .

The cardiac glycoside ouabain is an inhibitor of ATP1A1 that has been used to treat congestive heart failure in European countries [41] . The PHB inhibitor rocaglamide is a flavagline from an *Aglaia* tree used in traditional Chinese medicine [42] that has potent anticancer activity [43] . To examine whether pharmacological inhibition of ATP1A1 or PHB inhibited LCMV multiplication, we pretreated human (A549 and HEK 293T), nonhuman primate (Vero E6), and rodent (murine L929 and hamster BHK-21) cells with ouabain or rocaglamide and infected them with rLCMV/eGFP (S1 Fig). Ouabain treatment resulted in a strong dose-dependent inhibition of eGFP expression in infected human and nonhuman primate cells, but did not affect eGFP expression intensity in infected rodent cells (S1A Fig) . This finding is consistent with rodents expressing an ATP1A1 allele that is resistant to ouabain inhibition [44] .

Likewise, we observed a dose-dependent rocaglamide inhibition of eGFP expression in all cell lines infected with rLCMV/eGFP (S1B Fig) . Consistent with these findings, production of infectious LCMV progeny was reduced by treatment with either ouabain or rocaglamide (Fig 5A) within a concentration range that had minimal impact on cell viability (Fig 5B) . To examine the correlation between efficacy and cytotoxicity of these compounds, we determined their therapeutic index (TI = CC 50 /IC 50 (Fig 5Bi) ; whereas rocaglamide had TIs of >105 (CC 50 > 1000 nM, IC 50 = 9.51 nM) and 10.3 (CC 50 = 100 nM, IC 50 = 9.75 nM) in A549 and Vero E6 cells, respectively (Fig 5Bii) . Moreover, the ATP1A1 antagonist inhibitor, bufalin, also exhibited robust anti-LCMV activity with TIs of 8.92 (CC 50 Heat shock protein HSP 90- α HSP90AA1 P07900 8 10 9

Endoplasmic HSP90B1 P14625 9 9 9

Large neutral amino acids transporter small subunit 1 SLC7A5 Q01650 6 12 9

Keratin, type II cuticular Hb1 KRT81 Q14533 10 8 9

Putative helicase MOV-10 MOV10 Q9HCE1 8 10 9

Microtubule-associated protein 1B MAP1B P46821 6 11 8.5

Spectrin beta chain, rocaglamide (100 nM) (Fig 5C) , further supporting a specific anti-LCMV activity of ouabain and rocaglamide that was not due to reduced cell viability.

To gain insights about the mechanisms by which ouabain and rocaglamide exert their anti-LCMV activity, we examined effects of these compounds on distinct steps of the LCMV life cycle. First, we asked whether ouabain and rocaglamide affected cell entry of LCMV. We conducted time-of-addition experiments in which we treated cells with ouabain or rocaglamide prior to virus inoculation (-1.5 h), at the time of inoculation (0 h), or 1.5 h pi (+1.5 h) (Fig 6A) .

In some samples, we used ammonium chloride starting at 4 h pi to block multiple rounds of virus infection. The timing of compound addition did not significantly change the number of eGFP-positive cells, indicating that neither ouabain nor rocaglamide inhibited cell entry of LCMV. The number of eGFP + cells in ouabain-treated cells was reduced at all time-of-addition points compared to vehicle dimethyl sulfoxide (DMSO)-treated cells, but was similar to that observed in ammonium chloride-treated cells. Thus, ouabain did not inhibit LCMV RNA replication and gene expression, but rather a late step of the LCMV life cycle. In contrast, rocaglamide treatment resulted in a negligible number of eGFP + cells, indicating that rocaglamide inhibited virus RNA replication and gene transcription.

To further investigate the effect of ouabain and rocaglamide on virus RNA synthesis, we infected A549 cells with a recombinant single-cycle infectious LCMV expressing eGFP (rLCMVΔGPC/eGFP) and treated cells with either ouabain or rocaglamide. Seventy-two hours later, we determined percent normalized eGFP expression in infected cells (Fig 6B) . Consistent with our results from the time-of-addition experiment, ouabain did not affect reporter eGFP expression. However, rocaglamide reduced eGFP expression, confirming inhibitory effect of rocaglamide on virus RNA synthesis.

We also examined the effect of ouabain and rocaglamide on the late step of the arenavirus life cycle, Z-mediated budding. For this experiment, we transfected cells with Z-Strep-and Z-FLAG (DYKDDDDK epitope)-expressing plasmids from LCMV and LASV, respectively. At 24 h post-transfection, we removed the tissue culture supernatant (TCS) and washed extensively transfected cells to eliminate already budded Z. We cultured the transfected cells in the presence or absence of ouabain or rocaglamide. At 24 h, we determined by WB levels of Z protein in both whole cell lysates and associated with virus-like particles (VLPs) collected from TCS. Treatment with rocaglamide, but not with ouabain, caused a reduction in both LCMV and LASV Z budding efficiency (Fig 6C and 6D) . The reproducibility of these findings was confirmed based on results from four independent experiments (Fig 6E) . We also examined whether ouabain could interfere with a step of assembly of infectious progeny that was not captured by the Z budding assay through two additional experiments. The first experiment involved the use of a newly generated single-cycle infectious recombinant LCMV expressing the reporter gene ZsGreen (scrLCMV/ZsG-P2A-NP) to infect (MOI = 0.1) A549 cells (1 st infection). These cells were subsequently transfected with a plasmid expressing LCMV GPC. After 24 h, we used TCS to infect a fresh cell monolayer (2 nd infection) and identified infected cells based on ZsGreen expression. To assess the effect of ouabain on de novo assembly of infectious progeny we determined normalized

ratios (2nd / 1st infection) of ZsGreen + cells (Fig 6F) . The second experiment involved infection (MOI = 0.1) of cells with WT LCMV, and 48 h later we washed infected cells three times to remove the extracellular infectious progeny produced during the first 48 h of infection. Then, fresh media containing ouabain or DMSO vehicle control were added, and 24 h later we determined virus titers in TCS (Fig 6G) . Results from both experiments consistently showed that ouabain did not inhibit assembly de novo of extracellular infectious virus.

Combination therapy can significantly alleviate the problem posed by the rapid emergence of drug-resistant variants commonly observed during monotherapy strategies to control RNA virus infections. Since ouabain and rocaglamide inhibited different steps of the LCMV life cycle, we examined whether their use in combination results in a synergistic anti-LCMV effect. For this experiment, we infected A549 cells with rLCMV/eGFP and treated them with ouabain and rocaglamide using different concentrations and combinations. At 48 h pi, we determined percent eGFP expression (Fig 7) . Combination treatment with ouabain and rocaglamide resulted in synergistic anti-LCMV activity that was enhanced under conditions using higher concentrations of ouabain and lower concentrations of rocaglamide.

We next asked whether the ATP1A1 and PHB host-cell factors contributed also to multiplication of viral hemorrhagic fever-causing LASV. We treated A549 cells with ouabain, bufalin, or rocaglamide and inoculated the treated cells with recombinant LASV expressing eGFP (rLASV/eGFP). eGFP expression was examined 48 h later. Similar to rLCMV infection, LASV multiplication was restricted in ouabain-, bufalin-, or rocaglamide-treated cells at concentrations minimally impacting cell viability, although their IC₅₀ values were slightly higher than those found with the LCMV infection system (Fig 5B and S2 Fig) as ouabain had IC₅₀ of 9.34 nM, bufalin had IC₅₀ of 1.66 nM and rocaglamide had IC₅₀ of 37.0 nM (Fig 8) . We also tested the effect of compounds targeting ATP1A1 and PHB on multiplication of JUNV. Consistent with our results obtained with LCMV and LASV, ouabain, bufalin, and rocaglamide strongly suppressed JUNV multiplication (S3 Fig). These findings indicate that ATP1A1 and PHB function as pro-viral factors of a range of mammarenaviruses.

We identified ATP1A1 and PHB as novel host-cell proteins that contribute to efficient multiplication of mammarenaviruses. Our approach using a recombinant LCMV expressing NP with an affinity tag facilitated defining the NP interactome in the context of LCMV infection. Recently, using an NP-specific monoclonal antibody (mAb) to precipitate NP and

associated cellular protein partners in a mammarenavirus NP interactome, King et al. identified 348 host proteins that associated with LCMV NP [45]. We found 99 common proteins between our filtered LCMV NP interactome of 171 proteins and the LCMV NP interactome documented by King et al. Differences in both experimental conditions and analysis methodologies used to generate the LCMV NP interactome documented by King et al. and ours likely h post-transfection, cells were washed with fresh media to eliminate Z-mediated production of VLPs in the absence of compound treatment, and cultured for another 24 h in fresh media in the presence of ouabain or Roc-A at the indicated concentrations. VLPs present in TCS were collected by ultracentrifugation, and cell lysates were prepared. Z protein expression in VLPs and cell lysates were determined by western blots using antibodies to Strep-tag (C) and FLAG-tag (D). Budding efficiency for each sample was estimated by dividing the signal intensity of the Z protein associated with VLPs by that of Z detected in the cell lysate. Numbers on the bottom of panel C correspond to LCMV Z budding efficiencies determined in a representative experiment. Results shown in panel E correspond to the average and SD from four independent experiments including the one shown in panel D. The mean budding efficiency of DMSO treated samples was set to 100%. Data represent mean $\pm$ SD of four independent experiments. (F) Effect of ouabain on incorporation of viral glycoprotein into virions. 293T cells seeded (4.0×10^5 cells/well) in a 12-well plate and cultured overnight were infected (MOI = 0.1) with scrLCMV/ZsG (1st infection) for 2 h and subsequently transfected with 0.5 μ g of pC-GPC. At 24 h pi, cells were washed with fresh medium to eliminate infectious virus particle produced in the absence of compound treatment, and cultured for another 24 h in fresh media in the presence of ouabain at 40 nM (OUA). At 48 h pi, TCS was collected and used to infect fresh monolayer of BHK-21 cells (2nd infection) seeded (4.0×10^5 cells/well) in a 12-well plate 1 day before the infection, and 293T cell lysate was prepared. 24 h later, BHK-21 cell lysate was prepared. ZsGreen signal intensity was measured by a fluorescent plate reader. GP-incorporation efficiency was estimated by dividing ZsGreen signal intensity in BHK-21 cell lysate (2nd) by that in 293T cell lysate (1st). The mean GP-incorporation efficiency of DMSO treated samples was set to 100%. Data represent means $\pm$ SD from three independent experiments. contributed to the observed differences between data sets. Despite progress in the implementation of global proteomics-based screens to identify virus-host protein-protein interactions, overlap between datasets for the same viral system is usually limited. However, the substantial overlap of 99 of the 171 NP-interacting proteins from both studies supports the overall reliability of both systems. We used results of the eGFP-Strep interactome, determined in r3LCMV/eGFP-Strep-infected cells, as a control to filter out non-specific NP interactions, which may have resulted in a higher degree of stringency than in the study by King et al for selection of NP-interacting candidates. The combined information provided by the NP interactome reported by King et al. and the one

we present in this work, will facilitate future studies to further characterize the functional and biological implications of NP-host cell interacting proteins.

All tested mammarenavirus NPs, with exception of the NP from TCRV, blocked IRF-3-dependent IFN-I induction [25, 46]. The anti-IFN activity of NP was mapped to its C-terminal part and linked to the 3'-5' exonuclease domain present with the NP C-terminal part [30]. Inhibitor-B kinase ϵ (IKK ϵ) was identified as an NP-binding protein using plasmid-mediated overexpression in transfected cells [47], and the NP-IKK ϵ binding affinity correlated with NP's ability to inhibit IFN-I induction [47]. We, as well as the work by King et al. [45], did not detect this NP-IKK ϵ interaction. This discrepancy may have been caused by very low expression of IKK ϵ in LCMV-infected cells, which prevented detection of IKK ϵ by LC-MS/MS. Alternatively, NP-IKK ϵ interaction could possibly be temporarily regulated and take place at early times pi, but could be mostly absent at 48 h pi, the time at which we prepared the cell lysates for our proteomics studies. Future studies comparing the NP interactome at different times during infection will contribute to a better understanding of the dynamics of NP/hostcell protein interactions.

Na⁺/K⁺-ATPase is a well-characterized membrane ion transporter and is composed of two functional subunits (α and β) and one regulatory γ subunit [48]. ATP1A1 represents one of four α subunits [49, 50]. Recent evidence has suggested that the Na⁺/K⁺-ATPase is involved in multiple cell signaling pathways that are independent of its ion-pumping function [51]. Cardiac glycoside inhibitors of the Na⁺/K⁺-ATPase (NKA), so-called cardiotonic steroids (CST; e.g., ouabain, bufalin), have been shown to inhibit multiplication of different viruses including Ebola virus [35], coronaviruses [36], herpes simplex virus 1 [52, 53], CHIKV [54], human immunodeficiency virus 1 (HIV-1) [55], adenovirus [56] and porcine reproductive and respiratory syndrome virus 1 [57].

Different mechanisms are likely to contribute to the antiviral activity of CSTs, including altered cell functions modulated by the signaling activity of Na⁺/K⁺-ATPase [58]. Thus, a low concentration of ouabain induces a conformational change in ATP1A1 that results in activation and release of proto-oncogene tyrosine protein kinase, Src, from ATP1A1, followed by activation of as yet unknown downstream signaling that inhibits, for instance, cell entry of murine hepatitis virus (MHV) [59]. However, our results indicated that ouabain did not interfere with LCMV cell entry. In addition, treatment with the Src inhibitor 4-amino-5-(4-methylphenyl)-7-(t-butyl)pyrazolo [3,4-d] pyrimidine (PP1) did not counteract the anti-

LCMV activity of ouabain (S4 Fig). Nevertheless, ATP1A1-mediated Src signaling could plausibly contribute to the inhibitory effect of ouabain on JUNV multiplication as similarly to that observed with MHV. Moreover, cell entry of JUNV occurs also by clathrin-mediated endocytosis [60], a process affected by Src signaling. Ouabain has been clinically used in several European countries for the management of congestive heart failure, whereas bufalin has been tested in clinical trials for cancer treatments [61], and the CST digoxin has been FDA-approved since 1997 to treat heart failure and atrial fibrillation. Hence, opportunities for the repurposing CSTs have potential as therapeutics to treat infections caused by viral hemorrhagic fever-causing arenaviruses.

The PHB inhibitor, rocaglamide, appeared to interfere with LCMV RNA synthesis and budding, but did not affect LCMV cell entry. In contrast, PHB was reported to be a cell entry receptor for DENV-2 [39] and CHIKV [40]. On the other hand, PHB did not act as a virus cell entry receptor for HCV. Rather, PHB contributed to HCV cell entry through binding to cellular RAF (c-Raf; proto-oncogene serine/threonine-protein kinase) and subsequent Harvey rat sarcoma proto-oncogene (HRas) activation that induces a signal transduction pathway required for epidermal growth factor receptor (EGFR)-mediated HCV cell entry [37]. In addition, siRNA-mediated kd of PHB decreased production of H5N1 FLUAV [38]. These findings indicate that PHB is involved in different steps of the life cycle of a variety of viruses, and thereby an attractive target for the development of broad-spectrum antiviral drugs.

Rocaglate is a group of natural compounds, which includes rocaglamide, that inhibits protein synthesis by targeting the ATP-dependent DEAD-box RNA helicase eukaryotic initiation factor 4A (eIF4A) and exerts anti-tumor activity [62, 63]. The rocaglate compound, silvestrol, inhibits Ebola virus multiplication likely by interfering with the role of eIF4A in viral protein translation [64].

While we focused on two host proteins, ATP1A1 and PHB, in this study, our proteomics approach also identified several NP-interacting host-cell proteins whose kd expression via siRNA resulted in increased LCMV multiplication. These proteins, which included MAP1B, might have anti-LCMV activity. MAP1B has been shown to bind to nonstructural proteins 1 (NS1) and 2 (NS2) of human respiratory syncytial virus (HRSV) [34]. NS1 and NS2 of HRSV antagonizes host IFN-I response by reducing the levels of TNF receptor associated factor (TRAF3), IKK ϵ (NS1), and STAT2 (NS2) [65]. NS2-MAP1B interaction interfered with HRSV

NS2's ability to reduce levels of STAT2, whereas the role of NS1-MAP1B interaction remains to be determined [34] . Examining the role of NP-MAP1B interaction in modulating NP's ability to inhibit induction of type I IFN is of interest.

We identified among the NP-interacting host cell proteins the RNA helicase Moloney leukemia virus 10 (MOV10), which has been reported to be an antiviral factor for FLUAV [66] , retroviruses [67] [68] [69] [70] [71] , and DENV-2 [72] . We did not observe increased LCMV multiplication in cells subjected to siRNA-mediated kd of MOV10, a finding that would question an anti-LCMV activity of MOV10. However, we consider that LCMV has already optimal multiplication in A549 cells and further increases may occur only under rather unique conditions. MOV10 was shown to enhance IRF-3-mediated IFN-I induction following SeV infection through a tank binding kinase 1 (TBK1)-independent and IKK ϵ -dependent manner. This finding was further supported by demonstrating MOV10-IKK ϵ interaction by co-immunoprecipitation studies [73] . We documented that the anti-IFN activity of mammarenavirus NP correlated with its ability to interact with IKK ϵ [47] . Whether NP-MOV10 interaction prevents MOV10 from enhancing IRF-3-mediated IFN-I induction remains to be determined.

Several members of the mammalian chaperonin-containing T-complex (CCT) were identified as prominent hits in our NP interactome. The mammalian CCT is critical for folding of many proteins with important functions in diverse cellular processes [74] , and may protect complex protein topologies within its central cavity during biosynthesis and folding [75] . The contribution of CCT members to NP assembly into a nucleocapsid structure could account for their presence in the NP, but not eGFP, interactome. Interestingly, members of the CCT have been implicated in multiplication of different viruses including rabies virus [76, 77] , HCV [78] and FLUAV [79] . However, the role of these CCT proteins in virus multiplication remains unknown and may involve functions other than acting as molecular chaperones.

Previous studies documented the presence of several components of the of eIF4F, including 4A, within viral replication-transcription complexes (RTC) detected in cells infected with LCMV [80] and TCRV [81] . These findings, together with the detection of a number of ribosomal proteins in the NP interactome, suggest that translation of viral mRNAs may take place within RTC. However, rocaglamide interference with the activity of eIF4A within the viral RTC might contribute to its anti-LCMV activity.

In this work, we documented the generation of rLCMV/Strep-NP and its use to define the NP-interactome in infected cells. We presented evidence that ATP1A1 and PHB contribute to efficient multiplication of mammarenaviruses using genetics and pharmacological inhibition of the genes. Consistent with our findings, bioinformatic analysis revealed that the protein network associated with ATP1A1 and PHB involves host cell proteins with functions in biological processes that have been implicated in virus multiplication (S5 Fig). The overall experimental approach described here can facilitate the identification of biologically relevant NP-interacting host-cell proteins. Future studies elucidating the roles of pro-and antiviral host-cell factors identified in this study in mammarenavirus multiplication will advance our understanding of the multiple functions of NP and uncover novel cellular targets for the development of antimammarenaviral drugs. In addition, by identifying proviral host-cell factors, drugs that are already approved can be repurposing as therapeutics to combat human pathogenic mammarenavirus infections.

Baby hamster kidney BHK-21 (American Type Culture Collection, ATCC, CCL-10), house mouse L929 (ATCC CCL-1), grivet Vero E6 (ATCC CRL-1586), human A549 (ATCC CCL-185), and human HEK 293T (ATCC CRL-3216) cells were grown in Dulbecco's modified Eagle's medium (Thermo Fisher Scientific, Waltham, MA) containing 10% heat-inactivated fetal bovine serum, 2 mM of L-glutamine, 100 mg/ml of streptomycin, and 100 U/ml of penicillin at 37 ° C and 5% CO₂.

WT recombinant LCMVs, Armstrong (rLCMV ARM) and clone-13 (rLCMV Cl-13) strains, were generated as described [30, 82, 83]. Generation of rLCMV/NP(D382A) and SeV, strain Cantell, was described [30, 82, 83]. An rLCMV lacking GPC and expressing eGFP (rLCMVΔGPC/eGFP) was generated by reverse genetics using procedures previously described [84]. rLCMV/Strep-NP and r3LCMV/eGFP-Strep were generated by reverse genetics using similar procedures to generate WT rLCMV and tri-segmented LCMV (r3LCMV) expressing eGFP [30]. For the generation of these novel rLCMV, we created pol1S Cl-13 plasmids that directed Pol1-mediated intracellular synthesis of recombinant LCMV S genome RNA species coding for Strep-tagged NP or eGFP, respectively (Fig 1A and 1B). The rLCMV expressing eGFP (rLCMV/eGFP) was generated as described [85], and the rLCMV expressing ZsGreen (rLCMV/ZsG) instead of eGFP was generated by reverse genetics using similar procedures to generate rLCMV/eGFP. Generation of rLASV expressing eGFP (rLASV/eGFP) will be described elsewhere. A tri-segmented recombinant live-

attenuated Candid #1 strain of JUNV expressing eGFP (r3JUNV/eGFP) was generated as described [86] . For the generation of a novel single cycle rLCMV expressing ZsGreen (scrLCMV/ZsG-P2A-NP), a pol1S plasmid was created by omitting GPC open reading frame (ORF) from pol1S plasmid used for the generation of rLCMV/ZsG. scrLCMV/ZsG-P2A-NP was rescued by reverse genetics using similar procedures to generate rLCMVΔGPC/eGFP [84] .

LCMV titers were determined by immunofocus forming assay (IFFA) as described [87] . Briefly, 10-fold serial virus dilutions were used to infect Vero E6 cell monolayers in a 96-well plate, and at 20 h pi, cells were fixed with 4% paraformaldehyde (PFA) in phosphate-buffered saline (PBS). After cell permeabilization by treatment with dilution buffer (DB) (0.3% Triton X-100 in PBS-containing 3% bovine serum albumin [BSA]), cells were stained with a rat mAb to NP (VL-4, Bio X Cell, West Lebanon, NH) conjugated with Alexa Fluor 488 (VL-4-AF488, Protein Labeling Kit, Life Technologies, Carlsbad, CA). VSV titers were determined by a plaque assay.

Total cell lysates were prepared in PD lysis buffer (+) (250 mM of NaCl, 50 mM of Tris-HCl [pH = 7.5], 0.5% TritonX-100, 10% glycerol, 1 mM of MgCl₂ , 1 μM of CaCl₂ , 1 μM of ZnCl₂) and clarified by centrifugation at 21,130 x g at 4 °C for 10 min. Clarified lysates were mixed at a 1:1 ratio with loading buffer (100 mM of Tris [pH 6.8], 20% 2-mercaptoethanol, 4% SDS, 0.2% bromophenol blue, 20% glycerol) and boiled for 5 min. Proteins samples were fractionated by SDS-PAGE using 4-20% gradient polyacrylamide gels (Mini-PROTEAN TGX gels 4-20%, Bio-Rad, Hercules, CA), and proteins were transferred by electroblotting onto polyvinylidene difluoride membranes (Immobilin Transfer Membranes, Millipore, Billerica, MA). To detect Strep-tagged proteins, membranes were reacted with mouse monoclonal antibodies to Strep (QIAGEN, Germantown, MD), eGFP (Takara Bio USA, Mountain View, CA), GP 2 (We33/ 36), ATP1A1 (ThermoFisher Scientific, Rockford, IL), PHB (Abcam, Cambridge, MA) or rabbit polyclonal antibodies to α-tubulin (Cell Signaling Technologies, Danvers, MA) or glyceraldehyde-3-phosphate dehydrogenase (GAPDH, Millipore), respectively, followed by incubation with appropriate horseradish peroxidase-conjugated anti-mouse or anti-rabbit immunoglobulin G (IgG) antibodies (Jackson ImmunoResearch Laboratories, West Grove, PA). SuperSignal West Pico or Femto chemiluminescent substrate (Thermo Fisher Scientific) was used to elicit chemiluminescent signals that were visualized using ImageQuant LAS 4000 Imager (GE Healthcare Bio-Sciences, Pittsburgh, PA).

Pull down of strep-tagged proteins from infected cell lysate. A549 cells prepared in six 15-cm dishes (approximately 1.0×10^8 cells in total) were infected with either rLCMV/Strep-NP or rLCMV/eGFP at an MOI of 0.1. At 48 h pi, cells were washed three times with ice-cold PBS, scraped into fresh ice-cold PBS, and centrifuged at $400 \times g$ at 4°C for 10 min. Supernatant was removed, and cells were lysed with 12 ml of PD lysis buffer (+) supplemented with halt protease and phosphatase inhibitor cocktail (Thermo Fisher Scientific) and $5 \mu\text{g/ml}$ of deoxyribonuclease I (Worthington Biochemical Corporation, Lakewood, NJ). Lysate was clarified by centrifugation at $3,900 \times g$ at 4°C for 30 min to remove cell debris. Clarified cell lysate was then incubated with strep-tactin sepharose resin (QIAGEN) at 4°C . After 2 h of incubation, the resin was washed three times with PD lysis buffer (+) and once with PD lysis buffer without TritonX-100 (PD lysis buffer [-]). After the centrifugation at $1,600 \times g$ and 4°C for 5 min, the last wash buffer was removed, and protein complexes associated with the resin were eluted into 2 ml of PD lysis buffer (-) containing 2.5 mM of desthiobiotin. The eluate was then subjected to TCA precipitation followed by trypsin digestion.

Multidimensional protein identification technology microcolumn. A MudPIT microcolumn was prepared by first creating a Kasil frit at one end of an un-deactivated 250- μm outside diameter (OD) capillary tube (interior diameter of 360 μm) (Agilent Technologies, Inc., Santa Clara, CA). The Kasil frit was prepared by briefly dipping a 20-30-cm capillary tube in 300 μl of Kasil 1624 potassium silicate well-mixed solution (PQ Corporation, Malvern, PA) and 100 μl of formamide, curing at 100°C for 4 h, and cutting the frit to a length of ≈ 2 mm. Strong cation exchange particles (SCX Luna, 5- μm diameter, 125 $\text{\AA}$ pores, Phenomenex, Torrance, CA) were packed in-house from particle slurries in methanol to 2.5 cm. Reversed phase particles (2 cm, C18 Aqua, 3- μm diameter, 125 $\text{\AA}$ pores, Phenomenex) were then successively packed onto the capillary tube using the same method as SCX loading.

MudPIT analysis. An analytical reversed-phase liquid chromatography column was generated by pulling a 100- μm (interior diameter (ID) of 360 μm) OD capillary tube (Polymicro Technologies, Phoenix, AZ) to 5- μm ID tip. Reversed-phase particles (Luna C18, 3- μm diameter, 125 $\text{\AA}$ pores, Phenomenex) were packed directly into the pulled column at 5.5 mPa until 15 cm long. The column was further packed, washed, and equilibrated at 10 mPa with buffer B (80% acetonitrile, 0.1% formic acid) followed by buffer A (5% acetonitrile and 0.1% formic acid). MudPIT and analytical columns were assembled using a zero-dead

volume union (Upchurch Scientific, Oak Harbor, WA). LC-MS/MS analysis was performed with an Agilent high-pressure LC pump (Agilent) and linear quadrupole ion dual cell trap Orbitrap Velos (Thermo) using an in-house built electrospray stage. Electrospray was performed directly from the analytical column by applying the electrospray ionization (ESI) voltage at a tee (150 μ m ID, Upchurch Scientific) directly downstream of a 1:1,000 split flow to reduce the flow rate to 300 nL/min through the columns. MudPIT experiments (10-step) were performed in which each step corresponds to 0, 10, 20, 40, 50, 60, 70, 80, 90, and 100% buffer C (500 mM of ammonium acetate, 0.1% formic acid, and 5% acetonitrile) and was run for 3 min at the beginning of a 110-min gradient.

Data analysis. Protein and peptide identification were performed with Integrated Proteomics Pipeline-IP2 (Integrated Proteomics Applications, San Diego, CA. <http://www.integratedproteomics.com/>) using ProLuCID and DTASelect2 algorithms. DTASelect parameters were -p 2 -y 1-tryptstat-pfp .01 -extra-pl-DB-dm-in. Spectrum raw files were extracted into ms2 files from raw files using open source RawExtract 1.9.9 (Scripps Research Institute, La Jolla, CA; <http://fields.scripps.edu/downloads.php>), and the tandem mass spectra were searched against a human protein database (UniprotKB). To accurately estimate peptide probabilities and false discovery rates, we used a decoy database containing the reversed sequences of all the proteins appended to the target database. Tandem mass spectra were matched to sequences using the ProLuCID algorithm with a 600-ppm peptide mass tolerance. ProLuCID searches were done on an Intel Xeon cluster processor running under the Linux operating system. The search space included half and fully tryptic peptide candidates that fell within the mass tolerance window with no miscleavage constraint. Carbamidomethylation (+57.02146 Da) of cysteine was considered as a static modification. siRNA screening A549 cells (1,000 cells/well) in a 384-well plate were reverse transfected with 0.5 pmol of siRNA pool (S2 Table) targeting each gene using 0.1 μ L of Lipofectamine RNAiMAX (Thermo Fisher Scientific) (final siRNA concentration was 10 nM), followed by incubation at 37 °C and 5% CO₂. At 72 h post-transfection, cells were infected (MOI = 0.05) with rLCMV/ZsG. siRNA target host-cell proteins were selected based on availability of validated siRNA sequences. The siRNAs we used to examine the effects on LCMV multiplication of knockdown expression of NP-interacting host cell protein candidate hits corresponded to the Genome-wide ON TARGET-Plus (OTP) Human siRNA library (18,301 genes, 4 siRNAs/gene; Dharmacon, Lafayette, CO).

A549 cells (3.0×10^4 cells/well) were reverse transfected in a 24-well plate with 6 pmol of siRNA pools targeting each gene using 1 μ l of Lipofectamine RNAiMAX (final siRNA concentration is 10 nM). At 72 h post-transfection, total cell lysate was prepared in modified lysis A buffer (25 mM Tris-HCl [pH = 8.0], 50 mM NaCl, 1% Triton X-100, 1.25% sodium deoxycholate) and clarified by centrifugation at $21,130 \times g$ at 4°C for 10 min. The total protein concentration of clarified cell lysate was measured by Pierce BCA Protein Assay Kit (Thermo Fisher Scientific). The same amount of protein from each sample was subjected to SDS-PAGE, and the protein expression of siRNA-targeted genes was analyzed by western blots.

Cells infected with eGFP-or ZsGreen-expressing rLCMV were fixed with 4% PFA in PBS. After cell permeabilization by treatment with DB, cells were stained with 4',6-diamidino-2-phenylindole (DAPI). Green fluorescence (eGFP or ZsGreen) and DAPI signals were measured by a fluorescent plate reader (Synergy H4 Hybrid Multi-Mode Microplate Reader, Bio-Tek, Winooski, VT).

Mock-and virus-infected cells were fixed with 4% PFA. After cell permeabilization and blocking by treatment with DB containing 1% normal goat serum, cells were incubated with primary mouse anti ATP1A1 or PHB antibody followed by secondary anti-mouse IgG antibody conjugated with Alexa Fluore 568 (anti-mouse IgG-AF568). Subsequently, cells were stained with VL-4-AF488. In some samples, primary antibody against ATP1A1 or PHB was omitted to determine background fluorescence. To visualize nuclei, DAPI Fluoromount-G (SouthernBiotech, Birmingham, AL) was used to mount coverslips on a slide glass. Stained cells were observed under a confocal microscope (LSM 710, Zeiss) and data analyzed by ZEN software (Zeiss). Co-localization analysis was performed on a pixel by pixel basis using Zen software (Zeiss). Eight green (NP-positive) cells were marked and every pixel in the marked area was plotted in the scatter diagram based on its intensity level from each channel. Thresholds for green and red channels were determined using mock-infected cells stained with VL-4-AF488 (anti-NP) and anti-mouse IgG antibody conjugated with Alexa Fluor 568, without using anti-ATP1A1 or -PHB antibodies. Each pixel was assigned a value of 1. Co-localization coefficients (CC) (or non-weighted CC) were determined by dividing the sum of both green-and red-positive pixels by the sum of green positive pixels. This calculation was repeated for eight individual cells. To assess the specificity of co-localization, we determined weighted CC by taking into consideration the brightness of each channel signal. Comparison of non-weighted and weighted CC allowed us to determine whether brighter pixels were present in the co-localized regions compared

to the non-co-localized regions. p values were determined by a two-tailed paired t test using GraphPad Prism software.

A549 or Vero E6 cells seeded (2.0×10^4 cells/well) in a 96-well plate and cultured overnight were treated with 3-fold serial compound dilutions at 37 °C and 5% CO₂ for 2 h, followed by infection with rLCMV/eGFP (MOI = 0.01). Compounds were present to study endpoint. At 48 h pi, cells were fixed with 4% PFA in PBS, and eGFP expression was examined by a fluorescent plate reader (Synergy H4 Hybrid Multi-Mode Microplate Reader, BioTek). Mean values obtained with DMSO-treated and rLCMV/eGFP-infected cells were set to 100%. The IC₅₀ concentrations were determined using GraphPad Prism.

A549 or Vero E6 cells seeded in a 96-well plate (2.0×10^4 cells/well) and cultured overnight were treated with 3-fold serial compound dilutions and cultured at 37 °C and 5% CO₂ for 48 h. Then, CellTiter 96 AQueous one solution reagent (Promega, Madison, WI) was added. Thereafter, the assay was performed according to the manufacturer's recommendations, and the absorbance (490 nm) was obtained using an enzyme-linked immunosorbent assay (ELISA) reader (SPECTRA max plus 384; Molecular Devices, Sunnyvale, CA). Mean values obtained with DMSO-treated cells were set to 100%. The CC₅₀ concentrations were determined using GraphPad Prism.

A plasmid expressing C-terminus Strep-tagged Z protein (pC-LCMV-Z-Strep) was generated using similar procedure to generate a plasmid expressing C-terminus FLAG-tagged LASV Z protein (pC-LASV-Z-FLAG), and the budding assay was performed as previously described [88]. Cells (HEK 293T) in a 12-well plate were transfected with 0.5 µg of empty pCAGGS vector or pC-LCMV-Z-Strep or pC-LASV-Z-FLAG using Lipofectamine 2000. At 5 h post-transfection, media were replaced with fresh media and incubated at 37 °C and 5% CO₂ for 19 h. Then the cells were three times washed with fresh medium. After the removal of the last wash medium, cells were cultured in fresh medium containing ouabain (30 or 40 nM) or rocaglamide (50 or 100 nM) or equivalent concentration of DMSO, and 24 h later, virion-like particle (VLP)-containing TCS and cells were collected. Total cell lysate was prepared by lysing the cells with lysis buffer (1% NP-40, 50 mM of Tris-HCl [pH 8.0], 62.5 mM NaCl, 0.4% sodium deoxycholate). After clarification of TCS from cell debris by centrifugation at 400 x g and 4 °C for 10 min, VLPs were collected by ultracentrifugation at 100,000 x g and 4 °C for 30 min through a 20% sucrose cushion. VLPs were resuspended in

PBS, and Z expression in total cell lysate and TCS (containing VLPs) were analyzed by western blots.

A549 cells infected with rLCMV/eGFP were harvested using Accutase cell detachment solution (Innovative Cell Technologies, San Diego, CA) and fixed with 4% PFA in PBS. eGFP expression was examined by flow cytometry using a BD LSR II (Becton Dickinson), and data were analyzed with FlowJo (Tree Star, Inc., Ashland, OR).

293T cells seeded (4.0×10^5 cells/well) in a 12-well plate and cultured overnight were infected with scrLCMV/ZsG-P2A-NP for 2 h and subsequently transfected with 0.5 μ g of pC-GPC. At 24 h pi, cells were three times washed with fresh media to eliminate infectious virus particle produced in the absence of compound treatment, and cultured for another 24 h in fresh media in the presence of 40 nM of ouabain or vehicle control (DMSO). At 48 h pi, TCS was collected and used to infect fresh monolayers of BHK-21 cells seeded (4.0×10^5 cells/well) in a 12-well plate 1 day before the infection, and 293T cell lysate was prepared. 24 h later, BHK-21 cell lysate was prepared. Total cell lysate was prepared in cell lysis buffer (150 mM of NaCl, 50 mM of Tris-HCl [pH = 7.5], 0.5% [NP-40], 1 mM of EDTA) and clarified by centrifugation at $21,130 \times g$ at 4°C for 10 min. ZsGreen signal intensity in clarified cell lysate was measured by a fluorescent plate reader (Synergy H4 Hybrid Multi-Mode Microplate Reader, BioTek).

A549 cells seeded (2×10^5 cells/well) in a 96-well plate and cultured overnight were treated with combinations of different concentrations of ouabain and rocaglamide for 2 h and then infected (MOI = 0.01) with rLCMV/eGFP. Compounds were present in the culture medium throughout the experiment. At 48 h pi, cells were fixed, permeabilized by treatment with DB, and stained with DAPI. eGFP and DAPI signals were measured by a fluorescent plate reader (Synergy H4 Hybrid Multi-Mode Microplate Reader, BioTek). eGFP readouts were normalized by DAPI readouts, and normalized data were used to analyze synergistic effect of the two compounds by the MacSynergy II program [89] .

Data were analyzed for p values by a two-tailed unpaired t test using GraphPad Prism software.

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Viruses and Evolution – Viruses First? A Personal Perspective

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Abstract: The discovery of exoplanets within putative habitable zones revolutionized astrobiology in recent years. It stimulated interest in the question about the origin of life and its evolution. Here, we discuss what the roles of viruses might have been at the beginning of life and during evolution. Viruses are the most abundant biological entities on Earth. They are present everywhere, in our surrounding, the oceans, the soil and in every living being. Retroviruses contributed to about half of our genomic sequences and to the evolution of the mammalian placenta. Contemporary viruses reflect evolution ranging from the RNA world to the DNA-protein world. How far back can we trace their contribution? Earliest replicating and evolving entities are the ribozymes or viroids fulfilling several criteria of life. RNA can perform many aspects of life and influences our gene expression until today. The simplest structures with non-protein-coding information may represent models of life built on structural, not genetic information. Viruses today are obligatory parasites depending on host cells. Examples of how an independent lifestyle might have been lost include mitochondria, chloroplasts, Rickettsia and others, which used to be autonomous bacteria and became intracellular parasites or endosymbionts, thereby losing most of their genes. Even in vitro the loss of genes can be recapitulated all the way from coding to non-coding RNA. Furthermore, the giant viruses may indicate that there is no

sharp border between living and non-living entities but an evolutionary continuum. Here, it is discussed how viruses can lose and gain genes, and that they are essential drivers of evolution. This discussion may stimulate the thinking about viruses as early possible forms of life. Apart from our view “viruses first”, there are others such as “proteins first” and “metabolism first.”

Text: *Mycoplasma mycoides* by systematic deletion of individual genes resulted in a synthetic minimal genome of 473 genes (Hutchison et al., 2016) . Can one consider simpler living entities?

There are elements with zero genes that fulfill many criteria for early life: ribozymes, catalytic RNAs closely related to viroids. They were recovered in vitro from 10¹⁵ molecules (aptamers), 220 nucleotides in length, by 10 rounds of selection. Among the many RNA species present in this collection of quasispecies RNAs were catalytically active members, enzymatically active ribozymes. The sequence space for 220-mer RNAs is about 3×10^{132} (Eigen, 1971; Wilson and Szostak, 1999; Brackett and Dieckmann, 2006) .

The selected ribozymes were able to replicate, cleave, join, and form peptide bonds. They can polymerize progeny chemically, allow for mutations to occur and can evolve. One molecule serves as catalyst, the other one as substrate. Replication of ribozymes was demonstrated in the test tube (Lincoln and Joyce, 2009) . Ribozymes can form peptide bonds between amino acids (Zhang and Cech, 1997) . Thus, small peptides were available by ribozyme activity. Consequently, an RNA modification has been proposed as peptide nucleic acid (PNA), with more stable peptide bonds instead of phosphodiester bonds (Zhang and Cech, 1997; Joyce, 2002) . Replication of RNA molecules can be performed chemically from RNA without polymerase enzymes. In addition, deoxyribozymes can form from ribonucleotides (Wilson and Szostak, 1999) . Thus, DNA can arise from RNA chemically, without the key protein enzyme, the reverse transcriptase.

An entire living world is possible from non-coding RNA (ncRNA) before evolution of the genetic code and protein enzymes. Ribozymes naturally consist of circular single-stranded RNAs (Orgel, 2004) . They lack the genetic triplet code and do not encode proteins. Instead, they exhibit structural information by hairpin-loops that form hydrogen bonds between incomplete double strands, and loops free to interact with other molecules. They represent

a quasispecies in which many species of RNA may form, such as ribozymes, tRNA-like molecules, and other ncRNAs. RNAs within such a pool can bind amino acids. Ninety different amino acids have been identified on the Murchison meteorite found in Australia, while on Earth only about 20 of them are used for protein synthesis (Meierhenrich, 2008) . Where formation of ribozymes occurred on the early Earth is a matter of speculation. The hydrothermal vents such as black smokers in the deep ocean are possibilities where life may have started (Martin et al., 2008) . There, temperature gradients and clay containing minerals such as magnesium or manganese are available. Pores or niches offer possibilities for concentration of building blocks, which is required for chemical reactions to occur. Interestingly, also ice is a candidate for ribozyme formation and chemical reactions. Ice crystals displace the biomolecules into the liquid phase, which leads to concentration, creating a quasicellular compartmentalization where de novo synthesis of nucleotide precursors is promoted. There, RNA and ribozymes can emerge, which are capable of self-replication (Attwater et al., 2010) .

tRNA-amino acid complexes can find RNAs as "mRNAs." Such interactions could have contributed to the evolution of the genetic code. This sequence of events can lead to primitive ribosome precursors. Ribozymes are the essential catalytic elements in ribosomes: "The ribosome is a ribozyme" (Cech, 2000) , supplemented with about a hundred scaffold proteins later during evolution. The proteins have structural functions and contribute indirectly to enzymatic activity. Are these ribosomebound ribozymes fossils from the early Earth? Small peptides can be formed by ribozymes before ribosomes evolved, whereby single or dimeric amino acids may originate from the universe (Meierhenrich, 2008) .

Small peptides with basic amino acids can increase the catalytic activity of ribozymes as shown in vitro (Müller et al., 1994) . Such proteins are known as RNA-binding proteins from RNA viruses that protect the RNA genome, with motifs such as RAPRKKG of the nucleocapsid NCp7 of HIV (Schmalzbauer et al., 1996) . Peptides can enhance the catalytic activity of ribozymes up to a 100-fold (Müller et al., 1994) . Such peptides of RNA viruses serve as chaperones that remove higher ordered RNA structures, allowing for more efficient interaction of RNA molecules and increasing transcription rates of RNA polymerases (Müller et al., 1994) . Ribonucleoproteins may have also been functionally important during the evolution of ribosomes (Harish and Caetano-Anolles, 2012) .

These pre-ribosomal structures are also similar to precursorlike structures of retroviruses. Reverse transcription can be performed by ribozymes chemically. This action does not necessarily require a protein polymerase such as the reverse transcriptase. Similarly, deoxyribonucleotides can arise by removal of an oxygen without the need of a protein enzyme (a reductase) as today, and allow for DNA polymerization (Wilson and Szostak, 1999; Joyce, 2002) . The same elements of the precursors for ribosomes are also building blocks of retroviruses, which may have a similar evolutionary origin (Moelling, 2012 (Moelling, , 2013 . tRNAs serve as primers for the reverse transcriptase, and the sequence of promoters of transposable elements are derived from tRNAs (Lander et al., 2001) . The ribozymes developed into more complex self-cleaving group II introns with insertion of genes encoding a reverse transcriptase and additional proteins (Moelling and Broecker, 2015; Moelling et al., 2017) (Figure 1) .

It came as a surprise that the genomes of almost all species are rich in ncDNA, transcribed into ncRNAs but not encoding proteins, as evidenced, for instance, by the "Encyclopedia of DNA Elements" (ENCODE) project. ncDNA amounts to more than 98% of the human DNA genome (Deveson et al., 2017) . Higher organisms tend to have more non-coding information, which allows for more complex modes of gene regulation. The ncRNAs are regulators of the protein-coding sequences. Highly complex organisms such as humans typically have a high number of ncRNA and regulatory mechanisms. ncRNA can range from close to zero in the smallest bacteria such as *Pelagibacter ubique* to about 98% in the human genome.

RNA viruses such as the retrovirus HIV harbor ncRNAs for gene regulation such as the trans-activating response element (TAR), the binding site for the Tat protein for early viral gene expression. Tat has a highly basic domain comprising mostly Lys and Arg residues, resembling other RNA binding proteins. ncRNA also serves on viral RNA genomes as ribosomal entry sites, primer binding sites or packaging signals. DNA synthesis depends on RNA synthesis as initial event, with RNA primers as starters for DNA replication, inside of cells as FIGURE 1 | A compartment is shown with essential components of life as discussed in the text. Non-coding RNA (ncRNA), ribozymes or viroids, can perform many steps for life without protein-coding genes but only by structural information. Individual amino acids are indicated as black dots and may be available on Earth from the universe. DNA may have existed before retroviruses. The compartment can be interpreted as pre-virus or pre-cell. Viroid, green; RNA, red; DNA, black.

well as during retroviral replication, proving a requirement of RNA (Flint, 2015) .

The number of mammalian protein-coding genes is about 20,000. Surprisingly, this is only a fifth of the number of genes of bread wheat (Appels et al., 2018) . Tulips, maize and other plants also have larger genomes, indicating that the number of genes does not necessarily reflect the complexity of an organism. What makes these plant genomes so large, is still an open question. Could the giant genomes possibly be the result to breeding of plants by farmers or gardeners?

According to Szostak there are molecules which appear like relics from the RNA world such as acetyl-CoA or vitamin B12, both of which are bound to a ribonucleotide for no obvious reason -was it "forgotten" to be removed? (Roberts and Szostak, 1997; Szostak et al., 2001; Szostak, 2011) . Perhaps the connected RNA serves as structural stabilizer. Lipid vesicles could have formed the first compartments and enclosed ribozymes, tRNAs with selected amino acids, and RNA which became mRNA. Is this a pre-cell or pre-virus (Figure 1) ? Patel et al. (2015) demonstrated that the building blocks of life, ribonucleotides, lipids and amino acids, can be formed from C, H, O, P, N, S in a "one pot" synthesis. This study can be regarded as a follow-up study of the classical Urey-Miller in vitro synthesis of biomolecules (Miller, 1953; Miller and Urey, 1959) . Transition from the RNA to the DNA world was promoted by the formation of the reverse transcriptase. The enzyme was first described in retroviruses but it is almost ubiquitous and found in numerous cellular species, many of which with unknown functions (Simon and Zimmerly, 2008; Lescot et al., 2016) . It is an important link between the RNA and the DNA worlds. The name reverse transcriptase is historical and irritating because it is the "real" transcriptase during the transition from the RNA to the DNA world. Similarly, the ribonuclease H (RNase H) is an essential enzyme of retroviruses (Mölling et al., 1971) . The RNase H turned out to be one of the five most frequent and ancient proteins (Ma et al., 2008) that belongs to a superfamily of more than sixty different unique representatives and 152 families with numerous functions (Majorek et al., 2014) .

Some of the many tRNAs can become loaded with amino acids. There are viruses containing tRNA-like structures (TLS), resembling these early RNAs (Dreher, 2009) . The TLS of these viruses typically bind to a single amino acid. TLS-viruses include plant viruses, such as Turnip yellow mosaic virus, in Peanut clump virus, Tobacco mosaic virus (TMV),

and Brome mosaic virus. Only half a tRNA is found in Narnaviruses of fungi. The amino acids known to be components of tRNA-like viruses are valine, histidine and tyrosine. The structures were also designated as "mimicry, " enhancing translation (Dreher, 2009 (Dreher, , 2010 . They look like "frozen" precursor-like elements for protein synthesis. This combination of a partial tRNA linked to one amino acid can be interpreted as an evolutionary early step toward protein synthesis, trapped in a viral element.

Ribozymes are related to the protein-free viroids. Viroids are virus-like elements that belong to the virosphere, the world of viruses (Chela-Flores, 1994) . Viroids lack protein coats and therefore were initially not designated as viruses but virus-like viroids when they were discovered in 1971 by Theodor Diener. He described viroids as "living fossils" (Diener, 2016) (Figure 2) .

From infected potatoes, Diener isolated the Potato spindle tuber viroid (PSTVd) whose genome was about a 100-fold smaller than those of viruses known at that time. The viroids known today are ranging from 246 to 467 nucleotides. They contain circular single-stranded RNA, are protein-free and self-replicating with no genetic information, but only structural

FIGURE 2 | Viroids are hairpin-loop structures and are shown schematically and as electron micrograph. Viroids are, like ribozymes, without genetic information and play major biological roles today in plant diseases, in carnation flowers, in liver cancer, as catalyst of protein synthesis in ribosomes and as circular regulatory RNAs, as "sponges" for other regulatory RNAs.

information in the form of hairpin-loops (Riesner et al., 1979) . They can generate copies of themselves in the appropriate environment. They were designated as the "frontiers of life" (Flores et al., 2014) .

The knowledge of virus composition was based on TMV and its crystallization by Wendell Stanley in 1935 (Pennazio and Roggero, 2000) . The genome of TMV is protein-coding singlestranded RNA of about 6,400 nucleotides that is enclosed by a rod-like protein coat. Viroids, in contrast, do not encode proteins and lack coats but they are closely related to viruses. Viroids can lose their autonomy and rely on host RNA polymerases to replicate, are capable of infecting plants and many are economically important pathogens. There are two families, the nucleus-replicating Pospiviroidae such as PSTVd and the chloroplast-

replicating Avsunviroidae like the Avocado sunblotch viroid (ASBVd). Their replication requires host enzymes. Thus, autonomy is replaced by dependence on host enzymes and an intracellular lifestyle.

Most viroids are often enzymatically active ribozymes -yet they are examples that this trait can get lost as a result of changing environmental conditions. Loss of ribozyme activity is a functional, not a genetic loss. Only the nuclear variants, the Pospiviroidae, can lose their ribozyme activity and use the cellular RNase III enzyme for their replication. In contrast, the Avsunviroidae are still active hammerhead ribozymes. Thus, inside the nucleus of a host cell, the enzymatic RNA function can become unnecessary. Not genes, but a function, the catalytic activity, gets lost.

Viroids did apparently not gain genes but cooperated for a more complex lifestyle. For example, Carnation small viroid-like RNA (CarSV RNA) cooperates with a retrovirus and is accompanied by a homologous DNA generated by a reverse transcriptase. This enzyme presumably originates from a pararetrovirus of plants. Pararetroviruses package virus particles at a different stage during replication than retroviruses, the DNA, not the RNA. This unique combination between two viral elements has so far only been detected with CarSV in carnation flowers (Flores et al., 2005 (Flores et al., , 2014 . Why did such a cooperation evolve -perhaps by breeding gardeners? RNA is sensitive to degradation; therefore, genetic increase and growth of the genome may not be favorable energetically - at least not in plants. Gain of function is, in this case, cooperation.

The circular RNA (circRNA) is related to ribozymes/viroids as a chief regulator of other regulatory RNAs, a "sponge" absorbing small RNAs. Micro RNAs (miRNAs) are post-transcriptional regulators that are affected by the presence of circRNAs. circRNAs were detected in human and mouse brains and testes as well as in plants. They can bind 70 conserved miRNAs in a cell and amount up to 25,000 molecules (Hansen et al., 2013) . Their structure is reminiscent of catalytically active ribozymes.

There is an exceptional viroid that gained coding information and entered the human liver (Taylor, 2009) . The viroid is known as hepatitis delta virus (HDV). It has the smallest genome of any known animal virus of about 1,680 nucleotides. It has properties typical of viroids, since it contains circRNA, forms similar hairpin-loops and replicates in the nucleus

using host enzymes. Two polymerases have to redirect their specificity from DNA to RNA to generate the HDV genome and antigenome. Both of them have ribozyme activity. In contrast to other ribozymes, HDV encodes a protein, the hepatitis delta antigen (HDVAg) that occurs in two forms, the small-HDVAg (24 kDa) supporting replication and the large-HDVAg (27 kDa) that helps virion assembly. The gene was presumably picked up from the host cell by recombination of HDV's mRNA intermediate with a host mRNA. Transmission depends on a helper virus, the Hepatitis B virus (HBV), which delivers the coat (Taylor, 2009)

) Does packaging by a helper virus protect the genome and thereby allow for a larger viroid to exist?

In plants, viroids may not be able to become bigger possibly due to their sensitivity to degradation -but they cannot become much smaller either. Only a single viroid is known that is completely composed of protein-coding RNA with triplets (AbouHaidar et al., 2014). Viroids and related replicating RNAs are error-prone replicating units and the error frequency imposes a certain minimal size onto them, as they would otherwise become extinct. This mechanism has been described as "error catastrophe," which prevents survival (Eigen, 1971 (Eigen, , 2013 . The viroids and related RNAs are the smallest known replicons. Smaller ones would become extinct in the absence of repair systems.

In summary, RNA can catalyze many reactions. Protein enzymes which may have evolved later have higher catalytic activities. Ribozymes are carriers of information, but do not require coding genes. Information is stored in their sequence and structure. Thus, replication of an initial RNA is followed by flow of information, from DNA to RNA to protein, as described the Central Dogma (Crick, 1968) . Even an information flow from protein to DNA has been described for some archaeal proteins (Béguin et al., 2015) . The DNA-protein world contains numerous ncRNAs with key functions. ncRNA may serve as a model compound for the origin of life on other planets. Hereby not the chemical composition of this molecule is of prime relevance, but its simplicity and multifunctionality. Furthermore, RNA is software and hardware in a single molecule, which makes it unique in our world. There are other scenarios besides the here discussed "virus-first," such as "protein-first", "metabolism-first" or the "lipid world" (Segré et al., 2001; Andras and Andras, 2005; Vasas et al., 2010; Moelling, 2012) . Some of these alternative concepts were built on phylogenomics, the reconstruction of the tree of life by genome sequencing (Delsuc et al., 2005) . Surprisingly, it was Sir Francis Crick, one of the discoverers of the DNA double-helix, who stated that he would not be surprised about a world completely built of RNA. A similar prediction was made by Walter Gilbert (Crick, 1968; Gilbert, 1986) . What a vision! Our

world was almost 50 years later defined as "RNAprotein" world (Altman, 2013) . One can speculate our world was built of ribozymes or viroids, which means "viruses first."

ncRNAs appear as relics from the past RNA world, before DNA, the genetic code and proteins evolved. However, ncRNA is essential in our biological DNA world today. It is possible to produce such ncRNA today in the test tube by loss of genic information from protein-coding RNA. This reduction to ncRNA was demonstrated in vitro with phage RNA. Phage Q β genomic RNA, 4,217 nucleotides in length, was incubated in the presence of Q β replicase, free nucleotides and salts, a rich milieu in the test tube. The RNA was allowed to replicate by means of the Q β replicase. Serial transfer of aliquots to fresh medium led to ever faster replication rates and reduction of genomic size, down to 218 nucleotides of ncRNA in 74 generations. This study demonstrated that, depending on environmental conditions, an extreme gene reduction can take place. This experiment performed in 1965 was designated as "Spiegelman's Monster." Coding RNA became replicating ncRNA (Spiegelman et al., 1965; Kacian et al., 1972) ! Manfred Eigen extended this experiment and demonstrated further that a mixture containing no RNA to start with but only ribonucleotides and the Q β replicase can under the right conditions in a test tube spontaneously generate self-replicating ncRNA. This evolved into a form similar to Spiegelman's Monster. The presence of the replicase enzyme was still necessary in these studies. Furthermore, a change in enzyme concentration and addition of short RNAs or an RNA intercalator influenced the arising RNA population (Sumper and Luce, 1975; Eigen, 2013) . Thus, the complexity of genomes depends on the environment: poor conditions lead to increased complexity and rich environments to reduced complexity.

The process demonstrated in this experiment with viral components indicates that reversion to simplicity, reduction in size, loss of genetic information and speed in replication can be major forces of life, even though this appears to be like a reversion of evolution. The experiment can perhaps be generalized from the test tube to a principle, that the most successful survivors on our planet are the viruses and microorganisms, which became the most abundant entities. Perhaps life can start from there again.

These studies raise the question of how RNA molecules can become longer, if the small polymers become smaller and smaller, replicate faster and outcompete longer ones. This may be overcome by heat flow across an open pore in submerged rocks, which concentrates replicating oligonucleotides from a constant feeding flow and selection for

longer strands. This has been described for an increase from 100 to 1,000 nucleotides in vitro. RNA molecules shorter than 75 nucleotides will die out (Kreysing et al., 2015) . Could a poor environment lead to an increase of complexity? This could be tested. Ribozymes were shown to grow in size by uptake of genes, as demonstrated for HDV (Taylor, 2009).

An interesting recent unexpected example supporting the notion that environmental conditions influence genetic complexity, is the human gut microbiome. Its complexity increases with diverse food, while uniform rich food reduces its diversity and may lead to diseases such as obesity. Colonization of the human intestinal tract starts at birth. A few dozen bacterial and viral/phage species are conserved between individuals (core sequences) as a stable composition (Broecker et al., 2016c . Dysbiosis has been observed in several chronic diseases and in obesity, a loss of bacterial richness and diversity. Nutrition under affluent conditions with sugar-rich food contributes to obesity, which results in a significant reduction of the complexity of the microbiome. This reduction is difficult to revert (Cotillard et al., 2013; Le Chatelier et al., 2013) . The gut microbiome in human patients with obesity is reminiscent of the gene reduction described in the Spiegelman's Monster experiment: reduction of genes in a rich environment.

The reduction of the complexity of the microbiome is in part attributed to the action of phages, which under such conditions, defined as stress, lyse the bacteria. Fecal microbiota transplantation can even be replaced by soluble fractions containing phages or metabolites from the donor without bacteria (Ott et al., 2017) . Analogously, the most highly complex microbiomes are found in indigenous human tribes in Africa, which live on a broad variety of different nutrients. It is a slow process, though, to increase gut microbiota complexity by diverse nutrition. The obesity-associated microbiota that survive are fitter and more difficult to counteract. Urbanization and westernization of the diet is associated with a loss of microbial biodiversity, loss of microbial organisms and genes (Segata, 2015) .

To understand the mechanism and driving force for genome reduction, deletion rates were tested by insertion of an indicator gene into the *Salmonella enterica* genome. The loss of the indicator gene was monitored by serial passage in rich medium. After 1,000 generations about 25% of the deletions caused increased bacterial fitness. Deletions resulted in smaller genomes with reduced or absence of DNA repair genes (Koskiniemi et al., 2012) . Gene loss conferred a higher fitness to the bacteria under these experimental conditions.

The recently discovered mimiviruses and other giant viruses are worth considering for understanding the evolution of life with respect to the contribution of viruses. Their hosts are, for example, *Acanthamoeba*, *Chlorella*, and *Coccolithus* algae (*Emiliana huxleyi*), but also corals or sponges as discussed more recently. Mimiviruses were first discovered in cooling water towers in Bradford, United Kingdom in 2003 with about 1,000 genes, most of which unrelated to previously known genes. Mimiviruses have received attention because they contain elements that were considered hallmarks of living cells, not of viruses, such as elements required for protein synthesis, tRNAs and amino acid transferases. The mimiviruses harbor these building blocks as incomplete sets not sufficient for independent protein synthesis as bacteria or archaea can perform, preventing them from leading an autonomous life (La Scola et al., 2003 Scola et al., , 2008 . They are larger than some bacteria. Giant viruses can be looked at as being on an evolutionary path toward a cellular organism. Alternatively, they may have evolved from a cellular organism by loss of genetic information (Nasir and Caetano-Anolles, 2015) . Giant viruses have frequently taken up genes from their hosts by horizontal gene transfer (HGT) (La Scola et al., 2008; Nasir and Caetano-Anolles, 2015; Colson et al., 2018) . A graph on genome sizes shows that mimiviruses and bacteria overlap in size, indicating a continuous transition between viruses and bacteria and between living and non-living worlds (based on Holmes, 2011) (Figure 3) . Other giant viruses, such as megaviruses, were discovered in the ocean of Chile with 1,120 genes. Most recently the Klosneuvirus was identified in the sewage of the monastery Klosterneuburg in Austria in 2017 with 1.57 million (mio) basepairs (Mitch, 2017) . Pithovirus sibericum is the largest among giant viruses discovered to date with a diameter of 1.5 microns, a genome of 470,000 bp with 467 putative genes, 1.6 microns in length, and it is presumably 30,000 years old as it was recovered from permafrost in Siberia (Legendre et al., 2014) . The smaller Pandoraviruses with 1 micron in length have five times larger genomes, 2,500,000 bp (Philippe et al., 2013) (Figure 3) .

The giant viruses can even be hosts to smaller viruses, the virophages, reminiscent of bacteriophages, the viruses of bacteria. These virophages such as Sputnik are only 50 nm in size with 18,343 bp of circular dsDNA and 21 predicted protein-coding genes. They replicate in viral factories and consume the resources of the mimivirus, thereby destroying it. Some, virophages can even integrate into the genome of the cellular host and can be reactivated when the host is infected by giant viruses. Thus, giant viruses suggest that viruses are close to living entities or may have been alive (La Scola et al., 2008; Fischer and Hackl, 2016) . In biology it is common to distinguish between living and dead matter by the ability to synthesize proteins and replicate autonomously. The giant viruses may be considered as missing link between the two, because they harbor "almost" the protein

synthesis apparatus. The transition from living to the non-living world is continuous, not separated by a sharp borderline (Figure 3) .

Viruses are not considered alive by most of the scientific community and as written in textbooks, because they cannot replicate autonomously. Yet some of the giant viruses are equipped with almost all components of the protein synthesis machinery close to bacteria suggesting that they belong to the living matter (Schulz et al., 2017) . The ribozymes may have been the earliest replicating entity. Perhaps also other viruses were initially more independent of the early Earth than they are today. As described in Figure 1 there may have been initially no major difference between an early virus or an early cell. Only later viruses may have given up their autonomous replication and became parasites -as has been described for some bacteria (see below).

Efforts have been made to identify the smallest living cell that is still autonomously replicating. Among the presumably smallest naturally occurring bacteria is *Pelagibacter ubique* of the SAR11 clade of bacteria (Giovannoni, 2017) , which was discovered in 1990. It is an alpha-proteobacterium with 1,389 genes present ubiquitously in all oceans. It can reach up to 10²⁸ free living cells in total and represents about 25% of microbial plankton cells. Very little of its DNA is non-coding. It harbors podophage-type phages, designated as "pelagiphage" (Zhao et al., 2013) . This small bacterium was designated as the most common organism on the planet. Why is it so successful? This autonomous bacterium is smaller than some parasitic giant viruses. Craig Venter, who first succeeded in sequencing the human genome, tried to minimize the putative smallest genome of a living species, from *Mycoplasma mycoides*, a parasitic bacterium that lives in ruminants (Gibson et al., 2008 (Gibson et al., , 2010 . His group synthesized a genome of 531,000 bp with 473 genes, 149 of them (32%) with unknown functions (Hutchison et al., 2016) . Among the smallest parasitic living organisms is *Nanoarchaeum equitans*. It is a thermophile archaeon which lives at 80 °C and at pH 6 with 2% salt (Huber et al., 2003) . Its genome has a size of 490,000 bp and encodes 540 genes. *N. equitans* is an obligate symbiont of a bigger archaeon, *Ignicoccus* riding on it as on a horse, hence the name (Huber et al., 2003) .

The world of viruses covers a range of three logs in size of their genomes: from zero genes to about 2,500 genes amounting to about 2,500,000 bp of DNA. The zero-gene viroids are about 300 bases in length (Figure 3) .

The virosphere is the most successful reservoir of biological entities on our planet in terms of numbers of particles, speed of replication, growth rates, and sequence space. There are about 10^{33} viruses on our planet and they are present in every single existing species (Suttle, 2005) .

There is no living species without viruses! Viruses also occur freely in the oceans, in the soil, in clouds up to the stratosphere and higher, to at least 300 km in altitude. They populate the human intestine, birth canal, and the outside of the body as protective layer against microbial populations. Microbes contain phages that are activated during stress conditions such as lack of nutrients, change in temperatures, lack of space and other changes of environmental conditions.

One of the most earth-shaking papers of this century was the publication of the human genome sequence (Lander et al., 2001) . About half, possibly even two-thirds of the sequence are composed of more or less complete endogenous retroviruses (ERVs) and related retroelements (REs) (de Koning et al., 2011) . REs amplify via copy-and-paste mechanisms involving a reverse transcriptase step from an RNA intermediate into DNA. In addition, DNA transposable elements (TEs) move by a cut-and-paste mechanism. The origin of REs is being discussed as remnants of ancient retroviral germline infections that became evolutionarily fixed in the genome. About 450,000 human ERV (HERV) elements constitute about 8% of the human genome consisting of hallmark retroviral elements like the gag, pol, env genes and flanking long terminal repeats (LTR) that act as promoters (Lander et al., 2001) . Howard Temin, one of the discoverers of the reverse transcriptase, in 1985 already described endogenous retrovirus-like elements, which he estimated to about 10% of the human and mouse genome sequence (Temin, 1985) . The actual number is about 45% as estimated today (Lander et al., 2001) . In some genes such as the Protein Kinase Inhibitor B (PKIB) gene we determined about 70% retrovirus-related sequences (Moelling and Broecker, 2015) . Is there a limit? Could it have been 100%? Retroviruses are estimated to have entered the lineage of the mammalian genome 550 million years ago (MYA) (Hayward, 2017) . Older ERV sequences may exist but are unrecognizable today due to the accumulation of mutations.

ERVs undergo mutations, deletions or homologous recombination events with large deletions and can become as short as solo LTR elements, which are a few hundred bp in length -the left-overs from full-length retroviral genomes of about 10,000 bp. The LTR

promoters can deregulate neighboring genes. Homologous recombination events may be considered as gene loss or gene reduction events. It is the assumption that the ERVs, which were no longer needed for host cell defense, were no longer selected for by evolution and consequently deleted as unnecessary consumers of energy.

Eugene Koonin points out that infection and integration are unique events occurring at a fast pace, while loss and gene reduction may take much longer time frames (Wolf and Koonin, 2013) .

A frequent gene reduction of eukaryotic genomes is the loss of the viral envelope protein encoded by the env gene. Without a coat, retroviruses can no longer leave the cell and infect other cells. They lose mobility and become obligatory intracellular elements. Helper viruses can supply envelope proteins in trans and mobilize the viruses. TEs or REs can be regarded as examples of coat-free intracellular virus relics -or could it have been the other way round, perhaps precursors of full-length retroviruses?

These elements can be amplified intracellularly and modify the host genomes by integration with the potential danger of gene disruption and genetic changes. REs can lead to gene duplications and pseudogene development, with one copy for stable conservation of acquired functions and the other one for innovations (Cotton and Page, 2005) . Such duplications constitute large amounts of mammalian genomes (Zhang, 2003) . Retroviruses have an RNase H moiety duplication, one of which serves as a catalytically inactive linker between the RT polymerase and the enzymatically active RNase H (Xiong and Eickbush, 1990; Malik and Eickbush, 2001; Moelling and Broecker, 2015; Moelling et al., 2017) . This gene duplication dates back to 500 mio years (Cotton and Page, 2005) .

Gene duplications are a common cause of cancer, which often occurs only in the genome of the cancer cell itself, less affecting offsprings. Myc, Myb, ErbB2, Ras, and Raf are oncogenes amplified in diverse types of human cancers (Vogelstein and Kinzler, 2002) . The ability of retroviruses to integrate makes them distinct from endosymbionts which stay separate. Yet the net result is very similar, acquisition of new genetic information, which is transmitted to the next generation, if the germline is infected and endogenization of the virus occurred.

Viral integration is not limited to eukaryotic cells but also a mechanism in prokaryotes for maintenance of the lysogenic state of phages inside bacteria.

Also, for other eukaryotic viruses such as HBV, the envelope surface antigen BHsAg can be deleted, which leads to an obligatory intracellular life style for the virus, which especially in the presence of HCV promotes cancer (Yang et al., 2016) .

HIV has been shown to rapidly lose one of its auxiliary genes, *nef*, originally for negative factor. The gene was lost within a rather low number of passages of the virus grown under tissue culture conditions by selection for high virus titer producing cells. Deletion of *nef* resulted in a significant increase of the virus titer in culture -hence the name. The *nef* gene product was of no need inside tissue culture cells, rather it was inhibitory for replication. However, it is essential for pathogenicity in animals, and subsequently *nef* was reinterpreted as "necessary factor" (Flint, 2015) .

Also, the human hosts of HIV can lose a significant terminal portion of a seven transmembrane receptor in lymphocytes, the primary target cell for HIV entry and for virus uptake. This molecule, the CCR5 cytokine receptor is truncated by 32 carboxy-terminal amino acids (CCR5-32), disabling the receptor functionally. The allele frequency of the mutant CCR5-32 mutant is about 10% in the European population, making these people resistant to HIV infections (Solloch et al., 2017) . This gene loss in Europeans has been shown to make the individuals resistant not only against HIV infection but also against malaria. This may have been the selective pressure in the past before HIV/AIDS arose. No side effect for humans lacking this gene has been described (Galvani and Slatkin, 2003) .

Viruses have been proven to be drivers of evolution (Villarreal and Witzany, 2010) , including the human genome, which by at least 45% is composed of sequences related to retroviruses. In addition, endogenized retroviruses supplied the syncytin genes that are essential for the development of the mammalian placenta, and allowed the growth of embryos without its rejection by the maternal immune system (Dupressoir et al., 2012) . Thus, the same property which causes immunodeficiency in HIV-infected patients and leads to AIDS causes syncytia formation, cell fusion after infection by a retrovirus. Viruses have also been proposed to be at the origin of the evolution of adaptive immunity

(Villarreal, 2009). Thus, viruses shaped genomes by supplying essential genes and mechanisms.

Endogenization of retroviruses has occurred in the mammalian genomes for at least 550 mio years (Hayward, 2017) . If the integrated ERVs did not provide any selective advantage, they deteriorated and accumulated mutations with loss of function. This was directly proven by reconstruction of an infectious retrovirus from the consensus sequence of 9 defective endogenous virus sequences, designated as Phoenix. The virus was expressed from a constructed synthetic DNA clone in cell culture and formed virus particles identified by high resolution microscopic analysis (Dewannieux and Heidmann, 2013) .

The koalas in Australia are currently undergoing endogenization of a retrovirus (koala retrovirus, KoRV) in "real time" and demonstrate possible consequences for immunity. In the early 1900s, some individuals were transferred to islands, including Kangaroo Island, close to the Australian mainland for repopulation purposes, as koalas were threatened to become extinct. Today, the majority of the koala population is infected by KoRV, which is closely related to the Gibbon ape leukemia virus (GALV). Yet, koalas isolated on Kangaroo Island are KoRV negative, which allows dating the introduction of KoRV into the koala population to about one hundred years ago. Many of the infected koalas fell ill and died, yet some populations became resistant within about 100 years, corresponding to about 10 generations. The koalas likely developed resistance due to the integrated DNA proviruses. The retrovirus is transmitted as exogenous as well as endogenous virus, similar to the Jaagsiekte sheep retrovirus (JSRV), whereby the endogenized viruses protect with a viral gene product, such as Env, against de novo infections by "superinfection exclusion" (Tarlinton, 2012) . The contribution of retroviruses to the antiviral defense is striking, since all retroviral genes have analogous genes in the siRNA/RNAi defense mechanism of eukaryotic cells (Moelling et al., 2006) .

Retroviruses can protect against infection by other related viruses, for example, by expressing Env proteins that block cell surface receptors (Villarreal, 2011) . A comparable mechanism protects bacterial cells against DNA phages, by integrated phage DNA fragments that are transcribed into mRNA and hybridize to incoming new DNA phages and thereby lead to their destruction by hybrid-specific nucleases, CRISPR/Cas immunity (Charpentier and Doudna, 2013) . It is often not realized that immunity acquisition in bacteria and mammalian cells follow analogous mechanisms (Figure 4) .

Integration of retroviruses normally occurs in somatic cells after infection as an obligatory step during the viral life cycle. Infection of germline cells can lead to transmission to the next generation and ultimately result in inherited resistance. Endogenized retroviruses likely caused resistance FIGURE 4 | Viruses protect against viruses: retroviruses protect a cell against a new infection by a similar virus designated as "superinfection exclusion" or viral interference. This is mediated by viral gene products such as proteins or nucleic acids. Similarly, phages protect against phages: superinfection of bacteria is prevented by CRISPR/Cas RNA originating from previous infections. The mechanisms of defense against viruses and phages are analogous. Protection by viruses or phages against superinfections represents cellular defense and acquired immunity. The four examples are discussed in the text.

to the exogenous counterparts. Similarly, resistance to Simian Immune Deficiency virus (SIV) in some monkey species may be explained by endogenization (Li et al., 2017 (Li et al., , 2018 . In the case of phages and their prokaryotic hosts the mechanism is described as CRISPR/Cas, which follow analogous principles of "endogenization" of incoming genetic material for subsequent exclusion.

One may speculate that HIV may also eventually become endogenized into the human genome. There is some evidence that HIV can infect human germline cells and can be transmitted to the embryonic genome (Wang et al., 2011) . How long this may take is not known -10 generations?

The loss of function of ERVs can occur by mutations, deletions of the env or other genes and ultimately all coding genes by homologous recombination, leaving behind only one LTR. The number of retrovirus-like elements add up to about 450,000, corresponding to 8% of the human genome (Lander et al., 2001; Cordaux and Batzer, 2009). The promoter regions were analyzed for their contribution to cancer by activating neighboring genes -as a consequence of a former retrovirus infection. Indeed, activated cellular genes by "downstream promotion" were identified in animal studies with activation of the myc gene as one of many examples, leading to chronic, not acute development of cancer (Ott et al., 2013) . As a general mechanism for human cancer today the LTRs are, however, not identified as a major culprit. Most of the ERVs we find today have been integrated during evolution in introns or other regions where their presence is relatively harmless. Did the

other ones result in death of the carriers which disappeared? The effects of LTRs on the expression levels of neighboring host genes was studied with the endogenous human virus, HERV-K, as a possible cause of cancer, but this appears not to be a general phenomenon (Broecker et al., 2016b) . As shown for the koalas, ERVs can confer immunity to viral infections (Feschotte and Gilbert, 2012) .

A related ERV, HERV-H, was shown to produce an RNA that keeps early embryonic cells pluripotent and even revert adult cells to regain pluripotency (Grow et al., 2015) . Thus, the role of ERVs may be more complex than we presently know.

Transposable elements and REs that lost the ability of cellular transmission by deletion of the coat protein majorly contribute to genetic complexity of host cells. They are "locked" inside the cells and are major drivers of the increase of genetic complexity (Cordaux and Batzer, 2009) . One could speculate that these intracellular elements are replicationincompetent retroviruses lacking coats (Lander et al., 2001) . Bats transmit viruses such as Ebola and SARS coronavirus without suffering from disease (Beltz, 2018) . Even RNA viruses such as Bornaviruses have been shown to integrate by illegitimate reverse transcription, possibly also supplying immunity against superinfection (Katzourakis and Gifford, 2010) .

There are two prominent events that significantly contributed to the success of life and the formation of cells. Both of them are associated with gene reduction. This phenomenon may play a role for the evolution of viruses from autonomous to parasitic lifestyles. In the 1960s Lynn Margulis proposed an extracellular origin for mitochondria (Margulis, 1970 (Margulis, , 1993) . An ancestral cell, perhaps an archaeon, was infected by an anaerobic bacterium, which gave rise to mitochondria. Similarly, cyanobacteria formed the chloroplasts in modern plant cells. Mitochondria arose around 1.45 billion years ago (BYA) (Embley and Martin, 2006) . Mitochondria and chloroplasts are the most striking examples for a change in lifestyle from autonomous bacteria to endosymbionts. This transition is often considered as extremely rare and a hallmark of evolution of life on our planet. However, there are many other obligate intracellular parasites such as Rickettsia, Chlamydia trachomatis, Coxiella burnetii (the causative agent of Q fever), Mycobacterium leprae, M. tuberculosis, and M. mycoides (Beare et al., 2006) .

The change of lifestyle of the endosymbionts in the two cases of mitochondria and chloroplasts is striking. Both of them drastically reduced their genetic make-up. Mitochondria contain less than 37 genes, left from the original about 3,000 genes. Is endogenization of retroviruses, the ERVs, which are integrated into germline cells, related to endosymbiosis? Are these endosymbionts models for the transition from autonomous lifestyle to a parasitic life-which may have taken place with viruses? A more recent typical example for a reductive evolution are Rickettsia. These bacteria were assumed for some time to be viruses because of their obligatory intracellular parasitic existence. Rickettsia have evolved from autonomously replicating bacteria. Reductive evolution of endosymbionts can yield bacteria with tiny genomes on the expense of autonomous extracellular life. Their genomes are 1.11 mio bp in length with about 834 protein-coding genes, and loss of 24% by reductive evolution (Ogata et al., 2001) . Rickettsia may have some relationship with cyanobacteria, which are considered as the major symbionts.

Can one speculate that viruses may have been autonomous entities initially? Viroids may have undergone transition from autonomy to parasites, just as shown for mitochondria, chloroplasts or Rickettsia? To which extent have viruses been autonomous and independent of cellular metabolisms originally -and contributed to the origin of cells? Could they only later have lost their autonomy and become parasitic?

Viruses are minimalistic in their composition and must have undergone stringent gene reductions (Flint, 2015) . How small can their genomes become? Most coding RNA viruses still contain regulatory elements, ncRNA at the 3 and 5 terminal regions for ribosomal entry, protein synthesis, transcriptional regulation, and others.

A subgroup of retroviruses is an interesting example in respect to simultaneous loss and gain of genetic information. The oncogenic retroviruses or tumorviruses can recombine with cellular genes which under the promoters of retroviruses can become oncogenes and drivers of cancer. About a hundred oncogenes have been selected for in the laboratories and studied over decades for understanding the molecular mechanisms of cancer. Selection for growth advantages of the host cells led to the discovery of the fastest growth-promoting oncogenes we know today, such as Ras, Raf, ErbB or Myc, which are in part successful targets for anticancer drugs (Moelling et al., 1984) .

These oncogenes were in most cases taken up by the retroviruses at the expense of structural (gag), replicating (pol) or envelope (env) genes, and are often expressed as fusion proteins with Gag. Thus, oncogenic retroviruses are obligatory intracellular defective viruses and were selected for in the laboratory by researchers for the oncogenes with the most potent growth promoting ability. They need the supply of replicatory genes in trans from co-infecting helper viruses to infect other cells (Flint, 2015) . Retroviruses are able to pick up cellular genes, transfer and integrate them into neighboring cells. Some strains of Rous sarcoma virus maintain replication competent when carrying the cell-derived src (for sarcoma) oncogene encoding a protein of 536 amino acids that apparently can fit into the retroviral particle along with the full-size viral genome (Broecker et al., 2016a) . Spatial reasons may have influenced the formation of oncogenic retroviruses and limited their size and thereby led to their defective phenotypes.

There are indications that the uncontrolled activity of (retro)transposons in germline cells can result in diseases such as male infertility -presumably by "error catastrophe," caused by too many transposition events. In mammals, piRNAs tame transposon activity by means of the RNase H activity of PIWI proteins during spermatogenesis (Girard et al., 2006) .

Only a minority of viruses are pathogens; most of them do not cause diseases. On the contrary, they are most important as drivers of evolution, as transmitters of genetic material, as innovative agents. In particular, the RNA viruses are the most innovative ones. Some of them are pathogenic and dangerous, such as HIV or influenza virus, or viroids in plants. RNA viruses are able to change so rapidly that the host immune system is unable to counteract the infection. Pathogenicity arises when environmental conditions change, for instance, when a virus enters a new organism or species.

Increase of cellular complexity by viruses is an important feature of evolution. Such major evolutionary changes are recently taken as arguments against the evolutionary theory by Charles Darwin who considered gradual changes, small increments by mutations as the main basis for selection and evolution. New criticism is addressing this thinking, considering larger changes as evolutionary drivers. Such changes arise by many complex phenomena such as endosymbiosis, infection by prokaryotes, viruses and fungi, recombination of genes, HGT, infections, sex. Dramatic changes such as endosymbiosis or pathogen infections extend Darwin's concept of evolution.

There are numerous examples for the contribution of viruses to the evolution of life since at least as long as 550 MYA (Hayward, 2017) . But genetic noise through random mutations does not allow us to go back to the origin of life. It may not be impossible that the earliest compartment was indistinguishable, either a pre-cell or a pre-virus. By analogy one may speculate that at some point autonomous viruses gave up independence for an obligatory intracellular life -as has been described for mitochondria and chloroplasts but also intracellular bacteria such as Rickettsia. This speculation is based on the concept that early life must have started simple and with high genetic variability and then became more complex. But complexity can be given up for a less energy consuming lifestyle with small genomes and high speed of replication (Moelling, 2012 (Moelling, , 2013 . Therefore, the question may be repeated: "Are viruses our oldest ancestors?" Some fossil life can be partially reproduced in vitro by Spiegelman's Monster and Eigen's follow-up experiments, explaining the great surviving potential of simple ncRNA.

Viruses can be pathogens, but their recognition as primarily causing diseases is wrong. This notion is based on the history of viruses in medicine, as explained in a book entitled "Viruses: More Friends Than Foes" (Moelling, 2017) . The scenario described here focuses on viruses as drivers of evolution.

The early RNA world gained interest 20-30 years ago as evidenced by the references provided above. Surprisingly, there are scientists who still believe in the "pansperm hypothesis" and think that retroviruses are of extraterrestrial origin (Steele et al., 2018) . The recent interest in the origin of life arose from the newly discovered exoplanets whose number increases daily -and which may be as numerous as 10^{25} . Thus, pure statistics make some people believe that there is extraterrestrial life.

The extraterrestrial life is mimicked in laboratories on Earth with many assumptions - perhaps this overview stimulates some thinking. The discussion presented here should be taken as concept about simple replicating and evolving entities possibly arising from different building blocks in other environments, with structure being more relevant than sequence.

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Chikungunya: A Potentially Emerging Epidemic?

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Abstract: Chikungunya virus is a mosquito-borne emerging pathogen that has a major health impact in humans and causes fever disease, headache, rash, nausea, vomiting, myalgia, and arthralgia. Indigenous to tropical Africa, recent large outbreaks have been reported in parts of South East Asia and several of its neighboring islands in 2005–07 and in Europe in 2007. Furthermore, positive cases have been confirmed in the United States in travelers returning from known outbreak areas. Currently, there is no vaccine or antiviral treatment. With the threat of an emerging global pandemic, the peculiar problems associated with the more immediate and seasonal epidemics warrant the development of an effective vaccine. In this review, we summarize the evidence supporting these concepts.

Text: Chikungunya virus (CHIKV), a mosquito-borne pathogen listed by National Institute of Allergy and Infectious Diseases (NIAID) as a Category C Priority Pathogen that causes Chikungunya fever (CHIKF), has been spreading throughout Asia, Africa, and parts of Europe in recent times [1, 2, 3] . CHIKV is an arthropod-borne virus (arbovirus) and is transmitted to humans primarily by *Aedes aegypti*, the infamous yellow fever propagator [4, 5] . CHIKV infection is marked by severe joint pain, contorting its victims into unusual postures [6] . The disease gets its name from the Kimakonde vernacular language of

Tanzania and Mozambique, and the word chikungunya means "that which contorts or bends up" and translates in Swahili to "the illness of the bended walker" [7, 8, 9] . In Africa, CHIKV is maintained in a sylvatic cycle among forest-dwelling *Aedes* spp. mosquitoes, wild primates, squirrels, birds, and rodents (Figure 1) [10] . In Asia, the disease is vectored by *Ae. aegypti* and *Ae. albopictus* [11] . Transmission in Asia occurs in an urban cycle whereby the mosquito spreads the disease from an infected human to an uninfected human, following an epidemiological pattern similar to dengue fever [12] .

The 2005-2006 epidemic of CHIKV in La Reunion islands in the Indian Ocean, spurred the discovery of a new vector species, *Ae. albopictus* [5] . Wrecking over one-third of the island's population, this epidemic peaked its devastation between January and February 2006, when over 46,000 cases came into light every week, including 284 deaths [5, 13] . *Ae. albopictus* is common in urban areas of the United States and is already flourishing in 36 states, raising grave concerns to the immunologically naive populace of the United States [14] .

Accordingly, this review elaborately details the epidemiology and global expansion of CHIKV, describes its clinical features and pathogenesis and its symptoms and complications, and finally nominates a possible vaccine approach against CHIKV infection.

CHIKV has been isolated into three genotypes based on phylogenetic studies. These genotypes, based on the gene sequences of an Envelope protein (E1), are Asian, East/Central/ South African, and West African [4, 11, 15] . Using phylogenetic models, Cherian et al. estimate that the Asian genotype of CHIKV emerged between 50 and 310 y ago, and the West and East African genotypes diverged between 100 and 840 y ago [15] . Since then, CHIKV has come a long way, with several mutations incorporated, and has continued to wreak epidemics in several regions. Recent activities of CHIKV include the Indian epidemic in 2005-2006, which was followed by a sudden explosion of cases in 2007. An estimated 1.3 million people across 13 states were reported to be infected in India [12, 16] , and CHIKV was also widespread in Malaysia, Sri Lanka, and Indonesia [17] . In July-August of 2007, CHIKV was reported in Italy, probably brought in by travelers from CHIKV-prone regions of India, Africa, and Indian Ocean islands such as Mauritius, Madagascar, and Seychelles. Few of the Italian isolates were found to have evolved from the Kerala isolate, which was associated with a A226V shift in E1 gene that represents a successful

evolutionary adaptation in the mosquito vector similar to the ones observed in Reunion Island [2, 18, 19] .

In recent times, with an increase in global travel, the risk for spreading CHIKV to non-endemic regions has heightened [1] . Several travelers have brought CHIKV home with them after visiting areas with actively infected populations [12, 20] . Such cases have been documented in European countries, Australia, Asia, and the United States [8, 21] . The United States has already reported at least twelve cases of travel-associated CHIKV, while France has reported 850 cases, and the United Kingdom 93 [8, 14] . Beyond this, CHIKV-infected travelers have also been diagnosed in Australia, Belgium, Canada, Czech Republic, French Guiana, Germany, Hong Kong, Italy, Japan, Kenya, Malaysia, Martinique, Norway, Switzerland, and Sri Lanka [21] . Some travelers were viremic, worrying public health officials about the spread of CHIKV to new areas [1, 8] .

The incubation time for CHIKV is relatively short, requiring only 2-6 d with symptoms usually appearing 4-7 d post-infection [22] . Vazeille et al. detected CHIKV in the salivary glands of *Ae. albopictus* only 2 d after infection [5] . Upon infection, CHIKF tends to present itself in two phases. The first stage is acute, while the second stage, experienced by most but not all, is persistent, causing disabling polyarthritis. Characteristics of the acute phase include an abrupt onset of fever, arthralgia, and in some cases, maculopapular rash [6, 23] . The acute phase causes such intense joint and muscular pain that makes movement very difficult and prostrates its victims [6, 20] .

Ninety-five percent of infected adults are symptomatic after infection, and of these, most become disabled for weeks to months as a result of decreased dexterity, loss of mobility, and delayed reaction. Eighteen months after disease onset, 40% of patients are found to still have anti-CHIKV IgM [6, 18, 23, 24] . The chronic stage of CHIKF is characterized by polyarthralgia that can last from weeks to years beyond the acute stage [6] . CHIKV has been shown to attack fibroblasts, explaining the involvement of muscles, joints, and skin connective tissues. The high number of nociceptive nerve endings found within the joints and muscle connective tissues can explain pain associated with CHIKF [25, 26] .

More than 50% of patients who suffer from severe CHIKF are over 65 y old, and more than 33% of them die. Most adults who suffer from severe CHIKF have underlying medical

conditions [6, 24, 27] . The other group that is disproportionately affected by severe CHIKV is children. Other complications associated with CHIKV, from most common to least common, include respiratory failure, cardiovascular decompensation, meningoencephalitis, severe acute hepatitis, severe cutaneous effects, other central nervous system problems, and kidney failure [6, 18, 20, 23, 24, 26, 27] .

CHIKV undertakes a complex replication cycle upon host infection (Figure 2), which makes its genome susceptible to mutations [28, 29] . For instance, *Ae. aegypti*, responsible for epidemics in Kenya, Comoros, and Seychelles, carried CHIKV with an alanine in the 226 position of the E1 gene (E1-A226) [4, 18] . However, when the virus struck La Reunion Islands, a decline in population of *Ae. aegypti*, due to massive dichlorodiphenyltrichloroethane usage and dearth of *Ae. albopictus* species' www.plosntds.org population, resulted in an ecological pressure, favoring replacement of alanine at position 226 with valine (E1-A226V) [5] . This mutation allowed CHIKV's secondary vector species, *Ae. albopictus*, to supplement *Ae. aegypti* as its primary vector [5] .

Within a year, the E1-A226V mutation was present in La Reunion Island, and *Ae. albopictus* apparently vectored the large epidemic infecting 34% of La Reunion Island's population [5] . All of the CHIKV strains isolated from Mayotte carried the E1-A226V mutation, and the mutation was also found in Madagascar in 2007 [5] . The E1-A226V mutation was not present at the beginning of the Indian Ocean Islands outbreak (before September 2005). However, more than 90% of later viral strains found there had incorporated the mutation (December-March 2006), indicating a genotype switch during the winter season [5, 18, 20] .

The E1-A226V mutation also enabled an increase in infectivity of *Ae. albopictus* when compared to its infectivity of *Ae. aegypti* [4, 11, 18, 30] , and with several factors taken together, *Ae. albopictus* has become the new preferred and more lethal vector for CHIKV [4, 5, 11] . In fact, Tsetsarkin et al. found that a Green Fluorescent Protein tagged E1-A226V virus was 100 times more infective to *Ae. albopictus* than it was to *Ae. aegypti* [4] . In all the Indian Ocean Islands, *Ae. albopictus* became the main vector for CHIKV within 1-2 y after CHIKV was introduced to the region [31] .

Of note is that *Ae. aegypti* has most likely been established in North America for over 300 y, while *Ae. albopictus* has been in many areas of the US, since 1985, primarily in Florida [32] and since then has expanded its range in the country. Reiskind et al. set out to determine if *Ae. aegypti* and *Ae. albopictus* mosquitoes captured in Florida were susceptible to CHIKV infection by a La Reunion isolate [32]. Each mosquito tested was highly susceptible to infection by a full-length infectious clone of the La Réunion Island isolate, CHIKV LR2006 OPY1 strain. Even though the *Ae. albopictus* strains were more susceptible to infection, overall ecology and differences in human biting patterns need to be studied further. Characteristically, there are two rounds of translation: (+) sense genomic RNA (49S9 = 11.7 kb) acts directly as mRNA and is partially translated (5' end) to produce non-structural proteins (nsp's). These proteins are responsible for replication and formation of a complementary (2) strand, the template for further (+) strand synthesis. Subgenomic mRNA (26S = 4.1 kb) replication occurs through the synthesis of full-length (2) intermediate RNA, which is regulated by nsp4 and p123 precursor in early infection and later by mature nsp's. Translation of the newly synthesized sub-genomic RNA results in production of structural proteins such as Capsid and protein E2-6k-E1 (from 3' end of genome). Assembly occurs at the cell surface, and the envelope is acquired as the virus buds from the cell and release and maturation almost simultaneously occurred. Replication occurs in the cytoplasm and is very rapid (<4 h) [28, 29].

doi:10.1371/journal.pntd.0000623.g002 www.plosntds.org to gain a more accurate understanding of a potential CHIKV epidemic in the US [32].

During the 7 d preceding birth, no human mother has been reported to transmit the disease vertically. However, about 50% of newborns delivered while the mother was infected with CHIKV contracted the disease from their mother, despite the method of delivery. Furthermore, there have been instances of CHIKV transmission from mother to fetus causing congenital illness and fetal death [33].

During the 2005-2006 La Reunion Island outbreaks, Ramful et al. discovered that mothers could transmit CHIKV to their progeny during the perinatal period (Day 24 to Day +1) [33, 34], and it is associated with a high degree of morbidity. By mean Day 4 of life, all of the neonates were symptomatic for CHIKV, exhibiting common CHIKF symptoms. Six neonates were confirmed to have contracted CHIKV and developed meningoencephalitis. Of those mothers who, during the La Reunion Island epidemic, were infected long before delivery, only three fetal deaths were reported [12, 33]. Ramful et al. theorized that mother-to-child transmission most likely happens transplacentally shortly before delivery [33]. A similar

study by Gerardin et al. reported nineteen cases of neonatal infection associated with intrapartum maternal viremia that progressed to develop encephalitis owing to vertical transmission from infected mothers [34] .

Clinical and epidemiological similarities with dengue fever make CHIKV diagnosis difficult, which may lead physicians to misdiagnose CHIKV as dengue fever; therefore, the incidence of CHIKV may actually be higher than currently believed (Table 1) [6, 12, 35] .

The amount of time elapsed since disease onset is the most critical parameter when choosing a diagnostic test. CHIKV can be detected and isolated by culturing with mosquito cells (C6/36), Vero cells (mammalian), or in mice [26] . However, this method can take at least a week and only achieves a high sensitivity during the viremic phase, which usually only lasts up to 48 h after the bite. Five days post-infection, the viral isolation approach has a low sensitivity but is still the preferred method for detecting the CHIKV strain [12, 26, 31, 35] . RT-PCR on the other hand is a faster and more sensitive method that can be used within the first week of disease onset [26] , and it is currently the most sensitive method for detecting and quantifying viral mRNA [4, 36] .

Classic serological detection, by assays such as ELISA [37] , immunofluorescence [5, 38] , complement binding, and haemagglutination inhibition [39] , constitutes the second diagnostic tool used for biological diagnosis of CHIKV infection. These proven techniques are useful for detection of Antigen in mosquitoes during epidemiological studies. These assays detect virus-specific IgM and IgG, however the sensitivity and specificity of these assays has been poorly characterized. Viral competence, or the potential of viral infection and transmission, is an important parameter that can be quantified by ELISA, viral culture, and PCR.

A study by Ng et al. showed biomarkers indicative of severe CHIKV infection [40] . They found decreased levels of RANTES and increased levels of Interleukin-6 (IL-6) and Interleukin-1b (IL-1b) that could be used for CHIKV detection in patients as indicators of CHIKV-driven cytokine storm. Couderc et al. demonstrate another cytokine, type-I IFN, as a key player in the progression to CHIKV infection [26] . Using an IFN- α /b null mouse model, they demonstrated evidence of muscles, joints, and skin as privileged CHIKV targets, which is consistent with human pathology. Although Ng et al. concluded that RANTES levels were

significantly suppressed in severe CHIKF patients [40] , interestingly, an increase in levels of RANTES has been observed in dengue infection [41] . Since the symptoms of CHIKF mimic those of dengue fever, results obtained from this study strongly suggest that RANTES could be a potential distinctive biomarker that differentiates between these two clinically similar diseases.

There are no approved antiviral treatments currently available for CHIKV [1, 3, 12, 42] . Currently, CHIKF is treated symptomatically, usually with non-steroidal anti-inflammatory drugs or steroids, bed rest, and fluids. Movement and mild exercise are thought to decrease stiffness and morning arthralgia, but heavy exercise may exacerbate rheumatic symptoms. Corticosteroids may be used in cases of debilitating chronic CHIKV infection. There is a debate about the appropriateness of chloroquine as treatment for unresolved, non-steroidal anti-inflammatory drug-resistant arthritis [43] . A study showed that viral production was www.plosntds.org drastically reduced at 16 h post-infection after treatment with 100 mM dec-RVKR-cmk (Decanoyl-Arg-Val-Lys-Arg-chloromethylketone), a furine inhibitor [42, 44] . Chloroquine acted by raising the pH, blocking low pH-dependent entry of virus into the cell. It is important to note that dec-RVKR-cmk or chloroquine only inhibited viral spreading from cell to cell, not CHIKV replication once it had entered the cell [43] . However, most would agree that the best weapon against CHIKV is prevention. A live CHIKV vaccine developed by the United States reached phase II clinical trial encompassing 59 healthy volunteers [45] . Eight percent of the volunteers experienced transient arthralgia, while 98% of the volunteers had seroconversion [45] . However, live CHIKV vaccines are still questionable. One cannot discount the risk of a live vaccine possibly inducing chronic rheumatism. Also, there is the question as to whether widespread use among the public could trigger mosquito transmission or lead to chronic infection or viral reversion [1] .

An alternative approach would be to produce a chimeric vaccine against CHIKV. Wang et al. developed a chimeric alphavirus vaccine that is uniformly attenuated and does not cause reactogenicity in mice [3] . Three different versions of this vaccine were made using three different backbone vectors: Venezuelan equine encephalitis virus (VEEV) attenuated vaccine strain T-83, naturally attenuated eastern equine encephalitis virus (EEEV), and attenuated Sindbis virus (SINV). In short, CHIKV structural proteins were engineered into the backbones of the aforementioned vaccines to produce the chimeras [3] . These chimeras were found to stimulate a strong humoral immunity, and even at doses of 5.3-5.8 log₁₀ PFU, they did not trigger reactogenicity. When vaccinated mice were challenged with CHIKV, neither adult nor neonatal mice gained weight, had fever, or displayed signs of

neurological illness. Upon comparison of the chimeras with the Army181/25 vaccine, the Army vaccine resulted in higher levels of viremia and replication in the joints of neonatal mice. Because the joints are known targets of CHIKV, Wang et al. noted their vaccine might avoid the negative reactogenic side effects of the Army vaccine. After being subcutaneously vaccinated with 5.3-5.8 log 10 PFU of the chimeric vaccines, mice produced strong neutralizing antibody titers. The VEEV and EEEV chimeras yielded higher neutralizing antibody titers than the SINV chimera without being more virulent. On top of this, the VEEV and EEEV CHIKV chimeras seemed to be more immunogenic than the Army vaccine despite the chimeras' lower viremia and replication in the joints of neonatal mice [3] .

Tiwari et al. [46] adopted a different strategy using formalin inactivated CHIKV in combination with alhydrogel (Aluminum Hydroxide) as an adjuvant. This study clearly suggests that this vaccine elicits both humoral and cell-mediated immune responses in mice, providing its immunogenic potential. A recent study by Couderc et al. [47] showed passive immunization as a potential treatment for CHIKV infection. Using purified immunoglobulin extracted from convalescent CHIKV patients, they demonstrated effective neutralizing activity against CHIKV infection both in vitro and in vivo. This thereby establishes a potential preventive and therapeutic approach to combat CHIKV infection. Pathogenesis studies conducted with related alpha virus, like RRV, have shown the role of macrophages in persistence on infection [48] . They also demonstrated the role of RRV-specific CD8 T cells in clearing viral load in infected patients, thereby warranting similar investigations with CHIKV and the importance of investigating a cell-mediated immune response-based vaccine against CHIKV [49] .

There are always certain risks associated with live attenuated or inactivated viral vaccines [50] . One way to avoid these potential problems is to construct a consensus-based DNA vaccine. DNA based vaccines have an improved safety profile as compared to live or attenuated vaccines [51, 52] . A consequence of CHIKV's rapid evolution is difficulty in constructing a vaccine that will be able to Figure 3 . Levels of CHIKV-specific IgG in mice immunized with CHIKV vaccines. Each group of C57BL/6 mice (n = 5) was immunized with 12.5 mg of pVax1 control vector or CHIKV vaccine plasmids as indicated at 0 and 2 wk. Mice were bled 2 wk after each immunization, and each group's serum pool was diluted to 1:100 and 1:500 for reaction with specific vaccine constructs. Serum was incubated for 1 h at 37°C on 96-well plates coated with 2 mg/ml of respective CHIKV peptides, and antibody was detected using anti-mouse IgG-HRP and OD was measured at 405 nm.

doi:10.1371/journal.pntd.0000623.g003 www.plosntds.org effectively protect large populations from multiple strains of the virus. One of the strengths of DNA consensus vaccines is its ability to induce cross-reactive immune responses against the three distinct phylogenetic groups of CHIKV. Also DNA-based vaccines can be produced more rapidly than protein-based vaccines.

Recently, Muthumani et al. constructed a vaccine that was shown to induce both humoral and cellular immunity in vivo in 3-4-wk-old female C57/BL6 mice [49] . These mice were immunized using an in vivo electroporation method to deliver the vaccine into the quadriceps muscle. The consensus construct was designed against E1, E2, and the core protein capsid. To design the construct, they aligned 21 sequences of CHIKV isolated between 1952 and 2006, using strains from differing countries, including La Reunion Island. The most common nucleotide among the sequences was chosen at each position to be used in the consensus construct, taking care not to alter the reading frame. They conducted codon and RNA optimization, added a strong Kozak sequence, and substituted signal peptide with an immunoglobulin E leader sequence to improve vaccine efficacy.

After immunizing the mice, spleens were harvested along with serum and tested to determine antibody titer. After three immunizations, consensus E1, E2, and C vaccines were shown to induce T-cell immune responses leading to strong IFN- γ responses and proliferation in C57/BL6 mice. Furthermore, when compared with control mice, immunized mice had higher total IgG levels as well as higher anti-E1 specific, anti-E2 specific, and anti-C specific IgG antibodies, suggesting a strong humoral immune response (Figure 3) and also specificity for the antigens encoded in the vaccine constructs (Figure 4). Because of its promising results and the need for a safer vaccine, this consensus DNA vaccine deserves further investigation. Determining longevity of protective effects of the vaccine and persistence of antibody and IFN- γ responses could be the next step of investigation. Challenged studies of immunized mice must also be carried out.

CHIKV mosquito-borne disease has caused massive outbreaks for at least half a century but is no longer confined to the www.plosntds.org developing nations. It began to encroach into the boundaries of the developing world. As a result, the NIAID has designated CHIKV as a Category C pathogen alongside the influenza and SARS-CoV viruses [3] . Realization of the potential severity of this disease is exigent; for instance, if used as a biological weapon, the world economy could be severely crippled; if enough members of the armed forces

were to become infected during a military deployment, military operations could be significantly affected. Efforts to monitor the disease will only provide minimal warning in a global society, and steps to prevent the morbidity and mortality associated with pandemic are imperative [21, 31] . Despite the gravity of its infectious potency and the fear of it being a potential biological weapon, there is currently no vaccine for CHIKV infections. Live attenuated vaccine trials were carried out in 2000, but funding for the project was discontinued. Newer approaches such as DNA vaccines appear promising over conventional strategies like live attenuated or inactivated virus and thus call for further investigation. Recent advances such as electroporation delivery and incorporation of adjuvants has boosted DNA vaccine efficacy [51, 53] . Despite the low antibody response to DNA vaccines, other numerous advantages have overshadowed these minor drawbacks (Table 2) , the most important one being the ability to induce both humoral and cellular immune responses [51, 54] .

Judging by recent success, such as the immunogenic construct developed by Muthumani et al., DNA vaccines could play a major role in combating CHIKV [49] . Vaccines are literally a critical component of CHIKV disease control and therefore research in this area is highly encouraged. The dramatic spread of dengue viruses (DENV) throughout tropical America since 1980 via the same vectors and human hosts underscores the risk to public health in the Americas. The adverse events associated with the current live vaccine are well documented [55] . Realizing these drawbacks, earnest efforts should be taken to develop new strategies to forestall further spread and complications.

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Health care workers indicate ill preparedness for Ebola Virus Disease outbreak in Ashanti Region of Ghana

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Abstract: **BACKGROUND:** The recent Ebola Virus Disease (EVD) epidemic that hit some countries in West Africa underscores the need to train front line high-risk health workers on disease prevention skills. Although Ghana did not record (and is yet to) any case, and several health workers have received numerous training schemes, there is no record of any study that assessed preparedness of healthcare workers (HCWs) regarding EVD and any emergency prone disease in Ghana. We therefore conducted a hospital based cross sectional study involving 101 HCWs from two facilities in Kumasi, Ghana to assess the level of preparedness of HCWs to respond to any possible EVD. **METHODS:** We administered a face-to-face questionnaire using an adapted WHO (2015) and CDC (2014) Checklist for Ebola Preparedness and assessed overall knowledge gaps, and preparedness of the Ghanaian HCWs in selected health facilities of the Ashanti Region of Ghana from October to December 2015. **RESULTS:** A total 92 (91.09%) HCWs indicated they were not adequately trained to handle an EVD suspected case. Only 25.74% (n = 26) considered their facilities sufficiently equipped to handle and manage EVD patients. When asked which disinfectant to use after attending to and caring for a suspected patient with EVD, only 8.91% (n = 9) could correctly identify the right disinfectant ($\chi^2 = 28.52$, $p = 0.001$). **CONCLUSION:** Our study demonstrates poor knowledge and ill preparedness and unwillingness of many HCWs to attend to EVD. Beyond knowledge acquisition, there is the need for more training from time to time to fully prepare HCWs to handle any possible EVD case.

Text: During the last outbreak of Ebola Virus Disease (EVD) and its consequential massive epidemic with very high mortality [1] , many health systems and services in West Africa were overwhelmed and disrupted. This was partly due to the poor and weak health systems coupled with unprepared and unskilled frontline healthcare workers (HCWs) compounded by poor understanding of the disease dynamics tied to lack of requisite resources. During

the early part of 2014, the emergence of EVD [1] in Guinea, jolted the health care systems of West African sub-region claiming over 9800 lives [2] including more than 491 (58.7%) deaths of HCWs from 839 infections [2] . This epidemic therefore reinforced the fact that HCWs are at high-risk of being infected with the disease in line with their core duties. Empirical data generated during and after the epidemic indicated how unprepared most HCWs were in the face of the crisis. Studies in Nigeria, Guinea and India indicate the low level of knowledge, negative attitude and sub-standard practices that can be eliminated through continued training of HCWs as well as provision of needed and adequate resources in their line of duties [3] [4] [5] [6] .

The countries worst hit were Liberia, Sierra Leone, Guinea and several other countries with imported cases [7] . Like most West African nations, Ghana was on high alert and was number one on the list of countries deemed to be at high risk of EVD. Thus, the country tried to make some preparations in the wake of the epidemic [8] . The government with support from donor organizations such as the World Health Organization (WHO), Médecins sans frontières (MSF) put in place resources for training of health professionals and some level of retooling of hospitals and preparedness of health workers in the face of any possible emergency scenarios. Various HCWs received both theoretical and practical training on how to manage infected and affected persons. These training sessions took the form of onsite and off site coaching as well as workshops. Simulation exercises were also conducted to bring to bear how HCWs would react during EVD emergency scenarios. For example, the German government through the Kumasi Centre for Collaborative Research in Tropical Medicine organized hands on training for several West African nationals on sample taking, sample testing and donning and doffing personal protective equipment (<http://kccr.org>). More importantly, there was the construction of three treatment centres and as well, a standby ambulance service for transportation of confirmed cases to the treatment centres. Incidentally, Ghana did not record any case in the wake of the epidemic and has so far not recorded any case of EVD. However, the response of HCWs to the scenarios identified several gaps. Following a series of training for HCWs, one could easily assume that health care workers are adequately prepared and equipped with the requisite knowledge and skills to deal with any possible EVD outbreak. It is unclear for example to what extent these exercises were practiced and for how long they were made a part of routine hospital activities. One therefore wonders how well prepared HCWs within Ghana are to responding not only to EVD but other epidemic prone diseases (EPDs) requiring a concerted approach to preparedness and management.

Even though some resources have been invested in response to the EVD scare in Ghana, there has not been any assessment on the preparedness of health workers in the face of any possible emergency scenarios. Simply providing the tools such as medications, personnel protective equipment (PPE) and other logistics is no panacea for adequately and appropriately responding to EPDs. Consequently, if healthcare staff lack the basic knowledge, practical and organizational skills to plan and respond to such emergency situations, it would not only spell doom for themselves but also for the general population as was the case of the recent epidemic in West Africa. It is important for example to understand the dynamics of what will propel a HCW to be willing to put his or her life in the line of duty for a case of EVD. It is therefore critical to understand current preparedness of the healthcare worker in Ghana in order to make recommendations for future training and planning for any epidemics situation. The need for Ghana to therefore have empirical data on general emergency preparedness to determine and understand knowledge gaps, and to assess knowledge versus practice in a bid to provide direction for policy cannot be overemphasized. In view of this, we therefore assessed the level of preparedness, readiness and knowledge of EVD emergency response among a population of healthcare workers (HCWs) in the Kumasi Metropolis of Ashanti Region, Ghana.

We conducted a hospital based cross-sectional study among healthcare workers at the Kumasi South and Suntreso Government Hospitals designated as "advanced Ebola holding unit" and "Ebola standing team" respectively, in the Kumasi Metropolis. The Kumasi South and Suntreso hospitals have an average monthly Out Patient Department (OPD) attendance of about 20,603 and 11,712 patients respectively and health staff of approximately 450 each. Similar to most facilities, there are more females nurses than males.

We organized a day's training for our research assistants on how to use Personal Digital Assistant device (PDAs) Samsung Galaxy note 8 GT-N5100 (Samsung Electronics Co. Ltd., Seoul, Korea) in capturing data.

The original version of the questionnaire was pretested, with five healthcare workers who were similar in their characteristics to the members of the study population but outside the area of jurisdiction and study to ensure validity and measurement bias. The questionnaire was revised based on the suggestions and comments (mainly on how the questions had

been constructed) obtained from the pilot. This was the final and validated data capturing tool used during the study.

At both facilities, we contacted the Medical Superintendents to obtain permission to attend their morning meetings to explain the aims and objectives of the work to HCWs. During this time, HCWs were given the opportunity to ask questions. Two field assistants were stationed at each of the study sites for data capture. Some of the questions asked included the organism responsible for EVD, the mode of transmission of the disease, HCW preparedness to handle any EVD case and among other things early clinical features of the infection.

In estimating the sample size for this study, previous data from the hospital indicates that there are approximately 900 HCWs at the two facilities. Assuming a 95% confidence interval and if 70% of these HCWs would come into contact with an EVD suspected case, allowing an error rate of 10%, approximately 87 HCWs would provide a default study power of 80% and an alpha of 5%. With approximately a non-response rate of 15% allowing us to sample 101 HCWs from the two facilities providing emergency services within the Ashanti Region of Ghana.

Any healthcare worker attending directly to patients in emergency situation was therefore eligible for inclusion in the study. Our sampling frame consisted of a list of a total of 200. From this list, we then took a systematic random sample of all eligible health workers to represent the sample size. After obtaining written informed consent indicated by signature and or thumbprint of participants, we then administered the questionnaires within the two facilities.

We used the WHO (2015) and CDC (2014) Checklist for Ebola Preparedness that provides practical and specific suggestions to ensure that health facilities are able to help their personnel detect possible Ebola cases, protect personnel, and respond appropriately [9, 10] . This checklist included facility evaluation, knowledge and preparedness of HCWs. Based on these checklists we developed a questionnaire to ascertain the overall knowledge and preparedness of Ghanaian HCWs on EVD outbreak. Our questionnaire was administered from a PDA and recorded each participant's demographics, preparedness, form of compensation HCWs think would be appropriate when taking care of EVD case,

and knowledge of EVD during the period October to December 2015. Answers to these questions were needed from HCWs to determine information access on EVD among HCWs, their knowledge about EVD and the form of compensation HCWs think would be appropriate when taking care of EVD case among others.

Data were collected electronically using tablets for cloud storage through CommCare ODK version 2.27.2, aggregated into Microsoft Excel file, exported into STATA version 14 and analyzed. Descriptive statistics was used to summarize the distribution of various variables into tables and figures. Categorical variables were analyzed using chisquare tests and logistic regression for associations.

Background of the study participants Table 1 shows the background characteristics of the study participants. A total of 101 study participants were interviewed, of which 85 (84.16%) were females. Respondents were categorized into three main groups by occupation: Nurses (76.24%), Medical Doctors (19.80%) and Physician Assistants (PA) (3.96%). Majority (54.46%) of the respondents were married. A total 52.48% (53) had been practicing their profession for less than 5 years ($SD = 9.22 \pm 10.52$ years). At both facilities, 75.25% (76) of the respondents had been working in the facility for less than 5 years ($SD = 4.04 \pm 4.07$ years). Table 2 shows the participants knowledge and awareness of EVD. Of the 101 HCWs interviewed, 83.17% ($n = 84$) correctly identified the cause of EVD, 13.86% ($n = 14$) did not know the cause, while 2.97% ($n = 3$) incorrectly labeled the cause to be a bacterium. Even though one (0.99%) Doctor and 16 (15.84%) Nurses were unable to correctly identify the cause; no group was significantly likely to incorrectly label the cause of EVD ($\chi^2 = 5.41$, $p = 0.247$).

A total of 72 (71.29%) HCWs indicated media especially radio as the main source of information when asked where they first heard of EVD. This was significantly more than other sources ($\chi^2 = 45.44$, $p < 0.05$). When asked which biosafety level laboratory (BSL) is required to test sample from suspected patient with EVD, a total 19 (18.81%) indicated BSL-3 of which 11 (10.89%) were Medical Doctors, while 8 (7.92) and 1 (0.99%) were Nurses and Physician Assistants, respectively. A further 76 (75.25%), of which 9 (8.91%) were doctors, 62 (61.39%) Nurses When asked which disinfectant to use after attending to and caring for a suspected patient with EVD, only 8.91% ($n = 9$) could correctly identify bleach (0.5% Sodium Hypochlorite) which disinfectant to use ($\chi^2 = 28.52$, $p = 0.001$).

Preparedness for an EVD outbreak by HCW category Table 3 shows the levels of preparedness of HCWs to handle and manage EVD outbreak. When HCWs were asked if they considered their facilities sufficiently equipped to handle and manage EVD patients, 25.74% (n = 26) responded in the affirmative, while 54.46% (55) indicated otherwise. Of this, 14 (13.86%) were Medical Doctors, 39 (38.61%) Nurses and 2 (1.98%) were PA ($\chi^2 = 2.66$, $p = 0.62$). If they became accidentally infected with EVD after attending to a patient with EVD, 98 (97.03%) of those surveyed indicated they would accept to be isolated ($\chi^2 = 4.69$, $p = 0.321$). Meanwhile, 44.55% (n = 45) of HCWs would willingly attend to an EVD suspected patient ($\chi^2 = 8.03$, $p = 0.09$).

A total of 92 (91.09%) HCWs surveyed indicated they were not adequately trained to handle an EVD suspected case. When asked to rate their competence in handling an EVD suspected patient, 18.81% (n = 19) indicated they had little confidence and competence, while 6.93% (n = 7) indicated they were extremely confident to handle a suspected case of EVD ($\chi^2 = 13.09$, $p = 0.11$).

Beyond EVD, we asked our survey population to name other epidemic prone diseases. Of the total number of HCWs interviewed, 56.43% (57/101) mentioned epidemic diseases of bacteria origin such as tuberculosis and cholera. A further 33.70% (34/101) named diseases of viral origin such as SARS, Flu, HIV, Lassa fever and dengue, while 9.90% (10) mentioned others referring to malaria. When asked the form of compensation HCWs thought would be appropriate when taking care of an Ebola suspected patient, responses given included personal insurance (32/101), family compensation in case of death (31/101), money (30/101) and awards (8/101) while others also suggested job promotion (7/101), and others (18/101).

Our survey population recommended the provision of logistics and training as two key issues in the way forward in adequately preparing HCWs towards any epidemic prone diseases.

Many issues surrounding the preparedness of HCWs have been extensively discussed globally especially in the aftermath of the Severe Acute Respiratory Syndrome Coronavirus (SARS-CoV) and the Middle East Respiratory Syndrome (MERS)-CoV epidemic. While it is on record that the recent EVD outbreak recorded very high mortality among HCWs, to the

best of our knowledge, only few studies have addressed these issues in anticipation of an EVD outbreak particularly in countries not hit by the EVD epidemic and especially in sub Saharan Africa, such a study is almost non-existent. Our study therefore assessed how prepared HCWs are in the face of a possible EVD epidemic.

The results of this survey showed that more than half (54.46%) HCWs indicated that their facilities were not ready to handle EVD cases. Nearly 92% indicated they were not adequately trained to handle an EVD suspected case and it is not surprising that less than 50% indicated they would willingly attend to a suspected patient. Moreover, nearly a third of HCWs would also want insurance for themselves and their families in case they were infected with EVD.

These results are clearly indicative of how ill-prepared the HCWs surveyed are in the face a potentially life threatening epidemic prone diseases, such as EVD in Ghana. In this study, only 25.7% of HCWs said their facility was sufficiently equipped to handle an EVD outbreak. Such low ratings of the hospitals by majority of HCWs is a mark of lack of confidence in their facilities preparedness and this may actually indicate a real lack of preparedness and readiness of the hospitals to handle not only EVD cases but potentially other epidemic prone diseases. Alternatively, it could also mean that HCWs were probably unaware of preparatory work and retooling of their facilities to handle EVD outbreak situation.

Willingness to work during outbreaks and emergencies is deemed a sense of duty even in the face of risk. In this study, less than 50% of HCWs indicated their willingness to work in the event of an EVD outbreak. Additionally, over one third indicated various forms of compensation for themselves or families in case of death or while taking care of an EVD case. This implies that if HCWs are assured or guaranteed that they and or their families would be taken care of in case of death or while taking care of an EVD case, they will willingly work in the face of any emergency scenario. The assumption is that HCWs would willingly work in the face of an infectious diseases emergency and respond appropriately; however, there are evidences of HCWs avoiding this "sacred duty" in caring for patients and would leave patients vulnerable in times of crisis [11] . In order to prevent HCWs from being infected while obliged to work even in the face of personal risk as required by their codes of ethics and professionalism, it is imperative to ensure that appropriate conventional standards, guarantees and effective public health practices are met to enable HCWs

respond to such outbreaks so that they are not infected and or affected despite the risks they might face and continue to face [12] . Thus, appropriate training of HCWs as indicated by those surveyed during the study, coupled with retooling of some health facilities preparation is very critical in ensuring that they are equipped with the needed knowledge and tools needed to work with in the face of any epidemic.

General knowledge of EVD is crucial to adequately respond to and care for patients. Nearly 17% of our study population could not identify that EVD as caused by a virus. Arguably, infection control measures would be difficult and problematic for such HCWs. Less than 10% could correctly identify 0.5% Sodium Hypochlorite as the best disinfectant out of the many options provided. This strongly contradicts a similar study in Conakry conducted during the peak of the epidemic where 68% of HCWs knew the correct concentration of disinfectant [5] . While not trying to compare these two scenarios, this information may be vital in the realization that knowledge of HCWs in infection prevention and control measures is critical in their line of duty.

This study showed that most HCWs first heard of EVD through the media especially radio. This establishes the crucial role media plays in informing the general populace in such disease outbreaks. In Ghana, there are over 350 media outlets (radio and television put together) and majority of households either own a radio, television or have access to internet. Notwithstanding the media pluralism, it is still incumbent upon health institutions and facilities to organize special training on any emerging infectious disease that occurs globally to update the knowledge of HCWs.

Isolation is a key public health measure to prevent the spread of infectious diseases. In this study, over 97% of HCW indicated their willingness to comply and accept to be isolated in case they became infected after attending to suspected EVD patient. However, a small proportion of HCWs surveyed stated that they would be very unhappy, and this could ultimately affect compliance. Isolation is one of the oldest methods of controlling communicable disease outbreaks for patients [13] . However, it is worthy of note that less than 50% said they would be willing to attend to an EVD suspected patient and we suspected that this could be related to fear of personal safety [14] . Emergency response from an epidemic prone disease from an exotic virulent virus or pathogen will naturally spark some level of fear and skepticism among any group of individuals especially when their knowledge about the dynamics of the disease outbreak is low. There are stories of

HCWs who have avoided the responsibility of treating patients [15] and this was apparent in the HIV/AIDS and Severe Acute Respiratory Syndrome-Coronavirus (SARS-CoV) during the 1980s and 2003, respectively where the fear of contact with suspected and infected patients gripped some HCWs [16, 17] . In the long run, this fear would likely affect their confidence and commitment to professionalism.

The results of this study point to the fact that knowledge and the provision of tools such as personnel protective equipment (PPE) and other logistics alone is not good enough strategy. There might be the need to as well address issues related to myth, and culture as well as assurances of upkeep should one be infected. The general outlook one's country's devotion to their health staff might be a contributory factor in all of this and cannot be ignored. However, getting HCWs inspired and feel safe in caring for such highly infectious disease outbreaks is critical. During our study, HCWs indicated various forms of compensation to be paid to them should they be affected in the case of EVD attack.

This study had some inherent limitations. This was an exploratory study and our sample size was limited. Therefore, while not trying to generalize the results, we are of the opinion that this may be a reflection of HCWs in general. Additionally, since our study focused mainly on two health facilities, we are again careful in extrapolating these to other to reflect other facilities. Moreover, since this has not been a real experience, and a questionnaire-based survey, responses may not accurately reflect real-life experiences in the event of an EVD epidemic. Despite these limitations, the need for training was strong among HCWs. The results further demonstrate the ill-preparedness of health facilities, and the large proportion of HCWs unwillingness to attend to a suspected case of EVD. This thus calls for concerted efforts of health institutions and facilities to fully equip and prepare HCWs with the requisite tools and knowledge and ensuring competency to handle any epidemic prone disease.

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RNA sequencing-based analysis of the spleen transcriptome following infectious bronchitis virus infection of chickens selected for different mannose-binding lectin serum concentrations

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Abstract: **BACKGROUND:** Avian infectious bronchitis is a highly contagious disease of the upper-respiratory tract caused by infectious bronchitis virus (IBV). Understanding the molecular mechanisms involved in the interaction between innate and adaptive immune responses to IBV infection is a crucial element for further improvements in strategies to control IB. To this end, two chicken lines, selected for high (L10H line) and low (L10L line) serum concentration of mannose-binding lectin (MBL) were studied. In total, 32 birds from each line were used. Sixteen birds from each line were infected with IBV and sixteen were left uninfected. Eight uninfected and infected birds from each line were euthanized at 1 and 3 weeks post infection. RNA sequencing was performed on spleen samples from all 64 birds and differential gene expression analysis was performed for four comparisons: L10L line versus L10H line for uninfected birds at weeks 1 and 3, respectively, and in the same way for infected birds. Functional analysis was performed using Gene Ontology (GO) Immune System Process terms specific for *Gallus gallus*. **RESULTS:** Comparing uninfected L10H and L10L birds, we identified 1698 and 1424 differentially expressed (DE) genes at weeks 1 and 3, respectively. For the IBV-infected birds, 1934 and 866 DE genes were identified between the two lines at weeks 1 and 3, respectively. The two most enriched GO terms emerging from the comparison of uninfected birds between the two lines were “Lymphocyte activation involved in immune response” and “Somatic recombination of immunoglobulin genes involved in immune response” at weeks 1 and 3, respectively. When comparing IBV-infected birds between the two lines, the most enriched GO terms were “Alpha-beta T cell activation” and “Positive regulation of leukocyte activation” at weeks 1

and 3, respectively. **CONCLUSIONS:** Healthy birds from the two lines showed significant differences in expression profiles for subsets of adaptive and innate immunity-related genes, whereas comparison of the IBV-infected birds from the two lines showed differences in expression of immunity-related genes involved in T cell activation and proliferation. The observed transcriptome differences between the two lines indicate that selection for MBL had influenced innate as well as adaptive immunity. **ELECTRONIC SUPPLEMENTARY MATERIAL:** The online version of this article (doi:10.1186/s12864-016-2403-1) contains supplementary material, which is available to authorized users.

Text: Conclusions: Healthy birds from the two lines showed significant differences in expression profiles for subsets of adaptive and innate immunity-related genes, whereas comparison of the IBV-infected birds from the two lines showed differences in expression of immunity-related genes involved in T cell activation and proliferation. The observed transcriptome differences between the two lines indicate that selection for MBL had influenced innate as well as adaptive immunity.

Keywords: IBV, Coronavirus, Infectious bronchitis, Chicken, RNA sequencing, Transcriptome, Spleen, Mannose-binding lectin, Immune response

Background Avian infectious bronchitis (IB) is an acute and highly contagious disease of the upper-respiratory tract caused by the infectious bronchitis virus (IBV). The virus is a member of the Coronaviridae family and has numerous serotypes and strains. Rapid replication combined with high mutation rate and recombination are the main causes of the observed high diversity [1]. The respiratory tract is the primary target organ and entry point for the virus, before further spread to kidneys and gonads. The most common symptoms of IB are related to the respiratory tract and include gasping, coughing, sneezing, tracheal rales, and nasal discharge [2]. Feed conversion and average daily gain are affected in broilers, and infection is often followed by secondary bacterial infections. In layers, IBV causes a reduction in egg production and egg quality. Today, IB is one of the most economically important diseases in the poultry industry [2]. Infection outbreaks are controlled by a combination of strict management practices and vaccination. The strict management practices, which include the maintenance of the housing temperature and ventilation, are essential, because IBV is highly contagious and spreads very fast. Live attenuated and inactivated vaccines are widely used for control and prevention of IBV infection [3, 4]. As there is little or no cross-protection between different serotypes/variants of the virus, hence vaccines should contain serotypes present in a particular area in order to induce adequate protection [1]. New multi-strain vaccines with the optimal antigen combination

and optimal adjuvants are therefore required for future IBV control. Understanding the molecular mechanisms involved in the interaction between innate and adaptive immune responses to IBV infection is a crucial element for further improvements of the vaccines.

IBV infection induces a wide range of immune responses in chickens. An innate immune response is activated during the initial stages of infection in the mucosal lining of the trachea following binding of IBV virions to receptors on epithelial cells [5]. Activation of this innate immune response may be initiated by Toll-like receptor (TLR) signaling upon IBV recognition [6, 7]. In addition, rapid activation of natural killer (NK) cells has been observed one day after IBV infection [8] as well as increased macrophage numbers in lungs and trachea after primary IBV infection [9]. In the case of the adaptive immune responses, T lymphocyte subpopulations are actively involved in the early stages of IBV clearance [7, 10] exhibiting rapid activation upon IBV infection [6]. Furthermore, studies have shown that cytotoxic T lymphocytes (CTL) play an important role in responding to primary infections with IBV [10, 11]. In addition to T cell responses, IBV specific antibodies, of all three antibody classes present in chickens, have been reported [12] [13] [14]. A specific local antibody response in avian infectious bronchitis is characteristic for the response to a secondary infection [15]. The innate and adaptive immune systems are strongly interconnected, which is also seen in the response to IBV infection, and the connection possibly involves the serum collectin, mannose-binding lectin (MBL) as a key player [16].

Two chicken lines which were selected for high and low MBL serum concentrations (designated L10H and L10L, respectively), were used in the present study. Selective breeding has been performed for 14 generations using the combination of two strains (67.5 % UM-B19 chickens and 33.5 % White Cornish) as a starting population, as described by Juul-Madsen et al. [17]. The final result was two divergent lines, with mean MBL serum concentrations of 33.4 µg/ml for the L10H line and 7.6 µg/ml for the L10L line, respectively [18, 19]. The mean MBL serum concentration for 14 different chicken lines representing both broilers and layers is around 6 µg/ml, but varies from 0.4 to 37.8 µg/ml in normal healthy chickens with protein produced in the liver as the main source of circulating MBL [17]. In chickens, a positive correlation between MBL serum concentrations and the severity of several infections, such as infections caused by IBV [19], *Escherichia coli* [20] and *Pasteurella multocida* [21], has been observed. Chicken MBL binds to IBV [16, 22], therefore it is possible that MBL facilitates innate responses such as opsono-phagocytosis, complement activation or virus neutralization, in the early stages of IBV infection. In mammals MBL has also been shown to influence induction of adaptive immunity [23]. In

support of the role of MBL in response to IBV, Kjaerup et al. [18] observed considerable differences in cellular adaptive immune parameters in response to an IBV infection between lines L10L and L10H. Furthermore, birds from L10H line exhibited lower viral loads and less severe damage of tracheal cilia following the IBV infection in comparison to birds from the L10L line.

The aim of this study was to characterize the spleen transcriptome of healthy birds from the two lines selected for serum MBL, and to investigate differences in molecular mechanisms behind the development of systemic adaptive immunity between the L10L and L10H lines infected with IBV.

The experimental timeline and sampling time points are as illustrated in Fig. 1 and a full description of the experimental infection is reported by Kjaerup et al. [18]. The birds were infected at 3 weeks of age and from day 2 post-infection (p.i.), showed clinical signs characteristic of IBV infection, including sneezing and labored breathing. Viral loads in tracheal swabs were assessed for all birds as reported in the previous paper published on the experimental infection study [18]. No virus was detected in the uninfected birds at any time point throughout the experiment. Viral genomes were detected in swabs from infected birds from day 1 to 8 p.i. Notably, significantly lower viral loads ($p < 0.03$) were observed in birds from line L10H in comparison to infected birds from line L10L [18].

Detection and quantification of splenic gene expression RNA sequencing data were produced from eight infected and eight uninfected birds from each of the two lines at two sampling occasions, as described in the materials and methods section. All samples passed quality control measures for raw and trimmed sequenced reads except for individual no. 46, which was removed due to a very low number of sequenced reads. For the remaining birds, an average of over 37 million reads were obtained per sample for the 63 samples analyzed, with 81 % of the reads mapping to the chicken genome reference sequence, as described in the materials and methods section (See summary statistics with the number of mapped and total reads is presented in Additional file 1: Table S1). In total, 17,113 expressed genes were identified. After filtering genes with fewer than one read per million in eight samples [24] (genes which would not achieve statistical significance for differential expression), the final list contained 11,292 expressed genes. Before performing the differential gene expression analysis, further multivariate analysis was carried out on the raw and normalized gene count data to identify any discrepancies.

Multi-dimensional scaling (MDS) plot on expressed genes between the two lines, L10H and L10L, showed that they differ considerably in their transcriptome profiles for both uninfected and IBV-infected birds [See Additional file 2: Figure S1]. Moreover, inter-individual variation in gene expression at week 1 was considerably higher than that observed at week 3 for both uninfected and IBV-infected birds [See Additional file 2: Figure S1].

Birds 22 and 47 were separated from the rest on the MDS plot [See Additional file 2: Figure S1]. However, inspection of raw sequence data and mapping parameters did not identify any technical problems which would explain the observed out-grouping of these birds. In addition an interclass principal component analysis (PCA) was performed using raw and normalized gene counts. The interclass PCA revealed that the birds 22 and 47 were placed outside the 95 % confidence intervals of their respective treatments [See Additional file 3: Figure S2]. However, the PCA did not identify any gene having extreme count profiles which may have contributed to the transcriptome dispersion of birds 22 and 47 with respect to their treatment groups. Although there was no clear technical or biological explanation for their out-grouping, these samples were removed from further analysis.

Differential gene expression analysis was performed to compare the two chicken lines (L10L and L10H) at two time points for uninfected (C1 and C2, see Fig. 1) and IBV-infected birds (C3 and C4, see Fig. 1). A large number of genes were differentially expressed (DE) between L10L and L10H lines at weeks 1 and 3, for both uninfected and IBV-infected birds (see Table 1 , see Fig. 1).

We identified 1,698 and 1,424 DE genes for the uninfected birds between lines L10L and L10H at weeks 1 and 3, respectively (see Table 1). In total 692 genes had higher expression in L10H line and 1,006 had higher expression in line L10L for the uninfected birds at week 1 [See Additional file 4: Table S2] and 774 genes had higher expression in L10H line and 650 genes had higher expression in L10L line for uninfected birds at week 3 [See Additional file 5: Table S3].

Comparing IBV-infected L10H and L10L birds, we identified 1,934 and 866 DE genes at weeks 1 and 3, respectively (see Table 1). In total 931 genes had higher expression in line

L10H and 1,003 had higher expression in line L10L at week 1 and at week 3, 508 had higher expression in line L10H and 358 had higher expression in line L10L (Table 1 , Additional file 6: Table S4 and Additional file 7: Table S5).

There were also status-related changes in gene expression as shown in the Venn diagram (Fig. 2) . At week 1, the total number of DE genes in uninfected birds Fig. 2). Out of 3,011 (1077 + 621 + 1313) DE genes for both uninfected and infected birds between the two lines only 621 (~20 %) were common for two comparisons (Fig. 2). At week 3, the total number of DE genes in uninfected birds between the two lines was 1424 (883 + 541) (Table 1 , Fig. 2) which was higher comparing to 866 (541 + 325) in infected birds between the two lines (Table 1 , Fig. 2). When comparing the uninfected and infected birds between the two lines, 541 (~30 %) genes were common out of total of 1749 (883 + 541+ 325) DE genes for both comparisons (Fig. 2) .

Moreover, we also performed differential gene expression analysis to compare two time points (week 1 and week 3) in the two chicken lines (L10L and L10H) for uninfected (C5 and C6, see Fig. 1) and IBV-infected birds (C7 and C8, see Fig. 1). Finally, differential gene expression analysis was also conducted to compare the two infection states (uninfected and IBV-infected) at two time points for the L10L chicken line (C9 and C10, see Table S6 , Additional file 9: Table S7 , Additional file 10: Table S8 , Additional file 11: Table S9 , Additional file 12: Table S10 , Additional file 13: Table S11 , Additional file 14: Table S12 and Additional file 15: Table S13].

An enrichment gene set analysis was carried out to identify over-represented Gene Ontology (GO) "Immune System Process" terms using the lists of DE genes from comparisons between uninfected and infected birds from the two lines at 1 and 3 weeks p.i. The most enriched GO Immune System terms between the two lines when comparing uninfected birds from the two lines and then infected from the two lines are shown in Fig. 3 .

GO Immune System terms associated with genes that were differentially expressed between the two lines for uninfected birds at week 1 were "Lymphocyte activation involved in immune response" (GO:0002285), "Activation of innate immune response" (GO:0002218), "Lymphocyte mediated immunity" (GO:0002449), and "Leukocyte

Comparison between the two lines, L10H and L10L, uninfected and infected birds at two time points (weeks 1 and 3). Comparisons C1 -C4 correspond to differential gene expression comparisons presented in Fig. 1 For uninfected birds at week 3, the most enriched GO Immune System terms were "Somatic recombination of immunoglobulin genes involved in immune response" (GO:0002204) and the "Adaptive immune response based on somatic recombination of immune receptors built from immunoglobulin superfamily" (GO:0002460) [See Fig. 3 , See Additional file 17: Figure S4]. In total, 47 DE genes mapped to GO Immune System terms in this comparison (Fig. 4) . Among the DE genes that had a higher expression in the line L10H at week 3, in the uninfected group, were IL7, FKBP1B, FAS and PTPN22, which were also seen differentially expressed between lines at week 1 [See Fig. 4 , Additional file 17: Figure S4].

Comparing infected birds from the two lines, at week 1, "Alpha-beta T cell activation" (GO:0046631), "Activation of innate immune response" (GO:0002218) and "Leukocyte differentiation" (GO:0002521) functions were the three most enriched GO Immune System terms (Fig. 3) . CXCR4 (Chemokine receptor 4), PTPN22 and FAS were among the most highly expressed genes in L10H [See Fig. 4 , Additional file 18: Figure S5].

The major GO Immune System term that was strongly enriched for in the infected birds at week 3 was "Positive regulation of leukocyte activation" (GO:0002696) [See Figure S6].

The present study used two lines, L10L and L10H, which have been divergently selected for high and low MBL serum concentration for 14 generations, Fig. 3 Functional map of differentially expressed genes enriched for GO Immune System terms. The top categories of the GO Immune System terms associated with differentially expressed (DE) genes. All categories were statistically significant (adjusted p-value < 0.001). The chart fragments represent the number of genes associated with the terms as a proportion of the total number of genes within the respective GO term. Terms which have not been grouped are shown in grey respectively. These two lines have earlier been extensively used for immunological studies and exhibit differences in immunological parameters after being challenged with several pathogens [18, 22, 25] .

The spleen is a secondary lymphoid organ where innate and adaptive immune responses can be efficiently mounted. In addition, the avian spleen is considered to play a very

important immunological role because avian lymphatic vessels and lymph nodes are poorly developed [26] . The transcriptome differences in the spleen between the two lines, L10L and L10H, for uninfected (healthy) and IBV-infected birds were investigated, focusing on the differential expression of immune-related genes within significantly enriched immune-related GO terms. Large differences in transcriptome profiles were observed between birds from the two lines, both uninfected (healthy) and following the experimental IBV challenge [See Additional file 2: Figure S1]. This suggests Fig. 4 Differentially expressed genes associated with the GO Immune System term. Heatmap representation of the differentially expressed (DE) genes associated with the GO Immune System terms for the four comparisons between the two lines, L10H and L10L, uninfected and infected groups at two time points, weeks 1 and 3. The heat map is constructed using the average values of counts per millions for each group that selection for MBL serum levels in the two lines had a much wider effect which goes beyond the expression of the MBL gene [27] . The observed transcriptome differences can probably be attributed to correlated response to selection [28] or random genetic drift [29] . Correlated selection occurs when a trait is affected by selection on a another trait and is dependent on a genetic correlation between the two traits, which is well known in animal breeding [30] . Alternatively, random genetic drift could contribute to the observed differences, considering that the founder population of the two lines was small [17] .

Focusing on the expression of immune-related genes: at week 1, the uninfected birds showed differences in the expression of genes involved in both adaptive immunity and innate immunity-related pathways [See Additional file 16: Figure S3]. In addition, the line L10H had a lower expression for the subset of the innate immune genes, TYRO3, TRAF3 and TLR7 at week 1 [See Additional file 16: Figure S3 and Fig. 4]. TYRO3 encodes tyrosineprotein kinase receptor 3 (TYRO3) which is involved in inhibition of TLR signaling pathways and TLR-induced cytokine signaling pathways. These two pathways influence immune-related processes, including cell proliferation/survival, cell adhesion and migration and inhibits the innate inflammatory response to pathogens [31] . TRAF3 encodes a cytoplasmic signaling protein, which plays a critical role in the regulation of antiviral response and viral evasion [32] [33] [34] . TLR7 was also among the DE innate immunity-related genes with higher expression in the line L10H. Chicken TLR7 has been shown to play a part in the response to IBV infections [9, 35] .

In addition to the differences in the expression profiles for a subset of innate immune genes, the two lines also differed in their expression of adaptive immune genes. Uninfected

birds from L10H had a higher gene expression compared to the line L10L, for TGFB3, IL7, FKBP1B, FAS and PTPN22. These genes are known to be involved in a wide range of adaptive immune processes. In humans, reducing the TGF- β signaling on T cells has been shown to increase the function of CD8 T cells in an indirect way which results in the rapid elimination of viruses, enabling the creation of an effective memory response [36] . Similarly, human IL-7 plays a key role in the survival of both naïve [37] and memory [38, 39] CD4 and CD8 T cells. Moreover, an in vitro study showed that inhibition of FKBP1B and other cyclophilins blocked the replication of different Coronaviruses, including IBV [40] . In analogy, uninfected birds from the L10H line had a higher expression of IL7, FKBP1B, FAS and PTPN22.

Generally the uninfected (healthy) birds from the two lines exhibit different expression profiles for this subset of innate and adaptive immune genes probably resulting from the divergent selection for the MBL serum concentration. Selection in animal breeding have been shown to have an extensive effect on a variety of traits including immunological [30] . Moreover, correlated response to selection has been observed in the case where selection was performed on less complex traits such as testosterone levels [41] . Finally, different expression profiles for the subset of innate and adaptive immune in the uninfected birds from the two lines might be due to the balance in the effect of MBL serum levels. High levels of human MBL have been mostly reported as beneficial while in case of intracellular parasitic disease the effect of MBL serum level might be opposite [42] . The results indicate that selection for MBL serum levels might lead to favoring specific modes of immune responses depending on the MBL function.

Large differences in the expression patterns were seen between the two lines following infection with IBV and these differences involved adaptive immunity-related pathways which are associated with "Alpha-beta T cell activation" (GO:0046631), and "Activation of innate immune response" (GO:0002218) [See Additional file 18: Figure S5] . The observed enrichment for GO terms related to T cell activation is in accordance with a previous study of these lines that showed that the IBV-specific T cells are present in large numbers in the spleen after IBV infection [43] . At week 1 post infection CXCR4, FAS and PTPN22 showed higher expression in line L10H. The CXC chemokine receptors are expressed on both effector and memory T cells and play a key role in the homeostasis of memory T cells [44] . FAS has been shown to be upregulated in the kidney of chickens challenged with IBV [45] . The Fas/FasL pathway is an important pathway of killing for cytotoxic T cells [46] . Similarly,

PTPN22 has been shown to be differentially expressed in chickens following pathogen challenge, and in particular following infection with *Escherichia coli* [47] .

Additionally VCAM1 and JMJD6 were among the adaptive immunity-related genes DE between lines at week 3 [See Fig. 4 , Additional file 19: Figure S6]. The VCAM1 gene is known to be involved in the activation of T cells [48] . Furthermore, a recent study demonstrated that JMJD6 regulates proliferation of memory T cells during a viral infection [49] which is of great interest considering that had higher expression in the IBVinfected birds from L10H line at week 3.

The results show that the two lines differ greatly in the expression of adaptive immunity-related genes following infection, which may imply the presence of different modes of gene regulation. The sampling times were chosen to access responses both in the effector phase (week 1) and memory phase (week 3) of the adaptive immune response to IBV. In accordance, at 1 week, post infection subsets of genes actively involved in T cell proliferation show differences between the lines. Also, at week 3 immune-related gene expression profiles in response to IBV infection that differ between the lines are more related to maintenance of T cell memory. MBL is known to be involved in regulation of dendritic cell maturation as well as cytokine production [50] . Dendritic cells, which are the main antigen presenting cells and are actively involved in regulation of adaptive immune responses, possess the receptors for MBL in mammals [23] . Therefore, the two lines selected for different MBL serum concentration may display differences in adaptive immune responses and development of adaptive immunity as a result of differences in response to cytokine signaling from dendritic cells. Other studies of the two lines, L10L and L10H, have shown that they differ in disease response parameters after being challenged with different pathogens. The L10L line has been associated with increased viral replication in the airway after an infectious bronchitis virus (IBV) infection [18] , reduced growth rate after an *Escherichia coli* infection [20] and greater intestinal colonization after *Salmonella* Infantis infection [51] . In the present experiment, significantly lower viral loads ($p < 0.03$) were observed in birds from line L10H in comparison to infected birds from line L10L [18] . Furthermore, L10H birds in the present study exhibited a less severe damage of tracheal cilia following the IBV infection in comparison to the L10L line (unpublished data). In the current experiment phenotypic differences in additional traits connected to adaptive immunity were observed, including numbers of circulating B cells and cytotoxic T cells [18] . Based on these observations it seems that selection for high MBL serum concentration allows birds to cope better after being infected with a range of pathogens. Therefore, the

observed differences in the expression profiles for the adaptive and innate immune-related genes are a reflection of differences in disease resistance and immune responses between the lines L10L and L10H.

In conclusion, large differences in the spleen transcriptome between the two chicken lines, L10L and L10H, were observed in both uninfected (healthy) and IBV-infected birds. The uninfected birds from the two lines showed differences in expression profiles for a subset of both adaptive and innate immunity-related genes, which may represent differences in preparedness to respond to an infection. Following infection with IBV, the two lines showed large differences in expression of genes involved in the adaptive cellular immune response such as T cell activation and proliferation pathways and hence their ability to respond to the infection, which is reflected in the difference in pathogen load seen between the two lines.

This study is a follow-up of the experiment performed by Kjaerup et al. [18] which characterized the cellular and humoral immune response of the two chicken lines, L10H and L10L, divergently selected for MBL serum concentrations following IBV infection. In total, 96 birds were used in the experimental study originating from the two Aarhus University inbred lines, L10H and L10L [19]. All 96 birds were reared together in a biosecure IBV-free environment until they were 3 weeks of age and then allocated to two different groups with 24 birds from each line in each group (uninfected and infected). The birds were transferred to a biosafety level 2 facility and placed in isolators. Two isolators contained uninfected chickens and two isolators contained infected chickens. Each isolator having an equal mix of the two lines as described by Kjaerup et al. [18].

The virulent IBV-M41 strain was used for the infection (a kind gift from Dr. med. vet. Hans C. Philipp at the Lohmann Animal Health GmbH, Cuxhaven, Germany). The virus had been passaged twice in specific pathogen-free embryonated eggs. The IBV inocula were prepared in phosphate-buffered saline (PBS) immediately before use and contained $2 \times 10^{5.2}$ EID₅₀ /200 µl of IBV-M41 virus. The first and the second group (the uninfected groups) were mock-infected with 200 µl PBS per bird. The third and fourth groups (the infected groups) received 200 µl of IBV-M41. The inocula were given half nasally and half orally to mimic the natural infection routes of IBV in the chicken. Chickens were fed diets that met or exceeded the National Research Council requirements. Feed and water were provided ad libitum. The birds were monitored daily for clinical signs of disease and disease

parameters were measured as reported by Kjaerup et al. [18] . None of the individuals received antibiotic therapy during the experimental period. The study was carried out under strict ethical approval and monitoring (see the statement at the end of the Materials and Methods section).

For this study 64 spleen samples were harvested and used for RNA sequencing. The birds were sacrificed 1 and 3 weeks post infection by cervical dislocation and spleen samples were collected. At both time points, eight samples from the two lines, L10H and line L10L, from each group (uninfected and infected) were collected as illustrated in Fig. 1 . After collection, spleens were sectioned (triangular cross-sectional slice from upper part) and identical samples from each chicken were immediately placed in RNeasy® Stabilization Solution (Qiagen Inc., Austin, Texas) that were incubated at 4°C overnight and then transferred to -20°C the following day.

Tissue samples were homogenized on a TissueLyzer LT (Qiagen, Hilden, Germany). Total RNA was extracted with the RNeasy Kit (Catalog ID 74104, Qiagen, Venlo, Netherlands) according to the manufacturer's instructions. The quality of the 64 total RNA samples was verified using a 2200 TapeStation RNA Screen Tape device (Agilent, Santa Clara, CA, USA) and the concentration ascertained using an ND-1000 spectrophotometer (Nano-Drop, Wilmington, DE).

Libraries were prepared with the Illumina TruseqRNA sample prep kit (Catalog ID FC-122-1001, Illumina, San Diego, USA) following the manufacturer's protocol and evaluated with the Agilent Tape Station 2200. Libraries were quantified by Picogreen and then normalized to 10 nM as recommended by Illumina for cluster generation on the HiSeq2000. Equimolar amounts of each library were mixed before NaOH denaturation.

The Illumina Truseq PE cluster kit v3 (Catalog ID PE-401-3001) was used to generate clusters on the grafted Illumina Flowcell and the hybridized libraries were sequenced on six lines of a Flowcell on the HiSeq2000 with 100 cycles of a paired-end sequencing module using the Truseq SBS kit v3 (Catalog ID FC-401-3001).

Quality control, mapping of RNA sequencing reads and counting mapped reads Initial control quality was assessed by the FastQC software version 0.11.3 [52] . Raw reads were then trimmed for low quality bases using the Trimmomatic tool version 0.32 [53] applying minimum Phred quality score >10 averaged across the sliding window of five bases. Furthermore, all reads with the length below 40 bp were removed.

The trimmed reads were mapped to the Gallus gallus reference genome (Gallus_gallus-4.0, release 80 [54]) using a spliced aligner TopHat2 version 2.014 [55] . The Gallus gallus gene annotation used for mapping was retrieved from Ensembl database version 80 (www.ensembl.org). The mapping quality was assessed using a set of Python scripts within the RSeQC toolkit [56] . The quality control assessment included inspection of the read coverage over the full gene body in order to assess if reads coverage was uniform and if there was any 5' or 3' bias as well as how the mapped reads were distributed over genome features.

Gene count estimation was performed using the HTSeq-count tool in 'union' mode. The HTSeq-count is a Python script within the HTSeq framework, version 0.7.1, which is an open source toolkit that allows the input of raw counts from aligned reads to be annotated with gene names based on genomic features [57] .

The read counts obtained were used to estimate gene expression and identify differentially expressed (DE) genes. This was achieved using Bioconductor package edgeR version 3.10.0 [58] and limma version 3.24.5 [59] following a previously described protocol [24] . Before performing statistical analysis, genes with low levels of expression were filtered out using a threshold of least one read per million in n of the samples, where n is the size of the smallest group of replicates, which in this case was eight.

In order to account for technical and biological effects reads counts were normalized using the "calc-NormFactors" function implemented in the edgeR package. This function normalizes the data by finding a set of scaling factors for the library sizes that minimizes the log-fold changes between the samples. The scale factors were computed using the trimmed mean of M-values (TMM) between samples [58] . Common and tag-wise dispersion estimates were calculated with the Cox-Reid profile adjusted likelihood method

in order to correct for the technical and biological variation when fitting the multivariate negative binomial model [60] .

Multidimensional scaling (MDS) was implemented in the edgeR package, to assess similarity of the samples visually. The MDS plot was created in order to visualize the relationship between samples and identify possible outliers [58] . MDS is based on comparing the relationship between all pairs of samples by applying a counts specific pairwise distance measure [58] . Possible outliers were further investigated using principal component analysis (PCA) to remove samples which fell outside a 95 % confidence ellipse.

A design matrix was created in order to specify the factors that were expected to affect the expression level. The matrix was constructed to fit the saturated model where each treatment combination was considered separately. Eight treatment combinations were considered as illustrated in Fig. 1 : uninfected birds from the line L10H at week 1, uninfected birds from the line L10L at week 1, infected birds from the line L10H at week 1, infected birds from the line L10L at week 1, uninfected birds from the line L10H at week 3, uninfected birds from the line L10L at week 3, infected birds from the line L10H at week 3, infected birds from the line L10L at week 3.

A generalized linear model likelihood ratio test, specifying the difference of interest, was used to test for differential expression between these treatment combinations. The differential expression analysis was performed comparing the log-fold differences in gene counts between two lines (L10H and L10L) at different time points (weeks 1 and 3) and for different status (uninfected and infected) separately (Fig. 1). Benjamini Hochberg false discovery rates (FDR) for a transcriptome-wide experiment were calculated to correct for multiple testing [61] . All genes with an FDR-adjusted p-value <0.05 were considered individual genes of interest and were retained for further analysis.

Functional analysis of the DE genes was performed using the Cytoscape version 3.2.1 [62, 63] with the ClueGo version 2.1.7 plug-in [64] to enrich the annotation and enrichment of the differentially expressed (DE) genes for four comparisons (C1-C4, see Fig. 1). ClueGO determines the distribution of the target genes across the GO (Gene Ontology) terms and pathways: this study focused on. The p-value was calculated using right-sided hypergeometric tests and Benjamini-Hochberg adjustment was used for multiple test

correction. An adjusted pvalue of 0.001 indicated a statistically significant deviation from the expected distribution, and that the corresponding GO terms and pathways were enriched for the target genes. The association strength between the terms was calculated using a corrected kappa statistic of 0.4. The network created represented the terms as nodes which were linked based on a 0.4 kappa score level. The size of the nodes reflected the enrichment significance of the terms. The network was automatically laid out using the Organic layout algorithm supported by Cytoscape. The functional groups were created by iterative merging of initially defined groups based on the predefined kappa score threshold. Only functional groups represented by their most significant term were visualized in the network providing an insightful view of their interrelations [64] .

The experimental procedures were conducted under the protocols approved by the Danish Animal Experiments Inspectorate and complied with the Danish Ministry of Justice Law no. 382 (June 10, 1987) and Acts 739 (December 6, 1988) and 333 (May 19, 1990) concerning animal experimentation and care of experimental animals. The license to conduct the animal experiment was obtained by Helle R. Juul-Madsen. comparison between uninfected birds at week 1. The GO Immune System terms were identified as nodes and linked based on their kappa score level (≥ 0.4) and p-value < 0.001 . Functionally related groups partially overlapped. The GO terms are labelled in colors according to hierarchical clustering of GO terms. Terms which have not been grouped are shown in grey. The colour pie charts of the GO Immune system nodes show the gene proportion associated with the respective term. (PDF 60 kb)

Additional file 17: Figure S4 . Network representation of enriched GO Immune System terms of differentially expressed (DE) genes for comparison between uninfected birds at week 3. The GO Immune System terms were identified as nodes and linked based on their kappa score level (≥ 0.4) and p-value < 0.001 . Functionally related groups partially overlapped. The GO terms are labelled in colours according to hierarchical clustering of GO terms. Terms which have not been grouped are shown in grey. The color pie charts of the GO Immune system nodes show the gene proportion associated with the respective term. (PDF 55 kb) Additional file 18: Figure S5 . Network representation of enriched GO Immune System terms of differentially expressed (DE) genes for comparison between infected birds at week 1. The GO Immune System terms were identified as nodes and linked based on their kappa score level (≥ 0.4) and p-value < 0.001 . Functionally related groups partially overlapped. The GO terms are labelled in colors according to hierarchical clustering of GO terms. Terms which have not been grouped are shown in grey. The color

pie charts of the GO Immune system nodes show the gene proportion associated with the respective term. (PDF 70 kb) Additional file 19: Figure S6 . Network representation of enriched GO Immune System terms of differentially expressed (DE) genes for comparison between infected birds at week 3. The GO Immune System terms were identified as nodes and linked based on their kappa score level (≥ 0.4) and p-value < 0.001 . Functionally related groups partially overlapped. The GO terms are labelled in colors according to hierarchical clustering of GO terms. Terms which have not been grouped are shown in grey. The color pie charts of the GO Immune system nodes show the gene proportion associated with the respective term. (PDF 30 kb)

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Accelerated viral dynamics in bat cell lines, with implications for zoonotic emergence

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Abstract: Bats host virulent zoonotic viruses without experiencing disease. A mechanistic understanding of the impact of bats' virus hosting capacities, including uniquely constitutive immune pathways, on cellular-scale viral dynamics is needed to elucidate zoonotic emergence. We carried out virus infectivity assays on bat cell lines expressing

induced and constitutive immune phenotypes, then developed a theoretical model of our in vitro system, which we fit to empirical data. Best fit models recapitulated expected immune phenotypes for representative cell lines, supporting robust antiviral defenses in bat cells that correlated with higher estimates for within-host viral propagation rates. In general, heightened immune responses limit pathogen-induced cellular morbidity, which can facilitate the establishment of rapidly-propagating persistent infections within-host. Rapidly-transmitting viruses that have evolved with bat immune systems will likely cause enhanced virulence following emergence into secondary hosts with immune systems that diverge from those unique to bats.

Text: Bats have received much attention in recent years for their role as reservoir hosts for emerging viral zoonoses, including rabies and related lyssaviruses, Hendra and Nipah henipaviruses, Ebola and Marburg filoviruses, and SARS coronavirus (Calisher et al., 2006; Wang and Anderson, 2019) . In most non-Chiropteran mammals, henipaviruses, filoviruses, and coronaviruses induce substantial morbidity and mortality, display short durations of infection, and elicit robust, long-term immunity in hosts surviving infection (Nicholls et al., 2003; Hooper et al., 2001; Mahanty and Bray, 2004) . Bats, by contrast, demonstrate no obvious disease symptoms upon infection with pathogens that are highly virulent in non-volant mammals (Schountz et al., 2017) but may, instead, support viruses as longterm persistent infections, rather than transient, immunizing pathologies (Plowright et al., 2016) .

Recent research advances are beginning to shed light on the molecular mechanisms by which bats avoid pathology from these otherwise virulent pathogens (Brook and Dobson, 2015) . Bats leverage a suite of species-specific mechanisms to limit viral load, which include host receptor sequence incompatibilities for some bat-virus combinations (Ng et al., 2015; Takadate et al., 2020) and constitutive expression of the antiviral cytokine, IFN- α , for others (Zhou et al., 2016) . Typically, the presence of viral RNA or DNA in the cytoplasm of mammalian cells will induce secretion of type I interferon proteins (IFN- α and IFN- β), which promote expression and translation of interferon-stimulated genes (ISGs) in neighboring cells and render them effectively antiviral (Stetson and Medzhitov, 2006) . In some bat cells, the transcriptomic blueprints for this IFN response are expressed constitutively, even in the absence of stimulation by viral RNA or DNA (Zhou et al., 2016) . In non-flying mammals, constitutive IFN expression would likely elicit widespread inflammation and concomitant immunopathology upon viral infection, but bats support unique adaptations to combat inflammation (Zhang et al., 2013; Ahn et al., 2019; Xie et al.,

2018; Pavlovich et al., 2018) that may have evolved to mitigate metabolic damage induced during flight (Kacprzyk et al., 2017) . The extent to which constitutive IFN- α expression signifies constitutive antiviral defense in the form of functional IFN- α protein remains unresolved. In bat cells constitutively expressing IFN- α , some protein-stimulated, downstream ISGs appear to be also constitutively expressed, but additional ISG induction is nonetheless possible following viral challenge and stimulation of IFN- β (Zhou et al., 2016; Xie et al., 2018) . Despite recent advances in molecular understanding of bat viral tolerance, the consequences of this unique bat immunity on within-host virus dynamics- and its implications for understanding zoonotic emergence-have yet to be elucidated.

The field of 'virus dynamics' was first developed to describe the mechanistic underpinnings of long-term patterns of steady-state viral load exhibited by patients in chronic phase infections with HIV, who appeared to produce and clear virus at equivalent rates (Nowak and May, 2000; Ho et al., 1995) . Models of simple target cell depletion, in which viral load is dictated by a bottom-eLife digest Bats can carry viruses that are deadly to other mammals without themselves showing serious symptoms. In fact, bats are natural reservoirs for viruses that have some of the highest fatality rates of any viruses that people acquire from wild animals -including rabies, Ebola and the SARS coronavirus.

Bats have a suite of antiviral defenses that keep the amount of virus in check. For example, some bats have an antiviral immune response called the interferon pathway perpetually switched on. In most other mammals, having such a hyper-vigilant immune response would cause harmful inflammation. Bats, however, have adapted anti-inflammatory traits that protect them from such harm, include the loss of certain genes that normally promote inflammation. However, no one has previously explored how these unique antiviral defenses of bats impact the viruses themselves. Now, Brook et al. have studied this exact question using bat cells grown in the laboratory. The experiments made use of cells from one bat species -the black flying fox -in which the interferon pathway is always on, and another -the Egyptian fruit bat -in which this pathway is only activated during an infection. The bat cells were infected with three different viruses, and then Brook et al. observed how the interferon pathway helped keep the infections in check, before creating a computer model of this response. The experiments and model helped reveal that the bats' defenses may have a potential downside for other animals, including humans. In both bat species, the strongest antiviral responses were countered by the virus spreading more quickly from cell to cell. This suggests that bat immune defenses may drive the evolution of faster

transmitting viruses, and while bats are well protected from the harmful effects of their own prolific viruses, other creatures like humans are not.

The findings may help to explain why bats are often the source for viruses that are deadly in humans. Learning more about bats' antiviral defenses and how they drive virus evolution may help scientists develop better ways to predict, prevent or limit the spread of viruses from bats to humans. More studies are needed in bats to help these efforts. In the meantime, the experiments highlight the importance of warning people to avoid direct contact with wild bats. up resource supply of infection-susceptible host cells, were first developed for HIV (Perelson, 2002) but have since been applied to other chronic infections, including hepatitis-C virus (Neumann et al., 1998) , hepatitis-B virus (Nowak et al., 1996) and cytomegalovirus (Emery et al., 1999) . Recent work has adopted similar techniques to model the within-host dynamics of acute infections, such as influenza A and measles, inspiring debate over the extent to which explicit modeling of top-down immune control can improve inference beyond the basic resource limitation assumptions of the target cell model (Baccam et al., 2006; Pawelek et al., 2012; Saenz et al., 2010; Morris et al., 2018) .

To investigate the impact of unique bat immune processes on in vitro viral kinetics, we first undertook a series of virus infection experiments on bat cell lines expressing divergent interferon phenotypes, then developed a theoretical model elucidating the dynamics of within-host viral spread. We evaluated our theoretical model analytically independent of the data, then fit the model to data recovered from in vitro experimental trials in order to estimate rates of within-host virus transmission and cellular progression to antiviral status under diverse assumptions of absent, induced, and constitutive immunity. Finally, we confirmed our findings in spatially-explicit stochastic simulations of fitted time series from our mean field model. We hypothesized that top-down immune processes would overrule classical resource-limitation in bat cell lines described as constitutively antiviral in the literature, offering a testable prediction for models fit to empirical data. We further predicted that the most robust antiviral responses would be associated with the most rapid within-host virus propagation rates but also protect cells against virus-induced mortality to support the longest enduring infections in tissue culture.

We first explored the influence of innate immune phenotype on within-host viral propagation in a series of infection experiments in cell culture. We conducted plaque assays on six-well plate monolayers of three immortalized mammalian kidney cell lines: [1]

Vero (African green monkey) cells, which are IFN-defective and thus limited in antiviral capacity (Desmyter et al., 1968) ; [2] RoNi/7.1 (*Rousettus aegyptiacus*) cells which demonstrate idiosyncratic induced interferon responses upon viral challenge (Kuzmin et al., 2017; Arnold et al., 2018; Biesold et al., 2011; Pavlovich et al., 2018) ; and [3] PaKiT01 (*Pteropus alecto*) cells which constitutively express IFN- α (Zhou et al., 2016; Crameri et al., 2009) . To intensify cell line-specific differences in constitutive immunity, we carried out infectivity assays with GFP-tagged, replication-competent vesicular stomatitis Indiana viruses: rVSV-G, rVSV-EBOV, and rVSV-MARV, which have been previously described (Miller et al., 2012; Wong et al., 2010) . Two of these viruses, rVSV-EBOV and rVSV-MARV, are recombinants for which cell entry is mediated by the glycoprotein of the bat-evolved filoviruses, Ebola (EBOV) and Marburg (MARV), thus allowing us to modulate the extent of structural, as well as immunological, antiviral defense at play in each infection. Previous work in this lab has demonstrated incompatibilities in the NPC1 filovirus receptor which render PaKiT01 cells refractory to infection with rVSV-MARV (Ng and Chandrab, 2018, Unpublished results) , making them structurally antiviral, over and above their constitutive expression of IFN- α . All three cell lines were challenged with all three viruses at two multiplicities of infection (MOI): 0.001 and 0.0001. Between 18 and 39 trials were run at each cell-virus-MOI combination, excepting rVSV-MARV infections on PaKiT01 cells at MOI = 0.001, for which only eight trials were run (see Materials and methods; Figure 1 -figure supplements 1-3, Supplementary file 1).

Because plaque assays restrict viral transmission neighbor-to-neighbor in two-dimensional cellular space (Howat et al., 2006) , we were able to track the spread of GFP-expressing virus-infected cells across tissue monolayers via inverted fluorescence microscopy. For each infection trial, we monitored and re-imaged plates for up to 200 hr of observations or until total monolayer destruction, processed resulting images, and generated a time series of the proportion of infectious-cell occupied plate space across the duration of each trial (see Materials and methods). We used generalized additive models to infer the time course of all cell culture replicates and construct the multi-trial dataset to which we eventually fit our mechanistic transmission model for each cell line-virus-specific combination (Figure 1; Figure 1 -figure supplements 1-5).

All three recombinant vesicular stomatitis viruses (rVSV-G, rVSV-EBOV, and rVSV-MARV) infected Vero, RoNi/7.1, and PaKiT01 tissue cultures at both focal MOIs. Post-invasion, virus spread rapidly across most cell monolayers, resulting in virus-induced epidemic extinction. Epidemics were less severe in bat cell cultures, especially when infected with

the recombinant filoviruses, rVSV-EBOV and rVSV-MARV. Monolayer destruction was avoided in the case of rVSV-EBOV and rVSV-MARV infections on PaKiT01 cells: in the former, persistent viral infection was maintained throughout the 200 hr duration of each experiment, while, in the latter, infection was eliminated early in the time series, preserving a large proportion of live, uninfected cells across the duration of the experiment. We assumed this pattern to be the result of immune-mediated epidemic extinction (Figure 1) . Patterns from $\text{MOI} = 0.001$ were largely recapitulated at $\text{MOI} = 0.0001$, though at somewhat reduced total proportions (Figure 1-figure supplement 5) .

A theoretical model fit to in vitro data recapitulates expected immune phenotypes for bat cells We next developed a within-host model to fit to these data to elucidate the effects of induced and constitutive immunity on the dynamics of viral spread in host tissue (Figure 1). The compartmental within-host system mimicked our two-dimensional cell culture monolayer, with cells occupying five distinct infection states: susceptible (S), antiviral (A), exposed (E), infectious (I), and dead (D). We modeled exposed cells as infected but not yet infectious, capturing the 'eclipse phase' of viral integration into a host cell which precedes viral replication. Antiviral cells were immune to viral infection, in accordance with the 'antiviral state' induced from interferon stimulation of ISGs in tissues adjacent to infection (Stetson and Medzhitov, 2006) . Because we aimed to translate available data into modeled processes, we did not explicitly model interferon dynamics but instead scaled the rate of cell progression from susceptible to antiviral (r) by the proportion of exposed cells (globally) in the system. In systems permitting constitutive immunity, a second rate of cellular acquisition of antiviral status (r') additionally scaled with the global proportion of susceptible cells in the model. Compared with virus, IFN particles are small and highly diffusive, justifying this global signaling assumption at the limited spatial extent of a six-well plate and maintaining consistency with previous modeling approximations of IFN signaling in plaque assay (Howat et al., 2006) .

To best represent our empirical monolayer system, we expressed our state variables as proportions (P_S , P_A , P_E , P_I , and P_D), under assumptions of frequency-dependent transmission in a wellmixed population (Keeling and Rohani, 2008) , though note that the inclusion of P_D (representing the proportion of dead space in the modeled tissue) had the functional effect of varying transmission with infectious cell density. This resulted in the following system of ordinary differential equations:

We defined 'induced immunity' as complete, modeling all cells as susceptible to viral invasion at disease-free equilibrium, with defenses induced subsequent to viral exposure through the term r . By contrast, we allowed the extent of constitutive immunity to vary across the parameter range of $\sigma > 0$, defining a 'constitutive' system as one containing any antiviral cells at disease-free equilibrium. In fitting this model to tissue culture data, we independently estimated both r and σ ; as well as the cell-to-cell transmission rate, b , for each cell-virus combination. Since the extent to which constitutively-expressed IFN- α is constitutively translated into functional protein is not yet known for bat hosts (Zhou et al., 2016), this approach permitted our tissue culture data to drive modeling inference: even in PaKiT01 cell lines known to constitutively express IFN- α , the true constitutive extent of the system (i.e. the quantity of antiviral cells present at disease-free equilibrium) was allowed to vary through estimation of σ : For the purposes of model-fitting, we fixed the value of c , the return rate of antiviral cells to susceptible status, at 0. The small spatial scale and short time course (max 200 hours) of our experiments likely prohibited any return of antiviral cells to susceptible status in our empirical system; nonetheless, we retained the term c in analytical evaluations of our model because regression from antiviral to susceptible status is possible over long time periods in vitro and at the scale of a complete organism (Radke et al., 1974; Rasmussen and Farley, 1975; Samuel and Knutson, 1982).

Before fitting to empirical time series, we undertook bifurcation analysis of our theoretical model and generated testable hypotheses on the basis of model outcomes. From our within-host model system (Equation 1-5), we derived the following expression for R_0 , the pathogen basic reproduction number (Supplementary file 2):

Pathogens can invade a host tissue culture when $R_0 > 1$. Rapid rates of constitutive antiviral acquisition (σ) will drive $R_0 < 1$: tissue cultures with highly constitutive antiviral immunity will be therefore resistant to virus invasion from the outset. Since, by definition, induced immunity is stimulated following initial virus invasion, the rate of induced antiviral acquisition (r) is not incorporated into the equation for R_0 ; while induced immune processes can control virus after initial invasion, they cannot prevent it from occurring to begin with. In cases of fully induced or absent immunity ($\sigma \neq 0$), the R_0 equation thus reduces to a form typical of the classic SEIR model:

At equilibrium, the theoretical, mean field model demonstrates one of three infection states: endemic equilibrium, stable limit cycles, or no infection (Figure 2). Respectively,

these states approximate the persistent infection, virus-induced epidemic extinction, and immune-mediated epidemic extinction phenotypes previously witnessed in tissue culture experiments (Figure 1). Theoretically, endemic equilibrium is maintained when new infections are generated at the same rate at which infections are lost, while limit cycles represent parameter space under which infectious and susceptible populations are locked in predictable oscillations. Endemic equilibria resulting from cellular regeneration (i.e. births) have been described in vivo for HIV (Coffin, 1995) and in vitro for herpesvirus plaque assays (Howat et al., 2006) , but, because they so closely approach zero, true limit cycles likely only occur theoretically, instead yielding stochastic extinctions in empirical time series.

Bifurcation analysis of our mean field model revealed that regions of no infection (pathogen extinction) were bounded at lower threshold (Branch point) values for b , below which the pathogen was unable to invade. We found no upper threshold to invasion for b under any circumstances (i.e. b high enough to drive pathogen-induced extinction), but high b values resulted in Hopf bifurcations, which delineate regions of parameter space characterized by limit cycles. Since limit cycles so closely approach zero, high b s recovered in this range would likely produce virus-induced epidemic extinctions under experimental conditions. Under more robust representations of immunity, with higher values for either or both induced (r) and constitutive (ρ) rates of antiviral acquisition, Hopf bifurcations occurred at increasingly higher values for b , meaning that persistent infections could establish at higher viral transmission rates (Figure 2). Consistent with our derivation for R_0 , we found that the Branch point threshold for viral invasion was independent of changes to the induced immune parameter (r) but saturated at high values of ρ that characterize highly constitutive immunity (Figure 3) .

We next fit our theoretical model by least squares to each cell line-virus combination, under absent, induced, and constitutive assumptions of immunity. In general, best fit models recapitulated expected outcomes based on the immune phenotype of the cell line in question, as described in the general literature (Table 1). Ironically, the induced immune model offered a slightly better fit than the constitutive to rVSV-MARV infections on the PaKiT01 cell line (the one cell line-virus combination for which we know a constitutively antiviral cell-receptor incompatibility to be at play). Because constitutive immune assumptions can prohibit pathogen invasion ($R_0 < 1$), model fits to this time series under constitutive assumptions were handicapped by overestimations of ρ , which prohibited pathogen invasion. Only by incorporating an exceedingly rapid rate of induced antiviral

acquisition could the model guarantee that initial infection would be permitted and then rapidly controlled. In all panel (A) plots, the rate of induced immune antiviral acquisition (r) was fixed at 0.01. Panel (B) depicts dynamics under variably induced immunity, ranging from absent (left: $r=0$) to high (right: $r=1$). In all panel (B) plots, the rate of constitutive antiviral acquisition (ρ) was fixed at 0.0001. Branch point curves are represented as solid lines and Hopf curves as dashed lines. White space indicates endemic equilibrium (persistence), gray space indicates limit cycles, and black space indicates no infection (extinction). Other parameter values for equilibrium analysis were fixed at: $b = .025$, $m = .001$, $s = 1/6$, $c = 0$. Special points from bifurcations analyses are listed in Supplementary file 3.

In fitting our theoretical model to in vitro data, we estimated the within-host virus transmission rate (b) and the rate(s) of cellular acquisition to antiviral status (r or $r + \rho$) (Table 1 ; Supplementary file 4). Under absent immune assumptions, r and ρ were fixed at 0 while b was estimated; under induced immune assumptions, ρ was fixed at 0 while r and b were estimated; and under constitutive immune assumptions, all three parameters (r , ρ , and b) were simultaneously estimated for each cell-virus combination. Best fit parameter estimates for MOI=0.001 data are visualized in conjunction with b and $b - \rho$ bifurcations in (r) and (B) the constitutive immunity rate of antiviral acquisition (ρ). Panels show variation in the extent of immunity, from absent (left) to high (right). Branch point curves are represented as solid lines and Hopf curves as dashed lines. White space indicates endemic equilibrium (persistence), gray space indicates limit cycling, and black space indicates no infection (extinction). Other parameter values for equilibrium analysis were fixed at: $b = .025$, $m = .001$, $s = 1/6$, $a = 1/6$, $c = 0$. Special points from bifurcations analyses are listed in Supplementary file 3. space corresponding to theoretical limit cycles, consistent with observed virus-induced epidemic extinctions in stochastic tissue cultures.

In contrast to Vero cells, the induced immunity model offered the best fit to all RoNi/7.1 data, consistent with reported patterns in the literature and our own validation by qPCR (Table 1; Arnold et al., 2018; Kuzmin et al., 2017; Biesold et al., 2011; Pavlovich et al., 2018) . As in Vero cell trials, we estimated highest b values for rVSV-G infections on RoNi/7.1 cell lines but here recovered higher b estimates for rVSV-MARV than for rVSV-EBOV. This reversal was balanced by a higher estimated rate of acquisition to antiviral status (r) for rVSV-EBOV versus rVSV-MARV. In general, we observed that more rapid rates of antiviral acquisition (either induced, r , constitutive, ρ , or both) correlated with higher transmission rates (b). When offset by r , b values estimated for RoNi/7.1 infections maintained the same

amplitude as those estimated for immune-absent Vero cell lines but caused gentler epidemics and reduced cellular mortality (Figure 1) . RoNi/7.1 parameter estimates localized in the region corresponding to endemic equilibrium for the deterministic, theoretical model (Figure 4) , yielding less acute epidemics which nonetheless went extinct in stochastic experiments.

Finally, rVSV-G and rVSV-EBOV trials on PaKiT01 cells were best fit by models assuming constitutive immunity, while rVSV-MARV infections on PaKiT01 were matched equivalently by models assuming either induced or constitutive immunity-with induced models favored over constitutive in AIC comparisons because one fewer parameter was estimated (Figure 1-figure supplements 4-5; Supplementary file 4). For all virus infections, PaKiT01 cell lines yielded b estimates a full order of magnitude higher than Vero or RoNi/7.1 cells, with each b balanced by an immune response (either r , or r combined with γ) also an order of magnitude higher than that recovered for the other cell lines (Figure 4 ; Table 1). As in RoNi/7.1 cells, PaKiT01 parameter fits localized in the region corresponding to endemic equilibrium for the deterministic theoretical model. Because constitutive immune processes can actually prohibit initial pathogen invasion, constitutive immune fits to rVSV-MARV infections on PaKiT01 cell lines consistently localized at or below the Branch point threshold for virus invasion ($R_0 \leq 1$). During model fitting for optimization of γ , any parameter tests of γ values producing $R_0 < 1$ resulted in no infection and, consequently, produced an exceedingly poor fit to infectious time series data. In all model fits assuming constitutive immunity, across all cell lines, antiviral contributions from γ prohibited virus from invading at all. The induced immune model thus produced a more parsimonious recapitulation of these data because virus invasion was always permitted, then rapidly controlled.

In order to compare the relative contributions of each cell line's disparate immune processes to epidemic dynamics, we next used our mean field parameter estimates to calculate the initial 'antiviral rate'-the initial accumulation rate of antiviral cells upon virus invasion for each cell-virus-MOI combination-based on the following equation:

where P_E was calculated from the initial infectious dose (MOI) of each infection experiment and P_S was estimated at disease-free equilibrium:

Because β and γ both contribute to this initial antiviral rate, induced and constitutive immune assumptions are capable of yielding equally rapid rates, depending on parameter fits. Indeed, under fully induced immune assumptions, the induced antiviral acquisition rate (r) estimated for rVSV-MARV infection on PaKiT01 cells was so high that the initial antiviral rate exceeded even that estimated under constitutive assumptions for this cell-virus combination (Supplementary file 4). In reality, we know that NPC1 receptor incompatibilities make PaKiT01 cell lines constitutively refractory to rVSV-MARV infection (Ng and Chandrab, 2018, Unpublished results) and that PaKiT01 cells also constitutively express the antiviral cytokine, IFN- α . Model fitting results suggest that this constitutive expression of IFN- α may act more as a rapidly inducible immune response following virus invasion than as a constitutive secretion of functional IFN- α protein. Nonetheless, as hypothesized, PaKiT01 cell lines were by far the most antiviral of any in our study—with initial antiviral rates estimated several orders of magnitude higher than any others in our study, under either induced or constitutive assumptions (Table 1 ; Supplementary file 4). RoNi/7.1 cells displayed the second-most-pronounced signature of immunity, followed by Vero cells, for which the initial antiviral rate was essentially zero even under forced assumptions of induced or constitutive immunity (Table 1 ; Supplementary file 4).

Using fitted parameters for β and γ , we additionally calculated R_0 , the basic reproduction number for the virus, for each cell line-virus-MOI combination (Table 1 ; Supplementary file 4). We found that R_0 was essentially unchanged across differing immune assumptions for RoNi/7.1 and Vero cells, for which the initial antiviral rate was low. In the case of PaKiT01 cells, a high initial antiviral rate under either induced or constitutive immunity resulted in a correspondingly high estimation of β (and, consequently, R_0) which still produced the same epidemic curve that resulted from the much lower estimates for β and R_0 paired with absent immunity. These findings suggest that antiviral immune responses protect host tissues against virus-induced cell mortality and may facilitate the establishment of more rapid within-host transmission rates.

Total monolayer destruction occurred in all cell-virus combinations excepting rVSV-EBOV infections on RoNi/7.1 cells and rVSV-EBOV and rVSV-MARV infections on PaKiT01 cells. Monolayer destruction corresponded to susceptible cell depletion and epidemic turnover where $R_{\text{effective}}$ (the product of R_0 and the proportion susceptible) was reduced below one (Figure 5). For rVSV-EBOV infections on RoNi/7.1, induced antiviral cells safeguarded remnant live cells, which birthed new susceptible cells late in the time series. In rVSV-EBOV and rVSV-MARV infections on PaKiT01 cells, this antiviral protection halted the

epidemic (Figure 5 ; $R_{\text{effective}} < 1$) before susceptibles fully declined. In the case of rVSV-EBOV on PaKiT01, the birth of new susceptibles from remnant live cells protected by antiviral status maintained late-stage transmission to facilitate long-term epidemic persistence. Importantly, under fixed parameter values for the infection incubation rate (s) and infection-induced mortality rate (a), models were unable to reproduce the longer-term infectious time series captured in data from rVSV-EBOV infections on PaKiT01 cell lines without incorporation of cell births, an assumption adopted in previous modeling representations of IFN-mediated viral dynamics in tissue culture (Howat et al., 2006) . In our experiments, we observed that cellular reproduction took place as plaque assays achieved confluency. Finally, because the protective effect of antiviral cells is more clearly observable spatially, we confirmed our results by simulating fitted time series in a spatially-explicit, stochastic reconstruction of our mean field model. In spatial simulations, rates of antiviral acquisition were fixed at fitted values for r and β derived from mean field estimates, while transmission rates (b) were fixed at values ten times greater than those estimated under mean field conditions, accounting for the intensification of parameter thresholds permitting pathogen invasion in local spatial interactions (see Materials and methods; Videos 1-3; Figure 5-figure supplement 3; Supplementary file 5; Webb et al., 2007) . In immune capable time series, spatial antiviral cells acted as 'refugia' which protected live cells from infection as each initial epidemic wave 'washed' across a cell monolayer. Eventual birth of new susceptibles from these living refugia allowed for sustained epidemic transmission in cases where some infectious cells persisted at later timepoints in simulation (Videos 1-3; Figure 5-figure supplement 3).

Bats are reservoirs for several important emerging zoonoses but appear not to experience disease from otherwise virulent viral pathogens. Though the molecular biological literature has made great progress in elucidating the mechanisms by which bats tolerate viral infections (Zhou et al., 2016; Ahn et al., 2019; Xie et al., 2018; Pavlovich et al., 2018; Zhang et al., 2013) , the impact of unique bat immunity on virus dynamics within-host has not been well-elucidated. We used an innovative combination of in vitro experimentation and within-host modeling to explore the impact of unique bat immunity on virus dynamics. Critically, we found that bat cell lines demonstrated a signature of enhanced interferon-mediated immune response, of either constitutive or induced form, which allowed for establishment of rapid within-host, cell-to-cell virus transmission rates (b). These results were supported by both data-independent bifurcation analysis of our mean field theoretical model, as well as fitting of this model to viral infection time series established in bat cell culture. Additionally, we demonstrated that the antiviral state induced by the interferon pathway protects live cells from mortality in tissue culture, resulting in in vitro

epidemics of extended duration that enhance the probability of establishing a long-term persistent infection. Our findings suggest that viruses evolved in bat reservoirs possessing enhanced IFN capabilities could achieve more rapid within-host transmission rates without causing pathology to their hosts. Such rapidly-reproducing viruses would likely generate extreme virulence upon spillover to hosts lacking similar immune capacities to bats.

To achieve these results, we first developed a novel, within-host, theoretical model elucidating the effects of unique bat immunity, then undertook bifurcation analysis of the model's equilibrium properties under immune absent, induced, and constitutive assumptions. We considered a cell line to be constitutively immune if possessing any number of antiviral cells at disease-free equilibrium but allowed the extent of constitutive immunity to vary across the parameter range for β , the constitutive rate of antiviral acquisition. In deriving the equation for R_0 , the basic reproduction number, which defines threshold conditions for virus invasion of a tissue ($R_0 > 1$), we demonstrated how the invasion threshold is elevated at high values of constitutive antiviral acquisition, β . Constitutive immune processes can thus prohibit pathogen invasion, while induced responses, by definition, can only control infections post-hoc. Once thresholds for pathogen invasion have been met, assumptions of constitutive immunity will limit the cellular mortality (virulence) incurred at high transmission rates. Regardless of mechanism (induced or constitutive), interferon-stimulated antiviral cells appear to play a key role in maintaining longer term or persistent infections by safeguarding susceptible cells from rapid infection and concomitant cell death. Fitting of our model to in vitro data supported expected immune phenotypes for different bat cell lines as described in the literature. Simple target cell models that ignore the effects of immunity best recapitulated infectious time series derived from IFN-deficient Vero cells, while models assuming induced immune processes most accurately reproduced trials derived from RoNi/7.1 (*Rousettus aegyptiacus*) cells, which possess a standard virus-induced IFN-response. In most cases, models assuming constitutive immune processes best recreated virus epidemics produced on PaKiT01 (*Pteropus alecto*) cells, which are known to constitutively express the antiviral cytokine, IFN- α (Zhou et al., 2016). Model support for induced immune assumptions in fits to rVSV-MARV infections on PaKiT01 cells suggests that the constitutive IFN- α expression characteristic of *P. alecto* cells may represent more of a constitutive immune priming process than a perpetual, functional, antiviral defense. Results from mean field model fitting were additionally confirmed in spatially explicit stochastic simulations of each time series.

As previously demonstrated in within-host models for HIV (Coffin, 1995; Perelson et al., 1996; Nowak et al., 1995; Bonhoeffer et al., 1997; Ho et al., 1995), assumptions of simple target-cell depletion can often provide satisfactory approximations of viral dynamics, especially those reproduced in simple in vitro systems. Critically, our model fitting emphasizes the need for incorporation of top-down effects of immune control in order to accurately reproduce infectious time series derived from bat cell tissue cultures, especially those resulting from the robustly antiviral PaKiT01 *P. alecto* cell line. These findings indicate that enhanced IFN-mediated immune pathways in bat reservoirs may promote elevated within-host virus replication rates prior to cross-species emergence. We nonetheless acknowledge the limitations imposed by in vitro experiments in tissue culture, especially involving recombinant viruses and immortalized cell lines. Future work should extend these cell culture studies to include measurements of multiple state variables (i.e. antiviral cells) to enhance epidemiological inference.

The continued recurrence of Ebola epidemics across central Africa highlights the importance of understanding bats' roles as reservoirs for virulent zoonotic disease. The past decade has borne witness to emerging consensus regarding the unique pathways by which bats resist and tolerate highly virulent infections (Brook and Dobson, 2015; Xie et al., 2018; Zhang et al., 2013; Ahn et al., 2019; Zhou et al., 2016; Ng et al., 2015; Pavlovich et al., 2018). Nonetheless, an understanding of the mechanisms by which bats support endemic pathogens at the population level, or promote the evolution of virulent pathogens at the individual level, remains elusive. Endemic maintenance of infection is a defining characteristic of a pathogen reservoir (Haydon et al., 2002), and bats appear to merit such a title, supporting long-term persistence of highly transmissible viral infections in isolated island populations well below expected critical community sizes (Peel et al., 2012). Researchers debate the relative influence of population-level and within-host mechanisms which might explain these trends (Plowright et al., 2016), but increasingly, field data are difficult to reconcile without acknowledgement of a role for persistent infections (Peel et al., 2018; Brook et al., 2019). We present general methods to study cross-scale viral dynamics, which suggest that within-host persistence is supported by robust antiviral responses characteristic of bat immune processes. Viruses which evolve rapid replication rates under these robust antiviral defenses may pose the greatest hazard for cross-species pathogen emergence into spillover hosts with immune systems that differ from those unique to bats.

All experiments were carried out on three immortalized mammalian kidney cell lines: Vero (African green monkey), RoNi/7.1 (*Rousettus aegyptiacus*) (Kühl et al., 2011; Biesold et al., 2011) and PaKiT01 (*Pteropus alecto*) (Crameri et al., 2009). The species identifications of all bat cell lines was confirmed morphologically and genetically in the publications in which they were originally described (Kühl et al., 2011; Biesold et al., 2011; Crameri et al., 2009). Vero cells were obtained from ATCC.

Monolayers of each cell line were grown to 90% confluency ($\sim 9 \times 10^5$ cells) in 6-well plates. Cells were maintained in a humidified 37 °C, 5% CO₂ incubator and cultured in Dulbecco's modified Eagle medium (DMEM) (Life Technologies, Grand Island, NY), supplemented with 2% fetal bovine serum (FBS) (Gemini Bio Products, West Sacramento, CA), and 1% penicillin-streptomycin (Life Technologies). Cells were tested monthly for mycoplasma contamination while experiments were taking place; all cells assayed negative for contamination at every testing.

Previous work has demonstrated that all cell lines used are capable of mounting a type I IFN response upon viral challenge, with the exception of Vero cells, which possess an IFN- β deficiency (Desmyter et al., 1968; Rhim et al., 1969; Emeny and Morgan, 1979). RoNi/7.1 cells have been shown to mount idiosyncratic induced IFN defenses upon viral infection (Pavlovich et al., 2018; Kuzmin et al., 2017; Arnold et al., 2018; Kühl et al., 2011; Biesold et al., 2011), while PaKiT01 cells are known to constitutively express the antiviral cytokine, IFN- α (Zhou et al., 2016). This work is the first documentation of IFN signaling induced upon challenge with the particular recombinant VSVs outlined below. We verified known antiviral immune phenotypes via qPCR. Results were consistent with the literature, indicating a less pronounced role for interferon defense against viral infection in RoNi/7.1 versus PaKiT01 cells.

Replication-capable recombinant vesicular stomatitis Indiana viruses, expressing filovirus glycoproteins in place of wild type G (rVSV-G, rVSV-EBOV, and rVSV-MARV) have been previously described (Wong et al., 2010; Miller et al., 2012). Viruses were selected to represent a broad range of anticipated antiviral responses from host cells, based on a range of past evolutionary histories between the virus glycoprotein mediating cell entry and the host cell's entry receptor. These interactions ranged from the total absence of evolutionary history in the case of rVSV-G infections on all cell lines to a known receptor-level cell entry incompatibility in the case of rVSV-MARV infections on PaKiT01 cell lines.

To measure infectivities of rVSVs on each of the cell lines outlined above, so as to calculate the correct viral dose for each MOI, NH₄Cl (20 mM) was added to infected cell cultures at 1-2 hr postinfection to block viral spread, and individual eGFP-positive cells were manually counted at 12-14 hr post-infection.

Previously published work indicates that immortalized kidney cell lines of *Rousettus aegyptiacus* (RoNi/7.1) and *Pteropus alecto* (PaKiT01) exhibit different innate antiviral immune phenotypes through, respectively, induced (Biesold et al., 2011; Pavlovich et al., 2018; Köhl et al., 2011; Arnold et al., 2018) and constitutive (Zhou et al., 2016) expression of type I interferon genes. We verified these published phenotypes on our own cell lines infected with rVSV-G, rVSV-EBOV, and rVSV-MARV via qPCR of IFN- α and IFN- β genes across a longitudinal time series of infection.

Specifically, we carried out multiple time series of infection of each cell line with each of the viruses described above, under mock infection conditions and at MOIs of 0.0001 and 0.001-with the exception of rVSV-MARV on PaKiT01 cell lines, for which infection was only performed at MOI = 0.0001 due to limited viral stocks and the extremely low infectivity of this virus on this cell line (thus requiring high viral loads for initial infection). All experiments were run in duplicate on 6well plates, such that a typical plate for any of the three viruses had two control (mock) wells, two MOI = 0.0001 wells and two MOI = 0.001 wells, excepting PaKiT01 plates, which had two control and four MOI = 0.0001 wells at a given time. We justify this PaKiT01 exemption through the expectation that IFN- α expression is constitutive for these cells, and by the assumption that any expression exhibited at the lower MOI should also be present at the higher MOI.

For these gene expression time series, four 6-well plates for each cell line-virus combination were incubated with virus for one hour at 37 °C. Following incubation, virus was aspirated off, and cell monolayers were washed in PBS, then covered with an agar plaque assay overlay to mimic conditions under which infection trials were run. Plates were then harvested sequentially at timepoints of roughly 5, 10, 15, and 20 hr post-infection (exact timing varied as multiple trials were running simultaneously). Upon harvest of each plate, agar overlay was removed, and virus was lysed and RNA extracted from cells using the Zymo Quick RNA Mini Prep kit, according to the manufacturer's instructions and including the step for cellular DNA digestion. Post-extraction, RNA quality was verified via

nanodrop, and RNA was converted to cDNA using the Invitrogen Superscript III cDNA synthesis kit, according to the manufacturer's instructions. cDNA was then stored at 4 °C and as a frozen stock at -20 °C to await qPCR.

We undertook qPCR of cDNA to assess expression of the type I interferon genes, IFN- α and IFN β , and the housekeeping gene, b-Actin, using primers previously reported in the literature (Supplementary file 6) . For qPCR, 2 ml of each cDNA sample was incubated with 7 ml of deionized water, 1 ml of 5 U forward/reverse primer mix and 10 ml of iTaq Universal SYBR Green, then cycled on a QuantStudio3 Real-Time PCR machine under the following conditions: initial denaturation at 94 °C for 2 min followed by 40 cycles of: denaturation at 95 °C (5 s), annealing at 58 °C (15 s), and extension at 72 °C (10 s).

We report simple Δ -Ct values for each run, with raw Ct of the target gene of interest (IFN- α or IFN- β) subtracted from raw Ct of the b-Actin housekeeping gene in Figure 1 -figure supplement 6. Calculation of fold change upon viral infection in comparison to mock using the Δ - Δ -Ct method (Livak and Schmittgen, 2001) was inappropriate in this case, as we wished to demonstrate constitutive expression of IFN- α in PaKiT01 cells, whereby data from mock cells was identical to that produced from infected cells.

After being grown to ~90% confluency, cells were incubated with pelleted rVSVs expressing eGFP (rVSV-G, rVSV-EBOV, rVSV-MARV). Cell lines were challenged with both a low (0.0001) and high (0.001) multiplicity of infection (MOI) for each virus. In a cell monolayer infected at a given MOI (m), the proportion of cells (P), infected by k viral particles can be described by the Poisson distribution: $P(k) = \frac{e^{-m} m^k}{k!}$, such that the number of initially infected cells in an experiment equals: $1 - e^{-m}$. We assumed that a ~90% confluent culture at each trial's origin was comprised of $\sim 9 \times 10^5$ cells and conducted all experiments at MOIs of 0.0001 and 0.001, meaning that we began each trial by introducing virus to, respectively, ~81 or 810 cells, representing the state variable 'E' in our theoretical model. Low MOIs were selected to best approximate the dynamics of mean field infection and limit artifacts of spatial structuring, such as premature epidemic extinction when growing plaques collide with plate walls in cell culture.

Six-well plates were prepared with each infection in duplicate or triplicate, such that a control well (no virus) and 2-3 wells each at MOI 0.001 and 0.0001 were incubated

simultaneously on the same plate. In total, we ran between 18 and 39 trials at each cell-virus-MOI combination, excepting r-VSV-MARV infections on PaKiT01 cells at MOI = 0.001, for which we ran only eight trials due to the low infectivity of this virus on this cell line, which required high viral loads for initial infection. Cells were incubated with virus for one hour at 37 ° C. Following incubation, virus was aspirated off, and cell monolayers were washed in PBS, then covered with a molten viscous overlay (50% 2X MEM/Lglutamine; 5% FBS; 3% HEPES; 42% agarose), cooled for 20 min, and re-incubated in their original humidified 37 ° C, 5% CO₂ environment.

After application of the overlay, plates were monitored periodically using an inverted fluorescence microscope until the first signs of GFP expression were witnessed (~6-9.5 hr post-infection, depending on the cell line and virus under investigation). From that time forward, a square subset of the center of each well (comprised of either 64-or 36-subframes and corresponding to roughly 60% and 40% of the entire well space) was imaged periodically, using a CellInsight CX5 High Content Screening (HCS) Platform with a 4X air objective (ThermoFisher, Inc, Waltham, MA). Microscope settings were held standard across all trials, with exposure time fixed at 0.0006 s for each image. One color channel was imaged, such that images produced show GFP-expressing cells in white and non-GFP-expressing cells in black (Figure 1-figure supplement 1) .

Wells were photographed in rotation, as frequently as possible, from the onset of GFP expression until the time that the majority of cells in the well were surmised to be dead, GFP expression could no longer be detected, or early termination was desired to permit Hoechst staining.

In the case of PaKiT01 cells infected with rVSV-EBOV, where an apparently persistent infection established, the assay was terminated after 200+ hours (8+ days) of continuous observation. Upon termination of all trials, cells were fixed in formaldehyde (4% for 15 min), incubated with Hoechst stain (0.0005% for 15 min) (ThermoFisher, Inc, Waltham, MA), then imaged at 4X on the CellInsight CX5 High Content Screening (HCS) Platform. The machine was allowed to find optimal focus for each Hoechst stain image. One color channel was permitted such that images produced showed live nuclei in white and dead cells in black.

Hoechst stain colors cellular DNA, and viral infection is thought to interfere with the clarity of the stain (Dembowski and DeLuca, 2015) . As such, infection termination, cell fixation, and Hoechst staining enables generation of a rough time series of uninfected live cells (i.e. susceptible + antiviral cells) to complement the images which produced time series of proportions infectious. Due to uncertainty over the exact epidemic state of Hoechst-stained cells (i.e. exposed but not yet infectious cells may still stain), we elected to fit our models only to the infectious time series derived from GFP-expressing images and used Hoechst stain images as a post hoc visual check on our fit only (Figure 5 ; Figure 5 -figure supplements 1-2).

Images recovered from the time series above were processed into binary ('infectious' vs. 'non-infectious' or, for Hoechst-stained images, 'live' vs. 'dead') form using the EBImage package (Pau et al., 2010) in R version 3.6 for Macintosh, after methods further detailed in Supplementary file 7. Binary images were then further processed into time series of infectious or, for Hoechst-stained images, live cells using a series of cell counting scripts. Because of logistical constraints (i.e. many plates of simultaneously running infection trials and only one available imaging microscope), the time course of imaging across the duration of each trial was quite variable. As such, we fitted a series of statistical models to our processed image data to reconstruct reliable values of the infectious proportion of each well per hour for each distinct trial in all cell line-virus-MOI combinations (Figure 1

To derive the expression for R_0 , the basic pathogen reproductive number in vitro, we used Next Generation Matrix (NGM) techniques (Diekmann et al., 1990; Heffernan et al., 2005) , employing Wolfram Mathematica (version 11.2) as an analytical tool. R_0 describes the number of new infections generated by an existing infection in a completely susceptible host population; a pathogen will invade a population when $R_0 > 1$ (Supplementary file 2). We then analyzed stability properties of the system, exploring dynamics across a range of parameter spaces, using MatCont (version 2.2) (Dhooge et al., 2008) for Matlab (version R2018a) (Supplementary file 3).

The birth rate, b , and natural mortality rate, m , balance to yield a population-level growth rate, such that it is impossible to estimate both b and m simultaneously from total population size data alone. As such, we fixed b at .025 and estimated m by fitting an infection-absent version of our mean field model to the susceptible time series derived via Hoechst staining of control wells for each of the three cell lines (Figure 1-figure supplement

7) . This yielded a natural mortality rate, m , corresponding to a lifespan of approximately 121, 191, and 84 hours, respectively, for Vero, RoNi/7.1, and PaKiT01 cell lines (Figure 1-figure supplement 7) . We then fixed the virus incubation rate, s , as the inverse of the shortest observed duration of time from initial infection to the observation of the first infectious cells via fluorescent microscope for all nine cell line-virus combinations (ranging 6 to 9.5 hours). We fixed a , the infection-induced mortality rate, at $1/6$, an accepted standard for general viral kinetics (Howat et al., 2006) , and held c , the rate of antiviral cell regression to susceptible status, at 0 for the timespan (<200 hours) of the experimental cell line infection trials.

We estimated cell line-virus-MOI-specific values for b , r , and γ by fitting the deterministic output of infectious proportions in our mean field model to the full suite of statistical outputs of all trials for each infected cell culture time series (Figure 1-figure supplements 2-3) . Fitting was performed by minimizing the sum of squared differences between the deterministic model output and cell line-virus-MOI-specific infectious proportion of the data at each timestep. We optimized parameters for $\text{MOI} = 0.001$ and 0.0001 simultaneously to leverage statistical power across the two datasets, estimating a different transmission rate, b , for trials run at each infectious dose but, where applicable, estimating the same rates of r and γ across the two time series. We used the differential equation solver `lsoda()` in the R package `deSolve` (Soetaert et al., 2010) to obtain numerical solutions for the mean field model and carried out minimization using the 'Nelder-Mead' algorithm of the `optim()` function in base R. All model fits were conducted using consistent starting guesses for the parameters, b ($b = 3$), and where applicable, r ($r = 0.001$) and γ ($\gamma = 0.001$). In the case of failed fits or indefinite Hessians, we generated a series of random guesses around the starting conditions and continued estimation until successful fits were achieved.

All eighteen cell line-virus-MOI combinations of data were fit by an immune absent ($\gamma = r = 0$) version of the theoretical model and, subsequently, an induced immunity ($\gamma = 0$; $r > 0$) and constitutive immunity ($\gamma > 0$; $r > 0$) version of the model. Finally, we compared fits across each cell line-virus-MOI combination via AIC. In calculating AIC, the number of fitted parameters in each model (k) varied across the immune phenotypes, with one parameter (b) estimated for absent immune assumptions, two (b and r) for induced immune assumptions, and three (b , r , and γ) for constitutive immune assumptions. The sample size (n) corresponded to the number of discrete time steps across all empirical infectious trials to which the model was fitted for each cell-line virus combination. All fitting and model

comparison scripts are freely available for download at the following FigShare repository: DOI: 10.6084/m9.figshare.8312807.

Finally, we verified all mean field fits in a spatial context, in order to more thoroughly elucidate the role of antiviral cells in each time series. We constructed our spatial model in C++ implemented in R using the packages Rcpp and RcppArmadillo (Eddelbuettel and Francois, 2011; Eddelbuettel and Sanderson, 2017) . Following Nagai and Honda (2001) and Howat et al. (2006) , we modeled this system on a two-dimensional hexagonal lattice, using a ten-minute epidemic timestep for cell state transitions. At the initialization of each simulation, we randomly assigned a duration of natural lifespan, incubation period, infectivity period, and time from antiviral to susceptible status to all cells in a theoretical monolayer. Parameter durations were drawn from a normal distribution centered at the inverse of the respective fixed rates of m , s , a , and c , as reported with our mean field model. Transitions involving the induced (r) and constitutive (β) rates of antiviral acquisition were governed probabilistically and adjusted dynamically at each timestep based on the global environment. As such, we fixed these parameters at the same values estimated in the mean field model, and multiplied both r and β by the global proportion of, respectively, exposed and susceptible cells at a given timestep.

In contrast to antiviral acquisition rates, transitions involving the birth rate (b) and the transmission rate (β) occurred probabilistically based on each cell's local environment. The birth rate, b , was multiplied by the proportion of susceptible cells within a six-neighbor circumference of a focal dead cell, while β was multiplied by the proportion of infectious cells within a thirty-six neighbor vicinity of a focal susceptible cell, thus allowing viral transmission to extend beyond the immediate nearestneighbor boundaries of an infectious cell. To compensate for higher thresholds to cellular persistence and virus invasion which occur under local spatial conditions (Webb et al., 2007) , we increased the birth rate, b , and the cell-to-cell transmission rate, β , respectively, to six and ten times the values used in the mean field model (Supplementary file 4) . We derived these increases based on the assumption that births took place exclusively based on pairwise nearest-neighbor interactions (the six immediately adjacent cells to a focal dead cell), while viral transmission was locally concentrated but included a small (7.5%) global contribution, representing the thirty-six cell surrounding vicinity of a focal susceptible. We justify these increases and derive their origins further in Supplementary file 5.

We simulated ten stochastic spatial time series for all cell-virus combinations under all three immune assumptions at a population size of 10,000 cells and compared model output with data in . Transparent reporting form Data availability All data generated or analysed during this study are included in the manuscript and supporting files. All images and code used in this study have been made available for download at the following Figshare

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CDC Summary 21 MAR 2020,

<https://www.cdc.gov/coronavirus/2019-ncov/cases-updates/summary.html>

This is a rapidly evolving situation and CDC will provide updated information and guidance as it becomes available.

Updated March 21, 2020

CDC is responding to a pandemic of respiratory disease spreading from person-to-person caused by a novel (new) coronavirus. The disease has been named “coronavirus disease 2019” (abbreviated “COVID-19”). This situation poses a serious public health risk. The federal government is working closely with state, local, tribal, and territorial partners, as well as public health partners, to respond to this situation. COVID-19 can cause mild to severe illness; most severe illness occurs in older adults.

Situation in U.S.

Different parts of the country are seeing different levels of COVID-19 activity. The United States nationally is in the initiation phase of the pandemic. States in which community spread is occurring are in the acceleration phase. The duration and severity of each pandemic phase can vary depending on the characteristics of the virus and the public health response.

CDC and state and local public health laboratories are testing for the virus that causes COVID-19. View CDC's Public Health Laboratory Testing map.

All 50 states have reported cases of COVID-19 to CDC.

U.S. COVID-19 cases include:

Imported cases in travelers

Cases among close contacts of a known case

Community-acquired cases where the source of the infection is unknown.

Twenty-seven U.S. states are reporting some community spread of COVID-19.

View latest case counts, deaths, and a map of states with reported cases.

CDC Recommends

Everyone can do their part to help us respond to this emerging public health threat:

On March 16, the White House announced a program called "15 Days to Slow the Spread,"pdf iconexternal icon which is a nationwide effort to slow the spread of COVID-19 through the implementation of social distancing at all levels of society.

Older people and people with severe chronic conditions should take special precautions because they are at higher risk of developing serious COVID-19 illness.

If you are a healthcare provider, use your judgment to determine if a patient has signs and symptoms compatible with COVID-19 and whether the patient should be tested. Factors to consider in addition to clinical symptoms may include:

Does the patient have recent travel from an affected area?

Has the patient been in close contact with someone with COVID-19 or with patients with pneumonia of unknown cause?

Does the patient reside in an area where there has been community spread of COVID-19?

If you are a healthcare provider or a public health responder caring for a COVID-19 patient, please take care of yourself and follow recommended infection control procedures.

People who get a fever or cough should consider whether they might have COVID-19, depending on where they live, their travel history or other exposures. More than half of the

U.S. is seeing some level of community spread of COVID-19. Testing for COVID-19 may be accessed through medical providers or public health departments, but there is no treatment for this virus. Most people have mild illness and are able to recover at home without medical care.

For people who are ill with COVID-19, but are not sick enough to be hospitalized, please follow CDC guidance on how to reduce the risk of spreading your illness to others. People who are mildly ill with COVID-19 are able to isolate at home during their illness.

If you have been in China or another affected area or have been exposed to someone sick with COVID-19 in the last 14 days, you will face some limitations on your movement and activity. Please follow instructions during this time. Your cooperation is integral to the ongoing public health response to try to slow spread of this virus.

COVID-19 Emergence

COVID-19 is caused by a coronavirus. Coronaviruses are a large family of viruses that are common in people and many different species of animals, including camels, cattle, cats, and bats. Rarely, animal coronaviruses can infect people and then spread between people such as with MERS-CoV, SARS-CoV, and now with this new virus (named SARS-CoV-2).

The SARS-CoV-2 virus is a betacoronavirus, like MERS-CoV and SARS-CoV. All three of these viruses have their origins in bats. The sequences from U.S. patients are similar to the one that China initially posted, suggesting a likely single, recent emergence of this virus from an animal reservoir.

Early on, many of the patients at the epicenter of the outbreak in Wuhan, Hubei Province, China had some link to a large seafood and live animal market, suggesting animal-to-person spread. Later, a growing number of patients reportedly did not have exposure to animal markets, indicating person-to-person spread. Person-to-person spread was subsequently reported outside Hubei and in countries outside China, including in the United States. Some international destinations now have ongoing community spread with the virus that causes COVID-19, as do some parts of the United States. Community spread means some people have been infected and it is not known how or where they became exposed. Learn more about the spread of this newly emerged coronavirus.

Severity

The complete clinical picture with regard to COVID-19 is not fully known. Reported illnesses have ranged from very mild (including some with no reported symptoms) to severe, including illness resulting in death. While information so far suggests that most COVID-19 illness is mild, a report^{external icon} out of China suggests serious illness occurs in 16% of cases. Older people and people of all ages with severe chronic medical conditions — like heart disease, lung disease and diabetes, for example — seem to be at higher risk of developing serious COVID-19 illness. A CDC Morbidity & Mortality Weekly Report that looked at severity of disease among COVID-19 cases in the United States by age group found that 80% of deaths were among adults 65 years and older with the highest percentage of severe outcomes occurring in people 85 years and older.

Learn more about the symptoms associated with COVID-19.

COVID-19 Pandemic

A pandemic is a global outbreak of disease. Pandemics happen when a new virus emerges to infect people and can spread between people sustainably. Because there is little to no pre-existing immunity against the new virus, it spreads worldwide.

The virus that causes COVID-19 is infecting people and spreading easily from person-to-person. Cases have been detected in most countries worldwide and community spread is being detected in a growing number of countries. On March 11, the COVID-19 outbreak was characterized as a pandemic by the WHO^{external icon}.

This is the first pandemic known to be caused by the emergence of a new coronavirus. In the past century, there have been four pandemics caused by the emergence of novel influenza viruses. As a result, most research and guidance around pandemics is specific to influenza, but the same premises can be applied to the current COVID-19 pandemic. Pandemics of respiratory disease follow a certain progression outlined in a “Pandemic Intervals Framework.” Pandemics begin with an investigation phase, followed by recognition, initiation, and acceleration phases. The peak of illnesses occurs at the end of the acceleration phase, which is followed by a deceleration phase, during which there is a decrease in illnesses. Different countries can be in different phases of the pandemic at any point in time and different parts of the same country can also be in different phases of a pandemic.

There are ongoing investigations to learn more. This is a rapidly evolving situation and information will be updated as it becomes available.

Risk Assessment

Risk depends on characteristics of the virus, including how well it spreads between people; the severity of resulting illness; and the medical or other measures available to control the impact of the virus (for example, vaccines or medications that can treat the illness) and the relative success of these. In the absence of vaccine or treatment medications, nonpharmaceutical interventions become the most important response strategy. These are community interventions that can reduce the impact of disease.

The risk from COVID-19 to Americans can be broken down into risk of exposure versus risk of serious illness and death.

Risk of exposure:

The immediate risk of being exposed to this virus is still low for most Americans, but as the outbreak expands, that risk will increase. Cases of COVID-19 and instances of community spread are being reported in a growing number of states.

People in places where ongoing community spread of the virus that causes COVID-19 has been reported are at elevated risk of exposure, with the level of risk dependent on the location.

Healthcare workers caring for patients with COVID-19 are at elevated risk of exposure.

Close contacts of persons with COVID-19 also are at elevated risk of exposure.

Travelers returning from affected international locations where community spread is occurring also are at elevated risk of exposure, with level of risk dependent on where they traveled.

Risk of Severe Illness:

Early information out of China, where COVID-19 first started, shows that some people are at higher risk of getting very sick from this illness. This includes:

Older adults, with risk increasing by age.

People who have serious chronic medical conditions like:

Heart disease

Diabetes

Lung disease

CDC has developed guidance to help in the risk assessment and management of people with potential exposures to COVID-19.

What May Happen

More cases of COVID-19 are likely to be identified in the United States in the coming days, including more instances of community spread. CDC expects that widespread transmission of COVID-19 in the United States will occur. In the coming months, most of the U.S. population will be exposed to this virus.

Widespread transmission of COVID-19 could translate into large numbers of people needing medical care at the same time. Schools, childcare centers, and workplaces, may experience more absenteeism. Mass gatherings may be sparsely attended or postponed. Public health and healthcare systems may become overloaded, with elevated rates of hospitalizations and deaths. Other critical infrastructure, such as law enforcement, emergency medical services, and sectors of the transportation industry may also be affected. Healthcare providers and hospitals may be overwhelmed. At this time, there is no vaccine to protect against COVID-19 and no medications approved to treat it. Nonpharmaceutical interventions will be the most important response strategy to try to delay the spread of the virus and reduce the impact of disease.

CDC Response

Global efforts at this time are focused concurrently on lessening the spread and impact of this virus. The federal government is working closely with state, local, tribal, and territorial partners, as well as public health partners, to respond to this public health threat.

Highlights of CDC's Response

CDC established a COVID-19 Incident Management System on January 7, 2020. On January 21, CDC activated its Emergency Operations Center to better provide ongoing support to the COVID-19 response.

The U.S. government has taken unprecedented steps with respect to travel in response to the growing public health threat posed by this new coronavirus:

Foreign nationals who have been in China, Iran, the United Kingdom, Ireland and any one of the 26 European countries in the Schengen Area within the past 14 days cannot enter the United States.

U.S. citizens, residents, and their immediate family members who have been any one of those countries within in the past 14 days can enter the United States, but they are subject to health monitoring and possible quarantine for up to 14 days.

People at higher risk of serious COVID-19 illness avoid cruise travel and non-essential air travel.

CDC has issued additional specific travel guidance related to COVID-19.

CDC has issued clinical guidance, including:

Clinical Guidance for Management of Patients with Confirmed Coronavirus Disease (COVID-19).

Infection Prevention and Control Recommendations for Patients, including guidance on the use of personal protective equipment (PPE) during a shortage.

CDC also has issued guidance for other settings, including:

Preparing for COVID-19: Long-term Care Facilities, Nursing Homes

Discontinuation of Home Isolation for Persons with COVID-19

CDC has deployed multidisciplinary teams to support state health departments in case identification, contact tracing, clinical management, and public communications.

CDC has worked with federal partners to support the safe return of Americans overseas who have been affected by COVID-19.

An important part of CDC's role during a public health emergency is to develop a test for the pathogen and equip state and local public health labs with testing capacity.

CDC developed an rRT-PCR test to diagnose COVID-19.

As of the evening of March 17, 89 state and local public health labs in 50 states, the District of Columbia, Guam, and Puerto Rico have successfully verified and are currently using CDC COVID-19 diagnostic tests.

Commercial manufacturers are now producing their own tests.

CDC has grown the COVID-19 virus in cell culture, which is necessary for further studies, including for additional genetic characterization. The cell-grown virus was sent to NIH's BEI Resources Repository  for use by the broad scientific community.

CDC also is developing a serology test for COVID-19.

Other Available Resources

The following resources are available with information on COVID-19

World Health Organization, Coronavirus 

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Identifying Locations with Possible Undetected Imported Severe Acute Respiratory Syndrome Coronavirus 2 Cases by Using Importation Predictions,

https://wwwnc.cdc.gov/eid/article/26/7/20-0250_article

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Research

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Abstract

Cases of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infection exported from mainland China could lead to self-sustained outbreaks in other countries. By February 2020, several countries were reporting imported SARS-CoV-2 cases. To contain the virus, early detection of imported SARS-CoV-2 cases is critical. We used air travel volume estimates from Wuhan, China, to international destinations and a generalized linear regression model to identify locations that could have undetected imported cases. Our model can be adjusted to account for exportation of cases from other locations as the virus spreads and more information on importations and transmission becomes available. Early detection and appropriate control measures can reduce the risk for transmission in all locations.

A novel coronavirus, later named severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), was identified in December 2019 in the city of Wuhan, capital of Hubei Province, China, where cases were first confirmed (1). During December 2019–February 2020, the number of confirmed cases increased drastically. Model estimates suggested that >75,000 persons were infected by January 25, 2020, and the epidemic had a doubling time of ≈ 6 days (2). By the end of January 2020, travel restrictions were implemented for Wuhan and neighboring cities. Nonetheless, the virus spread from Wuhan to other cities in China and outside the country. By February 4, 2020, a total of 23 locations outside mainland China reported cases, 22 of which reported imported cases; Spain reported a case caused by secondary transmission (3).

Most cases imported to other locations have been linked to recent travel history from China (3), suggesting that air travel plays a major role in exportation of cases to locations outside of China. To prevent other cities and countries from becoming epicenters of the

SARS-CoV-2 epidemic, substantial targeted public health interventions are required to detect cases and control local spread of the virus. We collected estimates of air travel volume from Wuhan to 194 international destinations. We then identified 49 countries that had a score of $>49.2/100$ on category 2, Early Detection and Reporting of Epidemics of Potential International Concern, of the Global Health Security (GHS) Index (4). We assumed these locations would be proficient at detecting SARS-CoV-2 and reporting confirmed imported cases, which we refer to as imported-and-reported cases. We ran a generalized linear regression model on this subset; based on the results, we generated predictions for the remainder of the sample. Using these predictions, we identified locations that might not be detecting imported cases.

Methods

To identify locations reporting fewer than predicted imported SARS-CoV-2 infected cases, we fit a model to data from 49 locations outside mainland China with high surveillance capacity according to the GHS Index (4). Among these, 17 had high travel connectivity to Wuhan and 32 have low connectivity to Wuhan. We considered locations to be countries without any position on territorial claims. We performed a Poisson regression by using the cumulative number of imported-and-reported SARS-CoV-2 cases in these 49 countries and the estimated number of daily airline passengers from the Wuhan airport. We then compared predictions from this model with imported-and-reported cases across 194 locations from the GHS Index, excluding China as the epicenter of the outbreak.

The model requires data on imported-and-reported cases of SARS-CoV-2 infection, daily air travel volume, and surveillance capacity. We obtained data on imported-and-reported cases aggregated by destination from the World Health Organization technical report issued February 4, 2020 (3). We assumed a case count of 0 for locations not listed. We used February 4 as the cutoff for cumulative imported-and-reported case counts because exported cases from Hubei Province dropped rapidly after this date (3), likely because of travel restrictions for the province implement on January 23. We defined imported-and-reported cases as those with known travel history from China; of those, 83% had a travel history from Hubei Province and 17% traveled from unknown locations in China (3). We excluded reported cases likely caused by transmission outside of China or cases in which the transmission source was still under investigation (3). In addition, we excluded Hong Kong, Macau, and Taiwan from our model because locally transmitted and imported cases were not disaggregated in these locations.

We obtained data on daily air travel from a network-based modeling study (S. Lai et al., unpub. data, <https://doi.org/10.1101/2020.02.04.20020479>External Link) that reported monthly air travel volume estimates for the 27 locations outside mainland China that are most connected to Wuhan. These estimates were calculated from International Air Travel Association data from February 2018, which includes direct and indirect flight itineraries from Wuhan. For these 27 locations, estimated air travel volumes are >6 passengers/day. We assumed that travel volumes for locations not among the most connected are censored by a detection limit. We used a common method of dealing with censored data from environmental sampling (5), or metabolomics (6), to set the daily air travel volume to half the minimum previously reported. Therefore, we used 3 passengers/day for estimated travel volumes for the 167 locations from the GHS Index not listed by Lai et al. We tested the robustness of our results by using a set of alternative values of 0.1, 1, and 6 passengers/day for the censored data.

We defined high surveillance locations as those with a GHS Index for category 2 above the 75th quantile. We assessed the number of high surveillance locations, those with 0 imported-and-reported cases, and low surveillance locations, those with case counts >1 (Table).

For our model, we assumed that the cumulative imported-and-reported case counts across 49 high surveillance locations follow a Poisson distribution from the beginning of the epidemic until February 4, 2020. Then the expected case count is linearly proportional to the daily air travel volume in the following formula: where i denotes location, C_i denotes the imported-and-reported case count in a location, λ_i denotes the expected case count in a location, β denotes the regression coefficient, and x_i denotes the daily air travel volume of a location. The Poisson model assumes cases are independent and that the variance is equal to the expected case count. Imported-and-reported cases likely meet the independence assumption because the value excludes cases with local transmission. We also checked the robustness of our results by using an over dispersed model with a negative binomial likelihood. We computed the p value of the overdispersion parameter as shown in Gelman and Hill (7).

Thumbnail of Regression plot of locations with possible undetected imported cases of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) by air travel volume from

Wuhan, China. Air travel volume measured in number of persons/day. No. cases refers to possible undetected imported SARS-CoV-2 cases. Solid line indicates the expected imported-and-reported case counts for locations. Dashed lines represent 95% prediction interval bounds smoothed for all locations. Purple dots indicate location

Figure 1. Regression plot of locations with possible undetected imported cases of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) by air travel volume from Wuhan, China. Air travel volume measured in number of...

We used R version 3.6.1 (<https://www.r-project.org>External Link) to compute , the maximum likelihood estimate of β , and the expected imported-and-reported case count given high surveillance (Figure 1). We also computed the 95% prediction interval (PI) bounds under this model of high surveillance for all 194 values of daily air travel volume (Figure 1). First, we generated a bootstrapped dataset by sampling n locations with replacement among high surveillance locations. Then, we reestimated β by using the bootstrapped dataset. Finally, we simulated imported-and-reported case counts for all 194 locations under our model by using the estimate of β from the bootstrapped dataset. We repeated the 3 steps 50,000 times to generate 50,000 simulated imported-and-reported case counts for each of the locations computed to the lower and upper PI bounds (PI 2.5%–97.5%). We smoothed the 95% PI bounds by using ggplot2 in R (8). We fit the imported-and-reported case counts of the 49 high surveillance locations to the model and plotted these alongside 145 locations with low surveillance capacity (Figure 1). We noted some overlap between high and low surveillance locations (Figure 1).

Thumbnail of Analyses of imported-and-reported cases and daily air travel volume using a model to predict locations with potentially undetected cases of severe acute respiratory virus 2 (SARS-CoV-2). Air travel volume measured in number of persons/day. No. cases refers to possible undetected imported SARS-CoV-2 cases. Solid line shows the expected imported-and-reported case counts based on our model fitted to high surveillance locations, indicated by purple dots. Dashed lines indicate the 95% pr

Figure 2. Analyses of imported-and-reported cases and daily air travel volume using a model to predict locations with potentially undetected cases of severe acute respiratory virus 2 (SARS-CoV-2). Air travel volume measured in...

To assess the robustness of our results we ran 8 additional regression analyses by implementing a series of changes to the analysis. The changes included the following: set the daily air travel volume to 0.1, 1, or 6 passengers/day for locations not listed by Lai et al. (unpub. data, <https://doi.org/10.1101/2020.02.04.20020479>External Link) (Figure 2, panels A–C); removed all locations not listed by Lai et al. before fitting (Figure 2, panel D); defined high surveillance locations by using a more lenient GHS Index criterion, 50th quantile (Figure 2, panel E), and a more stringent criterion, 95th quantile (Figure 2, panel F); excluded Thailand from the model because it is a high-leverage point (Figure 2, panel G); or used an overdispersed Poisson likelihood with a negative-binomial likelihood (Figure 2, panel H). We provide code for these analyses on GitHub (<https://github.com/c2-d2/cov19flightimport>External Link).

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Results

We found that daily air travel volume positively correlates with imported-and-reported case counts of SARS-CoV-2 infection among high surveillance locations (Figure 1). We noted that increasing flight volume by 31 passengers/day is associated with 1 additional expected imported-and-reported case. In addition, Singapore and India lie above the 95% PI in our model; Singapore had 12 more imported-and-reported cases (95% PI 6–17 cases) than expected and India had 3 (95% PI 1–3 cases) more than expected. Thailand has a relatively high air travel volume compared with other locations, but it lies below the 95% PI, reporting 16 (95% PI 1–40 cases) fewer imported-and-reported cases than expected under the model. Indonesia lies below the PI and has no imported-and-reported cases, but the expected case count is 5 (95% PI 1–10 cases) in our model. Across all 8 robustness regression analyses, we consistently observed that Singapore lies above the 95% PI and Thailand and Indonesia lie below (Figure 2). India remains above the 95% PI in all robustness analyses except when we used the more stringent GHS Index, 95th quantile, for fitting; then India lies on the upper bound of the 95% PI (Figure 2, panel F).

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Discussion

We aimed to identify locations with likely undetected or underdetected imported cases of SARS-CoV-2 by fitting a model to the case counts in locations with high surveillance capacity and Wuhan-to-location air travel volumes. Our model can be adjusted to account for exportation of cases from locations other than Wuhan as the outbreak develops and more information on importations and self-sustained transmission becomes available. One key advantage of this model is that it does not rely on estimates of incidence or prevalence in the epicenter of the outbreak. Also, we intentionally used a simple generalized linear model. The linearity of the expected case count means that we have only 1 regression coefficient in the model and no extra parameters. The Poisson likelihood then captures the many 0-counts observed for less highly connected locations but also describes the slope between case-count and flight data among more connected locations. We believe this model provides the most parsimonious phenomenologic description of the data.

According to our model, locations above the 95% PI of imported-and-reported cases could have higher case-detection capacity. Locations below the 95% PI might have undetected cases because of expected imported-and-reported case counts under high surveillance. Underdetection of cases could increase the international spread of the outbreak because the transmission chain could be lost, reducing opportunities to deploy case-based control strategies. We recommend rapid strengthening of outbreak surveillance and control efforts in locations below the 95% PI lower bound, particularly Indonesia, to curb potential local transmission. Early detection of cases and implantation of appropriate control measures can reduce the risk for self-sustained transmission in all locations.

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MERS coronavirus: diagnostics, epidemiology and transmission

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Abstract: The first known cases of Middle East respiratory syndrome (MERS), associated with infection by a novel coronavirus (CoV), occurred in 2012 in Jordan but were reported retrospectively. The case first to be publicly reported was from Jeddah, in the Kingdom of Saudi Arabia (KSA). Since then, MERS-CoV sequences have been found in a bat and in many dromedary camels (DC). MERS-CoV is enzootic in DC across the Arabian Peninsula

and in parts of Africa, causing mild upper respiratory tract illness in its camel reservoir and sporadic, but relatively rare human infections. Precisely how virus transmits to humans remains unknown but close and lengthy exposure appears to be a requirement. The KSA is the focal point of MERS, with the majority of human cases. In humans, MERS is mostly known as a lower respiratory tract (LRT) disease involving fever, cough, breathing difficulties and pneumonia that may progress to acute respiratory distress syndrome, multiorgan failure and death in 20 % to 40 % of those infected. However, MERS-CoV has also been detected in mild and influenza-like illnesses and in those with no signs or symptoms. Older males most obviously suffer severe disease and MERS patients often have comorbidities. Compared to severe acute respiratory syndrome (SARS), another sometimes- fatal zoonotic coronavirus disease that has since disappeared, MERS progresses more rapidly to respiratory failure and acute kidney injury (it also has an affinity for growth in kidney cells under laboratory conditions), is more frequently reported in patients with underlying disease and is more often fatal. Most human cases of MERS have been linked to lapses in infection prevention and control (IPC) in healthcare settings, with approximately 20 % of all virus detections reported among healthcare workers (HCWs) and higher exposures in those with occupations that bring them into close contact with camels. Sero-surveys have found widespread evidence of past infection in adult camels and limited past exposure among humans. Sensitive, validated reverse transcriptase real-time polymerase chain reaction (RT-rtPCR)-based diagnostics have been available almost from the start of the emergence of MERS. While the basic virology of MERS-CoV has advanced over the past three years, understanding of the interplay between camel, environment, and human remains limited. ELECTRONIC SUPPLEMENTARY MATERIAL: The online version of this article (doi:10.1186/s12985-015-0439-5) contains supplementary material, which is available to authorized users.

Text: An email from Dr Ali Mohamed Zaki, an Egyptian virologist working at the Dr Soliman Fakeeh Hospital in Jeddah in the Kingdom of Saudi Arabia (KSA) announced the first culture of a new coronavirus to the world. The email was published on the website of the professional emerging diseases (ProMED) network on 20 th September 2012 [1] (Fig. 1) and described the first reported case, a 60 year old man from Bisha in the KSA. This information led to the rapid discovery of a second case of the virus, this time in an ill patient in the United Kingdom, who had been transferred from Qatar for care [2] . The new virus was initially called novel coronavirus (nCoV) and subsequently entitled the Middle East respiratory syndrome coronavirus (MERS-CoV). As of 2 nd of September 2015, there have been 1,493 detections of viral RNA or virus-specific antibodies across 26 countries

(Additional file 1: Figure S1) confirmed by the World Health Organization (WHO), with over a third of the positive people dying (at least 527, 35 %) [3] .

Since that first report, a slow discovery process over the following two to three years revealed a virus that had infected over 90 % of adult dromedary camels (DC; *Camelus dromedarius*) in the KSA [4] , also DCs across the Arabian Peninsula and parts of Africa that are a source of DC imports for the KSA [5] . To date, MERS-CoV has not been detected in DCs tested in zoos or herds from other parts of the world [6] [7] [8] [9] . Occasionally, virus is transmitted from infected DCs to exposed humans. Subsequent transmission to other humans requires relatively close and prolonged exposure [10] .

The first viral isolate was patented and concerns were raised that this would restrict access to both the virus and to viral diagnostics [11, 12] . However, sensitive, validated reverse transcriptase real-time polymerase chain reaction (RT-rtPCR)-based diagnostics were quickly described and virus was made freely available subject to routine biosafety considerations [13] . Subsequent epidemiology and research has identified the cell receptor as exopeptidase dipeptidyl peptidase 4 (DPP4; also called CD26); that MERS-CoV has a broad tropism, replicating better in some cells lines and eliciting a more proinflammatory response than SARS-CoV; is widespread in DCs; has the potential to infect other animals and that MERS kills its human host more often than SARS did (20-40 % versus 9 % for SARS [14]) [15] [16] [17] [18] [19] .

In humans, overt disease was given the name Middle East respiratory syndrome, with the acronym MERS. From intermittent animal-to-human spill-over events, the MERS-CoV spreads sporadically among people, causing more severe disease among older adults, especially males, with pre-existing diseases. The spread of MERS-CoV among humans has often been associated with outbreaks in hospitals, with around 20 % of all cases to date involving healthcare workers (HCWs).

Although DCs appear to suffer the equivalent of a 'common cold' from MERS-CoV infection, in humans, the virus can be a more serious and opportunistic pathogen associated with the death of up to 40 % of reported cases. It has yet to be established whether infections thought to have been acquired from an animal source produce a more severe outcome than those spread between humans [20] . Studies have established that

the mean incubation period for MERS is five to six days, ranging from two to 16 days, with 13 to 14 days between when illness begins in one person and subsequently spreads to another [21] [22] [23] [24] . Among those with progressive illness, the median time to death is 11 to 13 days, ranging from five to 27 days [23, 24] . Fever and gastrointestinal symptoms may form a prodrome, after which symptoms decline, only to be followed by a more severe systemic and respiratory syndrome [25, 26] .

The first WHO case definition [27] defined probable cases of MERS based on the presence of febrile illness, cough and requirement for hospitalization with suspicion of lower respiratory tract (LRT) involvement. It also included roles for contact with a probable or confirmed case or for travel or residence within the Arabian Peninsula. If strictly adhered to, only the severe syndrome would be subject to laboratory testing, which was the paradigm early on [21] . From July 2013, the revised WHO case definition included the importance of seeking out and understanding the role of asymptomatic cases and from June 2014, the WHO definition more clearly stated that a confirmed case included any person whose sample was RT-PCR positive for MERS-CoV, or who produced a seroconversion, irrespective of clinical signs and symptoms. [28] [29] [30] Apart from the WHO and the KSA Ministry of Health reports, asymptomatic or subclinical cases of MERS-CoV infection were documented in the scientific literature although not always as often as occurred early on [31, 32] . The KSA definition of a case became more strict on 13 th May 2014, relying on the presence of both clinical features and laboratory confirmation [33] . Testing of asymptomatic people was recommended against from December 2014 [34] , reinforced by a case definition released by the KSA Ministry of Health in June 2015 [35] . The KSA has been the source of 79 % of human cases. Severe MERS is notable for its impact among older men with comorbid diseases including diabetes mellitus, cirrhosis and various lung, renal and cardiac conditions [36] [37] [38] . Interestingly in June 2015, an outbreak in South Korea followed a similar distribution [39, 40] . Among laboratory confirmed cases, fever, cough and upper respiratory tract (URT) signs and symptoms usually occur first, followed within a week by progressive LRT distress and lymphopaenia [37] . Patients often present to a hospital with pneumonia, or worse, and secondary bacterial infections have been reported [37, 41] . Disease can progress to acute respiratory distress syndrome and multiorgan system failure [37] . MERS has reportedly killed approximately 35 % of all reported cases, 42 % of cases in the KSA, yet only 19 % of cases in South Korea, where mortality ranged from 7 % among younger age groups to 40 % among those aged 60 years and above [42] ; all may be inflated values with asymptomatic or mild infections sometimes not sought or not reported [34] . General supportive care is key to managing severe cases [43] . Children under the age of 14 years are rarely reported to be positive for

MERS-CoV, comprising only 1.1 % (n = 16) of total reported cases. Between 1 st September 2012 and 2 nd December 2013, a study described the then tally of paediatric cases in the KSA, which stood at 11 (two to 16 years of age; median 13 years); nine were asymptomatic (72 %) and one infant died [44] . In Amman, Jordan, 1,005 samples from hospitalized children under the age of two years with fever and/or respiratory signs and symptoms were tested but none were positive for MERS-CoV RNA, despite being collected at a similar time to the first known outbreak of MERS-CoV in the neighbouring town of Al-Zarqa [45] . A second trimester stillbirth occurred in a pregnant woman during an acute respiratory illness and while not RT-rtPCR positive, the mother did subsequently develop antibodies to MERS-CoV, suggestive of recent infection [46] . Her exposure history to a MERS-CoV RT-rtPCR positive relative and an antibody-reactive husband, her incubation period and her symptom history met the WHO criteria for being a probable MERS-CoV case [46] .

Diagnostic methods were published within days of the ProMED email announcing the first MERS case [47] , including several now gold standard in-house RT-rtPCR assays (Fig. 2) as well as virus culture in Vero and LLC-MK2 cells [18, 47, 48] . A colorectal adenocarcinoma (Caco-2) epithelial cell line has since been recommended for isolation of infections MERS-CoV [49] . We previously [18] .). Open reading frames are indicated as yellow rectangles bracketed by terminal untranslated regions (UTR; grey rectangles). FS-frame-shift. Predicted regions encompassing recombination break-points are indicated by orange pills. Created using Geneious v8.1 [211] and annotated using Adobe Illustrator. Beneath this is a schematic depicting the location of RT-PCR primers (blue arrows indicate direction) and oligoprobes (green rectangles) used in the earliest RT-rtPCR screening assays and conventional, semi-nested (three primers) RT-PCR confirmatory sequencing assays [47, 48] . Publication order is noted by first [27 th September 2012; red] and second [6 th December 2012; orange] coloured rectangles; both from Corman et al. [47, 48] Those assays recommended by the WHO are highlighted underneath by yellow dots [53] . The NSeq reverse primer has consistently contained one sequence mismatch with some MERS-CoV variants. An altered version of that from Mackay IM, Arden KE. Middle East respiratory syndrome: An emerging coronavirus infection tracked by the crowd. *Virus Res* 2015 Vol 202:60-88 with permission from Elsevier [5] reviewed the broad tropism of MERS-CoV [5] . However, as is well described, cell culture is a slow, specialised and insensitive method [50] while PCR-based techniques are the preferred method for MERS-CoV detection.

The first open reading frames (ORF 1a and 1b; Fig. 2) have become a key diagnostic and taxonomic target for CoV species identification. With less than 80 % identity between the

amino acid sequence of MERS ORF 1ab and betacoronavirus relatives, *Tylonycteris bat HKU4* and *Pipistrellus bat HKU5*, it can be concluded that it is a novel and distinct virus. MERS-CoV is predicted to encode ten open reading frames with 5' and 3' untranslated regions [51] . The structural proteins include the spike (S), envelope (E), membrane (M) and nucleocapsid (N) [52] . The products of ORF1a and ORF1b are predicted to encode nonstructural proteins.

The majority of specimen testing to date has employed validated RT-rtPCR assays shown to be sensitive and specific [47, 48, 53] . The RealStar® kit uses these WHO recommended assays [54] . The target sequences of these screening assays have not changed among genomes examined until at least mid-2015 (IMM observation). Other RT-rtPCR assays have been developed and validated for use as laboratory-based diagnostic tools [55] [56] [57] . Additionally, loop-mediated [58, 59] or recombinase polymerase [60] isothermal assays have been designed for field deployment.

The detection of MERS-CoV antigen has not been common to date but the combination of short turnaround time from test to result, high throughput and identification of viral proteins makes this an attractive option. Detection of viral proteins rather than viral RNA indicates the likely presence of infectious virus. The first rapid immunochromatographic tool described could detect recombinant MERS-CoV nucleocapsid protein from DC nasal swabs with 94 % sensitivity and 100 % specificity compared to RT-rtPCR [61] . A different approach used a monoclonal antibody-based capture ELISA targeting the MERS-CoV nucleocapsid protein with a sensitivity of 10³ TCID₅₀ and 100 % specificity [62] .

Demonstration of a seroconversion to a MERS-CoV infection meets the current WHO definition of a case so optimized and thoroughly validated sero-assays employed alongside good clinical histories are useful to both identify prior MERS-CoV infection and help support transmission studies. Because serology testing is, by its nature, retrospective, it is usual to detect a viral footprint, in the form of antibodies, in the absence of any signs or symptoms of disease and often in the absence of any viral RNA [63] .

Strategic, widespread sero-surveys of humans using samples collected after 2012 are infrequent. Much of the Arabian Peninsula and all of the Horn of Africa lack baseline data describing the proportion of the community who may have been infected by a MERS-CoV.

However, sero-surveys have had widespread use in elucidating the role of DCs as a transmission source for MERS-CoV. Because of the identity shared between DC and human MERS-CoV (see Molecular epidemiology: using genomes to understand outbreaks), serological assays for DC sero-surveys should be transferrable to human screening with minimal re-configuration. Also, no diagnostically relevant variation in neutralization activity have been found from among a range of circulating tested MERS-CoV isolates and sera, so whole virus or specific protein-based sero-assays should perform equivalently in detecting serological responses to the single MERS-CoV serotype [49] . The development of robust serological assays requires reliable panels of wellcharacterized animal or human sera, including those positive for antibodies specific to MERS-CoV, as well as to likely sources of cross-reaction [64] . Obtaining these materials was problematic and slowed the development and commercialization of antibody detection assays for human testing [64] . A number of commercial ELISA kits, immunofluorescent assays (IFA) kits, recombinant proteins and monoclonal antibodies have been released [31, [65] [66] [67] [68] . Initially, conventional IFAs were used for human sero-surveys. These relied on MERS-CoV-infected cell culture as an antigen source, detecting the presence of human anti-MERS-CoV IgG, IgM or neutralizing antibodies in human samples [18, 48, 69] . No sign of MERS-CoV antibodies was found among 2,400 sera from patients visiting Hospital in Jeddah, from 2010 through 2012, prior to the description of MERS-CoV [18] . Nor did IFA methods detect any sign of prior MERS-CoV infection among a small sample of 130 healthy blood donors from another Hospital in Jeddah (collected between Jan and Dec 2012) [70] . Of 226 slaughterhouse workers, only eight (3.5 %) were positive by IFA, and those sera could not be confirmed by virus neutralization (NT) test. The study indicated that HCoV-HKU1 was a likely source of crossreactive antigen in the whole virus IFA [70] . Whole virus MERS-CoV IFA also suffered from some cross-reactivity with convalescent SARS patient sera and this could not be resolved by an NT test which was also cross-reactive [71] . IFA using recombinant proteins instead of whole-virus IFA, has been shown to be a more specific tool [31] . Since asymptomatic zoonoses have been posited [72] , an absence of antibodies to MERS-CoV among some humans who have regular and close contact with camels may reflect the rarity of actively infected animals at butcheries, a limited transmission risk associated with slaughtering DCs [70] , a pre-existing cross-protective immune status or some other factor(s) resulting in a low risk of disease and concurrent seroconversion developing after exposure in this group. IFA using recombinant proteins instead.

Some sero-assays have bypassed the risks of working with infectious virus by creating transfected cells expressing recombinant portions of the MERS-CoV nucleocapsid and spike proteins [48, 73] , or using a recombinant lentivirus expressing MERS-CoV spike

protein and luciferase [74, 75] . A pseudo particle neutralization (ppNT) assay has seen widespread use in animal studies and was at least as sensitive as the traditional microneutralization (MNT) test. [10, 74, [76] [77] [78]] Studies using small sample numbers and ppNT found no evidence of MERS-CoV neutralizing antibody in sera from 158 children with LRT infections between May 2010 and May 2011, 110 sera from 19 to 52 year old male blood donors and 300 selfidentified animal workers from the Jazan Region of the KSA during 2012 [79, 80] . Similarly, a study of four herdsmen in contact with an infected DC herd in Al-Ahsa, eight people who had intermittent contact with the herd, 30 veterinary surgeons and support staff who were not exposed to the herd, three unprotected abattoir workers in Al-Ahsa and 146 controls who were not exposed to DCs in any professional role, found none with serological evidence of past MERS-CoV infection using the ppNT assay [10] . A delay in the neutralizing antibody response to MERS-CoV infection was associated with increased disease severity in South Korea cases with most responses detectable by week three of illness while others, even though disease was severe, did not respond for four or more weeks [81] . The implications for our ability to detect any response in mild or asymptomatic cases was not explored but may be a significant factor in understanding exposure in the wider community.

A Jordanian outbreak of acute LRT disease in a hospital in 2012 was retrospectively found to be associated with MERS-CoV infection, initially using RT-rtPCR, but subsequently, and on a larger scale, through positivity by ELISA and IFA or MNT test. [46, 82, 83] This outbreak predated the first case of MERS in the KSA. The ELISA used a recombinant nucleocapsid protein from the group 2 betacoronavirus bat-CoV HKU5 to identify antibodies against the equivalent crossreactive MERS-CoV protein [71] . It was validated using 545 sera collected from people with prior HCoV-OC43, HCoV-229E, SARS-CoV, HCoV-NL63, HRV, HMPV or influenza A(H1N1) infections but was reportedly less specific than the recombinant IFA discussed above. It was still considered an applicable tool for screening large sample numbers [82] . A protein microarray expressing the S1 protein subunit has also been validated and widely used for DC testing [5, 84] . Detection of MERS-CoV infection using ELISA or S1 subunit protein microarray [84] is usually followed by confirmatory IFA and/ or a plaque-reduction neutralization (PRNT) [69, 70, 85] or MNT test. [74, 85, 86] This confirmatory process aims to ensure the antibodies detected are able to specifically neutralize the intended virus and are not more broadly reactive to other coronaviruses found in DCs (bovine CoV, BCoV) or humans (HCoV-OC43, HCoV-229E, HCoV-NL63, HCoV-HKU1, SARS-CoV). In the largest study of human sera, a tiered diagnostic process assigned both recombinant IFA and recombinant ELISA positive sera to 'stage 1' seropositivity. A stage 2 seropositive result additionally required a suitably titred PRNT result [87] . The study

found 15 sera collected in 2012 to 2013 from 10,009 (0.2 %) people in 13 KSA provinces contained MERS-CoV antibodies, but significantly higher proportions in occurred in camel shepherds (two of 87; 2.3 %) and slaughterhouse workers (five of 140; 3.6 %) [87] . Contemporary surveys are needed.

MERS-CoV does not appear to be easily transmitted from DCs to humans, or perhaps it is [72] , but generally does not trigger a detectable immune response if only mild disease or asymptomatic infection results. Serology assays are in need of further validation in this area so care is required when moving newly developed diagnostic serology algorithms from a research setting to one that informs public health decisions. This was reinforced when a false positive US case, purported to have been infected after a handshake and two face-to-face meetings, did not withstand further confirmatory analysis using a more specific, NT assay and was subsequently retracted [88, 89] .

The WHO recommends sampling from the LRT for MERS-CoV RT-rtPCR testing, especially when sample collection is delayed by a week or more after onset of symptoms. [53] LRT samples are also best for attempting isolation of infectious virus, although the success of culture is reduced when disease persists [49] . Recommended sample types include bronchoalveolar lavage (BAL), tracheal/tracheobronchial aspirate, pleural fluid and sputum [53, 90] . Fresh samples yield better diagnostic results than refrigerated material [69] and if delays in testing of ≥ 72 h are likely, samples (except for blood) should be frozen at -70°C [90] . If available, lung biopsy or autopsy tissues can also be tested [53] . The URT is a less invasive and more convenient sampling site however, and an oropharyngeal and throat swab or a nasopharyngeal aspirate/wash are recommended when URT sampling is to be conducted [90] . Paired sera, collected two to three weeks apart are preferable for serological testing while a single sample is suggested to be sufficient if collected two weeks after onset of disease or a single serum collected during the first 10-12 days if conducting RT-rtPCR [53, 90] . Human urine and stool have been found to contain MERS-CoV RNA 12 to 26 days after symptom onset [25, 69, 91] and are listed as samples that should be considered [53, 90] . In two cases that arrived in the Netherlands, urine was RT-rtPCR negative but faeces was weakly positive and sera were RT-rtPCR positive for five days or more [25] . The finding of MERS-CoV viral RNA in serum provides an avenue for retrospective PCR-based studies if respiratory samples are unavailable [83] . RNAemia may also correlate with disease severity; signs of virus were cleared from the serum of a recovered patient, yet lingered until the death of another [92] .

Clinically suspected MERS cases may return negative results by RT-rtPCR. Data have shown one or more negative URT samples may be contradicted by further URT sampling or the use of LRT samples, which is preferred [2, 43, 93] . Higher viral loads occur in the LRT compared to the URT. [22, 69, 88, 94] This fits with the observation that the majority of disease symptoms are reported to manifest as systemic and LRT disease [21] . However, on occasion, even LRT specimens from MERS cases may initially be negative, only to later become positive by RT-PCR [95] . This may be due to poor sampling when a cough is absent or non-productive or because the viral load is low [95] . Despite this both the largest human MERS-CoV studies [32, [96] [97] [98] and smaller ones [22, 25, 99] , use samples from the URT. It is then noteworthy that one study reported an association between higher loads in the URT and worse clinical outcome including intensive care and death [94] . At writing, no human data exist to define whether the virus replicates solely or preferentially in the LRT or URT, or replicates in other human tissues in vivo although MERS-CoV RNA has been detected from both the URT and LRT in a macaque monkey model [100] .The distribution of DPP4 in the human upper airways is also not well described.

Individual human case studies report long periods of viral shedding, sometimes intermittently and not necessarily linked to the presence of disease symptoms. [25, 69, 99, 101] In one instance, a HCW shed viral RNA for 42 days in the absence of disease [99] . It is an area of high priority to better understand whether such cases are able to infect others. Over three quarters of MERS cases shed viral RNA in their LRT specimens (tracheal aspirates and sputum) for at least 30 days, while only 30 % of contacts were still shedding RNA in their URT specimens [91, 102] .

In the only study to examine the effect of sample type on molecular analysis, 64 nasopharyngeal aspirates (NPA; an URT sample), 30 tracheal aspirates, 13 sputa and three BAL were examined. The tracheal aspirates and BAL returned the highest viral load values followed by NPA and sputum. Unsurprisingly, higher viral loads generally paralleled whole genome sequencing and culture success and, in NPA testing, were significantly correlated with severe disease and death [49, 94, 103] . This study demonstrated the importance of LRT sampling for whole genome sequencing.

When tested, samples positive for MERS-CoV are often negative for other pathogens [2, 25, 93, 104] . However, many studies make no mention of additional testing for endemic

human respiratory viruses [21, 23, 73, 105] . When viruses are sought, they have included human herpesvirus (HHV), rhinoviruses (HRV), enteroviruses (EV), respiratory syncytial virus (RSV), parainfluenzavirus types 1, 2 and 3 (PIVs),influenzaviruses (IFVs), endemic HCoVs, adenoviruses (AdVs) metapneumovirus (MPV) and influenza A\H1N1 virus; co-detections with MERS-CoV have been found on occasion [2, 22, 37, 69, 97] . Bacterial testing is sometimes included (for example, for Legionella and Pneumococcus) but the impact of bacterial co-presence is also unclear [22, [104] [105] [106] . Further testing of the LRT sample from the first MERS case used IFA to screen for some viruses (negative for IFV, PIVs, RSV and AdVs) and RT-PCR for others (negative for AdV, EVs, MPV and HHVs) [18] . RT-PCR also detected MERS-CoV. The WHO strongly recommends testing for other respiratory pathogens [53] but with this recommendation often discounted, there are limited data to address the occurrence and impact of co-infections or alternative viral diagnoses among both MERS cases and their contacts. Little is known of other causes of MERS-like pneumonia in the KSA or of the general burden of disease due to the known classical respiratory viruses.

Testing of adult pilgrims performing the Hajj in 2012 to 2014 has not detected any MERS-CoV. In 2012, nasal swabs from 154 pilgrims collected prior to leaving for or departing from the KSA were tested [47] . In 2013, testing was significantly scaled up with 5,235 nasopharyngeal swabs from 3,210 incoming pilgrims and 2,025 swabs from outgoing pilgrims tested [98] . It should be noted that most pilgrims arrived from MERS-free countries. A further 114 swabs were taken from pilgrims with influenza-like illness [96, 107] . In earlier Hajj gatherings, it was found that influenza viruses circulated widely, whilst other viruses, often rhinoviruses, circulated more selectively, interpreted as indicating their importation along with foreign pilgrims. [107] [108] [109] Over time, increased influenza vaccination has been credited for a fall in the prevalence of influenza like illnesses among Hajj pilgrims. [110] A LRT sample is often not collected for these studies [98, 107, 109] , so false negative findings are a possibility although little is known about the initial site of MERS-CoV infection and replication; it may have been assumed it was the LRT because disease was first noticed there but the URT may be the site of the earliest replication.

In Jeddah between March and July 2014 (hereafter called the Jeddah-2014 outbreak; Fig. 3), there was a rapid increase in MERS cases, accompanied by intense screening; approximately 5,000 samples from in and around the region were tested in a month yielding around 140 MERS-CoV detections (~3 % prevalence) [111] . Among 5,065 individuals sampled and tested across the KSA between October 2012 and September 2013,108 (2.1

%) detections were made in a hospital-centric population which included hospitalized cases (n = 2,908; 57.4 %), their families (n = 462; 9.1 %) and associated HCWs (n = 1,695; 33.5 %) [32] . Among the detections, 19 (17.8 %) were HCWs and 10 (9.3 %) were family contacts [32] .

The 2-3 % prevalence of active MERS-CoV infections is not dissimilar to the hospital-based prevalence of other human CoVs. [112] However, the proportion of deaths among those infected with MERS-CoV is much higher than that known for the HCoVs NL63, HKU1, 229E or OC43 in other countries, and even above that for SARS-CoV; it is not a virus that could reasonably be described as a "storm in a teacup". It is the low transmission rate that has prevented worldwide spread, despite many "opportunities".

Very early in the MERS outbreak, some animals were highly regarded as either the reservoir or intermediate host(s) of MERS-CoV with three of the first five cases having contact with DCs [73, 113, 114] . Today, animal MERS-CoV infections must be reported to the world organization for animal health as an emerging disease [115] . A summary of the first MERS cases reported by the WHO defined animal contact with humans as being direct and within 10 days prior to symptom onset [20] . This definition made no specific allowance for acquisition from DCs through a droplet-based route, which is very likely route for acquisition of a virus that initially and predominantly causes respiratory disease [23] . Camels are known to produce high levels of MERS-CoV RNA in their URT and lungs [116] . Providing support for a droplet transmission route and perhaps indicating the presence of RNA in smaller, drier droplet nuclei, MERS-CoV RNA was identified in a high volume air sample collected from a barn housing an infected DC [117] . The precise source from which humans acquire MERS-CoV remains poorly studied but it seems likely that animal and human behavioural factors may play roles (Fig. 3) [118] . These factors may prove important for human cases who do not describe any DC contact [119] nor any contact with a confirmed case. Whether the WHO definition of animal contact is sufficient to identify exposure to this respiratory virus remains unclear. Wording focuses on consumption of DC products but does not specifically ascribe risk to a droplet route for acquisition of MERS-CoV from DC [120] . Some MERS patients are listed in WHO disease notices as being in proximity to DCs or farms, but the individuals have not described coming into contact with the animals. No alternative path for acquiring infection is reported in many of these instances. What constitutes a definition of "contact" during these interviews has been defined for one study [72] . Despite this lack of clarity, the WHO consider that evidence linking MERS-CoV transmission between DCs to humans is irrefutable (Fig. 4) [120] .

The possibility that bats were an animal host of MERS-CoV was initially widely discussed because of the existing diversity of coronaviruses known to reside among them [121] [122] [123] [124] . Conclusive evidence supporting bats as a source for human infections by MERS-CoV has yet to be found, but bats do appear to host ancestral representatives [53, 125] . However, these are not variants of the same virus nor always within the same phylogenetic lineage as MERS-CoV; they are each a genetically distinct virus. Bat-to-human infection by MERS-CoV is a purely speculative event. The only piece of MERS-CoV-specific evidence pointing to bats originates from amplification of a 190 nt fragment of the RNA-dependent RNA polymerase gene of the MERS-CoV genome, identified in a faecal pellet from an insectivorous Emballonuridae bat, *Taphozous perforatus* found in Bisha, the KSA [121] . While very short, the sequence of the fragment defined it as a diagnostic discovery. Subsequently a link to DCs was reported [85] and that link has matured into a verified association [38, 126] (Fig. 4) .

(See figure on previous page.) Fig. 3 Monthly detections of MERS-CoV (blue bars) and of cases who died (red bars) with some dates of interest marked for 2012 to 4 th September 2015. An approximation of when DC calving season [128] and when recently born DCs are weaned is indicated. Spring (green) and summer (orange) in the Arabian Peninsula are also shaded. Note the left-hand y-axis scale for 2014 and 2015 which is greater than for 2012/13. Sources of these public data include the WHO, Ministries of Health and FluTrackers [207] [208] [209] . Earlier and subsequent versions of this chart are maintained on a personal blog [210] . Modified and reprinted from Mackay IM, Arden KE. Middle East respiratory syndrome: An emerging coronavirus infection tracked by the crowd. *Virus Res* 2015 Vol 202:60-88 with permission from Elsevier [5] DCs, which make up 95 % of all camels, have a central presence in the Arabian Peninsula where human-DC contact ranges from little to close [119] . Contact may be commonplace and could occur in variety of ways (Fig. 4a) . There are several large well-attended festivals, races, sales and parades which feature DCs and DCs are also kept and bred close to populated areas in the KSA [127, 128] . DC milk and meat are widely consumed and the older DC is an animal of ritual significance after the Hajj pilgrimage [129] . However, MERS-CoV infection frequency is reportedly much lower than is the widespread and frequent habit of eating, drinking and preparing DC products. Daily ingestion of fresh unpasteurized DC milk is common among the desert Bedouin and many others in the KSA. DC urine is also consumed or used for supposed health benefits. Despite camel butchery being a local occupation, neither butchers nor other at-risk groups are identifiable among MERS cases; this may simply be a

reporting issue rather than an unexplainable absence of MERS. A small case-control study published in 2015 identified direct DC contact, and not ingestion of products, to be associated with onset of MERS [38] .

The first sero-survey of livestock living in the Middle East region was conducted during 2012-2013 [85] . DCs were sampled from a mostly Canary Island-born herd and from Omani DCs (originally imported from the Horn of Africa) [85] . A neutralising antibody assay found only 10 % of strongly seropositive Canary Island [5] . b Camel-to-human infections appear to be infrequent, while human-to-human spread of infection is regularly facilitated by poor IPC in healthcare settings where transmission is amplified, accounting for the bulk of cases. There are human MERS cases that do not fall into either category of source and it is unclear if these acquired infection through some entirely separate route, or from cases that escaped diagnosis. c Hypothetical ways in which subclinical (when infection may not meet a previously defined clinical threshold of signs and/or symptoms) or asymptomatic (no obvious signs or measured, noticed or recalled symptoms of illness) MERS-CoV infection may be implicated in transmission DC sera could neutralise MERS-CoV while all Omani DC sera had high levels of specific MERS-CoV neutralizing antibody [85] . This indicated that DCs had in the past been infected by MERS-CoV, or a very similar virus.

Since this study, a host of peer-reviewed reports have looked at both DCs and other animals, and the possibility that they may host MERS-CoV infection. Seropositive DCs have been found throughout the Arabian Peninsula including Oman, the KSA, Qatar, Jordan, the United Arab Emirates (UAE), Kuwait as well as Sudan, Somalia, Egypt, Tunisia, Nigeria, Kenya and Ethiopia in Africa and the Canary Islands [85, [130] [131] [132] [133] [134] . Other animals tested include sheep, cows, pigs, horses, donkeys, mules, birds, water buffalo, goats, Bactrian camels, llamas and guanaco (south American camelids) but none had detectable neutralising antibody against MERS-CoV [4, 74, 78, 85, 86, 135, 136] . No virology or serology studies of human samples from areas in Africa where there are camels with a history of MERS-CoV have been reported to date. However, an absence of unexplained pneumonia that may be attributable to MERS-CoV infection may not signal the absence of virus among humans in each country but simply reflect a lack of expensive epidemiology studies conducted by resource-poor countries. It is thus unclear whether MERS-CoV, or an antigenically related CoV, is an unrecognized pathogen in these regions, perhaps circulating for even longer than it has been known in the Arabian Peninsula [133] .

MERS-CoV RNA has also been detected in DC samples, and recovery of infectious virus has also been achieved from DC samples [4, 77, 117, 132, [137][138][139][140][141]. From some of these, full or majority length genomes of MERS-CoV have been sequenced [77, 137, 138]. DC versions of MERS-CoV were found to be as similar to each other, as were variants detected from different humans over time and across distance.

Antibody screening assays have also detected crossreactive antibodies in sera. These were identified as such by screening sera against similar viruses, for example BCoV or HCoV-OC43 (as an antigenic facsimile for BCoV). It is possible that other MERS-CoV-like viruses also reside within DCs, but this does not detract from the definitive finding of MERS-CoV genetic sequences in both DCs and humans [117, 142, 143].

Screening studies have shown that juvenile DCs are more often positive for virus or viral RNA while older DCs are more likely to be seropositive and RNA or virus negative [76, 77, 144]. In adult DCs, MERS-CoV RNA has been detected among animals with pre-existing antibody, suggesting re-infection is possible [77, 144]. Viral loads among positive DCs can be very high [4, 76, 77, 139, 144] and DCs have been found positive both when ill with URT respiratory signs [77, 117, 142, 145] or when apparently healthy [137]. These findings indicate DCs host natural MERS-CoV infections. Furthermore, stored DC sera have revealed signs of MERS-CoV in DCs which date back over three decades (the earliest collected in 1983) [4, 133, 135]. Older sera have not been tested and so precisely how long DCs have been afflicted by MERS-CoV, whether the virus is enzootic among them, introduced to them decades or centuries ago from bats in Africa or the Arabian Peninsula, or they are the subject of regular but short-lived viral incursions from an as yet unknown host, cannot be answered.

Researchers sought to determine a direction for infection; were DCs transmitting virus to humans or were humans infecting DCs? At a Qatari site, a farm owner and his employee became ill in mid-October 2013 and tested positive for MERS-CoV RNA in a sputum and throat swab sample, respectively. RT-rtPCRs found MERS-CoV RNA in 11 of 14 positive DC nasal swabs at the farm; six (43 %) positive by two or more assays [138]. The results indicated a recent outbreak had occurred in this herd; the first indication of MERS-CoV RNA found within DCs with a temporal association to human infections. Three positive DC samples were confirmed by sequencing a 358 nt portion of the spike gene; these sequences were identical to each other, again with close homology to other human and DC

MERS-CoV sequences [138] . The DCs and human contacts yielded ORF1a and ORF4b sequences differing by only a single nucleotide each, clustering closely with the Hafr-Al-Batin_1_2013 variant [138] . Subsequent case studies found evidence of a concurrent human and DC infection and the direction of that infection was inferred to be from the ill DCs and to their human owners [117, 142, 146] . Partial genome sequences indicated that a human and a MERS-CoV RT-rtPCR positive DC had been infected by a variant of the same virus, harbouring the same distinct pattern of nucleotide polymorphisms. [142] All nine DC in the owner's herd, serially sampled, reacted in a recombinant S1 antigen ELISA, with the two animals that had been RT-rtPCR positive showing a small, verifiable rise in antibody titre [142] . A rise in titre theoretically begins 10 to 21 days after DC infection [142] . The authors suggested that the rise in titre in DC sera which occurred alongside a declining RNA load, while the patient was actively ill and hospitalized, indicated that the DCs were infected first followed by the owner [117, 142] . BCoV antibodies were also present, and rising in one of the two RT-rtPCR positive animals but no animal's antibodies could neutralise BCoV infection [142] .

Camel calving season occurs in the winter months (between late October and late February; Fig. 3) and this may be a time when there is increased risk to humans of spill-over due to new infections among naïve DC populations [128] . What role maternal camel antibody might play in delaying infection of calves remains unknown [128, 142] . Juvenile DCs appear to host active infection more often than adult DCs and thus the sacrificial slaughter of DCs, which must be five years of age or older (termed a thane), may not be accompanied by significant risk of exposure to infection. In contrast to earlier results, slaughterhouse workers who kill both younger and older DCs, may be an occupational group with significantly higher incidence of seropositivity to MERS-CoV when animals have active MERS-CoV infections [129, 139, [147] [148] [149] . Expanded virological investigations of African DCs may lead to more seropositive animals and geographic areas in which humans may be at risk. It is possible that there are areas where humans already harbour MERS-CoV infections that have not been identified because of an absence of laboratory surveillance. Virological investigations of bats may lead to findings of ancestral viruses and viral 'missing links' and identifying any other animal sources of zoonotic spread is important to inform options for reducing human exposures [56, 76] .

Infectious MERS-CoV added to DC, goat or cow milk and stored at 4°C could be recovered at least 72 h later and, if stored at 22°C, recovery was possible for up to 48 h [150] . MERS-CoV titre decreased somewhat when recovered from milk at 22°C but pasteurization

completely ablated MERS-CoV infectivity [150] . In a subsequent study, MERS-CoV RNA was identified in the milk, nasal secretion and faeces of DCs from Qatar [151] .

A single study has examined the ability of MERS-CoV to survive in the environment [150] . Plastic or steel surfaces were inoculated with 10⁶ TCID₅₀ of MERS-CoV at different temperature and relative humidity (RH) and virus recovery was attempted in cell culture. At high ambient temperature (30°C) and low RH (30 %) MERS-CoV remained viable for 24 h [150] . By comparison, a well known and efficiently transmitted respiratory virus, influenza A virus, could not be recovered in culture beyond four hours under any conditions [150] . Aerosol experiments found MERS-CoV viability only decreased 7 % at low RH at 20°C. In comparison, influenza A virus decreased by 95 % [150] . MERS-CoV survival is inferior to that previously demonstrated for SARS-CoV [152] . For context, pathogenic bacteria can remain viable and airborne for 45 min in a coughed aerosol and can spread 4 m. MERS-CoV's ability to remain viable over long time periods gives it the capacity to thoroughly contaminate a room's surfaces when occupied by an infected and symptomatic patient [153] . Whether MERS-CoV can remain adrift and infectious for extended periods (truly airborne) remains unknown. Such findings expand our understanding of the possibilities for droplets to transmit respiratory viruses in many settings, including hospital waiting rooms, emergency departments, treatment rooms, open intensive care facilities and private patient rooms. The nature and quality of air exchange, circulation and filtration are important variables in risk measurement and reduction as is the use of negative pressure rooms to contain known cases. Droplet spread between humans is considered the mechanism of human-to-human transmission and the need for droplet precautions was emphasized after the Al-Ahsa hospital, the KSA and the South Korean outbreaks [21, 23, 154, 155] . By extrapolation, aerosol-generating events involving DCs (urination, defecation, and preparation and consumption of DC products) should be factored into risk measurement and reduction efforts and messaged using appropriate context. The provision of evidence supporting the best formulation of personal protective equipment to be worn by HCWs who receive, manage or conduct procedures on infectious cases remains a priority.

MERS-CoV was found and characterized because of its apparent association with severe, and therefore more obvious, illness in humans; we were the canaries in the coal mine. Sero-assays and prospective cohort studies have yet to determine the extent to which milder or asymptomatic cases contribute to MERS-CoV transmission chains. However, transmission of MERS-CoV is defined as sporadic (not sustained), intra-familial, often

healthcare associated, inefficient and requiring close and prolonged contact [22, 31, 63, 93, 97, 102, 156] In a household study, 14 of 280 (5 %) contacts of 26 MERS-CoV positive index patients were RNA or antibody positive; the rate of general transmission, even in outbreaks is around 3 % [31] . It seems that the majority of human cases of MERS-CoV, even when numbers appear to increase suddenly, do not readily transmit to more than one other human so to date, the localized epidemic of MERS-CoV has not been self-sustaining [157] [158] [159] [160] [161] . That is to say, the basic reproduction number (R_0) -the average number of infections caused by one infected individual in a fully susceptible population has been close to one throughout various clusters and outbreaks. If R_0 was greater than 1, a sustained increase in case numbers would be expected. Some R_0 calculations may be affected by incomplete case contact tracing, limited community testing and how a case is defined. That MERS has had a constant presence in the Arabian Peninsula since 2012 is due to ongoing, sporadic spill-over events from DCs amplified by poorly controlled hospital outbreaks.

The first known MERS human-to-human transmission event was one characterized by acute LRT disease in a healthcare setting in Jordan. In stark contrast, a sero-survey of HCW who were sometimes in close and prolonged contact with the first, fatal MERS-CoV case in 2012 [162] , found none of the HCW had seroconverted four months later, despite an absence of eye protection and variable compliance with required PPE standards [162] .

Early on in the MERS story, samples for testing were mostly collected from patients with severe illness and not those with milder acute respiratory tract infections. Contacts of confirmed MERS cases were often observed for clinical illness, but not tested. These omissions may have confounded our understanding of MERS-CoV transmission and biased early data towards higher numbers of seriously ill and hospitalized patients, inflating the apparent proportion of fatal cases. Case-control studies were not a focus. As testing paradigms changed and contacts were increasingly tested, more asymptomatic and mild infections were recognized [163] .

A rise in the cases termed asymptomatic (which enlarge the denominator for calculations of the proportion of fatal cases, defined in [164]) resulted in a drop in the proportion of fatal cases during the Jeddah-2014 outbreak. Historically, such rises are consistent with changing definitions and laboratory responses and clinical management of a newly discovered virus infection that was first noted only among the severely ill. Upon follow-up,

over three-quarters of such MERS-CoV RNA positive people did recall having one or more symptoms at the time, despite being reported as asymptomatic [165] raising some question over the reliability of other reported data.

The proportion of fatal MERS cases within the KSA compared to outside the KSA, as well as the age, and sex distribution change in different ways when comparing MERS outbreaks. Approximately 43 % of MERS cases (549 of 1277) in the KSA were fatal between 2012 and December 2015 while 21 % (72 of 330) died among those occurring outside of the KSA. The total number of male cases always outnumber females and the proportion of male deaths is always greater than the proportion of females who die. However the proportion of male deaths from total males with MERS is a similar figure to that for females. In the KSA, there is a greater proportion of younger males among cases and deaths than were observed from the 2015 South Korean or the Jeddah-2014 outbreaks (Additional file 2: Figure S2). Why these aspects have differed may be due to differences in the time to presentation and diagnosis, the nature and quality of supportive care, the way a person became infected (habits, exposure to a human or zoonotic source, viral load, route of infection) or the extent to which different populations are burdened by underlying diseases [40] .

As a group, HCWs comprised 16 % of MERS cases in the KSA and South Korea. It is apparent that the weekly proportion of infected HCWs increases alongside each steep rise in overall detections (Fig. 5) . In May 2013, the WHO published guidelines for IPC during care of probable or confirmed cases of MERS-CoV infection in a healthcare setting [166] . This is explainable because to date, each case rise has been intimately associated with healthcare-facility related outbreaks [118] . These rises in MERS-CoV detections can decrease the average age during each event because HCWs are usually younger than inpatients with MERS. Healthcare facilities have been a regular target for suggested improvements aimed at improving infection prevention and control (IPC) procedures [115, 118] .

Most of the analysis of MERS-CoV genetics has been performed using high throughput or "deep" sequencing methods for complete genome deduction [167] [168] [169] . MERS-CoV was the first subject of such widespread use of deep sequencing to study an emerging viral outbreak with global reach. The technique can produce genomic [207] [208] [209] . Earlier and subsequent versions of this chart are maintained on a personal blog [210] length coverage in a single experiment with highly repetitious measurement of each nucleotide

position [52, 140] . Despite assays having been published early on, subgenomic sequencing, once the mainstay of viral outbreak studies, has less often been published during MERS-CoV characterization [48] . As more genomes from both humans and DCs have been characterized, two clades have become apparent; A and B (Fig. 6) . Clade A contains only human-derived MERS-CoV genomes from Jordan, while Clade B comprises the majority of human and camel genomes deduced thus far [168] .

Two studies during 2015, one looking at Jeddah-2014 MERS-CoV variants and another looking at a variant exported from South Korea to China, have now identified signs of genetic recombination among MERS-CoV variants. While human and camel whole genome sequences have retained >99 % identity with each other, members of genetically distinct lineages can and do swap genetic material when suitable conditions and coinfections co-occur [170] [171] [172] . Shared identity implies that the major source for human acquisition is the DC, rather than another animal, although more testing of other animal species is needed to confirm that conclusion. Over a month, a DC virus sequenced on different occasions did not change at all indicating a degree of genomic stability in its host, supporting that DCs are the natural, rather than intermediate, host for the MERS-CoV we know today [77] . To date, recombination has been localised to breakpoints near the boundary between ORF1a and ORF1b regions, within the spike gene [170] and in the ORF1b region (Fig. 2) [172] . It is not unexpected that recombination should occur since it is well known among other CoVs [124] and because the majority of MERS-CoV whole genomes collected from samples spanning three years (2012-2015) and from humans, camels and different countries have shown close genetic identity to each other, with just enough subtle variation to support outbreak investigations so long as whole genome sequencing is applied [52, 77, 135, 138, 168, [173] [174] [175] .

Changes in genome sequence may herald alterations to virus transmissibility, replication, persistence, lethality or response to future drugs. If we have prior knowledge of the impact of genetic changes because of thorough characterization studies, we can closely Fig. 6 The genetic relationship between MERS-CoV nucleotide sequences (downloaded from GenBank using the listed accession numbers and from virological.org [212]). This neighbour joining tree was created in MEGA v6 using an alignment of human and DCderived MERS-CoV sequences (Geneious v8.1 [211]). Clades are indicated next to dark (Clade A) or pale (Clade B) blue vertical bars. Camel icons denote genomes from DCs. Healthcare or community outbreaks are boxed and labelled using previously described schemes [212, 213] monitor the genomic regions and better understand any changes in

transmission or disease patterns as they occur. Genetic mutations noted during the largest of human outbreaks, Jeddah-2014, did not impart any major replicative or immunomodulatory changes when compared to earlier viral variants in vitro [156, 176] . However, we understand very little of the phenotypic outcomes that result from subtle genetic change in MERS-CoV genomes. To date no clinical relevance or obvious in vivo changes to viral replication, shedding or transmission has been reported or attributed to mutations or to new recombinant viruses [156] . But vigilance and larger, more contemporary and in vivo studies are needed.

Genome sequence located to a distinct clade were identified from an Egyptian DC that was probably imported from Sudan. This does not fit into either of the current clades [125, 168, 177] . A virus sequenced from a *Neoromicia capensis* bat was more closely related to MERS-CoV than other large bat-derived sequences had been to that point, but the genome of a variant of a MERS-CoV has yet to be discovered and deduced from any bat [125] .

Analyses of MERS-CoV genomes have shown that most single nucleotide differences among variants were located in the last third of the genome (Fig. 2) , which encodes the spike protein and accessory proteins [168] . At least nine MERS-CoV genomes contained amino acid substitutions in the receptor binding domain (RBD) of the spike protein and codons 158 (N-terminal region), 460 (RBD), 1020 (in heptad repeat 1), 1202 and 1208 bear investigation as markers of adaptive change [140, 169] . The spike protein had not changed in the recombinant MERS-CoV genome identified in China in 2015 but was reported to have varied at a higher rate than that for complete MERS-CoV genomes, among South Korean variants [172, 178] . This highlights that subgenomic regions may not always contain enough genetic diversity to prove useful for differentiating viral variants. Despite this, one assay amplifying a 615 nucleotide fragment of the spike S2 domain gene for Sanger sequencing agreed with the results generated by the sequencing of a some full genomes and was useful to define additional sequence groupings [177] .

Genomic sequence can also be used to define the geographic boundaries of a cluster or outbreak and monitor its progress, based on the similarity of the variants found among infected humans and animals when occurring together, or between different sites and times (Fig. 6) [169] . This approach was employed when defining the geographically constrained MERS hospital outbreak in Al-Ahsa, which occurred between 1 st April and 23 rd May 2013, as well as clusters in Buraidah and a community outbreak in Hafr Al-Batin, the

KSA. Genomic sequencing identified that approximately 12 MERS-CoV detections from a community outbreak in Hafr Al-Batin between June and August 2013 may have been triggered by an index case becoming infected through DC contact [175] . Sequencing MERS-CoV genomes from the 2013 Al-Ahsa hospital outbreak indicated that multiple viral variants contributed to the cases but that most were similar enough to each other to be consistent with human-to-human transmission. Molecular epidemiology has revealed otherwise hidden links in transmission chains encompassing a period of up to five months [179] . However, most outbreaks have not continued for longer than two to three months and so opportunities for the virus to adapt further to humans through co-infection and sustained serial passage have been rare [169] . In Riyadh-2014, genetic evidence supported the likelihood of multiple external introductions of virus, implicating a range of healthcare facilities in an event that otherwise looked contiguous [23, 168, 179] . Riyadh is a nexus for camel and human travel and has had more MERS cases than any other region of the KSA to date but also harbours a wide range of MERS-CoV variants [128, 167, 179] . However the South Korean outbreak originated from a single infected person, resulting in three to four generations of cases [180, 181] . Studies of this apparently recombinant viral variant did not find an increased evolutionary rate and no sign of virus adaptation thus the outbreak seems to have been driven by circumstance rather than circumstance together with mutation [181] .

For many MERS cases detected outside the Arabian Peninsula, extensive contact tracing has been performed and the results described in detail. Contact tracing is essential to contain the emergence and transmission of a new virus and today it is supported by molecular epidemiology. Although it is an expensive and time consuming process, contact tracing can identify potential new infections and through active or passive monitoring, react more rapidly if disease does develop. Results of contact tracing to date have found that onward transmission among humans is an infrequent event. For example, there were 83 contacts, both symptomatic and asymptomatic, of a case treated in Germany who travelled from the UAE but no sign of virus or antibody were found in any of them [73] . The very first MERS case had made contact with 56 HCWs and 48 others, but none developed any indication of infection [162] . In a study of 123 contacts of a case treated in France, only seven matched the definition for a possible case and were tested; one who had shared a 20 m² hospital room while in a bed 1.5 m away from the index case for a prolonged period was positive [26] . None of the contacts of the first two MERS cases imported into the USA in 2014 contained any MERS-CoV footprint [182] and none of the 131 contacts of two travellers returning to the Netherlands developed MERS-CoV antibodies or tested RNA positive [25, 183] . Analyses of public data reveal many likely instances of nosocomial

acquisition of infection in the Arabian Peninsula and these data may be accompanied by some details noting contact with a known case or facility. One example identified the likely role of a patient with a subclinical infection, present in a hospital during their admission for other reasons, as the likeliest index case triggering a family cluster [93] . Contact tracing was a significant factor in the termination of a 2015 outbreak involving multiple South Korean hospitals [184] . Such studies demonstrate the necessity of finding and understanding a role for mild and asymptomatic cases, together with restricting close contact or prolonged exposure of infected people to others, especially older family members and friends with underlying disease (Fig. 4c) .

The hospital-associated outbreak in Jeddah in 2014 was the largest and most rapid accumulation of MERS-CoV detections to date. The greatest number of MERS-CoV detections of any month on record occurred in Jeddah in April. The outbreak was mostly (>60 % of cases) associated with human-to-human spread within hospital environments and resulted from a lack of, or breakdown in, infection prevention and control [37, 185, 186] . A rise in fatalities followed the rapid increase in case numbers.

In 2015 two large outbreaks occurred. South Korea was the site of the first large scale outbreak outside the Arabian Peninsula and produced the first cases in both South Korea and China, occurring between May and July 2015. This was closely followed by a distinct outbreak in Ar Riyad province in the KSA which appeared to come under control in early November.

After staying in Bahrain for two weeks, a 68 year old male (68 M) travelled home to South Korea via Qatar, arriving free of symptoms on the 4 th May 2015 [187] . He developed fever, myalgia and a cough nearly a week later (11 th). He visited a clinic as an outpatient between the 12 th and 15 th of May and was admitted to Hospital A on the 15 th [188] . He was discharged from Hospital A on the 17 th then visited and was admitted to the emergency department of Hospital B on the 18 th . During this second stay, a sputum sample was taken and tested positive for MERS-CoV on the 20 th [187, 188] , triggering transfer to the designated isolation treatment facility. Over a period of 10 days, the index case was seen at three different hospitals, demonstrating a key feature of "hospital shopping" that shaped the South Korean outbreak. Approximately 34 people were infected during this time [187] . In total 186 cases were generated in this outbreak, all linked through a single transmission chain to 68 M; 37 cases died [189] . In South Korea, the national

health insurance system provides for relatively low cost medical care, defraying some costs by making family members responsible for a portion of the ministrations of the sick, resulting in them sometimes staying for long periods in the rooms that often have more than four beds in them [24]. Other factors thought to have enabled this outbreak included unfamiliarity of local clinicians with MERS, ease with which the public can visit and be treated by tertiary hospitals, the custom of visiting sick friends and relatives in hospitals, the hierarchical nature of Korean society, crowded emergency rooms, poor IPC measures, a lack of negative pressure isolation rooms and poor inter-hospital communication of patient disease histories [24, [190] [191] [192]. All of the reported transmission occurred across three or four generations and apart from one unknown source, were all hospital-acquired [24, 120, 181, [193] [194] [195]. Few clinical details about these cases have been reported to date and detail on transmission and contact tracing is minimal. The hospitals involved were initially not identified, governmental guidance and actions produced confusing messages and there was very limited communication at all early on which resulted in unnecessary concern, distrust and a distinct economic impact [191, [196] [197] [198]. Early in the outbreak, an infected traveller, the son of an identified case in South Korea, passed through Hong Kong on his way to China where he was located, isolated and cared for in China [91, 199, 200]. No contacts became ill. The outbreak was brought under control in late July/ early August [201] after improved IPC measures were employed, strong contact tracing monitoring and quarantine, expanded laboratory testing, hospitals were better secured, specialized personnel were dispatched to manage cases and international cooperation increased [202, 203]. A review of public data showed that, as for MERS in the KSA, older age and the presence of underlying disease were significantly associated with a fatal outcome in South Korea. [40] Even though R_0 is <1 , super-spreading events facilitated by circumstances created in healthcare settings and characterized by cluster sizes over 150, such as this one, are not unexpected from MERS-CoV infection [204]. The dynamic of an outbreak depends on the R_0 and an individual's viral shedding patterns, contact type and frequency, hospital procedures and population structure and density [204].

In the region of Ar Riyad, including the capital city of Riyadh, a hospital based cluster began, within a single hospital, from late June 2015 [205]. By mid-September there had been approximately 170 cases reported but the outbreak appeared to have been brought under control in November.

It became apparent early on that MERS-CoV spread relatively ineffectively from human-to-human. Despite ongoing and possibly seasonal introduction of virus to the human

population via infected DCs and perhaps other animals yet to be identified, the vast majority of MERS-CoV transmission has occurred from infected to uninfected humans in close and prolonged contact through circumstances created by poor infection control in health care settings. This opportunistic virus has had its greatest impact on those with underlying diseases and such vulnerable people, sometimes suffering multiple comorbidities, have been most often associated with hospitals, creating a perfect storm of exposure, transmission and mortality. It remains unclear if this group are uniquely affected by MERS-CoV or if other respiratory virus infections, including those from HCoVs, produce a similarly serious impact. In South Korea, a single imported case created an outbreak of 185 cases and 36 deaths that had a disproportionate impact on economic performance, community behaviour and trust in government and the health care system. Household human-to-human transmission occurs but is also limited. Educational programs will be essential tools for combatting the spread of MERS-CoV both within urban and regional communities and for the health care setting.

Vigilance remains important for containment since MERS-CoV is a virus with a genetic makeup that has been observed for only three years and is not stable. Among all humans reported to be infected, nearly 40 % have died. Continued laboratory testing, sequencing, analysis, timely data sharing and clear communication are essential for such vigilance to be effective. Global alignment of case definitions would further aid accurate calculation of a case fatality ratio by including subclinical case numbers. Whole genome sequencing has been used extensively to study MERS-CoV travel and variation and although it remains a tool for experts, it appears to be the best tool for the job.

MERS and SARS have some clinical similarities but they also diverge significantly [206] . Defining characteristics include the higher PFC among MERS cases (above 50 % in 2013 and currently at 30-40 %; well above the 9 % of SARS) and the higher association between fatal MERS and older males with underlying comorbidities. For the viruses, MERS-CoV has a broader tropism, grows more rapidly in vitro, more rapidly induces cytopathogenic change, triggers distinct transcriptional responses, makes use of a different receptor, induces a more proinflammatory state and has a delayed innate antiviral response compared to SARS-CoV.

There appears to be a 2-3 % prevalence of MERS-CoV in the KSA with a 5 % chance of secondary transmission within the household. There is an increased risk of infection

through certain occupations at certain times and a much greater chance for spread to other humans during circumstances created by humans, which drives more effective transmission than any R_0 would predict on face value. Nonetheless, despite multiple mass gatherings that have afforded the virus many millions of opportunities to spread, there have remarkably been no reported outbreaks of MERS or MERS-CoV during or immediately after these events. There is no evidence that MERS-CoV is a virus of pandemic concern. Nonetheless, hospital settings continue to describe MERS cases and outbreaks in the Arabian Peninsula. As long as we facilitate the spread of MERS-CoV among our most vulnerable populations, the world must remain on alert for cases which may be exported more frequently when a host country with infected camel reservoirs is experiencing human clusters or outbreaks.

The MERS-CoV appears to be an enzootic virus infecting the DC URT with evidence of recent genetic recombination. It may once have had its origins among bats, but evidence is lacking and the relevance of that to today's ongoing epidemic is academic. Thanks to quick action, the sensitive and rapid molecular diagnostic tools required to achieve rapid and sensitive detection goal have been in place and made widely available since the virus was reported in 2012. RT-PCR testing of LRT samples remains the gold standard for MERS-CoV confirmation. Serological tools continue to emerge but they are in need of further validation using samples from mild and asymptomatic infections and a densely sampled cohort study to follow contacts of new cases may address this need. Similarly, the important question of whether those who do shed MERS-CoV RNA for extended periods are infectious while appearing well, continues to go unanswered. It is even unclear just how many 'asymptomatic' infections have been described and reported correctly which in turn raises questions about the reliability of other clinical data collection to date. While the basic virology of MERS-CoV has advanced over the course of the past three years, understanding what is happening in, and the interplay between, camel, environment and human is still in its infancy.

Additional file 1: Figure S1 . The

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On the Coronavirus (COVID-19) Outbreak and the Smart City Network: Universal Data Sharing Standards Coupled with Artificial Intelligence (AI) to Benefit Urban Health Monitoring and Management

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Abstract: As the Coronavirus (COVID-19) expands its impact from China, expanding its catchment into surrounding regions and other countries, increased national and international measures are being taken to contain the outbreak. The placing of entire cities in ‘lockdown’ directly affects urban economies on a multi-lateral level, including from social and economic standpoints. This is being emphasised as the outbreak gains ground in other countries, leading towards a global health emergency, and as global collaboration is sought in numerous quarters. However, while effective protocols in regard to the sharing of health data is emphasised, urban data, on the other hand, specifically relating to urban health and safe city concepts, is still viewed from a nationalist perspective as solely benefiting a nation’s economy and its economic and political influence. This perspective paper, written one month after detection and during the outbreak, surveys the virus outbreak from an urban standpoint and advances how smart city networks should work towards enhancing standardization protocols for increased data sharing in the event of outbreaks or disasters, leading to better global understanding and management of the same.

Text: The novel Coronavirus outbreak, (previously known as the 2019-nCoV and later renamed COVID-19 during the writing of this manuscript) is leading to the closure of entire cities in China, and causing stringent measures to be taken in others. While in distant

different continents, far from China where the virus was first reported, places are being placed on high alert. In Wuhan, where the virus broke, schools, roads and markets have been shut down [1] . The same is true in Hong Kong, Beijing and Hubei Province amongst surrounding areas, as precautionary measures are being emphasized to ensure that the spread of the virus is minimized, and complete and accurate information on the virus is being obtained [2] . However, the rate of spread of the virus and the uncertainties surrounding the entire situation has led the World Health Organization (WHO) on 30 January 2020 to declare the Coronavirus outbreak a 'Global Public Health Emergency'. WHO determined, however, not to declare the outbreak a 'Public Health Emergency of International Concern' (PHEIC) which is a higher level of declaration. A PHEIC is defined as "an extraordinary event which is determined to constitute a public health risk to other States through the international spread of disease and to potentially require a coordinated international response" whose scope may include: serious, sudden, unusual or unexpected; carries implications for public health beyond the affected State's national border; and may require immediate international action [3] .

With the world having experienced some notable influenza pandemics in the past, a Global Initiative on Sharing All Influenza Data (GISAID) platform [4] was established and was instrumental in the rapid sharing of information by the Chinese scientists regarding the emergence of the COVID-19 virus. Through this platform, scientists from other regions were observed to gain access to information and are, subsequently, able to act in a much faster capacity; like in the case of scientists from the Virus Identification Laboratory based at Doherty Institute, Australia, who managed to grow a similar virus in the laboratory after accessing the data shared by the Chinese scientists [5] .

Beyond the aspect of pandemic preparedness and response, the case of COVID-19 virus and its spread provide a fascinating case study for the thematics of urban health. Here, as technological tools and laboratories around the world share data and collectively work to devise tools and cures, similar efforts should be considered between smart city professionals on how collaborative strategies could allow for the maximization of public safety on such and similar scenarios. This is valid as smart cities host a rich array of technological products [6, 7] that can assist in early detection of outbreaks; either through thermal cameras or Internet of Things (IoT) sensors, and early discussions could render efforts towards better management of similar situations in case of future potential outbreaks, and to improve the health fabric of cities generally. While thermal cameras are not sufficient on their own for the detection of pandemics -like the case of the COVID-19,

the integration of such products with artificial intelligence (AI) can provide added benefits. The fact that initial screenings of temperature is being pursued for the case of the COVID-19 at airports and in areas of mass convergence is a testament to its potential in an automated fashion. Kamel Boulos et al. [8] supports that data from various technological products can help enrich health databases, provide more accurate, efficient, comprehensive and real-time information on outbreaks and their dispersal, thus aiding in the provision of better urban fabric risk management decisions.

The above improvements in the healthcare sector can only be achieved if different smart city products are fashioned to support standardized protocols that would allow for seamless communication between themselves. Weber and Podnar Žarko [9] suggest that IoT devices in use should support open protocols, and at the same time, the device provider should ensure that those fashioned uphold data integrity and safety during communication and transmission. Unfortunately, this has not been the case and, as Vermesan and Friess [10] explain, most smart city products use proprietary solutions that are only understood by the service providers. This situation often creates unnecessary fragmentation of information rendering only a partial integrated view on the dynamics of the urban realm. With restricted knowledge on emergent trends, urban managers cannot effectively take decisions to contain outbreaks and adequately act without compromising the social and economic integrity of their city. This paper, inspired by the case of the COVID-19 virus, explores how urban resilience can be further achieved, and outlines the importance of seeking standardization of communication across and between smart cities.

With the advent of the digital age and the plethora of Internet of Things (IoT) devices it brings, there has been a substantial rise in the amount of data gathered by these devices in different sectors like transport, environment, entertainment, sport and health sectors, amongst others [11]. To put this into perspective, it is believed that by the end of 2020, over 2314 exabytes (1 exabyte = 1 billion gigabytes) of data will be generated globally [12] from the health sector. Stanford Medicine [12] acknowledges that this increase, especially in the medical field, is witnessing a proportional increase due to the increase in sources of data that are not limited to hospital records. Rather, the increase is being underpinned by drawing upon a myriad and increasing number of IoT smart devices, that are projected to exponentially increase the global healthcare market to a value of more than USD \$543.3 billion by 2025 [13]. However, while the potential for the data market is understood, such issues like privacy of information, data protection and sharing, and obligatory requirements of healthcare management and monitoring, among others, are critical. Moreover, in the

present case of the Coronavirus outbreak, this ought to be handled with care to avoid jeopardizing efforts already in place to combat the pandemic. On the foremost, since these cut across different countries, which are part of the global community and have their unique laws and regulations concerning issues mentioned above, it is paramount to observe them as per the dictate of their source country's laws and regulations; hence, underlining the importance of working towards not only the promoting of data through its usage but also the need for standardized and universally agreed protocols.

While the significance of such data in advancing efficiency, productivity and processes in different sectors is being lauded, there are criticisms arising as to the nature of data collection, storage, management and accessibility by only a small group of users. The latter particularly includes select ICT corporations that are also located in specific geographies [6, [14] [15] [16] [17] . These criticisms are justified, as in recent years, big data is seen as the new 'gold rush' of the 21st century and limiting its access means higher economic returns and increased influence and control at various scales to those who control data. These associated benefits with big data are clearly influencing geopolitical standings, in both corporate and conventional governance realms, and there is increased competition between powerful economies to ensure that they have the maximum control of big data. As case in point is the amount of 'push and pull' that has arisen from Huawei's 5G internet planned rollout [18] . Though the latter service offers unprecedented opportunities to increase internet speeds, and thereby influence the handling of big data, countries like the U.S. and some European countries that are key proponents and players in global political, economic and health landscapes, are against this rollout, arguing that it is a deceptive way of gathering private data under the guise of espionage. On this, it has been noted that the issue of data control and handling by a few corporations accords with their principles of nationalism, and that these work for their own wellbeing as well as to benefit the territories they are registered in. Therefore, geopolitical issues are expected on the technological front as most large data-rich corporations are located in powerful countries that have influence both economically, health-wise and politically [19] [20] [21] . Such are deemed prized tokens on the international landscape, and it is expected that these economies will continue to work towards their predominant control as much as possible. On the health sector, the same approach is being upheld where critical information and data are not freely shared between economies as that would be seen to be benefiting other in-competition economies, whereas different economies would cherish the maximization of benefits from such data collections.

In addition to the obvious deep-rooted social issues related to nationalism, other challenges include the increasing movement of people globally that is being enhanced by reduced costs and higher speed. In particular, these challenges are more pronounced when it comes to public health. This is because most of the health-related data collected not only can compromise local nations, but also captures those of travelers. In such cases, in a bid to improve the health status of a nation, it becomes paramount to factor in data from other regions necessitating unhindered sharing of this data.

Such data-sharing truth is emphasized in situations like the recent case of Coronavirus outbreak threatening the global health environment, facilitated by air transportation. The virus was first reported in Wuhan, China, and in a matter of three weeks (by 17th January 2020) over 300 cases were confirmed in that region, and 10 days later (26th January 2020), a total of 2014 cases of Coronavirus have been reported, with 684 of those being confirmed, and with 29 reported outside China. The fatalities from the virus stands at 56 as of 26th January 2020 [22] . The virus had then been confirmed in various countries including Taiwan, South Korea, Japan, Thailand, France, the United States, Singapore and Vietnam [23] .

In the above case, though major cities are known to prepare themselves for potential outbreaks, their health policies and protocols are observed to diverge from one another. Thus, without a global collaborative approach, progress towards working for a cure and universally acceptable policy approach can take longer. Such fears, of a lack of international collaboration, were highlighted by the World Health Organization (WHO) during an emergency meeting in Geneva on 22nd January 2020 to determine whether the virus outbreak had reached a level warranting international emergency concern. However, WHO was satisfied that China was being proactive in this case, unlike in 2002, when China withheld information on the outbreak for far too long, causing delays in addressing the epidemic [3] . As in this instance, it is the opinion in this paper that if there was seamless collaboration and seamless sharing of data between different cities, it would not warrant such a high-level meeting to result in action, and instead, a decision could have been made much earlier. On this, the saddest part is that some global cities are less prepared to handle the challenges posed by this type of outbreak for lack of information on issues like symptoms of the virus, the protective measures to be taken, and the treatment procedures that an infected person should be processed through, amongst other issues.

The timely response by stakeholders in regard to this new outbreak are commendable compared to previous cases. The latter includes the Severe Acute Respiratory Syndrome (SARS) outbreak in 2002 that took substantial time (from November 2002 to April 2003) to identify and be dealt with [24] ; the Ebola outbreak in West Africa in 2013 that took months to determine; and the Zika Virus that was first reported in 2014 before being successfully identified in 2015.

With the Coronavirus (COVID-19) , it took only 17 days (31st December 2019 to 17th January 2020) to be identified. The sharing of data has also been quicker, as immediately after the virus' genetic sequence was discovered, Chinese scientists were able to share the information with the WHO, thus helping in its identification and enabling the auctioning of precautionary measures in other countries. Latest technological tools have also allowed for the receipt of information in realtime, in contrast to traditional epidemiological approaches that would have required months to identify the outbreak type [25] . Similarly, though substantial data and information on the disease has been shared, Wetsman [26] acknowledges that there is a lack of some vital information, like the ease of spread of the virus from person-to-person, and this is a key to containing the disease as interactions between people from different parts of the globe are still active. This hindrance can be made further possible as many cities advance in their smart and safe city model implementation towards constructing sufficient soft and hard urban infrastructures equipped with, for example, thermal imagery sensors to allow for early detections. However, while that is the case, data access to many is a challenge because the information is often seen as being sensitive for national security reasons, whilst at the same time, acknowledging that a virus outbreak is an equal threat to both national security and the economy.

The outbreak of any disease has significant impacts on local economies across the globe. For instance, when SARS (Severe Acute Respiratory Syndrome) (SARS-CoV) broke in China in 2002, it was estimated, that the Asian region incurred tremendous negative impacts socially, health-wise and economically, potentially amounting to Asian regional economy losses of between USD \$12-18 billion from tourism, travel and retail sales industries alone [27] . The Zika virus outbreak, spread by daytime-active Aedes mosquitoes, is estimated to have cost equator-belt local economies in affected areas between USD \$7 and USD \$18 billion [28] . The Ebola virus (or Ebola hemorrhagic fever (EHF)) caused an estimated loss of USD \$2.2 billion in GDP in three West African economies (Guinea, Liberia and Sierra Leone) in 2015 alone [29] . In regard to the current epidemic of Coronavirus, though it is too early

to quantify or project its impacts on the global economy, there are fears that it may take the precedent of other outbreaks where billions of dollars will be lost. The foundations for this escalating loss can be witnessed in the rapid growth of travel bans being enacted by some countries and their international airports, especially specifically restricting people from visiting the affected regions in China and their growth into general non-Chinese travel movements. On this, noting that the outbreak came almost on the eve of the Lunar New Year celebrations, and that it had been estimated that over 400 million people were expected to travel in different parts of the world and China to observe this festivity, the majority have had to reconsider their options as to flights, hotels and entertainment events due to service provider cancellations [30] . Those who had already booked their flights are expected to receive their refunds following the directive by the Civil Aviation Administration of China, however, this move has already affected the share value of Chinese airline companies [30] .

The above impacts demonstrate that the issues of virus outbreaks transcend urban safety and impacts upon all other facets of our urban fabric. Therefore, it becomes paramount to ensure that the measures taken to contain a virus transcend nationalist agendas where data and information sharing is normally restricted, to a more global agenda where humanity and global order are encouraged. With such an approach, it would be easier to share urban health data across geographies to better monitor emerging health threats in order to provide more economic stability, thereby ensuring no disruptions on such sectors like tourism and travel industries, amongst others. This is possible by ensuring collaborative, proactive measures to control outbreak spread and thus, human movements. This would remove fears on travelers, and would have positive impacts upon the tourism industry, that has been seen to bear the economic brunt whenever such outbreaks occur. This can be achieved by ensuring that protocols on data sharing are calibrated to remove all hurdles pertaining to sharing of information. On this, Lawpoolsri et al. [31] posits that such issues, like transparency, timelessness of sharing and access and quality of data, should be upheld so that continuous monitoring and assessment can be pursued.

Virus outbreaks in recent years have shown that, in the urban realm, data, including health data, can be sourced from diverse places. Presently, in the case of Coronavirus (COVID-19) outbreak, data is being collected from airports through screening and monitoring, through the use of smart sensors installed in airport infrastructures and from personnel working in those air/seaports. For instance, it has been reported that in the U.S.A., screening is being

carried out at 20 different airports to ensure that possible affected people are intercepted for quarantine at the point of entry. Beside airports, as reported by Buckley and May [2], data is also being collected at bus terminals, market places (in Wuhan), subways, and also in health facilities where patients are taken for further medical attention. Such is prevalent especially in China, and other Asian regions where cases of the virus have been recorded and confirmed.

In addition to these methods, other smart city data sources include the application of terminal tracking systems that are mostly emphasized in Safe City concepts, where, at the point of entry or departure, relevant data is collected and analyzed. Li et al. [32] highlights that sensors installed in such locations have the potential to receive and distribute data in real-time to digital infrastructures within the network, and their interconnectedness in the network renders them extremely efficient in providing real-time updates on different issues. Urban areas are also known to be amassed with numerous Urban Health sensors, some of which are wearable. Though these are not specifically fashioned to track the present case of virus outbreak, they are able to track other related parameters like heartbeat, blood pressure, body temperature and others variables, that when analyzed can offer valuable insights. Loncar-Turukalo et al. [33] hail these devices for their role in transforming the health care sector especially by allowing for Connected Health (CH) care, where data collected from them can be analyzed and provide insightful information on the health scenario in any given area. Vashist et al. [34] further highlight how emerging features such as spatiotemporal mapping, remote monitoring and management, and enhanced cloud computing capabilities can emanate from such endeavours, leading to better urban management potential.

While it is true that the basic source of medical data is generally sourced from general practitioners or medical laboratories-a fact that has also been affirmed in the case of the current epidemic-this paper explores how data sourced from an urban perspective can contribute to the medical narrative. The conviction to dwell on the urban realm in this manuscript is based on the fact that the current epidemic (COVID-19) is transmitted majorly through human-to-human contact, and in most cases, especially where the spread is reported in a different country, the first point of contact is an urban area, where large groups of people convene, like airports or subway stations. In most cases, such facilities, which are mostly based in urban areas, are observed to have installed surveillance technologies to ensure that anyone showing any symptoms of the disease are identified and quarantined. However, even in such cases, as underlined in the present manuscript,

the need for anonymizing medical data is emphasized to ensure that the use of current technologies does not breach data privacy and security requirements, across different geographies. In this case, novel technologies like Blockchain technologies and quantum cryptography can aid in the discussion and be made to integrate with data collecting technologies. This would render an increased wealth of data from both the medical field and smart city operators, while ensuring privacy and security; hence, aiding in providing relevant information for better informed decisions.

However, despite the indisputable roles that installed devices play in providing relevant health information, their data communication aspect needs to be reviewed. First, communications are seen to be geography-restricted (restricted to a given location), such that they seldom expand or communicate with their like, installed beyond their restricted areas. Secondly, these devices are usually sourced and installed by separate corporations that maintain unique and specific standards for data processing and sharing, and accordingly, tying cities to the sole usage of their product(s). Such strategies are adopted as private corporations try to maximize their economic gains, since the digital solution market is a lucrative one and is expected to continue growing and expanding [6, 7] .

For its current application, the standardization of protocols as elaborated in this manuscript need to be pursued to ensure that there is seamless sharing of information and data. By doing this, it is expected that issues like burdens of collecting data, accuracy and other complexity that are experienced (when systems are fragmented) are reduced or eliminated altogether. The standardization can be achieved by, for example, ensuring that all the devices and systems are linked into a single network, like was done in the U.S., where all the surveillance of healthcare were combined into the National Healthcare Safety Network (NHSN) [35] . The fact that cities are increasingly tuning on the concept of Smart Cities and boasting an increased adoption rate of technological and connected products, existing surveillance networks can be re-calibrated to make use of those new sets of databases. Appropriate protocols however have to be drafted to ensure effective actions while ensuring privacy and security of data and people.

With scenarios like the present Coronavirus (COVID-19) outbreak, that not only impacts upon the economic status of cities, but also affects their social standing, it becomes imperative to emphasize the adoption of universal standards for data sharing. Such a move could have far reaching impact across cities and territories especially in positively

combating outbreaks and disasters in a quicker, safer and standardized way, such that when the cure is discovered, the results can be replicated in various parts of the globe. With a collaborated data sharing protocol, it would be possible to have a larger dataset resulting in increased processing capabilities especially with technologies that are powered by artificial intelligence (AI) tools. Through this way, as noted by Jiang et al. [36] and Allam [37], it would be possible to facilitate early detection, achieve better diagnosis and provide better urban management decisions for increased efficiency for virus containment.

An example of how beneficial collaboration and sharing of data can be occurred during the 2014 Ebola outbreak in West Africa where scientists, health workers and clinicians, amongst other stakeholders from around the world, openly worked together and were able to contain the spread of this pandemic [38]. On this front, Boué et al. [39] highlight that levels of trust and transparency need to be reviewed and enhanced to facilitate unfettered data generation and sharing. Such could lead to an even earlier detection scenario of future virus outbreaks, and in the better curative management of the same, without minimal compromise on urban functions and on an urban economy.

Furthermore, in cases of emergencies like the current outbreak of COVID-19 and any other, the need for observance of regulatory practices and international healthcare guidelines are paramount. This would ensure that both healthcare professionals and the general populace are informed, protected and remain within the prescribed rules and regulations. As noted by the WHO [40], the healthcare guidelines and regulatory practices are advanced to also ensure that the health risk in question is reduced together with its consequences. In the current era of technological advancement, such regulations and guidelines are paramount as they have potential to lead to positive or negative outcomes. The position of this paper is to advance that it is now possible to integrate technologies like the use of smart devices through IoT networks and wearable devices, data from mobile apps and others to help users to share information with accredited and certified health professionals, and in this case, improve the outcomes for better cross disciplinary and more resilient protocols and policies.

--- END OF ARTICLE ---

--- START OF ARTICLE 116 ---

The Battle Against Coronavirus Disease 2019 (COVID-19): Emergency Management and Infection Control in a Radiology Department

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The Battle Against Coronavirus Disease 2019 (COVID-19): Emergency Management and Infection Control in a Radiology Department

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Authors' contributions

ZXH: conceived the study and drafted the manuscript.

ZS: conceived the study and drafted the manuscript.

ZLL: The member of the emergency management and infection control team (EMICT) and was involved in the formulation of the measures.

WXC: The member of the EMICT and was involved in the formulation of the measures.

LHZ: The member of the EMICT and was involved in the formulation of the measures.

LPD: The member of the EMICT and was involved in the formulation of the

measures.

BS: Leader of the EMICT, conceived the study and reviewed the manuscript.

All authors read and approved the final manuscript.

The authors declare no conflict of interest.

The authors declare that they had full access to all of the data in this study and the authors take complete responsibility for the integrity of the data and the accuracy of the data analysis

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The Battle Against Novel Coronavirus Pneumonia (COVID-19): Emergency Management and Infection Control in a Radiology Department

Abstract

Objective: To describe the strategy and the emergency management and infection control procedure of our radiology department during the COVID-19 outbreak.

Methods: We set up emergency management and sensing control teams. The team formulated

various measures: reconfiguration of the radiology department, personal protection and training

of staff, examination procedures for patients suspected of or confirmed with COVID-19 as well

as patients without an exposure history or symptoms. Those with suspected or confirmed COVID-19 infection were scanned in the designated fever-CT unit.

Results: From January 21, 2020 to March 9, 2020, 3,083 people suspected of or confirmed with

COVID-19 underwent fever-CT examinations. Including initial examinations and reexaminations, the total number of fever-CT examinations numbered 3,340. As a result of our

precautions, none of the staff of the radiology department were infected with COVID-19.

Conclusion: Strategic planning and adequate protections can help protect patients and staff

against a highly infectious disease while maintaining function at a high volume capacity.

Keywords: Coronavirus, COVID-19, novel coronavirus pneumonia, infection control

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Introduction

The whole world has been closely focusing on an outbreak of respiratory disease caused by a

novel coronavirus that was first reported in Wuhan, China, on December 31, 2019, and that continues to spread. On February 11, 2020, the World Health Organization (WHO) named the

disease “coronavirus disease 2019” (COVID-19).

As of 24:00 on March 11, 2020, the National Health Commission (NHC) had received reports

of 80,793 confirmed cases and 3,169 deaths on the Chinese mainland. There remain 14,831

confirmed cases (including 4,257 in serious condition) and 253 suspected cases still hospitalized. To date, 677,243 people have been identified as having had close contact with

infected patients of whom 13,701 are under medical observation [1]. Outside China, 44,067 laboratory-confirmed cases and 1,440 deaths have occurred in 117 countries /territories/areas

according to the WHO [2]. COVID-19 poses significant threats to international health. Like the

flu, COVID-19 is thought to spread mainly from person-to-person between people who are in

close contact with one another through respiratory droplets produced when an infected person

coughs or sneezes. In light of the infectious nature of this disease, healthcare workers are at

high risk of infection of COVID-19. In China, healthcare workers account for 1,716 confirmed

cases of COVID-19, including six deaths [3].

Computed tomography (CT) can play a role in both diagnosing and categorizing COVID-19 on the basis of case definitions issued by the WHO and the treatment guidelines from the NHC [4]. Suspected patients having the virus may undergo chest CT. Isolation and barrier procedures are necessary to protect both the department staff and other patients in the

hospital. Note should be made that due to overlap of imaging findings with other respiratory

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diseases, CT is not helpful as a screening tool. But it can help identify the degree of pulmonary

involvement and disease course.

Our hospital is a national regional medical center with 4,300 beds and a tertiary referral center in Sichuan province. The initial response started on January 21, 2020, after transmission

of COVID-19 was confirmed to be human-to-human on January 20, 2020. The first suspected

case of COVID-19 in Sichuan province was reported on January 21, 2020. The Sichuan provincial government immediately launched the first-level response to major public health

emergencies. On the same day, our hospital was designated to care for Sichuan province patients with COVID-19.

This article describes the emergency management procedure of our radiology department for situations involving severe infectious diseases, such as COVID-19, and the

infection-protection experience of the department staff.

Methods

The hospital provided personal protective equipment (medical protective clothing, surgical cap, N95 mask, gloves, face shields, and goggles) to all its healthcare staff, erected

three medical tents (fever tents) for screening of fever cases in the parking lot of the emergency

department, planned an examination route and examination area for patients suspected of harboring the virus, and placed confirmed patients in an isolation ward. “Fever” was the colloquial term used to designate suspected COVID-19 based on symptoms such as a fever or

with an epidemiological history of a potential exposure as well as those with confirmed COVID-19 referred for treatment. Further, during outbreak, emergency and outpatient patients

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without fever were asked for information such as epidemiological history and sent to fever tents

as long as they met suspected criteria.

The radiology department has 65 diagnostic radiologists and 161 other staff members (trained technologists, nurses, engineers, and support staff). The equipment of the radiology

department includes 12 magnetic resonance (MR) scanners, 14 CT scanners, 15 digital subtraction angiography (DSA) systems, 32 sets of digital radiography (DR) systems (including nine mobile bedside DR sets), and 130 imaging diagnostic workstations for picture

archiving and communication systems (PACS). Most of the equipment is distributed among four buildings at the hospital main campus. 4 CT scanners, 4 MR scanners, 1 DR are located on

the first floor of the first inpatient building, and 9 DR and 8 DSA are located on the second floor. 1 CT and 1 MR scanner are located in the third inpatient building. 1 CT and 1 MR scanner

are located in the sixth inpatient building. 2 CT scanners, 2 MR scanners and 7 DSA are located

in the technical building. The rest of the equipment is located in the seventh inpatient building

in the branch campus.

The first inpatient building, located next to the emergency department, was reconfigured to handle cases of COVID-19. Fever tents were set up by the emergency department in the emergency department parking lot to separate normal emergency patients from patients with

symptoms or exposure history suspicious of COVID-19. We established separate means of access between fever tents and between the fever examination area of the radiology department

to avoid cross-contamination.

The emergency management and infection control measures, as described below and implemented in the radiology department during the outbreak, have been approved by the

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infection control committee of hospital. These measures are in accordance with relevant laws

and regulations, in order to protect patients as well as the staff.

Radiology Emergency Management and Infection Control Team (EMICT)

The radiology department director chaired the EMICT. Its members include the deputy director, chief technologist, head nurse, equipment engineer supervisor, and infection control

nurse of the radiology department. Team responsibilities included (1) coordination between the

hospital's management and planning of infection control and radiology departments; (2) collection of the most up-to-date protection-related information to educate and train staff in the department; (3) reallocation of staff according to the actual situation; (4) establishment of the CT procedures for patients with COVID-19; and (5) establishment of an emergency management plan for the radiology department to ensure that the department would run normally.

Suspected patients

The suspected patients were identified according to the Diagnosis and Treatment Program of the Novel Coronavirus Pneumonia of the NHC [5], mainly based on epidemiological history.

Reconfiguration of the radiology department

The radiology department was divided into four areas [6]: contaminated, semicontaminated, buffer, and clean areas (Figure 1). The contaminated area is connected to the fever clinic and includes the fever accessway, the CT examination room, and the DR examination room for confirmed and suspected cases. One CT scanner and one DR system closest to the emergency department are designated the fever-CT and fever-DR to examine patients with suspected and confirmed COVID-19. There is a separate dedicated access between the contaminated area and the fever screening tents. The semicontaminated area includes the fever-CT control room, fever-DR control room, and other patient examination access areas. The buffer zone includes access areas for medical personnel and a dressing area for technologists. The clean area

includes the administrative office and the diagnostic room.

The contaminated area was isolated from other areas using physical barricades.

Directional signs were newly installed to guide patients and staff.

Personal protection and training of staff

For providing care for patients with confirmed and suspected COVID-19, all hospital staff are required to wear complete personal protective equipment [7]: medical protective clothing,

surgical cap, N95 mask, gloves, face shields, and goggles. Wearing and removing of the equipment must be performed in accordance with the procedures and under the supervision of

the infection control nurse.

Because staff members working in the contaminated area are under much situational pressure, periodically taking time off could lower their physical and mental stress levels. The

technologists on fever-CT duty shifts are provided a break once a week for four hours. In addition, the health of staff in the contaminated area must be monitored closely for the symptoms of COVID-19. Pregnant staff must be assigned to the clean area.

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The EMICT formulates and continually updates guidelines and educates all staff for West China Hospital of Sichuan University. The EMICT training for staff is mainly involves documents regarding infection control and CT findings of COVID-19 and maintains an EMICT

WeChat group for West China Hospital of Sichuan University. WeChat is the most widely used

social media app in China. The EMICT releases the latest national and hospital-based information regarding COVID-19, guidance documents, and other notices from the hospital and radiology department in the WeChat group on a daily basis. Staff can also report to the

EMICT in the WeChat group any time. Protocols for each modality and infection control instructions are posted on the walls in all examination rooms. The EMICT periodically reminds

staff to undertake personal measures to reduce infection, such as wearing masks at all instances

in the radiology department and N95 masks if working in the contaminated area; not touching

the mask and the eyes; practicing hand hygiene; facing away from colleagues when eating, drinking, and talking; and not using personal cell phones while on duty.

In addition, the chief thoracic radiologist provided lectures on all radiologists and technologists on typical CT findings of COVID-19 infection using materials developed in Wuhan, the epicenter of the outbreak in China.

CT examination procedures

There are two sets of procedures for CT examination: the fever-CT procedure and routine CT

procedure for those not suspected of COVID-19.

The fever-CT procedure for suspected or confirmed COVID-19 (Figure 2)

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Before the fever-CT technologist operates the equipment, he or she should wear personal protective equipment according to three-level protection standard [8]. Before the CT examination of patients with suspected and confirmed COVID-19 begins, the fever tent or isolation ward notifies the radiologist in advance. The fever-CT technologist checks the equipment and prepares to disinfect the imaging equipment immediately after the examination.

The patient enters the fever-CT waiting area through the fever access area. If the patient can get onto and off the examination table by themselves, the patient is allowed to do so. If the

patient cannot get onto or off the examination table independently, the person accompanying

the patient assists the patient, rather than the technologist. The technologist checks the patient

information and, using an intercom system in the examination room, asks the patient to remove

any metal ornaments on the neck and chest. Also, by intercom, the technologist trains the patient to hold his or her breath during the examination.

The technologist uses a low-dose chest CT protocol to scan the patient. After scanning, the original images are reconstructed as 1 mm-thick layers. The technologist browses the images to

ensure that their quality meets the diagnostic requirements and then guides the patient to leave

through the fever access area. The disposable sheets for patient examination are changed after

each patient. The equipment is disinfected according to the procedure below.

To protect themselves, the technologists assigned to the fever-CT wear N95 mask and other personal protection as established by the EMICT.

The CT procedure for regular patients (figure.3)

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Some patients with COVID-19 have no symptoms, and they may call at the general clinic for other reasons. The following CT procedure is applicable under these circumstances:

When the patient makes an appointment for examination, the staff asks the patient about their epidemiological history, symptoms, and signs. If suspected criteria are met, the patient

will be sent to the fever tent for further screening. When a patient presents to the radiology department entrance, his/her temperature is measured. If the temperature is higher than 37.2 , °C

the patient is sent to the fever tent for further investigation.

Those with no exposure history, suspicious symptoms or fever are screened in one of the non-contaminated CT scanners. The technologists assigned to these scanners wear surgical

masks. All patients and the person accompanying them are required to wear surgical masks.

After the CT examination, the technologist browses the images quickly. If the CT appearance is

typical of lung infection, the technologist immediately reports it to the chest radiologist on duty

and asks the patient to wait in the CT examination room. If the chest radiologist does not suspect COVID-19 infection, the patient can leave the CT examination room. If the chest radiologist does suspect COVID-19 infection, the technologist immediately reports it to the EMICT and sends the patient to the fever tent. The floor and equipment in the CT examination

room are disinfected according to regulations, and air disinfection is conducted for 30 min before examining other patients. These CT scanners are considered noncontaminated (not fever-CTs) after these sterilization procedures.

Fever-DR examination procedure

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The COVID-19 guideline of the NHC does not recommend chest DR because its ability in diagnosing COVID-19 is limited. At our hospital, we only use mobile DR units to provide bedside examination for critically ill patients. The technologist operating the mobile DR wears personal protective equipment according to the three-level protection standard and sterilizes the mobile DR according to the ward management requirements as described below.

Equipment and environment disinfection procedures

Routine disinfection procedure [9]

- 1) Object surface disinfection: Object surface is wiped with 1000mg/L chlorine-containing disinfectant, wipe twice with 75% ethanol for non-corrosion resistance, once /4 hours.
- 2) Equipment disinfection: The equipment in the contaminated area are wiped with 2000mg/L chlorine-containing disinfectant. The DR and CT gantry in the contaminated area are wiped with 75% ethanol. The equipment in the buffer area is wiped with 500-1000mg/L chlorine-containing disinfectant or alcohol-containing disposable disinfectant wipes twice a day.
- 3) Air disinfection: Turning off all central air conditioners to prevent air contamination with each other. Polluted area: open the door for ventilation, each time more than 30 minutes, once /4 hours; The air sterilizer is continuously sterilized or the ultraviolet ray is continuously used in the unmanned state for 60 minutes, four times a day, remembered to close the inner shielding door when air disinfection. Other ambient air is sprayed with 1000mg/L chlorine-containing disinfectant and ventilated twice a day
- 4) Ground disinfection: The ground is wiped with 1000mg/L chlorine-containing disinfectant, once /4 hours.
- 5) When contaminated, disinfect at any time. In case of visible contamination, disposable absorbent materials should be used first to completely remove the pollutants, and then a cloth soaked with 2000mg/L chlorine-containing disinfectant should be used for 30 minutes before wiping.

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Fever-CT disinfection procedures after examination

In addition to the above, disinfect the examination bed and ground with chlorinated disinfectant

containing 2000mg/L [10].

Noncontaminated CT disinfection procedures after suspected COVID-19 case examination

In addition to the above routine disinfection procedure, air disinfection is conducted for 30 min

before examining other patients.

Results

From January 21, 2020 when screening for epidemiological history or symptoms suspicious for COVID-19, to March 9, 2020, our hospital screened a total of 7,203 individuals and confirmed 24 cases of COVID-19. Of these, 3,083 people underwent fever-CT examinations. Including the initial examination and reexamination, the total number of fever CT examination numbered 3,340. The fever-CT scanned a patient approximately every 21.5 minutes. As a result of our precautions, none of the staff of the radiology department developed symptoms suspicious for COVID-19. The fever-CT technologist, with the highest probability of exposure, remains PCR negative.

Discussion

It has been 17 years since the severe acute respiratory syndrome (SARS) epidemic, the last national spread of severe infectious disease, broke out. Currently, the Chinese people are panicking again. The speed and extent by which COVID-19 has spread in 2 months are

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unprecedented, beyond those of SARS, and this has been aided by its contagious nature and

rapid spread via droplets and contact. The droplet mode of transmission means that a person can

be infected easily by means of casual contact or even fomites on contaminated environmental

surfaces. Another theory has yet to be proved: aerosol propagation.

How radiology departments respond to any infectious disease outbreak is determined primarily by the estimated risk of cross-infection to the staff and other patients.

Appropriate

precautions taken only by staff in direct contact with patients may be adequate when the risk is

low. The strongest measures need to be implemented to limit the spread of the disease when the

risk is high. With severe infectious diseases such as COVID-19, the highest level of infection

control measures must be implemented; these include providing adequate standard protective

equipment, training staff, and instituting proper emergency plans.

Once a contagious infectious disease has been identified, the EMICT must consider four main areas of response: data gathering, collaboration, needs assessment, and expert advice [10].

Data gathering includes dissemination of up-to-date case definitions and information about

confirmatory tests to all staff with direct patient contact to allow appropriate barrier precautions

to be taken. All typical and atypical imaging features of the disease should be made known to

all radiologists to assist in recognition of the disease on images and to allow accurate reporting

of these findings. We have stored images of all probable cases of COVID-19 in the PACS so that these images were readily available for any radiologist to review, and images from previous imaging studies are also available for comparison.

Collaboration with the radiology departments of other hospitals is very important because patients may initially present to different centers, depending on geographic location and travel

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distance. These patients may be few in number at a single hospital, but if data from patients at

several hospitals are available, a more accurate overall understanding of both imaging features

and epidemiology can be achieved. Dissemination of this information to all healthcare facilities

will also lead to early recognition of the disease, and appropriate isolation measures may be

instituted.

The Internet and social media apps, especially WeChat, have been used for distribution of medical information, and because the exchange of information regarding infectious disease

outbreaks is almost instantaneous, it is an indispensable tool for radiologists. In fact, within a

month of the outbreak, the hospital that received the most infected patients from the source of

the outbreak made a PowerPoint presentation of the CT manifestations of COVID-19, which was shared via WeChat and disseminated across the country in a very short time. Subsequently,

COVID-19-teaching PowerPoint presentations from various hospitals appeared and were quickly shared via WeChat.

Our diagnostic process is limited as chest CT alone is not diagnostic of COVID-19 because of lack of imaging specificity. But when combined with other epidemiological, clinical, laboratory and virus nucleic acid information, typical chest CT imaging findings are helpful for making the diagnosis. In our opinion, the major role of chest CT is to understand the

extent and dynamic evolution of lung lesions induced by COVID-19. The reasons why we adopted the low-dose chest CT scan protocol are as follows: low-dose chest CT has been widely used in the screening of early lung cancer. It is well known that many early lung cancers

are ground-glass opacities (GGO), so we believe that low-dose screening is also applicable for

COVID-19. In addition, considering the rapid development of COVID-19, many CT

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examinations may be conducted in the same individual to monitor disease progress. Low-dose

scanning can reduce the radiation damage to patients.

Although the processes we established minimized the exposure of hospital staff, ancillary personnel and other patients, it remains limited as follows. Sichuan province is not the center of

the epidemic. The number of patients with COVID-19 whom we have treated has not been high, and most cases are from other provinces of China. However, we believe that our experience in management, the reconfiguration of our radiology department, and the workflow

changes implemented in the current COVID-19 situation are useful for other radiology departments that must prepare for dealing with patients with COVID-19. While no radiology personnel developed symptoms suspicious for or were confirmed as having COVID-19, there

may be asymptomatic personnel.

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Figure Legends

Figure 1. Diagram of the layout of our radiology department was divided into four areas:
contaminated
(shaded in black), semicontaminated (shaded in dark gray), buffer (shaded in light gray),
and clean areas
(shaded in white). The contaminated area was separated from other areas by barriers.

Figure 2. Diagram shows CT protocol for suspected and confirmed patients with COVID-19.

Figure 3. Diagram shows CT protocol for regular patients.

Abbreviations:

COVID-19: coronavirus disease 2019

CT: computed tomography

DR: digital radiography

EMICT: emergency management and infection control team

NHC: National Health Commission

PACS: picture archiving and communication system

SARS: severe acute respiratory syndrome

Sentence Summary

With severe infectious diseases such as COVID-19, the highest level of infection control measures must be implemented, collaboration with the radiology departments of other hospitals be needed, and social media be employed.

Take-home points

1. To response to a community infection emergency, a special emergency management team

needs to be setup at the departmental level to implement infection containment and control procedures that continues to allow the imaging examination and imaging diagnosis of those with suspected infection, and to prevent intra-departmental spreading of infection (EMICT).

2. Infection control measures, such as reconfiguration of department areas, personal protection and anti-infection training of all staff, standardized procedures including

contact minimization for chest CT and DR examinations, and timely disinfection of CT and DR examination rooms, should be implemented properly.

3. If there are more than one scanner in a hospital, only one of them should be assigned to suspected cases.

--- END OF ARTICLE ---

--- START OF ARTICLE 117 ---

Identification of COVID-19 Can be Quicker through Artificial Intelligence framework using a Mobile Phone-Based Survey in the Populations when Cities/Towns Are Under Quarantine

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Abstract: We are proposing to use machine learning algorithms to be able to improve possible case identifications of COVID-19 more quicker when we use a mobile phone-based web survey. This will also reduce the spread in the susceptible populations.

Text: Emerging and novel pathogens are a significant problem for global public health and technology can assist in faster identification of possible cases to bring timely interventions.

This is especially true for viral diseases that are easily and readily transmissible and have asymptomatic infectivity periods. The novel Coronavirus (SARSCoV2) described in December 2019 has resulted in major quarantines throughout the world, including major cities, villages and public areas throughout China [1] [2] [3] to prevent further spread. As of February 25th 2020, the World Health Organization's situational data indicates that there were about 77780 confirmed cases, including 2666 deaths due to COVID-19, including cases in 25 countries [4] . The majority of the deaths reported so far have been in China only [5] .

Organization have issued interim guidelines in order to protect the population, and to attempt to prevent the further spread of COVID-19 from infected individuals [6] .

Since cities and villages throughout China are unable to accommodate such large numbers of infected individuals, and be able to maintain the quarantined. China has built several new hospitals in an attempt to manage the infected individuals [7] . It is imperative that we evaluate novel models in an attempt to control the rapidly spreading virus [8] .

In order to reduce the time to identification of a person under investigation (PUI) for the COVID-19 infection, and the rapid isolation of this individual, we propose to collect the basic travel history along with the more common manifestations using a phone-based online survey. Such collected data can be used to assist in the preliminary screening and early identification of possible COVID-19 infected individuals. Thousands of data points are able to be collected and processed through an artificial intelligence (AI) framework which can ultimately evaluate individuals that may be infected and stratify them into no-risk, minimal-risk, moderate-risk, and high-risk of being infected with the virus. The identification of the high-risk cases can then be quarantined earlier, thus decreasing the chance of spread. Table 1 is inserted here.

See Appendix I for the details on the steps involved in data collection on all the respondents independent of whether or not they think they are infected. The AI algorithm described in Appendix II is to identify possible case identifications and send alerts to the nearest health clinic as well as to the respondent for an immediate health visit, we call this as an "alert for health check recommendation for COVID-2019. In case the respondent is unable to commute to the health center, the health department can then send an alert to a

mobile health unit so they can then do door-to-door assessments and even testing for the virus. This generates alert for mobile health check recommendation for 2019-nCoV (MHCRC). If a respondent does not have an immediate risk of having symptoms or signs related to the viral infection, then the AI-based health alert will be sent to the respondent to notify them that there is no current risk of COVID-2019. Figure 1 summarizes the outcomes of data collection and identification of possible cases. The data recorded in step 5 of the algorithm using signs and symptoms will be collected prior to both the groups who have received alerts HCRC or MHCRC (for possible identification and assessment) and NCRC (for non-identified respondents).

These are explained in steps (iii) and (iv) in the Appendix II. The extended analysis proposed will help to understand if there is any association with different sociodemographic variables and the manifestations such as fever and signs and lower respiratory infections, including cough and SOB in individuals defined as either with and without possible infection.

Applications of AI and deep learning argued to be useful tools in assisting diagnosis and treatment decision making [10] [11] . There were studies which promoted disease detection through AI models [12] [13] [14] [15] . Use of mobile phones [16] [17] [18] [19] and web based portals [20] [21] have been tested successfully in health related data collection. However, one need to apply such techniques in a timely way for faster results. Apart from cost-effectiveness, the proposed modeling will be of great assistance in identifying and controlling when populations are closed due to virus spread. In addition to these, our proposed algorithm can be easily extended to identify individuals who might have any mild symptoms and signs.

We have developed our data collection criteria based on CDC's Flowchart to Identify and Assess 2019 Novel Coronavirus [9] and added additional variables for the extended utility of our efforts in identifying infected and controlling the spread (see Table 1).

Let be the outputs recorded during the data collection steps 1 (ii) If the set of identifiers, , for is equal to one of the elements of the set then send HCRC or MHCRC to that respondent, else proceed to the test criteria (iv).

If is equal to one of the elements of the set , for then the respondent will be sent an NCRC alert.

(iv)

If is equal to one of the elements of the set , then the respondent will be sent an NCRC alert.

Comparison of test criteria results of (iii) and (iv) with their corresponding geographic and socio-demographic details will yield further investigations of signs and symptoms Suppose we define two events and using the sets and as below:

: out of responded cases are identified through the algorithm out of have responded to the survey.

The conditional probability of the event given the event , say, is computed as

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Overview of the 3rd isirv-Antiviral Group Conference – advances in clinical management

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Abstract: This review highlights the main points which emerged from the presentations and discussions at the 3rd isirv-Antiviral Group Conference - advances in clinical management. The conference covered emerging and potentially pandemic influenza viruses and discussed novel/pre-licensure therapeutics and currently approved antivirals and vaccines for the control of influenza. Current data on approved and novel treatments for non-influenza respiratory viruses such as MERS-CoV, respiratory syncytial virus (RSV) and rhinoviruses and the challenges of treating immunocompromised patients with respiratory infections was highlighted.

Text: Recurrent infections by influenza and other respiratory viruses contribute enormously to the burden of human disease and (emergent) sporadic zoonotic infections, such as by influenza H7N9 and H5N1 and Middle East respiratory syndrome (MERS) coronavirus, pose a constant threat of a new global epidemic. Despite extensive knowledge of the viruses and their interaction with the host, there is little in our armoury of vaccines and therapeutics to combat this perpetual onslaught. The 3rd isirv Antiviral Group conference on Influenza and Other Respiratory Virus Infections: Advances in Clinical Management, convened in Tokyo, Japan on 4-6 June 2014, attracted 188 clinicians, public health specialists and medical scientists from 34 countries to present their recent research and discuss various aspects of the impact of respiratory viruses in different patient groups/settings and in different regions of the world. The programme 1 focused on the latest advances in the mitigation and clinical management of influenza and other respiratory virus disease, and the successful use of antivirals (and vaccines) against seasonal and pandemic influenza, particularly in Japan, as well as the development/assessment of novel antiviral agents. This overview highlights some of the main points which emerged from the presentations (both oral and poster) and associated discussion.

In recent years, an increasing number of cases of novel animal influenza A viruses infecting humans have been reported. These include multiple avian influenza A virus subtypes, in particular H5N1 and H7N9, and swine-origin H3N2v. Reasons for this increase include both social factors, for example, increased human populations living in close proximity to

animals and increased surveillance and diagnostic testing. Risk assessment tools have been developed by many authorities around the world (e.g. the European CDC, US CDC, USAID and WHO) to assist in predicting the likelihood that a particular virus will emerge and its associated impact, as well as prioritising the development of candidate human vaccine viruses. 2 These tools allow continual reassessment as new data become available and provide an objective, transparent process with which to make resource allocation and pandemic planning decisions.

In February 2013, China detected the first human cases of H7N9 infection in severely ill patients with pneumonia. 3 As of May 22, 2014, there have been 446 confirmed H7N9 cases in China resulting in 163 deaths. 4 The cases have occurred mainly during two waves (weeks [8] [9] [10] [11] [12] [13] [14] [15] [16] [17] [18] 2013 and week 40, 2013week 20, 2014), 4 of which 85% had prior exposure to poultry or contaminated live poultry markets. The median time from poultry exposure to disease onset was 5 days, whereas the median time from illness onset to hospital admission, ARDS development, antiviral therapy and death was 5, 6, 8, 7 and 14 days, respectively. 5 Closure of live poultry markets has markedly reduced the risk of H7N9 infection. Across nine areas in the two most affected provinces in China, modelling analysis estimated that the effectiveness of market closure was 97% (95% CI: 89%, 100%). 6 A retrospective serological study of blood specimens taken in January-May and October-November in 2012 from 1544 subjects who worked in live poultry markets, farms, slaughter houses or kept backyard poultry revealed no evidence of H7N9 infection, 7 indicating widespread population susceptibility and lack of prior circulation of antigenically related viruses. Multiple family clusters have been reported, but no sustained human-to-human transmission, with studies demonstrating a very low detection of virus or specific antibody in close contacts (0.34% and 0.2% in the first and second waves, respectively) and healthcare workers of positive cases. 8 In both the first and the second waves, the majority of the patients hospitalised with H7N9 infection were older men (median age, 62 and 58 years, respectively with an overall male/female ratio of 2.2:1) and the case fatality was similar (32% and 39%, respectively). Pre-existing medical conditions occurred in >60% of these cases. The prominent clinical features on admission were those of a severe influenza syndrome with fever, cough, fatigue and dyspnoea, while the most striking laboratory findings were marked lymphopenia and thrombocytopenia. Elevated cytokine levels have been observed in patients and such excessive cytokine responses may contribute to the clinical severity of H7N9 infection. 9 Originating from reassortment events involving at least three avian influenza viruses, H7N9 viruses with multiple genotypes continue to emerge on a more frequent basis in 2014. Many viruses isolated from humans contain the E627K amino acid substitution in the

polymerase PB2 component, associated with mammalian adaptation, and the G186V and Q226L substitutions in the haemagglutinin (HA) that are associated with dual receptor binding to both $\alpha 2,3$ and $\alpha 2,6$ -linked sialic acid receptors. 10 All H7N9 viruses from the outbreak to date are antigenically similar to the (original) candidate vaccine strain. Prototype inactivated whole particle H7N9 vaccines have been investigated in macaques and shown to induce good antibody responses that significantly reduced the number of days of virus shedding in experimentally infected animals. Treatment of H7N9-infected patients with neuraminidase inhibitors (NAIs), including intravenous (IV) peramivir or zanamivir, 11 appears to have been beneficial even when therapy was started late, although emergence of oseltamivir resistance has been associated with poor clinical outcomes. 12 All H7N9 viruses are amantadine-resistant due to the S31N substitution in the M2 ion channel protein, while viruses containing the R292K substitution in neuraminidase (NA), which confers resistance to both oseltamivir and peramivir (>1000-fold rise in IC₅₀) and reduced susceptibility to zanamivir and laninamivir (50- and 25-fold rises in IC₅₀, respectively), have been reported in six cases. Two of these patients with severe H7N9 infection requiring extracorporeal membrane oxygenation (ECMO) also received systemic corticosteroid treatment leading to treatment failure and a poor clinical outcome. 12 The replication and transmission of H7N9 viruses containing the R292K NA mutation have been shown to be comparable to those of wild-type H7N9 viruses in guinea pigs, 13 and in ferrets following both contact and non-contact exposure. However, the wild-type virus did outgrow the R292K-resistant strain in some ferrets over the course of the infection. 14 Interestingly, the R292K variant appeared to be the dominant virus in ferret lung lobes, while in nasal turbinates, the wild-type virus was predominant. Therefore, it appears that the R292K mutation causes less fitness loss in H7N9 virus than in seasonal H3N2 viruses. 13

A new reassortant genotype of H5N1 containing the HA and NA genes from clade 1.1.2 and the internal genes from clade 2.3.2.1 emerged during 2013 and was associated with the highest number of cases ($n = 26$) and deaths ($n = 14$) in Cambodia. 15 Globally since 2003, there have been 650 confirmed H5N1 cases and 386 deaths reported in humans, 16 with most infections in the last 2 years being in children. Human-to-human transmission remains extremely rare based on virological and serological data from analysis of close contacts of confirmed cases in Cambodia, Thailand and Vietnam. 17 A study of household transmission patterns in Indonesia has shown that the overall household attack rate was 18.3% and the secondary attack rate was 5.5%, independent of household size. 18 Oseltamivir therapy appears to reduce mortality when administered within 8 days of H5N1 illness onset, although earlier treatment is more effective, highlighting the need for early

patient diagnosis. 19 Early initiation of oseltamivir was particularly effective in reducing mortality in H5N1 patients without respiratory failure (odds ratio, 0.17; $P = 0.04$), whereas those requiring ventilatory support at the time of oseltamivir initiation were more likely to die. 20 A study of the risk factors for mortality related to H5N1 identified age, country, per capita government health expenditure and delay from symptom onset to hospitalisation as the key parameters, highlighting the importance of early diagnosis, treatment and supportive care. 21 High-dose systemic corticosteroids (SC) are associated with worse outcomes in H5N1 patients. 22 One human case of avian H5N6 was recently detected in China; the virus was a reassortant that contained seven genes from H5N1 and the NA gene from an H6N6 virus circulating in ducks. 23 China has also reported the detection of three human infections, two fatal, with avian H10N8 viruses that contain the internal genes from H9N2, as does H7N9. 24 Like the H7N9 virus, the H10N8 virus has low pathogenicity in poultry and is therefore difficult to detect in birds.

Burden in target populations Pregnant women and infants have an increased risk of complications following influenza infection. Globally, significant numbers of pregnant women died during the 2009-2010 pandemic, but no maternal mortality occurred in Japan. 25 Through education campaigns directed at pregnant women and healthcare professionals, 67% of pregnant women were vaccinated against H1N1pdm09 resulting in an infection rate among pregnant women in Japan of 3.5% compared to the overall infection rate in the population of 12%. 26 Of those pregnant women who were infected with H1N1pdm09 in Japan, 95% were treated with antivirals, and importantly, 88% of those were treated within 2 days of symptom onset. 25 In Mongolia, a prospective cohort study during 2013-2014 found that influenza-like illness (ILI) was detected in 17.9% of pregnant women, of whom the majority tested positive for influenza A, with substantially lower influenza B and respiratory syncytial virus (RSV) infection. 27 During the same period, ILI was detected in 30.9% of infants <6 months of age, with an even spread of influenza A, influenza B and RSV. 27 The influenza burden in children in a rural Indian community was found to be substantial with 11.6% of ILI cases being caused by influenza A or B viruses. 28 These findings underscore the importance of maternal immunisation. 29 **Transmission patterns** Sequence analysis of influenza viruses isolated from students on a Singapore University Campus provided insights into the chain of transmission, showing that 62% of 32 viruses were highly similar, demonstrating that the majority of transmission was occurring on the university campus rather than from infections outside. 30 The effectiveness of surgical masks, hand hygiene and health education investigated in households in Hong Kong and Bangkok detected no significant difference in attack rate in cohorts using one of these interventions. 31 Further analysis of the data enabled some insights into the relative

importance of aerosol, large droplet and contact transmission within the households. For influenza A infections, aerosol transmission appeared to be the most common route, whereas contact transmission caused the highest number of influenza B infections. 31

Use and effectiveness Neuraminidase inhibitors are commonly used for the treatment of influenza in Japan, typically following a positive result from a point-of-care (POC) test. During the 2009-2010 pandemic, over 20 million POC test kits were shipped to hospitals and clinics in Japan to enable rapid diagnosis, and 89% of treated cases were administered NAIs within 48 hours of symptom onset. 32 In Japan during 2013, oseltamivir and laninamivir each represented 40% of NAIs used, while zanamivir (15%) and peramivir (5%) use was considerably less. NAI effectiveness has been assessed in numerous observational studies in Japan. Oseltamivir effectiveness is significantly reduced in patients with delayed treatment, and duration of fever and viral shedding is longer in treated patients with influenza B compared to influenza A virus infections. 33, 34 The reduced effectiveness against influenza B viruses was also observed in zanamivir 33, 35 and laninamivir 35 trials.

To determine whether NAIs reduced mortality during the 2009-2010 pandemic, data were compiled on 29 234 patients hospitalised with confirmed A(H1N1)pdm09 infection. 36 Compared with no treatment, NAI treatment was associated with significantly reduced mortality, with early treatment also showing a reduced risk of mortality compared to late treatment. Although there was no significant clinical effect when comparing late treatment with no treatment in hospitalised patients, there was a significant benefit in treating patients who arrive late into intensive care units. 36 In a household prophylaxis study, inhaled laninamivir given for either 2 or 3 days reduced the illness rate within households to 3.9% and 3.7%, respectively, compared to 16.9% in households given a placebo. 37 A ferret model of oseltamivir prophylaxis has shown that while morbidity was significantly reduced, the prophylaxis regimes did not prevent infection nor significantly reduce virus load. 38 Resistance Although all four NAIs are sialic acid analogues, they have subtle differences in chemical structure and binding properties. Consequently, resistance patterns vary across NAIs. The most commonly detected NA substitution causing NAI resistance in N1-containing influenza viruses is H275Y, which confers resistance to oseltamivir and peramivir, but not to zanamivir and laninamivir. This resistance mutation became fixed in seasonal H1N1 viruses circulating in 2008-2009. 39 A late 2013 cluster of H1N1pdm09 viruses containing the H275Y substitution was detected in 38 (39%) of 97 H1N1pdm09 viruses from community patients not receiving NAIs in Sapporo, Japan, 40

reminiscent of a similar cluster of oseltamivir-resistant H1N1pdm09 viruses in community patients in Australia in 2011. 41, 42 Importantly, both sets of viruses contained permissive NA mutations (V241I and N369K) that have been shown in ferret studies to offset the destabilising and negative effect of the H275Y NA mutation. 43 In hospitalised influenza patients being treated with intravenous zanamivir, next-generation sequencing has been utilised to identify minor resistant virus populations. A total of five NA substitutions were identified in different viruses, including E119K and E119D; however, all apart from E119D were present in such low proportions that they could not be detected by Sanger 'population' sequencing methods. 44 The effects of various mutations in catalytic and framework residues of influenza B NA were investigated using reverse genetics and a range of functional assays. Four substitutions (D198E, I222T, H274Y and N294S) conferred reduced susceptibility to oseltamivir, while three substitutions (E119A, D198Y and R371K) caused highly reduced inhibition by oseltamivir, zanamivir and peramivir. 45 Two of these variants (H274Y, E119A) had in vitro replication fitness comparable to the NAi-susceptible viruses. To date, these substitutions have only been detected on rare occasions in circulating influenza viruses.

An intravenous formulation of zanamivir showed both virological and clinical effectiveness without safety concerns in patients hospitalised with influenza in Japan. 46 A range of new adamantane derivatives have good antiviral activity in vitro and in animal models against H1N1pdm09 and H3N2 viruses that contain the S31N M2 ion channel substitution that confers resistance to amantadine. 47 Favipiravir is a novel pyrazinamide molecule that inhibits replication of various RNA viruses, including influenza types A, B and C (including oseltamivir-resistant strains), and has recently been licensed in Japan for the control of novel or reemerging influenza viruses. Its triphosphate metabolite is an RNA polymerase inhibitor which disrupts virus genome replication; synergy with oseltamivir has been demonstrated in pre-clinical models. 48 A phase II study in the US has shown that a twice-daily regimen decreased the titre and time to cessation of virus shedding, and had a significant benefit in reducing clinical symptoms (NCT01068912; www.clinicaltrials.gov). Subsequent phase III studies are currently ongoing (NCT02008344 and NCT02026349).

A neutralising monoclonal antibody (MHAA4549A) which binds to the HA stalk of influenza A viruses in both group 1 and group 2 HA subtypes has been effective when given up to 72 hours post-infection in mice and ferrets infected with H5N1. 49 Phase I and IIa trials in humans showed that the antibody was well tolerated, had a mean half-life of 21.9 days and was effective as therapy at high doses in experimentally infected volunteers

(NCT01877785). Upcoming placebo-controlled phase IIb trials will target hospitalised influenza patients requiring oxygen and compare the combination of the monoclonal antibody with oseltamivir to oseltamivir monotherapy (NCT01980966). Other broadly neutralising antibodies against multiple clades of H5N1 have been generated by glycan masking of key HA antigenic residues to direct antibody responses to the more conserved stem region of the HA. 50 FluPep, a novel peptide that prevents virus entry into cells, has been shown in mouse studies to be effective in reducing virus titres in lungs, inflammatory cytokines and mortality. 51 Fludase (DAS-181) is a host-targeted therapeutic agent that removes sialic acid from cellular receptors in the respiratory tract, thus preventing influenza virus binding. Delivered topically, it is effective in animal models of lethal H5N1 and H7N9 infection, including a NA1-resistant R292K H7N9 variant. 52 In a phase 2 RCT, inhaled DAS181 reduced pharyngeal viral replication in uncomplicated influenza but did not reduce nasal virus loads or improve clinical outcomes. 53 Another receptor-targeted approach is the development of multivalent sialic acid-binding proteins; 54 a single administration 7 days pre-infection resulted in the protection of 80-100% of mice from lethal H7N9 challenge. 55 Apart from blocking sialic acid, the compound appears to stimulate the expression of pro-inflammatory mediators, thereby 'preparing' the immune system for subsequent influenza infection. When delivered 24 hours post-infection, protection was, however, only 20-40%. 55 Although drug resistance is considered less likely to occur with host-directed therapies, escape mutants have developed rapidly following exposure to a host-directed vacuolar ATPase-inhibiting drug. 56 Furthermore, following serial passage of different viruses in the presence of bafilomycin A1, two HA mutations were selected (A19T and S210N) which resulted in reduced drug susceptibility and increased virulence in mice. 56

Multiple influenza vaccine effectiveness (IVE) studies have used the control test negative design approach to estimate IVE during early and late phases of influenza seasons, the 2009 pandemic, and by age or target groups. Typically, IVE estimates range from 40% to 60% each season. 57 Future studies will investigate IVE with respect to the type of influenza vaccine used, whether IVE differs between the start and end of the season and the effect of previous vaccination. In Japan in 2013/14, IVE for influenza A in children aged 1-5 years averaged 72% (95% CI 64-79), dropped to 48% (95% CI 31-61) in children aged 6-12 years and was not apparent against influenza B in any age group (À1%, 95% CI À19 to 14). 58 Influenza vaccine effectiveness is known to be lower in adults over 65 years of age, a group that accounts for >60% of seasonal influenza-related hospitalisations and >90% of influenza-related deaths. In an effort to improve IVE in the elderly, recent RCTs have investigated the use of adjuvants, intradermal injection and higher doses of antigen. While

the use of AS03-adjuvanted influenza vaccine was only moderately superior to non-adjuvanted vaccine in the elderly, 59 the use of a high-dose vaccine containing four times the standard level of HA (60 lg per virus) did result in improved effectiveness compared to the standard dose vaccine. 60 For vaccine manufacturers, generating high-growth reassortants of certain circulating viruses can be challenging. A recent study used random mutagenesis of PR8 and selection of high-growth clones in MDCK and Vero cells to derive a high-growth version of PR8. 61 Reassortment of the highgrowth PR8 virus with the HA/NA of either H5N1, H7N9 or seasonal influenza viruses showed that yields significantly exceeded equivalent reassortants that contained the internal genes of the 'normal' PR8 virus. 61

As of July 2014, the number of confirmed cases of MERS-CoV has exceeded 830, with at least 288 associated deaths. 62 The majority of cases have involved patients with comorbidities (76%) and are predominately males (63%) with a median age of 47. 63, 64 Fewer than 25% of patients have reported contact with animals including dromedary camels, which have been shown to be one likely animal reservoir based on sero-positivity and detection of MERS-CoV. 65 More than 25% of the infections have been in healthcare workers, and the large number of nosocomial infections is likely due to inadequate infection control in hospitals plus enhanced surveillance that has detected a substantial number of mild or asymptomatic infections. 63 Outside hospital, the burden of disease is likely to be larger than has been reported. 66 Serological analysis of several UK patients found a rapid rise in antibodies from day 10, and that titres were maintained for at least 300 days post-infection. Anti-S (spike glycoprotein) antibodies are responsible for virus neutralisation. Importantly for serological analyses, patients who experience only mild disease may mount only a modest serological response. 67 Sequential samples from three cases involved in a chain of transmission were extensively analysed using next-generation sequencing. 68 Various minority variants were detected, of which some were transient while others were transmitted, and there was evidence of variation in frequency of some variants in different body compartments.

Various therapeutic options have been investigated for the treatment of MERS-CoV, but no therapy of proven value currently exists. The use of SC was associated with adverse outcome in SARS 69 and is not recommended for MERS-CoV. Many agents have shown inhibitory effects against MERS-CoV in cell culture including interferon α ribavirin, cyclosporine A, mycophenolic acid, chloroquine and lopinavir. 70 Interferons, lopinavir, mycophenolate, possibly alisporivir and combinations are reasonable choices for testing in controlled clinical trials. Exploratory post hoc metaanalysis of studies related to SARS and

severe influenza has shown a significant reduction in mortality following convalescent plasma treatment compared to placebo or no therapy (odds ratio 0.25; 95% CI 0.14-0.45).⁷¹ Thus, the early use of virus-specific neutralising antibodies in the form of convalescent plasma and monoclonal or polyclonal neutralising antibodies for treatment of MERS-CoV has the highest likelihood of clinical benefit.⁶⁴ Modalities with risks likely to exceed benefits include SC, ribavirin monotherapy and IVIG.⁷²

Respiratory syncytial virus (RSV) Respiratory syncytial virus disproportionately impacts children in low-income countries.⁷³ Almost all children will have been infected with RSV by their 2nd birthday, and it is the number one cause of hospitalisation of infants in the US, causing 10 times more infant deaths than influenza.⁷⁴ In addition, RSV infects 3-10% of adults annually and accounts for 5-15% of community acquired pneumonia (CAP) and 9-10% of hospitalisations,⁷⁵ a burden of disease that approaches that caused by influenza. In a study, conducted in Hong Kong, of 607 hospitalised adults with RSV, 40% had pneumonia and 70% required supplementary oxygen; mortality rates and duration of hospital stay were similar to those observed for influenza patients.⁷⁵ Approximately, 15% of hospitalised RSV patients had bacterial superinfections. Although corticosteroids were used to treat 38% of patients, treatment had no benefit on clinical outcome, and instead increased bacterial secondary infections and caused a longer duration of illness.⁷⁵ RSV replication appears prolonged in patients with comorbidities and LRT complications.

Palivizumab prophylaxis of premature infants of <6 months of age has been shown to reduce hospitalisation due to RSV by 55%.⁷⁶ Preventing RSV during infancy has been associated with reduction of wheezing later in life.⁷⁷ Trials of other monoclonal antibodies have typically shown that they do not achieve superiority compared to palivizumab and therefore do not achieve licensure. Furthermore, treatment with neutralising monoclonal antibodies does not appear to reduce virus load or disease severity in hospitalised infants.⁷⁸ Alternative options for RSV therapy to be assessed in future clinical trials include inhaled nanobodies, aerosolised peptides, nucleoside analogues and RNA-interference molecules.

Human rhinoviruses (HRV) usually cause mild acute respiratory infections, but on occasions can also cause more severe respiratory infections, including exacerbations of asthma and COPD. Of 115 Japanese children with asthma, a respiratory virus was detected in 86%, of which HRV (n = 36) or RSV (n = 47) were most common.⁷⁹ Ex vivo bronchial

epithelial cells from people with asthma are more susceptible to HRV infection, due to deficient induction of IFN- β and IFN- λ . In a study of 147 asthmatics on inhaled corticosteroid therapy, with a history of virus-associated exacerbations, patients were randomised to 14-day treatment with inhaled IFN- β or placebo within 24 hours of developing cold symptoms. Patients who received IFN- β had enhanced morning peak expiratory flow recovery, reduced need for additional treatment and boosted innate immunity as assessed by blood and sputum biomarkers. In an exploratory analysis of a subset of more difficult-to-treat asthma ($n = 27$ IFN- β ; $n = 31$ placebo), worsening of symptoms increased significantly in the placebo group, but was prevented by IFN- β ($P = 0.004$).⁸⁰ A picornavirus-specific antiviral, vapendavir, was found to reduce symptom scores, lower bronchodilator puffer use and reduce viral load in asthma patients with an URTI due to HRV.⁸¹

Diagnostics Point-of-care tests that can deliver a result in 15 minutes have been available in many countries for the last decade, but while having good specificity, the sensitivity has typically been poor, ranging from 10% to 80% compared to PCR or culture. Their use in emergency departments of hospitals can result in reduced unnecessary antibiotic use and an increased likelihood of discharge. Newer immunofluorescence-based POC tests with improved sensitivity are being developed. In addition, the Quidel Sofia POC test may be linked via the internet such that results can be reported in real-time to central databases. Other POC tests are using photographic silver amplification immunochromatography technology to increase sensitivity.⁸² PCR remains the gold standard for virus diagnostics with an ability to be rapid, sensitive, specific and to identify a wide range of pathogens via different assays. The ability to multiplex multiple pathogen targets allows costs to be reduced in a diagnostic setting. New closed-system technologies which involve only minimal hands-on time (a few minutes) and that conduct both automated nucleic acid extraction and PCR for multiple pathogens are now available, but are currently limited for clinical diagnostic purposes due to low-throughput capabilities.⁸³ Next-generation sequencing technologies and PCR-based analyses with increasing sensitivity both offer considerable scope in diagnosis, although our current understanding of the clinical impact of pathogens at low levels or the presence of variants as minor virus populations is limited. Providing low-cost, sensitive assays for diagnosing respiratory virus infections in low/middle-income countries is challenging, but has the potential to improve treatment and avoid unnecessary antibiotic use in these regions.

Repurposed drugs for respiratory viral infections Nitazoxanide (NTX) is an antiparasitic agent approved for *Giardia* and *Cryptosporidium* infections that also inhibits replication in vitro of influenza and other respiratory viruses. 84 Treatment with NTX 600 mg twice daily for 5 days was associated with a reduction in the duration of symptoms in participants with acute uncomplicated influenza. 85 In a subset analysis of 238 patients with no confirmed virus infection, treatment with NTX 600 mg also led to a shorter time to alleviation of symptoms in comparison to placebo (88Á4 versus 105Á7 hours, $P = 0.02$). 86 Systemic corticosteroids for respiratory virus infections A review of prospective observational studies has shown that SC increased the risks of mortality and morbidity (e.g. secondary infections, hospital-acquired pneumonia) in severe infection due to influenza A(H1N1)pdm09 especially with delayed antiviral therapy. 87 During SARS infections, a higher risk of avascular necrosis and prolonged virus shedding were observed in patients who had received high-dose SC therapy. 88 It is therefore important to avoid the use of high-dose SC in severe respiratory viral infections outside the context of clinical trials. Larger trials are needed to resolve the uncertainty regarding the effect of early SC therapy in ARDS. Low-dose SC is indicated for management of refractory septic shock, 89 and a short course of SC is indicated for acute exacerbations of obstructive airway diseases (asthma, COPD) 90 Current evidence does not support a clinically relevant effect of systemic or inhaled glucocorticoids on admission or length of hospitalisation for acute viral bronchiolitis in infants and young children. 91

Respiratory syncytial virus, influenza viruses, parainfluenza (PIV) viruses and adenoviruses (AdVs) cause the most serious disease in immunocompromised hosts, but other respiratory viruses are becoming increasingly appreciated as a cause of both upper and lower respiratory tract disease. The potential for these viruses to cause lower respiratory tract infections (LRTI) after transplantation varies. Human metapneumovirus infections have similar outcomes to RSV infection in hematopoietic stem cell transplant (HSCT) recipients, including potentially severe and fatal pneumonia. HRV and coronavirus infections are very frequent in transplant recipients, but severe lower respiratory tract disease is uncommon.

In a prospective study of 112 lung transplant recipients, the virus infection rates upon screening, routine and emergency visits were 14%, 15% and 34%, respectively. Picornaviruses were identified most frequently in nasopharyngeal (85/140; 61%) and BAL specimens (20/34; 59%). Asymptomatic virus carriage, mainly of picornaviruses, was found at 10% of screening visits. Infections were associated with transient lung function

loss and high calcineurin inhibitor blood levels. The hospitalisation rate was 50% for influenza and PIV and 16.9% for other viruses. Acute rejection was not associated with virus infection. 92 The risk factors for severe LRTI among transplant recipients include early onset post-transplant (<3 m), steroid boluses, young children (<1 year), chronic GVHD, lymphopenia/lymphodepletion and allogeneic HSCT patients. 92 Influenza

Immunocompromised patients with influenza exhibit more complications, longer virus shedding and more antiviral resistance, while often demonstrating milder clinical symptoms and signs on initial clinical assessment. 93 Influenza A (H1N1)pdm09 viruses have the potential for rapid emergence of oseltamivir resistance and causing severe morbidity, particularly in immunocompromised patients with lymphopenia and delayed antiviral therapy. 94 Influenza viraemia may serve as a marker for overall poor outcome with increased risk of progression to LRTI, hypoxaemia, respiratory failure and death. Influenza RNA in blood (viraemia) was detected in nine of 79 (11.4%) HSCT recipients with influenza. Among patients with LRTI, viraemia was associated with increased hazards of overall as well as influenza-associated death (hazard ratio 3.5, 1.1-12). 95 In 143 HSCT recipients with documented seasonal influenza infection, treatment with high-dose corticosteroids was associated with a trend towards prolonged virus shedding [(OR), 3.3; 95% CI 1.0-11; P = 0.05], whereas antiviral therapy initiated to treat upper respiratory tract infection (URTI) was associated with fewer cases of LRTI (OR, 0.04; 95% CI, 0-0.2; P < 0.01) and fewer hypoxaemia episodes (OR, 0.3; 95% CI, 0.1-0.9; P = 0.03). 96 In view of the risks of prolonged replication and drug resistance emergence, 97 a longer duration and a higher NAI dose may be beneficial. Early therapy was consistently demonstrated to have improved outcomes. [98] [99] [100] [101] Other treatment options under study are the use of triple combination therapy with amantadine, oseltamivir and ribavirin, 102 or intravenous peramivir 103 or zanamivir. 104 Therapy of influenza in lung transplant recipients is associated with a reduced risk of developing bronchiolitis obliterans syndrome. 105 Parainfluenza DAS181 is inhibitory for PIV and influenza viruses, including those resistant to the amantadine and NAIs 106 and may be effective in treating immunocompromised patients with severe PIV lung disease. 107 In a study of four severely immunocompromised children with PIV disease treatment with DAS181 for 5-10 days, by dry powder inhalation or nebulisation, was well tolerated. Transient increase in serum alkaline phosphatase, liver function and coagulation tests were observed, but nasal wash virus loads were reduced in all patients within 1 week with improved clinical features. 108

In a RCT of lung transplant recipients with RSV infection, the incidence of new or progressive bronchiolitis obliterans syndrome at day 90 was significantly reduced in 16 patients who received a small interfering RNA against the RSV N gene (ALN-RSV01)

compared with placebo (n = 8) (6.3% versus 50%, P = 0.027). 109 In a larger follow-up multicentre phase IIb study, treatment with ALN-RSV01 showed a greater than eightfold reduced risk in developing bronchiolitis obliterans syndrome at day 180. 110

Adenovirus is a serious, often fatal infection in immunocompromised patients, especially in HSCT recipients. The control of AdV is mostly T-cell mediated, and therefore, patients who have received T-cell suppressive regimens are at an increased risk for AdV infection. The annual incidence of AdV infections in HSCT recipients ranges from 5% to 50%, and is increasing, likely due to increased use of T-cell-depleted allografts and cord blood as source. The mortality rate is up to 80%. 111 Brincidofovir (BCV; formerly CMX-001) is an orally bioavailable lipid-conjugate of cidofovir (CDV) that provides high intracellular concentrations of CDV diphosphate with a long intracellular half-life (up to 4-6.5 days). BCV is 65-fold more potent against AdV than CDV in vitro with a low risk of myeloid or nephrotoxicity, but gastrointestinal side effects are more common. 112 In a retrospective study of 13 immunocompromised patients given BCV for AdV disease after failing or intolerance to i.v. cidofovir nine patients (69.2%) demonstrated a virological response (VR), which was defined as a 99% drop from baseline or undetectable AdV DNA in serum by week 8. Patients with VR had longer survival than those without VR (median 196 days versus 54.5 days; P = 0.04). 113

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Detectable 2019-nCoV viral RNA in blood is a strong indicator for the further clinical severity

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Abstract: The novel coronavirus (2019-nCoV) infection caused pneumonia. we retrospectively analyzed the virus presence in the pharyngeal swab, blood, and the anal swab detected by real-time PCR in the clinical lab. Unexpectedly, the 2019-nCoV RNA was readily detected in the blood (6 of 57 patients) and the anal swabs (11 of 28 patients). Importantly, all of the 6 patients with detectable viral RNA in the blood cohort progressed to severe symptom stage, indicating a strong correlation of serum viral RNA with the disease severity (p -value = 0.0001). Meanwhile, 8 of the 11 patients with anal swab virus-positive was in severe clinical stage. However, the concentration of viral RNA in the anal swab (Ct value = 24 ± 39) was higher than in the blood (Ct value = 34 ± 39) from patient 2, suggesting that the virus might replicate in the digestive tract. Altogether, our results confirmed the presence of virus RNA in extra-pulmonary sites.

Text: The 2019 novel coronavirus (2019-nCoV), originally outbreaking from Wuhan China, has transmitted in an extremely short period to 25 countries and infected over 31 000 individuals as of Feb 06, 2020, causing an international alarm. Basic scientific research has achieved significantly in the investigation of viral origination [1, 2], transmission and evolution [3], and unprecedented public health control actions in China have been activated and effectively prevented the otherwise dramatic spread. The 2019-nCoV virus seems more infectious in its public transmission capacity compared to the well-known 2003 SARS virus in spite of the unavailability of convincingly scientific evidence. The mechanism of viral transmission is still worthy of further exploration.

Currently, one urgent and critical challenge is to treat infected patients and save their lives. Several studies have roughly described the overall clinical features of 2019-nCoV patients [4, 5]. However, the more specific and classified clinical characteristics of the infected patients still require further investigation, particularly for those with severe symptoms, which is roughly estimated to be approximately 15-20 percent of totally confirmed cases based on the local data in our hospital. Clinically, for those severe patients, the main

symptoms of 2019-nCoV pneumonia are fever, decreased white blood cell and lymphocyte count, increased C reaction protein and abnormally expressed cytokines [6] .

One remaining question to be resolved is whether the 2019-nCoV virus can replicate in extra-pulmonary sites, which might account for the deteriorated clinical manifestation. In this study, we investigated whether the patients with severe clinical symptoms exhibited special profiles of virus replication or/and distribution compared to those only with mild symptoms.

Patients, who were confirmed to be infected by the 2019-nCoV virus, were firstly enrolled in or transferred to Guangzhou Eighth People's Hospital for treatment purposes. This study followed the guideline of the Ethics Committee of Guangzhou Eighth People's Hospital. All blood, pharyngeal swab, and anal swab samples were collected for diagnostic purposes in the laboratory and our study added no extra burden to patients. Viral RNA was extracted with Nucleic Acid Isolation Kit (Da'an Gene Corporation, Cat: DA0630) on an automatic workstation Smart 32 (Da'an Gene Corporation) following the guidelines. Real-time reverse transcriptional polymerase chain reaction (RT-PCR) reagent (Da'an Gene cooperation, Cat DA0930) was employed for viral detection per the protocol. In brief, two PCR primer and probe sets, which target orf1ab (FAM reporter) and N (VIC reporter) genes separately, were added in the same reaction tube. Positive and negative controls were included for each batch of detection. Samples were considered to be viral positive when either or both set(s) gave a reliable signal(s).

All patients had pneumonia-based diseases but with diversified clinical manifestation. To simplify data analysis, the patients were only classified as either mild or severe clinical symptom groups based on the guideline newly released by Chinese government. Patients who were with at least one of the following symptom should be diagnosed to be severe case, 1) distress of respiratory with respiratory rate $\geq 30/\text{min}$; 2) Oxygen saturation $\leq 93\%$ in the rest state, and 3) arterial oxygen tension (PaO_2) over inspiratory oxygen fraction (FIO_2) of less than 300 mm Hg. In the blood detection cohort (Figure 1 (A)), patients who had at least one serum sample measurement with the PCR method were included. In the 57, 6 cases were detected to be blood positive, all of them (100%) were severe in symptom requiring special care attention, and the blood of the rest 51 cases was without detectable virus in the blood, only 12 of them (23.5%) were severe cases. The ratio of severe symptoms between these two groups was significantly different (p value = 0.0001). In the

anal swab cohort (Figure 1 (B)), 11 of 28 cases were detected to be anal swab positive, 8 of them (72.7%) were with severe symptoms, which was significantly higher than that 4 (23.5%) of the rest 17 cases without detectable virus in anal were severe cases.

Fortunately, two cases with detectable virus both in blood and anal swab cohort were recorded. Patient 1 (Figure 2 (A)) was admitted to ICU after enrollment evaluation and was highly suspected infection with 2019-nCoV because of his recent travelling from Wuhan and of confirmed pneumonia by radiographic diagnosis with 5-day fever and 1-day continuous dry coughing. He was then confirmed to be infected by the 2019-nCoV virus on illness day 6 by CDC. High concentrations of the viral RNA were detected in the pharyngeal swabs on illness days 5 (Ct = 17 + 25), 7, 8 (Ct = 25 + 26), and 11 (Ct = 15 + 25). In the blood, no viral RNA was detected on day 5 but the sample on day 6 gave a weak positive signal (Ct = Neg+39), and then the signal was gone again on day 8. On day 9, a low level of viral RNA (Ct = 36 + 41) was detected again in the blood. On day 12, the blood lost signal again. A high concentration of virus RNA (Ct = 23 + 27) was detected in the anal sample on day 13, on the day the 2019-nCoV virus was not detected in the pharyngeal swab. Unfortunately, he was transferred out to another hospital after an emergency expert consultation.

Patient 2 (Figure 2 (B)), who had a clear infection history and started fever 5-day ago and dry coughing 2-day ago, was admitted with clinically highly suspect of 2019-nCoV infection, considering the radiographical diagnosis which indicated clear pneumonia in the bilateral lung lobes. The virus was detected in his blood on illness day 7 (Ct = 34 + 36) and 8 (Ct = 38 + 38). His infection was also informed by the CDC on day 8. Because his disease advanced very fast, he was transferred to the ICU ward for special medical care requirements on day 9, on which day high titers of virus (Ct = 25 + 36) were detected in the pharyngeal sample. Importantly, virus RNA was detected in all pharyngeal (Ct = 23 + 24), blood (Ct = 34 + 39) and anal (Ct = 24 + 29) samples on day 10. He was transferred out to another hospital after an emergency expert consultation.

Finally, we described here the four patients with detectable serum viral RNA. Patient 3 (Figure 3(A)) was transferred to the ICU directly on illness day 11 because of his severe condition, the 2019-nCoV virus was laboratory detected both in pharyngeal (Ct = 30 + 30) and blood samples (Ct = 37 + 39) on day 12, And his infection was confirmed by CDC on day 13. Pharyngeal samples were PCR positive on days 14 and 17 and became negative on day 22. Patient 4 (Figure 3(B)) was transferred to the ICU ward on the illness day 6 with a

CDC confirmation. His disease advanced pretty fast and became severe on day 7 and he was transferred to ICU after his blood sample was detected to be virus-positive (Ct = 32 + 37). On day 9, he was transferred out. Patient 5 (Figure 3(C)) was admitted on illness day 4 and his blood sample was virus-positive (Ct = 38 + Neg) on day 6. Her disease progressed rapidly to a severe stage within the next 3 days. Patient 6 (Figure 3 (D)) with a clear history of virus infection was confirmed to be infected on infection day 7. Viral RNA was detected in his blood sample on day 9, one day ahead of his transfer into ICU. As his condition worsens, he was transferred out on day 13.

In this retrospective study, we analyzed the PCR data of virus detection in different tissues in our laboratory. Firstly, our observation indicated that the presence of viral RNA outside of the respiratory tract might herald the severity of the disease and alarm the requirement of special care. In the blood test cohort, all the 6 infected patients were in (or later progressed to) severe disease stage when serum viral RNA became detectable, which showed a significant difference compared to the blood negative group ($p = 0.0001$). Patient 2 (Figure 2(B)), 5 (Figure 3 (C)) and 6 (Figure 3(D)) all had detectable viral RNA in the serum before they progressed to the clinical severe symptom stage. Unfortunately, we missed the earlier time points of patient 1 (Figure 2(A)) and 3 (Figure 3(A)) who were directly admitted to ICU on transfer to our hospital because of severe condition, of patient 4 (Figure 3(B)) who had serum sample collected one day post the diagnosis of severe illness. We, fortunately, observed high serum viral load in serum within their severe illness stage. In the anal swab cohort, we found that the presence of virus RNA in the anal digestive tract was also positively correlated with disease severity ($p = 0.0102$). The 3 patients detected with anal virus RNA but in mild stage should be monitored whether they will progress to the severe stage. We have summarized the information of approximately 70 percent of the patients in Guangzhou city, and the study represented nearly the whole picture of this region. However, the virus outbreak in such an emergence, allowing no delay in waiting for more patients to further confirm the findings.

Secondly, a high concentration of viral RNA in anal swabs suggested the digestive tract might be one extrapulmonary site for virus replication. For patient 1, a high concentration of viral RNA (Ct = 23 + 27, on day 13) was detected in anal swab but not in pharyngeal (the same day) and blood (1 d ahead). For patient 2, higher concentrations of viral RNAs were detected in anal swab (Ct = 24 + 39) and pharyngeal swab (Ct = 23 + 24) than in the blood (Ct = 34 + 39) on the same day. Angiotensin-converting enzyme 2 (ACE2) still is one of the receptors for 2019-nCoV attachment and entry [2] . Intensive structural analysis of the S

protein of 2019-nCoV with the SARS-Coronavirus suggested that several critical residues in the viral spike protein might confer favourable interaction with human ACE2 [7] . Of note, ACE2 is also abundantly present in humans in the epithelia of the small intestine besides the respiratory tract and is ubiquitously present in endothelial cells [8] , which might provide possible routes of transmission, and might account for the high transmission capacity of the new virus. We propose that rampant coronavirus replication in pulmonary alveolus results in the breakdown of the alveolar vessel and the subsequent virus leakage into the blood flow, through which the virus is disseminated across the whole body. Then the virus succeeds in establishing reinfection in the digestive tract by using the highly expressed ACE2 receptor, which exacerbated the disease vice versa. Bat originated coronavirus was found to replicate in the swine digestive tract recently, also suggesting the potential replication possibility in the human digestive tract [9] . Nevertheless, confirmation of virus transmission through the digestive tract warrants further virus isolation from the anal swab in high safety level lab.

Unfortunately, in our study, we did not collect stool samples from patients and did not pursue viral RNA in the stool. But we believe the existence of virus RNA in the stool samples from these patients because that a large amount of viral RNA was detected in anal swabs and that viral RNA had also been detected in a case reported from the United States [10] . Also, we didn't collect sputum and bronchoalveolar lavage fluid for virus detection because that the dry coughing characteristic of patients infected with 2019-nCoV prevents producing enough amount of sputum and that bronchoalveolar lavage fluid collection requires a sophisticated operation which increases virus exposure possibility of care providers to high concentrations of virus-containing aerosol.

In summary, we find that the presence of viral RNA in the blood and anal swab is positively correlated with the severe disease stage and that early monitoring of virus RNA in blood and the digestive tract on top of the respiratory tract might benefit the disease prediction.

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'Tiny Iceland' preparing for Ebola in a globalized world

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Abstract: Background: The Ebola epidemic in West Africa caused global fear and stirred up worldwide preparedness activities in countries sharing borders with those affected, and in geographically far-away countries such as Iceland. Objective: To describe and analyse Ebola preparedness activities within the Icelandic healthcare system, and to explore the perspectives and experiences of managers and frontline health workers. Methods: A qualitative case study, based on semi-structured interviews with 21 staff members in the national Ebola Treatment Team, Emergency Room at Landspítali University Hospital, and managers of the response team. Results: Contextual factors such as culture and demography influenced preparedness, and contributed to the positive state of mind of participants, and ingenuity in using available resources for preparedness. While participants believed they were ready to take on the task of Ebola, they also had doubts about the chances of Ebola ever reaching Iceland. Yet, factors such as fear of Ebola and the perceived stigma associated with caring for a potentially infected Ebola patient, influenced the preparation process and resulted in plans for specific precautions by staff to secure the safety of their families. There were also concerns about the teamwork and lack of commitment by some during training. Being a ‘tiny’ nation was seen as both an asset and a weakness in the preparation process. Honest information sharing and scenario-based training contributed to increased confidence amongst participants in the response plans. Conclusions: Communication and training were important for preparedness of health staff in Iceland, in order to receive, admit, and treat a patient suspected of having Ebola, while doubts prevailed on staff capacity to properly do so. For optimal preparedness, likely

scenarios for future global security health threats need to be repeatedly enacted, and areas plagued by poverty and fragile healthcare systems require global support.

Text: Global health; prevention and control; public policy; qualitative evaluation; emergency responders; communicable diseases; emerging; fear Background On 8 August 2014, the World Health Organization declared the Ebola epidemic in West Africa as a Public Health Emergency of International Concern (PHEIC) under the International Health Regulations (IHR) [1] . All three of the worst affected countries were to address the emerging epidemic challenge without staff, stuff, space and systems [2] [3] [4] . With the epidemic seemingly out of control, and a proportionately high number of doctors, nurses, and midwives succumbing to Ebola [5] , there was a growing fear of transmission beyond the region. In breach of WHO recommendations and guidelines [6] , flights were cancelled and cross-border movement curtailed [7] . The epidemic caused public concern outside West Africa [8] , as fear and racism found fertile ground [9] [10] [11] , and in an effort to stop the international spread of the disease, all states were advised to be prepared to detect, investigate, and manage Ebola cases [1] .

Preparedness as part of disaster risk reduction is defined as 'the knowledge and capacities developed by governments, response and recovery organizations, communities and individuals to effectively anticipate, respond to, and recover from the impacts of likely, imminent or current disasters' [12] . Yet, preparedness is also enveloped in and influenced by the socio-cultural dimension at the individual, organizational, and national levels, and measures to manage outbreaks are not always accepted or accommodated by the communities to which they are applied [13] . An analysis of eight European countries' preparedness plans since 2009 for countering a future influenza A (H1N1) pandemic revealed that the way plans were framed varied considerably, and '[told] us something about how the different countries want pandemics and preparedness to be understood by the public' [14] . More research was encouraged into cultural and social structures in the respective countries.

In Iceland, information about the Ebola epidemic in West Africa came from several sources. The Directorate of Health (DH) first reported on the epidemic on 8 April 2014 [15] . In Icelandic media, the rapid progress of the Ebola epidemic in West Africa was increasingly highlighted, and exported Ebola cases to Spain, USA, and elsewhere, were widely covered. Fear of a global epidemic was rife, and in media and online discussions,

doubts were raised about the Icelandic health system's capacity to take care of a patient with Ebola [16] [17] [18] , despite its ranking as one of the best in the world [16] .

On 11 August 2014, three days after WHO declared PHEIC because of Ebola, DH encouraged Icelandic citizens to avoid visits to the area, if possible, and reported that the national epidemic preparedness plan was being activated for Ebola [19] . It was elaborated by a team that involved the Chief Epidemiologist at the DH, Landspítali University Hospital (LSH), the Department of Civil Protection and Emergency Management (DCPEM), and the seven Primary Healthcare Regional Organizations in the country at the time. Key external partners were the European Centre for Disease Prevention and Control (ECDC) and WHO, in addition to Nordic collaborators in epidemic preparedness [20] . At the same time, it was regarded as highly unlikely that Ebola Virus Disease (EVD) would spread in the country [21] . Recognized scenarios included the possible appearance of an infected person in need of treatment, who could be either an Icelandic citizen who had visited or worked in one of the affected West African countries, or a person with signs of EVD on a trans-Atlantic flight in the navigation area controlled by Icelandic authorities [22] [23] [24] [25] . On 3 November 2014, the plan was put to the test when a foreign airline made a non-scheduled landing at Keflavík International Airport due to fear of EVD in one passenger from South Africa. Parked in a closed-off area, a physician in full Personal Protective Equipment (PPE) entered the plane, but quickly ruled out Ebola [26] .

Irrespective of good or bad overall performance, health systems are tested in times of crisis, such as epidemics. Here, the aim is to describe and analyse the process of establishing preparedness plans for Ebola in Iceland, with a specific focus on the perspectives and experiences of managers and frontline health workers involved in the process.

This study is part of a larger study on the impact that the global threat of the Ebola epidemic had in Iceland [16, 27] . Qualitative case study methodology was applied, perceiving the preparedness planning and training process as the case with clear boundaries of the initiation, process, and wrap-up of preparedness planning and training. The study was conducted in April-May 2016, and the interviewed participants were administrators and frontline health professionals central to the case, so as to explore their perspectives and experiences concerning Ebola preparedness [28, 29] . Staff in managerial positions were contacted by one of the authors (GG) for permission to interview them

based on their role in the preparedness plan. To identify potential interviewees in the Ebola Treatment Team (ETT), the director of the team listed relevant email contacts. Those who responded positively were subsequently invited for an interview, conducted in Icelandic by one of the authors (ÍEH), a physiotherapist. In case interviewees suggested other potential participants, they were invited through email to participate. A similar methodology was applied to identify participants from the Emergency Room (ER). They were included in order to represent frontline health workers who worked in the only ER in Reykjavík, where persons exposed to EVD were most likely to first seek care in case of acute illness.

Three separate interview guides were developed one each for managers, ETT, and ER respectively (see supplementary material). The interviews included open questions probing the role of their institution in preparedness, the experience of the training process, challenges encountered or expected, and any dilemmas that they may have experienced in relation to the preparedness plan. The recruitment of participants was concluded when saturation was reached. Each interview was recorded and took about 20 to 60 minutes; they were then transcribed and analysed using thematic analysis. The data material was read through repeatedly, sorted, and categorized, based on the participants' priorities in the representation of their views. From this exercise, three broad themes were inductively identified that corresponded to critical perspectives introduced by the participants.

Permission to conduct the study was granted by Iceland's National Bioethics Committee (VSN- and Landspítali University Hospital (LSH 13-16, 4 February 2016) . Reporting on the results was guided by the COREC guidelines [30] ; however, to ensure anonymity of the respondents within the small community of staff who took part in the preparedness activities, participant information is not associated to quotations.

The Icelandic Ebola Preparedness Plan included the establishment of an ETT within LSH [31] , and the preparatory activities engaged more than two hundred staff across all of its departments. The ETT consisted of about 50 healthcare professionals who had volunteered to participate, including 11 doctors and 28 nurses, a few laboratory technicians, radiologists, and auxiliary nurses. They attended special training sessions focused on protocols for admission and treatment of a patient with EVD, the donning/doffing of PPE, and personal protective measures during patient care. A new provisory unit was designed to be set up on the ground floor to minimize the risk of infection spreading to other units

within the hospital, with two rooms specifically identified for the care of a patient with EVD [31] .

Managers' accounts of this period elaborated the complexity of preparedness planning in terms of the involved institutions, actors, procedures and requirement of the plan. One manager concluded:

You get no discount. You can never go the shorter way. There was always something that surprised you. We thought this was a lot like a three headed monster, so when you chopped off one of its heads, three other emerged, every solution was followed by more problems.

The health professionals who volunteered to join ETT did so for different reasons. Ebola preparedness was 'a job that had to be done', and 'someone had to do it'. Some referred to ethical or professional obligations: This is just a part of being a nurse, to encounter situations that can be dangerous to you or someone else, but you have made this decision and you deal with it. Some connected their decision to their 'action gene' or 'addiction to taking risks', while others said they had already raised their kids and had years of experience, including work with other epidemics, such as HIV. Yet, the practice of volunteering in the preparation was questioned. One participant said:

We learned that we could not rely on volunteers ... when you work in an infectious disease department you cannot choose what infections you want to work with.

ER staff indicated that for them working in the ER was enough of a risk to take, no reason to expose oneself even more by joining the ETT, and appreciated that others had volunteered.

All participants noted that co-operation and communication had generally functioned well during the preparedness planning, with information flowing both ways. Short communication lines within the healthcare system were perceived as both a strength and a weakness; a strength, insofar as people knew each other, but a weakness because of the uneven burden of workload. Staff of the ETT and in the ER felt they had been well-informed,

and that openness and honesty had characterized the planning and diminished their initial fear. Those in managerial positions had listened and taken their opinions into consideration. One said:

They were honest, no one was hiding anything, everything was on the table, no one tried to make things more appealing and say that everything would be OK, they just told us about things as they were.

Both management and participants from the ETT and ER expressed their ambiguity in terms of trust, doubt, and fear. Participants conveyed trust in the health system and their own role as health professionals, while at the same time admitting to facing formidable challenges during the elaboration of the preparedness plan. Facilities for isolation and treatment of patients with Ebola were less than perfect:

We assessed how we could use the department ... and change it in just a few hours into some kind of an isolation unit that we could possibly use.

Some compared this short-term isolation facility to a 'camping site', as the facilities were too provisional and not comparable to those found elsewhere. There was also doubt about how many Ebola patients LSH would be able to care for: 'Maybe one or two patients, barely more'.

Respondents believed that the training and education of the members of the ETT and ER had been satisfactory. They felt that it had been proportionate to the risk, while some were concerned about the lack of staff. Nonetheless, there were contradictions on the division of labour among the professionals, exemplified by different ideas on how to proceed if a patient suspected of having an EVD came in an ambulance to the LSH for treatment. Almost all participants stated that they were ready to do their part in the Ebola response, or 'as ready as [we] could be'.

There were diverse opinions on what it meant to be ready: to treat one confirmed case of Ebola, one suspected case, or more EVD patients? When asked if Ebola was a real threat to

the country, participants usually referred to how easy it was to travel the globe: 'Yeah, why not, the world is getting smaller'. Although Ebola was thought of as a real danger by many, some participants expressed difficulty in taking their training seriously, doubting that Ebola would ever reach Iceland. One respondent said:

People were dedicated in the beginning, but when the news appeared that Ebola was receding, that diminished, and I never felt like this formally ended.

Participants described their relief that nothing really happened, while emphasizing the need to experience a real situation to evaluate the preparedness efforts. One participant said that 'a little bit more seriousness [would have been] needed in the PPE practices'.

It was taken as a manifestation of fear that some of the staff in the communicable disease department of the LSH refused to take part in the ETT. When describing their fears, ETT members frequently connected it to their working conditions. Many of them were afraid that they would not get the best PPE, others that they would not do the donning/doffing correctly and, lastly, they were worried about work performance while in the PPE. One participant said:

What bothered most of us was how uncomfortable the PPE was and I think that made people nervous: "How will I manage working in this for hours?"

Another described the donning/doffing process like a 'complicated ballroom dance'. Moreover, participants were afraid of 'unknown territories', that is, they did not know the hospital ward, they were supposed to work in, and some team members had no recent experience of clinical work. One participant said: I didn't think these [non-clinical] people belonged in the team, because this is a very clinical environment in addition to having to be in this costume [PPE] with the risk of becoming infected by mistake.

Those with non-clinical background were, however, aware of their limitations: I realized that I would not be the one in the front, I would not be managing patients directly.

The importance ascribed to teamwork was evident in relation to fear. Participants described fear of working with people they had not worked with before:

The weakest link in the preparation was that even though I knew their faces, I had never worked with them.

Another issue was no-show by some team members in training sessions or in lectures: This is team-work, one does this and the other one does this, [we] help each other. Then you don't want to be working with someone who didn't show up.

There were a lot of doctors who just dropped in, dropped out, and then dropped in again. I asked myself: Are these individuals ... ready to take this on?

Participants in the ETT mentioned the precautions they took or intended to take to cope with their feelings of fear, should Ebola emerge in Iceland. A major precaution was planning to avoid contact with the family while working with Ebola patients. One participant said: 'You thought ... about your children at school ... parents in the neighbourhood ...' if they knew (s)he was working with an Ebola patient. For them, it was important they would have access to special accommodation in case of clinical EVD work 'so I wouldn't be exposing anyone or creating hysteria'. ETT members mentioned the extra insurance offered as a prerequisite for taking part in the team. 'The normal insurance for LHS staff would not cover everything if we were to become sick or even lose our lives.' Amongst ER staff, the matter of insurance did seem to be less of an issue compared to the ETT. One respondent said: 'You are used to being at risk by many disease threats'. Furthermore, the issue of higher salaries and risk commission came up in the interviews, but overall did not matter as much to the participants as the insurance, or assurance of accommodation in case of need.

Characteristics associated with Iceland and the Icelandic people were referred to repeatedly by participants. The concept 'Tiny Iceland' was often mentioned and emerged with positive and negative connotations. 'Tiny Iceland' referred to the size of the country and population and its perceived capability to still 'get the job done'. even though compromises had to be made. Comparing how Iceland handled its responsibilities differently from other countries of a larger size was often brought up, both with pride in

Iceland as a strong independent nation, and with insecurities about its capacity in comparison to other countries. It was pointed out that since the preparedness process was in the hands of a few people, everyone knew their role. As one administrator said: This little hospital system, as complicated as it might seem every day, gives you the chance to just pick up the phone and call the one in charge.

Being a small population presents challenges regarding resources, infrastructure, and specialized medical training to comply with standards of international actors. Notions of Icelanders as resilient in spite of shortcomings were common; referring to the experience of preparedness planning and training, one health staff said:

It was very much the Icelandic way, we'll manage, we'll work it out, and there was so much ingenuity. This notion of a particular Icelandic approach to coping, in spite of shortcomings, was also detected more generally, as in the statement:

Would it have worked? Yes, it would have worked. Would it have been optimal? We cannot say, it would have been optimal; we can say, it would have been sufficient.

In contrast to this, there were concerns about whether Icelandic aid workers falling ill in Ebolaaffected countries should be transferred to Iceland or to hospitals in other Nordic countries with better isolation units. Some of the participants trusted that patients with EVD would not be transferred to Iceland. One participant stated: You heard that Norwegians were criticized for transferring their aid worker from Africa to Norway. We don't know what would have happened if they would have transferred an Icelander into the country.

We don't have good enough isolation unitsyou are not supposed to send patients to a hospital that is less than 100%. I thought there was assurance in that.

During the devastating Ebola epidemic in West Africa that spread to neighbouring sub-Saharan countries, North America, and Europe [32] , preparedness plans were widely elaborated and later evaluated. Evaluations have, for example, been conducted in 11

African countries close to the epidemic [33] , in the EU region [34, 35] , and the US [36] . Here we present data from a qualitative case study on the process, and experiences with establishing a preparedness plan for Ebola in Iceland in 2014. Interviews with staff who were engaged, either as administrators or frontline healthcare workers, alert us to the manner in which geographic, demographic, cultural, and organizational characteristics shaped the response. The results show that the process of establishing and training for preparedness was permeated by ambiguities of pride and pragmatism, trust, doubts, and fear.

'Getting the job done' (theme 1) refers to the multitude of tasks and considerations that surrounds and feeds into the preparedness plan itself and are necessary for successful planning and implementation. Using the metaphors of 'hard core' and 'soft periphery', Langley and Denis [37] emphasize the importance of relatively 'peripheral' concerns and processes for planning and implementation of new interventions. The hard core represents the actual intervention or goal, e.g. implementation of a preparedness plan. The soft periphery refers to all the contextually important networking, negotiations, and agreements necessary to deliver the hard core. If the soft periphery is neglected, it will cause multiple challenges in the implementation process, and the benefit of the hard core, the intervention itself, may not transpire as anticipated. Due attention to the soft periphery may, however, considerably promote the delivery of an innovation, and secure support from important stakeholders. In our data, one manager speaks of the preparedness process as dealing with a three-headed monster where every solution was followed by new problems. The data indicate that the process of dealing with 'the three headed monster' was given due attention as a means to successfully develop Iceland's preparedness plan. Comprehensive consultations and the involvement of many associated institutions were mentioned. Still ambiguity remained with some staff in terms of division of responsibilities and tasks e.g. when transporting a patient potentially infected with Ebola from the airport to the hospital, and other such activities.

During epidemics, rumours, gossip, and unreliable information on the news and social media spread rapidly, resulting in so-called 'infodemics' [38] . The West African Ebola epidemic was covered widely by media [39] , and the fear of Ebola reached every corner of the world, exemplified by travel bans from affected countries, and trade barriers [40] , in contrast to the ongoing epidemic in the Democratic Republic of Congo [41, 42] . In our second theme, trust, doubt, and fear of health workers were represented. Although all intentions were good, concerns remained about the suitability and safety of the isolation

ward, the PPE, and other tools, as well as adequate engagement of colleagues who might potentially work alongside them, in case an Ebola patient came to Iceland. The foreignness of putting on, removing, and working from within a PPE and the trustworthiness of available PPE were mentioned. In preparedness efforts in other countries, scarcity of resources in relation to manpower demand and problems with training and protocols involving PPE were common challenges [35]. Similar problems were encountered in Iceland. Provisory treatment facility had to be designed, called 'camping site' by some, in contrast to facilities found elsewhere [43]. Further, the ETT was established based on voluntary recruitment rather than on the staff's assigned roles within the healthcare system, a procedure that was deemed less than optimal. The members of the ETT pointed out that they had never worked together as a team under circumstances that demanded strict adherence to infectious control procedures. This eroded trust, compounded by the laissez-faire attitude of some of its members during the preparation exercises, possibly due to other competing tasks in a busy hospital and insufficient resources that hampered full participation [44]. Further, it was a constraint that simulation exercises were not an option, found to be an important element in preparation for epidemics [35]. This might have resulted in less than optimal staff protection for those who would have been in direct contact with an infected patient, as reported during the SARS epidemic in Canada [45, 46].

Anthropological work on emergency preparedness emphasizes the connectedness between health professionals, technological devices, and knowledge as a prerequisite for successful preparedness. Wolf and Hall present preparedness efforts as a form of governance that involves human bodies (those of health professionals), clinical architectures (e.g. isolation wards), and technical artefacts (gloves, protective suits, disinfectants, etc.) [47]. During preparedness training and implementation, 'nursing bodies are transformed into instruments of preparedness', and become part of infrastructural arrangements. Health professionals are, here, both vulnerable and powerful tools in the management of contamination. The authors argue that successful planning, training, and implementation of a preparedness plan require such intrinsic connectedness. In the case of Ebola preparedness in Iceland, health professionals draw our attention to dilemmas of connectedness, and their assessment of the fact that these shortcomings might hamper the mobilization of 'preparedness within the human body' that is, the embodied experience, routine, and tacit knowledge which Wolf and Hall state are key to successful implementation. Repeated enactment of receiving and treating a patient with Ebola within experienced and trustful teams would probably enhance such embodiment, provided that there is justified trust in the involved technology. In addition, repetition would also

strengthen the 'soft periphery' of preparedness, and divisions of responsibilities would be clearer manifested.

In the third theme, we observe how notions of the 'Icelandic way' help participants make sense of ambiguities about Ebola preparedness. Loftsdóttir explored how people negotiated the imagination of the local and the global during the 2008 economic crisis in Iceland [48]. Notions of the intrinsic character of Iceland, and of being Icelandic, serve to underscore certain points and explain positive and negative experiences with the preparedness plan. Iceland is far away from the continents, but still connected through global needs for policy, risk of contamination, and dependency in terms of collaboration, in emergencies emerging from elsewhere. In our study, participants highlighted the importance of believing in oneself and the 'Icelandic way of doing things,' summed up in the paraphrase 'þetta reddast' (things always have a way of working out in the end). The preparedness plan had to be completed, and adapted to Iceland's particular global situation.

In the 21st century, the world has faced new epidemic threats, such as SARS, and old scourges such as the plague have resurfaced [38]. One of the main findings on Ebola preparedness measures in the EU was that measures taken were based on past preparedness and experience of other epidemics, such as SARS and H1N1 [35]. Further, key stakeholders within each country found their measures to have been adequate for dealing with a single case of Ebola, as was the case in Iceland. A preparedness plan for pandemic influenzae in Iceland was elaborated in 2006 activated in response to the H1N1 epidemic in 2009 and revised in 2016 [49]. During the elaboration of these plans, communication among the different levels of the healthcare system and supporting agencies, such as the DCPEM, had been clearly defined, and proved to be useful in the preparedness for Ebola. Further, as found important in preparedness activities for pandemic influenzae elsewhere [44], honesty, transparency in communication, and sharing of information from managers to front-line health professionals, was found to be critical. It gave a feeling of being involved, and mitigated the fear that is so frequently encountered during epidemics [38].

Iceland was far away from the epicentre of the Ebola epidemic in West Africa. Yet this case study shows that health professionals felt the strain of possibly having to treat one or more patients with EVD. Their situation stands in sharp contrast to the situation in the three

worst affected West African countries that lacked staff, stuff, space, and systems to effectively address the challenge of EVD. Although Icelandic health professionals had trust in the national healthcare system, and in their own capacity, doubt and fear influenced the reflections on preparedness planning of both administrators and healthcare staff. References to national identity and the characteristic of an 'Icelandic approach' to handling challenges assisted participants in coming to terms with the experienced shortcomings of the preparedness plan, and underscored the pride in the ingenuity applied in the process. These references negotiate the role and character of the nation of Iceland, and its role in a globalized world, as both a small and isolated nation on one hand, and a central and capable one, on the other.

The experienced ambiguity needs attention in a health system and among healthcare staff that have to act resolutely and unfailingly, should they be placed in charge of containing contamination. This study points to the necessity of repeatedly re-enacting, as realistically as possible, the likely scenarios of receiving and treating one or more patients infected with Ebola (or other contagious global health threats) as a routine matter. This would assist in the identification of overlooked 'soft periphery' concerns, and promote embodied preparedness among teams of health care staff on the frontline. Geir Gunnlaugsson conceptualized the study, and took part in all necessary steps towards its completion, such as analysis and interpretation of data, and writing the manuscript for submission. Íris Eva Hauksdóttir collected and analysed the data as part of a master thesis work conducted under the supervision of all three co-authors, revised the manuscript, and approved the final version. Ib Bygbjerg took part in the interpretation of data, revision of the manuscript, and approved the final version. Britt Pinkowski Tersbøl took part in designing interview tools and in the thematic analysis of interview data, interpretation, revision of the manuscript, and approved the final version.

Dr. Gunnlaugsson reports he was the Chief Medical Officer (CMO) for Iceland, Directorate of Health, in the period 2010-2014. Other authors report no conflict of interest.

The study was reported to the Data Protection Authority and approved by the National Bioethics Committee in Iceland (number VSI-). Subsequently, the study was approved by the University Hospital Ethical Committee on 4 February 2016 (number LSH [13] [14] [15] [16]). Participants signed an informed consent form before taking part in the study.

Not applicable.

The manuscript builds on the work of Íris Eva Hauksdóttir towards a MSc in Global Health, Section of Global Health, Department of Public Health, Copenhagen University, Denmark.

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Recent Progress in Studies of Arterivirus- and Coronavirus-Host Interactions

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Abstract: Animal coronaviruses, such as infectious bronchitis virus (IBV), and arteriviruses, such as porcine reproductive and respiratory syndrome virus (PRRSV), are able to manifest highly contagious infections in their specific native hosts, thereby arising in critical economic damage to animal industries. This review discusses recent progress in studies of virus-host interactions during animal and human coronavirus and arterivirus infections, with emphasis on IBV-host cell interactions. These interactions may be directly involved in viral replication or lead to the alteration of certain signaling pathways, such as cell stress response and innate immunity, to facilitate viral replication and pathogenesis.

Text: Coronaviruses, together with arteriviruses and toroviruses, belong in the order Nidovirales, a group of large, non-segmented, positive sense and single stranded RNA animal viruses that produce an extensive 3'-nested set of subgenomic mRNAs for transcription during infection [1] (Table 1) . Nidoviruses such as avian infectious bronchitis coronavirus (IBV), human coronavirus 229E (HCoV-229E), equine arteritis virus (EAV) and the porcine reproductive and respiratory syndrome arterivirus (PRRSV) are important pathogens of both human and animals [2] [3] [4] , and are commonly associated with mild respiratory and enteric diseases, although they are also known to cause more OPEN ACCESS critical lower respiratory tract illness, such as the severe acute respiratory syndrome coronavirus (SARS-CoV) epidemic that occurred in 2003 [5] .

As these viruses infect livestock, coronaviral and arteriviral infections in farms have resulted in large-scale economic losses in farming nations, and are therefore of exceptional veterinary research value. Coronaviruses in fowls, as exemplified by the highly contagious IBV in chickens, can be highly lethal to young chicks. IBV is the etiological agent of infectious bronchitis, an avian disease that is mainly associated with upper respiratory and urogenital tract infections in adult chickens, and to a lesser extent, nephrogenic infections, or inflammation of the kidneys [6, 7] . The impact of IBV infection is also emphatically increased as a consequence of its enhancement of diseases associated with lethal co-infections by bacteria and mycoplasmas [8, 9] . Domestic animals such as cats and dogs are also susceptible to coronaviruses. Feline coronaviruses, especially feline infectious peritonitis virus (FIPV)-a mutation of Feline Enteric Coronavirus (FECV), may induce lethal diseases in cats [10] , while canine coronavirus (CCoV) infections, which cause canine enteric illness in dogs, are prevalent as well [11] . In larger livestock like pigs and cattle, on the other hand, coronaviruses and arteriviruses typically establish enteric infections. An infection or outbreak can cause severe economic losses from the death of young offspring, lifelong impact on the yield of animal produce such as eggs and milk, weight losses and the general health of the population. Bovine coronavirus (BCV), for example, causes Winter Dysentery (WD) in adult cows and diarrhea in young calves [12] .

The pathogenicity of these viruses is typically species-dependent, as is the severity of infection; they infect mainly their natural hosts and/or species that are closely related. Certain virus infections, however, can cross the species barrier, with the prime example being the zoonotic SARS-CoV, a novel coronavirus that is thought to have originated from bats before it adapted to its intermediate host, civet cats, and finally to humans [13] . Bat colonies, which are scattered worldwide, are widely known to play host to a variety of

coronaviral and adenoviral pathogens while acting as natural wildlife reservoirs of these viruses [14] [15] [16] .

Coronavirus infections are also generally tissue-specific-the Transmissible Gastroenteritis Coronavirus (TGEV), for example, affects mainly the gastrointestinal tract [17] that may lead to the onset of fatal watery diarrhoea and severe dehydration in pigs [18] , while human coronaviruses mostly cause respiratory infections [4] .

With respect to their significance to the economy, vaccines have also been developed for many of these viruses in a bid to prevent localized infections from progressing into serious outbreaks. This has, however, proven to be a hard battle as the vaccines are unable to provide complete cross-protection among the various serotypes of each virus [19] .

During infection, the virus replicates in the host cytosol amidst a myriad of host signaling pathways and systems such that interaction between the virus and the host systems is inevitable. Virus infection and the consequent host cell response also involve complicated interaction between various host cellular and viral networks. Virus-host interplay occurs at multiple points during the virus replication cycle, from entry to exit. The nature of such interactions can range from a simple exploitation of existing host machinery to destructive interactions that modulate the host environment to the advantage of the virus while inhibiting host activities. One of the most important interactions between virus and host is the modulation of host cell environment, such that the latter is converted into one in which the virus can replicate successfully. Viruses also regulate the differential expression of host genes, as well as various host antiviral defense mechanisms, for more efficient replication.

Previous studies on the infection of different hosts by nidoviruses, for example, have shown various modifications in host innate immune and stress responses, cell cycle, autophagy and cell death pathways [20] [21] [22] [23] , all of which will be discussed in this review. The significance of host components being used to supplement the gene-poor virus in various processes cannot be dismissed, for although they typically serve as enhancers, they could also become major pathogenicity factors.

A number of stimuli can precipitate apoptotic events, including cell homeostatic imbalance such as cell stress, and the binding of ligands to cell surface "death" receptors; these in turn trigger the onset of major apoptotic pathways: the extrinsic or intrinsic pathway [24] .

The extrinsic pathway can be induced by several cytokine "death" receptors from the tumor necrosis factor (TNF) family, such as Fas (Apo1/CD95) [25] . Upon recruitment of their respective ligands, they form complexes that subsequently bind death effector domain (DED)-containing pro-Cysteine Aspartyl-Specific Proteases (pro-caspases), in particular pro-caspase-8, where the activation and consequent oligomerization of which further serves as a signal for downstream activations, thus pledging the doomed cell towards its own death [26] . The intrinsic pathway, on the other hand, is activated by the release of cytochrome c from the mitochondria into the cytoplasm [27] . In the cytosol, cytochrome c binds the apoptotic protease-activating factor (Apaf1); together, they form an apoptosome that leads to the release of active caspase 9.

Apoptotic mitochondrial events are also regulated primarily through the activation of pro-survival and pro-apoptotic proteins [28] . The Bcl-2 family of proteins constitutes a critical control point in the regulation of apoptosis. They form three major protein subgroups: the Bcl-2 homology (BH) 3, or BH3-only proteins [e.g., BH3-interacting domain death agonist (Bid), BCL2-associated agonist of cell death (Bad)], Bax-like proteins [e.g., BCL-2-antagonist/killer 1 (Bak), BCL-2-associated X (Bax)] and the Bcl-2-like factors [e.g., Myeloid cell leukemia-1 (Mcl-1), Bcl-extra large (Bcl-X L)] [29] . BH3-only and Bax-like proteins are essential initiators of apoptosis while the Bcl-2-like proteins are pro-survival factors that safeguard the cells against apoptosis.

Both caspase 8, from the extrinsic pathway, and caspase 9, from the intrinsic pathway, have been observed to activate the main effector caspase 3, which in turn activates a caspase cascade to eventually evoke the morphological hallmarks of apoptosis such as DNA fragmentation [30, 31] .

A third apoptotic pathway, induced by prolonged endoplasmic reticulum (ER) stress, has also been shown to activate multiple downstream apoptotic targets, including rodent caspase 12, growth arrest and DNA damage-inducible gene 153 (GADD153), also known as

the transcription factor C/EBP homologous protein (CHOP) as well as activation of the pro-apoptotic c-Jun NH2-terminal kinase (JNK) [32] . Human pro-inflammatory caspase 4, a nearly identical paralogue of the rodent form of caspase 12, has also been shown to possess comparable roles in ER-stressed apoptosis [33] . JNK activation is mediated by ER transmembrane protein kinases, while CHOP is triggered by ER stress at the transcriptional level [32] . The downstream apoptotic activities of both JNK and CHOP have also been postulated, at least in part, to be connected with the Bcl-2 family of proteins (Bak and Bax) for recruitment to the ER and subsequent initiation of apoptosis in response to stress [32] .

As viruses depend on the host cells they infect in order to reproduce, apoptosis is often employed as an important host antiviral defense mechanism that, as a protective measure, leads to the abortion of virus infection such that viral productivity and persistent infectivity is consequently limited [34] . In many cases, p53 and the Bcl-2 family of proteins have been shown to be the main mediators that induce the beleaguered cell to undergo self-induced death at various stages of the infection cycle [35] .

Moreover, host endosomal membranes are forced to undergo conformational changes for the fusion of virus and host cell membranes during virus uncoating; membrane integrity is also antagonized during the process of virus disassembly. As such, these drastic alterations to membranes may elicit downstream pro-death signals that prompt infected cells to commit suicide [36] .

However, certain viruses have evolved strategies to both counteract and induce apoptosis in order to maximize the production of virus progeny and promote its spread to neighbouring cells. An increasing number of known viruses from different families, including arteriviruses, have been found to induce apoptosis during their infection cycle, which may possibly contribute to the cytotoxicity associated with virus infections, especially during late stages of infection [37] . Membrane-bound cell fragments are also produced as apoptotic bodies to be phagocytosed by surrounding cells. This provides an excellent method for a virus to disperse its progeny without eliciting host immune response [38] .

While more comprehensive work needs to be done to paint a clearer picture of how coronaviruses and arteriviruses regulate apoptosis during infection, recent reports have

suggested the possible activation of more than one apoptotic pathway during infection. EAV, an arterivirus that is prevalent among global horse populations and which may induce abortions in pregnant mares [39], has been shown to activate apoptosis through the initiation of caspase-8-dependent mechanisms, which is followed by mitochondria-dependent caspase-9 activation mechanisms [40]. PPRSV, a virus that causes respiratory tract problems in young pigs and has a commercially significant impact on swine industries worldwide as a result of reproductive impairment in breeding livestock [41], has also been implicated to regulate both apoptosis and necrosis during infection [42, 43]. Replication of TGEV in porcine kidney cells, as well as that of canine coronavirus type II (CCoV-II) in canine fibrosarcoma cells, on the other hand, has been reported to induce apoptosis both through Fas/FasL-activation and mitochondrial-dependent pathways [44, 45].

Figure 1. Viral genes and the activation of apoptosis. Extrinsic signals from receptors (e.g., Fas ligand) culminate in the activation of caspase 8, which activates the effector caspase 3 while intrinsic signaling requires the participation of the mitochondria in releasing cytochrome c (shown as circles labeled "C") to activate caspase 9 for the downstream activation of caspase 3. Key proteins in the intrinsic apoptosis signaling pathway are p53, both pro-apoptotic (e.g., Bax, Bak) and anti-apoptotic proteins (e.g., Bcl-2, Bcl-X L) from the Bcl-2 family. Viral genes (see yellow boxes), which act at multiple points along the different signaling pathways, target both the extrinsic and intrinsic apoptosis signaling pathways and enhances the pro-apoptotic effect brought upon by virus infection. Anti-apoptotic proteins (black oval) listed are key anti-apoptotic members from the Bcl-2 family of proteins. Other pathways triggered by coronavirus infections, such as ER stress and DNA damage response, activate apoptotic signaling pathways as well.

Although IBV is an avian virus, it can acclimate to primate cells and has been demonstrated to conquer the host species barrier and become zoonotic to infect both human and animal cells [46, 47]. IBV also triggers apoptosis during the late stages of its cytolytic infection cycle. Specifically, IBV-induced apoptosis has been shown to involve Bcl-2 family of proteins [48], through caspase-dependent [49] and p53-independent [50] pathways in cultured mammalian cells. The modulation of Bcl-2 family proteins during IBV infection has also been postulated to be under the regulation of signaling pathways such as ER stress and Mitogen-Activated Protein Kinase/Extracellular signal-Regulated Kinase (MAPK/ERK) pathways [48]. The effects of these pathways on the regulation of other host defenses will be discussed in later paragraphs.

While viral genes can also manipulate the induction of apoptosis to the benefit of viruses, only two have been reported in the case of coronaviruses. The unique SARS-CoV encoded protein, 7a, was discovered to have caspase-dependent, pro-apoptotic functions and may function as a plausible source of virus-derived apoptotic signal [51], while TGEV accessory gene 7, present only in coronaviruses classified under genus β 1 [52], thwarts virulence and host-induced antiviral mechanisms through the negative modulation of downstream caspase-dependent apoptotic pathways [53] (Figure 1).

The maintenance of apoptosis is also important in the establishment and governance of immune responses in a cell. A loss in the control of apoptosis leads to an imbalance in cell homeostasis, which ultimately affects immune sensitivity [54]. The presence of apoptotic cells, particularly in existence with infectious agents, may also lead to the mobilization and initiation of innate immune defenses [55]. This crosstalk between apoptosis and innate immunity is therefore of considerable importance during pathogenic infection and can be manipulated by both host and pathogen, either as a form of immune defense or immune evasion, respectively [56].

When the host immune system is exposed to viral pathogens, it reacts straightaway by triggering a diverse array of defense mechanisms in order to establish a more efficacious shield. The first line of defense is the mounting of an innate immune response, as characterized by the increased production of type I interferons (IFN- α and IFN- β) and other inflammatory cytokines. These, in turn, choreograph the expression of downstream IFN-stimulated genes (ISGs) and activate several signaling pathways, all of which collaboratively lead to the induction of a protective antiviral state and, subsequently, the inhibition of both viral replication and proliferation [57].

The cytokine family of interferons is dedicated to the conveyance of the presence of infection, as well as the expedition of numerous connections among the cells that provide protection against, or eradication of, foreign pathogens. Other than interfering with viral progeny production in host cells hence the name 'interferon'-IFNs also induce Natural Killer (NK) cells and macrophages to 'kill' or engulf infected cells, increase antigen presentation to thymus (T) cell lymphocytes for rapid recognition of infected cells and bring about virus resistance to new uninfected cells [58]. IFNs are conventionally classified into three types: Type I (IFN- α , IFN- β and IFN- ω), Type II (IFN- γ) and the more recently identified Type III (IFN- λ 1, IFN- λ 2 and IFN- λ 3) [59, 60]. Over time, mammalian hosts have gradually developed a

multitude of cellular sensors for the detection of viral infection, and it is the involvement and operation of these cellular protein receptors that eventually leads, through an intricate network of pathways, to the expression of type 1 IFNs. Major receptor systems that conduct immune surveillance and trigger the production and subsequent release of type I IFNs are known as pattern recognition receptors (PPRs), which detect viral infection through the identification of various pathogen-associated molecular patterns (PAMPs); PPR families include the toll-like receptor (TLR), RIG-like helicase (RLH) and Nucleotide-binding oligomerization domain (NOD)-like receptor (NLRs) families [61] .

The TLR family is mainly made up of transmembrane proteins, which conduct surveillance from the cell surface, as well as cellular compartments such as the endosome or ER, constantly scanning the extracellular environment for PAMPs that can be derived from a wide range of microorganisms, including viruses and bacteria. The expression of TLRs appear to be cell-specific and is mainly found in antigen-presenting cells such as dendritic cells (DCs), monocytes and macrophages, as well as on B cells [62] . TLRs recognize a wide variety of PAMPs, and the recognition of these ligands can be converted into specific intracellular responses through the direct interaction of the TLR toll-interleukin 1 receptor (TIR) domain with one of its cytoplasmic TIR-containing signaling adaptor molecules, such as myeloid differentiation primary response protein 88 (MyD88) and TIR domain-containing adaptor inducing IFN β (TRIF) [63] . Viral recognition by TLRs has been detailed in several reports [64, 65] . In particular, the activation of TLR-3, -4, -7, -8, and 9 can also culminate in type I IFN production [65, 66] . TLR3, as a general viral sensor, detects mainly through double stranded RNA (dsRNA), a replication intermediate of both DNA and RNA viruses; TLR4 recognizes envelope proteins from viruses such as mouse mammary tumor virus (MMTV); TLR7 and TLR8 have been identified to recognize ssRNA viruses like influenza and vesicular stomatitis virus (VSV); TLR9 detects dsDNA viruses such as herpesviruses [63] .

NLRs are stimulated by microbial agonists and collaborate with TLRs to evoke intracellular immune responses through MAPK and caspase signaling cascades upon sensing bacterial components [67] . NLRs are also known to sense both PAMPs from infectious agents and DAMPs (danger-associated molecular patterns) that arise as a result of insult or injury to the cell, as well as those derived from the environment [68] . Although the direct binding of virus-derived PAMPs to

NLRs is yet to be reported, structural and functional studies of the C-terminal domain of NLRX1, a mitochondrial member of the NLR family, has highlighted its ability to bind both ssRNA and dsRNA, implying that some NLRs may be capable of binding viral RNA directly as well [69] . As such, the notion of NLR-mediated recognition of coronaviruses is, therefore, probable and could be further investigated.

The RLH family of purely cytoplasmic PRRs is made up of the following: retinoic acid inducible gene-I (RIG-I or DDX58), melanoma differentiation-associated gene-5 (MDA5 or IFIH1), and laboratory of genetics and physiology 2 (LGP2). RIG-I and MDA5 are PRRs with two N-terminal caspase-recruitment domains (CARDs) followed by a DExD/H box RNA helicase domain; LGP2 lacks the signaling caspase recruitment domains but shares a helicase domain of similar homology and is thought to serve as a regulator of the former [70] . RIG-I and MDA-5 both sense cytoplasmic dsRNA, which the host recognizes as 'non-self', via the N-terminal CARDs [71] . However, the two PRRs each sense distinct PAMPs, depending on the length of viral dsRNA, from different RNA viruses. In addition to long dsRNAs (>2 kb) such as the synthetic dsRNA analogue poly-inosinic poly-cytidylic acid [poly(I:C)], MDA5 also recognizes picornaviruses and noroviruses [72, 73] . RIG-I, on the other hand, responds to paramyxoviruses, flaviviruses, orthomyxoviruses and rhabdoviruses [61, 74, 75] . This is through its recognition of a variety of ligands such as relatively short dsRNA (19-mer to 1 kb) and ssRNA (single stranded RNA), both preferably in the presence of a 5'-triphosphate end, full-length RNA viral genomes, the presence of secondary structures such as poly-uridine motifs within 5'-triphosphate genome termini or longer RNA sequences without 5'-triphosphates, such as 3' untranslated regions (UTRs) of the genome [76] [77] [78] [79] [80] .

While the elicitation of TLRs and RLRs by PAMPs trigger their distinct signaling cascades through divergent downstream effectors at varying efficacies, they ultimately cross paths at the juncture of transcriptional activation of interferon regulatory factor 3 (IRF3) [81] , IRF7 [82] and nuclear factor kappa-light-chain-enhancer of activated B cells (NF- κ B) [83] [84] [85] [86] , all of which translocate to the nucleus and activate the transcription of both type I interferons (IFN α and IFN β) and inflammatory cytokines that eventually culminates in the concerted induction and development of adaptive antiviral immune response.

The expression and induction of interferons from the cells occurs in response to viral insults and tumor growth [87] . Type I interferons, the major group of cytokines in innate

anti-virus defense, bind a specific cell surface heteromeric receptor, the interferon- α/β receptor (IFNAR), which composed of two subunits, IFNAR1 and IFNAR2 [88] . The best characterized type I IFNs can be classified into two major groups, the immediate-early genes (such as IFN β) that are triggered by the initial response to virus infection, and the delayed-set (such as IFN α subtypes) that rely on a secondary de novo protein synthesis pathway [89] . IFN expression is regulated by IFN regulatory factors. In particular, IRF3 and IRF7 play vital roles in activating innate immune response through their respective antiviral response [66] .

IRF3 has been functionally characterized to consist of a nuclear export signal (NES), a DNA-binding domain (DBD), a C-terminal IRF association domain (IAD), as well as a number of phosphorylation sites as well as two auto-inhibitory domains that prevents a constitutive activation of the NES, DBD and IAD [90] . Normally found in an inactive cytoplasmic form, IRF3 is phosphorylated as a consequence of virus infection. This activation signal exposes the DBD and IAD and results in the dimerization of IRF3, either as a homodimer or as a heterodimer with IRF7, allowing the activated IRF3 to form a complex with the transcriptional co-activator cAMP-response element-binding protein (CREBP) to translocate to the nucleus and bind to DNA to trigger the transcription and expression of immediate-early IFNs, which signals the JAK-STAT (Janus kinase-Signal Transducer and Activator of Transcription) pathway through the binding of the IFNAR [90] . This leads to the formation of a STAT1-STAT2 heterodimer that teams up with the interferon regulatory DNA binding factor IRF9, which together constitute an activated heterotrimeric factor, the interferon-stimulated gene transcription factor ISGF3, that, through the recognition and binding of specific ISREs, induces downstream expression of innate immunity genes for host defense against virus invasion [91] .

In contrast to the constitutive expression of IRF3, IRF7 is involved in the positive feedback regulation of IFN production [89] . Expressed only in minute amounts, the induction of IRF7, via virus-induced activation of ISGF3, results in either homo-dimerization or hetero-dimerization with IRF3 and is subsequently followed by nuclear translocation for the activation of both IFN α and IFN β genes [92, 93] .

Coronaviruses and arteriviruses have evolved multiple strategies to avoid elimination from the host. These tactics range from the prevention of detection to inhibition of antiviral

responses mounted by the host immune system. All these activities involve virus-host interactions at different levels (Figure 2) .

The modulation of SARS-CoV pathogenesis, for example, was reported to be independent of all three types (Types I, II and III) of interferon signaling mechanisms, with SARS-CoV ORF 3b, ORF 6, and nucleocapsid proteins identified to interfere with interferon signaling through various mechanisms [94] . However, STAT1 has been shown by the same group to be crucial in activating innate immune signaling pathways during SARS-CoV infection, with a secondary role in the prevention of uncontrolled cell reproduction [95] . In contrast, PRRSV appears to be receptive to both IFN- α (porcine IFN- α , or Ad5-pIFN- α) and - β (recombinant swine beta interferon, or swIFN-beta) dose-dependent treatment [96, 97] ; however, pigs infected with PRRSV do not evoke significant IFN responses, with little or no IFN- α and IFN- β production [98, 99] . Upstream signaling pathways that may potentially lead to the inhibition of IRF3 activation include the interference of PRRSV with RIG-I signaling events through inactivation of the RIG-I downstream signaling adaptor, MAVS (mitochondrial antiviral signaling protein) [100] . Recent studies have also suggested PRRSV infection inhibits type I IFN production and signaling processes through the impairment of STAT1/STAT2 nuclear translocation [101] .

One of the most indispensable adaptor proteins in the RLH signaling pathway is MAVS. Otherwise known as virus-induced signaling adaptor (VISA), interferon γ (IFN γ) promoter stimulator-1 (IPS-1) or CARD-adaptor inducing IFN- β (Cardif), it was discovered independently by four different research groups in 2005 [102] [103] [104] [105] , and contains an N-terminal CARD-like domain, which interacts with the CARD domains of RIG-I and MDA5, and a C-terminal transmembrane (TM) region that lodges the protein in the mitochondrial membrane, thus pioneering the connection between mitochondria and innate immunity. A deficiency in, or the cleavage of, MAVS from the mitochondrial membrane results in an arrested antiviral immune response, highlighting the pivotal role MAVS plays in mitochondriamediated innate antiviral immunity [106] . MAVS can also induce apoptosis independently of Type I IFN activation, with SARS-CoV non-structural protein 15 identified as an inhibitor of MAVS-induced apoptosis to escape host antiviral immune responses [107] .

Coronaviruses have devised a number of cell type-specific strategies to inhibit type I IFN responses. One such strategy that these viruses have adopted is to avoid detection of its

newly synthesized mRNAs in the cytoplasm through the encoding of a 2'-O-methylase (non-structural protein nsp16) that creates a 5'-cap structure analogous to the cellular mRNAs on their mRNAs, thereby escaping detection by Mda5 [108, 109] . More importantly, this also highlights the significance of mRNA cap alterations, as such 2'-O-methylation, as pertinent molecular signals for the discernment between self and non-self mRNA [108] . However, during the course of viral transcription and replication, uncapped double-stranded RNA intermediates are generated and these may serve as ligands for the RIG-I and/or Mda5. Mouse hepatitis virus (MHV) infection was first reported to activate type I IFN responses in various cell types such as macrophages and microglia in the brains of infected animals, and the virus has also been shown in the same report to be recognized specifically by MDA5 in mouse macrophages [110] . In other reports, both RIG-I and Mda5 have been implicated in the detection of mouse hepatitis virus (MHV) infection, particularly in mouse oligodendrocytes [111] , although conflicting evidence has been reported as well, especially since it is not understood how cellular helicases could bind viral replicative/transcriptive intermediates when the latter should have been isolated by the double membrane vesicles [112] . MHV infection can also delay IFN β -activated ISG induction; however this phenomenon is limited to certain cell types and is observed only when the infection occurs before IFN β exposure [113] . As such, the general sensing of coronaviruses appears to be cell-type specific, with Type I IFNs derived from plasmacytoid dendritic cells (pDCs), conventional DCs and macrophages being particularly essential in curbing the pathogenesis of mouse coronavirus infection [114] .

Detection by the PRRs would activate signaling cascades leading to the production of type I IFNs, resulting in the establishment of an antiviral state. Thus, coronaviruses do not just avoid detection by the host immune system, which appears to be the main strategy of ensuring successful replication [115] ; some viruses also encode proteins that function to disrupt the downstream signaling cascades at various points, preventing the establishment of an effective antiviral state when detection of the viral PAMPs has occurred. Indeed, the N protein has also been shown to interfere with the 2', 5'-oligoadenylate synthetase/RNaseL (2'-5'OAS) activation that occurs downstream of IFN induction and which leads to the inhibition of global translation shutdown [116] . This activity is in addition to its suppression of IFN β induction through the binding of viral RNAs that prevents their detection [117] .

In addition to the processing of polyproteins, the papain-like proteinase 2 (PLP2) domain of nsp3 has been shown to possess de-ubiquitinating activity [118] [119] [120] . PLP2 of nsp3 de-ubiquitinates TANK-binding kinase 1 (TBK1), the activating kinase for IRF3 and IRF3

itself, and sequesters the hypo-phosphorylated TBK1-IRF3 complex in the cytoplasm as well. This prevents IRF3 nuclear translocation [121, 122] , thereby inhibiting the transcription of type I interferons [123] . The activation of TLRs-3 and/or -7 as well as cytoplasmic helicases RIG-I and/or MDA-5 triggers signaling pathways resulting in the synthesis of type I IFNs, inflammatory cytokines and ISGs which acts in concert to establish an antiviral state. The activation of both 2'-5' OAS and PKR results in global degradation of cellular RNA and inhibition of translation, which may inhibit viral propagation. Coronaviruses encode many proteins (see yellow boxes) that target multiple steps in the innate immune response mounted by the host cells, ensuring its successful replication in the host.

In vitro activation of chicken splenocytes and peripheral blood leukocytes with IBV has also resulted in an increase in chicken interferon gamma (chIFN- γ) production as a form of cell-mediated immune response [124] . This appears to be a polyclonal, non-specific stimulation as chIFN- γ production levels are also elevated in IBV-stimulated chicken splenocytes which lack prior exposure to IBV, when compared to control un-stimulated cells, as well as in cells exposed to inactivated IBV [124] .

Such ineffectual host innate responses could lead to poor cellular responses, which may result in a delay in pathogen clearance and persistent viral infection in compromised pigs, which would present significant advantages to the virus for the subsequent release of viral progeny and spread.

The onset of coronavirus-induced apoptosis is intricately linked with other host antiviral innate defenses. Infections with viruses often result in cell cycle arrests and the activation of unfolded protein response (UPR) due to ER stress, both of which may be accompanied by the parallel activation of apoptosis in the infected cells.

During virus replication, newly translated viral proteins accumulate in the ER, which may also cause stress that leads to activation of the UPR. MHV S protein has been shown to activate three UPR transducers, inositol-requiring enzyme 1 (IRE1), activating transcription factor 6 (ATF6) and protein kinase RNA-like endoplasmic reticulum kinase (PERK) [125] . In particular, PERK activation leads to the activation of p38 MAPK [126] , which stimulates virus replication [127, 128] . SARS-CoV, on the other hand, has been shown to attenuate

the IRE1 signaling pathway, which generally led to the down-regulation of stress responses and UPR, and further decreased apoptotic signaling in infected cells as well, to the likely benefit of the virus for better viral progeny production and release [129] .

Coronavirus-induced apoptosis during the late stages of infection is also partially regulated by the p38 MAPK signaling pathway, which in turn up-regulates production of pro-inflammatory cytokines, interleukin (IL)-6 and IL-8, in infected host cells and mount an immune response against virus infection in these cells [130, 131] .

However, while pro-inflammatory cytokines are up-regulated at the transcriptional level during coronavirus infection, there is minimal to moderate up-regulation at the translational level, leading to the hypothesis that the interaction between IBV spike (S) protein and host eukaryotic initiation factor 3 (eIF3) modulates host gene expression, especially genes involved in innate immunity that are activated during coronavirus infection [132] . This also agrees with the hypothesis that, notwithstanding the presence of virus-induced ER stress responses, a significant decline in host mRNA translation in infected cells impedes the expression of ER stress proteins despite elevated mRNA concentrations of the former [133] . Moreover, the negative modulation of p38 MAPK occurs through an up-regulation of the dual-specificity phosphatases 1 (DUSP1) feedback loop during IBV infection, which reduces cytokine production by dephosphorylation of phosphor-threonine and phosphor-tyrosine residues on activated p38 MAPKs [130] .

PRRSV infection of porcine macrophages, too, appears to be under the control of the MAPK/ERK pathway, through various signaling pathways activated by the latter that regulate a variety of cellular processes [134] . In particular, chemical inhibition of the ERK pathway resulted in an attenuation of PRRSV infection during the early stages of infection post-virus attachment, during which a notable reduction in viral sub-genomic mRNA synthesis and translation, as well as in progeny virus release, was observed [134] . An increase in IL-10 production during the early stages of infection has also been observed in pigs infected with PRRSV. The up-regulation of IL-10, an important cytokine in the attenuation of innate and adaptive immune responses, is advantageous for PRRSV to sustain for a longer period of time in the host, and the PRRSV N protein has been suggested as the viral protein responsible for mediating IL-10 induction [135] .

Likewise, the expression of pro-inflammatory mediators, IL-1², IL-6, IL-8, and TNF- α , are up-regulated during EAV infection of equine endothelial cells and macrophages, with virulent EAV strains activating sufficiently greater amounts of these cytokines, especially TNF- α , than avirulent EAV strains [136] . Moreover, the expression of pro-inflammatory cytokines, IFN- α , TNF- α , IL-1 and IL-6, is also up-regulated during porcine respiratory coronavirus and TGEV infection, and the differential modulation of which are postulated to function as crucial mediators of viral respiratory diseases in pigs [41] . In particular, early type I IFN production in coronavirus-infected pigs effectuated immunomodulatory responses, which suggests the potential ability of IFN inducers as an effective antiviral strategy to control the rate of coronavirus infections in swine populations [137] .

RANTES (regulated upon activation, normal T cell expressed and secreted) is a known member of the pro-inflammatory CC chemokines family that function in the modulation of the migration of inflammatory cells such as monocytes and Natural Killer cells to sites of infection, particularly as a form of host antiviral immune response during virus infection [138] . Like coronaviruses, PRRSV infection also triggers the activation of pro-inflammatory cytokines and chemokines, including RANTES [139] , which may be fundamental in initiating the pathological conditions associated with the arterivirus. The activation of RANTES transcription also requires the participation of adaptor molecules from the TLR signaling pathway, such as MyD88, TRIF and TNF receptor-associated factor 6 (TRAF6) [139] .

Interestingly, respiratory PRRSV/PRCV viral co-infections that commonly occurs in pigs, on the other hand, frequently leads to severely attenuated innate and adaptive immune responses in order to extend the pathogenicity of PRRSV-thus down-regulating innate immunity-and renders the host more vulnerable to subsequent infections by PRCV, or other respiratory viruses, that instead up-regulates innate immunity, possibly due to further, more severe damage to pulmonary cells and tissue and which may ultimately lead to a more critical form of pneumonia [140] .

Other types of host innate defense against virus infection include processes such as autophagy, a vital physiological process in which cells degrade their own organelles in response to a severe lack of nutrients or other cellular stresses, so as to facilitate the removal of damaged cellular components and prevent the build-up of unwanted products in the cells [141] . Autophagy is also known to mount crucial innate immune responses

against invading pathogens, with degradation of the latter through autophagy; on the other hand, autophagosomes may instead help promote virus infection by bringing together viral replicase proteins [142]. Multiple host-derived cytokines have emerged to play differential roles in regulating the onset of this essential mechanism as well. Of note are the T-helper cells (Th1) group of cytokines, including IFN- γ , TNF- α , IL-1, IL-2 and IL-6, which have the ability to induce autophagy, while Th2 cytokines such as IL-4, IL-10 and IL-13 have been shown to inhibit autophagy [143].

Recent studies have shown the ability of IBV to induce autophagy during infection independently of cellular stress or nutrient deprivation, and this can be made possible through the functions of IBV nsp 6, as well as in nsp6 orthologs of mammalian coronaviruses such as MHV and arteriviruses such as PRRSV, although a direct link between IBV infection and autophagosome formation could not be established [22]. As such, the induction of autophagy that has previously been reported during coronavirus and arterivirus replication may instead be an example of an innate defense tactic against infection to remove unwanted viral particles, while nsp6-as well as its respective orthologs-may alternatively modify adaptive immune strategies by targeting the breakdown of immunomodulatory proteins synthesized by the ER in autophagosomes [22].

Viruses may also exploit the host cell cycle to benefit their own replication [144] (Figure 3). Cell cycle regulation typically involves mechanisms critical to cell sustainability, such as the surveillance and correction of genetic damage and the impediment of unrestrained cell division. Cyclins and cyclin-dependent kinases (CDKs) are well-known regulatory proteins, and the activation of which will dictate a cell's progress through the cell cycle. Briefly, activated heterodimers consisting of both cyclins and CDKs as the regulatory and catalytic subunits, respectively, will alter the phosphorylation state of various target proteins for progression into the next stage of the cell cycle. CDK inhibitors include tumor suppressors such as the cip/kip (CDK interacting protein/Kinase inhibitory protein) and the INK4a/ARF (Inhibitor of Kinase 4/Alternative Reading Frame) family of genes that thwart progression to the next stage of the cell cycle [145].

The inactivation of CDKs can also be a reversible process through the phosphorylation of essential residues Tyr 15 and Thr 14, both of which are located within the CDK ATP-binding loop [146]. Dual-specificity phosphatases of the Cdc25 (cell division cycle 25) family are responsible for the dephosphorylation of Tyr 15 and Thr 14 [147]. These Cdc25 proteins

are, in turn, inactivated by Chk1/Chk2-mediated phosphorylation; this inactivation averts Cdk dephosphorylation and impedes its subsequent activation [148] . Chk1/Chk2 activation after DNA damage and/or DNA replication hindrance is dependent on the ataxia-telangiectasia mutated (ATM) and ATM/Rad3-related (ATR) protein kinases in mammalian cells [149, 150] .

IBV infection of cultured cells results in cell cycle arrest, at both S and G2/M phases, to boost viral replication and to enhance the production of viral proteins as well. This p53-independent growth inhibitory outcome is catalyzed by regulation of the expression of multiple cell cycle regulatory genes such as corresponding CDK complexes [50] and the down-regulation of G1 phase regulatory cyclins D1 and D2 [151] , as well as through systemic modulation of ATR-dependent cellular DNA damage response [152] . Specifically, the interaction of coronavirus nsp13 with the p125 subunit of DNA polymerase δ was discovered to trigger DNA replication stress in cells during IBV infection, which eventually led to cell cycle arrest at the S phase [152] .

It has also been observed that the N protein of several coronaviruses can localize in the nucleolus where it may perturb cell cycle activities of the host cell for the benefit of viral mRNA synthesis [153] [154] [155] [156] . IBV N, for example, appears to target CDK2, cyclins A and D1 for proteasomemediated degradation [50, 157] and cause the accumulation of hypophosphorylated retinoblastoma (pRB), resulting in the downregulation of CDK1, cyclins E and B1 [50] .

The regulation of host protein synthesis, especially at the initiation stage, is often a regular viral objective for the extensive reduction of host protein translation so as to construct the most favorable condition for viral replication and progeny assembly. Protein kinase R (PKR) is a prevalent serine/threonine protein kinase that can be induced by interferon in its latent state, and whose activation is dependent by the presence of dsRNA products, is one such viral target. PKR plays an important role in the cellular anti-viral response pathway, and becomes activated via auto-phosphorylation upon binding to virus-derived dsRNA, which then subsequently leads to host translation inhibition through phosphorylation of the alpha subunit of eukaryotic initiation factor 2, eIF2alpha [158] .

The dephosphorylation of eIF2 α is established by cellular protein phosphatase-1 (PP1), which functions to regulate various cellular processes through the physical interaction of its catalytic subunit (PP1c) with regulatory proteins such as GADD34/CHOP that is induced by DNA damage signals [159, 160] .

In the case of IBV, for example, eIF2 α phosphorylation was significantly suppressed in both human and animal cells during infection. The dephosphorylation of eIF2 α during IBV infection appears to be induced by the up-regulation of GADD34 expression, which, together with the simultaneous attenuation of PKR auto-phosphorylation following IBV infection in these cells, serves as a viral regulatory tactic in boosting coronavirus replication while managing the delicate balance of de novo protein translation along with the identification of IBV nsp2 as a potential, albeit weak, mediator in the dose-dependent inhibition of PKR activation [161] . TGEV, too, suppresses both cellular RNA degradation and eIF2 α phosphorylation during infection through an interaction between TGEV protein 7 and PP1 that regulates host antiviral responses and extends the period of viral progeny dispersal as well [53] .

A subunit of the eukaryotic initiation factor 3, eIF3f, has also been discovered to modulate host translation inhibitory effects through physical interaction with coronavirus spike protein [132] . As this inhibition takes place during the late stages of the coronavirus replication cycle, translation of virus-induced transcripts are largely affected, especially pro-inflammatory cytokines and chemokines. This could account for the fact that while IL-6 mRNA expression is up-regulated during the early stages of coronavirus infection, insubstantial increase in IL-6 protein expression was observed [132] . The inhibition of host protein synthesis through the interaction between coronavirus spike protein and eIF3f may therefore have a significant impact on the modulation of coronavirus pathogenicity.

Host proteins are also known to play a role in the virus life cycle, especially during viral RNA replication and transcription (Figure 4) . The most well studied host protein that interacts with the coronavirus genome is heterogeneous nuclear ribonucleoprotein A1 (hnRNP A1), a nuclear protein, whose biological function is to regulate alternative splicing of cellular RNAs [162, 163] . The hnRNP A1 has been shown to bind to both negative-sense leader sequence and negative-sense intergenic (IG) sequence of MHV [164] . The formation of a RNP complex among hnRNP A1, negative-sense leader and IG sequences has also been demonstrated [165] . In addition to its ability to interact with the coronavirus RNA, hnRNP

A1 was also found to interact and co-localize with N protein [166, 167], an important player in coronavirus RNA synthesis [168, 169]. It has also been highlighted that hnRNP A1 may be required to recruit other cellular proteins to the replicase complex [170].

To ascertain the involvement of cellular factors in TGEV RNA synthesis, TGEV 3' and 5' genome ends were used as baits for RNA affinity protein purification [171]. Of the ten cellular proteins pulled down with either genome end, poly(A)-binding protein (PABP), hnRNP Q, and glutamyl-prolyl-tRNA synthetase (EPRS) were confirmed to enhance TGEV infection through their respective interactions with the TGEV 3' end, while glyceraldehyde 3-phosphate dehydrogenase (GAPDH)-originally employed as a control-was discovered, surprisingly, to have a diminishing effect on TGEV infection instead [171].

For arteriviruses, the common leader sequence of EAV sub-genomic viral RNAs, too, possesses the ability to interact several cellular proteins from the cytoplasmic fractions of Vero cells, likely for the modulation of EAV RNA synthesis [172].

Recent reports, based on a yeast-based three-hybrid system to identify RNA-RNA-binding regulatory protein interactions [173], and using the 5'-UTR (untranslated region) of SARS coronavirus as bait, identified zinc finger CCHC-type and RNA-binding motif 1 (MADP1) as a potential cellular protein that interacts with SARS-CoV, and this was eventually established via an in vitro pull-down assay with the 5'-UTR of IBV [174]. MADP1 was also shown by the same authors to play a role in the early stages of the coronavirus replication cycle, with the RNA recognition motif in the N-terminal region of the protein interacting with stem loop 1 of the IBV 5'-UTR. Specifically, this protein translocates from the nucleus to the cytoplasm of the cell during IBV infection and co-localizes, in part, with viral replicase/transcriptase complexes (RTCs) in order to enhance viral replication and coronavirus RNA synthesis [174].

Interaction with viral RNA is not the only way by which cellular proteins can take part in viral RNA synthesis; protein-protein interactions with the viral replicase complex can modulate this process as well. This was confirmed in coronaviruses with a yeast two-hybrid screen that was performed using IBV nsp14 as a bait protein, which ultimately led to the discovery of DDX1, an ATP-dependent RNA helicase in the DExD/H helicase family, as an interacting partner that translocates from the nucleus to the cytoplasm and enhances IBV replication

in cultured mammalian cells through subcellular colocalization with nsp14, an exonuclease, during infection [175] .

Another yeast two-hybrid screen, with IBV membrane (M) protein as the bait, identified beta-actin as an interacting partner, leading to the suggestion that actin filaments may possess the ability to participate in virion assembly and budding during the coronavirus replication cycle [176] . This interaction may possibly lead to the incorporation of actin into the mature virion. In fact, recent proteomic analysis of purified IBV particles through two-dimensional gel electrophoresis and subsequent mass spectrometry have confirmed, among others, an abundance of actin within the virus particles, further cementing the hypothesis that cytoskeletal elements play crucial roles in IBV replication [177] .

Retinoblastoma tumor suppressor proteins are also interacting partners of coronavirus nsp15, an endoribonuclease, which may result in alterations to the cell cycle that impact coronavirus infection and viral progeny release [178] . Similarly, with nsp1², the functional proteolytic product of PRRSV nsp1, as a bait protein, cellular poly(C)-binding proteins 1 and 2 (PCBP1 and PCBP2, respectively) have been identified interacting partners of the former, with specific functions in modulating PRRSV replication and RNA synthesis [179] . .

Host proteins and the coronavirus life cycle. Virus particle attaches onto the host cell via cellular receptors on the surface and enters. Entry is followed by the uncoating of the ribonucleocapsid to expose the positive-sense genomic RNA which is translated by the host ribosomes to yield the viral replication complex. The viral replication complex continues with viral transcription and genome replication and, with the aid of host proteins such as hnRNPA1, yields a nested set of positive-sense sub-genomic sized mRNAs as well as the full-length virus genome. Sub-genomic sized mRNAs are translated by host ribosomes into viral structural (S, E, M, N) and accessory proteins. The N protein packages the positive-sense genomic RNA into a ribonucleocapsid and is assembled into the virus particles with the help of ²-actin. The newly formed virus particles undergo maturation when passing through the Golgi and exit the host cell via exocytosis.

The relationship between a virus and its host is complex; the virus must evolve evasive strategies to avoid detection and immunological defense mechanisms from the host while the host must develop various lines of defense in order to combat viral invasion.

As highlighted in this review, the diverse virus-host interactions established during animal coronavirus and arterivirus infections may have direct implications on viral replication itself, or they may also lead to the modification of numerous signaling pathways, such as cellular stress and/or host antiviral innate immune response, as a means to expedite viral replication and pathogenesis.

The intricate network between a virus and its host is a complicated affair that involves many players-derived from both virus and host-which are pitted against one another in the battle for ascendancy; certain host processes are readily assimilated and manipulated by the virus in its bid to impede host antiviral responses while the host is itself armored with several lines of defense mechanisms and numerous antiviral factors to combat viral invasion to prevent its spread.

While much progress has been made in the elucidation of the various regulatory pathways that, under the most propitious of conditions, will allow virus and host to co-exist in an uneasy truce, a clearer understanding of the complex interplay between the virus and its host could give new insights into the role of main players, such as those that govern the onset of apoptosis, or type I interferons and their regulators, and lead to the discovery of novel and/or universal targets for the therapeutic mediation against pathogenic infection.

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A mathematical model for simulating the phase-based transmissibility of a novel coronavirus

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Abstract: **Background** As reported by the World Health Organization, a novel coronavirus (2019-nCoV) was identified as the causative virus of Wuhan pneumonia of unknown etiology by Chinese authorities on 7 January, 2020. The virus was named as severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) by International Committee on Taxonomy of Viruses on 11 February, 2020. This study aimed to develop a mathematical model for calculating the transmissibility of the virus. **Methods** In this study, we developed a Bats-Hosts-Reservoir-People transmission network model for simulating the potential transmission from the infection source (probably be bats) to the human infection. Since the Bats-Hosts-Reservoir network was hard to explore clearly and public concerns were focusing on the transmission from Huanan Seafood Wholesale Market (reservoir) to people, we simplified the model as Reservoir-People (RP) transmission network model. The next generation matrix approach was adopted to calculate the basic reproduction number (R_0) from the RP model to assess the transmissibility of the SARS-CoV-2. **Results** The value of R_0 was estimated of 2.30 from reservoir to person and 3.58 from person to person which means that the expected number of secondary infections that result from introducing a single infected individual into an otherwise susceptible population was 3.58. **Conclusions** Our model showed that the transmissibility of SARS-CoV-2 was higher than the Middle East respiratory syndrome in the Middle East countries, similar to severe acute respiratory syndrome, but lower than MERS in the Republic of Korea.

Text: On 31 December 2019, the World Health Organization (WHO) China Country Office was informed of cases of pneumonia of unknown etiology (unknown cause) detected in Wuhan City, Hubei Province of China, and WHO reported that a novel coronavirus (2019-nCoV), which was named as severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) by International Committee on Taxonomy of Viruses on 11 February, 2020, was identified as the causative virus by Chinese authorities on 7 January [1] . It is reported that the virus might be bat origin [2] , and the transmission of the virus might related to a seafood market (Huanan Seafood Wholesale Market) exposure [3, 4] . The genetic features and some clinical findings of the infection have been reported recently [4] [5] [6] . Potentials for international spread via commercial air travel had been assessed [7] . Public health concerns are being paid globally on how many people are infected and suspected.

Therefore, it is urgent to develop a mathematical model to estimate the transmissibility and dynamic of the transmission of the virus. There were several researches focusing on mathematical modelling [3, 8] . These researches focused on calculating the basic reproduction number (R_0) by using the serial intervals and intrinsic growth rate [3, 9, 10] , or using ordinary differential equations and Markov Chain Monte Carlo methods [8] . However, the bat origin and the transmission route from the seafood market to people were not considered in the published models.

In this study, we developed a Bats-Hosts-Reservoir-People (BHRP) transmission network model for simulating the potential transmission from the infection source (probably be bats) to the human infection. Since the Bats-Hosts-Reservoir network was hard to explore clearly and public concerns were focusing on the transmission from Huanan Seafood Wholesale Market (reservoir) to people, we simplified the model as Reservoir-People (RP) transmission network model, and R_0 was calculated based on the RP model to assess the transmissibility of the SARS-CoV-2.

The reported cases of SARS-CoV-2, which have been named as COVID-19, were collected for the modelling study from a published literature [3] . As reported by Li et al. [3] , the onset date of the first case was on 7 December, 2020, and the seafood market was closed on 1 January, 2020 [11] . The epidemic curve from 7 December, 2019 to 1 January, 2020 was collected for our study, and the simulation time step was 1 day. fourth-order Runge-Kutta method, with tolerance set at 0.001, was used to perform curve fitting. While the curve fitting is in progress, Berkeley Madonna displays the root mean square deviation between the data and best run so far. The coefficient of determination (R^2) was employed to assess the goodness-of-fit. SPSS 13.0 (IBM Corp., Armonk, NY, USA) was employed to calculate the R^2 .

The Bats-Hosts-Reservoir-People (BHRP) transmission network model

The BHRP transmission network model was posted to bioRxiv on 19 January, 2020 [12] . We assumed that the virus transmitted among the bats, and then transmitted to unknown hosts (probably some wild animals). The hosts were hunted and sent to the seafood market which was defined as the reservoir of the virus. People exposed to the market got the risks

of the infection (Fig. 1) . The BHRP transmission network model was based on the following assumptions or facts:

a) The bats were divided into four compartments: susceptible bats (S_B), exposed bats (E_B), infected bats (I_B), and removed bats (R_B). The birth rate and death rate of bats were defined as n_B and m_B . In this model, we set $\Lambda_B = n_B \times N_B$ as the number of the newborn bats where N_B refer to the total number of bats. The incubation period of bat infection was defined as $1/\omega_B$ and the infectious period of bat infection was defined as $1/\gamma_B$. The S_B will be infected through sufficient contact with I_B , and the transmission rate was defined as β_B . b) The hosts were also divided into four compartments: susceptible hosts (S_H), exposed hosts (E_H), infected hosts (I_H), and removed hosts (R_H). The birth rate and death rate of hosts were defined as n_H and m_H . In this model, we set $\Lambda_H = n_H \times N_H$ where N_H refer to the total number of hosts. The incubation period of host infection was defined as $1/\omega_H$ and the infectious period of host infection was defined as $1/\gamma_H$. The S_H will be infected through sufficient contact with I_B and I_H , and the transmission rates were defined as β_{BH} and β_H , respectively. c) The SARS-CoV-2 in reservoir (the seafood market) was denoted as W . We assumed that the retail purchases rate of the hosts in the market was a , and that the prevalence of SARS-CoV-2 in the purchases was I_H/N_H , therefore, the rate of the SARS-CoV-2 in W imported form the hosts was $aW I_H/N_H$ where N_H was the total number of hosts. We also assumed that symptomatic infected people and asymptomatic infected people could export the virus into W with the rate of μP and $\mu' P$, although this assumption might occur in a low probability. The virus in W will subsequently leave the W compartment at a rate of ϵW , where $1/\epsilon$ is the lifetime of the virus. d) The people were divided into five compartments:

susceptible people (S_P), exposed people (E_P), symptomatic infected people (I_P), asymptomatic infected people (A_P), and removed people (R_P) including recovered and death people. The birth rate and death rate of people were defined as n_P and m_P . In this model, we set $\Lambda_P = n_P \times N_P$ where N_P refer to the total number of people. The incubation period and latent period of human infection was defined as $1/\omega_P$ and $1/\omega'_P$. The infectious period of I_P and A_P was defined as $1/\gamma_P$ and $1/\gamma'_P$. The proportion of asymptomatic infection was defined as δ_P . The S_P will be infected through sufficient contact with W and I_P , and the transmission rates were defined as β_W and β_P , respectively. We also assumed that the transmissibility of A_P was κ times that of I_P , where $0 \leq \kappa \leq 1$.

The parameters of the BHRP model were shown in Table 1 .

We assumed that the SARS-CoV-2 might be imported to the seafood market in a short time. Therefore, we added the further assumptions as follows:

a) The transmission network of Bats-Host was ignored. b) Based on our previous studies on simulating importation [13, 14] , we set the initial value of W as following impulse function:

In the function, n , t_0 and t_i refer to imported volume of the SARS-CoV-2 to the market, start time of the simulation, and the interval of the importation.

Therefore, the BHRP model was simplified as RP model and is shown as follows:

During the outbreak period, the natural birth rate and death rate in the population was in a relative low level. However, people would commonly travel into and out from Wuhan City mainly due to the Chinese New Year holiday. Therefore, n_P and m_P refer to the rate of people traveling into Wuhan City and traveling out from Wuhan City, respectively.

In the model, people and viruses have different dimensions. Based on our previous research [15] , we therefore used the following sets to perform the normalization:

In the normalization, parameter c refers to the relative shedding coefficient of A_P compared to I_P . The normalized RP model is changed as follows:

The transmissibility of the SARS-CoV-2 based on the RP model

In this study, we used the R_0 to assess the transmissibility of the SARS-CoV-2. Commonly, R_0 was defined as the expected number of secondary infections that result from

introducing a single infected individual into an otherwise susceptible population [13, 16, 17]. If $R_0 > 1$, the outbreak will occur. If $R_0 < 1$, the outbreak will toward an end. In this study, R_0 was deduced from the RP model by the next generation matrix approach [18]. The multiple of the transmissibility of A P to that of I P.

The parameters were estimated based on the following facts and assumptions:

a) The mean incubation period was 5.2 days (95% confidence interval [CI]: 4.1-7.0) [3]. We set the same value (5.2 days) of the incubation period and the latent period in this study. Thus, $\omega_P = \omega'_P = 0.1923$. b) There is a mean 5-day delay from symptom onset to detection/hospitalization of a case (the cases detected in Thailand and Japan were hospitalized from 3 to 7 days after onset, respectively) [19] [20] [21]. The duration from illness onset to first medical visit for the 45 patients with illness onset before January 1 was estimated to have a mean of 5.8 days (95% CI: 4.3-7.5) [3]. In our model, we set the infectious period of the cases as 5.8 days. Therefore, $\gamma_P = 0.1724$. c) Since there was no data on the proportion of asymptomatic infection of the virus, we simulated the baseline value of proportion of 0.5 ($\delta_P = 0.5$). d) Since there was no evidence about the transmissibility of asymptomatic infection, we assumed that the transmissibility of asymptomatic infection was 0.5 times that of symptomatic infection ($\kappa = 0.5$), which was the similar value as influenza [22]. We assumed that the relative shedding rate of A P compared to I P was 0.5. Thus, $c = 0.5$. e) Since 14 January, 2020, Wuhan City has strengthened the body temperature detection of passengers leaving Wuhan at airports, railway stations, long-distance bus stations and passenger terminals. As of January 17, a total of nearly 0.3 million people had been tested for body temperature [23]. In Wuhan, there are about 2.87 million mobile population [24]. We assumed that there was 0.1 million people moving out to Wuhan City per day since January 10, 2020, and we believe that this number would increase (mainly due to the winter vacation and the Chinese New Year holiday) until 24 January, 2020. This means that the 2.87 million would move out from Wuhan City in about 14 days. Therefore, we set the moving volume of 0.2 million per day in our model. Since the population of Wuhan was about 11 million at the end of 2018 [25], the rate of people traveling out from Wuhan City would be 0.018 (0.2/11) per day. However, we assumed that the normal population mobility before January 1 was 0.1 times as that after January 10. Therefore, we set the rate of people moving into and moving out from Wuhan City as 0.0018 per day ($n_P = m_P = 0.0018$).

f) The parameters b_P and b_W were estimated by fitting the model with the collected data.
 g) At the beginning of the simulation, we assumed that the prevalence of the virus in the market was $1/100000$.
 h) Since the SARS-CoV-2 is an RNA virus, we assumed that it could be died in the environment in a short time, but it could be stay for a longer time (10 days) in the unknown hosts in the market. We set $\varepsilon = 0.1$.

In this study, we assumed that the incubation period ($1/\omega_P$) was the same as latent period ($1/\omega'_P$) of human infection, thus $\omega_P = \omega'_P$. Based on the equations of RP model, we can get the disease free equilibrium point as: In the matrix:

By the next generation matrix approach, we can get the next generation matrix and R_0 for the RP model:

The R_0 of the normalized RP model is shown as follows:

Our modelling results showed that the normalized RP model fitted well to the reported SARS-CoV-2 cases data ($R^2 = 0.512$, $P < 0.001$) (Fig. 2). The value of R_0 was estimated of 2.30 from reservoir to person, and from person to person and 3.58 from person to person which means that the expected number of secondary infections that result from introducing a single infected individual into an otherwise susceptible population was 3.58.

In this study, we developed RP transmission model, which considering the routes from reservoir to person and from person to person of SARS-CoV-2 respectively. We used the models to fit the reported data in Wuhan City, China from published literature [3]. The simulation results showed that the R_0 of SARS-CoV-2 was 3.58 from person to person. There was a research showed that the R_0 of SARS-CoV-2 was 2.68 (95% CI: 2.47-2.86) [8]. Another research showed that the R_0 of SARS-CoV-2 was 2.2 (95% CI: 1.4-3.9) [3]. The different values might be due to the different methods. The methods which Li et al. employed were based on the epidemic growth rate of the epidemic curve and the serial interval [3]. Our previous study showed that several methods could be used to calculate the R_0 based on the epidemic growth rate of the epidemic curve and the serial interval, and different methods might result in different values of R_0 [26]. Our results also showed that the R_0 of SARS-CoV-2 was 2.30 from reservoir to person which was lower than that of

person to person. This means that the transmission route was mainly from person to person rather than from reservoir to person in the early stage of the transmission in Wuhan City. However, this result was based on the limited data from a published literature, and it might not show the real situation at the early stage of the transmission.

Researches showed that the R_0 of severe acute respiratory syndrome (SARS) was about 2.7-3.4 or 2-4 in Hong Kong, China [27, 28]. Another research found that the R_0 of SARS was about 2.1 in Hong Kong, China, 2.7 in Singapore, and 3.8 in Beijing, China [29]. Therefore, we believe that the commonly acceptable average value of the R_0 of SARS might be 2.9 [30]. The transmissibility of the Middle East respiratory syndrome (MERS) is much lower than SARS. The reported value of the R_0 of MERS was about 0.8-1.3 [31], with the inter-human transmissibility of the disease was about 0.6 or 0.9 in Middle East countries [32]. However, MERS had a high transmissibility in the outbreak in the Republic of Korea with the R_0 of 2.5-7.2 [33, 34]. Therefore, the transmissibility of SARS-CoV-2 might be higher than MERS in the Middle East countries, similar to SARS, but lower than MERS transmitted in the Republic of Korea.

To contain the transmission of the virus, it is important to decrease R_0 . According to the equation of R_0 deduced from the simplified RP model, R_0 is related to many parameters. The mainly parameters which could be changed were b_P , b_W , and γ . Interventions such as wearing masks and increasing social distance could decrease the b_P , the intervention that close the seafood market could decrease the b_W , and shorten the duration from symptoms onset to be diagnosed could decrease $1/\gamma$. All these interventions could decrease the effective reproduction number and finally be helpful to control the transmission.

Since there are too many parameters in our model, several limitations exist in this study. Firstly, we did not use the detailed data of the SARS-CoV-2 to perform the estimation instead of using the data from literatures [3]. We simulated the natural history of the infection that the proportion of asymptomatic infection was 50%, and the transmissibility of asymptomatic infection was half of that of symptomatic infection, which were different to those of MERS and SARS. It is known that the proportion of asymptomatic infection of MERS and SARS was lower than 10%. Secondly, the parameters of population mobility were not from an accurate dataset. Thirdly, since there was no data of the initial prevalence of the virus in the seafood market, we assumed the initial value of 1/100 000. This

assumption might lead to the simulation been under-or over-estimated. In addition, since we did not consider the changing rate of the individual's activity (such as wearing masks, increasing social distance, and not to travel to Wuhan City), the estimation of importation of the virus might not be correct. All these limitations will lead to the uncertainty of our results. Therefore, the accuracy and the validity of the estimation would be better if the models fit the first-hand data on the population mobility and the data on the natural history, the epidemiological characteristics, and the transmission mechanism of the virus.

By calculating the published data, our model showed that the transmissibility of SARS-CoV-2 might be higher than MERS in the Middle East countries, similar to SARS, but lower than MERS in the Republic of Korea. Since the objective of this study was to provide a mathematical model for calculating the transmissibility of SARS-CoV-2, the R_0 was estimated based on limited data which published in a literature. More data were needed to estimate the transmissibility accurately.

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Potential Maternal and Infant Outcomes from (Wuhan) Coronavirus 2019-nCoV Infecting Pregnant Women: Lessons from SARS, MERS, and Other Human Coronavirus Infections

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Abstract: In early December 2019 a cluster of cases of pneumonia of unknown cause was identified in Wuhan, a city of 11 million persons in the People's Republic of China. Further investigation revealed these cases to result from infection with a newly identified coronavirus, termed the 2019-nCoV. The infection moved rapidly through China, spread to Thailand and Japan, extended into adjacent countries through infected persons travelling by air, eventually reaching multiple countries and continents. Similar to such other coronaviruses as those causing the Middle East respiratory syndrome (MERS) and severe acute respiratory syndrome (SARS), the new coronavirus was reported to spread via natural aerosols from human-to-human. In the early stages of this epidemic the case fatality rate is estimated to be approximately 2%, with the majority of deaths occurring in special populations. Unfortunately, there is limited experience with coronavirus infections during pregnancy, and it now appears certain that pregnant women have become infected during the present 2019-nCoV epidemic. In order to assess the potential of the Wuhan 2019-nCoV to cause maternal, fetal and neonatal morbidity and other poor obstetrical outcomes, this communication reviews the published data addressing the epidemiological and clinical effects of SARS, MERS, and other coronavirus infections on pregnant women and their infants. Recommendations are also made for the consideration of pregnant women in the design, clinical trials, and implementation of future 2019-nCoV vaccines.

Text: Coronaviruses are spherical, enveloped, and the largest of positive-strand RNA viruses. They have a wide host range, including birds, farm animals, pets, camels, and bats, in which they primarily cause respiratory and gastrointestinal disease. Belonging to the order Nidovirales, family Coronaviridae, and the subfamily Orthocoronaviridae there are four genera of coronaviruses-Alphacoronavirus, Betacoronavirus, Deltacoronavirus, and Gammacoronavirus [1] [2] [3] [4] .

In humans, they are a cause of mild illnesses including the common colds occurring in children and adults, and were believed to be of modest medical importance. However, two zoonotic coronaviruses-including the severe acute respiratory syndrome coronavirus (SARS-CoV) and Middle East respiratory syndrome coronavirus (MERS-CoV)-can produce severe lower respiratory In the beginning of December 2019, a cluster of persons with a pneumonia of unknown cause was identified in Wuhan, the capital of Hubei Province and a large city of approximately 11 million persons located in the central region of the People's Republic of China [7, 8] . Between 8 and 18 December 2019 there were 7 cases of pneumonia identified whose clinical features resembled that of a viral pneumonia. The outbreak was initially believed to be linked to the Wuhan Huanan (South China) Seafood

Wholesale Market. This market, termed a "wet" market, sells a variety of seafood, cuts of meat, and both live and dead animals in over one thousand stalls in constant close contact; however, whether this market was the origin of the outbreak remains unknown [9] . On 31 December 2019, the Chinese Center for Disease Control and Prevention (China CDC) sent a rapid response team to Hubei to work alongside health personnel from the provincial and Wuhan city health departments to conduct an epidemiologic investigation. As the disease was spreading through secondary and tertiary cases, the World Health Organization (WHO) China Country Office was informed on 31 December 2019 of the occurrence of these cases of pneumonia of unknown etiology. During the period from 31 December 2019 to 3 January 2020, 44 patients with pneumonia of unknown etiology were reported by the Chinese authorities to the WHO. On 7 January 2020 investigators in China identified the etiological agent of the epidemic as a previously unknown coronavirus, and it was given the designation 2019-nCoV (for 2019 novel coronavirus) [8] . Analysis of the clinical features of 41 hospitalized patients with laboratory-confirmed 2019-nCoV infection revealed that 30 were men (73%); less than one-half had underlying co-morbid conditions (13; 32%) which included diabetes (8, 20%) , hypertension (6, 15%), and cardiovascular disease (6; 15%); and the average age was 49.0 years old. The most common symptoms at the beginning of their illness included fever (40, 98%) , cough (31, 76%) , and fatigue or myalgia (18, 44%) , sputum production (11, 28%) , and headache (3, 8%) [10] . Among these 41 initial cases of 2019-nCoV infection there were 12 patients (32%) who developed acute respiratory distress syndrome (ARDS), 13 (32%) required intensive care and 6 (15%) died. During the first weeks of January the infection spread rapidly through China and extended to adjacent countries where cases began to appear-13 January in Thailand, 15 January in Japan, 20 January in the Republic of Korea, and Taiwan and the United States on 21 January [11] . Infected travelers, mostly via commercial air travel, are known to have been responsible for introducing the virus outside of Wuhan. The new coronavirus continued to spread throughout multiple countries and continents, and by 9 February 2020 the WHO reported 37,251 confirmed cases in China that resulted in 812 deaths, surpassing the number of deaths that occurred during the 2002-2003 SARS epidemic. An additional 307 cases of 2019-nCoV infection have occurred among 24 other countries outside of China [12] . (Figure 1) At the meeting of the Emergency Committee of the WHO on 30 January, the novel coronavirus 2019 epidemic was declared a Public Health Emergency of International Concern (PHEIC) [11, 13] .

Viruses 2020, 12, 194 3 of 16 epidemic. An additional 307 cases of 2019-nCoV infection have occurred among 24 other countries outside of China [12] . (Figure 1) At the meeting of the Emergency Committee of the WHO on 30 January, the novel coronavirus 2019 epidemic

was declared a Public Health Emergency of International Concern (PHEIC) [11, 13]. This newly recognized coronavirus, producing a disease that has been termed COVID-19, is rapidly spreading throughout China, has crossed international borders to infect persons in neighboring countries, and humans infected by the virus are travelling via commercial airlines to other continents. It is certain that 2019-nCoV will infect women who are pregnant, leaving the question open as to whether the novel coronavirus will have a similar or different effect on them compared with SARS-CoV and MERS-CoV. In order to address the potential obstetrical outcomes of infection to both mother and infant, the present communication describes the current state of knowledge regarding the effects of other coronavirus infections in pregnancy.

Pneumonia arising from any infectious etiology is an important cause of morbidity and mortality among pregnant women. It is the most prevalent non-obstetric infectious condition that occurs during pregnancy [14] [15] [16]. In one study pneumonia was the 3rd most common cause of indirect maternal death [17]. Approximately 25 percent of pregnant women who develop pneumonia will need to be hospitalized in critical care units and require ventilatory support [16]. Although bacterial pneumonia is a serious disease when it occurs in pregnant women, even when the agent(s) are susceptible to antibiotics, viral pneumonia has even higher levels of morbidity and mortality during pregnancy [18]. As with other infectious diseases, the normal maternal physiologic changes that accompany pregnancy-including altered cell-mediated immunity [19] and changes in pulmonary function-have been hypothesized to affect both susceptibility to and clinical severity of pneumonia [20] [21] [22]. This has been evident historically during previous epidemics. The case fatality rate (CFR) for pregnant women infected with influenza during the 1918-1919 pandemic was 27%-even higher when exposure occurred during the 3rd trimester and upwards of 50% if pneumonia supervened [23]. During the 1957-1958 Asian flu epidemic, 10% of all deaths occurred in pregnant women, and their CFR was twice as high as that of infected women who were not pregnant [24]. The most common adverse obstetrical outcomes associated with maternal pneumonias from all causes include This newly recognized coronavirus, producing a disease that has been termed COVID-19, is rapidly spreading throughout China, has crossed international borders to infect persons in neighboring countries, and humans infected by the virus are travelling via commercial airlines to other continents. It is certain that 2019-nCoV will infect women who are pregnant, leaving the question open as to whether the novel coronavirus will have a similar or different effect on them compared with SARS-CoV and MERS-CoV. In order to address the potential obstetrical outcomes of infection to both mother and infant, the present

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The SARS epidemic began quietly at the turn of the 21st century. In November 2002, a cook in Guangdong Province, China, died from an unidentified illness. He had worked at a restaurant in which meat from wild animals was served. On 27 November 2002 Chinese-language media and internet reports were picked up by Canada's Global Public Health Intelligence Network (GPHIN) that indicated a flu-like illness was occurring in China [25, 26] . Unfortunately, the reports were not translated, and China failed to report the occurrence of this illness to the World Health Organization (WHO) until February 2003. The disease spread to other countries where it primarily infected healthcare workers. One of these was Dr. Carlo Urbani, a WHO physician investigating a patient with the new disease in Hanoi. He recognized that the pneumonia was probably caused by a new, highly infectious agent, and rapidly notified the WHO. He contracted the SARS-CoV while there, became febrile and later died after traveling to Thailand to attend a conference. On 12

March 2003, WHO issued a global alert regarding the disease that was occurring primarily among health care workers in Hanoi, Vietnam and Hong Kong. The disease continued to spread, and by 31 July 2003 there were 8422 probable cases, leading to 916 deaths in 29 countries, with the majority of cases occurring in mainland China and Hong Kong. Approximately 30% of infections occurred in healthcare workers. By the termination of the epidemic the global CFR was 11% [27] .

Although there were relatively few documented cases of SARS occurring during pregnancy, several case reports and small clinical studies have described the clinical effects in pregnant women and their infants. In reviewing these reports describing pregnant women with SARS in China it is possible, and perhaps even probable, that some of the same patients were included in more than one publication. However, even if this is the case, there is no doubt that SARS coronavirus infection was found to be associated with severe maternal illness, maternal death, and spontaneous abortion [19, [28] [29] [30] [31] . Martha Anker, an expert in statistics formerly with the WHO and the University of Massachusetts, estimated that more than 100 cases of SARS-CoV infection occurred in pregnant women, which warrants closer inspection [27] .

The clinical outcomes among pregnant women with SARS in Hong Kong were worse than those occurring in infected women who were not pregnant [32] . Wong et al. [29] evaluated the obstetrical outcomes from a cohort of pregnant women who developed SARS in Hong Kong during the period of 1 February to 31 July 2003. Four of the 7 women (57%) that presented during the 1st trimester sustained spontaneous miscarriages, likely a result of the hypoxia that was caused by SARS-related acute respiratory distress. Among the 5 women who presented after 24 weeks gestation, 4 had preterm deliveries (80%).

A case-control study to determine the effects of SARS on pregnancy compared 10 pregnant and 40 non-pregnant women with the infection at the Princess Margaret Hospital in Hong Kong [27, 33] . There were 3 deaths among the pregnant women with SARS (maternal mortality rate of 30%) and no deaths in the non-pregnant group of infected women ($P = 0.006$). Renal failure ($P = 0.006$) and disseminated intravascular coagulopathy ($P = 0.006$) developed more frequently in pregnant SARS patients when compared with the non-pregnant SARS group. Six pregnant women with SARS required admission to the intensive care unit (ICU) (60%) and 4 required endotracheal intubation (40%), compared with a

12.5% intubation rate ($P = 0.065$) and 17.5% ICU admission rate ($P = 0.012$) in the non-pregnant group.

Maxwell et al. [32] reported 7 pregnant women infected with SARS-CoV who were followed at a designated SARS unit-2 of the 7 died (CFR of 28%), and 4 (57%) required ICU hospitalization and mechanical ventilation. In contrast, the mortality rate was less than 10% and mechanical ventilation rate less than 20% among non-pregnant, age-matched counterparts who were not infected with SARS-CoV. Two women with SARS recovered and maintained their pregnancy but had infants with IUGR. Among the live newborn infants, none had clinical or laboratory evidence for SARS-CoV infection. The new mothers who had developed SARS were advised not to breastfeed to prevent possible vertical transmission of the virus.

Zhang et al. [34] described SARS-CoV infections in 5 primagravidas from Guangzhou, China at the height of the SARS epidemic. Two of the mothers became infected in the 2nd trimester, and 3 developed infection in the 3rd trimester. Two of the pregnant women had hospital-acquired SARS infections, and the other 3 were community-acquired. All 5 pregnant women had fever and abnormal chest radiographs; 4 had cough; 4 developed hypoalbuminemia; 3 had elevated alanine aminotransferase levels (ALT), 3 had chills or rigor, 2 had decreased lymphocytes, and 2 had decreased platelets. One pregnant woman required intensive care, but all recovered and there were no maternal deaths. The 5 infants were clinically evaluated, and none had evidence of SARS.

Two pregnant women with SARS were reported from the United States. In a detailed case report, Robertson et al. [35] described a 36-year-old pregnant woman with an intermittent cough of approximately 10 days duration and no fever. While travelling in Hong Kong during the 2003 epidemic, she was exposed at her hotel to a person subsequently known to be infected with SARS-CoV. At 19 weeks gestation she developed fever, anorexia, headache, increasing cough, weakness, and shortness of breath. Upon returning to the United States she was hospitalized with pneumonia. Obstetrical ultrasounds revealed a low-lying placenta (placenta previa) but were otherwise normal. Following her discharge home and clinical recovery, she was found to have antibodies to SARS-CoV. She underwent cesarean section at 38 weeks gestation because of the placenta previa and a healthy baby girl was delivered [35, 36]. The placenta was interpreted as being normal. At 130 days post-maternal illness, maternal serum and whole blood, swabs from maternal nasopharynx and

rectum, post-delivery placenta, umbilical cord blood, amniotic fluid, and breast milk were collected for analysis-no viral RNA was detected in specimens tested by reverse transcriptase polymerase chain reaction (RT-PCR). Antibodies to SARS-CoV were detected from maternal serum, umbilical cord blood, and breast milk by enzyme immunoassay (EIA) and indirect immunofluorescence assay. No clinical specimens (except for cord blood) were available for testing from the infant. The second case in the USA occurred in a 38-year-old woman who had travelled to Hong Kong at 7 weeks gestation where she was exposed to SARS-CoV in the same hotel as the aforementioned American woman [37]. Following her return to the United States, her husband developed the clinical onset of SARS, and 6 days later she became ill with fever, myalgia, chills, headache, coryza, and a productive cough with shortness of breath and wheezing. Following her hospitalization for SARS she recovered, serum samples taken on days 28 and 64 post-onset of illness were positive for antibodies to SARS-CoV by enzyme immunoassay and immunofluorescent assays. Her pregnancy continued and was unremarkable except for developing elevated glucose levels. A cesarean section that was performed at 36 weeks gestation due to preterm rupture of membranes and fetal distress resulted in a healthy baby boy. At the time of delivery, the mother's serum samples were positive for antibodies to SARS-CoV, but samples taken of umbilical cord blood and placenta were negative. Breast milk sampled 12 and 30 days after delivery were also negative for SARS-CoV antibodies. Specimens evaluated from maternal blood, stool, and nasopharynx samples, as well as umbilical cord blood of the infant, were all negative for coronavirus RNA by RT-PCR. Neonatal stool samples obtained on days-of-life 12 and 30 were also negative for viral RNA.

From Canada, Yudin et al. [38] reported a 33-year-old pregnant woman who was admitted to the hospital at 31 weeks gestation with a fever, dry cough, and abnormal chest radiograph demonstrating patchy infiltrates. She had acquired SARS from contact with an infected family member. Following a 21-day stay in the hospital, during which she did not require ventilatory support, her convalescent antibody titers were positive for coronavirus infection. She had a normal labor and delivery and her newborn girl had no evidence of infection.

In a study of 5 liveborn neonates who were delivered to women infected with SARS-CoV during the Hong Kong epidemic, results from multiple tests-including serial RT-PCR assays, viral culture, and paired neonatal serological titers-were negative for SARS-CoV [39]. None of the 5 neonates developed any clinical signs or symptoms of respiratory infection or compromise.

Fortunately, there were no cases of vertical transmission identified among pregnant women infected with SARS-CoV during the 2002-2003 Asian epidemic [27, 30, 31, 39, 40] , and with the exception of a small cluster of cases that recurred in late 2003, no new cases of SARS have occurred.

In the only reported study of the placental pathology of mothers with SARS, Ng et al. [41] reported the findings from 7 pregnant women infected with SARS-CoV. In the case of 2 women who were convalescing from SARS-CoV infection during the 1st trimester of pregnancy, the placentas were found to be normal. Three placentas were delivered from pregnancies in which the mothers had acute SARS-CoV infection-these were abnormal and demonstrated increased subchorionic and intervillous fibrin, a finding that can be associated with abnormal maternal blood flow to the placenta. In the placentas of 2 women who were convalescing from SARS-CoV infection in the 3rd trimester of pregnancy the placentas were highly abnormal. They showed extensive fetal thrombotic vasculopathy with areas of avascular chorionic villi-chronic findings of fetal vascular malperfusion. These 2 pregnancies also were complicated by oligohydramnios and had poor obstetrical outcomes-both infants had developed IUGR. It is interesting that villitis, the microscopic finding of inflammation of the chorionic villi that is the histologic hallmark of many maternal hematogenous infections that are transmitted through the placenta to the fetus, was not identified in any of these placentas.

Similar to other coronavirus infections, SARS-CoV is easily spread from person-to-person via respiratory droplets and secretions as well as through nosocomial contacts [42, 43] . In addition to transmission of SARS-CoV through natural aerosols from infected patients, it was found that in Hong Kong the SARS-CoV could also be transmitted by mechanical aerosols [44] . Environmental factors had an important role when it was discovered that during the Amoy Gardens housing estate outbreak as many as two-thirds of infected persons had diarrhea, SARS-CoV was excreted in their stools, and that aerosols arising from the flushing of toilets could transmit the virus [44] . Healthcare facilities were also an important source of new SARS infections during the 2002-2003 epidemic, and healthcare workers were also at high risk for acquiring the infection.

In order to address the safety issues for the obstetrical management and delivery of pregnant women with SARS, guidelines were prepared by the Canadian Task Force on

Preventive Health Care and the Society of Obstetricians and Gynaecologists of Canada [45] . These recommendations include:

1.

"All hospitals should have infection control systems in place to ensure that alerts regarding changes in exposure risk factors for SARS or other potentially serious communicable diseases are conveyed promptly to clinical units, including the labour and delivery unit.

At times of SARS outbreaks, all pregnant patients being assessed or admitted to the hospital should be screened for symptoms of and risk factors for SARS.

Upon arrival in the labour triage unit, pregnant patients with suspected and probable SARS should be placed in a negative pressure isolation room with at least 6 air exchanges per hour. All labour and delivery units caring for suspected and probable SARS should have available at least one room in which patients can safely labour and deliver while in need of airborne isolation.

If possible, labour and delivery (including operative delivery or Caesarean section) should be managed in a designated negative pressure isolation room, by designated personnel with specialized infection control preparation and protective gear. 5.

Either regional or general anaesthesia may be appropriate for delivery of patients with SARS.

Neonates of mothers with SARS should be isolated in a designated unit until the infant has been well for 10 days, or until the mother's period of isolation is complete. The mother should not breastfeed during this period. 7.

A multidisciplinary team, consisting of obstetricians, nurses, pediatricians, infection control specialists, respiratory therapists, and anaesthesiologists, should be identified in each unit and be responsible for the unit organization and implementation of SARS management protocols. 8.

Staff caring for pregnant SARS patients should not care for other pregnant patients. Staff caring for pregnant SARS patients should be actively monitored for fever and other symptoms of SARS. Such individuals should not work in the presence of any SARS symptoms within 10 days of exposure to a SARS patient. 9.

All health care personnel, trainees, and support staff should be trained in infection control management and containment to prevent spread of the SARS virus. 10. Regional health authorities in conjunction with hospital staff should consider designating specific facilities or health care units, including primary, secondary, or tertiary health care centers, to care for patients with SARS or similar illnesses."

Middle East respiratory syndrome (MERS) was first reported in September 2012 in Saudi Arabia, following isolation of MERS-CoV from a male patient who died months earlier from severe pneumonia and multiple organ failure [1] . In the 8 years since then, there have been more than 2494 confirmed cases of MERS resulting in upwards of 858 deaths globally [46] . While 27 countries have reported cases of MERS, approximately 80% of confirmed cases originated in Saudi Arabia [47] . To date, all known cases of MERS can be linked to travel or residence in countries along the Arabian Peninsula-that is, Bahrain; Iraq; Iran; Israel, the West Bank, and Gaza; Jordan; Kuwait; Lebanon; Oman; Qatar, Saudi Arabia; Syria; the United Arab Emirates (UAE); and Yemen [48] . The largest documented outbreak outside of this region occurred in 2015 in the Republic of Korea, in which 186 infections occurred, resulting in 38 deaths [49] . The index case in this outbreak reportedly returned from the Arabian Peninsula just prior to onset of illness [50] .

MERS-CoV is characterized by sporadic zoonotic transmission events as well as spread between infected patients and close contacts (i.e., intra-familial transmission) [51] . Nosocomial outbreaks in health care settings-the result of poor infection control and prevention-are widely recognized as the hallmark of MERS [1] . Superspreading events have been recorded in healthcare settings in Jordan, Al Hasa, Jeddah, Abu Dhabi and South

Korea [47, [52] [53] [54] [55] . Like other coronaviruses, MERS-CoV can be spread through person-to-person contact, likely via infected respiratory secretions [48] . Transmission dynamics, however, are otherwise poorly understood [1] . Bats are believed to be the natural reservoir of MERS-CoV, and dromedary camels can have the virus and have been suggested as possible intermediary hosts as well as a source of infection to humans [2, 56, 57] .

There are no clinical or serological reports of perinatal transmission of MERS, though vertical transmission has been reported for non-coronavirus respiratory viruses including influenza and respiratory syncytial virus (RSV) [58] . Researchers have not yet discovered ongoing transmission of MERS-CoV within communities outside of health care settings.

The clinical presentation of MERS varies from asymptomatic to severe pneumonia with acute respiratory distress syndrome (ARDS), septic shock, and multiple organ failure, often resulting in death. Most patients with MERS develop severe acute respiratory illness accompanied by fever, cough, and shortness of breath [50] . Progression to pneumonia is swift-usually within the first week -and at least one-third of patients also present with gastrointestinal symptoms [1] . MERS progresses much more rapidly to respiratory failure and has a higher case fatality rate than SARS [1] . Unlike SARS, however, infection with MERS-CoV is generally mild in healthy individuals but more severe in immunocompromised patients and people with underlying comorbidities [1] . The overall CFR of MERS is approximately 34.4% [46] . Most fatalities have been associated with pre-existing medical conditions like chronic lung disease, diabetes, and renal failure, as well as weakened immune systems [59] , making such individuals high risk. As a result of the immunological changes that occur during pregnancy, women who are pregnant are included in this high-risk group. Pregnant women may develop severe disease and fatal maternal and/or fetal outcomes as a result of MERS-CoV infection; however, little is known of the pathophysiology of this infection during pregnancy.

Limited data exists on the prevalence and clinical features of MERS during pregnancy, birth, and the postnatal period. It is likely, however, that the immunological changes that normally occur in pregnancy may alter susceptibility to the MERS-CoV and the severity of clinical illness [60] . Pregnant women infected with SARS-CoV, a related coronavirus, appear to have increased morbidity and mortality when compared to non-pregnant women, suggesting that MERS-CoV could also lead to severe clinical outcomes in

pregnancy. To date, however, very few pregnancy-associated cases (n = 11) have been documented, with 91% having adverse clinical outcomes.

Between November 2012 and February 2016, there were 1308 cases of MERS reported by the Saudi Arabia Ministry of Health (MoH). Of these, 5 patients were pregnant, according to a retrospective study by Assiri et al. [47], and all resulted in adverse outcomes. Patient ages ranged from 27 to 34 years, with occurrence of exposure in either the 2nd or 3rd trimester. All 5 cases received intensive care. Two women died and there were 2 cases of perinatal death-1 stillbirth and 1 neonatal death shortly after emergency cesarean section. These instances of severe maternal and perinatal outcomes are consistent with other reports of MERS-CoV infection in pregnant women, as well as outcomes associated with SARS-CoV infection. The authors of the retrospective study concede that unreported cases of MERS in pregnancy are likely due to lack of routine pregnancy testing [47]. They conclude that pregnancy testing for women of reproductive age should be considered for those who test positive for MERS-CoV, to contribute to overall understanding of pathogenesis and epidemiological risk. Additionally, 2 of the 5 patients were healthcare workers, which corresponds with existing knowledge of higher risk of exposure to MERS-CoV in healthcare settings.

In a separate case report of MERS occurring in pregnancy, Alserehi et al. [58] described a 33-year-old critical care nurse who became infected during the 3rd trimester in the midst of a large hospital outbreak. In the days following hospital admission, she developed respiratory failure necessitating mechanical ventilation and administration of dexamethasone as prophylaxis for the fetus. Following an emergency cesarean section at 32 weeks gestation, she was transferred to the intensive care unit (ICU) and later recovered. The preterm but otherwise healthy infant was kept in the neonatal unit for observation and later released along with his mother. In contrast to other reported cases, this patient had a successful outcome, perhaps due to the timing of MERS-CoV exposure, her young age, the use of steroids, and differences in immune response.

Alfaraj et al. [61] described 2 cases of maternal infection with MERS-CoV at the Prince Mohammed Bin Abdulaziz Hospital (PMAH) in Saudi Arabia. Maternal infection in both cases was confirmed by nasopharyngeal swab testing by RT-PCR. One patient was a 29-year-old woman at 6 weeks gestation with no underlying medical conditions. The second patient, a 39-year-old at 24 weeks gestation, had several comorbidities, including end

stage renal disease, hypertension, and hemodialysis. This woman presented to the hospital after contact with a MERS-CoV-infected person during an active outbreak. Both patients later tested negative for MERS-CoV and were subsequently discharged. The younger patient delivered a healthy, full-term infant. The status of the other delivery is unknown. Neither fetus was tested for MERS-CoV.

According to Payne et al. [62], epidemiologic investigation of the 2012 MERS outbreak in Zarqa, Jordan, revealed that a 2nd trimester stillbirth (5 months gestational age) had occurred as a result of maternal exposure to MERS-CoV. The mother experienced fever, fatigue, headache and cough, concurrently with vaginal bleeding and abdominal pain. On the 7th day of symptoms, she had a fetal death. The mother was confirmed to have antibody to MERS-CoV, and she self-reported having had unprotected contact with family members who later tested positive for the virus. This was the first documented occurrence of stillbirth during maternal infection with MERS-CoV.

On 24 November 2013, a 32-year-old pregnant woman in the United Arab Emirates (UAE) developed ARDS following admission to the ICU after suspected community-acquired pneumonia advanced to respiratory failure and hypotension [60]. Later that day, her baby was delivered by caesarean section and subsequent Apgar scores were within healthy range. The next day, RT-PCR evaluation revealed that the mother was positive for MERS-CoV. Despite rigorous intervention, including oral ribavirin-peginterferon- α therapy and ventilator support, the woman continued to deteriorate, developed septic shock, and died. While the outcome for this mother was fatal, Malik et al. noted that virus shedding ceased during therapy with ribavirin and peginterferon- α and radiographic evidence indicated clinical improvement before her death [58]. More research is needed to determine safety, efficacy, and dosage of these therapies in the general population but also in pregnant women. While few data exist on the effects of these treatments in pregnant humans, ribavirin is generally contraindicated during pregnancy [58].

Outside of the Middle East the only confirmed case of MERS in pregnancy occurred in 2015 in South Korea. Jeong et al. [49] reported that a 39-year-old patient was exposed during the 3rd trimester following contact with a patient having MERS. Despite abrupt vaginal bleeding and rupture of membranes, the patient recovered fully and delivered a healthy infant at 37 weeks and 5 days gestation. Subsequent testing of the infant's blood did not detect any IgG, IgM, or IgA antibodies to MERS-CoV.

The mean maternal age of the 11 confirmed maternal SARS cases described above was 33.2 years, with a mean gestational age of 26.3 weeks. The source of infection in 2 of the cases was attributed to contact with family members who tested positive for MERS-CoV, unknown in 3 cases, likely due to animal exposure in 1 case, and 6 were healthcare-associated (2 of these patients were healthcare workers). Six patients required intensive care and 3 died. Of those who died, 2 were exposed to MERS-CoV in the 3rd trimester, and 1 was exposed during the 2nd trimester. The infant death rate for all 11 cases was 27%. Fetal survival did not appear to correlate with the timing of maternal infection and gestational age; however, more data are needed to draw conclusions about this relationship. According to Alfaraj et al. [61], the CFR for the 11 infected women-also 27%-was not statistically different from the overall CFR of MERS in the general population (35%) ($P = 0.75$). Only 1 case resulted in both maternal and fetal death.

Similar to SARS in pregnancy, more research is needed to understand the pathogenesis and epidemiology of MERS in pregnancy including the relationship between the timing of maternal infection, gestational age of the fetus, the effects of comorbid factors, and the occurrence of adverse outcomes. Few studies documented the presence of MERS-CoV antibodies in the umbilical cord or neonatal blood, making it difficult to assess perinatal transmission. As such, future studies should involve the collection of samples from relevant specimens including amniotic fluid, placenta, and umbilical cord [49].

MERS prevention should be high priority for high-risk exposures such as healthcare workers, pregnant women and individuals working with camels, camel meat-milk processors and in abattoirs [57]. Since 2013, the Saudi Arabia MoH has recommended that pregnant women postpone travel to Saudi Arabia for the Hajj and Umrah [47]. To further reduce risk of exposure among pregnant women, additional measures such as avoiding contact with camels and sick persons-particularly in healthcare settings-are also recommended. Pregnant women who present with symptoms of pneumonia, influenza-like illness (ILI), or sepsis on the Arabian Peninsula may also benefit from MERS-CoV screening to expedite early diagnosis and improve disease management [60].

While multiple agents have been used to treat MERS, none have been tested in large clinical studies. Available data are limited to the use of combination therapies of interferon

and other agents in case reports and case series [63] . A prospective or randomized study may prove difficult given the sporadic nature of MERS-CoV outbreaks.

Due to a gap in research on the treatment of MERS in pregnancy, there are no therapeutic options currently recommended for pregnant women [58] . Therapies under development and testing may be considered inappropriate for pregnant women due to the unknown potential for teratogenic effects. For example, during the 2003 SARS outbreak, ribavirin was administered to pregnant women with severe cases of the disease, but ribavirin therapy has been documented to increase the risk of teratogenic effects in newborns [58] .

The Alphacoronaviruses HCoV 229E and NL63, as well as the Betacoronaviruses HKU 1 and OC43, can infect humans and cause the common cold. In order to investigate the potential maternal-fetal transmission of human coronaviruses during pregnancy, Gagneur et al. [64, 65] evaluated 3 types of maternal-infant paired specimens that included maternal vaginal and respiratory specimens that were obtained during labor, as well as gastric samples from the newborn infants. These specimens were evaluated for the presence of HCoV 229E, OC-43, NL63 and HKU 1 using RT-PCR methodology. Between the period from July 2003 to August 2005 the authors examined 159 mother-infant dyads. Human coronaviruses were identified in 12 samples (HCoV 229E: 11; HKU 1 : 1) from 7 mother-child pairs. In 3 mother-infant dyads only maternal respiratory samples were positive; in 2 other pairs all 3 of the samples tested positive for human coronavirus; in 1 case only the maternal vaginal and newborn gastric samples were positive; and in another case the maternal vaginal sample alone was positive. There were no signs of clinical infection in any of the 3 neonates that had positive gastric samples for human coronavirus.

It is beyond the scope of this communication to discuss the various technical challenges inherent in developing a safe and efficacious vaccine for coronavirus infections in humans. There are clearly challenges to this endeavor-protective antibodies to coronaviruses are not long-lasting, tissue damage has been reported to occur as a result of exposure to SARS-CoV, development of animal models that closely resemble human infection are limited, and the extensive time and expense necessary to perform clinical trials in humans, to name a few [66] [67] [68] .

It is vitally important that pregnant women be considered in the design, clinical trial, and implementation of vaccine candidates for 2019-nCoV. In examining the history of vaccine design, it is clear that the needs of pregnant women have rarely been prioritized in either the preclinical development or the clinical trial phases of production. Today, pregnant women are usually excluded from experimental trial of drugs and vaccines that do not target obstetric conditions [69] . Excluding pregnant women and their infants from participation in vaccine development and implementation undermines ethical principles of justice-fairness, equity, and maximization of benefit-and potentially places their health at risk during outbreaks and other health emergencies [69] [70] [71] .

On 23 January 2020 the Coalition for Epidemic Preparedness Innovations (CEPI) announced three programs to develop a vaccine against the novel Wuhan coronavirus. The Chief Executive Officer of CEPI, Richard Hatchett, said [72] :

"Given the rapid global spread of the nCoV-2019 virus the world needs to act quickly and in unity to tackle this disease. Our intention with this work is to leverage our work on the MERS coronavirus and rapid response platforms to speed up vaccine development."

The novel coronavirus is the first epidemic disease to emerge since the formation of CEPI in Davos in 2017. CEPI was created with the express intent to enable speedy research and development of vaccines against emerging pathogens. In May 2017, WHO released the Target Product Profile (TPP) for MERS-CoV vaccines, following the prioritization of MERS-CoV as one of eight priority pathogens for prevention of epidemics [73] . CEPI and partners aim to use existing platforms-that is, the existing "backbone" that can be adapted for use against new pathogens-that are currently in preclinical development for MERS-CoV vaccine candidates. Following the WHO declaration on 30 January that the current 2019-nCoV outbreak is a public health emergency of international concern (PHEIC), global health organizations and researchers will be further mobilized-bolstered by new mechanisms for action and greater resources-to stop the spread of disease.

A critical question that must be answered at this stage-with a clear view of the potential deleterious effects of a new coronavirus in pregnancy-is will maternal immunization be a priority in research and development? As of the PHEIC declaration, 12 groups have announced that they are developing new vaccines against 2019-nCoV and seven others

announced initiatives to develop new therapies [74] . Safe testing of experimental vaccines in a pregnant population is difficult and, as a result, vaccines are not typically developed with pregnant women in mind. To date, very few clinical trials for vaccines have proactively included pregnant women [75] , and the exclusion of pregnant and lactating women from receiving the rVSV-ZEBOV vaccine through 3 Ebola virus epidemics serves as a recent example [69] [70] [71] . Given the potential severity in pregnancy, as demonstrated by this review of maternal infections of SARS and MERS, women who are pregnant should be considered a priority population in all efforts to prepare for and prevent infection by novel coronaviruses.

On 5 February 2020 it was reported by multiple media outlets that a newborn infant delivered during the epidemic in Wuhan had tested positive for 2019-nCoV at the Wuhan Children's Hospital in Hubei Province 30 hours following its birth. According to the official Xinhua news agency, the infant was delivered on 2 February to a mother who had tested positive for the virus. Reports have stated that the infant had stable vital signs, no fever or cough, but had shortness of breath together with abnormal chest radiographs and abnormalities of liver function [76] [77] [78] . Dr. Zeng Lingkong, Chief Physician at the Neonatal Medicine Department of the hospital, said [78] , "This reminds us to pay attention to mother-to-child being a possible route of coronavirus transmission"

The hospital also provided information about a previous case of a baby that had been delivered on 13 January 2020. Following its birth, the infant's nanny was diagnosed with 2019-nCoV, and the mother was diagnosed days later [76] . On 29 January the baby began to develop symptoms. According to Dr. Zeng Lingkong [76] , "Whether it was the baby's nanny who passed the virus to the mother who passed it to the baby, we cannot be sure at the moment. But we can confirm that the baby was in close contact with patients infected with the new coronavirus, which says newborns can also be infected"

In considering whether these and future cases of neonatal infection are acquired prior to delivery, it is important to remember that newborn infants can acquire an infection in other ways beyond intrauterine maternal-fetal transmission. In some cases, viral infection can be acquired when the infant passes through the birth canal during a vaginal delivery or through post-partum breast feeding, although these mechanisms would be highly unusual for a respiratory virus. Neonatal infection from respiratory viruses can occur after delivery through such mechanisms as inhalation of the agent through aerosols produced by

coughing from the mother, relatives or healthcare workers or other sources in the hospital environment. Based upon past experience with pregnant women who developed MERS and SARS, and realizing that the numbers are limited, there has never been confirmed intrauterine coronavirus transmission from mother to fetus. Discussing the most recent baby to be diagnosed with the 2019-nCoV infection, Dr. Stephen Morse, an epidemiologist at the Mailman School of Public Health at Columbia University stated [77] , "It's more likely that the baby contracted the virus from the hospital environment, the same way healthcare workers get infected by the patients they treat," "It's quite possible that the baby picked it up very conventionally-by inhaling virus droplets that came from the mother coughing."

And according to Dr. Paul Hunter, Professor of Medicine at the University of East Anglia [79] , "As far as I am aware there is currently no evidence that the novel coronavirus can be transmitted in the womb. When a baby is born vaginally it is exposed to the mother's gut microbiome, therefore if a baby does get infected with coronavirus a few days after birth we currently cannot tell if the baby was infected in the womb or during birth."

There is limited knowledge regarding coronavirus infections that occur during pregnancy-what is known has, for the most part, been the result of epidemics resulting from two different diseases, SARS and MERS. These previous experiences with coronavirus infections in pregnancy indicates that these agents are capable of causing adverse clinical outcomes including life-threatening maternal disease that in some cases requires hospitalization, intensive care and ventilatory support. Both of these coronaviruses can result in maternal death in a small but significant number of cases, but the specific risk factors for a fatal outcome during pregnancy have not been clarified. Coronaviruses can also result in adverse outcomes for the fetus and infant including intrauterine growth restriction, preterm delivery, admission to the ICU, spontaneous abortion and perinatal death. Unlike some viral infections, notably Ebola virus [70] and Zika virus [80] , the likelihood of intrauterine maternal-fetal transmission of coronaviruses is low-there have been no documented cases of vertical transmission occurring with either SARS or MERS. It remains to be seen during the current Wuhan 2019-nCoV epidemic how this newly-emergent coronavirus affects pregnant women and their infants, as well as which factors may modulate obstetrical disease and outcomes including the timing of maternal coronavirus exposure by gestational age, the effects of medications or other treatment regimens, differences in host immune responses, occurrence of coexisting medical and obstetrical conditions, and other covariables. However, pregnant women should be considered to be at high risk for developing severe infection during this current outbreak of

2019-nCoV. Additional clinical research on the treatment of SARS, MERS, and the new coronavirus 2019-nCoV is necessary if we are to understand the potential risks and benefits of novel therapies and new vaccines in pregnancy. This research will be critical in improving the care, and even saving the lives, of pregnant women in the current as well as future outbreaks.

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Backcalculating the Incidence of Infection with COVID-19 on the Diamond Princess

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Abstract: To understand the time-dependent risk of infection on a cruise ship, the Diamond Princess, I estimated the incidence of infection with novel coronavirus (COVID-19). The epidemic curve of a total of 199 confirmed cases was drawn, classifying individuals into passengers with and without close contact and crew members. A backcalculation method was employed to estimate the incidence of infection. The peak time of infection was seen for the time period from 2 to 4 February 2020, and the incidence has abruptly declined afterwards. The estimated number of new infections among passengers without close contact was very small from 5 February on which a movement restriction policy was imposed. Without the intervention from 5 February, it was predicted that the cumulative incidence with and without close contact would have been as large as 1373 (95% CI: 570,

2176) and 766 (95% CI: 587, 946) cases, respectively, while these were kept to be 102 and 47 cases, respectively. Based on an analysis of illness onset data on board, the risk of infection among passengers without close contact was considered to be very limited. Movement restriction greatly reduced the number of infections from 5 February onwards.

Text: An outbreak of novel coronavirus disease (COVID-19) has occurred on a cruise ship, the Diamond Princess [1]. The primary case remains unknown, but the index case, defined as the first identified case, is a passenger who started coughing from 19 January 2020 on board, disembarking the ship in Hong Kong on 25 January. As the case was diagnosed on 1 February, the ship was requested to remain in the ocean near Yokohama from 3 February onwards. Subsequently, the movement of all passengers was restricted on board from 5 February, for a matter of 14 days of quarantine. Out of a total of 3711 persons (consisting of 2666 passengers and 1045 crew members), 199 symptomatic cases have been diagnosed on board as of 24 February, and additional asymptomatic infections and symptomatic cases after disembarkation have also been reported.

One of the critical issues in infectious disease epidemiology is that the time of infection event is seldom directly observable. For this reason, the time of infection needs to be statistically estimated, employing a backcalculation method [2]. Using a sophisticated statistical model with doubly intervalcensored likelihood and right truncation with an exponential growth of cases, the mean incubation period has been estimated to be about 5.0 days [3]. To understand the time-dependent risk of infection throughout the course of outbreak and estimate the effectiveness of the quarantine measure from 5 to 19 February 2020, I aimed to estimate the incidence of infection with COVID-19 and also predict the likely number of infections prevented by the quarantine measure.

I analyzed the epidemic curve, ct , on day t , illustrated by the number of confirmed cases by the date of illness onset. The confirmatory diagnosis was made, using the reverse transcriptase polymerase chain reaction (RT-PCR). The date of illness onset was defined as the first date of fever. In addition to the date of illness onset, cases were classified by contact history inside the cabin and also by the type of membership, i.e., crew or passenger. Close contact was defined as having at least one cabinmate who was confirmed by RT-PCR.

We estimate the number of cases by time of infection, it. Using the probability mass function of the incubation period of length s , f_s , the incidence of infection is known to satisfy

where $E(.)$ represents the expected value. As for f_s , it is known that the mean and standard deviation are 5.0 and 3.0 days, respectively, best fitted by lognormal distribution [3]. Employing a step function, the incidence of infection was statistically estimated via a maximum likelihood method. The estimation was implemented independently by the history of contact and type of membership.

Regarding the real-time forecasting, we employed the so-called Richards model, an analogue to the generalized logistic model [4, 5]:

where I_t is the cumulative incidence on day t , Z is the cumulative incidence at the end of the outbreak, s is the parameter that governs the flexibility of the logistic curve, a is the early growth rate of cases and t_i is the inflection point of the cumulative incidence curve. Assuming that the cumulative incidence is Gaussian distributed, four unknown parameters were estimated. The Richards model was fitted to two different datasets, i.e., (i) the dataset of the entire course of the epidemic and (ii) the dataset by 4 February 2020. The latter dataset corresponds to the time period without any impact of movement restriction that was in place from 5 February onwards. Figure 1 shows the epidemic curve by contact history and type of membership. The highest incidence of illness onset was observed on 7 February. The epidemic curve in a latter half period was dominated by crew members whose movement was not strictly controlled due to the need to continue service on the ship. The second dominating group was passengers with close contact history. The last illness onset date on board of a passenger without close contact was on 14 February. Estimating the incidence of infection, the peak incidence was identified for the period from 2 to 4 February among passengers both with and without close contact (Figure 2). The incidence of infection abruptly dropped after 5 February, the date of movement restriction. Among passengers without close contact, the incidence was estimated to be zero, except for 8-10 February 2020, during which 0.98 persons (95% confidence intervals (CI): 0, 7.74) per day were estimated to have been infected. The epidemic peak among crew members was seen for the period from 8 to 10 February 2020. Figure 3 compares the cumulative incidence with and without movement restriction policy from 5 February. In the presence of intervention, the cumulative incidence among passengers with and without close contact

and crew members were 102, 47 and 48 cases, respectively, as of 24 February 2020. These were well realized by the Richards model. Without intervention from 5 February onwards, it was predicted that the cumulative incidence with and without close contact would have been 1373 (95% CI: 570, 2176) and 766 (95% CI: 587, 946) cases, respectively.

A large outbreak of COVID-19 occurred on a cruise ship. Estimating the incidence, the peak time of infection was shown to have been from 2 to 4 February, and the incidence abruptly declined afterwards. The estimated number of new infections among passengers without close contact was very small from 5 February, on which the movement restriction policy was imposed, and at most there was, on average, one case of infection per day from 8 to 10 February. Other than continued exposure among crew members, the estimated incidence in this study indicates that the movement restriction policy from 5 February 2020 was highly successful in greatly reducing the number of secondary transmissions on board. Based on an analysis of illness onset data on board (and before the disembarkation of a large number of passengers), the risk of infection among passengers without close contact was considered to be very limited. Among disembarked passengers, symptomatic cases have started to be reported on the ground in and outside of Japan. In particular, cases arising from passengers without close contact indicate a possible pathway of infection via mechanisms that were not covered by the abovementioned analysis that relied on symptomatic cases. Although the transmission via direct human-to-human contact was prevented by movement restrictions, the role of other modes of transmission, e.g., environmental and asymptomatic transmissions, should be further explored.

The author declares no conflict of interest.

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--- START OF ARTICLE 125 ---

The Extent of Transmission of Novel Coronavirus in Wuhan, China, 2020

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Abstract: A cluster of pneumonia cases linked to a novel coronavirus (2019-nCoV) was reported by China in late December 2019. Reported case incidence has now reached the hundreds, but this is likely an underestimate. As of 24 January 2020, with reports of thirteen exportation events, we estimate the cumulative incidence in China at 5502 cases (95% confidence interval: 3027, 9057). The most plausible number of infections is in the order of thousands, rather than hundreds, and there is a strong indication that untraced exposures other than the one in the epidemiologically linked seafood market in Wuhan have occurred.

Text: Since the announcement of a cluster of pneumonia cases of unknown etiology in Wuhan, Hubei Province, China, was made on 31 December 2019, many rapid virological, clinical, and epidemiological research responses have taken place [1, 2]. The causative agent of the pneumonia is suggested to be a novel coronavirus (2019-nCoV) of the same lineage (but genetically distinct) from the coronavirus causing severe acute respiratory syndrome (SARS) [1]. Cases in the initial cluster reported a common exposure—a seafood market in Wuhan where wild animals were served at a restaurant—indicating that a point-source zoonotic (animal-to-human) route was likely the main mode of transmission for those cases [2].

Although early reports from Wuhan [3] stated that (i) there were only tens of cases in the cluster and (ii) no human-to-human transmission was directly observed, the scientific community was alert to the possibility that the novel coronavirus would spread to other geographic locations—including other countries—via direct human-to-human transmission. In early January, the outbreak began to escalate rapidly with hundreds of cases now confirmed along with the presence of a few household clusters [4] [5] [6] [7].

As of 24 January 2020, the cumulative incidence in China is 830 cases, of which 549 cases were diagnosed in Hubei, 26 in Beijing, 20 in Shanghai, and 53 in Guangdong. Additionally, twenty-six deaths have been linked to the outbreak [6, 8] , and thirteen cases were exported to Japan, Singapore, South Korea, Taiwan, Thailand, Vietnam and the United States as of 22 January 2020. Considering that enhanced surveillance has been underway in these importing countries, case ascertainment has been perhaps better in exported case data.

Using a spatial back-calculation method and analyzing exported cases, we estimate the cumulative incidence of 2019-nCoV cases in China in real time, allowing us to update and discuss the extent of transmission at the source. Table 1 shows the incidence of exported cases by date of hospitalization and report. Due to the initial difficulty of diagnosis in the absence of established primer for polymerase chain reaction testing, the time lag between hospitalization and reporting was longer for early cases compared with that of more recent cases. Among the seven locations reporting importation, the total volume of inbound passengers from China was $m = 63.1$ million per year in 2017 [9] , of which $100q = 2.1\%$ were from Wuhan [10] , a home of $n = 19.0$ million people as the catchment population of Wuhan airport. Two other locations with confirmed cases, i.e., Macau and Hong Kong, were excluded from the analysis, because it is commutable by land transportation and the first case in Hong Kong was indeed not via airtravel. As we already know from elsewhere [11] [12] [13] , given the observed cumulative count of c exported cases, we have a balance equation of the cumulative risk of infection:

where T is the sum of incubation and infectious periods, and here is assumed to be 3.2 and 9.3 days [14] , respectively, assuming that these periods are similar to those of other coronaviruses, and thus, $T = 12.5$ days. The estimated incidence in China is then given by p_n . With an ad-hoc assumption that the data are generated following the binomial sampling process among travelers from Wuhan, the cumulative incidence is then estimated using a maximum likelihood method. Table 1 also shows the estimated incidence in China. The first exportation event in Thailand suggests 423 cases with the upper confidence limit of 1863 cases. The estimated cumulative incidence has grown as additional cases have been reported. As of 24 January 2020, with reports of thirteen exportation events, the cumulative incidence in China is estimated at 5502 cases (95% confidence interval: 3027, 9057).

Our latest estimate is comparable to a preliminary report posted by a research group at Imperial College London (ICL) on their own homepage on 22 January 2020 [26] that estimated the incidence based on three importation events at 4000 cases (95% CI: 1000, 9700). Possible reasons for the slight difference include (i) the number of travelers in the previous study was derived from airline passenger data [27] and (ii) the assumed length of T was different. Two other estimates have also been published: a preliminary study by a Northeastern University group estimated 1250 cases (95% CI: 350, 3000) as of 17 January 2020 [28] and a University of Hong Kong group estimated 1343 cases (95% CI: 547, 3446) as of 17 January 2020 [29]. The former study from the United States assumes that the catchment area population is 10 million (we use 11.1 million).

The number of reported 2019-nCoV infections continues to grow as surveillance and detection methods improve. Our estimate and others [26, 28, 29] agree that the actual number of cases is likely in the order of thousands, rather than hundreds, and there is a strong indication that untraced exposures other than that of the originally linked seafood market in Wuhan have occurred. Such exposures are expected to include human-to-human transmission, but the levels of transmissibility have yet to be quantified. It is still plausible that a substantial number of human infections arose from animal-to-human exposures, such as was the case during the first outbreak of highly pathogenic influenza (H7N9) in China, 2013, and the human-to-human transmissibility has yet to be quantified in an explicit manner.

Despite initially restricting what information on the outbreak was shared publicly, the Chinese government has begun to respectfully provide updates on the situation on a daily basis. This encourages the real-time release of information by means of regularly updated situation reports, including epidemiological information with dates of exposure, illness onset, and hospitalization among cases.

For researchers to be able to contribute to control efforts by improving situation awareness via an explicit risk assessment, it is crucial that detailed epidemiological data are posted to a public domain in real-time. Such datasets should include not only a deidentified line list of cases but also updates on the infection status of traced contacts. Information on exposure period and illness onset can assist with the estimation of important natural history parameters such as the incubation period. It is critical for the public health

community and the public at large to understand more about the process of case ascertainment, including the current case definition and reporting system mechanisms.

The authors declare no conflicts of interest.

--- END OF ARTICLE ---

--- START OF ARTICLE 126 ---

The human viral challenge model: accelerating the evaluation of respiratory antivirals, vaccines and novel diagnostics

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Abstract: The Human Viral Challenge (HVC) model has, for many decades, helped in the understanding of respiratory viruses and their role in disease pathogenesis. In a controlled setting using small numbers of volunteers removed from community exposure to other infections, this experimental model enables proof of concept work to be undertaken on novel therapeutics, including vaccines, immunomodulators and antivirals, as well as new diagnostics. Crucially, unlike conventional phase 1 studies, challenge studies include evaluable efficacy endpoints that then guide decisions on how to optimise subsequent field studies, as recommended by the FDA and thus licensing studies that follow. Such a

strategy optimises the benefit of the studies and identifies possible threats early on, minimising the risk to subsequent volunteers but also maximising the benefit of scarce resources available to the research group investing in the research. Inspired by the principles of the 3Rs (Replacement, Reduction and Refinement) now commonly applied in the preclinical phase, HVC studies allow refinement and reduction of the subsequent development phase, accelerating progress towards further statistically powered phase 2b studies. The breadth of data generated from challenge studies allows for exploration of a wide range of variables and endpoints that can then be taken through to pivotal phase 3 studies. We describe the disease burden for acute respiratory viral infections for which current conventional development strategies have failed to produce therapeutics that meet clinical need. The Authors describe the HVC model's utility in increasing scientific understanding and in progressing promising therapeutics through development. The contribution of the model to the elucidation of the virus-host interaction, both regarding viral pathogenicity and the body's immunological response is discussed, along with its utility to assist in the development of novel diagnostics. Future applications of the model are also explored. ELECTRONIC SUPPLEMENTARY MATERIAL: The online version of this article (10.1186/s12931-018-0784-1) contains supplementary material, which is available to authorized users.

Text: Acute respiratory infections (ARIs) manifest as Upper (URI) or Lower (LRI) respiratory tract infections and may move between the two compartments; ARIs represent the most common infectious diseases and are predominantly of viral aetiology. The global burden of ARI is substantial with significant morbidity and mortality occurring in children, the elderly and immunocompromised [1] .

In the UK alone during the period 2014-2015, respiratory disease caused an estimated 15,800 excess winter deaths [2] . In the USA, influenza and respiratory syncytial virus (RSV) cause substantial mortality especially among people aged 65 and older [3] .

However, although deaths in the industrialised world are widely reported, developing countries feel the burden particularly; out of an estimated 1.9 million child deaths from ARIs in 2000, 70% of those deaths occurred in Africa and south-east Asia [4] . The Millennium Summit at the United Nations in 2000 led to the setting up of the Millennium Development Goals.

A study reported the progress made in meeting those goals in 40 developing countries; it concluded that the prevalence of ARI was 13%, health expenditure and per capita gross domestic product is directly associated with the prevalence of the disease [5] .

Viral heterogeneity associated with ARIs is well established [6] . In the past, human rhinovirus (HRV) has been identified as the virus most frequently associated with respiratory illness with 30-50% of infections annually on average, and up to 80% of upper respiratory infections during the autumn outbreaks [7] . After HRVs, coronaviruses (CoV), influenza, respiratory syncytial virus (RSV) and parainfluenza viruses (PIV) are the next most frequent.

More recently an evaluation of illness in 6,266 children under ten years of age in Australia, South East Asia and Latin America emphasised both the viral heterogeneity and the impact of ARI. Of the 2,421 children who experienced 3,717 individual influenza-like illness (ILI) episodes, rhinovirus/enterovirus was most prevalent (41. 5%). Influenza followed this (15.8%), adenovirus (ADV) (9.8%), PIV and RSV (both 9.7%), CoV (5.6%), human metapneumovirus (HMPV) (5.5%) and human bocavirus (HBoV) (2.0%). The percentage of children missing school or childcare was between 21.4% for HBoV and 52.1% for influenza [8] .

We have compared the data from the two reports one from 2003 [7] and the other in 2017 [8] and found that the reports, despite being separated by 14 years, were similar, with the single exception of HBoV, discovered in 2005 (Table 1) , which we discuss later.

Feng et al. [9] described in detail the distribution of ARIs causing hospitalisation by age group: they observed that RSV was predominantly observed in the young and elderly, and influenza although significant in the young was noticeably more predominant in the elderly. Interestingly they observed that co-detection of viruses tended to occur more commonly in the younger age groups, particularly those under the age of five.

Rhinovirus (the "common" cold) HRV infections, often considered trivial can significantly contribute to missed days from work and school, though infections are typically self-limiting [7] . HRV infections throughout the year and in many cases, manifest with

symptoms such as nasal congestion, rhinorrhoea, sneezing, sore throat, and cough. HRV is known to be the primary cause of ARI and a severe contributing factor in exacerbations of atopic disease, e.g., asthma as well other conditions such as chronic obstructive pulmonary disease (COPD) [10] [11] [12] [13] .

HRV infections are associated with significant economic implications as well as being an important contributor to sinusitis, otitis media, bronchitis and primary pneumonia [14] [15] [16] . HRV is a considerable cause of morbidity in specific at-risk groups such as infants, the elderly, immunocompromised, and, as already mentioned, chronic respiratory diseases such as asthma, COPD and cystic fibrosis. At present, HRV is considered the number one cause of asthma exacerbations [15] [16] [17] [18] [19] .

Asthma is a complex disease, characterised by chronic airway inflammation, and a history of respiratory symptoms such as wheeze, shortness of breath, chest tightness and cough. Over time these symptoms can vary in their intensity [20] . Each year over 300 million people worldwide are affected by asthma: approximately 250,000 people die as a result. Many deaths are due to suboptimal long-term medical care and delay in obtaining help during severe exacerbations of the disease [21] . Treatments to prevent worsening of symptoms and other therapies for mild to moderate asthma that avert relapse, i.e., the symptoms worsen again when the treatment stops, are significant unmet medical needs.

The human challenge model has been used to investigate the viral pathogenicity [22] [23] [24] [25] [26] and recent publications on the asthma challenge model have focused on how the asthmatic host responds to HRV infection. Work is ongoing as to susceptibility to viral induced asthma worsening [27, 28] innate immune dysregulation [29] and induction of innate, and type 2 responses in nasal and bronchial epithelial secretions [30] . The pathogenesis of rhinoviral infection, along with other ARIs, in exacerbations of airway disease, has been investigated extensively. Impaired host responses to virus infection, a better understanding of the mechanisms of abnormal immune responses and the potential to develop novel therapeutic targets for virus-induced exacerbations have all used the HVC model [12, [31] [32] [33] [34] .

Despite previous research work on multiple small molecule antivirals, such as pleconaril which have been tested using both the experimental challenge model and field studies [35]

[36] [37] , there is currently no licensed treatment for HRV infections Other compounds have been tested against HRV, such as Vapendavir (BTA798) which prevented the release of viral RNA into the target cell and demonstrated a reduction in peak viral load in the HVC model [38] . A subsequent study in asthmatics was completed and although not published the compound did have a limited effect [39] .

Pirodavir an intranasal capsid-binding molecule reached phase 3 clinical trials for HRV prevention and treatment in the 1990s. Although the compound decreased viral replication and shedding, it failed to show a significant reduction in the duration or severity of symptoms [40, 41] .

A Protease inhibitor, rupintrivir that prevents cleavage of viral proteins required for replication was tested in an HRV challenge trial. Rupintrivir was well tolerated and reduced viral loads and respiratory symptoms [36] . However, in studies of natural infection, it did not significantly affect viral loads or symptom severity [42] .

Treatments such as zinc-containing products are now widely discredited as demonstrated by the withdrawal of a Cochrane report and JAMA editorial [43] [44] [45] .

Current treatment of HRV infections primarily consists of over-the-counter (OTC) medicines to manage symptoms. There is also no licensed vaccine, and while there has been some progress on developing multivalent vaccines [46] , development in this area is hampered by the sheer number of serotypes that need to be covered (at present over 160). Despite HRV being associated with up to 50% of adult asthma exacerbations and up to 80% of childhood exacerbations, there are no HRV-specific asthma therapies [34] .

As we better understand the interaction between the virus and the host, new therapies such as the monoclonal antibodies (anti-IgE [omalizumab] and anti-IL-5 [mepolizumab]) along with small molecules carefully targeting specific immune signalling pathways, HRV-specific prophylactic treatment may become practical [47] [48] [49] [50] .

In order to prevent exacerbations, the design of new therapeutics could potentially improve efficacy by both directly acting to inhibit viral replication and alleviate the symptoms of asthma and COPD [51] .

Influenza virus is a well-known human pathogen and can cause severe morbidity and mortality, particularly in older patients, those with co-morbidities and in the immunocompromised. In 2009, the first pandemic virus of the 21 st century hospitalised 195,000 to 403,000 in the US alone resulting in 8,870 to 18,300 deaths by mid-2010 [52] . A World Health Organization (WHO) global pooled analysis of 70,000 laboratory-confirmed hospitalised H1N1 pandemic patients from 19 countries revealed that of the 9,700 patients admitted to intensive care units, 2,500 died, and that morbid obesity might be a risk factor for hospitalisation and/or death [52] . Obesity was confirmed as a factor associated with a higher likelihood of admission to hospital in influenzainfected patients [53] .

The 2009 pandemic was considered mild. However, the classic W shaped age distribution curve of infection for a pandemic virus was observed. That is high mortality in the very young and the old, but an additional spike in death amongst the "young and healthy". The pandemic, as did previous outbreaks, occurred in successive waves, but despite national policies favouring the use of antiviral drugs, few patients received these before admission to hospital, and many were given antibiotics [54] . The lack of real, or perceived, "real world" efficacy of currently available antivirals leads to the overuse of antibiotics and the subsequent problems that may arise [55] [56] [57] .

The yearly seasonal morbidity and mortality of influenza results in hospitalisation and death mainly among the high-risk groups. Each year epidemics of seasonal influenza are estimated to result in about 3 to 5 million cases of severe illness, and about 290,000 to 650,000 deaths worldwide [58] .

In first world / industrialised countries, most deaths associated with influenza occur among people age 65 or older [59] . Clinics and hospitals, in many countries, can be overwhelmed during peak illness periods, and there can be substantial economic cost [60]

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The virus itself has been well characterised, and the two surface proteins, the haemagglutinin (HA) and the neuraminidase (NA) are important in both vaccine and antiviral development [61] .

The effects of seasonal influenza epidemics in developing countries are not fully known, but research estimates that 99% of deaths in children under five years of age with influenza-related lower respiratory tract infections are found in developing countries [59, 62] .

Currently, vaccines and antivirals exist for the prevention and treatment of influenza, but both have limitations in efficacy due to the rapid evolution of the virus as it mutates on a yearly basis and the sudden unexpected emergence of pandemic influenza strains.

The effectiveness of recent annual influenza vaccines (to date mostly based on the HA, and rarely the NA surface glycoproteins) has languished between 37% and 70% over successive influenza seasons. In particular, the failure of the vaccine across the winter season of 2014-2015, where the overall adjusted effectiveness was 23% [95% confidence interval 14, 31] [63] is memorable. In a mismatched year, the mortality rate is increased in the most at-risk populations [64, 65] . The problem of ensuring that the seasonal vaccine is correctly matched to the upcoming circulating strain highlights the need for rapid development of inter-seasonal/universal vaccines and also the need for a way of testing their efficiency rapidly and accurately before the lengthy and expensive mass production is engaged which takes many months [66, 67] .

Antiviral drugs exist of which currently the NA inhibitor oseltamivir is most commonly used. This is active against all known NA subtypes of influenza, and one would, therefore, assume against all influenza strains. They may have decreasing effect with the emergence of resistant influenza strains in which NA protein changes preventing efficient oseltamivir binding and thus its ability to inhibit the essential activity of the viral NA. For example, one genetic mutation known as 'H275Y' a substitution of histidine for tyrosine at NA position 275 - confers an evolutionary advantage to the virus including the 2009 H1N1 influenza [68] . During the 2013-2014 influenza season, 59 (1.2%) of 1,811 influenza A(H1N1) pdm09 virus isolates in 20 of 50 US states had the H275Y oseltamivir resistance substitution. No isolates were resistant to zanamivir [69] . Although animal studies have demonstrated

limited transmission of mutant viruses [70, 71] , it is thought that the rise of oseltamivir resistance may be due to community transmission [72, 73] rather than the H275Y mutation becoming fixed in the viral genome.

A systematic review and meta-analysis of published data from 2000 onwards concluded that most RSV-associated child deaths occur particularly in preterm infants and in infants up to 1-year of age [62, 74] . An effective maternal RSV vaccine or monoclonal antibody could have a substantial effect on disease burden in this age group [75] .

The RSV-specific monoclonal antibody palivizumab is approved for prevention of serious LRI caused by RSV in susceptible infants. Economic benefit in a UK health setting has not been shown due to the high cost and lack of benefit on serious outcomes [76] . A single-centre cohort study of 22 infants showed no difference in treatment outcomes for patients receiving palivizumab when compared to patients only receiving "standard of care" treatment [77] . Despite the lack of evidence for clinical benefit, post-licensure data supports the use of palivizumab for reducing RSV-associated hospitalisations in premature infants under 33 weeks and in children with chronic lung and heart diseases [78] . Importantly, palivizumab resistant mutant virus has rarely been isolated in clinical specimens [79] .

The RSV treatment ribavirin is limited due to difficulty with aerosol delivery, cost and potential harm to healthcare workers, despite off-label treatment of immunocompromised patients being reasonably successful. In the immunocompromised, therapy with a concomitant immunoglobulin or palivizumab has had mixed results, probably due to the difficulty of knowing when to initiate treatment [80] .

Despite the call for the accelerated development of prevention and treatment strategies for an effective RSV vaccine for children [81] , research has stalled for decades since the death in the 1960s of two subjects in a clinical study. These subjects were infected with a community-acquired RSV infection after receiving the US National Institutes for Health (NIH's) formalin-inactivated, alum-precipitated RSV candidate vaccine.

In contrast to influenza for which vaccines to date have shown themselves to be moderately effective but in need of improvement, RSV vaccines require substantially more research. There is currently no licensed vaccine for RSV; the most advanced candidate vaccine recently failed to show efficacy in a field study [82] . Effective treatments are urgently required.

RSV is, even amongst healthcare professionals, considered a childhood disease and other confounders have obscured the understanding of the consequences of RSV in adults.

RSV is poorly understood as a disease in the elderly [83] , and while the morbidity and mortality in children are of importance, it has been clearly shown that RSV has a comparable health burden to influenza in the elderly [84] .

As an example, a recent study was conducted on adult (≥ 18 years) patients admitted to an emergency department with suspected ARI during 2013-2015 (N = 3743). Multiplex PCR was used to diagnose the cause of the respiratory infection. Eighty-seven patients were identified with RSV. A comparator group with influenza (n=312) was utilised. Based on a 20-day all-cause mortality endpoint, adult patients were less likely to be diagnosed with RSV than with flu (2.3 vs 8.3%, respectively), also they were older, often diagnosed with pneumonia, COPD, hypoxemia, and bacterial co-infection. RSV infection in the elderly was significantly associated with a greater risk of death than seasonal influenza, adjusted for potential confounders and comorbidities. [85]

The clinical significance of viral/bacterial co-infections has long been a controversial topic. While severe bacterial pneumonia following influenza infection has been well described, associations are less clear among infections caused by viruses common in young children; secondary infections due to other viruses are less well understood and has been reviewed by others [86] . Although assessing the overall contribution of bacteria to disease severity is complicated by the presence of many confounding factors in clinical studies, understanding the role of viral/bacterial co-infections in defining the outcome of paediatric ARI may potentially reveal novel treatment and prevention strategies, improving patient outcomes [33, [86] [87] [88] [89] [90] [91] [92] [93] [94] [95] .

A recent (2017) publication considered the role of bacterial colonisation with *Streptococcus pneumoniae*, *Haemophilus influenzae* and *Moraxella catarrhalis* during symptomatic and asymptomatic viral upper respiratory infection in the nasopharynx of 4 to 7-year-old children during URI and when well. Using a multiplex PCR, virus was detected in about 80% of upper respiratory tract infections (URIs) in children and is also detectable in the nasopharynx of 30% of asymptomatic children. All three bacteria "levels" were higher during acute URI visits compared to asymptomatic surveillance visits by the children. Of note, however, is that even during asymptomatic follow-up visits, if the virus was present, all bacteria were detected at higher levels [96] .

It is worth noting that the presence of confounding infections, can mask the importance of the primary aetiology. Taylor et al. [8] report the incidence of HBoV following its identification in 2005 from the respiratory tract samples of children, as an important respiratory pathogen in children. However, the role of this virus on its own as a pathogen of significance was initially unclear, co-infection with other viruses or bacteria was common and confounding.

Moesker et al. [97] studied whether HBoV alone could cause acute respiratory infections in children. Using Next Generation Sequencing (NGS), they were able to exclude co-infections amongst those admitted to intensive care unit and studied HBoV viral loads. Of the 990 children who tested positive for a respiratory virus by RT-PCR, HBoV and RSV were detected in 178 and 366 of the children respectively. Forty-nine HBoV-positive patients and 72 RSV-positive patients were admitted to the intensive care. Seven HBoV-infected cases with severe ARI had no other co-infection (7/49, 14%). Importantly, these children did not have another detectable virus as determined by highly sensitive NGS. Also, they had much higher HBoV loads than other patients positive for HBoV, i.e., those with a co-infection. Although small, this study provides strong support that HBoV can cause serious ARI in children with no viral and bacterial co-infections.

The history of the human viral challenge model Since Sir Edward Jenner performed the first documented HVC study with smallpox on the 14 th of May 1796 the usefulness of such studies has been apparent [98] . More than a century later, Sir Christopher Andrews returned from the US in 1931 he had observed the use of chimpanzees in the study of influenza. The funding for similar work in the UK was insufficient, and therefore Sir Christopher enrolled students from St Bartholomew's Hospital in London. He explained the

next best thing would be a "Bart's" student as "they were cheaper than chimpanzees". Over 100 students immediately enrolled, but continued their studies and were not isolated in the same way the chimpanzees had been in the USA [99] . Unfortunately the investigators believed that the symptoms observed may not have been due to the challenge virus, but other respiratory infections acquired in the community, thus confounding the studies. A year later the UK's Medical Research Council (MRC) terminated the work.

After the conclusion of World War II, the withdrawal of the US troops from the UK left the American Red Cross 'Harvard Hospital' Field Unit on Salisbury plain. The hospital became the Common Cold Unit (CCU) led by Dr David Tyrell, from 1946, volunteers were inoculated by instilling small quantities of the virus into their noses [100] . The CCU housed healthy volunteers in relative isolation from other people, thereby reducing the risk of contact with community-acquired sources of infection or from them passing on the virus to members of the public. The unit was eventually closed in 1989; during four decades of research, it attracted 20,000 volunteers. Its research contributed to a better understanding of respiratory viruses, viral lifecycle, possible vaccines [101] as well as the first licensed antiinfluenza compound amantadine [102] .

The use of healthy volunteers in the HVC model provided, and still offers, a unique opportunity to describe the viral lifecycle. Investigators know with certainty the time of infection, nasal virus shedding can be measured, symptoms recorded prospectively, and participants are selected with low pre-existing immunity to the challenge virus to ensure a statistically significant infection rate with a small number of volunteers. Thus, such studies can maximise the safety and efficacy data obtained while minimising the risk to study volunteers and limited research funding.

Although serum IgG, for influenza virus, was traditionally measured via the HAI assay, as the entry criteria for volunteers into studies, micro neutralisation assays are used for RSV and HRV. Other work does suggest screening for antibodies to the NA influenza surface protein should be considered [103] or T-cell responses to internal proteins [104] should be considered.

After the closure of the CCU experimental infection studies continued in the USA using small motels and hotels replacing the huts on Salisbury Plain. These studies contributed to

the significant development of the new NA inhibitors during the 1990s, including the inhaled drug zanamivir and the orally available drug oseltamivir [105] [106] [107] [108] [109] [110] [111] [112] [113] [114] .

Studies however also continued in the UK, specifically the University of Southampton who performed important work in atopic volunteers, demonstrating they had more severe colds when experimentally challenged with rhinovirus, than non-atopic controls [115] .

The experimental A/Texas H1N1 influenza virus that was used successfully during the 1990s was implicated in the development of myocarditis in an experimentally infected subject, although a causal link was never demonstrated [116] . However, this incident halted work in the USA for a substantial period.

Most, if not all, challenge viruses are manufactured according to Good Manufacturing Practice (GMP) standard. Although controlled nasal inoculation differs from naturally occurring infection in which exposure to variable quantities of the virus may occur at various mucosal sites -the developed HVC model used in challenge studies mimics natural disease as far as possible [25, 117, 118] .

We have described the production of a new GMP stock of virus using an HRV-16 isolate from an 18-year-old experimentally infected healthy female volunteer, provided by colleagues from University of Virginia Children's Hospital, USA. Importantly, the clinical sample was provided with the appropriate medical history and consent of the donor. We manufactured this new HRV-16 stock by minimal passage in a WI-38 cell line, to reduce the risk of mutations during the Good Manufacturing Practice process. Having first subjected the stock to rigorous adventitious agent testing and determining the virus suitability for human use, we conducted an initial "safety and pathogenicity" clinical study in adult volunteers in a dedicated clinical quarantine facility in London [118] .

Our group started HVC studies in the UK in 2001, and since then we have conducted multiple studies with over 2,500 volunteers inoculated with influenza, respiratory syncytial virus (RSV) or human rhinovirus (HRV), and provided numerous proofs of concept [119] [120] [121] .

The human viral challenge model: shortening the drug development pathway for ARIs
Influenza, RSV and HRV infection have similar symptomatology, but this differs in severity and predominance of upper, lower or systemic symptoms as has been described by the Center for Disease Control [122] . However, it is not easy to diagnose between the different aetiologies of ARIs, and better diagnostics are needed [123] .

Symptoms are common to each infection and manifest on a gradient. Generally, but far from always, influenza infection is more likely to result in a patient feeling so unwell as to take to their bed and have a fever, than RSV, an HRV, CoV or other common cold virus infection, during which daily life is usually less impacted.

A variety of animal models exist to research respiratory viruses such as influenza [124] [125] [126] , RSV [127] [128] [129] [130] [131] [132] [133] [134] [135] [136] [137] , HRV [22, [138] [139] [140] . No single animal offers a platform for all respiratory viruses that infect humans, and different animal models exist for the same virus, which can give different, often conflicting results.

In addition, the principles of the 3Rs (Replacement, Reduction and Refinement) were developed over 50 years ago to provide guidance and ensure humane animal research. Over time they have become national and international legislation/regulations. The policies of organisations that fund or conduct animal research include these principles as part of the condition of funding [141] .

The shared symptomatology of respiratory viruses requires a single standard research platform that can be used to evaluate respiratory disease pathogenesis and the efficacy of candidate therapeutics. The use of a dedicated, purpose-built 24 en-suite bedroom isolation facility in which carefully screened volunteers can be safely inoculated with challenge viruses and intensively monitored may help reduce the use of animals while providing a single consistent research platform with standardised evaluable endpoints for respiratory virus research. Also, we have used a standardised diary card across our studies, which allows for comparison of the symptoms that each virus causes and the efficacy of the therapeutic being tested. We have included a copy of the diary card in the Additional file 1.

It is difficult to evaluate the efficacy of a specific antiviral therapeutic "in the field" due to the presence of circulating community co-infections of differing microbial aetiology. The HVC model allows the opportunity to study a virus in isolation. HVC studies and field studies are complementary research stratagems necessary for the development of effective ARI therapeutics.

In contemporary HVC trials, (Fig. 1) healthy volunteers are administered an investigational therapeutic either before (prophylaxis trials) or after (treatment trials) inoculation with the specific challenge strain of the virus. The viruses used in the HVC model are not attenuated and produce symptoms consistent with clinically observed ARI [25, 117, 118] . Each virus is propagated under GMP conditions, with a minimal number of passages from the isolates to the challenge stocks [118, 142] . The few mutations that occur within the virus are rapidly selected out due to a genetic bottleneck, with the consequence that the virus in the human host is considered wild-type [143] . The similarity between virus recovered from the inoculated host and the originator reference virus strain provides assurance that the model disease process is closely aligned with the reference virus strain and is not altered nor attenuated.

There are limited licensed therapeutic options against respiratory viruses, highlighting a significant unmet medical need. A model such as the HVC allows the rapid evaluation of novel therapeutics. The model shortens both preclinical and early clinical development phases by providing a better understanding of the host and pathogen's initial interaction and has the potential to make the necessary vaccines and medicines more rapidly available than traditional development approaches otherwise might.

Shortening the traditional development pathway through the early use of a Proof of Concept (PoC) study that incorporates the HVC model (Fig. 2) provides essential evaluable endpoints. Unlike conventional phase 1 studies which rarely include any assessment of efficacy, almost all HVC studies include evaluable efficacy endpoints such as reduction in AUC viral load (mainly recovered from upper respiratory tract samples such as nasal wash or nasopharyngeal swab), volunteer self-reported symptoms, peak symptom score, total symptom score amongst others. Small numbers of subjects often in the order of 30-45 per treatment group-are typically included in these rapid to execute short duration studies. The resulting safety and pharmacokinetic (PK) and pharmacodynamic (PD) data in controlled

conditions, guide decisions on whether or not to progress to field studies, providing a most valuable set of data immediately after, or even as part of, the conventional phase 1 safety study.

The HVC model also opens a different development route alongside traditional phase 1 allowing rapid progress to statistically powered phase 2b studies that will generate the efficacy data needed to support licensing, while still providing suitable safety data. The FDA guidance on developing influenza therapeutics [144] states that challenge trials cannot take the place of efficacy (phase 2) trials. The guidance states; "...Challenge trials can provide useful exposure-response and safety information, as well as an opportunity to demonstrate pharmacological antiviral activity in humans under controlled conditions outside the influenza season. Specifically, data from challenge trials can contribute to dose selection for phase 2b and phase 3 trials, and provide the opportunity to explore the effects of different times of drug initiation relative to virus exposure...".

Challenge trial refinements are closing the gap between the experimental infection model and the natural infection setting. The HVC study duration of several weeks is shorter than a field-based phase 2 study that waits for a natural outbreak of the virus and the duration of which can be several months/years. These studies save Fig. 1 The Human Viral Challenge Model. The study typically consists of inputs, such as the volunteers, their selection criteria, isolation in quarantine and exposure to a GMP virus. There are two treatment options; a vaccination/prophylaxis with an antiviral or b treatment with an antiviral. Outputs from the study, summarised on the right, such as virus symptoms, virus shedding etc. X is the number of days before virus exposure vaccination may occur. Y is the number of days post virus exposure that a volunteer may be followed for development time when the transition between phases is fully optimised.

Importantly, unlike traditional phase 1b/phase 2 studies, HVC studies are not dependent on a natural outbreak of infection, which can occur at random, and for which the exact time of infection may not be apparent. They provide evaluable endpoints, comparative PD and PK data, along with additional biomarker data on product performance in humans. It must, however, be stated that most often such studies enrol otherwise healthy young adults which imply that the outcome of the infection in the placebo group may be seen as mild to moderate, to some extent. The safety of volunteers has to remain the priority of investigators.

The HRV/HVC model can be a potent tool, not just to study HRV infection and disease, but also to investigate the mechanisms of exacerbation in patients with chronic respiratory disease and to conduct efficacy studies for new therapies.

Human challenge studies with HRV have been shown to produce infection in over 90% of serologically susceptible subjects and result in a clinical syndrome that is comparable to that reported with natural colds [145, 146] . Symptoms usually appear within 24 hours and peak at 48-72 hours after inoculation. Virus shedding follows a pattern similar to that of their symptoms. In recent times, several hundred inoculations of adult subjects have been reported and have established this as a safe and effective method in which to study HRV-related disease in both healthy and asthmatic subjects [145] .

These studies have provided a knowledge base to further develop the HRV experimental model and provide a controlled and useful tool to develop new therapies for the disease areas associated with HRV infection. New treatments for asthma and COPD are urgently needed, and small animal models of asthma are poorly predictive of efficacy. Most drugs that are effective in these animal models are not found to be effective in later stages of development in humans. Models that more closely follow clinical features of human asthma and COPD are needed [32, [147] [148] [149] [150] [151]].

We have already described current influenza antiviral drugs that can shorten disease and reduce the severity of symptoms if taken early enough after infection, and their prophylactic use can decrease the risk of infection; their utility has been debated however [152] .

The two main classes of currently effective antiinfluenza drugs are the NA inhibitors, such as zanamivir (Relenza™), oseltamivir (Tamiflu™), peramivir (Rapivab™) [153] and M2 inhibitors, although drug resistance makes this class unusable [154] .

The HVC model has recently been used extensively to evaluate new classes of antiviral compounds against influenza, including those such as experimental monoclonal antibodies targeting epitopes within the highly conserved and exposed part of the M2 viral

surface Fig. 2 The role of the HVC model in the clinical development pathway. Short duration proof of concept studies, which incorporate the HVC model, typically include small numbers of subjects. The resulting safety and, particularly, efficacy data can more accurately guide decisions on whether to expose a larger number of subjects to promising candidate therapeutics in field studies than conventional phase 1 safety data alone otherwise might protein [155, 156] the conserved stalk of the HA [157] and small molecule antiviral drugs that target the viral polymerase, e.g. favipiravir [158] .

The HVC model allows for the rapid evaluation of novel therapeutic compounds which may be difficult to evaluate in the field, due to the nature of "at risk" groups, e.g. paediatrics. Specifically, and given the described historical experience with RSV vaccines, it is important that benefit can first be demonstrated in a healthy population.

In the past, unlike influenza and HRV, the HVC model has not been routinely used with RSV. Recently, however, there are several antiviral therapeutics that have reached an advanced stage of development using the model.

We had for some time wished to restart the HVC/RSV studies at the University of London, the two significant challenges that had stalled antiviral development for RSV presented a considerable research need. In association with the DeVincenzo lab at the University of Tennessee and the biotech company Alnylam, we set about designing possibly the first HVC/RSV study.

Alnylam pioneered the use of RNA interference (RNAi) which is a natural mechanism that regulates protein expression and is mediated by small interfering RNAs (siRNA). Working with both groups, we manufactured an RSV Type A virus to GMP standard and titrated it in 35 human volunteers who we divided into five groups, each which was intranasally inoculated with increasing titre (3.0-5.4 log plaque-forming units/person) of the challenge virus. Intranasally. Overall, in this new model, 77% of volunteers consistently shed virus. Infection rate, viral loads, disease severity, and safety were similar between cohorts and were unrelated to the quantity of RSV received.

Symptoms began soon after initial viral detection, peaked in severity near when viral load peaked and subsided as viral loads slowly declined. We concluded that regardless of the titre administered once infections were established the viral load drove illness. We saw no adverse events linked to the virus [25] . Using this new model we conducted an HVC clinical study and demonstrated for the first time that an RNAi had significant antiviral activity against human RSV infection -this established the first-ever proof of concept for an RNAi therapeutic in humans adults [159] .

An editorial in the American Journal of Respiratory and Critical Care Medicine, described the utility of the HVC/RSV model saying; "This model permits the relatively quick and efficient study of new therapeutics in humans and assists in making critical decisions whether to advance a product into costly human trials in populations at highest risk for disease; children, elderly or immunocompromised patients. This constitutes a major and welcome advance in the field of RSV." [81] It is notable that two compounds that have distinct modes of action have recently been evaluated using the HVC model.

First-in-class nucleoside analogue ALS-008176, the efficacy of which was first demonstrated in the HVC model, is currently under evaluation in hospitalised infants [160, 161] . The HVC trial was of randomised, double-blind design, and studied healthy adults inoculated with RSV Memphis 37B [25] . A total of 62 participants received ALS-008176 or placebo for five days after confirmation of RSV infection by PCR (tested twice daily post inoculation). The primary endpoint was the area under the curve (AUC) for viral load post infection. More rapid RSV clearance and a greater reduction in viral load, with accompanying improvements in the severity of clinical disease, were demonstrated in the groups treated with ALS-008176 when compared to the placebo group [160] . Intensive sampling allowed for any potential mutations associated with resistance to be rapidly identified. No such resistant mutations were observed [160] .

An RSV-entry inhibitor, GS-5806, a second molecule, first-in-[its]-class was also evaluated. Among the 54 subjects that received active treatment, lower viral load, lower total mucus weight and a lower AUC symptom score were highly significant when compared to placebo [119] . Based on these challenge study data, this therapeutic is now also progressing into potentially pivotal field studies [162] .

An essential element of design in both studies was the timing of the first administration of therapeutic postexperimental virus inoculation; the timing was dependent on the detection of virus in nasal wash samples post inoculation of challenge virus by a rapid PCR assay [163], rather than at an arbitrary time point. Subsequently the therapeutic was administered every 12 hours. Careful dose timing, at a clinically relevant point of detection, contributed to the positive outcomes of both studies. It is also believed that by using this "triggered dosing" model, it better mimicked what would happen in a clinical setting as symptoms are known to appear soon after the onset of virus shedding.

The HVC model is not limited to novel antiviral compounds but is also important for the evaluation of novel vaccines. Influenza vaccine performance in recent years raises questions about the most appropriate correlates of protection.

Unlike field studies, HVC studies are useful tools for assessing the correlates of protection, vital for vaccine development [103, 104, 164]. Specifically, the importance of the humoral and cellular responses has been highlighted along with the pre-existing T-cell immunity for other respiratory viruses [104].

A recent publication describes the use of the HVC model to demonstrate the efficacy of a novel intranasal proteosome-adjuvanted trivalent inactivated influenza vaccine (P-TIV). In two separate studies, selected subjects who were naïve to A/Panama/2007/1999 (H3N2) virus, were dosed via nasal spray with one of three regimens of P-TIV or placebo. Together, the studies evaluated one or two doses, 15 µg or 30 µg, either once only or twice 14 days apart (1 x 30 µg, 2 x 30 µg, 2 x 15 µg) and subjects were challenged with A/Panama/2007/1999 (H3N2) virus. Immune responses to the vaccine antigens were measured by haemagglutination inhibition (HAI) assay and nasal wash secretory IgA (sIgA) antibodies. Vaccine efficacy was observed ranging from 58% to 82%, comparable to traditional vaccines. The studies also demonstrate that protection against illness associated with evidence of influenza infection significantly correlated with pre-challenge HAI (serum IgG) titres ($p = 0.0003$) and mucosal IgA ($p \leq 0.0001$) individually, and HAI ($p = 0.028$) and sIgA ($p = 0.0014$) together. HAI and sIgA levels were inversely related to rates of illness. These studies demonstrated the efficacy of this novel intranasal vaccine and answered some important questions concerning true correlates of protection against influenza infection which will help drive future vaccine design. As well as achieving its

primary aims, it revealed valuable insights into the correlates of protection and will, we hope, aid future vaccine design [164] .

An inter-seasonal or universal influenza vaccine is desperately needed; it will save many lives, whether in those unexpected years when the recommended composition is not matched, or when a pandemic occurs, as it did in 2009. The significance of the 1918 pandemic [165, 166] makes it very clear; up to 100 million people died. A universal vaccine is one that can be prepared for the unexpected, a virus that occurs due to the reassortment of viral genes from different host species.

The HVC model is possibly the only way to initially test such a universal vaccine.

A universal candidate could generate an immune response against the highly conserved virus ion channel protein M2, [167] [168] [169] [170] , although no vaccine has been shown to be effective in this regard; monoclonal antibodies alone have, the HVC model showed their efficacy [156] . Alternatively, a vaccine may target the conserved stalk of the HA protein [104, 171] , or elicit a T-cell response to the internal proteins [172] [173] [174] [175] . All are possibilities that have been and can be explored more efficiently using the HVC model.

Although HVC studies provide PoC, researchers, as we have shown, have employed regulatory design standards typical of later phase efficacy studies.

With the development of molecular technology, it is now possible to refine the statistical analysis by stratifying the subjects based on their immune profile. For instance, it is now possible to assess whether a subject is carrying other known respiratory pathogens (bacteria, viruses etc.) and if there is a possible impact on the set of results from the volunteer. Subjects often consent for further analysis of their samples, which allows a valuable biobank of samples to be built for further testing. Moving forward, such samples will allow the use of the HVC model to understand further what happens when a virus infects a person.

It is worth noting that the HVC model is not limited to PoC work on potential therapeutic agents; it is also extensively being used for research purposes, upon which improved treatments for respiratory viruses can be built. In recent years it has been used to demonstrate "gene switching signatures" that could form part of a diagnostic that would reveal infected individuals before they become symptomatic, in the early stages of infection; this could be vitally important in the event of a pandemic [176, 177] .

Also, the HVC model has been used to allow a comparison of the relative disease dynamics of different respiratory viruses [24] and to provide a better understanding of the interaction of the virus and the human host [26, 178, 179] .

The HVC model has increased our understanding of the viral life cycle and disease pathogenesis in a tightly controlled setting using small numbers of volunteers. Each volunteer is isolated from each other, and the wider community, ensuring that the disease under consideration is the only one of interest.

The applicability of the virus used to challenge volunteers in the HVC model to a virus that an individual might become exposed to in the "real world" is significant. Whether challenge trials are feasible is dependent on the availability of adequately safety-tested challenge virus strains that are of known provenance.

The HVC model provides certain knowledge of the character of the virus; the exact time point of infection; measurability of nasal virus shedding; prospective recording of symptoms and pre-selection of participants for viral challenge who are sero-suitable. This ensures that a statistically significant rate of infection is achieved with the minimal number of volunteers, thus optimising the risk-benefit ratio that supports the determination of therapeutic efficacy.

Crucial to HVC study design is the timing of administration of the first dose of product under investigation to determine optimal effectiveness, not just in the challenge study itself, but in both later stage clinical studies and final clinical use.

The HVC model is an important tool in drug development, in particular with regard to acute respiratory infections. It can accelerate the development of therapeutics that address multiple unmet medical needs. It helps in the understanding of the relationship between a virus and its human host and offers the potential for the development of early-stage diagnostics. It contributes towards identifying new areas for therapeutic intervention. Possibly, and arguably, more importantly, it can ensure that scarce medical resources are directed towards later stage clinical development in an evidence-based manner, and promising therapeutic opportunities are prioritised.

A careful and targeted study design process is a crucial step towards the successful outcome of a challenge trial, because almost all parameters, can be either controlled or at least known (either pre-or post-hoc). Furthermore, results from such trials can be used to make commercial decisions and can lead to major publications, expanding the collective understanding of the scientific community.

Samples from such experiments are of immense value to researchers for the understanding of host interaction mechanisms and the development and validation of therapeutics. Utilisation of consistently collected historical data from HVC studies informs the accurate design and powering of subsequent studies.

HVC studies have been successful in providing proof of concept for DNA vaccines, T-cell vaccines, intranasal vaccines, monoclonal antibodies and small molecules against a range of important respiratory viruses.

It is also encouraging to see that the HVC model is now expanding into further patient populations such as the elderly, asthmatics and those with other conditions such as chronic obstructive pulmonary disease.

An expanding archive of data from preceding studies is an invaluable asset to assist in the selection of volunteers, decide on appropriate endpoints and refine future field study designs.

This allows for safer, statistically sound and more rapidly delivered research. drafted the initial version of this manuscript with author RLW. hVIVO was responsible for overall management of this work and verified the accuracy of the data presented. Other non-author contributors included Ben Murdoch of hVivo who provided figures. hVivo would like to thank the volunteers without whose altruism the human viral challenge studies conducted at hVivo over many years would not have been possible.

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Author RLW conceived the strategy for this paper. Author RLW and professional medical writer Samina Hamilton drafted the article (see 'Acknowledgements'). Authors RLW and AG critically reviewed the complete article for important intellectual content. Authors RLW and AG had full authority over the choice of the journal and approved the final article. Author RLW is a guarantor for the paper and takes overall responsibility for this publication. All other authors contributed to the writing and review of this manuscript.

Ethics approval and consent to participate All clinical studies were described received appropriate Ethical Committee approval, including informed consent of volunteers.

All authors declare that they are employees of hVivo and as such, have provided or do provide ethical professional clinical research services to academic, biotechnology, or pharmaceutical clients. A patent (patent applications 14/366602 (US) 12813946.6 (EP) application is in progress regarding specific utilisation of the HVC model. This does not alter the authors' adherence to International Society for Medical Publication Professionals (ISMPP) 'Good Publication Practice for Communicating Company-Sponsored Medical Research: GPP3'.

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Epidemiological research priorities for public health control of the ongoing global novel coronavirus (2019-nCoV) outbreak

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Abstract: Infections with 2019-nCoV can spread from person to person, and in the earliest phase of the outbreak the basic reproductive number was estimated to be around 2.2, assuming a mean serial interval of 7.5 days [2]. The serial interval was not precisely estimated, and a potentially shorter mean serial interval would have corresponded to a slightly lower basic reproductive number. Control measures and changes in population behaviour later in January should have reduced the effective reproductive number. However, it is too early to estimate whether the effective reproductive number has been reduced to below the critical threshold of 1 because cases currently being detected and reported would have mostly been infected in mid- to late-January. Average delays between infection and illness onset have been estimated at around 5–6 days, with an upper limit of around 11–14 days [2,5], and delays from illness onset to laboratory confirmation added a further 10 days on average [2].

Text: It is now 6 weeks since Chinese health authorities announced the discovery of a novel coronavirus (2019-nCoV) [1] causing a cluster of pneumonia cases in Wuhan, the major transport hub of central China. The earliest human infections had occurred by early December 2019, and a large wet market in central Wuhan was linked to most, but not all, of the initial cases [2]. While evidence from the initial outbreak investigations seemed to suggest that 2019-nCoV could not easily spread between humans [3], it is now very clear that infections have been spreading from person to person [2]. We recently estimated that

more than 75,000 infections may have occurred in Wuhan as at 25 January 2020 [4] , and increasing numbers of infections continue to be detected in other cities in mainland China and around the world. A number of important characteristics of 2019-nCoV infection have already been identified, but in order to calibrate public health responses we need improved information on transmission dynamics, severity of the disease, immunity, and the impact of control and mitigation measures that have been applied to date.

Infections with 2019-nCoV can spread from person to person, and in the earliest phase of the outbreak the basic reproductive number was estimated to be around 2.2, assuming a mean serial interval of 7.5 days [2] . The serial interval was not precisely estimated, and a potentially shorter mean serial interval would have corresponded to a slightly lower basic reproductive number. Control measures and changes in population behaviour later in January should have reduced the effective reproductive number. However, it is too early to estimate whether the effective reproductive number has been reduced to below the critical threshold of 1 because cases currently being detected and reported would have mostly been infected in mid-to late-January. Average delays between infection and illness onset have been estimated at around 5-6 days, with an upper limit of around 11-14 days [2, 5] , and delays from illness onset to laboratory confirmation added a further 10 days on average [2] .

Chains of transmission have now been reported in a number of locations outside of mainland China. Within the coming days or weeks it will become clear whether sustained local transmission has been occurring in other cities outside of Hubei province in China, or in other countries. If sustained transmission does occur in other locations, it would be valuable to determine whether there is variation in transmissibility by location, for example because of different behaviours or control measures, or because of different environmental conditions. To address the latter, virus survival studies can be done in the laboratory to confirm whether there are preferred ranges of temperature or humidity for 2019-nCoV transmission to occur.

In an analysis of the first 425 confirmed cases of infection, 73% of cases with illness onset between 12 and 22 January reported no exposure to either a wet market or another person with symptoms of a respiratory illness [2] . The lack of reported exposure to another ill person could be attributed to lack of awareness or recall bias, but China's health minister publicly warned that pre-symptomatic transmission could be occurring [6] . Determining

the extent to which asymptomatic or pre-symptomatic transmission might be occurring is an urgent priority, because it has direct implications for public health and hospital infection control. Data on viral shedding dynamics could help in assessing duration of infectiousness. For severe acute respiratory syndrome-related coronavirus (SARS-CoV), infectivity peaked at around 10 days after illness onset [7] , consistent with the peak in viral load at around that time [8] . This allowed control of the SARS epidemic through prompt detection of cases and strict isolation. For influenza virus infections, virus shedding is highest on the day of illness onset and relatively higher from shortly before symptom onset until a few days after onset [9] . To date, transmission patterns of 2019-nCoV appear more similar to influenza, with contagiousness occurring around the time of symptom onset, rather than SARS.

Transmission of respiratory viruses generally happens through large respiratory droplets, but some respiratory viruses can spread through fine particle aerosols [10] , and indirect transmission via fomites can also play a role. Coronaviruses can also infect the human gastrointestinal tract [11, 12] , and faecal-oral transmission might also play a role in this instance. The SARS-CoV superspreading event at Amoy Gardens where more than 300 cases were infected was attributed to faecal-oral, then airborne, spread through pressure differentials between contaminated effluent pipes, bathroom floor drains and flushing toilets [13] . The first large identifiable superspreading event during the present 2019-nCoV outbreak has apparently taken place on the Diamond Princess cruise liner quarantined off the coast of Yokohama, Japan, with at least 130 passengers tested positive for 2019-nCoV as at 10 February 2020 [14] . Identifying which modes are important for 2019-nCoV transmission would inform the importance of personal protective measures such as face masks (and specifically which types) and hand hygiene.

The first human infections were identified through a surveillance system for pneumonia of unknown aetiology, and all of the earliest infections therefore had Modelling studies incorporating healthcare capacity and processes pneumonia. It is well established that some infections can be severe, particularly in older adults with underlying medical conditions [15, 16] , but based on the generally mild clinical presentation of 2019-nCoV cases detected outside China, it appears that there could be many more mild infections than severe infections. Determining the spectrum of clinical manifestations of 2019-nCoV infections is perhaps the most urgent research priority, because it determines the strength of public health response required. If the seriousness of infection is similar to the 1918/19 Spanish influenza, and therefore at the upper end of severity scales in influenza pandemic

plans, the same responses would be warranted for 2019-nCoV as for the most severe influenza pandemics. If, however, the seriousness of infection is similar to seasonal influenza, especially during milder seasons, mitigation measures could be tuned accordingly.

Beyond a robust assessment of overall severity, it is also important to determine high risk groups. Infections would likely be more severe in older adults, obese individuals or those with underlying medical conditions, but there have not yet been reports of severity of infections in pregnant women, and very few cases have been reported in children [2] .

Those under 18 years are a critical group to study in order to tease out the relative roles of susceptibility vs severity as possible underlying causes for the very rare recorded instances of infection in this age group. Are children protected from infection or do they not fall ill after infection? If they are naturally immune, which is unlikely, we should understand why; otherwise, even if they do not show symptoms, it is important to know if they shed the virus. Obviously, the question about virus shedding of those being infected but asymptomatic leads to the crucial question of infectivity. Answers to these questions are especially pertinent as basis for decisions on school closure as a social distancing intervention, which can be hugely disruptive not only for students but also because of its knock-on effect for child care and parental duties. Very few children have been confirmed 2019-nCoV cases so far but that does not necessarily mean that they are less susceptible or that they could not be latent carriers. Serosurveys in affected locations could inform this, in addition to truly assessing the clinical severity spectrum.

Another question on susceptibility is regarding whether 2019-nCoV infection confers neutralising immunity, usually but not always, indicated by the presence of neutralising antibodies in convalescent sera. Some experts already questioned whether the 2019-nCoV may behave similarly to MERS-CoV in cases exhibiting mild symptoms without eliciting neutralising antibodies [17] . A separate question pertains to the possibility of antibody-dependent enhancement of infection or of disease [18, 19] . If either of these were to be relevant, the transmission dynamics could become more complex.

A wide range of control measures can be considered to contain or mitigate an emerging infection such as 2019-nCoV. Internationally, the past week has seen an increasing number

of countries issue travel advisories or outright entry bans on persons from Hubei province or China as a whole, as well as substantial cuts in flights to and from affected areas out of commercial considerations. Evaluation of these mobility restrictions can confirm their potential effectiveness in delaying local epidemics [20] , and can also inform when as well as how to lift these restrictions.

If and when local transmission begins in a particular location, a variety of community mitigation measures can be implemented by health authorities to reduce transmission and thus reduce the growth rate of an epidemic, reduce the height of the epidemic peak and the peak demand on healthcare services, as well as reduce the total number of infected persons [21] . A number of social distancing measures have already been implemented in Chinese cities in the past few weeks including school and workplace closures. It should now be an urgent priority to quantify the effects of these measures and specifically whether they can reduce the effective reproductive number below 1, because this will guide the response strategies in other locations. During the 1918/19 influenza pandemic, cities in the United States, which implemented the most aggressive and sustained community measures were the most successful ones in mitigating the impact of that pandemic [22] .

Similarly to international travel interventions, local social distancing measures should be assessed for their impact and when they could be safely discontinued, albeit in a coordinated and deliberate manner across China such that recrudescence in the epidemic curve is minimised. Mobile telephony global positioning system (GPS) data and location services data from social media providers such as Baidu and Tencent in China could become the first occasion when these data inform outbreak control in real time.

At the individual level, surgical face masks have often been a particularly visible image from affected cities in China. Face masks are essential components of personal protective equipment in healthcare settings, and should be recommended for ill persons in the community or for those who care for ill persons. However, there is now a shortage of supply of masks in China and elsewhere, and debates are ongoing about their protective value for uninfected persons in the general community.

The Table summarises research gaps to guide the public health response identified.

In conclusion, there are a number of urgent research priorities to inform the public health response to the global spread of 2019-nCoV infections. Establishing robust estimates of the clinical severity of infections is probably the most pressing, because flattening out the surge in hospital admissions would be essential if there is a danger of hospitals becoming overwhelmed with patients who require inpatient care, not only for those infected with 2019-nCoV but also for urgent acute care of patients with other conditions including those scheduled for procedures and operations. In addressing the research gaps identified here, there is a need for strong collaboration of a competent corps of epidemiological scientists and public health workers who have the flexibility to cope with the surge capacity required, as well as support from laboratories that can deliver on the ever rising demand for diagnostic tests for 2019-nCoV and related sequelae. The readiness survey by Reusken et al. in this issue of Eurosurveillance testifies to the rapid response and capabilities of laboratories across Europe should the outbreak originating in Wuhan reach this continent [23] .

In the medium term, we look towards the identification of efficacious pharmaceutical agents to prevent and treat what may likely become an endemic infection globally. Beyond the first year, one interesting possibility in the longer term, perhaps borne of wishful hope, is that after the first few epidemic waves, the subsequent endemic re-infections could be of milder severity. Particularly if children are being infected and are developing immunity hereafter, 2019-nCoV could optimistically become the fifth human coronavirus causing the common cold.

None declared.

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Respiratory Viral Infections in Exacerbation of Chronic Airway Inflammatory Diseases:
Novel Mechanisms and Insights From the Upper Airway Epithelium

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Abstract: Respiratory virus infection is one of the major sources of exacerbation of chronic airway inflammatory diseases. These exacerbations are associated with high morbidity and even mortality worldwide. The current understanding on viral-induced exacerbations is that viral infection increases airway inflammation which aggravates disease symptoms. Recent advances in in vitro air-liquid interface 3D cultures, organoid cultures and the use of novel human and animal challenge models have evoked new understandings as to the mechanisms of viral exacerbations. In this review, we will focus on recent novel findings that elucidate how respiratory viral infections alter the epithelial barrier in the airways, the upper airway microbial environment, epigenetic modifications including miRNA modulation, and other changes in immune responses throughout the upper and lower airways. First, we reviewed the prevalence of different respiratory viral infections in causing exacerbations in chronic airway inflammatory diseases. Subsequently we also summarized how recent models have expanded our appreciation of the mechanisms of viral-induced exacerbations. Further we highlighted the importance of the virome within the airway microbiome environment and its impact on subsequent bacterial infection. This review consolidates the understanding of viral induced exacerbation in chronic airway inflammatory diseases and indicates pathways that may be targeted for more effective management of chronic inflammatory diseases.

Text: The prevalence of chronic airway inflammatory disease is increasing worldwide especially in developed nations (GBD 2015 Chronic Respiratory Disease Collaborators, 2017 Guan et al., 2018) . This disease is characterized by airway inflammation leading to complications such as coughing, wheezing and shortness of breath. The disease can

manifest in both the upper airway (such as chronic rhinosinusitis, CRS) and lower airway (such as asthma and chronic obstructive pulmonary disease, COPD) which greatly affect the patients' quality of life (Calus et al., 2012; Bao et al., 2015) . Treatment and management vary greatly in efficacy due to the complexity and heterogeneity of the disease. This is further complicated by the effect of episodic exacerbations of the disease, defined as worsening of disease symptoms including wheeze, cough, breathlessness and chest tightness (Xepapadaki and Papadopoulos, 2010) . Such exacerbations are due to the effect of enhanced acute airway inflammation impacting upon and worsening the symptoms of the existing disease (Hashimoto et al., 2008; Viniol and Vogelmeier, 2018) . These acute exacerbations are the main cause of morbidity and sometimes mortality in patients, as well as resulting in major economic burdens worldwide. However, due to the complex interactions between the host and the exacerbation agents, the mechanisms of exacerbation may vary considerably in different individuals under various triggers. Acute exacerbations are usually due to the presence of environmental factors such as allergens, pollutants, smoke, cold or dry air and pathogenic microbes in the airway (Gautier and Charpin, 2017; Viniol and Vogelmeier, 2018) . These agents elicit an immune response leading to infiltration of activated immune cells that further release inflammatory mediators that cause acute symptoms such as increased mucus production, cough, wheeze and shortness of breath. Among these agents, viral infection is one of the major drivers of asthma exacerbations accounting for up to 80-90% and 45-80% of exacerbations in children and adults respectively (Grissell et al., 2005; Xepapadaki and Papadopoulos, 2010; Jartti and Gern, 2017; Adeli et al., 2019) . Viral involvement in COPD exacerbation is also equally high, having been detected in 30-80% of acute COPD exacerbations (Kherad et al., 2010; Jafarinejad et al., 2017; Stolz et al., 2019) . Whilst the prevalence of viral exacerbations in CRS is still unclear, its prevalence is likely to be high due to the similar inflammatory nature of these diseases (Rowan et al., 2015; Tan et al., 2017) . One of the reasons for the involvement of respiratory viruses' in exacerbations is their ease of transmission and infection (Kutter et al., 2018) . In addition, the high diversity of the respiratory viruses may also contribute to exacerbations of different nature and severity (Busse et al., 2010; Costa et al., 2014; Jartti and Gern, 2017) . Hence, it is important to identify the exact mechanisms underpinning viral exacerbations in susceptible subjects in order to properly manage exacerbations via supplementary treatments that may alleviate the exacerbation symptoms or prevent severe exacerbations.

While the lower airway is the site of dysregulated inflammation in most chronic airway inflammatory diseases, the upper airway remains the first point of contact with sources of exacerbation. Therefore, their interaction with the exacerbation agents may directly

contribute to the subsequent responses in the lower airway, in line with the "United Airway" hypothesis. To elucidate the host airway interaction with viruses leading to exacerbations, we thus focus our review on recent findings of viral interaction with the upper airway. We compiled how viral induced changes to the upper airway may contribute to chronic airway inflammatory disease exacerbations, to provide a unified elucidation of the potential exacerbation mechanisms initiated from predominantly upper airway infections.

Despite being a major cause of exacerbation, reports linking respiratory viruses to acute exacerbations only start to emerge in the late 1950s (Pattemore et al., 1992) ; with bacterial infections previously considered as the likely culprit for acute exacerbation (Stevens, 1953; Message and Johnston, 2002) . However, with the advent of PCR technology, more viruses were recovered during acute exacerbations events and reports implicating their role emerged in the late 1980s (Message and Johnston, 2002) . Rhinovirus (RV) and respiratory syncytial virus (RSV) are the predominant viruses linked to the development and exacerbation of chronic airway inflammatory diseases (Jartti and Gern, 2017) . Other viruses such as parainfluenza virus (PIV), influenza virus (IFV) and adenovirus (AdV) have also been implicated in acute exacerbations but to a much lesser extent (Johnston et al., 2005; Oliver et al., 2014; Ko et al., 2019) . More recently, other viruses including bocavirus (BoV), human metapneumovirus (HMPV), certain coronavirus (CoV) strains, a specific enterovirus (EV) strain EV-D68, human cytomegalovirus (hCMV) and herpes simplex virus (HSV) have been reported as contributing to acute exacerbations . The common feature these viruses share is that they can infect both the upper and/or lower airway, further increasing the inflammatory conditions in the diseased airway (Mallia and Johnston, 2006; Britto et al., 2017) .

Respiratory viruses primarily infect and replicate within airway epithelial cells . During the replication process, the cells release antiviral factors and cytokines that alter local airway inflammation and airway niche (Busse et al., 2010) . In a healthy airway, the inflammation normally leads to type 1 inflammatory responses consisting of activation of an antiviral state and infiltration of antiviral effector cells. This eventually results in the resolution of the inflammatory response and clearance of the viral infection (Vareille et al., 2011; Braciale et al., 2012) . However, in a chronically inflamed airway, the responses against the virus may be impaired or aberrant, causing sustained inflammation and erroneous infiltration, resulting in the exacerbation of their symptoms (Mallia and Johnston, 2006; Dougherty and Fahy, 2009; Busse et al., 2010; Britto et al., 2017; Linden et al., 2019) . This is usually further compounded by the increased susceptibility of chronic airway inflammatory disease

patients toward viral respiratory infections, thereby increasing the frequency of exacerbation as a whole (Dougherty and Fahy, 2009; Busse et al., 2010; Linden et al., 2019). Furthermore, due to the different replication cycles and response against the myriad of respiratory viruses, each respiratory virus may also contribute to exacerbations via different mechanisms that may alter their severity. Hence, this review will focus on compiling and collating the current known mechanisms of viral-induced exacerbation of chronic airway inflammatory diseases; as well as linking the different viral infection pathogenesis to elucidate other potential ways the infection can exacerbate the disease. The review will serve to provide further understanding of viral induced exacerbation to identify potential pathways and pathogenesis mechanisms that may be targeted as supplementary care for management and prevention of exacerbation. Such an approach may be clinically significant due to the current scarcity of antiviral drugs for the management of viral-induced exacerbations. This will improve the quality of life of patients with chronic airway inflammatory diseases.

Once the link between viral infection and acute exacerbations of chronic airway inflammatory disease was established, there have been many reports on the mechanisms underlying the exacerbation induced by respiratory viral infection. Upon infecting the host, viruses evoke an inflammatory response as a means of counteracting the infection. Generally, infected airway epithelial cells release type I (IFN α/β) and type III (IFN λ) interferons, cytokines and chemokines such as IL-6, IL-8, IL-12, RANTES, macrophage inflammatory protein 1 α (MIP-1 α) and monocyte chemoattractant protein 1 (MCP-1) (Wark and Gibson, 2006; Matsukura et al., 2013). These, in turn, enable infiltration of innate immune cells and of professional antigen presenting cells (APCs) that will then in turn release specific mediators to facilitate viral targeting and clearance, including type II interferon (IFN γ), IL-2, IL-4, IL-5, IL-9, and IL-12 (Wark and Gibson, 2006; Singh et al., 2010; Braciale et al., 2012). These factors heighten local inflammation and the infiltration of granulocytes, T-cells and B-cells (Wark and Gibson, 2006; Braciale et al., 2012). The increased inflammation, in turn, worsens the symptoms of airway diseases.

Additionally, in patients with asthma and patients with CRS with nasal polyp (CRSwNP), viral infections such as RV and RSV promote a Type 2-biased immune response (Becker, 2006; Jackson et al., 2014; Jurak et al., 2018). This amplifies the basal type 2 inflammation resulting in a greater release of IL-4, IL-5, IL-13, RANTES and eotaxin and a further increase in eosinophilia, a key pathological driver of asthma and CRSwNP (Wark and Gibson, 2006; Singh et al., 2010; Chung et al., 2015; Dunican and Fahy, 2015). Increased eosinophilia, in

turn, worsens the classical symptoms of disease and may further lead to life-threatening conditions due to breathing difficulties. On the other hand, patients with COPD and patients with CRS without nasal polyp (CRSsNP) are more neutrophilic in nature due to the expression of neutrophil chemoattractants such as CXCL9, CXCL10, and CXCL11 (Cukic et al., 2012; Brightling and Greening, 2019) . The pathology of these airway diseases is characterized by airway remodeling due to the presence of remodeling factors such as matrix metalloproteinases (MMPs) released from infiltrating neutrophils (Linden et al., 2019) . Viral infections in such conditions will then cause increase neutrophilic activation; worsening the symptoms and airway remodeling in the airway thereby exacerbating COPD, CRSsNP and even CRSwNP in certain cases (Wang et al., 2009; Tacon et al., 2010; Linden et al., 2019) .

An epithelial-centric alarmin pathway around IL-25, IL-33 and thymic stromal lymphopoietin (TSLP), and their interaction with group 2 innate lymphoid cells (ILC2) has also recently been identified (Nagarkar et al., 2012; Hong et al., 2018; Allinne et al., 2019) . IL-25, IL-33 and TSLP are type 2 inflammatory cytokines expressed by the epithelial cells upon injury to the epithelial barrier (Gabryelska et al., 2019; Roan et al., 2019) . ILC2s are a group of lymphoid cells lacking both B and T cell receptors but play a crucial role in secreting type 2 cytokines to perpetuate type 2 inflammation when activated (Scanlon and McKenzie, 2012; Li and Hendriks, 2013) . In the event of viral infection, cell death and injury to the epithelial barrier will also induce the expression of IL-25, IL-33 and TSLP, with heighten expression in an inflamed airway (Allakhverdi et al., 2007; Goldsmith et al., 2012; Byers et al., 2013; Shaw et al., 2013; Beale et al., 2014; Jackson et al., 2014; Uller and Persson, 2018; Ravanetti et al., 2019) . These 3 cytokines then work in concert to activate ILC2s to further secrete type 2 cytokines IL-4, IL-5, and IL-13 which further aggravate the type 2 inflammation in the airway causing acute exacerbation (Camelo et al., 2017) . In the case of COPD, increased ILC2 activation, which retain the capability of differentiating to ILC1, may also further augment the neutrophilic response and further aggravate the exacerbation (Silver et al., 2016) . Interestingly, these factors are not released to any great extent and do not activate an ILC2 response during viral infection in healthy individuals (Yan et al., 2016; Tan et al., 2018a) ; despite augmenting a type 2 exacerbation in chronically inflamed airways (Jurak et al., 2018) . These classical mechanisms of viral induced acute exacerbations are summarized in Figure 1 .

As integration of the virology, microbiology and immunology of viral infection becomes more interlinked, additional factors and **FIGURE 1 | Current understanding of viral induced**

exacerbation of chronic airway inflammatory diseases. Upon virus infection in the airway, antiviral state will be activated to clear the invading pathogen from the airway. Immune response and injury factors released from the infected epithelium normally would induce a rapid type 1 immunity that facilitates viral clearance. However, in the inflamed airway, the cytokines and chemokines released instead augmented the inflammation present in the chronically inflamed airway, strengthening the neutrophilic infiltration in COPD airway, and eosinophilic infiltration in the asthmatic airway. The effect is also further compounded by the participation of Th1 and ILC1 cells in the COPD airway; and Th2 and ILC2 cells in the asthmatic airway.

Frontiers in Cell and Developmental Biology | www.frontiersin.org mechanisms have been implicated in acute exacerbations during and after viral infection (Murray et al., 2006) . Murray et al. (2006) has underlined the synergistic effect of viral infection with other sensitizing agents in causing more severe acute exacerbations in the airway. This is especially true when not all exacerbation events occurred during the viral infection but may also occur well after viral clearance (Kim et al., 2008; Stolz et al., 2019) in particular the late onset of a bacterial infection (Singanayagam et al., 2018 (Singanayagam et al., , 2019a . In addition, viruses do not need to directly infect the lower airway to cause an acute exacerbation, as the nasal epithelium remains the primary site of most infections. Moreover, not all viral infections of the airway will lead to acute exacerbations, suggesting a more complex interplay between the virus and upper airway epithelium which synergize with the local airway environment in line with the "united airway" hypothesis (Kurai et al., 2013) . On the other hand, viral infections or their components persist in patients with chronic airway inflammatory disease (Kling et al., 2005; Wood et al., 2011; Ravi et al., 2019) . Hence, their presence may further alter the local environment and contribute to current and future exacerbations. Future studies should be performed using metagenomics in addition to PCR analysis to determine the contribution of the microbiome and mycobioime to viral infections. In this review, we highlight recent data regarding viral interactions with the airway epithelium that could also contribute to, or further aggravate, acute exacerbations of chronic airway inflammatory diseases.

Patients with chronic airway inflammatory diseases have impaired or reduced ability of viral clearance (Hammond et al., 2015; McKendry et al., 2016; Akbarshahi et al., 2018; Gill et al., 2018; Wang et al., 2018; Singanayagam et al., 2019b) . Their impairment stems from a type 2-skewed inflammatory response which deprives the airway of important type 1 responsive CD8 cells that are responsible for the complete clearance of virusinfected cells

(Becker, 2006; McKendry et al., 2016) . This is especially evident in weak type 1 inflammation-inducing viruses such as RV and RSV (Kling et al., 2005; Wood et al., 2011; Ravi et al., 2019) . Additionally, there are also evidence of reduced type I (IFN β) and III (IFN λ) interferon production due to type 2-skewed inflammation, which contributes to imperfect clearance of the virus resulting in persistence of viral components, or the live virus in the airway epithelium (Contoli et al., 2006; Hwang et al., 2019; Wark, 2019) . Due to the viral components remaining in the airway, antiviral genes such as type I interferons, inflammasome activating factors and cytokines remained activated resulting in prolong airway inflammation (Wood et al., 2011; Essaidi-Laziosi et al., 2018) . These factors enhance granulocyte infiltration thus prolonging the exacerbation symptoms. Such persistent inflammation may also be found within DNA viruses such as AdV, hCMV and HSV, whose infections generally persist longer (Imperiale and Jiang, 2015) , further contributing to chronic activation of inflammation when they infect the airway (Yang et al., 2008; Morimoto et al., 2009; Imperiale and Jiang, 2015; Lan et al., 2016; Tan et al., 2016; Kowalski et al., 2017) . With that note, human papilloma virus (HPV), a DNA virus highly associated with head and neck cancers and respiratory papillomatosis, is also linked with the chronic inflammation that precedes the malignancies (de Visser et al., 2005; Gillison et al., 2012; Bonomi et al., 2014; Fernandes et al., 2015) . Therefore, the role of HPV infection in causing chronic inflammation in the airway and their association to exacerbations of chronic airway inflammatory diseases, which is scarcely explored, should be investigated in the future. Furthermore, viral persistence which lead to continuous expression of antiviral genes may also lead to the development of steroid resistance, which is seen with RV, RSV, and PIV infection (Chi et al., 2011; Ford et al., 2013; Papi et al., 2013) . The use of steroid to suppress the inflammation may also cause the virus to linger longer in the airway due to the lack of antiviral clearance (Kim et al., 2008; Hammond et al., 2015; Hewitt et al., 2016; McKendry et al., 2016; Singanayagam et al., 2019b) . The concomitant development of steroid resistance together with recurring or prolong viral infection thus added considerable burden to the management of acute exacerbation, which should be the future focus of research to resolve the dual complications arising from viral infection.

On the other end of the spectrum, viruses that induce strong type 1 inflammation and cell death such as IFV (Yan et al., 2016; Guibas et al., 2018) and certain CoV (including the recently emerged COVID-19 virus) (Tao et al., 2013; Yue et al., 2018; Zhu et al., 2020) , may not cause prolonged inflammation due to strong induction of antiviral clearance. These infections, however, cause massive damage and cell death to the epithelial barrier, so much so that areas of the epithelium may be completely absent post infection (Yan et al., 2016; Tan et al., 2019) . Factors such as RANTES and CXCL10, which recruit immune cells

to induce apoptosis, are strongly induced from IFV infected epithelium (Ampomah et al., 2018; Tan et al., 2019) . Additionally, necroptotic factors such as RIP3 further compounds the cell deaths in IFV infected epithelium . The massive cell death induced may result in worsening of the acute exacerbation due to the release of their cellular content into the airway, further evoking an inflammatory response in the airway (Guibas et al., 2018) . Moreover, the destruction of the epithelial barrier may cause further contact with other pathogens and allergens in the airway which may then prolong exacerbations or results in new exacerbations. Epithelial destruction may also promote further epithelial remodeling during its regeneration as viral infection induces the expression of remodeling genes such as MMPs and growth factors . Infections that cause massive destruction of the epithelium, such as IFV, usually result in severe acute exacerbations with non-classical symptoms of chronic airway inflammatory diseases. Fortunately, annual vaccines are available to prevent IFV infections (Vasileiou et al., 2017; Zheng et al., 2018) ; and it is recommended that patients with chronic airway inflammatory disease receive their annual influenza vaccination as the best means to prevent severe IFV induced exacerbation.

Another mechanism that viral infections may use to drive acute exacerbations is the induction of vasodilation or tight junction opening factors which may increase the rate of infiltration. Infection with a multitude of respiratory viruses causes disruption of tight junctions with the resulting increased rate of viral infiltration. This also increases the chances of allergens coming into contact with airway immune cells. For example, IFV infection was found to induce oncostatin M (OSM) which causes tight junction opening (Pothoven et al., 2015; Tian et al., 2018) . Similarly, RV and RSV infections usually cause tight junction opening which may also increase the infiltration rate of eosinophils and thus worsening of the classical symptoms of chronic airway inflammatory diseases (Sajjan et al., 2008; Kast et al., 2017; Kim et al., 2018) . In addition, the expression of vasodilating factors and fluid homeostatic factors such as angiopoietin-like 4 (ANGPTL4) and bactericidal/permeabilityincreasing fold-containing family member A1 (BPIFA1) are also associated with viral infections and pneumonia development, which may worsen inflammation in the lower airway (Akram et al., 2018) . These factors may serve as targets to prevent viral-induced exacerbations during the management of acute exacerbation of chronic airway inflammatory diseases.

Another recent area of interest is the relationship between asthma and COPD exacerbations and their association with the airway microbiome. The development of chronic airway inflammatory diseases is usually linked to specific bacterial species in the

microbiome which may thrive in the inflamed airway environment (Diver et al., 2019) . In the event of a viral infection such as RV infection, the effect induced by the virus may destabilize the equilibrium of the microbiome present (Molyneaux et al., 2013; Kloefer et al., 2014; Kloefer et al., 2017; Jubinville et al., 2018; van Rijn et al., 2019) . In addition, viral infection may disrupt biofilm colonies in the upper airway (e.g., *Streptococcus pneumoniae*) microbiome to be released into the lower airway and worsening the inflammation (Marks et al., 2013; Chao et al., 2014) . Moreover, a viral infection may also alter the nutrient profile in the airway through release of previously inaccessible nutrients that will alter bacterial growth (Siegel et al., 2014; Mallia et al., 2018) . Furthermore, the destabilization is further compounded by impaired bacterial immune response, either from direct viral influences, or use of corticosteroids to suppress the exacerbation symptoms (Singanayagam et al., 2018 (Singanayagam et al., , 2019a Wang et al., 2018; Finney et al., 2019) . All these may gradually lead to more far reaching effect when normal flora is replaced with opportunistic pathogens, altering the inflammatory profiles (Teo et al., 2018) . These changes may in turn result in more severe and frequent acute exacerbations due to the interplay between virus and pathogenic bacteria in exacerbating chronic airway inflammatory diseases (Wark et al., 2013; Singanayagam et al., 2018) . To counteract these effects, microbiome-based therapies are in their infancy but have shown efficacy in the treatments of irritable bowel syndrome by restoring the intestinal microbiome (Bakken et al., 2011) . Further research can be done similarly for the airway microbiome to be able to restore the microbiome following disruption by a viral infection.

Viral infections can cause the disruption of mucociliary function, an important component of the epithelial barrier. Ciliary proteins  FIGURE 2 | Changes in the upper airway epithelium contributing to viral exacerbation in chronic airway inflammatory diseases. The upper airway epithelium is the primary contact/infection site of most respiratory viruses. Therefore, its infection by respiratory viruses may have far reaching consequences in augmenting and synergizing current and future acute exacerbations. The destruction of epithelial barrier, mucociliary function and cell death of the epithelial cells serves to increase contact between environmental triggers with the lower airway and resident immune cells. The opening of tight junction increasing the leakiness further augments the inflammation and exacerbations. In addition, viral infections are usually accompanied with oxidative stress which will further increase the local inflammation in the airway. The dysregulation of inflammation can be further compounded by modulation of miRNAs and epigenetic modification such as DNA methylation and histone modifications that promote dysregulation in inflammation. Finally, the change in the local airway environment and inflammation promotes growth of pathogenic bacteria that may replace the airway

microbiome. Furthermore, the inflammatory environment may also disperse upper airway commensals into the lower airway, further causing inflammation and alteration of the lower airway environment, resulting in prolonged exacerbation episodes following viral infection.

Viral specific trait contributing to exacerbation mechanism (with literature evidence)
Oxidative stress ROS production (RV, RSV, IFV, HSV)

As RV, RSV, and IFV were the most frequently studied viruses in chronic airway inflammatory diseases, most of the viruses listed are predominantly these viruses. However, the mechanisms stated here may also be applicable to other viruses but may not be listed as they were not implicated in the context of chronic airway inflammatory diseases exacerbation (see text for abbreviations).

that aid in the proper function of the motile cilia in the airways are aberrantly expressed in ciliated airway epithelial cells which are the major target for RV infection (Griggs et al., 2017) . Such form of secondary cilia dyskinesia appears to be present with chronic inflammations in the airway, but the exact mechanisms are still unknown (Peng et al., , 2019 Qiu et al., 2018) . Nevertheless, it was found that in viral infection such as IFV, there can be a change in the metabolism of the cells as well as alteration in the ciliary gene expression, mostly in the form of down-regulation of the genes such as dynein axonemal heavy chain 5 (DNAH5) and multiciliate differentiation And DNA synthesis associated cell cycle protein (MCIDAS) (Tan et al., 2018b . The recently emerged Wuhan CoV was also found to reduce ciliary beating in infected airway epithelial cell model (Zhu et al., 2020) . Furthermore, viral infections such as RSV was shown to directly destroy the cilia of the ciliated cells and almost all respiratory viruses infect the ciliated cells (Jumat et al., 2015; Yan et al., 2016; Tan et al., 2018a) . In addition, mucus overproduction may also disrupt the equilibrium of the mucociliary function following viral infection, resulting in symptoms of acute exacerbation (Zhu et al., 2009) . Hence, the disruption of the ciliary movement during viral infection may cause more foreign material and allergen to enter the airway, aggravating the symptoms of acute exacerbation and making it more difficult to manage. The mechanism of the occurrence of secondary cilia dyskinesia can also therefore be explored as a means to limit the effects of viral induced acute exacerbation.

MicroRNAs (miRNAs) are short non-coding RNAs involved in post-transcriptional modulation of biological processes, and implicated in a number of diseases (Tan et al., 2014) . miRNAs are found to be induced by viral infections and may play a role in the modulation of antiviral responses and inflammation (Gutierrez et al., 2016; Deng et al., 2017; Feng et al., 2018) . In the case of chronic airway inflammatory diseases, circulating miRNA changes were found to be linked to exacerbation of the diseases (Wardzynska et al., 2020) . Therefore, it is likely that such miRNA changes originated from the infected epithelium and responding immune cells, which may serve to further dysregulate airway inflammation leading to exacerbations. Both IFV and RSV infections has been shown to increase miR-21 and augmented inflammation in experimental murine asthma models, which is reversed with a combination treatment of anti-miR-21 and corticosteroids (Kim et al., 2017) . IFV infection is also shown to increase miR-125a and b, and miR-132 in COPD epithelium which inhibits A20 and MAVS; and p300 and IRF3, respectively, resulting in increased susceptibility to viral infections (Hsu et al., 2016 (Hsu et al., , 2017 . Conversely, miR-22 was shown to be suppressed in asthmatic epithelium in IFV infection which lead to aberrant epithelial response, contributing to exacerbations (Moheimani et al., 2018) . Other than these direct evidence of miRNA changes in contributing to exacerbations, an increased number of miRNAs and other non-coding RNAs responsible for immune modulation are found to be altered following viral infections (Globinska et al., 2014; Feng et al., 2018; Hasegawa et al., 2018) . Hence non-coding RNAs also presents as targets to modulate viral induced airway changes as a means of managing exacerbation of chronic airway inflammatory diseases. Other than miRNA modulation, other epigenetic modification such as DNA methylation may also play a role in exacerbation of chronic airway inflammatory diseases. Recent epigenetic studies have indicated the association of epigenetic modification and chronic airway inflammatory diseases, and that the nasal methylome was shown to be a sensitive marker for airway inflammatory changes (Cardenas et al., 2019; Gomez, 2019) . At the same time, it was also shown that viral infections such as RV and RSV alters DNA methylation and histone modifications in the airway epithelium which may alter inflammatory responses, driving chronic airway inflammatory diseases and exacerbations (McErlean et al., 2014; Pech et al., 2018; Caixia et al., 2019) . In addition, Spalluto et al. (2017) also showed that antiviral factors such as IFN γ epigenetically modifies the viral resistance of epithelial cells. Hence, this may indicate that infections such as RV and RSV that weakly induce antiviral responses may result in an altered inflammatory state contributing to further viral persistence and exacerbation of chronic airway inflammatory diseases (Spalluto et al., 2017) .

Finally, viral infection can result in enhanced production of reactive oxygen species (ROS), oxidative stress and mitochondrial dysfunction in the airway epithelium (Kim et al., 2018; Mishra et al., 2018; Wang et al., 2018) . The airway epithelium of patients with chronic airway inflammatory diseases are usually under a state of constant oxidative stress which sustains the inflammation in the airway (Barnes, 2017; van der Vliet et al., 2018) . Viral infections of the respiratory epithelium by viruses such as IFV, RV, RSV and HSV may trigger the further production of ROS as an antiviral mechanism Aizawa et al., 2018; Wang et al., 2018) . Moreover, infiltrating cells in response to the infection such as neutrophils will also trigger respiratory burst as a means of increasing the ROS in the infected region. The increased ROS and oxidative stress in the local environment may serve as a trigger to promote inflammation thereby aggravating the inflammation in the airway (Tiwari et al., 2002) . A summary of potential exacerbation mechanisms and the associated viruses is shown in Figure 2 and Table 1 .

While the mechanisms underlying the development and acute exacerbation of chronic airway inflammatory disease is extensively studied for ways to manage and control the disease, a viral infection does more than just causing an acute exacerbation in these patients. A viral-induced acute exacerbation not only induced and worsens the symptoms of the disease, but also may alter the management of the disease or confer resistance toward treatments that worked before. Hence, appreciation of the mechanisms of viral-induced acute exacerbations is of clinical significance to devise strategies to correct viral induce changes that may worsen chronic airway inflammatory disease symptoms. Further studies in natural exacerbations and in viral-challenge models using RNA-sequencing (RNA-seq) or single cell RNA-seq on a range of time-points may provide important information regarding viral pathogenesis and changes induced within the airway of chronic airway inflammatory disease patients to identify novel targets and pathway for improved management of the disease. Subsequent analysis of functions may use epithelial cell models such as the air-liquid interface, in vitro airway epithelial model that has been adapted to studying viral infection and the changes it induced in the airway (Yan et al., 2016; Boda et al., 2018; Tan et al., 2018a) . Animal-based diseased models have also been developed to identify systemic mechanisms of acute exacerbation (Shin, 2016; Gubernatorova et al., 2019; Tanner and Single, 2019) . Furthermore, the humanized mouse model that possess human immune cells may also serves to unravel the immune profile of a viral infection in healthy and diseased condition (Ito et al., 2019; Li and Di Santo, 2019) . For milder viruses, controlled in vivo human infections can be performed for the best mode of verification of the associations of the virus with the proposed mechanism of viral induced acute exacerbations . With the advent of suitable diseased models, the

verification of the mechanisms will then provide the necessary continuation of improving the management of viral induced acute exacerbations.

In conclusion, viral-induced acute exacerbation of chronic airway inflammatory disease is a significant health and economic burden that needs to be addressed urgently. In view of the scarcity of antiviral-based preventative measures available for only a few viruses and vaccines that are only available for IFV infections, more alternative measures should be explored to improve the management of the disease. Alternative measures targeting novel viral-induced acute exacerbation mechanisms, especially in the upper airway, can serve as supplementary treatments of the currently available management strategies to augment their efficacy. New models including primary human bronchial or nasal epithelial cell cultures, organoids or precision cut lung slices from patients with airways disease rather than healthy subjects can be utilized to define exacerbation mechanisms. These mechanisms can then be validated in small clinical trials in patients with asthma or COPD. Having multiple means of treatment may also reduce the problems that arise from resistance development toward a specific treatment.

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Interferon-Induced Transmembrane Protein 3 Inhibits Hantaan Virus Infection, and Its Single Nucleotide Polymorphism rs12252 Influences the Severity of Hemorrhagic Fever with Renal Syndrome

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Abstract: Hantaan virus (HTNV) causes hemorrhagic fever with renal syndrome (HFRS). Previous studies have identified interferon-induced transmembrane proteins (IFITMs) as an interferon-stimulated gene family. However, the role of IFITMs in HTNV infection is unclear. In this study, we observed that IFITM3 single nucleotide polymorphisms (SNP) rs12252 C allele and CC genotype associated with the disease severity and HTNV load in the plasma of HFRS patients. In vitro experiments showed that the truncated protein produced by the rs12252 C allele exhibited an impaired anti-HTNV activity. We also proved that IFITM3 was able to inhibit HTNV infection in both HUVEC and A549 cells by overexpression and RNAi assays, likely via a mechanism of inhibiting virus entry demonstrated by binding and entry assay. Localization of IFITM3 in late endosomes was also observed. In addition, we demonstrated that the transcription of IFITM3 is negatively regulated by an lncRNA negative regulator of interferon response (NRIR). Taken together, we conclude that IFITM3, negatively regulated by NRIR, inhibits HTNV infection, and its SNP rs12252 correlates with the plasma HTNV load and the disease severity of patients with HFRS.

Text: associates with the severity of disease, indicating the importance of viremia in the pathogenesis of HFRS (2) . Therefore, further studies of host factors limiting HTNV infection and influencing antiviral response as well as disease progression are clinically significant and timely. The human family of interferon-induced transmembrane proteins (IFITMs) was discovered 25 years ago to consist of interferon-stimulated genes (ISGs) (3) . This family includes five members, namely, IFITM1, 2, 3, 5, and 10, among which IFITM1, 2, and 3 possess antiviral activity (4) . Different IFITM proteins have different antiviral spectrum (5) . For example, IFITM3 has been shown to prevent influenza virus infection in vitro and in mice (6, 7) , and it also inhibits multiple viruses, including filoviruses, rhabdoviruses, flaviviruses, and even Ebola and Zika virus (7) (8) (9) (10) (11) . The antiviral mechanism of IFITM3 is thought to be the restriction of viral entry into cells (4, 12) . Single nucleotide polymorphisms (SNPs) are single nucleotide variations in a genetic sequence that occur at an appreciable frequency in the population. Several SNPs has been identified in IFITM3, among which the rs12252 site with C allele results in a N-terminal truncation of IFITM3 protein, leading to impaired inhibition of influenza virus in vitro (13, 14) . Notably, the frequencies of rs12252 C allele and CC genotype correlate with disease severity in patients

infected with influenza virus (13, 15) . HTNV has been shown to induce a type I interferon response (though in later time postinfection) (16, 17) . While overexpression of IFITM1, 2, and 3 in Vero E6 cells has been reported to inhibit HTNV infection (18) , however, the effect of IFITMs on HTNV infection in human cell lines and its role in HFRS still remain unknown.

LncRNA comprises a group of non-coding RNAs longer than 200 nt that function as gene regulators. Some lncRNAs have been shown to play a role in innate immunity (19) . Among them, negative regulator of interferon response (NRIR) (lncRNA NRIR, also known as lncRNA-CMPK2) is a non-coding ISG that negatively regulates IFITM1 and Mx1 expression in HCV infection (20) . Notably, IFITM3 is largely homologous to IFITM1, but the role of NRIR in the regulation of IFITM3 in HTNV infection remains unclear.

In the present study, we investigate the effect of IFITM3 on the replication of HTNV and its role in the development of HFRS in humans. We provide primary evidence suggesting that IFITM3, regulated by NRIR, can inhibit HTNV infection and its SNP rs12252 correlates with the disease severity and viral load in patients with HFRS. This study expands our understanding of the antiviral activity of IFITM3 and enriches our knowledge of innate immune responses to HTNV infection.

This study was conducted in accordance with the recommendations of the biomedical research guidelines involving human participants established by the National Health and Family Planning Commission of China. The Institutional Ethics Committee of Tangdu Hospital approved this study. All subjects gave written informed consent in accordance with the Declaration of Helsinki. Before inclusion, all participants were informed of the study objectives and signed the consent form before blood samples and medical records were obtained.

Sixty-nine HFRS patients admitted into the Department of Infectious Diseases, Tangdu Hospital between October 2014 and March 2016 were enrolled in this study. All patients were Han Chinese. The diagnosis of HFRS was made based on typical symptoms and signs as well as positive IgM and IgG antibodies against HTNV in the serum assessed by enzyme linked immunosorbent assay (ELISA) in our department. The classification of HFRS severity and the exclusion criteria were described as follows (21) : white blood cells (WBC),

platelets (PLT), blood urea nitrogen (BUN), serum creatinine (Scr), and heteromorphic lymphocytes that were tested by the Department of Clinical Laboratory (shown in Table 1).

According to clinical symptoms and signs, such as fever, effusion, hemorrhage, edema, and renal function, the severity of HFRS can be classified as previously described (21): (1) mild patients were identified with mild renal failure without an obvious oliguric stage; (2) moderate patients were those with obvious symptoms of uremia, effusion (bulbar conjunctiva), hemorrhage (skin and mucous membrane), and renal failure with a typical oliguric stage; (3) severe patients had severe uremia, effusion (bulbar conjunctiva and either peritoneum or pleura), hemorrhage (skin and mucous membrane), and renal failure with oliguria (urine output, 50-500 ml/day) for ≤ 5 days or anuria (urine output, < 50 ml/day) for ≤ 2 days; and (4) critical patients exhibited ≥ 1 of the following signs during the illness: refractory shock, visceral hemorrhage, heart failure, pulmonary edema, brain edema, severe secondary infection, and severe renal failure with oliguria (urine output, 50-500 ml/day) for > 5 days, anuria (urine output, < 50 ml/day) for > 2 days, or a BUN level of > 42.84 mmol/l. Due to the sample quantity required for SNP typing, the mild and moderate patients were assessed together in the mild group, and we combined severe and critical patients as severe group.

The exclusion criteria for this study were patients with: (1) any other kidney disease, (2) diabetes mellitus, (3) autoimmune disease, (4) hematological disease, (5) cardiovascular disease, (6) viral hepatitis (types A, B, C, D, or E), or (7) any other liver disease. In addition, no patients received corticosteroids or other immunomodulatory drugs during the study period (21) .

Genomic DNA was extracted from the peripheral blood of patients using the PureGene DNA Isolation kit (Gentra Systems, Minneapolis, MN, USA). The region encompassing the human IFITM3 rs12252 were amplified by PCR (forward primer, 5'-GGAAACTGTTGAGAAACCGAA-3' and reverse primer, 5'-CATACGCACCTTCACGGAGT-3'). The PCR products were purified and sequenced using an Applied Biosystems 3730xl DNA Analyzer (Thermo Scientific, Waltham, MA, USA). The allele frequencies and genotypes of healthy Han Chinese and other groups were obtained from the 1,000 genomes project ([http:// www.1000genomes.org](http://www.1000genomes.org)).

The HTNV load in plasma samples (collected during the acute phase) from 24 age- and sex-matched HFRS patients with different genotypes were measured using previously reported methods (2). Briefly, viral RNA was extracted from the plasma of HFRS patients using Purelink Viral RNA/DNA Kits (Invitrogen, Carlsbad, CA, USA). The SuperScript III Platinum One-Step Quantitative RT-PCR System kit (Invitrogen, Carlsbad, CA, USA) was employed for the real-time RT-PCR assay. The primers and probe (provided by Sangon Biotech, Shanghai, China) were as follows: forward, 5'-TACAGAGGGAAATCAATGCC-3', reverse, 5'-TGTTCAACTCATCTGGATCCTT-3', and probe, 5'-(FAM) ATCCCTCACCTTCTGCCTGGCTATC (TAMRA)-3'. The synthetic S segment of the HTNV standard strain 76-118 RNA transcript was used as the quantitative calibrator. The external standard was the culture supernatant of Vero E6 cells infected with HTNV 76-118, which was quantified using synthetic quantitative calibrators. For each experiment, one aliquot of calibrated 76-118 standard was extracted in parallel with the clinical samples and serially 10-fold diluted with concentrations ranging from 10.56 to 2.56 log₁₀ copies/ml. PCR was performed using an iQ5 Cyclor (Bio-Rad, Hercules, CA, USA) with following conditions: 42°C for 15 min, 95°C for 2 min, and 50 cycles of 15 s at 95°C, 30 s at 53°C, and 30 s at 72°C. Fluorescence was read during the 72°C step of the final segment of every cycling program.

HUVEC cells (ScienCell Research Laboratories, Carlsbad, CA, USA) were grown in ECM BulletKit (ScienCell Research Laboratories, Carlsbad, CA, USA) in a 5% CO₂ incubator. A549 cells (ATCC Cat# CRM-CCL-185, RRID:CVCL_0023) were grown in our laboratory in DMEM with 10% FBS (Thermo Scientific, Waltham, MA, USA) in a 5% CO₂ incubator. Cells were used within passage 10 after primary culture. HTNV strain 76-118 was cultured in Vero E6 cells (ATCC Cat# CRL-1586, RRID:CVCL_0574) in our laboratory and titrated using an immunofluorescence staining assay for HTNV nucleocapsid protein (NP) as previously described (22). The TCID₅₀ was 10⁵ /ml, which was calculated using the Reed-Muench method.

The recombinant human IFN- α 2a was obtained from PBL Interferon Source (Piscataway, NJ, USA) and dissolved in the buffer provided by the manufacturer (composition not disclosed). HUVEC and A549 cells were infected by incubation with HTNV as indicated moi at 37°C for 60 mins. Subsequently, the virus solution was removed and fresh medium added to the cell culture.

Cells were transfected with lentiviral vectors of c-myc-tagged IFITM1, IFITM2, IFITM3, and IFITM3 NΔ21 (purchased from GENECHM, Shanghai, China) at a moi of 10. Puromycin (2 µg/ ml for HUVEC and 6 µg/ml for A549 cells) was used to create cell lines stably expressing IFITMs. Cells were transfected with control (scrambled) short interfering RNA (siRNA), IFITM1 siRNA, IFITM2 siRNA, or IFITM3 siRNA (10 nM) using Lipofectamine 3000 transfection reagent (Invitrogen, Carlsbad, CA, USA). SiRNAs were purchased from Origene (Rockville, MD, USA), and the sequences were not disclosed.

Total RNA was extracted using TRIzol reagent (Invitrogen, Carlsbad, CA, USA), and cDNA was synthesized using the K1622 kit (Thermo Scientific, Waltham, MA, USA). Quantitative realtime PCR (qPCR) was performed using SYBR Premix Ex Taq II (Takara Biotechnology Co., Dalian, China) with a Bio-Rad iQ5 cycler (Bio-Rad, Hercules, CA, USA). β-actin was used as the reference gene. The primers (Sangon Biotech, Shanghai, China) were as follows: IFITM1 (forward, 5'-ACTCCGTGAAGTCTAGGGACA-3' and reverse, 5'-TGTCACAGAGCCGAATACCAG-3'); IFITM2 (forward, 5'-ATCCCGGTAACCCGATCAC-3' and reverse, 5'-CTTCCTGTCCCTAGACTTCAC-3'); IFITM3 (forward, 5'-GGTCTTCGCTGGACACCAT-3' and reverse, 5'-TGTCCCTAGACTTCACGGAGTA-3'); IFITM3 pre-mRNA (forward, 5'-CATAGCACGCGGCTCT CAG-3' and reverse, 5'-CGTCGCCAACCATCTTCCTG-3'); HTNV S segment (forward, 5'-GCCTGGAGACCATCTGA AAG-3' and reverse, 5'-AGTATCGGGACGACAAAGGA-3'); β-actin (forward, 5'-GCTACGTCGCCCTGGACTTC-3' and reverse, 5'-GTCATAGTCCGCCTAGAAGC-3'); NRIR (forward, 5'-ATGGTTTTCTGGTGCCTTG-3' and reverse, 5'-GGAGGTTAGAGGTGTCTGCTG-3'); NRAV (forward, 5'-TCACTACTGCCCCAGGATCA-3' and reverse, 5'-GGTGGTCACAGGACTCATGG-3').

For detection of miR-130a, cDNA was synthesized using the TaqMan microRNA reverse transcription kit (Invitrogen, Carlsbad, CA, USA) with a specific primer in gene-specific TaqMan assay kit (000454, Invitrogen, Carlsbad, CA, USA). MiR-130a level was determined using the gene-specific TaqMan assay kit (000454, Invitrogen, Carlsbad, CA, USA). U6 (001973, Invitrogen, Carlsbad, CA, USA) was used as an endogenous control (23) . Because the pre-mRNA levels can represent the initial transcription rate (24) , the primers used to detect the pre-mRNA of IFITM3 were designed targeting the intron of IFITM3 as previously described (25) . IFITM3 has two exons and one intron. For qPCR of IFITM3 pre-mRNA, the forward primers were positioned in the intron, and the reverse primer was positioned at the beginning of the second exon. For qPCR of IFITM3 mRNA, the forward primers were positioned in the first exon, and the reverse primer was positioned at the beginning of the

second exon (24) . Because the basal expression of IFITM3 is low in A549 cells, we detected IFITM3 mRNA and pre-mRNA in A549 cells following IFN- α 2a treatment (20 IU/ml for 12 h) after the overexpression of NRIR.

Cell lysates were prepared using Radio Immunoprecipitation Assay (RIPA) buffer (Sigma-Aldrich, St. Louis, MO, USA). Equal amounts of protein (20 μ g protein/lane) were electrophoresed on a 10%-SDS-polyacrylamide gel and electrophoretically transferred to a polyvinylidene difluoride membrane (Millipore, Billerica, MA, USA). After blocking with 5% bovine serum albumin in Trisbuffered saline at room temperature for 1 h, the membranes were incubated with antibodies against IFITM1 (Proteintech Group Cat# 60074-1-Ig Lot# RRID:AB_2233405), IFITM2, IFITM3 (Proteintech Group Cat# 66081-1-Ig Lot# RRID:AB_11182821), and β -actin (Proteintech, Wuhan, Hubei, China) or HTNV NP (provided by the Department of Microbiology, The Fourth Military Medical University) overnight at 4°C. The membranes were then washed and incubated with HRP-conjugated IgG antibody (Cell Signaling Technology, Danvers, MA, USA) for 1 h at room temperature. The blots were developed using an enhanced chemiluminescence detection kit (Millipore, Billerica, MA, USA) and visualized using X-ray film. The blot densities were analyzed using the Quantity One software (Bio-Rad, Hercules, CA, USA). In addition, the RIPA buffer contains 50mM Tris (pH = 7.4), 150 mM NaCl, 1% Triton X-100, 1% sodium deoxycholate, 0.1% SDS. Protease inhibitor cocktail (Roche, Basel, Switzerland) was added before use.

The cells were cultured on glass coverslips (Millipore, Billerica, MA, USA) until they were semi-confluence and then incubated with HTNV for 60 min (moi = 1). At the indicated times post-HTNV infection, the cells were fixed with 4% PFA, incubated with 0.3% Triton X-100 (Sigma-Aldrich, St. Louis, MO, USA), and blocked with 5% BSA for 1 h. Following incubation with a mouse monoclonal antibody against c-myc-tag (Sigma-Aldrich, St. Louis, MO, USA, Sigma-Aldrich Cat# M5546), IFITM3, lysosome-associated membrane glycoprotein 1 (LAMP1, Cell Signaling Technology, Danvers, MA, USA), or HTNV NP at 37°C for 2 h, the cells were washed and incubated with anti-rabbit Ig conjugated to Alexa 555 and anti-mouse Ig conjugated to Alexa 488 (Abcam, Cambridge, MA, USA) secondary antibodies at room temperature for 1 h. The nuclei were counterstained with DAPI. An Olympus BX51 fluorescence microscope system and FV1000 confocal microscopy system (Olympus, Tokyo, Japan) were used to capture the images. hTnV binding and entry assay Cells transduced with IFITM3 or the empty vector were detached and washed extensively with cold PBS. The cells and HTNV were pre-chilled on ice for 30 min, mixed at a moi of 1 and incubated at 4°C for 1 h with rotation. Part of cells were washed extensively with ice-cold

PBS and harvested for binding assay. Another part of cells were switched to 37°C for 2 h to allow HTNV entry. The HTNV that remained on the cell surface was removed by treatment with proteinase K (0.1 mg/ml, Thermo Scientific, Waltham, MA, USA). To achieve direct entry of HTNV into cells by virus-plasma membrane fusion as a positive control, cells were pre-chilled on ice for 10 min with 20 mM NH₄Cl. Adsorption of HTNV (moi = 1) was performed at 4°C for 1 h. The cells were then washed, and fusion of the virus with the plasma membrane was triggered by incubation in low pH medium (20 mM sodium succinate, pH = 5.5) for 10 min at 37°C. Infection was followed by incubation for 2 h at 37°C in the presence of 20 mM NH₄Cl (26). qPCR analysis of the HTNV S segment was conducted to evaluate the influence of IFITMs on HTNV cell adhesion and HTNV entry.

All data were expressed as the mean ± SEM. Statistical analyses were performed using GraphPad Prism 5 (GraphPad Software, La Jolla, CA, USA). For association analysis of the rs12252 allele and genotype, Fisher's exact test was used. Independent samples t-tests were used for normally distributed data. Differences among groups were determined by one-way analysis of variance (ANOVA) with repeated measures, followed by Bonferroni's post hoc test. $P < 0.05$ was considered statistically significant.

The iFiTM3 snP rs12252 c allele and cc genotype associated with severe hFrS Disease and a higher Plasma hTnV load

To determine the clinical significance of IFITM3 SNP in HTNV infection, the relationship between rs12252 SNP and the severity of HFRS in 69 patients were examined. We sequenced 300 bp of the IFITM3 locus encompassing SNP rs12252 in all enrolled patients. Then, we stratified these patients into mild and severe, based on the clinical assessment as described in Section "Material and Methods." We found a significantly higher frequency of the C allele among severe HFRS patients compared with the healthy Han Chinese in the 1,000 genomes sequence database (68.29 vs. 52.16%, $P = 0.0076$). The frequency of rs12252 C in severe patients was also higher than those mild patients (68.29 vs. 46.43%, $P = 0.013$, Figures 1A,B; Table 2). These data suggest that harboring rs12252 C allele increases the risk of suffering severe disease in HTNV-infected individuals, with an odds ratio (95% CI) of 2.124 (1.067-4.230). For genotypes, 43.90% of the severe patients carried the CC genotype, a significantly higher frequency than the control Han Chinese per 1,000 genomes sequence database (26.92% CC genotype, $P = 0.03$) as well as mildly infected patients (14.29%, $P = 0.02$, Figures 1A,B; Table 2). However, mildly ill individuals did not

exhibit a Fisher's exact test was used to test the association between rs12252 allele/genotype and HFRS severity. (c) The plasma HTNV load in CC genotype patients and CT/TT genotype patients, tested by qRCR analysis. Each symbol represents one individual patient. Independent samples t-test was used to test the difference of HTNV load between groups. * $P < 0.05$, ** $P < 0.01$. significantly different genotype frequencies compared with the Han Chinese population. In addition, we also found that patients with CC genotype had higher plasma viral load in acute phase (Figure 1C) . These results support the notion that the normal function of IFITM3 plays a critical role in the immune response to HTNV infection in vivo, which has a substantial influence on the clinical manifestation of HFRS.

Previous studies reveal that the truncated IFITM3 protein produced by SNP rs12252 C allele (Figure 2A , the missing part stands for the truncated 21 amino acids from N-terminal of IFITM3, the intramembrane helix, and transmembrane helix was presented as boxes) leads to an impaired anti-influenza activity (14) . To test the functional significance of this polymorphism in HTNV infection, we transfected the majority T or minority C variant IFITM3 alleles that produce full-length or N-terminally truncated (NΔ21) proteins (Figure 2A) with c-myc-tag to HUVEC and A549 cell using lentivirus vectors (Figure 2B) . Then, we challenged the cells with HTNV at moi = 1 for 24 h and found that cells with the minority C variant were more susceptible to HTNV infection with higher expression of HTNV S segment (Figure 2C) and more positive of HTNV NP (Figure S3 in Supplementary Material). Indeed, compared with the mock (empty vector)-infected control, the NΔ21 protein almost lost the ability to inhibit HTNV infection in both HUVEC and A549 cells (Figures 2C,D ; Figure S3 in Supplementary Material).

To determine the role of HTNV infection in inducing IFITMs, qPCR as well as Western blot of IFITMs were conducted in HUVEC and A549 cells (Figures 3A,B ; Figure S1 in Supplementary Material). While we observed only a moderate upregulation of IFITM1, 2, and 3 mRNA and protein in HUVECs after more than 24 h postinfection; IFITM1, 2, and 3 mRNA, however, were only transiently upregulated in A549 cells and caused no significant change in protein level.

We knocked down the IFITM1, 2, and 3 expression by transfection of their siRNAs individually. The effect of siRNAs on the expression of target IFITMs was tested by qPCR in HUVECs (Figure S2 in Supplementary Material), and the effect of the best oligo against each IFITMs (IFITM1C, IFITM2A, IFITM3B) was tested by Western blot in A549 (Figure 4A)

and HUVEC cells (Figure 4B) . To assess the role of IFITMs in anti-HTNV effect of IFN- α 2a, IFITM1, 2, and 3 were knocked down respectively by transfecting the above-tested oligoes for 12 h, followed by IFN- α 2a treatment (20 IU/ml for another 12 h). The cells were then challenged with HTNV (moi = 1) for 24 h. The HTNV S segment and NP levels were significantly suppressed in both HUVEC and A549 cells in response to IFN- α 2a treatment.

Notably, knockdown of IFITM3 significantly restored the levels of HTNV S segment and NP in HUVEC and A549 cells. Knockdown of IFITM1 also partially restored the HTNV level in A549 cells (Figures 4C,D) . These results demonstrate that

To assess the anti-HTNV effects of IFITMs, we tested the effect of overexpressed IFITM1, 2, and 3 on HTNV infection. c-myc-tagged IFITM1, 2, and 3 were expressed in both HUVEC and A549 cells (Figure 5A) , and the cells were then challenged with HTNV (moi = 1) for 24 h. The HTNV S segment and NP levels were suppressed by IFITM3 overexpression in HUVEC cells (Figures 5B-D) . They were also suppressed by expressing IFITM1 and IFITM3 in A549 cells (Figures 5B-D) .

The inhibitory effect of IFITM3 was further confirmed by immunofluorescence analysis of HTNV NP (Figure S3 in Supplementary Material). These results were in accordance with the above-described RNAi results. To determine whether IFITM3 inhibited HTNV binding or entry, HUVEC and A549 cells were incubated with HTNV (moi = 1) at 4°C for 1 h, unbound virus was washed away, and HTNV RNA collected at this time point represents HTNV bound to the cell surface. After virus binding, the cells were shifted to 37°C for 2 h to allow HTNV internalization, and HTNV RNA collected at this time point represents cell-internalized virus. As a positive control for inhibition of virus entry, we incubated a parallel group of cells with HTNV at pH = 5.5 as described in Section "Materials and Methods." Expression of IFITM3 did not affect HTNV binding (Figure 6A) but significantly suppressed HTNV entry in both HUVEC and A549 cells (Figure 6B) . IFITM3 Was Partially localized to LAMP1 + late endosomes in the host cells

To elucidate the mechanism of IFITM3 function, we investigated the subcellular localization of IFITM3 in the host cells. IFITM3 was found partially localized to LAMP1 + late endosomes in HUVECs analyzed by confocal microscopy (Figure 6C) . The co-localization of IFITM3 and LAMP1 + late endosomes had also been found in A549 cells (27) . Because the transfer into

LAMP1 + late endosomes is a necessary step for HTNV entry (28) , this result provides an evidence for the anti-HTNV mechanism of IFITM3.

LncRNA-and microRNA-mediated regulation of IFITM3 has been reported in several studies. We tested the change of previously reported regulators of IFITMs, such as NRAV, NRIR, and miR-130a after HTNV infection, among which NRIR was the only changed one (downregulated) after HTNV infection (Figure 7A ; Figure S4 in Supplementary Material) in HUVEC. However, the expression of NRIR was unchanged in A549 cells. We overexpressed NRIR in HUVEC and A549 cells using the pcDNA3.1 vector (Figure 7B) . Importantly, overexpression of NRIR significantly suppressed IFITM3 mRNA and pre-mRNA levels and facilitated HTNV infection in HUVEC and A549 cells (Figures 7C-E) . These data suggest that lncRNA NRIR is a negative regulator of IFITM3 transcription.

Hantaan virus is an enveloped, negative-sense RNA virus from the genus Hantavirus within the family Bunyaviridae. It causes HFRS, which is an important threat to public health worldwide. It is also a potential weapon for biological terrorism. Reservoir animals, usually rodents, are asymptomatic during persistent infection. Unlike in rodents, Hantavirus infection leads to HFRS and Hantavirus pulmonary syndrome (HPS) in humans (21) . The major clinical characteristics of HFRS include fever, hemorrhage, hypotension, and renal injury (1, 21) , causing severe manifestations and death in some cases. The current standard of care for HFRS relies on symptomatic and supportive treatment. It has been confirmed that the plasma viral load is associated with the severity of HFRS, implicating the importance of viremia in the pathogenesis of HFRS (2). However, no direct antiviral medications are currently available for this illness. Interferon is the key molecule for the antiviral response and has been used as an antiviral medicine in many diseases. It has been reported that HTNV infection induces a late type I interferon response (16) . However, the set of ISGs required for IFN-mediated inhibition of HTNV has not yet been identified. Therefore, identification of ISGs that are effective against HTNV is an attractive strategy to identify novel therapeutic targets.

In this study, we demonstrated a significantly high frequency of the rs12252 C allele and CC genotype among HFRS patients with severe illness compared with mildly infected individuals and the healthy Han Chinese. The rs12252 C allele and CC genotype are also found to be associated with higher plasma viral load in the early stage of HFRS. We also discovered that HTNV infection induces IFITMs, and the truncated IFITM3 produced by

rs12252 C allele exhibits significantly decreased anti-HTNV activity. Interestingly, IFITM3 is found to restrict HTNV infection with a mechanism of cellular entry inhibition. Indeed, IFITM3 is localized to the late endosome in the host cells, which is a necessary structure for HTNV entry. In addition, we find that HTNV infection downregulated lncRNA NRIR 48 h post infection, which negatively regulates the transcription of IFITM3. Collectively, these results suggest that IFITM3, regulated by NRIR, inhibits HTNV infection, and its SNP rs12252 correlates with the disease severity and viral load in patients with HFRS.

The antiviral properties of IFITM proteins were identified in 2009 in an RNAi screen for host factors that influence influenza virus replication (29) . IFITM1, 2, and 3 have been demonstrated to possess antiviral activity in several studies. Everitt et al. demonstrated that the severity of influenza virus infection was greatly increased in IFITM3-knockout mice compared with wild-type animals (15) . Different IFITM members have also been confirmed to inhibit the cellular entry of multiple virus families (including filoviruses, rhabdoviruses, and flaviviruses) (7, (9) (10) (11) 30) . For example, HIV-1 and HCV infection are inhibited by IFITM1 (31) (32) (33) (34) . It is commonly believed that IFITMs restrict viral infection at the stage of cellular entry (12) . Recent studies suggested that the cellular location of different IFITMs may influence the range of viruses restricted by each protein (5) . IFITM1 prevents HCV entry because it colocalizes with CD81 on the cell membrane, interrupting the endocytosis of HCV particles (32) , whereas IFITM3 confines influenza virus in acidified endosomal compartments (27) . Notably, retrovirus subvirus particles (ISVPs), which do not require endosomal acidification for entry, are not inhibited by IFITM3 expression, suggesting that IFITM3 may function at the stage of endosomal entry (35) . Studies utilizing cell-cell fusion assays have suggested that IFITM3 blocks the entry of enveloped virus by preventing the fusion of the viral membrane with a limiting membrane of the host cell, either the plasma membrane and/or the endosomal membranes. The results obtained using two-photon laser scanning and fluorescence lifetime imaging (FLIM) suggest that IFITM proteins may reduce membrane fluidity and increase the spontaneous positive curvature in the outer leaflet of membranes (36) . In the present study, we demonstrated that IFN- α 2a (20 U/ ml) significantly inhibited HTNV infection, siRNA-mediated depletion of IFITM3 alone significantly mitigated the antiviral effect of IFN- α 2a in both HUVEC and A549 cells, whereas depletion of IFITM1 alone alleviated the antiviral effect of IFN- α 2a in A549 cells. Overexpression of IFITM3 inhibited HTNV infection to HUVEC and A549 cells. IFITM1 overexpression was also effective in inhibition of HTNV in A549 cells. All these results suggest that IFITM3 is an important control factor under natural infection of HTNV. Our results also demonstrate that the effectiveness of IFITM3 is cell type-independent, which is in accordance with the results from similar viruses, such as RVFV (18) . Binding and entry

assays, conducted by controlling the temperature and pH, showed that IFITM3 did not significantly influence HTNV binding but inhibited HTNV entry into HUVEC and A549 cells. Indeed, IFITM3 partially localizes to the late endosome of the host cells, which is a necessary site for the HTNV entry. However, we failed in tracking the transportation of HTNV in infected cells possibly due to the lack of fluorescence-labeled virus. In addition, IFITM1 also suppressed HTNV infection in A549 cells. The mechanism underlying anti-HTNV effect of IFITM1 remains undetermined and deserves to be further explored.

According to a recent study on the three-dimensional structure of IFITM3, there is a C-terminal transmembrane α -helix and a two-N-terminal intramembrane α -helices (shown in Figure 2A as black boxes) in IFITM3 (14). There are two splice variants that differ by the presence or absence of the first N-terminal 21 amino acids (deleted part, shown in Figure 2A as red dotted line). Several SNPs including 13 non-synonymous, 13 synonymous, 1 in-frame stop, and 1 splice site acceptor altering have been reported in the translated IFITM3 sequence (15, 29). Among them, the rare SNP rs12252C allele of IFITM3 truncates the protein as described above, leading to a reduced inhibition of influenza virus infection in A549 cells (15). We demonstrated that truncated IFITM3 protein also loses the ability to inhibit HTNV infection in vitro. In Northern European patients hospitalized with seasonal influenza or pandemic influenza A virus, increased homozygosity of the minor C allele of SNP rs12252 in IFITM3 was observed (37). In Chinese patients infected with influenza A (H1N1) virus, there was also an increased frequency of the C allele and CC genotype of SNP rs12252 (13). In the present study, we observed an increased frequency of the C allele and CC genotype of SNP rs12252 in severely infected HFRS patients compared with healthy control and mildly affected patients. Patients carrying the CC genotype also had higher plasma viral loads compared with those with the CT/TT genotype. Given the impaired function of the IFITM3 protein produced by the C mutation, and the fact that enrichment of the rs12252 C allele in patients with severe disease and the higher viral load in patients with the CC genotype, this finding suggests that IFITM3 plays a pivotal role in the anti-HTNV response in vivo. We speculate that the much higher level of CC allele at healthy population of Han Chinese compared with Caucasians may place the Chinese at a higher risk for developing severe illness upon HTNV infection, which needs further investigation.

LncRNAs are a group of non-coding RNAs longer than 200 nt that function as gene regulators, playing a role in regulating multiple cellular functions, including the innate immunity. For example, lncRNA NEAT1 is reported to be upregulated by influenza virus or PolyI:C stimulation, which promotes IL-8 expression (38). lncRNA NRAV has been shown to

negatively regulate the initial transcription of IFITM3 and Mx1 by affecting the histone modification of these genes (25) . lncRNA NRIR is a non-coding ISG, which has been reported to negatively regulate IFITM1 and Mx1 expression in HCV infection (20) . Mir-130a was also reported as a regulator of IFITM1 (23) . In this analysis, lncRNA NRIR was downregulated in HUVECs after HTNV infection for 48 h, overexpression of NRIR negatively regulates the initial transcription of IFITM3, evidenced by the decreased pre-mRNA as well as mRNA levels. NRIR overexpression also facilitated HTNV infection. These results indicate that the downregulation of NRIR after HTNV infection is possibly involved in the activation of innate immune responses against HTNV infection. We have also evaluated other potential regulators of IFITM3 before we choose NRIR for further study. Another lncRNA that can regulate IFITM3, i.e., NRAV (NR_038854), remained unchanged after HTNV infection (Figures S4A,B in Supplementary Material). Additionally, miR-130a, which potentially regulate IFITM3, was also unaltered after HTNV infection (Figures S4C,D in Supplementary Material).

In conclusion, this study revealed a critical role for IFITM3 in HTNV infection. We demonstrated, for the first time to our knowledge, that IFITM3 is a newly identified anti-HTNV ISG; its expression is negatively regulated by NRIR; and its antiviral activity seems via a mechanism of inhibiting virus entry into the host cells. In addition, we discovered that the IFITM3 SNP rs12252 C allele and CC genotype correlates with the plasma HTNV load and the severity of HFRS; and the rs12252 C allele produces a truncated IFITM3 protein (NΔ21) that attenuates its anti-HTNV function. These results provide new insights into the role of IFITM3 in regulating innate immunity against HTNV infection, which is the basis for identifying new targets to develop novel agent against this worldwide infectious disease. aUThOr cOnTribUTiOns ZX-y, BP-y, YC-t, and MH-w performed the experiments; WP-z, BX-f, LY-f, ZY, and JZ-s designed the research; HC-x, YW, and WX analyzed the data; TK and ZC-m provided clinical data; ZX-y and BP-y wrote the paper.

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Estimating the number of infections and the impact of non-pharmaceutical interventions on COVID-19 in 11 European countries

30 March 2020 Imperial College COVID-19 Response Team

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Summary

Following the emergence of a novel coronavirus (SARS-CoV-2) and its spread outside of China, Europe

is now experiencing large epidemics. In response, many European countries have implemented

unprecedented non-pharmaceutical interventions including case isolation, the closure of schools and

universities, banning of mass gatherings and/or public events, and most recently, widescale social

distancing including local and national lockdowns.

In this report, we use a semi-mechanistic Bayesian hierarchical model to attempt to infer the impact

of these interventions across 11 European countries. Our methods assume that changes in the

reproductive number— a measure of transmission - are an immediate response to these interventions

being implemented rather than broader gradual changes in behaviour. Our model estimates these

changes by calculating backwards from the deaths observed over time to estimate transmission that

occurred several weeks prior, allowing for the time lag between infection and death.

One of the key assumptions of the model is that each intervention has the same effect on the

reproduction number across countries and over time. This allows us to leverage a greater amount of

data across Europe to estimate these effects. It also means that our results are driven strongly by the

data from countries with more advanced epidemics, and earlier interventions, such as Italy and Spain.

We find that the slowing growth in daily reported deaths in Italy is consistent with a significant impact

of interventions implemented several weeks earlier. In Italy, we estimate that the effective reproduction number, R_t , dropped to close to 1 around the time of lockdown (11th March), although

with a high level of uncertainty.

Overall, we estimate that countries have managed to reduce their reproduction number. Our

estimates have wide credible intervals and contain 1 for countries that have implemented all

interventions considered in our analysis. This means that the reproduction number may be above or

below this value. With current interventions remaining in place to at least the end of March, we

estimate that interventions across all 11 countries will have averted 59,000 deaths up to 31 March

[95% credible interval 21,000-120,000]. Many more deaths will be averted through ensuring that

interventions remain in place until transmission drops to low levels. We estimate that, across all 11

countries between 7 and 43 million individuals have been infected with SARS-CoV-2 up to 28th March,

representing between 1.88% and 11.43% of the population. The proportion of the population infected

to date — the attack rate - is estimated to be highest in Spain followed by Italy and lowest in Germany

and Norway, reflecting the relative stages of the epidemics.

Given the lag of 2-3 weeks between when transmission changes occur and when their impact can be

observed in trends in mortality, for most of the countries considered here it remains too early to be

certain that recent interventions have been effective. If interventions in countries at earlier stages of

their epidemic, such as Germany or the UK, are more or less effective than they were in the countries

with advanced epidemics, on which our estimates are largely based, or if interventions have improved

or worsened over time, then our estimates of the reproduction number and deaths averted would

change accordingly. It is therefore critical that the current interventions remain in place and trends in

cases and deaths are closely monitored in the coming days and weeks to provide reassurance that

transmission of SARS-CoV-2 is slowing.

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1 Introduction

Following the emergence of a novel coronavirus (SARS-CoV-2) in Wuhan, China in December 2019 and

its global spread, large epidemics of the disease, caused by the virus designated COVID-19, have

emerged in Europe. In response to the rising numbers of cases and deaths, and to maintain the

capacity of health systems to treat as many severe cases as possible, European countries, like those in

other continents, have implemented or are in the process of implementing measures to control their

epidemics. These large-scale non-pharmaceutical interventions vary between countries but include

social distancing (such as banning large gatherings and advising individuals not to socialize outside

their households), border closures, school closures, measures to isolate symptomatic individuals and

their contacts, and large-scale lockdowns of populations with all but essential internal travel banned.

Understanding firstly, whether these interventions are having the desired impact of controlling the

epidemic and secondly, which interventions are necessary to maintain control, is critical given their

large economic and social costs.

The key aim of these interventions is to reduce the effective reproduction number, R_t , of the infection,

a fundamental epidemiological quantity representing the average number of infections, at time t , per

infected case over the course of their infection. If it is maintained at less than 1, the incidence of new

infections decreases, ultimately resulting in control of the epidemic. If R_t is greater than 1, then

infections will increase (dependent on how much greater than 1 the reproduction number is) until the

epidemic peaks and eventually declines due to acquisition of herd immunity.

In China, strict movement restrictions and other measures including case isolation and quarantine

began to be introduced from 23rd January, which achieved a downward trend in the number of

confirmed new cases during February, resulting in zero new confirmed indigenous cases in Wuhan by

March 19th. Studies have estimated how R_t changed during this time in different areas of China from

around 2-4 during the uncontrolled epidemic down to below 1, with an estimated 7-9 fold decrease

in the number of daily contacts per person.^{1,2} Control measures such as social distancing, intensive

testing, and contact tracing in other countries such as Singapore and South Korea have successfully

reduced case incidence in recent weeks, although there is a risk the virus will spread again once control

measures are relaxed.^{3'4}

The epidemic began slightly later in Europe, from January or later in different regions.⁵ Countries have

implemented different combinations of control measures and the level of adherence to government

recommendations on social distancing is likely to vary between countries, in part due to different

levels of enforcement.

Estimating reproduction numbers for SARS-CoV-2 presents challenges due to the high proportion of

infections not detected by health systems⁷ and regular changes in testing policies, resulting in

different proportions of infections being detected over time and between countries. Most countries

so far only have the capacity to test a small proportion of suspected cases and tests are reserved for

severely ill patients or for high-risk groups (e.g. contacts of cases). Looking at case data, therefore,

gives a systematically biased view of trends.

An alternative way to estimate the course of the epidemic is to back-calculate infections from

observed deaths. Reported deaths are likely to be more reliable, although the early focus of most

surveillance systems on cases with reported travel histories to China may mean that some early deaths

will have been missed. Whilst the recent trends in deaths will therefore be informative, there is a time

lag in observing the effect of interventions on deaths since there is a 2-3-week period between

infection, onset of symptoms and outcome.

In this report, we fit a novel Bayesian mechanistic model of the infection cycle to observed deaths in

11 European countries, inferring plausible upper and lower bounds (Bayesian credible intervals) of the

total populations infected (attack rates), case detection probabilities, and the reproduction number

over time (R_t). We fit the model jointly to COVID-19 data from all these countries to assess whether

there is evidence that interventions have so far been successful at reducing R_t below 1, with the strong

assumption that particular interventions are achieving a similar impact in different countries and that

the efficacy of those interventions remains constant over time. The model is informed more strongly

by countries with larger numbers of deaths and which implemented interventions earlier, therefore

estimates of recent R_t in countries with more recent interventions are contingent on similar

intervention impacts. Data in the coming weeks will enable estimation of country-specific R_t with

greater precision.

Model and data details are presented in the appendix, validation and sensitivity are also presented in

the appendix, and general limitations presented below in the conclusions.

2 Results

The timing of interventions should be taken in the context of when an individual country's epidemic

started to grow along with the speed with which control measures were implemented. Italy was the

first to begin intervention measures, and other countries followed soon afterwards (Figure 1). Most

interventions began around 12th-14th March. We analyzed data on deaths up to 28th March, giving a

2-3-week window over which to estimate the effect of interventions. Currently, most countries in our

study have implemented all major non-pharmaceutical interventions.

For each country, we model the number of infections, the number of deaths, and R_t , the effective

reproduction number over time, with R_t changing only when an intervention is introduced (Figure 2-

12). R_t is the average number of secondary infections per infected individual, assuming that the

interventions that are in place at time t stay in place throughout their entire infectious period. Every

country has its own individual starting reproduction number R_t before interventions take place.

Specific interventions are assumed to have the same relative impact on R_t in each country when they

were introduced there and are informed by mortality data across all countries.

Figure I: Intervention timings for the 11 European countries included in the analysis. For further details see Appendix 8.6.

2.1 Estimated true numbers of infections and current attack rates

In all countries, we estimate there are orders of magnitude fewer infections detected (Figure 2) than

true infections, mostly likely due to mild and asymptomatic infections as well as limited testing

capacity. In Italy, our results suggest that, cumulatively, 5.9 [1.9-15.2] million people have been

infected as of March 28th, giving an attack rate of 9.8% [3.2%-25%] of the population (Table 1). Spain

has recently seen a large increase in the number of deaths, and given its smaller population, our model

estimates that a higher proportion of the population, 15.0% (7.0 [18-19] million people) have been

infected to date. Germany is estimated to have one of the lowest attack rates at 0.7% with 600,000

[240,000-1,500,000] people infected.

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Table I: Posterior model estimates of percentage of total population infected as of 28th March 2020.

Country % of total population infected (mean [95% credible intervall])

Austria 1.1% [0.36%-3.1%]

Belgium 3.7% [1.3%-9.7%]
Denmark 1.1% [0.40%-3.1%]
France 3.0% [1.1%-7.4%]
Germany 0.72% [0.28%-1.8%]
Italy 9.8% [3.2%-26%]
Norway 0.41% [0.09%-1.2%]
Spain 15% [3.7%-41%]
Sweden 3.1% [0.85%-8.4%]
Switzerland 3.2% [1.3%-7.6%]
United Kingdom 2.7% [1.2%-5.4%]

2.2 Reproduction numbers and impact of interventions

Averaged across all countries, we estimate initial reproduction numbers of around 3.87 [3.01-4.66],

which is in line with other estimates.¹⁸ These estimates are informed by our choice of serial interval

distribution and the initial growth rate of observed deaths. A shorter assumed serial interval results in

lower starting reproduction numbers (Appendix 8.4.2, Appendix 8.4.6). The initial reproduction

numbers are also uncertain due to (a) importation being the dominant source of new infections early

in the epidemic, rather than local transmission (b) possible under-ascertainment in deaths particularly

before testing became widespread.

We estimate large changes in R_t in response to the combined non-pharmaceutical interventions. Our

results, which are driven largely by countries with advanced epidemics and larger numbers of deaths

(e.g. Italy, Spain), suggest that these interventions have together had a substantial impact on

transmission, as measured by changes in the estimated reproduction number R_t . Across all countries

we find current estimates of R_t to range from a posterior mean of 0.97 [0.14-2.14] for Norway to a

posterior mean of 2.64 [1.40-4.18] for Sweden, with an average of 1.43 across the 11 country posterior

means, a 64% reduction compared to the pre-intervention values. We note that these estimates are

contingent on intervention impact being the same in different countries and at different times. In all

countries but Sweden, under the same assumptions, we estimate that the current reproduction

number includes 1 in the uncertainty range. The estimated reproduction number for Sweden is higher,

not because the mortality trends are significantly different from any other country, but as an artefact

of our model, which assumes a smaller reduction in R_t because no full lockdown has been ordered so

far. Overall, we cannot yet conclude whether current interventions are sufficient to drive R_t below 1

(posterior probability of being less than 1.0 is 44% on average across the countries). We are also

unable to conclude whether interventions may be different between countries or over time.

There remains a high level of uncertainty in these estimates. It is too early to detect substantial

intervention impact in many countries at earlier stages of their epidemic (e.g. Germany, UK, Norway).

Many interventions have occurred only recently, and their effects have not yet been fully observed

due to the time lag between infection and death. This uncertainty will reduce as more data become

available. For all countries, our model fits observed deaths data well (Bayesian goodness of fit tests).

We also found that our model can reliably forecast daily deaths 3 days into the future, by withholding

the latest 3 days of data and comparing model predictions to observed deaths (Appendix 8.3).

The close spacing of interventions in time made it statistically impossible to determine which had the

greatest effect (Figure 1, Figure 4). However, when doing a sensitivity analysis (Appendix 8.4.3) with

uninformative prior distributions (where interventions can increase deaths) we find similar impact of

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interventions, which shows that our choice of prior distribution is not driving the effects we see in the

main analysis.

Figure 2: Country-level estimates of infections, deaths and R_t . Left: daily number of infections, brown

bars are reported infections, blue bands are predicted infections, dark blue 50% credible interval (CI),

light blue 95% CI. The number of daily infections estimated by our model drops immediately after an

intervention, as we assume that all infected people become immediately less infectious through the

intervention. Afterwards, if the R_t is above 1, the number of infections will start growing again.

Middle: daily number of deaths, brown bars are reported deaths, blue bands are predicted deaths, CI

as in left plot. Right: time-varying reproduction number R_t , dark green 50% CI, light green 95% CI.

Icons are interventions shown at the time they occurred.

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Table 2: Total forecasted deaths since the beginning of the epidemic up to 31 March in our model

and in a counterfactual model (assuming no intervention had taken place). Estimated averted deaths

over this time period as a result of the interventions. Numbers in brackets are 95% credible intervals.

2.3 Estimated impact of interventions on deaths

Table 2 shows total forecasted deaths since the beginning of the epidemic up to and including 31

March under our fitted model and under the counterfactual model, which predicts what would have

happened if no interventions were implemented (and $R_t = R_0$ i.e. the initial reproduction number

estimated before interventions). Again, the assumption in these predictions is that intervention

impact is the same across countries and time. The model without interventions was unable to capture

recent trends in deaths in several countries, where the rate of increase had clearly slowed (Figure 3).

Trends were confirmed statistically by Bayesian leave-one-out cross-validation and the widely

applicable information criterion assessments —WAIC).

By comparing the deaths predicted under the model with no interventions to the deaths predicted in

our intervention model, we calculated the total deaths averted up to the end of March. We find that,

across 11 countries, since the beginning of the epidemic, 59,000 [21,000-120,000] deaths have been

averted due to interventions. In Italy and Spain, where the epidemic is advanced, 38,000 [13,000-

84,000] and 16,000 [5,400-35,000] deaths have been averted, respectively. Even in the UK, which is

much earlier in its epidemic, we predict 370 [73-1,000] deaths have been averted.

These numbers give only the deaths averted that would have occurred up to 31 March. If we were to

include the deaths of currently infected individuals in both models, which might happen after 31

March, then the deaths averted would be substantially higher.

Figure 3: Daily number of confirmed deaths, predictions (up to 28 March) and forecasts (after) for (a)

Italy and (b) Spain from our model with interventions (blue) and from the no interventions counterfactual model (pink); credible intervals are shown one week into the future. Other countries

are shown in Appendix 8.6.

03/0 25% 50% 75% 100%

(no effect on transmissibility) (ends transmissibility)

Relative % reduction in R_t .

Figure 4: Our model includes five covariates for governmental interventions, adjusting for whether

the intervention was the first one undertaken by the government in response to COVID-19 (red) or

was subsequent to other interventions (green). Mean relative percentage reduction in R_t is shown

with 95% posterior credible intervals. If 100% reduction is achieved, $R_t = 0$ and there is no more

transmission of COVID-19. No effects are significantly different from any others, probably due to the

fact that many interventions occurred on the same day or within days of each other as shown in

Figure 1.

3 Discussion

During this early phase of control measures against the novel coronavirus in Europe, we analyze trends

in numbers of deaths to assess the extent to which transmission is being reduced.

Representing the

COVID-19 infection process using a semi-mechanistic, joint, Bayesian hierarchical model, we can

reproduce trends observed in the data on deaths and can forecast accurately over short time horizons.

We estimate that there have been many more infections than are currently reported. The high level

of under-ascertainment of infections that we estimate here is likely due to the focus on testing in

hospital settings rather than in the community. Despite this, only a small minority of individuals in

each country have been infected, with an attack rate on average of 4.9% [1.9%-11%] with considerable

variation between countries (Table 1). Our estimates imply that the populations in Europe are not

close to herd immunity ("50-75% if R_0 is 2-4). Further, with R_t values dropping substantially, the rate

of acquisition of herd immunity will slow down rapidly. This implies that the virus will be able to spread

rapidly should interventions be lifted. Such estimates of the attack rate to date urgently need to be

validated by newly developed antibody tests in representative population surveys, once these become available.

We estimate that major non-pharmaceutical interventions have had a substantial impact on the time-

varying reproduction numbers in countries where there has been time to observe intervention effects

on trends in deaths (Italy, Spain). If adherence in those countries has changed since that initial period,

then our forecast of future deaths will be affected accordingly: increasing adherence over time will

have resulted in fewer deaths and decreasing adherence in more deaths. Similarly, our estimates of

the impact of interventions in other countries should be viewed with caution if the same interventions

have achieved different levels of adherence than was initially the case in Italy and Spain.

Due to the implementation of interventions in rapid succession in many countries, there are not

enough data to estimate the individual effect size of each intervention, and we discourage attributing

associations to individual intervention. In some cases, such as Norway, where all interventions were

implemented at once, these individual effects are by definition unidentifiable. Despite this, while

individual impacts cannot be determined, their estimated joint impact is strongly empirically justified

(see Appendix 8.4 for sensitivity analysis). While the growth in daily deaths has decreased, due to the

lag between infections and deaths, continued rises in daily deaths are to be expected for some time.

To understand the impact of interventions, we fit a counterfactual model without the interventions

and compare this to the actual model. Consider Italy and the UK - two countries at very different stages

in their epidemics. For the UK, where interventions are very recent, much of the intervention strength

is borrowed from countries with older epidemics. The results suggest that interventions will have a

large impact on infections and deaths despite counts of both rising. For Italy, where far more time has

passed since the interventions have been implemented, it is clear that the model without interventions does not fit well to the data, and cannot explain the sub-linear (on the logarithmic scale)

reduction in deaths (see Figure 10).

The counterfactual model for Italy suggests that despite mounting pressure on health systems,

interventions have averted a health care catastrophe where the number of new deaths would have

been 3.7 times higher (38,000 deaths averted) than currently observed. Even in the UK, much earlier

in its epidemic, the recent interventions are forecasted to avert 370 total deaths up to 31 of March.

4 Conclusion and Limitations

Modern understanding of infectious disease with a global publicized response has meant that

nationwide interventions could be implemented with widespread adherence and support. Given

observed infection fatality ratios and the epidemiology of COVID-19, major non-pharmaceutical

interventions have had a substantial impact in reducing transmission in countries with more advanced

epidemics. It is too early to be sure whether similar reductions will be seen in countries at earlier

stages of their epidemic. While we cannot determine which set of interventions have been most

successful, taken together, we can already see changes in the trends of new deaths. When forecasting

3 days and looking over the whole epidemic the number of deaths averted is substantial. We note that

substantial innovation is taking place, and new more effective interventions or refinements of current

interventions, alongside behavioral changes will further contribute to reductions in infections. We

cannot say for certain that the current measures have controlled the epidemic in Europe; however, if

current trends continue, there is reason for optimism.

Our approach is semi-mechanistic. We propose a plausible structure for the infection process and then

estimate parameters empirically. However, many parameters had to be given strong prior distributions or had to be fixed. For these assumptions, we have provided relevant citations to

previous studies. As more data become available and better estimates arise, we will update these in

weekly reports. Our choice of serial interval distribution strongly influences the prior distribution for

starting R_0 . Our infection fatality ratio, and infection-to-onset-to-death distributions strongly

influence the rate of death and hence the estimated number of true underlying cases.

We also assume that the effect of interventions is the same in all countries, which may not be fully

realistic. This assumption implies that countries with early interventions and more deaths since these

interventions (e.g. Italy, Spain) strongly influence estimates of intervention impact in countries at

earlier stages of their epidemic with fewer deaths (e.g. Germany, UK).

We have tried to create consistent definitions of all interventions and document details of this in

Appendix 8.6. However, invariably there will be differences from country to country in the strength of

their intervention — for example, most countries have banned gatherings of more than 2 people when

implementing a lockdown, whereas in Sweden the government only banned gatherings of more than

10 people. These differences can skew impacts in countries with very little data. We believe that our

uncertainty to some degree can cover these differences, and as more data become available,

coefficients should become more reliable.

However, despite these strong assumptions, there is sufficient signal in the data to estimate changes

in R , (see the sensitivity analysis reported in Appendix 8.4.3) and this signal will stand to increase with

time. In our Bayesian hierarchical framework, we robustly quantify the uncertainty in our parameter

estimates and posterior predictions. This can be seen in the very wide credible intervals in more recent

days, where little or no death data are available to inform the estimates. Furthermore, we predict

intervention impact at country-level, but different trends may be in place in different parts of each

country. For example, the epidemic in northern Italy was subject to controls earlier than the rest of

the country.

5 Data

Our model utilizes daily real-time death data from the ECDC (European Centre of Disease Control),

where we catalogue case data for 11 European countries currently experiencing the epidemic: Austria,

Belgium, Denmark, France, Germany, Italy, Norway, Spain, Sweden, Switzerland and the United

Kingdom. The ECDC provides information on confirmed cases and deaths attributable to COVID-19.

However, the case data are highly unrepresentative of the incidence of infections due to underreporting as well as systematic and country-specific changes in testing.

We, therefore, use only deaths attributable to COVID-19 in our model; we do not use the ECDC case

estimates at all. While the observed deaths still have some degree of unreliability, again due to

changes in reporting and testing, we believe the data are of sufficient fidelity to model. For population

counts, we use UNPOP age-stratified counts.¹⁰

We also catalogue data on the nature and type of major non-pharmaceutical interventions. We looked

at the government webpages from each country as well as their official public health division/information webpages to identify the latest advice/laws being issued by the government and

public health authorities. We collected the following:

School closure ordered: This intervention refers to nationwide extraordinary school closures which in

most cases refer to both primary and secondary schools closing (for most countries this also includes

the closure of other forms of higher education or the advice to teach remotely). In the case of Denmark

and Sweden, we allowed partial school closures of only secondary schools. The date of the school

closure is taken to be the effective date when the schools started to be closed (if this was on a Monday,

the date used was the one of the previous Saturdays as pupils and students effectively stayed at home

from that date onwards).

Case-based measures: This intervention comprises strong recommendations or laws to the general

public and primary care about self—isolation when showing COVID-19-like symptoms. These also

include nationwide testing programs where individuals can be tested and subsequently self—isolated.

Our definition is restricted to nationwide government advice to all individuals (e.g. UK) or to all primary

care and excludes regional only advice. These do not include containment phase interventions such

as isolation if travelling back from an epidemic country such as China.

Public events banned: This refers to banning all public events of more than 100 participants such as

sports events.

Social distancing encouraged: As one of the first interventions against the spread of the COVID-19

pandemic, many governments have published advice on social distancing including the recommendation to work from home wherever possible, reducing use of public transport and all other

non-essential contact. The dates used are those when social distancing has officially been recommended by the government; the advice may include maintaining a recommended physical

distance from others.

Lockdown decreed: There are several different scenarios that the media refers to as lockdown. As an

overall definition, we consider regulations/legislations regarding strict face-to-face social interaction:

including the banning of any non-essential public gatherings, closure of educational and

public/cultural institutions, ordering people to stay home apart from exercise and essential tasks. We

include special cases where these are not explicitly mentioned on government websites but are

enforced by the police (e.g. France). The dates used are the effective dates when these legislations

have been implemented. We note that lockdown encompasses other interventions previously

implemented.

First intervention: As Figure 1 shows, European governments have escalated interventions rapidly,

and in some examples (Norway/Denmark) have implemented these interventions all on a single day.

Therefore, given the temporal autocorrelation inherent in government intervention, we include a

binary covariate for the first intervention, which can be interpreted as a government decision to take

major action to control COVID-19.

A full list of the timing of these interventions and the sources we have used can be found in Appendix

8.6.

6 Methods Summary

A Visual summary of our model is presented in Figure 5 (details in Appendix 8.1 and 8.2).

Replication

code is available at

<https://github.com/ImperiaCollegeLondon/covid19model/releases/tag/v1.0>

We fit our model to observed deaths according to ECDC data from 11 European countries.

The

modelled deaths are informed by an infection-to-onset distribution (time from infection to the onset

of symptoms), an onset-to-death distribution (time from the onset of symptoms to death), and the

population-averaged infection fatality ratio (adjusted for the age structure and contact patterns of

each country, see Appendix). Given these distributions and ratios, modelled deaths are a function of

the number of infections. The modelled number of infections is informed by the serial interval

distribution (the average time from infection of one person to the time at which they infect another)

and the time-varying reproduction number. Finally, the time-varying reproduction number is a

function of the initial reproduction number before interventions and the effect sizes from interventions.

Figure 5: Summary of model components.

Following the hierarchy from bottom to top gives us a full framework to see how interventions affect

infections, which can result in deaths. We use Bayesian inference to ensure our modelled deaths can

reproduce the observed deaths as closely as possible. From bottom to top in Figure 5, there is an

implicit lag in time that means the effect of very recent interventions manifest weakly in current

deaths (and get stronger as time progresses). To maximise the ability to observe intervention impact

on deaths, we fit our model jointly for all 11 European countries, which results in a large data set. Our

model jointly estimates the effect sizes of interventions. We have evaluated the effect of our Bayesian

prior distribution choices and evaluate our Bayesian posterior calibration to ensure our results are

statistically robust (Appendix 8.4).

7 Acknowledgements

Initial research on covariates in Appendix 8.6 was crowdsourced; we thank a number of people

across the world for help with this. This work was supported by Centre funding from the UK Medical

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8 Appendix: Model Specifics, Validation and Sensitivity Analysis

8.1 Death model

We observe daily deaths D_{am} for days $t \in 1, \dots, n$ and countries $m \in 1, \dots, p$. These daily deaths are

modelled using a positive real-valued function $\text{dam} = E(D_{am})$ that represents the expected number

of deaths attributed to COVID-19. D_{am} is assumed to follow a negative binomial distribution with

The expected number of deaths (D_{am}) in a given country on a given day is a function of the number of

infections C occurring in previous days.

At the beginning of the epidemic, the observed deaths in a country can be dominated by deaths that

result from infection that are not locally acquired. To avoid biasing our model by this, we only include

observed deaths from the day after a country has cumulatively observed 10 deaths in our model.

To mechanistically link our function for deaths to infected cases, we use a previously estimated COVID-

19 infection-fatality-ratio ifr (probability of death given infection)⁹ together with a distribution of times

from infection to death T_E . The ifr is derived from estimates presented in Verity et al¹¹ which assumed

homogeneous attack rates across age-groups. To better match estimates of attack rates by age

generated using more detailed information on country and age-specific mixing patterns, we scale

these estimates (the unadjusted ifr, referred to here as ifr') in the following way as in previous work.⁴

Let C_a be the number of infections generated in age-group a , N_a the underlying size of the population

in that age group and $AR_a = 2 C_a / N_a$ the age-group-specific attack rate. The adjusted ifr is then given

by: $ifr_a = ifr_{ref} \cdot AR_{50-59} / AR_a$, where AR_{50-59} is the predicted attack-rate in the 50-59 year age-group after

incorporating country-specific patterns of contact and mixing. This age-group was chosen as the

reference as it had the lowest predicted level of underreporting in previous analyses of data from the

Chinese epidemic⁴. We obtained country-specific estimates of attack rate by age, AR_a , for the 11

European countries in our analysis from a previous study which incorporates information on contact

between individuals of different ages in countries across Europe.¹² We then obtained overall ifr

estimates for each country adjusting for both demography and age-specific attack rates.

Using estimated epidemiological information from previous studies,¹¹ we assume TE to be the sum of

two independent random times: the incubation period (infection to onset of symptoms or infection-

to-onset) distribution and the time between onset of symptoms and death (onset-to-death). The

infection-to-onset distribution is Gamma distributed with mean 5.1 days and coefficient of variation

0.86. The onset-to-death distribution is also Gamma distributed with a mean of 18.8 days and a

coefficient of variation 0.45. ifrm is population averaged over the age structure of a given country. The

infection-to-death distribution is therefore given by:

$$um \sim ifrm \sim (\text{Gamma}(5.1, 0.86) + \text{Gamma}(18.8, 0.45))$$

Figure 6 shows the infection-to-death distribution and the resulting survival function that integrates

to the infection fatality ratio.

Figure 6: Left, infection-to-death distribution (mean 23.9 days). Right, survival probability of infected

individuals per day given the infection fatality ratio (1%) and the infection-to-death distribution on

the left.

Using the probability of death distribution, the expected number of deaths dam_t , on a given day t , for

country, m , is given by the following discrete sum:

The number of deaths today is the sum of the past infections weighted by their probability of death,

where the probability of death depends on the number of days since infection.

8.2 Infection model

The true number of infected individuals, C , is modelled using a discrete renewal process. This approach

has been used in numerous previous studies^{13,16} and has a strong theoretical basis in stochastic

individual-based counting processes such as Hawkes process and the Bellman-Harris process.”¹⁸ The

renewal model is related to the Susceptible-Infected-Recovered model, except the renewal is not

expressed in differential form. To model the number of infections over time we need to specify a serial

interval distribution g with density $g(T)$, (the time between when a person gets infected and when

they subsequently infect another other people), which we choose to be Gamma distributed:

$g \sim \text{Gamma}(6.5, 0.62)$.

The serial interval distribution is shown below in Figure 7 and is assumed to be the same for all countries.

Figure 7: Serial interval distribution g with a mean of 6.5 days.

Given the serial interval distribution, the number of infections $E_{m,t}$ on a given day t , and country, m ,

is given by the following discrete convolution function:

$$E_{m,t} = \sum_{\tau=0}^{t-1} C_{m,\tau} I_{m,t-\tau}$$

$$C_{m,\tau} = R_{m,\tau} \sum_{s=0}^{\tau} P_{m,s} I_{m,t-s}$$

where, similar to the probability of death function, the daily serial interval is discretized by

$$P_{m,s} = \frac{1}{\Delta t} \int_{(s-0.5)\Delta t}^{(s+0.5)\Delta t} p(t) dt$$

$$P_{m,s} = \frac{1}{\Delta t} \int_{(s-0.5)\Delta t}^{(s+0.5)\Delta t} p(t) dt$$

$$I_{m,t} = \sum_{s=0}^{t-1} C_{m,s} I_{m,t-s} + \sum_{s=0}^{t-1} O_{m,s} I_{m,t-s}$$

Infections today depend on the number of infections in the previous days, weighted by the discretized

serial interval distribution. This weighting is then scaled by the country-specific time-Varying

reproduction number, $R_{m,t}$, that models the average number of secondary infections at a given time.

The functional form for the time-Varying reproduction number was chosen to be as simple as possible

to minimize the impact of strong prior assumptions: we use a piecewise constant function that scales

$R_{m,t}$ from a baseline prior $R_{0,m}$ and is driven by known major non-pharmaceutical interventions

occurring in different countries and times. We included 6 interventions, one of which is constructed

from the other 5 interventions, which are timings of school and university closures ($k=1$), self-isolating

if ill ($k=2$), banning of public events ($k=3$), any government intervention in place ($k=4$), implementing

a partial or complete lockdown ($k=5$) and encouraging social distancing and isolation ($k=6$). We denote

the indicator variable for intervention $k \in 1, 2, 3, 4, 5, 6$ by $I_{k,t}^m$, which is 1 if intervention k is in place

in country m at time t and 0 otherwise. The covariate "any government intervention" ($k=4$) indicates

if any of the other 5 interventions are in effect, i.e. $I_{4,t}^m$ equals 1 at time t if any of the interventions

$k \in 1, 2, 3, 4, 5$ are in effect in country m at time t and equals 0 otherwise. Covariate 4 has the interpretation of indicating the onset of major government intervention. The effect of each intervention is assumed to be multiplicative. $R_{0,t}^m$ is therefore a function of the intervention indicators

$I_{k,t}^m$ in place at time t in country m :

$R_{0,t}^m = R_{0,m} \exp\left(-\sum_{k=1}^6 \beta_k I_{k,t}^m\right)$

The exponential form was used to ensure positivity of the reproduction number, with $R_{0,m}$ constrained to be positive as it appears outside the exponential. The impact of each intervention on

$R_{0,t}^m$ is characterised by a set of parameters $\beta_1, \dots, \beta_6$, with independent prior distributions chosen

to be

$\text{ock} \sim \text{Gamma}(.5, 1)$.

The impacts ock are shared between all m countries and therefore they are informed by all available

data. The prior distribution for R_0 was chosen to be

$R_{0,m} \sim \text{Normal}(2.4, \text{IKI})$ with $K \sim \text{Normal}(0, 0.5)$,

Once again, K is the same among all countries to share information.

We assume that seeding of new infections begins 30 days before the day after a country has

cumulatively observed 10 deaths. From this date, we seed our model with 6 sequential days of

infections drawn from $\text{cl}'_m, \dots, 66'_m \sim \text{EXponential}(T)$, where $T \sim \text{Exponential}(0.03)$. These seed

infections are inferred in our Bayesian posterior distribution.

We estimated parameters jointly for all 11 countries in a single hierarchical model. Fitting was done

in the probabilistic programming language Stan,¹⁹ using an adaptive Hamiltonian Monte Carlo (HMC)

sampler. We ran 8 chains for 4000 iterations with 2000 iterations of warmup and a thinning factor 4

to obtain 2000 posterior samples. Posterior convergence was assessed using the Rhat statistic and by

diagnosing divergent transitions of the HMC sampler. Prior-posterior calibrations were also performed

(see below).

8.3 Validation

We validate accuracy of point estimates of our model using cross-validation. In our cross-validation

scheme, we leave out 3 days of known death data (non-cumulative) and fit our model. We forecast

what the model predicts for these three days. We present the individual forecasts for each day, as

well as the average forecast for those three days. The cross-validation results are shown in the Figure

8.

Figure 8: Cross-Validation results for 3-day and 3-day aggregated forecasts

Figure 8 provides strong empirical justification for our model specification and mechanism. Our

accurate forecast over a three-day time horizon suggests that our fitted estimates for R_t are appropriate and plausible.

Along with from point estimates we all evaluate our posterior credible intervals using the Rhat

statistic. The Rhat statistic measures whether our Markov Chain Monte Carlo (MCMC) chains have

converged to the equilibrium distribution (the correct posterior distribution). Figure 9 shows the Rhat

statistics for all of our parameters

Figure 9: Rhat statistics - values close to 1 indicate MCMC convergence.

Figure 9 indicates that our MCMC have converged. In fitting we also ensured that the MCMC sampler

experienced no divergent transitions - suggesting non pathological posterior topologies.

8.4 SensitivityAnalysis

8.4.1 Forecasting on log-linear scale to assess signal in the data

As we have highlighted throughout in this report, the lag between deaths and infections means that

it takes time for information to propagate backwards from deaths to infections, and ultimately to R_t .

A conclusion of this report is the prediction of a slowing of R_t in response to major interventions. To

gain intuition that this is data driven and not simply a consequence of highly constrained model

assumptions, we show death forecasts on a log-linear scale. On this scale a line which curves below a

linear trend is indicative of slowing in the growth of the epidemic. Figure 10 to Figure 12 show these

forecasts for Italy, Spain and the UK. They show this slowing down in the daily number of deaths. Our

model suggests that Italy, a country that has the highest death toll of COVID-19, will see a slowing in

the increase in daily deaths over the coming week compared to the early stages of the epidemic.

We investigated the sensitivity of our estimates of starting and final R_t to our assumed serial interval

distribution. For this we considered several scenarios, in which we changed the serial interval

distribution mean, from a value of 6.5 days, to have values of 5, 6, 7 and 8 days.

In Figure 13, we show our estimates of R_0 , the starting reproduction number before interventions, for

each of these scenarios. The relative ordering of the $R_t=0$ in the countries is consistent in all settings.

However, as expected, the scale of $R_t=0$ is considerably affected by this change — a longer serial

interval results in a higher estimated $R_t=0$. This is because to reach the currently observed size of the

epidemics, a longer assumed serial interval is compensated by a higher estimated R_0 .

Additionally, in Figure 14, we show our estimates of R_t at the most recent model time point, again for

each of these scenarios. The serial interval mean can influence R_t substantially, however, the posterior

credible intervals of R_t are broadly overlapping.

Figure 13: Initial reproduction number R_0 for different serial interval (SI) distributions (means

between 5 and 8 days). We use 6.5 days in our main analysis.

Figure 14: R_t on 28 March 2020 estimated for all countries, with serial interval (SI) distribution means

between 5 and 8 days. We use 6.5 days in our main analysis.

8.4.3 Uninformative prior sensitivity on or

We ran our model using implausible uninformative prior distributions on the intervention effects,

allowing the effect of an intervention to increase or decrease R_t . To avoid collinearity, we ran 6

separate models, with effects summarized below (compare with the main analysis in Figure 4). In this

series of univariate analyses, we find (Figure 15) that all effects on their own serve to decrease R_t .

This gives us confidence that our choice of prior distribution is not driving the effects we see in the

main analysis. Lockdown has a very large effect, most likely due to the fact that it occurs after other

interventions in our dataset. The relatively large effect sizes for the other interventions are most likely

due to the coincidence of the interventions in time, such that one intervention is a proxy for a few

others.

Figure 15: Effects of different interventions when used as the only covariate in the model.

8.4.4

To assess prior assumptions on our piecewise constant functional form for R_t we test using a

nonparametric function with a Gaussian process prior distribution. We fit a model with a Gaussian

process prior distribution to data from Italy where there is the largest signal in death data. We find

that the Gaussian process has a very similar trend to the piecewise constant model and reverts to the

mean in regions of no data. The correspondence of a completely nonparametric function and our

piecewise constant function suggests a suitable parametric specification of R_t .

Nonparametric fitting of R_f using a Gaussian process:

8.4.5 Leave country out analysis

Due to the different lengths of each European countries' epidemic, some countries, such as Italy have

much more data than others (such as the UK). To ensure that we are not leveraging too much

information from any one country we perform a "leave one country out" sensitivity analysis, where

we rerun the model without a different country each time. Figure 16 and Figure 17 are examples for

results for the UK, leaving out Italy and Spain. In general, for all countries, we observed no significant

dependence on any one country.

Figure 16: Model results for the UK, when not using data from Italy for fitting the model. See the

Figure 17: Model results for the UK, when not using data from Spain for fitting the model. See caption

of Figure 2 for an explanation of the plots.

8.4.6 Starting reproduction numbers vs theoretical predictions

To validate our starting reproduction numbers, we compare our fitted values to those theoretically

expected from a simpler model assuming exponential growth rate, and a serial interval distribution

mean. We fit a linear model with a Poisson likelihood and log link function and extracting the daily

growth rate r . For well-known theoretical results from the renewal equation, given a serial interval

distribution $g(r)$ with mean m and standard deviation 5 , given $a = mZ/S^2$ and $b = m/SZ$, and

a

subsequently $R_0 = (1 + \%)$. Figure 18 shows theoretically derived R_0 along with our fitted

estimates of $R_{t=0}$ from our Bayesian hierarchical model. As shown in Figure 18 there is large

correspondence between our estimated starting reproduction number and the basic reproduction

number implied by the growth rate r .

R_0 (red) vs $R(FO)$ (black)

Figure 18: Our estimated R_0 (black) versus theoretically derived R_u (red) from a log-linear regression fit.

8.5 Counterfactual analysis — interventions vs no interventions

Figure 19: Daily number of confirmed deaths, predictions (up to 28 March) and forecasts (after) for

all countries except Italy and Spain from our model with interventions (blue) and from the no

interventions counterfactual model (pink); credible intervals are shown one week into the future.

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8.6 Data sources and Timeline of Interventions

Figure 1 and Table 3 display the interventions by the 11 countries in our study and the dates these

interventions became effective.

Table 3: Timeline of Interventions.

Country	Type	Event	Date effective
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	School closure		
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	ordered Nationwide school closures.	20	14/3/2020
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	Public events		
--	---------------	--	--

	banned Banning of gatherings of more than 5 people.	21	10/3/2020
--	---	----	-----------

	Banning all access to public spaces and gatherings		
--	--	--	--

	Lockdown of more than 5 people. Advice to maintain 1m		
--	---	--	--

	ordered distance.	22	16/3/2020
--	-------------------	----	-----------

	Social distancing		
--	-------------------	--	--

	encouraged Recommendation to maintain a distance of 1m.	22	16/3/2020
--	---	----	-----------

	Case-based		
--	------------	--	--

	Austria measures Implemented at lockdown.	22	16/3/2020
--	---	----	-----------

	School closure		
--	----------------	--	--

	ordered Nationwide school closures.	23	14/3/2020
--	-------------------------------------	----	-----------

	Public events All recreational activities cancelled regardless of		
--	---	--	--

	banned size.	23	12/3/2020
--	--------------	----	-----------

	Citizens are required to stay at home except for		
--	--	--	--

	Lockdown work and essential journeys. Going outdoors only		
--	---	--	--

	ordered with household members or 1 friend.	24	18/3/2020
--	---	----	-----------

Public transport recommended only for essential

Social distancing journeys, work from home encouraged, all public encouraged places e.g. restaurants closed.²³ 14/3/2020

Case-based Everyone should stay at home if experiencing a Belgium measures cough or fever.²⁵ 10/3/2020

School closure Secondary schools shut and universities (primary ordered schools also shut on 16th).²⁶ 13/3/2020

Public events Bans of events >100 people, closed cultural banned institutions, leisure facilities etc.²⁷ 12/3/2020

Lockdown Bans of gatherings of >10 people in public and all ordered public places were shut.²⁷ 18/3/2020

Limited use of public transport. All cultural

Social distancing institutions shut and recommend keeping encouraged appropriate distance.²⁸ 13/3/2020

Case-based Everyone should stay at home if experiencing a Denmark measures cough or fever.²⁹ 12/3/2020

School closure

ordered Nationwide school closures.³⁰ 14/3/2020

Public events

banned Bans of events >100 people.³¹ 13/3/2020

Lockdown Everybody has to stay at home. Need a self-ordered authorisation form to leave home.³² 17/3/2020

Social distancing

encouraged Advice at the time of lockdown.³² 16/3/2020

Case-based

France measures Advice at the time of lockdown.³² 16/03/2020

School closure

ordered Nationwide school closures.³³ 14/3/2020

Public events No gatherings of >1000 people. Otherwise

banned regional restrictions only until lockdown.³⁴ 22/3/2020

Lockdown Gatherings of > 2 people banned, 1.5 m

ordered distance.³⁵ 22/3/2020

Social distancing Avoid social interaction wherever possible

encouraged recommended by Merkel.³⁶ 12/3/2020

Advice for everyone experiencing symptoms to

Case-based contact a health care agency to get tested and

Germany measures then self—isolate.³⁷ 6/3/2020

School closure

ordered Nationwide school closures.³⁸ 5/3/2020

Public events

banned The government bans all public events.³⁹ 9/3/2020

Lockdown The government closes all public places. People

ordered have to stay at home except for essential travel.⁴⁰ 11/3/2020

A distance of more than 1m has to be kept and

Social distancing any other form of alternative aggregation is to be

encouraged excluded.⁴⁰ 9/3/2020

Case-based Advice to self—isolate if experiencing symptoms

Italy measures and quarantine if tested positive.⁴¹ 9/3/2020

Norwegian Directorate of Health closes all

School closure educational institutions. Including childcare

ordered facilities and all schools.⁴² 13/3/2020

Public events The Directorate of Health bans all non-necessary
banned social contact.⁴² 12/3/2020

Lockdown Only people living together are allowed outside
ordered together. Everyone has to keep a 2m distance.⁴³ 24/3/2020

Social distancing The Directorate of Health advises against all
encouraged travelling and non-necessary social contacts.⁴² 16/3/2020

Case-based Advice to self—isolate for 7 days if experiencing a
Norway measures cough or fever symptoms.⁴⁴ 15/3/2020

ordered Nationwide school closures.⁴⁵ 13/3/2020

Public events

banned Banning of all public events by lockdown.⁴⁶ 14/3/2020

Lockdown

ordered Nationwide lockdown.⁴³ 14/3/2020

Social distancing Advice on social distancing and working remotely
encouraged from home.⁴⁷ 9/3/2020

Case-based Advice to self—isolate for 7 days if experiencing a
Spain measures cough or fever symptoms.⁴⁷ 17/3/2020

School closure

ordered Colleges and upper secondary schools shut.⁴⁸ 18/3/2020

Public events

banned The government bans events >500 people.⁴⁹ 12/3/2020

Lockdown

ordered No lockdown occurred. NA

People even with mild symptoms are told to limit

Social distancing social contact, encouragement to work from

encouraged home.⁵⁰ 16/3/2020

Case-based Advice to self—isolate if experiencing a cough or

Sweden measures fever symptoms.⁵¹ 10/3/2020

School closure

ordered No in person teaching until 4th of April.⁵² 14/3/2020

Public events

banned The government bans events >100 people.⁵² 13/3/2020

Lockdown

ordered Gatherings of more than 5 people are banned.⁵³ 2020-03-20

Advice on keeping distance. All businesses where

Social distancing this cannot be realised have been closed in all

encouraged states (kantons).⁵⁴ 16/3/2020

Case-based Advice to self—isolate if experiencing a cough or

Switzerland measures fever symptoms.⁵⁵ 2/3/2020

Nationwide school closure. Childminders,

School closure nurseries and sixth forms are told to follow the

ordered guidance.⁵⁶ 21/3/2020

Public events

banned Implemented with lockdown.⁵⁷ 24/3/2020

Gatherings of more than 2 people not from the

Lockdown same household are banned and police

ordered enforceable.⁵⁷ 24/3/2020

Social distancing Advice to avoid pubs, clubs, theatres and other

encouraged public institutions.⁵⁸ 16/3/2020

Case-based Advice to self—isolate for 7 days if experiencing a

UK measures cough or fever symptoms.⁵⁹ 12/3/2020

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Knowledge, Attitudes and Practices (KAP) related to the Pandemic (H1N1) 2009 among
Chinese General Population: a Telephone Survey

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Abstract: BACKGROUND: China is at greatest risk of the Pandemic (H1N1) 2009 due to its huge population and high residential density. The unclear comprehension and negative attitudes towards the emerging infectious disease among general population may lead to unnecessary worry and even panic. The objective of this study was to investigate the Chinese public response to H1N1 pandemic and provide baseline data to develop public

education campaigns in response to future outbreaks. METHODS: A close-ended questionnaire developed by the Chinese Center for Disease Control and Prevention was applied to assess the knowledge, attitudes and practices (KAP) of pandemic (H1N1) 2009 among 10,669 responders recruited from seven urban and two rural areas of China sampled by using the probability proportional to size (PPS) method. RESULTS: 30.0% respondents were not clear whether food spread H1N1 virus and. 65.7% reported that the pandemic had no impact on their life. The immunization rates of the seasonal flu and H1N1 vaccine were 7.5% and 10.8%, respectively. Farmers and those with lower education level were less likely to know the main transmission route (cough or talk face to face). Female and those with college and above education had higher perception of risk and more compliance with preventive behaviors. Relationships between knowledge and risk perception (OR = 1.69; 95%CI 1.54-1.86), and knowledge and practices (OR = 1.57; 95%CI 1.42-1.73) were found among the study subjects. With regard to the behavior of taking up A/H1N1 vaccination, there are several related factors found in the current study population, including the perception of life disturbed (OR = 1.29; 95%CI 1.11-1.50), the safety of A/H1N1 vaccine (OR = 0.07; 95%CI 0.04-0.11), the knowledge of free vaccination policy (OR = 7.20; 95%CI 5.91-8.78), the state's priority vaccination strategy (OR = 1.33; 95%CI 1.08-1.64), and taking up seasonal influenza vaccine behavior (OR = 4.69; 95%CI 3.53-6.23). CONCLUSIONS: This A/H1N1 epidemic has not caused public panic yet, but the knowledge of A/H1N1 in residents is not optimistic. Public education campaign may take the side effects of vaccine and the knowledge about the state's vaccination strategy into account.

Text: At the end of March 2009, an outbreak of novel influenza A (H1N1) (here after called A/H1N1) infection occurred in Mexico, followed by ongoing spread to all over the world in a short period [1] . On June 11 2009, the World Health Organization raised its pandemic alert level to the highest level, phase 6 [2] , meaning that the A/H1N1 flu had spread in more than two continents and reached pandemic proportions. As of June 13, 2010, it had caused over 18,172 deaths in more than 214 countries and overseas territories or communities [3] . Most illness, especially the severe illness and deaths, had occurred among healthy young adults, which was markedly different from the disease pattern seen during epidemics of seasonal influenza [4, 5] .

China is highly susceptible to A/H1N1 because of its huge population and high residential density, besides the high infectiousness of this novel influenza virus. After the first imported case reported on May 11, 2009 , the confirmed cases were reported in various

provinces of China [6] . By the late of October 2009, A/H1N1 cases had increased dramatically, with 44,981 cases and 6 deaths confirmed at the end of October 2009. The A/H1N1 infection rate peaked in November 2009, when approximately 1500 new cases of A/H1N1 were being confirmed each day. By the end of this month, a total of 92,904 cases and 200 deaths had resulted from A/ H1N1-related causes [7] . The Chinese government has taken a series of preventive measures according to WHO guidelines, including the promotion of public knowledge about flu through mass media, patient isolation, quarantine of close contact person, and free vaccinations to population at high risk (e.g. young children, healthcare workers, and people with chronic disease) [8] . However, there were few public reports on the assessment of the effect of these policies and the level of knowledge, attitude and practice (KAP) associating with A/H1N1 among general population.

It is well-known that confused comprehension and negative attitude towards the emerging communicable disease may lead to unnecessary worry and chaos, even excessive panic which would aggravate the disease epidemic [9] . For instance, during SARS epidemic from 2002 to 2004, the misconceptions and the excessive panic of Chinese public to SARS led the public resistant to comply with the suggested preventive measures such as avoiding public transportation, going to hospital when they were sick, which contributed to the rapid spread of SARS and resulted in a more serious epidemic situation, making China one of the worst affected countries with over 5327 cases and 439 deaths [10, 11] . In addition, the panic of infectious disease outbreak could cause huge economic loss, for example the economic loss of SARS has been estimated at \$30-\$100 billion in US, though less than 10,000 persons were infected [12] . SARS experience has demonstrated the importance of monitoring the public perception in disease epidemic control, which may affect the compliance of community to the precautionary strategies. Understanding related factors affecting people to undertake precautionary behavior may also help decision-makers take appropriate measures to promote individual or community health. Therefore, it is important to monitor and analyze the public response to the emerging disease.

To investigate community responses to A/H1N1 in China, we conducted this telephone survey to describe the knowledge, attitudes and practices of A/H1N1 among general population in China and put forward policy recommendations to government in case of future similar conditions.

This study was performed in seven urban regions (Beijing, Shanghai, Wuhan, Jingzhou, Xi'an, Zhengzhou, Shenzhen cities) and two rural areas (Jingzhou and Zhengzhou counties) of China with over one million people in each region. Regarding the urban sites, Beijing as the capital of China locates in the northeast; Shanghai is a municipality in the east of China; Wuhan (the provincial capital of Hubei) and Zhengzhou (the provincial capital of Henan province) are both in the centre of China; Xi'an in the northwest of China is the provincial capital of Shanxi province; and Shenzhen of the Guangdong province is in the southeast of China. As for the rural sites, Jingzhou county and Zhengzhou county, from Hubei and Henan provinces, respectively, both locate in the centre of China.

This current study was carried out in three phases during the pandemic peak season of A/H1N1. The first phase was from 30 November 2009 to 27 December 2009, the second from 4 January 2010 to 24 January 2010, and the third from 24 February to 25 March in 2010.

A two-stage proportional probability to size (PPS) sampling method was used in each phase. In stage I, about 30% of administrative regions in each study site were selected as primary sample units (PSUs) for cluster sampling. In stage II, telephone numbers were sampled randomly, of which the first four digitals were obtained from each PSU's post office as initial number and the other three or four digitals were obtained from random number generated by Excel 2003. Then each family was chosen as per unit (excluding school, hotel public or cell phone etc.) and at least 400 families in each site at each phase were selected finally. If the family was selected repeatedly or refused to answer the questionnaire, we added one to the last digit of phone number and dial again. If the line was busy or of no response, we would dial three times and then give up this phone number if there was still no respondent.

Anonymous telephone interviews were conducted from 6:30 pm to 10:00 pm so as to avoid over-presenting the non-work population by well-trained interviewers with Bachelor degree of Epidemiology. The Questionnaire to Survey the Level of Knowledge, Attitude and Practice in Different Stages of A/H1N1 Pandemic by Telephone was designed by the Chinese Centre for Disease Control and Prevention (China CDC, Beijing). The majority of the questions were closed-ended and variables in the questionnaire were categorical, except age. The inclusion criteria of subjects were: age ≥ 18 and proper communication skills. There were seven questions related to the knowledge of A/H1N1, four referred to the attitude, and five

concerning about the practice in this questionnaire (See additional file 1: The Questionnaire to Survey the Level of Knowledge, Attitude and Practice in Different Stages of H1N1 Pandemic by Telephone in China).

This study was approved by the institutional review board of the Tongji Medical College of Huazhong University of Science and Technology. All respondents were informed consent. We respected their wishes whether to accept our survey and promised to protect their secrets.

All data were entered into computer using Epidata V.3.1 and were analyzed in SPSS statistical software V.12. Chi-square test was applied to compare the immunization rates of the seasonal flu and A/H1N1 vaccine. The associations between the socio-demographic factors and the KAP regarding A/H1N1 were firstly investigated by using univariate odds ratios (OR) and then stepwise logistic regression modeling applied. Adjusting for such background variables including gender, age, level of education, occupation, region, and survey wave, stepwise multivariate logistic regression models were applied to investigate the impact factors associated with the risk perception of A/H1N1, A/H1N1 vaccination uptake and the compliance with suggested preventive measures (avoid crowd places/wash hand frequently/keep distance from people with influenza-like symptoms). For the purposes of analysis, the factor knowledge about the main modes of transmission was divided into two groups according to whether the respondents knew both cough and talk face-to-face can spread A/H1N1. Odds ratios and respective 95% confidence intervals (CI) were obtained from the logistic regression analysis. P values lower than 0.05 were judged to be statistically significant.

A total of 88541 telephone numbers were dialed. Except 65323 invalid calls (including vacant numbers, fax numbers, busy tone numbers and non-qualified respondents whose age <18 and whose phones were from school, hotel or other public places), 23218 eligible respondents were identified. Among these respondents, 12360 completed the interview. Therefore, the response rate was 46.8%. Excluding missing, and logical erroneous data, 10669 questionnaires in total were eligible for analysis. The baseline characteristics of the respondents were presented in Table1. The mean age of all respondents was 41.47 years (over range: 18-90 year) . Of all respondents, 54.4% were female, and 42.4% had received college or above education (Table 1) .

The overall KAP related to A/H1N1 was reported in Table 2 . As to knowledge, 75.6% of all respondents knew that influenza could be transmitted by coughing and sneezing, and 61.9% thought that talking face-to-face was the transmission route, whereas 30.0% believed the transmission could be through food. Less than one third of respondents knew that virus could be transmitted by handshaking and indirect hand contact (26.8% and 22.3%, respectively). Multiple logistic regression analysis showed that those with middle school (OR = 1.71; 95%CI 1.48-1.98), or having an education level of college and above (OR = 2.16; 95%CI 1.83-2.54) were more likely to know the transmission routes comparing with other people. Comparing with students, teachers (OR = 1.46; 95%CI 1.09-1.96) were more likely to answer the above questions Table 3 and Table 4). Regarding the A/H1N1vaccination, 69.9% respondents believed that the occurrence rate of adverse reactions caused by A/H1N1 vaccination was fairly low and they were not afraid of taking up vaccination. Most residents (96.1%) thought that the state's vaccination strategy was reasonable.

About half of the respondents (42.9%) had avoided going to crowded places during the past two weeks of our survey. In case people nearby held influenza-like symptoms such as fever or cough, 56.9% increased the frequency of hand-washing and 57.4% would stay away from them. Multiple logistic regression analysis indicated compliance with the preventive practices were more likely to be taken by those who were females (OR = 1. Table 3 and Table 4). The immunization rates of the seasonal flu and A/ H1N1 in respondents were 7.5% and 10.8% respectively. The multivariate stepwise models further showed that except the health care workers (OR = 1.52; 95%CI 1.09-2.11), residents in other occupations (OR = 0.06-0.67) were less likely to take up the A/H1N1 vaccination comparing with students (in Table 3). Adjusting for the background covariates the knowledge about the free vaccination policy (OR = 7.20; 95%CI 5.91-8.78) and the state's initial vaccination strategy(OR = 1.33; 95%CI 1.08-1.64), perception of daily life disturbed (OR = 1.29; 95% CI 1.11-1.50), practice of injecting the seasonal influenza vaccine (OR = 4.69; 95%CI 3.53-6.23) were significantly associated with behavior of taking up the A/H1N1 vaccination positively (in Table 5), and the adverse reaction of A/H1N1 vaccine negatively influenced people's practice (OR = 0.07; 95%CI 0.04-0.11).

Novel A/H1N1 has caused pandemic in this century. It is important to encourage the public to adopt precautionary behaviors, which is based on the correct knowledge of the epidemic and appropriate response among residents. Many studies have examined the various levels of KAP about infectious disease outbreaks, such as SARS, avian influenza

[13] [14] [15] . Some studies have been reported specifically on community responses to A/H1N1 in Australia and Europe [16, 17] . But through literature search, we haven't found any public reports on KAP regarding A/H1N1 among Chinese population until now. Therefore, we conducted this large population-based survey (10669 respondents) to investigate community responses to A/H1N1 and to provide baseline data to government for preventive measures in case of future outbreaks.

Unless people have basic knowledge about the modes of transmission, they respond appropriately during an outbreak [16] . It has been proved that influenza is transmitted through person to person via respiratory secretions [18] . Most residents in our survey recognized that OR m : odds ratio obtained from stepwise multivariate logistics regression analysis using univariately significant variables as candidate variables and adjusting for region; NU: not significant in the univariate analysis; *: $P < 0.05$; †: $P < 0.01$; ‡: $P < 0.0001$.

the risk of getting infected would increase when an infected person coughed or sneezed in close distance. This may be due to the previous experience of SARS and avian flu. Multivariate analysis results showed that workers and farmers with lower education level were less likely to have this knowledge, which indicated that the contents and forms of propaganda should be more understandable and acceptable. A large proportion of residents in our survey overlooked the indirect hand contact and hand-shaking transmission route and about one third of public misconceived that A/H1N1 was food borne, which was associated with the previous knowledge of avian flu and the new A/H1N1 flu in the general population. The confusion with avian flu might mislead some residents to believe that the A/H1N1 virus is fatal and cause public panic [19] . Therefore, it is important for the government and health authorities to provide continuously updated information of the emerging disease through televisions, newspapers, radios, and Internet. There are regional differences in the perception of A/H1N1. For example, the public in Hong Kong did not perceive a high likelihood of having a local A/H1N1 outbreak [19] , but Malaysians were particularly anxious about the pandemic [20] . The current study shows that emotional distress was relatively mild in China as few residents worried about being infected (25.1%). This phenomenon may also be related to the previous experience of the SARS epidemic, as well as the open epidemic information. A survey in Korean university showed that women perceived higher illness severity and personal susceptibility to A/ H1N1 infection, which had been reconfirmed in our study [21] . Logistic regression analysis results suggested that women with higher educational level had higher perception of risk. As time went by, the knowledge about the main transmission route increased, but the risk perception of being

infected in residents decreased, suggesting the positive effect of government policy regarding A/H1N1 infection prevention, as well as the promotion of the media.

The previous study presented various results of influencing factors on the the compliance with the preventive practices. The study in Saudi showed that older men with better education were more likely to take preventive practices [9] ; female students in Korean washed hands more frequently during the peak pandemic period of A/ H1N1 [21] ; in another pandemic study in USA, younger people was found to have greater uptake of recommended behaviors but not for gender [16] . We found female with higher education took more precautionary behaviors, but office staffs and farmers took less comparing with students. While such differences could result from study population demographics, profound differences may also exist in the knowledge of A/H1N1 and the perceptions of recommended behaviors in those countries. Adjusting for the background factors, the multivariate logistic regression showed the possible relationship between knowledge and risk perception, knowledge and practices (odd ratios were 1.57 and 2.09, respectively), which indicated that good knowledge is important to enable individuals to have better attitudes and practices in influenza risk reduction. Similar findings were observed in other studies performed during A/ H1N1 pandemic in Singapore [22] and during SARS pandemic in Hong Kong [13] . Therefore, it is important to focus on inculcating the correct knowledge to individuals as it will influence both attitudes and practices. Injecting vaccination is an effective measure to prevent infectious disease [23] . In China, the seasonal influenza vaccination is not included in the national immunization program and must be purchased by recipients. Those who are above 60 years old, the pupil and children in kindergarten, and people with chronic diseases are recommended to get inoculation. Data provided by China CDC in 2009 showed that the immunization rate of the seasonal flu in Chinese population was below 2% [24] , which was much lower than 7.5% in our study ($P < 0.0001$). This phenomenon is partly due to the state's prior vaccination strategy for population at high risk such as students, teachers, healthcare workers and people with chronic disease, as well as the confusion between seasonal flu vaccine and A/H1N1 vaccine in residents. People who couldn't access the A/H1N1 vaccine may take up seasonal flu vaccine as preventive behaviors. The A/ H1N1 vaccine was not available in China until the middle of September 2009. All populations at high risk above three years old were invited for vaccination free of charge [25] . A survey among 868 European travelers showed 14.2% participants were vaccinated against pandemic influenza A/H1N1 [26] , higher than 10.8% in our study ($P < 0.01$). Our study also showed students and health care workers were more likely to take up, which may be due to the prior vaccination strategy. Multivariate stepwise logistic regression analysis, which allowed us to adjust for background factors, further

showed the perceived risk of infection and the knowledge about the main modes of transmission related to A/H1N1 vaccination were insignificantly, similar results seen in Lau's study [8] . Therefore, the vaccination rate of A/H1N1 is not expected to increase even if the virus becomes more prevalent or the knowledge of its transmission mode improved. Additionally, the behavior of taking up A/H1N1 vaccine was associated with perceptions of vaccine's safety and influence on daily life by A/H1N1 as well as the knowledge about the free vaccination policy and the state's initial vaccination strategy. This suggests that improving the safety of vaccine, the acceptability of side effect and the knowledge about the state's strategy related to A/H1N1 vaccination in residents may be helpful to promote A/H1N1 vaccination in the general population. The cross-sectional telephone survey adopted in the study has some limitations. We were unable to interview the people who did not have phones and the depth of the questionnaire was largely limited because questions and pre-existing answers could not be too long and complex. In addition, the telephone response rate was 46.8%, which means more than half of the interviewees rejected or didn't finish the survey. It was impossible to compare the difference between respondents and nonrespondents due to the lack of their basic information.

This A/H1N1 epidemic has not caused public panic yet, but the knowledge of A/H1N1 in residents is not optimistic as most of them confused the transmission route of A/H1N1. There are many factors influencing the KAP related to A/H1N1. Female with higher educational level had higher perceived risk of infection and took more precautionary behaviors. Public education campaign may take the side effects of vaccine and the knowledge about the state's vaccination strategy into account. The data collected in this survey could be used as baseline data to monitor public perceives and behaviors in the event of future outbreak of infectious disease in China.

Additional file 1: Questionnaire. The Questionnaire to Survey the Level of Knowledge, Attitude and Practice in Different Stages of H1N1 Pandemic by Telephone in China.

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Molecular and serological investigation of 2019-nCoV infected patients: implication of multiple shedding routes

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Abstract: In December 2019, a novel coronavirus (2019-nCoV) caused an outbreak in Wuhan, China, and soon spread to other parts of the world. It was believed that 2019-nCoV was transmitted through respiratory tract and then induced pneumonia, thus molecular diagnosis based on oral swabs was used for confirmation of this disease. Likewise, patient will be released upon two times of negative detection from oral swabs. However, many coronaviruses can also be transmitted through oral–fecal route by infecting intestines. Whether 2019-nCoV infected patients also carry virus in other organs like intestine need to be tested. We conducted investigation on patients in a local hospital who were infected with this virus. We found the presence of 2019-nCoV in anal swabs and blood as well, and more anal swab positives than oral swab positives in a later stage of infection, suggesting shedding and thereby transmitted through oral–fecal route. We also showed serology test can improve detection positive rate thus should be used in future epidemiology. Our report provides a cautionary warning that 2019-nCoV may be shed through multiple routes.

Text: Coronaviruses (CoVs) belong to the subfamily Orthocoronavirinae in the family Coronaviridae and the order Nidovirales. A human coronavirus (SARS-CoV) caused the severe acute respiratory syndrome coronavirus (SARS) outbreak in 2003. Most recently, an SARS-related CoV was implicated as the etiological agent responsible for the outbreak in Wuhan, central China. This outbreak is estimated to have started on 12th December 2019 and 17,332 laboratory confirmed cases with 361 deaths as of 3rd February 2020 in China [1] . The virus has spread to 23 other countries by travellers from Wuhan [1] . Typical

symptoms are fever, malaise, shortness of breath and in severe cases, pneumonia [2] [3] [4] . The disease was first called unidentified viral pneumonia.

We quickly identified the etiological agent, termed 2019-nCoV (virus name designated by the World Health Organization). The newly identified virus is an SARS-related virus (SARSr-CoV) but shares only 74.5% genome identity to SARS-CoV [2] . We developed molecular detection tools based on viral spike genes. Our previous studies indicate that qPCR method can be used for the detection of 2019-nCoV in oral swabs or in bronchoalveolar lavage fluid (BALF) [5] .

Additionally, we developed IgM and IgG detection methods using a cross-reactive nucleocapsid protein (NP) from another SARSr-CoV Rp3 [6] , which is 92% identical to 2019-nCoV NP. Using these serological tools, we demonstrate viral antibody titres increase in patients infected with 2019-nCoV [5] .

Like SARS-CoV, 2019-nCoV induced pneumonia through respiratory tract by clinical observation. Therefore, the presence of viral antigen in oral swabs was used as detection standard for 2019-nCoV. Similarly, two times of oral swabs negative in a 24-h interval was considered as viral clearance by patients officially.

Here we launched an investigation of 2019-nCoV in a Wuhan hospital, aiming to investigate the other possible transmission route of this virus.

Human samples, including oral swabs, anal swabs and blood samples were collected by Wuhan pulmonary hospital with the consent from all patients and approved by the ethics committee of the designated hospital for emerging infectious diseases. Two investigations were performed. In the first investigation, we collected samples from 39 patients, 7 of which were in severe conditions. In the second investigation, we collected samples from 139 patients, yet their clinical records were not available. We only showed patients who were viral nucleotide detection positive. Patients were sampled without gender or age preference unless where indicated. For swabs, 1.5 ml DMEM+2% FBS medium was added in each tube. Supernatant was collected after 2500 rpm, 60 s vortex and 15-30 min standing. Supernatant from swabs were added to lysis buffer for RNA extraction. Serum

was separated by centrifugation at 3000 g for 15 min within 24 h of collection, followed by 56°C 30 min inactivation, and then stored at 4°C until use.

Whenever commercial kits were used, manufacturer's instructions were followed without modification. RNA was extracted from 200 µl of samples with the High Pure Viral RNA Kit (Roche). RNA was eluted in 50 µl of elution buffer and used as the template for RT-PCR. QPCR detection method based on 2019-nCoV S gene can be found in the previous study [5]. In brief, RNA extracted from above used in qPCR by HiScript® II One Step qRT-PCR SYBR® Green Kit (Vazyme Biotech Co., Ltd). The 20 µl qPCR reaction mix contained 10 µl 2× One Step SYBR Green Mix, 1 µl One Step SYBR Green Enzyme Mix, 0.4 µl 50 × ROX Reference Dye 1, 0.4 µl of each primer (10 µM) and 2 µl template RNA. Amplification was performed as follows: 50°C for 3 min, 95°C for 30 s followed by 40 cycles consisting of 95°C for 10 s, 60°C for 30 s, and a default melting curve step in an ABI 7500 machine.

In-house anti-SARSr-CoV IgG and IgM ELISA kits were developed using SARSr-CoV Rp3 NP as antigen, which shared above 90% amino acid identity to all SARSr-CoVs, as reported previously [5]. For IgG test, MaxiSorp Nunc-immuno 96 well ELISA plates were coated (100 ng/well) overnight with recombinant NP. Human sera were used at 1:20 dilution for 1 h at 37°C. An anti-Human IgG-HRP conjugated monoclonal antibody (Kyab Biotech Co., Ltd, Wuhan, China) was used at a dilution of 1:40,000. The OD value (450-630) was calculated. For IgM test, Maxi-Sorp Nunc-immuno 96 well ELISA plates were coated (500 ng/well) overnight with anti-human IgM (µ chain). Human sera were used at 1:100 dilution for 40 min at 37°C, followed by anti-Rp3 NP-HRP conjugated (Kyab Biotech Co., Ltd, Wuhan, China) at a dilution of 1:4000. The OD value (450-630) was calculated.

In the first investigation, we aimed to test whether viral positive can be found in anal swab and blood as well as oral swabs. We conducted a molecular investigation to patients in Wuhan pulmonary hospital, who were detected as oral swabs positive for 2019-nCoV upon admission. We collected blood, oral swabs and anal swabs for 2019-nCoV qPCR test using previously established method [5].

We found 15 patients who still carry virus following days of medical treatments. Of these patients, 8 were oral swabs positive (53.3%), 4 were anal swabs positive (26.7%), 6 blood positives (40%) and 3 serum positives (20%). Two patients were positive by both oral swab

and anal swab, yet none of the blood positive was also swabs positive. Not surprisingly, all serum positives were also whole serum positive (Table 1). In summary, viral nucleotide can be found in anal swab or blood even if it cannot be detected in oral swabs. It should be noted that although swabs may be negative, the patient might still be viremic.

We then did another investigation to find out the dynamic changes of viral presence in two consecutive studies in both oral and anal swabs in another group of patients. The target patients were those who received around 10 days of medical treatments upon admission. We tested for both viral antibody and viral nucleotide levels by previously established method [5] . We showed that both IgM and IgG titres were relatively low or undetectable in day 0 (the day of first sampling). On day 5, an increase of viral antibodies can be seen in nearly all patients, which was normally considered as a transition from earlier to later period of infection (Figure 1 and supplementary table 1). IgM positive rate increased from 50% (8/16) to 81% (13/16), whereas IgG positive rate increased from 81% (13/16) to 100% (16/16). This is in contrast to a relatively low detection positive rate from molecular test (below). For molecular detection, we found 8 oral swabs positive (50%) and 4 anal swabs (25%) in these 16 people on day 0. On day 5, we were only able to find 4 oral swabs positive (25%). In contrast, we found 6 anal swabs positive (37.5%). When counting all swab positives together, we found most of the positives came from oral swab (8/10, 80%) on day 0. However, this trend appears to change on day 5. We found more (6/8, 75%) anal swab positive than oral swab positive (4/8, 50%). Another observation is the reoccurrence of virus in 6 patients who were detected negative on day 0. Of note, 4 of these 6 viral positives were from anal swabs (Table 2) . These data suggested a shift from more oral positive during early period (as indicated by antibody titres) to more anal positive during later period might happen.

Within 1 month of the 2019-nCoV disease outbreak, we rapidly developed molecular and serological detection tools. This is the first molecular and serological study on this virus after the initial identification of 2019-NCoV from 7 patients diagnosed with unidentified viral pneumonia [5] . We detected the virus in oral swabs, anal swabs and blood, thus infected patients can potentially shed this pathogen through respiratory, fecal-oral or body fluid routes. In addition, we successfully applied serology test a large population and showed which could greatly improved detection positive rate.

We show that the current strategy for the detection of viral RNA in oral swabs used for 2019-nCoV diagnosis is not perfect. The virus may be present in anal swabs or blood of patients when oral swabs detection negative. In SARS-CoV and MERS-CoV infected patients, intestinal infection was observed at later stages of infection [7] [8] [9] . However, patients infected with 2019-nCoV may harbour the virus in the intestine at the early or late stage of disease. It is also worth to note none of the patients with viremia blood had positive swabs. These patients would likely be considered as 2019-nCoV negative through routine surveillance, and thus pose a threat to other people. In contrast, we found viral antibodies in near all patients, indicating serology should be considered for 2019-nCoV epidemiology. A possible shift from oral positive during early infection to anal swab positive during late infection can be observed. This observation implied that we cannot discharge a patient purely based on oral swabs negative, who may still shed the virus by oral-fecal route. Above all, we strongly suggest using viral IgM and IgG serological test to confirm an infection, considering the unreliable results from oral swabs detection.

In summary, we provide a cautionary warning that 2019-nCoV may be transmitted through multiple routes. Both molecular and serological tests are needed to definitively confirm a virus carrier.

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--- START OF ARTICLE 133 ---

The Long Noncoding RNA NEAT1 Exerts Antihantaviral Effects by Acting as Positive Feedback for RIG-I Signaling

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Abstract: Hantavirus infection, which causes zoonotic diseases with a high mortality rate in humans, has long been a global public health concern. Over the past decades, accumulating evidence suggests that long noncoding RNAs (lncRNAs) play key regulatory roles in innate immunity. However, the involvement of host lncRNAs in hantaviral control remains uncharacterized. In this study, we identified the lncRNA NEAT1 as a vital antiviral modulator. NEAT1 was dramatically upregulated after Hantaan virus (HTNV) infection, whereas its downregulation in vitro or in vivo delayed host innate immune responses and aggravated HTNV replication. Ectopic expression of NEAT1 enhanced beta interferon (IFN- β) production and suppressed HTNV infection. Further investigation suggested that NEAT1 served as positive feedback for RIG-I signaling. HTNV infection activated NEAT1 transcription through the RIG-I–IRF7 pathway, whereas NEAT1 removed the transcriptional inhibitory effects of the splicing factor proline- and glutamine-rich protein (SFPQ) by relocating SFPQ to paraspeckles, thus promoting the expression of RIG-I and DDX60. RIG-I and DDX60 had synergic effects on IFN production. Taken together, our findings demonstrate that NEAT1 modulates the innate immune response against HTNV infection, providing another layer of information about the role of lncRNAs in controlling viral infections. **IMPORTANCE** Hantaviruses have attracted worldwide attention as archetypal emerging pathogens. Recently, increasing evidence has highlighted long noncoding RNAs (lncRNAs) as key regulators of innate immunity; however, their roles in hantavirus infection remain unknown. In the present work, a new unexplored function of lncRNA NEAT1 in controlling HTNV replication was found. NEAT1 promoted interferon (IFN) responses by acting as positive feedback for RIG-I signaling. This lncRNA was induced by HTNV through the RIG-I–IRF7 pathway in a time- and dose-dependent manner and promoted HTNV-induced IFN production by facilitating RIG-I and DDX60 expression. Intriguingly, NEAT1 relocated SFPQ and formed paraspeckles after HTNV infection, which might reverse inhibitive effects of SFPQ on the transcription of RIG-I and DDX60. To the best of our knowledge, this is the first study to address the regulatory role of the lncRNA NEAT1 in host innate immunity after HTNV infection. In summary, our findings provide additional insights regarding the role of lncRNAs in controlling viral infections.

Text: glycoprotein (GP), and viral RNA-dependent polymerase protein (RdRp), respectively. Humans become infected by inhaling contaminated aerosols or by coming into contact with rodent excreta, and they develop two severe acute diseases, namely, hemorrhagic fever with renal syndrome (HFRS) and hantavirus pulmonary syndrome (HPS) (2) . Hantavirus infection affects up to 100,000 to 200,000 humans annually, with fulminant HFRS cases most represented in China (3) . Chinese HFRS cases, mainly caused by Hantaan virus (HTNV) infection, account for approximately 90% of all global cases, with a mortality rate ranging from 0.1 to 15% (4) . Since there is neither an effective therapeutic nor FDA-licensed vaccine, further understanding of host immune responses against hantaviral infection is of great significance for global public health and safety. The innate immune system, characterized by interferon (IFN) responses and immunocyte activation, provides the initial defense against viral invasions. Cellular pathogen recognition receptors (PRRs), including Toll-like receptors (TLRs) and RIG-I like receptors (RLRs), can detect distinct pathogen-associated molecular patterns (PAMPs) and trigger the expression of IFNs and cytokines. RIG-I has been shown to recognize hantaviral invasion, but its regulatory process remains unclear (5) . Long noncoding RNAs (lncRNAs) have emerged as important modulators of gene expression. lncRNA nuclear paraspeckle assembly transcript 1 (NEAT1) is an essential architectural constituent of paraspeckles in the mammalian nucleus, interacting with Drosophila DBHS RNA-binding proteins such as the splicing factor proline- and glutamine-rich protein (SFPQ) and the non-POU domain-containing, octamer-binding protein (NONO/p54) (6) . To date, two isoform transcripts of the NEAT1 gene have been identified, namely, the 3.7-kb NEAT1-1 (MEN) and the 23-kb NEAT1-2 (MEN^L) (Fig. 1A) . A large amount of research has shown that NEAT1 is associated with oncogenesis and tumor progression (7) (8) (9) , promoting cancer formation in mice by dampening oncogene-dependent activation of p53 (10) . Nevertheless, studies assessing the function of NEAT1 in viral infections are scarce.

Here, the human umbilical vein endothelial cell (HUVEC) transcriptome was analyzed after HTNV infection by digital gene expression (DGE) profiling, and lncRNA NEAT1 was found to be remarkably upregulated by viral infection. Silencing NEAT1 in vitro or in vivo suppressed host immune responses and aggravated HTNV infection, whereas NEAT1 overexpression in vitro enhanced beta interferon (IFN- β) production and inhibited HTNV replication. Further investigation showed that NEAT1 promoted RIG-I and DDX60 expression by relocating SFPQ and removing the transcriptional inhibitory effects of SFPQ, which are critical for IFN responses against HTNV infection. We also found that RIG-I signaling, rather than TLR3 and TLR4, accounted for the elevation of HTNV-induced NEAT1. Taken together, our findings

provide novel insights into the lncRNA-mediated regulatory mechanism of host innate defense against HTNV infection.

To explore the potential role of long noncoding RNAs in host innate immune responses, DGE analysis of HUVECs for whole-genome profiling was performed at 24 h post-HTNV infection. As shown in Fig. 1B, the NEAT1 level in the HTNV group was higher than that in the mock group ($P \leq 6.86 \times 10^{-13}$, false discovery rate [FDR] $\leq 9.75 \times 10^{-12}$) or the 60 Co-inactivated HTNV group ($P \leq 1.75 \times 10^{-14}$, FDR $\leq 3.10 \times 10^{-13}$); however, the difference between the 60 Co-inactivated HTNV group and the mock group was not significant ($P \leq 0.21034$, FDR ≤ 0.58211). To confirm the profiling results, two primer pairs from the published literature (11), one recognizing both NEAT1-1 and NEAT1-2 and the other specific for NEAT1-2 (Fig. 1A), were applied to quantify NEAT1 RNA isoforms by quantitative real-time PCR (qRT-PCR). It has been reported that NEAT1-2 rather than NEAT1-1 plays a key regulatory role in paraspeckle formation (11), and we also found that elevated NEAT1 levels depend on live HTNV infection rather than 60 Co-inactivated HTNV stimulation (Fig. 1C). Fluorescence in situ hybridization (FISH) with probes specific for NEAT1-2 was performed with HUVECs, and the results confirmed increased NEAT1-2 expression and the aggregation of NEAT1-2 in the nucleus at 24 and 48 h postinfection (hpi) (Fig. 1D).

To further investigate whether NEAT1 expression was altered in other cell lines, HEK293, HeLa, and A549 cells were used. All these cells could be infected by HTNV (Fig. 1E and F) and generated hantavirus progeny (Fig. 1G). Similar to the data obtained from HUVECs, NEAT1 was indeed upregulated by HTNV at a multiplicity of infection (MOI) of 1 beginning at 24 hpi in HUVECs and A549, HEK293, and HeLa cells, and the increasing tendency occurred in a time-dependent manner (Fig. 1H). Of note, the NEAT1 elevation at 2 hpi might have been unrelated to the virus but resulted in cellular stress responses. Besides, NEAT1 expression increased from an MOI of 0.1 to 1, indicating that the elevation occurred in a viral dose-dependent manner (Fig. 1I).

NEAT1-2 and not NEAT1-1 suppresses HTNV replication in HUVECs. The abovedescribed data showed that HTNV infection increased NEAT1, and we wondered how NEAT1 could reciprocally influence HTNV replication. The small interfering RNA (siRNA) transfection efficiency in HUVECs was confirmed by flow cytometry, and NEAT1 expression was significantly decreased, as assessed by qRT-PCR after RNA interference (RNAi) (Fig. 2A).

Of note, si-NEAT1 targets both NEAT1-1 and NEAT1-2, whereas the stealth siRNA NEAT1-2 (st-NEAT1-2) is specific for NEAT1-2. Compared with the cells transfected with control siRNA (negative control [NC]), HUVECs with si-NEAT1 could dramatically promote HTNV NP production, and NP expression seemed to be related to the amount of applied si-NEAT1 (Fig. 2B). Intriguingly, depletion of NEAT1-2 alone could mimic the antiviral effects of simultaneous NEAT1-1 and NEAT1-2 silencing (Fig. 2C), indicating that NEAT1-2 was critical for the antiviral responses. Consistent with those data, the expressions of HTNV mRNA of S segment (Fig. 2D) and HTNV titers (Fig. 2E) were increased after NEAT1 silencing.

On the other hand, plasmids, of which pCMV-NEAT1-1 is transcribed into the 3.7-kb NEAT1-1 (MEN) and pCMV-NEAT1-2 is transcribed into the 2-to 3-kb NEAT1-2 (MEN^N), were applied to directly investigate the role of NEAT1 in HTNV infection (Fig. 2F). Surprisingly, we found NEAT1-1 overexpression restricted NEAT1-2 transcription (Fig. 2F). Overexpression of NEAT1 with both pCMV-NEAT1-1 and pCMV-NEAT1-2 could conspicuously repress HTNV NP expression, and NP expression seemed to be associated with the transfected plasmids (Fig. 2G). Furthermore, overexpression of NEAT1-2 instead of NEAT1-1 could efficiently suppress HTNV replication (Fig. 2H). NEAT1-1 upregulation even aggravated HTNV infection (Fig. 2H), which may be the result of downregulation of NEAT1-2. Consistently, through analysis of viral load detected by qRT-PCR and the 50% tissue culture infective dose (TCID₅₀) test by ELISA, we found that expression of HTNV-specific mRNA (Fig. 2I) and HTNV titers (Fig. 2J) were limited in HUVECs in which NEAT1-2 was ectopically expressed in comparison to those transfected with control vector or pCMV-NEAT1-1. These data further showed that NEAT1-2 and not NEAT1-1 suppresses HTNV replication in HUVECs.

Alteration of NEAT1-2 affects HTNV-induced IFN expression in HUVECs. IFN- γ production or pretreatment at an early infection stage plays an important role in limiting HTNV infection, while IFN- γ treatment after 24 hpi exerts little antiviral effect (12, 13). It has been reported that the GnT of hantaviruses suppressed IFN- γ expression of host cells at an early stage of infection (14). Here, we also found that HUVECs could not efficiently produce IFN- γ until 12 hpi at an MOI of 0.1 or until 24 hpi at an MOI of 1 (Fig. 3A), which indicated that high doses of HTNV could hamper prompt IFN responses. Notably, enhanced NEAT1-2 transcription appeared at 8 hpi at an MOI of 0.1 or at 20 hpi at an MOI of 1 (Fig. 3B), suggesting that NEAT1-2 expression increased just before IFN production. We found that

expression of endogenous IFN- λ mRNA was much lower in cells transfected with si-NEAT1-2 at MOIs of both 0.1 (Fig. 3C) and 1 (Fig. 3D) than in those transfected with control siRNA (NC). In contrast, overexpression of NEAT1 in HUVECs increased IFN- λ expression after HTNV infection (MOI $\leq$ 1) at 24 and 48 hpi (Fig. 3E). More importantly, HUVECs transfected with pCMV-NEAT1-2 conspicuously increased IFN- λ gene expression compared with those cells with vector plasmids at 12 hpi (MOI $\leq$ 1), demonstrating that NEAT1-2 overexpression accelerated robust IFN responses in host cells against HTNV infection. With a dual luciferase reporter system Twenty-four hours after transfection, the cells expressing FAM were calculated by flow cytometry. Right, HUVECs in six-well plates were transfected with NC sequences, si-NEAT1, or the stealth siRNA NEAT1-2 (st-NEAT1-2) (3 g). Twenty-four hours after transfection, the cells were infected with HTNV at an MOI of 1. At 48 hpi, the NEAT1 expression levels were measured by qRT-PCR. Values are means $\pm$ SD (n $\leq$ 3; *, P $\leq$ 0.01; **, P $\leq$ 0.001; ***, P $\leq$ 0.0001; Student's t test, compared with the NC group). NS, nonsignificant. (B) HUVECs in six-well plates were transfected with NC sequences (the amount of Si-NEAT1-2 is considered 0 g) or increasing amounts of si-NEAT1 (0.1, 0.5, 1, and 3 g). Twenty-four hours after transfection, the cells were infected with HTNV at an MOI of 0.1 for 48 h. The expression of HTNV NP was measured by Western blotting. (C) HUVECs were treated as described for panel A, right, but at an MOI of 0.1. In-cell Western (ICW) analysis for HTNV NP was performed at 48 hpi. The ICW for HTNV NP staining is shown on the left, while the relative intensity of fluorescence (NP/ λ -actin) was analyzed using Student's t test. (n $\leq$ 4; *, P $\leq$ 0.01; **, P $\leq$ 0.001; Student's t test, compared with the NC group). (D) HUVECs were treated as described for panel A, right, but at an MOI of 0.1. The expression of HTNV S segment was measured by qRT-PCR. Values are means $\pm$ SD (n $\leq$ 3; *, P $\leq$ 0.01; Student's t test, compared with the NC group). (E) HUVECs were treated as described for panel A, right, but at an MOI of 0.1. The propagated HTNV was acquired at 72 hpi, and viral titers were detected by TCID₅₀ with ELISA in Vero E6 cells. Values are means $\pm$ SD (n $\leq$ 3; *, P $\leq$ 0.01; Student's t test, compared with the NC group). (F) Left, HUVECs in six-well plates were transfected with vectors or pGFP (3 g). Twenty-four hours after transfection, the cells expressing green fluorescent protein (GFP) were calculated by flow cytometry. Right, HUVECs in six-well plates were transfected with control plasmids (vector), pCMV-NEAT1-1, or pCMV-NEAT1-2 (3 g). Twenty-four hours after transfection, the cells were infected with HTNV at an MOI of 1. At (Continued on next page) NEAT1 Promotes Innate Antiviral Responses Journal of Virology maintaining IFN- λ promoters, we found NEAT1-2 silencing or overexpression could inhibit or increase the promoter activity of the IFN- λ gene after HTNV infection, respectively, whereas silencing NEAT1-2 or ectopically expressing NEAT1-2 without HTNV infection could not inhibit or enhance IFN- λ expression (Fig. 3F). These results showed that NEAT1-2 regulated HTNV-induced IFN- λ expression.

To explore whether the antihantavirus effects of NEAT1 were caused by IFN- λ alteration, a series of compensatory experiments was designed. In NEAT1-2 knockdown HUVECs, the addition of IFN- λ at 12 hpi could efficiently block HTNV NP production (MOI $\leq$ 0.1), and such phenomena were also determined by the amount of applied IFN- λ (Fig. 3G). In addition, in cells with high NEAT1-2 expression, treatment with neutralizing antibodies (NABs) of IFN- α and IFN- λ could counteract the antiviral effects of NEAT1-2 (MOI $\leq$ 1), and the compensatory effects were dependent on the magnitude of the NABs. Together these results demonstrated that NEAT1-2 especially enhanced the host antihantaviral innate immune responses by regulating IFN- λ signaling.

RIG-I and DDX60 regulated by NEAT1-2 facilitate HTNV-induced IFN- λ production. PRRs maintain a vital role in the promotion of IFN responses, and we conjectured that NEAT1 might amplify IFN responses by modulating these molecules. TLR3, TLR4, and RIG-I have been shown to recognize HTNV infection (5, 15, 16) . DDX60 was recently reported as an important activator of RIG-I, but the antiviral effects of DDX60 remain a subject of debate (17, 18) , Here, we found that multiple Toll-like receptors like TLR1, TLR2, TLR3, and TLR4, as well as MDA5, were increased after HTNV infection, but none of them were influenced by silencing NEAT1-2 (Fig. 4A). The upregulated RIG-I and DDX60 were blocked in the cells with low NEAT1-2 expression after HTNV infection (Fig. 4A). HUVECs with declining NEAT1-2 expression showed gradually decreasing expression of RIG-I and DDX60 (Fig. 4B) , and increasing NEAT1-2 transcription was found to activate RIG-I and DDX60 production accordingly (Fig. 4C). These data indicated that NEAT1-2 could positively modulate RIG-I and DDX60 expression, while the role of RIG-I and DDX60 upon HTNV infection is obscure.

We then found that RIG-I and DDX60 colocalized after HTNV infection (Fig. 4D), implying that RIG-I and DDX60 might collaborate with each other to exert antiviral effects. To verify the antiviral role of RIG-I and DDX60, we designed a series of siRNAs targeting RIG-I and DDX60, and we selected the si-RIG-I-2 and siRNA-DDX60-1 with the highest knockdown efficiency by qRT-PCR in HUVECs (data not shown). Simultaneously knocking down RIG-I and DDX60 significantly promoted HTNV NP expression (Fig. 4E), and knockdown of both of them could greatly affect IFN- λ expression (Fig. 4F and G). Ectopic expression of either RIG-I or DDX60 inhibited viral replication, whereas overexpression of both resulted in superior antiviral effects (Fig. 4H), indicating that efficient anti-HTNV responses might depend on the interactive effects of DDX60 and RIG-I. More importantly, RIG-I or/and

DDX60 overexpression enhanced HTNV-induced IFN- λ expression, and they had synergistic effects on IFN- λ production (Fig. 4I and J). Consequently, NEAT1 might regulate IFN- λ production by upregulating RIG-I and DDX60, and thus we were interested in how NEAT1 regulated RIG-I and DDX60 expression.

SFPQ, which is relocated by NEAT1 HTNV infection, regulates the expression of RIG-I and DDX60. NEAT1 was found to interact with SFPQ by RNA immunoprecipitation (RIP) after HTNV infection (Fig. 5A) , indicating that modulatory effects of NEAT1 48 hpi, the NEAT1 expression levels were measured by qRT-PCR. Values are means $\pm$ SD (n $\leq$ 3; *, P $\leq$ 0.01; **, P $\leq$ 0.001; ***, P $\leq$ 0.0001; Student's t test, compared with the vector group). (G) HUVECs in six-well plates were transfected with control plasmids (vector, the amount of pCMV-NEAT1-1 plus pCMV-NEAT1-2 is considered 0 g) or increasing amounts of pCMV-NEAT1-1 plus pCMV-NEAT1-2 (0.05 $\pm$ 0.05 g, 0.25 $\pm$ 0.25 g, 0.5 $\pm$ 0.5 g, 1.5 $\pm$ 1.5 g, respectively). Twenty-four hours after transfection, the cells were infected with HTNV at an MOI of 1 for 48 h. The expression of HTNV NP was measured by Western blotting. (H) HUVECs were treated as described for panel F, right. In-Cell Western (ICW) analysis for HTNV NP was performed at 48 hpi. The ICW for HTNV NP staining is shown on the left, while the relative intensity of fluorescence (NP/ λ -actin) was analyzed using Student's t test. (n $\leq$ 4; *, P $\leq$ 0.01; **, P $\leq$ 0.001; Student's t test, compared with the vector group). (H) HUVECs were treated as described for panel F, right. The expression of HTNV S segment was measured by qRT-PCR. Values are means $\pm$ SD (n $\leq$ 3; *, P $\leq$ 0.01; **, P $\leq$ 0.001; Student's t test, compared with the vector group). (I) HUVECs were treated as described for panel F, right. The propagated HTNV was acquired at 72 hpi, and viral titers were detected by TCID₅₀ with ELISA in Vero E6 cells. Values are means $\pm$ SD (n $\leq$ 3; *, P $\leq$ 0.01; Student's t test, compared with the vector group). might be involved in SFPQ. Interestingly, the protein level of SFPQ, as well as another paraspeckle-forming constituent, NONO, remained unchanged after HTNV infection (Fig. 5B) or after NEAT1 overexpression and knockdown (Fig. 5C). However, SFPQ became centralized rather than diffuse in the nucleus after HTNV infection (Fig. 5D) . The enhanced interaction of SFPQ and NONO indicated excess formation of paraspeckles in the nucleus (Fig. 5E) and relocalization of SFPQ. SFPQ knockdown could inhibit HTNV replication (Fig. 5F and G), which might have been related to the increase in RIG-I (Fig. 5H) and DDX60 (Fig. 5I). SFPQ has been suggested to bind to the promoter region of RIG-I and DDX60 (11), thus preventing the expression of RIG-I and DDX60. Taken together, the above results suggested that NEAT1 might relocate SFPQ from the RIG-I signaling is crucial for NEAT1 expression after HTNV infection. Elevated NEAT1 exerts antiviral effects by modulating the innate immune response, yet it is unclear how HTNV triggers NEAT1

transcription. Interestingly, overexpression of the S or M segment of HTNV in HEK293 cells failed to induce NEAT1 expression, suggesting that NEAT1 transcription was closely related to live viral replication (Fig. 6A) . Of note, the upregulation of NEAT1 by HTNV could not be reversed by applying IFN-I neutralizing antibodies (Fig. 6B) . Meanwhile, NEAT1 expression could not be induced by stimulation with different types of IFNs (Fig. 6C, D, and E) or cytokines (Fig. 6F and G) .

We conjectured that NEAT1 expression was related to the activation of PRRs. By knocking down several PRRs, we found that the RIG-I and TLR4 pathways played important roles in HTNV-induced NEAT1 upregulation (Fig. 7A) . Using RIG-I-and TLR4-deficient cell lines which could be well infected by HTNV (Fig. 7B) , RIG-I was confirmed to be indispensable for NEAT1 induction after HTNV infection (Fig. 7C) . Moreover, using STAT1 as a positive control, we found that the transcription factor IRF7, rather than IRF3 and p65, translocated into the nucleus in HTNV-infected HUVECs at 2 dpi (Fig. 7D) . Furthermore, IRF7 knockdown blocked HTNV-induced NEAT1 upregulation (Fig. 7E) . Therefore, HTNV caused transcriptional activation of the NEAT1 gene, probably via the RIG-I-IRF7 pathway.

NEAT1 silencing has profound effects on innate immune responses after HTNV infection in mice. Although cell-based experiments revealed that NEAT1-2 is a crucial regulator of innate antihantaviral responses, its function *in vivo* has remained unclear. To address this question, we intravenously injected siRNAs targeting mouse NEAT1-2 at 1 day before HTNV infection. NEAT1-2 expression levels in the liver, kidney, and spleen were reduced at 2 dpi (Fig. 8A) . Previous studies have shown that NEAT1 knockout does not affect physiological processes except *potentia generandi* in mice; hence, we assessed its role under pathological conditions. Body weight loss in NEAT1-2-depleted mice was observed from 2 dpi to 5 dpi, and the IFN production in serum was remarkably decreased in the NEAT1-2 silenced group than those in the NC group at 3 dpi (Fig. 8B) . As expected, NEAT1-2 knockdown mice showed considerably higher HTNV NP levels in the liver, spleen, and kidney at 3 dpi (Fig. 8C) . Moreover, the virus titers in related organs were higher in the NEAT1-2 silenced group than in the NC group (Fig. 8D) . In addition, reduced inflammatory cell filtration but increased tissue injury was found in NEAT1-2 knockdown mice during the early stage of infection (Fig. 8E) . Infiltration of macrophages in the spleen was attenuated (Fig. 8F) , and the activation of macrophages was also suppressed (by flow cytometry; data not shown). Moreover, CD8 α IFN- γ α T cells were reduced in the spleens of NEAT1-2 knockdown mice in comparison to those in the NC group at 3 dpi (Fig. 8G) . Nevertheless, NEAT1-2 silencing had no effect on the production of neutralizing antibodies at 7 dpi (data

not shown). The above-described findings indicated that NEAT1-2 depletion might influence multiple aspects of the innate immune response in HTNV-infected mice.

Innate immunity is a phylogenetically ancient and conserved system that counteracts invading microbes, the regulatory mechanism of which is sophisticated and complex. Long noncoding RNAs, which were once considered dark materials in the mammalian genome, have been shown to exert vital modulatory effects on host innate immunity (19). In this report, we first demonstrated that NEAT1 was induced by HTNV through the RIG-I-IRF7 pathway and served as positive feedback for RIG-I signaling.

Using DGE analysis, we observed upregulated NEAT1 and confirmed its alteration in To further determine the function of NEAT1 after HTNV infection in vivo, mice were injected intravenously with si-NEAT1-2 (1 g/g) or nontarget control siRNA (NC) (1 g/g); 1 day later, they were infected with HTNV (100 LD₅₀) by intramuscular injection. (A) To maintain high knockdown efficiency, siRNAs were injected intravenously every other day. (A) The knockdown efficiency was assessed by qRT-PCR in kidney, liver, and spleen samples at 2 dpi (n = 6 in each group). (B) The effects of NEAT1 on HTNV virulence in mice were determined by body weight loss from 0 to 10 dpi (left panel, n = 10 in each group). The IFN- γ in sera of different groups was measured by ELISA at 3dpi (right panel, n = 8 in each group). Values are means $\pm$ SD (*, P < 0.01; **, P < 0.001; Student's t test, compared with the NC+HTNV group). (C) Mice were sacrificed at 3 dpi, and livers, spleens, and kidneys were collected for ELISA detection of HTNV NP titers (upper panels, n = 8 in each group) and qRT-PCR to assess HTNV S segment levels (bottom panels, n = 8 in each group) at 3 dpi. Values are means $\pm$ SD (*, P < 0.01; **, P < 0.001; Student's t test, compared with the NC+HTNV group). (D) NEAT1 effects on HTNV infection kinetics at 3 dpi were determined by testing the HTNV titers in livers, spleens, and kidneys. Values are means $\pm$ SD (n = 8; *, P < 0.01; **, P < 0.001; Student's t test, compared with the NC+HTNV group). (E) Hematoxylin and eosin (H&E) staining for mouse liver, spleen, or kidney specimens was performed (3 dpi, n = 8 in each group). After HTNV infection, livers in the NC group showed inflammatory cell infiltration in certain regions, while those in the si-NEAT1-2 group showed slight acute viral hepatitis. Spleens in the NC group showed lymph node hyperplasia, while those in the si-NEAT1-2 group were severely congestive. Kidneys in the NC group also showed inflammatory cell infiltration, while those in the si-NEAT1-2 group had moderate interstitial congestion. (F) Macrophage infiltration in spleens was analyzed by detecting CD11b and F4/80 by flow cytometry at 3 dpi, and the results obtained for four mice in each group are presented. (G) CD3 $\pm$ CD8 $\pm$ IFN- γ $\pm$ T cells were analyzed by flow cytometry at 3 dpi, and

the results obtained for three mice in each group are presented. different cell lines. To assess its effects on HTNV replication, NEAT1 was silenced both in vitro and in vivo, which resulted in increased HTNV infection and suppressed innate immune responses. Further analysis indicated that NEAT1 might interact with SFPQ and regulate DDX60 and RIG-I expression. By virtue of RNAi, the RIG-I-IRF7 pathway was confirmed to be necessary for HTNV-triggered NEAT1 elevation.

Recently, large-scale transcriptomic studies identified numerous noncoding transcripts in the mammalian genome, which were speculated to influence diverse biological processes. Among these noncoding RNAs (ncRNAs), long noncoding RNAs (lncRNAs) emerged as important regulators of gene expression and are closely related to the activation of the host innate immune system. TLR2 controls lncRNA-COX2 expression in a MyD88- and NF- κ B-dependent manner, whereas lncRNA-COX either promotes interleukin 6 (IL-6) secretion or represses ISG15 and CCL5 expression (20). TLR2 activation or tumor necrosis factor alpha (TNF- α) stimulation induces transcription of the lncRNA THRIL, the downregulation of which impairs TNF- α and IL-6 secretion (21). TLR4 signaling in response to lipopolysaccharide (LPS) induces lncRNA IL-1 β -eRNA and IL-1 β -RBT46, the knockdown of which attenuates IL-1 β and CXCL8 release (22). The lncRNA Lethe, triggered by TNF- α and IL-1 β , acts as a negative feedback regulator of NF- κ B signaling (23).

The roles of lncRNAs in host-virus interactions have been progressively unveiled. Various viruses, such as influenza virus (IAV), coronavirus, enterovirus, human immunodeficiency virus (HIV), hepatitis B virus (HBV), hepatitis C virus (HCV), Japanese encephalitis virus (JEV), and rabies virus, have been reported to activate the transcription of different lncRNAs in host cells (11, (24) (25) (26)). Importantly, multiple lncRNAs have been shown to affect the IFN response in recent years and have gradually become hot spots in the field of antiviral research. NeST was shown to enhance IFN- γ production, controlling the susceptibility of mice to persistent Theiler's virus infection as well as resistance to *Salmonella enterica* serovar Typhimurium infection (27). Both CMPK2 and NRAV were identified as negative regulators of IFN immune reactions. CMPK2, induced by IFN- α or HCV infection, suppresses various ISGs, the knockdown of which dramatically blocks HCV replication (26). NRAV inhibits some critical ISGs, such as IFITM3 and Mx1, the depletion of which suppresses IAV replication both in vitro and in vivo (25). Numerous lncRNAs, including lnc-ISG15 and ISR2, respond to IFNs such as ISGs, although their actual function requires further investigation (28). Considering the poor evolutionary conservation but

rapid divergence of lncRNAs, their functions may be highly species and virus specific. Though considerable progress has been achieved to demonstrate the antiviral effects of lncRNAs on model viruses, there are no published reports assessing the role of lncRNAs in hantaviral infection. NEAT1 has been reported to interact with *Drosophila* DBHS RNA-binding proteins (e.g., SFPQ, NONO-p54nrb, and PSPC1), recruiting them to paraspeckles, a nuclear substructure found in all cultured and primary cells except embryonic stem cells (24). The versatile function of NEAT1 is rapidly progressing in multiple areas of biology. NEAT1 has been reported to be involved in the pathogenesis of multiple types of cancer (7) (8) (9). NEAT1 also participates in neurodegenerative diseases such as Huntington's disease (29) and seems to potentially contribute to the elevated production of a number of cytokines and chemokines in patients with systemic lupus erythematosus (SLE) (30). Furthermore, poly I:C can activate NEAT1 transcription through the TLR3 pathway, whereas NEAT1 positively regulates IL-8 transcription and potentially affects the expression of multiple ISGs after poly I:C stimulation (11). In addition, NEAT1 has been reported to suppress the export of Rev-dependent instability element (INS)containing HIV-1 mRNAs from the nucleus to the cytoplasm, thus inhibiting HIV replication (24). However, the role of NEAT1 in hantaviral infection remains unclear.

In this report, NEAT1 has been identified as an important regulator of the host innate immune system against HTNV infection. Elevated NEAT1 promotes IFN secretion, most likely by enhancing RIG-I and DDX60 expression. DDX60, a DEXD/H box RNA helicase similar to *Saccharomyces cerevisiae* Ski2, is induced after viral infection (31). DDX60 recognizes viral RNA and activates endogenous RIG-I, thereby promoting the RIG-I signaling-related IFN response. However, the antiviral effects of DDX60 seem to vary among viruses (17). We found that NEAT1-regulated DDX60 was involved in IFN production in response to HTNV infection. In HTNV-infected cells, double-stranded RNA (dsRNA) could not be detected, and it is unclear how host PRRs, especially RIG-I, recognize HTNV invasion (5). Here, considering the interaction of RIG-I and DDX60 and the effect of DDX60 on IFN- α production, we hypothesize that DDX60 might mediate RIG-I signaling activation upon HTNV infection, which requires further investigation.

Of note, we applied multiple cell lines to explore the role of NEAT1 during HTNV infection. HTNV primarily targets vascular endothelial cells in vivo and contributes to the increased vascular permeability and coagulation disorders in HFRS; hence, HUVECs are the most common in vitro cell model to study host innate immunity against HTNV infection or viral pathogenesis (32). EVC-304 cells are also endovascular cells, whereas EVC-304 TLR4 Δ

cells are TLR4-deleted cells, both of which have been used for HTNV infection related studies (15, 33) . A549 cells were once used to isolate HTNV, and they were confirmed to be a mature model of infection (34) (35) (36) (37) . Additionally, Huh 7.0 and Huh 7.5 (RIG-I Δ) cells used in our study have been reported to be infected by HTNV by Lee et al. (5) and can be used as a cell model to study immune responses against HTNV replication (38, 39) . Additionally, HEK293 (40) and HeLa (41) cells have been reported to be infected by HTNV. Using qRT-PCR, Western blotting, and immunofluorescence assays, we have also shown that both HEK293 and HeLa cells can be infected by HTNV.

To study the molecular mechanism underlying the effect of NEAT1 on IFN expression and HTNV infection, it may be suitable to use HEK293 and HeLa cells as a cell model, especially under conditions in which HTNV NP can be detected using Western blot or immunofluorescence analyses.

In experiments to assess the effect of NEAT1 on the control of hantaviruses, In-Cell Western (ICW) analysis was applied to qualify HTNV NP production. Alterations in the relative fluorescence intensity of NP after silencing or overexpressing NEAT1-2 did not seem to be as remarkable as qRT-PCR or Western blot analysis results. The NP spotted and exhibited in the ICW results forms obvious stains that mimic PFU. However, the specific values scanned and analyzed by the ICW assay reflect only the fluorescence intensity of the integral well instead of the number of spots. As a consequence, the intensity represented the quantity of NP production but could not directly indicate the virulence, which was better shown by plaque-forming assays. The RNAi studies in vivo are encouraging (Fig. 8) , but the NC used by our group was not mutated si-NEAT1-2 (i.e., same sense strand, but with a point mutation in the targeting strand). The results would be more compelling if the control mice had been treated with the mutated si-NEAT1-2.

One major finding of our study is that the lncRNA NEAT1 serves as positive feedback for RIG-I signaling. After observing that NEAT1 can regulate IFN expression by HTNV infection, we were interested in the function of NEAT1. We noticed that silencing NEAT1-2 or ectopically expressing NEAT1-2 could not inhibit or enhance IFN expression without HTNV infection (Fig. 3F) , which indicated that NEAT1-2 could not directly affect IFN- α expression. This finding excludes the possibility that NEAT1-2 directly promoted IFN- α and that IFN- α promoted the expression of PRRs such as RIG-I. Thereafter, NEAT1 was found to modulate HTNV-induced RIG-I and DDX60 expression. Recent findings have shown that RIG-I

signaling is essential for an efficient polyfunctional T cell response during IAV infection (42). Indeed, we found that the function of T cells was suppressed after NEAT1-2 depletion in our animal experiments (Fig. 8G), which might be due to the disrupted RIG-I signaling in NEAT1-2 silenced T cells.

In conclusion, this is the first study to describe the role of NEAT1 in HTNV infection. HTNV infection induced NEAT1 expression through the RIG-I-IRF7 pathway, while NEAT1 displayed positive feedback for RIG-I signaling. NEAT1 relocated SFPQ from the potential promoter region of several antiviral genes to the paraspeckles, removing the transcriptional inhibitory effects of SFPQ. This phenomenon would facilitate the expression of DDX60 and RIG-I, thus promoting IFN responses and suppressing HTNV infection (Fig. 9). nontarget control (i.e., negative control [NC]) or targeted RIG-I and DDX60 were designed by Gene-Pharma as follows: NC, 5'-UUCUUCGAACGUGUCACGUTT-3'; si-RIG-I-1, 5'-GCCCAUUUAAACCAAGAAATT-3', si-RIG-I-2, 5'-GGUGGAGGAUAUUUGAACUTT-3', and si-RIG-I-3, 5'-CCCAACGAUAUCAUUUCUTT-3'; si-DDX60-1, 5'-GUCCAGGUGUCAGUUUGAUTT-3', si-DDX60-2, 5'-CCGAAGUGAAGAAGGUAAATT-3', and si-DDX60-3, 5'-GAUGGAUGCUAGGAAAUAUTT-3'. The pCMV-NEAT1-1 and pCMV-NEAT1-2 plasmids, which transcribe NEAT1-1 and NEAT1-2, respectively, were provided by Nakagawa Shinichi (10). The Flag-RIG-I and pUNO-DDX60 plasmids were purchased from Invitrogen.

Reagents. Mouse monoclonal antibody (MAb) 1A8 for the HTNV nucleocapsid protein (NP) was produced as previously described (43). Abs against RIG-I, IRF3, IRF7, p65, and STAT1, as well as the neutralizing antibodies against IFN- α and IFN- γ , were purchased from Abcam (Cambridge, MA, USA). Phorbol myristate acetate (PMA) and Ab against DDX60 were purchased from Sigma-Aldrich, Inc. (St. Louis, MO, USA). Abs against SFPQ, NONO, GAPDH (glyceraldehyde-3-phosphate dehydrogenase), and β -actin were purchased from Protein Tech, Inc. (Wuhan, China). The Abs targeting CD11b, F4/80, CD3, CD8, IFN- γ , inducible nitric oxide synthase (iNOS), and CD206 for flow cytometry were purchased from BD Biosciences (San Jose, CA, USA). IFN- α , - γ , and - β , TNF- α , and IL-1 β were from PeperoTech (Rocky Hill, NJ). ELISA kits for IFN- γ detection were manufactured by R&D Systems, Inc. (Minneapolis, MN, USA).

RNA extraction and quantitative real-time PCR (qRT-PCR) analysis. Total cellular RNAs were extracted with RNAiso (TaKaRa, Dalian, China), the concentration of which was measured using a NanoDrop 1000 spectrophotometer. Reverse transcription (RT) was then performed with PrimeScript RT master mix (TaKaRa) according to the instructions provided by the manufacturer. Each cDNA was denatured at 95°C for 5 min and amplified for 40 cycles of 15 s at 98°C, 30 s at 58°C, and 30 s at 72°C using a LightCycler 96 (Roche, Basel, Switzerland). The mRNA expression level of each target gene was normalized to the respective β -actin and analyzed. The qRT-PCR primer sequences for NEAT1, NEAT1-2, IFN- β , HTNV S segment, RIG-I, DDX60, β -actin, and GAPDH were obtained from previous reports (24, 45). The methods used to quantify HTNV RNA load have been described by our group previously (46). DGE analysis and lncRNA sequencing. HUVECs with a confluence of 80% in 6 wells were mock infected or infected with live or 60 Co-inactivated HTNV at an MOI of 1. RNAs were extracted as previously described at 24 hpi, and the quality was analyzed using FastQC software by the Beijing Genomics Institute (BGI, Shenzhen, China). Digital gene expression (DGE) tags were annotated to the human transcriptome (Ensembl version 58) by mapping the reads to the sequence flanking NlaIII restriction sites on both coding and noncoding strands. Tags matching more than one gene region were discarded. Tag counts were normalized to TPM (transcripts per million) by dividing the raw tag count by the total number of tags from each library and multiplying by 1 million. To avoid the possible noise signal from high-throughput sequencing, the genes with average TPM of less than 1 in these three states were excluded. In this study, an absolute fold change of no less than 1.5 and a false discovery rate (FDR) of less than 0.001 were used to define the differentially expressed genes. Genes were selected as differentially expressed using a P value threshold of 0.01. Genes were selected as differentially expressed using a P value threshold of 0.01.

FISH and immunofluorescence assays (IFA). Fluorescence in situ hybridization (FISH) was performed with a FISH kit (Ribobio Co.) according to the manufacturer's instructions. In brief, cells were fixed with 4% paraformaldehyde (PFA) for 10 min at room temperature and permeabilized with 0.5% Triton X-100 for 15 min at room temperature. Prehybridization was performed with lncRNA FISH probe mix at 37°C for 30 min, and then hybridization was performed by adding NEAT1-2 FISH probe mix and incubating the mixture at 37°C overnight. After washing with 4 $\times$, 2 $\times$, and 1 $\times$ SSC, the cell nuclei were stained with DAPI (4',6-diamidino-2-phenylindole). Finally, the samples were observed using a BX60 fluorescence microscope (Olympus, Tokyo, Japan).

IFA was performed after FISH or independently. The cells were fixed with 4% PFA for 10 min and permeabilized with 0.1% Triton X-100 for 15 min. Primary Abs were added and incubated at 37°C for 2 h. After five washes with DPBS, secondary Cy3-or fluorescein isothiocyanate (FITC)-conjugated goat anti-rabbit or goat anti-mouse IgG (Sangon, Shanghai, China) was added and incubated at 37°C for 2 h. Cell nuclei were stained with DAPI. Finally, the samples were observed using a BX60 fluorescence microscope (Olympus).

Cells were washed twice with ice-cold DPBS and lysed with 1x SDS protein loading buffer (50 mM Tris, 2% SDS, 10% glycerol, 2% 2-mercaptoethanol, and 0.1% bromophenol blue). The samples were then boiled at 95°C for 10 min. The lysates were resolved by 10%, 12%, or 15% SDS-PAGE and transferred to polyvinylidene fluoride (PVDF) membranes (Millipore). The membranes were incubated with the primary antibodies, followed by secondary antibodies labeled with infrared dyes (Li-Cor Biosciences, Lincoln, NE, USA). The signals on the PVDF membrane were visualized using an Odyssey infrared imaging system (Li-Cor Biosciences, Lincoln, NE, USA).

ICW assay. The In-Cell Western (ICW) assay was performed using an Odyssey imaging system (Li-Cor) according to the manufacturer's instructions. HUVECs were either mock transfected or transfected with NC sequences, si-NEAT1, st-NEAT1-2, vector plasmids, pCMV-NEAT1-1, or pCMV-NEAT1-2 and grown in 96-well plates (2 x 10⁴ cells/well). Twenty-four hours posttransfection, the cells were either infected or mock infected with HTNV at an MOI of 1. At 48 hpi, HUVECs were washed twice with ice-cold DPBS, fixed with 4% PFA for 10 min, and permeabilized with 1.0% Triton X-100 for 15 min. Cells were added with Li-Cor Odyssey blocking solution at room temperature for 30 min and incubated at 4°C overnight with mouse IgG MAb 1A8 against HTNV NP together with rabbit IgG antibody against α -actin, both of which were diluted in PBS containing 3% bovine serum albumin (BSA; HyClone). Subsequently, the cells were washed and stained with goat anti-mouse IgG IRDye 800 antibody (1:5,000; Li-Cor) and goat anti-rabbit

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First cases of coronavirus disease 2019 (COVID-19) in the WHO European Region, 24 January to 21 February 2020

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Abstract: In the WHO European Region, COVID-19 surveillance was implemented 27 January 2020. We detail the first European cases. As at 21 February, nine European countries reported 47 cases. Among 38 cases studied, 21 were linked to two clusters in Germany and France, 14 were infected in China. Median case age was 42 years; 25 were male. Late detection of the clusters' index cases delayed isolation of further local cases. As at 5 March, there were 4,250 cases.

Text: In the WHO European Region, COVID-19 surveillance was implemented 27 January 2020. We detail the first European cases. As at 21 February, nine European countries reported 47 cases. Among 38 cases studied, 21 were linked to two clusters in Germany and

France, 14 were infected in China. Median case age was 42 years; 25 were male. Late detection of the clusters' index cases delayed isolation of further local cases. As at 5 March, there were 4,250 cases.

A cluster of pneumonia of unknown origin was identified in Wuhan, China, in December 2019 [1] . On 12 January 2020, Chinese authorities shared the sequence of a novel coronavirus termed severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) isolated from some clustered cases [2] . Since then, the disease caused by SARS-CoV-2 has been named coronavirus disease 2019 (COVID -19) . As at 21 February 2020, the virus had spread rapidly mostly within China but also to 28 other countries, including in the World Health Organization (WHO) European Region [3] [4] [5] .

Here we describe the epidemiology of the first cases of COVID-19 in this region, excluding cases reported in the United Kingdom (UK), as at 21 February 2020. The study includes a comparison between cases detected among travellers from China and cases whose infection was acquired due to subsequent local transmission.

On 27 January 2020, the European Centre for Disease Prevention and Control (ECDC) and the WHO Regional Office for Europe asked countries to complete a WHO standard COVID-19 case report form for all confirmed and probable cases according to WHO criteria [6] [7] [8] . The overall aim of surveillance at this time was to support the global strategy of containment of COVID-19 with rapid identification and follow-up of cases linked to affected countries in order to minimise onward transmission. The surveillance objectives were to: describe the key epidemiological and clinical characteristics of COVID-19 cases detected in Europe; inform country preparedness; and improve further case detection and management. Data collected included demographics, history of recent travel to affected areas, close contact with a probable or confirmed COVID-19 case, underlying conditions, signs and symptoms of disease at onset, type of specimens from which the virus was detected, and clinical outcome. The WHO case definition was adopted for surveillance: a confirmed case was a person with laboratory confirmation of SARS-CoV-2 infection (ECDC recommended two separate SARS-CoV-2 RT-PCR tests), irrespective of clinical signs and symptoms, whereas a probable case was a suspect case for whom testing for SARS-CoV-2 was inconclusive or positive using a pan-coronavirus assay [8] . By 31 January 2020, 47 laboratories in 31 countries, including 38 laboratories in 24 European Union and European Economic Area (EU/EEA) countries, had diagnostic capability for SARS-CoV-2 available

(close to 60% of countries in the WHO European Region), with cross-border shipment arrangements in place for many of those lacking domestic testing capacity. The remaining six EU/EEA countries were expected to have diagnostic testing available by mid-February [9] .

As at 09:00 on 21 February 2020, 47 confirmed cases of COVID-19 were reported in the WHO European Region and one of these cases had died [4] . Data on 38 of these cases (i.e. all except the nine reported in the UK) are included in this analysis.

The first three cases detected were reported in France on 24 January 2020 and had onset of symptoms on 17, 19 and 23 January respectively [10] . The first death was reported on 15 February in France. As at 21 February, nine countries had reported cases (Figure) : Belgium (1), Finland (1), France (12), Germany (16), Italy (3), Russia (2), Spain (2), Sweden (1) and the UK (9 -not included further).

The place of infection (assessed at national level based on an incubation period presumed to be up to 14 days [11] , travel history and contact with probable or confirmed cases as per the case definition) was reported for 35 cases (missing for three cases), of whom 14 were infected in China (Hubei province: 10 cases; Shandong province: one case; province not reported for three cases). The remaining 21 cases were infected in Europe. Of these, 14 were linked to a cluster in Bavaria, Germany, and seven to a cluster in Haute-Savoie, France [12, 13] . Cases from the Bavarian cluster were reported from Germany and Spain, whereas cases from the Haute-Savoie cluster were reported from France All but two cases were hospitalised (35 of 37 where information on hospitalisation was reported), although it is likely that most were hospitalised to isolate the person rather than because of severe disease. The time from onset of symptoms to hospitalisation (and isolation) ranged between 0 and 10 days with a mean of 3.7 days (reported for 29 cases). The mean number of days to hospitalisation was 2.5 days for cases imported from China, but 4.6 days for those infected in Europe. This was mostly a result of delays in identifying the index cases of the two clusters in France and Germany. In the German cluster, for example, the first three cases detected locally were hospitalised in a mean of 5.7 days, whereas the following six took only a mean of 2 days to be hospitalised.

Symptoms at the point of diagnosis were reported for 31 cases. Two cases were asymptomatic and remained so until tested negative. The asymptomatic cases were tested as part of screening following repatriation and during contact tracing respectively. Of the remaining 29, 20 reported fever, 14 reported cough and eight reported weakness. Additional symptoms reported included headaches (6 cases), sore throat (2), rhinorrhoea (2), shortness of breath (2), myalgia (1), diarrhoea (1) and nausea (1). Fever was reported as the sole symptom for nine cases. In 16 of 29 symptomatic cases, the symptoms at diagnosis were consistent with the case definition for acute respiratory infection [16], although it is possible that cases presented additional symptoms after diagnosis and these were not reported.

Data on pre-existing conditions were reported for seven cases; five had no pre-existing conditions while one was reported to be obese and one had pre-existing cardiac disease. No data on clinical signs e.g. dyspnea etc. were reported for any of the 38 cases.

All hospitalised cases had a benign clinical evolution except four, two reported in Italy and two reported in France, all of whom developed viral pneumonia. All three cases who were aged 65 years or over were admitted to intensive care and required respiratory support and one French case died. The case who died was hospitalised for 21 days and required intensive care and mechanical ventilation for 19 days. The duration of hospitalisation was reported for 16 cases with a median of 13 days (range: 8-23 days). As at 21 February 2020, four cases were still hospitalised.

All cases were confirmed according to specific assays targeting at least two separate genes (envelope (E) gene as a screening test and RNA-dependent RNA polymerase (RdRp) gene or nucleoprotein (N) gene for confirmation) [8, 17]. The specimen types tested were reported for 27 cases: 15 had positive nasopharyngeal swabs, nine had positive throat swabs, three cases had positive sputum, two had a positive nasal swab, one case had a positive nasopharyngeal aspirate and one a positive endotracheal aspirate.

As at 09:00 on 21 February, few COVID-19 cases had been detected in Europe compared with Asia. However the situation is rapidly developing, with a large outbreak recently identified in northern Italy, with transmission in several municipalities and at least two

deaths [18] . As at 5 March 2020, there are 4,250 cases including 113 deaths reported among 38 countries in the WHO European region [19] .

In our analysis of early cases, we observed transmission in two broad contexts: sporadic cases among travellers from China (14 cases) and cases who acquired infection due to subsequent local transmission in Europe (21 cases). Our analysis shows that the time from symptom onset to hospitalisation/case isolation was about 3 days longer for locally acquired cases than for imported cases. People returning from affected areas are likely to have a low threshold to seek care and be tested when symptomatic, however delays in identifying the index cases of the two clusters in France and Germany meant that locally acquired cases took longer to be detected and isolated. Once the exposure is determined and contacts identified and quarantined (171 contacts in France and 200 in Germany for the clusters in Haute-Savoie and Bavaria, respectively), further cases are likely to be rapidly detected and isolated when they develop symptoms [15, 20] . In the German cluster, for example, the first three cases detected locally were hospitalised in a mean of 5.7 days, whereas the following six were hospitalised after a mean of 2 days. Locally acquired cases require significant resources for contact tracing and quarantine, and countries should be prepared to allocate considerable public health resources during the containment phase, should local clusters emerge in their population. In addition, prompt sharing of information on cases and contacts through international notification systems such as the International Health Regulations (IHR) mechanism and the European Commission's European Early Warning and Response System is essential to contain international spread of infection.

All of the imported cases had a history of travel to China. This was consistent with the epidemiological situation in Asia, and supported the recommendation for testing of suspected cases with travel history to China and potentially other areas of presumed ongoing community transmission. The situation has evolved rapidly since then, however, and the number of countries reporting COVID-19 transmission increased rapidly, notably with a large outbreak in northern Italy with 3,089 cases reported as at 5 March [18, 19] . Testing of suspected cases based on geographical risk of importation needs to be complemented with additional approaches to ensure early detection of local circulation of COVID-19, including through testing of severe acute respiratory infections in hospitals irrespective of travel history as recommended in the WHO case definition updated on 27 February 2020 [21] .

The clinical presentation observed in the cases in Europe is that of an acute respiratory infection. However, of the 31 cases with information on symptoms, 20 cases presented with fever and nine cases presented only with fever and no other symptoms. These findings, which are consistent with other published case series, have prompted ECDC to include fever among several clinical signs or symptoms indicative for the suspected case definition.

Three cases were aged 65 years or over. All required admission to intensive care and were tourists (imported cases). These findings could reflect the average older age of the tourist population compared with the local contacts exposed to infection in Europe and do not allow us to draw any conclusion on the proportion of severe cases that we could expect in the general population of Europe. Despite this, the finding of older individuals being at higher risk of a severe clinical course is consistent with the evidence from Chinese case series published so far although the majority of infections in China have been mild [22, 23] .

This preliminary analysis is based on the first reported cases of COVID-19 cases in the WHO European Region. Given the small sample size, and limited completeness for some variables, all the results presented should be interpreted with caution.

With increasing numbers of cases in Europe, data from surveillance and investigations in the region can build on the evidence from countries in Asia experiencing more widespread transmission particularly on disease spectrum and the proportion of infections with severe outcome [22] . Understanding the infection-severity is critical to help plan for the impact on the healthcare system and the wider population. Serological studies are vital to understand the proportion of cases who are asymptomatic. Hospital-based surveillance could help estimate the incidence of severe cases and identify risk factors for severity and death. Established hospital surveillance systems that are in place for influenza and other diseases in Europe may be expanded for this purpose. In addition, a number of countries in Europe are adapting and, in some cases, already using existing sentinel primary care based surveillance systems for influenza to detect community transmission of SARS-CoV-2. This approach will be used globally to help identify evidence of widespread community transmission and, should the virus spread and containment no longer be deemed feasible, to monitor intensity of disease transmission, trends and its geographical spread.

Additional research is needed to complement surveillance data to build knowledge on the infectious period, modes of transmission, basic and effective reproduction numbers, and effectiveness of prevention and case management options also in settings outside of China. Such special studies are being conducted globally, including a cohort study on citizens repatriated from China to Europe, with the aim to extrapolate disease incidence and risk factors for infection in areas with community transmission. Countries together with ECDC and WHO, should use all opportunities to address these questions in a coordinated fashion at the European and global level.

provided input to the outline, multiple versions of the manuscript and gave approval to the final draft.

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Frontiers in antiviral therapy and immunotherapy

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Abstract: nan

Text: Globally, recent decades have witnessed a growing disjunction, a 'Valley of Death' ^{1,2} no less, between broadening strides in fundamental biomedical research and their incommensurate reach into the clinic. Plumbing work on research funding and development pipelines through recent changes in the structure of government funding, ² new public and private joint ventures and specialist undergraduate and postgraduate courses now aim to incorporate pathways to translation at the earliest stages. Reflecting this shift, the number of biomedical research publications targeting 'translational' concepts has increased exponentially, up 1800% between 2003 and 2014 ³ and continuing to rise rapidly up to the present day. Fuelled by the availability of new research technologies, as well as changing disease, cost and other pressing issues of our time, further growth in this exciting space will undoubtedly continue. Despite recent advances in the therapeutic control of immune function and viral infection, current therapies are often challenging to develop, expensive to deploy and readily select for resistance-conferring mutants. Shaped by the hostvirus immunological 'arms race' and tempered in the forge of deep time, the biodiversity of our world is increasingly being harnessed for new biotechnologies and therapeutics. Simultaneously, a shift towards host-oriented antiviral therapies is currently underway. In this Clinical & Translational Immunology Special Feature, I illustrate a strategic vision integrating these themes to create new, effective, economical and robust antiviral therapies and immunotherapies, with both the realities and the opportunities afforded to researchers working in our changing world squarely in mind.

Opening this CTI Special Feature, I outline ways these issues may be solved by creatively leveraging the so-called 'strengths' of viruses. Viral RNA polymerisation and reverse transcription enable resistance to treatment by conferring extraordinary genetic diversity. However, these exact processes ultimately restrict viral infectivity by strongly limiting virus genome sizes and their incorporation of new information. I coin this evolutionary dilemma the 'information economy paradox'. Many viruses attempt to resolve this by manipulating multifunctional or multitasking host cell proteins (MMHPs), thereby maximising host subversion and viral infectivity at minimal informational cost. ⁴ I argue this exposes an 'Achilles Heel' that may be safely targeted via host-oriented therapies to impose devastating informational and fitness barriers on escape mutant selection. Furthermore, since MMHPs are often conserved targets within and between virus families, MMHP-targeting therapies may exhibit both robust and broadspectrum antiviral efficacy. Achieving this through drug repurposing will break the vicious cycle of escalating therapeutic development costs and trivial escape mutant selection, both quickly and in multiple places. I also discuss alternative posttranslational and RNA-based antiviral approaches,

designer vaccines, immunotherapy and the emerging field of neo-virology. 4 I anticipate international efforts in these areas over the coming decade will enable the tapping of useful new biological functions and processes, methods for controlling infection, and the deployment of symbiotic or subclinical viruses in new therapies and biotechnologies that are so crucially needed.

Upon infection, pathogens stimulate expression of numerous host inflammatory factors that support recruitment and activation of immune cells. On the flip side, this same process also causes immunopathology when prolonged or deregulated. 5 In their contribution to this Special Feature, Yoshinaga and Takeuchi review endogenous RNA-binding proteins (RBPs) that post-transcriptionally control expression of crucial inflammatory factors in various tissues and their potential therapeutic applications. 6 These RBPs include tristetraprolin and AUF1, which promote degradation of AU-rich element (ARE)-containing mRNA; members of the Roquin and Regnase families, which respectively promote or effect degradation of mRNAs harbouring stem-loop structures; and the increasingly apparent role of the RNA methylation machinery in controlling inflammatory mRNA stability. These activities take place in various subcellular compartments and are differentially regulated during infection. In this way, mRNA-destabilising RBPs constitute a 'brake' on the immune system, which may ultimately be toggled therapeutically. I anticipate continued efforts in this area will lead to new methods of regaining control over inflammation in autoimmunity, selectively enhancing immunity in immunotherapy, and modulating RNA synthesis and virus replication during infection.

Another mRNA under post-transcriptional regulation by Regnase-1 and Roquin is Furin, which encodes a conserved proprotein convertase crucial in human health and disease. Furin, along with other PCSK family members, is widely implicated in immune regulation, cancer and the entry, maturation or release of a broad array of evolutionarily diverse viruses including human papillomavirus (HPV), influenza (IAV), Ebola (EboV), dengue (DenV) and human immunodeficiency virus (HIV). Here, Braun and Sauter review the roles of furin in these processes, as well as the history and future of furin-targeting therapeutics. 7 They also discuss their recent work revealing how two IFN-cinducible factors exhibit broad-spectrum inhibition of IAV, measles (MV), zika (ZikV) and HIV by suppressing furin activity. 8 Over the coming decade, I expect to see an ever-finer spatiotemporal resolution of host-oriented therapies to achieve safe, effective and broad-spectrum yet costeffective therapies for clinical use.

The increasing abundance of affordable, sensitive, high-throughput genome sequencing technologies has led to a recent boom in metagenomics and the cataloguing of the microbiome of our world. The MinION nanopore sequencer is one of the latest innovations in this space, enabling direct sequencing in a miniature form factor with only minimal sample preparation and a consumer-grade laptop computer. Nakagawa and colleagues here report on their latest experiments using this system, further improving its performance for use in resource-poor contexts for meningitis diagnoses. ⁹ While direct sequencing of viral genomic RNA is challenging, this system was recently used to directly sequence an RNA virus genome (IAV) for the first time. ¹⁰ I anticipate further improvements in the performance of such devices over the coming decade will transform virus surveillance efforts, the importance of which was underscored by the recent EboV and novel coronavirus (nCoV / COVID-19) outbreaks, enabling rapid deployment of antiviral treatments that take resistance-conferring mutations into account.

Decades of basic immunology research have provided a near-complete picture of the main armaments in the human antiviral arsenal. Nevertheless, this focus on mammalian defences and pathologies has sidelined examination of the types and roles of viruses and antiviral defences that exist throughout our biosphere. One case in point is the CRISPR/Cas antiviral immune system of prokaryotes, which is now repurposed as a revolutionary gene-editing biotechnology in plants and animals. ¹¹ Another is the ancient lineage of nucleocytoplasmic large DNA viruses (NCLDVs), which are emerging human pathogens that possess enormous genomes of up to several megabases in size encoding hundreds of proteins with unique and unknown functions. ¹² Moreover, hundreds of human- and avian-infective viruses such as IAV strain H5N1 are known, but recent efforts indicate the true number may be in the millions and many harbour zoonotic potential. ¹³ It is increasingly clear that host-virus interactions have generated truly vast yet poorly understood and untapped biodiversity. Closing this Special Feature, Watanabe and Kawaoka elaborate on neo-virology, an emerging field engaged in cataloguing and characterising this biodiversity through a global consortium. ¹⁴ I predict these efforts will unlock a vast wealth of currently unexplored biodiversity, leading to biotechnologies and treatments that leverage the host-virus interactions developed throughout evolution.

When biomedical innovations fall into the 'Valley of Death', patients who are therefore not reached all too often fall with them. Being entrusted with the resources and expectation to

conceive, deliver and communicate dividends to society is both cherished and eagerly pursued at every stage of our careers. Nevertheless, the road to research translation is winding and is built on a foundation of basic research. Supporting industry-academia collaboration and nurturing talent and skills in the Indo-Pacific region are two of the four pillars of the National Innovation and Science Agenda. 2 These frame Australia's Medical Research and Innovation Priorities, which include antimicrobial resistance, global health and health security, drug repurposing and translational research infrastructure, 15 capturing many of the key elements of this CTI Special Feature. Establishing durable international relationships that integrate diverse expertise is essential to delivering these outcomes. To this end, NHMRC has recently taken steps under the International Engagement Strategy 16 to increase cooperation with its counterparts overseas. These include the Japan Agency for Medical Research and Development (AMED), tasked with translating the biomedical research output of that country. Given the reciprocal efforts at accelerating bilateral engagement currently underway, 17 the prospects for new areas of international cooperation and mobility have never been more exciting nor urgent. With the above in mind, all contributions to this CTI Special Feature I have selected from research presented by fellow invitees to the 2018 Awaji International Forum on Infection and Immunity (AIFI) and 2017 Consortium of Biological Sciences (ConBio) conferences in Japan. Both Australia and Japan have strong traditions in immunology and related disciplines, and I predict that the quantity, quality and importance of our bilateral cooperation will accelerate rapidly over the short to medium term. By expanding and cooperatively leveraging our respective research strengths, our efforts may yet solve the many pressing disease, cost and other sustainability issues of our time.

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Haunted with and hunting for viruses

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Abstract: nan

Text: pecially with next-generation sequencing (NGS) for new virus genome discovery, e.g., Ruben Donis et al. [10] sequenced a bat-derived influenza virus genome by using NGS in 2012, raising a serious question as to whether or not our seasonal or pandemic flu might have another reservoir host. Chen and colleagues [11] confirmed the SFTSV independently by using NGS. Indeed, metagenomics analysis has yielded a great deal of new viruses, especially from the environment. Our actively hunting for new viruses has made some significant contributions for our understanding of virus ecology, pathogenesis and interspecies transmission.

Science China Life Sciences has focused on this hot topic in the event of the H7N9 outbreak after a comprehensive overview of the topic addressing HPAIV H5N1 in 2009 in the journal [12] [13] [14] . In this issue, six groups have been invited to present their recent findings on the emerging viruses, in addition to a previous report on H7N9 [3] .

Shi [15] reviewed recent discoveries of new viruses or virus genomes from bat. Bat is believed to harbor many more viruses than we ever thought as a reservoir host or even a susceptible host [16] . After the SARS-CoV virus, we have been actively seeking for new coronaviruses from bat and have yielded many of them, including potential human infecting HKU-1, 4, 5 and 9 [17, 18] . Recent MERS-CoV infection is another example for severe disease caused by used-to-be-less pathogenic coronaviruses. Shi and colleagues [19] by using NGS have discovered many unknown animal viruses from bat, especially some important paramyxoviruses and reoviruses. Filovirus has also been identified in bat with potential severe outcomes. Lyssaviruses (with many genotypes, including rabies virus) in the Rhabdoviridae family have been linked with severe fatal human cases, even in the developed countries, including Australia, with the bites of bats in the city [20, 21] . The

potential roles of these viruses in bats for interspecies transmission are yet to be elucidated.

Tan and colleagues [22] specifically focused on the newly-emerged MERS-CoV. The virus was identified in 2012 in the Middle East with some exported cases to Europe. In 2013 the virus has been re-emerging and expanding its borders to more European countries. In the initial diagnosis, the pan-coronavirus real-time reverse transcription polymerase chain reaction (RT-PCR) assay played a very important role for the identification of the causative agents. By using this method, scientists detected an expected-size PCR fragment for the corresponding conserved region of ORF1b of the replicase gene of a coronavirus. This is another example that molecular biology methods played for the discovery of new pathogens. Soon the receptor used by MERS-CoV to enter the host cells was identified [23] and the molecular basis of the receptor binding to the virus was also elucidated recently [8]

Enterovirus has been known as serious human pathogens for a long time but their significance to the public health has been emphasized by the emergence of enterovirus 71 in 1998 as a serious pathogenic agents for children in Taiwan [24] and re-emerged in mainland China in 2008 [25] . In this issue, Duan and colleagues [26] summarized the findings of new enteroviruses by using NGS. Because of the application of new NGS technology they also challenged the Koch's postulates. A new model of Koch's postulates, named the metagenomic Koch's postulates, has provided guidance for the study of the pathogenicity of novel viruses. The review also provided a detailed description of the NGS and related molecular methods for the virus discovery followed by a list of new enteroviruses found in human feces. These include viruses in the family of Picornaviridae, Parvoviridae, Circoviridae, Astroviridae and Polyomaviridae.

Yu Xue-Jie and colleagues [27] reviewed the new bunyavirus, SFTSV, identified in China. As the virus discoverers, they have overviewed the whole process of the discovery, which is helpful and meaningful for the new virus discoveries in the future. The disease caused by SFTSV, with a CFR of 12%, had been in China for a couple of years before the causative agent was finally identified. There are still a lot of questions remained unknown for this new virus and vigorous studies are in great need. The transmission route of the virus has not been clarified but tick as vector is suspected. Domestic and wild animals, e.g., goats, boars, cattle and dogs, are believed to be the virus-amplifying hosts. Therefore the effective

control measures are still under evaluation. Vaccines protecting the SFTSV infection are under its way in Chinese Center for Disease Control and Prevention. Recently a similar virus has been identified in both Japan and USA (a new name of Heartland virus was proposed for the US virus) [9] .

In addition to new viruses infecting human beings, some new viruses infecting animals but their public health significance needing to be further evaluated, have also been discovered. The new flavivirus, duck egg-drop syndrome virus (DEDSV), is a good example. Su and colleagues [28] reviewed the characterization of the DEDSV and its disease form in this issue. The virus was found closely-related to a long-time-known virus, Tembusu virus [29, 30] . Initially, the disease was only found in egg-raising ducks but soon it was found in pigeons, chickens and geese [31, 32] . Yet the transmission vector, though mosquitoes are suspected, has not been identified. Due to the public health concerns of its related viruses, potential human infection of DEDSV should be evaluated.

Research on insect viruses is reviving in recent years. In this issue, Zhou and colleagues [33] reviewed the newly-identified insect viruses in China. Insects are the largest group of animals on the Earth therefore they also carry many more viruses. Studies on these viruses can provide useful knowledge for our understanding about animal or human infecting viruses. More importantly, modification and application of insect-infecting viruses can be used as effective biologicals for the control of insect pest. The new viruses identified include Wuhan nodavirus (WhNV), a member of family Nodaviridae; *Dendrolimus punctatus* tetravirus (DpTV), a new member of the genus Omegatetravirus of the family Alphanodaviridae; *Ectropis obliqua* picorna-like virus (EoV), a positive-strand RNA virus causing a lethal granulosis infection in the larvae of the tea looper (*Ectropis obliqua*), the virus a member of the Flaviridae family.

While we are enjoying ourselves with the civilization of modern societies, the ecology has ever been changing. Human beings encounter more ecology-climate-changing problems, including the zoonotic pathogens. We have to face some unknown pathogenic agents passively. To get ourselves well prepared we also ought to actively hunt for unknown pathogens. Prediction and pre-warning can only be realized by knowing more about the unknown. This is especially true for infectious agents.

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1918 Influenza: the Mother of All Pandemics

Jeffery K. Taubenberger¹ and David M. Morens¹

The “Spanish” influenza pandemic of 1918—1919, which caused ~50 million deaths worldwide, remains an ominous warning to public health. Many questions about its origins, its unusual epidemiologic features, and the basis of its pathogenicity remain unanswered. The public health implications of the pandemic therefore remain in doubt even as we now grapple with the feared emergence of a pandemic caused by H5N1 or other virus. However, new information about the 1918 virus is emerging, for example, sequencing of the entire genome from archival autopsy tissues. But, the viral genome alone is unlikely to provide answers to some critical questions. Understanding the 1918 pandemic and its implications for future pandemics requires careful experimentation and in-depth historical analysis.

"Curiouser and curiouser/" cried Alice

Lewis Carroll, *Alice's Adventures in Wonderland*, 1865

An estimated one third of the world's population (or ≥ 500 million persons) were infected and had clinically apparent illnesses (1,2) during the 1918/1919 influenza pandemic. The disease was exceptionally severe. Case-fatality rates were $>2.5\%$, compared to $<0.1\%$ in other influenza pandemics (3,4). Total deaths were estimated at ≥ 50 million (5,7) and were arguably as high as 100 million (7).

The impact of this pandemic was not limited to 1918/1919. All influenza A pandemics since that time, and indeed almost all cases of influenza A worldwide (excluding human infections from avian Viruses such as H5N1 and H7N7), have been caused by descendants of the 1918 Virus, including "drifted" H1N1 Viruses and reassorted H2N2 and H3N2 Viruses. The latter are composed of key genes from the 1918 Virus, updated by subsequently incorporated avian influenza genes that code for novel surface

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proteins, making the 1918 Virus indeed the “mother” of all pandemics.

In 1918, the cause of human influenza and its links to avian and swine influenza were unknown. Despite clinical and epidemiologic similarities to influenza pandemics of 1889, 1847, and even earlier, many questioned whether such an explosively fatal disease could be influenza at all. That question did not begin to be resolved until the 1930s, when closely related influenza Viruses (now known to be H1N1 Viruses) were isolated, first from pigs and shortly thereafter from humans. Seroepidemiologic studies soon linked both of these viruses to the 1918 pandemic (8). Subsequent research indicates that descendants of the 1918 Virus still persists enzootically in pigs. They probably also circulated continuously in humans, undergoing gradual antigenic drift and causing annual epidemics, until the 1950s. With the appearance of a new H2N2 pandemic strain in 1957 (“Asian flu”), the direct H1N1 Viral descendants Of the 1918 pandemic strain disappeared from human circulation entirely, although the related lineage persisted enzootically in pigs. But in 1977, human H1N1 Viruses suddenly “reemerged” from a laboratory freezer (9). They continue to circulate endemically and epidemically.

Thus in 2006, 2 major descendant lineages of the 1918 H1N1 Virus, as well as 2 additional reassortant lineages, persist naturally: a human epidemic/endemic H1N1 lineage, a porcine enzootic H1N1 lineage (so-called classic swine flu), and the reassorted human H3N2 Virus lineage, which like the human H1N1 Virus, has led to a porcine H3N2 lineage. None of these Viral descendants, however, approaches the pathogenicity of the 1918 parent Virus. Apparently, the porcine H1N1 and H3N2 lineages uncommonly infect humans, and the human H1N1 and H3N2 lineages have both been associated with substantially lower rates of illness and death than the virus of 1918. In fact, current H1N1 death rates are even lower than those for H3N2 lineage strains (prevalent from 1968 until the present). H1N1 Viruses descended from the 1918 strain, as well as H3N2 Viruses, have now been cocirculating worldwide for 29 years and show little evidence of imminent extinction.

Trying To Understand What Happened

By the early 1990s, 75 years of research had failed to answer a most basic question about the 1918 pandemic: why was it so fatal? No Virus from 1918 had been isolated, but all of its apparent descendants caused substantially milder human disease. Moreover, examination of mortality

data from the 1920s suggests that within a few years after 1918, influenza epidemics had settled into a pattern of annual epidemicity associated with strain drifting and substantially lowered death rates. Did some critical Viral genetic event produce a 1918 Virus of remarkable pathogenicity and then another critical genetic event occur soon after the 1918 pandemic to produce an attenuated H1N1 Virus?

In 1995, a scientific team identified archival influenza autopsy materials collected in the autumn of 1918 and began the slow process of sequencing small Viral RNA fragments to determine the genomic structure of the causative influenza Virus (10). These efforts have now determined the complete genomic sequence of 1 Virus and partial sequences from 4 others. The primary data from the above studies (11717) and a number of reviews covering different aspects of the 1918 pandemic have recently been published ([8720] and confirm that the 1918 Virus is the likely ancestor of all 4 of the human and swine H1N1 and H3N2 lineages, as well as the “extinct” H2N2 lineage. No known mutations correlated with high pathogenicity in other human or animal influenza Viruses have been found in the 1918 genome, but ongoing studies to map Virulence factors are yielding interesting results. The 1918 sequence data, however, leave unanswered questions about the origin of the Virus (19) and about the epidemiology of the

pandemic.

When and Where Did the 1918 Influenza Pandemic Arise?

Before and after 1918, most influenza pandemics developed in Asia and spread from there to the rest of the world. Confounding definite assignment of a geographic point of origin, the 1918 pandemic spread more or less simultaneously in 3 distinct waves during an 12-month period in 1918-1919, in Europe, Asia, and North America (the first wave was best described in the United States in March 1918). Historical and epidemiologic data are inadequate to identify the geographic origin of the Virus (21), and recent phylogenetic analysis of the 1918 Viral genome does not place the Virus in any geographic context ([9]).

Although in 1918 influenza was not a nationally reportable disease and diagnostic criteria for influenza and pneumonia were vague, death rates from influenza and pneumonia in the United States had risen sharply in 1915 and 1916 because of a major respiratory disease epidemic beginning in December 1915 (22). Death rates then dipped slightly in 1917. The first pandemic influenza wave appeared in the spring of 1918, followed in rapid succession by much more fatal second and third waves in the fall

and winter of 1918/1919, respectively (Figure 1). Is it possible that a poorly-adapted H1N1 Virus was already beginning to spread in 1915, causing some serious illnesses but not yet sufficiently fit to initiate a pandemic? Data consistent with this possibility were reported at the time from European military camps (23), but a counter argument is that if a strain with a new hemagglutinin (HA) was causing enough illness to affect the US national death rates from pneumonia and influenza, it should have caused a pandemic sooner, and when it eventually did, in 1918, many people should have been immune or at least partially immunoprotected. “Herald” events in 1915, 1916, and possibly even in early 1918, if they occurred, would be difficult to identify.

The 1918 influenza pandemic had another unique feature, the simultaneous (or nearly simultaneous) infection of humans and swine. The Virus of the 1918 pandemic likely expressed an antigenically novel subtype to which most humans and swine were immunologically naive in 1918 (12,20). Recently published sequence and phylogenetic analyses suggest that the genes encoding the HA and neuraminidase (NA) surface proteins of the 1918 Virus were derived from an avianlike influenza Virus shortly before the start of the pandemic and that the precursor Virus had not circulated widely in humans or swine in the few

decades before (12,15, 24). More recent analyses of the other gene segments of the Virus also support this conclusion. Regression analyses of human and swine influenza sequences obtained from 1930 to the present place the initial circulation of the 1918 precursor Virus in humans at approximately 1915-1918 (20). Thus, the precursor was probably not circulating widely in humans until shortly before 1918, nor did it appear to have jumped directly from any species of bird studied to date (19). In summary, its origin remains puzzling.

Were the 3 Waves in 1918—1919 Caused
by the Same Virus? If So, How and Why?

Historical records since the 16th century suggest that new influenza pandemics may appear at any time of year, not necessarily in the familiar annual winter patterns of interpandemic years, presumably because newly shifted influenza Viruses behave differently when they find a universal or highly susceptible human population. Thereafter, confronted by the selection pressures of population immunity, these pandemic Viruses begin to drift genetically and eventually settle into a pattern of annual epidemic recurrences caused by the drifted Virus variants.

Figure 1. Three pandemic waves: weekly combined influenza and pneumonia mortality, United Kingdom, 1918—1919 (21).

In the 1918-1919 pandemic, a first or spring wave began in March 1918 and spread unevenly through the United States, Europe, and possibly Asia over the next 6 months (Figure 1). Illness rates were high, but death rates in most locales were not appreciably above normal. A second or fall wave spread globally from September to November 1918 and was highly fatal. In many nations, a third wave occurred in early 1919 (21). Clinical similarities led contemporary observers to conclude initially that they were observing the same disease in the successive waves. The milder forms of illness in all 3 waves were identical and typical of influenza seen in the 1889 pandemic and in prior interpandemic years. In retrospect, even the rapid progressions from uncomplicated influenza infections to fatal pneumonia, a hallmark of the 1918/1919 fall and winter waves, had been noted in the relatively few severe spring wave cases. The differences between the waves thus seemed to be primarily in the much higher frequency of complicated, severe, and fatal cases in the last 2 waves.

But 3 extensive pandemic waves of influenza within 1 year, occurring in rapid succession, with only the briefest of quiescent intervals between them, was unprecedented. The occurrence, and to some extent the severity, of recur-

rent annual outbreaks, are driven by Viral antigenic drift, with an antigenic variant Virus emerging to become dominant approximately every 2 to 3 years. Without such drift, circulating human influenza Viruses would presumably disappear once herd immunity had reached a critical threshold at which further Virus spread was sufficiently limited. The timing and spacing of influenza epidemics in interpandemic years have been subjects of speculation for decades. Factors believed to be responsible include partial herd immunity limiting Virus spread in all but the most favorable circumstances, which include lower environmental temperatures and human nasal temperatures (beneficial to thermolabile Viruses such as influenza), optimal humidity, increased crowding indoors, and imperfect ventilation due to closed windows and suboptimal airflow.

However, such factors cannot explain the 3 pandemic waves of 1918-1919, which occurred in the spring-summer, summer—fall, and winter (of the Northern Hemisphere), respectively. The first 2 waves occurred at a time of year normally unfavorable to influenza Virus spread. The second wave caused simultaneous outbreaks in the Northern and Southern Hemispheres from September to November. Furthermore, the interwave periods were so brief as to be almost undetectable in some locales. Reconciling epidemiologically the steep drop in

cases in the first and second waves with the sharp rises in cases of the second and third waves is difficult. Assuming even transient postinfection immunity, how could susceptible persons be too few to sustain transmission at 1 point, and yet enough to start a new explosive pandemic wave a few weeks later? Could the Virus have mutated profoundly and almost simultaneously around the world, in the short periods between the successive waves? Acquiring Viral drift sufficient to produce new influenza strains capable of escaping population immunity is believed to take years of global circulation, not weeks of local circulation. And having occurred, such mutated Viruses normally take months to spread around the world.

At the beginning of other “off season” influenza pandemics, successive distinct waves within a year have not been reported. The 1889 pandemic, for example, began in the late spring of 1889 and took several months to spread throughout the world, peaking in northern Europe and the United States late in 1889 or early in 1890. The second recurrence peaked in late spring 1891 (more than a year after the first pandemic appearance) and the third in early 1892 (21). As was true for the 1918 pandemic, the second 1891 recurrence produced of the most deaths. The 3 recurrences in 1889-1892, however, were spread over >3 years, in contrast to 1918-1919, when the sequential waves seen

in individual countries were typically compressed into
z879 months.

What gave the 1918 Virus the unprecedented ability to generate rapidly successive pandemic waves is unclear. Because the only 1918 pandemic Virus samples we have yet identified are from second-wave patients ([6], nothing can yet be said about whether the first (spring) wave, or for that matter, the third wave, represented circulation of the same Virus or variants of it. Data from 1918 suggest that persons infected in the second wave may have been protected from influenza in the third wave. But the few data bearing on protection during the second and third waves after infection in the first wave are inconclusive and do little to resolve the question of whether the first wave was caused by the same Virus or whether major genetic evolutionary events were occurring even as the pandemic exploded and progressed. Only influenza RNAi positive human samples from before 1918, and from all 3 waves, can answer this question.

What Was the Animal Host

Origin of the Pandemic Virus?

Viral sequence data now suggest that the entire 1918 Virus was novel to humans in, or shortly before, 1918, and

that it thus was not a reassortant Virus produced from old existing strains that acquired 1 or more new genes, such as those causing the 1957 and 1968 pandemics. On the contrary, the 1918 Virus appears to be an avianlike influenza Virus derived in toto from an unknown source (17,19), as its 8 genome segments are substantially different from contemporary avian influenza genes. Influenza Virus gene sequences from a number of fixed specimens of wild birds collected circa 1918 show little difference from avian Viruses isolated today, indicating that avian Viruses likely undergo little antigenic change in their natural hosts even over long periods (24,25).

For example, the 1918 nucleoprotein (NP) gene sequence is similar to that of viruses found in wild birds at the amino acid level but very divergent at the nucleotide level, which suggests considerable evolutionary distance between the sources of the 1918 NP and of currently sequenced NP genes in wild bird strains (13,19). One way of looking at the evolutionary distance of genes is to compare ratios of synonymous to nonsynonymous nucleotide substitutions. A synonymous substitution represents a silent change, a nucleotide change in a codon that does not result in an amino acid replacement. A nonsynonymous substitution is a nucleotide change in a codon that results in an amino acid replacement. Generally, a Viral gene sub-

jected to immunologic drift pressure or adapting to a new host exhibits a greater percentage of nonsynonymous mutations, while a Virus under little selective pressure accumulates mainly synonymous changes. Since little or no selection pressure is exerted on synonymous changes, they are thought to reflect evolutionary distance.

Because the 1918 gene segments have more synonymous changes from known sequences of wild bird strains than expected, they are unlikely to have emerged directly from an avian influenza Virus similar to those that have been sequenced so far. This is especially apparent when one examines the differences at 4-fold degenerate codons, the subset of synonymous changes in which, at the third codon position, any of the 4 possible nucleotides can be substituted without changing the resulting amino acid. At the same time, the 1918 sequences have too few amino acid differences from those of wild-bird strains to have spent many years adapting only in a human or swine intermediate host. One possible explanation is that these unusual gene segments were acquired from a reservoir of influenza Virus that has not yet been identified or sampled. All of these findings beg the question: where did the 1918 Virus come from?

In contrast to the genetic makeup of the 1918 pandem-

ic Virus, the novel gene segments of the reassorted 1957 and 1968 pandemic Viruses all originated in Eurasian avian Viruses (26); both human Viruses arose by the same mechanism: reassortment of a Eurasian wild waterfowl strain with the previously circulating human H1N1 strain.

Proving the hypothesis that the Virus responsible for the 1918 pandemic had a markedly different origin requires samples of human influenza strains circulating before 1918 and samples of influenza strains in the wild that more closely resemble the 1918 sequences.

What Was the Biological Basis for
1918 Pandemic Virus Pathogenicity?

Sequence analysis alone does not offer clues to the pathogenicity of the 1918 Virus. A series of experiments are under way to model Virulence in Vitro and in animal models by using Viral constructs containing 1918 genes produced by reverse genetics.

Influenza Virus infection requires binding of the HA protein to sialic acid receptors on host cell surface. The HA receptor-binding site configuration is different for those influenza Viruses adapted to infect birds and those adapted to infect humans. Influenza Virus strains adapted to birds preferentially bind sialic acid receptors with 01 (273) linked

sugars (27729). Human-adapted influenza Viruses are thought to preferentially bind receptors with 01 (2%) linkages. The switch from this avian receptor configuration requires of the Virus only 1 amino acid change (30), and the HAs of all 5 sequenced 1918 Viruses have this change, which suggests that it could be a critical step in human host adaptation. A second change that greatly augments Virus binding to the human receptor may also occur, but only 3 of 5 1918 HA sequences have it (16).

This means that at least 2 H1N1 receptor-binding variants cocirculated in 1918: 1 with high—affinity binding to the human receptor and 1 with mixed-affinity binding to both avian and human receptors. No geographic or chronologic indication exists to suggest that one of these variants was the precursor of the other, nor are there consistent differences between the case histories or histopathologic features of the 5 patients infected with them. Whether the Viruses were equally transmissible in 1918, whether they had identical patterns of replication in the respiratory tree, and whether one or both also circulated in the first and third pandemic waves, are unknown.

In a series of in Vivo experiments, recombinant influenza Viruses containing between 1 and 5 gene segments of the 1918 Virus have been produced. Those constructs bearing the 1918 HA and NA are all highly pathogenic in

mice (31). Furthermore, expression microarray analysis performed on whole lung tissue of mice infected with the 1918 HA/NA recombinant showed increased upregulation of genes involved in apoptosis, tissue injury, and oxidative damage (32). These findings are unexpected because the Viruses with the 1918 genes had not been adapted to mice; control experiments in which mice were infected with modern human Viruses showed little disease and limited Viral replication. The lungs of animals infected with the 1918 HA/NA construct showed bronchial and alveolar epithelial necrosis and a marked inflammatory infiltrate, which suggests that the 1918 HA (and possibly the NA) contain Virulence factors for mice. The Viral genotypic basis of this pathogenicity is not yet mapped. Whether pathogenicity in mice effectively models pathogenicity in humans is unclear. The potential role of the other 1918 proteins, singularly and in combination, is also unknown. Experiments to map further the genetic basis of Virulence of the 1918 Virus in various animal models are planned. These experiments may help define the Viral component to the unusual pathogenicity of the 1918 Virus but cannot address whether specific host factors in 1918 accounted for unique influenza mortality patterns.

Why Did the 1918 Virus Kill So Many Healthy

Young Adults?

The curve of influenza deaths by age at death has historically, for at least 150 years, been U-shaped (Figure 2), exhibiting mortality peaks in the very young and the very old, with a comparatively low frequency of deaths at all ages in between. In contrast, age-specific death rates in the 1918 pandemic exhibited a distinct pattern that has not been documented before or since: a “W-shaped” curve, similar to the familiar U-shaped curve but with the addition of a third (middle) distinct peak of deaths in young adults 20–40 years of age. Influenza and pneumonia death rates for those 15–34 years of age in 1918–1919, for example, were 20 times higher than in previous years (35). Overall, nearly half of the influenza-related deaths in the 1918 pandemic were in young adults 20–40 years of age, a phenomenon unique to that pandemic year. The 1918 pandemic is also unique among influenza pandemics in that absolute risk of influenza death was higher in those <65 years of age than in those ≥65; persons <65 years of age accounted for >99% of all excess influenza-related deaths in 1918–1919. In comparison, the <65-year age group accounted for 36% of all excess influenza-related deaths in the 1957 H2N2 pandemic and 48% in the 1968 H3N2 pandemic (33).

A sharper perspective emerges when 1918 age-specific influenza morbidity rates (21) are used to adjust the W-

shaped mortality curve (Figure 3, panels, A, B, and C [35,37]). Persons 65 years of age in 1918 had a disproportionately high influenza incidence (Figure 3, panel A).

But even after adjusting age-specific deaths by age-specific clinical attack rates (Figure 3, panel B), a W-shaped curve with a case-fatality peak in young adults remains and is significantly different from U-shaped age-specific case-fatality curves typically seen in other influenza years, e.g., 1928-1929 (Figure 3, panel C). Also, in 1918 those 5 to 14 years of age accounted for a disproportionate number of influenza cases, but had a much lower death rate from influenza and pneumonia than other age groups. To explain this pattern, we must look beyond properties of the Virus to host and environmental factors, possibly including immunopathology (e.g., antibody-dependent infection enhancement associated with prior Virus exposures [38]) and exposure to risk cofactors such as coinfecting agents, medications, and environmental agents.

One theory that may partially explain these findings is that the 1918 Virus had an intrinsically high virulence, tempered only in those patients who had been born before 1889, e.g., because of exposure to a then-circulating Virus capable of providing partial immunoprotection against the 1918 Virus strain only in persons old enough (>35 years) to

have been infected during that prior era (35). But this theory would present an additional paradox: an obscure precursor Virus that left no detectable trace today would have had to have appeared and disappeared before 1889 and then reappeared more than 3 decades later.

Epidemiologic data on rates of clinical influenza by age, collected between 1900 and 1918, provide good evidence for the emergence of an antigenically novel influenza Virus in 1918 (21). Jordan showed that from 1900 to 1917, the 5- to 15-year age group accounted for 11% of total influenza cases, while the >65-year age group accounted for 6 % of influenza cases. But in 1918, cases in

Figure 2. “U-” and “W—” shaped combined influenza and pneumonia mortality, by age at death, per 100,000 persons in each age group, United States, 1911—1918. Influenza- and pneumonia-specific death rates are plotted for the interpandemic years 1911—1917 (dashed line) and for the pandemic year 1918 (solid line) (33,34).

Incidence male per 1 .nao persunslage group

Mortality per 1.000 persunslige group

+ Case—fataiity rale 1918—1919

Case fatality par 100 persons ill with P&I per age group

Figure 3. Influenza plus pneumonia (P&I) (combined) age-specific incidence rates per 1,000 persons per age group (panel A), death rates per 1,000 persons, ill and well combined (panel B), and case-fatality rates (panel C, solid line), US Public Health Service house-to-house surveys, 8 states, 1918 (36). A more typical curve of age-specific influenza case-fatality (panel C, dotted line) is taken from US Public Health Service surveys during 1928—1929 (37).

the 5 to 15-year-old group jumped to 25% of influenza cases (compatible with exposure to an antigenically novel Virus strain), while the >65-year age group only accounted for 0.6% of the influenza cases, findings consistent with previously acquired protective immunity caused by an identical or closely related Viral protein to which older persons had once been exposed. Mortality data are in accord. In 1918, persons >75 years had lower influenza and

pneumonia case-fatality rates than they had during the prepandemic period of 1911-1917. At the other end of the age spectrum (Figure 2), a high proportion of deaths in infancy and early childhood in 1918 mimics the age pattern, if not the mortality rate, of other influenza pandemics.

Could a 1918-like Pandemic Appear Again?

If So, What Could We Do About It?

In its disease course and pathologic features, the 1918 pandemic was different in degree, but not in kind, from previous and subsequent pandemics. Despite the extraordinary number of global deaths, most influenza cases in 1918 (>95% in most locales in industrialized nations) were mild and essentially indistinguishable from influenza cases today. Furthermore, laboratory experiments with recombinant influenza Viruses containing genes from the 1918 Virus suggest that the 1918 and 1918-like Viruses would be as sensitive as other typical Virus strains to the Food and Drug Administration-approved antiinfluenza drugs rimantadine and oseltamivir.

However, some characteristics of the 1918 pandemic appear unique: most notably, death rates were 5–20 times higher than expected. Clinically and pathologically, these high death rates appear to be the result of several factors, including a higher proportion of severe and complicated infections of the respiratory tract, rather than involvement of organ systems outside the normal range of the influenza Virus. Also, the deaths were concentrated in an unusually young age group. Finally, in 1918, 3 separate recurrences of influenza followed each other with unusual rapidity,

resulting in 3 explosive pandemic waves within a year's time (Figure 1). Each of these unique characteristics may reflect genetic features of the 1918 Virus, but understanding them will also require examination of host and environmental factors.

Until we can ascertain which of these factors gave rise to the mortality patterns observed and learn more about the formation of the pandemic, predictions are only educated guesses. We can only conclude that since it happened once, analogous conditions could lead to an equally devastating pandemic.

Like the 1918 Virus, H5N1 is an avian Virus (39), though a distantly related one. The evolutionary path that led to pandemic emergence in 1918 is entirely unknown, but it appears to be different in many respects from the current situation with H5N1. There are no historical data, either in 1918 or in any other pandemic, for establishing that a pandemic "precursor" Virus caused a highly pathogenic outbreak in domestic poultry, and no highly pathogenic avian influenza (HPAI) Virus, including H5N1 and a number of others, has ever been known to cause a major human epidemic, let alone a pandemic. While data bearing on influenza Virus human cell adaptation (e.g., receptor binding) are beginning to be understood at the molecular

level, the basis for Viral adaptation to efficient human-to-human spread, the chief prerequisite for pandemic emergence, is unknown for any influenza Virus. The 1918 Virus acquired this trait, but we do not know how, and we currently have no way of knowing whether H5N1 Viruses are now in a parallel process of acquiring human-to-human transmissibility. Despite an explosion of data on the 1918 Virus during the past decade, we are not much closer to understanding pandemic emergence in 2006 than we were in understanding the risk of H1N1 “swine flu” emergence in 1976.

Even with modern antiviral and antibacterial drugs, vaccines, and prevention knowledge, the return of a pandemic Virus equivalent in pathogenicity to the Virus of 1918 would likely kill >100 million people worldwide. A pandemic Virus with the (alleged) pathogenic potential of some recent H5N1 outbreaks could cause substantially more deaths.

Whether because of Viral, host or environmental factors, the 1918 Virus causing the first or ‘spring’ wave was not associated with the exceptional pathogenicity of the second (fall) and third (winter) waves. Identification of an influenza RNA-positive case from the first wave could point to a genetic basis for Virulence by allowing differ-

ences in Viral sequences to be highlighted. Identification of pre-1918 human influenza RNA samples would help us understand the timing of emergence of the 1918 Virus.

Surveillance and genomic sequencing of large numbers of animal influenza Viruses will help us understand the genetic basis of host adaptation and the extent of the natural reservoir of influenza Viruses. Understanding influenza pandemics in general requires understanding the 1918 pandemic in all its historical, epidemiologic, and biologic aspects.

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Estimating the Unreported Number of Novel Coronavirus (2019-nCoV) Cases in China in the First Half of January 2020: A Data-Driven Modelling Analysis of the Early Outbreak

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Abstract: **Background:** In December 2019, an outbreak of respiratory illness caused by a novel coronavirus (2019-nCoV) emerged in Wuhan, China and has swiftly spread to other parts of China and a number of foreign countries. The 2019-nCoV cases might have been under-reported roughly from 1 to 15 January 2020, and thus we estimated the number of unreported cases and the basic reproduction number, R_0 , of 2019-nCoV. **Methods:** We modelled the epidemic curve of 2019-nCoV cases, in mainland China from 1 December 2019 to 24 January 2020 through the exponential growth. The number of unreported cases was determined by the maximum likelihood estimation. We used the serial intervals (SI) of infection caused by two other well-known coronaviruses (CoV), Severe Acute Respiratory Syndrome (SARS) and Middle East Respiratory Syndrome (MERS) CoVs, as approximations of the unknown SI for 2019-nCoV to estimate R_0 . **Results:** We confirmed that the initial growth phase followed an exponential growth pattern. The under-reporting was likely to have resulted in 469 (95% CI: 403–540) unreported cases from 1 to 15 January 2020. The reporting rate after 17 January 2020 was likely to have increased 21-fold (95% CI: 18–25) in comparison to the situation from 1 to 17 January 2020 on average. We estimated the R_0 of 2019-nCoV at 2.56 (95% CI: 2.49–2.63). **Conclusion:** The under-reporting was likely to have occurred during the first half of January 2020 and should be considered in future investigation.

Text: A novel coronavirus (2019-nCoV) infected pneumonia infection, which is deadly [1], was first identified in Wuhan, China in December 2019 [2]. The virus causes a range of symptoms including fever, cough, and shortness of breath [3]. The cumulative number of reported cases slowly increased to cumulative 41 cases by 1 January 2020, and rapidly increased after 16 January 2020. As of 26 January 2020, the still ongoing outbreak had resulted in 2066 (618 of them are in Wuhan) confirmed cases and 56 (45 of them were in Wuhan) deaths in mainland China [4], and sporadic cases exported from Wuhan were

reported in Thailand, Japan, Republic of Korea, Hong Kong, Taiwan, Australia, and the United States, please see the World Health Organization (WHO) news release via <https://www.who.int/csr/don/en/> from 14 to 21 January 2020. Using the number of cases exported from Wuhan to other countries, a research group at Imperial College London estimated that there had been 4000 (95%CI: 1000-9700) cases in Wuhan with symptoms onset by 18 January 2020, and the basic reproduction number (R_0) was estimated at 2.6 (95%CI: 1.5-3.5) [5]. Leung et al. drew a similar conclusion and estimated the number of cases exported from Wuhan to other major cities in China [6], and the potentials of travel related risks of disease spreading was also indicated by [7].

Due to an unknown reason, the cumulative number of cases remained at 41 from 1 to 15 January 2020 according to the official report, i.e., no new case was reported during these 15 days, which appears inconsistent with the following rapid growth of the epidemic curve since 16 January 2020. We suspect that the 2019-nCoV cases were under-reported roughly from 1 to 15 January 2020. In this study, we estimated the number of unreported cases and the basic reproduction number, R_0 , of 2019-nCoV in Wuhan from 1 to 15 January 2020 based on the limited data in the early outbreak.

The time series data of 2019-nCoV cases in mainland China were initially released by the Wuhan Municipal Health Commission from 10 to 20 January 2020 [8], and later by the National Health Commission of China after 21 January 2020 [9]. The case time series data in December 2019 were obtained from a published study [3]. All cases were laboratory confirmed following the case definition by the national health commission of China [10]. We chose the data up to 24 January 2020 instead of to the present study completion date. Given the lag between timings of case confirmation and news release of new cases, the data of the most recent few days were most likely to be tentative, and thus they were excluded from the analysis to be consistent.

We suspected that there was a number of cases, denoted by ξ , under-reported from 1 to 15 January 2020. The cumulative total number of cases, denoted by C_i , of the i -th day since 1 December 2019 is the summation of the cumulative reported, c_i , and cumulative unreported cases, Ξ_i . We have $C_i = c_i + \Xi_i$, where c_i is observed from the data, and Ξ_i is 0 for i before 1 January and ξ for i after 15 January 2020. Following previous studies [11, 12], we modelled the epidemic curve, i.e., the C_i series, as an exponential growing Poisson process. Since the data from 1 to 15 January 2020 appeared constant due to unclear

reason(s), we removed these data from the fitting of exponential growth. The ξ and the intrinsic growth rate (γ) of the exponential growth were to be estimated based on the log-likelihood, denoted by ℓ , from the Poisson priors. The 95% confidence interval (95% CI) of ξ was estimated by the profile likelihood estimation framework with cutoff threshold determined by a Chi-square quantile [13], $\chi^2_{pr} = 0.95$, $df = 1$. With γ estimated, the basic reproduction number could be obtained by $R_0 = 1/M(-\gamma)$ with 100% susceptibility for 2019-nCoV presumed at this early stage. Here, the function $M(\cdot)$ was the Laplace transform, i.e., the moment generating function, of the probability distribution for the serial interval (SI) of the disease [11, 14], denoted by $h(k)$ and k is the mean SI. Since the transmission chain of 2019-nCoV remained unclear, we adopted the SI information from Severe Acute Respiratory Syndrome (SARS) and Middle East Respiratory Syndrome (MERS), which share the similar pathogen as 2019-nCoV [15] [16] [17]. We modelled $h(k)$ as Gamma distributions with mean of 8.0 days and standard deviation (SD) of 3.6 days by averaging the SI mean and SD of SARS, mean of 7.6 days and SD of 3.4 days [18], and MERS, mean of 8.4 days and SD of 3.8 days [19].

We were also interested in inferring the patterns of the daily number of cases, denoted by ε_i for the i -th day, and thus it is obviously that $C_i = C_{i-1} + \varepsilon_i$. A simulation framework was developed for the iterative Poisson process such that $E[$

denoted the expectation. The simulation was implemented starting from 1 January 2020 with a cumulative number of cases seed of 40, the same as reported on 31 December 2019. We conducted 1000 samples and calculated the median and 95% CI.

The number of 2019-nCoV unreported cases was estimated at 469 (95% CI: 403-540), see Figure 1a, which was significantly larger than 0. This finding implied the occurrence of under-reporting between 1 and 15 January 2020. After accounting for the effect of under-reporting, the R_0 was estimated at 2.56 (95% CI: 2.49-2.63), see Figure 1b, which is consistent with many existing online preprints with range from 2 to 4 [5, [20] [21] [22]. With the R_0 of 2.56 and ξ of 469, the exponential growing framework fitted the cumulative total number of cases (C_i) remarkably well, see Figure 1c iterative Poisson process such that

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Our estimation of R_0 rely on the SI of 2019-nCoV, which remains unknown as of 26 January 2020. In this work, we employed the SIs of SARS and MERS as approximations to that of 2019-nCoV. The determination of SI requires the knowledge of the chain of disease transmission that needs a sufficient number of patient samples and periods of time for follow-up [23], and thus this is unlikely to be achieved shortly. However, using SIs of SARS and MERS as approximation could provide an panels (a,b), the green shading area represents the 95% CI (on the horizontal axis), and the vertical green line represents the maximum likelihood estimate (MLE) of the number of unreported cases. With the MLE of R_0 at 2.56, panels (c,d) show the exponential growth fitting results of the cumulative number of cases (C_i) and the daily number of cases (ϵ_i) respectively. In panels (c,d), the gold squares are the reported cases, the blue bold curve represents the median of the fitting results, the dashed blue curves are the 95% CI of the fitting results, and the purple shading area represents the time window from 1 to 15 January 2020. In panel (c), the blue dots are the cumulative total, i.e., reported and unreported, number of cases. In panel (d), the grey curves are the 1000 simulation samples.

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The determination of SI requires the knowledge of the chain of disease transmission that needs a sufficient number of patient samples and periods of time for follow-up [23], and thus this is unlikely to be achieved shortly. However, using SIs of SARS and MERS as approximation could provide an insight into the transmission potential of 2019-nCoV at the early outbreak. We note that slightly varying the mean and SD of SI would not affect our main conclusions. The R_0 of 2019-nCoV was estimated at 2.56 (95% CI: 2.49-2.63), and it is generally in line with those of SARS, i.e., 2-5 [19, 24, 25], and MERS, i.e., 2.7-3.9 [26].

For the simulated daily number of cases (ε_i), see Figure 1d, we found that ε_i matched the observed daily number after 17 January 2020, but was significantly larger than the observations from 1 to 17 January 2020. This finding implied that under-reporting was likely to have occurred in the first half of January 2020. We estimated that the reporting rate after 17 January 2020 increased 21-fold (95% CI: [18] [19] [20] [21] [22] [23] [24] [25] compared to the situation from 1 to 17 January 2020 on average. One of the possible reasons was that the official diagnostic protocol was released by WHO on 17 January 2020 [27], and the diagnosis and reporting efforts of 2019-nCoV infections probably increased. Thereafter, the daily number of newly reported cases started increasing rapidly after 17 January 2020, see Figure 1d. We conducted additional sensitivity analysis by varying the starting date of the under-reporting time window, e.g., 1 January 2020 in the main results, from 2 December 2019 to 3 January 2020, and we report our estimates largely hold. The exact value of the reporting rate was difficult to determine due to lack of serological surveillance data. The reporting rate can be determined if serological surveillance data are available for a population; we would know who was infected (seropositive) and who was not (seronegative), with high confidence. The reporting rate is the ratio of reported cases over the number of seropositive individuals. It was statistically evident that increasing in reporting was likely, and thus it should be considered in the future investigation of this outbreak.

Previous preprint suggested cumulative cases of 1723 (95% CI: 427-4471) as of 12 January 2020, and 4000 (95% CI: 1000-9700) as of 18 January 2020 based on the aggregated international export cases [5]. Our analysis yielded cumulative cases of 280 (95% CI: 128-

613) as of 12 January 2020, and 609 (95% CI: 278-1333) as of 18 January 2020 based on the exponential growing mechanistic in the early outbreak. Although our estimate case number appeared to have a lower mean than those estimated by Imai et al. [5] , they are not statistically different. This study applied a different screening effort to detect the 2019-nCoV cases from that in Imai et al. [5] . Imai et al. assumed the average screening effort at overseas airports that covered travelers arriving from Wuhan. Whereas we assumed a constant screening effort applied in Wuhan at the same point of time, and then a number of cases (i.e., ξ) should have been reported yet failed to be reported in the first half of January 2020 due to all sorts of reasons. It is not surprising that different assumptions yielded different results, and this difference in screening effort also partly explained why the detected cases out of China mainly presented mild symptoms. Thus, it was reasonable that our estimates appeared lower than those estimated by Imai et al. [5] . It must be emphasized that such a gap in the knowledge would be resolved by serological survey study (for a large population to approximate the actual positive rate) or an explicit estimation of the actual reporting rate.

Under-reporting was likely to have occurred and resulted in 469 (95% CI: 403-540) unreported cases from 1 to 15 January 2020. The reporting rate after 17 January 2020 was likely to have increased 21-fold (95% CI: 18-25) compared with the situation from 1 to 17 January 2020 on average, and it should be considered in future investigation. We estimated the R_0 at 2019-nCoV to be 2.56 (95% CI: 2.49-2.63).

Author Contributions: All authors conceived the study, carried out the analysis, discussed the results, drafted the first manuscript. All authors have read and agreed to the published version of the manuscript.

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Genomic characterization of the 2019 novel human-pathogenic coronavirus isolated from a patient with atypical pneumonia after visiting Wuhan

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7067204/>

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Abstract: A mysterious outbreak of atypical pneumonia in late 2019 was traced to a seafood wholesale market in Wuhan of China. Within a few weeks, a novel coronavirus tentatively named as 2019 novel coronavirus (2019-nCoV) was announced by the World Health Organization. We performed bioinformatics analysis on a virus genome from a patient with 2019-nCoV infection and compared it with other related coronavirus genomes. Overall, the genome of 2019-nCoV has 89% nucleotide identity with bat SARS-like-CoVZXC21 and 82% with that of human SARS-CoV. The phylogenetic trees of their orf1a/b, Spike, Envelope, Membrane and Nucleoprotein also clustered closely with those of the bat, civet and human SARS coronaviruses. However, the external subdomain of Spike's receptor binding domain of 2019-nCoV shares only 40% amino acid identity with other SARS-related coronaviruses. Remarkably, its orf3b encodes a completely novel short protein. Furthermore, its new orf8 likely encodes a secreted protein with an alpha-helix, following with a beta-sheet(s) containing six strands. Learning from the roles of civet in SARS and camel in MERS, hunting for the animal source of 2019-nCoV and its more ancestral virus would be important for understanding the origin and evolution of this novel lineage B betacoronavirus. These findings provide the basis for starting further studies on the pathogenesis, and optimizing the design of diagnostic, antiviral and vaccination strategies for this emerging infection.

Text: Coronaviruses (CoVs) are enveloped, positive-sense, single-stranded RNA viruses that belong to the subfamily Coronavirinae, family Coronaviridae, order Nidovirales. There are four genera of CoVs, namely, Alphacoronavirus (α CoV), Betacoronavirus (β CoV), Deltacoronavirus (δ CoV), and Gammacoronavirus (γ CoV) [1]. Evolutionary analyses have shown that bats and rodents are the gene sources of most α CoVs and β CoVs, while avian

species are the gene sources of most δ CoVs and γ CoVs. CoVs have repeatedly crossed species barriers and some have emerged as important human pathogens. The best-known examples include severe acute respiratory syndrome CoV (SARS-CoV) which emerged in China in 2002-2003 to cause a large-scale epidemic with about 8000 infections and 800 deaths, and Middle East respiratory syndrome CoV (MERS-CoV) which has caused a persistent epidemic in the Arabian Peninsula since 2012 [2, 3] . In both of these epidemics, these viruses have likely originated from bats and then jumped into another amplification mammalian host [the Himalayan palm civet (*Paguma larvata*) for SARS-CoV and the dromedary camel (*Camelus dromedarius*) for MERS-CoV] before crossing species barriers to infect humans.

Prior to December 2019, 6 CoVs were known to infect human, including 2 α CoV (HCoV-229E and HKU-NL63) and 4 β CoV (HCoV-OC43 [

HCoV-OC43 and HCoV-HKU1 usually cause self-limiting upper respiratory infections in immunocompetent hosts and occasionally lower respiratory tract infections in immunocompromised hosts and elderly [4] . In contrast, SARS-CoV (lineage B β CoV) and MERS-CoV (lineage C β CoV) may cause severe lower respiratory tract infection with acute respiratory distress syndrome and extrapulmonary manifestations, such as diarrhea, lymphopenia, deranged liver and renal function tests, and multiorgan dysfunction syndrome, among both immunocompetent and immunocompromised hosts with mortality rates of $\sim 10\%$ and $\sim 35\%$, respectively [5, 6] . On 31 December 2019, the World Health Organization (WHO) was informed of cases of pneumonia of unknown cause in Wuhan City, Hubei Province, China [7] . Subsequent virological testing showed that a novel CoV was detected in these patients. As of 16 January 2020, 43 patients have been diagnosed to have infection with this novel CoV, including two exported cases of mild pneumonia in Thailand and Japan [8, 9] . The earliest date of symptom onset was 1 December 2019 [10] . The symptomatology of these patients included fever, malaise, dry cough, and dyspnea. Among 41 patients admitted to a designated hospital in Wuhan, 13 (32%) required intensive care and 6 (15%) died. All 41 patients had pneumonia with abnormal findings on chest computerized tomography scans [10] . We recently reported a familial cluster of 2019-nCoV infection in a Shenzhen family with travel history to Wuhan [11] . In the present study, we analyzed a 2019-nCoV complete genome from a patient in this familial cluster and compared it with the genomes of related β CoVs to provide insights into the potential source and control strategies.

The complete genome sequence of 2019-nCoV HKU-SZ-005b was available at GenBank (accession no. MN975262) (Table 1). The representative complete genomes of other related β CoVs strains collected from human or mammals were included for comparative analysis. These included strains collected from human, bats, and Himalayan palm civet between 2003 and 2018, with one 229E coronavirus strain as the outgroup.

Phylogenetic tree construction by the neighbour joining method was performed using MEGA X software, with bootstrap values being calculated from 1000 trees [12] . The percentage of replicate trees in which the associated taxa clustered together in the bootstrap test (1000 replicates) was shown next to the branches [13] . The tree was drawn to scale, with branch lengths in the same units as those of the evolutionary distances used to infer the phylogenetic tree. The evolutionary distances were computed using the Poisson correction method and were in the units of the number of amino acid substitutions per site [14] . All ambiguous positions were removed for each sequence pair (pairwise deletion option). Evolutionary analyses were conducted in MEGA X [15] . Multiple alignment was performed using CLUSTAL 2.1 and further visualized using BOX-SHADE 3.21. Structural analysis of orf8 was performed using PSI-blast-based secondary structure PREDiction (PSIPRED) [16] . For the prediction of protein secondary structure including beta sheet, alpha helix, and coil, initial amino acid sequences were input and analysed using neural networking and its own algorithm. Predicted structures were visualized and highlighted on the BOX-SHADE alignment. Prediction of transmembrane domains was performed using the TMHMM 2.0 server (<http://www.cbs.dtu.dk/services/TMHMM/>). Secondary structure prediction in the 5' -untranslated region (UTR) and 3' -UTR was performed using the RNAfold WebServer (<http://rna.tbi.univie.ac.at/cgi-bin/RNAWebSuite/RNAfold.cgi>) with minimum free energy (MFE) and partition function in Fold algorithms and Table 2 . Putative functions and proteolytic cleavage sites of 16 nonstructural proteins in orf1a/b as predicted by bioinformatics.

Putative function/domain Amino acid position Putative cleave site

complex with nsp3 and 6: DMV formation

complex with nsp3 and 4: DMV formation

short peptide at the end of orf1a basic options. The human SARS-CoV 5' -and 3' -UTR were used as references to adjust the prediction results.

The single-stranded RNA genome of the 2019-nCoV was 29891 nucleotides in size, encoding 9860 amino acids. The G + C content was 38%. Similar to other (Table 2). There are no remarkable differences between the orfs and nsps of 2019-nCoV with those of SARS-CoV (Table 3) . The major distinction between SARSr-CoV and SARS-CoV is in orf3b, Spike and orf8 but especially variable in Spike S1 and orf8 which were previously shown to be recombination hot spots.

Spike glycoprotein comprised of S1 and S2 subunits. The S1 subunit contains a signal peptide, followed by an N-terminal domain (NTD) and receptor-binding domain (RBD), while the S2 subunit contains conserved fusion peptide (FP), heptad repeat (HR) 1 and 2, transmembrane domain (TM), and cytoplasmic domain (CP). We found that the S2 subunit of 2019-nCoV is highly conserved and shares 99% identity with those of the two bat SARS-like CoVs (SL-CoV ZXC21 and ZC45) and human SARS-CoV (Figure 2). Thus the broad spectrum antiviral peptides against S2 would be an important preventive and treatment modality for testing in animal models before clinical trials [18] . Though the S1 subunit of 2019-nCoV shares around 70% identity to that of the two bat SARS-like CoVs and human SARS-CoV (Figure 3(A)), the core domain of RBD (excluding the external subdomain) are highly conserved (Figure 3(B)). Most of the amino acid differences of RBD are located in the external subdomain, which is responsible for the direct interaction with the host receptor. Further investigation of this soluble variable external subdomain region will reveal its receptor usage, interspecies transmission and pathogenesis. Unlike 2019-nCoV and human SARS-CoV, most known bat SARSr-CoVs have two stretches of deletions in the spike receptor binding domain (RBD) when compared with that of human SARS-CoV. But some Yunnan strains such as the WIV1 had no such deletions and can use human ACE2 as a cellular entry receptor. It is interesting to note that the two bat SARS-related coronavirus ZXC21 and ZC45, being closest to 2019-nCoV, can infect suckling rats and cause inflammation in the brain tissue, and pathological changes in lung & intestine. However, these two viruses could not be isolated in Vero E6 cells and were not investigated further. The two retained deletion sites in the Spike genes of ZXC21 and ZC45 may lessen their likelihood of jumping species barriers imposed by receptor specificity.

A novel short putative protein with 4 helices and no homology to existing SARS-CoV or SARS-r-CoV protein was found within Orf3b (Figure 4). It is notable that SARS-CoV deletion mutants lacking orf3b replicate to levels similar to those of wildtype virus in several cell types [19] , suggesting that orf3b is dispensable for viral replication in vitro. But orf3b may have a role in viral pathogenicity as Vero E6 but not 293T cells transfected with a construct expressing Orf3b underwent necrosis as early as 6 h after transfection and underwent simultaneous necrosis and apoptosis at later time points [20] . Orf3b was also shown to inhibit expression of IFN- β at synthesis and signalling [21] . Subsequently, orf3b homologues identified from three bat SARS-related-CoV strains were C-terminally truncated and lacked the C-terminal nucleus localization signal of SARS-CoV [22] . IFN antagonist activity analysis demonstrated that one SARS-related-CoV orf3b still possessed IFN antagonist and IRF3-modulating activities. These results indicated that different orf3b proteins display different IFN antagonist activities and this function is independent of the protein's nuclear localization, suggesting a potential link between bat SARS-related-CoV orf3b function and pathogenesis. The importance of this new protein in 2019-nCoV will require further validation and study.

Orf8 orf8 is an accessory protein found in the Betacoronavirus lineage B coronaviruses. Human SARS-CoVs isolated from early-phase patients, all civet SARS-CoVs, and other bat SARS-related CoVs contain fulllength orf8 [23] . However, a 29-nucleotide deletion,

Bat SL-CoV ZXC21 2018

Bat which causes the split of full length of orf8 into putative orf8a and orf8b, has been found in all SARS-CoV isolated from mid-and late-phase human patients [24] . In addition, we have previously identified two bat SARS-related-CoV (Bat-CoV YNLF_31C and YNLF_34C) and proposed that the original SARS-CoV full-length orf8 is acquired from these two bat SARS-related-CoV [25] . Since the SARS-CoV is the closest human pathogenic virus to the 2019-nCoV, we performed phylogenetic analysis and multiple alignments to investigate the orf8 amino acid sequences. The orf8 protein sequences used in the analysis derived from early phase SARS-CoV that includes full-length orf8 (human SARS-CoV GZ02), the mid-and late-phase SARS-CoV that includes the split orf8b (human SARS-CoV Tor2), civet SARS-CoV (paguma SARS-CoV), two bat SARS-related-CoV containing full-length orf8 (bat-CoV YNLF_31C and YNLF_34C), 2019-nCoV, the other two closest bat SARS-related-CoV to 2019-nCoV SL-CoV ZXC21 and ZC45), and bat SARS-related-CoV HKU3-1 (Figure

5(A)). As expected, orf8 derived from 2019-nCoV belongs to the group that includes the closest genome sequences of bat SARS-related-CoV ZXC21 and ZC45. Interestingly, the new 2019-nCoV orf8 is distant from the conserved orf8 or Figure 5(B)) which was shown to trigger intracellular stress pathways and activates NLRP3 inflammasomes [26] , but this is absent in this novel orf8 of 2019-nCoV. Based on a secondary structure prediction, this novel orf8 has a high possibility to form a protein with an alpha-helix, following with a betasheet(s) containing six strands (Figure 5(C)).

The genome of 2019-nCoV has overall 89% nucleotide identity with bat SARS-related-CoV SL-CoVZXC21 (MG772934.1), and 82% with human SARS-CoV BJ01 2003 (AY278488) and human SARS-CoV Tor2 (AY274119). The phylogenetic trees constructed using the amino acid sequences of orf1a/b and the 4 structural genes (S, E, M, and N) were shown (Figure 6(A-E)). For all these 5 genes, the 2019-nCoV was clustered with lineage B β CoVs. It was most closely related to the bat SARS-related CoVs ZXC21 and ZC45 found in Chinese horseshoe

As shown in Figure 7 (A-C), the SARS-CoV 5' -UTR contains SL1, SL2, SL3, SL4, S5, SL5A, SL5B, SL5C, SL6, SL7, and SL8. The SL3 contains trans-cis motif [27] . The SL1, SL2, SL3, SL4, S5, SL5A, SL5B, and SL5C structures were similar among the 2019-nCoV, human SARS-CoV and the bat SARS-related ZC45. In the 2019-nCoV, part of the S5 found was inside Figure 7 Continued the orf1a/b (marked in red), which was similar to SARS-CoV. In bat SARS-related CoV ZC45, the S5 was not found inside orf1a/b. The 2019-nCoV had the same SL6, SL7, and SL8 as SARS-CoV, and an additional stem loop. Bat SARS-related CoV ZC45 did not have the SARS-COV SL6-like stem loop. Instead, it possessed two other stem loops in this region. All three strains had similar SL7 and SL8. The bat SARS-like CoV ZC45 also had an additional stem loop between SL7 and SL8. Overall, the 5' -UTR of 2019-nCoV was more similar to that of SARS-CoV than the bat SARS-related CoV ZC 45. The biological relevance and effects of virulence of the 5' -UTR structures should be investigated further. The 2019-nCoV had various 3' -UTR structures, including BSL, S1, S2, S3, S4, L1, L2, L3, and HVR (Figure 7(D-F)). The 3' -UTR was conserved among 2019-nCoV, human SARS-CoV and SARS-related CoVs [27] .

In summary, 2019-nCoV is a novel lineage B Betacoronavirus closely related to bat SARS-related coronaviruses. It also has unique genomic features which deserves further investigation to ascertain their roles in viral replication cycle and pathogenesis. More

animal sampling to determine its natural animal reservoir and intermediate animal host in the market is important. This will shed light on the evolutionary history of this emerging coronavirus which has jumped into human after the other two zoonotic Betacoronaviruses, SARS-CoV and MERS-CoV.

--- END OF ARTICLE ---

--- START OF ARTICLE 140 ---

A rapid advice guideline for the diagnosis and treatment of 2019 novel coronavirus (2019-nCoV) infected pneumonia (standard version)

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Abstract: In December 2019, a new type viral pneumonia cases occurred in Wuhan, Hubei Province; and then named “2019 novel coronavirus (2019-nCoV)” by the World Health Organization (WHO) on 12 January 2020. For it is a never been experienced respiratory disease before and with infection ability widely and quickly, it attracted the world’s attention but without treatment and control manual. For the request from frontline clinicians and public health professionals of 2019-nCoV infected pneumonia management, an evidence-based guideline urgently needs to be developed. Therefore, we drafted this guideline according to the rapid advice guidelines methodology and general rules of WHO guideline development; we also added the first-hand management data of Zhongnan Hospital of Wuhan University. This guideline includes the guideline methodology, epidemiological characteristics, disease screening and population prevention, diagnosis, treatment and control (including traditional Chinese Medicine), nosocomial infection prevention and control, and disease nursing of the 2019-nCoV. Moreover, we also provide a whole process of a successful treatment case of the severe 2019-nCoV infected pneumonia and experience and lessons of hospital rescue for 2019-nCoV infections. This rapid advice guideline is suitable for the first frontline doctors and nurses, managers of hospitals and healthcare sections, community residents, public health persons, relevant researchers, and all person who are interested in the 2019-nCoV.

Text: In December 2019, the 2019 novel coronavirus (2019-nCoV) was discovered and identified in the viral pneumonia cases that occurred in Wuhan, Hubei Province, China; And then was named by the World Health Organization (WHO) on 12 January 2020. In the following month, the 2019-nCoV quickly spreading inside and outside of Hubei Province and even other countries. What's more, the sharp increase of the case number caused widespread panic among the people.

Medical professionals require an up-to-date guideline to follow when an urgent healthcare problem emerging. In response to the requests for reliable advice from frontline clinicians and public healthcare professionals managing 2019-nCoV pandemics, we developed this rapid advance guideline, involving disease epidemiology, etiology, diagnosis, treatment, nursing, and hospital infection control for clinicians, and also for public health workers and community residents.

This guideline was prepared in accordance with the methodology and general rules of WHO Guideline Development and the WHO Rapid Advice Guidelines [1, 2] .

This guideline development group is multidisciplinary and composed of individuals from health professionals and methodologists. Health professionals included frontline clinical doctors, nurses who work in departments of respiratory medicine, fever clinic, critical medicine, emergency, infectious disease, and experts of respiratory infectious disease and hospital management board. The methodologists included methodologists of guideline development, systematic review, and literature searching professionals.

This guideline is suitable for frontline doctors and nurses, managers of hospitals and healthcare sections, healthy community residents, personnel in public healthcare, relevant researchers, and all persons who are interested in the 2019-nCoV management.

This guideline is aimed to serve the healthcare professionals to tackle the suspected 2019-nCoV infected cases, confirmed 2019-nCoV infected cases, clustered 2019-nCoV infected cases, and those with close contacts or suspicious exposure to 2019-nCoV infected cases.

Oral inquiry for financial interests of relevant personal was conducted at the first meeting while to start this guideline. Relevant financial as well as nonfinancial interests were surveyed and disclosed and subsequently assessed in consensus conference in order to minimize potential bias in guideline development. Finally, there is no conflict of interests for all the personnel participating to prepare this guideline.

This guideline is a rapid guideline to responding to the emerging infectious disease of 2019-nCoV. Due to the urgent need and tight work schedule, we conducted no wide-range survey but a discussion meeting with front-line clinicians who managed patients with 2019-nCoV infections to finalize guideline topics and key questions.

2.6 Literature searching and preparation of evidence profiles 2.6.1 General notes

Considering the lack of direct evidence for this newly identified 2019-nCoV infection, we searched and referred to the guidelines that related to the SARS (Severe Acute Respiratory Syndrome), MERS (Middle East Respiratory Syndrome), and influenza. We also referred to

the guidelines which were newly-issued by the National Health Commission of People's Republic of China and WHO for 2019-nCoV. In addition, we have an independent literature searching team to search available indirect evidence from systematic reviews and/or RCTs (randomized controlled trials), which were for the treatment and/ or chemoprophylaxis of SARS, MERS, or other influenza virus infections.

If the existing evidence addressed topics or questions covered by the guideline, then its quality should be assessed. If there is a lack of higher-level quality evidence, our panel considered observational studies and case series. Because of the limited time, we did not perform new systematic review. We identified relevant literature up to 20 January 2020.

We searched the bibliographic databases: PubMed, Embase, and Cochrane library.

We also searched following websites: the WHO (<https://www.who.int/>), CDC (Centers for Disease Control and Prevention, <https://www.cdc.gov/>), NICE (National Institute for Health and Clinical Excellence, <https://www.nice.org.uk/>), National Health Commission of the People's Republic of China (<http://www.nhc.gov.cn/>), and National Administration of Traditional Chinese Medicine (<http://www.satcm.gov.cn/>).

As the 2019-nCoV is a newly identified pathogen responsible for the outbreak of the pandemic disease, there is no sufficient evidence to reveal the whole nature of this virus. In these situations, obtaining evidence from the experts who fighting the disease on the frontline can be efficient and the main source [3] .

Until to 24:00 on 29 January 2020, 11,500 persons were screened, and 276 were identified as suspected infectious victims, and 170 were diagnosed (including 33 in critical condition) for 2019-nCoV infection in Zhongnan Hospital of Wuhan University. During this process, frontline clinicians and nurses have accumulated valuable experience in the diagnosis, treatment and nursing for 2019-nCoV infected patients. Hence, these experiences were assessed and then used as "Expert Evidence" for our guideline development. We took interviews and group surveys to collect information on treatment evidence during the guideline panel's meeting, so that it could be integrated into the guideline panel in the summary of findings (see Additional files 1 and 2). Experts' evidence can be solicited by the

description of case reports, summaries, and reports of topics or questions of all cases they management.

We accorded to the Grading of Recommendations Assessment, Development and Evaluation (GRADE) basic approaches and rules [4, 5] , and particularly considered experts' evidence to assess the quality of a body of evidence to make recommendations.

The quality of evidence reflects whether the extent to which our confidence estimating the effect is adequate to support a particular recommendation. The level of evidence was categorized as "high quality", "moderate quality", "low quality", or "very low quality"; Recommendations were classified as "strong" or "weak."

The strong recommendation does not always mean there is sufficient intervention effectiveness. Besides the effectiveness of intervention, the forming of recommendations is based on the severity of the disease, patient willingness, safety, and economics [4] . See Tables 1 and 2 [4, 6] .

Before meetings, experts' evidence was collected systematically and available to panel members. Once the evidence has been identified and assessed, recommendations were formulated based on the evidence by a face-to-face meeting of panel members and supplemented by experts participating in the panel meeting.

Experts' evidence was highly valued in this guideline development. During the consensus process, if the evidence was agreed on by more than 70% frontline clinicians in the consensus meeting, it is considered as highquality evidence.

In specific recommendations, we used "should" or "strongly recommend" for strong recommendations; whereas, "suggest" or "consider" was used for weak ones.

This guideline was published in both Chinese and English versions at the same time. Due to space limitations, the current standard revision does not include evidence descriptions.

The full revision will be published in New Medicine (Chinese name: Yixue Xinzhi; <http://www.jnewmed.com/>), Volume 30 and Issue 1 2020 [7] .

Since December 2019, multiple cases occurring unexplainable pneumonia were successively reported in some hospitals in Wuhan city with a history of exposure to a large Hua'nan seafood market in Wuhan city, Hubei province, China. It has been confirmed to be an acute respiratory infection caused by a novel coronavirus. So far, the number of cases without a history of the Hua'nan seafood market exposure is increasing. In addition, clustered cases and confirmed cases without a history of travel to Wuhan emerged. Also, confirmed cases without clear exposure to the Wuhan seafood market have been found in many foreign countries or regions [8] .

At 24:00 on 26 January 2020, the National Health Commission of the People's Republic of China has recorded a total of 2744 confirmed cases of pneumonia with 2019-nCoV infection from 30 provinces (districts and cities), including 461 severe cases and 80 deaths. A total of 51 cases have been cured and discharged. At present, 5794 suspected cases were recorded, 32,799 with close contacts to the confirmed patients have been tracked, 583 people were released from medical observation that day, and 30,453 people were still undergoing medical observation. A total of confirmed cases were reported from Hong Kong, Macao and Taiwan of China: 8 cases in Hong Kong, 5 cases in Macao, and 4 cases in Taiwan. In addition, confirmed cases had been reported from abroad: 7 in Thailand, 4 in Australia, 4 in Singapore, 3 in France, 3 in Japan, 3 in Korea, 3 in

Wild animal, bats [10] is the most possible host of the 2019-nCoV. It requires further confirmation whether pneumonia infected by the 2019-nCoV is transmitted directly from bats or through an intermediate host. It is believed that clarifying the source of the virus will help determine zoonotic transmission patterns [11] .

Up to present, the main infection source was the patients who with pneumonia infected by the 2019-nCoV. Respiratory droplet transmission is the main route of transmission, and it can also be transmitted through contact [12] . Although many details, such as the source of the virus and its ability to spread between people remain unknown, an increasing number of cases show the signs of human-tohuman transmission [8, 13] .

The 2019-nCoV isolated from the lower respiratory tract of patients with unexplainable pneumonia in Wuhan, and it is a novel coronavirus belonging to the β genus. The 2019-nCoV has an envelope; its particles are round or oval, often polymorphic, with a diameter from 60 nm to 140 nm. Its genetic characteristics are significantly different from SARSr-CoV (SARS related coronaviruses) and MERSr-CoV (MERS related coronaviruses). Current research shows it has more than 85% homology with SARSr-CoV (bat-SL-CoVZC45). 2019-nCoV can be found in human respiratory epithelial cells 96 h after in vitro isolation and culture, while it takes about 6 days in VeroE6 or Huh-7 cell lines [12]. The source of the virus, the time span of the patients discharging infective virus, and the pathogenesis are still not clear [14].

No evidence of viral mutation has been found so far [14]. It is necessary to obtain much more clinically isolated viruses with time and geographical variety to assess the extent of the virus mutations, and also whether these mutations indicate adaptability to human hosts [11].

Based on currently epidemiological survey, the latency period is generally from 3 to 7 days, with a maximum of 14 days [10]. Unlike SARSr-CoV, 2019-nCoV is contagious during the latency period [15]. The evidence agreed on by more than 70% frontline clinicians in consensus meeting is viewed as high-quality evidence

The population is generally susceptible to the virus. The elderly and those with underlying diseases show more serious conditions after infection, and children and infants also get infected by the 2019-nCoV. From current knowledge of the cases, most patients have a good prognosis, the symptoms of children are relatively mild, and a few patients are in critical condition. Death cases are more frequently seen in the elderly and those with chronic underlying diseases [12]. The newest study including the first 41 confirmed cases admitted to Wuhan between 16 December 2019 and 2 January 2020 showed the median age of patients was 49 years; and the main underlying diseases were diabetes, hypertension, and cardiovascular diseases. Of them, 12 cases experienced acute respiratory distress syndrome (ARDS), 13 cases were admitted to the intensive care unit (ICU), and 6 cases died [16]. Patients with any 2 of the following clinical features and any epidemiological risk: (1) clinical features: fever, imaging features of pneumonia, normal or reduced white blood cell count, or reduced lymphocyte count in the early stages of the disease onset. (2) epidemiologic risk: a history of travel to or residence in Wuhan city,

China or other cities with continuous transmission of local cases in the last 14 days before symptom onset; contact with patients with fever or respiratory symptoms from Wuhan city, China or other cities with continuous transmission of local cases in the last 14 days before symptom onset; or epidemiologically connected to 2019-nCoV infections or clustered onsets [12] .

Those with one of the following pathogenic evidence is the confirmed case: (1) positive for the 2019-nCoV by the real-time PCR test for nucleic acid in respiratory or blood samples [17] . 2) viral gene sequencing shows highly homogeneity to the known 2019-nCoV in respiratory or blood samples [12] .

Suspected clustering cases are defined when one confirmed case and at the same time, one or more cases of fever or respiratory infection are found in a small area (such as a family, a construction site, a unit, etc.) within 14 days.

Under the above circumstances, 2 or more confirmed cases are found, and there is the possibility of human-to-human transmission due to close contact or infection due to co-exposure, then it is determined as a clustered case [8, 18] .

Those who have one of the following contacts after the onset of confirmed cases in the absence of effective protection [18] : (1) those who live, study, work, or have close contact with the confirmed cases, or other close contacts such as working closely with or sharing the same classroom or living in the same house with the confirmed case. (2) medical and nursing staffs and their family members living with them, who treated, nursed or visited the confirmed case, or other personnel who have similar close contact with the case, such as providing direct treatment or care of the case, visiting the case or staying in a closed environment where the cases are located; other patients or caregivers in the same room with the case. (3) people who have close contact with the patients in a same transport vehicle, including those who had taken care of the patients on the vehicle; the person who had companied the patients (family members, colleagues, friends, etc.); other passengers and traffic staff considered having likely close contact with the patients by investigation and evaluation. (4) other circumstances considered to be closely contacted with the person with close contact with the patients by the professional investigation and evaluation.

Persons with suspicious exposure are those who are exposed without effective protection to processing, sales, handling, distributing, or administrative management of wild animals, materials, and the environments that are positive for the 2019-nCoV test [18] .

Persons with close contacts and suspicious exposure should be advised to have a 14-day health observation period, which starts from the last day of contact with the 2019-nCoV infected patients or suspicious environmental exposure. Once they display any symptoms, especially fever, respiratory symptoms such as coughing, shortness of breath, or diarrhea, they should reach out for medical attention immediately [19] . Contact surveillance should be carried out for those who had exposed to accidental contact, low-level exposure to suspected or confirmed patients, i.e. checking any potential symptoms when carrying out daily activities [20] . See Table 3 for details [21] .

Patients with a suspected infection should be isolated, monitored, and diagnosed in hospital as soon as possible. Doctors should make recommendations based on the patient's situation. Patients with mild symptoms and suspected infection may consider in-home isolation and home care (weak recommendation). Suspected infected with severe symptoms and those who need to stay in hospital for observation by doctor's judgment should follow the isolation guidelines for suspected patients (see Tables 4 and 5 for details).

It should also be noted that: (1) whether the suspected patients should be given in-home isolation and care or not requires careful clinical evaluation and safety assessment by professionals. (2) if the suspected patients do not get improvement in the symptoms or even worsened in condition during home care, they need to go to the doctor for treatment. (3) during the period of home care, the patients' medication and symptoms should be closely recorded and their caregivers should also monitor their body temperature daily.

Throughout the period of home care, healthcare personnel should perform regular (e.g., daily) follow-up through face-to-face visits or phone interviews (ideally, if feasible) to follow the progress of symptoms and, if necessary, specific diagnostic tests should be conducted [14, 19, 21] .

International visitors should take routine precautions when entering and leaving the affected areas, including avoiding close contacts with people with acute respiratory infection, washing hands frequently, especially after contacting with the sick or their surrounding environment; following appropriate coughing etiquette; and avoiding close contact with live or dead farming animals or bats or other wild animals [22, 23] . Passengers should avoid unnecessary travel as possible.

If they had travelled to Hubei Province (especially Wuhan city) in the past 14 days and is in fever, cough or difficulty in breathing, they should: (1) see a doctor immediately; (2) call the doctor about his/her recent trips and symptoms before going to the doctor's office or emergency room; (3) avoid contact with others; (4) not to travel around; (5) cover mouth and nose with a tissue or sleeve (not hands) when coughing or sneezing; and (6) wash hands with soap and water for at least 20 s. If soap and water are not available, use alcohol-based hand sanitizers [24] .

The 2019-nCoV infected cases have symptoms like fever, fatigue, dry cough, dyspnea etc., with or without nasal congestion, runny nose or other upper respiratory symptoms [13, 16] . Despite the atypical symptoms were reported [25] , Nan-Shan Zhong, the academician of Chinese Academy of Engineering in an exclusive interview with Xinhua News Agency on 28 January 2020, pointed out that fever is still the typical symptom of 2019-nCoV infection.

Patients with mild symptoms may not present positive signs. Patients in severe condition may have shortness of breath, moist rales in lungs, weakened breath sounds, dullness in percussion, and increased or decreased tactile speech tremor, etc.

The imaging findings vary with the patient's age, immunity status, disease stage at the time of scanning, underlying diseases, and drug interventions. When walking on the road or waiting in the hospital, try to stay away from other people (at least 1 m away) and wear a mask.

The family members accompanying those for inspection should immediately follow the monitoring recommendations to close contacts, keep the respiratory hygiene and clean their hands properly.

The community or street hospital should be informed before the suspected contacts to the hospital. The vehicle used should be cleaned and disinfected with 500 mg/L chlorine-containing disinfectant, and the window should be opened for ventilation. The imaging features of lesions show: (1) dominant distribution (mainly subpleural, along the bronchial vascular bundles); (2) quantity (often more than three or more lesions, occasional single or double lesions); (3) shape (patchy, large block, nodular, lumpy, honeycomblake or grid-like, cord-like, etc.); (4) density (mostly uneven, a paving stones-like change mixed with ground glass density and interlobular septal thickening, consolidation and thickened bronchial wall, etc.); and (5) concomitant signs vary (air-bronchogram, rare pleural effusion and mediastinal lymph nodes enlargement, etc.).

Typical CT/X-ray imaging manifestation, including

(1) Multiple, patchy, sub-segmental or segmental groundglass density shadows in both lungs. They were classified as "paving stone-like" changes by fine-grid or small honeycomb-like thickening of interlobular septa. The thinner the CT scan layers, the clearer the ground-glass opacity and thickening of interlobular septa were displayed. A slightly high-density and ground-glass change with fuzzy edge in the fine-grid or small honeycomb-like thickening of interlobular septa were presented by the high-resolution computed tomography (HRCT), (Fig. 1: 45 cases, 54.2% in a total of 83 cases). The resolution of X-ray was worse lower than that of CT in the resolution, which was basically manifested as ground-glass When coughing or sneezing, it is necessary to wear a medical mask, or cover with a paper towel and bent elbow, and clean hands immediately after coughing and sneezing.

Strong 10 N95 masks should be worn in the same room with patients (preferred strategy).

Disposable surgical mask (alternative strategy). Use the mask in strict accordance with the instruction manual.

After washing hands with running water, dry them with a paper towel (preferred strategy).

Dry with a towel, and wash and disinfect the towel daily (alternative strategy).

Home caregivers 1 Clean and disinfect hands after contact with the patient, before leaving patient's room or the house, before and after eating, after using the toilet and after entering house from outside (for visible contaminant on hands, wash hands with running water then use hand disinfection).

Avoid direct contact with patient's secretions or discharges, especially oral or respiratory discharges; avoid direct contact with patient's feces. (2) Multiple, patchy or large patches of consolidation in both lungs, with a little grid-like or honeycombshaped interlobular septal thickening, especially in the middle and lower lobes (Fig. 3: 26 cases, 31.3% in a total of 83 cases). It was more common in the elderly or severe condition patients.

Atypical CT/X-ray imaging manifestation, including (1) Single, or multiple, or extensive subpleural grid-like or honeycomb-like thickening of interlobular septum, thickening of the bronchial wall, and tortuous and thick strand-like opacity. Several patchy consolidations, occasionally with a small amount pleural effusion or enlargement of mediastinal lymph nodes, can be seen (Fig. 4 : 6 cases, 7.2% in a total of 83 cases). This is mostly seen in the elderly.

(2) Single or multiple solid nodules or consolidated nodules in the center of lobule, surrounded by ground-glass opacities (Fig. 5 : 5 cases, 6.2% in a total of 83 cases).

Stage based on CT image The CT imaging demonstrates 5 stages according to the time of onset and the response of body to the virus, including:

(1) Ultra-early stage. This stage usually refers to the stage of patients without clinical manifestation, negative laboratory test but positive throat swab for 2019-nCoV (2) Early

stage. This stage refers to the period of 1-3 days after clinical manifestations (fever, cough, dry cough, etc.). The pathological process during this stage is dilatation and congestion of alveolar septal capillary, exudation of fluid in alveolar cavity and interlobular interstitial edema. It showed that single or multiple scattered patchy or agglomerated ground-glass opacities, separated by honeycomb-like or grid-like thickened of interlobular septa (Fig. 7 : 45 cases, 54.2% in a total of 83 cases).

It mainly should be distinguished from other known viral virus of pneumonia, such as influenza viruses, parainfluenza virus, adenovirus, respiratory syncytial virus, rhinovirus, human metapneumovirus, SARSr-CoV, etc.; and also from mycoplasma pneumonia, chlamydia pneumonia, and bacterial pneumonia. In addition, it should be distinguished from non-infectious diseases, such as vasculitis, dermatomyositis, and organizing pneumonia.

In the early stage of the disease, the total number of leukocytes decreased or keeps normal, with decreased lymphocyte count or increased or normal monocytes. High attention should be paid on the situation where the absolute value of lymphocyte is less than $0.8 \times 10^9 /L$, or the numbers of CD4 and CD8 T cells are significantly decreased, which generally recommend rechecking the blood routine changes after 3 days.

(1) Flu antigens. At present, routinely detected flu antigens are A, B, and H7N-subtypes. Sampling of throat swabs is conducive to early rapid screening for flu because of the fast test, but it has a relatively high false negative rate. (2) Respiratory virus nucleic acid. The detection of respiratory virus nucleic acid is commonly used to

In the early stage of this disease, the total number of leukocytes in peripheral blood was normal or decreased, and the lymphocyte count decreased. In some patients, liver enzyme (transaminase), creatine kinase (CK) and myoglobin increased. CRP, ESR, and IL-6 increased, and PCT was normal in most patients. The increased Ddimer occurred in severe cases. Compared with 120 healthy check-ups, the absolute value of lymphocyte (0.87 vs $2.13 \times 10^9 /L$), lymphocyte percentage (19.5% vs 33.7%), eosinophil percentage (0.13% vs 2.16%), and absolute value (0.0061 vs $0.1417 \times 10^9 /L$) in 2019-nCoV patients were significantly reduced ($P < 0.05$). The absolute number (4.2 vs $3.7 \times 10^9 /L$) and the percentage (72.0% vs 57.0%) increased in 2019-nCoV patients ($P < 0.05$). The percentage of

monocytes increased slightly (8.1% vs 6.8%), while the absolute number of monocytes did not change significantly (0.38 vs 0.44) $\times 10^9$ /L.

The next generation sequencing (NGS) and electron microscope technology play a role in the early diagnosis, but their diagnostic values have been weakened by the discovery of specific nucleic acid detection technology. In addition, NGS detection can tell whether the pathogen has mutated or not.

Suspected and confirmed cases need to be treated in designated hospitals with effective isolation and protection conditions. Suspected cases need to be treated separately in single room, confirmed cases are admitted to a same ward, and critical cases should be admitted to ICU as soon as possible.

(1) The patient should rest in bed, being monitored for vital signs (heart rate, pulse oxygen saturation, respiratory rate, blood pressure) and given supportive treatment to ensure sufficient energy intake and balance for water, electrolytes, acidbase levels and other internal environment factors (Strong recommendation).

(2) The patient should be monitored for blood routine, CRP, PCT, organ function (liver enzyme, bilirubin, myocardial enzyme, creatinine, urea nitrogen, Urine volume, etc.), coagulation function, arterial blood gas analysis and chest imaging (Strong recommendation). First, oxygen therapy is the choice for patients with severe respiratory infections, respiratory distress, hypoxemia or shock. The initial flow rate is 5 L/min, and the titration flow rate is to reach the target oxygen saturation (adults: $SpO_2 \geq 90\%$ in non-pregnant patients, $SpO_2 \geq 92-95\%$ in pregnant patients; children: $SpO_2 \geq 94\%$ in children with obstructive dyspnea, apnea, severe respiratory distress, central cyanosis, shock, coma or convulsions, and $\geq 90\%$ in other children).

Second, respiratory support should be given to patients with hypoxic respiratory failure and acute respiratory distress syndrome. HFNO or NIV can be selected when nasal cannula or mask oxygen therapy was ineffective or the patient had hypoxic respiratory failure. However, when patients had hypercapnia (acute exacerbation of chronic obstructive pulmonary disease, cardiogenic pulmonary edema), hemodynamic instability, multiple

organ failure, and abnormal mental status HFNO oxygen is not the routinely adopted measure. If respiratory failure cannot be improved or worsens continuously within a short time (1 h) after using HFNO or NIV, intubation should be performed immediately. Low tidal volume (4-8 ml/kg) and low suction pressure (platform pressure < 30cmH₂O) are used for invasive mechanical ventilation. It is suggested that positive endexpiratory pressure (PEEP) with high positive endexpiratory pressure should be used in patients with moderate or severe acute respiratory distress syndrome, and PEEP should be titrated according to FiO₂ to maintain SpO₂, in order to improve alveolar atelectasis and reduce alveolar hyper-expansion and pulmonary vascular resistance at the end of inspiration. For severe patients with ARDS, it is recommended to ventilate in prone position for more than 12 h/d.

should be considered for the patients with refractory hypoxemia that is difficult to be corrected by protective lung ventilation. (Strong recommendation).

(1) At present, there is no evidence from RCT to support specific drug treatment against the new coronavirus in suspected or confirmed cases. (2) The α-interferon atomization inhalation can be considered (5 million U per time for adults in sterile injection water, twice a day) (Weak recommendation); lopinavir/ritonavir orally, 2 capsules each time, twice a day, can be also considered (Weak recommendation).

Low-level evidences included retrospective cohort, historically controlled studies, case reports, and case series revealed that lopinavir/ritonavir alone or in combination with antivirals produced certain benefits in the treatment of SARS and MERS, such as reducing the incidence or mortality of ARDS [26] [27] [28] [29]. A recently systematic review showed that lopinavir/ritonavir's anti-coronavirus effect was mainly seen in its early application, for reducing patient mortality and reduced glucocorticoid consumption. However, if the early treatment window is missed, there will be no significant effect in their late application [30]. Real-world study stills need to further explore the clinical effects of its early use in 2019-nCoV infected pneumonia.

The effectiveness of the combined use of antivirals is still controversial [31] [32] [33] [34].

(1) Principles. Avoid blind or inappropriate use of antibacterial drugs, especially the combination of broad-spectrum antibacterial drugs. Enhancement of bacteriological surveillance should be performed and promptly given appropriate antibacterial drugs when it occurs secondary bacterial infection. (2) According to the clinical manifestations of patients, if the accompanying bacterial infection cannot be ruled out, mild patients can take antibacterial drugs against community-acquired pneumonia, such as amoxicillin, azithromycin, or fluoroquinolones; empirical antibacterial treatment in severe patients should cover all possible pathogens, deescalating therapy until the pathogenic bacteria are clarified.

The use of corticosteroids for severe ARDS is controversial; therefore, systemic use of glucocorticoids needs to be cautious. Methylprednisolone can be used as appropriate for patients with rapid disease progression or severe illness. According to the severity of the disease, 40 to 80 mg of methylprednisolone per day can be considered, and the total daily dose should not exceed 2 mg/kg (Weak recommendation). SARS management related researches showed that timely use of non-invasive continuous positive airway pressure and corticosteroids is an effective strategy for increased lung shadows and increased dyspnea. Appropriate use of glucocorticoids is able to significantly improve the clinical symptoms of patients with SARS, reduce the degree of disease progression, and accelerate the absorption of lung lesions; but it cannot shorten the length of hospital stay [35, 36] . Be cautious that hormone therapy has some incidence of adverse reactions [37] .

(1) Symptomatic treatment of fever. When the temperature is higher than 38.5 °C, ibuprofen can be used for Treat the patient based on syndrome differentiation individually. Prevention before illness is better than treatment after getting diseased.

(1) Community. Implement relevant national regulations and take great effort to keep away from contaminated materials, disinfect the environment, and improve the healthcare management. i Fumigation with moxa in the room, 1-5 g/m² for 30 min per day. ii Wearing perfumed Chinese herb bags (clove, fineleaf schizonepeta herb, *Perilla frutescens*, *Atractylodes lancea*, cinnamon, biond magnolia flower, *asarum sieboldii*, and *Elettaria cardamomum*, 2 g for each, crushed into powder and put it into bags for external use, change a new one every 10 days). iii Prescription of Chinese herbs for feet bath (vulgaris 10 g, carthamus 10 g, and dried ginger 6 g) Soaking the herbs in boiling water and bath the feet

into the medical liquid when the temperature is suitable. Soak feet for about 20 min. iv Prescription of Chinese herbs for prophylaxis:

Astragalus mongholicus 12 g, roasted rhizoma atractylodis macrocephalae 10 g, saposhnikovia divaricata 10 g, Cyrtomium fortunei 10 g, honeysuckle 10 g, dried tangerine or orange peel 6 g, eupatorium 10 g, and licorice 10 g. Taking the medicine above yielded decoction once a day for adults, and for 5 days as a treatment course. If for children, cutting the dose to half. v Medical tea: perilla leaf 6 g, agastache leaf 6 g, dried tangerine or orange peel 9 g, stewed amomum tsao-ko 6 g, and 3 slices of ginger. Soak the herbs in hot water and drink the water just like enjoying the tea. vi Chinese patent medicine: Huoxiang Zhengqi capsule or Huoxiang Zhengqi Shui (in half dose).

In medical observation period There are two clinical symptoms in this period, including:

(1) Clinical symptoms 1: hypodynamia accompanied by gastrointestinal upset. And the recommended Chinese patent medicine is the Huoxiang Zhengqi capsules (ball, liquid, or oral liquid).

(2) Clinical symptoms 2: hypodynamia and fever. And the recommended Chinese patent medicines is the Jinhua Qinggan granules, Lianhua Qingwen capsules (granules), Shufeng Jiedu capsules (granules), or Fangfeng Tongsheng pills (granules).

Clinical treatment period This period involving 7 stages, including:

(1) Early-stage, characterized as exterior syndrome of cold-dampness. In this stage, the clinical manifestations presents as follow: aversion to cold without sweating, headache and generalized heaviness, limb pain, glomus and fullness in the chest and diaphragm, thirst with no desire to drink, ungratifying loose stool, yellow urine, frequent micturition and yellow urine. The therapeutic logic is to dissipate cold and eliminate dampness. And the recommended prescription is the Huoxiang Zhengqi powder (Yin dampness injuring superficies case from the National Famous Traditional Chinese Medical Doctor Medical Cases); which comprises of perilla leaf 10 g, atractylodes lancea 15 g, radix angelicae

dahuricae 10 g, dried tangerine or orange peel 10 g, notopterygium root 10 g, agastache rugosus 10 g (end addition), mangnolia officinalis 10 g, saposhnikovia divaricata 10 g, poria peel 15 g, and Tetrapanax papyriferus 10 g above yielded decoction. In addition, the recommended Chinese patent medicine is Huoxiang Zhengqi capsules or Huoxiang Zhengqi Shui. (2) Early-stage, characterized as cold-dampness obstructing lung. In this stage, the clinical manifestations presents as follow: aversion to cold with or without fever, dry cough, dry throat, fatigue and hypodynamia, oppression in chest, epigastric fullness, or nausea, loose stool. The tongue is pale or reddish, the tongue fur is slimy white, and soggy pulse. Hence, the therapeutic logic is to dissipate cold and resolve obstruction. And the recommended prescriptions comprises of atractylodes lancea 15 g, dried tangerine or orange peel 10 g, mangnolia officinalis 10 g, agastache rugosus 10 g (end addition), amomum tsao-ko 6 g, ephedra herb 6 g, notopterygium root 10 g, ginger 10 g, areca-nut 10 g (end addition), periostracum cicada 10 g, bombyx batryticatus 10 g, and rhizoma curcumae longae 10 g above yielded decoction. (3) Middle-stage, characterized as epidemic toxin blocking the lung. In this stage, its clinical manifestations includes persistent fever or alternating cold and heat, cough with less phlegm, or yellow phlegm, abdominal distension and constipation; oppression in chest with anhelation, cough with wheezes, panting on exertion; or red tongue, slimy yellow fur or yellow dry fur, slippery and rapid pulse. Therefore, the therapeutic logic is clearing heat and detoxicating. And the recommended prescription comprises of almond 10 g, gypsum 30 g (predecoction), trichosanthes kirilowii 30 g, rhubarb 6 g (end addition), ephedra with honey fried 6 g, semen lepidii 10 g, peach kernel 10 g, amomum tsao-ko 6 g, arecanut 10 g, and atractylodes lancea 10 g above yielded decoction.

In addition, the recommended Chinese patent medicine is Xiyanping injection or Xuebijing injection. (4) Severe stage, characterized as heat toxin generating stasis. In this stage, the clinical manifestations is known as high fever, oppression in chest with anhelation, purple-black facial complexion, lips dark and swollen, obnubilation, crimson tongue, yellow dry fur, surging and fine rapid stringlike pulse. Thus, its therapeutic logic is detoxicating and dispersing blood stasis. The recommended prescription is three Yellows and Gypsum decoction, Shang Jiang Powder, and Toxin-Resolving Blood-quickening decoction. Its composition comprises of ephedra with honey fried 10 g, almond 10 g, gypsum 20-30 g, periostracum cicada 10 g, bombyx batryticatus 10 g, rhizoma curcumae longae 10 g, rhubarb stir-fried with wine 10 g, scutellaria baicalensis 10 g, coptis chinensis 5 g, phillyrin 15 g, angelica sinensis 10 g, peach kernel 10 g, radix paeoniae rubra 15 g, and rhizome of rehmannia 15 g above yielded decoction. The recommended Chinese patent medicines is the Xiyanping injection, Xuebijing injection, Qingkailing injection, or Angong Niu Huang pills.

(5) Severe-stage, characterized as inner blocking causing collapse. In this stage, the clinical manifestations include dyspnea, panting on exertion or need assisted ventilation, accompanied by coma, and agitation, cold limbs with cold sweating, dark purple tongue, thick or dry thick tongue fur, floating and rootless pulse. The therapeutic logic is rescuing from collapse by restoring Yang. Hence, the recommended prescription comprises of ginseng 15 g, aconitine 10 g (predecoction), and *Cornus officinalis* 15 g above yielded decoction, and both taken with fluid Suhexiang pills or Angong Niu Huang pills. The recommended Chinese patent medicines is Xuebijing injection, Shenfu injection, or Shengmai injection. (6) Recovery-stage, presents as lung and spleen Qi deficiency. Its clinical manifestations include shortness of breath, fatigue and hypodynamia, anorexia, nausea and vomiting, glomus and fullness, weak stools, ungratifying loose stool, pale tender-soft enlarged tongue, slimy white tongue fur. Therefore, therapeutic logic is to supplement the spleen and lung. The recommended prescription comprises of rhizoma pinellinae praeparata 9 g, dried tangerine or orange peel 10 g, *Codonopsis pilosula* 15 g, radix astragali preparata 30 g, *poria cocos* 15 g, agastache rugosus 10 g, and fructus amomi 6 g (end addition) above yielded decoction. In addition, the recommended Chinese patent medicines is pill of costus and amomum with six noble ingredients. (7) Recovery-stage, characterized as deficiency of Qi and Yin. The clinical manifestations of this stage is generalized heat with sweating, chest heat vexation, Qi counterflow with retching and vomiting, shortness of breath and lassitude of essence-spirit, red tongue and thin tongue fur, vacuous pulse. Hence, the therapeutic logics is boost Qi and nourish Yin. The recommended prescription is Zhuye Shigao decoction with cogongrass rhizome and rhizoma phragmitis; and the composition of this prescription includes bamboo leaf 15 g, gypsum 15 g (predecoction), *Codonopsis pilosula* 15 g, radix ophiopogonis 10 g, pinellia ternate 9 g, cogongrass rhizome 15-30 g, rhizoma phragmitis 20 g, licorice 10 g, and polished round-grained rice 30 g above yielded decoction. The recommended Chinese patent medicine: Shengmai Yin.

6.5 Treatment of severe patients 6.5.1 Hypoxemic respiratory failure and ARDS treatments

Treatment principle: treat the patients to improve the symptoms and underlying diseases, actively prevent potential complications and secondary infection; provide timely measures to support organ function.

(1) Hypoxic respiratory failure and severe ARDS. Give oxygen therapy immediately to patients with ARDS, and closely monitor them for signs of clinical deterioration, such as rapidly progressive respiratory failure. Consider severe hypoxemic respiratory failure when

standard oxygen therapy fails. Conservative fluid management can be adopted for ARDS patients without tissue hypoperfusion. Use vasoactive drugs to improve microcirculation. Empirical antibiotics targeting the suspected potential infection should be used as soon as possible, blind or improper combination of broad-spectrum antibiotics should be avoided. Unless for special reasons, the routine use of corticosteroids should be avoided. Glucocorticoids can be used in a short time (3-5 days) according to the degree of dyspnea and the progress of chest imaging if appropriate and the recommended dose is not more than the equivalent to 1-2 mg/kg methylprednisone per day. Provide intensive standard supportive care to the critically ill patients, including prevention of deep vein thrombosis and stress-induced gastrointestinal bleeding, blood glucose control and so on. Enteral nutrition can be provided. Supplemental nutrition with omega-3 fatty acids and antioxidants is not recommended. Inhaled or intravenous beta-adrenergic agonists are not recommended to promote alveolar fluid clearance and resolution of pulmonary edema.

(1) Recognize the septic shock. When infection is suspected or confirmed, and on the basis of full fluid resuscitation, vasoconstrictor drugs are still needed to maintain mean arterial pressure (MAP) ≥ 65 mmHg with lactate ≥ 2 mmol/L, the existence of septic shock should be considered. If lactate cannot be monitored for some reasons, the following three manifestations (changes in mental state, oliguria, poor peripheral perfusion and prolonged capillary filling time) should be considered as signs of a combination of infection and hypoperfusion. (2) In resuscitation from septic shock in adults, at least 30 ml/kg of isotonic crystalloid was considered for adults in the first 3 h. In resuscitation from septic shock in children, give 20 ml/kg as a rapid bolus and up to 40-60 ml/kg in first aid. (3) Isosmotic crystal solution is recommended for resuscitation. Do not use hypotonic crystalloids, starches, or gelatins for resuscitation in the first hour. Albumin may be considered as a resuscitation fluid, but this recommendation was based on low-quality evidence under certain conditions. (4) Administer vasoconstrictor is suggested when shock persists after fluid resuscitation, noradrenaline as the first choice. The initial blood pressure target is MAP ≥ 65 mmHg in adults and age-appropriate targets in children. can also be administered via intraosseous needles.

6.6 Condition evaluation and treatment effect evaluation 6.6.1 Criteria to withdraw ECLS

(1) Remove VV-ECMO. The oxygen concentration of the ECMO air-oxygen mixer has dropped to 21%, the air flow rate has dropped to 0, and the ventilator is not strong enough.

Lasting for 2-3 h, the respiratory rate is within 25 breaths/min, SpO₂ > 92%, PaCO₂ is normal, and withdrawal from VV-ECMO may be considered. (2) Remove VA-ECMO. The blood flow rate is reduced to the rate of (0.2 to 0.5 L / min) every 5 to 6 h from 3 L/min, and the hemodynamic condition is stable. The blood flow rate is reduced to 1.5 L/min within 24 h. If there is a bridging tube, the arteriovenous end can be connected with a bridging tube to form an ECMO circuit for self-circulation, so that the body's hemodynamics is driven by the heart. If hemodynamics is stable for at least 6 h, consider removing the machine.

When the patient is well aware, cough reflexes are obvious when sucking the sputum, the hemodynamics is stable, and the ventilator parameters are close to offline parameters, the spontaneous breathing test (SBT) is performed. After the SBT is passed, invasive breathing can be considered to remove the endotracheal tube.

Patients do not need advanced respiratory support (HFNO, NIV, MV, ECLS, etc.); stable hemodynamics and tissue perfusion; no significant impairment of organ function; and no need for organ support treatment (CRRT, artificial liver, etc.). Consider transferring the patient out of ICU procedure.

The body temperature returned to normal for more than 3 days; respiratory symptoms improved significantly; inflammation of the lungs showed obvious signs of absorption; and respiratory nucleic acid was negative for two consecutive times (one-day sampling time interval at least); and the patient can be released from isolation.

7 Prevent and control nosocomial infection 7.1 Restriction and isolation guidelines for patient/ suspected patients See Table 7 . (Strong recommendation).

According to the principles of standard prevention and tertiary protection, all personnel entering various zones should be evaluated using individual inventory tables according to the exposure risk level. Chose personal protective equipment of various levels is necessary. Personal protective equipment should be worn strictly in accordance with the instructions and only used for one time (The patient's home isolation scheme is shown in Table 5 .

Patients should monitor their body temperature and illness at home. If your body temperature continues to be higher than 38 °C, or your breath is getting worse, you should seek medical treatment timely.

In addition to taking protective measures, the home caregivers also should monitor their body temperature closely.

Mild patients generally use a nasal catheter and a mask for oxygen. Adjust the oxygen flow as appropriate according to the patient's condition and doctor's instruction, and Requirements to the medical staff request 6. Medical personnel enter the isolation area with proper self-protection through designated channels.

6.1 Medical staff should perform the personal protection practice under the Personal Protection Guidelines in Table 8 closely monitor the patient's breathing and blood oxygen saturation. If oxygen therapy fails to reach the expected effect, the nurse should analyze the cause comprehensively and be vigilant to notify the doctor.

Mild patients generally use antiviral drugs, antibacterial drugs (when bacterial infection exists), and symptomatic treatment. The doctor's advice should be followed accurately and timely. The adverse reactions of oseltamivir mainly include nausea, vomiting, diarrhea, abdominal pain and bronchitis, cough, etc. The adverse reactions of interferon are mainly flu-like symptoms such as fever, fatigue, myalgia, and headache, followed by mild suppression of bone marrow. Attention should be paid to identify the change of clinical manifestations or adverse drug reactions.

According to the patients' condition, provide highprotein, high-vitamin, carbohydrate-containing diets (e.g. eggs, fish, lean meat, milk, etc.) for enough nutrition to improve physical condition.

Take good care of the patient and respond to the patient's question timely. Positively encourage patients to reduce their anxiety and fear.

Dynamically monitor patients' vital signs, water-electrolytes balance, acid-base balance, and functions of various organs, monitor patients' infection indicators, and determine the occurrence of complications such as acute respiratory distress syndrome, septic shock, stress ulcers, and deep vein thrombosis.

The critically ill patients mainly use oxygen therapy such as HFNO, NIV and invasive mechanical ventilation. When using various oxygen treatments in a sequential manner, the airway and breathing circuit need to be kept open, and the effect of oxygen treatment needs to be monitored dynamically. At the same time, skincare products need to be used reasonably to avoid damage to the nose, face and lips by pressure. When using a high-flow nasal catheter to inhale oxygen, the oxygen flow and temperature and humidity should be adjusted appropriately. When using non-invasive mechanical ventilation, patient should receive relevant health education. Patients are instructed to inhale through the nose. The pressure is set from low to high and gradually reaches the target value. The human-machine coordination is maximized. The patient's consciousness and respiratory function are closely observed. Patients with artificial airway established should use a closed suction tube to reduce virus spread. Nurses should wear goggles or a face shield to avoid occupational exposure.

If the patient develops moderate to severe ARDS, invasive mechanical ventilation combined with a prone position need to be adopted. Standard operating procedure for prone position needs to be followed. At the same time, be cautious to prevent pressure ulcers, falling bed, tube slippage, and eye damage by pressure and other complications. Patients treated with ECMO should be monitored for the performance of the oxygenator. If the oxygenator changes its color to darker, indicating the possibility of coagulation, the doctor should be notified to adjust the heparin dose as necessary. The oxygenator should be replaced if necessary. The coagulation function need to be monitored dynamically, including the whole set of coagulation and DIC (disseminated intravascular coagulation), and the time of activating partial thromboplastin, etc., the patient should be closely observed for signs of bleeding, such as bruising on the skin and mucous membranes, bleeding in the nasal cavity, oral cavity, bloody sputum, hematuria, blood in the stool, swelling of the abdomen, moving dullness, and the size of bilateral pupils. Make sure that the ECMO pipelines are tightly connected and firmly fixed to prevent air embolism and pipeline slippage.

Perform oral care and skin care, assist the patient to use toilet, and take eyes on the indwelling tubes. Rules and regulations for aseptic operation and isolation should be strictly followed to prevent ventilator-related pneumonia, catheter-related sepsis, urinary catheter related urinary tract infections and other secondary infections.

Dynamically assess the patients' nutritional risks and timely nutritional support can be given if needed. For the patients who can eat, the diet rich in protein and carbohydrates is recommended. Those patients who cannot eat but are compatible with enteral nutrition should be given enteral nutrition as soon as possible. For the patients incompatible with enteral nutrition, parenteral nutrition should be given timely to meet energy requirement.

Psychological and humanistic care should be performed in high priority especially for the awake patients. Psychological techniques like mindfulness -based stress reduction can be adopted to relieve the patients' anxiety and panic by building up their optimistic confidence in overcoming the disease.

Our guideline has three major limitations: Firstly, time is so limited that we cannot fully consider all clinical issues for this emergency disease. Secondly, many evidences came from data search is indirect. Thirdly, because some recommendations are based on the evidence from existing guidelines and experts' experience, there are situations where strong recommendations were produced on the base of low-quality evidence or very low-quality evidence, so high-quality evidence, when they appear, is likely to change current recommendations.

Supplementary information accompanies this paper at <https://doi.org/10.1186/s40779-020-0233-6>.

Additional file 1. A successful treatment case of the severe 2019-nCoV infected pneumonia patient.

Additional file 2. Experience and lessons in hospital rescue for 2019-nCoV infections.

TB: tuberculosis; TNF: Tumor Necrosis Factor; WBC: White blood cells; WHO: World Health Organization

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Note from the editors: novel coronavirus (2019-nCoV)

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6988271/>

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Abstract: nan

Text: The situation has continued to evolve rapidly since then and just a few weeks later, as at 23 January, 614 laboratory-confirmed cases and 17 deaths have been reported [2] including some cases detected outside mainland China [3] . Meanwhile, on 7 January 2020, the novel coronavirus, currently named 2019-nCoV, was officially announced as the causative agent by Chinese authorities [3] . In order to support public health action, viral genome sequences were released by Chinese researchers on 10 January [4] and 2 days later, four further sequences were also made available on the Global Initiative on Sharing All Influenza Data (GISAID) (<https://www.gisaid.org/>). While more cases are being reported on a daily basis and there is evidence for some human-to-human transmission in China, a number of important questions remain unanswered. For example, there is no certainty

about the source of the outbreak, the transmissibility of the virus as well as the clinical picture and severity of the disease.

In this issue of Eurosurveillance, we are publishing two articles on different aspects of the newly emerged 2019-nCoV. One is a research article by Corman et al. on the development of a diagnostic methodology based on RT-PCR of the E and RdRp genes, without the need for virus material; the assays were validated in five international laboratories [5] . Before this publication, a description of the assays had already been made publically available on a dedicated WHO webpage [6] to support rapid development of laboratory testing capacities. The other is a rapid communication where researchers based in Hong Kong report on their attempt to estimate the severity among hospitalised cases of 2019-nCoV infection through modelling based on publically available information, mainly from Wuhan health authorities [7] . It also points out the need for more detailed information to make an informed evaluation of the situation as basis for public health decision-making.

Today, the WHO Director-General Tedros Adhanom Ghebreyesus, taking into consideration the deliberations of the members of the International Health Regulations (IHR) Emergency Committee on 2019-nCoV in their second meeting, decided not to declare a public health emergency of international concern.

International health organisations such as the European Centre for Disease Prevention and Control (ECDC) and the WHO are monitoring the situation and provide regular updates. ECDC has set up a dedicated webpage on which updates and risk assessments with focus on Europe are available: <https://www.ecdc.europa.eu/en/novel-coronavirus-china>.

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COVID-19 and smoking: A systematic review of the evidence

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7083240/>

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Abstract: COVID-19 is a coronavirus outbreak that initially appeared in Wuhan, Hubei Province, China, in December 2019, but it has already evolved into a pandemic spreading rapidly worldwide(1,2). As of 18 March 2020, a total number of 194909 cases of COVID-19 have been reported, including 7876 deaths, the majority of which have been reported in China (3242) and Italy (2505)(3). However, as the pandemic is still unfortunately under progression, there are limited data with regard to the clinical characteristics of the patients as well as to their prognostic factors(4). Smoking, to date, has been assumed to be possibly associated with adverse disease prognosis, as extensive evidence has highlighted the negative impact of tobacco use on lung health and its causal association with a plethora of respiratory diseases(5). Smoking is also detrimental to the immune system and its responsiveness to infections, making smokers more vulnerable to infectious diseases(6). Previous studies have shown that smokers are twice more likely than non-smokers to contract influenza and have more severe symptoms, while smokers were also noted to have higher mortality in the previous MERS-CoV outbreak(7,8). Given the gap in the evidence, we conducted a systematic review of studies on COVID-19 that included information on patients' smoking status to evaluate the association between smoking and COVID-19 outcomes including the severity of the disease, the need for mechanical ventilation, the need for intensive care unit (ICU) hospitalization and death. The literature search was conducted on 17 March 2020, using two databases (PubMed, ScienceDirect), with the search terms: ['smoking' OR 'tobacco' OR 'risk factors' OR 'smoker*'] AND ['COVID-19' OR 'COVID 19' OR 'novel coronavirus' OR 'sars cov-2' OR 'sars cov 2'] and included studies published in 2019 and 2020. Further inclusion criteria were that the studies were in English and referred to humans. We also searched the reference lists of the studies included. A total of 71 studies were retrieved through the search, of which 66 were excluded after full-text screening, leaving five studies that were included. All of the studies were conducted in China, four in Wuhan and one across provinces in mainland China. The

populations in all studies were patients with COVID-19, and the sample size ranged from 41 to 1099 patients. With regard to the study design, retrospective and prospective methods were used, and the timeframe of all five studies covered the first two months of the COVID-19 pandemic (December 2019, January 2020). Specifically, Zhou et al.(9) studied the epidemiological characteristics of 191 individuals infected with COVID-19, without, however, reporting in more detail the mortality risk factors and the clinical outcomes of the disease. Among the 191 patients, there were 54 deaths, while 137 survived. Among those that died, 9% were current smokers compared to 4% among those that survived, with no statistically significant difference between the smoking rates of survivors and non-survivors ($p=0.21$) with regard to mortality from COVID-19. Similarly, Zhang et al.(10) presented clinical characteristics of 140 patients with COVID-19. The results showed that among severe patients ($n=58$), 3.4% were current smokers and 6.9% were former smokers, in contrast to non-severe patients ($n=82$) among which 0% were current smokers and 3.7% were former smokers, leading to an OR of 2.23; (95% CI: 0.65–7.63; $p=0.2$). Huang et al.(11) studied the epidemiological characteristics of COVID-19 among 41 patients. In this study, none of those who needed to be admitted to an ICU ($n=13$) was a current smoker. In contrast, three patients from the non-ICU group were current smokers, with no statistically significant difference between the two groups of patients ($p=0.31$), albeit the small sample size of the study. The largest study population of 1099 patients with COVID-19 was provided by Guan et al.(12) from multiple regions of mainland China. Descriptive results on the smoking status of patients were provided for the 1099 patients, of which 173 had severe symptoms, and 926 had non-severe symptoms. Among the patients with severe symptoms, 16.9% were current smokers and 5.2% were former smokers, in contrast to patients with non-severe symptoms where 11.8% were current smokers and 1.3% were former smokers. Additionally, in the group of patients that either needed mechanical ventilation, admission to an ICU or died, 25.5% were current smokers and 7.6% were former smokers. In contrast, in the group of patients that did not have these adverse outcomes, only 11.8% were current smokers and 1.6% were former smokers. No statistical analysis for evaluating the association between the severity of the disease outcome and smoking status was conducted in that study. Finally, Liu et al.(13) found among their population of 78 patients with COVID-19 that the adverse outcome group had a significantly higher proportion of patients with a history of smoking (27.3%) than the group that showed improvement or stabilization (3.0%), with this difference statistically significant at the $p=0.018$ level. In their multivariate logistic regression analysis, the history of smoking was a risk factor of disease progression (OR=14.28; 95% CI: 1.58–25.00; $p=0.018$). We identified five studies that reported data on the smoking status of patients infected with COVID-19. Notably, in the largest study that assessed severity, there were higher percentages of current and former smokers among patients that needed ICU

support, mechanical ventilation or who had died, and a higher percentage of smokers among the severe cases(12). However, from their published data we can calculate that the smokers were 1.4 times more likely (RR=1.4, 95% CI: 0.98–2.00) to have severe symptoms of COVID-19 and approximately 2.4 times more likely to be admitted to an ICU, need mechanical ventilation or die compared to non-smokers (RR=2.4, 95% CI: 1.43–4.04). In conclusion, although further research is warranted as the weight of the evidence increases, with the limited available data, and although the above results are unadjusted for other factors that may impact disease progression, smoking is most likely associated with the negative progression and adverse outcomes of COVID-19.

Text: non-survivors ($p=0.21$) with regard to mortality from COVID-19. Similarly, Zhang et al. 10 presented clinical characteristics of 140 patients with COVID-19. The results showed that among severe patients ($n=58$), 3.4% were current smokers and 6.9% were former smokers, in contrast to non-severe patients ($n=82$) among which 0% were current smokers and 3.7% were former smokers, leading to an OR of 2.23; (95% CI: 0.65-7.63; $p=0.2$). Huang et al. 11 studied the epidemiological characteristics of COVID-19 among 41 patients. In this study, none of those who needed to be admitted to an ICU ($n=13$) was a current smoker. In contrast, three patients from the non-ICU group were current smokers, with no statistically significant difference between the two groups of patients ($p=0.31$), albeit the small sample size of the study. The largest study population of 1099 patients with COVID-19 was provided by Guan et al. 12 from multiple regions of mainland China. Descriptive results on the smoking status of patients were provided for the 1099 patients, of which 173 had severe symptoms, and 926 had non-severe symptoms. Among the patients with severe symptoms, 16.9% were current smokers and 5.2% were former smokers, in contrast to patients with non-severe symptoms where 11.8% were current smokers and 1.3% were former smokers. Additionally, in the group of patients that either needed mechanical ventilation, admission to an ICU or died, 25.5% were current smokers and 7.6% were former smokers. In contrast, in the group of patients that did not have these adverse outcomes, only 11.8% were current smokers and 1.6% were former smokers. No statistical analysis for evaluating the association between the severity of the disease outcome and smoking status was conducted in that study. Finally, Liu et al. 13 found among their population of 78 patients with COVID-19 that the adverse outcome group had a significantly higher proportion of patients with a history of smoking (27.3%) than the group that showed improvement or stabilization (3.0%), with this difference statistically significant at the $p=0.018$ level. In their multivariate logistic regression analysis, the history of smoking was a risk factor of disease progression (OR=14.28; 95% CI: 1.58-25.00; $p=0.018$).

We identified five studies that reported data on the smoking status of patients infected with COVID-19. Notably, in the largest study that assessed severity, there were higher percentages of current and former smokers among patients that needed ICU support, mechanical ventilation or who had died, and a higher percentage of smokers among the severe cases 12 . However, from their published data we can calculate that the smokers were 1.4 times more likely (RR=1.4, 95% CI: 0.98-2.00) to have severe symptoms of COVID-19 and approximately 2.4 times more likely to be admitted to an ICU, need mechanical ventilation or die compared to non-smokers (RR=2.4, 95% CI: 1.43-4.04).

In conclusion, although further research is warranted as the weight of the evidence increases, with the limited available data, and although the above results are unadjusted for other factors that may impact disease progression, smoking is most likely associated with the negative progression and adverse outcomes of COVID-19.

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2019-nCoV: The Identify-Isolate-Inform (3I) Tool Applied to a Novel Emerging Coronavirus

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7081861/>

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Abstract: 2019 Novel Coronavirus (2019-nCoV) is an emerging infectious disease closely related to MERS-CoV and SARS-CoV that was first reported in Wuhan City, Hubei Province, China in December 2019. As of January 2020, cases of 2019-nCoV are continuing to be reported in other Eastern Asian countries as well as in the United States, Europe, Australia, and numerous other countries. An unusually high volume of domestic and international travel corresponding to the beginning of the 2020 Chinese New Year complicated initial identification and containment of infected persons. Due to the rapidly rising number of cases and reported deaths, all countries should be considered at risk of imported 2019-nCoV. Therefore, it is essential for prehospital, clinic, and emergency department personnel to be able to rapidly assess 2019-nCoV risk and take immediate actions if indicated. The Identify-Isolate-Inform (3I) Tool, originally conceived for the initial detection and management of Ebola virus and later adjusted for other infectious agents, can be adapted for any emerging infectious disease. This paper reports a modification of the 3I Tool for use in the initial detection and management of patients under investigation for 2019-nCoV. After initial assessment for symptoms and epidemiological risk factors, including travel to affected areas and exposure to confirmed 2019-nCoV patients within 14 days, patients are classified in a risk-stratified system. Upon confirmation of a suspected 2019-nCoV case, affected persons must immediately be placed in airborne infection isolation and the appropriate public health agencies notified. This modified 3I Tool will assist emergency and primary care clinicians, as well as out-of-hospital providers, in effectively managing persons with suspected or confirmed 2019-nCoV.

Text: 2019 Novel Coronavirus (2019-nCoV) is a novel respiratory disease first reported in Wuhan, Hubei Province, China in December 2019. 1 Chinese health officials were originally investigating a sudden increase in cases of pneumonia which were later determined to be linked to 2019-nCoV. While most cases originated within mainland China, the disease spread to neighboring countries including Taiwan, Thailand, South Korea, and Japan, and later to the United States, Europe, and Australia. A near real-time updated tracking website for cases and locations worldwide, along with reported deaths is available. 2 Chinese health authorities have sequenced 2019-nCoV and freely shared its genetic profile online. 3, 4 Additionally, on January 28, 2020, an Australian laboratory reported growing the virus from a patient sample. As of January 30, 2020, there have been at least 9,776 persons infected and 213 verified deaths. 2 These numbers are likely underestimates due to the limited information available regarding incubation time, transmissibility, and virus origin. The What was the research question? Investigators adapted the "Identify, Isolate, Inform" (3I) Tool for use in suspected cases of 2019-nCoV.

What was the major finding of the study? A novel 2019-nCoV 3I Tool is designed for frontline clinicians in the management of suspected patients.

This 2019-nCoV 3I adaptation will aid healthcare providers most likely to encounter the disease in the containment and effective treatment of patients.

age distribution of these verified deaths is currently not available.

One preliminary, small-scale study of 41 patients in Wuhan China, reported 6 deaths (15% mortality) with a median age of 49.0 years. 5 Additionally, transmission of the virus has reportedly occurred in healthcare facilities in Wuhan City, raising concerns of spread to healthcare workers, as was seen during prior outbreaks of the novel coronaviruses, Middle Eastern Respiratory Syndrome (MERS) and Severe Acute Respiratory Syndrome (SARS). Due to the dynamic nature of the outbreak, exposure criteria may change depending on where new cases of 2019-nCoV are detected, the degree of transmissibility, and when additional information regarding the origin of the virus is discovered and reported. On January 15, 2020, the Centers for Disease Control and Prevention (CDC) confirmed the first known imported case of 2019-nCoV in the US state of Washington. The patient had recently returned from Wuhan City, where he likely contracted the disease. Chicago health authorities reported a second US case on January 24, 2020. This was quickly followed by additional imported cases reported in Orange and Los Angeles Counties, California on January 26, 2020. Additional suspected cases continue to be evaluated. On January 30, 2020, the CDC reported the first local transmission in the US between members in a household. On the same day, the World Health Organization declared 2019-nCoV to be a Public Health Emergency of International Concern (PHEIC). 6 On January 31, 2020, the US Department of Health and Human Services declared coronavirus a public health emergency. 7 Healthy individuals and those with mild illness may be asymptomatic, while others may have more pronounced symptoms of fever or lower respiratory illness. Upon identification of a suspected patient, that individual should immediately be isolated with airborne precautions. Further workup and laboratory confirmation can then proceed. Emergency physicians (EPs), emergency medical services (EMS) personnel, and other healthcare workers who encounter patients with suspected 2019-nCoV infection must inform the appropriate authorities, including but not limited to hospital infection control and local or state public health agencies.

Healthcare workers must follow on-going developments related to the outbreak, especially new information concerning detection and management. 8, 9 The 3I Tool outlined in this paper is consistent with current US CDC guidelines and can be applied in a variety of settings such as those in emergency departments, urgent-care clinics, physicians' offices, and prehospital settings. This paper will first briefly review 2019-nCoV and then present the novel 2019-nCoV 3I Tool as modified from its initial conception for Ebola virus disease 10,11 and later adapted for measles, 12 MERS, 13 mumps, 14 Zika virus disease, 15 hepatitis A, 16 pertussis, 17 and scabies. 18

Coronavirus 2019-nCoV infection commonly presents with signs and symptoms of pneumonia or as a nonspecific lower respiratory illness, with coughing or difficulty breathing accompanied by fever. 5, 19, 20 Fever and cough constitute the most common presentations. However, patients may have other respiratory symptoms, sore throat, nasal congestion, malaise, myalgia, and headache. Bilateral infiltrates may be seen on chest X-ray. Severe cases may present with sepsis and even shock. Conversely, some patients may present as only mildly ill or asymptomatic altogether. 21 To date, patients with underlying medical conditions and the elderly are more likely to become severely ill, require hospitalization, and ultimately die. 22 Early predictions for incubation time are between 2 and 14 days, based on data from similar coronaviruses. The 14-day criterion for epidemiological risk assumes the longest estimated incubation time. 23 In addition, the World Health Organization (WHO) has created its own interim case definition. 24

By definition, the main features of a novel virus, for example, how it is transmitted, will not be immediately known. However, as with the development of any 3I Tool, it is essential to understand specific characteristics of the disease. In the case of a novel virus such as 2019-CoV, this is challenging since information is rapidly evolving and the science is not yet fully understood. It is possible that the virus will undergo mutations over time that could substantially change its

The Identify-Isolate-Inform (3I) Tool Applied to a Novel Emerging Coronavirus Koenig et al. features. Nevertheless, an appreciation of the key concepts that drive evidence-based management is beneficial (Table 1) . Management guidance will likely change over time.

With the initial discovery of a new potential public health threat, it will likely be unclear how patients become sick. For example, rather than a contagion, there could be a contaminant or a toxin responsible for signs and symptoms. In this case, the possibility of an environmental toxin in the Wuhan Market was a consideration early on when limited to no human-to-human transmission was reported. The mode of transmission has implications for the types of personal protective equipment (PPE) needed to protect healthcare providers in the prehospital, clinic, and hospital settings. ²⁵ In addition, patients may need decontamination after exposure to certain toxins. ²⁶ Another important consideration for application of the 3I Tool is whether the disease is contagious prior to symptom onset (like measles) or only after symptoms develop (like Ebola). A January 30, 2020 letter to the New England Journal of Medicine describes a purported confirmed instance of transmission from an asymptomatic individual. Researchers state that, before symptom onset, the primary case infected two individuals, one of which infected two additional colleagues. ²⁷ Subsequent investigation suggested that the source patient did have mild symptoms and had taken an antipyretic, calling this reported asymptomatic transmission into question.

While quarantine may not be feasible and can have unintended consequences, ^{28, 29, 30} it is a public health tool that can be considered in cases when disease is transmissible before symptom onset. ³⁰ Conversely, if a disease is known not to be transmissible prior to symptom onset, asymptomatic exposed patients must be monitored, but do not require quarantine or isolation unless they develop symptoms.

Initially, it may be unclear whether an infectious agent occurred naturally or was deliberately or accidentally released. In this case, a BSL-4 laboratory studying coronaviruses was located approximately 32 kilometers away from the market where initial exposures were felt to occur. ³¹ Recall that in 2001, the anthrax letter attacks were initially thought to be naturally occurring. Once determined to be bioterrorism, management of the event was similar to that for a chemical exposure with a sudden impact, defined scene, and need for a rapid response and decontamination on site. This differed from the WHO's modeling predicting an aerosolized release that would result in an incubation period with 100,000 or more persons exposed rather than the 22 people who contracted anthrax in 2001. ³² By understanding the key features of a novel disease, healthcare workers can take evidence-based measures to protect themselves, optimize individual patient management, and prevent further disease spread.

It is currently unclear how 2019-nCoV is spread, but it is suspected to be transmitted through contact with infected respiratory secretions, like other known coronaviruses. There are instances of sustained human-to-human transmission across generations of cases, especially near the epicenter in Wuhan City. ²¹ Current evidence suggests that close contact with an infected person is a major factor in disease transmission. CDC defines "close contact" ³³ as being in or within two meters of an area with a confirmed patient or being directly exposed to infectious secretions without appropriate PPE. Healthcare facilities in China have reported spread from person to person. In addition, some mildly ill or potentially even asymptomatic patients may have a higher chance of spreading the disease to others as they may be less likely to seek medical care. ³⁴ The possibility that patients may be infectious prior to symptom onset further compounds the difficulty of containing the virus and effectively preventing transmission.

The current majority of 2019-nCoV cases have been within China and its bordering countries. ² Persons with recent travel (within 14 days) to Wuhan City or another region with widespread disease, or exposure to a patient under investigation, are considered to have an epidemiologic risk factor and should be assessed for signs and symptoms of a viral illness such as fever and respiratory symptoms. Coronavirus is a zoonotic virus

The Identify-Isolate-Inform (3I) Tool Applied to a Novel Emerging Coronavirus that is transmitted to humans via contact with infected animals. Preliminary reports suggest the disease may have originated in a seafood and live animal market in Wuhan City, but it is still unknown how or whether such transmission occurred.

Clinicians working with local public health departments must arrange to have specimens from patients under investigation (PUIs) sent to the CDC laboratory. At this time, the CDC has the only laboratory that can definitively test for 2019-nCoV, though laboratory testing capacity is being rapidly expanded. Polymerase chain reaction (PCR) assays conducted on samples from a patient's upper and lower respiratory tracts will be used to confirm potential cases. In addition, serum antibody titers can be analyzed for confirmation of infection or evidence of immunity. Up-to-date information about the needed specimens and handling requirements to test for 2019-nCoV are available on the CDC website. ³⁵

Like other related coronaviruses, patients with 2019-nCoV frequently present with non-specific symptoms resembling that of influenza. Physicians may consider differential diagnoses related to a wide variety of respiratory infections. In order to relate these symptoms to 2019-nCoV, it is imperative that the identification of a potential exposure event (epidemiologic risk factor) within 14 days of symptom onset is made so that a more focused work-up for 2019-nCoV can be completed. Although the likelihood of coinfection of 2019-nCoV and another respiratory virus is thought to be low, a positive finding of another respiratory pathogen does not exclude the diagnosis of 2019-nCoV. Many commercially available respiratory panels include "coronavirus" in the results, but neither a positive nor a negative finding on these panels should be used to include or exclude a diagnosis of 2019-nCoV.

Supportive care with appropriate infection control is the mainstay of current CDC treatment guidelines for 2019-nCoV. There are not yet any approved antiviral treatments for 2019-nCoV. Emergency Use Authorizations (EUA) for compassionate use cases may be forthcoming from the US federal government for normally unapproved treatments. Supportive treatment predominantly includes respiratory support, hydration, and antipyretics. General treatment for severe cases should focus on the preservation of vital organ function. In the future, antiviral medications may be available. If a secondary bacterial infection such as pneumonia develops, targeted antibiotics are indicated.

Prevention of 2019-nCoV transmission, like any other infectious agent, involves minimizing risk of exposure. Vaccines are under accelerated development and may be useful in the future for post-exposure prophylaxis. Healthcare personnel are at increased risk and should practice standard, droplet, and airborne precautions when encountering an infected person, a PUI, or any symptomatic close contacts. Healthcare workers handling specimens should also adhere to CDC guidelines and should not attempt to perform any virus isolation or characterization.

Fever screening has been implemented at numerous airports, including major international hubs within Asia and the US. The efficacy of this intervention is not well documented, however, as some infected persons may be afebrile and disease transmission might occur prior to symptom onset.²⁷ In addition, people can artificially lower their temperature readings, e.g., by applying ice to their foreheads.

As outlined above, admission criteria for 2019-nCoV are similar to that of other patients. If patients do not meet medical criteria for hospitalization, they may be discharged home with isolation precautions and continued observation. EPs must notify local public health authorities so appropriate monitoring and community protective measures can be instituted.

The Identify-Isolate-Inform (3I) Tool was initially developed for Ebola virus disease 10,11 and later adapted for measles, 12 MERS, 13 mumps, 14 Zika virus disease, 15 hepatitis A, 16 pertussis, 17 and scabies. 18 This novel tool for suspected 2019-nCoV patients (Figure 1) provides frontline clinicians with a simple algorithm to manage an emerging disease. Identification of exposed patients with an epidemiologic risk factor within 14 days of symptom onset is a crucial first step. An automatic prompt in the electronic health record can be useful in assisting clinicians with early identification of patients at risk. Case definitions promulgated by the WHO 24 and CDC 33 provide useful comprehensive definitions that have been incorporated into the 3I Tool. The 2019-nCoV Tool provides an accurate, summarized algorithm to immediately, and effectively manage suspected patients until additional resources can be consulted.

Patients who do not have an exposure risk or any symptoms may be triaged normally. However, before making patient contact, providers must first apply the Vital Sign Zero concept. 36 Vital Sign Zero is a preliminary, non-contact assessment (i.e., performed prior to touching a patient to take traditional vital signs) to first determine whether specific PPE is indicated before the examination commences. By taking the additional time to complete this assessment, risk of exposure and further transmission can be minimized. while in the treatment facility should be started and maintained to assist with the possibility of contact tracing. Following isolation, physicians should immediately inform the appropriate authorities. Patients who do not meet medical criteria for admission can be isolated at home during the evaluation phase. 37 Health department officials can help prevent transmission in isolated patients by providing in-home monitoring and implementing appropriate exposure-control measures.

Providers in the prehospital setting who have a high likelihood of encountering 2019-nCoV patients, such as those near international ports of entry, should adhere to established exposure control guidelines. 38 Along with appropriate PPE, providers should also carry thermometers to quantify any fever. In the US, providers should contact the appropriate

CDC quarantine station upon isolation of infected or suspected patients, especially those from Wuhan, China or other regions with widespread disease, who report symptoms in the last 14 days. As for other infectious diseases, assessing travel history is essential. Dispatch protocols have been instituted to facilitate identification of callers to 911 or the country-equivalent emergency number prior to prehospital personnel arrival. 39 In addition, CDC has promulgated EMS guidelines for prehospital PPE, transportation of PUIs, vehicle decontamination, and 911 Public Safety Answering Points (PSAPs) for 2019-nCoV. 40

2019-nCoV is an emerging infectious disease with rapidly evolving features, the full scope of which will be defined over time. Prior outbreaks of coronaviruses can help inform needed actions in the short term to assist with both treatment of individual patients and prevention of global disease spread. This adaptation of the Identify-Isolate-Inform Tool serves as a resource for healthcare workers who need to make clear, rapid assessments when confronted with potential patients. The concise nature of the 2019-nCoV 3I Tool allows for the rapid and effective management of a novel disease by healthcare providers.

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Responding to the COVID-19 pandemic in complex humanitarian crises

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7085188/>

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Abstract: nan

Text: Over 168 million people across 50 countries are estimated to need humanitarian assistance in 2020 [1] . Response to epidemics in complex humanitarian crises such as the recent cholera epidemic in Yemen and the Ebola epidemic in the Democratic Republic of Congo is a global health challenge of increasing scale [2] . The thousands of Yemeni and Congolese who have died in these years-long epidemics demonstrate the difficulty of combatting even well-known pathogens in humanitarian settings. The novel severe acute respiratory syndrome coronavirus-2 (SARS-CoV-2) may represent a still greater threat to those in complex humanitarian crises, which lack the infrastructure, support, and health systems to mount a comprehensive response. Poor governance, public distrust, and political violence may further undermine interventions in these settings.

Populations affected by humanitarian crises are expected to be particularly susceptible to COVID-19, the disease caused by SARS-CoV-2, due to displacement, crowded housing, malnutrition, inadequate water, sanitation, and hygiene (WASH) tools, and stigmatization. Disease outbreaks further reduce access to limited healthcare, which is increasingly disrupted by attacks on health facilities and the persistent overburdening of health systems. These situations escalate both the necessity and the difficulty of delivering accurate and actionable information to potentially affected populations [3] .

As the international community responds to SARS-CoV-2, public health authorities in humanitarian crises begin at a disadvantage to enact appropriate infection control to prevent transmission in healthcare settings, identify infectious cases, administer supportive care and novel treatments for the seriously ill, and trace contacts. These standard public health measures are particularly difficult to perform in humanitarian settings. For example, limited public health, laboratory, and primary care services represent a barrier to testing. Providing the limited healthcare worker cadre with appropriate training and personal protective equipment, and ensuring a continuous supply chain for such, is a challenge in all settings, exacerbated in complex humanitarian crises. Frequent displacement and limited contact information may prevent effective contact tracing. Finally, intractable structural challenges such as overcrowding limit the implementation of both quarantine of those exposed and isolation of those who are ill. Given these increased vulnerabilities, humanitarian crises should be viewed as a priority

for national and international bodies that seek to combat this unfolding pandemic. Resources must be identified to protect healthcare workers, develop and deploy rapid testing, improve surveillance, and enact quarantine and isolation of contacts and cases.

To mitigate the impact of COVID-19 on crises-affected populations, governments and agencies will implement the familiar, global evidence-based approaches for combatting respiratory viruses. Respiratory hygiene is a highly effective public health intervention, supported by evidence demonstrating that the spread of respiratory viruses, such as SARS-CoV-2, can be prevented by hand hygiene, safe cough practice, and social distancing [4] . Hand hygiene is a readily implemented behavior: the distribution of soap to households in humanitarian settings has been shown to increase handwashing by over 30% [5] . Furthermore, hand hygiene is an avenue of agency for protecting one's own health, consistent with the rights to dignity and to fully participate in decisions related to assistance in humanitarian crises. Widespread introduction of alcohol-based hand rubs is also possible in many resource-limited settings, with published protocols for local production [6] .

The Sphere Handbook, a collection of rights-based guidelines for humanitarian response, is the foremost authority on minimum standards for humanitarian assistance [7] . However, despite the indisputable evidence for the efficacy of hand hygiene for reducing both bacterial and viral pathogen transmission, humanitarian WASH standards are based on evidence pertaining to the prevention of illnesses transmitted by the faecal-oral route, with the focus on hand hygiene proximate to latrines [5, 8] . And yet, latrines in crisis settings are often shared and distant from residential shelters, conferring a high risk of gender-based violence [9] . Gender-based violence around latrines is an important deterrent for accessing latrine-adjacent handwashing stations, particularly for hand hygiene to prevent respiratory pathogen transmission.

Evidence-based guidelines alone in complex humanitarian crises may not suffice during the emergence of the current SARS-CoV-2 pandemic. Without the adaptation of existing standards, mitigation plans will fall short of health and human rights obligations in outbreak response. Crisis-affected community engagement is integral in pandemic planning, in order to maximize the real-world effectiveness of efficacious interventions. Transparent and credible information-sharing mechanisms are increasingly essential when pandemics threaten vulnerable populations [10] . Diplomacy bridging long-standing

mistrust of public health and biomedical interventions and facilitating engagement with contentious actors is a necessary component of effective health governance in complex crisis settings [2] . Interventions tailored to the needs of crisis-affected populations, delivered with transparent information, in the context of inclusive governance practices, are urgently needed in the global response to the COVID-19 pandemic.

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It is Unlikely That Influenza Viruses Will Cause a Pandemic Again Like What Happened in 1918 and 1919

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4019839/>

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Text: Influenza and influenza viruses are wellknown popular topics to medical professionals and the general public. Influenza viruses had caused a pandemic globally during 1918 and 1919, and that influenza pandemic had taken away more than 20 million people's lives in the world. However, in my opinion, it is unlikely that influenza viruses will again cause a pandemic on a level (both of the morbidity rate and the mortality rate) comparable to what happened in 1918 and 1919.

Influenza viruses very easily reassort, recombine, and point mutate in nature due to their segmented RNA genome structures, however, unlike highly pathogenic (virulent) viruses like rabies virus, Lassa fever virus, smallpox virus, eastern equine encephalitis virus, Ebola virus, Marburg virus, and human immunodeficiency virus 1 (HIV-1); most influenza viruses (wild types and mutants) are moderately pathogenic. The case fatality rates of some highly virulent viruses and related references are listed in Table 1 .

On November 11, 1918 , the fighting of World War I was stopped, and World War I was officially ended on June 28, 1919 with the signing of the Versailles Treaty. It is estimated that around 8.5-10 million soldiers lost their lives in World War I due to battle. The war also directly caused more than 6 million civilian deaths. Millions of people suffered from hunger and malnutrition during the war. Malnutrition weakened the human immune system and made a person more vulnerable to infectious diseases like tuberculosis and influenza, therefore, hunger and malnutrition were indirectly responsible for millions of deaths in the world in that period of time. For example, about 700,000 Germans died from malnutrition-related diseases in the years of 1914-1918. During the 1918-1919 influenza pandemic, between 21 and 25 million people died of influenza worldwide. Those people were killed both directly and indirectly by influenza virus infections. Many families were too poor to buy food and coal, and to afford health care expenses when their family members were ill. Influenza virus could infect all members of a family, and this could result in no one left to feed the fires, and to prepare food for the whole family, even if they had firewood, coal, and food left in their homes. Sadly, a large number of people died of influenza virus infections along with starvation, cold, and poor living conditions (8) .

In recent years, while hunger and malnutrition are not major and serious problems in some developed countries anymore, they are still very difficult to overcome in many developing countries. In these less-developed countries, there were approximately 925 million people who suffered from hunger; 125 million children were underweight; and 195 million children were stunted each year (9) . Nevertheless, in comparison to 1918 and 1919, currently, we have much better social and economic conditions and public health systems globally; and generally speaking, the majority of people in the world have better nutritional and educational statuses; better living and working conditions; therefore, better general health and immunity. Furthermore, in 1918 and 1919, physicians and nurses almost had nothing in their hands to help individuals who were infected by influenza viruses. Today, although we still do not have very effective, powerful, and practical anti-influenza drugs available, we at least have some improved, useful, and helpful anti-viral drugs like zanamivir, and

effective, convenient anti-cold medicines like Tylenol or Advil. We do not have a universal vaccine to prevent all influenza virus infections, but we can make effective vaccines to a specific influenza virus strain in a short time. Actually, in the United States of America, the influenza classed mortality rate declined from 10.2/100,000 in the 1940s to 0.56/100,000 in the 1990s; and the classed mortality rates of 1957-1958 and 1968-1969 influenza pandemics were not remarkably different from the non-pandemic seasons (10) .

Because of the above reasons, we can optimistically assume that even the same strain of influenza virus, which caused pandemic in 1918 and 1919, would not be able to kill millions of people and cause a pandemic comparable to the 1918-1919 pandemic again in the future.

Additionally, a significant number of viruses can cause influenza-like syndromes, such as rhinovirus, parainfluenza virus, adenovirus, coronavirus, respiratory syncytial virus, Coxsackie B virus, echovirus, and metapneumovirus (11, 12) . Some of the above-mentioned viruses like adenovirus and mutated coronavirus could cause problems that are comparable to influenza viruses (13, 14) .

The World Health Organization (WHO) mistakenly raised the level of influenza pandemic alert from phase 5 to the highest phase 6 on June 11, 2009 (15) . However, the truth was that most cases of H1N1 influenza A virus infections were mild, the symptomatic case fatality rate was only 0.005% in New Zealand (16) ; and in New York City, the case fatality rate was 0.0094-0.0147% for persons ≥ 65 years old, and for those of 0-17 years old, the case fatality rate was 0.0008-0.0012% (17) . Some researchers argued that it should not have been called an influenza pandemic in the first place if the clinical severity was considered (15, (18) (19) (20) . I believe it was unwise that we had paid too much www.frontiersin.org 23) . Not surprisingly, every year there would be some influenza patients and a few of them would die from the infections, as it is almost impossible to eliminate influenza viruses from the natural environment in many years. The severity of a viral infection is determined by both of the viral virulence (pathogenicity) and the host immunity. Some researchers' opinions on H7N9 avian influenza virus were incorrect and/or inadequate. They mainly focused on influenza viruses and worried about viral mutations, viral pathogenicity, viral adaptation, and transmission. They overestimated the negative part of socio-economic factors of the present east China: overcrowded population in the epidemic region; very busy national and international transportation and travel; a large

number of live poultry markets . . . but they underestimated the currently changed, developed, and improved positive part of socio-economic factors in China. The following factors might be used to explain why that H7N9 influenza A virus epidemic was limited and controlled in China, and only a few immunocompromised patients were killed by H7N9 influenza A virus. First, China has a relatively organized and effective public health system, there are four levels of (national, provincial, prefectural-level city, and county) centers for disease control and prevention all over China (24) . Second, physicians and nurses in China were prepared and knowledgeable of influenza virus infections. Third, samples from patients with suspected influenza virus infections were collected and sent to the local and national centers for disease control and prevention promptly. H7N9 influenza A viruses were isolated and identified very quickly. Thereby, they were able to diagnose, confirm, and report three cases of H7N9 influenza patients in the early stage of the epidemic (24, 25) . Fourth, health care and public health workers were protected properly. Consequently, none of the health professionals was infected by H7N9 influenza A virus in 2013. However, a surgeon died of H7N9 influenza in Shanghai, China in January of 2014 (26) . Fifth, they detected H7N9 influenza A viruses from the samples of chickens, pigeons, and the environment of live poultry markets in Shanghai (27) ; and closed the live poultry markets of the involved epidemic region quickly. Sixth, patients were isolated and treated timely in hospitals, 74% (1251/1689) of those close contacts of H7N9 influenza patients were monitored and observed. Thus, H7N9 influenza A virus could not spread to a bigger population (24) . Last but not least, we are connected to the Internet now, and it seems that our planet is much smaller today than the earlier days when we did not have the Internet, because communication and information exchange have become so fast, easy, and convenient presently. During that avian influenza epidemic, some influenza experts in the world shared/exchanged H7N9 influenza A virus information and provided professional consultations and suggestions efficiently and rapidly. All these public health routine practices and measures resulted in that H7N9 influenza epidemic being controlled and stopped in China (24) . I have to point out that the cases of diagnosed H7N9 avian influenza A virus infection might only be the tip of the iceberg. Aside from one laboratory confirmed asymptomatic case of H7N9 influenza A virus infection in Beijing (22), there were probably many undetected mild or asymptomatic cases of influenza A H7N9 infection. The reason is that most people usually think a common cold is a very common and normal occurrence, and they don't take flu-like illnesses seriously. In most situations, they would just stay home and take some medicines. Only those who have very severe flu-like symptoms would see doctors, and thereby be detected and diagnosed, accordingly the real case fatality rate should be much lower than the detected 32.14% (45/140, one case from Taiwan, and one case from Hong Kong) (22, 23).

Nowadays, we travel faster, and we travel more frequently and globally, and we have more complicated social activities and lifestyles, thereby increasing the chances of viral mutation; and we realize that influenza viruses are even easier to reassort, recombine, and mutate in nature than many other RNA viruses. However, we are now living in a technologically, economically, and socially much better and advanced society. I believe influenza virus infections are controllable and preventable, with the increased population health and immunity, with the WHO Global Influenza Surveillance and Response System, and with standard/routine epidemiological practices, and with new effective anti-viral agents and vaccines in production in the future. Now, I first predict that influenza viruses will unlikely again cause a pandemic on a level comparable to what happened in 1918 and 1919. Hopefully, one day we could consider a strategy to produce a universal vaccine that can prevent people from infections of all influenza virus strains, or we could produce some very effective anti-influenza virus drugs; then influenza would not be a problem anymore. We should learn lessons from the mistakes we made in the past. It is reasonable and necessary to be cautious about influenza viruses, but overreactions or catastrophic reactions should be avoided in the future. My opinion is anti-traditional; the purpose of this article is to influence public health policy, and to save some of the limited resources and money for more important diseases like heart diseases, cancer, diabetes, AIDS, hepatitises, and tuberculosis (15) .

Linting Song: conception of manuscript, drafting of manuscript, critical revision of manuscript, and final approval of manuscript.

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Ebola Virus Maintenance: If Not (Only) Bats, What Else?

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Abstract: The maintenance mechanisms of ebolaviruses in African forest ecosystems are still unknown, but indirect evidences point at the involvement of some bat species. Despite intense research, the main bat-maintenance hypothesis has not been confirmed yet. The alternative hypotheses of a non-bat maintenance host or a maintenance community including, or not, several bat and other species, deserves more investigation. However, African forest ecosystems host a large biodiversity and abound in potential maintenance hosts. How does one puzzle out? Since recent studies have revealed that several bat species have been exposed to ebolaviruses, the common denominator to these hypotheses is that within the epidemiological cycle, some bats species must be exposed to the viruses and infected by these potential alternative hosts. Under this constraint, and given the peculiar ecology of bats (roosting behaviour, habitat utilisation, and flight mode), we review the hosts and transmission pathways that can lead to bat exposure and infection to ebolaviruses. In contrast to the capacity of bats to transmit ebolaviruses and other pathogens to many hosts, our results indicate that only a limited number of hosts and pathways can lead to the transmission of ebolaviruses to bats, and that the alternative maintenance host, if it exists, must be amongst them. A list of these pathways is provided, along with protocols to prioritise and investigate these alternative hypotheses. In conclusion, taking into account the ecology of bats and their known involvement in ebolaviruses ecology drastically reduces the list of potential alternative maintenance hosts for ebolaviruses. Understanding the natural history of ebolaviruses is a health priority, and investigating these alternative hypotheses could complete the current effort focused on the role of bats.

Text: Ebolaviruses (EBVs), according to Kuhn et al. classification [1]) are single-strand RNA filoviruses that can induce a high mortality in some hosts, including apes and humans [2, 3] . The different ebolaviruses have caused localised but dramatic human outbreaks, mainly in Central Africa, in the last 40 years. The recent West African outbreak in 2013-2016 gave an outline of the pandemic potential of these pathogens [4, 5] .

Disentangling the complexity of maintenance hosts or communities in multi-host systems at the wildlife/livestock/human interface is a difficult task [16] [17] [18] . The maintenance of EBV in equatorial forests is yet to be understood. Some mammal species played a major role in triggering human outbreaks: apes such as chimpanzees (*Pan troglodytes troglodytes* and *P. t. verus*) and western lowland gorillas (*Gorilla gorilla gorilla*) were at the origin of several human outbreaks [10] [11] [12] , but have been found to be highly susceptible to EBV with potential drastic impact for their populations [12, 19] . EBOV PCR positive duiker carcasses (*Cephalophus* sp.) have also been found [20] . One would not expect such a high mortality (relative to their population density) of EBOV in maintenance hosts. However, these data indicate their possible involvement in the transmission function of EBOV, bridging the maintenance host with human populations during a spillover event [18] (Figure 1). The EBOV susceptibility and exposure (tested by virology, serology and/or PCR) of many other potential forest hosts, including invertebrates, birds, bats, monkeys, rodents, and other small mammals, have been tested in the field or experimentally with an interestingly large amount of negative results (e.g., [12, [21] [22] [23] [24] [25] [26]). A few monkey and bat individuals serologically positive to EBV antigen represent the only exceptions [12] .

Today, African bats are considered by many as the best candidates for acting as maintenance hosts for EBOV. Partial vRNA was sequenced from living specimens of three different bat species in Central Africa [23] , and antibodies against ebolavirus antigen have been detected in 9 bat species (8 frugivorous and 1 insectivorous) [3, 23, [27] [28] [29] [30] . Recently, a new ebolavirus species with an unknown pathogenic risk has also been isolated from two insectivorous bat species roosting inside a house [31] . Moreover, Swanepoel et al. showed that EBOV replicated in three species of experimentally infected bats (*Tadarida condylura*, *Tadarida pumila*, and *Epomophorus wahlbergi*), including virus isolated from faeces 21 days after experimental infection [22] . In addition, some bat species have been shown to act as maintenance hosts for multiple RNA viruses, including filoviruses (e.g., [32] [33] [34]). However, to date, no EBOV replicative strain has been isolated from healthy wild bats despite thousands of individuals tested [14, [23] [24] [25] 28, 34, 35] . Given the current knowledge, the main hypotheses for EBOV maintenance are

a single bat species as *Rousettus aegyptiacus* is considered the maintenance host for Marburg virus (Figure 1A1); or a network of interacting bat species creating a maintenance community for EBOV (Figure 1A2).

The bat system is complex. First, for its diversity: globally, they represent over 20% of the mammal diversity, forming the second largest mammalian order after rodents, and Africa hosts 317 known living species, 25% of the global bat diversity [36] . Secondly, bats have exceptional lifestyles that have already been reviewed, especially in relation to their role in disease ecology [33, [37] [38] [39] [40] [41] [42] [43] . They are unique mammal species regrouping such peculiar life history traits as their aerial life mode, their longevity, their gregarious and migration patterns, as well as their immune system. bridging the maintenance host with human populations during a spillover event [18] (Figure 1). The EBOV susceptibility and exposure (tested by virology, serology and/or PCR) of many other potential forest hosts, including invertebrates, birds, bats, monkeys, rodents, and other small mammals, have been tested in the field or experimentally with an interestingly large amount of negative results (e.g., [12, [21] [22] [23] [24] [25] [26]). A few monkey and bat individuals serologically positive to EBV antigen represent the only exceptions [12] .

Potential maintenance mechanisms of ebolaviruses in wildlife, according to current knowledge. Circles (plain or dotted) indicate a maintenance function play by the host(s); arrows represent infectious transmission pathways between hosts. Humans, non-human primates, and duikers are examples of known non-maintenance hosts, exposed occasionally to ebolavirus directly or indirectly through the main maintenance host. (A1) Main maintenance hypothesis: there is one bat Figure 1 . Potential maintenance mechanisms of ebolaviruses in wildlife, according to current knowledge. Circles (plain or dotted) indicate a maintenance function play by the host(s); arrows represent infectious transmission pathways between hosts. Humans, non-human primates, and duikers are examples of known non-maintenance hosts, exposed occasionally to ebolavirus directly or indirectly through the main maintenance host. (A1) Main maintenance hypothesis: there is one bat species maintaining each ebolavirus alone. Currently this is logically the most investigated hypothesis given the available data, and represents the maintenance mechanism for another filovirus, the Marburg virus, as currently understood. (A2) Several bat species are needed to create a maintenance community for Zaire ebolavirus (EBOV); each bat species cannot complete EBOV maintenance alone, as it requires interactions with the other species. (B) Alternate non-bat maintenance host hypothesis: if it exists, it is known that it can transmit ebolaviruses to some bat species. In this article, we review the

potential hosts and associated transmission pathways that link this host to bat species (red arrow). (C) The maintenance community hypothesis, in which several hosts are needed to maintain ebolaviruses (ellipses represent different scenarios of community maintenance). This could be one or more alternative hosts involving possibly bat species. By definition, if such an alternative host exists, there are infectious transmission pathways from this host towards bats that are reviewed here (red arrows).

Proving that a bat species maintains EBOV (e.g., [44, 45]), or that interconnected populations of different bat species create the cradle for EBOV maintenance in a specific ecosystem, is a difficult task. Finding a live virus in a healthy bat specimen would constitute a great step in proving that this particular species is part or the totality of the EBOV maintenance. However, this finding would also trigger new questions: does this species act alone to maintain EBOV, or do other sympatric bat species' populations create a maintenance community for EBOV? Is this EBOV maintenance system unique or ecosystem specific? Additionally, are other non-bat species involved in the maintenance? The road to identifying the maintenance host(s) of EBOV is still long.

The gaps in knowledge concerning the maintenance of EBOV and other EBV are therefore still significant. Available data indicates a systematic but weak signal in some bat species, a pattern in line with the main bat maintenance hypotheses, but not excluding as well alternative hypotheses as presented in Figure 1B ,C. If those alternative scenarios do not necessarily agree with the Occam's razor principle, they still cannot be ignored by the scientific community. African forest ecosystems host a high diversity of organisms relative to other ecosystems, and provide a rich pool of candidate species for playing a role in EBOV maintenance. EBOV specialists agree in calling for more integrated efforts across scientific fields, notably epidemiology, ecology, molecular biology, remote sensing modelling, and social sciences to test new hypotheses [39] . We provide, here, an ecological perspective on the EBOV multi-host system to provide a hypothesis-driven framework for future work. There is still a possibility that bats are not part of or that non-bat species are involved in the EBOV maintenance system and alternative scenarios should be considered and explored (Figure 1) [46] . These scenarios should be investigated, when possible, alongside bat-centred protocols, to confirm or invalidate the case for bats as EBOV maintenance hosts.

When a probability P is difficult or impossible to estimate, it is sometimes easier to estimate its inverse probability $(1-P)$, the probability that it does not happen. It would be tedious to quantitatively estimate probabilities in the case of ebolavirus maintenance given the current lack of information, but trying to define the components of this probability could help. Hence, instead of proving that bats are the maintenance host for EBOV, what if we consider that "bats are not the (only) maintenance host for EBOV"?

Here, we consider the scenario presented in Figure 1B ,C, namely, that bats are not the maintenance host for EBOV or that bat species are involved with alternative host(s) in the EBOV maintenance community. Current data and knowledge support both scenarios. Some bats are sometimes in contact with the virus and experience waves of exposure during outbreaks [27] . Once infected, bats could either be dead-end hosts, as some experimental studies suggest that some bat species cannot excrete the virus [47]); or they could transmit viruses to other hosts, such as primates including humans [6, 48, 49] as a bridge host, linking the maintenance host with humans. This means based on the definition of a bridge host [18] , that these bats must have been in contact, at some point in the epidemiological cycle, with the maintenance host (or another bridge host) to get the EBOV infection. Here, "contact" means infectious contact, and can be direct (e.g., physical) or indirect (e.g., through the environment). The search for alternative maintenance hosts for EBOV should, therefore, concentrate on hosts that can transmit the virus to bats. In other words, any host that could not transmit the virus to bats would be ineligible to be a maintenance host for EBOV. This holds for any host found exposed to EBOV (e.g., some duiker sp.) but the focus on bats is justified in the following section.

The ecology of most African bat species is largely unknown. It can still be summarised as follows: roosting in trees (hanging or in holes) or caves, flying, eating insects while flying (insectivorous bats)/eating fruits in trees (fruit bat), flying back and roosting in trees or caves; with biannual long-range migration or nomadic movements for some species [50] . A single bat can cover a large variety of habitats and even regions for those migrating. Therefore, the transmission pathways from bats to other animals through urine, saliva, birthing fluids, and placental material and/or guano could be important (see review on Ebola isolated from body tissues and fluids [51]). Predation is also a less known but potential transmission pathway from bats to predators [48, 52] . The range of potential species at risk of infection from bats is thus large [53] . However, the range of potential transmission pathways available for the maintenance or bridge host (under scenario B and C in Figure 1) to infect bats seems to be much more limited. For example, bats seldom use

the ground floor: transmission routes requiring direct contact or environmental transmission on the ground do not expose bats. In other terms, direct contacts with strictly ground-dwelling animals would be very unlikely. Four habitat types structure the various transmission pathways from the alternative host to bats (and each bat species will frequent only a fraction of these habitats: (i) open air while flying, for insectivorous bats also while feeding; (ii) surface water when drinking; (iii) cave roofs and walls as roost habitat; (iv) tree canopy for roosting or feeding. From these four habitats, potential transmission routes to infect bats from other hosts can be inferred (Table 1). In the following sections, the different transmission pathways that can link potential alternative hosts to bats are listed and discussed, along with examples of these alternative hosts.

Firstly, EBOV transmission to bats could occur through aerosol transmission in all four habitats. This means that the maintenance host would release, in bats' airspace, enough EBOV to contaminate bats. In theory, this would be possible in most bat environments, but we have discarded open-air transmission (e.g., in-flight bird to bat transmission) as the load of virus in the air cannot reach the levels that ensure infection. However, in the confined atmosphere of caves, bat to human transmission of rabies has been suspected [54] [55] [56] . EBOV and other filovirus particles seem to be able to persist for at least 90 min as aerosol [57, 71] , and experimental studies conducted on non-human primates (NHPs) by inoculating EBOV via the aerosol route were able to induce fatal disease 5 to 12 days post-inoculation [58] . Experimental airborne transmission of EBOV between animals from different species, e.g., from pigs to non-human primates, also seems possible [74] . In caves, the aerosol route might thus be possible. However, as bats tend to roost aggregated in groups and sometimes in large colonies, the ambient air may be saturated by bats' aerosols, rather than an alternative host. Air screening could be attempted in bat habitats but experimental aerosol transmission trials from alternative hosts to bats would be more efficient.

Bats are exposed to ectoparasitism [61] . If the biting invertebrate has previously bitten the alternative maintenance host, it could, in principle, infect bats. Hematophagous insects have been screened for EBOV during or after outbreaks with no conclusive results [26, 75] . However, absence of exposure during an outbreak does not mean that the host is not involved in the maintenance of the virus in-between outbreaks. For example, the process of amplification in disease ecology can involve different hosts than maintenance hosts. Little information is available on ticks in bats. Ticks have been suggested to be involved in the transmission of Crimean-Congo haemorrhagic fever-like viruses to bats [76] , and are

seriously considered as potential hosts for the transmission of other pathogens from non-bat hosts to bats. Mosquitos could also be a vessel for a vector-borne transmission of EBOV. Studies on mosquito blood meals have revealed that mosquito could feed on bats and other mammals [62, 63] . Bat flies appear to be highly bat-specific, adapted to their lifestyle [77] [78] [79] [80] and are involved in the transmission of pathogens [64] . However, this specificity would preclude interspecies pathogen transmission. Ectoparasitism provides a potential solid source of indirect contacts between the alternative maintenance host and bats. This transmission pathway should be explored much further, and ecological insights, including insect and bat behavioural ecology, will be necessary to target the right insect species within the diversity of available biting species, in the right habitat (e.g., tree canopy level, caves' roofs, when bats are immobile) at a proper time (e.g., nocturnal behaviour of bats) and season, when both hosts (i.e., the maintenance host and bats) can be fed upon by the vector. To our knowledge, such targeted protocols have not been implemented so far.

Insectivorous bats feed on insects that could be a source of EBOV [61] . This food-borne route has been little investigated as well. A recent study pointed out the role of insect-specific viruses in the evolution of numerous viral families, including mononegaviruses, which infect vertebrates [81] . There is a possibility that prey-insects are the maintenance host for EBOV [61] . Insect vectors, such as blood feeding insects (e.g., mosquitos) could also, in theory, transport viruses in their blood meal after a bite on an infected host. They have been suspected in other filovirus outbreaks in the past [82] . In theory, these insects preyed upon by bats could also link bats to any type of maintenance host they could feed on. Bats actively search for prey in many different habitats hosting hematophagous insects that feed on habitat-specific fauna. Moreover, Reiskind et al. suggested that blood fed female mosquitos are more susceptible to predation [66] . Leendertz et al. also suggested that the population dynamics of mayflies may act as a driver of EBOV emergence in mammals and humans [46] . Insectivorous bat diet analysis could, therefore, indicate the relative proportion of hematophagous insect fed upon by bats and their identity, in order to subsequently target these insect species for sampling.

The EBOV maintenance host could shed viable viruses in the environment where bats could get infected by environmental exposure. The most likely habitats where this can happen are tree canopies and holes, and cave roofs/walls used only by a fraction of hosts inhabiting forests. The probability of infection will be dependent on the capacity of the virus to survive in the environmental conditions available in the specific habitat. Therefore, a

better understanding of the capacity of EBOV to survive under different biotic and abiotic conditions is important to explore further (e.g., [71, 73]). These experimental approaches should consider the specific environmental conditions occurring in the tree canopy and cave roofs in terms of substrate, temperature, humidity and light properties.

One particular mechanism that has been put forward in the literature is the fruit-borne route concerning frugivorous bats in the tree canopy. The availability of fruits attracts fruit-eating animals, including birds, tree-dwelling mammals, and invertebrates. This behaviour can create a network of contacts between hosts, leading to several transmission pathways, and this interaction network can be denser during seasons with food resource limitations [23, 27] . Indirect contacts through faecal material, urine, or saliva left on fruits or branches could link the maintenance host with bats, in the same way that bats have been shown to be able to transmit other viruses (e.g., henipaviruses) through body fluids on fruit [33, 70, 83] . EBOV and filoviruses have been shown to persist for some time (3 to 7 days) in the environment, depending on the biotic and abiotic conditions [71] [72] [73] . In addition, EBV can be shed in some bat faeces [22] (but not all, [47]), and have been cultured from human urine and saliva [51] , hence, could also be transmitted from faeces, urine, and saliva from other species. This transmission route is therefore possible, but restrained to the fauna feeding at the same height as bats (or, technically, above). The hypothesis of fruits soiled with infected body fluids falling on the ground and opening a transmission pathway towards other ground-level foraging hosts (e.g., duikers) does not expose bats to the alternative maintenance hosts (e.g., [83]).

A relation between river systems and EBOV outbreaks has been suggested in Central Africa, with tributaries influencing the spatial distribution of cases [84] . If river systems can harbour specific biotic communities with potential alternative hosts, such as water-dependent vectors [46] , they can also represent, in remote forest ecosystems, the main transport pathways for people, providing a means for pathogens to spread through infected people or their hunted animals. Of course, in principle, while drinking, bats could get infected if the virus is present at the surface of the water. The capacity of EBOV to survive in the water has been the focus of a recent experimental study reporting an EBOV survival in water of 4 to 7 days between 21 and 27 °C [72] . Bats usually drink in open water, and not on the shores where viruses could be more concentrated by the presence of the maintenance host, for example. A dilution effect expected in open water, relative to some shallow water near the shores, would not favour such a transmission route *a priori*.

Tree and cave roosts could expose hanging and resting bats to direct contact with a potential maintenance host. However, as a first observation, the upside-down vertical position of bat roosting does not really favour disease transmission from an alternative host. For bat species roosting in tree-holes, the situation can be different as they can share temporally or directly their nest space with other animals [85] . Secondly, the density of bats roosting in caves prevents the presence of many other potential hosts in the cave roof (but, for example, snakes can predate on bats in caves). During their feeding behaviour, frugivorous bats could be in direct contact with other hosts attracted by the fruits. Their nocturnal habits will limit the diversity of host they can interact with. We are not aware of any extensive study on the network of potential contacts between bats and other animals during their roosting and feeding behaviour. The majority of studies investigated potential of infectious contact from bats to other organisms [53] . Novel technologies, such as camera traps equipped with nocturnal vision, could provide opportunities for more research on this topic.

As the ecology of most Africa bats is unknown, other opportunities exposing bat to potential maintenance hosts may be discovered in the future. For example, some bat species feed on fish [86] and, more recently, using stable isotopes of carbon and nitrogen as dietary tracers, it was demonstrated that a bat species, *Nyctalus lasiopterus*, was seasonally feeding on migrating Palearctic birds [87] , a feeding behaviour unknown until now. Failed predation on bats could also be a rare opportunity for infectious transmission [52] .

Considering the scenario B and C in Figure 1 , that bats are not the maintenance hosts of EBOV or that they are not the only host involved in the maintenance of EBOV, helps in focusing EBOV research protocols on a reduced range of potential transmission routes and potential alternative hosts interacting with bats in their specific and limited habitats. This means that if bats are not the maintenance hosts for EBOV, then there is only a limited number of candidate species to play the role of alternative maintenance hosts. This limited number of alternative maintenance hosts is defined by the ecology of bats that imposes on those alternative maintenance hosts only a few possible EBOV transmission pathways towards bats. From the biodiversity of African forest and the full web of interactions between species, a set of secondary hypotheses indicated in Table 1 can be tested through protocols presented to further investigate the role of different maintenance host

candidates for EBOV. The observation of this limited number of hosts calls for testing them, even if only to exclude them from the list of hypotheses and strengthen the main hypothesis. As warned above, the EBOV multi-host maintenance system could include a complex network of interacting bat species (Figure 1A2) and to proceed by elimination of alternative hypotheses may be a way to zoom-in on the maintenance community. The hypothesis of human playing a role in ebolavirus maintenance has not been addressed here, even if persistence of EBOV in previously infected humans has been recently proven [51] . This scenario would be more indicating of a change in the evolutionary trajectory of the pathogen (as moving from Step 4 to 5 in Figure 1 of Wolfe et al. [88]) than of the natural maintenance of ebolaviruses that is considered here.

In order for these protocols to be efficient and well designed, insights from behavioural ecology, plant phenology, and molecular biology (amongst other disciplines) will be necessary. Integrated approaches to health have been proposed recently and, in EBOV ecology, they should promote the integration of ecological sciences into health sciences that are usually at the forefront of epidemiological investigations. For example, a lot of sampling of potential alternative hosts has been undertaken during ebolaviruses outbreaks (e.g., [12, [21] [22] [23] [24] [25] [26]). These investigations concerned mainly the search for "what transmits ebolaviruses to people" as they were implemented during a human (or great ape) outbreak, and in the vicinity of outbreaks. This does not mean that they can automatically inform on "what maintains ebolaviruses". When looking for the maintenance host, investigations should also target the same and other alternative hosts during inter-outbreak periods with ecologically driven hypotheses. This is what is currently done for bats following the main maintenance hypothesis (e.g., [30]), but not often for alternative hosts. Experimental trials should also concentrate on the environmental conditions occurring in bat-specific habitats, which can be very different from human outbreak conditions.

The transmission routes towards bats represent interhost contacts of unknown intensity and frequency, and it would be difficult to compare their relative importance. However, one can prioritize some transmission routes based on the current knowledge. The insect food-borne and vector-borne routes of transmission need, surely, to be further investigated, as they can expose bats to numerous other hosts. Previous works on insects have mainly concentrated on sampling insects in the human outbreaks' surroundings (e.g., [26]). When searching for a maintenance host that can transmit EBOV to bats, protocols should concentrate on insects in interaction with known-exposed bat species. This would mean

combining bat behavioural ecology and arthropod capture protocols to detect their potential carriage of EBOV, as well as protocols exploring bat feeding habits (e.g., molecular detection of prey DNA in bat's guano) [65, 67] . For example, insect captures should be targeted where insects can bite bats, in caves or at canopy level, and not at ground level where bats may not occur. Studying host interaction networks at fruit feeding sites is also an interesting avenue to explore direct, environmental, and fruit-borne routes of transmission. Behavioural ecology could inform and help targeting protocols. Chimpanzees and monkeys can feed at the same height as bats. Some rodent species feed on fruits, but the selection of the arboricolous species feeding at the same height as bats can reduce the list drastically. Camera trap protocols could inform host interaction networks placing bat species in symmetric or asymmetric interactions with other potential alternative hosts.

Under field reality, and especially in rainforests, this list of protocols will need a carefully designed programme to be successful, rooted in interdisciplinarity. As bats, and especially those species that have been exposed to ebolaviruses, are the entry point of most of these alternative hypotheses (i.e., alternative host need to be in contact with bats), the behavioural and community ecology of targeted bat species will need to be locally understood. Data recorders, such as vector or camera traps, will need to be deployed where bats are currently roosting or feeding. This can be a difficult task. Understanding which feeding resources attract bats at a specific season requires a good understanding of indigenous and domesticated tree phenology (e.g., [89]). Prior to this work, a guano-based dietary analysis of the feeding behaviour of bats could help to map locally where and when bats will be present. Then, simultaneous protocols on bats and sympatric alternative hosts can be implemented, and a biological search for antibodies or antigens can be implemented. Combining protocols to test the main and alternative hypotheses could provide cost-effective and synergetic options.

To conclude, alternative hypotheses presented here should be explored alongside efforts to confirm bat species as maintenance hosts for EBOV. The ecology of those bat species already known to be exposed should be used to design protocols in order to target relevant alternative maintenance hosts. Given the number of species already involved/exposed to EBOV, the ecology of EBOV and its maintenance system can be expected to be complex, ecosystem dependent [46] , and dynamic, due to global changes [90] . The Ebola maintenance system, once isolated in the forests, is now interacting with humans and their

modified environments and will adapt to it. Aiming at this moving target will require out-of-the-box thinking and interdisciplinary collaboration.

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